

SECONDARY SCHOOL



Mathematics

For Class 10



R S Aggarwal

V Aggarwal

Secondary School Mathematics

FOR CLASS 10

(In accordance with the latest CBSE syllabus)

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Preface

We feel proud to present the revised edition of this well-acclaimed book, which is in accordance with the current Board examination pattern and includes the valuable feedback received from our esteemed readers.

Though revised, the book retains its qualities which have made it so popular among the teachers and students. It now has a large number of questions from the NCERT textbook and previous years CBSE board papers with full explanatory solutions. A large number of multiple-choice questions (MCQs) on all topics have also been included.

A separate exercise for Very-Short-Answer and Short-Answer Questions has also been given at the end of each chapter, to provide extra questions for ample practice.

We hope that the present edition will be of immense help to students who wish to sit for the CBSE class X board examination. We hope that we shall continue to receive invaluable feedback from teachers and students for the improvement of the book.

—Authors

Mathematics Syllabus

For Class 10

Unit I: Number Systems

1. Real Numbers (15 Periods)

Euclid's division lemma, Fundamental Theorem of Arithmetic—statements after reviewing work done earlier and after illustrating and motivating through examples, proofs of irrationality of $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. Decimal representation of rational numbers in terms of terminating/nonterminating recurring decimals.

Unit II: Algebra

1. Polynomials (7 Periods)

Zeros of a polynomial. Relationship between zeros and coefficients of quadratic polynomials. Statement and simple problems on division algorithm for polynomials with real coefficients.

2. Pair of Linear Equations in Two Variables (15 Periods)

Pair of linear equations in two variables and graphical method of their solution, consistency/inconsistency.

Algebraic conditions for number of solutions. Solution of a pair of linear equations in two variables algebraically—by substitution, by elimination and by cross multiplication method. Simple situational problems. Simple problems on equations reducible to linear equations.

3. Quadratic Equations (15 Periods)

Standard form of a quadratic equation $ax^2 + bx + c = 0$, ($a \neq 0$). Solutions of quadratic equations (only real roots) by factorisation, by completing the square and by using quadratic formula. Relationship between discriminant and nature of roots.

Situational problems based on quadratic equations related to day-to-day activities to be incorporated.

4. Arithmetic Progressions (8 Periods)

Motivation for studying Arithmetic Progression. Derivation of the n th term and sum of the first n terms of AP and their application in solving daily life problems.

Unit III: Coordinate Geometry

1. Lines (In two dimensions)

(14 Periods)

Review: Concepts of coordinate geometry, graphs of linear equations. Distance between two points. Section formula (internal division). Area of a triangle.

Unit IV: Geometry

1. Triangles

(15 Periods)

Definitions, examples, counterexamples of similar triangles.

1. **(Prove)** If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.
2. **(Motivate)** If a line divides two sides of a triangle in the same ratio, the line is parallel to the third side.
3. **(Motivate)** If in two triangles, the corresponding angles are equal, their corresponding sides are proportional and the triangles are similar.
4. **(Motivate)** If the corresponding sides of two triangles are proportional, their corresponding angles are equal and the two triangles are similar.
5. **(Motivate)** If one angle of a triangle is equal to one angle of another triangle and the sides including these angles are proportional, the two triangles are similar.
6. **(Motivate)** If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other.
7. **(Prove)** The ratio of the areas of two similar triangles is equal to the ratio of the squares on their corresponding sides.
8. **(Prove)** In a right triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.
9. **(Prove)** In a triangle, if the square on one side is equal to the sum of the squares on the other two sides, the angles opposite to the first side is a right angle.

2. Circles

(8 Periods)

Tangents to a circle at a point of contact.

1. **(Prove)** The tangent at any point of a circle is perpendicular to the radius through the point of contact.

2. **(Prove)** The lengths of tangents drawn from an external point to circle are equal.

3. Constructions (8 Periods)

1. Division of a line segment in a given ratio (internally).
2. Tangent to a circle from a point outside it.
3. Construction of a triangle similar to a given triangle.

Unit V: Trigonometry

1. Introduction to Trigonometry (10 Periods)

Trigonometric ratios of an acute angle of a right-angled triangle. Proof of their existence (well defined): motivate the ratios, whichever are defined at 0° and 90° . Values (with proofs) of the trigonometric ratios of 30° , 45° and 60° . Relationships between the ratios.

2. Trigonometric Identities (15 Periods)

Proof and applications of the identity $\sin^2 A + \cos^2 A = 1$. Only simple identities to be given. Trigonometric ratios of complementary angles.

3. Heights and Distances (8 Periods)

Simple problems on heights and distances. Problems should not involve more than two right triangles. Angles of elevation/depression should be only 30° , 45° and 60° .

Unit VI: Mensuration

1. Areas Related to Circles (12 Periods)

Motivate the area of a circle; area of sectors and segments of a circle. Problems based on areas and perimeter/circumference of the above-said plane figures. (In calculating area of segment of a circle, problems should be restricted to central angle of 60° , 90° and 120° only. Plane figures involving triangles, simple quadrilaterals and circle should be taken.)

2. Surface Areas and Volumes (12 Periods)

- (i) Surface areas and volumes of combinations of any two of the following: cubes, cuboids, spheres, hemispheres and right-circular cylinders/cones. Frustum of a cone.
- (ii) Problems involving converting one type of metallic solid into another and other mixed problems. (Problems with combination of not more than two different solids to be taken.)

Unit VII: Statistics and Probability

1. Statistics (18 Periods)

Mean, median and mode of grouped data (bimodal situations to be avoided). Cumulative frequency graph.

2. Probability (10 Periods)

Classical definition of probability. Simple problems on single events (not using set notation).



Weightage

MATHEMATICS

CLASS 10

Time: 3 Hours

Max. Marks: 80

The weightage or the distribution of marks over different dimensions of the question paper shall be as follows:

Weightage to Content/Subject Units

S. No.	Unit	Marks
1.	Number Systems • <i>Real Numbers</i>	6
2.	Algebra • <i>Polynomials</i> • <i>Linear Equations in Two Variables</i> • <i>Quadratic Equations</i> • <i>Arithmetic Progression</i>	20
3.	Coordinate Geometry • <i>Coordinate Geometry</i>	6
4.	Geometry • <i>Triangles</i> • <i>Circles</i> • <i>Constructions</i>	15
5.	Trigonometry • <i>Trigonometric Ratios</i> • <i>T-Ratios of Some Particular Angles</i> • <i>Trigonometric Identities</i> • <i>Trigonometric Ratios of Complementary Angles</i> • <i>Heights and Distances</i>	12
6.	Mensuration • <i>Perimeter and Area of Plane Figures</i> • <i>Area of Circle, Sector and Segment</i> • <i>Volume and Surface Areas of Solids</i>	10

7. Statistics and Probability	11
• <i>Mean, Median, Mode of Grouped Data, Cumulative Frequency Graph and Ogive</i> • <i>Probability</i>	
Total	80



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Real Numbers

In class IX we studied about real numbers, especially about irrational numbers. In this chapter, we shall continue our discussion on real numbers. We begin with two important results, namely Euclid's division lemma and the fundamental theorem of arithmetic.

LEMMA A lemma is a proven statement used for proving another statement.

EUCLID'S DIVISION LEMMA

For any two given positive integers a and b , there exist unique whole numbers q and r such that

$$a = bq + r, \text{ where } 0 \leq r < b. \quad [\text{CBSE 2009C}]$$

Here, we call a as *dividend*, b as *divisor*, q as *quotient* and r as *remainder*.

$$\text{Dividend} = (\text{divisor} \times \text{quotient}) + \text{remainder}.$$

Example Suppose we divide 117 by 14. Then, we get 8 as quotient and 5 as remainder.

Here *dividend* = 117, *divisor* = 14, *quotient* = 8 and *remainder* = 5.

Clearly, $117 = (14 \times 8) + 5$.

$$\begin{array}{r} 14 \overline{)117} (8 \\ \underline{-112} \\ 5 \end{array}$$

EXAMPLE 1 A number when divided by 73 gives 34 as quotient and 23 as remainder. Find the number.

SOLUTION Here divisor = 73, quotient = 34 and remainder = 23.

By Euclid's division lemma, we have

$$\begin{aligned} \text{dividend} &= (\text{divisor} \times \text{quotient}) + \text{remainder} \\ &= (73 \times 34) + 23 \\ &= (2482 + 23) = 2505. \end{aligned}$$

Hence, the required number is 2505.

ALGORITHM An algorithm is a series of well-defined steps which gives a method for solving a certain type of problem.

EUCLID'S DIVISION ALGORITHM It is a technique to compute the HCF of two given positive integers, say a and b with $a > b$, in the following steps.

Step 1. On dividing a by b , we get the quotient q and remainder r such that $a = bq + r$, where $0 \leq r < b$.

Step 2. If $r = 0$ then $\text{HCF}(a, b) = b$.

If $r \neq 0$ then apply the division lemma to b and r .

Step 3. Continue the process till the remainder is 0.

The last divisor will be the required HCF.

EXAMPLE 2 Use Euclid's algorithm to find the HCF of 272 and 1032.

SOLUTION We find $\text{HCF}(272, 1032)$ using the following steps.

Step 1. Since $1032 > 272$, we divide 1032 by 272 to get 3 as quotient and 216 as remainder.

So, by Euclid's division lemma, we get

$$1032 = 272 \times 3 + 216.$$

Step 2. Since the remainder $216 \neq 0$, we divide 272 by 216 to get 1 as quotient and 56 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$272 = 216 \times 1 + 56.$$

Step 3. Since the remainder $56 \neq 0$, we divide 216 by 56 to get 3 as quotient and 48 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$216 = 56 \times 3 + 48.$$

Step 4. Since the remainder $48 \neq 0$, we divide 56 by 48 to get 1 as quotient and 8 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$56 = 48 \times 1 + 8.$$

Step 5. Since the remainder $8 \neq 0$, we divide 48 by 8 to get 6 as quotient and 0 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$48 = 8 \times 6 + 0.$$

$$\begin{array}{r}
 272 \overline{)1032} 3 \\
 \underline{816} \\
 216 \overline{)272} 1 \\
 \underline{216} \\
 56 \overline{)216} 3 \\
 \underline{168} \\
 48 \overline{)56} 1 \\
 \underline{48} \\
 8 \overline{)48} 6 \\
 \underline{48} \\
 0
 \end{array}$$

The remainder has now become 0, so our procedure stops.

Hence, $\text{HCF}(272, 1032) = 8$.

Note $8 = \text{HCF}(48, 8) = \text{HCF}(56, 48) = \text{HCF}(216, 56) = \text{HCF}(272, 216) = \text{HCF}(1032, 272)$.

REMARK This method is also known as **successive division method**.

EXAMPLE 3 Use Euclid's algorithm to find $\text{HCF}(196, 38220)$.

SOLUTION We find $\text{HCF}(196, 38220)$, using the following steps.

Since $38220 > 196$, we divide 38220 by 196 to get 195 as quotient and 0 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$38220 = 196 \times 195 + 0.$$

Since the remainder is 0, so our procedure stops.

$$\therefore \text{HCF}(196, 38220) = 196.$$

$$\begin{array}{r} 196 \overline{)38220} (195 \\ \underline{196} \\ 1862 \\ \underline{1764} \\ 980 \\ \underline{980} \\ 0 \end{array}$$

EXAMPLE 4 Use Euclid's algorithm to find HCF of 1651 and 2032. Express the HCF in the form $1651m + 2032n$.

SOLUTION We find $\text{HCF}(1651, 2032)$ using the following steps.

Step 1. Since $2032 > 1651$, we divide 2032 by 1651 to get 1 as quotient and 381 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$2032 = 1651 \times 1 + 381. \quad \dots (i)$$

Step 2. Since the remainder $381 \neq 0$, we divide 1651 by 381 to get 4 as quotient and 127 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$1651 = 381 \times 4 + 127. \quad \dots (ii)$$

Step 3. Since the remainder $127 \neq 0$, we divide 381 by 127 to get 3 as quotient and 0 as remainder.

$\therefore$ by Euclid's division lemma, we get

$$381 = 127 \times 3 + 0. \quad \dots (iii)$$

The remainder is now 0, so our procedure stops.

$$\therefore \text{HCF}(1651, 2032) = 127.$$

$$\begin{array}{r} 1651 \overline{)2032} (1 \\ \underline{1651} \\ 381 \\ 381 \overline{)1651} (4 \\ \underline{1524} \\ 127 \\ 127 \overline{)381} (3 \\ \underline{381} \\ 0 \end{array}$$

Now, from (ii), we get

$$1651 = 381 \times 4 + 127$$

$$\Rightarrow 127 = 1651 - 381 \times 4$$

$$\Rightarrow 127 = 1651 - (2032 - 1651 \times 1) \times 4 \quad [\text{from (i)}]$$

$$\Rightarrow 127 = 1651 - 2032 \times 4 + 1651 \times 4$$

$$\Rightarrow 127 = 1651 \times 5 + 2032 \times (-4).$$

Hence, $m = 5$, $n = -4$.

SOME APPLICATIONS OF EUCLID'S DIVISION LEMMA

EXAMPLE 1 Show that every positive even integer is of the form $2m$ and that every positive odd integer is of the form $(2m + 1)$, where m is some integer.

SOLUTION Let n be an arbitrary positive integer.

On dividing n by 2, let m be the quotient and r be the remainder.

Then, by Euclid's division lemma, we have

$$n = 2m + r, \text{ where } 0 \leq r < 2.$$

$$\therefore n = 2m \text{ or } (2m + 1), \text{ for some integer } m.$$

Case I When $n = 2m$.

In this case, n is clearly even.

Case II When $n = 2m + 1$.

In this case, n is clearly odd.

Hence, every positive even integer is of the form $2m$ and every positive odd integer is of the form $(2m + 1)$ for some integer m .

EXAMPLE 2 Show that any positive integer is of the form $3m$ or $(3m + 1)$ or $(3m + 2)$ for some integer m .

SOLUTION Let n be an arbitrary positive integer.

On dividing n by 3, let m be the quotient and r be the remainder.

Then, by Euclid's division lemma, we have

$$n = 3m + r, \text{ where } 0 \leq r < 3.$$

$$\therefore n = 3m \text{ or } (3m + 1) \text{ or } (3m + 2), \text{ for some integer } m.$$

Thus, any positive integer is of the form $3m$ or $(3m + 1)$ or $(3m + 2)$ for some integer m .

EXAMPLE 3 Show that any positive odd integer is of the form $(4m + 1)$ or $(4m + 3)$ for some integer m .

SOLUTION Let n be an arbitrary odd positive integer.

On dividing n by 4, let m be the quotient and r be the remainder. So, by Euclid's division lemma, we have

$$n = 4m + r, \text{ where } 0 \leq r < 4.$$

$$\therefore n = 4m \text{ or } (4m + 1) \text{ or } (4m + 2) \text{ or } (4m + 3).$$

Clearly, $4m$ and $(4m + 2)$ are even and since n is odd, so $n \neq 4m$ and $n \neq (4m + 2)$.

$$\therefore n = (4m + 1) \text{ or } (4m + 3), \text{ for some integer } m.$$

Hence, any positive odd integer is of the form $(4m + 1)$ or $(4m + 3)$ for some integer m .

EXAMPLE 4 Show that every positive odd integer is of the form $(6m + 1)$ or $(6m + 3)$ or $(6m + 5)$ for some integer m .

SOLUTION Let n be a given positive odd integer.

On dividing n by 6, let m be the quotient and r be the remainder. Then, by Euclid's division lemma, we have

$$n = 6m + r, \text{ where } 0 \leq r < 6$$

$$\Rightarrow n = 6m + r, \text{ where } r = 0, 1, 2, 3, 4, 5$$

$$\Rightarrow n = 6m \text{ or } (6m + 1) \text{ or } (6m + 2) \text{ or } (6m + 3) \text{ or } (6m + 4) \text{ or } (6m + 5).$$

But, $n = 6m, (6m + 2), (6m + 4)$ give even values of n .

Thus, when n is odd, it is of the form $(6m + 1)$ or $(6m + 3)$ or $(6m + 5)$ for some integer m .

EXAMPLE 5 Using Euclid's division lemma, show that the square of any positive integer is either of the form $3m$ or $(3m + 1)$ for some integer m .

[CBSE 2008]

SOLUTION Let n be an arbitrary positive integer.

On dividing n by 3, let q be the quotient and r be the remainder. Then, by Euclid's division lemma, we have

$$n = 3q + r, \text{ where } 0 \leq r < 3.$$

$$\therefore n^2 = 9q^2 + r^2 + 6qr \quad \dots (i), \text{ where } 0 \leq r < 3.$$

Case I When $r = 0$.

Putting $r = 0$ in (i), we get

$$n^2 = 9q^2 = 3(3q^2) = 3m, \text{ where } m = 3q^2 \text{ is an integer.}$$

Case II When $r = 1$.

Putting $r = 1$ in (i), we get

$$\begin{aligned} n^2 &= (9q^2 + 1 + 6q) = 3(3q^2 + 2q) + 1 = 3(3q^2 + 2q) + 1 \\ &= 3m + 1, \text{ where } m = (3q^2 + 2q) \text{ is an integer.} \end{aligned}$$

Case III When $r = 2$.

Putting $r = 2$ in (i), we get

$$\begin{aligned} n^2 &= (9q^2 + 4 + 12q) = 3(3q^2 + 4q + 1) + 1 \\ &= 3m + 1, \text{ where } m = (3q^2 + 4q + 1) \text{ is an integer.} \end{aligned}$$

Hence, the square of any positive integer is of the form $3m$ or $(3m + 1)$ for some integer m .

EXAMPLE 6

Using Euclid's division lemma, show that the cube of any positive integer is of the form $9m$ or $(9m + 1)$ or $(9m + 8)$ for some integer m .

[CBSE 2009C]

SOLUTION

Let n be an arbitrary positive integer.

On dividing n by 3, let q be the quotient and r be the remainder. So, by Euclid's division lemma, we have

$$n = 3q + r, \text{ where } 0 \leq r < 3.$$

$$\begin{aligned} \therefore n^3 &= (3q + r)^3 = 27q^3 + r^3 + 9qr(3q + r) \\ &= (27q^3 + 27q^2r + 9qr^2) + r^3 \quad \dots \text{(i), where } 0 \leq r < 3. \end{aligned}$$

Case I When $r = 0$.

Putting $r = 0$ in (i), we get

$$n^3 = 27q^3 = 9(3q^3) = 9m, \text{ where } m = 3q^3 \text{ is an integer.}$$

Case II When $r = 1$.

Putting $r = 1$ in (i), we get

$$\begin{aligned} n^3 &= (27q^3 + 27q^2 + 9q) + 1 = 9q(3q^2 + 3q + 1) + 1 \\ &= 9m + 1, \text{ where } m = q(3q^2 + 3q + 1) \text{ is an integer.} \end{aligned}$$

Case III When $r = 2$.

Putting $r = 2$ in (i), we get

$$\begin{aligned} n^3 &= (27q^3 + 54q^2 + 36q) + 8 = 9q(3q^2 + 6q + 4) + 8 \\ &= 9m + 8, \text{ where } m = q(3q^2 + 6q + 4) \text{ is an integer.} \end{aligned}$$

Hence, the cube of any positive integer is of the form $9m$ or $(9m + 1)$ or $(9m + 8)$ for some integer m .

EXAMPLE 7 Show that one and only one out of n , $(n + 1)$ and $(n + 2)$ is divisible by 3, where n is any positive integer.

SOLUTION On dividing n by 3, let q be the quotient and r be the remainder.

Then, $n = 3q + r$, where $0 \leq r < 3$

$\Rightarrow n = 3q + r$, where $r = 0, 1$ or 2

$\Rightarrow n = 3q$ or $n = (3q + 1)$ or $n = (3q + 2)$.

Case I If $n = 3q$ then n is clearly divisible by 3.

Case II If $n = (3q + 1)$ then $(n + 2) = (3q + 3) = 3(q + 1)$, which is clearly divisible by 3.

In this case, $(n + 2)$ is divisible by 3.

Case III If $n = (3q + 2)$ then $(n + 1) = (3q + 3) = 3(q + 1)$, which is clearly divisible by 3.

In this case, $(n + 1)$ is divisible by 3.

Hence, one and only one out of n , $(n + 1)$ and $(n + 2)$ is divisible by 3.

EXAMPLE 8 Show that one and only one out of n , $n + 2$, $n + 4$ is divisible by 3, where n is any positive integer. [CBSE 2008C]

SOLUTION On dividing n by 3, let q be the quotient and r be the remainder.

Then, $n = 3q + r$, where $0 \leq r < 3$

$\Rightarrow n = 3q + r$, where $r = 0, 1, 2$

$\Rightarrow n = 3q$ or $n = 3q + 1$ or $n = 3q + 2$.

Case I If $n = 3q$ then n is divisible by 3.

Case II If $n = 3q + 1$ then $(n + 2) = 3q + 3 = 3(q + 1)$, which is divisible by 3.

So, in this case, $(n + 2)$ is divisible by 3.

Case III When $n = 3q + 2$ then $(n + 4) = 3q + 6 = 3(q + 2)$, which is divisible by 3.

So, in this case, $(n + 4)$ is divisible by 3.

Hence, one and only one out of n , $n + 2$, $n + 4$ is divisible by 3.

EXAMPLE 9 If n is an odd integer then show that $n^2 - 1$ is divisible by 8.

SOLUTION We know that every odd integer is of the form $4q + 1$ or $4q + 3$ for some integer q .

Case I When $n = 4q + 1$.

$$\begin{aligned}\text{Then, } n^2 - 1 &= (4q + 1)^2 - 1 \\ &= 16q^2 + 8q + 1 - 1 \\ &= 16q^2 + 8q \\ &= 8(q + 2q^2), \text{ which is divisible by 8.}\end{aligned}$$

Case II When $n = 4q + 3$.

$$\begin{aligned}\text{Then, } n^2 - 1 &= (4q + 3)^2 - 1 \\ &= 16q^2 + 24q + 9 - 1 \\ &= 16q^2 + 24q + 8 \\ &= 8(2q^2 + 3q + 1), \text{ which is divisible by 8.}\end{aligned}$$

Hence, if n is an odd integer then $n^2 - 1$ is divisible by 8.

EXERCISE 1A

1. What do you mean by Euclid's division lemma?
2. A number when divided by 61 gives 27 as quotient and 32 as remainder. Find the number.
3. By what number should 1365 be divided to get 31 as quotient and 32 as remainder?
4. Using Euclid's division algorithm, find the HCF of
(i) 405 and 2520 (ii) 504 and 1188 (iii) 960 and 1575.
5. Show that every positive integer is either even or odd.
6. Show that any positive odd integer is of the form $(6m + 1)$ or $(6m + 3)$ or $(6m + 5)$, where m is some integer.
7. Show that any positive odd integer is of the form $(4m + 1)$ or $(4m + 3)$, where m is some integer.
8. For any positive integer n , prove that $n^3 - n$ is divisible by 6.
9. Prove that if x and y are both odd positive integers then $x^2 + y^2$ is even but not divisible by 4.
10. Use Euclid's algorithm to find HCF of 1190 and 1445. Express the HCF in the form $1190m + 1445n$.

ANSWERS (EXERCISE 1A)

2. 1679 3. 43 4. (i) 45 (ii) 36 (iii) 15
10. 85; $m = -6$, $n = 5$

HINTS TO SOME SELECTED QUESTIONS

3. Let the required divisor be x .

Then, dividend = (divisor $\times$ quotient) + remainder

$$\Rightarrow 1365 = (x \times 31) + 32. \text{ Find } x.$$

5. Let n be an arbitrary positive integer.

On dividing n by 2, let m be the quotient and r be the remainder.

Then, $n = 2m + r$, where $0 \leq r < 2$ [by Euclid's division lemma].

$$\therefore n = 2m \text{ or } n = 2m + 1 \text{ for some integer } m.$$

So, n is either even or odd.

FUNDAMENTAL THEOREM OF ARITHMETIC *Every composite number can be uniquely expressed as a product of primes, except for the order in which these prime factors occurs.*

Examples (i) $12 = 2 \times 2 \times 3$ (ii) $69 = 3 \times 23$ (iii) $105 = 3 \times 5 \times 7$

(iv) $234 = 2 \times 3 \times 3 \times 13$ (v) $462 = 2 \times 3 \times 7 \times 11$

(vi) $651 = 3 \times 7 \times 31$

The above factorisation can easily be verified by actual division.

SOLVED EXAMPLES

EXAMPLE 1 Show that each of the following is a composite number:

(i) $5 \times 11 \times 13 + 13$

(ii) $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

SOLUTION We have

$$(i) 5 \times 11 \times 13 + 13 = 13 \times (5 \times 11 + 1) = (13 \times 56).$$

Clearly, it shows that the given number has more than two factors. Hence, it is a composite number.

$$(ii) 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 = 5 \times (6 \times 4 \times 3 \times 2 \times 1 + 1) \\ = (5 \times 145).$$

Clearly, it shows that the given number has more than two factors. Hence, it is a composite number.

EXAMPLE 2 Show that any number of the form 4^n , $n \in N$ can never end with the digit 0.

SOLUTION If 4^n ends with 0 then it must have 5 as a factor.

But, $4^n = (2^2)^n = 2^{2n}$ shows that 2 is the only prime factor of 4^n .

Also, we know from the fundamental theorem of arithmetic that the prime factorisation of each number is unique.

So, 5 is not a factor of 4^n .

Hence, 4^n can never end with the digit 0.

EXAMPLE 3 Show that any number of the form 6^n , where $n \in \mathbb{N}$ can never end with the digit 0.

SOLUTION If 6^n ends with 0 then it must have 5 as a factor.

But, $6^n = (2 \times 3)^n = (2^n \times 3^n)$ shows that 2 and 3 are the only prime factors of 6^n .

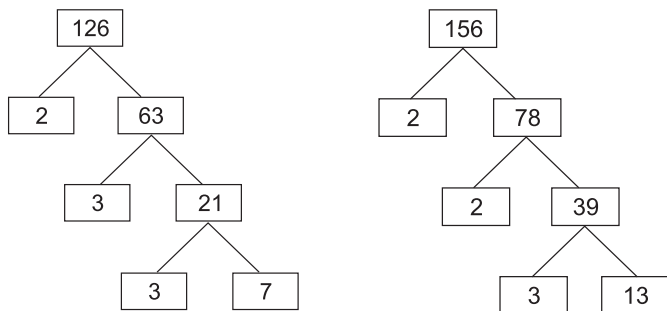
Also, we know from the fundamental theorem of arithmetic that the prime factorisation of each number is unique.

So, 5 is not a factor of 6^n .

Hence, 6^n can never end with the digit 0.

EXAMPLE 4 Find the HCF and LCM of 126 and 156 using prime factorisation method.

SOLUTION We have



$$\therefore 126 = (2 \times 3 \times 3 \times 7) = (2 \times 3^2 \times 7)$$

$$\text{and } 156 = (2 \times 2 \times 3 \times 13) = (2^2 \times 3 \times 13).$$

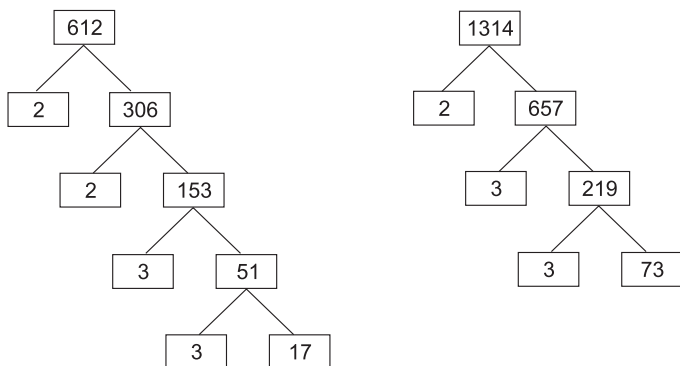
$$\begin{aligned} \therefore \text{HCF}(126, 156) &= \text{product of common terms with lowest power} \\ &= (2^1 \times 3^1) = (2 \times 3) = 6 \end{aligned}$$

and $\text{LCM}(126, 156) = \text{product of prime factors with highest power}$
 $= (2^2 \times 3^2 \times 7 \times 13) = (4 \times 9 \times 7 \times 13)$
 $= 3276.$

$\therefore \text{HCF} = 6$ and $\text{LCM} = 3276.$

EXAMPLE 5 Find the HCF and LCM of 612 and 1314 using prime factorisation method.

SOLUTION We have



$$\therefore 612 = (2 \times 2 \times 3 \times 3 \times 17) = (2^2 \times 3^2 \times 17)$$

$$\text{and } 1314 = (2 \times 3 \times 3 \times 73) = (2 \times 3^2 \times 73).$$

$$\therefore \text{HCF}(612, 1314) = \text{product of common terms with lowest power}$$

$$= (2 \times 3^2) = (2 \times 9) = 18$$

$$\text{and } \text{LCM}(612, 1314) = \text{product of prime factors with highest power}$$

$$= (2^2 \times 3^2 \times 17 \times 73) = (4 \times 9 \times 17 \times 73)$$

$$= 44676.$$

Hence, $\text{HCF} = 18$ and $\text{LCM} = 44676.$

AN IMPORTANT PROPERTY

Product of two given numbers = product of their HCF and LCM.

Thus, $(a \times b) = \text{HCF}(a, b) \times \text{LCM}(a, b).$

CAUTION The above result is true for two numbers only.

EXAMPLE 6 The HCF of two numbers is 23 and their LCM is 1449. If one of the numbers is 161, find the other.

SOLUTION For two numbers a and b , we know that

$$(a \times b) = \{\text{HCF}(a, b)\} \times \{\text{LCM}(a, b)\}.$$

Here $a = 161$, $\text{HCF} = 23$ and $\text{LCM} = 1449$.

And, we have to find b .

$$\therefore (161 \times b) = (23 \times 1449) \Rightarrow b = \frac{(23 \times 1449)}{161} = 207.$$

Hence, the other number is 207.

EXAMPLE 7 Given that $\text{HCF}(252, 594) = 18$, find $\text{LCM}(252, 594)$.

SOLUTION We have

$$\begin{aligned} \text{LCM} &= \frac{\text{product of two given numbers}}{\text{their HCF}} \\ &= \frac{(252 \times 594)}{18} = 8316. \end{aligned}$$

Hence, $\text{LCM}(252, 594) = 8316$.

EXAMPLE 8 Find the simplest form of $\frac{148}{185}$.

SOLUTION We find $\text{HCF}(148, 185)$, which is 37.

So, we divide the numerator and denominator of the given fraction by 37.

$$\therefore \frac{148}{185} = \frac{148 \div 37}{185 \div 37} = \frac{4}{5}.$$

Hence, the simplest form of the given fraction is $\frac{4}{5}$.

EXAMPLE 9 Find the HCF and LCM of 108, 120 and 252 using prime factorisation method.

SOLUTION By prime factorisation, we get

$$108 = (2^2 \times 3^3)$$

$$120 = (2^3 \times 3 \times 5)$$

$$252 = (2^2 \times 3^2 \times 7)$$

2	108	2	120	2	252
2	54	2	60	2	126
3	27	2	30	3	63
3	9	3	15	3	21
	3		5		7

$\text{HCF}(108, 120, 252) = \text{product of common terms with lowest power}$

$$= (2^2 \times 3) = (4 \times 3) = 12.$$

$$\begin{aligned}\text{LCM}(108, 120, 252) &= \text{product of prime factors with highest power} \\ &= (2^3 \times 3^3 \times 5 \times 7) = 7560.\end{aligned}$$

$$\therefore \text{HCF} = 12 \text{ and } \text{LCM} = 7560.$$

EXAMPLE 10 Find the largest number which divides 245 and 1037, leaving remainder 5 in each case.

SOLUTION Clearly, the required number divides $(245 - 5) = 240$ and $(1037 - 5) = 1032$ exactly.

So, the required number is $\text{HCF}(240, 1032)$.

$$\text{Now, } 240 = (2^4 \times 3 \times 5)$$

$$\text{and } 1032 = (2^3 \times 3 \times 43).$$

$$\therefore \text{HCF}(240, 1032) = (2^3 \times 3) = 24.$$

Hence, the required number is 24.

2 240	2 1032
2 120	2 516
2 60	2 258
2 30	3 129
3 15	43
5	

EXAMPLE 11 Find the largest number which divides 129 and 545, leaving remainders 3 and 5 respectively.

SOLUTION Clearly, the required number divides $(129 - 3) = 126$ and $(545 - 5) = 540$ exactly.

$$\therefore \text{required number} = \text{HCF}(126, 540).$$

$$\text{Now, } 126 = (2 \times 3 \times 3 \times 7) = (2 \times 3^2 \times 7)$$

$$\text{and } 540 = (2 \times 2 \times 3 \times 3 \times 3 \times 5)$$

$$= (2^2 \times 3^3 \times 5).$$

$$\therefore \text{HCF}(126, 540) = \text{product of common terms with lowest power}$$

$$= (2 \times 3^2) = (2 \times 9) = 18.$$

Hence, the required number is 18.

EXAMPLE 12 Find the largest number that will divide 398, 436 and 542, leaving remainders 7, 11 and 15 respectively.

SOLUTION Clearly, the required number divides $(398 - 7) = 391$, $(436 - 11) = 425$ and $(542 - 15) = 527$ exactly.

$$\therefore \text{required number} = \text{HCF}(391, 425, 527).$$

$$\text{Now, } 391 = (17 \times 23),$$

$$425 = (5^2 \times 17),$$

$$527 = (17 \times 31).$$

17 391	5 425
23	5 85
	17

$$\therefore \text{HCF}(391, 425, 527) = 17.$$

Hence, the required number is 17.

EXAMPLE 13 Two tanks contain 504 and 735 litres of milk respectively. Find the maximum capacity of a container which can measure the milk of either tank an exact number of times.

SOLUTION Resolving 504 and 735 into prime factors, we get

$$504 = (2^3 \times 3^2 \times 7)$$

$$\text{and } 735 = (5 \times 3 \times 7^2).$$

$$\therefore \text{HCF}(504, 735) = (3 \times 7) = 21.$$

$$\therefore \text{capacity of the required container} = 21 \text{ litres.}$$

$$\begin{array}{r|l} 2 & 504 \\ \hline 2 & 252 \\ \hline 2 & 126 \\ \hline 3 & 63 \\ \hline 3 & 21 \\ \hline & 7 \end{array} \quad \begin{array}{r|l} 5 & 735 \\ \hline 3 & 147 \\ \hline 7 & 49 \\ \hline & 7 \end{array}$$

EXAMPLE 14 An army contingent of 612 members is to march behind an army band of 48 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march?

SOLUTION Clearly, the maximum number of columns = HCF(612, 48).

$$\text{Now, } 612 = (2^2 \times 3^2 \times 17)$$

$$\text{and } 48 = (2^4 \times 3).$$

$$\therefore \text{HCF}(612, 48) = (2^2 \times 3) = (4 \times 3) = 12.$$

$$\therefore \text{required number of columns} = 12.$$

$$\begin{array}{r|l} 2 & 612 \\ \hline 2 & 306 \\ \hline 3 & 153 \\ \hline 3 & 51 \\ \hline & 17 \end{array} \quad \begin{array}{r|l} 2 & 48 \\ \hline 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline & 3 \end{array}$$

EXAMPLE 15 A sweetseller has 420 kaju burfis and 150 badam burfis. He wants to stack them in such a way that each stack has the same number, and they take up the least area of the tray. How many of these can be placed in each stack? How many stacks are formed?

SOLUTION Maximum number of burfis in each stack = HCF(420, 150).

$$\text{Now, } 420 = (2^2 \times 3 \times 5 \times 7)$$

$$\text{and } 150 = (2 \times 3 \times 5^2).$$

$$\therefore \text{HCF}(420, 150) = (2 \times 3 \times 5) = 30.$$

$$\therefore \text{maximum number of burfis in each stack} = 30.$$

$$\therefore \text{number of stacks} = \left(\frac{420}{30} + \frac{150}{30} \right) = (14 + 5) = 19.$$

$$\begin{array}{r|l} 2 & 420 \\ \hline 2 & 210 \\ \hline 3 & 105 \\ \hline 5 & 35 \\ \hline & 7 \end{array} \quad \begin{array}{r|l} 2 & 150 \\ \hline 3 & 75 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

EXAMPLE 16 *Ravi and Sikha drive around a circular sports field. Ravi takes 16 minutes to take one round, while Sikha completes the round in 20 minutes. If both start at the same point, at the same time and go in the same direction, after how much time will they meet at the starting point?*

SOLUTION Required number of minutes = LCM(16, 20).

Now, $16 = 2^4$ and $20 = (2^2 \times 5)$.

$$\therefore \text{LCM}(16, 20) = (2^4 \times 5) = (16 \times 5) = 80.$$

Hence, both will meet at the starting point after 80 minutes.

EXAMPLE 17 *In a school there are two sections, namely A and B, of class X. There are 30 students in section A and 28 students in section B. Find the minimum number of books required for their class library so that they can be distributed equally among students of section A or section B.*

SOLUTION Clearly, the required number of books are to be distributed equally among the students of section A or B.

So, the number of these books must be a multiple of 30 as well as that of 28.

Consequently, the required number is LCM(30, 28).

Now, $30 = 2 \times 3 \times 5$

and $28 = 2^2 \times 7$.

$$\begin{aligned}\therefore \text{LCM}(30, 28) &= \text{product of prime factors with highest power} \\ &= (2^2 \times 3 \times 5 \times 7) = (4 \times 3 \times 5 \times 7) = 420.\end{aligned}$$

Hence, the required number of books = 420.

EXERCISE 1B

1. Using prime factorisation, find the HCF and LCM of:

- | | | |
|---------------|---------------|-----------------|
| (i) 36, 84 | (ii) 23, 31 | (iii) 96, 404 |
| (iv) 144, 198 | (v) 396, 1080 | (vi) 1152, 1664 |

In each case, verify that:

$$\text{HCF} \times \text{LCM} = \text{product of given numbers.}$$

2. Using prime factorisation, find the HCF and LCM of:

- | | | |
|-----------------|-----------------|---------------------|
| (i) 8, 9, 25 | (ii) 12, 15, 21 | (iii) 17, 23, 29 |
| (iv) 24, 36, 40 | (v) 30, 72, 432 | (vi) 21, 28, 36, 45 |

3. The HCF of two numbers is 23 and their LCM is 1449. If one of the numbers is 161, find the other.
4. The HCF of two numbers is 145 and their LCM is 2175. If one of the numbers is 725, find the other.
5. The HCF of two numbers is 18 and their product is 12960. Find their LCM.
6. Is it possible to have two numbers whose HCF is 18 and LCM is 760? Give reason.

HINT HCF always divides LCM completely.

7. Find the simplest form of:

(i) $\frac{69}{92}$

(ii) $\frac{473}{645}$

(iii) $\frac{1095}{1168}$

(iv) $\frac{368}{496}$

8. Find the largest number which divides 438 and 606, leaving remainder 6 in each case.
9. Find the largest number which divides 320 and 457, leaving remainders 5 and 7 respectively.
10. Find the least number which when divided by 35, 56 and 91 leaves the same remainder 7 in each case.
11. Find the smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively.
12. Find the smallest number which when increased by 17 is exactly divisible by both 468 and 520.
13. Find the greatest number of four digits which is exactly divisible by 15, 24 and 36.
14. Find the largest four-digits number which when divided by 4, 7 and 13 leaves a remainder of 3 in each case.
15. Find the least number which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4 and 3.
16. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.
17. Find the least number which when divided by 20, 25, 35 and 40 leaves remainders 14, 19, 29 and 34 respectively.
18. In a seminar, the number of participants in Hindi, English and mathematics are 60, 84 and 108 respectively. Find the minimum number of rooms required, if in each room, the same number of participants are to be seated and all of them being in the same subject.

19. Three sets of English, mathematics and science books containing 336, 240 and 96 books respectively have to be stacked in such a way that all the books are stored subjectwise and the height of each stack is the same. How many stacks will be there?
20. Three pieces of timber 42 m, 49 m and 63 m long have to be divided into planks of the same length. What is the greatest possible length of each plank? How many planks are formed?
21. Find the greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm and 12 m 95 cm.
22. Find the maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets the same number of pens and the same number of pencils.
23. Find the least number of square tiles required to pave the ceiling of a room 15 m 17 cm long and 9 m 2 cm broad.
24. Three measuring rods are 64 cm, 80 cm and 96 cm in length. Find the least length of cloth that can be measured an exact number of times, using any of the rods.
25. An electronic device makes a beep after every 60 seconds. Another device makes a beep after every 62 seconds. They beeped together at 10 a.m. At what time will they beep together at the earliest?
26. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they all change simultaneously at 8 a.m. then at what time will they again change simultaneously?
27. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10, 12 minutes respectively. In 30 hours, how many times do they toll together?

ANSWERS (EXERCISE 1B)

1. (i) HCF = 12, LCM = 252 (ii) HCF = 1, LCM = 713
(iii) HCF = 4, LCM = 9696 (iv) HCF = 18, LCM = 1584
(v) HCF = 36, LCM = 11880 (vi) HCF = 128, LCM = 14976
2. (i) HCF = 1, LCM = 1800 (ii) HCF = 3, LCM = 420
(iii) HCF = 1, LCM = 11339 (iv) HCF = 4, LCM = 360
(v) HCF = 6, LCM = 2160 (vi) HCF = 1, LCM = 1260
3. 207 4. 435 5. 720

6. No, since HCF does not divide LCM exactly
7. (i) $\frac{3}{4}$ (ii) $\frac{11}{15}$ (iii) $\frac{15}{16}$ (iv) $\frac{23}{31}$
8. 24 9. 45 10. 3647 11. 204 12. 4663 13. 9720
14. 9831 15. 23 16. 4 17. 1394 18. 21 19. 14
20. 7 m, 22 planks 21. 35 cm 22. 91 23. 814 24. 9.6 m
25. 10 : 31 hrs 26. 8 : 7 : 12 hrs 27. 16 times

HINTS TO SOME SELECTED QUESTIONS

8. Required number = HCF(432, 600) = 24.
9. Required number = HCF(315, 450) = 45.
10. Required number = {LCM(35, 56, 91) + 7} = (3640 + 7) = 3647.
11. Here, (28 - 8) = 20 and (32 - 12) = 20.
 $\therefore$ required number = {LCM(28, 32) - 20} = (224 - 20) = 204.
12. Required number = {LCM(468, 520) - 17} = (4680 - 17) = 4663.
13. Greatest number of four digits = 9999.
 LCM(15, 24, 36) = 360.
 On dividing 9999 by 360, remainder = 279.
 $\therefore$ required number = (9999 - 279) = 9720.
14. Required number = (greatest number of 4-digits divisible by 4, 7 and 13) + 3.
15. LCM of 5, 6, 4 and 3 = 60.
 On dividing 2497 by 60, remainder = 37.
 $\therefore$ number to be added = (60 - 37) = 23.
16. Required number = HCF of (91 - 43), (183 - 91) and (183 - 43), i.e., HCF of 48, 92 and 140.
17. Here, (20 - 14) = 6, (25 - 19) = 6, (35 - 29) = 6, (40 - 34) = 6.
 $\therefore$ required number = (LCM of 20, 25, 35, 40) - 6.
18. Maximum number of participants in each room = HCF(60, 84, 108) = 12.
 Minimum number of rooms required = $\left(\frac{60}{12} + \frac{84}{12} + \frac{108}{12}\right) = 21$.
19. HCF(336, 240, 96) = 48. So, we make stacks of 48 books each.
 Number of stacks = $\left(\frac{336}{48} + \frac{240}{48} + \frac{96}{48}\right) = (7 + 5 + 2) = 14$.
20. Required length of each plank in metres = HCF(42 m, 49 m, 63 m) = 7 m.
 Number of planks = $\left(\frac{42}{7} + \frac{49}{7} + \frac{63}{7}\right) = (6 + 7 + 9) = 22$.
21. Required length = HCF(700 cm, 385 cm, 1295 cm) = 35 cm.

22. Maximum number of students = $\text{HCF}(1001, 910) = 91$.

23. Side of each square tile = $\text{HCF}(1517 \text{ cm}, 902 \text{ cm}) = 41 \text{ cm}$.

$$\text{Required number of tiles} = \left(\frac{1517 \times 902}{41 \times 41} \right) = 814.$$

24. Required length = $\text{LCM}(64 \text{ cm}, 80 \text{ cm}, 96 \text{ cm}) = 960 \text{ cm} = 9.6 \text{ m}$.

25. Interval of beeping together = $\text{LCM}(60 \text{ seconds}, 62 \text{ seconds}) = 1860 \text{ seconds}$
 $= \frac{1860}{60} \text{ min} = 31 \text{ min}.$

So, they will beep together again at 10 : 31 a.m.

26. Interval of change = $\text{LCM}(48 \text{ seconds}, 72 \text{ seconds}, 108 \text{ seconds}) = 432 \text{ seconds}$
 $= 7 \text{ min } 12 \text{ seconds}.$

Required time of simultaneous change = 8 : 7 : 12 hours.

27. LCM of 2, 4, 6, 8, 10, 12 = 120.

After every 2 hours they toll together.

$$\text{Required number of times} = \left(\frac{30}{2} + 1 \right) \text{ times} = 16 \text{ times}.$$

RATIONAL NUMBERS

NATURAL NUMBERS *Counting numbers 1, 2, 3, 4, ..., etc., are known as natural numbers.*

WHOLE NUMBERS *All counting numbers together with 0 form the collection of whole numbers.*

Thus, 0, 1, 2, 3, 4, 5, ..., etc., are whole numbers.

INTEGERS *All counting numbers, negatives of counting numbers and 0 form the collection of all integers.*

Thus, ..., -4, -3, -2, -1, 0, 1, 2, 3, ..., etc., are integers.

RATIONAL NUMBERS *The numbers of the form $\frac{p}{q}$, where p and q are integers, and $q \neq 0$ are called rational numbers.*

RATIONAL NUMBERS IN DECIMAL FORM *Every rational number when expressed in decimal form is expressible either in terminating or in nonterminating repeating decimal form.*

AN IMPORTANT OBSERVATION

To Test Whether a Given Rational Number is a Terminating or Repeating Decimal

Let x be a rational number whose simplest form is $\frac{p}{q}$, where p and q are integers and $q \neq 0$. Then,

(i) x is a terminating decimal only when q is of the form $(2^m \times 5^n)$ for some non-negative integers m and n .

(ii) x is a nonterminating repeating decimal, if $q \neq (2^m \times 5^n)$.

AN IMPORTANT TEST

Let $\frac{p}{q}$ be the simplest form of a given rational number.

(i) If $q = (2^m \times 5^n)$ for some non-negative integers m and n then $\frac{p}{q}$ is a *terminating decimal*.

(ii) If $q \neq (2^m \times 5^n)$ then $\frac{p}{q}$ is a *nonterminating repeating decimal*.

SOLVED EXAMPLES

EXAMPLE 1 Without actual division, show that each of the following rational numbers is a terminating decimal. Express each in decimal form.

(i) $\frac{31}{(2^2 \times 5^3)}$ (ii) $\frac{33}{50}$ (iii) $\frac{41}{1000}$ (iv) $\frac{17}{625}$

SOLUTION (i) The given number is $\frac{31}{(2^2 \times 5^3)}$.

Clearly, none of 2 and 5 is a factor of 31.

So, the given rational is in its simplest form.

Clearly, $(2^2 \times 5^3)$ is of the form $(2^m \times 5^n)$.

So, the given number is a terminating decimal.

$$\begin{aligned}\text{Now, } \frac{31}{(2^2 \times 5^3)} &= \frac{31 \times 2}{(2^3 \times 5^3)} = \frac{62}{(2 \times 5)^3} = \frac{62}{(10)^3} = \frac{62}{1000} \\ &= 0.062.\end{aligned}$$

(ii) The given number is $\frac{33}{50}$.

Now, $50 = (2 \times 5^2)$ and none of 2 and 5 is a factor of 33.

So, the given rational number is in its simplest form.

Clearly, $50 = (2 \times 5^2) = (2^m \times 5^n)$, where $m = 1$ and $n = 2$.

So, the given number is a terminating decimal.

$$\begin{aligned}\text{Now, } \frac{33}{50} &= \frac{33}{(2 \times 5^2)} = \frac{33 \times 2}{(2^2 \times 5^2)} = \frac{66}{(2 \times 5)^2} \\ &= \frac{66}{(10)^2} = \frac{66}{100} = 0.66.\end{aligned}$$

(iii) The given number is $\frac{41}{1000}$.

$$\text{Now, } 1000 = (8 \times 125) = (2^3 \times 5^3).$$

Clearly, none of 2 and 5 is a factor of 41.

So, the given number is in its simplest form.

Now, $1000 = (2^3 \times 5^3)$ which is of the form $(2^m \times 5^n)$, where $m = 3$ and $n = 3$.

So, the given number is a terminating decimal.

$$\text{And, } \frac{41}{1000} = 0.041.$$

(iv) The given number is $\frac{17}{625}$.

$$\text{And, } 625 = 5^4 \text{ and } 5 \text{ is not a factor of } 17.$$

So, the given number is in its simplest form.

Now, $625 = 5^4$ is of the form $(2^m \times 5^n)$, where $m = 0, n = 4$.

So, the given number is a terminating decimal.

$$\begin{aligned} \text{Now, } \frac{17}{625} &= \frac{17}{5^4} = \frac{17 \times 2^4}{5^4 \times 2^4} = \frac{17 \times 16}{(5 \times 2)^4} = \frac{272}{(10)^4} \\ &= \frac{272}{10000} = 0.0272. \end{aligned}$$

EXAMPLE 2

Without actual division, show that each of the following rational numbers is a nonterminating repeating decimal.

$$(i) \frac{121}{(2^3 \times 3^2 \times 7^5)} \quad (ii) \frac{17}{90} \quad (iii) \frac{53}{343} \quad (iv) \frac{66}{180}$$

SOLUTION

$$(i) \text{ Given number is } \frac{121}{(2^3 \times 3^2 \times 7^5)}.$$

Clearly, none of 2, 3 and 7 is a factor of 121.

So, the given rational number is in its simplest form.

$$\text{And, } (2^3 \times 3^2 \times 7^5) \neq (2^m \times 5^n).$$

$\therefore \frac{121}{(2^3 \times 3^2 \times 7^5)}$ is a nonterminating repeating decimal.

$$(ii) \text{ Given number is } \frac{17}{90}.$$

$$\text{And, } 90 = (2 \times 3^2 \times 5).$$

Clearly, none of 2, 3 and 5 is a factor of 17.

$\therefore \frac{17}{90}$ is in its simplest form.

Also, $90 = (2 \times 3^2 \times 5) \neq (2^m \times 5^n)$.

$\therefore \frac{17}{90}$ is a nonterminating repeating decimal.

(iii) Given number is $\frac{53}{343}$.

Now, $343 = 7^3$ and 7 is not a factor of 53.

$\therefore \frac{53}{343}$ is in its simplest form.

Also, $343 = 7^3 \neq (2^m \times 5^n)$.

$\therefore \frac{53}{343}$ is a nonterminating repeating decimal.

(iv) Given number is $\frac{66}{180}$ and $\text{HCF}(66, 180) = 6$.

$\therefore \frac{66}{180} = \frac{66 \div 6}{180 \div 6} = \frac{11}{30}$.

Now, $30 = (2 \times 3 \times 5)$ and none of 2, 3, 5 is a factor of 11.

$\therefore \frac{11}{30}$ is in its simplest form.

Also, $30 = (2 \times 3 \times 5) \neq (2^m \times 5^n)$.

$\therefore \frac{11}{30}$ and hence $\frac{66}{180}$ is a nonterminating repeating decimal.

EXAMPLE 3 The decimal expansion of the rational number $\frac{43}{2^4 \cdot 5^3}$, will terminate after how many places of decimals? [CBSE 2009]

SOLUTION We have

$$\frac{43}{2^4 \cdot 5^3} = \frac{43 \times 5}{2^4 \times 5^4} = \frac{215}{(2 \times 5)^4} = \frac{215}{10^4} = \frac{215}{10000} = 0.0215.$$

So, it will terminate after 4 places of decimals.

EXAMPLE 4 Express each of the following as a rational number in simplest form.

(a) $0.\overline{6}$ (b) $1.\overline{8}$ (c) $0.1\overline{6}$

SOLUTION (a) Let $x = 0.\overline{6}$. Then,

$$x = 0.666 \dots \quad \dots \text{ (i)}$$

$$\therefore 10x = 6.666 \dots \quad \dots \text{ (ii)}$$

On subtracting (i) from (ii), we get

$$9x = 6 \Rightarrow x = \frac{6}{9} = \frac{2}{3}.$$

$$\text{Hence, } 0.\overline{6} = \frac{2}{3}.$$

(b) Let $x = 1.\overline{8}$. Then,

$$x = 1.888 \dots \quad \dots (i)$$

$$\therefore 10x = 18.888 \dots \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$9x = 17 \Rightarrow x = \frac{17}{9} = 1\frac{8}{9}.$$

$$\text{Hence, } 1.\overline{8} = 1\frac{8}{9}.$$

(c) Let $x = 0.1\overline{6}$. Then,

$$x = 0.1666 \dots \quad \dots (i)$$

$$\therefore 10x = 1.6666 \dots \quad \dots (ii)$$

$$\text{And, } 100x = 16.6666 \dots \quad \dots (iii)$$

On subtracting (ii) from (iii), we get

$$90x = 15 \Rightarrow x = \frac{15}{90} = \frac{1}{6}.$$

$$\therefore 0.1\overline{6} = \frac{1}{6}.$$

EXAMPLE 5 Express $0.\overline{32}$ as a fraction in simplest form.

SOLUTION Let $x = 0.3\overline{2}$. Then,

$$x = 0.3232 \dots \quad \dots (i)$$

$$\therefore 100x = 32.3232 \dots \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$99x = 32 \Rightarrow x = \frac{32}{99}.$$

$$\text{Hence, } 0.\overline{32} = \frac{32}{99}.$$

EXAMPLE 6 Express $0.2\overline{54}$ as a fraction in simplest form.

SOLUTION Let $x = 0.2\overline{54}$. Then,

$$x = 0.2545454 \dots \quad \dots (i)$$

$$\therefore 10x = 2.545454 \dots \quad \dots (ii)$$

$$\text{and } 1000x = 254.545454 \dots \quad \dots (iii)$$

On subtracting (ii) from (iii), we get

$$990x = 252 \Rightarrow x = \frac{252}{990} = \frac{126}{495} = \frac{42}{165} = \frac{14}{55}.$$

$$\therefore 0.2\overline{54} = \frac{14}{55}.$$

EXAMPLE 7 Show that each of the following numbers is rational. What can you say about the prime factors of their denominators?

(i) 23.123456789

(ii) $32.\overline{123456789}$

SOLUTION (i) Clearly, the given number 23.123456789 is a terminating decimal. So, it is rational and the prime factors of its denominator are 2 or 5 or both.

(ii) Clearly, the given number $32.\overline{123456789}$ is a nonterminating repeating decimal. So, it is rational and the prime factors of its denominator are other than 2 or 5 also.

EXAMPLE 8 Decide whether the number $0.12012001200012 \dots$ is rational or not. Give reason to support your answer.

SOLUTION Clearly, the given number $0.12012001200012 \dots$ is a nonterminating and nonrepeating decimal. So, it is not rational.

EXERCISE 1C

1. Without actual division, show that each of the following rational numbers is a terminating decimal. Express each in decimal form.

(i) $\frac{23}{(2^3 \times 5^2)}$

(ii) $\frac{24}{125}$

(iii) $\frac{171}{800}$

(iv) $\frac{15}{1600}$

(v) $\frac{17}{320}$

(vi) $\frac{19}{3125}$

2. Without actual division, show that each of the following rational numbers is a nonterminating repeating decimal.

(i) $\frac{11}{(2^3 \times 3)}$

(ii) $\frac{73}{(2^2 \times 3^3 \times 5)}$

(iii) $\frac{129}{(2^2 \times 5^3 \times 7^2)}$

(iv) $\frac{9}{35}$

(v) $\frac{77}{210}$

(vi) $\frac{32}{147}$

(vii) $\frac{29}{343}$

(viii) $\frac{64}{455}$

3. Express each of the following as a fraction in simplest form.

(i) $0.\bar{8}$

(ii) $2.\bar{4}$

(iii) $0.\bar{24}$

(iv) $0.1\bar{2}$

(v) $2.2\bar{4}$

(vi) $0.3\overline{65}$

ANSWERS (EXERCISE 1C)

1. (i) 0.115 (ii) 0.192 (iii) 0.21375 (iv) 0.009375 (v) 0.053125
(vi) 0.00608

3. (i) $\frac{8}{9}$ (ii) $\frac{22}{9}$ (iii) $\frac{8}{33}$ (iv) $\frac{11}{90}$ (v) $\frac{101}{45}$ (vi) $\frac{181}{495}$

HINTS TO SOME SELECTED QUESTIONS

1. (i) $\frac{23}{(2^3 \times 5^2)} = \frac{23 \times 5}{(2^3 \times 5^3)} = \frac{115}{(10)^3} = \frac{115}{1000} = 0.115.$

(ii) $\frac{24}{125} = \frac{24}{5^3} \times \frac{2^3}{2^3} = \frac{24 \times 8}{(5 \times 2)^3} = \frac{192}{(10)^3} = \frac{192}{1000} = 0.192.$

(iii) $\frac{171}{800} = \frac{171}{8} \times \frac{1}{100} = \frac{21.375}{100} = 0.21375.$

(iv) $\frac{15}{1600} = \frac{15}{16} \times \frac{1}{100} = \frac{0.9375}{100} = 0.009375.$

(v) $\frac{17}{320} = \frac{17 \times 5}{320 \times 5} = \frac{85}{1600} = \frac{85}{16} \times \frac{1}{100} = \frac{5.3125}{100} = 0.053125.$

(vi) $\frac{19}{3125} = \frac{19 \times 8}{3125 \times 8} = \frac{152}{25000} = \frac{152}{25} \times \frac{1}{1000} = \frac{6.08}{1000} = 0.00608.$

2. (iv) $\frac{9}{35} = \frac{9}{(5 \times 7)}$

(v) $\frac{77}{210} = \frac{11}{30} = \frac{11}{(2 \times 3 \times 5)}$

(vi) $\frac{32}{147} = \frac{32}{(3 \times 7^2)}$

(vii) $\frac{29}{343} = \frac{29}{7^3}$

(viii) $\frac{64}{455} = \frac{64}{(5 \times 7 \times 13)}$

IRRATIONAL NUMBERS

IRRATIONAL NUMBERS The numbers which when expressed in decimal form are expressible as nonterminating and nonrepeating decimals are known as irrational numbers.

Examples Note that every nonterminating and nonrepeating decimal is irrational.

Type 1. (i) Clearly, 0.1010010001... is a nonterminating and nonrepeating decimal. So, it is irrational.

(ii) 0.2020020002, 0.3030030003..., etc., are all irrational.

(iii) 0.12112111211112... is irrational,

0.13113111311113... is irrational, and so on.

(iv) 0.232232223... is irrational,

0.343343334... is irrational, and so on.

Type 2. If m is a positive integer which is not a perfect square then $\sqrt{m}$ is irrational.

Thus, $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{10}, \sqrt{11}$, etc., are all irrational.

Type 3. If m is a positive integer which is not a perfect cube then $\sqrt[3]{m}$ is irrational.

Thus, $\sqrt[3]{2}, \sqrt[3]{3}, \sqrt[3]{5}, \sqrt[3]{6}$, etc., are all irrational.

Type 4. π is irrational, while $\frac{22}{7}$ is rational.

SOME RESULTS ON IRRATIONALS

THEOREM 1 Let p be a prime number and a be a positive integer. If p divides a^2 then show that p divides a .

PROOF Let p be a prime number and a be a positive integer such that p divides a^2 .

We know that every positive integer can be expressed as the product of primes.

Let $a = p_1 p_2 \dots p_n$, where $p_1, p_2, \dots, p_n$ are primes, not necessarily all distinct. Then,

$$a^2 = (p_1 p_2 \dots p_n)(p_1 p_2 \dots p_n)$$

$$\therefore a^2 = (p_1^2 p_2^2 \dots p_n^2).$$

Now, p divides a^2

$$\Rightarrow p \text{ is a prime factor of } a^2$$

$$\Rightarrow p \text{ is one of } p_1, p_2, \dots, p_n$$

[$\because$ prime factors of a^2 are $p_1, p_2, \dots, p_n$]

$$\Rightarrow p \text{ divides } a \quad [\because a = p_1 p_2 \dots p_n].$$

Thus, $(p \text{ divides } a^2) \Rightarrow (p \text{ divides } a)$.

Using the above result, we can prove the following.

THEOREM 2 Prove that $\sqrt{2}$ is irrational.

[CBSE 2008, '09]

PROOF If possible, let $\sqrt{2}$ be rational and let its simplest form be $\frac{a}{b}$.

Then, a and b are integers having no common factor other than 1, and $b \neq 0$.

$$\text{Now, } \sqrt{2} = \frac{a}{b} \Rightarrow 2 = \frac{a^2}{b^2} \quad [\text{on squaring both sides}]$$

$$\Rightarrow 2b^2 = a^2 \quad \dots (i)$$

$$\Rightarrow 2 \text{ divides } a^2 \quad [\because 2 \text{ divides } 2b^2]$$

$$\Rightarrow 2 \text{ divides } a$$

$$[\because 2 \text{ is prime and divides } b^2 \Rightarrow 2 \text{ divides } b].$$

Let $a = 2c$ for some integer c .

Putting $a = 2c$ in (i), we get

$$2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$$

$$\Rightarrow 2 \text{ divides } b^2 \quad [\because 2 \text{ divides } 2c^2]$$

$$\Rightarrow 2 \text{ divides } b$$

$$[\because 2 \text{ is prime and } 2 \text{ divides } b^2 \Rightarrow 2 \text{ divides } b].$$

Thus, 2 is a common factor of a and b .

But, this contradicts the fact that a and b have no common factor other than 1.

The contradiction arises by assuming that $\sqrt{2}$ is rational.

Hence, $\sqrt{2}$ is irrational.

THEOREM 3 Prove that $\sqrt{3}$ is irrational.

[CBSE 2008, '09C]

PROOF If possible, let $\sqrt{3}$ be rational and let its simplest form be $\frac{a}{b}$.

Then, a and b are integers having no common factor other than 1, and $b \neq 0$.

$$\text{Now, } \sqrt{3} = \frac{a}{b} \Rightarrow 3 = \frac{a^2}{b^2} \quad [\text{on squaring both sides}]$$

$$\Rightarrow 3b^2 = a^2 \quad \dots (i)$$

$$\Rightarrow 3 \text{ divides } a^2 \quad [\because 3 \text{ divides } 3b^2]$$

$$\Rightarrow 3 \text{ divides } a$$

$$[\because 3 \text{ is prime and } 3 \text{ divides } a^2 \Rightarrow 3 \text{ divides } a].$$

Let $a = 3c$ for some integer c .

Putting $a = 3c$ in (i), we get

$$3b^2 = 9c^2 \Rightarrow b^2 = 3c^2$$

$$\Rightarrow 3 \text{ divides } b^2 \quad [\because 3 \text{ divides } 3c^2]$$

$$\Rightarrow 3 \text{ divides } b$$

$$[\because 3 \text{ is prime and } 3 \text{ divides } b^2 \Rightarrow 3 \text{ divides } b].$$

Thus, 3 is a common factor of a and b .

But, this contradicts the fact that a and b have no common factor other than 1.

The contradiction arises by assuming that $\sqrt{3}$ is rational.

Hence, $\sqrt{3}$ is irrational.

THEOREM 4 Prove that $\sqrt{5}$ is irrational.

[CBSE 2008, '09]

PROOF If possible, let $\sqrt{5}$ be rational and let its simplest form be $\frac{a}{b}$.

Then, a and b are integers having no common factor other than 1, and $b \neq 0$.

$$\text{Now, } \sqrt{5} = \frac{a}{b} \Rightarrow 5 = \frac{a^2}{b^2} \quad [\text{on squaring both sides}]$$

$$\Rightarrow 5b^2 = a^2 \quad \dots (i)$$

$$\Rightarrow 5 \text{ divides } a^2 \quad [:\because 5 \text{ divides } 5b^2]$$

$$\Rightarrow 5 \text{ divides } a$$

$$[:\because 5 \text{ is prime and } 5 \text{ divides } a^2 \Rightarrow 5 \text{ divides } a].$$

Let $a = 5c$ for some integer c .

Putting $a = 5c$ in (i), we get

$$5b^2 = 25c^2 \Rightarrow b^2 = 5c^2$$

$$\Rightarrow 5 \text{ divides } b^2 \quad [:\because 5 \text{ divides } 5c^2]$$

$$\Rightarrow 5 \text{ divides } b$$

$$[:\because 5 \text{ is prime and } 5 \text{ divides } b^2 \Rightarrow 5 \text{ divides } b].$$

Thus, 5 is a common factor of a and b .

But, this contradicts the fact that a and b have no common factor other than 1.

The contradiction arises by assuming that $\sqrt{5}$ is rational.

Hence, $\sqrt{5}$ is irrational.

THEOREM 5 Prove that $\sqrt{11}$ is irrational.

PROOF If possible, let $\sqrt{11}$ be rational and let its simplest form be $\frac{a}{b}$.

Then, a and b are integers having no common factor other than 1, and $b \neq 0$.

$$\text{Now, } \sqrt{11} = \frac{a}{b} \Rightarrow 11 = \frac{a^2}{b^2} \quad [\text{on squaring both sides}]$$

$$\Rightarrow 11b^2 = a^2 \quad \dots (i)$$

$$\Rightarrow 11 \text{ divides } a^2 \quad [:\because 11 \text{ divides } 11b^2]$$

$$\Rightarrow 11 \text{ divides } a$$

$$[:\because 11 \text{ is prime and } 11 \text{ divides } a^2 \Rightarrow 11 \text{ divides } a].$$

Let $a = 11c$ for some positive integer c .

Putting $a = 11c$ in (i), we get

$$11b^2 = 121c^2 \Rightarrow b^2 = 11c^2$$

$$\Rightarrow 11 \text{ divides } b^2 \quad [\because 11 \text{ divides } 11c^2]$$

$$\Rightarrow 11 \text{ divides } b$$

$$[\because 11 \text{ is prime and } 11 \text{ divides } b^2 \Rightarrow 11 \text{ divides } b].$$

Thus, 11 is a common factor of a and b .

But, this contradicts the fact that a and b have no common factor other than 1.

The contradiction arises by assuming that $\sqrt{11}$ is rational.

Hence, $\sqrt{11}$ is irrational.

THEOREM 6 *If p is a prime number then prove that $\sqrt{p}$ is irrational.*

PROOF Let p be a prime number and if possible, let $\sqrt{p}$ be rational.

Let its simplest form be $\sqrt{p} = \frac{m}{n}$, where m and n are integers having no common factor other than 1, and $n \neq 0$.

$$\text{Then, } \sqrt{p} = \frac{m}{n} \Rightarrow p = \frac{m^2}{n^2} \quad [\text{on squaring both sides}]$$

$$\Rightarrow pn^2 = m^2 \quad \dots (i)$$

$$\Rightarrow p \text{ divides } m^2 \quad [\because p \text{ divides } pn^2]$$

$$\Rightarrow p \text{ divides } m$$

$$[\because p \text{ is prime and } p \text{ divides } m^2 \Rightarrow p \text{ divides } m].$$

Let $m = pq$ for some integer q .

Putting $m = pq$ in (i), we get

$$pn^2 = p^2q^2 \Rightarrow n^2 = pq^2$$

$$\Rightarrow p \text{ divides } n^2 \quad [\because p \text{ divides } pq^2]$$

$$\Rightarrow p \text{ divides } n$$

$$[\because p \text{ is prime and } p \text{ divides } n^2 \Rightarrow p \text{ divides } n].$$

Thus, p is a common factor of m and n .

But, this contradicts the fact that m and n have no common factor other than 1.

The contradiction arises by assuming that $\sqrt{p}$ is rational.

Hence, $\sqrt{p}$ is irrational.

THEOREM 7 *If a is rational and $\sqrt{b}$ is irrational then prove that $(a + \sqrt{b})$ is irrational.*

PROOF Let a be rational and $\sqrt{b}$ be irrational. Then, we have to prove that $(a + \sqrt{b})$ is irrational.

If possible, let $(a + \sqrt{b})$ be rational. Then,

$(a + \sqrt{b})$ is rational and a is rational

$\Rightarrow \{(a + \sqrt{b}) - a\}$ is rational [$\because$ difference of rationals is rational]

$\Rightarrow \sqrt{b}$ is rational.

This contradicts the fact that $\sqrt{b}$ is irrational.

The contradiction arises by assuming that $(a + \sqrt{b})$ is rational.

Hence, $(a + \sqrt{b})$ is irrational.

THEOREM 8 *If a is a nonzero rational and $\sqrt{b}$ is irrational then show that $a\sqrt{b}$ is irrational.*

PROOF Let a be a nonzero rational and let $\sqrt{b}$ be irrational.

Then, we have to show that $a\sqrt{b}$ is irrational.

If possible, let $a\sqrt{b}$ be rational.

Then, $a\sqrt{b} = \frac{p}{q}$, where p and q are nonzero integers, having no common factor other than 1.

Now, $a\sqrt{b} = \frac{p}{q} \Rightarrow \sqrt{b} = \frac{p}{aq}$ (i)

But, p and aq are both rational and $aq \neq 0$.

$\therefore \frac{p}{aq}$ is rational.

Thus, from (i), it follows that $\sqrt{b}$ is rational.

This contradicts the fact that $\sqrt{b}$ is irrational.

The contradiction arises by assuming that $a\sqrt{b}$ is rational.

Hence, $a\sqrt{b}$ is irrational.

SOLVED EXAMPLES

EXAMPLE 1 *Show that $(2 + \sqrt{3})$ is an irrational number.*

SOLUTION Let us assume, to the contrary, that $(2 + \sqrt{3})$ is rational.

Then, there exist co-primes a and b ($b \neq 0$) such that

$$(2 + \sqrt{3}) = \frac{a}{b}$$

$$\Rightarrow \sqrt{3} = \frac{a}{b} - 2 \Rightarrow \sqrt{3} = \frac{a - 2b}{b}.$$

Since a and b are integers, so $\frac{a - 2b}{b}$ is rational.

Thus, $\sqrt{3}$ is also rational.

But, this contradicts the fact that $\sqrt{3}$ is irrational. So, our assumption is incorrect.

Hence, $(2 + \sqrt{3})$ is irrational.

EXAMPLE 2 Show that $2\sqrt{3}$ is irrational.

SOLUTION Let us assume, to the contrary, that $2\sqrt{3}$ is rational.

Then, there exist co-primes a and b ($b \neq 0$) such that

$$2\sqrt{3} = \frac{a}{b} \Rightarrow \sqrt{3} = \frac{a}{2b}.$$

Since a and b are integers, so $\frac{a}{2b}$ is rational.

Thus, $\sqrt{3}$ is also rational.

But, this contradicts the fact that $\sqrt{3}$ is irrational. So, our assumption is incorrect.

Hence, $2\sqrt{3}$ is irrational.

EXAMPLE 3 Show that $\frac{1}{\sqrt{2}}$ is irrational.

SOLUTION Let us assume, to the contrary, that $\frac{1}{\sqrt{2}}$ is rational.

Then, there exist co-primes a and b ($b \neq 0$) such that

$$\frac{1}{\sqrt{2}} = \frac{a}{b} \Rightarrow \sqrt{2} = \frac{b}{a}.$$

Since a and b are integers, so $\frac{b}{a}$ is rational.

Thus, $\sqrt{2}$ is also rational.

But, this contradicts the fact that $\sqrt{2}$ is irrational. So, our assumption is incorrect.

Hence, $\frac{1}{\sqrt{2}}$ is irrational.

EXAMPLE 4 Prove that $(3 + 5\sqrt{2})$ is irrational.

SOLUTION Let us assume, to the contrary, that $(3 + 5\sqrt{2})$ is rational.

Then, there exist co-primes a and b ($b \neq 0$) such that

$$3 + 5\sqrt{2} = \frac{a}{b}$$

$$\Rightarrow 5\sqrt{2} = \frac{a}{b} - 3 = \frac{a - 3b}{b}$$

$$\Rightarrow \sqrt{2} = \frac{a - 3b}{5b}.$$

Since a and b are integers, so $\frac{a - 3b}{5b}$ is rational.

Thus, $\sqrt{2}$ is also rational.

But, this contradicts the fact that $\sqrt{2}$ is irrational. So, our assumption is incorrect.

Hence, $(3 + 5\sqrt{2})$ is irrational.

EXAMPLE 5 Prove that $(\sqrt{2} + \sqrt{3})$ is irrational.

SOLUTION Let us assume that $(\sqrt{2} + \sqrt{3})$ is irrational.

Then, there exist co-primes a and b such that

$$\sqrt{2} + \sqrt{3} = \frac{a}{b}$$

$$\Rightarrow \sqrt{3} = \frac{a}{b} - \sqrt{2}$$

$$\Rightarrow (\sqrt{3})^2 = \left(\frac{a}{b} - \sqrt{2}\right)^2 \quad \text{[squaring both sides]}$$

$$\Rightarrow 3 = \frac{a^2}{b^2} - \frac{2a}{b}\sqrt{2} + 2$$

$$\Rightarrow \frac{2a}{b}\sqrt{2} = \frac{a^2}{b^2} - 1$$

$$\Rightarrow \sqrt{2} = \frac{a^2 - b^2}{2ab}.$$

Since a and b are integers, so $\frac{a^2 - b^2}{2ab}$ is rational.

Thus, $\sqrt{2}$ is also rational.

But, this contradicts the fact that $\sqrt{2}$ is irrational. So, our assumption is incorrect.

Hence, $(\sqrt{2} + \sqrt{3})$ is irrational.

EXAMPLE 6 Prove that $\sqrt{p} + \sqrt{q}$ is irrational, where p and q are primes.

SOLUTION Let us assume that $\sqrt{p} + \sqrt{q}$ is rational.

Then, there exist co-primes a and b such that

$$\begin{aligned}\sqrt{p} + \sqrt{q} &= \frac{a}{b} \\ \Rightarrow \sqrt{p} &= \frac{a}{b} - \sqrt{q} \\ \Rightarrow (\sqrt{p})^2 &= \left(\frac{a}{b} - \sqrt{q}\right)^2 && \text{[squaring both sides]} \\ \Rightarrow p &= \frac{a^2}{b^2} - \frac{2a}{b}\sqrt{q} + q \\ \Rightarrow \frac{2a}{b}\sqrt{q} &= \frac{a^2}{b^2} + q - p \\ \Rightarrow \sqrt{q} &= (a^2 + b^2q - b^2p) \times \frac{b}{2a} = \frac{a^2b + b^3(q-p)}{2a}.\end{aligned}$$

Since a, b, p, q are integers so $\frac{a^2b + b^3(q-p)}{2a}$ is rational.

Thus, $\sqrt{q}$ is also rational.

But, q being prime, $\sqrt{q}$ is irrational.

Since, a contradiction arises so our assumption is incorrect.

Hence, $(\sqrt{p} + \sqrt{q})$ is irrational.

EXERCISE 1D

1. Define (i) rational numbers (ii) irrational numbers (iii) real numbers.

2. Classify the following numbers as rational or irrational:

(i) $\frac{22}{7}$ (ii) 3.1416 (iii) π (iv) $3.\overline{142857}$

(v) 5.636363 ... (vi) 2.040040004 ... (vii) 1.535335333 ...

(viii) $3.121221222 \dots$ (ix) $\sqrt{21}$ (x) $\sqrt[3]{3}$

3. Prove that each of the following numbers is irrational.

(i) $\sqrt{6}$ (ii) $(2 - \sqrt{3})$ [CBSE 2008]

(iii) $(3 + \sqrt{2})$ [CBSE 2009] (iv) $(2 + \sqrt{5})$ [CBSE 2008C]

(v) $(5 + 3\sqrt{2})$ [CBSE 2008] (vi) $3\sqrt{7}$

(vii) $\frac{3}{\sqrt{5}}$ (viii) $(2 - 3\sqrt{5})$ [CBSE 2010]

(ix) $(\sqrt{3} + \sqrt{5})$

4. Prove that $\frac{1}{\sqrt{3}}$ is irrational.

HINT $\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{1}{3} \cdot \sqrt{3}.$

5. (i) Give an example of two irrationals whose sum is rational.
 (ii) Give an example of two irrationals whose product is rational.

HINT (i) Take $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$.

(ii) Take $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$.

6. State whether the given statement is true or false.

- (i) The sum of two rationals is always rational.
 (ii) The product of two rationals is always rational.
 (iii) The sum of two irrationals is always an irrational.
 (iv) The product of two irrationals is always an irrational.
 (v) The sum of a rational and an irrational is irrational.
 (vi) The product of a rational and an irrational is irrational.

7. Prove that $(2\sqrt{3} - 1)$ is an irrational number. [CBSE 2010]

8. Prove that $(4 - 5\sqrt{2})$ is an irrational number. [CBSE 2010]

9. Prove that $(5 - 2\sqrt{3})$ is an irrational number. [CBSE 2010]

10. Prove that $5\sqrt{2}$ is irrational.

11. Prove that $\frac{2}{\sqrt{7}}$ is irrational.

HINT $\frac{2}{\sqrt{7}} = \left(\frac{2}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} \right) = \frac{2}{7} \cdot \sqrt{7}.$

ANSWERS (EXERCISE 1D)

2. (i) rational (ii) rational (iii) irrational (iv) rational
 (v) rational (vi) irrational (vii) irrational (viii) irrational
 (ix) irrational (x) irrational
6. (i) True (ii) True (iii) False (iv) False (v) True (vi) True

EXERCISE 1E

Very-Short-Answer Questions

1. State Euclid's division lemma.

2. State fundamental theorem of arithmetic.
3. Express 360 as product of its prime factors.
4. If a and b are two prime numbers then find $\text{HCF}(a, b)$.
5. If a and b are two prime numbers then find $\text{LCM}(a, b)$.
6. If the product of two numbers is 1050 and their HCF is 25, find their LCM.
7. What is a composite number?
8. If a and b are relatively prime then what is their HCF?
9. If the rational number $\frac{a}{b}$ has a terminating decimal expansion, what is the condition to be satisfied by b ? [CBSE 2008]
10. Simplify: $\frac{(2\sqrt{45} + 3\sqrt{20})}{2\sqrt{5}}$. [CBSE 2010]
11. Write the decimal expansion of $\frac{73}{(2^4 \times 5^3)}$. [CBSE 2009]
12. Show that there is no value of n for which $(2^n \times 5^n)$ ends in 5.
13. Is it possible to have two numbers whose HCF is 25 and LCM is 520?
14. Give an example of two irrationals whose sum is rational.
15. Give an example of two irrationals whose product is rational.
16. If a and b are relatively prime, what is their LCM?
17. The LCM of two numbers is 1200. Show that the HCF of these numbers cannot be 500. Why?

Short-Answer Questions

18. Express $0.\overline{4}$ as a rational number in simplest form.
19. Express $0.\overline{23}$ as a rational number in simplest form.
20. Explain why $0.15015001500015 \dots$ is an irrational number.
21. Show that $\frac{\sqrt{2}}{3}$ is irrational.
22. Write a rational number between $\sqrt{3}$ and 2.
23. Explain why $3.\overline{1416}$ is a rational number.

ANSWERS (EXERCISE 1E)

3. $(2^3 \times 3^2 \times 5)$ 4. 1 5. ab 6. 42 8. 1
9. $b = (2^m \times 5^n)$, where m and n are some non-negative integers
10. 6 11. 0.0365 13. No 16. ab 17. since 500 is not a factor of 1200
18. $\frac{4}{9}$ 19. $\frac{23}{99}$ 22. 1.8

HINTS TO SOME SELECTED QUESTIONS

7. A number having at least 3 factors is called a composite number.
11. $\frac{73}{(2^4 \times 5^3)} = \frac{73 \times 5}{(2^4 \times 5^4)} = \frac{365}{(2 \times 5)^4} = \frac{365}{(10)^4} = \frac{365}{10000} = 0.0365$.
12. $(2^n \times 5^n) = (2 \times 5)^n = 10^n$, which always ends in a zero.
13. HCF always divides the LCM completely.
14. $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$.
15. $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$.
20. Given number is nonterminating and nonrepeating decimal.
22. Clearly, $\sqrt{3} = 1.732 \dots$. So, we may take 1.8 as the required rational number between $\sqrt{3}$ and 2.
23. Clearly, it is a nonterminating repeating decimal.
-

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- Which of the following is a pair of co-primes?
 (a) (14, 35) (b) (18, 25) (c) (31, 93) (d) (32, 62)
- If $a = (2^2 \times 3^3 \times 5^4)$ and $b = (2^3 \times 3^2 \times 5)$ then $\text{HCF}(a, b) = ?$
 (a) 90 (b) 180 (c) 360 (d) 540
- HCF of $(2^3 \times 3^2 \times 5)$, $(2^2 \times 3^3 \times 5^2)$ and $(2^4 \times 3 \times 5^3 \times 7)$ is
 (a) 30 (b) 48 (c) 60 (d) 105
- LCM of $(2^3 \times 3 \times 5)$ and $(2^4 \times 5 \times 7)$ is
 (a) 40 (b) 560 (c) 1120 (d) 1680
- The HCF of two numbers is 27 and their LCM is 162. If one of the numbers is 54, what is the other number?
 (a) 36 (b) 45 (c) 9 (d) 81
- The product of two numbers is 1600 and their HCF is 5. The LCM of the numbers is
 (a) 8000 (b) 1600 (c) 320 (d) 1605
- What is the largest number that divides each one of 1152 and 1664 exactly?
 (a) 32 (b) 64 (c) 128 (d) 256
- What is the largest number that divides 70 and 125, leaving remainders 5 and 8 respectively?
 (a) 13 (b) 9 (c) 3 (d) 585

9. What is the largest number that divides 245 and 1029, leaving remainder 5 in each case?
- (a) 15 (b) 16 (c) 9 (d) 5
10. The simplest form of $\frac{1095}{1168}$ is
- (a) $\frac{17}{26}$ (b) $\frac{25}{26}$ (c) $\frac{13}{16}$ (d) $\frac{15}{16}$
11. Euclid's division lemma states that for any positive integers a and b , there exist unique integers q and r such that $a = bq + r$, where r must satisfy
- (a) $1 < r < b$ (b) $0 < r \leq b$ (c) $0 \leq r < b$ (d) $0 < r < b$
12. A number when divided by 143 leaves 31 as remainder. What will be the remainder when the same number is divided by 13?
- (a) 0 (b) 1 (c) 3 (d) 5
13. Which of the following is an irrational number?
- (a) $\frac{22}{7}$ (b) 3.1416
(c) $3.\overline{1416}$ (d) 3.141141114 ...
14. π is
- (a) an integer (b) a rational number
(c) an irrational number (d) none of these
15. $2.\overline{35}$ is
- (a) an integer (b) a rational number
(c) an irrational number (d) none of these
16. $2.13113111311113 \dots$ is
- (a) an integer (b) a rational number
(c) an irrational number (d) none of these
17. The number $3.24636363 \dots$ is
- (a) an integer (b) a rational number
(c) an irrational number (d) none of these
18. Which of the following rational numbers is expressible as a terminating decimal?
- (a) $\frac{124}{165}$ (b) $\frac{131}{30}$ (c) $\frac{2027}{625}$ (d) $\frac{1625}{462}$

19. The decimal expansion of the rational number $\frac{37}{2^2 \times 5}$ will terminate after
(a) one decimal place (b) two decimal places
(c) three decimal places (d) four decimal places
20. The decimal expansion of the number $\frac{14753}{1250}$ will terminate after
(a) one decimal place (b) two decimal places
(c) three decimal places (d) four decimal places
21. The number 1.732 is
(a) an irrational number (b) a rational number
(c) an integer (d) a whole number
22. a and b are two positive integers such that the least prime factor of a is 3 and the least prime factor of b is 5. Then, the least prime factor of $(a + b)$ is
(a) 2 (b) 3 (c) 5 (d) 8
23. $\sqrt{2}$ is
(a) a rational number
(b) an irrational number
(c) a terminating decimal
(d) a nonterminating repeating decimal
24. $\frac{1}{\sqrt{2}}$ is
(a) a fraction (b) a rational number
(c) an irrational number (d) none of these
25. $(2 + \sqrt{2})$ is
(a) an integer (b) a rational number
(c) an irrational number (d) none of these
26. What is the least number that is divisible by all the natural numbers from 1 to 10 (both inclusive)?
(a) 100 (b) 1260 (c) 2520 (d) 5040

ANSWERS (MCQ)

1. (b) 2. (b) 3. (c) 4. (d) 5. (d) 6. (c) 7. (c) 8. (a) 9. (b)
10. (d) 11. (c) 12. (d) 13. (d) 14. (c) 15. (b) 16. (c) 17. (b) 18. (c)
19. (b) 20. (d) 21. (b) 22. (a) 23. (b) 24. (c) 25. (c) 26. (c)

HINTS TO SOME SELECTED QUESTIONS

2. $\text{HCF}(a, b)$ = product of common terms with lowest power

$$= (2^2 \times 3^2 \times 5) = (4 \times 9 \times 5) = 180.$$

3. HCF = product of common terms with lowest power

$$= (2^2 \times 3 \times 5) = (4 \times 3 \times 5) = 60.$$

4. LCM = product of prime factors with highest power

$$= (2^4 \times 3 \times 5 \times 7) = (16 \times 3 \times 5 \times 7) = 1680.$$

5. Other number = $\frac{\text{HCF} \times \text{LCM}}{\text{given number}} = \frac{27 \times 162}{54} = 81.$

6. $\text{LCM} = \frac{\text{product of two numbers}}{\text{their HCF}} = \frac{1600}{5} = 320.$

7. Required number = $\text{HCF}(1152, 1664) = 128.$

8. Required number = $\text{HCF}\{(70 - 5), (125 - 8)\} = \text{HCF}(65, 117) = 13.$

9. Required number = $\text{HCF}\{(245 - 5), (1029 - 5)\} = \text{HCF}(240, 1024) = 16.$

10. $\text{HCF}\{1095, 1168\} = 73.$

$$\therefore \frac{1095}{1168} = \frac{1095 \div 73}{1168 \div 73} = \frac{15}{16}.$$

11. On dividing a by b , let q be the quotient and r be the remainder.

Then, we have

$$a = bq + r, \text{ where } 0 \leq r < b.$$

12. Let the given number when divided by 143 give q as quotient and 31 as remainder.

$$\text{Then, number} = 143q + 31 = \{13 \times 11q + 13 \times 2 + 5\} = 13 \times (11q + 2) + 5.$$

So, the same number when divided by 13 gives 5 as remainder.

13. 3.141141114 ... is a nonterminating, nonrepeating decimal. So, it is irrational.

14. π is an irrational number.

15. $2.\overline{35} = 2.353535\dots$, which is a repeating decimal.

$$\therefore 2.\overline{35} \text{ is rational.}$$

16. 2.13113111311113 ... is a nonterminating, nonrepeating decimal. So, it is irrational.

17. The number 3.24636363 ... is a nonterminating repeating decimal.

So, it is a rational number.

18. $\frac{2027}{625} = \frac{2027}{(5^4 \times 2^0)}.$

So, it is expressible as a terminating decimal.

19. $\frac{37}{2^2 \times 5} = \frac{37 \times 5}{2^2 \times 5^2} = \frac{185}{(2 \times 5)^2} = \frac{185}{(10)^2} = \frac{185}{100} = 1.85.$

So, it will terminate after 2 decimal places.

$$20. \frac{14753}{1250} \times \frac{8}{8} = \frac{14753 \times 8}{10000} = \frac{14753 \times 8}{(10)^4}.$$

So, it will terminate after 4 decimal places.

$$21. 1.732 = \frac{1732}{1000}, \text{ which is a rational number.}$$

22. Clearly, 2 is neither a factor of a nor that of b .

$\therefore a$ and b are both odd.

Hence, $(a + b)$ is even.

$\therefore$ least prime factor of $(a + b)$ is 2.

$$24. \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{1}{2} \cdot \sqrt{2}.$$

Here, $\frac{1}{2}$ is rational and $\sqrt{2}$ is irrational.

And, the product of a rational and an irrational is irrational.

$\therefore \frac{1}{2} \cdot \sqrt{2}$ and hence $\frac{1}{\sqrt{2}}$ is irrational.

25. The sum of a rational and an irrational is irrational.

Here, 2 is rational and $\sqrt{2}$ is irrational.

$\therefore (2 + \sqrt{2})$ is irrational.

26. Required number = LCM{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

$$= \text{LCM}\{1, 2, 3, 2^2, 5, 2 \times 3, 7, 2^3, 3^2, 2 \times 5\}$$

$$= (1 \times 2^3 \times 3^2 \times 5 \times 7) = (8 \times 9 \times 5 \times 7) = 2520.$$

SUMMARY OF RESULTS

1. Euclid's Division Lemma

Given positive integers a and b , there exist whole numbers q and r satisfying $a = bq + r$, where $0 \leq r < b$.

2. Euclid's Division Algorithm According to this, we find the HCF of two positive integer a and b with $a > b$ in following steps.

Step 1. Apply the division lemma to find q and r such that $a = bq + r$, where $0 \leq r < b$.

Step 2. If $r = 0$ then $\text{HCF} = b$. If $r \neq 0$, apply Euclid's division lemma to b and r .

Step 3. Continue the process till the remainder is zero. The divisor at this stage is $\text{HCF}(a, b)$.

3. The Fundamental Theorem of Arithmetic

Every composite number can be expressed as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

4. If p is prime and p divides a^2 then p divides a , where a is a positive integer.
5. To prove that $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}$, etc., are irrationals.
6. Let x be a rational number which can be expressed as a terminating decimal. If we put it in the simplest form $\frac{p}{q}$ then $q = (2^m \times 5^n)$ for some non-negative integers m and n .
7. Let $x = \frac{p}{q}$ be a rational number such that $q \neq (2^m \times 5^n)$ then x has a nonterminating repeating decimal expansion.
8. A number which can be expressed as a nonterminating and nonrepeating decimal is an irrational number.

$\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{10}, \dots, \pi, e$, etc., are all irrational numbers.

.....

TEST YOURSELF**MCQ**

1. The decimal representation of $\frac{71}{150}$ is
 - (a) a terminating decimal
 - (b) a nonterminating, repeating decimal
 - (c) a nonterminating and nonrepeating decimal
 - (d) none of these
2. Which of the following has a terminating decimal expansion?

(a) $\frac{32}{91}$	(b) $\frac{19}{80}$	(c) $\frac{23}{45}$	(d) $\frac{25}{42}$
---------------------	---------------------	---------------------	---------------------
3. On dividing a positive integer n by 9, we get 7 as remainder. What will be the remainder if $(3n - 1)$ is divided by 9?

(a) 1	(b) 2	(c) 3	(d) 4
-------	-------	-------	-------
4. $0.\overline{68} + 0.\overline{73} = ?$

(a) $1.\overline{41}$	(b) $1.\overline{42}$	(c) $0.\overline{141}$	(d) None of these
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Short-Answer Questions

5. Show that any number of the form $4^n, n \in \mathbb{N}$ can never end with the digit 0.

6. The HCF of two numbers is 27 and their LCM is 162. If one of the number is 81, find the other.
7. Examine whether $\frac{17}{30}$ is a terminating decimal.
8. Find the simplest form of $\frac{148}{185}$.
9. Which of the following numbers are irrational?

(a) $\sqrt{2}$	(b) $\sqrt[3]{6}$	(c) 3.142857	(d) $2.\bar{3}$
(e) π	(f) $\frac{22}{7}$	(g) 0.232332333...	(h) $5.274\bar{1}$
10. Prove that $(2 + \sqrt{3})$ is irrational.
11. Find the HCF and LCM of 12, 15, 18, 27.
12. Give an example of two irrationals whose sum is rational.
13. Give prime factorisation of 4620.
14. Find the HCF of 1008 and 1080 by prime factorisation method.
15. Find the HCF and LCM of $\frac{8}{9}$, $\frac{10}{27}$ and $\frac{16}{81}$.
16. Find the largest number which divides 546 and 764, leaving remainders 6 and 8 respectively.

Long-Answer Questions

17. Prove that $\sqrt{3}$ is an irrational number.
18. Show that every positive odd integer is of the form $(4q + 1)$ or $(4q + 3)$ for some integer q .
19. Show that one and only one out of n , $(n + 2)$ and $(n + 4)$ is divisible by 3, where n is any positive integer.
20. Show that $(4 + 3\sqrt{2})$ is irrational.

ANSWERS (TEST YOURSELF)

- | | | | | | | |
|--|--------|---|--------|--|-------|------------------|
| 1. (b) | 2. (b) | 3. (b) | 4. (b) | 6. 54 | 7. No | 8. $\frac{4}{5}$ |
| 9. $\sqrt{2}$, $\sqrt[3]{6}$, π , 0.232332333... | | | | 11. HCF = 3, LCM = 540 | | |
| 12. $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$ | | | | 13. $(2^2 \times 3 \times 5 \times 7 \times 11)$ | | |
| 14. 72 | | 15. HCF = $\frac{2}{81}$, LCM = $\frac{80}{9}$ | | 16. 108 | | |





Polynomials

POLYNOMIALS An expression of the form $p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where $a_n \neq 0$, is called a polynomial in x of degree n .

Here $a_0, a_1, a_2, \dots, a_n$ are real numbers and each power of x is a non-negative integer.

- EXAMPLES**
- (i) $3x + 5$ is a polynomial in x of degree 1.
 - (ii) $8x^2 - 5x + 3$ is a polynomial in x of degree 2.
 - (iii) $2y^3 + \frac{4}{9}y^2 - 5y + \sqrt{3}$ is a polynomial in y of degree 3.
 - (iv) $3z^4 - 5z^3 + 2z^2 - 8z + 1$ is a polynomial in z of degree 4.

REMARK Note that $(\sqrt{x} + 3), \frac{1}{(x+5)}, \frac{6}{(x^2 - 3x + 1)}$, etc., are not polynomials.

LINEAR POLYNOMIAL A polynomial of degree 1 is called a linear polynomial.

A linear polynomial is of the form $p(x) = ax + b$, where $a \neq 0$.

Thus, $(3x - 5), (\sqrt{2}x + 3), \left(x - \frac{5}{8}\right)$, etc., are all linear polynomials.

QUADRATIC POLYNOMIAL A polynomial of degree 2 is called a quadratic polynomial.

A quadratic polynomial is of the form $p(x) = ax^2 + bx + c$, where $a \neq 0$.

Thus, $(3x^2 + 5x - 8), (2x^2 - 2\sqrt{2}x + 6), (y^2 - 3y + \sqrt{3})$, etc., are all quadratic polynomials.

CUBIC POLYNOMIAL A polynomial of degree 3 is called a cubic polynomial. A cubic polynomial is of the form $p(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$.

Thus, $(2x^3 - 3x^2 + 8x + 1), (\sqrt{2}y^3 - 2y^2 + y - 8), (z^3 + 2z^2 - \sqrt{3}z + 3)$, etc., are all cubic polynomials.

BIQUADRATIC POLYNOMIAL A polynomial of degree 4 is called a biquadratic polynomial.

A biquadratic polynomial is of the form $p(x) = ax^4 + bx^3 + cx^2 + dx + e$, where $a \neq 0$.

Thus, $(2x^4 + 3x^3 - 5x^2 + 9x + 1), (4y^4 - 5y^3 + 6y^2 - 8y + 3)$, etc., are all biquadratic polynomials.

VALUE OF A POLYNOMIAL AT A GIVEN POINT

If $p(x)$ is a polynomial in x and if α is any real number then the value obtained by putting $x = \alpha$ in $p(x)$ is called the value of $p(x)$ at $x = \alpha$.

The value of $p(x)$ at $x = \alpha$ is denoted by $p(\alpha)$.

EXAMPLE Let $p(x) = 2x^2 - 3x + 5$. Then,

$$p(2) = \{2 \times 2^2 - 3 \times 2 + 5\} = (8 - 6 + 5) = 7,$$

$$p(-1) = \{2 \times (-1)^2 - 3 \times (-1) + 5\} = (2 + 3 + 5) = 10.$$

ZEROS OF A POLYNOMIAL A real number α is called a zero of the polynomial $p(x)$, if $p(\alpha) = 0$.

EXAMPLE Let $p(x) = x^2 - 2x - 3$. Find (i) $p(3)$ and (ii) $p(-1)$.
What do you conclude?

SOLUTION We have $p(x) = x^2 - 2x - 3$.

$$\therefore \text{ (i) } p(3) = (3^2 - 2 \times 3 - 3) = (9 - 6 - 3) = 0$$

$$\text{and (ii) } p(-1) = \{(-1)^2 - 2 \times (-1) - 3\} = (1 + 2 - 3) = 0.$$

This shows that 3 and -1 are the zeros of the polynomial $p(x)$.

RELATION BETWEEN THE ZEROS AND COEFFICIENTS OF A QUADRATIC POLYNOMIAL

Let α and β be the zeros of a quadratic polynomial

$$p(x) = ax^2 + bx + c, \text{ where } a \neq 0.$$

Then, $(x - \alpha)$ and $(x - \beta)$ are the factors of $p(x)$.

$$\therefore (ax^2 + bx + c) = k(x - \alpha)(x - \beta), \text{ where } k \text{ is a constant}$$

$$= k \cdot \{x^2 - (\alpha + \beta)x + \alpha\beta\}$$

$$= kx^2 - k(\alpha + \beta)x + k(\alpha\beta).$$

On comparing coefficients of like powers of x on both sides, we get

$$k = a, -k(\alpha + \beta) = b \text{ and } k(\alpha\beta) = c$$

$$\Rightarrow -a(\alpha + \beta) = b \text{ and } a(\alpha\beta) = c \quad [\because k = a]$$

$$\Rightarrow (\alpha + \beta) = \frac{-b}{a} \text{ and } \alpha\beta = \frac{c}{a}.$$

$$\therefore \text{ sum of zeros} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)},$$

$$\text{product of zeros} = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

SUMMARY

I. If α and β are the zeros of $p(x) = ax^2 + bx + c$, $a \neq 0$ then

$$(i) \alpha + \beta = \frac{-b}{a}$$

$$(ii) \alpha\beta = \frac{c}{a}.$$

II. A quadratic polynomial whose zeros are α and β is given by

$$p(x) = \{x^2 - (\alpha + \beta)x + \alpha\beta\}.$$

SOLVED EXAMPLES

EXAMPLE 1 Find the zeros of the polynomial $2x^2 + 5x - 12$ and verify the relationship between its zeros and coefficients.

SOLUTION Let the given polynomial be denoted by $f(x)$. Then,

$$\begin{aligned} f(x) &= 2x^2 + 5x - 12 \\ &= 2x^2 + 8x - 3x - 12 \\ &= 2x(x + 4) - 3(x + 4) \\ &= (x + 4)(2x - 3). \end{aligned}$$

$$\begin{aligned} \therefore f(x) = 0 &\Rightarrow (x + 4)(2x - 3) = 0 \\ &\Rightarrow x + 4 = 0 \text{ or } 2x - 3 = 0 \\ &\Rightarrow x = -4 \text{ or } x = \frac{3}{2}. \end{aligned}$$

So, the zeros of $f(x)$ are -4 and $\frac{3}{2}$.

$$\text{Sum of the zeros} = \left(-4 + \frac{3}{2}\right) = \frac{-5}{2} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)},$$

$$\text{product of the zeros} = (-4) \times \frac{3}{2} = \frac{-12}{2} = \frac{\text{constant term}}{(\text{coefficient of } x^2)}.$$

EXAMPLE 2 Find the zeros of the polynomial $6x^2 - 3 - 7x$ and verify the relationship between the zeros and the coefficients. [CBSE 2008]

SOLUTION Let the given polynomial be denoted by $f(x)$. Then,

$$\begin{aligned} f(x) &= 6x^2 - 3 - 7x \\ &= 6x^2 - 7x - 3 \quad [\text{in standard form}] \\ &= 6x^2 - 9x + 2x - 3 \\ &= 3x(2x - 3) + (2x - 3) \\ &= (2x - 3)(3x + 1). \end{aligned}$$

$$\begin{aligned}
 \therefore f(x) = 0 &\Rightarrow (2x-3)(3x+1) = 0 \\
 &\Rightarrow 2x-3 = 0 \text{ or } 3x+1 = 0 \\
 &\Rightarrow x = \frac{3}{2} \text{ or } x = -\frac{1}{3}.
 \end{aligned}$$

So, the zeros of $f(x)$ are $\frac{3}{2}$ and $-\frac{1}{3}$.

$$\text{Sum of zeros} = \left\{ \frac{3}{2} + \left(-\frac{1}{3}\right) \right\} = \left(\frac{3}{2} - \frac{1}{3} \right) = \frac{7}{6} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)}.$$

$$\text{Product of zeros} = \frac{3}{2} \times \left(-\frac{1}{3}\right) = -\frac{1}{2} = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

EXAMPLE 3 Find the zeros of the polynomial $f(x) = x^2 - 2$ and verify the relationship between its zeros and coefficients.

SOLUTION We have

$$\begin{aligned}
 f(x) &= (x^2 - 2) = \{x^2 - (\sqrt{2})^2\} = (x + \sqrt{2})(x - \sqrt{2}). \\
 \therefore f(x) = 0 &\Rightarrow (x + \sqrt{2})(x - \sqrt{2}) = 0 \\
 &\Rightarrow x + \sqrt{2} = 0 \text{ or } x - \sqrt{2} = 0 \\
 &\Rightarrow x = -\sqrt{2} \text{ or } x = \sqrt{2}.
 \end{aligned}$$

So, the zeros of $f(x)$ are $-\sqrt{2}$ and $\sqrt{2}$.

$$\text{Sum of zeros} = (-\sqrt{2} + \sqrt{2}) = 0 = \frac{0}{1} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)},$$

$$\text{product of zeros} = (-\sqrt{2}) \times (\sqrt{2}) = -2 = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

EXAMPLE 4 Obtain the zeros of the quadratic polynomial $\sqrt{3}x^2 - 8x + 4\sqrt{3}$ and verify the relation between its zeros and coefficients. [CBSE 2008C]

SOLUTION We have

$$\begin{aligned}
 f(x) &= \sqrt{3}x^2 - 8x + 4\sqrt{3} = \sqrt{3}x^2 - 6x - 2x + 4\sqrt{3} \\
 &= \sqrt{3}x(x - 2\sqrt{3}) - 2(x - 2\sqrt{3}) = (x - 2\sqrt{3})(\sqrt{3}x - 2). \\
 \therefore f(x) = 0 &\Rightarrow (x - 2\sqrt{3})(\sqrt{3}x - 2) = 0 \\
 &\Rightarrow (x - 2\sqrt{3}) = 0 \text{ or } (\sqrt{3}x - 2) = 0 \\
 &\Rightarrow x = 2\sqrt{3} \text{ or } x = \frac{2}{\sqrt{3}}.
 \end{aligned}$$

So, the zeros of $f(x)$ are $2\sqrt{3}$ and $\frac{2}{\sqrt{3}}$.

$$\text{Sum of zeros} = \left(2\sqrt{3} + \frac{2}{\sqrt{3}} \right) = \frac{8}{\sqrt{3}} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)},$$

$$\text{product of zeros} = \left(2\sqrt{3} \times \frac{2}{\sqrt{3}} \right) = \frac{4\sqrt{3}}{\sqrt{3}} = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

EXAMPLE 5 Find a quadratic polynomial, the sum and product of whose zeros are -5 and 6 respectively.

SOLUTION Let α and β be the zeros of the required polynomial $f(x)$.

Then, $(\alpha + \beta) = -5$ and $\alpha\beta = 6$.

$$\begin{aligned}\therefore f(x) &= x^2 - (\alpha + \beta)x + \alpha\beta \\ &= x^2 - (-5)x + 6 \\ &= x^2 + 5x + 6.\end{aligned}$$

Hence, the required polynomial is $f(x) = x^2 + 5x + 6$.

EXAMPLE 6 Find the quadratic polynomial, the sum of whose zeros is $\sqrt{2}$ and their product is -12 . Hence, find the zeros of the polynomial.

SOLUTION Let α and β be the zeros of the required polynomial $f(x)$.

Then, $(\alpha + \beta) = \sqrt{2}$ and $\alpha\beta = -12$.

$$\begin{aligned}\therefore f(x) &= x^2 - (\alpha + \beta)x + \alpha\beta \\ &= x^2 - \sqrt{2}x - 12.\end{aligned}$$

So, the required polynomial is $f(x) = x^2 - \sqrt{2}x - 12$.

$$\begin{aligned}\text{Now, } f(x) &= x^2 - \sqrt{2}x - 12 \\ &= x^2 - 3\sqrt{2}x + 2\sqrt{2}x - 12 \quad [\text{note it}] \\ &= x(x - 3\sqrt{2}) + 2\sqrt{2}(x - 3\sqrt{2}) \\ &= (x - 3\sqrt{2})(x + 2\sqrt{2}).\end{aligned}$$

$$\begin{aligned}\therefore f(x) = 0 &\Rightarrow (x - 3\sqrt{2})(x + 2\sqrt{2}) = 0 \\ &\Rightarrow x - 3\sqrt{2} = 0 \quad \text{or} \quad x + 2\sqrt{2} = 0 \\ &\Rightarrow x = 3\sqrt{2} \quad \text{or} \quad x = -2\sqrt{2}.\end{aligned}$$

Hence, the required polynomial is $f(x) = x^2 - \sqrt{2}x - 12$ whose zeros are $3\sqrt{2}$ and $-2\sqrt{2}$.

EXAMPLE 7 If the product of the zeros of the polynomial $(ax^2 - 6x - 6)$ is 4 , find the value of a . [CBSE 2009]

SOLUTION Let α and β be the zeros of the polynomial $(ax^2 - 6x - 6)$.

$$\text{Then, } \alpha\beta = \frac{\text{constant term}}{\text{coefficient of } x^2} = \frac{-6}{a}.$$

But, $\alpha\beta = 4$ (given).

$$\therefore \frac{-6}{a} = 4 \Rightarrow 4a = -6 \Rightarrow a = \frac{-6}{4} = \frac{-3}{2}.$$

Hence, $a = \frac{-3}{2}$.

EXAMPLE 8 *If one zero of the polynomial $(a^2 + 9)x^2 + 13x + 6a$ is reciprocal of the other, find the value of a .* [CBSE 2008]

SOLUTION Let one zero of the given polynomial be α .

Then, the other zero is $\frac{1}{\alpha}$.

$$\therefore \text{product of zeros} = \left(\alpha \times \frac{1}{\alpha}\right) = 1.$$

$$\text{But, product of zeros} = \frac{\text{constant term}}{\text{coefficient of } x^2} = \frac{6a}{(a^2 + 9)}.$$

$$\therefore \frac{6a}{(a^2 + 9)} = 1 \Rightarrow a^2 + 9 = 6a$$

$$\Rightarrow a^2 + 9 - 6a = 0$$

$$\Rightarrow (a - 3)^2 = 0$$

$$\Rightarrow a - 3 = 0 \Rightarrow a = 3.$$

Hence, $a = 3$.

EXAMPLE 9 *Find a quadratic polynomial whose zeros are 1 and -3 . Verify the relation between the coefficients and zeros of the polynomial.*

[CBSE 2008C]

SOLUTION Let $\alpha = 1$ and $\beta = -3$.

$$\text{Sum of zeros} = (\alpha + \beta) = 1 + (-3) = -2.$$

$$\text{Product of zeros} = \alpha\beta = 1 \times (-3) = -3.$$

So, the required polynomial is

$$\begin{aligned} x^2 - (\alpha + \beta)x + \alpha\beta &= x^2 - (-2)x + (-3) \\ &= x^2 + 2x - 3. \end{aligned}$$

$$\text{Sum of zeros} = -2 = \frac{-2}{1} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)},$$

$$\text{product of zeros} = -3 = \frac{-3}{1} = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

EXERCISE 2A

Find the zeros of the following quadratic polynomials and verify the relationship between the zeros and the coefficients:

1. $x^2 + 7x + 12$

2. $x^2 - 2x - 8$

3. $x^2 + 3x - 10$

4. $4x^2 - 4x - 3$

[CBSE 2008C]

5. $5x^2 - 4 - 8x$ [CBSE 2008]

6. $2\sqrt{3}x^2 - 5x + \sqrt{3}$

[CBSE 2011]

7. $2x^2 - 11x + 15$

8. $4x^2 - 4x + 1$

9. $x^2 - 5$
10. $8x^2 - 4$
11. $5y^2 + 10y$
12. $3x^2 - x - 4$
13. Find the quadratic polynomial whose zeros are 2 and -6 . Verify the relation between the coefficients and the zeros of the polynomial.
14. Find the quadratic polynomial whose zeros are $\frac{2}{3}$ and $-\frac{1}{4}$. Verify the relation between the coefficients and the zeros of the polynomial.
15. Find the quadratic polynomial, sum of whose zeros is 8 and their product is 12. Hence, find the zeros of the polynomial. [CBSE 2008]
16. Find the quadratic polynomial, the sum of whose zeros is 0 and their product is -1 . Hence, find the zeros of the polynomial.
17. Find the quadratic polynomial, the sum of whose zeros is $\left(\frac{5}{2}\right)$ and their product is 1. Hence, find the zeros of the polynomial.
18. Find the quadratic polynomial, the sum of whose roots is $\sqrt{2}$ and their product is $\frac{1}{3}$.
19. If $x = \frac{2}{3}$ and $x = -3$ are the roots of the quadratic equation $ax^2 + 7x + b = 0$ then find the values of a and b . [CBSE 2011]
20. If $(x + a)$ is a factor of the polynomial $2x^2 + 2ax + 5x + 10$, find the value of a . [CBSE 2009]
21. One zero of the polynomial $3x^3 + 16x^2 + 15x - 18$ is $\frac{2}{3}$. Find the other zeros of the polynomial.

ANSWERS (EXERCISE 2A)

1. $-4, -3$ 2. $4, -2$ 3. $-5, 2$ 4. $\frac{3}{2}, \frac{-1}{2}$ 5. $2, \frac{-2}{5}$
6. $\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{3}}$ 7. $3, \frac{5}{2}$ 8. $\frac{1}{2}, \frac{1}{2}$ 9. $\sqrt{5}, -\sqrt{5}$ 10. $\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}$
11. $0, -2$ 12. $\frac{4}{3}, -1$ 13. $x^2 + 4x - 12$ 14. $12x^2 - 5x - 2$
15. $x^2 - 8x + 12, \{6, 2\}$ 16. $(x^2 - 1), \{1, -1\}$ 17. $(2x^2 - 5x + 2), \left\{2, \frac{1}{2}\right\}$
18. $3x^2 - 3\sqrt{2}x + 1$ 19. $a = 3, b = -6$ 20. $a = 2$ 21. $-3, -3$

HINTS TO SOME SELECTED QUESTIONS

$$\begin{aligned} 5. \quad 5x^2 - 8x - 4 &= 5x^2 - 10x + 2x - 4 \\ &= 5x(x - 2) + 2(x - 2) = (x - 2)(5x + 2). \end{aligned}$$

$$\begin{aligned}
 6. \quad 2\sqrt{3}x^2 - 5x + \sqrt{3} &= 2\sqrt{3}x^2 - 3x - 2x + \sqrt{3} \\
 &= \sqrt{3}x(2x - \sqrt{3}) - (2x - \sqrt{3}) \\
 &= (2x - \sqrt{3})(\sqrt{3}x - 1).
 \end{aligned}$$

$$8. \quad 4x^2 - 4x + 1 = (2x - 1)^2.$$

$$9. \quad (x^2 - 5) = (x - \sqrt{5})(x + \sqrt{5}).$$

$$10. \quad 8x^2 - 4 = 4(2x^2 - 1) = 4(\sqrt{2}x - 1)(\sqrt{2}x + 1).$$

$$11. \quad 5y^2 + 10y = 5y(y + 2).$$

$$12. \quad 3x^2 - x - 4 = 3x^2 - 4x + 3x - 4 = x(3x - 4) + (3x - 4) = (3x - 4)(x + 1).$$

$$19. \quad \text{Sum of the roots} = \left(\frac{2}{3} - 3\right) = \frac{-7}{3}.$$

$$\text{Product of roots} = \frac{2}{3} \times (-3) = -2.$$

$$\therefore \left\{ \frac{-7}{a} = \frac{-7}{3} \text{ and } \frac{b}{a} = -2 \right\} \Rightarrow a = 3 \text{ and } b = -6.$$

$$20. \quad \text{Let } f(x) = 2x^2 + 2ax + 5x + 10.$$

Since $(x + a)$ is a factor of $f(x)$, we have $f(-a) = 0$.

$$\therefore 2(-a)^2 + 2a(-a) + 5(-a) + 10 = 0$$

$$\Rightarrow 2a^2 - 2a^2 - 5a + 10 = 0 \Rightarrow 5a = 10 \Rightarrow a = 2.$$

$$21. \quad \left(x - \frac{2}{3}\right) \text{ is a factor of the given polynomial and therefore, } (3x - 2) \text{ is also its factor.}$$

On dividing the given polynomial by $(3x - 2)$, we get $(x^2 + 6x + 9)$ as quotient, i.e., $(x + 3)^2 = 0 \Rightarrow x = -3$.

RELATION BETWEEN THE ZEROS AND COEFFICIENTS OF A CUBIC POLYNOMIAL

Let α, β and γ be the zeros of a cubic polynomial

$$p(x) = ax^3 + bx^2 + cx + d, \text{ where } a \neq 0.$$

Then, $(x - \alpha)$, $(x - \beta)$ and $(x - \gamma)$ are the factors of $p(x)$.

$$\begin{aligned}
 \therefore (ax^3 + bx^2 + cx + d) &= k(x - \alpha)(x - \beta)(x - \gamma) \text{ for some constant } k \\
 &= k\{x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - (\alpha\beta\gamma)\} \\
 &= kx^3 - k(\alpha + \beta + \gamma)x^2 + k(\alpha\beta + \beta\gamma + \gamma\alpha)x - k(\alpha\beta\gamma).
 \end{aligned}$$

Comparing coefficients of like powers of x on both sides, we get

$$k = a, -k(\alpha + \beta + \gamma) = b, k(\alpha\beta + \beta\gamma + \gamma\alpha) = c, -k(\alpha\beta\gamma) = d$$

$$\Rightarrow -a(\alpha + \beta + \gamma) = b, a(\alpha\beta + \beta\gamma + \gamma\alpha) = c, -a(\alpha\beta\gamma) = d \quad [\because k = a]$$

$$\Rightarrow (\alpha + \beta + \gamma) = \frac{-b}{a}, (\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a}, \alpha\beta\gamma = \frac{-d}{a}.$$

SUMMARY

I. If α , β and γ are the zeros of $p(x) = ax^3 + bx^2 + cx + d$ then

$$(i) (\alpha + \beta + \gamma) = \frac{-b}{a} \quad (ii) (\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a} \quad (iii) \alpha\beta\gamma = \frac{-d}{a}.$$

II. A cubic polynomial whose zeros are α , β and γ is given by

$$p(x) = \{x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma\}.$$

SOLVED EXAMPLES

EXAMPLE 1 Verify that 3, -1 and $-\frac{1}{3}$ are the zeros of the cubic polynomial $p(x) = 3x^3 - 5x^2 - 11x - 3$ and verify the relation between its zeros and coefficients.

SOLUTION The given polynomial is $p(x) = 3x^3 - 5x^2 - 11x - 3$.

$$\therefore p(3) = \{3 \times 3^3 - 5 \times 3^2 - 11 \times 3 - 3\} = (81 - 45 - 33 - 3) = 0;$$

$$p(-1) = \{3 \times (-1)^3 - 5 \times (-1)^2 - 11 \times (-1) - 3\} \\ = (-3 - 5 + 11 - 3) = 0;$$

$$\text{and } p\left(-\frac{1}{3}\right) = \left\{3 \times \left(-\frac{1}{3}\right)^3 - 5 \times \left(-\frac{1}{3}\right)^2 - 11 \times \left(-\frac{1}{3}\right) - 3\right\} \\ = \left\{3 \times \left(-\frac{1}{27}\right) - 5 \times \frac{1}{9} + \frac{11}{3} - 3\right\} = \left(-\frac{1}{9} - \frac{5}{9} + \frac{11}{3} - 3\right) \\ = \frac{(-1 - 5 + 33 - 27)}{9} = 0.$$

$$\therefore 3, -1 \text{ and } -\frac{1}{3} \text{ are the zeros of } p(x).$$

Let $\alpha = 3, \beta = -1$ and $\gamma = -\frac{1}{3}$. Then,

$$(\alpha + \beta + \gamma) = \left(3 - 1 - \frac{1}{3}\right) = \frac{5}{3} = \frac{-(\text{coefficient of } x^2)}{(\text{coefficient of } x^3)};$$

$$(\alpha\beta + \beta\gamma + \gamma\alpha) = \left(-3 + \frac{1}{3} - 1\right) = \frac{-11}{3} = \frac{(\text{coefficient of } x)}{(\text{coefficient of } x^3)};$$

$$\alpha\beta\gamma = \left\{3 \times (-1) \times \left(-\frac{1}{3}\right)\right\} = 1 = \frac{3}{3} = \frac{-(\text{constant term})}{(\text{coefficient of } x^3)}.$$

EXAMPLE 2 Find a cubic polynomial with the sum of its zeros, sum of the products of its zeros taken two at a time and the product of its zeros as 2, -7 and -14 respectively.

SOLUTION Let α, β, γ be the zeros of the required polynomial. Then,

$$\alpha + \beta + \gamma = 2, \alpha\beta + \beta\gamma + \gamma\alpha = -7 \text{ and } \alpha\beta\gamma = -14.$$

So, the required polynomial is

$$\begin{aligned} p(x) &= x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma \\ &= x^3 - 2x^2 - 7x - (-14) = x^3 - 2x^2 - 7x + 14. \end{aligned}$$

EXAMPLE 3 If the zeros of the polynomial $x^3 - 3x^2 + x + 1$ are $(a - b)$, a , $(a + b)$, find a and b .

SOLUTION Given polynomial is $f(x) = x^3 - 3x^2 + x + 1$.

Let $\alpha = (a - b)$, $\beta = a$ and $\gamma = (a + b)$. Then,

$$\alpha + \beta + \gamma = 3 \Rightarrow (a - b) + a + (a + b) = 3 \Rightarrow 3a = 3 \Rightarrow a = 1.$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 1 \Rightarrow a(a - b) + a(a + b) + (a + b)(a - b) = 1$$

$$\Rightarrow 3a^2 - b^2 = 1 \Rightarrow (3 \times 1^2) - b^2 = 1$$

$$\Rightarrow b^2 = 2 \Rightarrow b = \pm\sqrt{2}.$$

$$\therefore a = 1 \text{ and } b = \pm\sqrt{2}.$$

EXAMPLE 4 Find a cubic polynomial whose zeros are $3, \frac{1}{2}$ and -1 .

SOLUTION Let $\alpha = 3, \beta = \frac{1}{2}$ and $\gamma = -1$. Then,

$$(\alpha + \beta + \gamma) = \left(3 + \frac{1}{2} - 1\right) = \frac{5}{2},$$

$$(\alpha\beta + \beta\gamma + \gamma\alpha) = \left(\frac{3}{2} - \frac{1}{2} - 3\right) = \frac{-4}{2} = -2$$

$$\text{and } \alpha\beta\gamma = \left\{3 \times \frac{1}{2} \times (-1)\right\} = \frac{-3}{2}.$$

Hence, the required polynomial is

$$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma = x^3 - \frac{5}{2}x^2 - 2x + \frac{3}{2}.$$

Thus, $2x^3 - 5x^2 - 4x + 3$ is the desired polynomial.

DIVISION ALGORITHM FOR POLYNOMIALS

If $f(x)$ and $g(x)$ are any two polynomials with $g(x) \neq 0$ then we can find polynomials $q(x)$ and $r(x)$ such that

$$f(x) = q(x) \times g(x) + r(x),$$

where $r(x) = 0$ or $\{\text{degree of } r(x)\} < \{\text{degree of } g(x)\}$.

We may write it as

$$\text{Dividend} = (\text{Quotient} \times \text{Divisor}) + \text{Remainder}.$$

SOME MORE EXAMPLES

EXAMPLE 5 Divide $3 - x + 2x^2$ by $(2 - x)$ and verify the division algorithm.

SOLUTION First we write the terms of dividend and divisor in decreasing order of their degrees and then perform the division as shown below.

$$\begin{array}{r}
 -x + 2 \overline{) 2x^2 - x + 3} \quad (-2x - 3) \\
 \underline{2x^2 - 4x} \\
 3x + 3 \\
 \underline{3x - 6} \\
 9
 \end{array}$$

Clearly, degree (9) = 0 < degree $(-x + 2)$.

$\therefore$ quotient = $(-2x - 3)$ and remainder = 9

$\Rightarrow$ (quotient $\times$ divisor) + remainder

$$\begin{aligned}
 &= (-2x - 3)(-x + 2) + 9 \\
 &= 2x^2 - 4x + 3x - 6 + 9 \\
 &= 2x^2 - x + 3 \\
 &= \text{dividend.}
 \end{aligned}$$

Thus, (quotient $\times$ divisor) + remainder = dividend.

Hence, the division algorithm is verified.

EXAMPLE 6 Divide $5x^3 - 13x^2 + 21x - 14$ by $(3 - 2x + x^2)$ and verify the division algorithm.

SOLUTION First we write the given polynomials in standard form in decreasing order of degrees and then perform the division as shown below.

$$\begin{array}{r}
 x^2 - 2x + 3 \overline{) 5x^3 - 13x^2 + 21x - 14} \quad (5x - 3) \\
 \underline{5x^3 - 10x^2 + 15x} \\
 -3x^2 + 6x - 14 \\
 \underline{-3x^2 + 6x - 9} \\
 -5
 \end{array}$$

Clearly, degree $(-5) = 0 < \text{degree } (x^2 - 2x + 3)$.

$\therefore$ quotient = $(5x - 3)$ and remainder = -5

$\Rightarrow$ (quotient $\times$ divisor) + remainder

$$= (5x - 3)(x^2 - 2x + 3) - 5$$

$$\begin{aligned}
 &= 5x^3 - 10x^2 + 15x - 3x^2 + 6x - 9 - 5 \\
 &= 5x^3 - 13x^2 + 21x - 14 \\
 &= \text{dividend.}
 \end{aligned}$$

Thus, (quotient $\times$ divisor) + remainder = dividend.

Hence, the division algorithm is verified.

EXAMPLE 7 What real number should be subtracted from the polynomial $(3x^3 + 10x^2 - 14x + 9)$ so that $(3x - 2)$ divides it exactly? [CBSE 2009C]

SOLUTION On dividing $(3x^3 + 10x^2 - 14x + 9)$ by $(3x - 2)$, we get

$$\begin{array}{r}
 3x - 2 \overline{) 3x^3 + 10x^2 - 14x + 9} \quad (x^2 + 4x - 2) \\
 \underline{3x^3 - 2x^2} \\
 12x^2 - 14x + 9 \\
 \underline{12x^2 - 8x} \\
 -6x + 9 \\
 \underline{-6x + 4} \\
 5
 \end{array}$$

Required number to be subtracted = 5.

EXAMPLE 8 If the polynomial $(x^4 + 2x^3 + 8x^2 + 12x + 18)$ is divided by another polynomial $(x^2 + 5)$, the remainder comes out to be $(px + q)$. Find the values of p and q . [CBSE 2009]

SOLUTION Let $f(x) = (x^4 + 2x^3 + 8x^2 + 12x + 18)$ and $g(x) = (x^2 + 5)$.

On dividing $f(x)$ by $g(x)$, we get

$$\begin{array}{r}
 x^2 + 5 \overline{) x^4 + 2x^3 + 8x^2 + 12x + 18} \quad (x^2 + 2x + 3) \\
 \underline{x^4 + 5x^2} \\
 2x^3 + 3x^2 + 12x + 18 \\
 \underline{2x^3 + 10x} \\
 3x^2 + 2x + 18 \\
 \underline{3x^2 + 15} \\
 2x + 3
 \end{array}$$

Now, $px + q = 2x + 3 \Rightarrow p = 2$ and $q = 3$.

EXAMPLE 9 On dividing $(x^3 - 3x^2 + x + 2)$ by a polynomial $g(x)$, the quotient and remainder are $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

[CBSE 2009C]

SOLUTION Let $f(x) = (x^3 - 3x^2 + x + 2)$, $q(x) = (x - 2)$ and $r(x) = (-2x + 4)$.

Then, $f(x) = g(x) \cdot q(x) + r(x)$

$$\Rightarrow g(x) = \frac{\{f(x) - r(x)\}}{q(x)}. \quad \dots (i)$$

$$\begin{aligned} \text{Now, } \{f(x) - r(x)\} &= (x^3 - 3x^2 + x + 2) - (-2x + 4) \\ &= (x^3 - 3x^2 + 3x - 2). \end{aligned}$$

$$\therefore g(x) = \frac{(x^3 - 3x^2 + 3x - 2)}{(x - 2)}. \quad [\text{using (i)}]$$

On dividing $(x^3 - 3x^2 + 3x - 2)$ by $(x - 2)$, we get $g(x)$.

$$\begin{array}{r} x-2 \overline{) x^3 - 3x^2 + 3x - 2} \\ \underline{x^3 - 2x^2} \\ -x^2 + 3x - 2 \\ \underline{-x^2 + 2x} \\ x - 2 \\ \underline{x - 2} \\ \times \end{array}$$

$$\therefore g(x) = (x^2 - x + 1).$$

AN IMPORTANT NOTE

If α is a zero of the polynomial $f(x)$ then $(x - \alpha)$ is a divisor of $f(x)$.

EXAMPLE 10 It being given that 1 is a zero of the polynomial $(7x - x^3 - 6)$, find its other zeros.

SOLUTION Let $f(x) = -x^3 + 7x - 6$.

Since 1 is a zero of $f(x)$, so $(x - 1)$ is a factor of $f(x)$.

On dividing $f(x)$ by $(x - 1)$, we get

$$\begin{array}{r} x-1 \overline{) -x^3 + 7x - 6} \\ \underline{-x^3 + x^2} \\ -x^2 + 7x - 6 \\ \underline{-x^2 + x} \\ 6x - 6 \\ \underline{6x - 6} \\ \times \end{array}$$

$$\begin{aligned} \therefore f(x) &= (-x^3 + 7x - 6) = (x - 1)(-x^2 - x + 6) \\ &= -(x - 1)(x^2 + x - 6) = -(x - 1)(x^2 + 3x - 2x - 6) \\ &= (1 - x)[x(x + 3) - 2(x + 3)] = (1 - x)(x + 3)(x - 2). \end{aligned}$$

$$\begin{aligned}
 \therefore f(x) = 0 &\Rightarrow (1-x)(x+3)(x-2) = 0 \\
 &\Rightarrow (1-x) = 0 \text{ or } (x+3) = 0 \text{ or } (x-2) = 0 \\
 &\Rightarrow x = 1 \text{ or } x = -3 \text{ or } x = 2.
 \end{aligned}$$

Thus, the other zeros are -3 and 2 .

EXAMPLE 11 Obtain all zeros of the polynomial $(2x^3 - 4x - x^2 + 2)$, if two of its zeros are $\sqrt{2}$ and $-\sqrt{2}$. [CBSE 2008C]

SOLUTION The given polynomial is $f(x) = 2x^3 - x^2 - 4x + 2$.

Since $\sqrt{2}$ and $-\sqrt{2}$ are the zeros of $f(x)$, it follows that each one of $(x - \sqrt{2})$ and $(x + \sqrt{2})$ is a factor of $f(x)$.

Consequently, $(x - \sqrt{2})(x + \sqrt{2}) = (x^2 - 2)$ is a factor of $f(x)$.

On dividing $f(x) = 2x^3 - x^2 - 4x + 2$ by $(x^2 - 2)$, we get

$$\begin{array}{r}
 x^2 - 2 \overline{) 2x^3 - x^2 - 4x + 2} \quad (2x - 1) \\
 \underline{2x^3 - 4x} \\
 -x^2 + 2 \\
 \underline{-x^2 + 2} \\
 \hline
 x
 \end{array}$$

$$\begin{aligned}
 \therefore f(x) = 0 &\Rightarrow (x^2 - 2)(2x - 1) = 0 \\
 &\Rightarrow (x - \sqrt{2})(x + \sqrt{2})(2x - 1) = 0 \\
 &\Rightarrow (x - \sqrt{2}) = 0 \text{ or } (x + \sqrt{2}) = 0 \text{ or } (2x - 1) = 0 \\
 &\Rightarrow x = \sqrt{2} \text{ or } x = -\sqrt{2} \text{ or } x = \frac{1}{2}.
 \end{aligned}$$

Hence, all zeros of $f(x)$ are $\sqrt{2}$, $-\sqrt{2}$ and $\frac{1}{2}$.

EXAMPLE 12 If two zeros of the polynomial $(x^4 - 6x^3 - 26x^2 + 138x - 35)$ are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$, find other zeros.

SOLUTION Let $f(x) = x^4 - 6x^3 - 26x^2 + 138x - 35$.

Let $\alpha = (2 + \sqrt{3})$ and $\beta = (2 - \sqrt{3})$. Then,

$$(\alpha + \beta) = 4 \text{ and } \alpha\beta = (4 - 3) = 1.$$

So, the quadratic polynomial whose roots are α and β is given by

$$x^2 - (\alpha + \beta)x + \alpha\beta = (x^2 - 4x + 1).$$

$\therefore (x^2 - 4x + 1)$ is a factor of $f(x)$.

On dividing $f(x)$ by $(x^2 - 4x + 1)$, we get

$$\begin{array}{r}
 x^2 - 4x + 1 \overline{) x^4 - 6x^3 - 26x^2 + 138x - 35} \quad (x^2 - 2x - 35) \\
 \underline{x^4 - 4x^3 + x^2} \\
 -2x^3 - 27x^2 + 138x - 35 \\
 \underline{-2x^3 + 8x^2 - 2x} \\
 -35x^2 + 140x - 35 \\
 \underline{-35x^2 + 140x - 35} \\
 \times
 \end{array}$$

$$\therefore f(x) = (x^2 - 4x + 1)(x^2 - 2x - 35).$$

$\therefore$ the other two zeros of $f(x)$ are given by $(x^2 - 2x - 35) = 0$.

$$\text{Now, } x^2 - 2x - 35 = 0 \Rightarrow x^2 - 7x + 5x - 35 = 0$$

$$\Rightarrow x(x - 7) + 5(x - 7) = 0$$

$$\Rightarrow (x - 7)(x + 5) = 0$$

$$\Rightarrow x - 7 = 0 \text{ or } x + 5 = 0$$

$$\Rightarrow x = 7 \text{ or } x = -5.$$

Hence, the other two zeros of $f(x)$ are 7 and -5.

EXAMPLE 13 Obtain all zeros of $(3x^4 - 15x^3 + 13x^2 + 25x - 30)$, if two of its zeros are $\sqrt{\frac{5}{3}}$ and $-\sqrt{\frac{5}{3}}$. [CBSE 2009C]

SOLUTION The given polynomial is $f(x) = (3x^4 - 15x^3 + 13x^2 + 25x - 30)$.

Since $\sqrt{\frac{5}{3}}$ and $-\sqrt{\frac{5}{3}}$ are the zeros of $f(x)$, it follows that each one of $(x - \sqrt{\frac{5}{3}})$ and $(x + \sqrt{\frac{5}{3}})$ is a factor of $f(x)$.

$$\therefore \left(x - \frac{\sqrt{5}}{\sqrt{3}}\right)\left(x + \frac{\sqrt{5}}{\sqrt{3}}\right) = \left(x^2 - \frac{5}{3}\right) = \frac{(3x^2 - 5)}{3} \text{ is a factor of } f(x).$$

Consequently, $(3x^2 - 5)$ is a factor of $f(x)$.

On dividing $f(x)$ by $(3x^2 - 5)$, we get

$$\begin{array}{r}
 3x^2 - 5 \overline{) 3x^4 - 15x^3 + 13x^2 + 25x - 30} \quad (x^2 - 5x + 6) \\
 \underline{3x^4 - 5x^2} \\
 -15x^3 + 18x^2 + 25x - 30 \\
 \underline{-15x^3 + 25x} \\
 18x^2 - 30 \\
 \underline{18x^2 - 30} \\
 \times
 \end{array}$$

$$\begin{aligned}
 \therefore f(x) &= 3x^4 - 15x^3 + 13x^2 + 25x - 30 \\
 &= (3x^2 - 5)(x^2 - 5x + 6) \\
 &= (\sqrt{3}x + \sqrt{5})(\sqrt{3}x - \sqrt{5})(x - 2)(x - 3). \\
 \therefore f(x) = 0 &\Rightarrow (\sqrt{3}x + \sqrt{5}) = 0 \text{ or } (\sqrt{3}x - \sqrt{5}) = 0 \\
 &\text{or } (x - 2) = 0 \text{ or } (x - 3) = 0 \\
 &\Rightarrow x = -\sqrt{\frac{5}{3}} \text{ or } x = \sqrt{\frac{5}{3}} \text{ or } x = 2 \text{ or } x = 3.
 \end{aligned}$$

Hence, all zeros of $f(x)$ are $\sqrt{\frac{5}{3}}, -\sqrt{\frac{5}{3}}, 2$ and 3 .

EXERCISE 2B

- Verify that 3, -2, 1 are the zeros of the cubic polynomial $p(x) = x^3 - 2x^2 - 5x + 6$ and verify the relation between its zeros and coefficients.
- Verify that 5, -2 and $\frac{1}{3}$ are the zeros of the cubic polynomial $p(x) = 3x^3 - 10x^2 - 27x + 10$ and verify the relation between its zeros and coefficients.
- Find a cubic polynomial whose zeros are 2, -3 and 4.
- Find a cubic polynomial whose zeros are $\frac{1}{2}, 1$ and -3.
- Find a cubic polynomial with the sum, sum of the product of its zeros taken two at a time, and the product of its zeros as 5, -2 and -24 respectively.

Find the quotient and the remainder when

- $f(x) = x^3 - 3x^2 + 5x - 3$ is divided by $g(x) = x^2 - 2$.
- $f(x) = x^4 - 3x^2 + 4x + 5$ is divided by $g(x) = x^2 + 1 - x$.
- $f(x) = x^4 - 5x + 6$ is divided by $g(x) = 2 - x^2$.
- By actual division, show that $x^2 - 3$ is a factor of $2x^4 + 3x^3 - 2x^2 - 9x - 12$.
- On dividing $3x^3 + x^2 + 2x + 5$ by a polynomial $g(x)$, the quotient and remainder are $(3x - 5)$ and $(9x + 10)$ respectively. Find $g(x)$.

HINT $g(x) = \frac{(3x^3 + x^2 + 2x + 5) - (9x + 10)}{(3x - 5)}.$

- Verify division algorithm for the polynomials $f(x) = 8 + 20x + x^2 - 6x^3$ and $g(x) = 2 + 5x - 3x^2$.
- It is given that -1 is one of the zeros of the polynomial $x^3 + 2x^2 - 11x - 12$. Find all the zeros of the given polynomial.

13. If 1 and -2 are two zeros of the polynomial $(x^3 - 4x^2 - 7x + 10)$, find its third zero.
14. If 3 and -3 are two zeros of the polynomial $(x^4 + x^3 - 11x^2 - 9x + 18)$, find all the zeros of the given polynomial.
15. If 2 and -2 are two zeros of the polynomial $(x^4 + x^3 - 34x^2 - 4x + 120)$, find all the zeros of the given polynomial. [CBSE 2008]
16. Find all the zeros of $(x^4 + x^3 - 23x^2 - 3x + 60)$, if it is given that two of its zeros are $\sqrt{3}$ and $-\sqrt{3}$. [CBSE 2009C]
17. Find all the zeros of $(2x^4 - 3x^3 - 5x^2 + 9x - 3)$, it being given that two of its zeros are $\sqrt{3}$ and $-\sqrt{3}$.
18. Obtain all other zeros of $(x^4 + 4x^3 - 2x^2 - 20x - 15)$ if two of its zeros are $\sqrt{5}$ and $-\sqrt{5}$. [CBSE 2009C]
19. Find all the zeros of the polynomial $(2x^4 - 11x^3 + 7x^2 + 13x - 7)$, it being given that two of its zeros are $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$.

ANSWERS (EXERCISE 2B)

3. $x^3 - 3x^2 - 10x + 24$
4. $2x^3 + 3x^2 - 8x + 3$
5. $x^3 - 5x^2 - 2x + 24$
6. $q(x) = (x - 3), r(x) = (7x - 9)$
7. $q(x) = x^2 + x - 3, r(x) = 8$
8. $q(x) = -x^2 - 2, r(x) = -5x + 10$
10. $g(x) = x^2 + 2x + 1$
12. $-4, -1, 3$
13. 5
14. $1, -2, 3, -3$
15. $2, -2, -6, 5$
16. $\sqrt{3}, -\sqrt{3}, 4, -5$
17. $\sqrt{3}, -\sqrt{3}, 1, \frac{1}{2}$
18. $-1, -3$
19. $(3 + \sqrt{2}), (3 - \sqrt{2}), \frac{1}{2}, -1$

EXERCISE 2C

Very-Short-Answer Questions

1. If one zero of the polynomial $x^2 - 4x + 1$ is $(2 + \sqrt{3})$, write the other zero. [CBSE 2010]
2. Find the zeros of the polynomial $x^2 + x - p(p + 1)$. [CBSE 2011]
3. Find the zeros of the polynomial $x^2 - 3x - m(m + 3)$. [CBSE 2011]
4. If α, β are the zeros of a polynomial such that $\alpha + \beta = 6$ and $\alpha\beta = 4$ then write the polynomial. [CBSE 2010]
5. If one zero of the quadratic polynomial $kx^2 + 3x + k$ is 2 then find the value of k .
6. If 3 is a zero of the polynomial $2x^2 + x + k$, find the value of k . [CBSE 2010]

7. If -4 is a zero of the polynomial $x^2 - x - (2k + 2)$ then find the value of k .
[CBSE 2009]
8. If 1 is a zero of the polynomial $ax^2 - 3(a - 1)x - 1$ then find the value of a .
9. If -2 is a zero of the polynomial $3x^2 + 4x + 2k$ then find the value of k .
[CBSE 2010]
10. Write the zeros of the polynomial $x^2 - x - 6$.
[CBSE 2008]
11. If the sum of the zeros of the quadratic polynomial $kx^2 - 3x + 5$ is 1 , write the value of k .
12. If the product of the zeros of the quadratic polynomial $x^2 - 4x + k$ is 3 then write the value of k .
13. If $(x + a)$ is a factor of $(2x^2 + 2ax + 5x + 10)$, find the value of a . [CBSE 2010]
14. If $(a - b)$, a and $(a + b)$ are zeros of the polynomial $2x^3 - 6x^2 + 5x - 7$, write the value of a .
15. If $x^3 + x^2 - ax + b$ is divisible by $(x^2 - x)$, write the values of a and b .
16. If α and β are the zeros of the polynomial $2x^2 + 7x + 5$, write the value of $\alpha + \beta + \alpha\beta$.
[CBSE 2010]
17. State division algorithm for polynomials.
18. The sum of the zeros and the product of zeros of a quadratic polynomial are $-\frac{1}{2}$ and -3 respectively. Write the polynomial.

Short-Answer Questions

19. Write the zeros of the quadratic polynomial $f(x) = 6x^2 - 3$.
20. Write the zeros of the quadratic polynomial $f(x) = 4\sqrt{3}x^2 + 5x - 2\sqrt{3}$.
21. If α and β are the zeros of the polynomial $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of k .
22. If α and β are the zeros of the polynomial $f(x) = 6x^2 + x - 2$, find the value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$.
23. If α and β are the zeros of the polynomial $f(x) = 5x^2 - 7x + 1$, find the value of $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$.
24. If α and β are the zeros of the polynomial $f(x) = x^2 + x - 2$, find the value of $\left(\frac{1}{\alpha} - \frac{1}{\beta}\right)$.

25. If the zeros of the polynomial $f(x) = x^3 - 3x^2 + x + 1$ are $(a - b)$, a and $(a + b)$, find a and b .

ANSWERS (EXERCISE 2C)

1. $(2 - \sqrt{3})$ 2. $-(p + 1)$ and p 3. $(m + 3)$ and $-m$ 4. $x^2 - 6x + 4$
 5. $k = \frac{-6}{5}$ 6. $k = -21$ 7. $k = 9$ 8. $a = 1$ 9. $k = -2$
 10. 3 and -2 11. $k = 3$ 12. $k = 3$ 13. $a = 2$ 14. $a = 1$
 15. $a = 2$ and $b = 0$ 16. -1 18. $x^2 + \frac{1}{2}x - 3$ 19. $x = \frac{1}{\sqrt{2}}$ or $x = \frac{-1}{\sqrt{2}}$
 20. $x = \frac{-2}{\sqrt{3}}$ or $x = \frac{\sqrt{3}}{4}$ 21. $k = 6$ 22. $\frac{-25}{12}$ 23. 7 24. $\frac{-3}{2}$
 25. $a = 1$ and $b = \pm\sqrt{2}$

HINTS TO SOME SELECTED QUESTIONS

1. Let the other zero be α .

$$\text{Then, sum of zeros} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)} = \frac{-(-4)}{1} = 4.$$

$$\therefore (2 + \sqrt{3}) + \alpha = 4 \Rightarrow \alpha = (2 - \sqrt{3}).$$

$$\begin{aligned} 2. \quad x^2 + x - p(p + 1) &= x^2 + (p + 1)x - px - p(p + 1) \\ &= x\{x + (p + 1)\} - p\{x + (p + 1)\} \\ &= \{x + (p + 1)\}\{x - p\}. \end{aligned}$$

So, its zeros are $-(p + 1)$ and p .

$$\begin{aligned} 3. \quad x^2 - 3x - m(m + 3) &= x^2 - (m + 3)x + mx - m(m + 3) \\ &= x\{x - (m + 3)\} + m\{x - (m + 3)\} \\ &= \{x - (m + 3)\}\{x + m\}. \end{aligned}$$

So, its zeros are $(m + 3)$ and $-m$.

$$4. \text{ Required polynomial is } x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - 6x + 4.$$

$$5. \text{ Since } 2 \text{ is a zero of } kx^2 + 3x + k, \text{ we have } 4k + 6 + k = 0.$$

$$\therefore 5k = -6 \Rightarrow k = \frac{-6}{5}.$$

$$6. \text{ Since } 3 \text{ is a zero of } 2x^2 + x + k, \text{ we have } 18 + 3 + k = 0 \Rightarrow k = -21.$$

$$7. \text{ Since } -4 \text{ is a zero of } x^2 - x - (2k + 2), \text{ we have } 16 + 4 - 2k - 2 = 0.$$

$$\therefore 2k = 18 \Rightarrow k = 9.$$

$$8. \text{ Since } 1 \text{ is a zero of } ax^2 - 3(a - 1)x - 1, \text{ we have } a - 3(a - 1) - 1 = 0.$$

$$\therefore a - 3a + 3 - 1 = 0 \Rightarrow 2a = 2 \Rightarrow a = 1.$$

$$9. \text{ Since } -2 \text{ is a zero of } 3x^2 + 4x + 2k, \text{ we have } 12 - 8 + 2k = 0.$$

$$\therefore 2k = -4 \Rightarrow k = -2.$$

$$10. x^2 - x - 6 = x^2 - 3x + 2x - 6 = x(x-3) + 2(x-3) = (x-3)(x+2).$$

So, its zeros are 3 and -2.

$$11. \text{Sum of the zeros} = \frac{-(\text{coefficient of } x)}{(\text{coefficient of } x^2)} = \frac{3}{k}.$$

$$\therefore \frac{3}{k} = 1 \Rightarrow k = 3.$$

$$12. \text{Product of the zero} = \frac{\text{constant term}}{\text{coefficient of } x^2} = \frac{k}{1}. \text{ So, } k = 3.$$

$$13. \text{Let } f(x) = 2x^2 + 2ax + 5x + 10. \text{ Then, } f(-a) = 0.$$

$$\therefore 2a^2 - 2a^2 - 5a + 10 = 0 \Rightarrow 5a = 10 \Rightarrow a = 2.$$

$$14. (\alpha + \beta + \gamma) = \frac{-(\text{coefficient of } x^2)}{(\text{coefficient of } x^3)} = \frac{-(-6)}{2} = 3.$$

$$\therefore (a-b) + a + (a+b) = 3 \Rightarrow 3a = 3 \Rightarrow a = 1.$$

$$15. \text{On dividing } x^3 + x^2 - ax + b \text{ by } x^2 - x, \text{ we get}$$

$$\begin{array}{r} x^2 - x \overline{) x^3 + x^2 - ax + b} \\ \underline{x^3 - x^2} \\ 2x^2 - ax \\ \underline{2x^2 - 2x} \\ (2-a)x + b \end{array}$$

$$\therefore (2-a=0 \text{ and } b=0) \Rightarrow a=2 \text{ and } b=0.$$

$$16. \text{Clearly, } \alpha + \beta = \frac{-7}{2} \text{ and } \alpha\beta = \frac{5}{2}.$$

$$\therefore \alpha + \beta + \alpha\beta = \left(\frac{-7}{2} + \frac{5}{2}\right) = \frac{(-7+5)}{2} = \frac{-2}{2} = -1.$$

$$18. \alpha + \beta = \frac{-1}{2} \text{ and } \alpha\beta = -3.$$

$$\text{So, the polynomial is } x^2 - (\alpha + \beta)x + \alpha\beta = x^2 + \frac{1}{2}x - 3.$$

$$19. 6x^2 - 3 = 3(2x^2 - 1) = 3(\sqrt{2}x - 1)(\sqrt{2}x + 1).$$

$$\text{So, its zeros are } \frac{1}{\sqrt{2}} \text{ and } \frac{-1}{\sqrt{2}}.$$

$$20. f(x) = 4\sqrt{3}x^2 + 5x - 2\sqrt{3}$$

$$= 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} = 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2)$$

$$= (\sqrt{3}x + 2)(4x - \sqrt{3}).$$

$$\text{So, its zeros are } \frac{-2}{\sqrt{3}}, \frac{\sqrt{3}}{4}.$$

$$21. (\alpha + \beta) = 5 \text{ and } \alpha\beta = k. \text{ Also, } \alpha - \beta = 1 \text{ (given).}$$

$$\therefore (\alpha + \beta)^2 - (\alpha - \beta)^2 = 4\alpha\beta \Rightarrow 5^2 - 1^2 = 4\alpha\beta$$

$$\therefore 4\alpha\beta = (25 - 1) = 24 \Rightarrow \alpha\beta = 6.$$

Hence, $k = 6$.

$$22. \alpha + \beta = \frac{-1}{6} \text{ and } \alpha\beta = \frac{-2}{6} = \frac{-1}{3}.$$

$$\therefore \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) = \frac{(\alpha^2 + \beta^2)}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{1}{36} + \frac{2}{3} \right)}{\left(\frac{-1}{3} \right)} = \left(\frac{25}{36} \times \frac{3}{-1} \right) = \frac{-25}{12}.$$

$$23. \alpha + \beta = \frac{7}{5} \text{ and } \alpha\beta = \frac{1}{5}.$$

$$\therefore \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) = \frac{(\alpha + \beta)}{\alpha\beta} = \left(\frac{7}{5} \times \frac{5}{1} \right) = 7.$$

$$24. \alpha + \beta = -1 \text{ and } \alpha\beta = -2.$$

$$(\beta + \alpha)^2 - (\beta - \alpha)^2 = 4\alpha\beta$$

$$\Rightarrow (\beta - \alpha)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (-1)^2 - 4 \times (-2) = 9$$

$$\Rightarrow \beta - \alpha = 3.$$

$$\therefore \left(\frac{1}{\alpha} - \frac{1}{\beta} \right) = \frac{(\beta - \alpha)}{\alpha\beta} = \frac{3}{-2} = \frac{-3}{2}.$$

$$25. \alpha + \beta + \gamma = 3, \alpha\beta + \beta\gamma + \gamma\alpha = 1 \text{ and } \alpha\beta\gamma = -1$$

$$\therefore (a - b) + a + (a + b) = 3 \Rightarrow 3a = 3 \Rightarrow a = 1$$

$$\text{and } (a - b) \times a \times (a + b) = -1 \Rightarrow a(a^2 - b^2) = -1$$

$$\Rightarrow 1 - b^2 = -1 \Rightarrow b^2 = 2 \Rightarrow b = \pm\sqrt{2}.$$

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

1. Which of the following is a polynomial?

(a) $x^2 - 5x + 4\sqrt{x} + 3$

(b) $x^{3/2} - x + x^{1/2} + 1$

(c) $\sqrt{x} + \frac{1}{\sqrt{x}}$

(d) $\sqrt{2}x^2 - 3\sqrt{3}x + \sqrt{6}$

2. Which of the following is not a polynomial?

(a) $\sqrt{3}x^2 - 2\sqrt{3}x + 5$

(b) $9x^2 - 4x + \sqrt{2}$

(c) $\frac{3}{2}x^3 + 6x^2 - \frac{1}{\sqrt{2}}x - 8$

(d) $x + \frac{3}{x}$

3. The zeros of the polynomial $x^2 - 2x - 3$ are

(a) $-3, 1$

(b) $-3, -1$

(c) $3, -1$

(d) $3, 1$

4. The zeros of the polynomial $x^2 - \sqrt{2}x - 12$ are

(a) $\sqrt{2}, -\sqrt{2}$

(b) $3\sqrt{2}, -2\sqrt{2}$

(c) $-3\sqrt{2}, 2\sqrt{2}$

(d) $3\sqrt{2}, 2\sqrt{2}$

5. The zeros of the polynomial $4x^2 + 5\sqrt{2}x - 3$ are

(a) $-3\sqrt{2}, \sqrt{2}$

(b) $-3\sqrt{2}, \frac{\sqrt{2}}{2}$

(c) $\frac{-3\sqrt{2}}{2}, \frac{\sqrt{2}}{4}$

(d) none of these

6. The zeros of the polynomial $x^2 + \frac{1}{6}x - 2$ are
(a) $-3, 4$ (b) $\frac{-3}{2}, \frac{4}{3}$ (c) $\frac{-4}{3}, \frac{3}{2}$ (d) none of these
7. The zeros of the polynomial $7x^2 - \frac{11x}{3} - \frac{2}{3}$ are
(a) $\frac{2}{3}, \frac{-1}{7}$ (b) $\frac{2}{7}, \frac{-1}{3}$ (c) $\frac{-2}{3}, \frac{1}{7}$ (d) none of these
8. The sum and the product of the zeros of a quadratic polynomial are 3 and -10 respectively. The quadratic polynomial is
(a) $x^2 - 3x + 10$ (b) $x^2 + 3x - 10$ (c) $x^2 - 3x - 10$ (d) $x^2 + 3x + 10$
9. A quadratic polynomial whose zeros are 5 and -3 , is
(a) $x^2 + 2x - 15$ (b) $x^2 - 2x + 15$ (c) $x^2 - 2x - 15$ (d) none of these
10. A quadratic polynomial whose zeros are $\frac{3}{5}$ and $\frac{-1}{2}$, is
(a) $10x^2 + x + 3$ (b) $10x^2 + x - 3$ (c) $10x^2 - x + 3$ (d) $10x^2 - x - 3$
11. The zeros of the quadratic polynomial $x^2 + 88x + 125$ are
(a) both positive (b) both negative
(c) one positive and one negative (d) both equal
12. If α and β are the zeros of $x^2 + 5x + 8$ then the value of $(\alpha + \beta)$ is
(a) 5 (b) -5 (c) 8 (d) -8
13. If α and β are the zeros of $2x^2 + 5x - 9$ then the value of $\alpha\beta$ is
(a) $\frac{-5}{2}$ (b) $\frac{5}{2}$ (c) $\frac{-9}{2}$ (d) $\frac{9}{2}$
14. If one zero of the quadratic polynomial $kx^2 + 3x + k$ is 2 then the value of k is
(a) $\frac{5}{6}$ (b) $\frac{-5}{6}$ (c) $\frac{6}{5}$ (d) $\frac{-6}{5}$
15. If one zero of the quadratic polynomial $(k-1)x^2 + kx + 1$ is -4 then the value of k is
(a) $\frac{-5}{4}$ (b) $\frac{5}{4}$ (c) $\frac{-4}{3}$ (d) $\frac{4}{3}$
16. If -2 and 3 are the zeros of the quadratic polynomial $x^2 + (a+1)x + b$ then
(a) $a = -2, b = 6$ (b) $a = 2, b = -6$
(c) $a = -2, b = -6$ (d) $a = 2, b = 6$
17. If one zero of $3x^2 + 8x + k$ be the reciprocal of the other then $k = ?$
(a) 3 (b) -3 (c) $\frac{1}{3}$ (d) $\frac{-1}{3}$

18. If the sum of the zeros of the quadratic polynomial $kx^2 + 2x + 3k$ is equal to the product of its zeros then $k = ?$
(a) $\frac{1}{3}$ (b) $\frac{-1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{-2}{3}$
19. If α, β are the zeros of the polynomial $x^2 + 6x + 2$ then $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = ?$
(a) 3 (b) -3 (c) 12 (d) -12
20. If α, β, γ are the zeros of the polynomial $x^3 - 6x^2 - x + 30$ then $(\alpha\beta + \beta\gamma + \gamma\alpha) = ?$
(a) -1 (b) 1 (c) -5 (d) 30
21. If α, β, γ are the zeros of the polynomial $2x^3 + x^2 - 13x + 6$ then $\alpha\beta\gamma = ?$
(a) -3 (b) 3 (c) $\frac{-1}{2}$ (d) $\frac{-13}{2}$
22. If α, β, γ be the zeros of the polynomial $p(x)$ such that $(\alpha + \beta + \gamma) = 3$, $(\alpha\beta + \beta\gamma + \gamma\alpha) = -10$ and $\alpha\beta\gamma = -24$ then $p(x) = ?$
(a) $x^3 + 3x^2 - 10x + 24$ (b) $x^3 + 3x^2 + 10x - 24$
(c) $x^3 - 3x^2 - 10x + 24$ (d) None of these
23. If two of the zeros of the cubic polynomial $ax^3 + bx^2 + cx + d$ are 0 then the third zero is
(a) $\frac{-b}{a}$ (b) $\frac{b}{a}$ (c) $\frac{c}{a}$ (d) $\frac{-d}{a}$
24. If one of the zeros of the cubic polynomial $ax^3 + bx^2 + cx + d$ is 0 then the product of the other two zeros is
(a) $\frac{-c}{a}$ (b) $\frac{c}{a}$ (c) 0 (d) $\frac{-b}{a}$
25. If one of the zeros of the cubic polynomial $x^3 + ax^2 + bx + c$ is -1 then the product of the other two zeros is
(a) $a - b - 1$ (b) $b - a - 1$ (c) $1 - a + b$ (d) $1 + a - b$
26. If α, β be the zeros of the polynomial $2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = \frac{21}{4}$ then $k = ?$
(a) 3 (b) -3 (c) -2 (d) 2
27. On dividing a polynomial $p(x)$ by a nonzero polynomial $q(x)$, let $g(x)$ be the quotient and $r(x)$ be the remainder then $p(x) = q(x) \cdot g(x) + r(x)$, where
(a) $r(x) = 0$ always
(b) $\deg r(x) < \deg g(x)$ always
(c) either $r(x) = 0$ or $\deg r(x) < \deg g(x)$
(d) $r(x) = g(x)$

28. Which of the following is a true statement?

- (a) $x^2 + 5x - 3$ is a linear polynomial.
 (b) $x^2 + 4x - 1$ is a binomial.
 (c) $x + 1$ is a monomial.
 (d) $5x^3$ is a monomial.

ANSWERS (MCQ)

1. (d) 2. (d) 3. (c) 4. (b) 5. (c) 6. (b) 7. (a) 8. (c) 9. (c)
 10. (d) 11. (b) 12. (b) 13. (c) 14. (d) 15. (b) 16. (c) 17. (a) 18. (d)
 19. (b) 20. (a) 21. (a) 22. (c) 23. (a) 24. (b) 25. (c) 26. (d) 27. (c)
 28. (d)

HINTS TO SOME SELECTED QUESTIONS

1. Clearly, $\sqrt{2}x^2 - 3\sqrt{3}x + \sqrt{6}$ is a polynomial.
 2. Clearly, $x + \frac{3}{x}$ is not a polynomial.
 3. $x^2 - 2x - 3 = x^2 - 3x + x - 3$
 $= x(x-3) + (x-3) = (x-3)(x+1)$.
 $\therefore (x-3)(x+1) = 0 \Rightarrow x = 3$ or $x = -1$.
 4. $x^2 - \sqrt{2}x - 12 = x^2 - 3\sqrt{2}x + 2\sqrt{2}x - 12$
 $= x(x-3\sqrt{2}) + 2\sqrt{2}(x-3\sqrt{2}) = (x-3\sqrt{2})(x+2\sqrt{2})$.
 $\therefore x = 3\sqrt{2}$ or $x = -2\sqrt{2}$.
 5. $4x^2 + 5\sqrt{2}x - 3 = 4x^2 + 6\sqrt{2}x - \sqrt{2}x - 3$
 $= 2\sqrt{2}x(\sqrt{2}x+3) - (\sqrt{2}x+3) = (\sqrt{2}x+3)(2\sqrt{2}x-1)$.
 $\therefore x = \frac{-3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{-3\sqrt{2}}{2}$ or $x = \frac{1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{4}$.
 6. $x^2 + \frac{1}{6}x - 2 = \frac{6x^2 + x - 12}{6}$.
 $6x^2 + x - 12 = 6x^2 + 9x - 8x - 12 = 3x(2x+3) - 4(2x+3)$
 $= (2x+3)(3x-4)$
 $\therefore$ the zeros are $\frac{-3}{2}$ and $\frac{4}{3}$.
 7. $7x^2 - \frac{11x}{3} - \frac{2}{3} = \frac{21x^2 - 11x - 2}{3}$.
 Now, $21x^2 - 11x - 2 = 21x^2 - 14x + 3x - 2$
 $= 7x(3x-2) + (3x-2) = (3x-2)(7x+1)$.
 $\therefore$ the zeros are $\frac{2}{3}$, $-\frac{1}{7}$.

8. The required polynomial is

$$x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - 3x - 10.$$

9. $\alpha + \beta = 5 + (-3) = 2$, $\alpha\beta = 5 \times (-3) = -15$.

Required polynomial is $x^2 - 2x - 15$.

10. $\alpha + \beta = \left(\frac{3}{5} - \frac{1}{2}\right) = \frac{1}{10}$, $\alpha\beta = \frac{3}{5} \times \left(-\frac{1}{2}\right) = -\frac{3}{10}$.

Required polynomial is $x^2 - \frac{1}{10}x - \frac{3}{10}$, i.e., $10x^2 - x - 3$.

11. Let α and β be the zeros of the given polynomial.

Then, $\alpha + \beta = -88$ and $\alpha\beta = 125$.

This happens when α and β are both negative.

12. For $ax^2 + bx + c$, we have $\alpha + \beta = \frac{-b}{a}$.

For $x^2 + 5x + 8$, we have $\alpha + \beta = -5$.

13. For $ax^2 + bx + c$, we have $\alpha\beta = \frac{c}{a}$.

For $2x^2 + 5x - 9$, we have $\alpha\beta = \frac{-9}{2}$.

14. $x = 2$ satisfies $kx^2 + 3x + k = 0$.

$$\therefore 4k + 6 + k = 0 \Rightarrow 5k = -6 \Rightarrow k = \frac{-6}{5}.$$

15. $x = -4$ satisfies $(k-1)x^2 + kx + 1 = 0$.

$$\therefore 16(k-1) - 4k + 1 = 0 \Rightarrow 12k = 15 \Rightarrow k = \frac{5}{4}.$$

16. $\alpha + \beta = 3 + (-2) = 1$ and $\alpha\beta = 3 \times (-2) = -6$.

$$\therefore -(a+1) = 1 \Rightarrow a+1 = -1 \Rightarrow a = -2.$$

Also, $b = -6$.

17. $\alpha\beta = 1 \Rightarrow \frac{k}{3} = 1 \Rightarrow k = 3$.

18. $\alpha + \beta = \alpha\beta \Rightarrow \frac{-2}{k} = \frac{3k}{k} \Rightarrow \frac{-2}{k} = 3 \Rightarrow k = \frac{-2}{3}$.

19. $\alpha + \beta = -6$ and $\alpha\beta = 2$.

$$\therefore \left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = \frac{(\alpha + \beta)}{\alpha\beta} = \frac{-6}{2} = -3.$$

Note For $ax^3 + bx^2 + cx + d$, we have

$$(\alpha + \beta + \gamma) = \frac{-b}{a}, (\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a} \text{ and } \alpha\beta\gamma = \frac{-d}{a}.$$

20. $(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a} = \frac{-1}{1} = -1$.

21. $\alpha\beta\gamma = \frac{-d}{a} = \frac{-6}{2} = -3$.

22. $p(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$
 $= x^3 - 3x^2 - 10x + 24.$

23. Let $\alpha, 0, 0$ be the zeros of $ax^3 + bx^2 + cx + d$. Then,

$$\text{sum of zeros} = \frac{-b}{a} \Rightarrow \alpha + 0 + 0 = \frac{-b}{a} \Rightarrow \alpha = \frac{-b}{a}.$$

Hence, the third zero is $\frac{-b}{a}$.

24. Let $\alpha, \beta, 0$ be the zeros of $ax^3 + bx^2 + cx + d$. Then,

sum of the products of zeros, taken two at a time is given by

$$(\alpha\beta + \beta \times 0 + \alpha \times 0) = \frac{c}{a} \Rightarrow \alpha\beta = \frac{c}{a}.$$

$\therefore$ the product of the other two zeros is $\frac{c}{a}$.

25. Since -1 is a zero of $x^3 + ax^2 + bx + c$, we have

$$(-1)^3 + a \times (-1)^2 + b \times (-1) + c = 0 \Rightarrow a - b + c - 1 = 0 \Rightarrow c = 1 - a + b.$$

Also, product of all zeros is given by

$$\alpha\beta \times (-1) = -c \Rightarrow \alpha\beta = c \Rightarrow \alpha\beta = 1 - a + b.$$

26. $\alpha + \beta = \frac{-5}{2}$ and $\alpha\beta = \frac{k}{2}$.

$$\alpha^2 + \beta^2 + \alpha\beta = \frac{21}{4} \Rightarrow (\alpha + \beta)^2 - \alpha\beta = \frac{21}{4}$$

$$\Rightarrow \left(\frac{-5}{2}\right)^2 - \frac{k}{2} = \frac{21}{4}$$

$$\Rightarrow \frac{k}{2} = \left(\frac{25}{4} - \frac{21}{4}\right) = 1 \Rightarrow k = 2.$$

SUMMARY OF RESULTS

1. **I. Polynomial:** An expression of the form $p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where $a_n \neq 0$, is called a polynomial in x of degree n .

II. A polynomial is said to be *linear, quadratic, cubic and biquadratic* according as its degree is 1, 2, 3 and 4 respectively.

General form:

(i) linear equation: $ax + b$.

(ii) quadratic equation: $ax^2 + bx + c$.

(iii) cubic equation: $ax^3 + bx^2 + cx + d$.

(iv) biquadratic equation: $ax^4 + bx^3 + cx^2 + dx + e$.

III. **Value of a polynomial at a point:** The value of a polynomial $p(x)$ at $x = \alpha$ is obtained by putting $x = \alpha$ and it is denoted by $p(\alpha)$.

IV. **Zeros of a polynomial:** A real number α is called a zero of $p(x)$, if $p(\alpha) = 0$.

2. I. If α, β are the zeros of $p(x) = ax^2 + bx + c, a \neq 0$ then

$$(i) \alpha + \beta = \frac{-b}{a}, \quad (ii) \alpha\beta = \frac{c}{a}.$$

- II. If α and β are the zeros of a quadratic polynomial $p(x)$ then

$$p(x) = \{x^2 - (\alpha + \beta)x + \alpha\beta\}.$$

3. I. If α, β, γ are the zeros of $p(x) = ax^3 + bx^2 + cx + d, a \neq 0$ then

$$(i) \alpha + \beta + \gamma = \frac{-b}{a}, \quad (ii) \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad (iii) \alpha\beta\gamma = \frac{-d}{a}.$$

- II. If α, β and γ be the zeros of a polynomial $p(x)$ then

$$p(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma.$$

TEST YOURSELF

MCQ

- Zeros of $p(x) = x^2 - 2x - 3$ are
(a) 1, -3 (b) 3, -1 (c) -3, -1 (d) 1, 3
- If α, β, γ are the zeros of the polynomial $x^3 - 6x^2 - x + 30$ then the value of $(\alpha\beta + \beta\gamma + \gamma\alpha)$ is
(a) -1 (b) 1 (c) -5 (d) 30
- If α, β are the zeros of $kx^2 - 2x + 3k$ such that $\alpha + \beta = \alpha\beta$ then $k = ?$
(a) $\frac{1}{3}$ (b) $\frac{-1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{-2}{3}$
- It is given that the difference between the zeros of $4x^2 - 8kx + 9$ is 4 and $k > 0$. Then, $k = ?$
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{5}{2}$ (d) $\frac{7}{2}$

Short-Answer Questions

- Find the zeros of the polynomial $x^2 + 2x - 195$.
- If one zero of the polynomial $(a^2 + 9)x^2 + 13x + 6a$ is the reciprocal of the other, find the value of a .
- Find a quadratic polynomial whose zeros are 2 and -5.
- If the zeros of the polynomial $x^3 - 3x^2 + x + 1$ are $(a - b)$, a and $(a + b)$, find the values of a and b .
- Verify that 2 is a zero of the polynomial $x^3 + 4x^2 - 3x - 18$.
- Find the quadratic polynomial, the sum of whose zeros is -5 and their product is 6.

11. Find a cubic polynomial whose zeros are 3, 5 and -2.
12. Using remainder theorem, find the remainder when $p(x) = x^3 + 3x^2 - 5x + 4$ is divided by $(x - 2)$.
13. Show that $(x + 2)$ is a factor of $f(x) = x^3 + 4x^2 + x - 6$.
14. If α, β, γ are the zeros of the polynomial $p(x) = 6x^3 + 3x^2 - 5x + 1$, find the value of $\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)$.
15. If α, β are the zeros of the polynomial $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of k .
16. Show that the polynomial $f(x) = x^4 + 4x^2 + 6$ has no zero.

Long-Answer Questions

17. If one zero of the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is 3, find the other two zeros.
18. If two zeros of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ are $\sqrt{2}$ and $-\sqrt{2}$, find its other two zeros.
19. Find the quotient when $p(x) = 3x^4 + 5x^3 - 7x^2 + 2x + 2$ is divided by $(x^2 + 3x + 1)$.
20. Use remainder theorem to find the value of k , it being given that when $x^3 + 2x^2 + kx + 3$ is divided by $(x - 3)$ then the remainder is 21.

ANSWERS (TEST YOURSELF)

- | | | | | | |
|---------------------|-----------------------------|-------------------|---------------------------|---------------------------|------------|
| 1. (a) | 2. (a) | 3. (c) | 4. (c) | 5. -15, 13 | 6. $a = 3$ |
| 7. $x^2 + 3x - 10$ | 8. $a = 1, b = \pm\sqrt{2}$ | 9. $x^2 + 5x + 6$ | 10. $x^3 - 6x^2 - x + 30$ | 11. $x^3 - 6x^2 - x + 30$ | |
| 12. 14 | 13. 5 | 14. $k = 6$ | 15. 1, 2 | 16. $1, \frac{1}{2}$ | |
| 17. $3x^2 - 4x + 2$ | 18. $k = -9$ | | | | |





Linear Equations in Two Variables

LINEAR EQUATIONS IN TWO VARIABLES

An equation of the form $ax + by + c = 0$, where a, b, c are real numbers ($a \neq 0, b \neq 0$), is called a linear equation in two variables x and y .

Examples Each of the equations

(i) $3x - 4y + 2 = 0$

(ii) $2x + 5y = 9$

(iii) $0.4x + 0.3y = 2.7$

(iv) $\sqrt{2}x - \sqrt{3}y = 0$

is a linear equation in x and y .

SOLUTION OF A LINEAR EQUATION

We say that $x = \alpha$ and $y = \beta$ is a solution of $ax + by + c = 0$ if $a\alpha + b\beta + c = 0$.

EXAMPLE Show that $x = 3$ and $y = 2$ is a solution of $5x - 3y = 9$.

SOLUTION Substituting $x = 3$ and $y = 2$ in the given equation, we get

$$\text{LHS} = 5 \times 3 - 3 \times 2 = (15 - 6) = 9 = \text{RHS.}$$

$\therefore x = 3$ and $y = 2$ is a solution of $5x - 3y = 9$.

SIMULTANEOUS LINEAR EQUATIONS IN TWO VARIABLES

Two linear equations in two unknowns x and y are said to form a system of simultaneous linear equations if each of them is satisfied by the same pair of values of x and y .

Example Consider the system of linear equations

$$x + y = 10, x - y = 2.$$

By substitution, you will find that each of these equations is satisfied by the values $x = 6$ and $y = 4$.

Hence, the given equations form a system of simultaneous linear equations in x and y .

SOLUTION OF A GIVEN SYSTEM OF TWO SIMULTANEOUS EQUATIONS

A pair of values of x and y satisfying each of the equations in a given system of two simultaneous equations in x and y is called a solution of the system.

EXAMPLE 1 Show that $x = 5, y = 2$ is a solution of the system of linear equations
 $2x + 3y = 16, x - 2y = 1$.

SOLUTION The given equations are

$$2x + 3y = 16 \quad \dots (i)$$

$$x - 2y = 1. \quad \dots (ii)$$

Putting $x = 5$ and $y = 2$ in (i), we get

$$\text{LHS} = (2 \times 5 + 3 \times 2) = 16 = \text{RHS}.$$

Putting $x = 5$ and $y = 2$ in (ii), we get

$$\text{LHS} = (5 - 2 \times 2) = 1 = \text{RHS}.$$

Thus, $x = 5$ and $y = 2$ satisfy both (i) and (ii).

Hence, $x = 5, y = 2$ is a solution of the given system of equations.

EXAMPLE 2 Show that $x = 3, y = 2$ is not a solution of the system of linear equations $3x - 2y = 5, 2x + y = 7$.

SOLUTION The given equations are

$$3x - 2y = 5 \quad \dots (i)$$

$$2x + y = 7. \quad \dots (ii)$$

Putting $x = 3$ and $y = 2$ in (i), we get

$$\text{LHS} = (3 \times 3 - 2 \times 2) = 5 = \text{RHS}.$$

Putting $x = 3$ and $y = 2$ in (ii), we get

$$\text{LHS} = (2 \times 3 + 2) = 8 \neq \text{RHS}.$$

Thus, the values $x = 3, y = 2$ do not satisfy (ii).

Hence, $x = 3, y = 2$ is not a solution of the given system of equations.

CONSISTENT AND INCONSISTENT SYSTEMS OF LINEAR EQUATIONS

CONSISTENT SYSTEM OF LINEAR EQUATIONS

A system of two linear equations in two unknowns is said to be consistent if it has at least one solution.

INCONSISTENT SYSTEM OF LINEAR EQUATIONS

A system of two linear equations in two unknowns is said to be inconsistent if it has no solution at all.

Example Consider the system of linear equations

$$x + y = 3, 2x + 2y = 7.$$

Clearly, we cannot find values of x and y which may satisfy both the given equations simultaneously.

Hence, the given system is inconsistent.

SOLVING SIMULTANEOUS LINEAR EQUATIONS (GRAPHICAL METHOD)

METHOD Let the given system of linear equations be

$$a_1x + b_1y + c_1 = 0 \quad \dots (i)$$

$$a_2x + b_2y + c_2 = 0. \quad \dots (ii)$$

On the same graph paper, we draw the graph of each one of the given linear equations.

Each such graph is always a straight line.

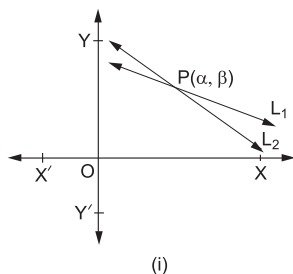
Let the lines L_1 and L_2 represent the graphs of (i) and (ii) respectively.

Now, the following cases arise.

Case I When the lines L_1 and L_2 intersect at a point

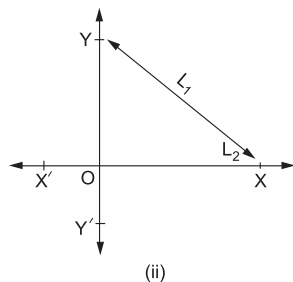
Let the graph lines L_1 and L_2 intersect at a point $P(\alpha, \beta)$, as shown in the adjoining figure.

Then, $x = \alpha, y = \beta$ is the unique solution of the given system of equations.



Case II When the lines L_1 and L_2 are coincident

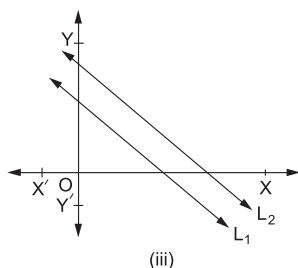
When the two graph lines L_1 and L_2 coincide, the given system of equations has *infinitely many solutions*.



Case III When the lines L_1 and L_2 are parallel

In this case, there is no common solution of the given system of equations.

Thus, in this case, the given system is *inconsistent*.

**SUMMARY**

A system of two linear equations in x and y has

- (i) a unique solution if the graph lines intersect at a point
- (ii) infinitely many solutions if the two graph lines coincide
- (iii) no solution if the two graph lines are parallel

SOLVED EXAMPLES**GRAPHS OF EQUATIONS HAVING UNIQUE SOLUTIONS**

Note Such graph lines intersect at a point.

EXAMPLE 1 Solve graphically the system of linear equations

$$x + 2y = 3, \quad 4x + 3y = 2.$$

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

Graph of $x + 2y = 3$

$$\begin{aligned} x + 2y = 3 &\Rightarrow 2y = (3 - x) \\ &\Rightarrow y = \frac{(3 - x)}{2}. \end{aligned} \quad \dots (i)$$

Putting $x = -3$ in (i), we get $y = 3$.

Putting $x = -1$ in (i), we get $y = 2$.

Putting $x = 1$ in (i), we get $y = 1$.

Thus, we have the following table for the equation $x + 2y = 3$.

x	-3	-1	1
y	3	2	1

Now, plot the points $A(-3, 3)$, $B(-1, 2)$ and $C(1, 1)$ on the graph paper.

Join AB and BC to get the graph line ABC . Extend it on both ways.

Thus, the line ABC is the graph of $x + 2y = 3$.

Graph of $4x + 3y = 2$

$$4x + 3y = 2 \Rightarrow 3y = (2 - 4x)$$

$$\Rightarrow y = \frac{(2 - 4x)}{3} \quad \dots \text{(ii)}$$

Putting $x = -4$ in (ii), we get $y = 6$.

Putting $x = -1$ in (ii), we get $y = 2$.

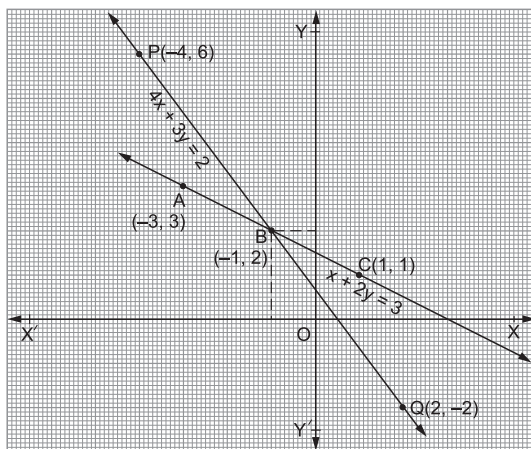
Putting $x = 2$ in (ii), we get $y = -2$.

Thus, we have the following table for the equation $4x + 3y = 2$.

x	-4	-1	2
y	6	2	-2

Now, on the same graph paper as above, plot the points $P(-4, 6)$ and $Q(2, -2)$. The point $B(-1, 2)$ has already been plotted. Join PB and BQ to get the line PBQ . Extend it on both ways.

Thus, the line PBQ is the graph of $4x + 3y = 2$.



The two graph lines intersect at the point $B(-1, 2)$.

$\therefore x = -1$ and $y = 2$ is the solution of the given system of equations.

EXAMPLE 2

Solve graphically the system of linear equations

$$4x - 5y + 16 = 0 \text{ and } 2x + y - 6 = 0.$$

Determine the vertices of the triangle formed by these lines and the x -axis. [CBSE 2006]

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

Graph of $4x - 5y + 16 = 0$

$$4x - 5y + 16 = 0 \Rightarrow 5y = 4x + 16$$

$$\Rightarrow y = \frac{(4x + 16)}{5} \quad \dots (i)$$

Putting $x = -4$ in (i), we get $y = 0$.

Putting $x = 1$ in (i), we get $y = 4$.

Putting $x = 6$ in (i), we get $y = 8$.

Thus, we have the following table for $4x - 5y + 16 = 0$.

x	-4	1	6
y	0	4	8

Now, plot the points $A(-4, 0)$, $B(1, 4)$ and $C(6, 8)$ on the graph paper.

Join AB and BC to get the graph line ABC . Extend it on both ways.

Thus, the line ABC is the graph of $4x - 5y + 16 = 0$.

Graph of $2x + y - 6 = 0$

$$2x + y - 6 = 0 \Rightarrow y = (6 - 2x) \quad \dots (ii)$$

Putting $x = 0$ in (ii), we get $y = 6$.

Putting $x = 2$ in (ii), we get $y = 2$.

Putting $x = 3$ in (ii), we get $y = 0$.

Thus, we have the following table for $2x + y - 6 = 0$.

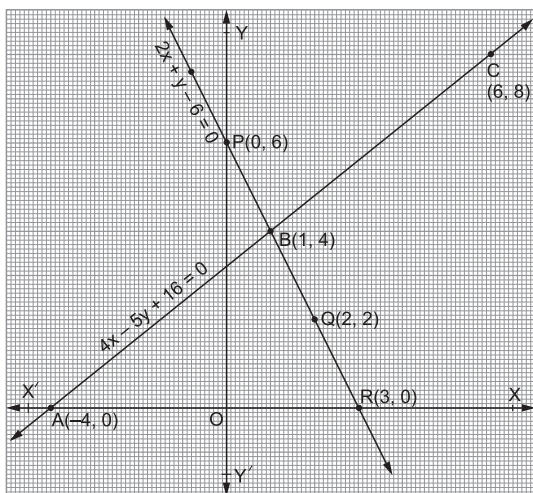
x	0	2	3
y	6	2	0

On the same graph paper as above, plot the points

$P(0, 6)$, $Q(2, 2)$ and $R(3, 0)$.

Join PQ and QR to get the graph line PQR . Extend it on both ways.

Thus, the line PQR is the graph of $2x + y - 6 = 0$.



The two graph lines ABC and PQR intersect at the point $B(1, 4)$.

$\therefore x = 1$ and $y = 4$ is the solution of the given system of equations.

These lines form $\triangle BAR$ with the x -axis, whose vertices are $B(1, 4)$, $A(-4, 0)$ and $R(3, 0)$.

EXAMPLE 3 Solve the following system of linear equations graphically:

$$4x - 5y - 20 = 0 \text{ and } 3x + 5y - 15 = 0.$$

Determine the vertices of the triangle formed by the lines representing the above equations and the y -axis. [CBSE 2004]

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

Graph of $4x - 5y - 20 = 0$

$$4x - 5y - 20 = 0 \Rightarrow 5y = (4x - 20)$$

$$\Rightarrow y = \frac{(4x - 20)}{5}. \quad \dots (i)$$

Putting $x = 0$ in (i), we get $y = -4$.

Putting $x = 2$ in (i), we get $y = -2.4$.

Putting $x = 5$ in (i), we get $y = 0$.

Thus, we have the following table for $4x - 5y - 20 = 0$.

x	0	2	5
y	-4	-2.4	0

Now, plot the points $A(0, -4)$, $B(2, -2.4)$ and $C(5, 0)$ on the graph paper.

Join AB and BC to get the graph line ABC . Extend it on both ways.

Thus, the line ABC is the graph of $4x - 5y - 20 = 0$.

Graph of $3x + 5y - 15 = 0$

$$3x + 5y - 15 = 0 \Rightarrow 5y = (15 - 3x)$$

$$\Rightarrow y = \frac{(15 - 3x)}{5} \quad \dots (ii)$$

Putting $x = -5$ in (ii), we get $y = 6$.

Putting $x = 0$ in (ii), we get $y = 3$.

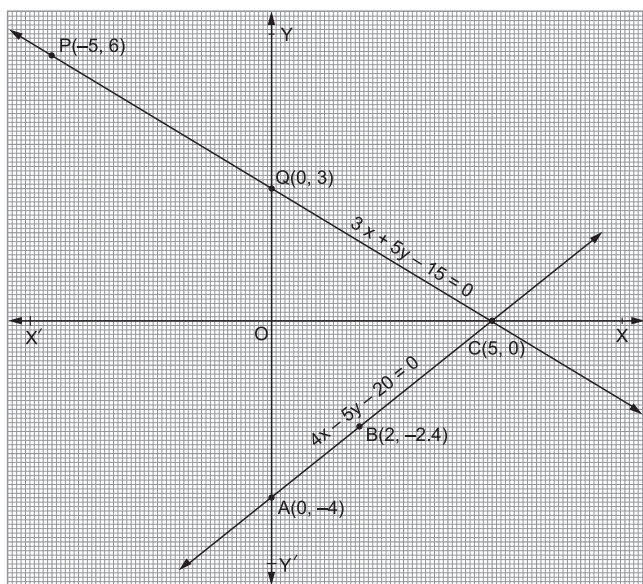
Putting $x = 5$ in (ii), we get $y = 0$.

Thus, we have the following table for $3x + 5y - 15 = 0$.

x	-5	0	5
y	6	3	0

On the same graph paper as above, plot the points $P(-5, 6)$ and $Q(0, 3)$. The third point $C(5, 0)$ has already been plotted.

Join PQ and QC to get the graph line PQC . Extend it on both ways.



Thus, the line PQC is the graph of $3x + 5y - 15 = 0$.

The two graph lines intersect at the point $C(5, 0)$.

$\therefore x = 5, y = 0$ is the solution of the given system of equations.

Clearly, the given equations are represented by the graph lines ABC and PQC respectively.

The vertices of $\triangle AQC$ formed by these lines and the y -axis are $A(0, -4)$, $Q(0, 3)$ and $C(5, 0)$.

EXAMPLE 4 Solve the following system of equations graphically:

$$3x + 2y - 11 = 0 \text{ and } 2x - 3y + 10 = 0. \quad [\text{CBSE 2006C}]$$

Shade the region bounded by these lines and the x -axis.

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

$$\text{Graph of } 3x + 2y - 11 = 0$$

$$\begin{aligned} 3x + 2y - 11 = 0 &\Rightarrow 2y = (11 - 3x) \\ &\Rightarrow y = \frac{(11 - 3x)}{2}. \end{aligned} \quad \dots (i)$$

Putting $x = -1$ in (i), we get $y = 7$.

Putting $x = 1$ in (i), we get $y = 4$.

Putting $x = 3$ in (i), we get $y = 1$.

Thus, we have the following table for $3x + 2y - 11 = 0$.

x	-1	1	3
y	7	4	1

Now, plot the points $A(-1, 7)$, $B(1, 4)$ and $C(3, 1)$ on the graph paper.

Join AB and BC to obtain the graph line ABC . Extend it on both ways.

Thus, the line ABC is the graph of the equation $3x + 2y - 11 = 0$.

$$\text{Graph of } 2x - 3y + 10 = 0$$

$$2x - 3y + 10 = 0 \Rightarrow 3y = 2x + 10 \Rightarrow y = \frac{(2x + 10)}{3}. \quad \dots (ii)$$

Putting $x = -2$ in (ii), we get $y = 2$.

Putting $x = 1$ in (ii), we get $y = 4$.

Putting $x = 4$ in (ii), we get $y = 6$.

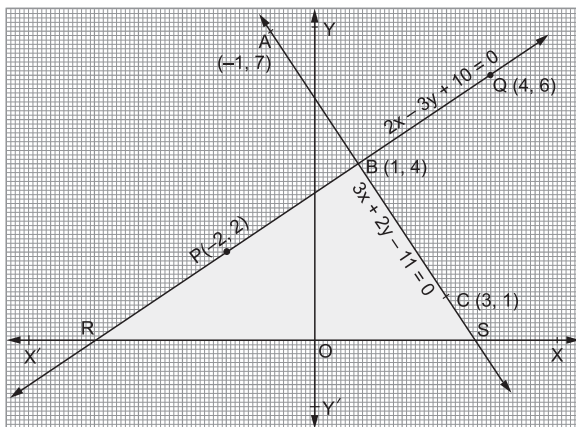
Thus, we have the following table for $2x - 3y + 10 = 0$.

x	-2	1	4
y	2	4	6

On the same graph paper as above, plot the points $P(-2, 2)$ and $Q(4, 6)$. The third point $B(1, 4)$ has already been plotted.

Now, join PB and BQ to obtain the graph line PBQ . Extend it on both ways.

Thus, the line PBQ is the graph of the equation $2x - 3y + 10 = 0$.



The two graph lines intersect at the point $B(1, 4)$.

$\therefore x = 1, y = 4$ is the solution of the given system of equations.

These graph lines intersect the x -axis at R and S .

The region bounded by these lines and the x -axis has been shaded.

The shaded region is the $\triangle BRS$ with $B(1, 4)$, $R(-5, 0)$ and $S(\frac{11}{3}, 0)$.

EXAMPLE 5

Solve the following system of linear equations graphically:

$$3x + y - 11 = 0, x - y - 1 = 0.$$

Shade the region bounded by these lines and the y -axis.

Find the coordinates of the points where the graph lines cut the y -axis.

[CBSE 2002C]

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

$$\text{Graph of } 3x + y - 11 = 0$$

$$3x + y - 11 = 0 \Rightarrow y = (11 - 3x). \quad \dots (i)$$

Putting $x = 2$ in (i), we get $y = 5$.

Putting $x = 3$ in (i), we get $y = 2$.

Putting $x = 5$ in (i), we get $y = -4$.

Thus, we have the following table for equation (i).

x	2	3	5
y	5	2	-4

On the graph paper, plot the points $A(2, 5)$, $B(3, 2)$ and $C(5, -4)$. Join AB and BC to get the graph line ABC .

Thus, the line ABC is the graph of the equation $3x + y - 11 = 0$.

$$\text{Graph of } x - y - 1 = 0$$

$$x - y - 1 = 0 \Rightarrow y = (x - 1). \quad \dots (ii)$$

Putting $x = -3$ in (ii), we get $y = -4$.

Putting $x = 0$ in (ii), we get $y = -1$.

Putting $x = 3$ in (ii), we get $y = 2$.

Thus, we have the following table for equation (ii).

x	-3	0	3
$-y$	-4	-1	2

On the same graph paper as above, plot the points $P(-3, -4)$ and $Q(0, -1)$. The third point $B(3, 2)$ is already plotted.

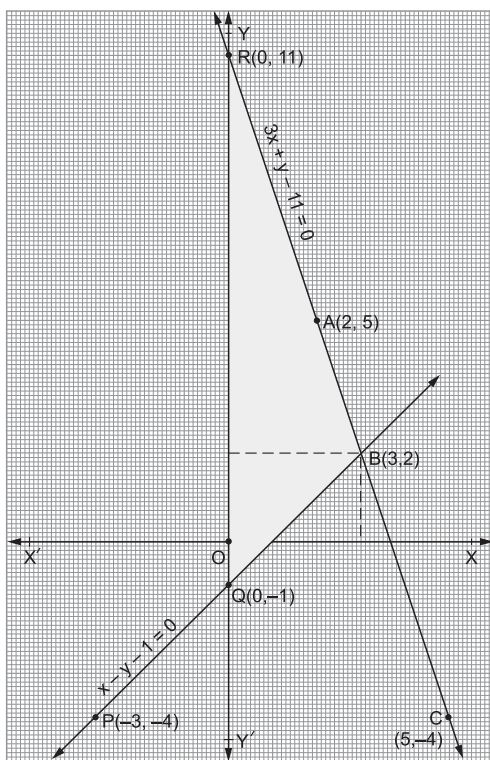
Join PQ and QB to get the graph line PQB .

Thus, line PQB is the graph of the equation $x - y - 1 = 0$.

The two graph lines intersect at the point $B(3, 2)$.

$\therefore x = 3, y = 2$ is the solution of the given system of equations.

The region bounded by these lines and the y -axis has been shaded.



On extending the graph lines on both sides, we find that these graph lines intersect the y -axis at the points $Q(0, -1)$ and $R(0, 11)$.

GRAPH OF EQUATIONS HAVING INFINITELY MANY SOLUTIONS

Note Such graph lines just coincide.

EXAMPLE 6 Show graphically that the system of equations

$$3x - y = 2, 9x - 3y = 6$$

has an infinite number of solutions.

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

Graph of $3x - y = 2$

$$3x - y = 2 \Rightarrow y = (3x - 2). \quad \dots (i)$$

Putting $x = -1$ in (i), we get $y = -5$.

Putting $x = 0$ in (i), we get $y = -2$.

Putting $x = 2$ in (i), we get $y = 4$.

Thus, we have the following table for $3x - y = 2$.

x	-1	0	2
y	-5	-2	4

Now, plot the points $A(-1, -5)$, $B(0, -2)$ and $C(2, 4)$ on the graph paper.

Join AB and BC to get the graph line ABC .

Extend the graph line ABC on both sides.

Thus, the line ABC is the graph of the equation $3x - y = 2$.

Graph of $9x - 3y = 6$

$$9x - 3y = 6 \Rightarrow 3y = (9x - 6)$$

$$\Rightarrow y = \frac{(9x - 6)}{3}. \quad \dots \text{(ii)}$$

Putting $x = -2$ in (ii), we get $y = -8$.

Putting $x = 1$ in (ii), we get $y = 1$.

Putting $x = 2$ in (ii), we get $y = 4$.

Thus, we have the following table for $9x - 3y = 6$.

x	-2	1	2
y	-8	1	4

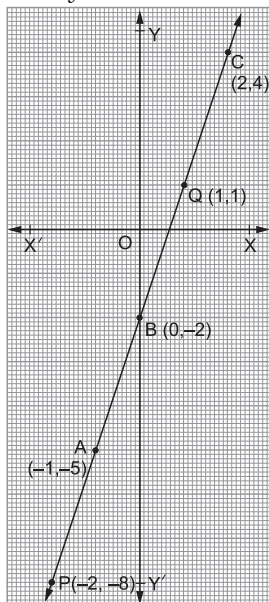
Now, plot the points $P(-2, -8)$ and $Q(1, 1)$ on the same graph paper.

The point $C(2, 4)$ has already been plotted.

Join PQ and QC to obtain the line PQC .

Thus, the line PQC is the graph of $9x - 3y = 6$.

Thus, we find that the two graph lines coincide. Hence, the given system of equations has an infinite number of solutions.



GRAPH OF EQUATIONS HAVING NO SOLUTION

Note Such graph lines are parallel.

EXAMPLE 7 Show graphically that the system of linear equations

$$2x - 3y = 5, 6y - 4x = 3$$

is inconsistent, i.e., has no solution.

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and the y -axis respectively.

$$\text{Graph of } 2x - 3y = 5$$

$$2x - 3y = 5 \Rightarrow 3y = (2x - 5)$$

$$\Rightarrow y = \frac{(2x - 5)}{3}. \quad \dots (i)$$

Putting $x = -2$ in (i), we get $y = -3$.

Putting $x = 1$ in (i), we get $y = -1$.

Putting $x = 4$ in (i), we get $y = 1$.

Thus, we have the following table for the equation $2x - 3y = 5$.

x	-2	1	4
y	-3	-1	1

Now, plot the points $A(-2, -3)$, $B(1, -1)$ and $C(4, 1)$ on the graph paper.

Join AB and BC to get the graph line ABC . Extend it on both ways.

Thus, the line ABC is the graph of the equation $2x - 3y = 5$.

$$\text{Graph of } 6y - 4x = 3$$

$$6y - 4x = 3 \Rightarrow 6y = (3 + 4x)$$

$$\Rightarrow y = \frac{(3 + 4x)}{6}. \quad \dots (ii)$$

Putting $x = -3$ in (ii), we get $y = \frac{-3}{2}$.

Putting $x = 0$ in (ii), we get $y = \frac{1}{2}$.

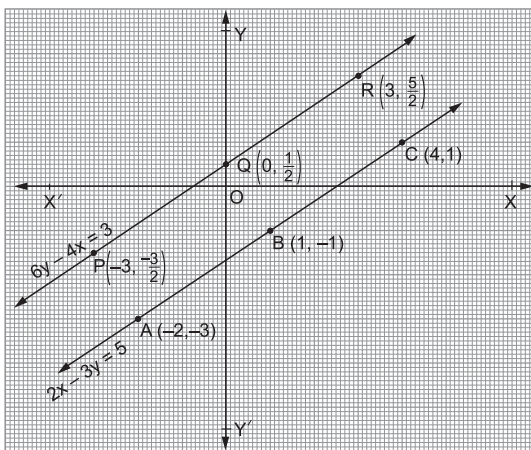
Putting $x = 3$ in (ii), we get $y = \frac{5}{2}$.

Thus, we have the following table for the equation $6y - 4x = 3$.

x	-3	0	3
y	$-\frac{3}{2}$	$\frac{1}{2}$	$\frac{5}{2}$

Now, plot the points $P(-3, -\frac{3}{2})$, $Q(0, \frac{1}{2})$ and $R(3, \frac{5}{2})$ on the same graph paper as above. Join PQ and QR to get the graph line PQR . Extend it on both ways.

Then, the line PQR is the graph of the equation $6y - 4x = 3$.



It is clear from the graph that the two graph lines are parallel and do not intersect even when produced.

Hence, the given system of equations has no solution.

EXAMPLE 8 Show graphically that the system of linear equations

$$x - y = 8, 3x - 3y = 16$$

is inconsistent, i.e., has no solution.

SOLUTION On a graph paper, draw a horizontal line $X'OX$ and a vertical line YOY' as the x -axis and y -axis respectively.

Graph of $x - y = 8$

The first equation is $x - y = 8$.

Now, $x - y = 8 \Rightarrow y = x - 8$ (i)

Putting $x = 3$ in (i), we get $y = -5$.

Putting $x = 4$ in (i), we get $y = -4$.

Putting $x = 5$ in (i), we get $y = -3$.

Thus, we have the following table for the equation $x - y = 8$.

x	3	4	5
y	-5	-4	-3

Now, we plot the points $A(3, -5)$, $B(4, -4)$ and $C(5, -3)$ on the graph paper. Join AB and BC to get the graph line ABC .

Graph of $3x - 3y = 16$

The second equation is $3x - 3y = 16$.

$$3x - 3y = 16 \Rightarrow 3y = 3x - 16 \Rightarrow y = \frac{(3x - 16)}{3}. \quad \dots (ii)$$

Putting $x = 2$ in (ii), we get $y = \frac{-10}{3} = -3\frac{1}{3} = -3.3$.

Putting $x = 3$ in (ii), we get $y = \frac{-7}{3} = -2\frac{1}{3} = -2.3$.

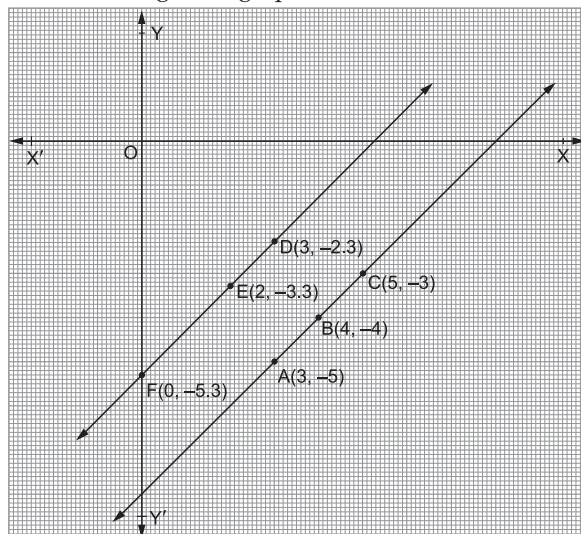
Putting $x = 0$ in (ii), we get $y = \frac{-16}{3} = -5\frac{1}{3} = -5.3$.

Thus, we have the following table for the equation $3x - 3y = 16$.

x	3	2	0
y	-2.3	-3.3	-5.3

Now, we plot $D(3, -2.3)$, $E(2, -3.3)$ and $F(0, -5.3)$ on the same graph paper as above.

Join DE and EF to get the graph line DEF .



It is clear from the graph that the lines ABC and DEF are parallel and do not meet when produced.

Hence, the given system of equations has no solution and therefore, it is inconsistent.

EXERCISE 3A

Solve each of the following systems of equations graphically:

- | | |
|---|---|
| 1. $2x + 3y = 2,$
$x - 2y = 8.$ [CBSE 2007] | 2. $3x + 2y = 4,$
$2x - 3y = 7.$ [CBSE 2006C] |
| 3. $2x + 3y = 8,$
$x - 2y + 3 = 0.$ [CBSE 2005] | 4. $2x - 5y + 4 = 0,$
$2x + y - 8 = 0.$ [CBSE 2005] |
| 5. $3x + 2y = 12,$
$5x - 2y = 4.$ [CBSE 2006] | 6. $3x + y + 1 = 0,$
$2x - 3y + 8 = 0.$ [CBSE 2007C] |
| 7. $2x + 3y + 5 = 0,$
$3x - 2y - 12 = 0.$ | 8. $2x - 3y + 13 = 0,$
$3x - 2y + 12 = 0.$ |
| 9. $2x + 3y - 4 = 0,$
$3x - y + 5 = 0.$ [CBSE 2004C] | 10. $x + 2y + 2 = 0,$
$3x + 2y - 2 = 0.$ |

Solve each of the following given systems of equations graphically and find the vertices and area of the triangle formed by these lines and the x-axis:

11. $x - y + 3 = 0, 2x + 3y - 4 = 0.$
12. $2x - 3y + 4 = 0, x + 2y - 5 = 0.$ [CBSE 2005]
13. $4x - 3y + 4 = 0, 4x + 3y - 20 = 0.$ [CBSE 2008]
14. $x - y + 1 = 0, 3x + 2y - 12 = 0.$ [CBSE 2002]
15. $x - 2y + 2 = 0, 2x + y - 6 = 0.$

Solve each of the following given systems of equations graphically and find the vertices and area of the triangle formed by these lines and the y-axis:

16. $2x - 3y + 6 = 0, 2x + 3y - 18 = 0.$ [CBSE 2004]
17. $4x - y - 4 = 0, 3x + 2y - 14 = 0.$ [CBSE 2006C]
18. $x - y - 5 = 0, 3x + 5y - 15 = 0.$ [CBSE 2009C]
19. $2x - 5y + 4 = 0, 2x + y - 8 = 0.$ [CBSE 2005]
20. $5x - y - 7 = 0, x - y + 1 = 0.$
21. $2x - 3y = 12, x + 3y = 6.$ [CBSE 2008]

Show graphically that each of the following given systems of equations has infinitely many solutions:

22. $2x + 3y = 6$, $4x + 6y = 12$.

[CBSE 2010]

23. $3x - y = 5$, $6x - 2y = 10$.

24. $2x + y = 6$, $6x + 3y = 18$.

25. $x - 2y = 5$, $3x - 6y = 15$.

Show graphically that each of the following given systems of equations is inconsistent, i.e., has no solution:

26. $x - 2y = 6$, $3x - 6y = 0$.

27. $2x + 3y = 4$, $4x + 6y = 12$.

28. $2x + y = 6$, $6x + 3y = 20$.

29. Draw the graphs of the following equations on the same graph paper:

$$2x + y = 2, 2x + y = 6.$$

Find the coordinates of the vertices of the trapezium formed by these lines. Also, find the area of the trapezium so formed. [HOTS]

HINT The line $2x + y = 2$ cuts the x -axis at $A(1, 0)$ and the y -axis at $B(0, 2)$.

The line $2x + y = 6$ cuts the x -axis at $C(3, 0)$ and the y -axis at $D(0, 6)$.

Area of trap. $ABDC = \text{ar}(\triangle OCD) - \text{ar}(\triangle OAB)$

$$= \left(\frac{1}{2} \times 3 \times 6\right) - \left(\frac{1}{2} \times 1 \times 2\right) = 8 \text{ sq units.}$$

ANSWERS (EXERCISE 3A)

1. $x = 4$, $y = -2$

2. $x = 2$, $y = -1$

3. $x = 1$, $y = 2$

4. $x = 3$, $y = 2$

5. $x = 2$, $y = 3$

6. $x = -1$, $y = 2$

7. $x = 2$, $y = -3$

8. $x = -2$, $y = 3$

9. $x = -1$, $y = 2$

10. $x = 2$, $y = -2$

11. $(x = -1, y = 2)$; $A(-1, 2)$, $B(-3, 0)$, $C(2, 0)$; $\text{ar}(\triangle ABC) = 5 \text{ sq units}$

12. $(x = 1, y = 2)$; $A(1, 2)$, $B(-2, 0)$, $C(5, 0)$; $\text{ar}(\triangle ABC) = 7 \text{ sq units}$

13. $(x = 2, y = 4)$; $A(2, 4)$, $B(-1, 0)$, $C(5, 0)$; $\text{ar}(\triangle ABC) = 12 \text{ sq units}$

14. $(x = 2, y = 3)$; $A(2, 3)$, $B(-1, 0)$, $C(4, 0)$; $\text{ar}(\triangle ABC) = 7.5 \text{ sq units}$

15. $(x = 2, y = 2)$; $A(2, 2)$, $B(-2, 0)$, $C(3, 0)$; $\text{ar}(\triangle ABC) = 5 \text{ sq units}$

16. $(x = 3, y = 4)$; $A(3, 4)$, $B(0, 2)$, $C(0, 6)$; $\text{ar}(\triangle ABC) = 6 \text{ sq units}$

17. $(x = 2, y = 4)$; $A(2, 4)$, $B(0, -4)$, $C(0, 7)$; $\text{ar}(\triangle ABC) = 11 \text{ sq units}$

18. $(x = 5, y = 0)$; $A(5, 0)$, $B(0, -5)$, $C(0, 3)$; $\text{ar}(\triangle ABC) = 20 \text{ sq units}$

19. $(x = 3, y = 2)$; $A(3, 2)$, $B(0, 0.8)$, $C(0, 8)$; $\text{ar}(\triangle ABC) = 10.8 \text{ sq units}$

20. $(x = 2, y = 3)$; $A(2, 3)$, $B(0, -7)$, $C(0, 1)$; $\text{ar}(\triangle ABC) = 8 \text{ sq units}$

21. $(x = 6, y = 0)$, $A(6, 0)$, $B(0, -4)$, $C(0, 2)$, $\text{ar}(\triangle ABC) = 18$ sq units.

29. 8 sq units

SOLVING SIMULTANEOUS LINEAR EQUATIONS (BY ALGEBRAIC METHODS)

(I) SUBSTITUTION METHOD

METHOD Suppose we are given two linear equations in x and y . For solving these equations by the substitution method, we proceed according to the following steps.

- Step 1. *Express y in terms of x in one of the given equations.*
- Step 2. *Substitute this value of y in terms of x in the other equation. This gives a linear equation in x .*
- Step 3. *Solve the linear equation in x obtained in Step 2.*
- Step 4. *Substitute this value of x in the relation taken in Step 1 to obtain a linear equation in y .*
- Step 5. *Solve the above linear equation in y to get the value of y .*

REMARK We may interchange the role of x and y in the above method.

SOLVED EXAMPLES

EXAMPLE 1 *Solve for x and y , using substitution method:*

$$2x + y = 7, \quad 4x - 3y + 1 = 0.$$

SOLUTION The given system of equations is

$$2x + y = 7 \quad \dots (i)$$

$$4x - 3y = -1. \quad \dots (ii)$$

From (i), we get $y = (7 - 2x)$.

Substituting $y = (7 - 2x)$ in (ii), we get

$$4x - 3(7 - 2x) = -1$$

$$\Rightarrow 4x - 21 + 6x = -1$$

$$\Rightarrow 10x = 20$$

$$\Rightarrow x = 2.$$

Substituting $x = 2$ in (i), we get

$$2 \times 2 + y = 7 \Rightarrow y = 7 - 4 = 3.$$

Hence, the solution is $x = 2, y = 3$.

EXAMPLE 2 Solve for x and y , using substitution method:

$$\frac{3x}{2} - \frac{5y}{3} = -2, \quad \frac{x}{3} + \frac{y}{2} = \frac{13}{6}.$$

SOLUTION The given system of equations may be written as

$$9x - 10y + 12 = 0 \quad \dots (i)$$

$$2x + 3y - 13 = 0. \quad \dots (ii)$$

From (ii), we get $y = \frac{13-2x}{3}$.

Substituting $y = \frac{13-2x}{3}$ in (i), we get

$$9x - \frac{10(13-2x)}{3} + 12 = 0$$

$$\Rightarrow 27x - 10(13-2x) + 36 = 0$$

$$\Rightarrow 27x - 130 + 20x + 36 = 0$$

$$\Rightarrow 47x - 94 = 0 \Rightarrow 47x = 94 \Rightarrow x = \frac{94}{47} = 2.$$

Substituting $x = 2$ in (i), we get

$$9 \times 2 - 10y + 12 = 0 \Rightarrow 10y = 30 \Rightarrow y = \frac{30}{10} = 3.$$

Hence, $x = 2$ and $y = 3$ is the required solution.

(II) ELIMINATION METHOD

METHOD In this method, we eliminate one of the unknown variables and proceed using the following steps.

- Step 1. Multiply the given equations by suitable numbers so as to make the coefficients of one of the unknown variables numerically equal.
- Step 2. If the numerically equal coefficients are opposite in sign then add the new equations.
Otherwise, subtract them.
- Step 3. The resulting equation is linear in one unknown variable.
Solve it to get the value of one of the unknown quantities.
- Step 4. Substitute this value in any of the given equations.
- Step 5. Solve it to get the value of the other unknown variable.

EXAMPLE 3 Solve for x and y using elimination method:

$$10x + 3y = 75, 6x - 5y = 11.$$

SOLUTION The given equations are

$$10x + 3y = 75 \quad \dots (i)$$

$$6x - 5y = 11. \quad \dots (ii)$$

Multiplying (i) by 5 and (ii) by 3, we get

$$50x + 15y = 375 \quad \dots (iii)$$

$$18x - 15y = 33. \quad \dots (iv)$$

Adding (iii) and (iv), we get

$$68x = 408 \Rightarrow x = \frac{408}{68} \Rightarrow x = 6.$$

Putting $x = 6$ in (i), we get

$$(10 \times 6) + 3y = 75 \Rightarrow 60 + 3y = 75$$

$$\Rightarrow 3y = 75 - 60 \Rightarrow 3y = 15 \Rightarrow y = 5.$$

$$\therefore x = 6, y = 5.$$

EXAMPLE 4 Solve for x and y :

$$11x + 15y + 23 = 0, 7x - 2y - 20 = 0.$$

SOLUTION The given equations are

$$11x + 15y = -23 \quad \dots (i)$$

$$7x - 2y = 20. \quad \dots (ii)$$

Multiplying (i) by 2 and (ii) by 15 and adding the results, we get

$$22x + 105x = -46 + 300$$

$$\Rightarrow 127x = 254$$

$$\Rightarrow x = \frac{254}{127} = 2.$$

Putting $x = 2$ in (i), we get

$$22 + 15y = -23$$

$$\Rightarrow 15y = -23 - 22$$

$$\Rightarrow 15y = -45 \Rightarrow y = \frac{-45}{15} = -3.$$

Hence, $x = 2$ and $y = -3$.

EXAMPLE 5 Solve for x and y :

[CBSE 2000, '04C, '05]

$$0.4x - 1.5y = 6.5,$$

$$0.3x + 0.2y = 0.9.$$

SOLUTION Multiplying each of the equations by 10, we get

$$4x - 15y = 65 \quad \dots (i)$$

$$3x + 2y = 9. \quad \dots (ii)$$

Multiplying (i) by 2 and (ii) by 15 and adding, we get

$$8x + 45x = 130 + 135$$

$$\Rightarrow 53x = 265$$

$$\Rightarrow x = \frac{265}{53} = 5.$$

Putting $x = 5$ in (ii), we get

$$15 + 2y = 9 \Rightarrow 2y = 9 - 15 \Rightarrow 2y = -6 \Rightarrow y = -3.$$

Hence, $x = 5$ and $y = -3$.EXAMPLE 6 Solve for x and y :

$$\frac{4}{x} + 3y = 8, \frac{6}{x} - 4y = -5. \quad \text{[CBSE 2010]}$$

SOLUTION Putting $\frac{1}{x} = u$, the given equations become:

$$4u + 3y = 8 \quad \dots (i)$$

$$6u - 4y = -5. \quad \dots (ii)$$

Multiplying (i) by 4 and (ii) by 3 and adding the results, we get

$$16u + 18u = 32 - 15$$

$$\Rightarrow 34u = 17$$

$$\Rightarrow u = \frac{17}{34} = \frac{1}{2}. \quad \dots (iii)$$

Putting $u = \frac{1}{2}$ in (i), we get

$$\begin{aligned} \left(4 \times \frac{1}{2}\right) + 3y &= 8 \Rightarrow 2 + 3y = 8 \\ &\Rightarrow 3y = 8 - 2 \Rightarrow 3y = 6 \Rightarrow y = 2. \end{aligned}$$

$$\text{Now, } u = \frac{1}{2} \Rightarrow \frac{1}{x} = \frac{1}{2} \Rightarrow x = 2.$$

Hence, $x = 2$ and $y = 2$.EXAMPLE 7 Solve for x and y :

$$\frac{2}{x} + \frac{3}{y} = 13, \frac{5}{x} - \frac{4}{y} = -2 \quad (x \neq 0 \text{ and } y \neq 0). \quad \text{[CBSE 2008C]}$$

SOLUTION Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, the given equations become

$$2u + 3v = 13 \quad \dots (i)$$

$$5u - 4v = -2. \quad \dots (ii)$$

Multiplying (i) by 4 and (ii) by 3 and adding the results, we get

$$8u + 15u = 52 - 6$$

$$\Rightarrow 23u = 46$$

$$\Rightarrow u = \frac{46}{23} = 2.$$

Putting $u = 2$ in (i), we get

$$(2 \times 2) + 3v = 13 \Rightarrow 3v = 13 - 4 = 9 \Rightarrow v = 3.$$

$$\text{Now, } u = 2 \Rightarrow \frac{1}{x} = 2 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2}.$$

$$\text{And, } v = 3 \Rightarrow \frac{1}{y} = 3 \Rightarrow 3y = 1 \Rightarrow y = \frac{1}{3}.$$

$$\text{Hence, } x = \frac{1}{2} \text{ and } y = \frac{1}{3}.$$

EXAMPLE 8 Solve for x and y :

$$\frac{1}{2x} - \frac{1}{y} = -1, \frac{1}{x} + \frac{1}{2y} = 8 \quad (x \neq 0, y \neq 0). \quad [\text{CBSE 2004C}]$$

SOLUTION Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, the given equations become

$$\frac{u}{2} - v = -1 \Rightarrow u - 2v = -2 \quad \dots (i)$$

$$u + \frac{v}{2} = 8 \Rightarrow 2u + v = 16. \quad \dots (ii)$$

Multiplying (ii) by 2 and adding the result with (i), we get

$$u + 4u = -2 + 32$$

$$\Rightarrow 5u = 30$$

$$\Rightarrow u = \frac{30}{5} = 6.$$

Putting $u = 6$ in (i), we get

$$6 - 2v = -2 \Rightarrow 2v = 8 \Rightarrow v = 4.$$

$$\text{Now, } u = 6 \Rightarrow \frac{1}{x} = 6 \Rightarrow 6x = 1 \Rightarrow x = \frac{1}{6}.$$

$$\text{And, } v = 4 \Rightarrow \frac{1}{y} = 4 \Rightarrow 4y = 1 \Rightarrow y = \frac{1}{4}.$$

$$\text{Hence, } x = \frac{1}{6} \text{ and } y = \frac{1}{4}.$$

EXAMPLE 9 Solve for x and y :

$$\frac{1}{7x} + \frac{1}{6y} = 3, \quad \frac{1}{2x} - \frac{1}{3y} = 5 \quad (x \neq 0, y \neq 0).$$

SOLUTION Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, the given equations become

$$\frac{u}{7} + \frac{v}{6} = 3 \Rightarrow 6u + 7v = 126 \quad \dots (i)$$

$$\frac{u}{2} - \frac{v}{3} = 5 \Rightarrow 3u - 2v = 30. \quad \dots (ii)$$

Multiplying (i) by 2 and (ii) by 7 and adding the results, we get

$$12u + 21u = 252 + 210$$

$$\Rightarrow 33u = 462$$

$$\Rightarrow u = \frac{462}{33} = 14.$$

Putting $u = 14$ in (i), we get

$$(6 \times 14) + 7v = 126$$

$$\Rightarrow 7v = 126 - 84 = 42 \Rightarrow v = \frac{42}{7} = 6.$$

$$\text{Now, } u = 14 \Rightarrow \frac{1}{x} = 14 \Rightarrow 14x = 1 \Rightarrow x = \frac{1}{14}.$$

$$\text{And, } v = 6 \Rightarrow \frac{1}{y} = 6 \Rightarrow 6y = 1 \Rightarrow y = \frac{1}{6}.$$

$$\text{Hence, } x = \frac{1}{14} \text{ and } y = \frac{1}{6}.$$

EXAMPLE 10 Solve for x and y :

$$\frac{3a}{x} - \frac{2b}{y} + 5 = 0, \quad \frac{a}{x} + \frac{3b}{y} - 2 = 0 \quad (x \neq 0, y \neq 0). \quad [\text{CBSE 2005C}]$$

SOLUTION Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, the given equations become

$$3au - 2bv = -5 \quad \dots (i)$$

$$au + 3bv = 2. \quad \dots (ii)$$

Multiplying (ii) by 3 and subtracting (i) from the result, we get

$$9bv + 2bv = 6 + 5$$

$$\Rightarrow 11bv = 11 \Rightarrow v = \frac{11}{11b} = \frac{1}{b}.$$

Putting $v = \frac{1}{b}$ in (ii), we get

$$au + 3 = 2 \Rightarrow au = -1 \Rightarrow u = \frac{-1}{a}.$$

$$\text{Now, } u = \frac{-1}{a} \Rightarrow \frac{1}{x} = \frac{-1}{a} \Rightarrow -x = a \Rightarrow x = -a.$$

$$\text{And, } v = \frac{1}{b} \Rightarrow \frac{1}{y} = \frac{1}{b} \Rightarrow y = b.$$

Hence, $x = -a$ and $y = b$.

EXAMPLE 11 Solve for x and y :

$$6x + 3y = 7xy, 3x + 9y = 11xy \quad (x \neq 0, y \neq 0).$$

SOLUTION On dividing each of the given equations by xy , we get

$$\frac{6}{y} + \frac{3}{x} = 7 \quad \dots \text{(i)}$$

$$\frac{3}{y} + \frac{9}{x} = 11. \quad \dots \text{(ii)}$$

Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$ in (i) and (ii), we get

$$6v + 3u = 7 \quad \dots \text{(iii)}$$

$$3v + 9u = 11. \quad \dots \text{(iv)}$$

Multiplying (iii) by 3 and subtracting (iv) from the result, we get

$$18v - 3v = 21 - 11$$

$$\Rightarrow 15v = 10$$

$$\Rightarrow v = \frac{10}{15} = \frac{2}{3}.$$

Putting $v = \frac{2}{3}$ in (iii), we get

$$\left(6 \times \frac{2}{3}\right) + 3u = 7 \Rightarrow 4 + 3u = 7 \Rightarrow 3u = 3 \Rightarrow u = 1.$$

$$\text{Now, } u = 1 \Rightarrow \frac{1}{x} = 1 \Rightarrow x = 1.$$

$$\text{And, } v = \frac{2}{3} \Rightarrow \frac{1}{y} = \frac{2}{3} \Rightarrow 2y = 3 \Rightarrow y = \frac{3}{2}.$$

Hence, $x = 1$ and $y = \frac{3}{2}$.

EXAMPLE 12 Solve the following system of equations for x and y :

$$\frac{5}{x-1} + \frac{1}{y-2} = 2, \frac{6}{x-1} - \frac{3}{y-2} = 1. \quad [\text{CBSE 2008C, '09}]$$

SOLUTION Putting $\frac{1}{x-1} = y$ and $\frac{1}{y-2} = v$, the given equations become

$$5u + v = 2 \quad \dots \text{(i)}$$

$$6u - 3v = 1. \quad \dots \text{(ii)}$$

Multiplying (i) by 3 and adding the result with (ii), we get

$$\begin{aligned} 15u + 6u &= 6 + 1 \\ \Rightarrow 21u &= 7 \\ \Rightarrow u &= \frac{7}{21} = \frac{1}{3}. \end{aligned}$$

Putting $u = \frac{1}{3}$ in (i), we get

$$\left(5 \times \frac{1}{3}\right) + v = 2 \Rightarrow \frac{5}{3} + v = 2 \Rightarrow v = 2 - \frac{5}{3} \Rightarrow v = \frac{1}{3}.$$

$$\text{Now, } u = \frac{1}{3} \Rightarrow \frac{1}{x-1} = \frac{1}{3} \Rightarrow x-1 = 3 \Rightarrow x = 4$$

[by cross multiplication].

$$\text{Similarly, } v = \frac{1}{3} \Rightarrow \frac{1}{y-2} = \frac{1}{3} \Rightarrow y-2 = 3 \Rightarrow y = 5$$

[by cross multiplication].

Hence, $x = 4$ and $y = 5$.

EXAMPLE 13 Solve for x and y :

$$\frac{15}{x-y} + \frac{22}{x+y} = 5, \frac{40}{x-y} + \frac{55}{x+y} = 13, x \neq y \text{ and } x \neq -y.$$

[CBSE 2008C]

SOLUTION Putting $\frac{1}{x-y} = u$ and $\frac{1}{x+y} = v$, the given equations become

$$15u + 22v = 5 \quad \dots \text{ (i)}$$

$$40u + 55v = 13. \quad \dots \text{ (ii)}$$

Multiplying (ii) by 2 and (i) by 5 and subtracting the results, we get

$$\begin{aligned} 80u - 75u &= 26 - 25 \\ \Rightarrow 5u &= 1 \\ \Rightarrow u &= \frac{1}{5}. \end{aligned}$$

Putting $u = \frac{1}{5}$ in (i), we get

$$\begin{aligned} \left(15 \times \frac{1}{5}\right) + 22v &= 5 \\ \Rightarrow 3 + 22v &= 5 \\ \Rightarrow 22v &= 2 \Rightarrow v = \frac{2}{22} = \frac{1}{11}. \end{aligned}$$

$$\text{Now, } u = \frac{1}{5} \Rightarrow \frac{1}{x-y} = \frac{1}{5} \Rightarrow x-y = 5. \quad \dots \text{ (iii)}$$

$$\text{And, } v = \frac{1}{11} \Rightarrow \frac{1}{x+y} = \frac{1}{11} \Rightarrow x+y = 11. \quad \dots \text{ (iv)}$$

On adding (iii) and (iv), we get $2x = 16 \Rightarrow x = 8$.

On subtracting (iii) from (iv), we get $2y = 6 \Rightarrow y = 3$.

Hence, $x = 8$ and $y = 3$.

EXAMPLE 14 Solve for x and y :

$$\frac{1}{2(2x+3y)} + \frac{12}{7(3x-2y)} = \frac{1}{2}, \quad \frac{7}{(2x+3y)} + \frac{4}{(3x-2y)} = 2,$$

where $(2x+3y) \neq 0$ and $(3x-2y) \neq 0$.

[CBSE 2004C]

SOLUTION Putting $\frac{1}{(2x+3y)} = u$ and $\frac{1}{3x-2y} = v$, the given equations become

$$\frac{u}{2} + \frac{12v}{7} = \frac{1}{2} \Rightarrow 7u + 24v = 7 \quad \dots (i)$$

$$\text{and } 7u + 4v = 2. \quad \dots (ii)$$

Subtracting (ii) from (i), we get

$$20v = 5 \Rightarrow v = \frac{5}{20} = \frac{1}{4}.$$

Putting $v = \frac{1}{4}$ in (i), we get

$$7u + \left(24 \times \frac{1}{4}\right) = 7$$

$$\Rightarrow 7u + 6 = 7 \Rightarrow 7u = 1 \Rightarrow u = \frac{1}{7}.$$

$$\text{Now, } u = \frac{1}{7} \Rightarrow \frac{1}{(2x+3y)} = \frac{1}{7} \Rightarrow 2x+3y = 7 \quad \dots (iii)$$

$$\text{and } v = \frac{1}{4} \Rightarrow \frac{1}{(3x-2y)} = \frac{1}{4} \Rightarrow 3x-2y = 4. \quad \dots (iv)$$

Multiplying (iii) by 2 and (iv) by 3 and adding the results, we get

$$4x + 9x = 14 + 12$$

$$\Rightarrow 13x = 26 \Rightarrow x = 2.$$

Putting $x = 2$ in (iii), we get

$$(2 \times 2) + 3y = 7 \Rightarrow 3y = (7 - 4) = 3 \Rightarrow y = 1.$$

Hence, $x = 2$ and $y = 1$.

EXAMPLE 15 Solve for x and y :

$$\frac{2}{\sqrt{x}} + \frac{3}{\sqrt{y}} = 2, \quad \frac{4}{\sqrt{x}} - \frac{9}{\sqrt{y}} = -1 \quad (x \neq 0, y \neq 0).$$

SOLUTION Putting $\frac{1}{\sqrt{x}} = u$ and $\frac{1}{\sqrt{y}} = v$, the given equations become

$$2u + 3v = 2 \quad \dots (i)$$

$$4u - 9v = -1. \quad \dots (ii)$$

Multiplying (i) by 3 and adding the result with (ii), we get

$$6u + 4u = 6 - 1$$

$$\Rightarrow 10u = 5$$

$$\Rightarrow u = \frac{5}{10} \Rightarrow u = \frac{1}{2}.$$

Putting $u = \frac{1}{2}$ in (i), we get

$$\left(2 \times \frac{1}{2}\right) + 3v = 2$$

$$\Rightarrow 1 + 3v = 2 \Rightarrow 3v = 1 \Rightarrow v = \frac{1}{3}.$$

$$\text{Now, } u = \frac{1}{2} \Rightarrow \frac{1}{\sqrt{x}} = \frac{1}{2} \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4.$$

$$\text{And, } v = \frac{1}{3} \Rightarrow \frac{1}{\sqrt{y}} = \frac{1}{3} \Rightarrow \sqrt{y} = 3 \Rightarrow y = 9.$$

Hence, $x = 4$ and $y = 9$.

EXAMPLE 16 Solve the following system of linear equations:

$$(a - b)x + (a + b)y = a^2 - 2ab - b^2,$$

$$(a + b)(x + y) = a^2 + b^2.$$

[CBSE 2004, '07C, '08]

SOLUTION The given equations may be written as

$$(a - b)x + (a + b)y = a^2 - 2ab - b^2 \quad \dots (i)$$

$$(a + b)x + (a + b)y = a^2 + b^2. \quad \dots (ii)$$

Subtracting (i) from (ii), we get

$$\{(a + b) - (a - b)\}x = 2ab + 2b^2$$

$$\Rightarrow 2bx = 2b(a + b) \Rightarrow x = \frac{2b(a + b)}{2b}$$

$$\Rightarrow x = (a + b).$$

Putting $x = (a + b)$ in (ii), we get

$$(a + b)^2 + (a + b)y = a^2 + b^2$$

$$\Rightarrow (a + b)y = (a^2 + b^2) - (a + b)^2$$

$$\Rightarrow (a + b)y = (a^2 + b^2) - (a^2 + b^2 + 2ab)$$

$$\Rightarrow (a + b)y = -2ab \Rightarrow y = \frac{-2ab}{(a + b)}.$$

$\therefore x = (a + b)$ and $y = \frac{-2ab}{(a + b)}$ is the required solution.

EXAMPLE 17 Solve for x and y :

$$ax + by - a + b = 0, bx - ay - a - b = 0. \quad [\text{CBSE 2000, '04C, '05}]$$

SOLUTION The given equations may be written as

$$ax + by = a - b \quad \dots (i)$$

$$bx - ay = a + b. \quad \dots (ii)$$

Multiplying (i) by a and (ii) by b and adding the results, we get

$$(a^2 + b^2)x = (a^2 + b^2) \Rightarrow x = 1.$$

Putting $x = 1$ in (i), we get

$$(a \times 1) + by = a - b$$

$$\Rightarrow a + by = a - b$$

$$\Rightarrow by = -b \Rightarrow y = \frac{-b}{b} = -1.$$

Hence, $x = 1$ and $y = -1$.

EXAMPLE 18 Solve the following system of linear equations:

$$2(ax - by) + (a + 4b) = 0, 2(bx + ay) + (b - 4a) = 0. \quad [\text{CBSE 2004}]$$

SOLUTION The given equations may be written as

$$2ax - 2by = -a - 4b \quad \dots (i)$$

$$2bx + 2ay = 4a - b. \quad \dots (ii)$$

Multiplying (i) by a and (ii) by b and adding, we get

$$(2a^2 + 2b^2)x = (-a^2 - b^2)$$

$$\Rightarrow 2(a^2 + b^2)x = -(a^2 + b^2) \Rightarrow x = \frac{-1}{2}.$$

Putting $x = \frac{-1}{2}$ in (i), we get

$$2a \times \left(\frac{-1}{2}\right) - 2by = -a - 4b$$

$$\Rightarrow -a - 2by = -a - 4b$$

$$\Rightarrow 2by = 4b \Rightarrow y = \frac{4b}{2b} = 2.$$

Hence, $x = \frac{-1}{2}$ and $y = 2$.

EXAMPLE 19 Solve for x and y :

$$\frac{ax}{b} - \frac{by}{a} = a + b, ax - by = 2ab. \quad [\text{CBSE 2000, '04C, '05, '07C, '09}]$$

SOLUTION The given equations may be written as

$$a^2x - b^2y = a^2b + ab^2 \quad \dots (i)$$

$$ax - by = 2ab. \quad \dots (ii)$$

Multiplying (ii) by b and subtracting the result from (i), we get

$$(a^2 - ab)x = a^2b - ab^2$$

$$\Rightarrow (a^2 - ab)x = b(a^2 - ab) \Rightarrow x = b.$$

Putting $x = b$ in (ii), we get

$$ab - by = 2ab \Rightarrow by = -ab \Rightarrow y = \frac{-ab}{b} = -a.$$

Hence, $x = b$ and $y = -a$.

EXAMPLE 20 Solve for x and y :

$$\sqrt{2}x - \sqrt{3}y = 0, \sqrt{5}x + \sqrt{2}y = 0. \quad [\text{CBSE 2000, '04C, '05}]$$

SOLUTION The given system of equations is

$$\sqrt{2}x - \sqrt{3}y = 0 \quad \dots (i)$$

$$\sqrt{5}x + \sqrt{2}y = 0. \quad \dots (ii)$$

Multiplying (i) by $\sqrt{2}$ and (ii) by $\sqrt{3}$ and adding, we get

$$(2 + \sqrt{15})x = 0 \Rightarrow x = 0.$$

Putting $x = 0$ in (ii), we get

$$\sqrt{2}y = 0 \Rightarrow y = 0.$$

Hence, $x = 0$ and $y = 0$.

A SPECIAL CASE When the coefficients of x and y in one equation are interchanged in the other.

METHOD Step 1. Add the given equations and from it obtain an equation of the form $x + y = a$.

Step 2. Subtract the given equations and from it obtain an equation of the form $x - y = b$.

Now, $x + y = a$ and $x - y = b$ can be solved easily.

EXAMPLE 21 Solve for x and y :

$$37x + 43y = 123, 43x + 37y = 117. \quad [\text{CBSE 2008}]$$

SOLUTION The given equations are

$$37x + 43y = 123 \quad \dots (i)$$

$$43x + 37y = 117. \quad \dots (ii)$$

Clearly, the coefficients of x and y in one equation are interchanged in the other.

Adding (i) and (ii), we get

$$(37 + 43)x + (43 + 37)y = (123 + 117)$$

$$\Rightarrow 80x + 80y = 240$$

$$\Rightarrow 80(x + y) = 240$$

$$\Rightarrow x + y = \frac{240}{80} \Rightarrow x + y = 3. \quad \dots \text{(iii)}$$

Subtracting (i) from (ii), we get

$$6x - 6y = -6$$

$$\Rightarrow 6(x - y) = -6 \Rightarrow x - y = -1 \quad \dots \text{(iv)}$$

Adding (iii) and (iv), we get $2x = 2 \Rightarrow x = 1$.

Subtracting (iv) from (iii), we get $2y = 4 \Rightarrow y = 2$.

Hence, $x = 1$ and $y = 2$.

EXAMPLE 22 Solve for x and y :

$$152x - 378y = -74, -378x + 152y = -604.$$

SOLUTION The given equations are

$$152x - 378y = -74 \quad \dots \text{(i)}$$

$$-378x + 152y = -604. \quad \dots \text{(ii)}$$

Clearly, the coefficients of x and y in one equation are interchanged in the other.

Adding (i) and (ii), we get

$$\{152 + (-378)\}x + \{(-378) + 152\}y = -(74 + 604)$$

$$\Rightarrow (-226)x + (-226)y = -678$$

$$\Rightarrow (-226)(x + y) = -678$$

$$\Rightarrow (x + y) = \frac{-678}{-226} \Rightarrow x + y = 3. \quad \dots \text{(iii)}$$

Subtracting (ii) from (i), we get

$$(152 + 378)x + (-378 - 152)y = (-74 + 604)$$

$$\Rightarrow 530x - 530y = 530$$

$$\Rightarrow 530(x - y) = 530 \Rightarrow x - y = 1. \quad \dots \text{(iv)}$$

Adding (iii) and (iv), we get $2x = 4 \Rightarrow x = 2$.

Subtracting (iv) from (iii), we get $2y = 2 \Rightarrow y = 1$.

Hence, $x = 2$ and $y = 1$.

SOME MORE EXAMPLES

EXAMPLE 23 Solve for x and y :

$$\frac{x+1}{2} + \frac{y-1}{3} = 8, \frac{x-1}{3} + \frac{y+1}{2} = 9. \quad [\text{CBSE 2007}]$$

SOLUTION The given equations may be written as

$$3(x+1) + 2(y-1) = 48 \Rightarrow 3x + 2y = 47 \quad \dots (i)$$

$$\text{and } 2(x-1) + 3(y+1) = 54 \Rightarrow 2x + 3y = 53. \quad \dots (ii)$$

Multiplying (i) by 2 and (ii) by 3 and subtracting, we get

$$(4-9)y = 94 - 159$$

$$\Rightarrow -5y = -65 \Rightarrow y = \frac{-65}{-5} \Rightarrow y = 13.$$

Putting $y = 13$ in (i), we get

$$3x + (2 \times 13) = 47 \Rightarrow 3x = (47 - 26) = 21$$

$$\Rightarrow x = \frac{21}{3} = 7.$$

Hence, $x = 7$ and $y = 13$.

EXAMPLE 24 Solve for x and y :

$$\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}, \frac{4}{x} + \frac{9}{y} = \frac{21}{xy} \quad (x \neq 0, y \neq 0).$$

SOLUTION Multiplying each equation throughout by xy , we get

$$2y + 3x = 9 \quad \dots (i)$$

$$4y + 9x = 21. \quad \dots (ii)$$

Multiplying (i) by 3 and subtracting (ii) from the result, we get

$$(6-4)y = (27-21) \Rightarrow 2y = 6 \Rightarrow y = 3.$$

Putting $y = 3$ in (i), we get

$$(2 \times 3) + 3x = 9 \Rightarrow 6 + 3x = 9 \Rightarrow 3x = 3 \Rightarrow x = 1.$$

Hence, $x = 1$ and $y = 3$.

EXAMPLE 25 Solve for x and y :

$$\frac{xy}{x+y} = \frac{6}{5}, \frac{xy}{y-x} = 6 \quad (x \neq 0, y \neq 0 \text{ and } x \neq y).$$

SOLUTION The given equations may be written as

$$\frac{x+y}{xy} = \frac{5}{6} \Rightarrow \frac{1}{y} + \frac{1}{x} = \frac{5}{6} \quad \dots (i)$$

$$\frac{y-x}{xy} = \frac{1}{6} \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{6} \quad \dots (ii)$$

Adding (i) and (ii), we get

$$\frac{2}{x} = \left(\frac{5}{6} + \frac{1}{6} \right) = \frac{6}{6} = 1 \Rightarrow x = 2.$$

Subtracting (ii) from (i), we get

$$\frac{2}{y} = \left(\frac{5}{6} - \frac{1}{6} \right) = \frac{4}{6} = \frac{2}{3} \Rightarrow y = 3.$$

Hence, $x = 2$ and $y = 3$.

EXAMPLE 26 Solve for x and y :

$$\frac{7x-2y}{xy} = 5, \frac{8x+7y}{xy} = 15 \quad (x \neq 0, y \neq 0).$$

SOLUTION We have

$$\frac{7x-2y}{xy} = 5 \Rightarrow \frac{7}{y} - \frac{2}{x} = 5 \quad \dots (i)$$

$$\frac{8x+7y}{xy} = 15 \Rightarrow \frac{8}{y} + \frac{7}{x} = 15. \quad \dots (ii)$$

Multiplying (i) by 7 and (ii) by 2 and adding the results, we get

$$\left(\frac{49}{y} + \frac{16}{y} \right) = (35 + 30)$$

$$\Rightarrow \frac{65}{y} = 65 \Rightarrow 65y = 65 \Rightarrow y = 1.$$

Putting $y = 1$ in (ii), we get

$$\frac{8}{1} + \frac{7}{x} = 15 \Rightarrow \frac{7}{x} = (15 - 8) = 7 \Rightarrow 7x = 7 \Rightarrow x = 1.$$

Hence, $x = 1$ and $y = 1$.

EXERCISE 3B

Solve for x and y :

1. $x + y = 3,$

2. $x - y = 3,$

$4x - 3y = 26.$

$\frac{x}{3} + \frac{y}{2} = 6.$

3. $2x + 3y = 0,$

$3x + 4y = 5.$

5. $3x - 5y - 19 = 0,$

$-7x + 3y + 1 = 0.$

7. $\frac{x}{2} - \frac{y}{9} = 6,$

$\frac{x}{7} + \frac{y}{3} = 5.$

9. $4x - 3y = 8,$

$6x - y = \frac{29}{3}.$

11. $2x + 5y = \frac{8}{3},$

$3x - 2y = \frac{5}{6}.$

13. $0.4x + 0.3y = 1.7,$

$0.7x - 0.2y = 0.8.$

15. $7(y + 3) - 2(x + 2) = 14,$

$4(y - 2) + 3(x - 3) = 2.$

17. $\frac{x + y - 8}{2} = \frac{x + 2y - 14}{3} = \frac{3x + y - 12}{11}$

HINT $a = b = c \Rightarrow a = b$ and $b = c.$

18. $\frac{5}{x} + 6y = 13,$

$\frac{3}{x} + 4y = 7 \quad (x \neq 0).$

20. $2x - \frac{3}{y} = 9,$

$3x + \frac{7}{y} = 2 \quad (y \neq 0).$

22. $\frac{9}{x} - \frac{4}{y} = 8,$

$\frac{13}{x} + \frac{7}{y} = 101 \quad (x \neq 0, y \neq 0).$

24. $\frac{1}{2x} + \frac{1}{3y} = 2,$

$\frac{1}{3x} + \frac{1}{2y} = \frac{13}{6} \quad (x \neq 0, y \neq 0).$

26. $x + y = 5xy,$

$3x + 2y = 13xy \quad (x \neq 0, y \neq 0).$

4. $2x - 3y = 13,$

$7x - 2y = 20.$

6. $2x - y + 3 = 0,$

$3x - 7y + 10 = 0.$

8. $\frac{x}{3} + \frac{y}{4} = 11,$

$\frac{5x}{6} - \frac{y}{3} = -7.$

10. $2x - \frac{3y}{4} = 3,$

$5x = 2y + 7.$

12. $2x + 3y + 1 = 0,$

$\frac{7 - 4x}{3} = y.$

14. $0.3x + 0.5y = 0.5,$

$0.5x + 0.7y = 0.74$

16. $6x + 5y = 7x + 3y + 1 = 2(x + 6y - 1)$

19. $x + \frac{6}{y} = 6,$

$3x - \frac{8}{y} = 5 \quad (y \neq 0). \quad [\text{CBSE 2007}]$

21. $\frac{3}{x} - \frac{1}{y} + 9 = 0,$

$\frac{2}{x} + \frac{3}{y} = 5 \quad (x \neq 0, y \neq 0).$

23. $\frac{5}{x} - \frac{3}{y} = 1,$

$\frac{3}{2x} + \frac{2}{3y} = 5 \quad (x \neq 0, y \neq 0).$

25. $4x + 6y = 3xy,$

$8x + 9y = 5xy \quad (x \neq 0, y \neq 0).$

27. $\frac{5}{x + y} - \frac{2}{x - y} = -1,$

$\frac{15}{x + y} + \frac{7}{x - y} = 10.$

28. $\frac{3}{x+y} + \frac{2}{x-y} = 2,$
 $\frac{9}{x+y} - \frac{4}{x-y} = 1.$
29. $\frac{5}{x+1} - \frac{2}{y-1} = \frac{1}{2},$
 $\frac{10}{x+1} + \frac{2}{y-1} = \frac{5}{2}, \quad x \neq -1 \text{ and } y \neq 1.$
30. $\frac{44}{x+y} + \frac{30}{x-y} = 10,$
 $\frac{55}{x+y} + \frac{40}{x-y} = 13.$
31. $\frac{10}{x+y} + \frac{2}{x-y} = 4,$
 $\frac{15}{x+y} - \frac{9}{x-y} = -2.$
32. $71x + 37y = 253,$
 $37x + 71y = 287. \quad [\text{CBSE 2007C}]$
33. $217x + 131y = 913,$
 $131x + 217y = 827.$
34. $23x - 29y = 98,$
 $29x - 23y = 110.$
35. $\frac{2x+5y}{xy} = 6,$
 $\frac{4x-5y}{xy} = -3.$
36. $\frac{1}{(3x+y)} + \frac{1}{(3x-y)} = \frac{3}{4},$
 $\frac{1}{2(3x+y)} - \frac{1}{2(3x-y)} = \frac{-1}{8}.$
37. $\frac{1}{2(x+2y)} + \frac{5}{3(3x-2y)} = \frac{-3}{2},$
 $\frac{5}{4(x+2y)} - \frac{3}{5(3x-2y)} = \frac{61}{60}.$
38. $\frac{2}{(3x+2y)} + \frac{3}{(3x-2y)} = \frac{17}{5},$
 $\frac{5}{(3x+2y)} + \frac{1}{(3x-2y)} = 2.$
39. $3(2x+y) = 7xy,$
 $3(x+3y) = 11xy \quad (x \neq 0 \text{ and } y \neq 0).$
40. $x+y = a+b,$
 $ax-by = a^2-b^2.$
41. $\frac{x}{a} + \frac{y}{b} = 2,$
 $ax-by = a^2-b^2. \quad [\text{CBSE 2005}]$
42. $px+qy = p-q,$
 $qx-py = p+q.$
43. $\frac{x}{a} - \frac{y}{b} = 0,$
 $ax+by = (a^2+b^2).$
44. $6(ax+by) = 3a+2b,$
 $6(bx-ay) = 3b-2a.$
45. $ax-by = a^2+b^2,$
 $x+y = 2a. \quad [\text{CBSE 2006C}]$
46. $\frac{bx}{a} - \frac{ay}{b} + a+b = 0,$
 $bx-ay+2ab = 0. \quad [\text{CBSE 2006}]$
47. $\frac{bx}{a} + \frac{ay}{b} = a^2+b^2,$
 $x+y = 2ab. \quad [\text{CBSE 2010}]$
48. $x+y = a+b,$
 $ax-by = a^2-b^2.$
49. $a^2x+b^2y = c^2,$
 $b^2x+a^2y = d^2.$
50. $\frac{x}{a} + \frac{y}{b} = a+b, \quad \frac{x}{a^2} + \frac{y}{b^2} = 2.$

ANSWERS (EXERCISE 3B)

1. $x = 5, y = -2$ 2. $x = 9, y = 6$ 3. $x = 15, y = -10$ 4. $x = 2, y = -3$

5. $x = -2, y = -5$ 6. $x = -1, y = 1$ 7. $x = 14, y = 9$ 8. $x = 6, y = 36$
 9. $x = \frac{3}{2}, y = \frac{-2}{3}$ 10. $x = 3, y = 4$ 11. $x = \frac{1}{2}, y = \frac{1}{3}$ 12. $x = 4, y = -3$
 13. $x = 2, y = 3$ 14. $x = 0.5, y = 0.7$ 15. $x = 5, y = 1$ 16. $x = 3, y = 2$
 17. $x = 2, y = 6$ 18. $x = \frac{1}{5}, y = -2$ 19. $x = 3, y = 2$ 20. $x = 3, y = -1$
 21. $x = \frac{-1}{2}, y = \frac{1}{3}$ 22. $x = \frac{1}{4}, y = \frac{1}{7}$ 23. $x = \frac{1}{2}, y = \frac{1}{3}$ 24. $x = \frac{1}{2}, y = \frac{1}{3}$
 25. $x = 3, y = 4$ 26. $x = \frac{1}{2}, y = \frac{1}{3}$ 27. $x = 3, y = 2$ 28. $x = \frac{5}{2}, y = \frac{1}{2}$
 29. $x = 4, y = 5$ 30. $x = 8, y = 3$ 31. $x = \frac{21}{8}, y = \frac{9}{8}$ 32. $x = 2, y = 3$
 33. $x = 3, y = 2$ 34. $x = 3, y = -1$ 35. $x = 1, y = 2$ 36. $x = 1, y = 1$
 37. $x = \frac{1}{2}, y = \frac{5}{4}$ 38. $x = 1, y = 1$ 39. $x = 1, y = \frac{3}{2}$ 40. $x = a, y = b$
 41. $x = a, y = b$ 42. $x = 1, y = -1$ 43. $x = a, y = b$ 44. $x = \frac{1}{2}, y = \frac{1}{3}$
 45. $x = a + b, y = a - b$ 46. $x = -a, y = b$ 47. $x = ab, y = ab$
 48. $x = a, y = b$ 49. $x = \frac{(a^2c^2 - b^2d^2)}{(a^4 - b^4)}, y = \frac{(a^2d^2 - b^2c^2)}{(a^4 - b^4)}$ 50. $x = a^2, y = b^2$

HINTS TO SOME SELECTED QUESTIONS

9. The second equation is $18x - 3y = 29$.
 11. Given equations are $6x + 15y = 8, 18x - 12y = 5$.
 12. $\frac{7-4x}{3} = y \Rightarrow 7 - 4x = 3y \Rightarrow 4x + 3y = 7$.
 13. Multiply each of the given equations throughout by 10.
 14. Multiply each of the given equations throughout by 10.
 15. The given equations are

$$2x - 7y = 3, 3x + 4y = 19.$$

16. $6x + 5y = 7x + 3y + 1 \Rightarrow x - 2y = -1$.

$$7x + 3y + 1 = 2x + 12y - 2 \Rightarrow 5x - 9y = -3.$$

17. $\frac{x+y-8}{2} = \frac{x+2y-14}{3} \Rightarrow 3(x+y-8) = 2(x+2y-14)$

$$\Rightarrow x - y = -4.$$

... (i)

$$\frac{x+2y-14}{3} = \frac{3x+y-12}{11} \Rightarrow 11(x+2y-14) = 3(3x+y-12)$$

$$\Rightarrow 2x + 19y = 118.$$

... (ii)

Now, solve (i) and (ii).

18. Multiply (i) by 3 and (ii) by 5 and subtract.
 19. Multiply (i) by 4 and (ii) by 3 and add.

23. Put $\frac{1}{x} = u$ and $\frac{1}{y} = v$ to get

$$5u - 3v = 1 \text{ and } \frac{3u}{2} + \frac{2v}{3} = 5 \text{ or } 9u + 4v = 30.$$

Now, solve $5u - 3v = 1$ and $9u + 4v = 30$.

24. Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, we get

$$\frac{u}{2} + \frac{v}{3} = 2 \Rightarrow 3u + 2v = 12 \quad \dots (i)$$

$$\frac{u}{3} + \frac{v}{2} = \frac{13}{6} \Rightarrow 2u + 3v = 13. \quad \dots (ii)$$

Add (i) and (ii). Subtract (ii) from (i).

25. Divide each equation throughout by xy to get $\frac{4}{y} + \frac{6}{x} = 3, \frac{8}{y} + \frac{9}{x} = 5$.

27. Put $\frac{1}{x+y} = u$ and $\frac{1}{x-y} = v$.

29. Put $\frac{1}{x+1} = u$ and $\frac{1}{y-1} = v$.

32. Add (i) and (ii) to get $108(x+y) = 540 \Rightarrow x+y = 5$.

Subtract (ii) from (i) to get $34(x-y) = -34 \Rightarrow x-y = -1$.

35. $\frac{2}{y} + \frac{5}{x} = 6, \frac{4}{y} - \frac{5}{x} = -3$. Now, add.

36. Put $\frac{1}{(3x+y)} = u$ and $\frac{1}{(3x-y)} = v$.

37. Put $\frac{1}{(x+2y)} = u$ and $\frac{1}{(3x-2y)} = v$.

39. $6x + 3y = 7xy \Rightarrow \frac{6}{y} + \frac{3}{x} = 7. \quad \dots (i)$

$$3x + 9y = 11xy \Rightarrow \frac{3}{y} + \frac{9}{x} = 11. \quad \dots (ii)$$

40. Multiply (i) by b and add (ii) to the result.

$$41. bx + ay = 2ab \quad \dots (i) \quad \text{and} \quad ax - by = a^2 - b^2. \quad \dots (ii)$$

Multiply (i) by b and (ii) by a and add.

42. Multiply (i) by p and (ii) by q and add.

$$43. bx - ay = 0 \quad \dots (i) \quad \text{and} \quad ax + by = (a^2 + b^2). \quad \dots (ii)$$

Multiply (i) by b and (ii) by a and add.

$$44. 6ax + 6by = 3a + 2b \quad \dots (i) \quad \text{and} \quad 6bx - 6ay = 3b - 2a. \quad \dots (ii)$$

Multiply (i) by a and (ii) by b and add.

45. Multiply (ii) by b and add the result with (i).

$$46. b^2x - a^2y + a^2b + ab^2 = 0 \quad \dots (i) \quad \text{and} \quad b^2x - aby + 2ab^2 = 0. \quad \dots (ii)$$

Now, subtract (ii) from (i).

$$47. b^2x + a^2y = a^3b + ab^3 \quad \dots (i) \quad \text{and} \quad x + y = 2ab. \quad \dots (ii)$$

Multiply (ii) by b^2 and subtract the result from (i).

48. Multiply (i) by b and add the result with (ii).

49. Multiply (i) by b^2 and (ii) by a^2 and subtract the results.

$$50. bx + ay = a^2b + ab^2 \quad \dots (i) \quad \text{and} \quad b^2x + a^2y = 2a^2b^2. \quad \dots (ii)$$

METHOD OF CROSS MULTIPLICATION

THEOREM *The system of two linear equations*

$$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0,$$

where $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, has a unique solution, given by

$$x = \frac{(b_1c_2 - b_2c_1)}{(a_1b_2 - a_2b_1)}, y = \frac{(c_1a_2 - c_2a_1)}{(a_1b_2 - a_2b_1)}.$$

PROOF The given equations are

$$a_1x + b_1y + c_1 = 0 \quad \dots (i)$$

$$a_2x + b_2y + c_2 = 0. \quad \dots (ii)$$

Multiplying (i) by b_2 , (ii) by b_1 and subtracting, we get

$$(a_1b_2 - a_2b_1)x = (b_1c_2 - b_2c_1)$$

$$\Rightarrow x = \frac{(b_1c_2 - b_2c_1)}{(a_1b_2 - a_2b_1)} \quad [\because \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow (a_1b_2 - a_2b_1) \neq 0].$$

Multiplying (ii) by a_1 , (i) by a_2 and subtracting, we get

$$(a_1b_2 - a_2b_1)y = (c_1a_2 - c_2a_1)$$

$$\Rightarrow y = \frac{(c_1a_2 - c_2a_1)}{(a_1b_2 - a_2b_1)} \quad [\because (a_1b_2 - a_2b_1) \neq 0].$$

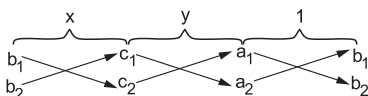
Hence, a unique solution exists, which is given by

$$x = \frac{(b_1c_2 - b_2c_1)}{(a_1b_2 - a_2b_1)}, y = \frac{(c_1a_2 - c_2a_1)}{(a_1b_2 - a_2b_1)}.$$

This is customarily written as

$$\frac{x}{(b_1c_2 - b_2c_1)} = \frac{y}{(c_1a_2 - c_2a_1)} = \frac{1}{(a_1b_2 - a_2b_1)}.$$

REMARK The following diagram helps in remembering the above solution.



Rule Numbers with downward arrows are multiplied first; and from this product, the product of numbers with upward arrows is subtracted.

SOLVED EXAMPLES

EXAMPLE 1 Solve the system of equations $2x + 3y = 17$, $3x - 2y = 6$ by the method of cross multiplication.

SOLUTION The given equations may be written as

$$2x + 3y - 17 = 0 \quad \dots (i)$$

$$3x - 2y - 6 = 0. \quad \dots (ii)$$

By cross multiplication, we have

$$\begin{aligned} \therefore \frac{x}{\{3 \times (-6) - (-2) \times (-17)\}} &= \frac{y}{\{(-17) \times 3 - (-6) \times 2\}} \\ &= \frac{1}{\{2 \times (-2) - 3 \times 3\}} \end{aligned}$$

$$\Rightarrow \frac{x}{(-18 - 34)} = \frac{y}{(-51 + 12)} = \frac{1}{(-4 - 9)}$$

$$\Rightarrow \frac{x}{-52} = \frac{y}{-39} = \frac{1}{-13}$$

$$\Rightarrow x = \frac{-52}{-13} = 4, y = \frac{-39}{-13} = 3.$$

Hence, $x = 4$ and $y = 3$ is the required solution.

EXAMPLE 2 Solve $4x - 7y + 28 = 0$, $5y - 7x + 9 = 0$.

SOLUTION The given equations are

$$4x - 7y + 28 = 0 \quad \dots (i)$$

$$-7x + 5y + 9 = 0. \quad \dots (ii)$$

By cross multiplication, we have

$$\therefore \frac{x}{\{(-7) \times 9 - 5 \times 28\}} = \frac{y}{\{28 \times (-7) - 9 \times 4\}} = \frac{1}{\{4 \times 5 - (-7) \times (-7)\}}$$

$$\Rightarrow \frac{x}{(-63-140)} = \frac{y}{(-196-36)} = \frac{1}{(20-49)}$$

$$\Rightarrow \frac{x}{-203} = \frac{y}{-232} = \frac{1}{-29}$$

$$\Rightarrow x = \left(\frac{-203}{-29}\right) = 7 \text{ and } y = \left(\frac{-232}{-29}\right) = 8.$$

Hence, $x = 7$ and $y = 8$ is the required solution.

EXAMPLE 3 Solve $\frac{2}{x} + \frac{3}{y} = 13$, $\frac{5}{x} - \frac{4}{y} = -2$, where $x \neq 0$ and $y \neq 0$.

SOLUTION Taking $\frac{1}{x} = u$ and $\frac{1}{y} = v$, the given equations become

$$2u + 3v - 13 = 0 \quad \dots (i)$$

$$5u - 4v + 2 = 0. \quad \dots (ii)$$

By cross multiplication, we have

$$\therefore \frac{u}{[3 \times 2 - (-4) \times (-13)]} = \frac{v}{[(-13) \times 5 - 2 \times 2]} = \frac{1}{[2 \times (-4) - 5 \times 3]}$$

$$\Rightarrow \frac{u}{-46} = \frac{v}{-69} = \frac{1}{-23}$$

$$\Rightarrow u = \left(\frac{-46}{-23}\right) = 2, v = \left(\frac{-69}{-23}\right) = 3$$

$$\Rightarrow \frac{1}{x} = 2, \frac{1}{y} = 3$$

$$\Rightarrow x = \frac{1}{2}, y = \frac{1}{3}.$$

Hence, $x = \frac{1}{2}$ and $y = \frac{1}{3}$ is the required solution.

EXAMPLE 4 Solve $ax + by = c$, $bx + ay = 1 + c$.

SOLUTION The given equations may be written as

$$ax + by - c = 0 \quad \dots (i)$$

$$bx + ay - (1 + c) = 0. \quad \dots (ii)$$

By cross multiplication, we have

$$\begin{aligned}\therefore \frac{x}{-b(1+c)+ac} &= \frac{y}{-cb+a(1+c)} = \frac{1}{(a^2-b^2)} \\ \Rightarrow x &= \frac{c(a-b)-b}{(a^2-b^2)} \text{ and } y = \frac{c(a-b)+a}{(a^2-b^2)} \\ \text{Hence, } x &= \frac{c(a-b)-b}{(a^2-b^2)} \text{ and } y = \frac{c(a-b)+a}{(a^2-b^2)}.\end{aligned}$$

EXERCISE 3C

Solve each of the following systems of equations by using the method of cross multiplication:

1. $x + 2y + 1 = 0,$
 $2x - 3y - 12 = 0.$
2. $3x - 2y + 3 = 0,$
 $4x + 3y - 47 = 0.$
3. $6x - 5y - 16 = 0,$
 $7x - 13y + 10 = 0.$
4. $3x + 2y + 25 = 0,$
 $2x + y + 10 = 0.$
5. $2x + 5y = 1,$
 $2x + 3y = 3.$
6. $2x + y = 35,$
 $3x + 4y = 65.$
7. $7x - 2y = 3,$
 $22x - 3y = 16.$
8. $\frac{x}{6} + \frac{y}{15} = 4,$
 $\frac{x}{3} - \frac{y}{12} = \frac{19}{4}.$
9. $\frac{1}{x} + \frac{1}{y} = 7,$
 $\frac{2}{x} + \frac{3}{y} = 17 \ (x \neq 0, y \neq 0).$
10. $\frac{5}{(x+y)} - \frac{2}{(x-y)} + 1 = 0,$
 $\frac{15}{(x+y)} + \frac{7}{(x-y)} - 10 = 0 \ (x \neq y, x \neq -y).$
11. $\frac{ax}{b} - \frac{by}{a} = a + b,$
 $ax - by = 2ab. \quad [\text{CBSE 2007C}]$
12. $2ax + 3by = (a + 2b),$
 $3ax + 2by = (2a + b).$
13. $\frac{a}{x} - \frac{b}{y} = 0, \frac{ab^2}{x} + \frac{a^2b}{y} = (a^2 + b^2),$ where $x \neq 0$ and $y \neq 0.$

ANSWERS (EXERCISE 3C)

1. $x = 3, y = -2$
2. $x = 5, y = 9$
3. $x = 6, y = 4$
4. $x = 5, y = -20$
5. $x = 3, y = -1$
6. $x = 15, y = 5$
7. $x = 1, y = 2$
8. $x = 18, y = 15$
9. $x = \frac{1}{4}, y = \frac{1}{3}$
10. $x = 3, y = 2$
11. $x = b, y = -a$
12. $x = \frac{(4a-b)}{5a}, y = \frac{(4b-a)}{5b}$
13. $x = a, y = b$

HINTS TO SOME SELECTED QUESTIONS

8. The given equations are $5x + 2y - 120 = 0$ and $4x - y - 57 = 0$.

9. Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, we get

$$u + v - 7 = 0 \text{ and } 2u + 3v - 17 = 0.$$

Now, solve for u and v .

10. Putting $\frac{1}{(x+y)} = u$ and $\frac{1}{(x-y)} = v$, we get

$$5u - 2v + 1 = 0 \text{ and } 15u + 7v - 10 = 0.$$

Now, solve for u and v .

11. The given equations are $a^2x - b^2y - (a^2b + ab^2) = 0$ and $ax - by - 2ab = 0$.

12. $2ax + 3by - (a + 2b) = 0$ and $3ax + 2by - (2a + b) = 0$.

13. Put $\frac{1}{x} = u$ and $\frac{1}{y} = v$. Then, $au - bv = 0$, $ab^2u + a^2bv - (a^2 + b^2) = 0$.

CONDITIONS FOR SOLVABILITY OF LINEAR EQUATIONS**CONSISTENT AND INCONSISTENT SYSTEMS OF LINEAR EQUATIONS**

A system of equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ is said to be *consistent* if it has at least one solution. On the other hand, the above system is said to be *inconsistent* if it has no solution at all.

CONDITIONS FOR SOLVABILITY OF LINEAR EQUATIONS

The system of a pair of linear equations

$$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0$$

(i) has a unique solution,

$$\text{if } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\text{and the solution is } x = \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \right), y = \left(\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right);$$

(ii) has an infinite number of solutions,

$$\text{if } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2};$$

(iii) has no solution (i.e., inconsistent),

$$\text{if } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}.$$

SOLVED EXAMPLES**EXAMPLE 1** Show that the system of equations

$$2x + 5y = 17, 5x + 3y = 14$$

has a unique solution. Find the solution.

SOLUTION The given system of equations is

$$2x + 5y - 17 = 0, 5x + 3y - 14 = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0,$$

where $a_1 = 2, b_1 = 5, c_1 = -17$ and $a_2 = 5, b_2 = 3, c_2 = -14$.

$$\therefore \frac{a_1}{a_2} = \frac{2}{5}, \frac{b_1}{b_2} = \frac{5}{3}, \frac{c_1}{c_2} = \frac{-17}{-14} = \frac{17}{14}.$$

$$\text{Thus, } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}.$$

Hence, the given system of equations has a unique solution.

By cross multiplication, we have

$$\begin{array}{ccccc} & x & & y & & 1 \\ 5 & & -17 & & 2 & & 5 \\ 3 & & -14 & & 5 & & 3 \end{array}$$

$$\therefore \frac{x}{(-70 + 51)} = \frac{y}{(-85 + 28)} = \frac{1}{(6 - 25)}$$

$$\Rightarrow \frac{x}{-19} = \frac{y}{-57} = \frac{1}{-19}$$

$$\Rightarrow x = \frac{-19}{-19} = 1 \text{ and } y = \frac{-57}{-19} = 3.$$

Hence, $x = 1$ and $y = 3$ is the required solution.**EXAMPLE 2** Find the values of k for which the system of equations

$$x - 2y = 3, 3x + ky = 1$$

has a unique solution.

SOLUTION The given system of equations is

$$x - 2y - 3 = 0, 3x + ky - 1 = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 1, b_1 = -2, c_1 = -3$ and $a_2 = 3, b_2 = k, c_2 = -1$.For a unique solution, we must have $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

$$\therefore \frac{1}{3} \neq \frac{-2}{k} \Rightarrow k \neq -6.$$

Hence, the given system of equations will have a unique solution for all real values of k , other than -6 .

EXAMPLE 3 Show that the system of equations

$$4x + 6y = 7, 12x + 18y = 21$$

has infinitely many solutions.

SOLUTION The given system of equations is

$$4x + 6y - 7 = 0, 12x + 18y - 21 = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 4, b_1 = 6, c_1 = -7$ and $a_2 = 12, b_2 = 18, c_2 = -21$.

$$\therefore \frac{a_1}{a_2} = \frac{4}{12} = \frac{1}{3}, \frac{b_1}{b_2} = \frac{6}{18} = \frac{1}{3} \text{ and } \frac{c_1}{c_2} = \left(\frac{-7}{-21}\right) = \frac{1}{3}.$$

$$\text{Thus, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

Hence, the given system of equations has infinitely many solutions.

EXAMPLE 4 Find the value of k for which the following pair of linear equations has infinitely many solutions:

$$2x - 3y = 7, (k + 1)x + (1 - 2k)y = (5k - 4). \quad [\text{CBSE 2008C}]$$

SOLUTION The given equations are

$$2x - 3y - 7 = 0,$$

$$(k + 1)x + (1 - 2k)y + (4 - 5k) = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 2, b_1 = -3, c_1 = -7$

and $a_2 = (k + 1), b_2 = (1 - 2k), c_2 = (4 - 5k)$.

Let the given system of equations have infinitely many solutions.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{2}{(k + 1)} = \frac{-3}{(1 - 2k)} = \frac{-7}{(4 - 5k)}$$

$$\begin{aligned} \Rightarrow \frac{2}{(k+1)} &= \frac{3}{(2k-1)} = \frac{7}{(5k-4)} \\ \Rightarrow \frac{2}{(k+1)} &= \frac{3}{(2k-1)} \text{ and } \frac{3}{(2k-1)} = \frac{7}{(5k-4)} \\ \Rightarrow 4k-2 &= 3k+3 \text{ and } 15k-12 = 14k-7 \\ \Rightarrow k &= 5 \text{ and } k = 5. \end{aligned}$$

Hence, $k = 5$.

EXAMPLE 5 Find the value of k for which the given system of equations has infinitely many solutions:

$$kx + 3y = k - 3, 12x + ky = k.$$

SOLUTION The given system of equations is

$$kx + 3y + (3 - k) = 0,$$

$$12x + ky - k = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = k, b_1 = 3, c_1 = (3 - k)$ and $a_2 = 12, b_2 = k, c_2 = -k$.

Let the given system of equations have infinitely many solutions.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{k}{12} = \frac{3}{k} = \frac{(3-k)}{-k}$$

$$\Rightarrow \frac{k}{12} = \frac{3}{k} = \frac{k-3}{k}$$

$$\Rightarrow \frac{k}{12} = \frac{3}{k} \text{ and } \frac{3}{k} = \frac{k-3}{k}$$

$$\Rightarrow k^2 = 36 \text{ and } k^2 - 6k = 0$$

$$\Rightarrow (k = 6 \text{ or } k = -6) \text{ and } k(k - 6) = 0$$

$$\Rightarrow (k = 6 \text{ or } k = -6) \text{ and } (k = 0 \text{ or } k = 6)$$

$$\Rightarrow k = 6.$$

Hence, the given system of equations has infinitely many solutions when $k = 6$.

EXAMPLE 6 Find the value of k for which the given system of equations has infinitely many solutions:

$$x + (k + 1)y = 5,$$

$$(k + 1)x + 9y + (1 - 8k) = 0.$$

[CBSE 2003]

SOLUTION The given system of equations is

$$x + (k+1)y - 5 = 0 \quad \dots (i)$$

$$(k+1)x + 9y + (1-8k) = 0. \quad \dots (ii)$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 1, b_1 = (k+1), c_1 = -5$

and $a_2 = (k+1), b_2 = 9, c_2 = (1-8k).$

$$\therefore \frac{a_1}{a_2} = \frac{1}{(k+1)}, \frac{b_1}{b_2} = \frac{(k+1)}{9} \text{ and } \frac{c_1}{c_2} = \frac{-5}{(1-8k)} = \frac{5}{(8k-1)}.$$

Let the given system of equations have infinitely many solutions.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{1}{(k+1)} = \frac{(k+1)}{9} = \frac{5}{(8k-1)}$$

$$\Rightarrow \frac{1}{(k+1)} = \frac{(k+1)}{9} \text{ and } \frac{(k+1)}{9} = \frac{5}{(8k-1)}$$

$$\Rightarrow (k+1)^2 = 9 \text{ and } (k+1)(8k-1) = 45$$

$$\Rightarrow (k+1 = 3 \text{ or } k+1 = -3) \text{ and } 8k^2 + 7k - 46 = 0$$

$$\Rightarrow (k = 2 \text{ or } k = -4) \text{ and } (k-2)(8k+23) = 0$$

$$\Rightarrow (k = 2 \text{ or } k = -4) \text{ and } \left(k = 2 \text{ or } k = \frac{-23}{8}\right)$$

$$\Rightarrow k = 2.$$

Hence, the given system of equations will have infinitely many solutions when $k = 2$.

EXAMPLE 7 Find the values of a and b for which the following pair of linear equations have an infinite number of solutions:

$$2x + 3y = 7, (a-b)x + (a+b)y = 3a + b - 2. \quad [\text{CBSE 2008C}]$$

SOLUTION The given equations are

$$2x + 3y - 7 = 0, \quad \dots (i)$$

$$(a-b)x + (a+b)y + (2-3a-b) = 0. \quad \dots (ii)$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 2, b_1 = 3, c_1 = -7$

and $a_2 = (a-b), b_2 = (a+b), c_2 = (2-3a-b).$

$$\therefore \frac{a_1}{a_2} = \frac{2}{(a-b)}, \frac{b_1}{b_2} = \frac{3}{(a+b)} \text{ and } \frac{c_1}{c_2} = \frac{-7}{(2-3a-b)} = \frac{7}{(3a+b-2)}.$$

Let the given system of equations have infinitely many solutions.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{2}{(a-b)} = \frac{3}{(a+b)} = \frac{7}{(3a+b-2)}$$

$$\Rightarrow \frac{2}{(a-b)} = \frac{3}{(a+b)} \text{ and } \frac{3}{(a+b)} = \frac{7}{(3a+b-2)}$$

$$\Rightarrow 2a+2b=3a-3b \text{ and } 9a+3b-6=7a+7b$$

$$\Rightarrow a-5b=0 \text{ and } 2a-4b=6$$

$$\Rightarrow a-5b=0 \text{ and } a-2b=3.$$

On solving $a-5b=0$ and $a-2b=3$, we get $a=5$ and $b=1$.

Hence, the required values are $a=5$ and $b=1$.

EXAMPLE 8

Find the values of m and n for which the following system of linear equations has infinitely many solutions:

$$3x+4y=12,$$

$$(m+n)x+2(m-n)y=(5m-1).$$

SOLUTION

The given system of equations is

$$3x+4y-12=0,$$

$$(m+n)x+2(m-n)y+(1-5m)=0.$$

These equations are of the form

$$a_1x+b_1y+c_1=0 \text{ and } a_2x+b_2y+c_2=0,$$

where $a_1=3$, $b_1=4$, $c_1=-12$

and $a_2=(m+n)$, $b_2=2(m-n)$, $c_2=(1-5m)$.

$$\therefore \frac{a_1}{a_2} = \frac{3}{(m+n)}, \frac{b_1}{b_2} = \frac{2}{(m-n)} \text{ and } \frac{c_1}{c_2} = \frac{12}{(5m-1)}.$$

Let the given system of equations have infinitely many solutions.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{3}{(m+n)} = \frac{2}{(m-n)} = \frac{12}{(5m-1)}$$

$$\Rightarrow \frac{3}{(m+n)} = \frac{2}{(m-n)} \text{ and } \frac{2}{(m-n)} = \frac{12}{(5m-1)}$$

$$\Rightarrow 3m-3n=2m+2n \text{ and } 10m-2=12m-12n$$

$$\Rightarrow m - 5n = 0 \quad \dots (i) \quad \text{and} \quad m - 6n + 1 = 0. \quad \dots (ii)$$

On solving (i) and (ii), we get $m = 5$ and $n = 1$.

Hence, the required values are $m = 5$ and $n = 1$.

EXAMPLE 9 Show that the system of equations

$$3x - 5y = 7, 6x - 10y = 3$$

has no solution.

SOLUTION The given system of equations is

$$3x - 5y - 7 = 0 \text{ and } 6x - 10y - 3 = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 3, b_1 = -5, c_1 = -7$ and $a_2 = 6, b_2 = -10, c_2 = -3$.

$$\therefore \frac{a_1}{a_2} = \frac{3}{6} = \frac{1}{2}, \frac{b_1}{b_2} = \frac{-5}{-10} = \frac{1}{2} \text{ and } \frac{c_1}{c_2} = \frac{-7}{-3} = \frac{7}{3}.$$

$$\text{Clearly, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}.$$

Hence, the given system of equations has no solution.

EXAMPLE 10 Find the value of k for which the system of equations

$$x + 2y = 5, 3x + ky - 15 = 0$$

has no solution.

SOLUTION The given system of equations is

$$x + 2y - 5 = 0 \quad \dots (i)$$

$$3x + ky - 15 = 0. \quad \dots (ii)$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 1, b_1 = 2, c_1 = -5$ and $a_2 = 3, b_2 = k, c_2 = -15$.

$$\therefore \frac{a_1}{a_2} = \frac{1}{3}, \frac{b_1}{b_2} = \frac{2}{k} \text{ and } \frac{c_1}{c_2} = \frac{-5}{-15} = \frac{1}{3}.$$

Let the given system of equations have no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\Rightarrow \frac{1}{3} = \frac{2}{k} \neq \frac{1}{3}$$

$$\Rightarrow \frac{1}{3} = \frac{2}{k} \text{ and } \frac{2}{k} \neq \frac{1}{3}$$

$$\Rightarrow k = 6 \text{ and } k \neq 6, \text{ which is impossible.}$$

Hence, there is no value of k for which the given system of equations has no solution.

EXAMPLE 11 Find the values of k for which the pair of linear equations

$$kx + 3y = k - 2 \text{ and } 12x + ky = k$$

has no solution.

[CBSE 2009]

SOLUTION The given equations are

$$kx + 3y + (2 - k) = 0 \text{ and } 12x + ky - k = 0.$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = k$, $b_1 = 3$, $c_1 = (2 - k)$ and $a_2 = 12$, $b_2 = k$, $c_2 = -k$.

$$\therefore \frac{a_1}{a_2} = \frac{k}{12}, \frac{b_1}{b_2} = \frac{3}{k} \text{ and } \frac{c_1}{c_2} = \frac{(2 - k)}{-k} = \frac{(k - 2)}{k}.$$

Let the given system of equations have no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\Rightarrow \frac{k}{12} = \frac{3}{k} \neq \frac{(k - 2)}{k}$$

$$\Rightarrow \frac{k}{12} = \frac{3}{k} \text{ and } \frac{3}{k} \neq \frac{(k - 2)}{k}$$

$$\Rightarrow k^2 = 36 \text{ and } k^2 - 2k \neq 3k$$

$$\Rightarrow k^2 = 36 \text{ and } k^2 - 5k \neq 0$$

$$\Rightarrow (k = 6 \text{ or } k = -6) \text{ and } k(k - 5) \neq 0.$$

Case 1. When $k = 6$.

In this case, $k(k - 5) = 6(6 - 5) = 6 \times 1 = 6 \neq 0$.

Case 2. When $k = -6$.

In this case, $k(k - 5) = (-6)(-6 - 5) = (-6) \times (-11) = 66 \neq 0$.

Thus, in each case, the given system has no solution.

Hence, $k = 6$ or $k = -6$.

EXAMPLE 12 Find the value of k for which the pair of linear equations

$$(3k + 1)x + 3y - 2 = 0, (k^2 + 1)x + (k - 2)y - 5 = 0$$

has no solution.

SOLUTION The given linear equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = (3k + 1)$, $b_1 = 3$, $c_1 = -2$

and $a_2 = (k^2 + 1)$, $b_2 = (k - 2)$, $c_2 = -5$.

$$\therefore \frac{a_1}{a_2} = \frac{(3k+1)}{(k^2+1)}, \frac{b_1}{b_2} = \frac{3}{(k-2)} \text{ and } \frac{c_1}{c_2} = \frac{-2}{-5} = \frac{2}{5}.$$

Let the given system of equations have no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\Rightarrow \frac{(3k+1)}{(k^2+1)} = \frac{3}{(k-2)} \neq \frac{2}{5}$$

$$\Rightarrow \frac{(3k+1)}{(k^2+1)} = \frac{3}{(k-2)} \quad \dots \text{ (i)} \quad \text{and} \quad \frac{3}{(k-2)} \neq \frac{2}{5}. \quad \dots \text{ (ii)}$$

$$\text{Now, } \frac{(3k+1)}{(k^2+1)} = \frac{3}{(k-2)} \Rightarrow (3k+1)(k-2) = 3(k^2+1)$$

$$\Rightarrow 3k^2 - 6k + k - 2 = 3k^2 + 3$$

$$\Rightarrow 5k = -5 \Rightarrow k = -1.$$

$$\text{When } k = -1, \text{ then clearly, } \frac{3}{(k-2)} \neq \frac{2}{5}.$$

Hence, the given system of equations has no solution when $k = -1$.

EXAMPLE 13 Find the value of k for which the following pair of linear equations has no solution:

$$3x + y = 1, (2k-1)x + (k-1)y = 2k+1.$$

SOLUTION

The given linear equations are

$$3x + y - 1 = 0 \quad \dots \text{ (i)}$$

$$(2k-1)x + (k-1)y - (2k+1) = 0. \quad \dots \text{ (ii)}$$

These equations are of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = 3, b_1 = 1, c_1 = -1$

and $a_2 = (2k-1), b_2 = (k-1), c_2 = -(2k+1).$

$$\therefore \frac{a_1}{a_2} = \frac{3}{(2k-1)}, \frac{b_1}{b_2} = \frac{1}{(k-1)}, \frac{c_1}{c_2} = \frac{-1}{-(2k+1)} = \frac{1}{(2k+1)}.$$

Let the given system of equations have no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\Rightarrow \frac{3}{(2k-1)} = \frac{1}{(k-1)} \neq \frac{1}{(2k+1)}$$

$$\Rightarrow \frac{3}{(2k-1)} = \frac{1}{(k-1)} \text{ and } \frac{1}{(k-1)} \neq \frac{1}{(2k+1)}$$

$$\Rightarrow 3k - 3 = 2k - 1 \text{ and } \frac{1}{(k-1)} \neq \frac{1}{(2k+1)}$$

$$\Rightarrow k = 2 \text{ and } \frac{1}{(k-1)} \neq \frac{1}{(2k+1)}.$$

Clearly, when $k = 2$, then $\frac{1}{(k-1)} \neq \frac{1}{(2k+1)}$

$$\left[\text{as } \frac{1}{(2-1)} \neq \frac{1}{(4+1)} \right].$$

Hence, the given system of equations will have no solution, when $k = 2$.

EXAMPLE 14 Find the values of k for which the system of equations

$$kx - y = 2, 6x - 2y = 3$$

has (i) a unique solution, (ii) no solution.

(iii) Is there a value of k for which the given system has infinitely many solutions?

SOLUTION The given system of equations is

$$kx - y - 2 = 0, 6x - 2y - 3 = 0.$$

This is of the form

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0,$$

where $a_1 = k, b_1 = -1, c_1 = -2$ and $a_2 = 6, b_2 = -2, c_2 = -3$.

(i) For a unique solution, we must have $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

$$\therefore \frac{k}{6} \neq \frac{-1}{-2} \Rightarrow \frac{k}{6} \neq \frac{1}{2} \Rightarrow k \neq 3.$$

Hence, the given system of equations will have a unique solution when $k \neq 3$.

(ii) For no solution, we must have $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$$\therefore \frac{k}{6} = \frac{-1}{-2} \neq \frac{-2}{-3}$$

$$\Rightarrow \frac{k}{6} = \frac{1}{2} \neq \frac{2}{3}$$

$$\Rightarrow \frac{k}{6} = \frac{1}{2} \text{ and } \frac{k}{6} \neq \frac{2}{3} \Rightarrow k = 3 \text{ and } k \neq 4.$$

Clearly, $k = 3$ also satisfies the condition $k \neq 4$.

Hence, the given system of equations will have no solution when $k = 3$.

(iii) For infinitely many solutions, we must have

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2},$$

i.e., $\frac{k}{6} = \frac{1}{2} = \frac{1}{3}$, which is never possible, as $\frac{1}{2} \neq \frac{1}{3}$.

Hence, there is no real value of k for which the given system of equations has infinitely many solutions.

HOMOGENEOUS SYSTEM OF EQUATIONS

The system of equations $a_1x + b_1y = 0$ and $a_2x + b_2y = 0$ has

- (i) the zero solution, $x = 0$ and $y = 0$ when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.
- (ii) an infinite number of nonzero solutions when $\frac{a_1}{a_2} = \frac{b_1}{b_2}$.

Note The homogeneous system of equations is always consistent.

EXAMPLE 15 Find the value of k for which the system of equations

$$3x + 5y = 0, kx + 10y = 0$$

has a nonzero solution.

SOLUTION The given equations are of the form

$$a_1x + b_1y = 0 \text{ and } a_2x + b_2y = 0,$$

where $a_1 = 3, b_1 = 5$ and $a_2 = k, b_2 = 10$.

$$\text{Then, } \frac{a_1}{a_2} = \frac{3}{k} \text{ and } \frac{b_1}{b_2} = \frac{5}{10} = \frac{1}{2}.$$

Let the given system of equations have a nonzero solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow \frac{3}{k} = \frac{1}{2} \Rightarrow k = 6.$$

Hence, the required value of k is 6.

EXERCISE 3D

Show that each of the following systems of equations has a unique solution and solve it:

1. $3x + 5y = 12, 5x + 3y = 4$.

2. $2x - 3y = 17, 4x + y = 13$.

3. $\frac{x}{3} + \frac{y}{2} = 3, x - 2y = 2$.

Find the value of k for which each of the following systems of equations has a unique solution:

4. $2x + 3y - 5 = 0, kx - 6y - 8 = 0$. 5. $x - ky = 2, 3x + 2y + 5 = 0$.

6. $5x - 7y - 5 = 0, 2x + ky - 1 = 0$. 7. $4x + ky + 8 = 0, x + y + 1 = 0$.

8. $4x - 5y = k$, $2x - 3y = 12$. 9. $kx + 3y = (k - 3)$, $12x + ky = k$.

10. Show that the system of equations

$$2x - 3y = 5, 6x - 9y = 15$$

has an infinite number of solutions.

11. Show that the system of equations

$$6x + 5y = 11, 9x + \frac{15}{2}y = 21$$

has no solution.

12. For what value of k does the system of equations

$$kx + 2y = 5, 3x - 4y = 10$$

have (i) a unique solution, (ii) no solution?

13. For what value of k does the system of equations

$$x + 2y = 5, 3x + ky + 15 = 0$$

have (i) a unique solution, (ii) no solution?

14. For what value of k does the system of equations

$$x + 2y = 3, 5x + ky + 7 = 0$$

have (i) a unique solution, (ii) no solution?

Also, show that there is no value of k for which the given system of equations has infinitely many solutions.

Find the value of k for which each of the following systems of linear equations has an infinite number of solutions:

15. $2x + 3y = 7$,

$$(k - 1)x + (k + 2)y = 3k.$$

[CBSE 2010]

16. $2x + (k - 2)y = k$,

$$6x + (2k - 1)y = (2k + 5).$$

[CBSE 2000C]

17. $kx + 3y = (2k + 1)$,

$$2(k + 1)x + 9y = (7k + 1).$$

[CBSE 2000C]

18. $5x + 2y = 2k$,

$$2(k + 1)x + ky = (3k + 4).$$

[CBSE 2003C]

19. $(k - 1)x - y = 5$,

$$(k + 1)x + (1 - k)y = (3k + 1).$$

[CBSE 2003]

20. $(k - 3)x + 3y = k$,

$$kx + ky = 12.$$

Find the values of a and b for which each of the following systems of linear equations has an infinite number of solutions:

21. $(a - 1)x + 3y = 2$,

$$6x + (1 - 2b)y = 6.$$

[CBSE 2002C]

22. $(2a - 1)x + 3y = 5,$
 $3x + (b - 1)y = 2.$ [CBSE 2001C]
23. $2x - 3y = 7,$
 $(a + b)x - (a + b - 3)y = 4a + b.$ [CBSE 2002]
24. $2x + 3y = 7,$
 $(a + b + 1)x + (a + 2b + 2)y = 4(a + b) + 1.$ [CBSE 2003]
25. $2x + 3y = 7,$
 $(a + b)x + (2a - b)y = 21.$ [CBSE 2001]
26. $2x + 3y = 7,$
 $2ax + (a + b)y = 28.$ [CBSE 2001]

Find the value of k for which each of the following systems of equations has no solution:

27. $8x + 5y = 9, kx + 10y = 15.$
28. $kx + 3y = 3, 12x + ky = 6.$
29. $3x - y - 5 = 0, 6x - 2y + k = 0$ ($k \neq 0$). [CBSE 2008]
30. $kx + 3y = k - 3, 12x + ky = k.$ [CBSE 2009]
31. Find the value of k for which the system of equations
 $5x - 3y = 0, 2x + ky = 0$
 has a nonzero solution.

ANSWERS (EXERCISE 3D)

1. $x = -1, y = 3$ 2. $x = 4, y = -3$ 3. $x = 6, y = 2$ 4. $k \neq -4$
5. $k \neq \frac{-2}{3}$ 6. $k \neq \frac{-14}{5}$ 7. $k \neq 4$
8. k is any real number 9. k is any real number other than 6 and -6
12. (i) $k \neq \frac{-3}{2}$ (ii) $k = \frac{-3}{2}$ 13. (i) $k \neq 6$ (ii) $k = 6$ 14. (i) $k \neq 10$ (ii) $k = 10$
15. $k = 7$ 16. $k = 5$ 17. $k = 2$ 18. $k = 4$ 19. $k = 3$ 20. $k = 6$
21. $a = 3, b = -4$ 22. $a = \frac{17}{4}, b = \frac{11}{5}$ 23. $a = -5, b = -1$ 24. $a = 3, b = 2$
25. $a = 5, b = 1$ 26. $a = 4, b = 8$ 27. $k = 16$ 28. $k = -6$
29. $k \neq -10$ 30. $k = -6$ 31. $k = \frac{-6}{5}$

HINTS TO SOME SELECTED QUESTIONS

9. $\frac{k}{12} \neq \frac{3}{k} \Rightarrow k^2 \neq 36 \Rightarrow k \neq 6 \text{ and } k \neq -6.$

So, for a unique solution, we may take any real value of k , other than 6 and -6 .

12. (i) $\frac{k}{3} \neq \frac{2}{-4} \Rightarrow \frac{k}{3} \neq \frac{-1}{2} \Rightarrow k \neq \frac{-3}{2}$.
 (ii) $\frac{k}{3} = \frac{2}{-4} \neq \frac{5}{10} \Rightarrow \frac{k}{3} = \frac{-1}{2}$ and $\frac{k}{3} \neq \frac{1}{2} \Rightarrow k = \frac{-3}{2}$.
13. (i) $\frac{1}{3} \neq \frac{2}{k} \Rightarrow k \neq 6$.
 (ii) $\frac{1}{3} = \frac{2}{k} \neq \frac{-5}{15} \Rightarrow \frac{1}{3} = \frac{2}{k} \neq \frac{-1}{3} \Rightarrow k = 6$.
14. (i) $\frac{1}{5} \neq \frac{2}{k} \Rightarrow k \neq 10$.
 (ii) $\frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = 10$.
 Clearly, $\frac{1}{5} = \frac{2}{k} = \frac{-3}{7}$ is never true, as $\frac{1}{5} = \frac{-3}{7}$ is false.
20. $\frac{k-3}{k} = \frac{3}{k} = \frac{-k}{-12} \Rightarrow \frac{k-3}{k} = \frac{3}{k}$ and $\frac{3}{k} = \frac{k}{12}$.
 $\therefore k^2 - 6k = 0$ and $k^2 = 36 \Rightarrow k(k-6) = 0$ and $k^2 = 36$
 $\Rightarrow (k=0 \text{ or } k=6) \text{ and } (k=6 \text{ or } k=-6)$
 $\Rightarrow k=6$.
21. $\frac{a-1}{6} = \frac{3}{1-2b} = \frac{-2}{-6} = \frac{1}{3}$
 $\Rightarrow \frac{a-1}{6} = \frac{1}{3}$ and $\frac{3}{1-2b} = \frac{1}{3}$
 $\Rightarrow 3a-3=6$ and $1-2b=9$
 $\Rightarrow 3a=9$ and $2b=-8 \Rightarrow a=3$ and $b=-4$.
28. $\frac{k}{12} = \frac{3}{k} \neq \frac{-3}{-6} \Rightarrow \frac{k}{12} = \frac{3}{k}$ and $\frac{3}{k} \neq \frac{1}{2}$
 $\Rightarrow k^2 = 36$ and $k \neq 6 \Rightarrow k = -6$.
30. $\frac{k}{12} = \frac{3}{k} \neq \frac{k-3}{k} \Rightarrow k^2 = 36$ and $k(k-6) \neq 0 \Rightarrow k = -6$.
-

WORD PROBLEMS

SOLVED EXAMPLES

PROBLEMS ON MONEY MATTERS

EXAMPLE 1 8 chairs and 5 tables for a classroom cost ₹ 10500, while 5 chairs and 3 tables cost ₹ 6450. Find the cost of each chair and that of each table.

SOLUTION Let the cost of each chair be ₹ x and that of each table be ₹ y .
 Then, $8x + 5y = 10500$ (i)
 and $5x + 3y = 6450$ (ii)
 On multiplying (ii) by 5 and (i) by 3 and subtracting the results, we get

$$25x - 24x = 32250 - 31500 \Rightarrow x = 750.$$

Putting $x = 750$ in (i), we get

$$8 \times 750 + 5y = 10500 \Rightarrow 6000 + 5y = 10500$$

$$\Rightarrow 5y = (10500 - 6000) = 4500$$

$$\Rightarrow y = 900.$$

$\therefore$ cost of each chair = ₹ 750 and cost of each table = ₹ 900.

EXAMPLE 2 *The coach of a cricket team buys 7 bats and 6 balls for ₹ 13200. Later, he buys 3 bats and 5 balls for ₹ 5900. Find the cost of each bat and each ball.*

SOLUTION Let the cost of each bat be ₹ x and the cost of each ball be ₹ y .

$$\text{Then, } 7x + 6y = 13200. \quad \dots (i)$$

$$\text{And, } 3x + 5y = 5900. \quad \dots (ii)$$

On multiplying (i) by 5, (ii) by 6 and subtracting the results, we get

$$35x - 18x = 66000 - 35400 \Rightarrow 17x = 30600$$

$$\Rightarrow x = \frac{30600}{17} = 1800.$$

Putting $x = 1800$ in (ii), we get

$$5400 + 5y = 5900 \Rightarrow 5y = 5900 - 5400$$

$$\Rightarrow 5y = 500$$

$$\Rightarrow y = 100.$$

$\therefore$ cost of each bat = ₹ 1800 and cost of each ball = ₹ 100.

EXAMPLE 3 *37 pens and 53 pencils together cost ₹ 955, while 53 pens and 37 pencils together cost ₹ 1115. Find the cost of a pen and that of a pencil.*

SOLUTION Let the cost of each pen be ₹ x and that of a pencil be ₹ y .

$$\text{Then, } 37x + 53y = 955. \quad \dots (i)$$

$$\text{And, } 53x + 37y = 1115. \quad \dots (ii)$$

Adding (i) and (ii), we get

$$90x + 90y = 2070 \Rightarrow 90(x + y) = 2070$$

$$\Rightarrow x + y = \frac{2070}{90}$$

$$\Rightarrow x + y = 23. \quad \dots (iii)$$

On subtracting (i) from (ii), we get

$$16x - 16y = 160 \Rightarrow 16(x - y) = 160$$

$$\Rightarrow (x - y) = 10. \quad \dots (iv)$$

Adding (iii) and (iv), we get $2x = 33 \Rightarrow x = \frac{33}{2} = 16.50$.

Subtracting (iv) from (iii), we get $2y = 13 \Rightarrow y = \frac{13}{2} = 6.50$.

$\therefore$ cost of each pen = ₹ 16.50 and cost of each pencil = ₹ 6.50.

EXAMPLE 4 *Taxi charges in a city consist of fixed charges and the remaining depending upon the distance travelled in kilometres. If a person travels 60 km, he pays ₹ 960, and for travelling 80 km, he pays ₹ 1260. Find the fixed charges and the rate per kilometre.*

SOLUTION Let the fixed charges be ₹ x and the other charges be ₹ y per km.

$$\text{Then, } x + 60y = 960. \quad \dots (i)$$

$$\text{And, } x + 80y = 1260. \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$20y = 300 \Rightarrow y = \frac{300}{20} \Rightarrow y = 15.$$

Putting $y = 15$ in (i), we get

$$x + (60 \times 15) = 960 \Rightarrow x = 960 - 900 \Rightarrow x = 60.$$

$\therefore$ fixed charges = ₹ 60 and the rate per km = ₹ 15 per km.

EXAMPLE 5 *A part of monthly hostel charges in a school is fixed and the remaining depends on the number of days one has taken food in the mess. When a student A takes food for 22 days, he has to pay ₹ 4250 as hostel charges, whereas a student B, who takes food for 28 days, pays ₹ 5150 as hostel charges. Find the fixed charges and the cost of food per day.*

SOLUTION Let the fixed charges be ₹ x per month and the cost of meals per day be ₹ y . Then, we have

$$x + 22y = 4250 \quad \dots (i)$$

$$\text{and } x + 28y = 5150. \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$6y = 900 \Rightarrow y = 150.$$

Putting $y = 150$ in (i), we get

$$x + (22 \times 150) = 4250$$

$$\Rightarrow x + 3300 = 4250$$

$$\Rightarrow x = 4250 - 3300 = 950.$$

$$\therefore x = 950 \text{ and } y = 150.$$

Hence, the fixed charges are ₹ 950 per month and the cost of food is ₹ 150 per day.

EXAMPLE 6 *The monthly incomes of A and B are in the ratio 8 : 7 and their expenditures are in the ratio 19 : 16. If each saves ₹ 5000 per month, find the monthly income of each.*

SOLUTION Let the monthly incomes of A and B be ₹ $8x$ and ₹ $7x$ respectively, and let their expenditures be ₹ $19y$ and ₹ $16y$ respectively.

Then, A's monthly savings = ₹ $(8x - 19y)$.

And, B's monthly savings = ₹ $(7x - 16y)$.

But, the monthly saving of each is ₹ 5000.

$$\therefore 8x - 19y = 5000 \quad \dots (i)$$

$$\text{and } 7x - 16y = 5000. \quad \dots (ii)$$

Multiplying (ii) by 19, (i) by 16 and subtracting the results, we get

$$(19 \times 7 - 16 \times 8)x = (19 \times 5000 - 16 \times 5000)$$

$$\Rightarrow (133 - 128)x = 5000 \times (19 - 16)$$

$$\Rightarrow 5x = 15000 \Rightarrow x = 3000.$$

$$\therefore \text{A's monthly income} = ₹ (8x) = ₹ (8 \times 3000) = ₹ 24000.$$

$$\text{And, B's monthly income} = ₹ (7x) = ₹ (7 \times 3000) = ₹ 21000.$$

EXAMPLE 7 *On selling a TV at 5% gain and a fridge at 10% gain, a shopkeeper gains ₹ 3250. But, if he sells the TV at 10% gain and the fridge at 5% loss, he gains ₹ 1500. Find the actual cost price of TV and that of the fridge.*

SOLUTION Let the cost price of the TV set be ₹ x and that of the fridge be ₹ y .

Then, total CP of TV and fridge = ₹ $(x + y)$.

$$\text{SP in first case} = ₹ \left(\frac{105x}{100} + \frac{110y}{100} \right) = ₹ \left(\frac{21x}{20} + \frac{11y}{10} \right).$$

$$\therefore \text{gain in this case} = ₹ \left\{ \left(\frac{21x}{20} + \frac{11y}{10} \right) - (x + y) \right\} = ₹ \left(\frac{x}{20} + \frac{y}{10} \right).$$

$$\text{So, } \frac{x}{20} + \frac{y}{10} = 3250 \Rightarrow x + 2y = 65000. \quad \dots (i)$$

$$\text{SP in second case} = ₹ \left(\frac{110x}{100} + \frac{95y}{100} \right) = ₹ \left(\frac{11x}{10} + \frac{19y}{20} \right).$$

$$\therefore \text{ gain in this case} = ₹ \left\{ \left(\frac{11x}{10} + \frac{19y}{20} \right) - (x + y) \right\} = ₹ \left(\frac{x}{10} - \frac{y}{20} \right).$$

$$\text{So, } \frac{x}{10} - \frac{y}{20} = 1500 \Rightarrow 2x - y = 30000. \quad \dots (ii)$$

Multiplying (ii) by 2 and adding the result with (i), we get

$$5x = 60000 + 65000 \Rightarrow 5x = 125000 \Rightarrow x = 25000.$$

Putting $x = 25000$ in (i), we get

$$25000 + 2y = 65000 \Rightarrow 2y = 65000 - 25000 = 40000$$

$$\Rightarrow y = 20000.$$

$$\therefore x = 25000 \text{ and } y = 20000.$$

Hence, the CP of the TV set is ₹ 25000 and that of the fridge is ₹ 20000.

EXAMPLE 8

A man invested an amount at 12% per annum simple interest and another amount at 10% per annum simple interest. He received an annual interest of ₹ 2600. But, if he had interchanged the amounts invested, he would have received ₹ 140 less. What amounts did he invest at the different rates?

SOLUTION

Let the amount invested at 12% be ₹ x and that invested at 10% be ₹ y . Then,

total annual interest

$$= ₹ \left(\frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100} \right) = ₹ \left(\frac{6x + 5y}{50} \right).$$

$$\therefore \frac{6x + 5y}{50} = 2600 \Rightarrow 6x + 5y = 130000. \quad \dots (i)$$

Again, the amount invested at 12% is ₹ y and that invested at 10% is ₹ x .

Total annual interest at the new rates

$$= ₹ \left(\frac{y \times 12 \times 1}{100} + \frac{x \times 10 \times 1}{100} \right) = ₹ \left(\frac{6y + 5x}{50} \right).$$

But, interest received at the new rates = ₹ $(2600 - 140) = ₹ 2460$.

$$\therefore \frac{6y + 5x}{50} = 2460 \Rightarrow 5x + 6y = 123000. \quad \dots (ii)$$

Adding (i) and (ii), we get

$$11x + 11y = 253000 \Rightarrow 11(x + y) = 253000$$

$$\Rightarrow x + y = 23000. \quad \dots \text{(iii)}$$

Subtracting (ii) from (i), we get

$$x - y = 7000. \quad \dots \text{(iv)}$$

Adding (iii) and (iv), we get $2x = 30000 \Rightarrow x = 15000$.

Putting $x = 15000$ in (i), we get

$$15000 + y = 23000 \Rightarrow y = 23000 - 15000 = 8000.$$

$$\therefore x = 15000 \text{ and } y = 8000.$$

Hence, the amount at 12% is ₹ 15000 and that at 10% is ₹ 8000.

EXAMPLE 9

Each one of A and B has some money. If A gives ₹ 30 to B then B will have twice the money left with A. But, if B gives ₹ 10 to A then A will have thrice as much as is left with B. How much money does each have?

SOLUTION

Suppose A and B have ₹ x and ₹ y respectively.

Case I When A gives ₹ 30 to B.

Then, money with A = ₹ $(x - 30)$

and money with B = ₹ $(y + 30)$.

$$\therefore (y + 30) = 2(x - 30) \Rightarrow 2x - y = 90. \quad \dots \text{(i)}$$

Case II When B gives ₹ 10 to A.

Then, money with A = ₹ $(x + 10)$

and money with B = ₹ $(y - 10)$.

$$\therefore (x + 10) = 3(y - 10) \Rightarrow x - 3y = -40. \quad \dots \text{(ii)}$$

On multiplying (ii) by 2 and subtracting the result from (i), we get

$$5y = 170 \Rightarrow y = \frac{170}{5} = 34.$$

Putting $y = 34$ in (i), we get

$$2x - 34 = 90 \Rightarrow 2x = (90 + 34) = 124 \Rightarrow x = \frac{124}{2} = 62.$$

$$\therefore x = 62 \text{ and } y = 34.$$

Hence, A has ₹ 62 and B has ₹ 34.

PROBLEMS ON NUMBERS

EXAMPLE 10 *The students of a class are made to stand in rows. If 4 students are extra in each row, there would be 2 rows less. If 4 students are less in each row, there would be 4 rows more. Find the number of students in the class.*

SOLUTION Let the number of rows be x and the number of students in each row be y .

Then, the total number of students = xy .

Case I *When there are 4 more students in each row.*

Then, number of students in each row = $(y + 4)$.

And, number of rows = $(x - 2)$.

Total number of students = $(x - 2)(y + 4)$.

$$\begin{aligned}\therefore (x - 2)(y + 4) &= xy \Rightarrow 4x - 2y = 8 \\ &\Rightarrow 2x - y = 4. \qquad \dots (i)\end{aligned}$$

Case II *When 4 students are removed from each row.*

Then, number of students in each row = $(y - 4)$.

And, number of rows = $(x + 4)$.

Total number of students = $(x + 4)(y - 4)$.

$$\begin{aligned}\therefore (x + 4)(y - 4) &= xy \Rightarrow 4y - 4x = 16 \\ &\Rightarrow 4(y - x) = 16 \\ &\Rightarrow (y - x) = 4. \qquad \dots (ii)\end{aligned}$$

Adding (i) and (ii), we get $x = 8$.

Putting $x = 8$ in (ii), we get

$$y - 8 = 4 \Rightarrow y = 12.$$

Thus, $x = 8$ and $y = 12$.

This shows that there are 8 rows and there are 12 students in each row.

Hence, the number of students in the class = $xy = 8 \times 12 = 96$.

EXAMPLE 11 *The sum of two numbers is 1000 and the difference between their squares is 256000. Find the numbers.*

SOLUTION Let the larger number be x and the smaller number be y .

$$\text{Then, } x + y = 1000 \qquad \dots (i)$$

$$\text{and } x^2 - y^2 = 256000. \qquad \dots (ii)$$

On dividing (ii) by (i), we get

$$\frac{x^2 - y^2}{x + y} = \frac{256000}{1000} \Rightarrow x - y = 256. \quad \dots \text{(iii)}$$

Adding (i) and (iii), we get

$$2x = 1256 \Rightarrow x = 628.$$

Substituting $x = 628$ in (i), we get $y = 372$.

Hence, the required numbers are 628 and 372.

EXAMPLE 12 *If three times the larger of two numbers is divided by the smaller one, we get 4 as the quotient and 3 as the remainder. Also, if seven times the smaller number is divided by the larger one, we get 5 as the quotient and 1 as the remainder. Find the numbers.*

SOLUTION Let the larger number be x and the smaller one be y .

We know that

$$\text{dividend} = (\text{divisor} \times \text{quotient}) + \text{remainder}.$$

Using the above result and the given conditions, we have

$$3x = 4y + 3 \Rightarrow 3x - 4y = 3 \quad \dots \text{(i)}$$

$$\text{and } 7y = 5x + 1 \Rightarrow 5x - 7y = -1. \quad \dots \text{(ii)}$$

Multiplying (i) by 5, (ii) by 3 and subtracting, we get $y = 18$.

Putting $y = 18$ in (i), we get

$$3x - (4 \times 18) = 3 \Rightarrow 3x - 72 = 3$$

$$\Rightarrow 3x = 75 \Rightarrow x = 25.$$

Hence, the required numbers are 25 and 18.

EXAMPLE 13 *The sum of two numbers is 8 and the sum of their reciprocals is $\frac{8}{15}$. Find the numbers.* [CBSE 2009]

SOLUTION Let the required numbers be x and y .

$$\text{Then, } x + y = 8. \quad \dots \text{(i)}$$

$$\text{And, } \frac{1}{x} + \frac{1}{y} = \frac{8}{15} \Rightarrow \frac{x+y}{xy} = \frac{8}{15}$$

$$\Rightarrow \frac{8}{xy} = \frac{8}{15} \quad [\text{using (i)}]$$

$$\Rightarrow xy = 15.$$

$$\therefore (x - y) = \sqrt{(x + y)^2 - 4xy}$$

$$= \sqrt{8^2 - 4 \times 15} = \sqrt{64 - 60} = \sqrt{4} = \pm 2.$$

Thus, we have

$$\left. \begin{array}{l} x + y = 8 \quad \dots (i) \\ x - y = 2 \quad \dots (ii) \end{array} \right\} \text{ or } \left\{ \begin{array}{l} x + y = 8 \quad \dots (iii) \\ x - y = -2 \quad \dots (iv) \end{array} \right.$$

On solving (i) and (ii), we get $x = 5$ and $y = 3$.

On solving (iii) and (iv), we get $x = 3$ and $y = 5$.

Hence, the required numbers are 5 and 3.

EXAMPLE 14 *The difference of two numbers is 4 and the difference of their reciprocals is $\frac{4}{21}$. Find the numbers.* [CBSE 2008]

SOLUTION Let the larger number be x and the smaller number be y .

$$\text{Then, } x - y = 4. \quad \dots (i)$$

$$\text{And, } \frac{1}{y} - \frac{1}{x} = \frac{4}{21} \quad \left[\because x > y \Rightarrow \frac{1}{y} > \frac{1}{x} \right]$$

$$\Rightarrow \frac{x - y}{xy} = \frac{4}{21}$$

$$\Rightarrow \frac{4}{xy} = \frac{4}{21} \Rightarrow xy = 21 \quad [\text{using (i)}].$$

$$\begin{aligned} \therefore (x + y) &= \sqrt{(x - y)^2 + 4xy} \\ &= \sqrt{4^2 + 4 \times 21} = \sqrt{16 + 84} = \sqrt{100} = \pm 10. \end{aligned}$$

Thus, we have

$$\left. \begin{array}{l} x - y = 4 \quad \dots (i) \\ x + y = 10 \quad \dots (ii) \end{array} \right\} \text{ or } \left\{ \begin{array}{l} x - y = 4 \quad \dots (iii) \\ x + y = -10 \quad \dots (iv) \end{array} \right.$$

On solving (i) and (ii), we get $x = 7$ and $y = -3$.

On solving (iii) and (iv), we get $x = -3$ and $y = -7$.

Hence, the required numbers are (7 and 3) or (-3 and -7).

EXAMPLE 15 *The sum of the digits of a two-digit number is 12. The number obtained by interchanging its digits exceeds the given number by 18. Find the number.* [CBSE 2006]

SOLUTION Let the ten's digit of the required number be x and the unit's digit be y . Then,

$$x + y = 12. \quad \dots (i)$$

Required number = $(10x + y)$.

Number obtained on reversing the digits = $(10y + x)$.

$$\begin{aligned}\therefore (10y + x) - (10x + y) &= 18 \Rightarrow 9y - 9x = 18 \\ &\Rightarrow y - x = 2. \quad \dots \text{ (ii)}\end{aligned}$$

On adding (i) and (ii), we get

$$2y = 14 \Rightarrow y = 7.$$

Putting $y = 7$ in (i), we get

$$x + 7 = 12 \Rightarrow x = 12 - 7 = 5.$$

$$\therefore x = 5 \text{ and } y = 7.$$

Hence, the required number is 57.

EXAMPLE 16 *The sum of a two-digit number and the number obtained by reversing the order of its digits is 99. If the digits differ by 3, find the number.*

[CBSE 2002]

SOLUTION Let the ten's and unit's digits of the required number be x and y respectively.

Then, the number = $(10x + y)$.

The number obtained on reversing the digits = $(10y + x)$.

$$\therefore (10y + x) + (10x + y) = 99 \Rightarrow 11(x + y) = 99 \Rightarrow x + y = 9.$$

Also, $(x - y) = \pm 3$.

Thus, we have

$$\begin{array}{lcl} \left. \begin{array}{l} x + y = 9 \quad \dots \text{ (i)} \\ x - y = 3 \quad \dots \text{ (ii)} \end{array} \right\} \text{ or } \left\{ \begin{array}{l} x + y = 9 \quad \dots \text{ (iii)} \\ x - y = -3 \quad \dots \text{ (iv)} \end{array} \right.\end{array}$$

From (i) and (ii), we get $x = 6, y = 3$.

From (iii) and (iv), we get $x = 3, y = 6$.

Hence, the required number is 63 or 36.

EXAMPLE 17 *Seven times a two-digit number is equal to four times the number obtained by reversing the order of its digits. If the difference between the digits is 3, find the number.*

SOLUTION Let the ten's and unit's digits of the required number be x and y respectively.

Then, the number = $(10x + y)$.

The number obtained by reversing the digits = $(10y + x)$.

$$\therefore 7(10x + y) = 4(10y + x) \Rightarrow 33(2x - y) = 0$$

$$\Rightarrow 2x - y = 0 \Rightarrow y = 2x \quad \dots \text{ (i)}$$

Thus, unit's digit = 2 times the ten's digit.

$\therefore$ (unit's digit) > (ten's digit) and so $y > x$.

$$\therefore y - x = 3. \quad \dots (ii)$$

Using (i) in (ii), we get $(2x - x) = 3 \Rightarrow x = 3$.

On substituting $x = 3$ in (i), we get $y = 2 \times 3 = 6$.

Hence, the required number is 36.

EXAMPLE 18 *A two-digit number is such that the product of its digits is 14. If 45 is added to the number, the digits interchange their places. Find the number.* [CBSE 2005]

SOLUTION Let the ten's and unit's digits of the required number be x and y respectively. Then, $xy = 14$.

Required number = $(10x + y)$.

Number obtained on reversing its digits = $(10y + x)$.

$$\therefore (10x + y) + 45 = (10y + x)$$

$$\Rightarrow 9(y - x) = 45 \Rightarrow y - x = 5. \quad \dots (i)$$

Now, $(y + x)^2 - (y - x)^2 = 4xy$

$$\Rightarrow (y + x) = \sqrt{(y - x)^2 + 4xy} = \sqrt{25 + 4 \times 14} = \sqrt{81}$$

$$\Rightarrow y + x = 9 \quad \dots (ii) \quad [\because \text{digits are never negative}].$$

On adding (i) and (ii), we get

$$2y = 14 \Rightarrow y = 7.$$

Putting $y = 7$ in (ii), we get

$$7 + x = 9 \Rightarrow x = 9 - 7 = 2.$$

$$\therefore x = 2 \text{ and } y = 7.$$

Hence, the required number is 27.

EXAMPLE 19 *A two-digit number is four times the sum of its digits and twice the product of its digits. Find the number.* [CBSE 2005]

SOLUTION Let the ten's digit of the required number be x and its unit's digit be y .

$$\text{Then, } 10x + y = 4(x + y) \Rightarrow 6x - 3y = 0 \Rightarrow 2x - y = 0. \quad \dots (i)$$

$$\text{Also, } 10x + y = 2xy. \quad \dots (ii)$$

Putting $y = 2x$ from (i) in (ii), we get

$$\begin{aligned} 10x + 2x &= 4x^2 \Rightarrow 4x^2 - 12x = 0 \\ &\Rightarrow 4x(x - 3) = 0 \\ &\Rightarrow x - 3 = 0 \Rightarrow x = 3 \quad [\because \text{ten's digit, } x \neq 0]. \end{aligned}$$

Putting $x = 3$ in (i), we get $y = 6$.

Thus, ten's digit = 3 and unit's digit = 6.

Hence, the required number is 36.

EXAMPLE 20 A fraction becomes $\frac{1}{3}$, if 2 is added to both of its numerator and denominator. If 3 is added to both of its numerator and denominator then it becomes $\frac{2}{5}$. Find the fraction. [CBSE 2009C]

SOLUTION Let the required fraction be $\frac{x}{y}$. Then,

$$\begin{aligned} \frac{x+2}{y+2} &= \frac{1}{3} \Rightarrow 3x+6 = y+2 \\ &\Rightarrow 3x - y = -4. \end{aligned} \quad \dots (i)$$

$$\begin{aligned} \text{Also, } \frac{x+3}{y+3} &= \frac{2}{5} \Rightarrow 5x+15 = 2y+6 \\ &\Rightarrow 5x - 2y = -9. \end{aligned} \quad \dots (ii)$$

Multiplying (i) by 2 and subtracting (ii) from the result, we get

$$6x - 5x = -8 + 9 \Rightarrow x = 1.$$

Putting $x = 1$ in (i), we get

$$3 - y = -4 \Rightarrow y = 7.$$

Thus, $x = 1$ and $y = 7$.

Hence, the required fraction is $\frac{1}{7}$.

EXAMPLE 21 The sum of numerator and denominator of a fraction is 3 less than twice the denominator. If each of the numerator and denominator is decreased by 1, the fraction becomes $\frac{1}{2}$. Find the fraction. [CBSE 2010]

SOLUTION Let the required fraction be $\frac{x}{y}$. Then,

$$(x + y) = 2y - 3 \Rightarrow x - y = -3. \quad \dots (i)$$

$$\begin{aligned} \text{And, } \frac{x-1}{y-1} &= \frac{1}{2} \Rightarrow 2x - 2 = y - 1 \\ &\Rightarrow 2x - y = 1. \end{aligned} \quad \dots (ii)$$

On subtracting (i) from (ii), we get $x = 4$.

Putting $x = 4$ in (i), we get $y = x + 3 = 4 + 3 = 7$.

$\therefore x = 4$ and $y = 7$.

Hence, the required fraction is $\frac{4}{7}$.

EXAMPLE 22 In a given fraction, if the numerator is multiplied by 2 and the denominator is reduced by 5, we get $\left(\frac{6}{5}\right)$. But if the numerator of the given fraction is increased by 8 and the denominator is doubled, we get $\left(\frac{2}{5}\right)$. Find the fraction.

SOLUTION Let the required fraction be $\frac{x}{y}$. Then,

$$\begin{aligned}\frac{2x}{y-5} &= \frac{6}{5} \Rightarrow 10x = 6(y-5) \\ &\Rightarrow 10x - 6y = -30 \\ &\Rightarrow 5x - 3y = -15 \quad \dots (i)\end{aligned}$$

$$\begin{aligned}\text{and } \frac{x+8}{2y} &= \frac{2}{5} \Rightarrow 5(x+8) = 4y \\ &\Rightarrow 5x - 4y = -40. \quad \dots (ii)\end{aligned}$$

On subtracting (ii) from (i), we get $y = 25$.

Putting $y = 25$ in (i), we get

$$5x - (3 \times 25) = -15 \Rightarrow 5x - 75 = -15 \Rightarrow 5x = 60 \Rightarrow x = 12.$$

$\therefore x = 12$ and $y = 25$.

Hence, the required fraction is $\frac{12}{25}$.

PROBLEMS ON AGES

EXAMPLE 23 Five years ago, A was thrice as old as B and ten years later A shall be twice as old as B. What are the present ages of A and B? [CBSE 2002C]

SOLUTION Let the present ages of B and A be x years and y years respectively. Then,

$$\begin{aligned}\text{B's age 5 years ago} &= (x - 5) \text{ years} \\ \text{and A's age 5 years ago} &= (y - 5) \text{ years.} \\ \therefore (y - 5) &= 3(x - 5) \Rightarrow 3x - y = 10. \quad \dots (i)\end{aligned}$$

$$\begin{aligned}\text{B's age 10 years hence} &= (x + 10) \text{ years.} \\ \text{A's age 10 years hence} &= (y + 10) \text{ years.} \\ \therefore (y + 10) &= 2(x + 10) \Rightarrow 2x - y = -10. \quad \dots (ii)\end{aligned}$$

On subtracting (ii) from (i), we get $x = 20$.

Putting $x = 20$ in (i), we get

$$(3 \times 20) - y = 10 \Rightarrow y = 60 - 10 = 50.$$

$$\therefore x = 20 \text{ and } y = 50.$$

Hence, B's present age = 20 years

and A's present age = 50 years.

EXAMPLE 24 *A man's age is three times the sum of the ages of his two sons. After 5 years, his age will be twice the sum of the ages of his two sons. Find the age of the man.* [CBSE 2003]

SOLUTION Let the present age of the man be x years and the sum of the present ages of his two sons be y years. Then,

$$x = 3y \Rightarrow x - 3y = 0. \quad \dots (i)$$

Man's age after 5 years = $(x + 5)$ years.

Sum of the ages of his two sons after 5 years = $(y + 5 + 5)$ years
= $(y + 10)$ years.

$$\therefore (x + 5) = 2(y + 10) \Rightarrow x - 2y = 15. \quad \dots (ii)$$

On subtracting (i) from (ii), we get $y = 15$.

Putting $y = 15$ in (i), we get $x = 45$.

$$\therefore x = 45 \text{ and } y = 15.$$

Hence, the present age of the man is 45 years.

PROBLEMS ON TIME AND DISTANCE

EXAMPLE 25 *A man travels 370 km, partly by train and partly by car. If he covers 250 km by train and the rest by car, it takes him 4 hours. But, if he travels 130 km by train and the rest by car, he takes 18 minutes longer. Find the speed of the train and that of the car.*

SOLUTION Let the speed of the train be x km/hr and that of the car be y km/hr.

Case I Distance covered by train = 250 km.

Distance covered by car = $(370 - 250)$ km = 120 km.

Time taken to cover 250 km by train = $\frac{250}{x}$ hours.

Time taken to cover 120 km by car = $\frac{120}{y}$ hours.

Total time taken = 4 hours.

$$\therefore \frac{250}{x} + \frac{120}{y} = 4 \Rightarrow \frac{125}{x} + \frac{60}{y} = 2. \quad \dots (i)$$

Case II Distance covered by train = 130 km.

Distance covered by car = $(370 - 130)$ km = 240 km.

Time taken to cover 130 km by train = $\frac{130}{x}$ hours.

Time taken to cover 240 km by car = $\frac{240}{y}$ hours.

Total time taken = $4\frac{18}{60}$ hours = $4\frac{3}{10}$ hours = $\frac{43}{10}$ hours.

$$\therefore \frac{130}{x} + \frac{240}{y} = \frac{43}{10} \Rightarrow \frac{1300}{x} + \frac{2400}{y} = 43. \quad \dots (ii)$$

Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, equations (i) and (ii) become

$$125u + 60v = 2 \quad \dots (iii) \quad \text{and} \quad 1300u + 2400v = 43. \quad \dots (iv)$$

On multiplying (iii) by 40 and subtracting (iv) from the result, we get

$$\begin{aligned} 5000u - 1300u &= 80 - 43 \Rightarrow 3700u = 37 \\ \Rightarrow u &= \frac{37}{3700} = \frac{1}{100} \Rightarrow \frac{1}{x} = \frac{1}{100} \Rightarrow x = 100. \end{aligned}$$

Putting $u = \frac{1}{100}$ in (iv), we get

$$\begin{aligned} \left(1300 \times \frac{1}{100}\right) + 2400v &= 43 \Rightarrow 2400v = 43 - 13 = 30 \\ \Rightarrow v &= \frac{30}{2400} = \frac{1}{80} \Rightarrow \frac{1}{y} = \frac{1}{80} \Rightarrow y = 80. \\ \therefore x &= 100 \text{ and } y = 80. \end{aligned}$$

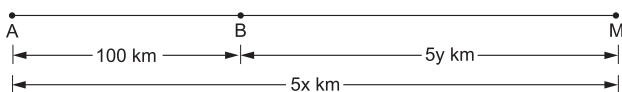
Hence, the speed of the train is 100 km/hr and that of the car is 80 km/hr.

EXAMPLE 26 Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars? [CBSE 2009]

SOLUTION Let the speeds of the cars from A and B be x km/hr and y km/hr respectively.

Case I When the two cars move in the same direction:

In this case, let them meet at M after 5 hours.



Then, $AM = 5x$ km.

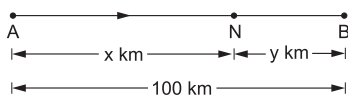
And, $BM = 5y$ km.

$$\therefore AM - BM = AB \Rightarrow 5x - 5y = 100$$

$$\Rightarrow 5(x - y) = 100 \Rightarrow x - y = 20. \quad \dots (i)$$

Case II When the two cars move in the opposite directions:

Let one car move from A to B and let the other move from B to A. Let them meet at N after 1 hour.



Then, $AN = x$ km and $BN = y$ km.

$$\therefore AN + BN = AB \Rightarrow x + y = 100. \quad \dots (ii)$$

Adding (i) and (ii), we get $2x = 120 \Rightarrow x = 60$.

Putting $x = 60$ in (ii), we get $60 + y = 100 \Rightarrow y = 40$.

$\therefore$ speed of the car from A = 60 km/hr,

and speed of the car from B = 40 km/hr.

EXAMPLE 27 A train covered a certain distance at a uniform speed. If the train had been 6 kmph faster, it would have taken 4 hours less than the scheduled time. And, if the train were slower by 6 kmph, it would have taken 6 hours more than the scheduled time. Find the length of the journey.

SOLUTION Let the original speed of the train be x kmph and let the time taken to complete the journey be y hours.

Length of whole journey = (xy) km.

Case I When speed = $(x + 6)$ kmph and time taken = $(y - 4)$ hours.

Length of total journey = $(x + 6)(y - 4)$ km.

$$\therefore xy = (x + 6)(y - 4) \Rightarrow xy = xy - 4x + 6y - 24$$

$$\Rightarrow 4x - 6y = -24 \Rightarrow 2x - 3y = -12. \quad \dots (i)$$

Case II When speed = $(x - 6)$ kmph and time taken = $(y + 6)$ hours.

Length of total journey = $(x - 6)(y + 6)$ km.

$$\therefore xy = (x-6)(y+6) \Rightarrow xy = xy + 6x - 6y - 36$$

$$\Rightarrow 6x - 6y = 36 \Rightarrow x - y = 6. \quad \dots (ii)$$

On multiplying (ii) by 3 and subtracting (i) from the result, we get

$$3x - 2x = 18 + 12 \Rightarrow x = 30.$$

On putting $x = 30$ in (ii), we get $y = 30 - 6 = 24$.

$$\therefore x = 30 \text{ and } y = 24.$$

So, length of journey = (xy) km = (30×24) km = 720 km.

PROBLEMS ON BOATS AND STREAM

EXAMPLE 28 *A man can row downstream 20 km in 2 hours, and upstream 4 km in 2 hours. Find his speed of rowing in still water. Also, find the speed of the current.*

SOLUTION Let the speed of the man in still water be x km/hr and let the speed of the current be y km/hr.

Speed downstream = $(x + y)$ km/hr.

Speed upstream = $(x - y)$ km/hr.

But, speed downstream = $\frac{\text{distance}}{\text{time}} = \frac{20}{2}$ km/hr = 10 km/hr.

And, speed upstream = $\frac{\text{distance}}{\text{time}} = \frac{4}{2}$ km/hr = 2 km/hr.

$$\therefore x + y = 10 \quad \dots (i) \quad \text{and} \quad x - y = 2 \quad \dots (ii).$$

Adding (i) and (ii), we get $2x = 12 \Rightarrow x = 6$.

Putting $x = 6$ in (i), we get $6 + y = 10 \Rightarrow y = 4$.

Thus, $x = 6$ and $y = 4$.

Hence, the speed of the man in still water is 6 km/hr and the speed of the current is 4 km/hr.

EXAMPLE 29 *A boat goes 16 km upstream and 24 km downstream in 6 hours. Also, it covers 12 km upstream and 36 km downstream in the same time. Find the speed of the boat in still water and that of the stream.*

SOLUTION Let the speed of the boat in still water be x km/hr and the speed of the stream be y km/hr. Then,

speed upstream = $(x - y)$ km/hr

and speed downstream = $(x + y)$ km/hr.

Time taken to cover 16 km upstream = $\frac{16}{(x - y)}$ hours.

Time taken to cover 24 km downstream = $\frac{24}{(x + y)}$ hours.

Total time taken = 6 hours.

$$\therefore \frac{16}{x - y} + \frac{24}{x + y} = 6. \quad \dots (i)$$

Again, time taken to cover 12 km upstream = $\frac{12}{(x - y)}$ hours.

Time taken to cover 36 km downstream = $\frac{36}{(x + y)}$ hours.

Total time taken = 6 hours.

$$\therefore \frac{12}{x - y} + \frac{36}{x + y} = 6. \quad \dots (ii)$$

Putting $\frac{1}{(x - y)} = u$ and $\frac{1}{(x + y)} = v$ in (i) and (ii), we get

$$16u + 24v = 6 \Rightarrow 8u + 12v = 3, \quad \dots (iii)$$

$$12u + 36v = 6 \Rightarrow 2u + 6v = 1. \quad \dots (iv)$$

On multiplying (iv) by 4 and subtracting (iii) from the result, we get

$$12v = 1 \Rightarrow v = \frac{1}{12}$$

$$\Rightarrow \frac{1}{x + y} = \frac{1}{12} \Rightarrow x + y = 12. \quad \dots (v)$$

On multiplying (iv) by 2 and subtracting the result from (iii), we get

$$4u = 1 \Rightarrow u = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x - y} = \frac{1}{4} \Rightarrow x - y = 4. \quad \dots (vi)$$

On adding (v) and (vi), we get $2x = 16 \Rightarrow x = 8$.

On subtracting (vi) from (v), we get $2y = 8 \Rightarrow y = 4$.

$\therefore$ speed of the boat in still water = 8 km/hr,

and speed of the stream = 4 km/hr.

PROBLEMS ON TIME AND WORK

EXAMPLE 30 8 men and 12 boys can finish a piece of work in 5 days, while 6 men and 8 boys can finish it in 7 days. Find the time taken by 1 man alone and that by 1 boy alone to finish the work.

SOLUTION Suppose 1 man alone can finish the work in x days and 1 boy alone can finish it in y days.

Then, 1 man's 1 day's work = $\frac{1}{x}$.

And, 1 boy's 1 day's work = $\frac{1}{y}$.

8 men and 12 boys can finish the work in 5 days

$$\Rightarrow (8 \text{ men's } 1 \text{ day's work}) + (12 \text{ boys' } 1 \text{ day's work}) = \frac{1}{5}$$

$$\Rightarrow \frac{8}{x} + \frac{12}{y} = \frac{1}{5}$$

$$\Rightarrow 8u + 12v = \frac{1}{5} \quad \dots \text{ (i), where } \frac{1}{x} = u \text{ and } \frac{1}{y} = v.$$

Again, 6 men and 8 boys can finish the work in 7 days

$$\Rightarrow (6 \text{ men's } 1 \text{ day's work}) + (8 \text{ boys' } 1 \text{ day's work}) = \frac{1}{7}$$

$$\Rightarrow \frac{6}{x} + \frac{8}{y} = \frac{1}{7}$$

$$\Rightarrow 6u + 8v = \frac{1}{7} \quad \dots \text{ (ii)}$$

On multiplying (i) by 3, (ii) by 4 and subtracting the results, we get

$$4v = \left(\frac{3}{5} - \frac{4}{7}\right) = \frac{1}{35} \Rightarrow v = \frac{1}{140} \Rightarrow \frac{1}{y} = \frac{1}{140} \Rightarrow y = 140.$$

On putting $v = \frac{1}{140}$ in (ii), we get

$$6u = \left(\frac{1}{7} - \frac{8}{140}\right) = \frac{12}{140} \Rightarrow u = \left(\frac{12}{140} \times \frac{1}{6}\right) = \frac{1}{70}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{70} \Rightarrow x = 70.$$

$\therefore$ one man alone can finish the work in 70 days,
and one boy alone can finish the work in 140 days.

PROBLEMS ON AREA

EXAMPLE 31 If the length of a rectangle is reduced by 5 units and its breadth is increased by 2 units then the area of the rectangle is reduced by 80 sq units. However, if we increase its length by 10 units and decrease

the breadth by 5 units, its area is increased by 50 sq units. Find the length and breadth of the rectangle.

SOLUTION Let the length and breadth of the rectangle be x units and y units respectively.

Then, area of the rectangle = xy sq units.

Case I When the length is reduced by 5 units and the breadth is increased by 2 units.

Then, new length = $(x - 5)$ units

and new breadth = $(y + 2)$ units.

$$\therefore \text{ new area} = (x - 5)(y + 2) \text{ sq units.}$$

$$\therefore xy - (x - 5)(y + 2) = 80 \Rightarrow 5y - 2x = 70. \quad \dots (i)$$

Case II When the length is increased by 10 units and the breadth is decreased by 5 units.

Then, new length = $(x + 10)$ units

and new breadth = $(y - 5)$ units.

$$\therefore \text{ new area} = (x + 10)(y - 5) \text{ sq units.}$$

$$\therefore (x + 10)(y - 5) - xy = 50$$

$$\Rightarrow 10y - 5x = 100 \Rightarrow 2y - x = 20. \quad \dots (ii)$$

On multiplying (ii) by 2 and subtracting the result from (i), we get $y = 30$.

Putting $y = 30$ in (ii), we get

$$(2 \times 30) - x = 20 \Rightarrow 60 - x = 20 \Rightarrow x = (60 - 20) = 40.$$

$$\therefore x = 40 \text{ and } y = 30.$$

Hence, length = 40 units and breadth = 30 units.

PROBLEMS ON GEOMETRY

EXAMPLE 32 In a $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Find the angles.

SOLUTION Let $\angle A = x^\circ$ and $\angle B = y^\circ$.

Then, $\angle C = 3\angle B = (3y)^\circ$.

Now, $\angle C = 2(\angle A + \angle B)$

$$\Rightarrow 3y = 2(x + y) \Rightarrow 2x - y = 0. \quad \dots (i)$$

We know that the sum of the angles of a triangle is 180° .

$$\begin{aligned}\therefore \angle A + \angle B + \angle C &= 180^\circ \Rightarrow x + y + 3y = 180 \\ &\Rightarrow x + 4y = 180. \quad \dots (ii)\end{aligned}$$

On multiplying (i) by 4 and adding the result with (ii), we get

$$8x + x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20.$$

Putting $x = 20$ in (i), we get $y = (2 \times 20) = 40$.

Thus, $x = 20$ and $y = 40$.

$$\therefore \angle A = 20^\circ, \angle B = 40^\circ \text{ and } \angle C = (3 \times 40^\circ) = 120^\circ.$$

EXAMPLE 33 Find the four angles of a cyclic quadrilateral ABCD in which $\angle A = (2x - 1)^\circ$, $\angle B = (y + 5)^\circ$, $\angle C = (2y + 15)^\circ$ and $\angle D = (4x - 7)^\circ$.

SOLUTION We know that the sum of the opposite angles of a cyclic quadrilateral is 180° .

$$\therefore \angle A + \angle C = 180^\circ \text{ and } \angle B + \angle D = 180^\circ.$$

$$\text{Now, } \angle A + \angle C = 180^\circ \Rightarrow (2x - 1) + (2y + 15) = 180$$

$$\Rightarrow 2(x + y) = 166$$

$$\Rightarrow x + y = 83. \quad \dots (i)$$

$$\text{And, } \angle B + \angle D = 180^\circ \Rightarrow (y + 5) + (4x - 7) = 180$$

$$\Rightarrow 4x + y = 182. \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$3x = 182 - 83 = 99 \Rightarrow x = 33.$$

Putting $x = 33$ in (i), we get

$$33 + y = 83 \Rightarrow y = (83 - 33) \Rightarrow y = 50.$$

Thus, $x = 33$ and $y = 50$.

$$\therefore \angle A = \{(2 \times 33) - 1\}^\circ = 65^\circ, \angle B = (50 + 5)^\circ = 55^\circ.$$

$$\angle C = \{(2 \times 50) + 15\}^\circ = 115^\circ \text{ and } \angle D = \{(4 \times 33) - 7\}^\circ = 125^\circ.$$

$$\text{Hence, } \angle A = 65^\circ, \angle B = 55^\circ, \angle C = 115^\circ \text{ and } \angle D = 125^\circ.$$

EXERCISE 3E

- 5 chairs and 4 tables together cost ₹ 5600, while 4 chairs and 3 tables together cost ₹ 4340. Find the cost of a chair and that of a table.
- 23 spoons and 17 forks together cost ₹ 1770, while 17 spoons and 23 forks together cost ₹ 1830. Find the cost of a spoon and that of a fork.

3. A lady has only 25-paisa and 50-paisa coins in her purse. If she has 50 coins in all totalling ₹ 19.50, how many coins of each kind does she have?
4. The sum of two numbers is 137 and their difference is 43. Find the numbers.
5. Find two numbers such that the sum of twice the first and thrice the second is 92, and four times the first exceeds seven times the second by 2.
6. Find two numbers such that the sum of thrice the first and the second is 142, and four times the first exceeds the second by 138.
7. If 45 is subtracted from twice the greater of two numbers, it results in the other number. If 21 is subtracted from twice the smaller number, it results in the greater number. Find the numbers.
8. If three times the larger of two numbers is divided by the smaller, we get 4 as the quotient and 8 as the remainder. If five times the smaller is divided by the larger, we get 3 as the quotient and 5 as the remainder. Find the numbers.
9. If 2 is added to each of two given numbers, their ratio becomes 1 : 2. However, if 4 is subtracted from each of the given numbers, the ratio becomes 5 : 11. Find the numbers.
10. The difference between two numbers is 14 and the difference between their squares is 448. Find the numbers.
11. The sum of the digits of a two-digit number is 12. The number obtained by interchanging its digits exceeds the given number by 18. Find the number. [CBSE 2006]
12. A number consisting of two digits is seven times the sum of its digits. When 27 is subtracted from the number, the digits are reversed. Find the number.
13. The sum of the digits of a two-digit number is 15. The number obtained by interchanging the digits exceeds the given number by 9. Find the number. [CBSE 2004]
14. A two-digit number is 3 more than 4 times the sum of its digits. If 18 is added to the number, the digits are reversed. Find the number.
15. A number consists of two digits. When it is divided by the sum of its digits, the quotient is 6 with no remainder. When the number is diminished by 9, the digits are reversed. Find the number. [CBSE 1999C]

16. A two-digit number is such that the product of its digits is 35. If 18 is added to the number, the digits interchange their places. Find the number. [CBSE 2006]
17. A two-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number. [CBSE 2006C]
18. The sum of a two-digit number and the number obtained by reversing the order of its digits is 121, and the two digits differ by 3. Find the number.
19. The sum of the numerator and denominator of a fraction is 8. If 3 is added to both of the numerator and the denominator, the fraction becomes $\frac{3}{4}$. Find the fraction. [CBSE 2003]
20. If 2 is added to the numerator of a fraction, it reduces to $\left(\frac{1}{2}\right)$ and if 1 is subtracted from the denominator, it reduces to $\left(\frac{1}{3}\right)$. Find the fraction.
21. The denominator of a fraction is greater than its numerator by 11. If 8 is added to both its numerator and denominator, it becomes $\frac{3}{4}$. Find the fraction.
22. Find a fraction which becomes $\left(\frac{1}{2}\right)$ when 1 is subtracted from the numerator and 2 is added to the denominator, and the fraction becomes $\left(\frac{1}{3}\right)$ when 7 is subtracted from the numerator and 2 is subtracted from the denominator.
23. The sum of the numerator and denominator of a fraction is 4 more than twice the numerator. If the numerator and denominator are increased by 3, they are in the ratio 2 : 3. Determine the fraction. [CBSE 2010]
24. The sum of two numbers is 16 and the sum of their reciprocals is $\frac{1}{3}$. Find the numbers. [CBSE 2005]
25. There are two classrooms A and B. If 10 students are sent from A to B, the number of students in each room becomes the same. If 20 students are sent from B to A, the number of students in A becomes double the number of students in B. Find the number of students in each room.
26. Taxi charges in a city consist of fixed charges and the remaining depending upon the distance travelled in kilometres. If a man travels

80 km, he pays ₹ 1330, and travelling 90 km, he pays ₹ 1490. Find the fixed charges and rate per km.

27. A part of monthly hostel charges in a college hostel are fixed and the remaining depends on the number of days one has taken food in the mess. When a student A takes food for 25 days, he has to pay ₹ 4500, whereas a student B who takes food for 30 days, has to pay ₹ 5200. Find the fixed charges per month and the cost of the food per day.
28. A man invested an amount at 10% per annum and another amount at 8% per annum simple interest. Thus, he received ₹ 1350 as annual interest. Had he interchanged the amounts invested, he would have received ₹ 45 less as interest. What amounts did he invest at different rates?
29. The monthly incomes of A and B are in the ratio 5 : 4 and their monthly expenditures are in the ratio 7 : 5. If each saves ₹ 9000 per month, find the monthly income of each.
30. A man sold a chair and a table together for ₹ 1520, thereby making a profit of 25% on chair and 10% on table. By selling them together for ₹ 1535, he would have made a profit of 10% on the chair and 25% on the table. Find the cost price of each.
31. Points A and B are 70 km apart on a highway. A car starts from A and another car starts from B simultaneously. If they travel in the same direction, they meet in 7 hours. But, if they travel towards each other, they meet in 1 hour. Find the speed of each car. [CBSE 2007C]
32. A train covered a certain distance at a uniform speed. If the train had been 5 kmph faster, it would have taken 3 hours less than the scheduled time. And, if the train were slower by 4 kmph, it would have taken 3 hours more than the scheduled time. Find the length of the journey.
33. Abdul travelled 300 km by train and 200 km by taxi taking 5 hours 30 minutes. But, if he travels 260 km by train and 240 km by taxi, he takes 6 minutes longer. Find the speed of the train and that of the taxi. [CBSE 2006C]
34. Places A and B are 160 km apart on a highway. One car starts from A and another from B at the same time. If they travel in the same direction, they meet in 8 hours. But, if they travel towards each other, they meet in 2 hours. Find the speed of each car. [CBSE 2009C]
35. A sailor goes 8 km downstream in 40 minutes and returns in 1 hour. Find the speed of the sailor in still water and the speed of the current.

36. A boat goes 12 km upstream and 40 km downstream in 8 hours. It can go 16 km upstream and 32 km downstream in the same time. Find the speed of the boat in still water and the speed of the stream.
37. 2 men and 5 boys can finish a piece of work in 4 days, while 3 men and 6 boys can finish it in 3 days. Find the time taken by one man alone to finish the work and that taken by one boy alone to finish the work.
38. The length of a room exceeds its breadth by 3 metres. If the length is increased by 3 metres and the breadth is decreased by 2 metres, the area remains the same. Find the length and the breadth of the room.
39. The area of a rectangle gets reduced by 8 m^2 , when its length is reduced by 5 m and its breadth is increased by 3 m. If we increase the length by 3 m and breadth by 2 m, the area is increased by 74 m^2 . Find the length and the breadth of the rectangle.
40. The area of a rectangle gets reduced by 67 square metres, when its length is increased by 3 m and breadth is decreased by 4 m. If the length is reduced by 1 m and breadth is increased by 4 m, the area is increased by 89 square metres. Find the dimensions of the rectangle.
41. A railway half ticket costs half the full fare and the reservation charge is the same on half ticket as on full ticket. One reserved first class ticket from Mumbai to Delhi costs ₹ 4150 while one full and one half reserved first class tickets cost ₹ 6255. What is the basic first class full fare and what is the reservation charge? [HOTS]
42. Five years hence, a man's age will be three times the age of his son. Five years ago, the man was seven times as old as his son. Find their present ages.
43. Two years ago, a man was five times as old as his son. Two years later, his age will be 8 more than three times the age of his son. Find their present ages. [CBSE 2008]
44. If twice the son's age in years is added to the father's age, the sum is 70. But, if twice the father's age is added to the son's age, the sum is 95. Find the ages of father and son.
45. The present age of a woman is 3 years more than three times the age of her daughter. Three years hence, the woman's age will be 10 years more than twice the age of her daughter. Find their present ages.
46. On selling a tea set at 5% loss and a lemon set at 15% gain, a crockery seller gains ₹ 7. If he sells the tea set at 5% gain and the lemon set at 10% gain, he gains ₹ 13. Find the actual price of each of the tea set and the lemon set. [HOTS]

47. A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Mona paid ₹ 27 for a book kept for 7 days, while Tanvy paid ₹ 21 for the book she kept for 5 days. Find the fixed charge and the charge for each extra day.
48. A chemist has one solution containing 50% acid and a second one containing 25% acid. How much of each should be used to make 10 litres of a 40% acid solution? [HOTS]
49. A jeweller has bars of 18-carat gold and 12-carat gold. How much of each must be melted together to obtain a bar of 16-carat gold, weighing 120 g? (Given: Pure gold is 24-carat). [HOTS]
50. 90% and 97% pure acid solutions are mixed to obtain 21 litres of 95% pure acid solution. Find the quantity of each type of acids to be mixed to form the mixture. [HOTS]
51. The larger of the two supplementary angles exceeds the smaller by 18° . Find them.
52. In a $\triangle ABC$, $\angle A = x^\circ$, $\angle B = (3x - 2)^\circ$, $\angle C = y^\circ$ and $\angle C - \angle B = 9^\circ$. Find the three angles.
53. In a cyclic quadrilateral $ABCD$, it is given that $\angle A = (2x + 4)^\circ$, $\angle B = (y + 3)^\circ$, $\angle C = (2y + 10)^\circ$ and $\angle D = (4x - 5)^\circ$. Find the four angles.

ANSWERS (EXERCISE 3E)

- | | | | | |
|--|--|------------------------|---------------------|---------------------|
| 1. ₹ 560, ₹ 700 | 2. ₹ 40, ₹ 50 | 3. 22, 28 | 4. 90, 47 | 5. 25, 14 |
| 6. 40, 22 | 7. 37, 29 | 8. 20, 13 | 9. 34, 70 | 10. 23, 9 |
| 11. 57 | 12. 63 | 13. 78 | 14. 35 | 15. 54 |
| 16. 57 | 17. 92 | | | |
| 18. 47 or 74 | 19. $\frac{3}{5}$ | 20. $\frac{3}{10}$ | 21. $\frac{25}{36}$ | 22. $\frac{15}{26}$ |
| 23. $\frac{5}{9}$ | 24. 12 and 4 | | | |
| 25. 100 in A and 80 in B | 26. fixed charges = ₹ 50, rate per km = ₹ 16 | | | |
| 27. fixed charges = ₹ 1000, cost of food per day = ₹ 140 | | | | |
| 28. ₹ 8500 at 10% p.a. and ₹ 6250 at 8% p.a. | 29. ₹ 30000, ₹ 24000 | | | |
| 30. ₹ 600, ₹ 700 | 31. 40 km/hr, 30 km/hr | 32. 1080 km | | |
| 33. 100 kmph, 80 kmph | 34. 50 kmph, 30 kmph | 35. 10 kmph, 2 kmph | | |
| 36. 6 kmph, 2 kmph | 37. 18 days, 36 days | | | |
| 38. length = 15 m, breadth = 12 m | 39. length = 19 m, breadth = 10 m | | | |
| 40. length = 28 m, breadth = 19 m | | | | |
| 41. first class full fare = ₹ 4090, reservation charges = ₹ 60 | | | | |
| 42. 40 years, 10 years | 43. 42 years, 10 years | 44. 40 years, 15 years | | |

45. 33 years, 10 years 46. tea set = ₹ 100, lemon set = ₹ 80
 47. ₹ 15, ₹ 3 per day 48. 50% solution = 6 litres, 25% solution = 4 litres
 49. 80 g, 40 g 50. 6 litres, 15 litres 51. 99° , 81°
 52. $\angle A = 25^\circ$, $\angle B = 73^\circ$, $\angle C = 82^\circ$
 53. $\angle A = 70^\circ$, $\angle B = 53^\circ$, $\angle C = 110^\circ$, $\angle D = 127^\circ$

HINTS TO SOME SELECTED QUESTIONS

3. Let the number of 25-p and 50-p coins be x and y respectively.
 Then, $x + y = 50$ and $\frac{x}{4} + \frac{y}{2} = \frac{1950}{100}$.
8. Let the larger number be x and the smaller number be y .
 Then, $3x = 4y + 8$ and $5y = 3x + 5$.
10. $x - y = 14$ and $x^2 - y^2 = 448$.
 $\therefore x + y = \frac{(x^2 - y^2)}{(x - y)} = \frac{448}{14} = 32$.
15. Let the ten's digit be x and the unit's digit be y . Then,
 $\frac{10x + y}{x + y} = 6$ and $10x + y - 9 = 10y + x$.
16. Let the ten's digit be x and the unit's digit be y . Then,
 $xy = 35$ and $10x + y + 18 = 10y + x \Rightarrow y - x = 2$.
 $\therefore (y + x)^2 - (y - x)^2 = 4xy \Rightarrow (y + x) = \sqrt{(y - x)^2 + 4xy} = \sqrt{144} = 12$.
 Now, solve $y - x = 2$ and $y + x = 12$.
18. Let the ten's digit be x and the unit's digit be y .
 Then, $(10x + y) + (10y + x) = 121 \Rightarrow 11(x + y) = 121 \Rightarrow x + y = 11$.
 $(x + y = 11, x - y = 3)$ or $(x + y = 11, y - x = 3)$.
21. Let the required fraction be $\frac{x}{y}$.
 Then, $x + 11 = y \Rightarrow x - y = -11$.
 And, $\frac{x + 8}{y + 8} = \frac{3}{4} \Rightarrow 4x + 32 = 3y + 24 \Rightarrow 4x - 3y = -8$.
23. Let the required fraction be $\frac{x}{y}$.
 Then, $x + y - 4 = 2x \Rightarrow y - x = 4$.
 $\frac{x + 3}{y + 3} = \frac{2}{3} \Rightarrow 3x + 9 = 2y + 6 \Rightarrow 2y - 3x = 3$.
24. $x + y = 16$ and $\frac{1}{x} + \frac{1}{y} = \frac{1}{3} \Rightarrow \frac{y + x}{xy} = \frac{1}{3} \Rightarrow xy = 16 \times 3 = 48$.
 $(x - y)^2 = (x + y)^2 - 4xy = (16)^2 - (4 \times 48) = 256 - 192 = 64$
 $\Rightarrow x - y = 8$ or $x - y = -8$.
 $\therefore (x + y = 16, x - y = 8)$ or $(x + y = 16, x - y = -8)$.
25. Let the number of students in room A and room B be x and y respectively. Then,
 $x - 10 = y + 10 \Rightarrow x - y = 20$ (i)

And, $(x + 20) = 2(y - 20) \Rightarrow x - 2y = -60$ (ii)

Now, solve (i) and (ii) to get x and y .

26. Let the fixed charges be ₹ x and rate per km be ₹ y . Then,

$$x + 80y = 1330 \text{ and } x + 90y = 1490.$$

27. Let the fixed charges be ₹ x and the charges for food be ₹ y per day. Then,

$$x + 25y = 4500 \text{ and } x + 30y = 5200.$$

28. Suppose he invested ₹ x at 10% p.a. and ₹ y at 8% p.a.

$$\frac{(x \times 10 \times 1)}{100} + \frac{(y \times 8 \times 1)}{100} = 1350 \Rightarrow 5x + 4y = 67500. \quad \dots (i)$$

$$\frac{(x \times 8 \times 1)}{100} + \frac{(y \times 10 \times 1)}{100} = 1350 - 45 \Rightarrow 4x + 5y = 65250. \quad \dots (ii)$$

Add (i) and (ii). Then, subtract (ii) from (i).

29. $5x - 7y = 9000$ and $4x - 5y = 9000$. Find x and y .

30. Let the CP of a chair be ₹ x and that of a table be ₹ y . Then,

$$\frac{125x}{100} + \frac{110y}{100} = 1520 \Rightarrow 25x + 22y = 30400. \quad \dots (i)$$

$$\frac{110x}{100} + \frac{125y}{100} = 1535 \Rightarrow 22x + 25y = 30700. \quad \dots (ii)$$

On adding (i) and (ii), we get $47(x + y) = 61100 \Rightarrow x + y = 1300$.

On subtracting (i) from (ii), we get $3(y - x) = 300 \Rightarrow y - x = 100$.

$$\therefore x = 600 \text{ and } y = 700.$$

31. Let the speed of the car from A be x kmph and that of the car from B be y kmph.

$$\text{Then, } 7x - 7y = 70 \Rightarrow x - y = 10. \quad \dots (i)$$

$$\text{And, } x + y = 70. \quad \dots (ii)$$

32. Let the original speed of the train be x kmph and let the time taken to complete the journey be y hours.

Then, length of the journey = xy km.

Case I Speed = $(x + 5)$ kmph and time taken = $(y - 3)$ hours.

$$\therefore xy = (x + 5)(y - 3) \Rightarrow 5y - 3x = 15. \quad \dots (i)$$

Case II Speed = $(x - 4)$ kmph and time taken = $(y + 3)$ hours.

$$\therefore xy = (x - 4)(y + 3) \Rightarrow 3x - 4y = 12. \quad \dots (ii)$$

From (i) and (ii), we get $x = 40$ and $y = 27$.

$$\therefore \text{length of the journey} = xy = 40 \times 27 \text{ km} = 1080 \text{ km}.$$

33. Let the speed of the train be x kmph and that of the taxi be y kmph.

$$\text{Then, } \frac{300}{x} + \frac{200}{y} = \frac{11}{2} \Rightarrow 600u + 400v = 11 \quad \dots (i), \text{ where } \frac{1}{x} = u \text{ and } \frac{1}{y} = v.$$

$$\text{And, } \frac{260}{x} + \frac{240}{y} = \frac{28}{5} \Rightarrow 325u + 300v = 7. \quad \dots (ii)$$

35. Let the speed of the sailor in still water be x kmph and the speed of the current be y kmph.

Then, speed downstream = $(x + y)$ km/hr.

And, speed upstream = $(x - y)$ km/hr.

$$\therefore \frac{8}{x+y} = \frac{40}{60} = \frac{2}{3} \Rightarrow x+y = 12 \quad \dots (i)$$

$$\text{and } \frac{8}{x-y} = 1 \Rightarrow x-y = 8. \quad \dots (ii)$$

Solve (i) and (ii).

37. Let 1 man's 1 day's work be $\frac{1}{x}$ and 1 boy's 1 day's work be $\frac{1}{y}$.

$$\text{Then, } \frac{2}{x} + \frac{5}{y} = \frac{1}{4} \Rightarrow 2u + 5v = \frac{1}{4}$$

$$\text{and } \frac{3}{x} + \frac{6}{y} = \frac{1}{3} \Rightarrow 3u + 6v = \frac{1}{3}.$$

38. Let length = x metres and breadth = y metres.

$$\text{Then, } x = y + 3 \text{ and } (x + 3)(y - 2) = xy.$$

$$\therefore x - y = 3 \text{ and } 3y - 2x = 6.$$

39. Let length = x metres and breadth = y metres. Then,

$$xy - (x - 5)(y + 3) = 8 \Rightarrow 3x - 5y = 7.$$

$$\text{And, } (x + 3)(y + 2) - xy = 74 \Rightarrow 2x + 3y = 68.$$

40. Let length = x metres and breadth = y metres. Then,

$$xy - (x + 3)(y - 4) = 67 \Rightarrow 4x - 3y = 55$$

$$\text{and } (x - 1)(y + 4) - xy = 89 \Rightarrow 4x - y = 93.$$

41. Let the basic first class full fare be ₹ x and the reservation charge be ₹ y . Then,

$$x + y = 4150. \quad \dots (i)$$

$$\text{And, } (x + y) + \left(\frac{1}{2}x + y\right) = 6255 \Rightarrow 3x + 4y = 12510. \quad \dots (ii)$$

42. Let the present ages of the man and his son be x years and y years respectively. Then,

$$x + 5 = 3(y + 5) \Rightarrow x - 3y = 10 \quad \dots (i)$$

$$\text{and } x - 5 = 7(y - 5) \Rightarrow x - 7y = -30. \quad \dots (ii)$$

43. Let the present ages of the man and his son be x years and y years respectively. Then,

$$(x - 2) = 5(y - 2) \Rightarrow x - 5y = -8 \quad \dots (i)$$

$$\text{and } (x + 2) = 3(y + 2) + 8 \Rightarrow x - 3y = 12. \quad \dots (ii)$$

44. Let father's age be x years and son's age be y years. Then,

$$x + 2y = 70 \text{ and } 2x + y = 95.$$

45. Let woman's present age be x years and daughter's present age be y years. Then,

$$x = 3y + 3 \Rightarrow x - 3y = 3$$

$$\text{and, } x + 3 = 2(y + 3) + 10 \Rightarrow x - 2y = 13.$$

46. Let the cost price of the tea set be ₹ x and that of the lemon set be ₹ y . Then,

$$\left(\frac{95x}{100} + \frac{115y}{100}\right) - (x + y) = 7 \Rightarrow \frac{(19x + 23y)}{20} - (x + y) = 7$$

$$\Rightarrow 3y - x = 140. \quad \dots (i)$$

$$\text{And, } \left(\frac{105x}{100} + \frac{110y}{100} \right) - (x + y) = 13 \Rightarrow \frac{(21x + 22y)}{20} - (x + y) = 13$$

$$\Rightarrow x + 2y = 260. \quad \dots \text{ (ii)}$$

Now, solve (i) and (ii).

47. Let the fixed charge be ₹ x and charge for each extra day be ₹ y .

$$\text{Then, } x + (7 - 3)y = 27 \Rightarrow x + 4y = 27 \quad \dots \text{ (i)}$$

$$\text{and, } x + (5 - 3)y = 21 \Rightarrow x + 2y = 21. \quad \dots \text{ (ii)}$$

48. Let x litres of 50% solution be mixed with y litres of 25% solution.

$$\text{Then, } x + y = 10 \quad \dots \text{ (i)}$$

$$\text{and, } \frac{50x}{100} + \frac{25y}{100} = \frac{40 \times 10}{100} \Rightarrow 2x + y = 16. \quad \dots \text{ (ii)}$$

49. Let x g of 18-carat gold be mixed with y g of 12-carat gold to get 120 g of 16-carat gold.

$$\text{Then, } x + y = 120. \quad \dots \text{ (i)}$$

$$\text{Gold \% in 18-carat gold} = \left(\frac{18}{24} \times 100 \right) \% = 75\%$$

$$\text{Gold \% in 12-carat gold} = \left(\frac{12}{24} \times 100 \right) \% = 50\%$$

$$\text{Gold \% in 16-carat gold} = \left(\frac{16}{24} \times 100 \right) \% = \frac{200}{3} \%$$

$$\therefore 75\% \text{ of } x + 50\% \text{ of } y = \frac{200}{3} \% \text{ of } 120$$

$$\Rightarrow \frac{75x}{100} + \frac{50y}{100} = \frac{200 \times 120}{3 \times 100} \Rightarrow 3x + 2y = 320. \quad \dots \text{ (ii)}$$

Now, solve (i) and (ii).

50. Let x litres of 90% pure solution be mixed with y litres of 97% pure solution to get 21 litres of 95% pure solution.

$$\text{Then, } x + y = 21 \quad \dots \text{ (i)}$$

$$\text{and } \frac{90x}{100} + \frac{97y}{100} = \frac{95 \times 21}{100} \Rightarrow 90x + 97y = 1995. \quad \dots \text{ (ii)}$$

Now, solve (i) and (ii).

51. $x - y = 18$ and $x + y = 180$. Find x and y .

$$52. \angle A + \angle B + \angle C = 180^\circ \Rightarrow x + (3x - 2) + y = 180 \Rightarrow 4x + y = 182. \quad \dots \text{ (i)}$$

$$\angle C - \angle B = 9^\circ \Rightarrow y - 3x + 2 = 9 \Rightarrow y - 3x = 7. \quad \dots \text{ (ii)}$$

From (i) and (ii), we get $7x = 175 \Rightarrow x = 25$.

Putting $x = 25$ in (i), we get $y = 82$.

$$\therefore \angle A = 25^\circ, \angle B = (3 \times 25 - 2)^\circ = 73^\circ, \angle C = 82^\circ.$$

53. We have, $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$.

$$\therefore (2x + 4) + (2y + 10) = 180 \Rightarrow x + y = 83 \quad \dots \text{ (i)}$$

$$\text{and } (y + 3) + (4x - 5) = 180 \Rightarrow 4x + y = 182. \quad \dots \text{ (ii)}$$

From (i) and (ii), we get

$$\therefore 3x = 99 \Rightarrow x = 33. \text{ And, } 33 + y = 83 \Rightarrow y = 50.$$

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EXERCISE 3F*Very-Short and Short-Answer Questions*

1. Write the number of solutions of the following pair of linear equations:

$$x + 2y - 8 = 0, 2x + 4y = 16. \quad [\text{CBSE 2009}]$$

2. Find the value of k for which the following pair of linear equations have infinitely many solutions:

$$2x + 3y = 7, (k - 1)x + (k + 2)y = 3k. \quad [\text{CBSE 2010}]$$

3. For what value of k does the following pair of linear equations have infinitely many solutions?

$$10x + 5y - (k - 5) = 0 \text{ and } 20x + 10y - k = 0.$$

4. For what value of k will the following pair of linear equations have no solution?

$$2x + 3y = 9, 6x + (k - 2)y = (3k - 2). \quad [\text{CBSE 2010}]$$

5. Write the number of solutions of the following pair of linear equations:

$$x + 3y - 4 = 0 \text{ and } 2x + 6y - 7 = 0.$$

6. Write the value of k for which the system of equations $3x + ky = 0$, $2x - y = 0$ has a unique solution.

7. The difference between two numbers is 5 and the difference between their squares is 65. Find the numbers.

8. The cost of 5 pens and 8 pencils is ₹ 120, while the cost of 8 pens and 5 pencils is ₹ 153. Find the cost of 1 pen and that of 1 pencil.

9. The sum of two numbers is 80. The larger number exceeds four times the smaller one by 5. Find the numbers.

10. A number consists of two digits whose sum is 10. If 18 is subtracted from the number, its digits are reversed. Find the number.

11. A man purchased 47 stamps of 20 p and 25 p for ₹ 10. Find the number of each type of stamps.

12. A man has some hens and cows. If the number of heads be 48 and number of feet be 140, how many cows are there?

13. If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, find the values of x and y .

14. If $\frac{x}{4} + \frac{y}{3} = \frac{5}{12}$ and $\frac{x}{2} + y = 1$ then find the value of $(x + y)$.

15. If $12x + 17y = 53$ and $17x + 12y = 63$ then find the value of $(x + y)$.

16. Find the value of k for which the system $3x + 5y = 0$, $kx + 10y = 0$ has a nonzero solution.
17. Find k for which the system $kx - y = 2$ and $6x - 2y = 3$ has a unique solution.
18. Find k for which the system $2x + 3y - 5 = 0$, $4x + ky - 10 = 0$ has an infinite number of solutions.
19. Show that the system $2x + 3y - 1 = 0$, $4x + 6y - 4 = 0$ has no solution.
20. Find k for which the system $x + 2y = 3$ and $5x + ky + 7 = 0$ is inconsistent.
21. Solve: $\frac{3}{x+y} + \frac{2}{x-y} = 2$ and $\frac{9}{x+y} - \frac{4}{x-y} = 1$.

ANSWERS (EXERCISE 3F)

1. infinitely many 2. $k = 7$ 3. $k = 10$ 4. $k = 11$ 5. 0
6. $k \neq \frac{-3}{2}$ 7. 9 and 4 8. ₹ 16, ₹ 5 9. 65 and 15 10. 64
11. 35 and 12 12. 22 13. $x = 1, y = 3$ 14. $\frac{3}{2}$ 15. 4
16. $k = 6$ 17. $k \neq 3$ 18. $k = 6$ 20. $k = \frac{2}{5}$ 21. $x = \frac{5}{2}, y = \frac{1}{2}$

HINTS TO SOME SELECTED QUESTIONS

1. Given equations are $x + 2y - 8 = 0$ and $2x + 4y - 16 = 0$.

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

So, there are infinitely many solutions.

2. $2x + 3y - 7 = 0$ and $(k-1)x + (k+2)y - 3k = 0$.

For infinitely many solutions, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

$$\therefore \frac{2}{(k-1)} = \frac{3}{(k+2)} \text{ and } \frac{3}{(k+2)} = \frac{7}{3k}$$

$$\Rightarrow 2k + 4 = 3k - 3 \text{ and } 9k = 7k + 14 \Rightarrow k = 7.$$

3. For infinitely many solutions, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

$$\therefore \frac{10}{20} = \frac{5}{10} = \frac{k-5}{k} \Rightarrow \frac{k-5}{k} = \frac{1}{2} \Rightarrow 2k - 10 = k \Rightarrow k = 10.$$

4. For no solution, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$$\therefore \frac{2}{6} = \frac{3}{k-2} \text{ and } \frac{2}{6} \neq \frac{9}{3k-2} \Rightarrow \frac{3}{k-2} = \frac{1}{3} \text{ and } \frac{9}{3k-2} \neq \frac{1}{3}$$

$$\Rightarrow k - 2 = 9 \text{ and } 3k - 2 \neq 27 \Rightarrow k = 11 \text{ and } k \neq \frac{29}{3}.$$

Hence, $k = 11$.

$$5. \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}.$$

Hence, the given system has no solution.

$$6. \text{ For unique solution, we have } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}.$$

$$\therefore \frac{3}{2} \neq \frac{k}{-1} \Rightarrow k \neq \frac{-3}{2}.$$

7. Let the required numbers be x and y . Then,

$$x - y = 5 \text{ and } x^2 - y^2 = 65 \Rightarrow \frac{x^2 - y^2}{x - y} = \frac{65}{5} = 13 \Rightarrow x + y = 13.$$

Solving $x - y = 5$ and $x + y = 13$, we get $x = 9$ and $y = 4$.

8. Let the cost of 1 pen be ₹ x and that of 1 pencil be ₹ y . Then,

$$5x + 8y = 120 \quad \dots (i) \text{ and } 8x + 5y = 153. \quad \dots (ii)$$

$$\text{Adding (i) and (ii), we get } 13(x + y) = 273 \Rightarrow x + y = 21. \quad \dots (iii)$$

$$\text{Subtracting (i) from (ii), we get } 3(x - y) = 33 \Rightarrow x - y = 11. \quad \dots (iv)$$

From (iii) and (iv), we get $x = 16$ and $y = 5$.

9. Let the required numbers be x and y such that $x > y$. Then,

$$x + y = 80 \quad \dots (i) \text{ and } x - 4y = 5. \quad \dots (ii)$$

Solve (i) and (ii).

10. Let the ten's digit be x and unit's digit be y . Then,

$$x + y = 10 \quad \dots (i) \text{ and } (10x + y) - 18 = (10y + x) \Rightarrow x - y = 2. \quad \dots (ii)$$

Solve (i) and (ii).

11. Let x stamps of 20 p and y stamps of 25 p be purchased. Then,

$$x + y = 47 \quad \dots (i) \text{ and } 20x + 25y = 1000 \Rightarrow 4x + 5y = 200. \quad \dots (ii)$$

Solve (i) and (ii).

12. Let there be x cows and y hens. Then,

$$x + y = 48 \quad \dots (i) \text{ and } 4x + 2y = 140 \Rightarrow 2x + y = 70. \quad \dots (ii)$$

Subtracting (i) from (ii), we get $x = 22$.

13. The given equations are

$$2y + 3x = 9 \quad \dots (i) \text{ and } 4y + 9x = 21. \quad \dots (ii)$$

Solve (i) and (ii).

$$14. \text{ Given equations are } 3x + 4y = 5 \quad \dots (i) \text{ and } x + 2y = 2. \quad \dots (ii)$$

Solve (i) and (ii) for x and y and find $(x + y)$.

15. Adding the given equations, we get

$$29(x + y) = 116 \Rightarrow x + y = \frac{116}{29} = 4.$$

$$16. \text{ The given system has a nonzero solution when } \frac{3}{k} = \frac{5}{10} \Rightarrow k = 6.$$

$$17. \text{ For a unique solution, we have } \frac{k}{6} \neq \frac{-1}{-2} \Rightarrow k \neq 3.$$

18. For an infinite number of solutions, we have

$$\frac{2}{4} = \frac{3}{k} = \frac{-5}{10} \Rightarrow k = 6.$$

19. Here $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. Hence, the given system has no solution.

20. The system will be inconsistent when $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$$\therefore \frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = \frac{2}{5}.$$

21. Put $\frac{1}{x+y} = u$ and $\frac{1}{x-y} = v$ to get $3u + 2v = 2$ and $9u - 4v = 1$.

This gives $u = \frac{1}{3}$ and $v = \frac{1}{2}$. So, $x + y = 3$ and $x - y = 2$.

Solve for x and y .

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

1. If $2x + 3y = 12$ and $3x - 2y = 5$ then

- (a) $x = 2, y = 3$ (b) $x = 2, y = -3$ (c) $x = 3, y = 2$ (d) $x = 3, y = -2$

2. If $x - y = 2$ and $\frac{2}{x+y} = \frac{1}{5}$ then

- (a) $x = 4, y = 2$ (b) $x = 5, y = 3$ (c) $x = 6, y = 4$ (d) $x = 7, y = 5$

3. If $\frac{2x}{3} - \frac{y}{2} + \frac{1}{6} = 0$ and $\frac{x}{2} + \frac{2y}{3} = 3$ then

- (a) $x = 2, y = 3$ (b) $x = -2, y = 3$ (c) $x = 2, y = -3$ (d) $x = -2, y = -3$

4. If $\frac{1}{x} + \frac{2}{y} = 4$ and $\frac{3}{y} - \frac{1}{x} = 11$ then

- (a) $x = 2, y = 3$ (b) $x = -2, y = 3$ (c) $x = \frac{-1}{2}, y = 3$ (d) $x = \frac{-1}{2}, y = \frac{1}{3}$

5. If $\frac{2x+y+2}{5} = \frac{3x-y+1}{3} = \frac{3x+2y+1}{6}$ then

- (a) $x = 1, y = 1$ (b) $x = -1, y = -1$ (c) $x = 1, y = 2$ (d) $x = 2, y = 1$

6. If $\frac{3}{x+y} + \frac{2}{x-y} = 2$ and $\frac{9}{x+y} - \frac{4}{x-y} = 1$ then

- (a) $x = \frac{1}{2}, y = \frac{3}{2}$ (b) $x = \frac{5}{2}, y = \frac{1}{2}$ (c) $x = \frac{3}{2}, y = \frac{1}{2}$ (d) $x = \frac{1}{2}, y = \frac{5}{2}$

7. If $4x + 6y = 3xy$ and $8x + 9y = 5xy$ then

- (a) $x = 2, y = 3$ (b) $x = 1, y = 2$ (c) $x = 3, y = 4$ (d) $x = 1, y = -1$

8. If $29x + 37y = 103$ and $37x + 29y = 95$ then

- (a) $x = 1, y = 2$ (b) $x = 2, y = 1$ (c) $x = 3, y = 2$ (d) $x = 2, y = 3$

9. If $2^{x+y} = 2^{x-y} = \sqrt{8}$ then the value of y is

- (a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) 0 (d) none of these

10. If $\frac{2}{x} + \frac{3}{y} = 6$ and $\frac{1}{x} + \frac{1}{2y} = 2$ then
(a) $x = 1, y = \frac{2}{3}$ (b) $x = \frac{2}{3}, y = 1$ (c) $x = 1, y = \frac{3}{2}$ (d) $x = \frac{3}{2}, y = 1$
11. The system $kx - y = 2$ and $6x - 2y = 3$ has a unique solution only when
(a) $k = 0$ (b) $k \neq 0$ (c) $k = 3$ (d) $k \neq 3$
12. The system $x - 2y = 3$ and $3x + ky = 1$ has a unique solution only when
(a) $k = -6$ (b) $k \neq -6$ (c) $k = 0$ (d) $k \neq 0$
13. The system $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution, when
(a) $k = 10$ (b) $k \neq 10$ (c) $k = \frac{-7}{3}$ (d) $k = -21$
14. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel then the value of k is
(a) $\frac{-5}{4}$ (b) $\frac{2}{5}$ (c) $\frac{3}{2}$ (d) $\frac{15}{4}$
15. For what value of k do the equations $kx - 2y = 3$ and $3x + y = 5$ represent two lines intersecting at a unique point?
(a) $k = 3$ (b) $k = -3$
(c) $k = 6$ (d) all real values except -6
16. The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has
(a) a unique solution (b) exactly two solutions
(c) infinitely many solutions (d) no solution
17. The pair of equations $2x + 3y = 5$ and $4x + 6y = 15$ has
(a) a unique solution (b) exactly two solutions
(c) infinitely many solutions (d) no solution
18. If a pair of linear equations is consistent then their graph lines will be
(a) parallel (b) always coincident
(c) always intersecting (d) intersecting or coincident
19. If a pair of linear equations is inconsistent then their graph lines will be
(a) parallel (b) always coincident
(c) always intersecting (d) intersecting or coincident
20. In a $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$, then $\angle B = ?$
(a) 20° (b) 40° (c) 60° (d) 80°
21. In a cyclic quadrilateral $ABCD$, it is being given that $\angle A = (x + y + 10)^\circ$, $\angle B = (y + 20)^\circ$, $\angle C = (x + y - 30)^\circ$ and $\angle D = (x + y)^\circ$. Then, $\angle B = ?$
(a) 70° (b) 80° (c) 100° (d) 110°

22. The sum of the digits of a two-digit number is 15. The number obtained by interchanging the digits exceeds the given number by 9. The number is
(a) 96 (b) 69 (c) 87 (d) 78
23. In a given fraction, if 1 is subtracted from the numerator and 2 is added to the denominator, it becomes $\frac{1}{2}$. If 7 is subtracted from the numerator and 2 is subtracted from the denominator, it becomes $\frac{1}{3}$. The fraction is
(a) $\frac{13}{24}$ (b) $\frac{15}{26}$ (c) $\frac{16}{27}$ (d) $\frac{16}{21}$
24. 5 years hence, the age of a man shall be 3 times the age of his son while 5 years earlier the age of the man was 7 times the age of his son. The present age of the man is
(a) 45 years (b) 50 years (c) 47 years (d) 40 years
25. The graphs of the equations $6x - 2y + 9 = 0$ and $3x - y + 12 = 0$ are two lines which are
(a) coincident
(b) parallel
(c) intersecting exactly at one point
(d) perpendicular to each other
26. The graphs of the equations $2x + 3y - 2 = 0$ and $x - 2y - 8 = 0$ are two lines which are
(a) coincident
(b) parallel
(c) intersecting exactly at one point
(d) perpendicular to each other
27. The graphs of the equations $5x - 15y = 8$ and $3x - 9y = \frac{24}{5}$ are two lines which are
(a) coincident
(b) parallel
(c) intersecting exactly at one point
(d) perpendicular to each other

ANSWERS (MCQ)

1. (c) 2. (c) 3. (a) 4. (d) 5. (a) 6. (b) 7. (c) 8. (a) 9. (c)
10. (b) 11. (d) 12. (b) 13. (a) 14. (d) 15. (d) 16. (d) 17. (d) 18. (d)
19. (a) 20. (b) 21. (b) 22. (d) 23. (b) 24. (d) 25. (b) 26. (c) 27. (a)

HINTS TO SOME SELECTED QUESTIONS

3. $4x - 3y = -1$, $3x + 4y = 18$. Solve.

4. Putting $\frac{1}{x} = u$ and $\frac{1}{y} = v$, we get $u + 2v = 4$ and $-u + 3v = 11$.

5. $\frac{2x+y+2}{5} = \frac{3x-y+1}{3} \Rightarrow 6x+3y+6 = 15x-5y+5 \Rightarrow 9x-8y = 1$ (i)

$\frac{3x-y+1}{3} = \frac{3x+2y+1}{6} \Rightarrow 18x-6y+6 = 9x+6y+3 \Rightarrow 9x-12y = -3$ (ii)

Solve (i) and (ii) to get $x = 1$ and $y = 1$.

6. Put $\frac{1}{x+y} = u$ and $\frac{1}{x-y} = v$ to get $3u + 2v = 2$ and $9u - 4v = 1$.

Solve for u and v to get $u = \frac{1}{3}$ and $v = \frac{1}{2}$.

$\therefore x + y = 3$ and $x - y = 2$.

7. Divide throughout by xy and put $\frac{1}{x} = u$ and $\frac{1}{y} = v$ to get

$4v + 6u = 3$... (i) and $8v + 9u = 5$ (ii)

This gives $u = \frac{1}{3}$ and $v = \frac{1}{4}$. Hence, $x = 3$ and $y = 4$.

8. Adding (i) and (ii), we get $66(x + y) = 198 \Rightarrow x + y = 3$.

Subtracting (ii) from (i), we get $8(y - x) = 8 \Rightarrow y - x = 1$.

9. $2^{x+y} = 2^{x-y} = 2^{3/2} \Rightarrow x + y = \frac{3}{2}$ and $x - y = \frac{3}{2}$. So, $y = 0$.

10. Put $\frac{1}{x} = u$ and $\frac{1}{y} = v$. Then, $2u + 3v = 6$... (i) and $u + \frac{1}{2}v = 2 \Rightarrow 2u + v = 4$ (ii)

11. For a unique solution, we have $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

$\therefore \frac{k}{6} \neq \frac{-1}{-2} \Rightarrow k \neq 3$.

12. For a unique solution, we have $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. So, $\frac{1}{3} \neq \frac{-2}{k} \Rightarrow k \neq -6$.

13. For no solution, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$\therefore \frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = 10$ and $k \neq \frac{-14}{3}$. Hence, $k = 10$.

14. For parallel lines, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$\therefore \frac{3}{2} = \frac{2k}{5} \neq \frac{-2}{1} \Rightarrow k = \frac{15}{4}$ and $k \neq -5 \Rightarrow k = \frac{15}{4}$.

15. For a unique intersecting point, we have $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

$\therefore \frac{k}{3} \neq \frac{-2}{1} \Rightarrow k \neq -6$. Hence, correct answer is (d).

16. Here, $\frac{a_1}{a_2} = \frac{1}{-3} = \frac{-1}{3}$, $\frac{b_1}{b_2} = \frac{2}{-6} = \frac{-1}{3}$ and $\frac{c_1}{c_2} = \frac{5}{1}$.

$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. So, the given system has no solution.

17. Here, $\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}$ and $\frac{c_1}{c_2} = \frac{-5}{-15} = \frac{1}{3}$.

$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. So, the given system has no solution.

18. If a pair of linear equations is consistent then their graph lines will be intersecting or coincident.

20. Let $C = 3B = 2(A + B) = x^\circ$.

Then, $C = x^\circ$, $B = \left(\frac{x}{3}\right)^\circ$ and $(A + B) = \left(\frac{x}{2}\right)^\circ$.

$(A + B) + C = 180^\circ \Rightarrow \frac{x}{2} + x = 180 \Rightarrow 3x = 360 \Rightarrow x = 120$.

$\therefore \angle B = \left(\frac{120}{3}\right)^\circ = 40^\circ$.

21. $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$.

This gives $x + y = 100$ and $x + 2y = 160$.

22. Let the ten's digit be x and the unit's digit be y . Then,

$x + y = 15 \quad \dots (i) \quad \text{and} \quad (10y + x) - (10x + y) = 9 \Rightarrow y - x = 1. \quad \dots (ii)$

23. Let the required fraction be $\frac{x}{y}$. Then,

$\frac{x-1}{y+2} = \frac{1}{2} \Rightarrow 2x - 2 = y + 2 \Rightarrow 2x - y = 4 \quad \dots (i)$

$\frac{x-7}{y-2} = \frac{1}{3} \Rightarrow 3x - 21 = y - 2 \Rightarrow 3x - y = 19. \quad \dots (ii)$

24. Let the present ages of the man and his son be x years and y years respectively. Then,

$(x + 5) = 3(y + 5) \Rightarrow x - 3y = 10 \quad \dots (i)$

$(x - 5) = 7(y - 5) \Rightarrow x - 7y = -30. \quad \dots (ii)$

25. $\frac{a_1}{a_2} = \frac{6}{3} = \frac{2}{1}$, $\frac{b_1}{b_2} = \frac{-2}{-1} = \frac{2}{1}$ and $\frac{c_1}{c_2} = \frac{9}{12} = \frac{3}{4}$.

$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

So, the system is inconsistent and hence the lines are parallel.

26. $\frac{a_1}{a_2} = \frac{2}{1}$, $\frac{b_1}{b_2} = \frac{3}{-2}$. So, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

Thus, the system has a unique solution and therefore the lines intersect exactly at one point.

27. The equations are $5x - 15y - 8 = 0$ and $15x - 45y - 24 = 0$.

$\therefore \frac{a_1}{a_2} = \frac{5}{15} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{-15}{-45} = \frac{1}{3}$ and $\frac{c_1}{c_2} = \frac{-8}{-24} = \frac{1}{3}$

$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ and therefore the system has infinitely many solutions.

Hence, the graph lines are coincident.

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TEST YOURSELF*MCQ*

- The graphic representation of the equations $x + 2y = 3$ and $2x + 4y + 7 = 0$ gives a pair of
 - parallel lines
 - intersecting lines
 - coincident lines
 - none of these
- If $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have an infinite number of solutions then
 - $a = 5, b = 1$
 - $a = -5, b = 1$
 - $a = 5, b = -1$
 - $a = -5, b = -1$
- The pair of equations $2x + y = 5$, $3x + 2y = 8$ has
 - a unique solution
 - two solutions
 - no solution
 - infinitely many solutions
- If $x = -y$ and $y > 0$, which of the following is wrong?
 - $x^2y > 0$
 - $x + y = 0$
 - $xy < 0$
 - $\frac{1}{x} - \frac{1}{y} = 0$

Short-Answer Questions

- Show that the system of equations $-x + 2y + 2 = 0$ and $\frac{1}{2}x - \frac{1}{2}y - 1 = 0$ has a unique solution.
- For what values of k is the system of equations $kx + 3y = k - 2$, $12x + ky = k$ inconsistent?
- Show that the equations $9x - 10y = 21$, $\frac{3x}{2} - \frac{5y}{3} = \frac{7}{2}$ have infinitely many solutions.
- Solve the system of equations $x - 2y = 0$, $3x + 4y = 20$.
- Show that the paths represented by the equations $x - 3y = 2$ and $-2x + 6y = 5$ are parallel.
- The difference between two numbers is 26 and one number is three times the other. Find the numbers.
- Solve: $23x + 29y = 98$, $29x + 23y = 110$.
- Solve: $6x + 3y = 7xy$ and $3x + 9y = 11xy$.
- Find the value of k for which the system of equations $3x + y = 1$ and $kx + 2y = 5$ has (i) a unique solution, (ii) no solution.
- In a $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Find the measure of each one of $\angle A$, $\angle B$ and $\angle C$.

15. 5 pencils and 7 pens together cost ₹ 195 while 7 pencils and 5 pens together cost ₹ 153. Find the cost of each one of the pencil and the pen.
16. Solve the following system of equations graphically:

$$2x - 3y = 1, 4x - 3y + 1 = 0.$$

Long-Answer Questions

17. Find the angles of a cyclic quadrilateral $ABCD$ in which $\angle A = (4x + 20)^\circ$, $\angle B = (3x - 5)^\circ$, $\angle C = (4y)^\circ$ and $\angle D = (7y + 5)^\circ$.
18. Solve for x and y : $\frac{35}{x+y} + \frac{14}{x-y} = 19$, $\frac{14}{x+y} + \frac{35}{x-y} = 37$.
19. If 1 is added to both the numerator and the denominator of a fraction, it becomes $\frac{4}{5}$. If, however, 5 is subtracted from both the numerator and the denominator, the fraction becomes $\frac{1}{2}$. Find the fraction.
20. Solve: $\frac{ax}{b} - \frac{by}{a} = a + b$, $ax - by = 2ab$.

ANSWERS (TEST YOURSELF)

1. (a) 2. (d) 3. (a) 4. (d) 5. $k = \pm 6$ 6. $x = 4, y = 2$
10. 39, 13 11. $x = 3, y = 1$ 12. $x = 1, y = \frac{3}{2}$ 13. (i) $k \neq 6$ (ii) $k = 6$
14. $\angle A = 20^\circ, \angle B = 40^\circ, \angle C = 120^\circ$ 15. ₹ 4 per pencil, ₹ 25 per pen
16. $x = -1, y = -1$ 17. $\angle A = 120^\circ, \angle B = 70^\circ, \angle C = 60^\circ, \angle D = 110^\circ$
18. $x = 4, y = 3$ 19. $\frac{7}{9}$ 20. $x = b, y = -a$



HISTORY

Babylonians were the first to solve the quadratic equations of the form $x^2 - px + q = 0$.

Brahmagupta (AD 598–665), an Indian mathematician gave an explicit formula to solve a quadratic equation of the form $ax^2 - bx = c$.

An Arab mathematician, Al-Khwarizmi in AD 800 also studied quadratic equations of various types.

An ancient Indian mathematician *Shridharacharya* derived the well-known *Quadratic Formula* for solving a quadratic equation $ax^2 + bx + c = 0$ by the method of completing the square. This is being used as the standard formula for solving such an equation.

According to this formula, the roots of $ax^2 + bx + c = 0$ are given by $\alpha = \frac{-b + \sqrt{D}}{2a}$ and $\beta = \frac{-b - \sqrt{D}}{2a}$, where $D = (b^2 - 4ac)$ is called the *discriminant* of this equation.

QUADRATIC EQUATION A quadratic equation in the variable x is an equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$.

ROOTS OF A QUADRATIC EQUATION A real number α is called a root of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ if $a\alpha^2 + b\alpha + c = 0$.

NOTE 1. If α is a root of $ax^2 + bx + c = 0$ then we say that

(i) $x = \alpha$ satisfies the equation $ax^2 + bx + c = 0$

or (ii) $x = \alpha$ is a solution of the equation $ax^2 + bx + c = 0$.

NOTE 2. The roots of a quadratic equation $ax^2 + bx + c = 0$ are called the *zeros* of the polynomial $ax^2 + bx + c$.

SOLVING A QUADRATIC EQUATION Solving a quadratic equation means finding its roots.

SOLVED EXAMPLES**EXAMPLE 1** Which of the following are quadratic equations?

- (i) $x^2 - 5x + 3 = 0$ (ii) $2x^2 - 3\sqrt{2}x + 6 = 0$
 (iii) $3x^2 - 2\sqrt{x} + 8 = 0$ (iv) $2x^2 - 3 = 0$
 (v) $x + \frac{1}{x} = x^2$ (vi) $x^2 + \frac{1}{x^2} = 4\frac{1}{4}$

SOLUTION

- (i) Clearly, $(x^2 - 5x + 3)$ is a quadratic polynomial.
 $\therefore x^2 - 5x + 3 = 0$ is a quadratic equation.
- (ii) Clearly, $(2x^2 - 3\sqrt{2}x + 6)$ is a quadratic polynomial.
 $\therefore 2x^2 - 3\sqrt{2}x + 6 = 0$ is a quadratic equation.
- (iii) $3x^2 - 2\sqrt{x} + 8$ is not of the form $ax^2 + bx + c = 0$.
 $\therefore 3x^2 - 2\sqrt{x} + 8 = 0$ is not a quadratic equation.
- (iv) $2x^2 - 3 = 0$ is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = 0$ and $c = -3$.
 $\therefore 2x^2 - 3 = 0$ is a quadratic equation.
- (v) $x + \frac{1}{x} = x^2 \Rightarrow x^2 + 1 = x^3 \Rightarrow x^3 - x^2 - 1 = 0$.
 And, $(x^3 - x^2 - 1)$ being a polynomial of degree 3, it is not quadratic.
 Hence, $x + \frac{1}{x} = x^2$ is not a quadratic equation.
- (vi) $x^2 + \frac{1}{x^2} = \frac{17}{4} \Rightarrow 4x^4 + 4 = 17x^2 \Rightarrow 4x^4 - 17x^2 + 4 = 0$.
 Clearly, $4x^4 - 17x^2 + 4$ is a polynomial of degree 4.
 $\therefore x^2 + \frac{1}{x^2} = \frac{17}{4}$ is not a quadratic equation.

EXAMPLE 2 Check whether the following are quadratic equations:

- (i) $(2x - 1)(x - 3) = (x + 4)(x - 2)$ (ii) $(x + 2)^3 = 2x(x^2 - 1)$
 (iii) $(x + 1)^3 = x^3 + x + 6$ (iv) $x(x + 3) + 6 = (x + 2)(x - 2)$

SOLUTION

We have

- (i) $(2x - 1)(x - 3) = (x + 4)(x - 2)$
 $\Rightarrow 2x^2 - 7x + 3 = x^2 + 2x - 8 \Rightarrow x^2 - 9x + 11 = 0$.
 This is of the form $ax^2 + bx + c = 0$, where $a = 1$, $b = -9$ and $c = 11$.
 Hence, the given equation is a quadratic equation.

$$(ii) (x+2)^3 = 2x(x^2-1)$$

$$\Rightarrow x^3 + 8 + 6x(x+2) = 2x^3 - 2x$$

$$\Rightarrow x^3 + 6x^2 + 12x + 8 = 2x^3 - 2x$$

$$\Rightarrow x^3 - 6x^2 - 14x - 8 = 0.$$

This is not of the form $ax^2 + bx + c = 0$.

Hence, the given equation is not a quadratic equation.

$$(iii) (x+1)^3 = x^3 + x + 6$$

$$\Rightarrow x^3 + 1 + 3x(x+1) = x^3 + x + 6$$

$$\Rightarrow 3x^2 + 2x - 5 = 0.$$

This is of the form $ax^2 + bx + c = 0$, where $a = 3$, $b = 2$ and $c = -5$.

Hence, the given equation is a quadratic equation.

$$(iv) x(x+3) + 6 = (x+2)(x-2)$$

$$\Rightarrow x^2 + 3x + 6 = x^2 - 4$$

$$\Rightarrow 3x + 10 = 0.$$

This is not of the form $ax^2 + bx + c = 0$.

Hence, the given equation is not a quadratic equation.

EXAMPLE 3 For the quadratic equation $2x^2 - 5x - 3 = 0$, show that

(i) $x = 3$ is its solution. (ii) $x = \frac{-1}{2}$ is its solution.

(iii) $x = 4$ is not its solution.

SOLUTION The given equation is $2x^2 - 5x - 3 = 0$.

(i) On substituting $x = 3$ in the given equation, we get

$$\text{LHS} = 2 \times 3^2 - 5 \times 3 - 3 = (18 - 15 - 3) = 0 = \text{RHS.}$$

$\therefore x = 3$ is a solution of $2x^2 - 5x - 3 = 0$.

(ii) On substituting $x = \frac{-1}{2}$ in the given equation, we get

$$\text{LHS} = 2 \times \left(\frac{-1}{2}\right)^2 - 5 \times \left(\frac{-1}{2}\right) - 3$$

$$= \left\{ 2 \times \frac{1}{4} + 5 \times \frac{1}{2} - 3 \right\}$$

$$= \left\{ \frac{1}{2} + \frac{5}{2} - 3 \right\} = 0 = \text{RHS.}$$

$\therefore x = \frac{-1}{2}$ is a solution of $2x^2 - 5x - 3 = 0$.

(iii) On substituting $x = 4$ in the given equation, we get

$$\text{LHS} = 2 \times 4^2 - 5 \times 4 - 3 = (32 - 20 - 3) = 9 \neq 0.$$

Thus, $\text{LHS} \neq \text{RHS}$.

$\therefore x = 4$ is not a solution of $2x^2 - 5x - 3 = 0$.

EXAMPLE 4 Show that $\sqrt{2}$ and $-2\sqrt{2}$ are the roots of the equation $x^2 + \sqrt{2}x - 4 = 0$.

SOLUTION The given equation is $x^2 + \sqrt{2}x - 4 = 0$.

Putting $x = \sqrt{2}$ in the given equation, we get

$$\text{LHS} = (\sqrt{2})^2 + (\sqrt{2} \times \sqrt{2}) - 4 = (2 + 2 - 4) = 0 = \text{RHS}.$$

$\therefore \sqrt{2}$ is a root of the given equation.

Putting $x = -2\sqrt{2}$ in the given equation, we get

$$\text{LHS} = (-2\sqrt{2})^2 + \sqrt{2} \times (-2\sqrt{2}) - 4 = (8 - 4 - 4) = 0 = \text{RHS}.$$

$\therefore -2\sqrt{2}$ is a root of the given equation.

Hence, $\sqrt{2}$ and $-2\sqrt{2}$ are the roots of the given equation.

SOLVING A QUADRATIC EQUATION BY FACTORISATION

Let the given quadratic equation be $ax^2 + bx + c = 0$, where $a \neq 0$. Let $(ax^2 + bx + c)$ be expressible as the product of two linear expressions, say $(px + q)$ and $(rx + s)$, where p, q, r, s are real numbers such that $p \neq 0$ and $r \neq 0$. Then,

$$\begin{aligned} ax^2 + bx + c = 0 &\Rightarrow (px + q)(rx + s) = 0 \\ &\Rightarrow (px + q) = 0 \quad \text{or} \quad (rx + s) = 0 \\ &\Rightarrow x = -\frac{q}{p} \quad \text{or} \quad x = -\frac{s}{r}. \end{aligned}$$

EXAMPLE 5 Solve: $(x + 2)(3x - 5) = 0$.

SOLUTION We have

$$\begin{aligned} (x + 2)(3x - 5) = 0 &\Rightarrow x + 2 = 0 \quad \text{or} \quad 3x - 5 = 0 \\ &\Rightarrow x = -2 \quad \text{or} \quad x = \frac{5}{3}. \end{aligned}$$

Hence, the roots of the given equation are -2 and $\frac{5}{3}$.

EXAMPLE 6 Solve: $5x^2 - 8x = 0$.

SOLUTION We have

$$\begin{aligned} 5x^2 - 8x = 0 &\Rightarrow x(5x - 8) = 0 \Rightarrow x = 0 \quad \text{or} \quad 5x - 8 = 0 \\ &\Rightarrow x = 0 \quad \text{or} \quad x = \frac{8}{5}. \end{aligned}$$

Hence, 0 and $\frac{8}{5}$ are the roots of the given equation.

REMARK In Class IX, we have learnt how to factorise a quadratic polynomial by splitting the middle term. We shall use it for finding the roots of a quadratic equation.

EXAMPLE 7 Solve: $6x^2 - x - 2 = 0$ by the factorisation method.

SOLUTION We write, $-x = -4x + 3x$ as $(-4x) \times 3x = -12x^2 = 6x^2 \times (-2)$.

$$\therefore 6x^2 - x - 2 = 0$$

$$\Rightarrow 6x^2 - 4x + 3x - 2 = 0 \Rightarrow 2x(3x - 2) + (3x - 2) = 0$$

$$\Rightarrow (3x - 2)(2x + 1) = 0 \Rightarrow 3x - 2 = 0 \text{ or } 2x + 1 = 0$$

$$\Rightarrow x = \frac{2}{3} \text{ or } x = -\frac{1}{2}.$$

Hence, $\frac{2}{3}$ and $-\frac{1}{2}$ are the roots of the given equation.

EXAMPLE 8 Solve: $8x^2 - 22x - 21 = 0$ by the factorisation method.

SOLUTION We write,

$$-22x = -28x + 6x \text{ as } 8x^2 \times (-21) = -168x^2 = (-28x) \times 6x.$$

$$\therefore 8x^2 - 22x - 21 = 0$$

$$\Rightarrow 8x^2 - 28x + 6x - 21 = 0 \Rightarrow 4x(2x - 7) + 3(2x - 7) = 0$$

$$\Rightarrow (2x - 7)(4x + 3) = 0 \Rightarrow 2x - 7 = 0 \text{ or } 4x + 3 = 0$$

$$\Rightarrow x = \frac{7}{2} \text{ or } x = -\frac{3}{4}.$$

Hence, $\frac{7}{2}$ and $-\frac{3}{4}$ are the roots of the given equation.

EXAMPLE 9 Solve: $6x^2 + 40 = 31x$.

SOLUTION The given equation may be written as $6x^2 - 31x + 40 = 0$.

We write,

$$-31x = -16x - 15x \text{ as } 6x^2 \times 40 = 240x^2 = (-16x) \times (-15x).$$

$$\therefore 6x^2 - 31x + 40 = 0$$

$$\Rightarrow 6x^2 - 16x - 15x + 40 = 0 \Rightarrow 2x(3x - 8) - 5(3x - 8) = 0$$

$$\Rightarrow (3x - 8)(2x - 5) = 0 \Rightarrow 3x - 8 = 0 \text{ or } 2x - 5 = 0$$

$$\Rightarrow x = \frac{8}{3} \text{ or } x = \frac{5}{2}.$$

Hence, $\frac{8}{3}$ and $\frac{5}{2}$ are the roots of the given equation.

EXAMPLE 10 Solve: $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$. [CBSE 2013]

SOLUTION Here, $4\sqrt{3} \times (-2\sqrt{3}) = -24$, $8 \times (-3) = -24$ and $8 + (-3) = 5$.
 $\therefore 4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0 \Rightarrow 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} = 0$
 $\Rightarrow 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2) = 0$
 $\Rightarrow (\sqrt{3}x + 2)(4x - \sqrt{3}) = 0 \Rightarrow \sqrt{3}x + 2 = 0$ or $4x - \sqrt{3} = 0$
 $\Rightarrow x = \frac{-2}{\sqrt{3}}$ or $x = \frac{\sqrt{3}}{4}$
 $\Rightarrow x = \left(\frac{-2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{-2\sqrt{3}}{3}$ or $x = \frac{\sqrt{3}}{4}$.

EXAMPLE 11 Solve: $4x^2 - 12x + 9 = 0$.

SOLUTION Here, $4 \times 9 = 36$, $(-6) \times (-6) = 36$, and $(-6) + (-6) = -12$.
 $\therefore 4x^2 - 12x + 9 = 0$
 $\Rightarrow 4x^2 - 6x - 6x + 9 = 0 \Rightarrow 2x(2x - 3) - 3(2x - 3) = 0$
 $\Rightarrow (2x - 3)(2x - 3) = 0 \Rightarrow (2x - 3)^2 = 0$
 $\Rightarrow 2x - 3 = 0 \Rightarrow x = \frac{3}{2}$.

Hence, $x = \frac{3}{2}$ is the repeated root of the given equation.

EXAMPLE 12 Solve the following equation by using factorisation method:

$$4x^2 - 4ax + (a^2 - b^2) = 0. \quad [\text{CBSE 2012, '15}]$$

SOLUTION We may write, $-4a = \{-2(a + b)\} + \{-2(a - b)\}$.

$$\text{Also, } \{-2(a + b)\} \times \{-2(a - b)\} = 4(a^2 - b^2).$$

$$\begin{aligned} \therefore 4x^2 - 4ax + (a^2 - b^2) &= 0 \\ \Rightarrow 4x^2 - 2(a + b)x - 2(a - b)x + (a^2 - b^2) &= 0 \\ \Rightarrow 2x\{2x - (a + b)\} - (a - b)\{2x - (a + b)\} &= 0 \\ \Rightarrow \{2x - (a + b)\} \times \{2x - (a - b)\} &= 0 \\ \Rightarrow 2x - (a + b) = 0 \text{ or } 2x - (a - b) &= 0 \\ \Rightarrow x = \frac{(a + b)}{2} \text{ or } x = \frac{(a - b)}{2}. \end{aligned}$$

Hence, $\frac{(a + b)}{2}$ and $\frac{(a - b)}{2}$ are the roots of the given equation.

EXAMPLE 13 Solve the following equation by using factorisation method:

$$9x^2 - 6b^2x - (a^4 - b^4) = 0. \quad [\text{CBSE 2015}]$$

SOLUTION We may write, $-6b^2 = 3(a^2 - b^2) - 3(a^2 + b^2)$.

$$\text{Also, } \{3(a^2 - b^2)\} \times \{-3(a^2 + b^2)\} = -9(a^4 - b^4)$$

$$\therefore 9x^2 - 6b^2x - (a^4 - b^4) = 0$$

$$\Rightarrow 9x^2 + 3(a^2 - b^2)x - 3(a^2 + b^2)x - (a^4 - b^4) = 0$$

$$\Rightarrow 3x\{3x + (a^2 - b^2)\} - (a^2 + b^2)\{3x + (a^2 - b^2)\} = 0$$

$$\Rightarrow \{3x + (a^2 - b^2)\} \times \{3x - (a^2 + b^2)\} = 0$$

$$\Rightarrow 3x + (a^2 - b^2) = 0 \text{ or } 3x - (a^2 + b^2) = 0$$

$$\Rightarrow x = \frac{(b^2 - a^2)}{3} \text{ or } x = \frac{(a^2 + b^2)}{3}.$$

Hence, $\frac{(b^2 - a^2)}{3}$ and $\frac{(a^2 + b^2)}{3}$ are the required roots of the given equation.

EXAMPLE 14 Solve: $\frac{x+3}{x+2} = \frac{3x-7}{2x-3}$.

SOLUTION By cross multiplication, we get

$$(x+3)(2x-3) = (x+2)(3x-7)$$

$$\Rightarrow 2x^2 + 3x - 9 = 3x^2 - x - 14 \Rightarrow x^2 - 4x - 5 = 0$$

$$\Rightarrow x^2 - 5x + x - 5 = 0 \Rightarrow x(x-5) + (x-5) = 0$$

$$\Rightarrow (x-5)(x+1) = 0 \Rightarrow x-5 = 0 \text{ or } x+1 = 0$$

$$\Rightarrow x = 5 \text{ or } x = -1.$$

Hence, 5 and -1 are the roots of the given equation.

EXAMPLE 15 Solve: $\frac{14}{x+3} - 1 = \frac{5}{x+1}$, $x \neq -3, -1$. [CBSE 2014]

SOLUTION The given equation may be written as

$$\frac{14}{x+3} - \frac{5}{x+1} = 1$$

$$\Rightarrow \frac{14(x+1) - 5(x+3)}{(x+3)(x+1)} = 1$$

$$\Rightarrow \frac{(9x-1)}{x^2+4x+3} = 1 \Rightarrow x^2+4x+3 = 9x-1$$

$$\Rightarrow x^2 - 5x + 4 = 0 \Rightarrow x^2 - 4x - x + 4 = 0$$

$$\Rightarrow x(x-4) - (x-4) = 0 \Rightarrow (x-4)(x-1) = 0$$

$$\Rightarrow x-4 = 0 \text{ or } x-1 = 0$$

$$\Rightarrow x = 4 \text{ or } x = 1.$$

Hence, 4 and 1 are the roots of the given equation.

EXAMPLE 16 Solve: $\frac{1}{(x+4)} - \frac{1}{(x-7)} = \frac{11}{30}$, $x \neq -4, 7$. [CBSE 2008, '10]

SOLUTION We have

$$\begin{aligned}\frac{1}{(x+4)} - \frac{1}{(x-7)} &= \frac{11}{30} \\ \Rightarrow \frac{(x-7) - (x+4)}{(x+4)(x-7)} &= \frac{11}{30} \Rightarrow \frac{-11}{(x^2-3x-28)} = \frac{11}{30} \\ \Rightarrow \frac{-1}{(x^2-3x-28)} &= \frac{1}{30} \quad [\text{on dividing both sides by } 11] \\ \Rightarrow (x^2-3x-28) &= -30 \quad [\text{by cross multiplication}] \\ \Rightarrow x^2-3x+2 &= 0 \Rightarrow x^2-2x-x+2 = 0 \\ \Rightarrow x(x-2) - (x-2) &= 0 \Rightarrow (x-2)(x-1) = 0 \\ \Rightarrow x-2 &= 0 \text{ or } x-1 = 0 \\ \Rightarrow x &= 2 \text{ or } x = 1.\end{aligned}$$

Hence, 2 and 1 are the roots of the given equation.

EXAMPLE 17 Solve: $\frac{1}{(a+b+x)} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$, $[x \neq 0, x \neq -(a+b)]$. [CBSE 2013]

SOLUTION We have

$$\begin{aligned}\frac{1}{(a+b+x)} &= \frac{1}{a} + \frac{1}{b} + \frac{1}{x} \\ \Rightarrow \frac{1}{(a+b+x)} - \frac{1}{x} &= \frac{1}{a} + \frac{1}{b} \Rightarrow \frac{x - (a+b+x)}{x(a+b+x)} = \frac{b+a}{ab} \\ \Rightarrow \frac{-(a+b)}{x(a+b+x)} &= \frac{(a+b)}{ab} \\ \Rightarrow \frac{-1}{x(a+b+x)} &= \frac{1}{ab} \quad [\text{on dividing both sides by } (a+b)] \\ \Rightarrow x(a+b+x) &= -ab \quad [\text{by cross multiplication}] \\ \Rightarrow x^2+ax+bx+ab &= 0 \Rightarrow x(x+a)+b(x+a) = 0 \\ \Rightarrow (x+a)(x+b) &= 0 \Rightarrow x+a = 0 \text{ or } x+b = 0 \\ \Rightarrow x &= -a \text{ or } x = -b.\end{aligned}$$

Hence, $-a$ and $-b$ are the roots of the given equation.

EXAMPLE 18 Solve: $\frac{x-2}{x-3} + \frac{x-4}{x-5} = 3\frac{1}{3}$, $x \neq 3, 5$. [CBSE 2014]

SOLUTION We have

$$\begin{aligned}\frac{x-2}{x-3} + \frac{x-4}{x-5} &= \frac{10}{3} \\ \Rightarrow \frac{(x-2)(x-5) + (x-4)(x-3)}{(x-3)(x-5)} &= \frac{10}{3}\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{(x^2 - 7x + 10) + (x^2 - 7x + 12)}{(x^2 - 8x + 15)} = \frac{10}{3} \\
&\Rightarrow \frac{(2x^2 - 14x + 22)}{(x^2 - 8x + 15)} = \frac{10}{3} \\
&\Rightarrow 3(2x^2 - 14x + 22) = 10(x^2 - 8x + 15) \quad [\text{by cross multiplication}] \\
&\Rightarrow 6x^2 - 42x + 66 = 10x^2 - 80x + 150 \\
&\Rightarrow 4x^2 - 38x + 84 = 0 \Rightarrow 2x^2 - 19x + 42 = 0 \\
&\Rightarrow 2x^2 - 12x - 7x + 42 = 0 \Rightarrow 2x(x - 6) - 7(x - 6) = 0 \\
&\Rightarrow (x - 6)(2x - 7) = 0 \Rightarrow x - 6 = 0 \text{ or } 2x - 7 = 0 \\
&\quad x = 6 \text{ or } x = \frac{7}{2}.
\end{aligned}$$

Hence, 6 and $\frac{7}{2}$ are the roots of the given equation.

EXAMPLE 19 Solve: $2\left(\frac{2x-1}{x+3}\right) - 3\left(\frac{x+3}{2x-1}\right) = 5, x \neq -3, \frac{1}{2}$. [CBSE 2014]

SOLUTION On putting $\frac{2x-1}{x+3} = y$, the given equation becomes

$$\begin{aligned}
2y - \frac{3}{y} = 5 &\Rightarrow 2y^2 - 3 = 5y \\
&\Rightarrow 2y^2 - 5y - 3 = 0 \Rightarrow 2y^2 - 6y + y - 3 = 0 \\
&\Rightarrow 2y(y - 3) + (y - 3) = 0 \Rightarrow (y - 3)(2y + 1) = 0 \\
&\Rightarrow y - 3 = 0 \text{ or } 2y + 1 = 0 \\
&\Rightarrow y = 3 \text{ or } y = -\frac{1}{2}.
\end{aligned}$$

Case I $y = 3 \Rightarrow \frac{2x-1}{x+3} = 3$
 $\Rightarrow 2x - 1 = 3(x + 3)$ [by cross multiplication]
 $\Rightarrow 2x - 1 = 3x + 9$
 $\Rightarrow x = -10$.

Case II $y = -\frac{1}{2} \Rightarrow \frac{2x-1}{x+3} = -\frac{1}{2}$
 $\Rightarrow 2(2x - 1) = -(x + 3)$
 $\Rightarrow 5x = -1$
 $\Rightarrow x = -\frac{1}{5}$.

Hence, -10 and $-\frac{1}{5}$ are the roots of the given equation.

EXAMPLE 20 Solve: $\frac{2}{(x+1)} + \frac{3}{2(x-2)} = \frac{23}{5x}$, $x \neq 0, -1, 2$. [CBSE 2015]

SOLUTION We have

$$\begin{aligned} \frac{2}{(x+1)} + \frac{3}{2(x-2)} &= \frac{23}{5x} \\ \Rightarrow \frac{4(x-2) + 3(x+1)}{2(x+1)(x-2)} &= \frac{23}{5x} \Rightarrow \frac{7x-5}{2(x^2-x-2)} = \frac{23}{5x} \\ \Rightarrow 5x(7x-5) &= 46(x^2-x-2) \quad [\text{by cross multiplication}] \\ \Rightarrow 35x^2 - 25x &= 46x^2 - 46x - 92 \\ \Rightarrow 11x^2 - 21x - 92 &= 0 \Rightarrow 11x^2 - 44x + 23x - 92 = 0 \\ \Rightarrow 11x(x-4) + 23(x-4) &= 0 \Rightarrow (x-4)(11x+23) = 0 \\ \Rightarrow x-4 = 0 \text{ or } 11x+23 &= 0 \\ \Rightarrow x = 4 \text{ or } x &= \frac{-23}{11}. \end{aligned}$$

Hence, 4 and $\frac{-23}{11}$ are the roots of the given equation.

EXAMPLE 21 Solve: $\frac{1}{(x+3)} + \frac{1}{(2x-1)} = \frac{11}{(7x+9)}$, $x \neq -3, \frac{1}{2}, \frac{-9}{7}$. [CBSE 2009C]

SOLUTION We have

$$\begin{aligned} \frac{1}{(x+3)} + \frac{1}{(2x-1)} &= \frac{11}{(7x+9)} \\ \Rightarrow \frac{(2x-1) + (x+3)}{(x+3)(2x-1)} &= \frac{11}{(7x+9)} \Rightarrow \frac{(3x+2)}{2x^2+5x-3} = \frac{11}{(7x+9)} \\ \Rightarrow (3x+2)(7x+9) &= 11(2x^2+5x-3) \quad [\text{by cross multiplication}] \\ \Rightarrow 21x^2 + 41x + 18 &= 22x^2 + 55x - 33 \\ \Rightarrow x^2 + 14x - 51 &= 0 \Rightarrow x^2 + 17x - 3x - 51 = 0 \\ \Rightarrow x(x+17) - 3(x+17) &= 0 \Rightarrow (x+17)(x-3) = 0 \\ \Rightarrow x+17 = 0 \text{ or } x-3 &= 0 \\ \Rightarrow x = -17 \text{ or } x &= 3. \end{aligned}$$

Hence, -17 and 3 are the roots of the given equation.

EXAMPLE 22 Solve: $5^{(x+1)} + 5^{(2-x)} = 5^3 + 1$.

SOLUTION We have

$$\begin{aligned} 5^{(x+1)} + 5^{(2-x)} &= 5^3 + 1 \\ \Rightarrow 5^x \cdot 5 + 5^2 \cdot 5^{-x} &= 126 \Rightarrow 5^x \cdot 5 + \frac{25}{5^x} = 126 \\ \Rightarrow 5y + \frac{25}{y} &= 126, \text{ where } 5^x = y \\ \Rightarrow 5y^2 - 126y + 25 &= 0 \Rightarrow 5y^2 - 125y - y + 25 = 0 \end{aligned}$$

$$\Rightarrow 5y(y-25) - (y-25) = 0 \Rightarrow (y-25)(5y-1) = 0$$

$$\Rightarrow y-25 = 0 \text{ or } 5y-1 = 0$$

$$\Rightarrow y = 25 \text{ or } y = \frac{1}{5}$$

$$\Rightarrow 5^x = 25 = 5^2 \text{ or } 5^x = 5^{-1}$$

$$\Rightarrow x = 2 \text{ or } x = -1.$$

Hence, 2 and -1 are the roots of the given equation.

EXERCISE 4A

1. Which of the following are quadratic equations in x ?

(i) $x^2 - x + 3 = 0$

(ii) $2x^2 + \frac{5}{2}x - \sqrt{3} = 0$

(iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

(iv) $\frac{1}{3}x^2 + \frac{1}{5}x - 2 = 0$

(v) $x^2 - 3x - \sqrt{x} + 4 = 0$

(vi) $x - \frac{6}{x} = 3$

(vii) $x + \frac{2}{x} = x^2$

(viii) $x^2 - \frac{1}{x^2} = 5$

(ix) $(x+2)^3 = x^3 - 8$

(x) $(2x+3)(3x+2) = 6(x-1)(x-2)$

(xi) $\left(x + \frac{1}{x}\right)^2 = 2\left(x + \frac{1}{x}\right) + 3$

2. Which of the following are the roots of $3x^2 + 2x - 1 = 0$?

(i) -1

(ii) $\frac{1}{3}$

(iii) $-\frac{1}{2}$

3. (i) Find the value of k for which $x=1$ is a root of the equation $x^2 + kx + 3 = 0$. Also, find the other root.

(ii) Find the values of a and b for which $x = \frac{3}{4}$ and $x = -2$ are the roots of the equation $ax^2 + bx - 6 = 0$.

4. Show that $x = -\frac{bc}{ad}$ is a solution of the quadratic equation

$$ad^2\left(\frac{ax}{b} + \frac{2c}{d}\right)x + bc^2 = 0.$$

[CBSE 2017]

Solve each of the following quadratic equations:

5. $(2x-3)(3x+1) = 0$

6. $4x^2 + 5x = 0$

7. $3x^2 - 243 = 0$

8. $2x^2 + x - 6 = 0$

9. $x^2 + 6x + 5 = 0$

10. $9x^2 - 3x - 2 = 0$

11. $x^2 + 12x + 35 = 0$

12. $x^2 = 18x - 77$

13. $6x^2 + 11x + 3 = 0$
14. $6x^2 + x - 12 = 0$
15. $3x^2 - 2x - 1 = 0$
16. $4x^2 - 9x = 100$
17. $15x^2 - 28 = x$
18. $4 - 11x = 3x^2$
19. $48x^2 - 13x - 1 = 0$
20. $x^2 + 2\sqrt{2}x - 6 = 0$
21. $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$ [CBSE 2017]
22. $\sqrt{3}x^2 + 11x + 6\sqrt{3} = 0$
23. $3\sqrt{7}x^2 + 4x - \sqrt{7} = 0$
24. $\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0$
25. $4\sqrt{6}x^2 - 13x - 2\sqrt{6} = 0$
26. $3x^2 - 2\sqrt{6}x + 2 = 0$ [CBSE 2010, '12]
27. $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ [CBSE 2011]
28. $x^2 - 3\sqrt{5}x + 10 = 0$ [CBSE 2011]
29. $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$ [CBSE 2015]
30. $x^2 + 3\sqrt{3}x - 30 = 0$ [CBSE 2015]
31. $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ [CBSE 2013]
32. $5x^2 + 13x + 8 = 0$ [CBSE 2013C]
33. $x^2 - (1 + \sqrt{2})x + \sqrt{2} = 0$
34. $9x^2 + 6x + 1 = 0$
35. $100x^2 - 20x + 1 = 0$
36. $2x^2 - x + \frac{1}{8} = 0$
37. $10x - \frac{1}{x} = 3$
38. $\frac{2}{x^2} - \frac{5}{x} + 2 = 0$
39. $2x^2 + ax - a^2 = 0$ [CBSE 2015]
40. $4x^2 + 4bx - (a^2 - b^2) = 0$ [CBSE 2015, '17]
41. $4x^2 - 4a^2x + (a^4 - b^4) = 0$ [CBSE 2015]
42. $x^2 + 5x - (a^2 + a - 6) = 0$ [CBSE 2015]
43. $x^2 - 2ax - (4b^2 - a^2) = 0$ [CBSE 2015]
44. $x^2 - (2b - 1)x + (b^2 - b - 20) = 0$ [CBSE 2015]
45. $x^2 + 6x - (a^2 + 2a - 8) = 0$ [CBSE 2015]
46. $abx^2 + (b^2 - ac)x - bc = 0$ [CBSE 2014]
47. $x^2 - 4ax - b^2 + 4a^2 = 0$ [CBSE 2012]
48. $4x^2 - 2(a^2 + b^2)x + a^2b^2 = 0$
49. $12abx^2 - (9a^2 - 8b^2)x - 6ab = 0$ [CBSE 2006]
50. $a^2b^2x^2 + b^2x - a^2x - 1 = 0$ [CBSE 2005]
51. $9x^2 - 9(a + b)x + (2a^2 + 5ab + 2b^2) = 0$ [CBSE 2009]
52. $\frac{16}{x} - 1 = \frac{15}{x+1}, x \neq 0, -1$ [CBSE 2014]
53. $\frac{4}{x} - 3 = \frac{5}{2x+3}, x \neq 0, \frac{-3}{2}$ [CBSE 2014]
54. $\frac{3}{x+1} - \frac{1}{2} = \frac{2}{3x-1}, x \neq -1, \frac{1}{3}$ [CBSE 2014]

55. (i) $\frac{1}{x-1} - \frac{1}{x+5} = \frac{6}{7}, x \neq 1, -5$
 (ii) $\frac{1}{2x-3} + \frac{1}{x-5} = 1\frac{1}{9}, x \neq \frac{3}{2}, 5$ [CBSE 2017]
56. $\frac{1}{2a+b+2x} = \frac{1}{2a} + \frac{1}{b} + \frac{1}{2x}$ [CBSE 2013]
57. $\frac{x+3}{x-2} - \frac{1-x}{x} = 4\frac{1}{4}, x \neq 2, 0$
58. $\frac{3x-4}{7} + \frac{7}{3x-4} = \frac{5}{2}, x \neq \frac{4}{3}$ [CBSE 2010]
59. (i) $\frac{x}{x-1} + \frac{x-1}{x} = 4\frac{1}{4}, x \neq 0, 1$
 (ii) $\frac{x-1}{2x+1} + \frac{2x+1}{x-1} = 2, x \neq -\frac{1}{2}, 1$ [CBSE 2017]
60. $\frac{x}{x+1} + \frac{x+1}{x} = 2\frac{4}{15}, x \neq 0, -1$ [CBSE 2014]
61. $\frac{x-4}{x-5} + \frac{x-6}{x-7} = 3\frac{1}{3}, x \neq 5, 7$ [CBSE 2014]
62. $\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}, x \neq 2, 4$ [CBSE 2017]
63. $\frac{1}{x-2} + \frac{2}{x-1} = \frac{6}{x}, x \neq 0, 1, 2$ [CBSE 2013]
64. (i) $\frac{1}{x+1} + \frac{2}{x+2} = \frac{5}{x+4}, x \neq -1, -2, -4$ [CBSE 2013C]
 (ii) $\frac{1}{x+1} + \frac{3}{5x+1} = \frac{5}{x+4}, x \neq -1, -\frac{1}{5}, -4$ [CBSE 2017]
65. $3\left(\frac{3x-1}{2x+3}\right) - 2\left(\frac{2x+3}{3x-1}\right) = 5, x \neq \frac{1}{3}, \frac{-3}{2}$ [CBSE 2014]
66. $3\left(\frac{7x+1}{5x-3}\right) - 4\left(\frac{5x-3}{7x+1}\right) = 11, x \neq \frac{3}{5}, \frac{-1}{7}$ [CBSE 2014]
67. $\left(\frac{4x-3}{2x+1}\right) - 10\left(\frac{2x+1}{4x-3}\right) = 3, x \neq \frac{-1}{2}, \frac{3}{4}$
68. $\left(\frac{x}{x+1}\right)^2 - 5\left(\frac{x}{x+1}\right) + 6 = 0, x \neq -1$
69. $\frac{a}{(x-b)} + \frac{b}{(x-a)} = 2, x \neq b, a$
70. $\frac{a}{(ax-1)} + \frac{b}{(bx-1)} = (a+b), x \neq \frac{1}{a}, \frac{1}{b}$
71. $3^{(x+2)} + 3^{-x} = 10$
72. $4^{(x+1)} + 4^{(1-x)} = 10$
73. $2^{2x} - 3 \cdot 2^{(x+2)} + 32 = 0$

ANSWERS (EXERCISE 4A)

1. (i), (ii), (iii), (iv), (vi), (ix)
2. -1 and $\frac{1}{3}$
3. (i) $k = -4$, other root $= 3$ (ii) $a = 4$, $b = 5$
5. $x = \frac{3}{2}$ or $x = -\frac{1}{3}$
6. $x = 0$ or $x = -\frac{5}{4}$
7. $x = 9$ or $x = -9$
8. $x = -2$ or $x = \frac{3}{2}$
9. $x = -5$ or $x = -1$
10. $x = \frac{2}{3}$ or $x = -\frac{1}{3}$
11. $x = -7$ or $x = -5$
12. $x = 11$ or $x = 7$
13. $x = -\frac{3}{2}$ or $x = -\frac{1}{3}$
14. $x = -\frac{3}{2}$ or $x = \frac{4}{3}$
15. $x = 1$ or $x = -\frac{1}{3}$
16. $x = \frac{25}{4}$ or $x = -4$
17. $x = -\frac{4}{3}$ or $x = \frac{7}{5}$
18. $x = -4$ or $x = \frac{1}{3}$
19. $x = \frac{1}{3}$ or $x = -\frac{1}{16}$
20. $x = \sqrt{2}$ or $x = -3\sqrt{2}$
21. $x = -4\sqrt{3}$ or $x = \frac{2}{\sqrt{3}}$
22. $x = -3\sqrt{3}$ or $x = -\frac{2}{\sqrt{3}}$
23. $x = \frac{-\sqrt{7}}{3}$ or $x = \frac{1}{\sqrt{7}}$
24. $x = \frac{13}{\sqrt{7}}$ or $x = -\sqrt{7}$
25. $x = \frac{2\sqrt{2}}{\sqrt{3}}$ or $x = \frac{-\sqrt{3}}{4\sqrt{2}}$
26. $x = \frac{\sqrt{2}}{\sqrt{3}}$ or $x = \frac{\sqrt{2}}{\sqrt{3}}$
27. $x = \sqrt{6}$ or $x = \frac{-\sqrt{2}}{\sqrt{3}}$
28. $x = \sqrt{5}$ or $x = 2\sqrt{5}$
29. $x = \sqrt{3}$ or $x = 1$
30. $x = 2\sqrt{3}$ or $x = -5\sqrt{3}$
31. $x = -\sqrt{2}$ or $x = \frac{-5}{\sqrt{2}}$
32. $x = -1$ or $x = \frac{-8}{5}$
33. $x = 1$ or $x = \sqrt{2}$
34. $x = \frac{-1}{3}$, $x = \frac{-1}{3}$
35. $x = \frac{1}{10}$, $x = \frac{1}{10}$
36. $x = \frac{1}{4}$, $x = \frac{1}{4}$
37. $x = \frac{1}{2}$ or $x = \frac{-1}{5}$
38. $x = 2$ or $x = \frac{1}{2}$
39. $x = -a$ or $x = \frac{a}{2}$
40. $x = \frac{-(a+b)}{2}$ or $x = \frac{a-b}{2}$
41. $x = \frac{a^2+b^2}{2}$ or $x = \frac{a^2-b^2}{2}$
42. $x = -(a+3)$ or $x = (a-2)$
43. $x = (a-2b)$ or $x = (a+2b)$
44. $x = (b-5)$ or $x = (b+4)$
45. $x = -(a+4)$ or $x = (a-2)$
46. $x = \frac{-b}{a}$ or $x = \frac{c}{b}$
47. $x = (2a+b)$ or $x = (2a-b)$
48. $x = \frac{a^2}{2}$ or $x = \frac{b^2}{2}$
49. $x = \frac{3a}{4b}$ or $x = \frac{-2b}{3a}$
50. $x = \frac{-1}{a^2}$ or $x = \frac{1}{b^2}$
51. $x = \frac{(a+2b)}{3}$ or $x = \frac{(2a+b)}{3}$
52. $x = 4$ or $x = -4$
53. $x = -2$ or $x = 1$

54. $x = 3$ or $x = 1$ 55. (i) $x = -6$ or $x = 2$ (ii) $x = \frac{37}{20}$ or $x = 6$
 56. $x = -a$ or $x = \frac{-b}{2}$ 57. $x = 4$ or $x = \frac{-2}{9}$ 58. $x = 6$ or $x = 2\frac{1}{2}$
 59. (i) $x = \frac{4}{3}$ or $x = \frac{-1}{3}$ (ii) $x = -2$ 60. $x = \frac{-5}{2}$ or $x = \frac{3}{2}$
 61. $x = 8$ or $x = 5\frac{1}{2}$ 62. $x = 5$ or $x = \frac{5}{2}$ 63. $x = 3$ or $x = \frac{4}{3}$
 64. (i) $x = 2$ or $x = \frac{-3}{2}$ (ii) $x = -\frac{11}{17}$ or $x = 1$ 65. $x = -7$ or $x = 0$
 66. $x = 1$ or $x = 0$ 67. $x = \frac{-4}{3}$ or $x = \frac{1}{8}$ 68. $x = \frac{-3}{2}$ or $x = -2$
 69. $x = (a+b)$ or $x = \frac{(a+b)}{2}$ 70. $x = \frac{(a+b)}{ab}$ or $x = \frac{2}{(a+b)}$
 71. $x = -2$ or $x = 0$ 72. $x = \frac{1}{2}$ or $x = \frac{-1}{2}$ 73. $x = 2$ or $x = 3$

HINTS TO SOME SELECTED QUESTIONS

16. $4x^2 - 9x - 100 = 0 \Rightarrow 4x^2 - 25x + 16x - 100 = 0$.
 17. $15x^2 - x - 28 = 0 \Rightarrow 15x^2 - 21x + 20x - 28 = 0$.
 18. $3x^2 + 11x - 4 = 0 \Rightarrow 3x^2 + 12x - x - 4 = 0$.
 19. $48x^2 - 13x - 1 = 0 \Rightarrow 48x^2 - 16x + 3x - 1 = 0$.
 20. $x^2 + 2\sqrt{2}x - 6 = 0 \Rightarrow x^2 + 3\sqrt{2}x - \sqrt{2}x - 6 = 0 \Rightarrow x(x + 3\sqrt{2}) - \sqrt{2}(x + 3\sqrt{2}) = 0$.
 21. $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0 \Rightarrow \sqrt{3}x^2 + 12x - 2x - 8\sqrt{3} = 0 \Rightarrow \sqrt{3}x(x + 4\sqrt{3}) - 2(x + 4\sqrt{3}) = 0$.
 22. $\sqrt{3}x^2 + 11x + 6\sqrt{3} = 0 \Rightarrow \sqrt{3}x^2 + 9x + 2x + 6\sqrt{3} = 0 \Rightarrow \sqrt{3}x(x + 3\sqrt{3}) + 2(x + 3\sqrt{3}) = 0$.
 23. $3\sqrt{7}x^2 + 4x - \sqrt{7} = 0 \Rightarrow 3\sqrt{7}x^2 + 7x - 3x - \sqrt{7} = 0 \Rightarrow \sqrt{7}x(3x + \sqrt{7}) - (3x + \sqrt{7}) = 0$.
 24. $\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0 \Rightarrow \sqrt{7}x^2 - 13x + 7x - 13\sqrt{7} = 0$
 $\Rightarrow x(\sqrt{7}x - 13) + \sqrt{7}(\sqrt{7}x - 13) = 0$.
 25. $4\sqrt{6}x^2 - 13x - 2\sqrt{6} = 0 \Rightarrow 4\sqrt{6}x^2 - 16x + 3x - 2\sqrt{6} = 0$
 $\Rightarrow 4\sqrt{2}x(\sqrt{3}x - 2\sqrt{2}) + \sqrt{3}(\sqrt{3}x - 2\sqrt{2}) = 0$.
 26. $3x^2 - 2\sqrt{6}x + 2 = 0 \Rightarrow 3x^2 - \sqrt{6}x - \sqrt{6}x + 2 = 0 \Rightarrow \sqrt{3}x(\sqrt{3}x - \sqrt{2}) - \sqrt{2}(\sqrt{3}x - \sqrt{2}) = 0$.
 27. $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0 \Rightarrow \sqrt{3}x^2 - 3\sqrt{2}x + \sqrt{2}x - 2\sqrt{3} = 0$
 $\Rightarrow \sqrt{3}x(x - \sqrt{6}) + \sqrt{2}(x - \sqrt{6}) = 0$.
 28. $x^2 - 3\sqrt{5}x + 10 = 0 \Rightarrow x^2 - 2\sqrt{5}x - \sqrt{5}x + 10 = 0 \Rightarrow x(x - 2\sqrt{5}) - \sqrt{5}(x - 2\sqrt{5}) = 0$.
 29. $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0 \Rightarrow x^2 - \sqrt{3}x - x + \sqrt{3} = 0 \Rightarrow x(x - \sqrt{3}) - (x - \sqrt{3}) = 0$.
 30. $x^2 + 3\sqrt{3}x - 30 = 0 \Rightarrow x^2 + 5\sqrt{3}x - 2\sqrt{3}x - 30 = 0 \Rightarrow x(x + 5\sqrt{3}) - 2\sqrt{3}(x + 5\sqrt{3}) = 0$.
 31. $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0 \Rightarrow \sqrt{2}x^2 + 5x + 2x + 5\sqrt{2} = 0 \Rightarrow x(\sqrt{2}x + 5) + \sqrt{2}(\sqrt{2}x + 5) = 0$.
 32. $5x^2 + 13x + 8 = 0 \Rightarrow 5x^2 + 5x + 8x + 8 = 0$.

$$34. 9x^2 + 6x + 1 = 0 \Rightarrow (3x + 1)^2 = 0 \Rightarrow 3x + 1 = 0.$$

$$35. 100x^2 - 20x + 1 = 0 \Rightarrow (10x - 1)^2 = 0 \Rightarrow 10x - 1 = 0.$$

$$36. 16x^2 - 8x + 1 = 0 \Rightarrow (4x - 1)^2 = 0 \Rightarrow 4x - 1 = 0.$$

$$37. 10x^2 - 3x - 1 = 0 \Rightarrow 10x^2 - 5x + 2x - 1 = 0 \Rightarrow 5x(2x - 1) + (2x - 1) = 0.$$

$$38. 2x^2 - 5x + 2 = 0 \Rightarrow 2x^2 - 4x - x + 2 = 0 \Rightarrow 2x(x - 2) - (x - 2) = 0.$$

$$39. 2x^2 + ax - a^2 = 0 \Rightarrow 2x^2 + 2ax - ax - a^2 = 0 \Rightarrow 2x(x + a) - a(x + a) = 0.$$

$$40. 4x^2 + 4bx + (b^2 - a^2) = 0 \Rightarrow 4x^2 + 2(b + a)x + 2(b - a)x + (b^2 - a^2) = 0.$$

$$41. 4x^2 - 4a^2x + (a^4 - b^4) = 0 \Rightarrow 4x^2 - 2(a^2 + b^2)x - 2(a^2 - b^2)x + (a^4 - b^4) = 0 \\ \Rightarrow 2x[2x - (a^2 + b^2)] - (a^2 - b^2)[2x - (a^2 + b^2)] = 0.$$

$$42. x^2 + 5x - (a + 3)(a - 2) = 0 \Rightarrow x^2 + (a + 3)x - (a - 2)x - (a + 3)(a - 2) = 0.$$

$$43. x^2 - 2ax + (a^2 - 4b^2) = 0 \Rightarrow x^2 - 2ax + (a + 2b)(a - 2b) = 0 \\ \Rightarrow x^2 - (a + 2b)x - (a - 2b)x + (a + 2b)(a - 2b) = 0.$$

$$44. x^2 - (2b - 1)x + (b - 5)(b + 4) = 0 \Rightarrow x^2 - (b - 5)x - (b + 4)x + (b - 5)(b + 4) = 0 \\ \Rightarrow x[x - (b - 5)] - (b + 4)[x - (b - 5)] = 0.$$

$$45. x^2 + 6x - (a + 4)(a - 2) = 0 \Rightarrow x^2 + (a + 4)x - (a - 2)x - (a + 4)(a - 2) = 0.$$

$$46. abx^2 + b^2x - acx - bc = 0 \Rightarrow bx(ax + b) - c(ax + b) = 0.$$

$$47. x^2 - 4ax + (2a + b)(2a - b) = 0 \Rightarrow x^2 - (2a + b)x - (2a - b)x + (4a^2 - b^2) = 0.$$

$$48. 4x^2 - 2a^2x - 2b^2x + a^2b^2 = 0 \Rightarrow 2x(2x - a^2) - b^2(2x - a^2) = 0.$$

$$49. 12abx^2 - 9a^2x + 8b^2x - 6ab = 0 \Rightarrow 3ax(4bx - 3a) + 2b(4bx - 3a) = 0.$$

$$50. b^2x(a^2x + 1) - (a^2x + 1) = 0.$$

$$51. 9x^2 - 9(a + b)x + (2a + b)(a + 2b) = 0$$

$$\Rightarrow 9x^2 - 3\{(2a + b) + (a + 2b)\}x + (2a + b)(a + 2b) = 0$$

$$\Rightarrow 9x^2 - 3(2a + b)x - 3(a + 2b)x + (2a + b)(a + 2b) = 0.$$

$$56. \frac{1}{2a + b + 2x} - \frac{1}{2x} = \frac{1}{2a} + \frac{1}{b}.$$

$$57. \frac{x(x + 3) - (1 - x)(x - 2)}{x(x - 2)} = \frac{17}{4} \Rightarrow \frac{(x^2 + 3x)(3x - x^2 - 2)}{(x^2 - 2x)} = \frac{17}{4}$$

$$\Rightarrow 4(2x^2 + 2) = 17(x^2 - 2x) \Rightarrow 9x^2 - 34x - 8 = 0 \Rightarrow 9x^2 - 36x + 2x - 8 = 0.$$

$$58. \text{Put } \frac{3x - 4}{7} = y. \text{ Then, } y + \frac{1}{y} = \frac{5}{2} \Rightarrow 2y^2 - 5y + 2 = 0.$$

$$65. \text{Putting } \frac{3x - 1}{2x + 3} = y, \text{ the given equation becomes}$$

$$3y - \frac{2}{y} = 5 \Rightarrow 3y^2 - 5y - 2 = 0 \Rightarrow 3y^2 - 6y + y - 2 = 0.$$

$$69. \text{The given equation is}$$

$$\left(\frac{a}{x - b} - 1\right) + \left(\frac{b}{x - a} - 1\right) = 0$$

$$\Rightarrow \frac{(a - x + b)}{(x - b)} + \frac{(a - x + b)}{(x - a)} = 0$$

$$\Rightarrow (a-x+b) \cdot \left[\frac{1}{(x-b)} + \frac{1}{(x-a)} \right] = 0$$

$$\Rightarrow (a-x+b) \left[\frac{2x-(a+b)}{(x-a)(x-b)} \right] = 0$$

$$\Rightarrow (a-x+b)[2x-(a+b)] = 0$$

$$\Rightarrow x = (a+b) \text{ or } x = \frac{(a+b)}{2}.$$

70. The given equation is

$$\left\{ \frac{a}{(ax-1)} - b \right\} + \left\{ \frac{b}{(bx-1)} - a \right\} = 0$$

$$\Rightarrow \frac{(a-abx+b)}{(ax-1)} + \frac{(a-abx+b)}{(bx-1)} = 0$$

$$\Rightarrow (a-abx+b) \cdot \left(\frac{1}{ax-1} + \frac{1}{bx-1} \right) = 0.$$

$$71. 3^x \times 3^2 + \frac{1}{3^x} = 10 \Rightarrow 9y + \frac{1}{y} = 10, \text{ where } y = 3^x.$$

$$72. 4^x \times 4 + \frac{4}{4^x} = 10 \Rightarrow 2y + \frac{2}{y} = 5, \text{ where } y = 4^x.$$

$$73. 2^{2x} - 3 \times 2^2 \times 2^x + 32 = 0 \Rightarrow y^2 - 12y + 32 = 0, \text{ where } 2^x = y$$

$$\Rightarrow y^2 - 8y - 4y + 32 = 0.$$

SOLVING QUADRATIC EQUATIONS BY COMPLETING THE SQUARE

METHOD In fact, we can convert any quadratic equation to the form $(ax+b)^2 - c^2 = 0$ and then we can easily find its roots.

SOLVED EXAMPLES

EXAMPLE 1 Solve the equation $x^2 - 10x - 2 = 0$ by the method of completing the square.

SOLUTION We have

$$x^2 - 10x - 2 = 0$$

$$\Rightarrow x^2 - 10x = 2$$

$$\Rightarrow x^2 - 2 \times x \times 5 + 5^2 = 2 + 5^2 \quad [\text{adding } 5^2 \text{ on both sides}]$$

$$\Rightarrow (x-5)^2 = (2+25) = 27$$

$$\Rightarrow x-5 = \pm\sqrt{27} = \pm 3\sqrt{3} \quad [\text{taking square root on both sides}]$$

$$\Rightarrow x-5 = 3\sqrt{3} \text{ or } x-5 = -3\sqrt{3}$$

$$\Rightarrow x = (5+3\sqrt{3}) \text{ or } x = (5-3\sqrt{3}).$$

Hence, $(5+3\sqrt{3})$ and $(5-3\sqrt{3})$ are the roots of the given equation.

EXAMPLE 2 Solve the equation $3x^2 - 5x + 2 = 0$ by the method of completing the square.

SOLUTION We have

$$\begin{aligned}
 & 3x^2 - 5x + 2 = 0 \\
 \Rightarrow & 9x^2 - 15x + 6 = 0 \quad [\text{multiplying each term by 3}] \\
 \Rightarrow & 9x^2 - 15x = -6 \\
 \Rightarrow & (3x)^2 - 2 \times 3x \times \frac{5}{2} + \left(\frac{5}{2}\right)^2 = -6 + \left(\frac{5}{2}\right)^2 \\
 & \hspace{15em} [\text{adding } \left(\frac{5}{2}\right)^2 \text{ on both sides}] \\
 \Rightarrow & \left(3x - \frac{5}{2}\right)^2 = \left(-6 + \frac{25}{4}\right) = \frac{(-24 + 25)}{4} = \frac{1}{4} = \left(\frac{1}{2}\right)^2 \\
 \Rightarrow & \left(3x - \frac{5}{2}\right) = \pm \frac{1}{2} \quad [\text{taking square root on both sides}] \\
 \Rightarrow & 3x - \frac{5}{2} = \frac{1}{2} \quad \text{or} \quad 3x - \frac{5}{2} = -\frac{1}{2} \\
 \Rightarrow & 3x = \left(\frac{1}{2} + \frac{5}{2}\right) = \frac{6}{2} = 3 \quad \text{or} \quad 3x = \left(-\frac{1}{2} + \frac{5}{2}\right) = \frac{4}{2} = 2 \\
 \Rightarrow & 3x = 3 \quad \text{or} \quad 3x = 2 \Rightarrow x = 1 \quad \text{or} \quad x = \frac{2}{3}.
 \end{aligned}$$

Hence, 1 and $\frac{2}{3}$ are the roots of the given equation.

EXAMPLE 3 Solve the equation $2x^2 + x - 4 = 0$ by the method of completing the square.

SOLUTION We have

$$\begin{aligned}
 & 2x^2 + x - 4 = 0 \\
 \Rightarrow & 4x^2 + 2x - 8 = 0 \quad [\text{multiplying both sides by 2}] \\
 \Rightarrow & 4x^2 + 2x = 8 \\
 \Rightarrow & (2x)^2 + 2 \times 2x \times \frac{1}{2} + \left(\frac{1}{2}\right)^2 = 8 + \left(\frac{1}{2}\right)^2 \\
 & \hspace{15em} [\text{adding } \left(\frac{1}{2}\right)^2 \text{ on both sides}] \\
 \Rightarrow & \left(2x + \frac{1}{2}\right)^2 = \left(8 + \frac{1}{4}\right) = \frac{33}{4} = \left(\frac{\sqrt{33}}{2}\right)^2 \\
 \Rightarrow & 2x + \frac{1}{2} = \pm \left(\frac{\sqrt{33}}{2}\right) \quad [\text{taking square root on both sides}] \\
 \Rightarrow & 2x + \frac{1}{2} = \frac{\sqrt{33}}{2} \quad \text{or} \quad 2x + \frac{1}{2} = -\frac{\sqrt{33}}{2}
 \end{aligned}$$

$$\Rightarrow 2x = \left(\frac{\sqrt{33}}{2} - \frac{1}{2} \right) = \frac{(\sqrt{33} - 1)}{2}$$

$$\text{or } 2x = \frac{-\sqrt{33}}{2} - \frac{1}{2} = \frac{-(\sqrt{33} + 1)}{2}$$

$$\Rightarrow x = \frac{(\sqrt{33} - 1)}{4} \text{ or } x = \frac{-(\sqrt{33} + 1)}{4}.$$

Hence, $\frac{-1 + \sqrt{33}}{4}$ and $\frac{-1 - \sqrt{33}}{4}$ are the roots of the given equation.

EXAMPLE 4 By using the method of completing the square, show that the equation $4x^2 + 3x + 5 = 0$ has no real roots.

SOLUTION We have

$$4x^2 + 3x + 5 = 0$$

$$\Rightarrow 4x^2 + 3x = -5$$

$$\Rightarrow (2x)^2 + 2 \times 2x \times \frac{3}{4} + \left(\frac{3}{4}\right)^2 = -5 + \left(\frac{3}{4}\right)^2$$

[adding $\left(\frac{3}{4}\right)^2$ on both sides]

$$\Rightarrow \left(2x + \frac{3}{4}\right)^2 = \left(-5 + \frac{9}{16}\right) = \frac{(-80 + 9)}{16} = \frac{-71}{16} < 0.$$

But, $\left(2x + \frac{3}{4}\right)^2$ cannot be negative for any real value of x .

So, there is no real value of x satisfying the given equation.

Hence, the given equation has no real roots.

EXAMPLE 5 Solve the equation $10x - \frac{1}{x} = 3$ by the method of completing the square.

SOLUTION We have

$$10x - \frac{1}{x} = 3$$

$$\Rightarrow 10x^2 - 1 = 3x$$

$$\Rightarrow 10x^2 - 3x = 1$$

$$\Rightarrow 100x^2 - 30x = 10 \quad [\text{multiplying each side by } 10]$$

$$\Rightarrow (10x)^2 - 2 \times 10x \times \frac{3}{2} + \left(\frac{3}{2}\right)^2 = 10 + \left(\frac{3}{2}\right)^2$$

[adding $\left(\frac{3}{2}\right)^2$ on both sides]

$$\Rightarrow \left(10x - \frac{3}{2}\right)^2 = \left(10 + \frac{9}{4}\right) = \frac{49}{4} = \left(\frac{7}{2}\right)^2$$

$$\Rightarrow 10x - \frac{3}{2} = \pm \frac{7}{2} \quad [\text{taking square root on both sides}]$$

$$\Rightarrow 10x - \frac{3}{2} = \frac{7}{2} \text{ or } 10x - \frac{3}{2} = -\frac{7}{2}$$

$$\Rightarrow 10x = \left(\frac{7}{2} + \frac{3}{2}\right) = \frac{10}{2} = 5 \text{ or } 10x = \left(-\frac{7}{2} + \frac{3}{2}\right) = \frac{-4}{2} = -2$$

$$\Rightarrow 10x = 5 \text{ or } 10x = -2 \Rightarrow x = \frac{5}{10} = \frac{1}{2} \text{ or } x = \frac{-2}{10} = \frac{-1}{5}.$$

Hence, $\frac{1}{2}$ and $\frac{-1}{5}$ are the roots of the given equation.

EXAMPLE 6 Solve the equation $a^2x^2 - 3abx + 2b^2 = 0$ by the method of completing the square.

SOLUTION We have

$$a^2x^2 - 3abx + 2b^2 = 0$$

$$\Rightarrow a^2x^2 - 3abx = -2b^2$$

$$\Rightarrow (ax)^2 - 2 \times (ax) \times \frac{3b}{2} + \left(\frac{3b}{2}\right)^2 = -2b^2 + \left(\frac{3b}{2}\right)^2$$

[adding $\left(\frac{3b}{2}\right)^2$ on both sides]

$$\Rightarrow \left(ax - \frac{3b}{2}\right)^2 = \left(-2b^2 + \frac{9b^2}{4}\right) = \frac{(-8b^2 + 9b^2)}{4} = \frac{b^2}{4} = \left(\frac{b}{2}\right)^2$$

$$\Rightarrow \left(ax - \frac{3b}{2}\right) = \pm \frac{b}{2} \quad [\text{taking square root on both sides}]$$

$$\Rightarrow \left(ax - \frac{3b}{2}\right) = \frac{b}{2} \text{ or } \left(ax - \frac{3b}{2}\right) = -\frac{b}{2}$$

$$\Rightarrow ax = \left(\frac{b}{2} + \frac{3b}{2}\right) = \frac{4b}{2} = 2b \text{ or } ax = \left(-\frac{b}{2} + \frac{3b}{2}\right) = \frac{2b}{2} = b$$

$$\Rightarrow x = \frac{2b}{a} \text{ or } x = \frac{b}{a}.$$

Hence, $\frac{2b}{a}$ and $\frac{b}{a}$ are the roots of the given equation.

EXAMPLE 7 Solve the equation $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$ by the method of completing the square.

SOLUTION We have

$$x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$$

$$\Rightarrow x^2 - (\sqrt{3} + 1)x = -\sqrt{3}$$

$$\Rightarrow x^2 - 2 \times x \times \left(\frac{\sqrt{3} + 1}{2}\right) + \left(\frac{\sqrt{3} + 1}{2}\right)^2 = -\sqrt{3} + \left(\frac{\sqrt{3} + 1}{2}\right)^2$$

$$\begin{aligned}
 & \Rightarrow \left\{ x - \frac{(\sqrt{3}+1)}{2} \right\}^2 = \left\{ \frac{(\sqrt{3}+1)^2}{4} - \sqrt{3} \right\} \quad \left[\text{adding } \left(\frac{\sqrt{3}+1}{2} \right)^2 \text{ on both sides} \right] \\
 & \Rightarrow \frac{(\sqrt{3}+1)^2 - 4\sqrt{3}}{4} = \left(\frac{\sqrt{3}-1}{2} \right)^2 \\
 & \Rightarrow \left\{ x - \frac{(\sqrt{3}+1)}{2} \right\} = \pm \frac{(\sqrt{3}-1)}{2} \quad \left[\text{taking square root on both sides} \right] \\
 & \Rightarrow x - \frac{(\sqrt{3}+1)}{2} = \frac{(\sqrt{3}-1)}{2} \quad \text{or} \quad x - \frac{(\sqrt{3}+1)}{2} = \frac{-(\sqrt{3}-1)}{2} \\
 & \Rightarrow x = \frac{(\sqrt{3}-1)}{2} + \frac{(\sqrt{3}+1)}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3} \\
 & \quad \text{or} \quad x = \frac{(-\sqrt{3}+1)}{2} + \frac{(\sqrt{3}+1)}{2} = \frac{2}{2} = 1 \\
 & \Rightarrow x = \sqrt{3} \quad \text{or} \quad x = 1.
 \end{aligned}$$

Hence, $\sqrt{3}$ and 1 are the roots of the given equation.

EXERCISE 4B

Solve each of the following equations by using the method of completing the square:

1. $x^2 - 6x + 3 = 0$
2. $x^2 - 4x + 1 = 0$
3. $x^2 + 8x - 2 = 0$
4. $4x^2 + 4\sqrt{3}x + 3 = 0$
5. $2x^2 + 5x - 3 = 0$
6. $3x^2 - x - 2 = 0$
7. $8x^2 - 14x - 15 = 0$
8. $7x^2 + 3x - 4 = 0$
9. $3x^2 - 2x - 1 = 0$
10. $5x^2 - 6x - 2 = 0$
11. $\frac{2}{x^2} - \frac{5}{x} + 2 = 0$
12. $4x^2 + 4bx - (a^2 - b^2) = 0$
13. $x^2 - (\sqrt{2} + 1)x + \sqrt{2} = 0$
14. $\sqrt{2}x^2 - 3x - 2\sqrt{2} = 0$
15. $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$
16. By using the method of completing the square, show that the equation $2x^2 + x + 4 = 0$ has no real roots.

ANSWERS (EXERCISE 4B)

1. $x = (3 + \sqrt{6})$ or $x = (3 - \sqrt{6})$
2. $x = (2 + \sqrt{3})$ or $x = (2 - \sqrt{3})$
3. $x = (-4 + 3\sqrt{2})$ or $x = (-4 - 3\sqrt{2})$
4. $x = \frac{-\sqrt{3}}{2}$ or $x = \frac{-\sqrt{3}}{2}$
5. $x = \frac{1}{2}$ or $x = -3$
6. $x = 1$ or $x = \frac{-2}{3}$
7. $x = \frac{5}{2}$ or $x = \frac{-3}{4}$

8. $x = -1$ or $x = \frac{4}{7}$ 9. $x = 1$ or $x = \frac{-1}{3}$
10. $x = \frac{3 + \sqrt{19}}{5}$ or $x = \frac{3 - \sqrt{19}}{5}$ 11. $x = 2$ or $x = \frac{1}{2}$
12. $x = \frac{-(a+b)}{2}$ or $x = \frac{(a-b)}{2}$ 13. $x = \sqrt{2}$ or $x = 1$
14. $x = \frac{-1}{\sqrt{2}}$ or $x = 2\sqrt{2}$ 15. $x = -\sqrt{3}$ or $x = \frac{-7}{\sqrt{3}}$
-

QUADRATIC FORMULA [SHRIDHARACHARYA'S RULE]

Consider the quadratic equation $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$. Then,

$$\begin{aligned}
 & ax^2 + bx + c = 0 \\
 \Rightarrow & ax^2 + bx = -c \\
 \Rightarrow & x^2 + \frac{b}{a} \cdot x = \frac{-c}{a} && \text{[dividing throughout by } a] \\
 \Rightarrow & x^2 + \frac{b}{a} \cdot x + \left(\frac{b}{2a}\right)^2 = \frac{-c}{a} + \left(\frac{b}{2a}\right)^2 && \text{[adding } \left(\frac{b}{2a}\right)^2 \text{ on both sides]} \\
 \Rightarrow & \left(x + \frac{b}{2a}\right)^2 = \left(\frac{-c}{a} + \frac{b^2}{4a^2}\right) \\
 \Rightarrow & \left(x + \frac{b}{2a}\right)^2 = \frac{(b^2 - 4ac)}{4a^2} \\
 \Rightarrow & \left(x + \frac{b}{2a}\right) = \frac{\pm \sqrt{b^2 - 4ac}}{2a}, \text{ when } (b^2 - 4ac) \geq 0 \\
 \Rightarrow & x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 \Rightarrow & x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.
 \end{aligned}$$

This is called the *quadratic formula* or *Shridharacharya's rule*.

Thus, $ax^2 + bx + c = 0$ has two roots α and β , given by

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

DISCRIMINANT For the equation $ax^2 + bx + c = 0$, the expression $D = (b^2 - 4ac)$ is called the *discriminant*.

AN IMPORTANT NOTE The roots of $ax^2 + bx + c = 0$ are real only when $(b^2 - 4ac) \geq 0$.

Taking $(b^2 - 4ac) = D$, the roots of $ax^2 + bx + c = 0$ are given by

$$\alpha = \frac{-b + \sqrt{D}}{2a} \quad \text{and} \quad \beta = \frac{-b - \sqrt{D}}{2a}.$$

SOLVED EXAMPLES

EXAMPLE 1 Show that the equation $9x^2 + 7x - 2 = 0$ has real roots and solve it.

SOLUTION The given equation is $9x^2 + 7x - 2 = 0$.

Comparing it with $ax^2 + bx + c = 0$, we get

$$a = 9, b = 7 \text{ and } c = -2.$$

$$\therefore D = (b^2 - 4ac) = (7^2 - 4 \times 9 \times (-2)) = 121 > 0.$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{121} = 11.$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{(-7 + 11)}{2 \times 9} = \frac{4}{18} = \frac{2}{9},$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{(-7 - 11)}{2 \times 9} = \frac{-18}{18} = -1.$$

Hence, the required roots are $\frac{2}{9}$ and -1 .

EXAMPLE 2 Show that the equation $x^2 + 6x + 6 = 0$ has real roots and solve it.

SOLUTION The given equation is $x^2 + 6x + 6 = 0$.

Comparing it with $ax^2 + bx + c = 0$, we get

$$a = 1, b = 6 \text{ and } c = 6.$$

$$\therefore D = (b^2 - 4ac) = (36 - 4 \times 1 \times 6) = 12 > 0.$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{12} = 2\sqrt{3}.$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{(-6 + 2\sqrt{3})}{2 \times 1} = \frac{(-6 + 2\sqrt{3})}{2} = (-3 + \sqrt{3}),$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{(-6 - 2\sqrt{3})}{2 \times 1} = \frac{(-6 - 2\sqrt{3})}{2} = (-3 - \sqrt{3}).$$

Hence, $(-3 + \sqrt{3})$ and $(-3 - \sqrt{3})$ are the roots of the given equation.

EXAMPLE 3 Show that the equation $2x^2 + 5\sqrt{3}x + 6 = 0$ has real roots and solve it.

SOLUTION The given equation is $2x^2 + 5\sqrt{3}x + 6 = 0$.

Comparing it with $ax^2 + bx + c = 0$, we get

$$a = 2, b = 5\sqrt{3} \text{ and } c = 6.$$

$$\therefore D = (b^2 - 4ac) = [(5\sqrt{3})^2 - 4 \times 2 \times 6] = (75 - 48) = 27 > 0.$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{27} = 3\sqrt{3}.$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{(-5\sqrt{3} + 3\sqrt{3})}{2 \times 2} = \frac{-2\sqrt{3}}{4} = \frac{-\sqrt{3}}{2},$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{(-5\sqrt{3} - 3\sqrt{3})}{2 \times 2} = \frac{-8\sqrt{3}}{4} = -2\sqrt{3}.$$

Hence, $\frac{-\sqrt{3}}{2}$ and $-2\sqrt{3}$ are the roots of the given equation.

EXAMPLE 4 Using quadratic formula, solve for x :

$$p^2x^2 + (p^2 - q^2)x - q^2 = 0. \quad [\text{CBSE 2014}]$$

SOLUTION The given equation is $p^2x^2 + (p^2 - q^2)x - q^2 = 0$.

Comparing it with $ax^2 + bx + c = 0$, we get

$$a = p^2, b = (p^2 - q^2) \text{ and } c = -q^2.$$

$$\begin{aligned} \therefore D &= (b^2 - 4ac) = (p^2 - q^2)^2 - 4 \times p^2 \times (-q^2) \\ &= (p^2 - q^2)^2 + 4p^2q^2 = (p^2 + q^2)^2 > 0. \end{aligned}$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = (p^2 + q^2).$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{-(p^2 - q^2) + (p^2 + q^2)}{2p^2} = \frac{2q^2}{2p^2} = \frac{q^2}{p^2},$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{-(p^2 - q^2) - (p^2 + q^2)}{2p^2} = \frac{-2p^2}{2p^2} = -1.$$

Hence, $\frac{q^2}{p^2}$ and -1 are the roots of the given equation.

EXAMPLE 5 Using quadratic formula, solve for x :

$$9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0. \quad [\text{CBSE 2009}]$$

SOLUTION The given equation is $9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0$.

This is of the form $Ax^2 + Bx + C = 0$, where

$$A = 9, B = -9(a+b) \text{ and } C = (2a^2 + 5ab + 2b^2).$$

$$\begin{aligned}
 \therefore D &= (B^2 - 4AC) = 81(a+b)^2 - 36(2a^2 + 5ab + 2b^2) \\
 &= 81(a^2 + b^2 + 2ab) - 36(2a^2 + 5ab + 2b^2) \\
 &= 9a^2 + 9b^2 - 18ab = 9(a^2 + b^2 - 2ab) \\
 &= 9(a-b)^2 \geq 0.
 \end{aligned}$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{9(a-b)^2} = 3(a-b).$$

$$\begin{aligned}
 \therefore \alpha &= \frac{-B + \sqrt{D}}{2A} = \frac{9(a+b) + 3(a-b)}{2 \times 9} = \frac{6(2a+b)}{18} = \frac{(2a+b)}{3}, \\
 \beta &= \frac{-B - \sqrt{D}}{2A} = \frac{9(a+b) - 3(a-b)}{2 \times 9} = \frac{6(a+2b)}{18} = \frac{(a+2b)}{3}.
 \end{aligned}$$

Hence, $\frac{(2a+b)}{3}$ and $\frac{(a+2b)}{3}$ are the roots of the given equation.

EXAMPLE 6 Using quadratic formula, solve for x :

$$abx^2 + (b^2 - ac)x - bc = 0. \quad [\text{CBSE 2014}]$$

SOLUTION The given equation is $abx^2 + (b^2 - ac)x - bc = 0$.

This is of the form $Ax^2 + Bx + C = 0$, where

$$A = ab, B = (b^2 - ac) \text{ and } C = -bc.$$

$$\begin{aligned}
 \therefore D &= (B^2 - 4AC) = (b^2 - ac)^2 + 4ab^2c \\
 &= b^4 + a^2c^2 - 2ab^2c + 4ab^2c \\
 &= b^4 + a^2c^2 + 2ab^2c = (b^2 + ac)^2 > 0.
 \end{aligned}$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = (b^2 + ac).$$

$$\begin{aligned}
 \therefore \alpha &= \frac{-B + \sqrt{D}}{2A} = \frac{-(b^2 - ac) + (b^2 + ac)}{2ab} = \frac{2ac}{2ab} = \frac{c}{b}, \\
 \beta &= \frac{-B - \sqrt{D}}{2A} = \frac{-(b^2 - ac) + (b^2 + ac)}{2ab} = \frac{-2b^2}{2ab} = \frac{-b}{a}.
 \end{aligned}$$

Hence, $\frac{c}{b}$ and $\frac{-b}{a}$ are the roots of the given equation.

EXAMPLE 7 Solve for x : $\frac{1}{x} - \frac{1}{(x-2)} = 3, x \neq 0, 2. \quad [\text{CBSE 2010}]$

SOLUTION The given equation may be written as

$$\begin{aligned}
 \frac{(x-2) - x}{x(x-2)} &= 3 \Rightarrow 3x(x-2) = -2 \\
 &\Rightarrow 3x^2 - 6x + 2 = 0. \quad \dots (i)
 \end{aligned}$$

This equation is of the form $ax^2 + bx + c = 0$, where $a = 3$, $b = -6$ and $c = 2$.

$$\therefore D = (b^2 - 4ac) = (-6)^2 - 4 \times 3 \times 2 = 36 - 24 = 12 > 0.$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{12} = 2\sqrt{3}.$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{6 + 2\sqrt{3}}{2 \times 3} = \frac{6 + 2\sqrt{3}}{6} = \frac{3 + \sqrt{3}}{3},$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{6 - 2\sqrt{3}}{2 \times 3} = \frac{6 - 2\sqrt{3}}{6} = \frac{3 - \sqrt{3}}{3}.$$

Hence, the required values of x are $\frac{(3 + \sqrt{3})}{3}$ and $\frac{(3 - \sqrt{3})}{3}$.

EXAMPLE 8 Solve for x : $\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}$, $x \neq 2, 4$.

[CBSE 2014]

SOLUTION

The given equation is

$$\frac{x-1}{x-2} + \frac{x-3}{x-4} = \frac{10}{3}$$

$$\Rightarrow \frac{(x-1)(x-4) + (x-3)(x-2)}{(x-2)(x-4)} = \frac{10}{3}$$

$$\Rightarrow \frac{(x^2 - 5x + 4) + (x^2 - 5x + 6)}{(x^2 - 6x + 8)} = \frac{10}{3} \Rightarrow \frac{2x^2 - 10x + 10}{x^2 - 6x + 8} = \frac{10}{3}$$

$$\Rightarrow 3(2x^2 - 10x + 10) = 10(x^2 - 6x + 8)$$

$$\Rightarrow 6x^2 - 30x + 30 = 10x^2 - 60x + 80$$

$$\Rightarrow 4x^2 - 30x + 50 = 0 \Rightarrow 2x^2 - 15x + 25 = 0. \quad \dots (i)$$

This equation is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -15$ and $c = 25$.

$$\therefore D = (b^2 - 4ac) = \{(-15)^2 - 4 \times 2 \times 25\} = (225 - 200) = 25 > 0.$$

So, the given equation has real roots.

$$\text{Now, } \sqrt{D} = \sqrt{25} = 5.$$

$$\therefore \alpha = \frac{-b + \sqrt{D}}{2a} = \frac{(15 + 5)}{2 \times 2} = \frac{20}{4} = 5,$$

$$\beta = \frac{-b - \sqrt{D}}{2a} = \frac{(15 - 5)}{2 \times 2} = \frac{10}{4} = \frac{5}{2}.$$

Hence, the required values of x are 5 and $\frac{5}{2}$.

EXERCISE 4C

Find the discriminant of each of the following equations:

1. (i) $2x^2 - 7x + 6 = 0$ (ii) $3x^2 - 2x + 8 = 0$
- (iii) $2x^2 - 5\sqrt{2}x + 4 = 0$ (iv) $\sqrt{3}x^2 + 2\sqrt{2}x - 2\sqrt{3} = 0$
- (v) $(x-1)(2x-1) = 0$ (vi) $1-x = 2x^2$

Find the roots of each of the following equations, if they exist, by applying the quadratic formula:

2. $x^2 - 4x - 1 = 0$ 3. $x^2 - 6x + 4 = 0$
4. $2x^2 + x - 4 = 0$ 5. $25x^2 + 30x + 7 = 0$
6. $16x^2 = 24x + 1$ 7. $15x^2 - 28 = x$
8. $2x^2 - 2\sqrt{2}x + 1 = 0$ 9. $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ [CBSE 2013, '17]
10. $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$ [CBSE 2011]
11. $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ [CBSE 2015]
12. $2x^2 + 6\sqrt{3}x - 60 = 0$ [CBSE 2011, '15]
13. $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$ [CBSE 2013]
14. $3x^2 - 2\sqrt{6}x + 2 = 0$ [CBSE 2012]
15. $2\sqrt{3}x^2 - 5x + \sqrt{3} = 0$ [CBSE 2011]
16. $x^2 + x + 2 = 0$ 17. $2x^2 + ax - a^2 = 0$ [CBSE 2015]
18. $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$ [CBSE 2015]

HINT $D = (\sqrt{3} + 1)^2 - 4\sqrt{3} = (\sqrt{3} - 1)^2$.

19. $2x^2 + 5\sqrt{3}x + 6 = 0$ 20. $3x^2 - 2x + 2 = 0$
21. $x + \frac{1}{x} = 3, x \neq 0$ 22. $\frac{1}{x} - \frac{1}{x-2} = 3, x \neq 0, 2$ [CBSE 2010]
23. $x - \frac{1}{x} = 3, x \neq 0$ [CBSE 2010]
24. $\frac{m}{n}x^2 + \frac{n}{m} = 1 - 2x$

HINT $\frac{m}{n}x^2 + 2x + \left(\frac{n}{m} - 1\right) = 0 \Rightarrow m^2x^2 + 2mnx + (n^2 - mn) = 0$.

25. $36x^2 - 12ax + (a^2 - b^2) = 0$ 26. $x^2 - 2ax + (a^2 - b^2) = 0$
27. $x^2 - 2ax - (4b^2 - a^2) = 0$ [CBSE 2015]
28. $x^2 + 6x - (a^2 + 2a - 8) = 0$ [CBSE 2015]
29. $x^2 + 5x - (a^2 + a - 6) = 0$ [CBSE 2015]
30. $x^2 - 4ax - b^2 + 4a^2 = 0$ [CBSE 2012]
31. $4x^2 - 4a^2x + (a^4 - b^4) = 0$ [CBSE 2015]

32. $4x^2 + 4bx - (a^2 - b^2) = 0$ [CBSE 2015]
 33. $x^2 - (2b - 1)x + (b^2 - b - 20) = 0$ [CBSE 2015]
 34. $3a^2x^2 + 8abx + 4b^2 = 0, a \neq 0$
 35. $a^2b^2x^2 - (4b^4 - 3a^4)x - 12a^2b^2 = 0, a \neq 0$ and $b \neq 0$ [CBSE 2006]
 36. $12abx^2 - (9a^2 - 8b^2)x - 6ab = 0$, where $a \neq 0$ and $b \neq 0$ [CBSE 2009]

ANSWERS (EXERCISE 4C)

1. (i) 1 (ii) -92 (iii) 18 (iv) 32 (v) 1 (vi) 9
 2. $x = (2 + \sqrt{5})$ or $x = (2 - \sqrt{5})$ 3. $x = (3 + \sqrt{5})$ or $x = (3 - \sqrt{5})$
 4. $x = \frac{(-1 + \sqrt{33})}{4}$ or $x = \frac{(-1 - \sqrt{33})}{4}$ 5. $x = \frac{(-3 + \sqrt{2})}{5}$ or $x = \frac{(-3 - \sqrt{2})}{5}$
 6. $x = \frac{(3 + \sqrt{10})}{4}$ or $x = \frac{(3 - \sqrt{10})}{4}$ 7. $x = \frac{7}{5}$ or $x = -\frac{4}{3}$
 8. $x = \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ 9. $x = -\sqrt{2}$ or $x = \frac{-5}{\sqrt{2}}$ 10. $x = \frac{2\sqrt{3}}{3}$ or $x = -4\sqrt{3}$
 11. $x = \sqrt{6}$ or $x = \frac{-\sqrt{2}}{\sqrt{3}}$ 12. $x = 2\sqrt{3}$ or $x = -5\sqrt{3}$ 13. $x = \frac{\sqrt{3}}{4}$ or $x = \frac{-2}{\sqrt{3}}$
 14. $x = \sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}$ 15. $x = \frac{\sqrt{3}}{2}$ or $x = \frac{1}{\sqrt{3}}$ 16. Do not exist
 17. $x = \frac{a}{2}$ or $x = -a$ 18. $x = 1$ or $x = \sqrt{3}$ 19. $x = \frac{-\sqrt{3}}{2}$ or $x = -2\sqrt{3}$
 20. Do not exist 21. $x = \frac{3 + \sqrt{5}}{2}$ or $x = \frac{3 - \sqrt{5}}{2}$
 22. $x = \frac{3 + \sqrt{3}}{3}$ or $x = \frac{3 - \sqrt{3}}{3}$ 23. $x = \frac{3 + \sqrt{13}}{2}$ or $x = \frac{3 - \sqrt{13}}{2}$
 24. $x = \frac{-n + \sqrt{mn}}{m}$ or $x = \frac{n - \sqrt{mn}}{m}$ 25. $x = \frac{(a + b)}{6}$ or $x = \frac{(a - b)}{6}$
 26. $x = (a + b)$ or $x = (a - b)$ 27. $x = (a + 2b)$ or $x = (a - 2b)$
 28. $x = (a - 2)$ or $x = -(4 + a)$ 29. $x = (a - 2)$ or $x = -(a + 3)$
 30. $x = (2a - b)$ or $x = (2a + b)$ 31. $x = \frac{1}{2}(a^2 + b^2)$ or $x = \frac{1}{2}(a^2 - b^2)$
 32. $x = \frac{1}{2}(a - b)$ or $x = -\frac{1}{2}(a + b)$ 33. $x = (b + 4)$ or $x = (b - 5)$
 34. $x = \frac{-2b}{3a}$ or $x = \frac{-2b}{a}$ 35. $x = \frac{4b^2}{a^2}$ or $x = \frac{-3a^2}{b^2}$ 36. $x = \frac{3a}{4b}$ or $x = \frac{-2b}{3a}$
-

NATURE OF THE ROOTS OF A QUADRATIC EQUATION

Let the given equation be $ax^2 + bx + c = 0$, where $a \neq 0$. Then, the *discriminant* is given by $D = (b^2 - 4ac)$. And, the roots of the given equation are

$$\alpha = \frac{-b + \sqrt{D}}{2a} \text{ and } \beta = \frac{-b - \sqrt{D}}{2a}.$$

Case I When $D > 0$

In this case, the roots are *real* and *distinct*. These roots are given by

$$\alpha = \frac{-b + \sqrt{D}}{2a} \text{ and } \beta = \frac{-b - \sqrt{D}}{2a}.$$

Case II When $D = 0$

In this case, the roots are *real* and *equal*.

$$\text{Each root} = \frac{(-b)}{2a}.$$

Case III When $D < 0$

In this case, the roots are *imaginary*, and we say that the given equation has no real roots.

SUMMARY

Value	Nature of roots	Roots
$D > 0$	Real and unequal	$\frac{-b \pm \sqrt{D}}{2a}$
$D = 0$	Real and equal	Each root = $\frac{-b}{2a}$
$D < 0$	No real roots	None

SOLVED EXAMPLES

EXAMPLE 1 Find the nature of the roots of the quadratic equation $4x^2 - 5x + 3 = 0$.

SOLUTION The given equation is $4x^2 - 5x + 3 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 4$, $b = -5$ and $c = 3$.

$$\therefore D = (b^2 - 4ac) = \{(-5)^2 - 4 \times 4 \times 3\} = (25 - 48) = -23 < 0.$$

Hence, the given equation has no real roots.

EXAMPLE 2 Show that the equation $2x^2 - 6x + 3 = 0$ has real roots and find these roots.

SOLUTION The given equation is $2x^2 - 6x + 3 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -6$ and $c = 3$.

$$\therefore D = (b^2 - 4ac) = [(-6)^2 - 4 \times 2 \times 3] = (36 - 24) = 12 > 0.$$

So, the given equation has real unequal roots.

Solving $2x^2 - 6x + 3 = 0$ by quadratic formula, we have

$$x = \frac{6 \pm \sqrt{36 - 4 \times 2 \times 3}}{(2 \times 2)} = \frac{6 \pm \sqrt{36 - 24}}{4} = \frac{6 \pm \sqrt{12}}{4}$$

$$\Rightarrow x = \frac{6 \pm 2\sqrt{3}}{4} \Rightarrow x = \frac{3 \pm \sqrt{3}}{2}.$$

So, $\frac{(3 + \sqrt{3})}{2}$ and $\frac{(3 - \sqrt{3})}{2}$ are the roots of the given equation.

EXAMPLE 3 Show that the equation $x^2 + ax - 4 = 0$ has real and distinct roots for all real values of a .

SOLUTION The given equation is $x^2 + ax - 4 = 0$.

This is of the form $Ax^2 + Bx + C = 0$, where $A = 1$, $B = a$ and $C = -4$.

$$\therefore D = (B^2 - 4AC) = \{a^2 - 4 \times 1 \times (-4)\} = (a^2 + 16) > 0 \text{ for all real values of } a.$$

Thus, $D > 0$ for all real values of a .

Hence, the given equation has real and distinct roots for all real values of a .

EXAMPLE 4 Find the nature of the roots of the quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$ and hence solve it.

SOLUTION The given equation is $3x^2 - 4\sqrt{3}x + 4 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 3$, $b = -4\sqrt{3}$ and $c = 4$.

$$\therefore D = (b^2 - 4ac) = \{(-4\sqrt{3})^2 - 4 \times 3 \times 4\} = (48 - 48) = 0.$$

This shows that the given quadratic equation has real and equal roots.

$$\text{Each root} = \frac{-b}{2a} = \frac{4\sqrt{3}}{6} = \frac{2\sqrt{3}}{3}.$$

Hence, $\frac{2\sqrt{3}}{3}$ and $\frac{2\sqrt{3}}{3}$ are the roots of the given equation.

EXAMPLE 5 Find the values of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two real equal roots.

SOLUTION The given equation is $2x^2 + kx + 3 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = k$ and $c = 3$.

$$\therefore D = (b^2 - 4ac) = (k^2 - 4 \times 2 \times 3) = (k^2 - 24).$$

For real and equal roots, we must have

$$D = 0 \Rightarrow k^2 - 24 = 0 \Rightarrow k^2 = 24 \Rightarrow k = \pm\sqrt{24} = \pm 2\sqrt{6}.$$

Hence, $2\sqrt{6}$ and $-2\sqrt{6}$ are the required values of k .

EXAMPLE 6 Find the value of k for which the roots of the quadratic equation $kx(x - 2) + 6 = 0$ are equal. [CBSE 2017]

SOLUTION The given equation is $kx^2 - 2kx + 6 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = k$, $b = -2k$ and $c = 6$.

$$\therefore D = (b^2 - 4ac) = (4k^2 - 4 \times k \times 6) = (4k^2 - 24k).$$

For equal roots, we must have

$$D = 0 \Rightarrow 4k^2 - 24k = 0 \Rightarrow 4k(k - 6) = 0 \Rightarrow k = 0 \text{ or } k = 6.$$

Now, $k = 0$, we get $6 = 0$, which is absurd.

$$\therefore k \neq 0 \text{ and hence } k = 6.$$

EXAMPLE 7 Find the value of k for which the quadratic equation $(k + 4)x^2 + (k + 1)x + 1 = 0$ has two real equal roots. [CBSE 2014]

SOLUTION The given equation is $(k + 4)x^2 + (k + 1)x + 1 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = (k + 4)$, $b = (k + 1)$ and $c = 1$.

$$\begin{aligned} \therefore D &= (b^2 - 4ac) = (k + 1)^2 - 4 \times (k + 4) \times 1 = (k + 1)^2 - 4(k + 4) \\ &= (k^2 + 1 + 2k - 4k - 16) = (k^2 - 2k - 15). \end{aligned}$$

For equal roots, we must have

$$\begin{aligned} D = 0 &\Rightarrow k^2 - 2k - 15 = 0 \\ &\Rightarrow k^2 - 5k + 3k - 15 = 0 \Rightarrow k(k - 5) + 3(k - 5) = 0 \\ &\Rightarrow (k - 5)(k + 3) = 0 \Rightarrow k - 5 = 0 \text{ or } k + 3 = 0 \\ &\Rightarrow k = 5 \text{ or } k = -3. \end{aligned}$$

Hence, the required value of k is 5 or -3.

EXAMPLE 8 Find the nonzero value of k for which the quadratic equation $kx^2 + 1 - 2(k - 1)x + x^2 = 0$ has equal roots. Hence, find the roots of the equation. [CBSE 2015]

SOLUTION The given equation is $(k+1)x^2 - 2(k-1)x + 1 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = (k+1)$, $b = -2(k-1)$ and $c = 1$.

$$\therefore D = (b^2 - 4ac) = \{4(k-1)^2 - 4 \times (k+1) \times 1\} = 4(k^2 - 3k).$$

For equal roots, we must have

$$D = 0 \Rightarrow 4(k^2 - 3k) = 0 \Rightarrow 4k(k-3) = 0 \Rightarrow k = 0 \text{ or } k = 3.$$

$\therefore$ the required nonzero value of k is 3.

Putting $k = 3$, the given equation becomes

$$4x^2 - 4x + 1 = 0 \Rightarrow (2x-1)^2 = 0 \Rightarrow (2x-1) = 0 \Rightarrow x = \frac{1}{2}.$$

Hence, the required roots are $\frac{1}{2}$ and $\frac{1}{2}$.

EXAMPLE 9 If -4 is a root of the equation $x^2 + px - 4 = 0$ and the equation $x^2 + px + q = 0$ has equal roots, find the values of p and q .

SOLUTION Since -4 is a root of the equation $x^2 + px - 4 = 0$, we have

$$(-4)^2 + p(-4) - 4 = 0 \Rightarrow 4p = (16 - 4) = 12 \Rightarrow p = 3.$$

Now, the roots of $x^2 + px + q = 0$ being equal, we have

$$p^2 - 4q = 0 \Rightarrow 3^2 - 4q = 0 \Rightarrow 4q = 9 \Rightarrow q = \frac{9}{4}.$$

Hence, $p = 3$ and $q = \frac{9}{4}$.

EXAMPLE 10 If -2 is a root of the equation $3x^2 + 7x + p = 0$, find the value of k so that the roots of the equation $x^2 + k(4x + k - 1) + p = 0$ are equal.

[CBSE 2015]

SOLUTION Since -2 is a root of the equation $3x^2 + 7x + p = 0$, we have

$$3 \times (-2)^2 + 7 \times (-2) + p = 0 \Rightarrow 12 - 14 + p = 0 \Rightarrow p = 2.$$

For $p = 2$, the other given equation becomes

$$x^2 + 4kx + k(k-1) + 2 = 0.$$

This is of the form $ax^2 + bx + c = 0$, where

$$a = 1, b = 4k \text{ and } c = (k^2 - k + 2).$$

$$\therefore D = (b^2 - 4ac) = \{16k^2 - 4 \times 1 \times (k^2 - k + 2)\} = (12k^2 + 4k - 8).$$

For equal roots we must have

$$D = 0 \Rightarrow 12k^2 + 4k - 8 = 0 \Rightarrow 4(3k^2 + k - 2) = 0$$

$$\Rightarrow 3k^2 + k - 2 = 0 \Rightarrow 3k^2 + 3k - 2k - 2 = 0$$

$$\Rightarrow 3k(k+1) - 2(k+1) = 0 \Rightarrow (k+1)(3k-2) = 0$$

$$\Rightarrow k+1=0 \text{ or } 3k-2=0 \Rightarrow k=-1 \text{ or } k=\frac{2}{3}.$$

Hence, the required value of k is -1 or $\frac{2}{3}$.

EXAMPLE 11 Prove that both the roots of the equation

$$(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$$

are real but they are equal only when $a = b = c$.

SOLUTION The given equation may be written as

$$3x^2 - 2x(a+b+c) + (ab+bc+ca) = 0.$$

$$\therefore D = 4(a+b+c)^2 - 12(ab+bc+ca)$$

$$= 4[(a+b+c)^2 - 3(ab+bc+ca)]$$

$$= 4(a^2+b^2+c^2-ab-bc-ca)$$

$$= 2(2a^2+2b^2+2c^2-2ab-2bc-2ca)$$

$$= 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$$

$$[\because (a-b)^2 \geq 0, (b-c)^2 \geq 0 \text{ and } (c-a)^2 \geq 0].$$

This shows that both the roots of the given equation are real.

For equal roots, we must have $D = 0$.

$$\text{Now, } D = 0 \Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$$\Rightarrow (a-b) = 0, (b-c) = 0 \text{ and } (c-a) = 0$$

$$\Rightarrow a = b = c.$$

Hence, the roots are equal only when $a = b = c$.

EXAMPLE 12 If the roots of the equation $(b-c)x^2 + (c-a)x + (a-b) = 0$ are equal, prove that $2b = a + c$. [CBSE 2002C, '06, '17]

SOLUTION Clearly, $x = 1$ satisfies the given equation.

Since its roots are equal, so 1 and 1 are its roots.

$$\therefore \text{ product of roots of the given equation } = (1 \times 1) = 1.$$

$$\text{But, product of roots } = \frac{a-b}{b-c}. \quad [\because \text{ product of roots } = \frac{C}{A}]$$

$$\therefore \frac{a-b}{b-c} = 1 \Rightarrow a-b = b-c \Rightarrow 2b = a+c.$$

Hence, $2b = a + c$.

EXAMPLE 13 Show that the equation $3x^2 + 7x + 8 = 0$ is not true for any real value of x .

SOLUTION The given equation is $3x^2 + 7x + 8 = 0$.

$$\therefore D = (7^2 - 4 \times 3 \times 8) = (49 - 96) = -47 < 0.$$

So, the given equation has no real roots.

Hence, the given equation is not true for any real value of x .

EXAMPLE 14 Show that the equation $2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$ has no real roots, when $a \neq b$.

SOLUTION The given equation is $2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$.

$$\begin{aligned}\therefore D &= 4(a + b)^2 - 8(a^2 + b^2) \\ &= -4(a^2 + b^2 - 2ab) = -4(a - b)^2 < 0, \text{ when } a - b \neq 0.\end{aligned}$$

So, the given equation has no real roots, when $a \neq b$.

EXAMPLE 15 Find the values of k for which the equation $x^2 + 5kx + 16 = 0$ has no real roots. [CBSE 2013C]

SOLUTION The given equation is $x^2 + 5kx + 16 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 1, b = 5k$ and $c = 16$.

$$\therefore D = (b^2 - 4ac) = (25k^2 - 4 \times 1 \times 16) = (25k^2 - 64).$$

Since, the given equation has no real root, we have

$$D < 0 \Rightarrow 25k^2 - 64 < 0 \Rightarrow 25k^2 < 64$$

$$\Rightarrow k^2 < \frac{64}{25} \Rightarrow k^2 < \left(\frac{8}{5}\right)^2$$

$$\Rightarrow \frac{-8}{5} < k < \frac{8}{5}.$$

Hence, the required real values of k are such that $\frac{-8}{5} < k < \frac{8}{5}$.

EXAMPLE 16 Find the values of k for which the given equation has real roots:

(i) $kx^2 - 6x - 2 = 0$ (ii) $3x^2 + 2x + k = 0$ (iii) $2x^2 + kx + 2 = 0$

SOLUTION (i) The given equation is $kx^2 - 6x - 2 = 0$.

$$\therefore D = [(-6)^2 - 4 \times k \times (-2)] = (36 + 8k).$$

The given equation will have real roots if $D \geq 0$.

$$\text{Now, } D \geq 0 \Rightarrow 36 + 8k \geq 0 \Rightarrow k \geq \frac{-36}{8} \Rightarrow k \geq \frac{-9}{2}.$$

(ii) The given equation is $3x^2 + 2x + k = 0$.

$$\therefore D = (2^2 - 4 \times 3 \times k) = (4 - 12k).$$

The given equation will have real roots if $D \geq 0$.

$$\text{Now, } D \geq 0 \Rightarrow 4 - 12k \geq 0 \Rightarrow 12k \leq 4 \Rightarrow k \leq \frac{1}{3}.$$

(iii) The given equation is $2x^2 + kx + 2 = 0$.

$$\therefore D = (k^2 - 4 \times 2 \times 2) = (k^2 - 16).$$

The given equation will have real roots if $D \geq 0$.

$$\text{Now, } D \geq 0 \Rightarrow (k^2 - 16) \geq 0 \Rightarrow k^2 \geq 16 \Rightarrow k \geq 4 \text{ or } k \leq -4.$$

EXAMPLE 17 Determine the positive value of p for which the equations $x^2 + 2px + 64 = 0$ and $x^2 - 8x + 2p = 0$ will both have real roots.

[CBSE 2013C]

SOLUTION Let D_1 and D_2 be the discriminants of the first and second given equations respectively.

For real roots, we must have $D_1 \geq 0$ and $D_2 \geq 0$.

Now, $D_1 \geq 0$ and $D_2 \geq 0$

$$\Rightarrow (4p^2 - 4 \times 64) \geq 0 \text{ and } (64 - 8p) \geq 0$$

$$\Rightarrow p^2 - 64 \geq 0 \text{ and } 64 - 8p \geq 0$$

$$\Rightarrow p^2 \geq 64 \text{ and } 8p \leq 64$$

$$\Rightarrow p \geq 8 \text{ and } p \leq 8 \quad [\because p \text{ is positive}]$$

$$\Rightarrow p = 8.$$

Hence, $p = 8$.

EXERCISE 4D

1. Find the nature of the roots of the following quadratic equations:

(i) $2x^2 - 8x + 5 = 0$

(ii) $3x^2 - 2\sqrt{6}x + 2 = 0$

(iii) $5x^2 - 4x + 1 = 0$

(iv) $5x(x - 2) + 6 = 0$

(v) $12x^2 - 4\sqrt{15}x + 5 = 0$

(vi) $x^2 - x + 2 = 0$

2. If a and b are distinct real numbers, show that the quadratic equation $2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$ has no real roots.

3. Show that the roots of the equation $x^2 + px - q^2 = 0$ are real for all real values of p and q .

4. For what values of k are the roots of the quadratic equation $3x^2 + 2kx + 27 = 0$ real and equal? [CBSE 2008C]

5. For what value of k are the roots of the quadratic equation $kx(x - 2\sqrt{5}) + 10 = 0$ real and equal? [CBSE 2013]

6. For what values of p are the roots of the equation $4x^2 + px + 3 = 0$ real and equal? [CBSE 2014]

7. Find the nonzero value of k for which the roots of the quadratic equation $9x^2 - 3kx + k = 0$ are real and equal. [CBSE 2014]
8. (i) Find the values of k for which the quadratic equation $(3k + 1)x^2 + 2(k + 1)x + 1 = 0$ has real and equal roots. [CBSE 2014]
(ii) Find the value of k for which the equation $x^2 + k(2x + k - 1) + 2 = 0$ has real and equal roots. [CBSE 2017]
9. Find the values of p for which the quadratic equation $(2p + 1)x^2 - (7p + 2)x + (7p - 3) = 0$ has real and equal roots. [CBSE 2014]
10. Find the values of p for which the quadratic equation $(p + 1)x^2 - 6(p + 1)x + 3(p + 9) = 0$, $p \neq -1$ has equal roots. Hence, find the roots of the equation. [CBSE 2015]
11. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find the value of k . [CBSE 2014]
12. If 3 is a root of the quadratic equation $x^2 - x + k = 0$, find the value of p so that the roots of the equation $x^2 + k(2x + k + 2) + p = 0$ are equal. [CBSE 2015]
13. If -4 is a root of the equation $x^2 + 2x + 4p = 0$, find the value of k for which the quadratic equation $x^2 + px(1 + 3k) + 7(3 + 2k) = 0$ has equal roots. [CBSE 2015]
14. If the quadratic equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, prove that $c^2 = a^2(1 + m^2)$. [CBSE 2014, '17]
15. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are real and equal, show that either $a = 0$ or $(a^3 + b^3 + c^3) = 3abc$. [CBSE 2017]
16. Find the values of p for which the quadratic equation $2x^2 + px + 8 = 0$ has real roots.
17. Find the value of α for which the equation $(\alpha - 12)x^2 + 2(\alpha - 12)x + 2 = 0$ has equal roots. [CBSE 2013]
18. Find the value of k for which the roots of $9x^2 + 8kx + 16 = 0$ are real and equal.
19. Find the values of k for which the given quadratic equation has real and distinct roots:
(i) $kx^2 + 6x + 1 = 0$ (ii) $x^2 - kx + 9 = 0$
(iii) $9x^2 + 3kx + 4 = 0$ (iv) $5x^2 - kx + 1 = 0$
20. If a and b are real and $a \neq b$ then show that the roots of the equation $(a - b)x^2 + 5(a + b)x - 2(a - b) = 0$ are real and unequal.

21. (i) If the roots of the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal, prove that $\frac{a}{b} = \frac{c}{d}$. [CBSE 2017]
- (ii) If $ad \neq bc$ then prove that the equation $(a^2 + b^2)x^2 + 2(ac + bd)x + (c^2 + d^2) = 0$ has no real roots. [CBSE 2017]
22. If the roots of the equations $ax^2 + 2bx + c = 0$ and $bx^2 - 2\sqrt{ac}x + b = 0$ are simultaneously real then prove that $b^2 = ac$.

ANSWERS (EXERCISE 4D)

1. (i) Real and unequal (ii) Real and equal (iii) Not real (iv) Not real
(v) Real and equal (vi) Not real
4. $k = 9$ or $k = -9$ 5. $k = 0$ or $k = 2$ 6. $p = 4\sqrt{3}$ or $p = -4\sqrt{3}$
7. $k = 4$ 8. (i) $k = 0$ or $k = 1$ (ii) $k = 2$ 9. $p = 4$ or $p = -\frac{4}{7}$
10. $p = 3$ or $p = -1$ 11. $k = \frac{49}{28}$ 12. $p = 12$ 13. $k = 2$ or $k = -\frac{10}{9}$
16. $p \geq 8$ or $p \leq -8$ 17. $\alpha = 14$ 18. $k = 3$ or $k = -3$
19. (i) $k < 9$ (ii) $k > 6$ or $k < -6$ (iii) $k > 4$ or $k < -4$ (iv) $k > 2\sqrt{5}$ or $k < -2\sqrt{5}$

HINTS TO SOME SELECTED QUESTIONS

2. $D = 4(a+b)^2 - 8(a^2 + b^2)$
 $= 4[(a+b)^2 - 2(a^2 + b^2)] = 4(-a^2 - b^2 + 2ab)$
 $= -4(a^2 + b^2 - 2ab) = -4(a-b)^2 < 0$.
3. $D = (p^2 + 4q^2) \geq 0$.
4. $D = 4k^2 - 324$. For real and equal roots, we must have, $D = 0$.
5. Given equation is $kx^2 - 2\sqrt{5}kx + 10 = 0$.
 $\therefore D = 20k^2 - 40k = 20k(k-2)$.
 For real and equal roots, we must have $D = 0$.
6. For real and equal roots, we must have $D = 0$.
 $\therefore D = 0 \Rightarrow p^2 - 48 = 0 \Rightarrow p^2 = 48 \Rightarrow p = \pm\sqrt{48} \Rightarrow p = 4\sqrt{3}$ or $-4\sqrt{3}$.
9. $D = (7p+2)^2 - 4(2p+1)(7p-3) = -7p^2 + 24p + 16$.
 $\therefore D = 0 \Rightarrow 7p^2 - 24p - 16 = 0 \Rightarrow 7p^2 - 28p + 4p - 16 = 0$
 $\Rightarrow 7p(p-4) + 4(p-4) = 0 \Rightarrow (p-4)(7p+4) = 0$.
15. $D = 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac)$
 Now, $D = 0 \Rightarrow (a^2 - bc)^2 - (c^2 - ab)(b^2 - ac) = 0$
 $\Rightarrow a^4 - 3a^2bc + ac^3 + ab^3 = 0$

$$\Rightarrow a(a^3 + b^3 + c^3 - 3abc) = 0$$

$$\Rightarrow a = 0 \text{ or } a^3 + b^3 + c^3 = 3abc.$$

16. $D = (p^2 - 64)$. So, $D \geq 0 \Rightarrow p^2 \geq 64 \Rightarrow p \geq 8 \text{ or } p \leq -8$.

19. (i) $36 - 4k > 0 \Rightarrow 36 > 4k \Rightarrow 4k < 36 \Rightarrow k < 9$.

(ii) $k^2 - 36 > 0 \Rightarrow k^2 > 36 \Rightarrow k > 6 \text{ or } k < -6$.

(iii) $9k^2 - 144 > 0 \Rightarrow k^2 > 16 \Rightarrow k > 4 \text{ or } k < -4$.

(iv) $k^2 - 20 > 0 \Rightarrow k^2 > 20 \Rightarrow k > 2\sqrt{5} \text{ or } k < -2\sqrt{5}$.

20. $D = 25(a+b)^2 + 8(a-b)^2$

$$= 17(a+b)^2 + 8[(a+b)^2 + (a-b)^2] = 17(a+b)^2 + 16(a^2 + b^2) > 0.$$

21. (i) $4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = 0$

$$\Rightarrow (a^2 + b^2)(c^2 + d^2) - (ac + bd)^2 = 0$$

$$\Rightarrow a^2d^2 + b^2c^2 - 2abcd = 0 \Rightarrow (ad - bc)^2 = 0 \Rightarrow ad - bc = 0$$

$$\Rightarrow ad = bc \Rightarrow \frac{a}{b} = \frac{c}{d}.$$

22. $4b^2 - 4ac \geq 0 \Rightarrow b^2 - ac \geq 0 \Rightarrow b^2 \geq ac$.

$$4ac - 4b^2 \geq 0 \Rightarrow ac - b^2 \geq 0 \Rightarrow b^2 \leq ac$$

Hence, $b^2 = ac$.

WORD PROBLEMS ON QUADRATIC EQUATIONS

SOLVED EXAMPLES

PROBLEMS ON NUMBERS

EXAMPLE 1 *If the sum of two natural numbers is 27 and their product is 182, find the numbers.*

SOLUTION Let the required numbers be x and $(27 - x)$. Then,

$$x(27 - x) = 182$$

$$\Rightarrow 27x - x^2 = 182 \Rightarrow x^2 - 27x + 182 = 0$$

$$\Rightarrow x^2 - 13x - 14x + 182 = 0$$

$$\Rightarrow x(x - 13) - 14(x - 13) = 0$$

$$\Rightarrow (x - 13)(x - 14) = 0$$

$$\Rightarrow x - 13 = 0 \text{ or } x - 14 = 0$$

$$\Rightarrow x = 13 \text{ or } x = 14.$$

Hence, the required numbers are 13 and 14.

EXAMPLE 2 *The sum of the squares of two consecutive odd numbers is 394. Find the numbers.* [CBSE 2014]

SOLUTION Let the required consecutive odd numbers be x and $(x + 2)$. Then,

$$x^2 + (x + 2)^2 = 394$$

$$\Rightarrow 2x^2 + 4x - 390 = 0 \Rightarrow x^2 + 2x - 195 = 0$$

$$\Rightarrow x^2 + 15x - 13x - 195 = 0$$

$$\Rightarrow x(x + 15) - 13(x + 15) = 0$$

$$\Rightarrow (x + 15)(x - 13) = 0$$

$$\Rightarrow x + 15 = 0 \text{ or } x - 13 = 0$$

$$\Rightarrow x = -15 \text{ or } x = 13$$

$$\Rightarrow x = 13. \quad [\text{rejecting } x = -15]$$

Hence, the required numbers are 13 and 15.

EXAMPLE 3 *The sum of the squares of two consecutive even numbers is 340. Find the numbers.* [CBSE 2014]

SOLUTION Let the required consecutive even numbers be x and $(x + 2)$. Then,

$$x^2 + (x + 2)^2 = 340$$

$$\Rightarrow 2x^2 + 4x - 336 = 0 \Rightarrow x^2 + 2x - 168 = 0$$

$$\Rightarrow x^2 + 14x - 12x - 168 = 0$$

$$\Rightarrow x(x + 14) - 12(x + 14) = 0$$

$$\Rightarrow (x + 14)(x - 12) = 0$$

$$\Rightarrow x + 14 = 0 \text{ or } x - 12 = 0$$

$$\Rightarrow x = -14 \text{ or } x = 12$$

$$\Rightarrow x = 12 \quad [\text{rejecting } x = -14]$$

Hence, the required numbers are 12 and 14.

EXAMPLE 4 *The sum of the squares of two consecutive multiples of 7 is 637. Find the multiples.* [CBSE 2014]

SOLUTION Let the required consecutive multiples of 7 be $7x$ and $7(x + 1)$.

$$\text{Then, } (7x)^2 + \{7(x + 1)\}^2 = 637$$

$$\Rightarrow 49x^2 + (7x + 7)^2 = 637 \Rightarrow 98x^2 + 98x - 588 = 0$$

$$\Rightarrow x^2 + x - 6 = 0 \Rightarrow x^2 + 3x - 2x - 6 = 0$$

$$\Rightarrow x(x+3) - 2(x+3) = 0 \Rightarrow (x+3)(x-2) = 0$$

$$\Rightarrow x+3 = 0 \text{ or } x-2 = 0$$

$$\Rightarrow x = -3 \text{ or } x = 2$$

$$\Rightarrow x = 2 \quad [\text{neglecting } x = -3]$$

Hence, the required numbers are (7×2) and (7×3) , i.e., 14 and 21.

EXAMPLE 5

The sum of two natural numbers is 9 and the sum of their reciprocals is $\frac{1}{2}$. Find the numbers. [CBSE 2012]

SOLUTION

Let the required natural numbers be x and $(9-x)$. Then,

$$\frac{1}{x} + \frac{1}{(9-x)} = \frac{1}{2}$$

$$\Rightarrow \frac{(9-x) + x}{x(9-x)} = \frac{1}{2}$$

$$\Rightarrow x(9-x) = 18 \quad [\text{by cross multiplication}]$$

$$\Rightarrow x^2 - 9x + 18 = 0 \Rightarrow x^2 - 6x - 3x + 18 = 0$$

$$\Rightarrow x(x-6) - 3(x-6) = 0 \Rightarrow (x-6)(x-3) = 0$$

$$\Rightarrow x-6 = 0 \text{ or } x-3 = 0 \Rightarrow x = 6 \text{ or } x = 3.$$

Hence, the required natural numbers are 6 and 3.

EXAMPLE 6

The difference of two natural numbers is 5 and the difference of their reciprocals is $\frac{1}{10}$. Find the numbers. [CBSE 2014]

SOLUTION

Let the required natural numbers be x and $(x-5)$. Then,

$$x > x-5 \Rightarrow \frac{1}{x} < \frac{1}{x-5} \Rightarrow \frac{1}{x-5} > \frac{1}{x}.$$

$$\therefore \frac{1}{x-5} - \frac{1}{x} = \frac{1}{10}$$

$$\Rightarrow \frac{x-(x-5)}{(x-5)x} = \frac{1}{10} \Rightarrow \frac{5}{(x-5)x} = \frac{1}{10} \quad [\text{by cross multiplication}]$$

$$\Rightarrow (x-5)x = 50 \Rightarrow x^2 - 5x - 50 = 0$$

$$\Rightarrow x^2 - 10x + 5x - 50 = 0 \Rightarrow x(x-10) + 5(x-10) = 0$$

$$\Rightarrow (x-10)(x+5) = 0$$

$$\Rightarrow x-10 = 0 \text{ or } x+5 = 0$$

$$\Rightarrow x = 10 \text{ or } x = -5$$

$$\Rightarrow x = 10 \quad [\because -5 \text{ is not a natural number}]$$

Hence, the required natural numbers are 10 and 5.

EXAMPLE 7 *The difference of squares of two numbers is 180. The square of the smaller number is 8 times the larger number. Find the two numbers.*

[CBSE 2012]

SOLUTION Let the required numbers be x and y such that $x > y$.

$$\text{Then, } x^2 - y^2 = 180. \quad \dots (i)$$

$$\text{And, } y^2 = 8x. \quad \dots (ii)$$

From (i) and (ii), we get

$$x^2 - 8x - 180 = 0$$

$$\Rightarrow x^2 - 18x + 10x - 180 = 0 \Rightarrow x(x - 18) + 10(x - 18) = 0$$

$$\Rightarrow (x - 18)(x + 10) = 0 \Rightarrow x - 18 = 0 \text{ or } x + 10 = 0$$

$$\Rightarrow x = 18 \text{ or } x = -10.$$

$$\text{Now, } x = 18 \Rightarrow y^2 = (8 \times 18) = 144$$

$$\Rightarrow y = 12 \text{ or } y = -12.$$

$$\text{Also, } x = -10 \Rightarrow y^2 = [8 \times (-10)] = -80, \text{ which is not possible.}$$

Hence, the numbers are (18 and 12) or (18 and -12).

EXAMPLE 8 *The numerator of a fraction is 3 less than its denominator. If 2 is added to both of its numerator and denominator then the sum of the new fraction and original fraction is $\frac{29}{20}$. Find the original fraction.*

[CBSE 2015]

SOLUTION Let the denominator of the required fraction be x .

Then, its numerator = $(x - 3)$.

So, the original fraction is $\frac{(x-3)}{x}$.

$$\therefore \frac{(x-3)+2}{x+2} + \frac{(x-3)}{x} = \frac{29}{20}$$

$$\Rightarrow \frac{(x-1)}{(x+2)} + \frac{(x-3)}{x} = \frac{29}{20}$$

$$\Rightarrow \frac{x(x-1) + (x-3)(x+2)}{(x+2)x} = \frac{29}{20}$$

$$\Rightarrow \frac{(x^2 - x) + (x^2 - x - 6)}{x^2 + 2x} = \frac{29}{20}$$

$$\Rightarrow 20(2x^2 - 2x - 6) = 29(x^2 + 2x)$$

$$\Rightarrow 40x^2 - 40x - 120 = 29x^2 + 58x$$

$$\Rightarrow 11x^2 - 98x - 120 = 0$$

$$\Rightarrow 11x^2 - 110x + 12x - 120 = 0$$

$$\begin{aligned}
 &\Rightarrow 11x(x-10) + 12(x-10) = 0 \\
 &\Rightarrow (x-10)(11x+12) = 0 \\
 &\Rightarrow x-10 = 0 \text{ or } 11x+12 = 0 \\
 &\Rightarrow x = 10 \text{ or } x = \frac{-12}{11} \\
 &\Rightarrow x = 10 \quad [\because \text{denominator of a fraction is never negative}] \\
 &\therefore \text{denominator} = 10 \text{ and numerator} = (10-3) = 7. \\
 &\text{Hence, the required fraction is } \frac{7}{10}.
 \end{aligned}$$

EXAMPLE 9 *The denominator of a fraction is one more than twice the numerator. If the sum of the fraction and its reciprocal is $2\frac{16}{21}$, find the fraction.*

SOLUTION Let the numerator of the required fraction be x .
 Then, its denominator = $(2x+1)$.

$$\begin{aligned}
 \therefore \text{fraction} &= \frac{x}{(2x+1)} \text{ and its reciprocal} = \frac{(2x+1)}{x}. \\
 \therefore \frac{x}{(2x+1)} + \frac{(2x+1)}{x} &= \frac{58}{21} \\
 \Rightarrow 21 \times [x^2 + (2x+1)^2] &= 58x(2x+1) \\
 \Rightarrow 21 \times [5x^2 + 4x + 1] &= 116x^2 + 58x \\
 \Rightarrow 11x^2 - 26x - 21 &= 0 \\
 \Rightarrow 11x^2 - 33x + 7x - 21 &= 0 \Rightarrow 11x(x-3) + 7(x-3) = 0 \\
 \Rightarrow (x-3)(11x+7) &= 0 \Rightarrow x-3 = 0 \text{ or } 11x+7 = 0 \\
 \Rightarrow x = 3 \text{ or } x &= \frac{-7}{11} \\
 \therefore x = 3 \quad [\because \text{numerator cannot be a negative fraction}]. \\
 \therefore \text{required fraction} &= \frac{x}{(2x+1)} = \frac{3}{7}.
 \end{aligned}$$

EXAMPLE 10 *A two-digit number is 5 times the sum of its digits and is also equal to 5 more than twice the product of its digits. Find the number.*

SOLUTION Let the tens and units digits of the required number be x and y respectively. Then,

$$10x + y = 5(x + y) \Rightarrow 4y = 5x \Rightarrow y = \frac{5x}{4} \quad \dots (i)$$

$$10x + y = 2xy + 5 \Rightarrow 10x + \frac{5x}{4} = 2x \times \frac{5x}{4} + 5 \quad [\text{using (i)}]$$

$$\Rightarrow \frac{45x}{4} = \frac{10x^2}{4} + 5 \Rightarrow 10x^2 - 45x + 20 = 0$$

$$\Rightarrow 2x^2 - 9x + 4 = 0 \Rightarrow 2x^2 - 8x - x + 4 = 0$$

$$\Rightarrow 2x(x-4) - (x-4) = 0$$

$$\Rightarrow (x-4)(2x-1) = 0$$

$$\Rightarrow x-4 = 0 \text{ or } 2x-1 = 0$$

$$\Rightarrow x = 4 \text{ or } x = \frac{1}{2}$$

$$\Rightarrow x = 4 \quad [\because \text{a digit cannot be a fraction}].$$

Putting $x = 4$ in (i), we get $y = 5$.

$$\therefore x = 4 \text{ and } y = 5.$$

Hence, the required number is 45.

EXAMPLE 11 *A two-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number.* [CBSE 2006C]

SOLUTION Let the tens and units digits of the required number be x and y respectively. Then,

$$xy = 18 \Rightarrow y = \frac{18}{x} \quad \dots (i)$$

$$\text{And, } (10x + y) - 63 = 10y + x$$

$$\Rightarrow 9x - 9y = 63 \Rightarrow x - y = 7 \quad \dots (ii)$$

Putting $y = \frac{18}{x}$ from (i) into (ii), we get

$$x - \frac{18}{x} = 7$$

$$\Rightarrow x^2 - 18 = 7x \Rightarrow x^2 - 7x - 18 = 0$$

$$\Rightarrow x^2 - 9x + 2x - 18 = 0 \Rightarrow x(x-9) + 2(x-9) = 0$$

$$\Rightarrow (x-9)(x+2) = 0 \Rightarrow x-9 = 0 \text{ or } x+2 = 0$$

$$\Rightarrow x = 9 \text{ or } x = -2$$

$$\Rightarrow x = 9 \quad [\because \text{a digit cannot be negative}].$$

Putting $x = 9$ in (i), we get $y = 2$.

Thus, the tens digit is 9 and the units digit is 2.

Hence, the required number is 92.

GENERAL PROBLEMS ON MONEY MATTERS

EXAMPLE 12 *A person on tour has ₹ 4200 for his expenses. If he extends his tour for 3 days, he has to cut down his daily expenses by ₹ 70. Find the original duration of the tour.* [CBSE 2008C]

SOLUTION Let the original duration of the tour be x days. Then,

$$\begin{aligned} \frac{4200}{x} - \frac{4200}{(x+3)} &= 70 \\ \Rightarrow 4200 \times \left[\frac{1}{x} - \frac{1}{(x+3)} \right] &= 70 \Rightarrow \frac{(x+3) - x}{x(x+3)} = \frac{70}{4200} \\ \Rightarrow x(x+3) &= 180 \Rightarrow x^2 + 3x - 180 = 0 \\ \Rightarrow x^2 + 15x - 12x - 180 &= 0 \Rightarrow x(x+15) - 12(x+15) = 0 \\ \Rightarrow (x+15)(x-12) &= 0 \Rightarrow x+15 = 0 \text{ or } x-12 = 0 \\ \Rightarrow x &= -15 \text{ or } x = 12 \\ \Rightarrow x &= 12 \quad [\because \text{number of days cannot be negative}]. \\ \therefore \text{original duration of the tour is } &12 \text{ days.} \end{aligned}$$

EXAMPLE 13 A bookseller buys a number of books for ₹ 1760. If he had bought 4 more books for the same amount, each book would have cost ₹ 22 less. How many books did he buy?

SOLUTION Let the bookseller buy x books for ₹ 1760.

$$\text{Then, cost of each book} = ₹ \frac{1760}{x}.$$

$$\text{Again, cost of } (x+4) \text{ books} = ₹ 1760$$

$$\therefore \text{cost of each book, now} = ₹ \frac{1760}{(x+4)}.$$

$$\therefore \frac{1760}{x} - \frac{1760}{(x+4)} = 22$$

$$\Rightarrow \frac{1}{x} - \frac{1}{(x+4)} = \frac{22}{1760} \Rightarrow \frac{(x+4) - x}{x(x+4)} = \frac{1}{80}$$

$$\Rightarrow \frac{4}{(x^2 + 4x)} = \frac{1}{80} \Rightarrow x^2 + 4x = 320$$

$$\Rightarrow x^2 + 4x - 320 = 0 \Rightarrow x^2 + 20x - 16x - 320 = 0$$

$$\Rightarrow x(x+20) - 16(x+20) = 0 \Rightarrow (x+20)(x-16) = 0$$

$$\Rightarrow x+20 = 0 \text{ or } x-16 = 0 \Rightarrow x = -20 \text{ or } x = 16$$

$$\Rightarrow x = 16 \quad [\because \text{number of books cannot be negative}]$$

Hence, the bookseller bought 16 books.

EXAMPLE 14 Some students planned a picnic. The total budget for hiring a bus was ₹ 1440. Later on, eight of these refused to go and instead paid their total share of money towards the fee of one economically weaker student of their class, and thus, the cost for each member who went for picnic, increased by ₹ 30.

(i) How many students attended the picnic?

(ii) How much money in total was paid towards the fee? Which value is reflected in this question? [CBSE 2013C]

SOLUTION

Let x students planned the picnic.

Then, $(x - 8)$ students attended the picnic.

Total bus charges = ₹ 1440.

$$\therefore \frac{1440}{(x-8)} - \frac{1440}{x} = 30$$

$$\Rightarrow \frac{1}{(x-8)} - \frac{1}{x} = \frac{30}{1440} \Rightarrow \frac{x - (x-8)}{(x-8)x} = \frac{1}{48}$$

$$\Rightarrow \frac{8}{(x^2 - 8x)} = \frac{1}{48} \Rightarrow x^2 - 8x = 384 \quad [\text{by cross multiplication}]$$

$$\Rightarrow x^2 - 8x - 384 = 0 \Rightarrow x^2 - 24x + 16x - 384 = 0$$

$$\Rightarrow x(x - 24) + 16(x - 24) = 0 \Rightarrow (x - 24)(x + 16) = 0$$

$$\Rightarrow x - 24 = 0 \text{ or } x + 16 = 0 \Rightarrow x = 24 \text{ or } x = -16$$

$$\Rightarrow x = 24 \quad [\because \text{number of students cannot be negative}]$$

Thus, 24 students planned the picnic.

(i) Number of students who attended the picnic = $(24 - 8) = 16$.

(ii) Share of 24 students = ₹ 1440

$$\text{Share of 8 students} = ₹ \left(\frac{1440}{24} \times 8 \right) = ₹ 480.$$

$\therefore$ money paid towards the fee = ₹ 480.

The value reflected in the given question is 'charity'.

EXAMPLE 15 The total cost of a certain length of a piece of wire is ₹ 200. If the piece was 5 metres longer and each metre of wire costs ₹ 2 less, the cost of the piece would have remained unchanged. How long is the piece and what is its original rate per metre? [CBSE 2015]

SOLUTION

Let the original length of the piece of wire be x metres.

Cost of this piece = ₹ 200.

$$\therefore \text{its original rate per metre} = ₹ \frac{200}{x}.$$

New length = $(x + 5)$ metres.

$$\text{New cost per metre} = ₹ \frac{200}{(x + 5)}.$$

$$\therefore \frac{200}{x} - \frac{200}{(x + 5)} = 2$$

$$\begin{aligned}
 &\Rightarrow \frac{1}{x} - \frac{1}{(x+5)} = \frac{2}{200} \Rightarrow \frac{(x+5) - x}{x(x+5)} = \frac{1}{100} \\
 &\Rightarrow \frac{5}{(x^2+5x)} = \frac{1}{100} \Rightarrow x^2+5x = 500 \\
 &\Rightarrow x^2+5x-500 = 0 \Rightarrow x^2+25x-20x-500 = 0 \\
 &\Rightarrow x(x+25)-20(x+25) = 0 \Rightarrow (x+25)(x-20) = 0 \\
 &\Rightarrow x+25 = 0 \text{ or } x-20 = 0 \Rightarrow x = -25 \text{ or } x = 20 \\
 &\Rightarrow x = 20 \quad [\because \text{length cannot be negative}] \\
 &\therefore \text{original length of wire} = 20 \text{ m.}
 \end{aligned}$$

$$\text{Original rate} = ₹ \frac{200}{20} \text{ per m} = ₹ 10 \text{ per m.}$$

EXAMPLE 16 ₹ 6500 were divided equally among a certain number of persons. Had there been 15 more persons, each would have got ₹ 30 less. Find the original number of persons.

SOLUTION Let the original number of persons be x .

Total amount to be divided = ₹ 6500.

$$\text{Share of each} = ₹ \frac{6500}{x}.$$

New number of persons = $(x+15)$.

$$\text{Now, share of each} = ₹ \frac{6500}{(x+15)}.$$

$$\therefore \frac{6500}{x} - \frac{6500}{(x+15)} = 30$$

$$\Rightarrow \frac{1}{x} - \frac{1}{(x+15)} = \frac{30}{6500} \Rightarrow \frac{(x+15) - x}{x(x+15)} = \frac{3}{650}$$

$$\Rightarrow \frac{15}{(x^2+15x)} = \frac{3}{650} \Rightarrow 3(x^2+15x) = 9750$$

$$\Rightarrow x^2+15x = 3250 \Rightarrow x^2+15x-3250 = 0$$

$$\Rightarrow x^2+65x-50x-3250 = 0 \Rightarrow x(x+65)-50(x+65) = 0$$

$$\Rightarrow (x+65)(x-50) = 0 \Rightarrow x+65 = 0 \text{ or } x-50 = 0$$

$$\Rightarrow x = -65 \text{ or } x = 50$$

$$\Rightarrow x = 50 \quad [\because \text{number of persons cannot be negative}]$$

Hence, the original number of persons is 50.

PROBLEMS ON AGES

EXAMPLE 17 A girl is twice as old as her sister. Four years hence, the product of their ages (in years) will be 160. Find their present ages. [CBSE 2010]

SOLUTION Let the present age of her sister be x years.
 Then, girl's present age = $2x$ years.
 Product of their ages 4 years hence = $(x + 4)(2x + 4)$.
 $\therefore (x + 4)(2x + 4) = 160$
 $\Rightarrow 2x^2 + 12x - 144 = 0 \Rightarrow x^2 + 6x - 72 = 0$
 $\Rightarrow x^2 + 12x - 6x - 72 = 0 \Rightarrow x(x + 12) - 6(x + 12) = 0$
 $\Rightarrow (x + 12)(x - 6) = 0 \Rightarrow x + 12 = 0$ or $x - 6 = 0$
 $\Rightarrow x = -12$ or $x = 6$
 $\Rightarrow x = 6$ [$\because$ the age cannot be negative]
 $\therefore$ sister's present age = 6 years and girl's present age = 12 years.

EXAMPLE 18 *The age of a man is twice the square of the age of his son. Eight years hence, the age of the man will be 4 years more than three times the age of his son. Find their present ages.* [CBSE 2009C]

SOLUTION Let the present age of the son be x years.
 Then, the present age of the man is $(2x^2)$ years.
 Age of the son 8 years hence = $(x + 8)$ years.
 Age of the man 8 years hence $(2x^2 + 8)$ years.
 $\therefore (2x^2 + 8) = 3(x + 8) + 4$
 $\Rightarrow 2x^2 - 3x - 20 = 0 \Rightarrow 2x^2 - 8x + 5x - 20 = 0$
 $\Rightarrow 2x(x - 4) + 5(x - 4) = 0 \Rightarrow (x - 4)(2x + 5) = 0$
 $\Rightarrow x - 4 = 0$ or $2x + 5 = 0$
 $\Rightarrow x = 4$ or $x = \frac{-5}{2}$
 $\Rightarrow x = 4$ [$\because$ age cannot be negative].
 $\therefore$ son's present age = 4 years, and
 man's present age = (2×4^2) years = 32 years.

EXAMPLE 19 *The sum of the reciprocals of Arun's ages (in years) 3 years ago and five years from now is $\frac{1}{3}$. Find his present age.*

SOLUTION Let Arun's present age be x years.
 Arun's age 3 years ago = $(x - 3)$ years.
 Arun's age 5 years hence = $(x + 5)$ years.
 $\therefore \frac{1}{(x - 3)} + \frac{1}{(x + 5)} = \frac{1}{3}$

$$\Rightarrow \frac{(x+5) + (x-3)}{(x-3)(x+5)} = \frac{1}{3} \Rightarrow (x-3)(x+5) = 6(x+1)$$

$$\Rightarrow x^2 + 2x - 15 = 6x + 6 \Rightarrow x^2 - 4x - 21 = 0$$

$$\Rightarrow x^2 - 7x + 3x - 21 = 0 \Rightarrow x(x-7) + 3(x-7) = 0$$

$$\Rightarrow (x-7)(x+3) = 0 \Rightarrow x-7 = 0 \text{ or } x+3 = 0$$

$$\Rightarrow x = 7 \text{ or } x = -3$$

$$\Rightarrow x = 7 \quad [\because \text{age cannot be negative}]$$

Hence, Arun's present age is 7 years.

EXAMPLE 20 *The sum of the ages of a man and his son is 45 years. Five years ago, the product of their ages (in years) was 124. Find their present ages.*

SOLUTION Let the present age of the man be x years.

Then, the present age of the son = $(45 - x)$ years.

Age of the man 5 years ago = $(x - 5)$ years.

Age of the son 5 years ago = $(45 - x - 5)$ years = $(40 - x)$ years.

$$\therefore (x-5)(40-x) = 124$$

$$\Rightarrow -x^2 + 45x - 200 = 124 \Rightarrow x^2 - 45x + 324 = 0$$

$$\Rightarrow x^2 - 36x - 9x + 324 = 0 \Rightarrow x(x-36) - 9(x-36) = 0$$

$$\Rightarrow (x-36)(x-9) = 0 \Rightarrow x-36 = 0 \text{ or } x-9 = 0$$

$$\Rightarrow x = 36 \text{ or } x = 9.$$

But, the man's age cannot be 9 years. So, we reject $x = 9$.

Hence, the man's present age = 36 years.

And, the son's present age = $(45 - 36)$ years = 9 years.

EXAMPLE 21 *7 years ago Varun's age was five times the square of Swati's age. 3 years hence, Swati's age will be two-fifth of Varun's age. Find their present ages.* [CBSE 2006C]

SOLUTION Let Swati's age 7 years ago be x years.

Then, Varun's age 7 years ago = $5x^2$ years.

Swati's present age = $(x + 7)$ years.

Varun's present age = $(5x^2 + 7)$ years.

Swati's age 3 years hence = $(x + 7 + 3)$ years = $(x + 10)$ years.

Varun's age 3 years hence = $(5x^2 + 7 + 3)$ years
 $= (5x^2 + 10)$ years.

$$\begin{aligned}
 \therefore (x + 10) &= \frac{2}{5}(5x^2 + 10) \\
 \Rightarrow 5x + 50 &= 10x^2 + 20 \Rightarrow 10x^2 - 5x - 30 = 0 \\
 \Rightarrow 2x^2 - x - 6 &= 0 \Rightarrow 2x^2 - 4x + 3x - 6 = 0 \\
 \Rightarrow 2x(x - 2) + 3(x - 2) &= 0 \Rightarrow (x - 2)(2x + 3) = 0 \\
 \Rightarrow x - 2 = 0 \text{ or } 2x + 3 = 0 &\Rightarrow x = 2 \text{ or } x = \frac{-3}{2} \\
 \Rightarrow x = 2 \quad [\because \text{age cannot be negative}] \\
 \therefore \text{Swati's present age} &= (2 + 7) \text{ years} = 9 \text{ years.} \\
 \text{Varun's present age} &= (5 \times 2^2 + 7) \text{ years} = 27 \text{ years.}
 \end{aligned}$$

PROBLEMS ON SPEED AND TIME

EXAMPLE 22 A bus travels at a certain average speed for a distance of 75 km and then travels a distance of 90 km at an average speed of 10 km/hr more than the first speed. If it takes 3 hours to complete the total journey, find its original speed. [CBSE 2015]

SOLUTION Let the original speed of the bus be x km/hr.

Time taken to cover 75 km = $\frac{75}{x}$ hours.

New speed = $(x + 10)$ km/hr.

Time taken to cover 90 km with new speed = $\frac{90}{(x + 10)}$ hours.

Total time taken to cover the whole journey = 3 hours.

$$\begin{aligned}
 \therefore \frac{75}{x} + \frac{90}{(x + 10)} &= 3 \\
 \Rightarrow \frac{75(x + 10) + 90x}{x(x + 10)} &= 3 \Rightarrow 165x + 750 = 3x^2 + 30x \\
 \Rightarrow 3x^2 - 135x - 750 &= 0 \Rightarrow x^2 - 45x - 250 = 0 \\
 \Rightarrow x^2 - 50x + 5x - 250 &= 0 \Rightarrow x(x - 50) + 5(x - 50) = 0 \\
 \Rightarrow (x - 50)(x + 5) &= 0 \Rightarrow x - 50 = 0 \text{ or } x + 5 = 0 \\
 \Rightarrow x = 50 \text{ or } x = -5 &\Rightarrow x = 50 \quad [\because \text{speed cannot be negative}] \\
 \text{Hence, the original speed of the bus} &= 50 \text{ km/hr.}
 \end{aligned}$$

EXAMPLE 23 In a flight of 2800 km, an aircraft was slowed down due to bad weather. Its average speed is reduced by 100 km/hr and time of flight increased by 30 minutes. Find the original duration of the flight. [CBSE 2012]

SOLUTION Let the original speed of the aircraft be x km/hr.

Time taken to cover 2800 km = $\frac{2800}{x}$ hours.

Reduced speed = $(x - 100)$ km/hr.

Time taken to cover 2800 km at this speed = $\frac{2800}{(x - 100)}$ hours.

$$\therefore \frac{2800}{(x - 100)} - \frac{2800}{x} = \frac{30}{60}$$

$$\Rightarrow \frac{1}{(x - 100)} - \frac{1}{x} = \frac{1}{2 \times 2800} \Rightarrow \frac{x - (x - 100)}{(x - 100)x} = \frac{1}{5600}$$

$$\Rightarrow \frac{100}{(x^2 - 100x)} = \frac{1}{5600} \Rightarrow x^2 - 100x - 560000 = 0$$

$$\Rightarrow x^2 - 800x + 700x - 560000 = 0$$

$$\Rightarrow x(x - 800) + 700(x - 800) = 0$$

$$\Rightarrow (x - 800)(x + 700) = 0$$

$$\Rightarrow x - 800 = 0 \text{ or } x + 700 = 0$$

$$\Rightarrow x = 800 \text{ or } x = -700$$

$$\Rightarrow x = 800 \quad [\because \text{speed cannot be negative}]$$

$\therefore$ original speed of the aircraft = 800 km/hr.

original duration of the flight = $\frac{2800}{800}$ hours

= 3 hours 30 minutes.

EXAMPLE 24 *An aeroplane left 30 minutes later than its scheduled time and in order to reach its destination 1500 km away in time, it had to increase its speed by 250 km/hr from its usual speed. Determine its usual speed.* [CBSE 2005C]

SOLUTION Let the usual speed be x km/hr.

Actual speed = $(x + 250)$ km/hr.

Time taken at usual speed = $\left(\frac{1500}{x}\right)$ hr.

Time taken at actual speed = $\left(\frac{1500}{x + 250}\right)$ hr.

Difference between the two times taken = $\frac{1}{2}$ hr.

$$\therefore \frac{1500}{x} - \frac{1500}{x + 250} = \frac{1}{2}$$

$$\Rightarrow \frac{1}{x} - \frac{1}{(x + 250)} = \frac{1}{3000} \Rightarrow \frac{(x + 250) - x}{x(x + 250)} = \frac{1}{3000}$$

$$\Rightarrow \frac{250}{(x^2 + 250x)} = \frac{1}{3000} \Rightarrow x^2 + 250x - 750000 = 0$$

$$\begin{aligned} \Rightarrow x^2 + 1000x - 750x - 750000 &= 0 \\ \Rightarrow x(x + 1000) - 750(x + 1000) &= 0 \Rightarrow (x + 1000)(x - 750) = 0 \\ \Rightarrow x + 1000 = 0 \text{ or } x - 750 = 0 &\Rightarrow x = -1000 \text{ or } x = 750 \\ \Rightarrow x = 750 \quad [\because \text{speed cannot be negative}] \end{aligned}$$

Hence, the usual speed of the aeroplane was 750 km/hr.

EXAMPLE 25 *If a man walks 1 km/hr faster than his usual speed then he covers a distance of 3 km in 15 minutes less time. Find his usual speed.*

[CBSE 2014]

SOLUTION Let the usual speed of the man be x km/hr.

Time taken to cover 3 km at usual speed = $\frac{3}{x}$ hours.

Actual speed of the man = $(x + 1)$ km/hr.

Time taken to cover 3 km at actual speed = $\frac{3}{(x + 1)}$ hours.

$$\therefore \frac{3}{x} - \frac{3}{(x + 1)} = \frac{15}{60} \Rightarrow \frac{3}{x} - \frac{3}{(x + 1)} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x} - \frac{1}{(x + 1)} = \frac{1}{12} \Rightarrow \frac{(x + 1) - x}{x(x + 1)} = \frac{1}{12}$$

$$\Rightarrow \frac{1}{(x^2 + x)} = \frac{1}{12} \Rightarrow x^2 + x - 12 = 0$$

$$\Rightarrow x^2 + 4x - 3x - 12 = 0 \Rightarrow x(x + 4) - 3(x + 4) = 0$$

$$\Rightarrow (x + 4)(x - 3) = 0 \Rightarrow x + 4 = 0 \text{ or } x - 3 = 0$$

$$\Rightarrow x = -4 \text{ or } x = 3 \Rightarrow x = 3 \quad [\because \text{speed cannot be negative}]$$

Hence, the usual speed of the man is 3 km/hr.

EXAMPLE 26 *A motorboat whose speed in still water is 24 km/hr, takes 1 hour more to go 32 km upstream than to return downstream to the same spot. Find the speed of the stream.*

[CBSE 2014]

SOLUTION Speed of the motorboat in still water = 24 km/hr.

Let the speed of the stream be x km/hr.

Then, speed upstream = $(24 - x)$ km/hr.

Speed downstream = $(24 + x)$ km/hr.

Time taken to go 32 km upstream = $\frac{32}{(24 - x)}$ hours.

Time taken to return 32 km downstream = $\frac{32}{(24 + x)}$ hours.

$$\therefore \frac{32}{(24 - x)} - \frac{32}{(24 + x)} = 1$$

$$\Rightarrow \frac{1}{(24-x)} - \frac{1}{(24+x)} = \frac{1}{32} \Rightarrow \frac{(24+x) - (24-x)}{(24-x)(24+x)} = \frac{1}{32}$$

$$\Rightarrow \frac{2x}{(576-x^2)} = \frac{1}{32} \Rightarrow 576 - x^2 = 64x$$

$$\Rightarrow x^2 + 64x - 576 = 0 \Rightarrow x^2 + 72x - 8x - 576 = 0$$

$$\Rightarrow x(x+72) - 8(x+72) = 0 \Rightarrow (x+72)(x-8) = 0$$

$$\Rightarrow x + 72 = 0 \text{ or } x - 8 = 0$$

$$\Rightarrow x = -72 \text{ or } x = 8$$

$$\Rightarrow x = 8 \quad [\because \text{speed of the stream cannot be negative}]$$

Hence, the speed of the stream is 8 km/hr.

EXAMPLE 27 A sailor can row a boat 8 km downstream and return back to the starting point in 1 hour 40 minutes. If the speed of the stream is 2 km/hr, find the speed of the boat in still water.

SOLUTION Let the speed of the boat in still water be x km/hr.

Speed of the stream = 2 km/hr.

$\therefore$ speed downstream = $(x + 2)$ km/hr,

speed upstream = $(x - 2)$ km/hr.

Time taken to cover 8 km downstream and return back to the starting point = $\frac{8}{(x+2)} + \frac{8}{(x-2)}$. But, this time is given as $1\frac{40}{60}$

hours = $1\frac{2}{3}$ hours = $\frac{5}{3}$ hours.

$$\therefore \frac{8}{x+2} + \frac{8}{x-2} = \frac{5}{3}$$

$$\Rightarrow \frac{1}{x+2} + \frac{1}{x-2} = \frac{5}{24} \Rightarrow \frac{(x-2) + (x+2)}{(x+2)(x-2)} = \frac{5}{24}$$

$$\Rightarrow \frac{2x}{(x^2-4)} = \frac{5}{24} \Rightarrow 5x^2 - 20 = 48x$$

$$\Rightarrow 5x^2 - 48x - 20 = 0 \Rightarrow 5x^2 - 50x + 2x - 20 = 0$$

$$\Rightarrow 5x(x-10) + 2(x-10) = 0 \Rightarrow (x-10)(5x+2) = 0$$

$$\Rightarrow x - 10 = 0 \text{ or } 5x + 2 = 0$$

$$\Rightarrow x = 10 \text{ or } x = \frac{-2}{5}$$

$$\Rightarrow x = 10 \quad [\because \text{speed of the boat cannot be negative}]$$

Hence, the speed of the boat in still water is 10 km/hr.

PROBLEMS ON TIME AND WORK

EXAMPLE 28 *A takes 6 days less than the time taken by B to finish a piece of work. If both A and B together can finish it in 4 days, find the time taken by B to finish the work.* [CBSE 2017]

SOLUTION Suppose B alone takes x days to finish the work and A alone can finish it in $(x - 6)$ days.

$$\text{B's 1 day's work} = \frac{1}{x}.$$

$$\text{A's 1 day's work} = \frac{1}{(x-6)}.$$

$$(\text{A} + \text{B})\text{'s 1 day's work} = \frac{1}{4}.$$

$$\therefore \frac{1}{x} + \frac{1}{(x-6)} = \frac{1}{4}$$

$$\Rightarrow \frac{(x-6) + x}{x(x-6)} = \frac{1}{4} \Rightarrow \frac{(2x-6)}{(x^2-6x)} = \frac{1}{4}$$

$$\Rightarrow x^2 - 6x = 8x - 24 \Rightarrow x^2 - 14x + 24 = 0$$

$$\Rightarrow x^2 - 12x - 2x + 24 = 0 \Rightarrow x(x-12) - 2(x-12) = 0$$

$$\Rightarrow (x-12)(x-2) = 0 \Rightarrow x-12 = 0 \text{ or } x-2 = 0$$

$$\Rightarrow x = 12 \text{ or } x = 2$$

$$\Rightarrow x = 12 \quad [\because x = 2 \Rightarrow (x-6) < 0]$$

Hence, B alone can finish the work in 12 days.

EXAMPLE 29 *Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.* [CBSE 2008]

SOLUTION Let the smaller tap fill the tank in x hours.

Then, the larger tap fills it in $(x - 10)$ hours.

Time taken by both together to fill the tank = $\frac{75}{8}$ hours.

$$\text{Part filled by the smaller tap in 1 hr} = \frac{1}{x}.$$

$$\text{Part filled by the larger tap in 1 hr} = \frac{1}{(x-10)}.$$

$$\text{Part filled by both the taps in 1 hr} = \frac{8}{75}.$$

$$\therefore \frac{1}{x} + \frac{1}{(x-10)} = \frac{8}{75}$$

$$\begin{aligned}
&\Rightarrow \frac{(x-10)+x}{x(x-10)} = \frac{8}{75} \Rightarrow \frac{(2x-10)}{x(x-10)} = \frac{8}{75} \\
&\Rightarrow 75(2x-10) = 8x(x-10) \quad [\text{by cross multiplication}] \\
&\Rightarrow 150x - 750 = 8x^2 - 80x \\
&\Rightarrow 8x^2 - 230x + 750 = 0 \Rightarrow 4x^2 - 115x + 375 = 0 \\
&\Rightarrow 4x^2 - 100x - 15x + 375 = 0 \Rightarrow 4x(x-25) - 15(x-25) = 0 \\
&\Rightarrow (x-25)(4x-15) = 0 \Rightarrow x-25 = 0 \text{ or } 4x-15 = 0 \\
&\Rightarrow x = 25 \text{ or } x = \frac{15}{4} \\
&\Rightarrow x = 25 \quad \left[\because x = \frac{15}{4} \Rightarrow (x-10) < 0 \right].
\end{aligned}$$

Hence, the time taken by the smaller tap to fill the tank = 25 hours.

And, the time taken by the larger tap to fill the tank
 = (25 - 10) hours = 15 hours.

PROBLEMS ON AREAS

EXAMPLE 30 The perimeter of a rectangular field is 82 m and its area is 400 m². Find the dimensions of the field.

SOLUTION We know that 2(length + breadth) = perimeter.

$$\therefore (\text{length} + \text{breadth}) = \frac{1}{2} \times \text{perimeter} = \frac{1}{2} \times 82 \text{ m} = 41 \text{ m}.$$

Let the length of the field be x metres.

Then, its breadth = $(41 - x)$ m.

$$\therefore \text{area of the field} = x(41 - x) \text{ m}^2 = (41x - x^2) \text{ m}^2.$$

But, area = 400 m² (given)

$$\therefore 41x - x^2 = 400$$

$$\Rightarrow x^2 - 41x + 400 = 0 \Rightarrow x^2 - 25x - 16x + 400 = 0$$

$$\Rightarrow x(x - 25) - 16(x - 25) = 0 \Rightarrow (x - 25)(x - 16) = 0$$

$$\Rightarrow x - 25 = 0 \text{ or } x - 16 = 0 \Rightarrow x = 25 \text{ or } x = 16.$$

$$\therefore \text{length of the field} = 25 \text{ m and breadth of the field} = 16 \text{ m}.$$

EXAMPLE 31 The diagonal of a rectangular field is 16 m more than the shorter side. If the longer side is 14 m more than the shorter side then find the lengths of the sides of the field. [CBSE 2015]

SOLUTION Let the shorter side of the field be x metres.

Then, longer side = $(x + 14)$ m.

And, diagonal = $(x + 16)$ m.

$$\therefore (x + 16)^2 - (x + 14)^2 = x^2$$

$$\Rightarrow (2x + 30) \times 2 = x^2 \Rightarrow x^2 - 4x - 60 = 0$$

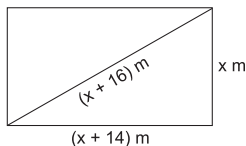
$$\Rightarrow x^2 - 10x + 6x - 60 = 0 \Rightarrow x(x - 10) + 6(x - 10) = 0$$

$$\Rightarrow (x - 10)(x + 6) = 0 \Rightarrow x - 10 = 0 \text{ or } x + 6 = 0$$

$$\Rightarrow x = 10 \text{ or } x = -6$$

$$\Rightarrow x = 10 \quad [\because \text{breadth cannot be negative}]$$

$$\therefore \text{breadth} = 10 \text{ m and length} = (10 + 14) \text{ m} = 24 \text{ m.}$$



EXAMPLE 32 A rectangular field is 20 m long and 14 m wide. There is a path of equal width all around it, having an area of 111 sq m. Find the width of the path.

SOLUTION Let the width of the path be x metres.

Length of the field including the path

$$= (20 + 2x) \text{ m.}$$

Breadth of the field including the path

$$= (14 + 2x) \text{ m.}$$

$$\text{Area of the field including the path} = (20 + 2x)(14 + 2x) \text{ m}^2.$$

$$\text{Area of the field excluding the path} = (20 \times 14) \text{ m}^2 = 280 \text{ m}^2.$$

$$\therefore \text{the area of the path} = [(20 + 2x)(14 + 2x) - 280] \text{ m}^2.$$

$$\therefore (20 + 2x)(14 + 2x) - 280 = 111$$

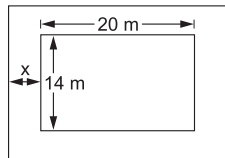
$$\Rightarrow 4x^2 + 68x - 111 = 0 \Rightarrow 4x^2 + 74x - 6x - 111 = 0$$

$$\Rightarrow 2x(2x + 37) - 3(2x + 37) = 0 \Rightarrow (2x + 37)(2x - 3) = 0$$

$$\Rightarrow x = -\frac{37}{2} \text{ or } x = \frac{3}{2}$$

$$\Rightarrow x = \frac{3}{2} = 1.5 \quad [\because \text{width can never be negative}].$$

Hence, the width of the path is 1.5 m.



EXAMPLE 33 Sum of the areas of two squares is 260 m^2 . If the difference of their perimeters is 24 m then find the sides of the two squares.

[CBSE 2009C]

SOLUTION Let the sides of the two squares be a metres and b metres.

Then, their areas are $(a^2) \text{ m}^2$ and $(b^2) \text{ m}^2$ respectively.

And, their perimeters are $(4a) \text{ m}$ and $(4b) \text{ m}$ respectively.

$$\begin{aligned}\therefore 4a - 4b &= 24 \Rightarrow 4(a - b) = 24 \\ &\Rightarrow a - b = 6 \Rightarrow b = (a - 6) \quad \dots (i)\end{aligned}$$

Sum of their areas = 260 m^2 .

$$\begin{aligned}\therefore a^2 + b^2 &= 260 \\ \Rightarrow a^2 + (a - 6)^2 &= 260 \quad [\text{using (i)}] \\ \Rightarrow 2a^2 - 12a - 224 &= 0 \Rightarrow a^2 - 6a - 112 = 0 \\ \Rightarrow a^2 - 14a + 8a - 112 &= 0 \\ \Rightarrow a(a - 14) + 8(a - 14) &= 0 \Rightarrow (a - 14)(a + 8) = 0 \\ \Rightarrow a - 14 = 0 \text{ or } a + 8 = 0 &\Rightarrow a = 14 \text{ or } a = -8 \\ \Rightarrow a = 14 \quad [\because \text{side of a square cannot be negative}]. \\ \therefore a = 14 \text{ and } b = (14 - 6) &= 8.\end{aligned}$$

Hence, the sides of the square are 14 m and 8 m respectively.

EXAMPLE 34 *The area of a right-angled triangle is 600 sq cm . If the base of the triangle exceeds the altitude by 10 cm , find the dimensions of the triangle.*

SOLUTION Let the altitude of the triangle be $x \text{ cm}$.

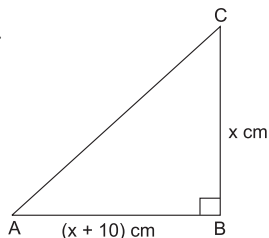
Then, its base = $(x + 10) \text{ cm}$.

$$\text{Area of the triangle} = \frac{1}{2}x(x + 10) \text{ cm}^2.$$

$$\begin{aligned}\therefore \frac{1}{2}x(x + 10) &= 600 \\ \Rightarrow x(x + 10) &= 1200 \\ \Rightarrow x^2 + 10x - 1200 &= 0 \\ \Rightarrow x^2 + 40x - 30x - 1200 &= 0 \\ \Rightarrow x(x + 40) - 30(x + 40) &= 0 \\ \Rightarrow (x + 40)(x - 30) &= 0 \\ \Rightarrow x = -40 \text{ or } x = 30 \\ \Rightarrow x = 30 \quad [\because \text{altitude cannot be negative}].\end{aligned}$$

Thus, the altitude of the triangle = 30 cm .

And, the base of the triangle = $(30 + 10) \text{ cm} = 40 \text{ cm}$.



EXAMPLE 35 *The hypotenuse of a right-angled triangle is 6 cm more than twice the shortest side. If the third side is 2 cm less than the hypotenuse, find the sides of the triangle.* [CBSE 2007]

SOLUTION Let the shortest side of the triangle be x cm.
 Then, its hypotenuse = $(2x + 6)$ cm.
 And, its third side = $(2x + 6 - 2)$ cm = $(2x + 4)$ cm.
 By Pythagoras' theorem, we have

$$(2x + 6)^2 = x^2 + (2x + 4)^2$$

$$\Rightarrow 4x^2 + 24x + 36 = 5x^2 + 16x + 16$$

$$\Rightarrow x^2 - 8x - 20 = 0$$

$$\Rightarrow x^2 - 10x + 2x - 20 = 0$$

$$\Rightarrow x(x - 10) + 2(x - 10) = 0 \Rightarrow (x - 10)(x + 2) = 0$$

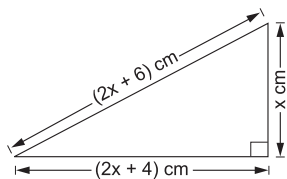
$$\Rightarrow x - 10 = 0 \text{ or } x + 2 = 0 \Rightarrow x = 10 \text{ or } x = -2$$

$$\Rightarrow x = 10 \quad [\because \text{the side of a triangle cannot be negative}]$$

$\therefore$ shorter side = 10 cm, longer side = $(2 \times 10 + 4)$ cm = 24 cm

and hypotenuse = $(2 \times 10 + 6)$ cm = 26 cm.

Hence, the sides of the given triangle are 10 cm, 24 cm and 26 cm.



EXERCISE 4E

PROBLEMS ON NUMBERS

1. The sum of a natural number and its square is 156. Find the number.
2. The sum of a natural number and its positive square root is 132. Find the number.
3. The sum of two natural numbers is 28 and their product is 192. Find the numbers.
4. The sum of the squares of two consecutive positive integers is 365. Find the integers.
5. The sum of the squares of two consecutive positive odd numbers is 514. Find the numbers.
6. The sum of the squares of two consecutive positive even numbers is 452. Find the numbers.
7. The product of two consecutive positive integers is 306. Find the integers.
8. Two natural numbers differ by 3 and their product is 504. Find the numbers.
9. Find two consecutive multiples of 3 whose product is 648.
10. Find two consecutive positive odd integers whose product is 483.
11. Find two consecutive positive even integers whose product is 288.

12. The sum of two natural numbers is 9 and the sum of their reciprocals is $\frac{1}{2}$. Find the numbers. [CBSE 2012]
13. The sum of two natural numbers is 15 and the sum of their reciprocals is $\frac{3}{10}$. Find the numbers. [CBSE 2005]
14. The difference of two natural numbers is 3 and the difference of their reciprocals is $\frac{3}{28}$. Find the numbers. [CBSE 2014]
15. The difference of two natural numbers is 5 and the difference of their reciprocals is $\frac{5}{14}$. Find the numbers. [CBSE 2014]
16. The sum of the squares of two consecutive multiples of 7 is 1225. Find the multiples.
17. The sum of a natural number and its reciprocal is $\frac{65}{8}$. Find the number.
18. Divide 57 into two parts whose product is 680.
19. Divide 27 into two parts such that the sum of their reciprocals is $\frac{3}{20}$.
20. Divide 16 into two parts such that twice the square of the larger part exceeds the square of the smaller part by 164.
21. Find two natural numbers, the sum of whose squares is 25 times their sum and also equal to 50 times their difference.
22. The difference of the squares of two natural numbers is 45. The square of the smaller number is four times the larger number. Find the numbers. [CBSE 2007]
23. Three consecutive positive integers are such that the sum of the square of the first and the product of the other two is 46. Find the integers. [CBSE 2010]
24. A two-digit number is 4 times the sum of its digits and twice the product of its digits. Find the number.
25. A two-digit number is such that the product of its digits is 14. If 45 is added to the number, the digits interchange their places. Find the number. [CBSE 2012]
26. The denominator of a fraction is 3 more than its numerator. The sum of the fraction and its reciprocal is $2\frac{9}{10}$. Find the fraction.
27. The numerator of a fraction is 3 less than its denominator. If 1 is added to the denominator, the fraction is decreased by $\frac{1}{15}$. Find the fraction. [CBSE 2012]

28. The sum of a number and its reciprocal is $2\frac{1}{30}$. Find the number.

SOME GENERAL PROBLEMS

29. A teacher on attempting to arrange the students for mass drill in the form of a solid square found that 24 students were left. When he increased the size of the square by one student, he found that he was short of 25 students. Find the number of students.
30. 300 apples are distributed equally among a certain number of students. Had there been 10 more students, each would have received one apple less. Find the number of students.
31. In a class test, the sum of Kamal's marks in mathematics and English is 40. Had he got 3 marks more in mathematics and 4 marks less in English, the product of the marks would have been 360. Find his marks in two subjects separately. [CBSE 2008C]
32. Some students planned a picnic. The total budget for food was ₹ 2000. But, 5 students failed to attend the picnic and thus the cost for food for each member increased by ₹ 20. How many students attended the picnic and how much did each student pay for the food? [CBSE 2010]
33. If the price of a book is reduced by ₹ 5, a person can buy 4 more books for ₹ 600. Find the original price of the book.
34. A person on tour has ₹ 10800 for his expenses. If he extends his tour by 4 days, he has to cut down his daily expenses by ₹ 90. Find the original duration of the tour.
35. In a class test, the sum of the marks obtained by P in mathematics and science is 28. Had he got 3 more marks in mathematics and 4 marks less in science, the product of marks obtained in the two subjects would have been 180. Find the marks obtained by him in the two subjects separately. [CBSE 2008]
36. A man buys a number of pens for ₹ 180. If he had bought 3 more pens for the same amount, each pen would have cost him ₹ 3 less. How many pens did he buy?
37. A dealer sells an article for ₹ 75 and gains as much per cent as the cost price of the article. Find the cost price of the article. [CBSE 2011]

PROBLEMS ON AGES

38. (i) One year ago, a man was 8 times as old as his son. Now, his age is equal to the square of his son's age. Find their present ages.

- (ii) A man is $3\frac{1}{2}$ times as old as his son. If the sum of the squares of their ages is 1325, find the ages of the father and the son. [CBSE 2017]
39. The sum of the reciprocals of Meena's ages (in years) 3 years ago and 5 years hence is $\frac{1}{3}$. Find her present age.
40. The sum of the ages of a boy and his brother is 25 years, and the product of their ages in years is 126. Find their ages.
41. The product of Tanvy's age (in years) 5 years ago and her age 8 years later is 30. Find her present age.
42. Two years ago, a man's age was three times the square of his son's age. In three years time, his age will be four times his son's age. Find their present ages.

PROBLEMS ON TIME AND DISTANCE

43. A truck covers a distance of 150 km at a certain average speed and then covers another 200 km at an average speed which is 20 km per hour more than the first speed. If the truck covers the total distance in 5 hours, find the first speed of the truck. [CBSE 2015]
44. While boarding an aeroplane, a passenger got hurt. The pilot showing promptness and concern, made arrangements to hospitalise the injured and so the plane started late by 30 minutes. To reach the destination, 1500 km away, in time, the pilot increased the speed by 100 km/hour. Find the original speed of the plane.
- Do you appreciate the values shown by the pilot, namely promptness in providing help to the injured and his efforts to reach in time?
- [CBSE 2013]
45. A train covers a distance of 480 km at a uniform speed. If the speed had been 8 km/hr less then it would have taken 3 hours more to cover the same distance. Find the usual speed of the train. [CBSE 2013C]
46. A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/hr more than the first speed. If it takes 3 hours to complete the total journey, what is its first speed? [CBSE 2015]
47. A train travels 180 km at a uniform speed. If the speed had been 9 km/hr more, it would have taken 1 hour less for the same journey. Find the speed of the train. [CBSE 2013]

48. A train covers a distance of 300 km at a uniform speed. If the speed of the train is increased by 5 km/hour, it takes 2 hours less in the journey. Find the original speed of the train. [CBSE 2017]
49. A passenger train takes 2 hours less for a journey of 300 km if its speed is increased by 5 km/hr from its usual speed. Find its usual speed. [CBSE 2007]
50. The distance between Mumbai and Pune is 192 km. Travelling by the Deccan Queen, it takes 48 minutes less than another train. Calculate the speed of the Deccan Queen if the speeds of the two trains differ by 20 km/hr. [CBSE 2003]
51. A motor boat whose speed in still water is 18 km/hr, takes 1 hour more to go 24 km upstream than to return to the same spot. Find the speed of the stream. [CBSE 2014]
52. The speed of a boat in still water is 15 km/hr. It goes 30 km upstream and returns back at the same point in 4 hours 30 minutes. Find the speed of the stream. [CBSE 2017]
53. A motorboat whose speed is 9 km/hr in still water, goes 15 km downstream and comes back in a total time of 3 hours 45 minutes. Find the speed of the stream. [CBSE 2007C]

PROBLEMS ON TIME AND WORK AND PIPES AND CISTERN

54. A takes 10 days less than the time taken by B to finish a piece of work. If both A and B together can finish the work in 12 days, find the time taken by B to finish the work.
55. Two taps running together can fill a tank in $3\frac{1}{13}$ hours. If one tap takes 3 hours more than the other to fill the tank then how much time will each tap take to fill the tank? [CBSE 2017]
56. Two pipes running together can fill a tank in $11\frac{1}{9}$ minutes. If one pipe takes 5 minutes more than the other to fill the tank separately, find the time in which each pipe would fill the tank separately. [CBSE 2010]
57. Two water taps together can fill a tank in 6 hours. The tap of larger diameter takes 9 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank. [CBSE 2011]

PROBLEMS ON AREA AND GEOMETRY

58. The length of a rectangle is twice its breadth and its area is 288 cm^2 . Find the dimensions of the rectangle.

59. The length of a rectangular field is three times its breadth. If the area of the field be 147 sq metres, find the length of the field.
60. The length of a hall is 3 metres more than its breadth. If the area of the hall is 238 sq metres, calculate its length and breadth.
61. The perimeter of a rectangular plot is 62 m and its area is 228 sq metres. Find the dimensions of the plot.
62. A rectangular field is 16 m long and 10 m wide. There is a path of uniform width all around it, having an area of 120 m^2 . Find the width of the path.
63. The sum of the areas of two squares is 640 m^2 . If the difference in their perimeters be 64 m, find the sides of the two squares. [CBSE 2008]
64. The length of a rectangle is thrice as long as the side of a square. The side of the square is 4 cm more than the width of the rectangle. Their areas being equal, find their dimensions.
65. A farmer prepares a rectangular vegetable garden of area 180 sq metres. With 39 metres of barbed wire, he can fence the three sides of the garden, leaving one of the longer sides unfenced. Find the dimensions of the garden.
66. The area of a right triangle is 600 cm^2 . If the base of the triangle exceeds the altitude by 10 cm, find the dimensions of the triangle.
67. The area of a right-angled triangle is 96 sq metres. If the base is three times the altitude, find the base.
68. The area of a right-angled triangle is 165 sq metres. Determine its base and altitude if the latter exceeds the former by 7 metres.
69. The hypotenuse of a right-angled triangle is 20 metres. If the difference between the lengths of the other sides be 4 metres, find the other sides.
70. The length of the hypotenuse of a right-angled triangle exceeds the length of the base by 2 cm and exceeds twice the length of the altitude by 1 cm. Find the length of each side of the triangle.
71. The hypotenuse of a right-angled triangle is 1 metre less than twice the shortest side. If the third side is 1 metre more than the shortest side, find the sides of the triangle.

ANSWERS (EXERCISE 4E)

- | | | | | |
|---------------|--------------|--------------|--------------|---------------|
| 1. 12 | 2. 121 | 3. 16 and 12 | 4. 13 and 14 | 5. 15 and 17 |
| 6. 14 and 16 | 7. 17 and 18 | 8. 24 and 21 | 9. 24 and 27 | 10. 21 and 23 |
| 11. 16 and 18 | 12. 3 and 6 | 13. 10 and 5 | 14. 7 and 4 | 15. 7 and 2 |

16. 21 and 28 17. 8 18. 17 and 40 19. 12 and 15
 20. 10 and 6 21. 30 and 10 22. 9 and 6 23. 4, 5, 6
 24. 36 25. 27 26. $\frac{2}{5}$ 27. $\frac{2}{5}$ 28. $\frac{5}{6}$ or $\frac{6}{5}$ 29. 600 30. 50
 31. (21 and 19) or (12 and 28) 32. 20, ₹ 100 33. ₹ 30 34. 20 days
 35. Either he got 12 marks in maths and 16 in science or he got 9 marks in maths and 19 in science
 36. 12 pens 37. ₹ 50
 38. (i) 49 years, 7 years (ii) 35 years, 10 years 39. 7 years
 40. 18 years and 7 years 41. 7 years 42. 29 years, 5 years
 43. 60 km/hr 44. 500 km/hr 45. 40 km/hr 46. 36 km/hr
 47. 36 km/hr 48. 25 km/hr 49. 25 km/hr 50. 80 km/hr
 51. 6 km/hr 52. 5 km/hr 53. 3 km/hr 54. 30 days
 55. 5 hours, 8 hours 56. 20 min, 25 min 57. 9 hours, 18 hours
 58. length = 24 cm, breadth = 12 cm 59. 21 m
 60. breadth = 14 m, length = 17 m 61. length = 19 m, breadth = 12 m
 62. 2 m 63. 24 m, 8 m 64. ($l = 18$ cm, $b = 2$ cm), side = 6 cm
 65. Either ($l = 24$ m, $b = 7.5$ m) or ($l = 15$ m, $b = 12$ m)
 66. 30 cm, 40 cm, 50 cm 67. 24 m 68. base = 15 m, altitude = 22 m
 69. 16 m, 12 m 70. base = 15 cm, altitude = 8 cm, hypotenuse = 17 cm
 71. 3 m, 4 m, 5 m

HINTS TO SOME SELECTED QUESTIONS

- $x + x^2 = 156 \Rightarrow x^2 + x - 156 = 0 \Rightarrow x^2 + 13x - 12x - 156 = 0.$
- $x + \sqrt{x} = 132 \Rightarrow y^2 + y = 132$, where $\sqrt{x} = y$
 $\therefore y^2 + y - 132 = 0 \Rightarrow y^2 + 12y - 11y - 132 = 0.$
- Let the required numbers be x and $(28 - x)$.
 Then, $x(28 - x) = 192.$
- $x^2 + (x+1)^2 = 365 \Rightarrow 2x^2 + 2x - 364 = 0 \Rightarrow x^2 + x - 182 = 0$
 $\therefore x^2 + 14x - 13x - 182 = 0.$
- $x^2 + (x+2)^2 = 514 \Rightarrow 2x^2 + 4x - 510 = 0 \Rightarrow x^2 + 2x - 255 = 0$
 $\Rightarrow x^2 + 17x - 15x - 255 = 0.$
- $x^2 + (x+2)^2 = 452 \Rightarrow 2x^2 + 4x - 448 = 0 \Rightarrow x^2 + 2x - 224 = 0$
 $\Rightarrow x^2 + 16x - 14x - 224 = 0.$

$$7. x(x+1) = 306 \Rightarrow x^2 + x - 306 = 0 \Rightarrow x^2 + 18x - 17x - 306 = 0.$$

$$8. x(x-3) = 504 \Rightarrow x^2 - 3x - 504 = 0 \Rightarrow x^2 - 24x + 21x - 504 = 0.$$

$$9. (3x)\{3(x+1)\} = 648 \Rightarrow x(x+1) = 72 \Rightarrow x^2 + 9x - 8x - 72 = 0.$$

$$10. x(x+2) = 483 \Rightarrow x^2 + 2x - 483 = 0 \Rightarrow x^2 + 23x - 21x - 483 = 0.$$

$$11. x(x+2) = 288 \Rightarrow x^2 + 2x - 288 = 0 \Rightarrow x^2 + 18x - 16x - 288 = 0.$$

14. Let the required numbers be x and $(x-3)$. Then,

$$\frac{1}{(x-3)} - \frac{1}{x} = \frac{3}{28}.$$

$$18. x(57-x) = 680 \Rightarrow x^2 - 57x + 680 = 0 \Rightarrow x^2 - 40x - 17x + 680 = 0.$$

$$19. \frac{1}{x} + \frac{1}{(27-x)} = \frac{3}{20} \Rightarrow \frac{27-x+x}{x(27-x)} = \frac{3}{20} \Rightarrow x^2 - 27x + 180 = 0$$

$$\Rightarrow x^2 - 15x - 12x + 180 = 0.$$

20. Let the two parts be x and $(16-x)$ such that x is larger. Then,

$$2x^2 - (16-x)^2 = 164 \Rightarrow x^2 + 32x - 420 = 0 \Rightarrow x^2 + 42x - 10x - 420 = 0.$$

21. Let the required natural numbers be a and b . Then,

$$a^2 + b^2 = 25(a+b) \text{ and } (a^2 + b^2) = 50(a-b).$$

$$\therefore 25(a+b) = 50(a-b) \Rightarrow a+b = 2(a-b) \Rightarrow a = 3b.$$

$$\therefore (3b)^2 + b^2 = 25 \times 4b \Rightarrow 10b^2 - 100b = 0$$

$$\Rightarrow 10b(b-10) = 0 \Rightarrow b = 10.$$

22. Let the required natural numbers be a and b . Then,

$$(a^2 - b^2 = 45 \text{ and } b^2 = 4a) \Rightarrow (a^2 - 4a - 45) = 0.$$

23. Let the required numbers be x , $(x+1)$ and $(x+2)$. Then,

$$x^2 + (x+1)(x+2) = 46 \Rightarrow 2x^2 + 3x - 44 = 0 \Rightarrow 2x^2 + 11x - 8x - 44 = 0$$

$$\Rightarrow (2x+11)(x-4).$$

$$x \neq \frac{-11}{2}, \text{ since } x \text{ is an integer.}$$

24. Let the tens digit be x and units digit be y . Then,

$$(10x+y) = 4(x+y) \Rightarrow 6x = 3y \Rightarrow y = 2x.$$

$$(10x+y) = 2xy \Rightarrow (10x+2x) = 4x^2 \Rightarrow 4x^2 - 12x = 0$$

$$\Rightarrow 4x(x-3) = 0 \Rightarrow x = 3 \quad [\because x \neq 0].$$

25. Let the tens digit be x and units digit be y . Then,

$$xy = 14 \text{ and } 10x + y + 45 = 10y + x$$

$$\Rightarrow xy = 14 \text{ and } 9(y-x) = 45 \Rightarrow xy = 14 \text{ and } y-x = 5$$

$$\Rightarrow x(x+5) = 14 \text{ and } y = (5+x).$$

26. Let the required fraction be $\frac{a}{(a+3)}$. Then,

$$\frac{a}{(a+3)} + \frac{(a+3)}{a} = \frac{29}{10} \Rightarrow 10[a^2 + (a+3)^2] = 29a(a+3)$$

$$\Rightarrow 9a^2 + 27a - 90 = 0 \Rightarrow a^2 + 3a - 10 = 0.$$

27. Let the required fraction in simplest form be $\frac{x-3}{x}$. Then,

$$\frac{x-3}{x} - \frac{x-3}{x+1} = \frac{1}{15} \Rightarrow (x-3) \left[\frac{1}{x} - \frac{1}{x+1} \right] = \frac{1}{15}$$

$$\therefore 15(x-3) = x(x+1) \Rightarrow x^2 - 14x + 45 = 0 \Rightarrow x = 5 \text{ or } x = 9.$$

So, the required fraction is $\frac{2}{5}$. [neglecting $\frac{6}{9}$]

28. Let the required number be x . Then, $x + \frac{1}{x} = \frac{61}{30}$.

$$\therefore 30x^2 - 61x + 30 = 0 \Rightarrow 30x^2 - 36x - 25x + 30 = 0.$$

29. Let there be x rows and the number of students in each row be x .

Then, total number of students = $(x^2 + 24)$.

$$x^2 + 24 = (x+1)^2 - 25.$$

30. Let the total number of students be x . Then,

$$\frac{300}{x} - \frac{300}{(x+10)} = 1 \Rightarrow x^2 + 10x - 3000 = 0$$

$$\Rightarrow (x-50)(x+60) = 0 \Rightarrow x = 50.$$

31. Suppose he gets x marks in mathematics and y marks in English. Then,

$$x + y = 40 \text{ and } (x+3)(y-4) = 360.$$

$$(x+3)(40-x-4) = 360 \Rightarrow (x+3)(36-x) = 360$$

$$\Rightarrow x^2 - 33x + 252 = 0 \Rightarrow (x-21)(x-12) = 0.$$

32. Let x students attended the picnic. Then, $(x+5)$ planned it.

$$\therefore \frac{2000}{x} - \frac{2000}{x+5} = 20 \Rightarrow \frac{1}{x} - \frac{1}{x+5} = \frac{20}{2000} = \frac{1}{100} \Rightarrow x^2 + 5x - 500 = 0.$$

$$\therefore x^2 + 25x - 20x - 500 = 0.$$

33. Let the original price per book be ₹ x . Then, reduced price = ₹ $(x-5)$.

$$\therefore \frac{600}{x-5} - \frac{600}{x} = 4 \Rightarrow \frac{1}{x-5} - \frac{1}{x} = \frac{4}{600} = \frac{1}{150}.$$

34. Let the original duration of the tour be x days. Then,

$$\frac{10800}{x} - \frac{10800}{(x+4)} = 90 \Rightarrow \frac{1}{x} - \frac{1}{x+4} = \frac{1}{120} \Rightarrow x^2 + 4x - 480 = 0$$

$$\therefore x^2 + 24x - 20x - 480 = 0.$$

35. Suppose P gets x marks in mathematics and y marks in science. Then,

$$x + y = 28 \Rightarrow y = 28 - x.$$

$$\text{And } (x+3)(y-4) = 180 \Rightarrow (x+3)(28-x-4) = 180$$

$$\therefore (x+3)(24-x) = 180 \Rightarrow x^2 - 21x + 108 = 0.$$

36. Suppose he bought x pens.

$$\therefore \frac{180}{x} - \frac{180}{(x+3)} = 3 \Rightarrow \frac{1}{x} - \frac{1}{x+3} = \frac{1}{60} \Rightarrow x(x+3) - 180 = 0$$

$$\therefore x^2 + 15x - 12x - 180 = 0.$$

37. Let the CP of the article be ₹ x . Then, gain = $x\%$.

$$\therefore \text{SP} = ₹ \left\{ \frac{(100+x)x}{100} \right\} = ₹ \left\{ \frac{x^2 + 100x}{100} \right\}$$

$$\Rightarrow \frac{x^2 + 100x}{100} = 75 \Rightarrow x^2 + 100x - 7500 = 0 \Rightarrow x^2 + 150x - 50x - 7500 = 0.$$

38. (i) Let the son's present age be x years.

Son's age 1 year ago = $(x - 1)$ years.

Man's age 1 year ago = $8(x - 1)$ years = $(8x - 8)$ years.

Man's present age = $(8x - 8 + 1)$ years = $(8x - 7)$ years.

$$\therefore (8x - 7) = x^2 \Rightarrow x^2 - 8x + 7 = 0.$$

39. Let Meena's present age be x years. Then,

$$\frac{1}{(x-3)} + \frac{1}{(x+5)} = \frac{1}{3} \Rightarrow \frac{(x+5) + (x-3)}{(x-3)(x+5)} = \frac{1}{3} \Rightarrow x^2 + 2x - 15 = 3(2x + 2)$$

$$\therefore x^2 - 4x - 21 = 0.$$

40. $x(25 - x) = 126 \Rightarrow x^2 - 25x + 126 = 0 \Rightarrow x^2 - 18x - 7x + 126 = 0.$

41. Let Tanvy's present age be x years.

$$\text{Then, } (x - 5)(x + 8) = 30 \Rightarrow x^2 + 3x - 70 = 0 \Rightarrow x^2 + 10x - 7x - 70 = 0.$$

42. Let son's age 2 years ago be x years.

Then, man's age 2 years ago = $3x^2$.

$$\therefore 3x^2 + 5 = 4(x + 5) \Rightarrow 3x^2 - 4x - 15 = 0 \Rightarrow 3x^2 - 9x + 5x - 15 = 0 \Rightarrow x = 3.$$

Son's age = $(x + 2)$ years = 5 years, man's age = $(3x^2 + 2)$ years = 29 years.

43. Let the first speed be x km/hr. Then,

$$\frac{150}{x} + \frac{200}{(x+20)} = 5 \Rightarrow 50 \left[\frac{3}{x} + \frac{4}{(x+20)} \right] = 5$$

$$\Rightarrow 10[3(x+20) + 4x] = x(x+20) \Rightarrow x^2 - 50x - 600 = 0.$$

44. Let the original speed of the plane be x km/hr. Then,

$$\frac{1500}{x} - \frac{1500}{(x+100)} = \frac{30}{60} \Rightarrow \frac{1}{x} - \frac{1}{(x+100)} = \frac{1}{3000} \Rightarrow x^2 + 100x - 300000 = 0$$

$$\Rightarrow x^2 + 600x - 500x - 300000 = 0.$$

His promptness is appreciable.

45. Let the usual speed of the train be x km/hr. Then,

$$\frac{480}{(x-8)} - \frac{480}{x} = 3 \Rightarrow \frac{1}{x-8} - \frac{1}{x} = \frac{3}{480} = \frac{1}{160}.$$

$$\Rightarrow 160[x - (x-8)] = x(x-8) \Rightarrow x^2 - 8x - 1280 = 0$$

$$\Rightarrow x^2 - 40x + 32x - 1280 = 0.$$

46. Let the first speed be x km/hr. Then,

$$\frac{54}{x} + \frac{63}{(x+6)} = 3 \Rightarrow 9 \left[\frac{6}{x} + \frac{7}{(x+6)} \right] = 3$$

$$\Rightarrow 3[6(x+6) + 7x] = x(x+6) \Rightarrow x(x+6) = 3(13x + 36)$$

$$\Rightarrow x^2 - 33x - 108 = 0.$$

47. Let the speed be x km/hr. Then,

$$\frac{180}{x} - \frac{180}{(x+9)} = 1$$

$$\Rightarrow x^2 + 9x - 1620 = 0 \Rightarrow x^2 + 45x - 36x - 1620 = 0.$$

$$49. \frac{300}{x} - \frac{300}{x+5} = 2 \Rightarrow \frac{1}{x} - \frac{1}{x+5} = \frac{1}{150} \Rightarrow x^2 + 5x - 750 = 0.$$

$$\Rightarrow x^2 + 30x - 25x - 750 = 0.$$

50. Let the speed of the Deccan queen be x km/hr. Then,

$$\frac{192}{(x-20)} - \frac{192}{x} = \frac{48}{60} \Rightarrow \frac{1}{x-20} - \frac{1}{x} = \frac{1}{240}.$$

$$\Rightarrow x^2 - 20x - 4800 = 0 \Rightarrow x^2 - 80x + 60x - 4800 = 0.$$

51. Let the speed of the stream be x km/hr.

Then, speed downstream = $(18 + x)$ km/hr

and speed upstream = $(18 - x)$ km/hr.

$$\therefore \frac{24}{(18-x)} - \frac{24}{(18+x)} = 1 \Rightarrow \frac{1}{(18-x)} - \frac{1}{(18+x)} = \frac{1}{24}$$

$$\Rightarrow x^2 + 48x - 324 = 0 \Rightarrow x^2 + 54x - 6x - 324 = 0.$$

52. Let the speed of the stream be x km/hr.

$$\text{Time to go 30 km upstream} = \frac{30 \text{ km}}{(15-x) \text{ km}} \text{ hr} = \frac{30}{(15-x)} \text{ hr.}$$

$$\text{Time to go 30 km downstream} = \frac{30 \text{ km}}{(15+x) \text{ km}} \text{ hr} = \frac{30}{(15+x)} \text{ hr.}$$

$$\therefore \frac{30}{(15-x)} + \frac{30}{(15+x)} = 4 \frac{1}{2} = \frac{9}{2}$$

$$\Rightarrow 30(15+x) + 30(15-x) = \frac{9}{2}(225 - x^2)$$

$$\Rightarrow 450 + 30x + 450 - 30x = \frac{9}{2}(225 - x^2)$$

$$\Rightarrow 200 = 225 - x^2 \Rightarrow x^2 = 25 \Rightarrow x = 5.$$

53. Let the speed of the stream be x km/hr.

Then, speed downstream = $(9 + x)$ km/hr

and speed upstream = $(9 - x)$ km/hr.

$$\therefore \frac{15}{9+x} + \frac{15}{9-x} = 3 \frac{45}{60} = 3 \frac{3}{4} = \frac{15}{4}$$

$$\Rightarrow \frac{1}{9+x} + \frac{1}{9-x} = \frac{1}{4} \Rightarrow 4[(9-x) + (9+x)] = (9+x)(9-x)$$

$$\Rightarrow 81 - x^2 = 72 \Rightarrow x^2 = 9 \Rightarrow x = 3.$$

54. Suppose B takes x days to finish the work. Then, A takes $(x - 10)$ days.

$$\therefore \frac{1}{x} + \frac{1}{(x-10)} = \frac{1}{12} \Rightarrow x^2 - 34x + 120 = 0 \Rightarrow x^2 - 30x - 4x + 120 = 0.$$

55. Let the faster pipe take x hr to fill it. Then, the other takes $(x + 3)$ hr.

$$\therefore \frac{1}{x} + \frac{1}{(x+3)} = \frac{13}{40} \Rightarrow \frac{(x+3) + x}{x(x+3)} = \frac{13}{40}$$

$$\Rightarrow 13x^2 - 41x - 120 = 0$$

$$\Rightarrow x = \frac{41 \pm \sqrt{1681 + 6240}}{26} = \frac{41 \pm \sqrt{7921}}{26} = \frac{(41 \pm 89)}{26} \Rightarrow x = 5.$$

56. Let the faster pipe take x min to fill it. Then, the other takes $(x + 5)$ min.

$$\therefore \frac{1}{x} + \frac{1}{(x+5)} = \frac{9}{100} \Rightarrow 100(2x+5) = 9(x^2+5x)$$

$$\Rightarrow 9x^2 - 155x - 500 = 0 \Rightarrow 9x^2 - 180x + 25x - 500 = 0.$$

HINT Quadratic formula may be used.

57. Let the faster tap take x hours to fill the tank.

Then, the slower tap takes $(x + 9)$ hours to fill it.

$$\therefore \frac{1}{x} + \frac{1}{(x+9)} = \frac{1}{6} \Rightarrow 6(2x+9) = x(x+9)$$

$$\Rightarrow x^2 - 3x - 54 = 0 \Rightarrow x^2 - 9x + 6x - 54 = 0.$$

58. $2x \times x = 288 \Rightarrow x^2 = 144 \Rightarrow x = 12$ cm.

59. $3x \times x = 147 \Rightarrow x^2 = 49 \Rightarrow x = 7$ m. So, length = 21 m.

60. $x(x+3) = 238 \Rightarrow x^2 + 3x - 238 = 0 \Rightarrow x^2 + 17x - 14x - 238 = 0.$

$\therefore$ breadth = 14 m and length = 17 m.

61. $2(l+b) = 62 \Rightarrow l+b = 31$. Let $b = x$ m. Then, $l = (31-x)$ m

$$\therefore (31-x)x = 228 \Rightarrow x^2 - 31x + 228 = 0 \Rightarrow x^2 - 19x - 12x + 228 = 0.$$

62. Let the width of the path be x m.

So, the length of the field with path = $(16+2x)$ m

and breadth of the field with path = $(10+2x)$ m.

$$\therefore (16+2x)(10+2x) - 16 \times 10 = 120$$

$$\Rightarrow 160 + 52x + 4x^2 - 160 = 120$$

$$\Rightarrow 4x^2 + 52x - 120 = 0 \Rightarrow x^2 + 13x - 30 = 0$$

$$\Rightarrow x^2 + 15x - 2x - 30 = 0 \Rightarrow x(x+15) - 2(x+15) = 0$$

$$\Rightarrow (x+15)(x-2) = 0 \Rightarrow x = -15 \text{ or } 2.$$

So, width of the path = 2 m.

63. Let the sides of the squares be a metres and b metres.

$$\text{Then, } 4a - 4b = 64 \Rightarrow 4(a-b) = 64 \Rightarrow (a-b) = 16.$$

$$\text{And, } a^2 + b^2 = 640 \Rightarrow a^2 + (a-16)^2 = 640$$

$$\Rightarrow 2a^2 - 32a - 384 = 0 \Rightarrow a^2 - 16a - 192 = 0$$

$$\Rightarrow a^2 - 24a + 8a - 192 = 0 \Rightarrow (a-24)(a+8) = 0 \Rightarrow a = 24.$$

$$\therefore a = 24 \text{ m, } b = 8 \text{ m.}$$

64. Let each side of the square be x cm.

Then, length of the rectangle = $3x$ cm.

Width of the rectangle = $(x-4)$ cm.

$$\therefore 3x(x-4) = x^2 \Rightarrow 2x^2 - 12x = 0 \Rightarrow 2x(x-6) = 0 \Rightarrow x = 6.$$

65. Let the length be x metres. Then, breadth = $\frac{180}{x}$ m.

$$\therefore x + \frac{180}{x} + \frac{180}{x} = 39 \Rightarrow x + \frac{360}{x} = 39$$

$$\Rightarrow x^2 - 39x + 360 = 0 \Rightarrow x^2 - 24x - 15x + 360 = 0$$

$$\Rightarrow (x-24)(x-15) = 0 \Rightarrow x = 24 \text{ or } x = 15.$$

Either ($l = 24$ m, $b = 7.5$ m) or ($l = 15$ m, $b = 12$ m).

66. Let the altitude be x cm. Then, base = $(x + 10)$ cm.

$$\therefore \frac{1}{2}(x+10)x = 600 \Rightarrow x^2 + 10x - 1200 = 0$$

$$\Rightarrow (x+40)(x-30) = 0 \Rightarrow x = 30.$$

$\therefore$ base = 40 cm, altitude = 30 cm.

$$\therefore \text{hypotenuse} = \sqrt{(40)^2 + (30)^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50 \text{ cm.}$$

67. Let the altitude be x metres. Then, base = $3x$ metres.

$$\therefore \frac{1}{2} \times 3x \times x = 96 \Rightarrow x^2 = 64 \Rightarrow x = 8 \text{ m. Hence, base} = 24 \text{ m.}$$

68. Let the base be x metres. Then, altitude = $(x + 7)$ m.

$$\therefore \frac{1}{2} \times x \times (x+7) = 165 \Rightarrow x^2 + 7x - 330 = 0$$

$$\Rightarrow x^2 + 22x - 15x - 330 = 0 \Rightarrow (x-15)(x+22) = 0 \Rightarrow x = 15.$$

69. Let the other sides be x metres and $(x - 4)$ metres.

$$\therefore x^2 + (x-4)^2 = (20)^2 \Rightarrow 2x^2 - 8x - 384 = 0$$

$$\Rightarrow x^2 - 4x - 192 = 0 \Rightarrow (x-16)(x+12) = 0 \Rightarrow x = 16.$$

70. Let the base be x cm. Then, hypotenuse = $(x + 2)$ cm.

$$(x+2) - (2 \times \text{altitude}) = 1 \Rightarrow 2 \times \text{altitude} = (x+1)$$

$$\Rightarrow \text{altitude} = \frac{1}{2}(x+1) \text{ cm.}$$

$$\therefore (x+2)^2 = x^2 + \frac{1}{4}(x+1)^2 \Rightarrow 4(x+2)^2 = 4x^2 + (x+1)^2$$

$$\Rightarrow 4(x^2 + 4x + 4) = 5x^2 + 2x + 1 \Rightarrow x^2 - 14x - 15 = 0$$

$$\Rightarrow (x-15)(x+1) = 0 \Rightarrow x = 15.$$

$\therefore$ base = 15 cm, hypotenuse = 17 cm, altitude = 8 cm.

71. Let the shortest side be x metres.

Then, its hypotenuse = $(2x - 1)$ m and third side = $(x + 1)$ m.

$$\therefore (2x-1)^2 = x^2 + (x+1)^2 \Rightarrow 2x^2 - 6x = 0 \Rightarrow 2x(x-3) = 0.$$

SUMMARY OF RESULTS

1. If α and β are the roots of a quadratic equation $ax^2 + bx + c = 0$ then

$$(i) \quad \alpha + \beta = \frac{-b}{a},$$

$$(ii) \quad \alpha\beta = \frac{c}{a}.$$

2. The roots of the equation $ax^2 + bx + c = 0$ are given by

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

3. For the equation $ax^2 + bx + c = 0$, $a \neq 0$ the *discriminant* is given by $D = (b^2 - 4ac)$.

4. Nature of Roots of quadratic equation $ax^2 + bx + c = 0, a \neq 0$.

Value of D	Nature of Roots	
(i) $D > 0$ and D is a perfect square.	Real, unequal and rational	$\frac{-b \pm \sqrt{D}}{2a}$
(ii) $D > 0$ and D is not a perfect square.	Real, unequal and irrational	$\frac{-b \pm \sqrt{D}}{2a}$
(iii) $D = 0$	Real and equal	Each = $\frac{-b}{2a}$
(iv) $D < 0$	Imaginary	None

TEST YOURSELF

MCQ

1. Which of the following is a quadratic equation?

(a) $x^2 - 3\sqrt{x} + 2 = 0$

(b) $x + \frac{1}{x} = x^2$

(c) $x^2 + \frac{1}{x^2} = 5$

(d) $2x^2 - 5x = (x - 1)^2$

2. Which of the following is a quadratic equation?

(a) $(x^2 + 1) = (2 - x)^2 + 3$

(b) $x^3 - x^2 = (x - 1)^3$

(c) $2x^2 + 3 = (5 + x)(2x - 3)$

(d) None of these

3. Which of the following is not a quadratic equation?

(a) $3x - x^2 = x^2 + 5$

(b) $(x + 2)^2 = 2(x^2 - 5)$

(c) $(\sqrt{2}x + 3)^2 = 2x^2 + 6$

(d) $(x - 1)^2 = 3x^2 + x - 2$

4. If $x = 3$ is a solution of the equation $3x^2 + (k - 1)x + 9 = 0$ then $k = ?$

(a) 11

(b) -11

(c) 13

(d) -13

5. If one root of the equation $2x^2 + ax + 6 = 0$ is 2 then $a = ?$

(a) 7

(b) -7

(c) $\frac{7}{2}$

(d) $-\frac{7}{2}$

6. The sum of the roots of the equation $x^2 - 6x + 2 = 0$ is

(a) 2

(b) -2

(c) 6

(d) -6

7. If the product of the roots of the equation $x^2 - 3x + k = 10$ is -2 then the value of k is

(a) -2

(b) -8

(c) 8

(d) 12

8. The ratio of the sum and product of the roots of the equation $7x^2 - 12x + 18 = 0$ is
 (a) 7 : 12 (b) 7 : 18 (c) 3 : 2 (d) 2 : 3
9. If one root of the equation $3x^2 - 10x + 3 = 0$ is $\frac{1}{3}$ then the other root is
 (a) $-\frac{1}{3}$ (b) $\frac{1}{3}$ (c) -3 (d) 3
10. If one root of $5x^2 + 13x + k = 0$ be the reciprocal of the other root then the value of k is
 (a) 0 (b) 1 (c) 2 (d) 5
11. If the sum of the roots of the equation $kx^2 + 2x + 3k = 0$ is equal to their product then the value of k is
 (a) $\frac{1}{3}$ (b) $-\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $-\frac{2}{3}$
12. The roots of a quadratic equation are 5 and -2. Then, the equation is
 (a) $x^2 - 3x + 10 = 0$ (b) $x^2 - 3x - 10 = 0$
 (c) $x^2 + 3x - 10 = 0$ (d) $x^2 + 3x + 10 = 0$
13. If the sum of the roots of a quadratic equation is 6 and their product is 6, the equation is
 (a) $x^2 - 6x + 6 = 0$ (b) $x^2 + 6x - 6 = 0$
 (c) $x^2 - 6x - 6 = 0$ (d) $x^2 + 6x + 6 = 0$
14. If α and β are the roots of the equation $3x^2 + 8x + 2 = 0$ then $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = ?$
 (a) $-\frac{3}{8}$ (b) $\frac{2}{3}$ (c) -4 (d) 4
15. The roots of the equation $ax^2 + bx + c = 0$ will be reciprocal of each other if
 (a) $a = b$ (b) $b = c$ (c) $c = a$ (d) none of these
16. If the roots of the equation $ax^2 + bx + c = 0$ are equal then $c = ?$
 (a) $-\frac{b}{2a}$ (b) $\frac{b}{2a}$ (c) $-\frac{b^2}{4a}$ (d) $\frac{b^2}{4a}$
17. If the equation $9x^2 + 6kx + 4 = 0$ has equal roots then $k = ?$
 (a) 2 or 0 (b) -2 or 0 (c) 2 or -2 (d) 0 only
18. If the equation $x^2 + 2(k+2)x + 9k = 0$ has equal roots then $k = ?$
 (a) 1 or 4 (b) -1 or 4 (c) 1 or -4 (d) -1 or -4
19. If the equation $4x^2 - 3kx + 1 = 0$ has equal roots then $k = ?$
 (a) $\pm \frac{2}{3}$ (b) $\pm \frac{1}{3}$ (c) $\pm \frac{3}{4}$ (d) $\pm \frac{4}{3}$

20. The roots of $ax^2 + bx + c = 0$, $a \neq 0$ are real and unequal, if $(b^2 - 4ac)$ is
(a) > 0 (b) $= 0$ (c) < 0 (d) none of these
21. In the equation $ax^2 + bx + c = 0$, it is given that $D = (b^2 - 4ac) > 0$. Then, the roots of the equation are
(a) real and equal (b) real and unequal
(c) imaginary (d) none of these
22. The roots of the equation $2x^2 - 6x + 7 = 0$ are
(a) real, unequal and rational (b) real, unequal and irrational
(c) real and equal (d) imaginary
23. The roots of the equation $2x^2 - 6x + 3 = 0$ are
(a) real, unequal and rational (b) real, unequal and irrational
(c) real and equal (d) imaginary
24. If the roots of $5x^2 - kx + 1 = 0$ are real and distinct then
(a) $-2\sqrt{5} < k < 2\sqrt{5}$ (b) $k > 2\sqrt{5}$ only
(c) $k < -2\sqrt{5}$ only (d) either $k > 2\sqrt{5}$ or $k < -2\sqrt{5}$
25. If the equation $x^2 + 5kx + 16 = 0$ has no real roots then
(a) $k > \frac{8}{5}$ (b) $k < \frac{-8}{5}$
(c) $\frac{-8}{5} < k < \frac{8}{5}$ (d) none of these
26. If the equation $x^2 - kx + 1 = 0$ has no real roots then
(a) $k < -2$ (b) $k > 2$
(c) $-2 < k < 2$ (d) none of these
27. For what values of k , the equation $kx^2 - 6x - 2 = 0$ has real roots?
(a) $k \leq \frac{-9}{2}$ (b) $k \geq \frac{-9}{2}$
(c) $k \leq -2$ (d) None of these
28. The sum of a number and its reciprocal is $2\frac{1}{20}$. The number is
(a) $\frac{5}{4}$ or $\frac{4}{5}$ (b) $\frac{4}{3}$ or $\frac{3}{4}$ (c) $\frac{5}{6}$ or $\frac{6}{5}$ (d) $\frac{1}{6}$ or 6
29. The perimeter of a rectangle is 82 m and its area is 400 m^2 . The breadth of the rectangle is
(a) 25 m (b) 20 m (c) 16 m (d) 9 m
30. The length of a rectangular field exceeds its breadth by 8 m and the area of the field is 240 m^2 . The breadth of the field is [CBSE 2014]
(a) 20 m (b) 30 m (c) 12 m (d) 16 m

31. The roots of the quadratic equation $2x^2 - x - 6 = 0$ are [CBSE 2012]
 (a) $-2, \frac{3}{2}$ (b) $2, \frac{-3}{2}$ (c) $-2, \frac{-3}{2}$ (d) $2, \frac{3}{2}$
32. The sum of two natural numbers is 8 and their product is 15. Find the numbers. [CBSE 2012]

Very-Short-Answer Questions

33. Show that $x = -3$ is a solution of $x^2 + 6x + 9 = 0$. [CBSE 2008]
34. Show that $x = -2$ is a solution of $3x^2 + 13x + 14 = 0$. [CBSE 2008]
35. If $x = \frac{-1}{2}$ is a solution of the quadratic equation $3x^2 + 2kx - 3 = 0$, find the value of k . [CBSE 2015]
36. Find the roots of the quadratic equation $2x^2 - x - 6 = 0$. [CBSE 2012]
37. Find the solution of the quadratic equation $3\sqrt{3}x^2 + 10x + \sqrt{3} = 0$. [CBSE 2009]
38. If the roots of the quadratic equation $2x^2 + 8x + k = 0$ are equal then find the value of k . [CBSE 2014]
39. If the quadratic equation $px^2 - 2\sqrt{5}px + 15 = 0$ has two equal roots then find the value of p . [CBSE 2015]
40. If 1 is a root of the equation $ay^2 + ay + 3 = 0$ and $y^2 + y + b = 0$ then find the value of ab . [CBSE 2012]
41. If one zero of the polynomial $x^2 - 4x + 1$ is $(2 + \sqrt{3})$, write the other zero. [CBSE 2010]
42. If one root of the quadratic equation $3x^2 - 10x + k = 0$ is reciprocal of the other, find the value of k . [CBSE 2014]
43. If the roots of the quadratic equation $px(x - 2) + 6 = 0$ are equal, find the value of p . [CBSE 2013]
44. Find the values of k so that the quadratic equation $x^2 - 4kx + k = 0$ has equal roots. [CBSE 2012]
45. Find the values of k for which the quadratic equation $9x^2 - 3kx + k = 0$ has equal roots. [CBSE 2014]

Short-Answer Questions

46. Solve: $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$. [CBSE 2015]
47. Solve: $2x^2 + ax - a^2 = 0$. [CBSE 2014]
48. Solve: $3x^2 + 5\sqrt{5}x - 10 = 0$. [CBSE 2014]
49. Solve: $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$. [CBSE 2014]

50. Solve: $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$. [CBSE 2014]
 51. Solve: $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$. [CBSE 2013]
 52. Solve: $4x^2 + 4bx - (a^2 - b^2) = 0$. [CBSE 2015]
 53. Solve: $x^2 + 5x - (a^2 + a - 6) = 0$. [CBSE 2015]
 54. $x^2 + 6x - (a^2 + 2a - 8) = 0$ [CBSE 2015]
 55. $x^2 - 4ax + 4a^2 - b^2 = 0$ [CBSE 2012]

ANSWERS (TEST YOURSELF)

1. (d) 2. (b) 3. (c) 4. (b) 5. (b) 6. (c) 7. (c) 8. (d) 9. (d)
 10. (d) 11. (d) 12. (b) 13. (a) 14. (c) 15. (c) 16. (d) 17. (c) 18. (a)
 19. (d) 20. (a) 21. (b) 22. (d) 23. (b) 24. (d) 25. (c) 26. (c) 27. (b)
 28. (a) 29. (c) 30. (c) 31. (b) 32. 3 and 5
 35. $k = \frac{-9}{4}$ 36. $x = 2$ or $x = \frac{-3}{2}$ 37. $x = -\sqrt{3}$ or $x = \frac{-1}{3\sqrt{3}}$ 38. $k = 8$
 39. $p = 3$ 40. $ab = 3$ 41. $(2 - \sqrt{3})$ 42. $k = 3$ 43. $p = 6$
 44. $k = 0$ or $k = \frac{1}{4}$ 45. $k = 0$ or $k = 1$ 46. $x = \sqrt{3}$ or $x = 1$
 47. $x = -a$ or $x = \frac{a}{2}$ 48. $x = -2\sqrt{5}$ or $x = \frac{\sqrt{5}}{3}$ 49. $x = -4\sqrt{3}$ or $x = \frac{2}{\sqrt{3}}$
 50. $x = \sqrt{6}$ or $x = \frac{\sqrt{2}}{\sqrt{3}}$ 51. $x = \frac{-2}{\sqrt{3}}$ or $x = \frac{\sqrt{3}}{4}$
 52. $x = \frac{-(b+a)}{2}$ or $x = \frac{(a-b)}{2}$ 53. $x = -(a+3)$ or $x = (a-2)$
 54. $x = -(a+4)$ or $x = (a-2)$ 55. $x = (2a+b)$ or $x = (2a-b)$

HINTS TO SOME SELECTED QUESTIONS

7. Given equation is $x^2 - 3x + (k - 10) = 0$.
 Product of roots $= (k - 10)$. So, $k - 10 = -2 \Rightarrow k = 8$.
 8. Required ratio $= \frac{12}{7} : \frac{18}{7} = 2 : 3$.
 9. Let the other root be α . Then, $\frac{1}{3} \times \alpha = \frac{3}{3} = 1 \Rightarrow \alpha = 3$.
 So, the other root is 3.
 10. Let the roots be α and $\frac{1}{\alpha}$. Then,
 product of roots $= \left(\alpha \times \frac{1}{\alpha}\right) = 1$. So, $\frac{k}{5} = 1 \Rightarrow k = 5$.

11. Sum of roots $= \frac{-2}{k}$ and product of roots $= \frac{3k}{k} = 3$.

$$\therefore \frac{-2}{k} = 3 \Rightarrow k = \frac{-2}{3}.$$

12. Sum of the roots $= 5 + (-2) = 3$, product of roots $= 5 \times (-2) = -10$.

$$\therefore x^2 - (\text{sum of roots})x + \text{product of roots} = 0.$$

Hence, $x^2 - 3x - 10 = 0$.

13. Required equation is $x^2 - 6x + 6 = 0$.

14. We have $\alpha + \beta = \frac{-8}{3}$ and $\alpha\beta = \frac{2}{3}$.

$$\therefore \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) = \frac{(\alpha + \beta)}{\alpha\beta} = \frac{-8}{3} \times \frac{3}{2} = -4.$$

15. Product of roots $= \frac{c}{a}$. Also, $\left(\alpha \times \frac{1}{\alpha} \right) = 1$.

$$\therefore \frac{c}{a} = 1 \Rightarrow c = a.$$

16. Since the roots are equal, we have $D = 0$.

$$\therefore b^2 - 4ac = 0 \Rightarrow 4ac = b^2 \Rightarrow c = \frac{b^2}{4a}.$$

17. Since the roots are equal, we have $D = 0$.

$$\therefore 36k^2 - 4 \times 9 \times 4 = 0 \Rightarrow 36k^2 = 144 \Rightarrow k^2 = 4 \Rightarrow k = 2 \text{ or } -2.$$

18. Since the roots are equal, we have $D = 0$.

$$\therefore 4(k+2)^2 - 36k = 0 \Rightarrow (k+2)^2 - 9k = 0$$

$$\therefore k^2 - 5k + 4 = 0 \Rightarrow k^2 - 4k - k + 4 = 0 \Rightarrow k(k-4) - (k-4) = 0$$

$$\Rightarrow (k-4)(k-1) = 0 \Rightarrow k = 4 \text{ or } k = 1.$$

19. Since the roots are equal, we have $D = 0$.

$$\therefore 9k^2 - 16 = 0 \Rightarrow k^2 = \frac{16}{9} \Rightarrow k = \frac{4}{3} \text{ or } k = -\frac{4}{3}.$$

20. The roots of $ax^2 + bx + c = 0$, $a \neq 0$ are real and unequal only when $(b^2 - 4ac) > 0$.

21. When $D > 0$, the roots of the given quadratic equation are real and unequal.

22. $D = (-6)^2 - 4 \times 2 \times 7 = (36 - 56) = -20 < 0$.

So, the roots of the given equation are imaginary.

23. Given equation is $2x^2 - 6x + 3 = 0$.

$$\therefore D = (-6)^2 - 4 \times 2 \times 3 = (36 - 24) = 12,$$

which is greater than 0 and not a perfect square.

$$\therefore \text{the roots are real, unequal and irrational.}$$

24. The roots of $5x^2 - kx + 1 = 0$ are real and distinct.

$$\therefore (k^2 - 4 \times 5 \times 1) > 0 \Rightarrow k^2 > 20 \Rightarrow k > \sqrt{20} \text{ or } k < -\sqrt{20}$$

$$\Rightarrow k > 2\sqrt{5} \text{ or } k < -2\sqrt{5}.$$

25. For no real roots, we must have $b^2 - 4ac < 0$.

$$\therefore (25k^2 - 4 \times 16) < 0 \Rightarrow 25k^2 < 64 \Rightarrow k^2 < \frac{64}{25} \Rightarrow \frac{-8}{5} < k < \frac{8}{5}.$$

26. For no real roots, we must have: $b^2 - 4ac < 0$.

$$k^2 - 4 < 0 \Rightarrow k^2 < 4 \Rightarrow -2 < k < 2.$$

27. For real roots, we must have $b^2 - 4ac \geq 0$.

$$\begin{aligned} \therefore (-6)^2 - 4 \times k \times (-2) &\geq 0 \Rightarrow 36 + 8k \geq 0 \\ &\Rightarrow 8k \geq -36 \Rightarrow k \geq \frac{-9}{2}. \end{aligned}$$

28. Let the required number be x . Then,

$$\begin{aligned} x + \frac{1}{x} &= \frac{41}{20} \Rightarrow 20x^2 - 41x + 20 = 0 \\ &\Rightarrow 20x^2 - 25x - 16x + 20 = 0 \\ &\Rightarrow 5x(4x - 5) - 4(4x - 5) = 0 \\ &\Rightarrow (4x - 5)(5x - 4) = 0 \\ &\Rightarrow x = \frac{5}{4} \text{ or } x = \frac{4}{5}. \end{aligned}$$

29. $2(l + b) = 82 \Rightarrow l + b = 41 \Rightarrow l = (41 - b)$.

$$\text{And, } lb = 400 \Rightarrow (41 - b)b = 400$$

$$\Rightarrow b^2 - 41b + 400 = 0 \Rightarrow b^2 - 25b - 16b + 400 = 0$$

$$\Rightarrow b(b - 25) - 16(b - 25) = 0$$

$$\Rightarrow (b - 25)(b - 16) = 0$$

$$\therefore b = 25 \text{ or } b = 16.$$

$$\text{But } b = 25 \Rightarrow l = (41 - 25) = 16 < b.$$

$$\therefore \text{breadth} = 16 \text{ m.}$$

30. Let the breadth of the field be x m. Then, length = $(x + 8)$ m.

$$\begin{aligned} \therefore (x + 8) \times x &= 240 \Rightarrow x^2 + 8x - 240 = 0 \Rightarrow x^2 + 20x - 12x - 240 = 0 \\ &\Rightarrow x(x + 20) - 12(x + 20) = 0 \\ &\Rightarrow (x + 20)(x - 12) = 0. \end{aligned}$$

32. $\alpha + \beta = 8$ and $\alpha\beta = 15$.

The quadratic equation whose roots are α and β is given by

$$x^2 - 8x + 15 = 0$$

$$\Rightarrow x^2 - 5x - 3x + 15 = 0$$

$$\Rightarrow x(x - 5) - 3(x - 5) = 0$$

$$\Rightarrow (x - 3)(x - 5) = 0$$

$$\therefore x = 3 \text{ or } x = 5.$$

33. Putting $x = -3$ in the given equation, we get

$$\text{LHS} = (-3)^2 + 6 \times (-3) + 9 = (9 - 18 + 9) = 0 = \text{RHS.}$$

Hence, $x = -3$ is a solution of $x^2 + 6x + 9 = 0$.

34. Putting $x = -2$ in the given equation, we get

$$\text{LHS} = 3 \times (-2)^2 + 13 \times (-2) + 14 = (12 - 26 + 14) = 0 = \text{RHS.}$$

Hence, $x = -2$ is a solution of $3x^2 + 13x + 14 = 0$.

35. Since $x = \frac{-1}{2}$ is a solution of $3x^2 + 2kx - 3 = 0$, we have

$$3 \times \left(\frac{-1}{2}\right)^2 + 2k \times \left(\frac{-1}{2}\right) - 3 = 0$$

$$\Rightarrow \frac{3}{4} - k - 3 = 0 \Rightarrow k = \left(\frac{3}{4} - 3\right) = \frac{-9}{4}.$$

36. $2x^2 - x - 6 = 0 \Rightarrow 2x^2 - 4x + 3x - 6 = 0 \Rightarrow 2x(x - 2) + 3(x - 2) = 0$
 $\Rightarrow (x - 2)(2x + 3) = 0 \Rightarrow x = 2 \text{ or } x = \frac{-3}{2}.$

37. $3\sqrt{3}x^2 + 10x + \sqrt{3} = 0 \Rightarrow 3\sqrt{3}x^2 + 9x + x + \sqrt{3} = 0$
 $\Rightarrow 3\sqrt{3}x(x + \sqrt{3}) + (x + \sqrt{3}) = 0 \Rightarrow (x + \sqrt{3})(3\sqrt{3}x + 1) = 0$
 $\Rightarrow x = -\sqrt{3} \text{ or } x = \frac{-1}{3\sqrt{3}}.$

38. $D = (8)^2 - 4 \times 2 \times k = (64 - 8k).$
 $\therefore D = 0 \Rightarrow 64 - 8k = 0 \Rightarrow 8k = 64 \Rightarrow k = 8.$

39. The given equation is of the form $ax^2 + bx + c = 0$, where

$$a = p, b = -2\sqrt{5}p \text{ and } c = 15.$$

$$\therefore D = (b^2 - 4ac) = (20p^2 - 60p).$$

$$\text{So, } D = 0 \Rightarrow 20p^2 - 60p = 0 \Rightarrow 20p(p - 3) = 0 \Rightarrow p = 3. \quad [\because p \neq 0]$$

40. Since 1 is a root of each of the equations $ay^2 + ay + 3 = 0$ and $y^2 + y + b = 0$, we have

$$a \times (1)^2 + a \times 1 + 3 = 0 \text{ and } 1^2 + 1 + b = 0$$

$$\Rightarrow a + a + 3 = 0 \text{ and } 2 + b = 0 \Rightarrow a = \frac{-3}{2} \text{ and } b = -2.$$

$$\therefore ab = \left(\frac{-3}{2}\right) \times (-2) = 3.$$

41. Let the other zero be α .

Sum of the zeros of $x^2 - 4x + 1$ is 4.

$$\therefore \alpha + (2 + \sqrt{3}) = 4 \Rightarrow \alpha = 4 - (2 + \sqrt{3}) = (2 - \sqrt{3}).$$

42. Let the roots of $3x^2 - 10x + k = 0$ be α and $\frac{1}{\alpha}$.

Then, product of roots = $\frac{k}{3}$.

$$\therefore \left(\alpha \times \frac{1}{\alpha}\right) = \frac{k}{3} \Rightarrow \frac{k}{3} = 1 \Rightarrow k = 3.$$

43. Given equation is $px^2 - 2px + 6 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = p, b = -2p$ and $c = 6$.

$$\therefore D = (b^2 - 4ac) = 4p^2 - 24p.$$

For equal roots, we have $D = 0$.

$$\therefore 4p^2 - 24p = 0 \Rightarrow 4p(p - 6) = 0 \Rightarrow p = 6. \quad [\because p \neq 0]$$

44. $D = (-4k)^2 - 4k = 16k^2 - 4k = 4k(4k - 1).$

For equal roots, we must have $D = 0$.

$$\text{Now, } D = 0 \Rightarrow 4k(4k-1) = 0 \Rightarrow k = 0 \text{ or } 4k-1 = 0 \Rightarrow k = 0 \text{ or } k = \frac{1}{4}.$$

$$45. D = (-3k)^2 - 9k = 9k^2 - 9k = 9k(k-1).$$

For equal roots, we must have $D = 0$.

$$\text{Now, } D = 0 \Rightarrow 9k(k-1) = 0 \Rightarrow k = 0 \text{ or } k-1 = 0 \Rightarrow k = 0 \text{ or } k = 1.$$

$$46. x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$$

$$\Rightarrow x^2 - \sqrt{3}x - x + \sqrt{3} = 0 \Rightarrow x(x - \sqrt{3}) - (x - \sqrt{3}) = 0$$

$$\Rightarrow (x - \sqrt{3})(x - 1) = 0 \Rightarrow x = \sqrt{3} \text{ or } x = 1.$$

$$47. 2x^2 + ax - a^2 = 0 \Rightarrow 2x^2 + 2ax - ax - a^2 = 0$$

$$\Rightarrow 2x(x + a) - a(x + a) = 0$$

$$\Rightarrow (x + a)(2x - a) = 0 \Rightarrow x = -a \text{ or } x = \frac{a}{2}.$$

$$48. 3x^2 + 5\sqrt{5}x - 10 = 0$$

$$\Rightarrow 3x^2 + 6\sqrt{5}x - \sqrt{5}x - 10 = 0$$

$$\Rightarrow 3x(x + 2\sqrt{5}) - \sqrt{5}(x + 2\sqrt{5}) = 0$$

$$\Rightarrow (x + 2\sqrt{5})(3x - \sqrt{5}) = 0 \Rightarrow x = -2\sqrt{5} \text{ or } x = \frac{\sqrt{5}}{3}.$$

$$49. \sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$$

$$\Rightarrow \sqrt{3}x^2 + 12x - 2x - 8\sqrt{3} = 0$$

$$\Rightarrow \sqrt{3}x(x + 4\sqrt{3}) - 2(x + 4\sqrt{3}) = 0$$

$$\Rightarrow (x + 4\sqrt{3})(\sqrt{3}x - 2) = 0 \Rightarrow x = -4\sqrt{3} \text{ or } x = \frac{2}{\sqrt{3}}.$$

$$50. \sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$$

$$\Rightarrow \sqrt{3}x^2 - 3\sqrt{2}x - \sqrt{2}x - 2\sqrt{3} = 0$$

$$\Rightarrow \sqrt{3}x(x - \sqrt{6}) - \sqrt{2}(x - \sqrt{6}) = 0$$

$$\Rightarrow (x - \sqrt{6})(\sqrt{3}x - \sqrt{2}) = 0 \Rightarrow x = \sqrt{6} \text{ or } x = \frac{\sqrt{2}}{\sqrt{3}}.$$

$$51. 4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$$

$$\Rightarrow 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} = 0$$

$$\Rightarrow 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x - 2) = 0$$

$$\Rightarrow (\sqrt{3}x + 2)(4x - \sqrt{3}) = 0 \Rightarrow x = \frac{-2}{\sqrt{3}} \text{ or } x = \frac{\sqrt{3}}{4}.$$

$$52. 4x^2 + 4bx + (b^2 - a^2) = 0$$

$$\Rightarrow 4x^2 + 2(b+a)x + 2(b-a)x + (b^2 - a^2) = 0$$

$$\Rightarrow 2x\{2x + (b+a)\} + (b-a)\{2x + (b+a)\} = 0$$

$$\Rightarrow \{2x + (b+a)\}\{2x + (b-a)\} = 0$$

$$\Rightarrow x = \frac{-(b+a)}{2} \text{ or } x = \frac{a-b}{0}.$$

$$53. x^2 + 5x - (a+3)(a-2) = 0$$

$$\Rightarrow x^2 + (a+3)x - (a-2)x - (a+3)(a-2) = 0$$

$$\Rightarrow x\{x + (a+3)\} - (a-2)\{x + (a+3)\} = 0$$

$$\Rightarrow \{x + (a+3)\}\{x - (a-2)\} = 0$$

$$\Rightarrow x = -(a+3) \text{ or } x = (a-2).$$

$$54. x^2 + 6x - (a+4)(a-2) = 0$$

$$\Rightarrow x^2 + (a+4)x - (a-2)x - (a+4)(a-2) = 0$$

$$\Rightarrow x\{x + (a+4)\} - (a-2)\{x + (a+4)\} = 0$$

$$\Rightarrow \{x + (a+4)\}\{x - (a-2)\} = 0$$

$$\Rightarrow x = -(a+4) \text{ or } x = (a-2).$$

$$55. x^2 - 4ax + (2a+b)(2a-b) = 0$$

$$\Rightarrow x^2 - (2a+b)x - (2a-b)x + (2a+b)(2a-b) = 0$$

$$\Rightarrow x\{x - (2a+b)\} - (2a-b)\{x - (2a+b)\} = 0$$

$$\Rightarrow \{x - (2a+b)\}\{x - (2a-b)\} = 0$$

$$\Rightarrow x = (2a+b) \text{ or } x = (2a-b).$$



S RAMANUJAN (1887–1920)

Srinivasa Aaiyengar Ramanujan was born in Erode, a small village near Chennai in 1887. His father was a clerk at a cloth shop and his mother was a housewife. On the basis of his good school work, he was admitted to the University of Madras. But, in first year he passed in mathematics and failed in other subjects. What earned him fame, reputation and success was his research work on numbers.

**HIS RESEARCH WORK AND CONTRIBUTION ON NUMBERS**

(i) In 1900, he began to work on *Geometric* and *Arithmetic Series*.

He gave new methods to solve *Cubic* and *Quadratic Equations*.

(ii) He investigated the series $\sum \frac{1}{n}$ and found *Bernoulli's Numbers*.

(iii) In 1908, he studied *Continued Fractions* and *Divergent Series*.

(iv) In 1911, he produced a research paper on *Bernoulli's Numbers*.

In 1918, he was given a fellowship by Trinity College, Cambridge for 7 years. He was a mathematical genius, despite his lack of university education.

He died at an early age of 33 years only.

SEQUENCE Some numbers arranged in a definite order, according to a definite rule, are said to form a sequence.

The number occurring at the n th place of a sequence is called its n th term, denoted by T_n or a_n .

Example Consider the rule, $T_n = (2n + 1)$.

Putting $n = 1, 2, 3, 4, 5, \dots$, we get

$T_1 = 3, T_2 = 5, T_3 = 7, T_4 = 9, T_5 = 11$, and so on.

Thus, the numbers 3, 5, 7, 9, 11, ... form a sequence.

In this sequence, the first term is 3, the second term is 5, the third term is 7, and so on.

ARITHMETIC PROGRESSION (AP)

A sequence in which each term differs from its preceding term by a constant is called an arithmetic progression, written as AP.

This constant is called the *common difference* of the AP.

EXAMPLE 1 Show that the progression 8, 11, 14, 17, 20, ... is an AP. Find its first term and the common difference.

SOLUTION The given progression is 8, 11, 14, 17, 20,
Clearly, $(11 - 8) = (14 - 11) = (17 - 14) = (20 - 17) = 3$ (constant).
Thus, each term differs from its preceding term by 3.
So, the given progression is an AP.
Its first term = 8 and common difference = 3.

EXAMPLE 2 Show that the progression 11, 6, 1, -4, -9, ... is an AP. Find its first term and the common difference.

SOLUTION Clearly, $(6 - 11) = (1 - 6) = (-4 - 1) = (-9 + 4) = -5$ (constant).
Thus, each term differs from its preceding term by -5.
So, the given progression is an AP.
Its first term = 11 and common difference = -5.

EXAMPLE 3 Show that each of the following progressions is an AP. Find the common difference and the next term of each.

(i) $\sqrt{7}, \sqrt{28}, \sqrt{63}, \dots$ [CBSE 2014] (ii) $\sqrt{18}, \sqrt{50}, \sqrt{98}, \dots$ [CBSE 2012]

SOLUTION (i) We may write the given terms as

$$\sqrt{7}, \sqrt{4 \times 7}, \sqrt{9 \times 7}, \dots$$

$$\text{i.e., } \sqrt{7}, 2\sqrt{7}, 3\sqrt{7}, \dots$$

$$\text{Clearly, } 2\sqrt{7} - \sqrt{7} = 3\sqrt{7} - 2\sqrt{7} = \sqrt{7}.$$

So, the given progression is an AP with common difference $\sqrt{7}$.

$$\text{Next term is } 4\sqrt{7} = \sqrt{16 \times 7} = \sqrt{112}.$$

(ii) We may write the given terms as

$$\sqrt{9 \times 2}, \sqrt{25 \times 2}, \sqrt{49 \times 2}, \dots$$

$$\text{i.e., } 3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, \dots$$

$$\text{Clearly, } (5\sqrt{2} - 3\sqrt{2}) = (7\sqrt{2} - 5\sqrt{2}) = 2\sqrt{2}.$$

So, the given progression is an AP with common difference $2\sqrt{2}$.

$$\text{It is clear that the next term is } 9\sqrt{2} = \sqrt{9 \times 9 \times 2} = \sqrt{162}.$$

EXAMPLE 4 Find a and b such that the numbers $a, 9, b, 25$ form an AP. [CBSE 2014]

SOLUTION Since the numbers $a, 9, b, 25$ form an AP, we have

$$9 - a = b - 9 = 25 - b.$$

$$\text{Now, } b - 9 = 25 - b \Rightarrow 2b = 34 \Rightarrow b = 17.$$

$$\text{And, } 9 - a = b - 9 \Rightarrow a + b = 18 \Rightarrow a + 17 = 18 \Rightarrow a = 1.$$

$$\text{Hence, } a = 1 \text{ and } b = 17.$$

ARITHMETIC SERIES

By adding the terms of an AP, we get the corresponding arithmetic series.

Example On adding the terms of the AP 3, 7, 11, 15, 19, ..., we get the arithmetic series $(3 + 7 + 11 + 15 + 19 + \dots)$.

TO FIND THE GENERAL TERM OF AN AP

THEOREM 1 If the first term of an AP is a and its common difference is d then show that its n th term is given by

$$T_n = a + (n - 1)d.$$

PROOF In the given AP, we have

$$\text{first term} = a \text{ and common difference} = d.$$

So, the given AP may be written as

$$a, a + d, a + 2d, a + 3d, a + 4d, \dots$$

In this AP, we have

$$\text{first term, } T_1 = a = a + (1 - 1)d;$$

$$\text{second term, } T_2 = a + d = a + (2 - 1)d;$$

$$\text{third term, } T_3 = a + 2d = a + (3 - 1)d;$$

$$\text{fourth term, } T_4 = a + 3d = a + (4 - 1)d;$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$\therefore$ n th term, $T_n = a + (n - 1)d$.

Hence, $T_n = a + (n - 1)d$.

NOTE The n th term of an AP is called its *general term*.

General term of an AP

In an AP with first term a and common difference d , the n th term is given by

$$T_n = a + (n - 1)d.$$

SOLVED EXAMPLES

EXAMPLE 1 Find the (i) n th term and (ii) 16th term of the AP 3, 5, 7, 9, 11,

SOLUTION The given AP is 3, 5, 7, 9, 11,

Its first term = 3 and common difference = $(5 - 3) = 2$.

$\therefore$ $a = 3$ and $d = (5 - 3) = 2$.

(i) Its n th term is given by

$$\begin{aligned} T_n &= a + (n - 1)d \\ &= 3 + (n - 1) \times 2 = (2n + 1) \quad [\because a = 3 \text{ and } d = 2]. \end{aligned}$$

$\therefore$ n th term = $(2n + 1)$.

(ii) 16th term of the given AP is

$$\begin{aligned} T_{16} &= a + (16 - 1)d = (a + 15d) \\ &= (3 + 15 \times 2) = 33 \quad [\because a = 3 \text{ and } d = 2]. \end{aligned}$$

$\therefore$ 16th term = 33.

EXAMPLE 2 Find the (i) n th term and (ii) 12th term of the AP 14, 9, 4, -1, -6,

SOLUTION The given AP is 14, 9, 4, -1, -6,

Its first term = 14 and common difference = $(9 - 14) = -5$.

$\therefore$ $a = 14$ and $d = -5$.

(i) The n th term of the AP is given by

$$\begin{aligned} T_n &= a + (n - 1)d \\ &= 14 + (n - 1) \times (-5) = (19 - 5n) \quad [\because a = 14 \text{ and } d = -5]. \end{aligned}$$

$\therefore$ n th term = $(19 - 5n)$.

(ii) 12th term of the given AP is

$$T_{12} = a + (12 - 1)d = (a + 11d)$$

$$= 14 + 11 \times (-5) = -41 \quad [\because a = 14 \text{ and } d = -5].$$

$$\therefore 12\text{th term} = -41.$$

EXAMPLE 3 Find the 105th term of the AP $4, 4\frac{1}{2}, 5, 5\frac{1}{2}, 6, \dots$

SOLUTION The given AP is $4, \frac{9}{2}, 5, \frac{11}{2}, 6, \dots$

$$\text{Its first term} = 4 \text{ and common difference} = \left(\frac{9}{2} - 4\right) = \frac{1}{2}.$$

$$\therefore a = 4 \text{ and } d = \frac{1}{2}.$$

The n th term of the AP is given by

$$T_n = a + (n - 1)d$$

$$\Rightarrow T_{105} = a + (105 - 1)d = (a + 104d)$$

$$4 + \left(104 \times \frac{1}{2}\right) = 56 \quad \left[\because a = 4 \text{ and } d = \frac{1}{2}\right].$$

Hence, 105th term = 56.

EXAMPLE 4 Find the 25th term of the AP $-5, -\frac{5}{2}, 0, \frac{5}{2}, \dots$ [CBSE 2015]

SOLUTION The given AP is $-5, -\frac{5}{2}, 0, \frac{5}{2}, \dots$

$$\text{Its first term} = -5 \text{ and common difference} = \left(\frac{5}{2} - 0\right) = \frac{5}{2}.$$

$$\therefore a = -5 \text{ and } d = \frac{5}{2}.$$

$$\Rightarrow T_{25} = a + (25 - 1)d$$

$$= a + 24d = (-5) + \left(24 \times \frac{5}{2}\right) = -5 + 60 = 55.$$

Hence, 25th term = 55.

EXAMPLE 5 If the n th term of an AP is $(5n - 2)$, find its (i) first term, (ii) common difference and (iii) 19th term.

SOLUTION $T_n = (5n - 2)$ (given)

$$\Rightarrow T_1 = (5 \times 1 - 2) = 3 \text{ and } T_2 = (5 \times 2 - 2) = 8.$$

Thus, we have

$$(i) \text{ first term} = 3.$$

$$(ii) \text{ common difference} = (T_2 - T_1) = (8 - 3) = 5.$$

$$\begin{aligned} \text{(iii) } 19\text{th term} &= a + (19 - 1)d, \text{ where } a = 3 \text{ and } d = 5 \\ &= (3 + 18 \times 5) = 93. \end{aligned}$$

EXAMPLE 6 If the seventh term of an AP is $\frac{1}{9}$ and its ninth term is $\frac{1}{7}$, find its 63rd term. [CBSE 2014]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$T_7 = \frac{1}{9} \Rightarrow a + 6d = \frac{1}{9} \quad \dots \text{(i)}$$

$$T_9 = \frac{1}{7} \Rightarrow a + 8d = \frac{1}{7} \quad \dots \text{(ii)}$$

On subtracting (i) from (ii), we get

$$2d = \left(\frac{1}{7} - \frac{1}{9}\right) = \frac{2}{63} \Rightarrow d = \left(\frac{1}{2} \times \frac{2}{63}\right) = \frac{1}{63}.$$

Putting $d = \frac{1}{63}$ in (i), we get

$$a + \left(6 \times \frac{1}{63}\right) = \frac{1}{9} \Rightarrow a + \frac{2}{21} = \frac{1}{9} \Rightarrow a = \left(\frac{1}{9} - \frac{2}{21}\right) = \left(\frac{7-6}{63}\right) = \frac{1}{63}.$$

Thus, $a = \frac{1}{63}$ and $d = \frac{1}{63}$.

$$\begin{aligned} \therefore T_{63} &= a + (63 - 1)d = (a + 62d) \\ &= \left(\frac{1}{63} + 62 \times \frac{1}{63}\right) = \left(\frac{1}{63} + \frac{62}{63}\right) = 1. \end{aligned}$$

Hence, 63rd term of the given AP is 1.

EXAMPLE 7 The sum of the 4th and 8th terms of an AP is 24 and the sum of its 6th and 10th terms is 44. Find the first three terms of the AP. [CBSE 2008, '09, '12]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$\begin{aligned} T_4 + T_8 = 24 &\Rightarrow (a + 3d) + (a + 7d) = 24 \\ &\Rightarrow 2a + 10d = 24 \\ &\Rightarrow a + 5d = 12 \quad \dots \text{(i)} \end{aligned}$$

$$\begin{aligned} \text{and } T_6 + T_{10} = 44 &\Rightarrow (a + 5d) + (a + 9d) = 44 \\ &\Rightarrow 2a + 14d = 44 \\ &\Rightarrow a + 7d = 22. \quad \dots \text{(ii)} \end{aligned}$$

On solving (i) and (ii), we get $a = -13$ and $d = 5$.

$\therefore$ first three terms of the given AP are $-13, -8$ and -3 .

EXAMPLE 8 Which term of the AP 5, 9, 13, 17, ... is 81? [CBSE 2005C]

SOLUTION In the given AP, we have

first term = 5 and common difference = $(9 - 5) = 4$.

$\therefore a = 5$ and $d = 4$.

Let its n th term be 81. Then,

$$\begin{aligned}T_n = 81 &\Rightarrow a + (n - 1)d = 81 \\&\Rightarrow 5 + (n - 1) \times 4 = 81 \quad [\because a = 5 \text{ and } d = 4] \\&\Rightarrow 4n = 80 \Rightarrow n = 20.\end{aligned}$$

Hence, the 20th term of the given AP is 81.

EXAMPLE 9 Which term of the AP 3, 15, 27, 39, ... will be 120 more than its 21st term? [CBSE 2009]

SOLUTION Here $a = 3$ and $d = (15 - 3) = 12$.

$\therefore$ 21st term is given by

$$T_{21} = a + (21 - 1)d = a + 20d = (3 + 20 \times 12) = 243.$$

Required term = $(243 + 120) = 363$.

Let it be n th term. Then,

$$\begin{aligned}T_n = 363 &\Rightarrow a + (n - 1)d = 363 \\&\Rightarrow 3 + (n - 1) \times 12 = 363 \\&\Rightarrow 12n = 372 \Rightarrow n = 31.\end{aligned}$$

Hence, 31st term is the required term.

EXAMPLE 10 Is 51 a term of the AP 5, 8, 11, 14, ...?

SOLUTION Here $a = 5$ and $d = (8 - 5) = 3$.

Let the n th term of the given AP be 51. Then,

$$\begin{aligned}T_n = 51 &\Rightarrow a + (n - 1)d = 51 \\&\Rightarrow 5 + (n - 1) \times 3 = 51 \quad [\because a = 5 \text{ and } d = 3] \\&\Rightarrow 3n = 49 \Rightarrow n = 16\frac{1}{3}.\end{aligned}$$

But, the number of terms cannot be a fraction.

$\therefore$ 51 is not a term of the given AP.

EXAMPLE 11 How many terms are there in the AP 7, 11, 15, ..., 139?

SOLUTION In the given AP, we have $a = 7$ and $d = (11 - 7) = 4$.

Suppose there are n terms in the given AP. Then,

$$\begin{aligned}T_n = 139 &\Rightarrow a + (n - 1)d = 139 \\&\Rightarrow 7 + (n - 1) \times 4 = 139 \\&\Rightarrow 4n = 136 \Rightarrow n = 34.\end{aligned}$$

Hence, there are 34 terms in the given AP.

EXAMPLE 12 Find the middle term of the AP 213, 205, 197, ..., 37. [CBSE 2015]

SOLUTION Here $a = 213$ and $d = (205 - 213) = -8$.

Let the given AP contain n terms. Then,

$$\begin{aligned}T_n = 37 &\Rightarrow a + (n - 1)d = 37 \\&\Rightarrow 213 + (n - 1) \times (-8) = 37 \\&\Rightarrow 221 - 8n = 37 \\&\Rightarrow 8n = 221 - 37 = 184 \Rightarrow n = 23.\end{aligned}$$

Thus, the given AP contains 23 terms.

$$\therefore \text{ its middle term} = \frac{1}{2}(23 + 1)\text{th term} = 12\text{th term.}$$

$$\begin{aligned}\text{So, middle term} &= T_{12} = (a + 11d) \\&= 213 + 11 \times (-8) = 213 - 88 = 125.\end{aligned}$$

Hence, the middle term is 125.

EXAMPLE 13 Which term of the AP 24, 21, 18, 15, ... is the first negative term?

SOLUTION Here $a = 24$ and $d = (21 - 24) = -3$.

Let the n th term of the given AP be the first negative term.

$$\begin{aligned}\text{Then, } T_n < 0 &\Rightarrow \{a + (n - 1)d\} < 0 \\&\Rightarrow \{24 + (n - 1) \times (-3)\} < 0 \\&\Rightarrow (27 - 3n) < 0 \Rightarrow 27 < 3n \\&\Rightarrow 3n > 27 \Rightarrow n > 9.\end{aligned}$$

$$\therefore n = 10.$$

Hence, the 10th term is the first negative term of the given AP.

EXAMPLE 14 For what value of n are the n th terms of the following two APs the same 13, 19, 25, ... and 69, 68, 67, ...? Also, find this term.

SOLUTION Let n th terms of the given progressions be t_n and T_n respectively.

The first AP is 13, 19, 25,

Let its first term be a and common difference be d . Then,

$$a = 13 \text{ and } d = (19 - 13) = 6.$$

So, its n th term is given by

$$t_n = a + (n - 1)d$$

$$\Rightarrow t_n = 13 + (n - 1) \times 6$$

$$\Rightarrow t_n = 6n + 7 \quad \dots (i)$$

The second AP is 69, 68, 67,

Let its first term be A and common difference be D . Then,

$$A = 69 \text{ and } D = (68 - 69) = -1.$$

So, its n th term is given by

$$T_n = A + (n - 1) \times D$$

$$\Rightarrow T_n = 69 + (n - 1) \times (-1)$$

$$\Rightarrow T_n = 70 - n \quad \dots (ii)$$

$$\text{Now, } t_n = T_n \Rightarrow 6n + 7 = 70 - n$$

$$\Rightarrow 7n = 63 \Rightarrow n = 9.$$

Hence, the 9th term of each AP is the same.

This term = $70 - 9 = 61$ [$\because T_n = (70 - n)$].

EXAMPLE 15 *If seven times the 7th term of an AP is equal to eleven times the 11th term then what will be its 18th term?* [CBSE 2017]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$7T_7 = 11T_{11}$$

$$\Rightarrow 7(a + 6d) = 11(a + 10d) \Rightarrow 7a + 42d = 11a + 110d$$

$$\Rightarrow 4a + 68d = 0 \Rightarrow a + 17d = 0$$

$$\Rightarrow a + (18 - 1)d = 0 \Rightarrow T_{18} = 0.$$

Hence, the 18th term of the given AP is zero.

EXAMPLE 16 *If the n th term of a progression be a linear expression in n then prove that this progression is an AP.*

SOLUTION Let the n th term of a given progression be given by

$$T_n = an + b, \text{ where } a \text{ and } b \text{ are constants.}$$

Then, $T_{n-1} = a(n-1) + b = [(an+b) - a]$.

$\therefore T_n - T_{n-1} = (an+b) - [(an+b-a)] = a$, which is a constant.

Hence, the given progression is an AP.

EXAMPLE 17 In a given AP if p th term is q and the q th term is p then show that the n th term is $(p+q-n)$. [CBSE 2008]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$T_p = a + (p-1)d \text{ and } T_q = a + (q-1)d.$$

Now, $T_p = q$ and $T_q = p$ (given).

$$\therefore a + (p-1)d = q \quad \dots (i)$$

$$\text{and } a + (q-1)d = p \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$(q-p)d = (p-q) \Rightarrow d = -1.$$

Putting $d = -1$ in (i), we get $a = (p+q-1)$.

Thus, $a = (p+q-1)$ and $d = -1$.

$$\begin{aligned} \therefore \text{nth term} &= a + (n-1)d = (p+q-1) + (n-1) \times (-1) \\ &= p+q-n. \end{aligned}$$

Hence, n th term $= (p+q-n)$.

EXAMPLE 18 If m times the m th term of an AP is equal to n times the n th term and $m \neq n$, show that its $(m+n)$ th term is zero. [CBSE 2004C, '08]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$T_m = a + (m-1)d \text{ and } T_n = a + (n-1)d.$$

$$\begin{aligned} \text{Now, } (m \cdot T_m) &= (n \cdot T_n) \Rightarrow m \cdot \{a + (m-1)d\} = n \cdot \{a + (n-1)d\} \\ &\Rightarrow a \cdot (m-n) + \{(m^2 - n^2) - (m-n)\} \cdot d = 0 \\ &\Rightarrow (m-n) \cdot \{a + (m+n-1)d\} \\ &\Rightarrow (m-n) \cdot T_{m+n} = 0 \\ &\Rightarrow T_{m+n} = 0 \quad [\because (m-n) \neq 0]. \end{aligned}$$

Hence, the $(m+n)$ th term is zero.

EXAMPLE 19 If the p th, q th and r th terms of an AP be a , b , c respectively then show that

$$a(q-r) + b(r-p) + c(p-q) = 0.$$

SOLUTION Let x be the first term and d be the common difference of the given AP. Then,

$$T_p = x + (p-1)d, T_q = x + (q-1)d \text{ and } T_r = x + (r-1)d.$$

$$\therefore x + (p-1)d = a \quad \dots (i)$$

$$x + (q-1)d = b \quad \dots (ii)$$

$$x + (r-1)d = c. \quad \dots (iii)$$

On multiplying (i) by $(q-r)$, (ii) by $(r-p)$ and (iii) by $(p-q)$, and adding, we get

$$\begin{aligned} & a(q-r) + b(r-p) + c(p-q) \\ &= x \cdot \{(q-r) + (r-p) + (p-q)\} \\ & \quad + d \cdot \{(p-1)(q-r) + (q-1)(r-p) + (r-1)(p-q)\} \\ &= (x \times 0) + (d \times 0) = 0. \end{aligned}$$

Hence, $a(q-r) + b(r-p) + c(p-q) = 0$.

EXAMPLE 20 If the m th term of an AP be $\frac{1}{n}$ and its n th term be $\frac{1}{m}$ then show that its (mn) th term is 1.

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$T_m = a + (m-1)d \text{ and } T_n = a + (n-1)d.$$

Now, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$ (given).

$$\therefore a + (m-1)d = \frac{1}{n} \quad \dots (i)$$

$$\text{and } a + (n-1)d = \frac{1}{m} \quad \dots (ii)$$

On subtracting (ii) from (i), we get

$$(m-n)d = \left(\frac{1}{n} - \frac{1}{m}\right) = \left(\frac{m-n}{mn}\right) \Rightarrow d = \frac{1}{mn}.$$

Putting $d = \frac{1}{mn}$ in (i), we get

$$a + \frac{(m-1)}{mn} = \frac{1}{n} \Rightarrow a = \left\{ \frac{1}{n} - \frac{(m-1)}{mn} \right\} = \frac{1}{mn}.$$

Thus, $a = \frac{1}{mn}$ and $d = \frac{1}{mn}$.

$$\begin{aligned} \therefore (mn)\text{th term} &= a + (mn-1)d \\ &= \left\{ \frac{1}{mn} + \frac{(mn-1)}{mn} \right\} \left[\because a = \frac{1}{mn} \right] \\ &= 1. \end{aligned}$$

Hence, the (mn) th term of the given AP is 1.

TO FIND THE n TH TERM FROM THE END OF AN AP

THEOREM 2 *If a be the first term, d be the common difference and l be the last term of a given AP then show that its n th term from the end is $\{l - (n - 1)d\}$.*

PROOF We may write the given AP as

$$a, (a + d), (a + 2d), \dots, (l - 2d), (l - d), l.$$

Thus, we have

$$\text{last term} = l = l - (1 - 1)d;$$

$$\text{2nd term from the end} = (l - d) = \{l - (2 - 1)d\};$$

$$\text{3rd term from the end} = (l - 2d) = \{l - (3 - 1)d\};$$

$$\text{4th term from the end} = (l - 3d) = \{l - (4 - 1)d\}.$$

$$\therefore \text{ } n\text{th term from the end} = \{l - (n - 1)d\}.$$

 n th term from the end of an AP

Let a be the first term, d be the common difference and l be the last term of an AP. Then,

$$n\text{th term from the end} = \{l - (n - 1)d\}.$$

EXAMPLE 21 Find the 11th term from the end of the AP 10, 7, 4, ..., -62.

SOLUTION We have

$$a = 10, d = (7 - 10) = -3, l = -62 \text{ and } n = 11.$$

$$\begin{aligned} \therefore \text{ 11th term from the end} &= \{l - (n - 1) \times d\} \\ &= \{-62 - (11 - 1) \times (-3)\} \\ &= (-62 + 30) = -32. \end{aligned}$$

Hence, the 11th term from the end of the given AP is -32.

WORD PROBLEMS

EXAMPLE 22 How many three-digit numbers are divisible by 7? [CBSE 2013]

SOLUTION All 3-digit numbers divisible by 7 are

$$105, 112, 119, \dots, 994.$$

Clearly, these numbers form an AP with

$$a = 105, d = (112 - 105) = 7 \text{ and } l = 994.$$

Let it contain n terms. Then,

$$T_n = 994 \Rightarrow a + (n - 1) \times d = 994$$

$$\Rightarrow 105 + (n - 1) \times 7 = 994$$

$$\Rightarrow 98 + 7n = 994 \Rightarrow 7n = 896 \Rightarrow n = 128.$$

Hence, there are 128 three-digit numbers divisible by 7.

EXAMPLE 23 *How many multiples of 4 lie between 10 and 250?* [CBSE 2011]

SOLUTION All multiples of 4 lying between 10 and 250 are

$$12, 16, 20, 24, \dots, 248.$$

Clearly, these numbers form an AP with

$$a = 12, d = (16 - 12) = 4 \text{ and } l = 248.$$

Let it contain n terms. Then,

$$T_n = 248 \Rightarrow a + (n - 1)d = 248$$

$$\Rightarrow 12 + (n - 1) \times 4 = 248 \Rightarrow 4n = 240 \Rightarrow n = 60.$$

Hence, there are 60 multiples of 4 lying between 10 and 250.

EXAMPLE 24 *A sum of ₹ 1000 is invested at 8% per annum simple interest. Calculate the interest at the end of 1, 2, 3, ... years. Is the sequence of interest an AP? Find the interest at the end of 30 years.*

SOLUTION Here $P = ₹ 1000$, $R = 8\%$ per annum. Let I_n be the SI at the end of n years.

$$\text{Then, } I_n = \frac{P \times R \times n}{100} = ₹ \left(\frac{1000 \times 8 \times n}{100} \right) = ₹ (80n). \quad \dots (i)$$

Putting $n = 1, 2, 3, \dots$ in (i), we get

$$I_1 = ₹ (80 \times 1) = ₹ 80, I_2 = ₹ (80 \times 2) = ₹ 160,$$

$$I_3 = ₹ (80 \times 3) = ₹ 240, \dots$$

Thus, $I_1 = ₹ 80, I_2 = ₹ 160, I_3 = ₹ 240, I_4 = ₹ 320, \dots$

Thus, ₹ 80, ₹ 160, ₹ 240, ₹ 320, ... form an AP with

$$a = 80 \text{ and } d = (160 - 80) = 80.$$

$$\therefore T_{30} = a + (30 - 1) \times d$$

$$= 80 + 29 \times 80 = 80 + 2320 = 2400.$$

Hence, the interest at the end of 30 years is ₹ 2400.

EXAMPLE 25 *Tanvy joined her job in a company in the year 2015 on a monthly salary of ₹ 40000 with an annual increment of ₹ 2500. In which year will she get ₹ 65000 as monthly salary?*

SOLUTION Monthly salary received by Tanvy in 2015, 2016, 2017, 2018, ... is respectively ₹ 40000, ₹ 42500, ₹ 45000, ₹ 47500,

This is an AP with $a = 40000$, $d = 2500$ and $l = 65000$.

Let the number of terms of this AP be n . Then,

$$\begin{aligned}T_n = 65000 &\Rightarrow a + (n - 1)d = 65000 \\&\Rightarrow 40000 + (n - 1) \times 2500 = 65000 \\&\Rightarrow (n - 1) \times 2500 = 65000 - 40000 = 25000 \\&\Rightarrow (n - 1) = \frac{25000}{2500} = 10 \Rightarrow n = 11.\end{aligned}$$

Thus, the 11th annual salary received by Tanvy will be ₹ 65000.

Thus, after 10 years, i.e., in the year 2025, her annual salary will be ₹ 65000.

EXAMPLE 26 In a new year, Reenu saved ₹ 50 in the first week and then increased her weekly savings by ₹ 17.50. If in the n th week, her weekly saving becomes ₹ 207.50, find the value of n .

SOLUTION Weekly savings by Reenu in successive weeks are ₹ 50, ₹ 67.50, ₹ 85, ₹ 102.50, ..., ₹ 207.50.

This is clearly an AP with $a = 50$, $d = 17.50$ and $l = 207.50$.

Let the number of terms of this AP be n . Then,

$$\begin{aligned}T_n = 207.50 &\Rightarrow a + (n - 1)d = 207.50 \\&\Rightarrow 50 + (n - 1) \times 17.50 = 207.50 \\&\Rightarrow (n - 1) \times 17.50 = 207.50 - 50 = 157.50 \\&\Rightarrow (n - 1) = \frac{157.50}{17.50} = \frac{15750}{1750} = 9 \Rightarrow n = 10.\end{aligned}$$

Hence, Reenu's weekly savings will be ₹ 207.50 in 10th week.

EXERCISE 5A

1. Show that each of the progressions given below is an AP. Find the first term, common difference and next term of each.

(i) 9, 15, 21, 27,

(ii) 11, 6, 1, -4,

(iii) $-1, \frac{-5}{6}, \frac{-2}{3}, \frac{-1}{2}, \dots$

(iv) $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \dots$

(v) $\sqrt{20}, \sqrt{45}, \sqrt{80}, \sqrt{125}, \dots$

2. Find:

- (i) the 20th term of the AP 9, 13, 17, 21, ...
- (ii) the 35th term of the AP 20, 17, 14, 11, ...
- (iii) the 18th term of the AP $\sqrt{2}, \sqrt{18}, \sqrt{50}, \sqrt{98}, \dots$
- (iv) the 9th term of the AP $\frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \dots$
- (v) the 15th term of the AP -40, -15, 10, 35, ...

3. (i) Find the 37th term of the AP $6, 7\frac{3}{4}, 9\frac{1}{2}, 11\frac{1}{4}, \dots$

(ii) Find the 25th term of the AP $5, 4\frac{1}{2}, 4, 3\frac{1}{2}, 3, \dots$

4. Find the value of p for which the numbers $2p - 1, 3p + 1, 11$ are in AP.
Hence, find the numbers. [CBSE 2017]

5. Find the n th term of each of the following APs:

(i) 5, 11, 17, 23, ...

(ii) 16, 9, 2, -5, ...

6. If the n th term of a progression is $(4n - 10)$ show that it is an AP. Find its
(i) first term, (ii) common difference, and (iii) 16th term.

7. How many terms are there in the AP 6, 10, 14, 18, ..., 174?

8. How many terms are there in the AP 41, 38, 35, ..., 8?

9. How many terms are there in the AP $18, 15\frac{1}{2}, 13, \dots, -47$?

10. Which term of the AP 3, 8, 13, 18, ... is 88?

11. Which term of the AP 72, 68, 64, 60, ... is 0?

12. Which term of the AP $\frac{5}{6}, 1, 1\frac{1}{6}, 1\frac{1}{3}, \dots$ is 3?

13. Which term of the AP 21, 18, 15, ... is -81?

14. Which term of the AP 8, 14, 20, 26, ... will be 72 more than its 41st term?
[CBSE 2017]

15. Which term of the AP 5, 15, 25, ... will be 130 more than its 31st term?
[CBSE 2006C]

16. If the 10th term of an AP is 52 and 17th term is 20 more than its 13th term, find the AP.
[CBSE 2009C, 2017]

17. Find the middle term of the AP 6, 13, 20, ..., 216. [CBSE 2015]

18. Find the middle term of the AP 10, 7, 4, ..., (-62). [CBSE 2009C]

19. Find the sum of two middle most terms of the AP $-\frac{4}{3}, -1, -\frac{2}{3}, \dots, 4\frac{1}{3}$.

[CBSE 2013C]

20. Find the 8th term from the end of the AP 7, 10, 13, ..., 184. [CBSE 2005]

21. Find the 6th term from the end of the AP 17, 14, 11, ..., (-40) . [CBSE 2005]
22. Is 184 a term of the AP 3, 7, 11, 15, ...?
23. Is -150 a term of the AP 11, 8, 5, 2, ...?
24. Which term of the AP 121, 117, 113, ... is its first negative term?
25. Which term of the AP 20, $19\frac{1}{4}$, $18\frac{1}{2}$, $17\frac{3}{4}$, ... is its first negative term?
[CBSE 2017]
26. The 7th term of an AP is -4 and its 13th term is -16 . Find the AP.
[CBSE 2014]
27. The 4th term of an AP is zero. Prove that its 25th term is triple its 11th term.
[CBSE 2005]
28. If the sixth term of an AP is zero then show that its 33rd term is three times its 15th term.
[CBSE 2017]
29. The 4th term of an AP is 11. The sum of the 5th and 7th terms of this AP is 34. Find its common difference.
[CBSE 2015]
30. The 9th term of an AP is -32 and the sum of its 11th and 13th terms is -94 . Find the common difference of the AP.
[CBSE 2015]
31. Determine the n th term of the AP whose 7th term is -1 and 16th term is 17.
[CBSE 2014]
32. If 4 times the 4th term of an AP is equal to 18 times its 18th term then find its 22nd term.
[CBSE 2012]
33. If 10 times the 10th term of an AP is equal to 15 times the 15th term, show that its 25th term is zero.
34. Find the common difference of an AP whose first term is 5 and the sum of its first four terms is half the sum of the next four terms. [CBSE 2012]
35. The sum of the 2nd and the 7th terms of an AP is 30. If its 15th term is 1 less than twice its 8th term, find the AP.
[CBSE 2014]
36. For what value of n , the n th terms of the arithmetic progressions 63, 65, 67, ... and 3, 10, 17, ... are equal?
[CBSE 2008, '17]
37. The 17th term of AP is 5 more than twice its 8th term. If the 11th term of the AP is 43, find its n th term.
[CBSE 2012]
38. The 24th term of an AP is twice its 10th term. Show that its 72nd term is 4 times its 15th term.
[CBSE 2013]
39. The 19th term of an AP is equal to 3 times its 6th term. If its 9th term is 19, find the AP.
[CBSE 2013]
40. If the p th term of an AP is q and its q th term is p then show that its $(p+q)$ th term is zero.

41. The first and last terms of an AP are a and l respectively. Show that the sum of the n th term from the beginning and the n th term from the end is $(a + l)$.
42. How many two-digit numbers are divisible by 6? [CBSE 2011]
43. How many two-digit numbers are divisible by 3? [CBSE 2012]
44. How many three-digit numbers are divisible by 9? [CBSE 2013]
45. How many numbers are there between 101 and 999, which are divisible by both 2 and 5? [CBSE 2014]
46. In a flower bed, there are 43 rose plants in the first row, 41 in the second, 39 in the third, and so on. There are 11 rose plants in the last row. How many rows are there in the flower bed?
47. A sum of ₹ 2800 is to be used to award four prizes. If each prize after the first is ₹ 200 less than the preceding prize, find the value of each of the prizes.
48. Find how many integers between 200 and 500 are divisible by 8. [CBSE 2017]

ANSWERS (EXERCISE 5A)

1. (i) $a = 9, d = 6, T_5 = 33$ (ii) $a = 11, d = -5, T_5 = -9$
 (iii) $a = -1, d = \frac{1}{6}, T_5 = \frac{-1}{3}$ (iv) $a = \sqrt{2}, d = \sqrt{2}, T_5 = \sqrt{50}$
 (v) $a = \sqrt{20}, d = \sqrt{5}, T_5 = \sqrt{180}$
2. (i) 85 (ii) -82 (iii) $35\sqrt{2}$ (iv) $4\frac{3}{4}$ (v) 310
3. (i) 69 (ii) -7 4. $p = 2; 3, 7, 11$ 5. (i) $(6n - 1)$ (ii) $(23 - 7n)$
6. (i) -6 (ii) 4 (iii) 54 7. 43 8. 12 9. 27 10. 18th
11. 19th 12. 14th 13. 35th 14. 53rd 15. 44th 16. 7, 12, 17, 22, ...
17. 111 18. -26 19. 3 20. 163 21. -25
22. No 23. No 24. 32nd 25. 28th 26. 8, 6, 4, 2, 0, ...
29. 3 30. -5 31. $T_n = (2n - 15)$ 32. 0 34. 2 35. 1, 5, 9, 13, ...
36. $n = 13$ 37. $T_n = (4n - 1)$ 39. 3, 5, 7, 9, ... 42. 15
43. 30 44. 100 45. 89 46. 17 47. ₹ 1000, ₹ 800, ₹ 600, ₹ 400 48. 37

HINTS TO SOME SELECTED QUESTIONS

1. (iv) Given progression is $\sqrt{2}, \sqrt{4 \times 2}, \sqrt{9 \times 2}, \sqrt{16 \times 2}, \dots$
 i.e., $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \dots$
 $T_5 = 5\sqrt{2} = \sqrt{5 \times 5 \times 2} = \sqrt{50}$.

(v) Given progression is $\sqrt{4 \times 5}, \sqrt{9 \times 5}, \sqrt{16 \times 5}, \sqrt{25 \times 5}, \dots$

i.e., $2\sqrt{5}, 3\sqrt{5}, 4\sqrt{5}, 5\sqrt{5}, \dots$

$$T_5 = 6\sqrt{5} = \sqrt{6 \times 6 \times 5} = \sqrt{180}.$$

$$3. \text{ (i) } a = 6, d = \left(\frac{31}{4} - 6\right) = \frac{7}{4}.$$

$$\text{(ii) } a = 5, d = \left(\frac{9}{2} - 5\right) = \frac{-1}{2}.$$

$$19. a = \frac{-4}{3}, d = \left(-1 + \frac{4}{3}\right) = \frac{1}{3} \text{ and } T_n = \frac{13}{3}.$$

$$\therefore T_n = \frac{13}{3} \Rightarrow a + (n-1)d = \frac{13}{3} \Rightarrow \frac{-4}{3} + (n-1) \times \frac{1}{3} = \frac{13}{3} \Rightarrow n = 18.$$

$\therefore$ sum of two middle most terms

$$= (T_9 + T_{10}) = (a + 8d) + (a + 9d) = (2a + 17d) = \left(\frac{-8}{3} + \frac{17}{3}\right) = \frac{9}{3} = 3.$$

$$22. \text{ Let } T_n = 184. \text{ Then, } a + (n-1)d = 184.$$

$$\therefore 3 + (n-1) \times 4 = 184 \Rightarrow n-1 = \frac{181}{4} \Rightarrow n = \left(\frac{181}{4} + 1\right) = \frac{185}{4} = 46\frac{1}{4}.$$

Thus, n is not a natural number.

So, 184 is not a term of the given AP.

$$24. \text{ Let } T_n < 0. \text{ Then, } 121 + (n-1) \times (-4) < 0.$$

$$\therefore -4n + 125 < 0 \Rightarrow 125 < 4n \Rightarrow 4n > 125 \Rightarrow n = 31\frac{1}{4}.$$

So, 32nd term is the first negative term.

$$27. a + 3d = 0 \Rightarrow a = -3d.$$

$$25\text{th term} = a + 24d = -3d + 24d = 21d.$$

$$3(11\text{th term}) = 3(a + 10d) = 3(-3d + 10d) = 21d.$$

Hence, 25th term = 3(11th term).

$$28. a + 5d = 0 \Rightarrow a = -5d.$$

$$33\text{th term} = a + 32d = -5d + 32d = 27d.$$

$$3(15\text{th term}) = 3(a + 14d) = 3(-5d + 14d) = 3 \times 9d = 27d.$$

$$29. a + 3d = 11, T_5 + T_7 = 34 \Rightarrow 2a + 10d = 34 \Rightarrow a + 5d = 17.$$

Solve $a + 3d = 11$ and $a + 5d = 17$.

$$32. 4(a + 3d) = 18(a + 17d) \Rightarrow 14a + 294d = 0 \Rightarrow a + 21d = 0 \Rightarrow T_{22} = 0.$$

$$34. \{5 + (5 + d) + (5 + 2d) + (5 + 3d)\} = \frac{1}{2}\{(5 + 4d) + (5 + 5d) + (5 + 6d) + (5 + 7d)\}$$

$$\Rightarrow 2(20 + 6d) = (20 + 22d) \Rightarrow 10d = 20 \Rightarrow d = 2.$$

$$35. T_2 + T_7 = 30 \Rightarrow (a + d) + (a + 6d) = 30 \Rightarrow 2a + 7d = 30. \quad \dots (i)$$

$$T_{15} = 2T_8 - 1 \Rightarrow (a + 14d) = 2(a + 7d) - 1 \Rightarrow a = 1.$$

Putting $a = 1$ in (i), we get $d = 4$.

$$36. T_n = t_n \Rightarrow 63 + (n-1) \times 2 = 3 + (n-1) \times 7 \Rightarrow 5n = 65 \Rightarrow n = 13.$$

$$37. T_{17} = 2T_8 + 5 \Rightarrow a + 16d = 2(a + 7d) + 5 \Rightarrow a - 2d = -5. \quad \dots (i)$$

Also, $a + 10d = 43$ (ii)

Solve (i) and (ii), we get $a = 3$ and $d = 4$.

38. $T_{24} = 2T_{10} \Rightarrow a + 23d = 2(a + 9d) \Rightarrow a - 5d = 0 \Rightarrow a = 5d$ (i)

$$T_{72} = a + 71d = 5d + 71d = 76d.$$

$$T_{15} = a + 14d = 5d + 14d = 19d.$$

$$\text{Hence, } T_{72} = 4 \times T_{15}.$$

40. $T_p = q \Rightarrow a + (p - 1)d = q$ (i)

$T_q = p \Rightarrow a + (q - 1)d = p$ (ii)

On subtracting (ii) from (i), we get $(p - q)d = (q - p) \Rightarrow d = -1$.

Putting $d = -1$ in (i), we get $a = p + q - 1$.

$$\therefore T_{p+q} = a + (p + q - 1)d = (p + q - 1) - (p + q - 1) = 0.$$

41. Required sum = $\{a + (n - 1)d\} + \{l - (n - 1)d\} = (a + l)$.

42. The given numbers are 12, 18, 24, 30, ..., 96.

Let their number be n . Then, $12 + (n - 1) \times 6 = 96$. Find n .

44. The given numbers are 108, 117, 126, ..., 990.

45. Clearly, each given number is divisible by 10.

So, these numbers are 110, 120, 130, ..., 990.

46. Let there be n rows. Then, the number of plants in various rows are 43, 41, 39, ..., 11 respectively.

Here, $a = 43$, $d = -2$ and $T_n = 11$. Find n .

47. Let the values of these prizes be ₹ x , ₹ $(x - 200)$, ₹ $(x - 400)$ and ₹ $(x - 600)$. Their sum is ₹ 2800.

ARITHMETIC MEAN

ARITHMETIC MEAN (AM) BETWEEN TWO NUMBERS

If a, A, b are in AP we say that A is the arithmetic mean between a and b and it is abbreviated as AM.

TO FIND THE ARITHMETIC MEAN BETWEEN TWO NUMBERS

Let the given numbers be a and b .

Let A be the AM between a and b . Then,

$$a, A, b \text{ are in AP} \Rightarrow (A - a) = (b - A)$$

$$\Rightarrow A = \frac{1}{2}(a + b).$$

$\therefore$ AM between a and $b = \frac{1}{2}(a + b)$.

Arithmetic mean between a and b is $\frac{1}{2}(a + b)$.

SOLVED EXAMPLES

EXAMPLE 1 Find the AM between

- (i) 13 and 19 (ii) $(a - b)$ and $(a + b)$

SOLUTION (i) AM between 13 and 19 = $\frac{1}{2}(13 + 19) = 16$.

(ii) AM between $(a - b)$ and $(a + b) = \frac{1}{2}[(a - b) + (a + b)] = a$.

EXAMPLE 2 If the numbers $(2n - 1)$, $(3n + 2)$ and $(6n - 1)$ are in AP, find n and hence find these numbers. [CBSE 2013C]

SOLUTION Since $(2n - 1)$, $(3n + 2)$ and $(6n - 1)$ are in AP, we have

$$(3n + 2) - (2n - 1) = (6n - 1) - (3n + 2)$$

$$\Rightarrow n + 3 = 3n - 3 \Rightarrow 2n = 6 \Rightarrow n = 3.$$

Hence, $n = 3$ and these numbers are 5, 11 and 17.

AN IMPORTANT RESULT

It is always convenient to make a choice of

- (i) 3 numbers in AP as $(a - d)$, a , $(a + d)$;
- (ii) 4 numbers in AP as $(a - 3d)$, $(a - d)$, $(a + d)$, $(a + 3d)$;
- (iii) 5 numbers in AP as $(a - 2d)$, $(a - d)$, a , $(a + d)$, $(a + 2d)$.

EXAMPLE 3 The sum of three numbers in AP is 21 and their product is 231. Find the numbers.

SOLUTION Let the required numbers be $(a - d)$, a and $(a + d)$.

$$\text{Then, } (a - d) + a + (a + d) = 21 \Rightarrow 3a = 21 \Rightarrow a = 7.$$

$$\text{And, } (a - d) \cdot a \cdot (a + d) = 231 \Rightarrow a(a^2 - d^2) = 231$$

$$\Rightarrow 7(49 - d^2) = 231 \quad [\because a = 7]$$

$$\Rightarrow 7d^2 = 343 - 231 = 112$$

$$\Rightarrow d^2 = 16 \Rightarrow d = \pm 4.$$

Thus, $a = 7$ and $d = \pm 4$.

Hence, the required numbers are (3, 7, 11) or (11, 7, 3).

EXAMPLE 4 Find four numbers in AP whose sum is 20 and the sum of whose squares is 120.

SOLUTION Let the required numbers be $(a - 3d)$, $(a - d)$, $(a + d)$ and $(a + 3d)$.

$$\text{Then, } (a - 3d) + (a - d) + (a + d) + (a + 3d) = 20$$

$$\Rightarrow 4a = 20 \Rightarrow a = 5.$$

$$\text{And, } (a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 120$$

$$\Rightarrow 4(a^2 + 5d^2) = 120$$

$$\Rightarrow (a^2 + 5d^2) = 30 \Rightarrow 25 + 5d^2 = 30 \quad [\because a = 5]$$

$$\Rightarrow 5d^2 = 5 \Rightarrow d^2 = 1 \Rightarrow d = \pm 1.$$

Thus, $a = 5$ and $d = \pm 1$.

Hence, the required numbers are (2, 4, 6, 8) or (8, 6, 4, 2).

EXERCISE 5B

- Determine k so that $(3k - 2)$, $(4k - 6)$ and $(k + 2)$ are three consecutive terms of an AP. [CBSE 2009C]
- Find the value of x for which the numbers $(5x + 2)$, $(4x - 1)$ and $(x + 2)$ are in AP.
- If $(3y - 1)$, $(3y + 5)$ and $(5y + 1)$ are three consecutive terms of an AP then find the value of y . [CBSE 2014]
- Find the value of x for which $(x + 2)$, $2x$, $(2x + 3)$ are three consecutive terms of an AP. [CBSE 2009C]
- Show that $(a - b)^2$, $(a^2 + b^2)$ and $(a + b)^2$ are in AP.
- Find three numbers in AP whose sum is 15 and product is 80.
HINT Let the numbers be $(a - d)$, a , $(a + d)$.
- The sum of three numbers in AP is 3 and their product is -35 . Find the numbers.
- Divide 24 in three parts such that they are in AP and their product is 440.
- The sum of three consecutive terms of an AP is 21 and the sum of the squares of these terms is 165. Find these terms.

10. The angles of a quadrilateral are in AP whose common difference is 10° . Find the angles.

HINT Let these angles be x° , $(x + 10)^\circ$, $(x + 20)^\circ$ and $(x + 30)^\circ$.
Their sum is 360° .

11. Find four numbers in AP whose sum is 28 and the sum of whose squares is 216.
12. Divide 32 into four parts which are the four terms of an AP such that the product of the first and the fourth terms is to the product of the second and the third terms as 7 : 15. [CBSE 2014]

HINT Let these parts be $(a - 3d)$, $(a - d)$, $(a + d)$ and $(a + 3d)$.

13. The sum of first three terms of an AP is 48. If the product of first and second terms exceeds 4 times the third term by 12. Find the AP.

[CBSE 2013C]

HINT Let these terms be $(a - d)$, a , $(a + d)$.

ANSWERS (EXERCISE 5B)

1. $k = 3$ 2. $x = 3$ 3. $y = 5$ 4. $x = 5$
 6. (2, 5, 8) or (8, 5, 2) 7. $(-5, 1, 7)$ or $(7, 1, -5)$ 8. (5, 8, 11) or (11, 8, 5)
 9. (4, 7, 10) or (10, 7, 4) 10. $75^\circ, 85^\circ, 95^\circ, 105^\circ$
 11. (4, 6, 8, 10) or (10, 8, 6, 4) 12. (2, 6, 10, 14) or (14, 10, 6, 2) 13. 7, 16, 25

SUM OF n TERMS OF AN AP

TO FIND THE SUM OF n TERMS OF AN AP

THEOREM 3 Prove that the sum of n terms of an AP in which first term = a , common difference = d and last term = l , is given by

$$S_n = \frac{n}{2}(a + l) \text{ and } S_n = \frac{n}{2} \cdot \{2a + (n - 1)d\}.$$

PROOF Consider an AP having n terms in which

first term = a , common difference = d and last term = l .

Then, $l = a + (n - 1)d$... (i) [$\because l = n$ th term]

Also, we may write the given AP as

$$a, (a + d), (a + 2d), \dots, (l - 2d), (l - d), l.$$

Let S_n be the sum of the first n terms of this AP. Then,

$$S_n = a + (a + d) + (a + 2d) + \dots + (l - 2d) + (l - d) + l \quad \dots \text{ (ii)}$$

Writing the above series in the reverse order, we get

$$S_n = l + (l - d) + (l - 2d) + \dots + (a + 2d) + (a + d) + a \quad \dots \text{(iii)}$$

Adding the corresponding terms of (ii) and (iii), we get

$$2S_n = (a + l) + (a + l) + (a + l) + \dots \text{ } n \text{ times} = n(a + l)$$

$$\Rightarrow S_n = \frac{n}{2}(a + l)$$

$$\Rightarrow S_n = \frac{n}{2}[a + a + (n - 1)d] = \frac{n}{2}[2a + (n - 1)d] \quad [\text{using (i)}].$$

$$\text{Hence, } S_n = \frac{n}{2}(a + l) \text{ and } S_n = \frac{n}{2}[2a + (n - 1)d].$$

SUMMARY

Sum of n terms of an AP $a, a + d, a + 2d, \dots, l$ is given by

$$S_n = \frac{n}{2}(a + l) \text{ or } S_n = \frac{n}{2} \times \{2a + (n - 1)d\}.$$

SOLVED EXAMPLES

EXAMPLE 1 Find the sum of first 24 terms of the AP 5, 8, 11, 14,

SOLUTION Here $a = 5$, $d = (8 - 5) = 3$ and $n = 24$.

Using the formula, $S_n = \frac{n}{2} \cdot \{2a + (n - 1)d\}$, we get

$$\begin{aligned} S_{24} &= \frac{24}{2} \cdot \{2 \times 5 + (24 - 1) \times 3\} \quad [\because a = 5, d = 3 \text{ and } n = 24] \\ &= 12 \times (10 + 69) = 948. \end{aligned}$$

Hence, the sum of first 24 terms of the given AP is 948.

EXAMPLE 2 Find the sum $25 + 28 + 31 + \dots + 100$. [CBSE 2006C]

SOLUTION Here $a = 25$, $d = (28 - 25) = 3$ and $l = 100$.

Let the total number of terms be n . Then,

$$\begin{aligned} T_n &= 100 \Rightarrow a + (n - 1)d = 100 \\ &\Rightarrow 25 + (n - 1) \times 3 = 100 \\ &\Rightarrow 3n + 22 = 100 \\ &\Rightarrow 3n = 78 \Rightarrow n = 26. \end{aligned}$$

$$\begin{aligned} \text{Required sum} &= \frac{n}{2} \cdot (a + l) \\ &= \frac{26}{2} \cdot (25 + 100) = 13 \times 125 = 1625. \end{aligned}$$

Hence, the required sum is 1625.

EXAMPLE 3 Find the sum $18 + 15\frac{1}{2} + 13 + \dots + (-49\frac{1}{2})$. [CBSE 2013]

SOLUTION We have, $\frac{31}{2} - 18 = 13 - \frac{31}{2} = \frac{-5}{2}$.
So, the given series is an arithmetic series.

Here, $a = 18$, $d = \frac{-5}{2}$ and $l = \frac{-99}{2}$.

Let the given series contain n terms. Then,

$$\begin{aligned} T_n = \frac{-99}{2} &\Rightarrow a + (n-1)d = \frac{-99}{2} \\ &\Rightarrow 18 + (n-1) \times \left(\frac{-5}{2}\right) = \frac{-99}{2} \\ &\Rightarrow 36 - 5n + 5 = -99 \\ &\Rightarrow -5n = -140 \Rightarrow n = 28. \end{aligned}$$

$$\begin{aligned} \therefore \text{required sum} &= \frac{n}{2}(a + l) \\ &= \frac{28}{2} \cdot \left\{18 - \frac{99}{2}\right\} = 7(36 - 99) = -441. \end{aligned}$$

Hence, the sum of the given series is -441 .

EXAMPLE 4 Find the sum of first n terms of an AP whose n th term is $(5n - 1)$.
Hence, find the sum of first 20 terms. [CBSE 2011]

SOLUTION The n th term of the given AP is given by, $T_n = (5n - 1)$.

Let a be the first term and d be the common difference of this AP. Then,

$$a = T_1 = (5 \times 1 - 1) = 4, T_2 = (5 \times 2 - 1) = 9.$$

$$\therefore d = (T_2 - T_1) = (9 - 4) = 5.$$

$$\begin{aligned} \therefore \text{sum of first } n \text{ terms} &= \frac{n}{2} \cdot \{2a + (n-1)d\} \\ &= \frac{n}{2} \times \{2 \times 4 + (n-1) \times 5\} = \frac{1}{2}n(5n + 3). \end{aligned}$$

$$\therefore \text{sum of first 20 terms} = \frac{1}{2} \times 20 \times (5 \times 20 + 3) = 1030.$$

Hence, $S_n = \frac{1}{2}n(5n + 3)$ and $S_{20} = 1030$.

EXAMPLE 5 If the sum of first n terms of an AP is $\frac{1}{2}(3n^2 + 7n)$ then find its n th term and hence, write its 20th term. [CBSE 2015]

SOLUTION We have, $S_n = \frac{1}{2}(3n^2 + 7n)$.

$$\therefore S_{n-1} = \frac{1}{2} \cdot \{3(n-1)^2 + 7(n-1)\} = \frac{1}{2}(3n^2 + n - 4).$$

$$\begin{aligned}\therefore T_n &= (S_n - S_{n-1}) = \frac{1}{2}(3n^2 + 7n) - \frac{1}{2}(3n^2 + n - 4) \\ &= \frac{1}{2}(6n + 4) = (3n + 2).\end{aligned}$$

$$\therefore \text{nth term} = (3n + 2).$$

$$\text{Hence, 20th term} = (3 \times 20 + 2) = 62.$$

EXAMPLE 6 How many terms of the AP 3, 5, 7, 9, ... must be added to get the sum 120?

SOLUTION Here, $a = 3$ and $d = (5 - 3) = 2$.

Let the required number of terms be n . Then,

$$\begin{aligned}S_n = 120 &\Rightarrow \frac{n}{2} \cdot \{2a + (n-1)d\} = 120 \\ &\Rightarrow \frac{n}{2} \cdot \{2 \times 3 + (n-1) \times 2\} = 120 \\ &\Rightarrow \frac{n}{2} \cdot (2n + 4) = 120 \Rightarrow n^2 + 2n - 120 = 0 \\ &\Rightarrow n^2 + 12n - 10n - 120 = 0 \\ &\Rightarrow n(n + 12) - 10(n + 12) = 0 \Rightarrow (n + 12)(n - 10) = 0 \\ &\Rightarrow n + 12 = 0 \text{ or } n - 10 = 0 \Rightarrow n = -12 \text{ or } n = 10 \\ &\Rightarrow n = 10 \quad [\because \text{number of terms cannot be negative}]\end{aligned}$$

Hence, the required number of terms is 10.

EXAMPLE 7 How many terms of the AP 17, 15, 13, 11, ... must be added to get the sum 72? Explain the double answer.

SOLUTION Here, $a = 17$ and $d = (15 - 17) = -2$.

Let the sum of n terms be 72. Then,

$$\begin{aligned}S_n = 72 &\Rightarrow \frac{n}{2} \cdot \{2a + (n-1)d\} = 72 \\ &\Rightarrow n \cdot \{2 \times 17 + (n-1) \times (-2)\} = 144 \\ &\Rightarrow n(36 - 2n) = 144 \Rightarrow 2n^2 - 36n + 144 = 0 \\ &\Rightarrow n^2 - 18n + 72 = 0 \Rightarrow n^2 - 12n - 6n + 72 = 0 \\ &\Rightarrow n(n - 12) - 6(n - 12) = 0 \Rightarrow (n - 12)(n - 6) = 0 \\ &\Rightarrow n = 6 \text{ or } n = 12.\end{aligned}$$

$$\therefore \text{sum of first 6 terms} = \text{sum of first 12 terms} = 72.$$

This means that the sum of all terms from 7th to 12th is zero.

EXAMPLE 8 *The first and the last terms of an AP are 7 and 49 respectively. If sum of all its terms is 420, find its common difference.* [CBSE 2014]

SOLUTION Let the given AP contain n terms.

Here $a = 7$, $l = 49$ and $S_n = 420$.

$$\therefore S_n = \frac{n}{2}(a + l)$$

$$\Rightarrow \frac{n}{2}(7 + 49) = 420 \Rightarrow \frac{n}{2} \times 56 = 420$$

$$\Rightarrow 28n = 420 \Rightarrow n = \frac{420}{28} = 15.$$

Thus, the given AP contains 15 terms and $T_{15} = 49$.

Let d be the common difference of the given AP. Then,

$$T_{15} = 49 \Rightarrow a + 14d = 49$$

$$\Rightarrow 7 + 14d = 49$$

$$\Rightarrow 14d = 42 \Rightarrow d = 3.$$

Hence, the common difference of the given AP is 3.

EXAMPLE 9 *The sum of the first 7 terms of an AP is 63 and the sum of its next 7 terms is 161. Find the 28th term of this AP.* [CBSE 2014]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then, using $S_n = \frac{n}{2} \cdot [2a + (n - 1)d]$, we get

$$S_7 = \frac{7}{2}(2a + 6d) \Rightarrow 7(a + 3d) = 63 \quad [\because S_7 = 63]$$

$$\Rightarrow a + 3d = 9. \quad \dots (i)$$

Clearly, the sum of first 14 terms $= 63 + 161 = 224$.

$$\therefore S_{14} = 224 \Rightarrow \frac{14}{2}(2a + 13d) = 224$$

$$\Rightarrow 7(2a + 13d) = 224$$

$$\Rightarrow 2a + 13d = 32. \quad \dots (ii)$$

Multiplying (i) by 2 and subtracting the result from (ii), we get

$$7d = 14 \Rightarrow d = 2.$$

Putting $d = 2$ in (i), we get $a = 9 - 6 = 3$.

Thus, $a = 3$ and $d = 2$.

$\therefore$ 28th term of this AP is given by

$$T_{28} = (a + 27d) = (3 + 27 \times 2) = 57.$$

Hence, the 28th term of the given AP is 57.

EXAMPLE 10 The 14th term of an AP is twice its 8th term. If its 6th term is -8 then find the sum of its first 20 terms. [CBSE 2015]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$\begin{aligned}T_{14} = 2 \times T_8 &\Rightarrow a + 13d = 2(a + 7d) \\&\Rightarrow a + d = 0. \quad \dots (i)\end{aligned}$$

$$\text{Also, } T_6 = -8 \Rightarrow a + 5d = -8. \quad \dots (ii)$$

On solving (i) and (ii), we get $a = 2$ and $d = -2$.

The sum of first 20 terms is given by

$$\begin{aligned}S_{20} &= \frac{n}{2} \cdot [2a + (n-1)d], \text{ where } n = 20 \\&= \left(\frac{20}{2}\right) \times \{(2 \times 2 + 19 \times (-2))\} \\&= 10 \times (4 - 38) = 10 \times (-34) = -340.\end{aligned}$$

Hence, the required sum is -340 .

EXAMPLE 11 The sum of 4th and 8th terms of an AP is 24 and the sum of its 6th and 10th terms is 44. Find the sum of first ten terms of the AP. [CBSE 2012]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$\begin{aligned}T_4 + T_8 = 24 &\Rightarrow (a + 3d) + (a + 7d) = 24 \\&\Rightarrow 2a + 10d = 24 \\&\Rightarrow a + 5d = 12. \quad \dots (i)\end{aligned}$$

$$\begin{aligned}\text{And, } T_6 + T_{10} = 44 &\Rightarrow (a + 5d) + (a + 9d) = 44 \\&\Rightarrow 2a + 14d = 44 \\&\Rightarrow a + 7d = 22. \quad \dots (ii)\end{aligned}$$

On solving (i) and (ii), we get $a = -13$ and $d = 5$.

$\therefore$ the sum of first 10 terms of the given AP is given by

$$\begin{aligned}S_{10} &= \left(\frac{10}{2}\right) \cdot (2a + 9d) \quad \left[\text{using } S_n = \frac{n}{2} \{(2a + (n-1)d)\}\right] \\&= 5 \times \{2 \times (-13) + 9 \times 5\} = 5(-26 + 45) = 5 \times 19 = 95.\end{aligned}$$

Hence, the sum of first 10 terms of the given AP is 95.

EXAMPLE 12 Sum of first 14 terms of an AP is 1505 and its first term is 10. Find its 25th term. [CBSE 2012]

SOLUTION Here $a = 10$ and let d be the common difference. Then,

$$\begin{aligned} S_{14} = 1505 &\Rightarrow \frac{n}{2}[2a + (n-1)d] = 1505, \text{ where } n = 14 \text{ and } a = 10 \\ &\Rightarrow \frac{14}{2} \cdot (20 + 13d) = 1505 \Rightarrow (20 + 13d) = \frac{1505}{7} = 215 \\ &\Rightarrow 13d = 195 \Rightarrow d = 15. \end{aligned}$$

Thus, $a = 10$ and $d = 15$.

$$\therefore T_{25} = (a + 24d) = (10 + 24 \times 15) = 370.$$

Hence, the 25th term is 370.

EXAMPLE 13 *In an AP of 50 terms, the sum of first 10 terms is 210 and the sum of its last 15 terms is 2565. Find the AP.* [CBSE 2014]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then, the sum of first n terms is given by

$$S_n = \frac{n}{2} \cdot \{2a + (n-1)d\}.$$

$$\begin{aligned} \therefore S_{10} &= \frac{10}{2} \cdot (2a + 9d) \Rightarrow 5(2a + 9d) = 210 \\ &\Rightarrow 2a + 9d = 42. \end{aligned} \quad \dots \text{ (i)}$$

Sum of last 15 terms $= (S_{50} - S_{35})$.

$$\begin{aligned} \therefore (S_{50} - S_{35}) &= 2565 \\ \Rightarrow \frac{50}{2}(2a + 49d) - \frac{35}{2}(2a + 34d) &= 2565 \\ \Rightarrow 25(2a + 49d) - 35(a + 17d) &= 2565 \\ \Rightarrow (50a - 35a) + (1225d - 595d) &= 2565 \\ \Rightarrow 15a + 630d = 2565 \Rightarrow a + 42d &= 171. \end{aligned} \quad \dots \text{ (ii)}$$

On solving (i) and (ii), we get $a = 3$ and $d = 4$.

Hence, the required AP is 3, 7, 11, 15, 19, ...

EXAMPLE 14 *The sum of first 6 terms of an AP is 42. The ratio of its 10th term to 30th term is 1 : 3. Find the first and the 13th term of the AP.*

[CBSE 2009]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$T_{10} = a + 9d \text{ and } T_{30} = a + 29d$$

$$\therefore \frac{T_{10}}{T_{30}} = \frac{1}{3} \Rightarrow \frac{a + 9d}{a + 29d} = \frac{1}{3}$$

$$\Rightarrow 3a + 27d = a + 29d$$

$$\Rightarrow 2a = 2d \Rightarrow a = d.$$

$$\text{Also, } S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\Rightarrow S_6 = \frac{6}{2} (2a + 5d)$$

$$= (6a + 15d) = (6a + 15a) \quad [\because d = a]$$

$$= 21a.$$

$$\text{But, } S_6 = 42 \quad (\text{given})$$

$$\therefore 21a = 42 \Rightarrow a = 2.$$

$$\text{Thus, } a = 2 \text{ and } d = 2.$$

$$\therefore 13\text{th term, } T_{13} = (a + 12d) = (2 + 12 \times 2) = 26.$$

Hence, the first term is 2 and the 13th term is 26.

EXAMPLE 15 If S_n denotes the sum of first n terms of an AP, prove that $S_{12} = 3(S_8 - S_4)$. [CBSE 2015]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$S_n = \frac{n}{2} \cdot [2a + (n-1)d].$$

$$\begin{aligned} \therefore 3(S_8 - S_4) &= 3 \left[\frac{8}{2} (2a + 7d) - \frac{4}{2} (2a + 3d) \right] \\ &= 3 [4(2a + 7d) - 2(2a + 3d)] = 6(2a + 11d) \\ &= \frac{12}{2} \cdot (2a + 11d) = S_{12}. \end{aligned}$$

$$\text{Hence, } S_{12} = 3(S_8 - S_4).$$

EXAMPLE 16 If the sum of first n , $2n$ and $3n$ terms of an AP be S_1 , S_2 and S_3 respectively then prove that $S_3 = 3(S_2 - S_1)$.

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$S_1 = \text{sum of first } n \text{ terms of the given AP,}$$

$$S_2 = \text{sum of first } 2n \text{ terms of the given AP,}$$

$$S_3 = \text{sum of first } 3n \text{ terms of the given AP.}$$

$$\therefore S_1 = \frac{n}{2} \cdot \{2a + (n-1)d\}, \quad S_2 = \frac{2n}{2} \cdot \{2a + (2n-1)d\}, \text{ and}$$

$$S_3 = \frac{3n}{2} \cdot \{2a + (3n-1)d\}$$

$$\begin{aligned}
 \Rightarrow 3(S_2 - S_1) &= 3 \cdot \left[\{2na + n(2n-1)d\} - \left\{na + \frac{1}{2}n(n-1)d\right\} \right] \\
 &= 3 \cdot \left[na + \frac{3}{2}n^2d - \frac{1}{2}nd \right] = \frac{3n}{2} \cdot [2a + 3nd - d] \\
 &= \frac{3n}{2} \cdot \{2a + (3n-1)d\} = S_3.
 \end{aligned}$$

Hence, $S_3 = 3(S_2 - S_1)$.

EXAMPLE 17 *If the sum of the first p terms of an AP is the same as the sum of its first q terms (where $p \neq q$) then show that the sum of its first $(p+q)$ terms is zero.*

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$\begin{aligned}
 S_p = S_q &\Rightarrow \frac{p}{2}[2a + (p-1)d] = \frac{q}{2}[2a + (q-1)d] \\
 &\Rightarrow (p-q)(2a) = (q-p)(q+p-1)d \\
 &\Rightarrow 2a = (1-p-q)d \quad \dots (i)
 \end{aligned}$$

Sum of the first $(p+q)$ terms of the given AP

$$\begin{aligned}
 &= \frac{(p+q)}{2} \cdot \{2a + (p+q-1)d\} \\
 &= \frac{(p+q)}{2} \cdot \{(1-p-q)d + (p+q-1)d\} \quad [\text{using (i)}] \\
 &= 0.
 \end{aligned}$$

EXAMPLE 18 *If the sum of the first m terms of an AP be n and the sum of its first n terms be m then show that the sum of its first $(m+n)$ terms is $-(m+n)$.*

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

$$\begin{aligned}
 S_m = n &\Rightarrow \frac{m}{2}[2a + (m-1)d] = n \\
 &\Rightarrow 2am + m(m-1)d = 2n. \quad \dots (i)
 \end{aligned}$$

$$\begin{aligned}
 \text{And, } S_n = m &\Rightarrow \frac{n}{2}[2a + (n-1)d] = m \\
 &\Rightarrow 2an + n(n-1)d = 2m. \quad \dots (ii)
 \end{aligned}$$

On subtracting (ii) from (i), we get

$$\begin{aligned}
 &2a(m-n) + [(m^2 - n^2) - (m-n)]d = 2(n-m) \\
 \Rightarrow &(m-n)[2a + (m+n-1)d] = 2(n-m) \\
 \Rightarrow &2a + (m+n-1)d = -2 \quad \dots (iii)
 \end{aligned}$$

Sum of the first $(m + n)$ terms of the given AP

$$\begin{aligned} &= \frac{(m+n)}{2} \cdot \{2a + (m+n-1)d\} \\ &= \frac{(m+n)}{2} \cdot (-2) = -(m+n) \text{ [using (iii)]}. \end{aligned}$$

Hence, the sum of first $(m + n)$ terms of the given AP is $-(m + n)$.

EXAMPLE 19 *The ratio of the sums of first m and first n terms of an AP is $m^2 : n^2$. Show that the ratio of its m th and n th terms is $(2m - 1) : (2n - 1)$.*

[CBSE 2017]

SOLUTION Let a be the first term and d be the common difference of the given AP. Then,

S_m = sum of first m terms of the given AP;

S_n = sum of first n terms of the given AP.

$$\begin{aligned} \frac{S_m}{S_n} = \frac{m^2}{n^2} &\Rightarrow \frac{\frac{m}{2}[2a + (m-1)d]}{\frac{n}{2}[2a + (n-1)d]} = \frac{m^2}{n^2} \\ &\Rightarrow \frac{2a + (m-1)d}{2a + (n-1)d} = \frac{m}{n} \\ &\Rightarrow 2an + mnd - nd = 2am + mnd - md \\ &\Rightarrow 2an - 2am = nd - md \\ &\Rightarrow 2a(n - m) = d(n - m) \Rightarrow 2a = d. \quad \dots (i) \end{aligned}$$

$$\begin{aligned} \therefore \frac{T_m}{T_n} &= \frac{a + (m-1)d}{a + (n-1)d} = \frac{a + (m-1) \cdot 2a}{a + (n-1) \cdot 2a} \quad [\text{from (i)}] \\ &= \frac{a + 2am - 2a}{a + 2an - 2a} = \frac{2am - a}{2an - a} = \frac{a(2m-1)}{a(2n-1)} = \frac{2m-1}{2n-1}. \end{aligned}$$

EXAMPLE 20 *The ratio of the 11th term to the 18th term of an AP is $2 : 3$. Find the ratio of the 5th term to the 21st term, and also the ratio of the sum of the first 5 terms to the sum of first 21 terms.*

[CBSE 2017]

SOLUTION Let a be the first term and d be the common difference of the given AP.

$$\begin{aligned} \text{Then, } \frac{T_{11}}{T_{18}} = \frac{2}{3} &\Rightarrow \frac{a + (11-1)d}{a + (18-1)d} = \frac{2}{3} \\ &\Rightarrow \frac{a + 10d}{a + 17d} = \frac{2}{3} \Rightarrow 3a + 30d = 2a + 34d \\ &\Rightarrow a = 4d. \quad \dots (i) \end{aligned}$$

$$\begin{aligned}
 \text{Ratio of 5th term to 21st term} &= \frac{T_5}{T_{21}} = \frac{a + (5-1)d}{a + (21-1)d} \\
 &= \frac{a + 4d}{a + 20d} = \frac{4d + 4d}{4d + 20d} \quad [\text{from (i)}] \\
 &= \frac{8d}{24d} = \frac{1}{3} = 1 : 3.
 \end{aligned}$$

Ratio of sum of first 5 terms to sum of first 21 terms

$$\begin{aligned}
 &= \frac{S_5}{S_{21}} = \frac{\frac{5}{2}[2a + (5-1)d]}{\frac{21}{2}[2a + (21-1)d]} = \frac{5(2a + 4d)}{21(2a + 20d)} \\
 &= \frac{10(a + 2d)}{42(a + 10d)} = \frac{10(4d + 2d)}{42(4d + 10d)} \quad [\text{from (i)}] \\
 &= \frac{60d}{588d} = \frac{60}{588} = \frac{5}{49} = 5 : 49.
 \end{aligned}$$

EXAMPLE 21 If the ratio of the sum of the first n terms of two APs is $(7n+1) : (4n+27)$ then find the ratio of their 9th terms. [CBSE 2017]

SOLUTION Let a_1 and a_2 be the first terms and d_1 and d_2 be the common difference of the two APs respectively.

Let S_n and S'_n be the sums of the first n terms of the two APs and T_n and T'_n be their n th terms respectively.

$$\begin{aligned}
 \text{Then, } \frac{S_n}{S'_n} &= \frac{7n+1}{4n+27} \Rightarrow \frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27} \\
 &\Rightarrow \frac{2a_1 + (n-1)d_1}{2a_2 + (n-1)d_2} = \frac{7n+1}{4n+27}. \quad \dots (i)
 \end{aligned}$$

To find the ratio of m th terms, we replace n by $(2m-1)$ in the above expression.

Replacing n by $(2 \times 9 - 1)$, i.e., 17 on both sides in (i), we get

$$\begin{aligned}
 \frac{2a_1 + (17-1)d_1}{2a_2 + (17-1)d_2} &= \frac{7 \times 17 + 1}{4 \times 17 + 27} \Rightarrow \frac{2a_1 + 16d_1}{2a_2 + 16d_2} = \frac{120}{95} \\
 &\Rightarrow \frac{a_1 + 8d_1}{a_2 + 8d_2} = \frac{24}{19} \\
 &\Rightarrow \frac{a_1 + (9-1)d_1}{a_2 + (9-1)d_2} = \frac{24}{19} \\
 &\Rightarrow \frac{T_9}{T'_9} = \frac{24}{19}.
 \end{aligned}$$

$\therefore$ required ratio = 24 : 19.

WORD PROBLEMS

EXAMPLE 22 Find the sum of all multiples of 7 lying between 500 and 900.

[CBSE 2012]

SOLUTION All multiples of 7 lying between 500 and 900 are

$$504, 511, 518, \dots, 896$$

This is an AP in which $a = 504$, $d = 7$ and $l = 896$.

Let the given AP contain n terms. Then,

$$\begin{aligned} T_n = 896 &\Rightarrow a + (n-1)d = 896 \Rightarrow 504 + (n-1) \times 7 = 896 \\ &\Rightarrow 497 + 7n = 896 \Rightarrow 7n = 399 \Rightarrow n = 57. \end{aligned}$$

$$\begin{aligned} \therefore \text{required sum} &= \frac{n}{2}(a + l) \\ &= \frac{57}{2} \cdot (504 + 896) = \left(\frac{57}{2} \times 1400\right) = 39900. \end{aligned}$$

Hence, the required sum is 39900.

EXAMPLE 23 Find the sum of all 3-digit natural numbers which are multiples of 11.

[CBSE 2009, '12]

SOLUTION All 3-digit natural numbers, which are multiples of 11 are given as 110, 121, 132, ..., 990.

This is an AP in which $a = 110$, $d = 11$ and $l = 990$.

Let the given AP contain n terms. Then,

$$\begin{aligned} T_n = 990 &\Rightarrow a + (n-1)d = 990 \Rightarrow 110 + (n-1) \times 11 = 990 \\ &\Rightarrow 99 + 11n = 990 \Rightarrow 11n = 891 \Rightarrow n = 81. \end{aligned}$$

$$\begin{aligned} \therefore \text{required sum} &= \frac{n}{2}(a + l) = \frac{81}{2} \times (110 + 990) \\ &= \left(\frac{81}{2} \times 1100\right) = 44550. \end{aligned}$$

Hence, the required sum is 44550.

EXAMPLE 24 Ramkali required ₹ 2500 after 12 weeks to send her daughter to school. She saved ₹ 100 in the first week and increased her weekly saving by ₹ 20 every week. Find whether she will be able to send her daughter to school after 12 weeks. What value is generated in the above situation?

[CBSE 2015]

SOLUTION The amounts saved by Ramkali in successive weeks are ₹ 100, ₹ 120, ₹ 140, ₹ 160, ... up to 12 terms.

These amounts form an AP in which $a = 100$, $d = 20$ and $n = 12$.

Using $S_n = \frac{n}{2} \times [2a + (n-1)d]$, we get

$$S_{12} = \frac{12}{2} \times [2 \times 100 + 11 \times 20] = (6 \times 420) = 2520 > 2500.$$

Thus, Ramkali will be able to deposit her daughter's fees and so she can send her to school.

The above situation shows that saving is a good habit as it helps preserve and collect money for a good cause.

EXAMPLE 25 200 logs are stacked in such a way that there are 20 logs in the bottom row, 19 in the next row, 18 in the next row, and so on. In how many rows 200 logs are placed and how many logs are there in the top row?

SOLUTION Let the required number of rows be n . Then,

$$20 + 19 + 18 + \dots \text{ to } n \text{ terms} = 200.$$

This is an arithmetic series in which

$$a = 20, d = (19 - 20) = -1 \text{ and } S_n = 200.$$

We know that $S_n = \frac{n}{2} \cdot \{2a + (n-1)d\}$.

$$\therefore \frac{n}{2} \cdot \{2 \times 20 + (n-1) \times (-1)\} = 200$$

$$\Rightarrow n(41 - n) = 400 \Rightarrow n^2 - 41n + 400 = 0$$

$$\Rightarrow n^2 - 25n - 16n + 400 = 0 \Rightarrow n(n - 25) - 16(n - 25) = 0$$

$$\Rightarrow (n - 25)(n - 16) = 0 \Rightarrow n - 25 = 0 \text{ or } n - 16 = 0$$

$$\Rightarrow n = 25 \text{ or } n = 16.$$

$$\text{Now, } T_{25} = (a + 24d) = 20 + 24 \times (-1) = -4.$$

This is meaningless as the number of logs cannot be negative.

So, we reject the value $n = 25$.

$\therefore n = 16$. Thus, there are 16 rows in the whole stack.

$$\text{Now, } T_{16} = (a + 15d) = 20 + 15 \times (-1) = 20 - 15 = 5.$$

Hence, there are 5 logs in the top row.

EXAMPLE 26 The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th year and 22600 in 9th year. Find the production during (i) first year (ii) 8th year (iii) first 6 years.

SOLUTION Let the production during first year be a and let d be the increase in production every year. Then,

$$T_6 = 16000 \Rightarrow a + 5d = 16000 \quad \dots (i)$$

$$\text{and } T_9 = 22600 \Rightarrow a + 8d = 22600. \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$3d = 6600 \Rightarrow d = 2200.$$

Putting $d = 2200$ in (i), we get

$$a + 5 \times 2200 = 16000 \Rightarrow a + 11000 = 16000$$

$$\Rightarrow a = 16000 - 11000 = 5000.$$

Thus, $a = 5000$ and $d = 2200$.

(i) Production during first year, $a = 5000$.

(ii) Production during 8th year is given by

$$T_8 = (a + 7d) = (5000 + 7 \times 2200) = (5000 + 15400) = 20400.$$

(iii) Production during first 6 years is given by

$$\begin{aligned} S_6 &= \frac{6}{2} \{2a + 5d\} = 3(2 \times 5000 + 5 \times 2200) \\ &= 3(10000 + 11000) = 63000. \end{aligned}$$

EXAMPLE 27 A spiral is made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1 cm, 1.5 cm, 2 cm, ... as shown in the given figure. What is the total length of such a spiral made up of 13 consecutive semicircles? (Take $\pi = \frac{22}{7}$)

SOLUTION Let $L_1, L_2, L_3, L_4, \dots, L_{13}$ be the lengths of semicircles of radii 0.5 cm, 1 cm, 1.5 cm, 2 cm, ... and $\frac{13}{2}$ cm respectively.

Then, we have

$$L_1 = (\pi \times 0.5) \text{ cm} = \frac{\pi}{2} \text{ cm},$$

$$L_2 = (\pi \times 1) \text{ cm} = 2\left(\frac{\pi}{2}\right) \text{ cm},$$

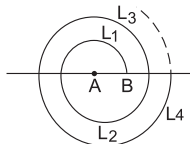
$$L_3 = (\pi \times 1.5) \text{ cm} = 3\left(\frac{\pi}{2}\right) \text{ cm},$$

$$L_4 = (\pi \times 2) \text{ cm} = 4\left(\frac{\pi}{2}\right) \text{ cm},$$

$$\dots \dots \dots \dots \dots \dots$$

$$\dots \dots \dots \dots \dots \dots$$

$$L_{13} = \left(\pi \times \frac{13}{2}\right) \text{ cm} = 13\left(\frac{\pi}{2}\right) \text{ cm}.$$



∴ total length of the spiral

$$\begin{aligned}
 &= L_1 + L_2 + L_3 + L_4 + \dots + L_{13} \\
 &= \left\{ \frac{\pi}{2} + 2\left(\frac{\pi}{2}\right) + 3\left(\frac{\pi}{2}\right) + 4\left(\frac{\pi}{2}\right) + \dots + 13\left(\frac{\pi}{2}\right) \right\} \text{ cm} \\
 &= \frac{\pi}{2} (1 + 2 + 3 + 4 + \dots + 13) \text{ cm} \\
 &= \frac{\pi}{2} \times \frac{13}{2} \times (1 + 13) \text{ cm} \qquad \left[S_n = \frac{n}{2} (a + l) \right] \\
 &= \left(\frac{1}{2} \times \frac{22}{7} \times \frac{13}{2} \times 14 \right) \text{ cm} = 143 \text{ cm}.
 \end{aligned}$$

Hence, the required length of the spiral is 143 cm.

EXAMPLE 28 A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. If the top and bottom rungs are 2.5 m apart, what is the length of the wood required for the rungs?

SOLUTION It is given that the top and bottom rungs are 250 cm apart and the gap between two consecutive rungs is 25 cm.

$$\therefore \text{ number of rungs} = \left(\frac{250}{25} + 1 \right) = 11.$$

The largest rung is 45 cm long and the smallest one is 25 cm long.

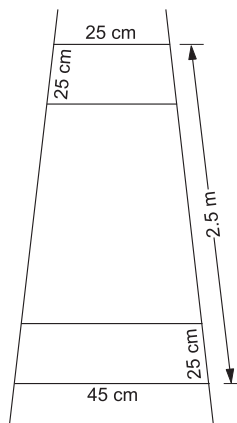
It is given that the rungs are decreasing uniformly in length from 45 cm at the bottom to 25 cm at the top.

So, the lengths of the rungs form an AP with $a = 45$ cm and $l = \text{length of 11th rung} = 25$ cm.

∴ length of the wood required to form 11 rungs

$$= \frac{n}{2} (a + l) \text{ cm} = \frac{11}{2} (45 + 25) \text{ cm} = \left(\frac{11}{2} \times 70 \right) \text{ cm} = 385 \text{ cm}.$$

Hence, the required length of the wood to form these rungs is 3.85 m.



EXAMPLE 29 The houses of a row in a colony are numbered consecutively from 1 to 49. Show that there is a value of x such that the sum of the numbers of the houses preceding the house numbered x is equal to the sum of the numbers of the houses following it. Find the value of x .

SOLUTION We are given an AP, namely

$$1, 2, 3, \dots, (x-1), x, (x+1), \dots, 49$$

such that $1 + 2 + 3 + \dots + (x-1) = (x+1) + (x+2) + \dots + 49$.

Thus, we have $S_{x-1} = S_{49} - S_x$ (i)

Using the formula, $S_n = \frac{n}{2}(a + l)$ in (i), we have

$$\frac{(x-1)}{2} \cdot \{1 + (x-1)\} = \frac{49}{2} \cdot (1 + 49) - \frac{x}{2} \cdot (1 + x)$$

$$\Rightarrow \frac{x(x-1)}{2} + \frac{x(x+1)}{2} = 1225$$

$$\Rightarrow 2x^2 = 2450 \Rightarrow x^2 = 1225 \Rightarrow x = \sqrt{1225} = 35.$$

Hence, $x = 35$.

EXAMPLE 30 Find the middle term of the sequence formed by all three-digit numbers which leave a remainder 3 when divided by 4. Also, find the sum of all numbers on both sides of the middle term. [CBSE 2015]

SOLUTION The sequence formed by the given numbers is

$$103, 107, 111, 115, \dots, 999.$$

This is an AP in which $a = 103$ and $d = (107 - 103) = 4$.

Let the total number of these terms be n . Then,

$$T_n = 999 \Rightarrow a + (n-1)d = 999$$

$$\Rightarrow 103 + (n-1) \times 4 = 999$$

$$\Rightarrow (n-1) \times 4 = 896 \Rightarrow (n-1) = 224 \Rightarrow n = 225.$$

$\therefore$ middle term = $\left(\frac{n+1}{2}\right)$ th term = $\left(\frac{225+1}{2}\right)$ th term = 113th term.

$$T_{113} = (a + 112d) = (103 + 112 \times 4) = 551.$$

$$\Rightarrow T_{112} = (551 - 4) = 547.$$

So, we have to find S_{112} and $(S_{225} - S_{113})$.

Using the formula $S_m = \frac{m}{2}(a + l)$ for each sum, we get

$$S_{112} = \frac{112}{2}(103 + 547) = (112 \times 325) = 36400.$$

$$\begin{aligned}
 (S_{225} - S_{113}) &= \frac{225}{2}(103 + 999) - \frac{113}{2}(103 + 551) \\
 &= (225 \times 551) - (113 \times 327) \\
 &= 123975 - 36951 = 87024.
 \end{aligned}$$

Sum of all numbers on LHS of the middle term is 36400.

Sum of all numbers on RHS of the middle term is 87024.

EXERCISE 5C

- Find the sum of each of the following APs:
 - 2, 7, 12, 17, ... to 19 terms.
 - 9, 7, 5, 3, ... to 14 terms.
 - 37, -33, -29, ... to 12 terms.
 - $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}, \dots$ to 11 terms.
 - 0.6, 1.7, 2.8, ... to 100 terms.
- Find the sum of each of the following arithmetic series:
 - $7 + 10\frac{1}{2} + 14 + \dots + 84$.
 - $34 + 32 + 30 + \dots + 10$.
 - $(-5) + (-8) + (-11) + \dots + (-230)$
 - $5 + (-41) + 9 + (-39) + 13 + (-37) + 17 + \dots + (-5) + 81 + (-3)$
- Find the sum of first n terms of an AP whose n th term is $(5 - 6n)$. Hence, find the sum of its first 20 terms.
- The sum of the first n terms of an AP is $(3n^2 + 6n)$. Find the n th term and the 15th term of this AP. [CBSE 2014]
- The sum of the first n terms of an AP is given by $S_n = (3n^2 - n)$. Find its
 - n th term, (ii) first term and (iii) common difference. [CBSE 2005C]
- The sum of the first n terms of an AP is $\left(\frac{5n^2}{2} + \frac{3n}{2}\right)$. Find the n th term and the 20th term of this AP. [CBSE 2006C]
 - The sum of the first n terms of an AP is $\left(\frac{3n^2}{2} + \frac{5n}{2}\right)$. Find its n th term and the 25th term. [CBSE 2006C]
- If m th term of an AP is $\frac{1}{n}$ and n th term is $\frac{1}{m}$ then find the sum of its first mn terms. [CBSE 2017]
- How many terms of the AP 21, 18, 15, ... must be added to get the sum 0?

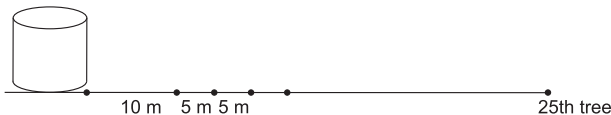
9. How many terms of the AP 9, 17, 25, ... must be taken so that their sum is 636? [CBSE 2017]
10. How many terms of the AP 63, 60, 57, 54, ... must be taken so that their sum is 693? Explain the double answer.
11. How many terms of the AP 20, $19\frac{1}{3}$, $18\frac{2}{3}$, ... must be taken so that their sum is 300? Explain the double answer.
12. Find the sum of all odd numbers between 0 and 50. [CBSE 20013C]
13. Find the sum of all natural numbers between 200 and 400 which are divisible by 7. [CBSE 2012]
14. Find the sum of first forty positive integers divisible by 6. [CBSE 2012]
15. Find the sum of the first 15 multiples of 8.
16. Find the sum of all multiples of 9 lying between 300 and 700.
17. Find the sum of all three-digit natural numbers which are divisible by 13. [CBSE 2011]
18. Find the sum of first 100 even natural numbers which are divisible by 5. [CBSE 2007C]
19. Find the sum of n terms of the following series:
$$\left(4 - \frac{1}{n}\right) + \left(4 - \frac{2}{n}\right) + \left(4 - \frac{3}{n}\right) + \dots$$
 [CBSE 2017]
20. In an AP, it is given that $S_5 + S_7 = 167$ and $S_{10} = 235$, then find the AP, where S_n denotes the sum of its first n terms. [CBSE 2015]
21. In an AP, the first term is 2, the last term is 29 and the sum of all the terms is 155. Find the common difference. [CBSE 2010]
22. In an AP, the first term is -4 , the last term is 29 and the sum of all its terms is 150. Find its common difference. [CBSE 2011]
23. The first and the last terms of an AP are 17 and 350 respectively. If the common difference is 9, how many terms are there and what is their sum?
24. The first and the last terms of an AP are 5 and 45 respectively. If the sum of all its terms is 400, find the common difference and the number of terms. [CBSE 2012, '14, '17]
25. In an AP, the first term is 22, n th term is -11 and sum of first n terms is 66. Find n and hence find the common difference. [CBSE 2008]
26. The 12th term of an AP is -13 and the sum of its first four terms is 24. Find the sum of its first 10 terms. [CBSE 2015]
27. The sum of the first 7 terms of an AP is 182. If its 4th and 17th terms are in the ratio 1 : 5, find the AP. [CBSE 2014]

28. The sum of the first 9 terms of an AP is 81 and that of its first 20 terms is 400. Find the first term and the common difference of the AP. [CBSE 2009]
29. The sum of the first 7 terms of an AP is 49 and the sum of its first 17 terms is 289. Find the sum of its first n terms. [CBSE 2008C]
30. Two APs have the same common difference. If the first terms of these APs be 3 and 8 respectively, find the difference between the sums of their first 50 terms. [CBSE 2011]
31. The sum of first 10 terms of an AP is -150 and the sum of its next 10 terms is -550 . Find the AP. [CBSE 2010]
32. The 13th term of an AP is 4 times its 3rd term. If its 5th term is 16, find the sum of its first 10 terms. [CBSE 2015]
33. The 16th term of an AP is 5 times its 3rd term. If its 10th term is 41, find the sum of its first 15 terms. [CBSE 2015]
34. (i) An AP 5, 12, 19, ... has 50 terms. Find its last term. Hence, find the sum of its last 15 terms. [CBSE 2015]
(ii) An AP 8, 10, 12, ... has 60 terms. Find its last term. Hence, find the sum of its last 10 terms. [CBSE 2015]
35. The sum of first n terms of two APs are in the ratio $(3n + 8) : (7n + 15)$. Find the ratio of their 12th terms.
36. The sum of the 4th and the 8th terms of an AP is 24 and the sum of its 6th and 10th terms is 44. Find the sum of its first 10 terms. [CBSE 2013C]
37. The sum of first m terms of an AP is $(4m^2 - m)$. If its n th term is 107, find the value of n . Also, find the 21st term of this AP. [CBSE 2013]
38. The sum of first q terms of an AP is $(63q - 3q^2)$. If its p th term is -60 , find the value of p . Also, find the 11th term of its AP. [CBSE 2013]
39. Find the number of terms of the AP $-12, -9, -6, \dots, 21$. If 1 is added to each term of this AP then find the sum of all terms of the AP thus obtained. [CBSE 2013]
40. Sum of the first 14 terms of an AP is 1505 and its first term is 10. Find its 25th term. [CBSE 2012]
41. Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.
42. In a school, students decided to plant trees in and around the school to reduce air pollution. It was decided that the number of trees that each section of each class will plant will be double of the class in which they are studying. If there are 1 to 12 classes in the school and each class has two sections, find how many trees were planted by students. Which value is shown in the question? [CBSE 2014]

43. In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato, and the other potatoes are placed 3 m apart in a straight line. There are 10 potatoes in the line. A competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket to drop it in, and he continues in the same way until all the potatoes are in the bucket. What is the total distance the competitor has to run?



44. There are 25 trees at equal distances of 5 m in a line with a water tank, the distance of the water tank from the nearest tree being 10 m. A gardener waters all the trees separately, starting from the water tank and returning back to the water tank after watering each tree to get water for the next. Find the total distance covered by the gardener in order to water all the trees.



45. A sum of ₹ 700 is to be used to give seven cash prizes to students of a school for their overall academic performance. If each prize is ₹ 20 less than its preceding prize, find the value of each prize.
46. A man saved ₹ 33000 in 10 months. In each month after the first, he saved ₹ 100 more than he did in the preceding month. How much did he save in the first month?
47. A man arranges to pay off a debt of ₹ 36000 by 40 monthly instalments which form an arithmetic series. When 30 of the instalments are paid, he dies leaving one-third of the debt unpaid. Find the value of the first instalment.
48. A contract on construction job specifies a penalty for delay of completion beyond a certain date as follows: ₹ 200 for the first day, ₹ 250 for the second day, ₹ 300 for the third day, etc., the penalty for each succeeding day being ₹ 50 more than for the preceding day. How much money the contractor has to pay as penalty, if he has delayed the work by 30 days?
49. A child puts one five-rupee coin of her savings in the piggy bank on the first day. She increases her saving by one five-rupee coin daily. If the piggy bank can hold 190 coins of five rupees in all, find the number of days she can contribute to put the five-rupee coins into it and find the total money she saved.

ANSWERS (EXERCISE 5C)

1. (i) 893 (ii) -56 (iii) -180 (iv) $\frac{33}{20}$ (v) 5505
2. (i) $1046\frac{1}{2}$ (ii) 286 (iii) -8930 (iv) 420 3. $S_n = n(2 - 3n)$, $S_{20} = -1160$
4. $T_n = (6n + 3)$, $T_{15} = 93$ 5. (i) $T_n = (6n - 4)$ (ii) $a = 2$ (iii) $d = 6$
6. (i) $T_n = (5n - 1)$, $T_{20} = 99$ (ii) $T_n = (3n + 1)$, $T_{25} = 76$ 7. $\frac{1 + mn}{2}$
8. 15 9. 12 10. $n = 21$ or $n = 22$, 22nd term is 0
11. $n = 25$ or $n = 36$, sum of last 11th terms is 0 12. 625 13. 8729
14. 4920 15. 960 16. 21978 17. 37674 18. 50500
19. $\frac{1}{2}(7n - 1)$ 20. 1, 6, 11, 16, ... 21. $d = 3$ 22. $d = 3$
23. $n = 38$ and $S_n = 6973$ 24. $d = \frac{8}{3}$ and $n = 16$ 25. $n = 12$, $d = -3$
26. $S_{10} = 0$ 27. 2, 10, 18, 26, ... 28. $a = 1$, $d = 2$ 29. $S_n = n^2$
30. 250 31. 3, -1, -5, -9, ... 32. $S_{10} = 175$ 33. $S_{15} = 495$
34. (i) $l = 348$, sum = 4485 (ii) $T_{60} = 126$, sum = 1170 35. 7 : 16
36. $S_{10} = 95$ 37. $n = 14$, $T_{21} = 163$ 38. $p = 21$, $T_{11} = 0$
39. $n = 12$, sum = 66 40. $T_{25} = 130$ 41. 5610 42. 312
43. 370 m 44. 3500 m 45. ₹ 160, ₹ 140, ₹ 120, ₹ 100, ₹ 80, ₹ 60, ₹ 40
46. ₹ 2850 47. ₹ 510 48. ₹ 27750 49. 19, ₹ 950

HINTS TO SOME SELECTED QUESTIONS

2. (i) $a = 7$, $d = \left(\frac{21}{2} - 7\right) = \frac{7}{2}$ and $l = 84$.
- $$T_n = 84 \Rightarrow a + (n - 1)d = 84 \Rightarrow 7 + (n - 1) \times \frac{7}{2} = 84 \Rightarrow n = 23.$$
- $$S_{23} = \frac{n}{2}(a + l) = \frac{23}{2}(7 + 84) = 1046\frac{1}{2}.$$
3. $T_n = (5 - 6n) \Rightarrow T_1 = (5 - 6) = -1$ and $T_2 = (5 - 6 \times 2) = -7$
- $$\therefore a = -1 \text{ and } d = (-7 + 1) = -6.$$
- $$S_n = \frac{n}{2} \times \{2 \times (-1) + (n - 1) \times (-6)\} = n(2 - 3n).$$
- $$\therefore S_{20} = 20 \times (2 - 3 \times 20) = -1160.$$
9. $4n^2 + 5n - 636 = 0 \Rightarrow 4n^2 + 53n - 48n - 636 = 0.$
10. $n^2 - 43n + 462 = 0 \Rightarrow n^2 - 21n - 22n + 462 = 0.$
11. $n^2 - 61n + 900 = 0 \Rightarrow n^2 - 25n - 36n + 900 = 0.$
12. Required sum = $(1 + 3 + 5 + 7 + \dots + 49).$

$$T_n = 49 \Rightarrow 1 + (n-1) \times 2 = 49 \Rightarrow n = 25.$$

$$\text{Use } S_n = \frac{n}{2}(a+l).$$

$$13. \text{ Required sum} = (203 + 210 + 217 + \dots + 399).$$

$$T_n = 399 \Rightarrow 203 + (n-1) \times 7 = 399 \Rightarrow n = 29.$$

$$\text{Use } S_n = \frac{n}{2}(a+l).$$

$$14. \text{ Required sum} = (6 + 12 + 18 + \dots + 240).$$

$$\text{Here } n = 40.$$

$$\text{Use } S_n = \frac{n}{2}(a+l).$$

$$15. \text{ Required sum} = (8 + 16 + 24 + \dots + 120).$$

$$\text{Here } n = 15. \text{ Use } S_n = \frac{n}{2}(a+l).$$

$$16. \text{ Required sum} = (306 + 315 + 324 + \dots + 693).$$

$$17. \text{ Required sum} = (104 + 117 + 130 + \dots + 988).$$

$$18. \text{ Required sum} = (10 + 20 + 30 + \dots + 1000).$$

$$19. \text{ Required sum} = (4 + 4 + \dots \text{ to } n \text{ terms}) - \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{n}{n}\right)$$

$$= 4n - \frac{n}{2}\left(\frac{1}{n} + \frac{n}{n}\right) \left[\text{sum} = \frac{n}{2}(a+l)\right]$$

$$= 4n - \frac{(1+n)}{2} = \frac{1}{2}(7n-1).$$

$$20. S_5 + S_7 = 167 \Rightarrow \frac{5}{2}(2a+4d) + \frac{7}{2}(2a+6d) = 167$$

$$\Rightarrow 12a + 31d = 167.$$

... (i)

$$S_{10} = 235 \Rightarrow \frac{10}{2}(2a+9d) = 235 \Rightarrow 2a+9d = 47.$$

... (ii)

Solve (i) and (ii), we get $a = 1$ and $d = 5$.

$$21. S_n = \frac{n}{2}(a+l) \Rightarrow \frac{n}{2}(2+29) = 155 \Rightarrow n = 10.$$

$$T_{10} = 29 \Rightarrow 2+9d = 29 \Rightarrow d = 3.$$

$$23. \text{ Let the number of terms be } n. \text{ Then,}$$

$$T_n = 350 \Rightarrow a + (n-1)d = 350 \Rightarrow 17 + (n-1) \times 9 = 350 \Rightarrow n = 38.$$

$$S_n = \frac{n}{2}(a+l).$$

$$24. S_n = \frac{n}{2}(a+l) \Rightarrow \frac{n}{2}(5+45) = 400 \Rightarrow n = 16.$$

$$T_{16} = 45 \Rightarrow a + 15d = 45 \Rightarrow 5 + 15d = 45 \Rightarrow d = \frac{8}{3}.$$

$$25. S_n = \frac{n}{2}(a+l) \Rightarrow \frac{n}{2}[22 + (-11)] = 66 \Rightarrow n = 12.$$

$$T_{12} = -11 \Rightarrow a + 11d = -11 \Rightarrow 22 + 11d = -11 \Rightarrow d = -3.$$

$$26. a + 11d = -13.$$

... (i)

$$\text{and } \frac{4}{2}(2a+3d) = 24 \Rightarrow 2a+3d = 12.$$

... (ii)

Solve (i) and (ii), we get $a = 9$ and $d = -2$.

Now, find S_{10} .

$$27. \frac{7}{2}(2a + 6d) = 182 \Rightarrow a + 3d = 26. \quad \dots (i)$$

$$\frac{a + 3d}{a + 16d} = \frac{1}{5} \Rightarrow 5a + 15d = a + 16d \Rightarrow 4a - d = 0. \quad \dots (ii)$$

Solve (i) and (ii), we get $a = 2$ and $d = 8$.

$$28. \frac{9}{2}(2a + 8d) = 81 \Rightarrow a + 4d = 9 \quad \dots (i)$$

$$\frac{20}{2}(2a + 19d) = 400 \Rightarrow 2a + 19d = 40. \quad \dots (ii)$$

$$30. \text{Required difference} = \frac{50}{2} \cdot [2 \times 8 + 49d] - \frac{50}{2} \cdot [2 \times 3 + 49d] = 400 - 150 = 250.$$

$$31. S_{10} = -150 \text{ and } S_{20} = (-150) + (-550) = -700.$$

$$32. T_{13} = 4 \times T_3 \Rightarrow a + 12d = 4(a + 2d) \Rightarrow 3a - 4d = 0. \quad \dots (i)$$

$$T_5 = 16 \Rightarrow a + 4d = 16. \quad \dots (ii)$$

Solve (i) and (ii), we get $a = 4$ and $d = 3$.

Now, find S_{10} .

$$34. (i) l = T_{50} = (5 + 49 \times 7) = 348.$$

$$\text{Required sum} = (S_{50} - S_{35}). \text{ Now, } T_{35} = (5 + 34 \times 7) = 243.$$

$$\therefore (S_{50} - S_{35}) = \frac{50}{2}(5 + 348) - \frac{35}{2}(5 + 243) = 8825 - 4340 = 4485.$$

$$(ii) l = T_{60} = (8 + 59 \times 2) = 126.$$

$$\text{Required sum} = (S_{60} - S_{50}). \text{ Now, } T_{50} = (8 + 49 \times 2) = 106.$$

$$\therefore (S_{60} - S_{50}) = \frac{60}{2}(8 + 126) - \frac{50}{2}(8 + 106) = 4020 - 2850 = 1170.$$

$$36. T_4 + T_8 = 24 \Rightarrow (a + 3d) + (a + 7d) = 24 \Rightarrow a + 5d = 12. \quad \dots (i)$$

$$T_6 + T_{10} = 44 \Rightarrow (a + 5d) + (a + 9d) = 44 \Rightarrow a + 7d = 22. \quad \dots (ii)$$

Solve (i) and (ii), we get $a = -13$ and $d = 5$.

$$37. S_m = (4m^2 - m) \Rightarrow S_{m-1} = 4(m-1)^2 - (m-1) = (4m^2 - 9m + 5).$$

$$\therefore T_m = (S_m - S_{m-1}) = (4m^2 - m) - (4m^2 - 9m + 5) = (8m - 5)$$

$$\Rightarrow T_n = (8n - 5) \Rightarrow 8n - 5 = 107 \Rightarrow n = 14.$$

$$\therefore T_{21} = (8 \times 21 - 5) = 163.$$

$$38. S_q = (63q - 3q^2) \Rightarrow S_{q-1} = 63(q-1) - 3(q-1)^2 = (69q - 3q^2 - 66).$$

$$\therefore T_q = (S_q - S_{q-1}) = -6q + 66 \Rightarrow T_p = -6p + 66.$$

$$\text{Now, } -6p + 66 = -60 \Rightarrow 6p = 126 \Rightarrow p = 21.$$

$$\text{Also, } T_{11} = (-6 \times 11) + 66 = 0.$$

$$39. \text{Here } a = -12, d = 3 \text{ and } l = 21.$$

$$T_n = 21 \Rightarrow a + (n-1)d = 21 \Rightarrow -12 + (n-1) \times 3 = 21 \Rightarrow n = 12.$$

$$\text{Required sum} = \frac{12}{2}(-12 + 21) + 12 = 54 + 12 = 66.$$

$$40. \frac{n}{2}(a+l) = 1505 \Rightarrow \frac{14}{2}(10+l) = 1505 \Rightarrow l = 205.$$

$$T_{14} = 205 \Rightarrow a + 13d = 205 \Rightarrow 10 + 13d = 205 \Rightarrow d = 15.$$

$$T_{25} = (a + 24d) = (10 + 24 \times 15) = 370.$$

41. Let a be the first term of the given AP.

$$\text{Then, } 14 - a = 18 - 14 \Rightarrow 14 - a = 4 \Rightarrow a = 10.$$

$$\therefore a = 10, d = 4 \text{ and } n = 51.$$

Now, find S_n .

$$42. \text{ Required sum} = 2 \times [2 + 4 + 6 + 8 + \dots + 24] = 2 \times \frac{12}{2}(2 + 24) = 312.$$

Planting trees helps in reducing air pollution.

43. Total distance run to pick up the 1st, 2nd, 3rd, ..., 10th potato

$$= [(2 \times 5) + 2 \times (5 + 3) + 2 \times (5 + 3 \times 2) + \dots + 2 \times (5 + 3 \times 9)] \text{ m}$$

$$= 2 \times [5 + 8 + 11 + \dots + 32] \text{ m} = 2 \times \frac{10}{2}(5 + 32) \text{ m} = 370 \text{ m}.$$

44. Total distance covered

$$= [(2 \times 10) + 2 \times (10 + 5 \times 1) + 2 \times (10 + 5 \times 2) + \dots + 2 \times (10 + 5 \times 24)] \text{ m}$$

$$= 2 \times [10 + 15 + 20 + \dots + 130] \text{ m} = 2 \times \frac{25}{2}(10 + 130) \text{ m} = 3500 \text{ m}.$$

45. Let the values of these prizes be ₹ x , ₹ $(x - 20)$, ₹ $(x - 40)$, ..., and ₹ $(x - 20 \times 6)$ respectively.

This is an AP with $a = x$, $l = (x - 120)$ and $n = 7$.

$$\therefore \text{ sum} = \frac{n}{2}(a+l) = \frac{7}{2}(x + x - 120) = 7(x - 60).$$

$$\Rightarrow 7x - 420 = 700 \Rightarrow 7x = 1120 \Rightarrow x = 160.$$

46. Suppose he saved ₹ x , ₹ $(x + 100)$, ₹ $(x + 200)$, ..., ₹ $(x + 900)$ in 1st, 2nd, 3rd, ... and 10th month respectively. Then,

$$x + (x + 100) + (x + 200) + \dots + (x + 900) = 33000$$

$$\Rightarrow 10x + (100 + 200 + 300 + \dots + 900) = 33000$$

$$\Rightarrow 10x + \frac{9}{2}(100 + 900) = 33000 \quad \left[\because S_n = \frac{n}{2}(a + l) \right]$$

$$\Rightarrow 10x = 33000 - 4500 = 28500 \Rightarrow x = 2850.$$

47. Let the first instalment be ₹ x and let it be increased by ₹ d each month. Then,

$$x + (x + d) + (x + 2d) + \dots + (x + 29d) = \frac{2}{3} \times 36000$$

$$\Rightarrow 30x + (1 + 2 + 3 + \dots + 29)d = 24000$$

$$\Rightarrow 30x + \frac{29}{2}(1 + 29)d = 24000 \Rightarrow 2x + 29d = 1600. \quad \dots \text{ (i)}$$

Also, $x + (x + d) + (x + 2d) + \dots + (x + 39d) = 36000$ gives

$$40x + \frac{39}{2}(1 + 39)d = 36000 \Rightarrow 2x + 39d = 1800. \quad \dots \text{ (ii)}$$

On solving (i) and (ii), we get $d = 20$ and $x = 510$.

48. Total penalty = ₹ $(200 + 250 + 300 + \dots \text{ up to } 30 \text{ terms})$.

$$T_{30} = (200 + 29 \times 50) = 1650.$$

$$\therefore \text{required sum} = ₹ \left\{ \frac{30}{2} (200 + 1650) \right\} = ₹ 27750.$$

.....

EXERCISE 5D

Very-Short and Short-Answer Questions

1. The first three terms of an AP are respectively $(3y - 1)$, $(3y + 5)$ and $(5y + 1)$, find the value of y . [CBSE 2014]
2. If k , $(2k - 1)$ and $(2k + 1)$ are the three successive terms of an AP, find the value of k . [CBSE 2014]
3. If 18 , a , $(b - 3)$ are in AP, then find the value of $(2a - b)$. [CBSE 2014]
4. If the numbers a , 9 , b , 25 form an AP, find a and b . [CBSE 2014]
5. If the numbers $(2n - 1)$, $(3n + 2)$ and $(6n - 1)$ are in AP, find the value of n and the numbers. [CBSE 2013C]
6. How many three-digit natural numbers are divisible by 7? [CBSE 2013]
7. How many three-digit natural numbers are divisible by 9? [CBSE 2013]
8. If the sum of first m terms of an AP is $(2m^2 + 3m)$ then what is its second term? [CBSE 2011]
9. What is the sum of first n terms of the AP a , $3a$, $5a$, ... [CBSE 2012]
10. What is the 5th term from the end of the AP 2 , 7 , 12 , ..., 47 ? [CBSE 2011]
11. If a_n denotes the n th term of the AP 2 , 7 , 12 , 17 , ..., find the value of $(a_{30} - a_{20})$. [CBSE 2011]
12. The n th term of an AP is $(3n + 5)$. Find its common difference. [CBSE 2009C]
13. The n th term of an AP is $(7 - 4n)$. Find its common difference. [CBSE 2008]
14. Write the next term of the AP $\sqrt{8}$, $\sqrt{18}$, $\sqrt{32}$, ... [CBSE 2008]
15. Write the next term of the AP $\sqrt{2}$, $\sqrt{8}$, $\sqrt{18}$, ... [CBSE 2008C]
16. Which term of the AP 21 , 18 , 15 , ... is zero? [CBSE 2008C]
17. Find the sum of first n natural numbers.
18. Find the sum of first n even natural numbers.
19. The first term of an AP is p and its common difference is q . Find its 10th term. [CBSE 2008]
20. If $\frac{4}{5}$, a , 2 are in AP, find the value of a . [CBSE 2009]
21. If $(2p + 1)$, 13 , $(5p - 3)$ are in AP, find the value of p . [CBSE 2009]

22. If $(2p - 1), 7, 3p$ are in AP, find the value of p . [CBSE 2009]
23. If the sum of first p terms of an AP is $(ap^2 + bp)$, find its common difference. [CBSE 2010]
24. If the sum of first n terms is $(3n^2 + 5n)$, find its common difference.
25. Find an AP whose 4th term is 9 and the sum of its 6th and 13th terms is 40. [CBSE 2011]
26. What is the common difference of an AP in which $a_{27} - a_7 = 84$? [CBSE 2017]
27. If $1 + 4 + 7 + 10 + \dots + x = 287$, find the value of x . [CBSE 2017]

ANSWERS (EXERCISE 5D)

1. $y = 5$ 2. $k = 3$ 3. 15 4. $a = 1, b = 17$
5. $n = 3, (5, 11, 17)$ 6. 128 7. 100 8. 9 9. $n^2 a$ 10. 27
11. 50 12. 3 13. -4 14. $\sqrt{50}$ 15. $\sqrt{32}$ 16. 8th
17. $\frac{1}{2}n(n+1)$ 18. $n(n+1)$ 19. $(p+9q)$ 20. $a = \frac{7}{5}$ 21. $p = 4$
22. $p = 3$ 23. $d = 2a$ 24. $d = 6$ 25. 3, 5, 7, 9, 11, ...
26. $d = 4.2$ 27. $x = 40$

HINTS TO SOME SELECTED QUESTIONS

1. We have, $(3y + 5) - (3y - 1) = (5y + 1) - (3y + 5)$
 $\Rightarrow 2y - 4 = 6 \Rightarrow 2y = 10 \Rightarrow y = 5$.
2. We have, $(2k - 1) - k = (2k + 1) - (2k - 1)$
 $\Rightarrow k - 1 = 2 \Rightarrow k = 3$.
3. We have, $a - 18 = (b - 3 - a) \Rightarrow 2a - b = (18 - 3) = 15$.
4. We have, $9 - a = b - 9 = 25 - b$
 $\therefore 2b = 34 \Rightarrow b = 17$.
 Also, $9 - a = 17 - 9 = 8 \Rightarrow a = 9 - 8 = 1$.
5. We have, $(3n + 2) - (2n - 1) = (6n - 1) - (3n + 2)$
 $\therefore n + 3 = 3n - 3 \Rightarrow 2n = 6 \Rightarrow n = 3$.
 The numbers are $(2 \times 3 - 1), (3 \times 3 + 2)$ and $(6 \times 3 - 1)$, i.e., 5, 11, 17.
6. These numbers are 105, 112, 119, 126, ..., 994.
 Let $T_n = 994$. Then, $105 + (n - 1) \times 7 = 994 \Rightarrow n = 128$.
7. These numbers are 108, 117, 126, ..., 999.
 Let $T_n = 999$. Then, $108 + (n - 1) \times 9 = 999 \Rightarrow n = 100$.

$$8. S_m = (2m^2 + 3m) \Rightarrow S_1 = 2 + 3 = 5 \text{ and } S_2 = 2 \times 4 + 3 \times 2 = 14.$$

$$\therefore T_2 = S_2 - S_1 = 14 - 5 = 9.$$

$$9. \text{ Required sum} = \frac{n}{2} [2a + (n-1) \times 2a] = n(na) = n^2 a.$$

$$10. n\text{th term from the end} = l - (n-1)d.$$

$$\therefore \text{ 5th term from the end} = 47 - 4 \times 5 = 47 - 20 = 27.$$

$$11. (a_{30} - a_{20}) = (2 + 29 \times 5) - (2 + 19 \times 5) = 147 - 97 = 50.$$

$$12. T_n = (3n + 5) \Rightarrow T_1 = (3 \times 1 + 5) = 8, T_2 = (3 \times 2 + 5) = 11.$$

$$\therefore d = (T_2 - T_1) = 3.$$

$$13. T_n = (7 - 4n) \Rightarrow T_1 = (7 - 4 \times 1) = 3, T_2 = (7 - 4 \times 2) = -1.$$

$$\therefore d = (T_2 - T_1) = (-1 - 3) = -4.$$

$$14. \text{ Given terms are } \sqrt{4 \times 2}, \sqrt{9 \times 2}, \sqrt{16 \times 2}, \dots, \text{ i.e., } 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \dots$$

$$\text{So, the next term is } 5\sqrt{2} = \sqrt{5 \times 5 \times 2} = \sqrt{50}.$$

$$15. \text{ Given terms are } \sqrt{2}, \sqrt{4 \times 2}, \sqrt{9 \times 2}, \dots, \text{ i.e., } \sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \dots$$

$$\text{The next term is } 4\sqrt{2} = \sqrt{4 \times 4 \times 2} = \sqrt{32}.$$

$$16. \text{ Let } T_n = 0. \text{ Then, } 21 + (n-1) \times (-3) = 0 \Rightarrow n = 8. \text{ So, 8th term is zero.}$$

$$17. S_n = 1 + 2 + 3 + \dots + n = \frac{n}{2}(1+n). \quad \left[S_n = \frac{n}{2}(a+l) \right]$$

$$18. S_n = 2 + 4 + 6 + \dots + 2n = \frac{n}{2}(2+2n) = n(n+1). \quad \left[S_n = \frac{n}{2}(a+l) \right]$$

$$19. \text{ Here } a = p \text{ and } d = q.$$

$$\therefore T_{10} = a + 9d = (p + 9q).$$

$$20. \text{ We have, } a - \frac{4}{5} = 2 - a \Rightarrow 2a = \left(2 + \frac{4}{5}\right) = \frac{14}{5} \Rightarrow a = \frac{7}{5}.$$

$$21. \text{ We have, } 13 - (2p + 1) = (5p - 3) - 13.$$

$$\therefore (5p - 3) + (2p + 1) = 26 \Rightarrow 7p = 28 \Rightarrow p = 4.$$

$$22. \text{ We have, } 7 - (2p - 1) = 3p - 7.$$

$$\therefore (2p - 1) + 3p = 14 \Rightarrow 5p = 15 \Rightarrow p = 3.$$

$$23. T_1 = S_1 = a + b \quad [\text{putting } p = 1].$$

$$\text{Now, } S_2 = (a \times 4 + b \times 2) = (4a + 2b).$$

$$\therefore T_2 = S_2 - S_1 = (4a + 2b) - (a + b) = (3a + b).$$

$$\therefore d = T_2 - T_1 = (3a + b) - (a + b) = 2a.$$

$$24. S_n = (3n^2 + 5n)$$

$$\Rightarrow T_1 = S_1 = (3 \times 1 + 5 \times 1) = 8.$$

$$\text{Now, } S_2 = 3 \times 4 + 5 \times 2 = 22.$$

$$\therefore T_2 = S_2 - S_1 = 22 - 8 = 14.$$

$$\Rightarrow d = T_2 - T_1 = 14 - 8 = 6.$$

$$25. T_4 = 9 \Rightarrow a + 3d = 9. \quad \dots (i)$$

$$T_6 + T_{13} = 40 \Rightarrow (a + 5d) + (a + 12d) = 40 \Rightarrow 2a + 17d = 40. \quad \dots (ii)$$

From (i) and (ii), we get $a = 3$ and $d = 2$.

$\therefore$ AP is 3, 5, 7, 9, 11, ...

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \dots$ is [CBSE 2013]
 (a) p (b) $-p$ (c) -1 (d) 1
- The common difference of the AP $\frac{1}{3}, \frac{1-3b}{3}, \frac{1-6b}{3}, \dots$ is [CBSE 2013]
 (a) $\frac{1}{3}$ (b) $-\frac{1}{3}$ (c) b (d) $-b$
- The next term of the AP $\sqrt{7}, \sqrt{28}, \sqrt{63}, \dots$ is [CBSE 2014]
 (a) $\sqrt{70}$ (b) $\sqrt{84}$ (c) $\sqrt{98}$ (d) $\sqrt{112}$
- If 4, x_1 , x_2 , x_3 , 28 are in AP then $x_3 = ?$
 (a) 19 (b) 23
 (c) 22 (d) Cannot be determined
- If the n th term of an AP is $(2n + 1)$ then the sum of its first three terms is [CBSE 2012]
 (a) $6n + 3$ (b) 15 (c) 12 (d) 21
- The sum of first n terms of an AP is $(3n^2 + 6n)$. The common difference of the AP is [CBSE 2014]
 (a) 6 (b) 9 (c) 15 (d) -3
- The sum of first n terms of an AP is $(5n - n^2)$. The n th term of the AP is
 (a) $(5 - 2n)$ (b) $(6 - 2n)$ (c) $(2n - 5)$ (d) $(2n - 6)$
- The sum of first n terms of an AP is $(4n^2 + 2n)$. The n th term of this AP is [CBSE 2014]
 (a) $(6n - 2)$ (b) $(7n - 3)$ (c) $(8n - 2)$ (d) $(8n + 2)$
- The 7th term of an AP is -1 and its 16th term is 17. The n th term of the AP is [CBSE 2014]
 (a) $(3n + 8)$ (b) $(4n - 7)$ (c) $(15 - 2n)$ (d) $(2n - 15)$

10. The 5th term of an AP is -3 and its common difference is -4 . The sum of its first 10 terms is [CBSE 2011]
(a) 50 (b) -50 (c) 30 (d) -30
11. The 5th term of an AP is 20 and the sum of its 7th and 11th terms is 64. The common difference of the AP is [CBSE 2015]
(a) 4 (b) 5 (c) 3 (d) 2
12. The 13th term of an AP is 4 times its 3rd term. If its 5th term is 16 then the sum of its first ten terms is [CBSE 2015]
(a) 150 (b) 175 (c) 160 (d) 135
13. An AP 5, 12, 19, ... has 50 terms. Its last term is [CBSE 2015]
(a) 343 (b) 353 (c) 348 (d) 362
14. The sum of first 20 odd natural numbers is [CBSE 2012]
(a) 100 (b) 210 (c) 400 (d) 420
15. The sum of first 40 positive integers divisible by 6 is [CBSE 2014]
(a) 2460 (b) 3640 (c) 4920 (d) 4860
16. How many two-digit numbers are divisible by 3? [CBSE 2012]
(a) 25 (b) 30 (c) 32 (d) 36
17. How many three-digit numbers are divisible by 9? [CBSE 2013]
(a) 86 (b) 90 (c) 96 (d) 100
18. What is the common difference of an AP in which $a_{18} - a_{14} = 32$?
(a) 8 (b) -8 (c) 4 (d) -4
19. If a_n denotes the n th term of the AP 3, 8, 13, 18, ... then what is the value of $(a_{30} - a_{20})$?
(a) 40 (b) 36 (c) 50 (d) 56
20. Which term of the AP 72, 63, 54, ... is 0?
(a) 8th (b) 9th (c) 10th (d) 11th
21. Which term of the AP 25, 20, 15, ... is the first negative term?
(a) 10th (b) 9th (c) 8th (d) 7th
22. Which term of the AP 21, 42, 63, 84, ... is 210?
(a) 9th (b) 10th (c) 11th (d) 12th
23. What is 20th term from the end of the AP 3, 8, 13, ..., 253?
(a) 163 (b) 158 (c) 153 (d) 148
24. $(5 + 13 + 21 + \dots + 181) = ?$
(a) 2476 (b) 2337 (c) 2219 (d) 2139

25. The sum of first 16 terms of the AP 10, 6, 2, ... is
 (a) 320 (b) -320 (c) -352 (d) -400
26. How many terms of the AP 3, 7, 11, 15, ... will make the sum 406?
 (a) 10 (b) 12 (c) 14 (d) 20
27. The 2nd term of an AP is 13 and its 5th term is 25. What is its 17th term?
 (a) 69 (b) 73 (c) 77 (d) 81
28. The 17th term of an AP exceeds its 10th term by 21. The common difference of the AP is
 (a) 3 (b) 2 (c) -3 (d) -2
29. The 8th term of an AP is 17 and its 14th term is 29. The common difference of the AP is
 (a) 3 (b) 2 (c) 5 (d) -2
30. The 7th term of an AP is 4 and its common difference is -4. What is its first term?
 (a) 16 (b) 20 (c) 24 (d) 28

ANSWERS (MCQ)

1. (c) 2. (d) 3. (d) 4. (c) 5. (b) 6. (a) 7. (b) 8. (c) 9. (d)
 10. (b) 11. (c) 12. (b) 13. (c) 14. (c) 15. (c) 16. (b) 17. (d) 18. (a)
 19. (c) 20. (b) 21. (d) 22. (b) 23. (b) 24. (d) 25. (b) 26. (c) 27. (b)
 28. (a) 29. (b) 30. (d)

HINTS TO SOME SELECTED QUESTIONS

1. $d = \left\{ \frac{1-p}{p} - \frac{1}{p} \right\} = \left(\frac{1-p-1}{p} \right) = \frac{-p}{p} = -1.$
2. $d = \left\{ \frac{1-3b}{3} - \frac{1}{3} \right\} = \left(\frac{1-3b-1}{3} \right) = \frac{-3b}{3} = -b.$
3. Given terms are $\sqrt{7}, \sqrt{4 \times 7}, \sqrt{9 \times 7}, \dots$, i.e., $\sqrt{7}, 2\sqrt{7}, 3\sqrt{7}, \dots$.
 So, the next term is $4\sqrt{7} = \sqrt{4 \times 4 \times 7} = \sqrt{112}.$
4. Given AP is 4, $x_1, x_2, x_3, 28$. Clearly, $a = 4$ and $T_5 = 28$.
 Now, $T_5 = 28 \Rightarrow a + 4d = 28 \Rightarrow 4 + 4d = 28 \Rightarrow 4d = 24 \Rightarrow d = 6.$
 $\therefore x_3 = T_4 = (a + 3d) = (4 + 3 \times 6) = 22.$
5. $(T_1 + T_2 + T_3) = (2 \times 1 + 1) + (2 \times 2 + 1) + (2 \times 3 + 1) = 3 + 5 + 7 = 15.$

6. Given $S_n = (3n^2 + 6n)$.

$$\therefore T_1 = S_1 = (3 \times 1^2 + 6 \times 1) = 9, S_2 = (3 \times 2^2 + 6 \times 2) = 24.$$

$$\therefore T_2 = (S_2 - S_1) = (24 - 9) = 15.$$

$$\text{Hence, } d = (T_2 - T_1) = 15 - 9 = 6.$$

7. $T_n = (S_n - S_{n-1}) = (5n - n^2) - \{5(n-1) - (n-1)^2\} = (5n - n^2) - (7n - n^2 - 6) = (6 - 2n).$

$$\begin{aligned} 8. T_n &= (S_n - S_{n-1}) = (4n^2 + 2n) - \{4(n-1)^2 + 2(n-1)\} \\ &= (4n^2 + 2n) - (4n^2 - 6n + 2) = (8n - 2). \end{aligned}$$

9. $T_7 = -1 \Rightarrow a + 6d = -1. \quad \dots (i)$

$T_{16} = 17 \Rightarrow a + 15d = 17. \quad \dots (ii)$

On solving (i) and (ii), we get $a = -13$ and $d = 2$.

$$\therefore T_n = a + (n-1)d = -13 + (n-1) \times 2 = (2n - 15).$$

10. $a + 4d = -3$ and $d = -4$. So, $a = 13$.

$$\therefore S_{10} = \frac{10}{2}[2a + 9d] = 5[2 \times 13 + 9 \times (-4)] = -50.$$

11. $T_5 = 20 \Rightarrow a + 4d = 20. \quad \dots (i)$

$(T_7 + T_{11}) = 64 \Rightarrow (a + 6d) + (a + 10d) = 64 \Rightarrow a + 8d = 32. \quad \dots (ii)$

On solving (i) and (ii), we get $d = 3$.

12. $T_{13} = 4 \times T_3 \Rightarrow a + 12d = 4(a + 2d) \Rightarrow 3a - 4d = 0. \quad \dots (i)$

$T_5 = 16 \Rightarrow a + 4d = 16. \quad \dots (ii)$

On solving (i) and (ii), we get $a = 4$ and $d = 3$.

$$S_{10} = \frac{10}{2}(2a + 9d) = 5(2 \times 4 + 9 \times 3) = 175.$$

13. 50th term $= a + (n-1)d = (5 + 49 \times 7) = 348.$

14. $S_{20} = 1 + 3 + 5 + 7 + \dots$ up to 20 terms.

Here $a = 1, d = 2$. So, $T_{20} = (a + 19d) = (1 + 19 \times 2) = 38 = l.$

$$\therefore S_{20} = \frac{n}{2}(a + l) = \frac{20}{2}(1 + 39) = 400.$$

15. Required sum $= 6 + 12 + 18 + \dots + 240 = \frac{40}{2}(6 + 240) = 4920.$

16. Two-digit numbers divisible by 3 are 12, 15, 18, ..., 99.

Let $T_n = 99$. Then, $12 + (n-1) \times 3 = 99 \Rightarrow (n-1) \times 3 = 87 \Rightarrow n = 30.$

17. Three-digit numbers divisible by 9 are 108, 117, 126, ..., 999.

Let $T_n = 999$. Then, $108 + (n-1) \times 9 = 999.$

$$\therefore (n-1) \times 9 = 891 \Rightarrow (n-1) = 99 \Rightarrow n = 100.$$

18. $a_{18} - a_{14} = 32 \Rightarrow (a + 17d) - (a + 13d) = 32 \Rightarrow 4d = 32 \Rightarrow d = 8.$

19. Here $a = 3$ and $d = 5$.

$$\therefore (a_{30} - a_{20}) = (3 + 29 \times 5) - (3 + 19 \times 5) = 148 - 98 = 50.$$

20. Let the n th term of 72, 63, 54, ... be 0. Then,

$$T_n = 0 \Rightarrow 72 + (n-1) \times (-9) = 0 \Rightarrow 9n = 81 \Rightarrow n = 9.$$

21. Let $T_n < 0$. Then, $25 + (n-1) \times (-5) < 0$.

$$\therefore 30 < 5n \Rightarrow 5n > 30 \Rightarrow n > 6.$$

So, the required term is 7th.

22. Let $T_n = 210$. Then, $21 + (n-1) \times 21 = 210$

$$\therefore (n-1) \times 21 = 189 \Rightarrow n-1 = 9 \Rightarrow n = 10.$$

23. 20th term from the end $= (253 - 19 \times 5) = (253 - 95) = 158$. $[l - (n-1)d]$

24. Let the number of terms be n . Then, $T_n = 181$.

$$\therefore 5 + (n-1) \times 8 = 181 \Rightarrow (n-1) \times 8 = 176 \Rightarrow (n-1) = 22 \Rightarrow n = 23.$$

$$\therefore \text{sum} = \frac{n}{2}(a+l) = \frac{23}{2}(5+181) = 23 \times 93 = 2139.$$

25. $l = T_{16} = 10 + 15 \times (-4) = -50$.

$$\therefore \text{sum} = \frac{n}{2}(a+l) = \frac{16}{2}(10-50) = 8 \times (-40) = -320.$$

26. Let $S_n = 406$. Then, $\frac{n}{2}[2 \times 3 + (n-1) \times 4] = 406$

$$\Rightarrow \frac{n}{2}(6+4n-4) = 406 \Rightarrow \frac{n}{2}(4n+2) = 406$$

$$\Rightarrow n(2n+1) = 406 \Rightarrow 2n^2 + n - 406 = 0.$$

$$\therefore n = \frac{-1 \pm \sqrt{1+3248}}{4} = \frac{-1 + \sqrt{3249}}{4} \quad [\text{neglecting negative value}]$$

$$\Rightarrow n = \frac{-1+57}{4} = \frac{56}{4} = 14.$$

27. $a+d = 13$... (i) and $a+4d = 25$

... (ii).

From (i) and (ii), we get $a = 9$ and $d = 4$.

$$\therefore T_{17} = (a+16d) = (9+16 \times 4) = 73.$$

28. $(T_{17} - T_{10}) = 21 \Rightarrow (a+16d) - (a+9d) = 21 \Rightarrow 7d = 21 \Rightarrow d = 3$.

29. $(T_{14} - T_8) = (29-17) = 12 \Rightarrow (a+13d) - (a+7d) = 12 \Rightarrow 6d = 12 \Rightarrow d = 2$.

30. $a+6d = 4 \Rightarrow a+6 \times (-4) = 4 \Rightarrow a = 28$.



DISTANCE BETWEEN TWO POINTS

THEOREM 1 *The distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by the formula*

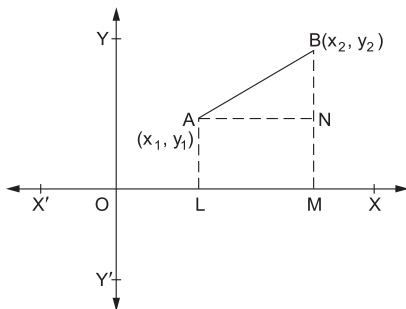
$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

PROOF Let $X'OX$ and YOY' be the x -axis and y -axis respectively.

Then, O is the origin.

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the given points.

Draw $AL \perp OX$, $BM \perp OX$ and $AN \perp BM$.



Now, $OL = x_1$, $OM = x_2$, $AL = y_1$ and $BM = y_2$.

$$\therefore AN = LM = (OM - OL) = (x_2 - x_1),$$

$$BN = (BM - NM) = (BM - AL) = (y_2 - y_1) \quad [\because NM = AL].$$

From right $\triangle ANB$, by Pythagoras' theorem, we have

$$AB^2 = (AN^2 + BN^2)$$

$$\Rightarrow AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\Rightarrow AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Hence, the distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

COROLLARY *The distance of the point $P(x, y)$ from the origin $O(0, 0)$ is given by*

$$OP = \sqrt{(x - 0)^2 + (y - 0)^2}, \text{ i.e., } OP = \sqrt{x^2 + y^2}.$$

SUMMARY

1. The distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2. The distance of the point $P(x, y)$ from the origin $O(0, 0)$ is given by

$$OP = \sqrt{x^2 + y^2}.$$

SOLVED EXAMPLES

EXAMPLE 1 Find the distance between the points $A(7, 13)$ and $B(10, 9)$.

SOLUTION The given points are $A(7, 13)$ and $B(10, 9)$.

Then, $x_1 = 7$, $y_1 = 13$ and $x_2 = 10$, $y_2 = 9$.

$$\begin{aligned}\therefore AB &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(10 - 7)^2 + (9 - 13)^2} = \sqrt{3^2 + (-4)^2} \\ &= \sqrt{9 + 16} = \sqrt{25} = 5 \text{ units.}\end{aligned}$$

EXAMPLE 2 Find the distance between the points $P(-4, 7)$ and $Q(2, -5)$.

SOLUTION The given points are $P(-4, 7)$ and $Q(2, -5)$.

Then, $x_1 = -4$, $y_1 = 7$ and $x_2 = 2$, $y_2 = -5$.

$$\begin{aligned}\therefore PQ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(2 - (-4))^2 + (-5 - 7)^2} = \sqrt{6^2 + (-12)^2} \\ &= \sqrt{36 + 144} = \sqrt{180} = \sqrt{36 \times 5} = 6\sqrt{5} \text{ units.}\end{aligned}$$

EXAMPLE 3 Find the distance of the point $P(6, -6)$ from the origin.

SOLUTION Let $P(6, -6)$ be the given point and $O(0, 0)$ be the origin.

$$\begin{aligned}\text{Then, } OP &= \sqrt{(6 - 0)^2 + (-6 - 0)^2} = \sqrt{6^2 + (-6)^2} \\ &= \sqrt{36 + 36} = \sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2} \text{ units.}\end{aligned}$$

EXAMPLE 4 Find the values of y for which the distance between the points $A(3, -1)$ and $B(11, y)$ is 10 units. [CBSE 2011]

SOLUTION We have

$$\begin{aligned}AB &= 10 \Rightarrow AB^2 = 100 \\ \Rightarrow (11 - 3)^2 + (y + 1)^2 &= 100 \\ \Rightarrow 8^2 + (y + 1)^2 &= 100 \\ \Rightarrow (y + 1)^2 &= 100 - 64 = 36 = 6^2\end{aligned}$$

$$\Rightarrow y + 1 = \pm 6$$

$$\Rightarrow y + 1 = 6 \text{ or } y + 1 = -6$$

$$\Rightarrow y = 5 \text{ or } y = -7.$$

Hence, the required values of y are 5 and -7 .

EXAMPLE 5 If the point $P(k-1, 2)$ is equidistant from the points $A(3, k)$ and $B(k, 5)$, find the values of k . [CBSE 2014]

SOLUTION It is being given that $P(k-1, 2)$ is equidistant from the points $A(3, k)$ and $B(k, 5)$. So, we have

$$PA = PB \Rightarrow PA^2 = PB^2$$

$$\Rightarrow (k-1-3)^2 + (2-k)^2 = (k-1-k)^2 + (2-5)^2$$

$$\Rightarrow (k-4)^2 + (2-k)^2 = (-1)^2 + (-3)^2$$

$$\Rightarrow 2k^2 - 12k + 20 = 1 + 9 \Rightarrow 2k^2 - 12k + 10 = 0$$

$$\Rightarrow k^2 - 6k + 5 = 0 \Rightarrow k^2 - 5k - k + 5 = 0$$

$$\Rightarrow k(k-5) - (k-5) = 0 \Rightarrow (k-5)(k-1) = 0$$

$$\Rightarrow k = 1 \text{ or } k = 5.$$

Hence, the required values of k are 1 and 5.

EXAMPLE 6 Find the relation between x and y such that the point $P(x, y)$ is equidistant from the points $A(1, 4)$ and $B(-1, 2)$. [CBSE 2012]

SOLUTION We have

$$PA = PB \Rightarrow PA^2 = PB^2$$

$$\Rightarrow (x-1)^2 + (y-4)^2 = (x+1)^2 + (y-2)^2$$

$$\Rightarrow x^2 + y^2 - 2x - 8y + 17 = x^2 + y^2 + 2x - 4y + 5$$

$$\Rightarrow 4x + 4y - 12 = 0 \Rightarrow x + y = 3 \Rightarrow y = 3 - x.$$

Hence, $y = 3 - x$ is the desired relation between x and y .

TWO IMPORTANT NOTES

(i) Any point on the x -axis is of the form $(x, 0)$.

(ii) Any point on the y -axis is of the form $(0, y)$.

EXAMPLE 7 Find those points on x -axis, each of which is at a distance of 5 units from the point $A(5, -3)$. [CBSE 2011]

SOLUTION Let the required point on the x -axis be $P(x, 0)$. Then,

$$PA = 5 \Rightarrow PA^2 = 25$$

$$\Rightarrow (x-5)^2 + (0+3)^2 = 25$$

$$\Rightarrow x^2 - 10x + 25 + 9 = 25 \Rightarrow x^2 - 10x + 9 = 0$$

$$\Rightarrow x^2 - x - 9x + 9 = 0 \Rightarrow x(x-1) - 9(x-1) = 0$$

$$\Rightarrow (x-1)(x-9) = 0 \Rightarrow x-1 = 0 \text{ or } x-9 = 0$$

$$\Rightarrow x = 1 \text{ or } x = 9.$$

Hence, the required points on the x -axis are $B(1, 0)$ and $C(9, 0)$.

EXAMPLE 8 Find those points on the y -axis, each of which is at a distance of 13 units from the point $A(-5, 7)$.

SOLUTION Let the required point on the y -axis be $P(0, y)$. Then,

$$PA = 13 \Rightarrow PA^2 = 169$$

$$\Rightarrow (0+5)^2 + (y-7)^2 = 169$$

$$\Rightarrow y^2 - 14y + 74 = 169 \Rightarrow y^2 - 14y - 95 = 0$$

$$\Rightarrow y^2 - 19y + 5y - 95 = 0 \Rightarrow y(y-19) + 5(y-19) = 0$$

$$\Rightarrow (y-19)(y+5) = 0 \Rightarrow y-19 = 0 \text{ or } y+5 = 0$$

$$\Rightarrow y = 19 \text{ or } y = -5.$$

Hence, the required points on the y -axis are $B(0, 19)$ and $C(0, -5)$.

EXAMPLE 9 Find the point on x -axis which is equidistant from the points $(5, -2)$ and $(-3, 2)$. [CBSE 2009]

SOLUTION Let the given points be $A(5, -2)$ and $B(-3, 2)$ and let the required point be $P(x, 0)$. Then,

$$PA = PB \Rightarrow PA^2 = PB^2$$

$$\Rightarrow (x-5)^2 + (0+2)^2 = (x+3)^2 + (0-2)^2$$

$$\Rightarrow (x-5)^2 + 4 = (x+3)^2 + 4$$

$$\Rightarrow (x-5)^2 - (x+3)^2 = 0$$

$$\Rightarrow (x^2 - 10x + 25) - (x^2 + 6x + 9) = 0$$

$$\Rightarrow -16x + 16 = 0$$

$$\Rightarrow 16x = 16 \Rightarrow x = 1.$$

Hence, the required point is $P(1, 0)$.

EXAMPLE 10 Find the point on y -axis which is equidistant from the points $(-5, 2)$ and $(9, -2)$. [CBSE 2009C]

SOLUTION Let the given points be $A(-5, 2)$ and $B(9, -2)$, and let the required point be $P(0, y)$. Then,

$$PA = PB \Rightarrow PA^2 = PB^2$$

$$\Rightarrow (0+5)^2 + (y-2)^2 = (0-9)^2 + (y+2)^2$$

$$\Rightarrow 5^2 + (y-2)^2 = (-9)^2 + (y+2)^2$$

$$\Rightarrow 25 + (y-2)^2 = 81 + (y+2)^2$$

$$\Rightarrow (y+2)^2 - (y-2)^2 = 25 - 81$$

$$\Rightarrow 8y = -56 \Rightarrow y = -7.$$

Hence, the required point is $P(0, -7)$.

EXAMPLE 11 Find the coordinates of the point equidistant from three given points $A(5, 1)$, $B(-3, -7)$ and $C(7, -1)$. [CBSE 2006]

SOLUTION Let the required point be $P(x, y)$. Then,

$$PA = PB = PC \Rightarrow PA^2 = PB^2 = PC^2$$

$$\Rightarrow PA^2 = PB^2 \text{ and } PB^2 = PC^2$$

$$\Rightarrow \begin{cases} (x-5)^2 + (y-1)^2 = (x+3)^2 + (y+7)^2, \\ (x+3)^2 + (y+7)^2 = (x-7)^2 + (y+1)^2 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + y^2 - 10x - 2y + 26 = x^2 + y^2 + 6x + 14y + 58, \\ x^2 + y^2 + 6x + 14y + 58 = x^2 + y^2 - 14x + 2y + 50. \end{cases}$$

$$\text{Now, } x^2 + y^2 - 10x - 2y + 26 = x^2 + y^2 + 6x + 14y + 58$$

$$\Rightarrow 16x + 16y = -32 \Rightarrow x + y = -2. \quad \dots (i)$$

$$\text{And, } x^2 + y^2 + 6x + 14y + 58 = x^2 + y^2 - 14x + 2y + 50$$

$$\Rightarrow 20x + 12y = -8 \Rightarrow 5x + 3y = -2. \quad \dots (ii)$$

Multiplying (i) by 5 and subtracting (ii) from the result, we get

$$2y = -8 \Rightarrow y = -4.$$

Putting $y = -4$ in (i), we get $x = 2$.

$$\therefore x = 2 \text{ and } y = -4.$$

Hence, the required point is $P(2, -4)$.

EXAMPLE 12 Points $A(-1, y)$ and $B(5, 7)$ lie on a circle with centre $O(2, -3y)$. Find the values of y . Hence, find the radius of the circle. [CBSE 2014]

SOLUTION Since the radii of a circle are equal, we have

$$OA = OB \Rightarrow OA^2 = OB^2$$

$$\Rightarrow (2+1)^2 + (-3y-y)^2 = (2-5)^2 + (-3y-7)^2$$

$$\Rightarrow 3^2 + (-4y)^2 = (-3)^2 + (3y+7)^2$$

$$\Rightarrow 9 + 16y^2 = 9 + 9y^2 + 49 + 42y$$

$$\Rightarrow 7y^2 - 42y - 49 = 0$$

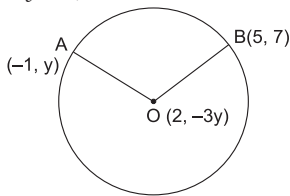
$$\Rightarrow y^2 - 6y - 7 = 0$$

$$\Rightarrow y^2 - 7y + y - 7 = 0$$

$$\Rightarrow y(y-7) + (y-7) = 0 \Rightarrow (y-7)(y+1) = 0$$

$$\Rightarrow y-7 = 0 \text{ or } y+1 = 0 \Rightarrow y = 7 \text{ or } y = -1.$$

$\therefore$ centre is either $(2, 3)$ or $(2, -21)$.



CASE I When centre is $(2, 3)$

$$\begin{aligned}\text{In this case, radius} &= \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{3^2 + 4^2} \\ &= \sqrt{9+16} = \sqrt{25} = 5 \text{ units.}\end{aligned}$$

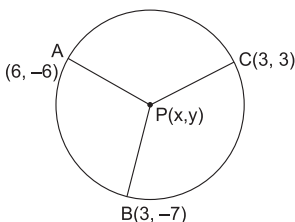
CASE II When centre is $(2, -21)$

$$\begin{aligned}\text{In this case, radius} &= \sqrt{(5-2)^2 + \{7 - (-21)\}^2} \\ &= \sqrt{3^2 + (28)^2} = \sqrt{9+784} \\ &= \sqrt{793} = 28.1 \text{ units.}\end{aligned}$$

EXAMPLE 13 Find the centre of a circle passing through the points $(6, -6)$, $(3, -7)$ and $(3, 3)$.

SOLUTION Let $P(x, y)$ be the centre of a circle passing through the points $A(6, -6)$, $B(3, -7)$ and $C(3, 3)$. Then,

$$PA^2 = PB^2 \text{ and } PB^2 = PC^2$$



$$\begin{aligned}\Rightarrow & \begin{cases} (x-6)^2 + (y+6)^2 = (x-3)^2 + (y+7)^2, \\ (x-3)^2 + (y+7)^2 = (x-3)^2 + (y-3)^2 \end{cases} \\ \Rightarrow & \begin{cases} x^2 + y^2 - 12x + 12y + 72 = x^2 + y^2 - 6x + 14y + 58, \\ x^2 + y^2 - 6x + 14y + 58 = x^2 + y^2 - 6x - 6y + 18 \end{cases} \\ \Rightarrow & 6x + 2y = 14 \text{ and } 20y = -40 \\ \Rightarrow & 3x + y = 7 \text{ and } y = -2 \\ \Rightarrow & x = 3 \text{ and } y = -2.\end{aligned}$$

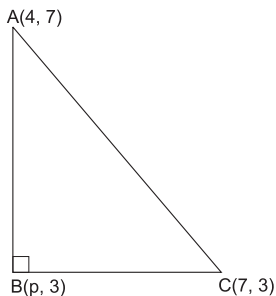
Hence, the centre of the circle is $P(3, -2)$.

EXAMPLE 14 The points $A(4, 7)$, $B(p, 3)$ and $C(7, 3)$ are the vertices of a right triangle, right angled at B . Find the value of p . [CBSE 2015]

SOLUTION Since $\triangle ABC$ is a right angled at B , we have

$$\begin{aligned}AB^2 + BC^2 &= AC^2 \\ \Rightarrow & \{(p-4)^2 + (3-7)^2\} + \{(7-p)^2 + (3-3)^2\} = (7-4)^2 + (3-7)^2 \\ \Rightarrow & (p-4)^2 + (-4)^2 + (7-p)^2 + 0 = 3^2 + (-4)^2\end{aligned}$$

$$\begin{aligned}
 &\Rightarrow 2p^2 - 22p + (16 + 49 + 16) = 9 + 16 \\
 &\Rightarrow 2p^2 - 22p + 56 = 0 \\
 &\Rightarrow p^2 - 11p + 28 = 0 \\
 &\Rightarrow p^2 - 7p - 4p + 28 = 0 \\
 &\Rightarrow p(p-7) - 4(p-7) = 0 \\
 &\Rightarrow (p-7)(p-4) = 0 \\
 &\Rightarrow p-7 = 0 \text{ or } p-4 = 0 \\
 &\Rightarrow p = 7 \text{ or } p = 4.
 \end{aligned}$$



When $p = 7$ then the points B and C coincide and so no triangle is formed in this case.

So, $p \neq 7$. Hence, $p = 4$.

EXAMPLE 15 Show that the points (a, a) , $(-a, -a)$ and $(-\sqrt{3}a, \sqrt{3}a)$ are the vertices of an equilateral triangle. Find its area. [CBSE 2015]

SOLUTION Let the given points be $A(a, a)$, $B(-a, -a)$ and $C(-\sqrt{3}a, \sqrt{3}a)$. Then, $AB^2 = (-a - a)^2 + (-a - a)^2 = (-2a)^2 + (-2a)^2 = 8a^2$,

$$\begin{aligned}
 BC^2 &= (-\sqrt{3}a + a)^2 + (\sqrt{3}a + a)^2 \\
 &= (a - \sqrt{3}a)^2 + (a + \sqrt{3}a)^2 = 2(a^2 + 3a^2) = 8a^2
 \end{aligned}$$

$$\begin{aligned}
 \text{and } AC^2 &= (-\sqrt{3}a - a)^2 + (\sqrt{3}a - a)^2 = (\sqrt{3}a + a)^2 + (\sqrt{3}a - a)^2 \\
 &= 2(3a^2 + a^2) = 8a^2.
 \end{aligned}$$

$$\therefore AB = BC = AC = \sqrt{8a^2} = 2\sqrt{2}a.$$

Hence, $\triangle ABC$ is equilateral and each of its sides is $2\sqrt{2}a$.

Area of equilateral $\triangle ABC$

$$\begin{aligned}
 &= \left\{ \frac{\sqrt{3}}{4} \times (\text{side})^2 \right\} \text{ sq units} = \left\{ \frac{\sqrt{3}}{4} \times (2\sqrt{2}a)^2 \right\} \text{ sq units} \\
 &= \left\{ \frac{\sqrt{3}}{4} \times 8a^2 \right\} \text{ sq units} = 2\sqrt{3}a^2 \text{ sq units.}
 \end{aligned}$$

EXAMPLE 16 Prove that the points $A(-3, 0)$, $B(1, -3)$ and $C(4, 1)$ are the vertices of an isosceles right-angled triangle. Find the area of this triangle.

SOLUTION Let $A(-3, 0)$, $B(1, -3)$ and $C(4, 1)$ be the given points. Then,

$$AB = \sqrt{[1 - (-3)]^2 + (-3 - 0)^2} = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5 \text{ units,}$$

$$BC = \sqrt{(4 - 1)^2 + [1 - (-3)]^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \text{ units}$$

$$\text{and } AC = \sqrt{[4 - (-3)]^2 + (1 - 0)^2} = \sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2} \text{ units.}$$

Thus, $AB = BC = 5$ units.

$\therefore \triangle ABC$ is isosceles.

Also, $(AB^2 + BC^2) = (5^2 + 5^2) = 50$ and

$$AC^2 = (5\sqrt{2})^2 = 50.$$

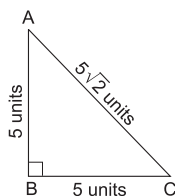
Thus, $AB^2 + BC^2 = AC^2$.

This shows that $\triangle ABC$ is right-angled at B .

In $\triangle ABC$, we have

base = $BC = 5$ units and height = $AB = 5$ units.

$$\begin{aligned}\therefore \text{ area of } \triangle ABC &= \left(\frac{1}{2} \times \text{base} \times \text{height} \right) = \left(\frac{1}{2} \times 5 \times 5 \right) \text{ sq units} \\ &= 12.5 \text{ sq units.}\end{aligned}$$



PROPERTIES OF VARIOUS TYPES OF QUADRILATERALS

A quadrilateral is a

- (i) *rectangle* if its opposite sides are equal and the diagonals are equal.
- (ii) *square* if all its sides are equal and the diagonals are equal.
- (iii) *parallelogram* if its opposite sides are equal.
- (iv) *parallelogram but not a rectangle* if its opposite sides are equal and the diagonals are not equal.
- (v) *rhombus but not a square* if all its sides are equal and the diagonals are not equal.

EXAMPLE 17 Show that the points $(1, 1)$, $(-1, 5)$, $(7, 9)$ and $(9, 5)$ taken in that order are the vertices of a rectangle. Also, find the area of the rectangle. [CBSE 2009C]

SOLUTION Let $A(1, 1)$, $B(-1, 5)$, $C(7, 9)$ and $D(9, 5)$ be the vertices of quad. $ABCD$. Then,

$$\begin{aligned}AB^2 &= (-1 - 1)^2 + (5 - 1)^2 \\ &= (-2)^2 + 4^2 = 4 + 16 = 20.\end{aligned}$$

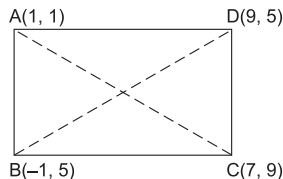
$$\begin{aligned}\therefore AB &= \sqrt{20} \text{ units} = \sqrt{4 \times 5} \text{ units} \\ &= 2\sqrt{5} \text{ units.}\end{aligned}$$

$$\begin{aligned}BC^2 &= (7 + 1)^2 + (9 - 5)^2 \\ &= 8^2 + 4^2 = 64 + 16 = 80.\end{aligned}$$

$$\therefore BC = \sqrt{80} \text{ units} = \sqrt{16 \times 5} \text{ units} = 4\sqrt{5} \text{ units.}$$

$$\begin{aligned}CD^2 &= 9 - 7^2 + 5 - 9^2 \\ &= 2^2 + (-4)^2 = 4 + 16 = 20.\end{aligned}$$

$$\therefore CD = \sqrt{20} \text{ units} = \sqrt{4 \times 5} \text{ units} = 2\sqrt{5} \text{ units.}$$



$$\begin{aligned}AD^2 &= (9-1)^2 + (5-1)^2 \\&= 8^2 + 4^2 = 64 + 16 = 80.\end{aligned}$$

$$\therefore AD = \sqrt{80} \text{ units} = \sqrt{16 \times 5} \text{ units} = 4\sqrt{5} \text{ units}.$$

Thus, $AB = CD = 2\sqrt{5}$ units and $BC = AD = 4\sqrt{5}$ units.

$$\begin{aligned}\text{Also, } AC^2 &= (7-1)^2 + (9-1)^2 \\&= 6^2 + 8^2 = 36 + 64 = 100.\end{aligned}$$

$$\therefore AC = \sqrt{100} \text{ units} = 10 \text{ units}.$$

$$\text{And, } BD^2 = (9+1)^2 + (5-5)^2 = 10^2 + 0^2 = 100.$$

$$\therefore BD = \sqrt{100} \text{ units} = 10 \text{ units}.$$

$$\therefore \text{diagonal } AC = \text{diagonal } BD.$$

Thus, $ABCD$ is a quadrilateral whose opposite sides are equal and the diagonals are equal.

Hence, quad. $ABCD$ is a rectangle.

$$\begin{aligned}\text{Area of rectangle } ABCD &= \text{length} \times \text{breadth} \\&= 4\sqrt{5} \times 2\sqrt{5} \text{ sq units} \\&= 40 \text{ sq units}.\end{aligned}$$

EXAMPLE 18 Show that the points $A(3, 5)$, $B(6, 0)$, $C(1, -3)$ and $D(-2, 2)$ are the vertices of a square $ABCD$. [CBSE 2012]

SOLUTION Let $A(3, 5)$, $B(6, 0)$, $C(1, -3)$ and $D(-2, 2)$ be the angular points of a quadrilateral $ABCD$. Join AC and BD .

$$\text{Now, } AB = \sqrt{(6-3)^2 + (0-5)^2}$$

$$= \sqrt{3^2 + (-5)^2}$$

$$= \sqrt{9 + 25} = \sqrt{34} \text{ units,}$$

$$BC = \sqrt{(1-6)^2 + (-3-0)^2} = \sqrt{(-5)^2 + (-3)^2}$$

$$= \sqrt{25 + 9} = \sqrt{34} \text{ units,}$$

$$CD = \sqrt{(-2-1)^2 + (2+3)^2} = \sqrt{(-3)^2 + 5^2}$$

$$= \sqrt{9 + 25} = \sqrt{34} \text{ units,}$$

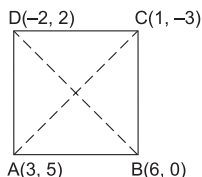
$$\text{and } DA = \sqrt{(3+2)^2 + (5-2)^2} = \sqrt{5^2 + 3^2}$$

$$= \sqrt{25 + 9} = \sqrt{34} \text{ units.}$$

Thus, $AB = BC = CD = DA$.

$$\text{Diag. } AC = \sqrt{(1-3)^2 + (-3-5)^2} = \sqrt{(-2)^2 + (-8)^2}$$

$$= \sqrt{4 + 64} = \sqrt{68} = 2\sqrt{17} \text{ units.}$$



$$\begin{aligned}
 \text{Diag. } BD &= \sqrt{(-2-6)^2 + (2-0)^2} \\
 &= \sqrt{(-8)^2 + 2^2} = \sqrt{64+4} \\
 &= \sqrt{68} = 2\sqrt{17} \text{ units.}
 \end{aligned}$$

$\therefore$ diag. $AC = \text{diag. } BD$.

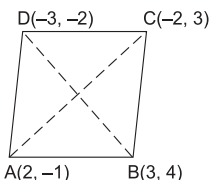
Thus, $ABCD$ is a quadrilateral in which all sides are equal and the diagonals are equal.

Hence, quad. $ABCD$ is a square.

EXAMPLE 19 If $A(2, -1)$, $B(3, 4)$, $C(-2, 3)$ and $D(-3, -2)$ be four points in a plane, show that $ABCD$ is a rhombus but not a square. Find the area of the rhombus. [CBSE 2013]

SOLUTION

Let $A(2, -1)$, $B(3, 4)$, $C(-2, 3)$ and $D(-3, -2)$ be the angular points of a quad. $ABCD$. Join AC and BD . Then,



$$\begin{aligned}
 AB &= \sqrt{(3-2)^2 + (4+1)^2} \\
 &= \sqrt{1^2 + 5^2} = \sqrt{26} \text{ units,}
 \end{aligned}$$

$$BC = \sqrt{(-2-3)^2 + (3-4)^2} = \sqrt{(-5)^2 + (-1)^2} = \sqrt{26} \text{ units,}$$

$$CD = \sqrt{(-3+2)^2 + (-2-3)^2} = \sqrt{(-1)^2 + (-5)^2} = \sqrt{26} \text{ units,}$$

$$DA = \sqrt{(-3-2)^2 + (-2+1)^2} = \sqrt{(-5)^2 + (-1)^2} = \sqrt{26} \text{ units.}$$

$\therefore AB = BC = CD = DA = \sqrt{26} \text{ units.}$

$$\begin{aligned}
 \text{Now, } AC &= \sqrt{(-2-2)^2 + (3+1)^2} = \sqrt{(-4)^2 + 4^2} \\
 &= \sqrt{32} = 4\sqrt{2} \text{ units}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } BD &= \sqrt{(-3-3)^2 + (-2-4)^2} = \sqrt{(-6)^2 + (-6)^2} \\
 &= \sqrt{72} = 6\sqrt{2} \text{ units.}
 \end{aligned}$$

$\therefore$ diagonal $AC \neq$ diagonal BD .

Thus, $ABCD$ is a quadrilateral in which all sides are equal but diagonals are not equal.

$\therefore ABCD$ is a rhombus but not a square.

$$\begin{aligned}
 \therefore \text{ area of the rhombus } ABCD &= \frac{1}{2} \times (\text{product of diagonals}) \\
 &= \left(\frac{1}{2} \times AC \times BD \right) \\
 &= \left(\frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} \right) \text{ sq units} \\
 &= 24 \text{ sq units.}
 \end{aligned}$$

Hence, the area of rhombus $ABCD$ is 24 sq units.

COLLINEAR POINTS Three points A, B, C are said to be collinear if they lie on the same straight line.

TEST FOR COLLINEARITY OF THREE POINTS

In order to show that three given points A, B, C are collinear, we find the distances AB, BC and AC . If the sum of any two of these distances is equal to the third distance then the given points are collinear.

EXAMPLE 20 Prove that the points $A(1, 1), B(-2, 7)$ and $C(3, -3)$ are collinear.

SOLUTION Let $A(1, 1), B(-2, 7)$ and $C(3, -3)$ be the given points. Then,

$\xrightarrow{\quad B \qquad \qquad A \qquad \qquad C \quad}$

$$AB = \sqrt{(-2-1)^2 + (7-1)^2} = \sqrt{(-3)^2 + 6^2} = \sqrt{45} = 3\sqrt{5} \text{ units,}$$

$$BC = \sqrt{(3+2)^2 + (-3-7)^2} = \sqrt{5^2 + (-10)^2} = \sqrt{125} = 5\sqrt{5} \text{ units,}$$

$$AC = \sqrt{(3-1)^2 + (-3-1)^2} = \sqrt{2^2 + (-4)^2} = \sqrt{20} = 2\sqrt{5} \text{ units.}$$

$$\therefore AB + AC = (3\sqrt{5} + 2\sqrt{5}) \text{ units} = 5\sqrt{5} \text{ units} = BC.$$

Thus, $AB + AC = BC$.

Hence, the given points A, B, C are collinear.

EXERCISE 6A

- Find the distance between the points:
 - $A(9, 3)$ and $B(15, 11)$
 - $A(7, -4)$ and $B(-5, 1)$
 - $A(-6, -4)$ and $B(9, -12)$
 - $A(1, -3)$ and $B(4, -6)$
 - $P(a+b, a-b)$ and $Q(a-b, a+b)$
 - $P(a \sin \alpha, a \cos \alpha)$ and $Q(a \cos \alpha, -a \sin \alpha)$
- Find the distance of each of the following points from the origin:
 - $A(5, -12)$
 - $B(-5, 5)$
 - $C(-4, -6)$.
- Find all possible values of x for which the distance between the points $A(x, -1)$ and $B(5, 3)$ is 5 units.
- Find all possible values of y for which the distance between the points $A(2, -3)$ and $B(10, y)$ is 10 units.
- Find the values of x for which the distance between the points $P(x, 4)$ and $Q(9, 10)$ is 10 units. [CBSE 2011]
- If the point $A(x, 2)$ is equidistant from the points $B(8, -2)$ and $C(2, -2)$, find the value of x . Also, find the length of AB .
- If the point $A(0, 2)$ is equidistant from the points $B(3, p)$ and $C(p, 5)$, find the value of p . Also, find the length of AB . [CBSE 2014]

8. Find the coordinates of the point on x -axis which is equidistant from the points $(-2, 5)$ and $(2, -3)$. [CBSE 2017]
9. Find points on the x -axis, each of which is at a distance of 10 units from the point $A(11, -8)$.
10. Find the point on the y -axis which is equidistant from the points $A(6, 5)$ and $B(-4, 3)$.
11. If the point $P(x, y)$ is equidistant from the points $A(5, 1)$ and $B(-1, 5)$, prove that $3x = 2y$. [CBSE 2005]
12. If $P(x, y)$ is a point equidistant from the points $A(6, -1)$ and $B(2, 3)$, show that $x - y = 3$. [CBSE 2008C]
13. Find the coordinates of the point equidistant from three given points $A(5, 3)$, $B(5, -5)$ and $C(1, -5)$. [CBSE 2006]
14. If the points $A(4, 3)$ and $B(x, 5)$ lie on a circle with the centre $O(2, 3)$, find the value of x . [CBSE 2009]
HINT $OA^2 = OB^2$.
15. If the point $C(-2, 3)$ is equidistant from the points $A(3, -1)$ and $B(x, 8)$, find the values of x . Also, find the distance BC . [CBSE 2013C]
16. If the point $P(2, 2)$ is equidistant from the points $A(-2, k)$ and $B(-2k, -3)$, find k . Also, find the length of AP . [CBSE 2014]
17. (i) If the point (x, y) is equidistant from the points $(a+b, b-a)$ and $(a-b, a+b)$, prove that $bx = ay$. [CBSE 2011]
(ii) If the distances of $P(x, y)$ from $A(5, 1)$ and $B(-1, 5)$ are equal then prove that $3x = 2y$. [CBSE 2017]
18. Using the distance formula, show that the given points are collinear:
(i) $(1, -1)$, $(5, 2)$ and $(9, 5)$ (ii) $(6, 9)$, $(0, 1)$ and $(-6, -7)$
(iii) $(-1, -1)$, $(2, 3)$ and $(8, 11)$ (iv) $(-2, 5)$, $(0, 1)$ and $(2, -3)$.
19. Show that the points $A(7, 10)$, $B(-2, 5)$ and $C(3, -4)$ are the vertices of an isosceles right triangle. [CBSE 2007, '13]
20. Show that the points $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$ are the vertices of an isosceles right triangle.
21. If $A(5, 2)$, $B(2, -2)$ and $C(-2, t)$ are the vertices of a right triangle with $\angle B = 90^\circ$ then find the value of t . [CBSE 2015]
22. Prove that the points $A(2, 4)$, $B(2, 6)$ and $C(2 + \sqrt{3}, 5)$ are the vertices of an equilateral triangle. [CBSE 2013C]
23. Show that the points $(-3, -3)$, $(3, 3)$ and $(-3\sqrt{3}, 3\sqrt{3})$ are the vertices of an equilateral triangle. [CBSE 2012]
24. Show that the points $A(-5, 6)$, $B(3, 0)$ and $C(9, 8)$ are the vertices of an isosceles right-angled triangle. Calculate its area.

25. Show that the points $O(0, 0)$, $A(3, \sqrt{3})$ and $B(3, -\sqrt{3})$ are the vertices of an equilateral triangle. Find the area of this triangle.
26. Show that the following points are the vertices of a square:
- (i) $A(3, 2)$, $B(0, 5)$, $C(-3, 2)$ and $D(0, -1)$ [CBSE 2008C]
 - (ii) $A(6, 2)$, $B(2, 1)$, $C(1, 5)$ and $D(5, 6)$ [CBSE 2006]
 - (iii) $A(0, -2)$, $B(3, 1)$, $C(0, 4)$ and $D(-3, 1)$
27. Show that the points $A(-3, 2)$, $B(-5, -5)$, $C(2, -3)$ and $D(4, 4)$ are the vertices of a rhombus. Find the area of this rhombus.
- HINT** Area of a rhombus $= \frac{1}{2} \times (\text{product of its diagonals})$.
28. Show that the points $A(3, 0)$, $B(4, 5)$, $C(-1, 4)$ and $D(-2, -1)$ are the vertices of a rhombus. Find its area.
29. Show that the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$ and $D(7, 3)$ are the vertices of a rhombus. Find its area.
30. Show that the points $A(2, 1)$, $B(5, 2)$, $C(6, 4)$ and $D(3, 3)$ are the angular points of a parallelogram. Is this figure a rectangle?
31. Show that $A(1, 2)$, $B(4, 3)$, $C(6, 6)$ and $D(3, 5)$ are the vertices of a parallelogram. Show that $ABCD$ is not a rectangle.
32. Show that the following points are the vertices of a rectangle:
- (i) $A(-4, -1)$, $B(-2, -4)$, $C(4, 0)$ and $D(2, 3)$
 - (ii) $A(2, -2)$, $B(14, 10)$, $C(11, 13)$ and $D(-1, 1)$
 - (iii) $A(0, -4)$, $B(6, 2)$, $C(3, 5)$ and $D(-3, -1)$
33. Show that $\triangle ABC$ with vertices $A(-2, 0)$, $B(0, 2)$ and $C(2, 0)$ is similar to $\triangle DEF$ with vertices $D(-4, 0)$, $F(4, 0)$ and $E(0, 4)$. [CBSE 2017]
34. Show that $\triangle ABC$, where $A(22, 0)$, $B(2, 0)$, $C(0, 2)$ and $\triangle PQR$ where $P(-4, 0)$, $Q(4, 0)$, $R(0, 4)$ are similar triangles. [CBSE 2017]

ANSWERS (EXERCISE 6A)

1. (i) 10 units (ii) 13 units (iii) 17 units
- (iv) $3\sqrt{2}$ units (v) $2\sqrt{2}b$ units (vi) $\sqrt{2}a$ units
2. (i) 13 units (ii) $5\sqrt{2}$ units (iii) $5\sqrt{2}$ units
3. $x = 8$ or $x = 2$ 4. $y = 3$ or $y = -9$ 5. $x = 1$ or $x = 17$
6. $x = 5$, $AB = 5$ units 7. $p = 1$, $AB = \sqrt{10}$ units 8. $(-2, 0)$
9. $(5, 0)$ and $(17, 0)$ 10. $(0, 9)$ 13. $P(3, -1)$
14. $x = 2$ 15. $x = 2$ or $x = -6$, $BC = \sqrt{41}$ units
16. $(k = -3 \Rightarrow AP = \sqrt{41})$, $(k = -1 \Rightarrow AP = 5)$ 21. $t = 1$

24. 50 sq units

25. $3\sqrt{3}$ sq units

27. 45 sq units

28. 24 sq units

29. 3 sq units

30. No

HINTS TO SOME SELECTED QUESTIONS

15. $AC = BC \Rightarrow AC^2 = BC^2$

$$\Rightarrow (3 + 2)^2 + (-4)^2 = (x + 2)^2 + 5^2$$

$$\Rightarrow (x + 2)^2 = 16 \Rightarrow x + 2 = 4 \text{ or } x + 2 = -4$$

$$\Rightarrow x = 2 \text{ or } x = -6.$$

Distance between $B(2, 8)$ and $C(-2, 3) = \sqrt{4^2 + 5^2} = \sqrt{41}$ units.Distance between $B(-6, 8)$ and $C(-2, 3) = \sqrt{4^2 + (-5)^2} = \sqrt{41}$ units.

16. $PA = PB \Rightarrow PA^2 = PB^2$

$$\Rightarrow (2 + 2)^2 + (2 - k)^2 = (2 + 2k)^2 + (2 + 3)^2$$

$$\Rightarrow (2 - k)^2 - (2 + 2k)^2 = 25 - 16 = 9$$

$$\Rightarrow (4 + k^2 - 4k) - (4 + 4k^2 + 8k) = 9$$

$$\Rightarrow 3k^2 + 12k + 9 = 0$$

$$\Rightarrow k^2 + 4k + 3 = 0 \Rightarrow (k + 3)(k + 1) = 0$$

$$\Rightarrow k = -3 \text{ or } k = -1.$$

Case 1. When $k = -3$, we have

$$AP = \text{distance between } A(-2, -3) \text{ and } P(2, 2) = \sqrt{41} \text{ units.}$$

Case 2. When $k = -1$, we have

$$AP = \text{distance between } A(-2, -1) \text{ and } P(2, 2) = \sqrt{25} = 5 \text{ units.}$$

17. (i) It is given that the point $P(x, y)$ is equidistant from the points $A(a + b, b - a)$ and $B(a - b, a + b)$. Then,

$$PA = PB \Rightarrow PA^2 = PB^2$$

$$\Rightarrow \{x - (a + b)\}^2 + \{y - (b - a)\}^2 = \{x - (a - b)\}^2 + \{y - (a + b)\}^2$$

$$\Rightarrow \{(x - a) + b\}^2 - \{(x - a) - b\}^2 = \{(y - b) + a\}^2 - \{(y - b) - a\}^2$$

$$\Rightarrow 4(x - a)b = 4(y - b)a \Rightarrow (x - a)b = (y - b)a \Rightarrow bx = ay.$$

21. $AB^2 + BC^2 = AC^2 \Rightarrow 25 + \{16 + (t + 2)^2\} = 49 + (t - 2)^2$

$$\Rightarrow (t + 2)^2 - (t - 2)^2 = 8 \Rightarrow 4 \times t \times 2 = 8 \Rightarrow t = 1.$$

23. Let the given points be $A(-3, -3)$, $B(3, 3)$ and $C(-3\sqrt{3}, 3\sqrt{3})$. Then,

$$AB^2 = (6^2 + 6^2) = 72,$$

$$BC^2 = (-3\sqrt{3} - 3)^2 + (3\sqrt{3} - 3)^2 = (-3)^2(\sqrt{3} + 1)^2 + 3^2(\sqrt{3} - 1)^2$$

$$= 9 \times [(\sqrt{3} + 1)^2 + (\sqrt{3} - 1)^2] = 9 \times 2 \times [(\sqrt{3})^2 + 1^2] = 72,$$

$$AC^2 = (-3\sqrt{3} + 3)^2 + (3\sqrt{3} + 3)^2 = (-3)^2 \times (\sqrt{3} - 1)^2 + 3^2 \times (\sqrt{3} + 1)^2$$

$$= 9 \times [(\sqrt{3} - 1)^2 + (\sqrt{3} + 1)^2] = 9 \times 2 \times [(\sqrt{3})^2 + 1^2] = 72.$$

$$\therefore AB = BC = AC = \sqrt{72} = 6\sqrt{2} \text{ units.}$$

25. Here, $a = \sqrt{12}$. So, area = $\left\{ \frac{\sqrt{3}}{4} \times a^2 \right\} = \left(\frac{\sqrt{3}}{4} \times 12 \right) = 3\sqrt{3}$ sq units.

27. Diagonal $AC = \sqrt{(-5)^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$, and

diagonal $BD = \sqrt{9^2 + 9^2} = \sqrt{162} = 9\sqrt{2}$.

$$\therefore \text{area} = \left(\frac{1}{2} \times AC \times BD \right) = \left(\frac{1}{2} \times 5\sqrt{2} \times 9\sqrt{2} \right) \text{sq units} = 45 \text{ sq units.}$$

33. Find the length of sides AB , BC and CA of $\triangle ABC$ and of sides DE , EF and FD of $\triangle DEF$.

Then, show that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$.

SECTION FORMULA

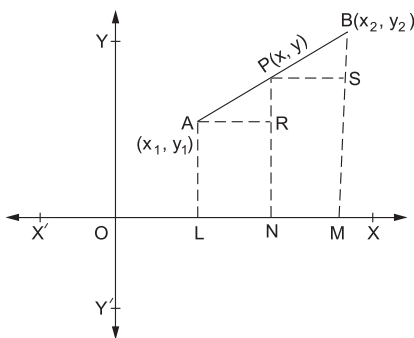
THEOREM 2 (Section formula) *The coordinates of the point $P(x, y)$ which divides the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m : n$ are given by*

$$x = \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}.$$

PROOF Let $X'OX$ and YOY' be the coordinate axes.

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the end points of the given line segment AB .

Let $P(x, y)$ be the point which divides AB in the ratio $m : n$.



Then, $\frac{AP}{PB} = \frac{m}{n}$.

Draw $AL \perp OX$; $BM \perp OX$; $PN \perp OX$; $AR \perp PN$; and $PS \perp BM$.

Now, $AR = LN = ON - OL = (x - x_1)$;

$$PS = NM = OM - ON = (x_2 - x);$$

$$PR = PN - RN = PN - AL = (y - y_1);$$

$$BS = BM - SM = BM - PN = (y_2 - y).$$

Clearly, $\triangle ARP$ and $\triangle PSB$ are similar and, therefore, their sides are proportional.

$$\begin{aligned}
 \therefore \quad \frac{AP}{PB} &= \frac{AR}{PS} = \frac{PR}{BS} \\
 \Rightarrow \quad \frac{m}{n} &= \frac{x - x_1}{x_2 - x} = \frac{y - y_1}{y_2 - y} \\
 \Rightarrow \quad \frac{m}{n} &= \frac{x - x_1}{x_2 - x} \quad \text{and} \quad \frac{m}{n} = \frac{y - y_1}{y_2 - y} \\
 \Rightarrow \quad mx_2 - mx &= nx - nx_1 \quad \text{and} \quad my_2 - my = ny - ny_1 \\
 \Rightarrow \quad (m+n)x &= (mx_2 + nx_1) \quad \text{and} \quad (m+n)y = (my_2 + ny_1) \\
 \Rightarrow \quad x &= \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}.
 \end{aligned}$$

Hence, the coordinates of P are $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$.

COROLLARY (Midpoint formula) *The coordinates of the midpoint M of a line segment AB with end points $A(x_1, y_1)$ and $B(x_2, y_2)$ are*

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

PROOF Let M be the midpoint of the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$.

Then, M divides AB in the ratio $1 : 1$.

So, by the section formula, the coordinates of M are

$$\left(\frac{1 \cdot x_2 + 1 \cdot x_1}{1+1}, \frac{1 \cdot y_2 + 1 \cdot y_1}{1+1} \right), \text{ i.e., } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Hence, the coordinates of the midpoint of AB are

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

SUMMARY

- (i) If $P(x, y)$ divides the join of $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m : n$ then

$$x = \frac{mx_2 + nx_1}{(m+n)} \quad \text{and} \quad y = \frac{my_2 + ny_1}{(m+n)}.$$

- (ii) The midpoint of the join of $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

SOLVED EXAMPLES

EXAMPLE 1 Find the coordinates of the point which divides the line segment joining the points $A(4, -3)$ and $B(9, 7)$ in the ratio $3 : 2$.

SOLUTION The end points of AB are $A(4, -3)$ and $B(9, 7)$.

$$\therefore (x_1 = 4, y_1 = -3) \text{ and } (x_2 = 9, y_2 = 7).$$

Also, $m = 3$ and $n = 2$.



Let the required point be $P(x, y)$.

By the section formula, we have

$$x = \frac{mx_2 + nx_1}{(m+n)}, \quad y = \frac{my_2 + ny_1}{(m+n)}$$

$$\Rightarrow x = \frac{(3 \times 9 + 2 \times 4)}{(3+2)}, \quad y = \frac{[3 \times 7 + 2 \times (-3)]}{(3+2)}$$

$$\Rightarrow x = 7, y = 3.$$

Hence, the required point is $P(7, 3)$.

EXAMPLE 2 Find the coordinates of the midpoint of the line segment joining the points $A(-5, 4)$ and $B(7, -8)$.

SOLUTION Let $M(x, y)$ be the midpoint of AB . Then,

$$x = \frac{[(-5) + 7]}{2} = 1 \text{ and } y = \frac{[4 + (-8)]}{2} = -2.$$

Hence, the required point is $M(1, -2)$.

EXAMPLE 3 Find the coordinates of the points of trisection of the line segment joining the points $A(-5, 6)$ and $B(4, -3)$. [CBSE 2009C]

SOLUTION Let P and Q be the points of trisection of AB .



Then, P divides AB in the ratio $1 : 2$.

So, the coordinates of P are

$$P\left(\frac{1 \times 4 + 2 \times (-5)}{1+2}, \frac{1 \times (-3) + 2 \times 6}{1+2}\right), \text{ i.e., } P(-2, 3).$$

Also, Q divides AB in the ratio $2 : 1$.

So, the coordinates of Q are

$$Q\left(\frac{2 \times 4 + 1 \times (-5)}{2+1}, \frac{2 \times (-3) + 1 \times 6}{2+1}\right), \text{ i.e., } Q(1, 0).$$

Hence, the points of trisection of AB are $P(-2, 3)$ and $Q(1, 0)$.

EXAMPLE 4 Find the coordinates of a point P on the line segment joining $A(1, 2)$ and $B(6, 7)$ such that $AP = \frac{2}{5} AB$. [CBSE 2015]

SOLUTION $AP = \frac{2}{5} AB \Rightarrow \frac{AP}{AB} = \frac{2}{5}$

$$\Rightarrow \frac{AP}{AP + PB} = \frac{2}{5} \Rightarrow 5AP = 2AP + 2PB$$

$$\Rightarrow 3AP = 2PB \Rightarrow \frac{AP}{PB} = \frac{2}{3} \Rightarrow AP:PB = 2:3.$$

So, P divides AB in the ratio $2:3$.

$\therefore$ coordinates of P are

$$\left(\frac{2 \times 6 + 3 \times 1}{2 + 3}, \frac{2 \times 7 + 3 \times 2}{2 + 3} \right) = \left(\frac{15}{5}, \frac{20}{5} \right) = (3, 4).$$

EXAMPLE 5

Point P divides the line segment joining the points $A(2, 1)$ and $B(5, -8)$ such that $\frac{AP}{AB} = \frac{1}{3}$. If P lies on the line $2x - y + k = 0$, find the value of k . [CBSE 2010]

SOLUTION

$$\frac{AP}{AB} = \frac{1}{3} \Rightarrow \frac{AP}{AP + PB} = \frac{1}{3}$$

$$\Rightarrow 3AP = AP + PB \Rightarrow 2AP = PB$$

$$\Rightarrow \frac{AP}{PB} = \frac{1}{2} \Rightarrow AP:PB = 1:2.$$

So, P divides AB in the ratio $1:2$.

$\therefore$ coordinates of P are

$$\left(\frac{1 \times 5 + 2 \times 2}{1 + 2}, \frac{1 \times (-8) + 2 \times 1}{1 + 2} \right) = \left(\frac{9}{3}, \frac{-6}{3} \right) = (3, -2).$$

Since $P(3, -2)$ lies on $2x - y + k = 0$, we have

$$2 \times 3 - (-2) + k = 0 \Rightarrow 6 + 2 + k = 0 \Rightarrow k = -8.$$

Hence, the required value of k is -8 .

EXAMPLE 6

In what ratio does the point $P(2, -5)$ divide the line segment joining $A(-3, 5)$ and $B(4, -9)$?

SOLUTION

Let the required ratio be $k:1$.

Then, by the section formula, the coordinates of P are

$$P\left(\frac{4k - 3}{k + 1}, \frac{-9k + 5}{k + 1} \right).$$

$$\therefore \frac{4k - 3}{k + 1} = 2 \text{ and } \frac{-9k + 5}{k + 1} = -5 \quad [\because P(2, -5) \text{ is given}]$$

$$\Rightarrow 4k - 3 = 2k + 2 \text{ and } -9k + 5 = -5k - 5$$

$$\Rightarrow 2k = 5 \text{ and } 4k = 10$$

$$\Rightarrow k = \frac{5}{2} \text{ in each case.}$$

So, the required ratio is $\frac{5}{2} : 1$, which is $5 : 2$.

Hence, P divides AB in the ratio $5 : 2$.

EXAMPLE 7

Find the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$. Also, find the value of x .

[CBSE 2014]

SOLUTION

Let the required ratio be $k : 1$.

Then, by section formula, the coordinates of P are

$$P\left(\frac{4k+12}{k+1}, \frac{-3k+5}{k+1}\right).$$

But, this point is given as $P(x, 2)$.

$$\therefore \frac{-3k+5}{k+1} = 2 \Rightarrow -3k+5 = 2k+2 \Rightarrow 5k = 3 \Rightarrow k = \frac{3}{5}.$$

So, the required ratio is $3 : 5$.

Putting $k = \frac{3}{5}$ in P , we get

$$x = \frac{\left\{4 \times \frac{3}{5} + 12\right\}}{\left(\frac{3}{5} + 1\right)} = \frac{72}{8} = 9.$$

Hence, $x = 9$.

EXAMPLE 8

Find the ratio in which the point $P(11, y)$ divides the line segment joining the points $A(15, 5)$ and $B(9, 20)$. Also, find the value of y .

[CBSE 2013C]

SOLUTION

Let the required ratio be $k : 1$.

Then, by the section formula, the coordinates of P are

$$P\left(\frac{9k+15}{k+1}, \frac{20k+5}{k+1}\right).$$

But, this point is given as $P(11, y)$.

$$\therefore \frac{9k+15}{k+1} = 11 \Rightarrow 9k+15 = 11k+11 \Rightarrow 2k = 4 \Rightarrow k = 2.$$

So, the required ratio is $2 : 1$.

Putting $k = 2$ in P , we get

$$y = \frac{20 \times 2 + 5}{(2+1)} = \frac{45}{3} = 15.$$

Hence, $y = 15$.

EXAMPLE 9 If the point $P(-1, 2)$ divides the line segment joining the points $A(2, 5)$ and B in the ratio $3 : 4$, find the coordinates of B .

SOLUTION Let the coordinates of B be (x_1, y_1) .
 Then, P divides the join of $A(2, 5)$ and $B(x_1, y_1)$ in the ratio $3 : 4$.

So, by the section formula, the coordinates of P are

$$P\left(\frac{3x_1 + 8}{3 + 4}, \frac{3y_1 + 20}{3 + 4}\right), \text{ i.e., } P\left(\frac{3x_1 + 8}{7}, \frac{3y_1 + 20}{7}\right).$$

But, the coordinates of P are $(-1, 2)$.

$$\therefore \frac{3x_1 + 8}{7} = -1 \text{ and } \frac{3y_1 + 20}{7} = 2$$

$$\Rightarrow 3x_1 + 8 = -7 \text{ and } 3y_1 + 20 = 14$$

$$\Rightarrow 3x_1 = -15 \text{ and } 3y_1 = -6$$

$$\Rightarrow x_1 = -5 \text{ and } y_1 = -2.$$

Hence, the coordinates of B are $(-5, -2)$.

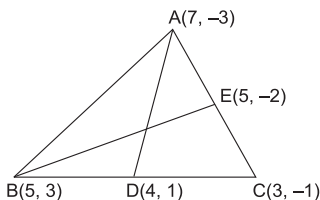
EXAMPLE 10 Find the lengths of the medians AD and BE of $\triangle ABC$ whose vertices are $A(7, -3)$, $B(5, 3)$ and $C(3, -1)$. [CBSE 2014]

SOLUTION Since D and E are the midpoints of BC and AC respectively, so their coordinates are

$$D\left(\frac{5+3}{2}, \frac{3+(-1)}{2}\right)$$

$$\text{and } E\left(\frac{7+3}{2}, \frac{-3-1}{2}\right),$$

i.e., $D(4, 1)$ and $E(5, -2)$.



$$\therefore AD = \sqrt{(7-4)^2 + (-3-1)^2} = \sqrt{3^2 + (-4)^2}$$

$$= \sqrt{9+16} = \sqrt{25} = 5 \text{ units.}$$

$$\therefore BE = \sqrt{(5-5)^2 + (-2-3)^2} = \sqrt{0+(-5)^2} = \sqrt{25} = 5 \text{ units.}$$

EXAMPLE 11 The three vertices of a parallelogram $ABCD$ taken in order are $A(3, -4)$, $B(-1, -3)$ and $C(-6, 2)$. Find the coordinates of the fourth vertex D . [CBSE 2013]

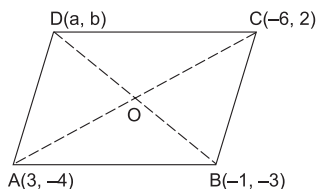
SOLUTION Let $A(3, -4)$, $B(-1, -3)$ and $C(-6, 2)$ be the three given vertices of a parallelogram $ABCD$ and let its fourth vertex be $D(a, b)$.

Join AC and BD .

Let AC and BD intersect at the point O .

We know that the diagonals of a ||gm bisect each other.

So, O is the midpoint of each one of AC and BD .



Midpoint of AC is $\left(\frac{3+(-6)}{2}, \frac{-4+2}{2}\right)$, i.e., $O\left(\frac{-3}{2}, -1\right)$

and midpoint of BD is $O\left(\frac{-1+a}{2}, \frac{-3+b}{2}\right)$.

$$\therefore \frac{-1+a}{2} = \frac{-3}{2} \Rightarrow -1+a = -3 \Rightarrow a = -2$$

$$\text{and } \frac{-3+b}{2} = -1 \Rightarrow -3+b = -2 \Rightarrow b = 1.$$

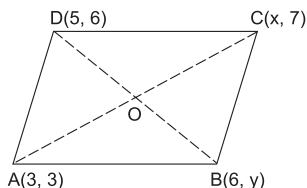
Hence, the fourth vertex of ||gm $ABCD$ is $D(-2, 1)$.

EXAMPLE 12

If $(3, 3)$, $(6, y)$, $(x, 7)$ and $(5, 6)$ are the vertices of a parallelogram taken in order, find the values of x and y . [CBSE 2011]

SOLUTION

Let $A(3, 3)$, $B(6, y)$, $C(x, 7)$ and $D(5, 6)$ be the vertices of a ||gm $ABCD$. Join AC and BD , intersecting each other at the point O .



We know that the diagonals of ||gm bisect each other.

Therefore, O is the midpoint of AC as well as that of BD .

$$\therefore \text{midpoint of } AC \text{ is } \left(\frac{x+3}{2}, \frac{7+3}{2}\right), \text{ i.e., } \left(\frac{x+3}{2}, 5\right)$$

$$\text{and midpoint of } BD \text{ is } \left(\frac{5+6}{2}, \frac{6+y}{2}\right), \text{ i.e., } \left(\frac{11}{2}, \frac{6+y}{2}\right).$$

But, these points coincide at the point O .

$$\therefore \frac{x+3}{2} = \frac{11}{2} \text{ and } \frac{6+y}{2} = 5$$

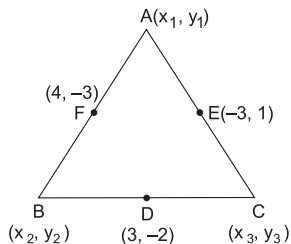
$$\Rightarrow x+3 = 11 \text{ and } 6+y = 10$$

$$\Rightarrow x = 8 \text{ and } y = 4.$$

Hence, $x = 8$ and $y = 4$.

EXAMPLE 13 Let $D(3, -2)$, $E(-3, 1)$ and $F(4, -3)$ be the midpoints of the sides BC , CA and AB respectively of $\triangle ABC$. Then, find the coordinates of the vertices A , B and C .

SOLUTION Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of the given $\triangle ABC$, and let $D(3, -2)$, $E(-3, 1)$ and $F(4, -3)$ be the midpoints of BC , CA and AB respectively.



D is the midpoint of BC

$$\Rightarrow \frac{x_2 + x_3}{2} = 3, \frac{y_2 + y_3}{2} = -2$$

$$\Rightarrow \begin{cases} x_2 + x_3 = 6 & \dots (i) \\ y_2 + y_3 = -4. & \dots (ii) \end{cases}$$

E is the midpoint of CA

$$\Rightarrow \frac{x_1 + x_3}{2} = -3, \frac{y_1 + y_3}{2} = 1$$

$$\Rightarrow \begin{cases} x_1 + x_3 = -6 & \dots (iii) \\ y_1 + y_3 = 2. & \dots (iv) \end{cases}$$

F is the midpoint of AB

$$\Rightarrow \frac{x_1 + x_2}{2} = 4, \frac{y_1 + y_2}{2} = -3$$

$$\Rightarrow \begin{cases} x_1 + x_2 = -8 & \dots (v) \\ y_1 + y_2 = -6. & \dots (vi) \end{cases}$$

Adding (i), (iii) and (v), we get

$$2(x_1 + x_2 + x_3) = 8 \Rightarrow (x_1 + x_2 + x_3) = 4. \dots (vii)$$

Using (i), (iii) and (v) with (vii), we get

$$x_1 = -2, x_2 = 10 \text{ and } x_3 = -4.$$

Adding (ii), (iv) and (vi), we get

$$2(y_1 + y_2 + y_3) = -8 \Rightarrow y_1 + y_2 + y_3 = -4. \dots (viii)$$

Using (ii), (iv) and (vi) with (viii), we get

$$y_1 = 0, y_2 = -6 \text{ and } y_3 = 2.$$

Hence, the vertices of $\triangle ABC$ are $A(-2, 0)$, $B(10, -6)$ and $C(-4, 2)$.

EXAMPLE 14 The coordinates of one end point of a diameter AB of a circle are $A(4, -1)$ and the coordinates of the centre of the circle are $C(1, -3)$. Find the coordinates of B .

SOLUTION Let $C(1, -3)$ be the centre of the given circle and let $A(4, -1)$ and $B(a, b)$ be the two end points of the given diameter AB .
Then, clearly C is the midpoint of AB .

$$\therefore \text{ the coordinates of } C \text{ are } C\left(\frac{4+a}{2}, \frac{-1+b}{2}\right).$$

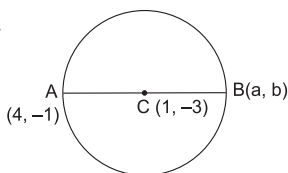
But, the coordinates of C are $(1, -3)$.

$$\therefore \frac{4+a}{2} = 1 \text{ and } \frac{-1+b}{2} = -3$$

$$\Rightarrow 4+a = 2 \text{ and } -1+b = -6$$

$$\Rightarrow a = -2 \text{ and } b = -5.$$

Hence, the coordinates of B are $B(-2, -5)$.



CENTROID OF A TRIANGLE *The point of intersection of the medians of a triangle is called its centroid.*

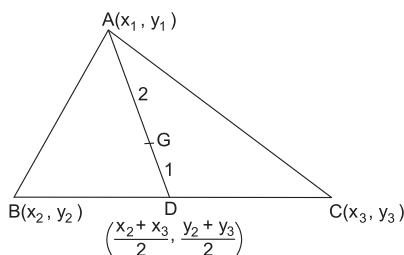
TO FIND THE COORDINATES OF THE CENTROID OF A TRIANGLE

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a $\triangle ABC$.

Let D be the midpoint of BC .

$$\text{Then, the coordinates of } D \text{ are } D\left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right).$$

Let $G(x, y)$ be the centroid of $\triangle ABC$.



Then, G divides AD in the ratio $2 : 1$.

$$\therefore x = \frac{\left\{ \frac{2(x_2+x_3)}{2} + x_1 \right\}}{(2+1)} = \frac{(x_1+x_2+x_3)}{3},$$

$$y = \frac{\left\{ \frac{2(y_2+y_3)}{2} + y_1 \right\}}{(2+1)} = \frac{(y_1+y_2+y_3)}{3}.$$

$\therefore$ the coordinates of G are $G\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.

EXAMPLE 15 Find the centroid of $\triangle ABC$ whose vertices are $A(-3, 0)$, $B(5, -2)$ and $C(-8, 5)$.

SOLUTION Here, $(x_1 = -3, y_1 = 0)$, $(x_2 = 5, y_2 = -2)$ and $(x_3 = -8, y_3 = 5)$.

Let $G(x, y)$ be the centroid of $\triangle ABC$. Then,

$$x = \frac{1}{3}(x_1 + x_2 + x_3) = \frac{1}{3}(-3 + 5 - 8) = -2,$$

$$y = \frac{1}{3}(y_1 + y_2 + y_3) = \frac{1}{3}(0 - 2 + 5) = 1.$$

Hence, the centroid of $\triangle ABC$ is $G(-2, 1)$.

EXAMPLE 16 Two vertices of a $\triangle ABC$ are given by $A(6, 4)$ and $B(-2, 2)$, and its centroid is $G(3, 4)$. Find the coordinates of the third vertex C of $\triangle ABC$. [CBSE 2005C]

SOLUTION It is being given that the two vertices of $\triangle ABC$ are $A(6, 4)$ and $B(-2, 2)$. Let its third vertex be $C(x, y)$.

Then, the coordinates of the centroid of $\triangle ABC$ are

$$G\left(\frac{6 + (-2) + x}{3}, \frac{4 + 2 + y}{3}\right), \text{ i.e., } G\left(\frac{4 + x}{3}, \frac{6 + y}{3}\right).$$

But, it is given that the centroid of $\triangle ABC$ is $G(3, 4)$.

$$\therefore \frac{4 + x}{3} = 3 \text{ and } \frac{6 + y}{3} = 4$$

$$\Rightarrow 4 + x = 9 \text{ and } 6 + y = 12.$$

$$\Rightarrow x = 5 \text{ and } y = 6.$$

Hence, the coordinates of the third vertex is $C(5, 6)$.

EXERCISE 6B

- Find the coordinates of the point which divides the join of $A(-1, 7)$ and $B(4, -3)$ in the ratio $2 : 3$.
 - Find the coordinates of the point which divides the join of $A(-5, 11)$ and $B(4, -7)$ in the ratio $7 : 2$.
- Find the coordinates of the points of trisection of the line segment joining the points $A(7, -2)$ and $B(1, -5)$. [CBSE 2017]
- If the coordinates of points A and B are $(-2, -2)$ and $(2, -4)$ respectively, find the coordinates of the point P such that $AP = \frac{3}{7}AB$, where P lies on the line segment AB . [CBSE 2015]

4. Point A lies on the line segment PQ joining $P(6, -6)$ and $Q(-4, -1)$ in such a way that $\frac{PA}{PQ} = \frac{2}{5}$. If the point A also lies on the line $3x + k(y + 1) = 0$, find the value of k . [CBSE 2015]
5. Points P, Q, R and S divide the line segment joining the points $A(1, 2)$ and $B(6, 7)$ in five equal parts. Find the coordinates of the points P, Q and R . [CBSE 2014]
6. Points P, Q and R in that order are dividing a line segment joining $A(1, 6)$ and $B(5, -2)$ in four equal parts. Find the coordinates of P, Q and R . [CBSE 2009C]
7. The line segment joining the points $A(3, -4)$ and $B(1, 2)$ is trisected at the points $P(p, -2)$ and $Q\left(\frac{5}{3}, q\right)$. Find the values of p and q . [CBSE 2005]
8. Find the coordinates of the midpoint of the line segment joining
(i) $A(3, 0)$ and $B(-5, 4)$ (ii) $P(-11, -8)$ and $Q(8, -2)$.
9. If $(2, p)$ is the midpoint of the line segment joining the points $A(6, -5)$ and $B(-2, 11)$, find the value of p . [CBSE 2010]
10. The midpoint of the line segment joining $A(2a, 4)$ and $B(-2, 3b)$ is $C(1, 2a + 1)$. Find the values of a and b .
11. The line segment joining $A(-2, 9)$ and $B(6, 3)$ is a diameter of a circle with centre C . Find the coordinates of C .
12. Find the coordinates of a point A , where AB is a diameter of a circle with centre $C(2, -3)$ and the other end of the diameter is $B(1, 4)$.
13. In what ratio does the point $P(2, 5)$ divide the join of $A(8, 2)$ and $B(-6, 9)$?
14. Find the ratio in which the point $P\left(\frac{3}{4}, \frac{5}{12}\right)$ divides the line segment joining the points $A\left(\frac{1}{2}, \frac{3}{2}\right)$ and $B(2, -5)$. [CBSE 2015]
15. Find the ratio in which the point $P(m, 6)$ divides the join of $A(-4, 3)$ and $B(2, 8)$. Also, find the value of m . [CBSE 2004]
16. Find the ratio in which the point $(-3, k)$ divides the join of $A(-5, -4)$ and $B(-2, 3)$. Also, find the value of k . [CBSE 2007]
17. In what ratio is the line segment joining $A(2, -3)$ and $B(5, 6)$ divided by the x -axis? Also, find the coordinates of the point of division.

18. In what ratio is the line segment joining the points $A(-2, -3)$ and $B(3, 7)$ divided by the y -axis? Also, find the coordinates of the point of division. [CBSE 2006C]
19. In what ratio does the line $x - y - 2 = 0$ divide the line segment joining the points $A(3, -1)$ and $B(8, 9)$? [CBSE 2007]
20. Find the lengths of the medians of a $\triangle ABC$ whose vertices are $A(0, -1)$, $B(2, 1)$ and $C(0, 3)$.
21. Find the centroid of $\triangle ABC$ whose vertices are $A(-1, 0)$, $B(5, -2)$ and $C(8, 2)$. [CBSE 2003]
22. If $G(-2, 1)$ is the centroid of a $\triangle ABC$ and two of its vertices are $A(1, -6)$ and $B(-5, 2)$, find the third vertex of the triangle.
23. Find the third vertex of a $\triangle ABC$ if two of its vertices are $B(-3, 1)$ and $C(0, -2)$, and its centroid is at the origin. [CBSE 2004C]
24. Show that the points $A(3, 1)$, $B(0, -2)$, $C(1, 1)$ and $D(4, 4)$ are the vertices of a parallelogram $ABCD$.
25. If the points $P(a, -11)$, $Q(5, b)$, $R(2, 15)$ and $S(1, 1)$ are the vertices of a parallelogram $PQRS$, find the values of a and b .
26. If three consecutive vertices of a parallelogram $ABCD$ are $A(1, -2)$, $B(3, 6)$ and $C(5, 10)$, find its fourth vertex D .
27. In what ratio does y -axis divide the line segment joining the points $(-4, 7)$ and $(3, -7)$? [CBSE 2012]
28. If the point $P\left(\frac{1}{2}, y\right)$ lies on the line segment joining the points $A(3, -5)$ and $B(-7, 9)$ then find the ratio in which P divides AB . Also, find the value of y . [CBSE 2010]
29. Find the ratio in which the line segment joining the points $A(3, -3)$ and $B(-2, 7)$ is divided by x -axis. Also, find the point of division. [CBSE 2014]
30. The base QR of an equilateral triangle PQR lies on x -axis. The coordinates of the point Q are $(-4, 0)$ and origin is the midpoint of the base. Find the coordinates of the points P and R . [CBSE 2015]
31. The base BC of an equilateral triangle ABC lies on y -axis. The coordinates of point C are $(0, -3)$. The origin is the midpoint of the base. Find the coordinates of the points A and B . Also, find the coordinates of another point D such that $ABCD$ is a rhombus. [CBSE 2015]
32. Find the ratio in which the point $P(-1, y)$ lying on the line segment joining points $A(-3, 10)$ and $B(6, -8)$ divides it. Also, find the value of y . [CBSE 2013]

33. $ABCD$ is a rectangle formed by the points $A(-1, -1)$, $B(-1, 4)$, $C(5, 4)$ and $D(5, -1)$. If P , Q , R and S be the midpoints of AB , BC , CD and DA respectively, show that $PQRS$ is a rhombus.
34. The midpoint P of the line segment joining the points $A(-10, 4)$ and $B(-2, 0)$ lies on the line segment joining the points $C(-9, -4)$ and $D(-4, y)$. Find the ratio in which P divides CD . Also find the value of y .
[CBSE 2014]
35. A line intersects the y -axis and x -axis at the points P and Q respectively. If $(2, -5)$ is the midpoint of PQ then find the coordinates of P and Q .
[CBSE 2017]
36. In what ratio does the point $\left(\frac{24}{11}, y\right)$ divide the line segment joining the points $P(2, -2)$ and $Q(3, 7)$? Also, find the value of y .
[CBSE 2017]
37. The midpoints of the sides BC , CA and AB of a $\triangle ABC$ are $D(3, 4)$, $E(8, 9)$ and $F(6, 7)$ respectively. Find the coordinates of the vertices of the triangle.
[CBSE 2017]
38. If two adjacent vertices of a parallelogram are $(3, 2)$ and $(-1, 0)$ and the diagonals intersect at $(2, -5)$ then find the coordinates of the other two vertices.
[CBSE 2017]

ANSWERS (EXERCISE 6B)

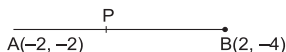
1. (i) $(1, 3)$ (ii) $(2, -3)$ 2. $(5, -3), (3, -4)$ 3. $P\left(\frac{-2}{7}, \frac{-20}{7}\right)$ 4. $k = 2$
5. $P(2, 3), Q(3, 4), R(4, 5)$ 6. $P(2, 4), Q(3, 2), R(4, 0)$
7. $p = \frac{7}{3}, q = 0$ 8. (i) $(-1, 2)$ (ii) $\left(\frac{-3}{2}, -5\right)$ 9. $p = 3$ 10. $a = 2, b = 2$
11. $C(2, 6)$ 12. $A(3, -10)$ 13. $3 : 4$ 14. $1 : 5$ 15. $3 : 2, m = \frac{-2}{5}$
16. $2 : 1, k = \frac{2}{3}$ 17. $(1 : 2), (3, 0)$ 18. $(2 : 3), (0, 1)$ 19. $2 : 3$
20. $AD = \sqrt{10}$ units, $BE = 2$ units, $CF = \sqrt{10}$ units 21. $G(4, 0)$
22. $C(-2, 7)$ 23. $A(3, 1)$ 25. $a = 4, b = 3$ 26. $D(3, 2)$
27. $4 : 3$ 28. $1 : 3, y = \frac{-3}{2}$ 29. $(3 : 7), \left(\frac{3}{2}, 0\right)$
30. $P(0, 4\sqrt{3})$ or $P(0, -4\sqrt{3})$ and $R(4, 0)$
31. $A(3\sqrt{3}, 0)$ or $A(-3\sqrt{3}, 0)$ and $B(0, 3)$;
 $[a(3\sqrt{3}, 0), b(0, 3), d(3\sqrt{3}, -6)]$ or $[a(-3\sqrt{3}, 0), b(0, 3), d(-3\sqrt{3}, -6)]$
32. $2 : 7, y = 6$ 34. $3 : 2, y = 6$ 35. $P(0, -10), Q(4, 0)$ 36. $2 : 9; y = -\frac{4}{11}$
37. $A(11, 12), B(1, 2), C(5, 8)$ 38. $(1, -12); (5, -10)$

HINTS TO SOME SELECTED QUESTIONS

$$3. \frac{AP}{AB} = \frac{3}{7} \Rightarrow \frac{AP}{AB} = \frac{7}{3} \Rightarrow \frac{AP + PB}{AP} = \frac{7}{3}$$

$$\Rightarrow \frac{AP}{AP} + \frac{PB}{AP} = \frac{7}{3} \Rightarrow 1 + \frac{PB}{AP} = \frac{7}{3}$$

$$\Rightarrow \frac{PB}{AP} = \left(\frac{7}{3} - 1\right) = \frac{4}{3} \Rightarrow \frac{AP}{PB} = \frac{3}{4} \Rightarrow AP : PB = 3 : 4.$$



$\therefore$ coordinates of P are $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$, where $m = 3, n = 4$,

$$\text{i.e., } P\left(\frac{3 \times 2 + 4 \times (-2)}{3+4}, \frac{3 \times (-4) + 4 \times (-2)}{3+4}\right) = P\left(\frac{-2}{7}, \frac{-20}{7}\right).$$

$$4. \frac{PA}{PQ} = \frac{2}{5} \Rightarrow \frac{PQ}{PA} = \frac{5}{2} \Rightarrow \frac{PA + AQ}{PA} = \frac{5}{2}$$

$$\Rightarrow 1 + \frac{AQ}{PA} = \frac{5}{2} \Rightarrow \frac{AQ}{PA} = \left(\frac{5}{2} - 1\right) = \frac{3}{2}$$

$$\Rightarrow \frac{PA}{AQ} = \frac{2}{3} \Rightarrow PA : AQ = 2 : 3.$$



$\therefore$ coordinates of A are $\left(\frac{2 \times (-4) + 3 \times 6}{2+3}, \frac{2 \times (-1) + 3 \times (-6)}{2+3}\right) = A(2, -4).$

Since the point A(2, -4) lies on the line $3x + k(y + 1) = 0$, we have

$$(3 \times 2) + k(-4 + 1) = 0 \Rightarrow 3k = 6 \Rightarrow k = 2.$$

5. P divides AB in the ratio 1 : 4.

Q divides AB in the ratio 2 : 3.

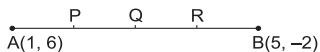
R divides AB in the ratio 3 : 2.



6. P divides AB in the ratio 1 : 3.

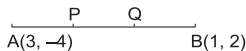
Q divides AB in the ratio 2 : 2 = 1 : 1.

R divides AB in the ratio 3 : 1.



7. P divides AB in the ratio 1 : 2.

This point is $P\left(\frac{1 \times 1 + 2 \times 3}{1+2}, \frac{1 \times 2 + 2 \times (-4)}{1+2}\right).$



$$\therefore \left(\frac{7}{3}, -2\right) \equiv (p, -2) \Rightarrow p = \frac{7}{3}.$$

Q divides AB in the ratio 2 : 1. So, we get

$$Q\left(\frac{2 \times 1 + 1 \times 3}{2+1}, \frac{2 \times 2 + 1 \times (-4)}{2+1}\right), \text{ i.e., } \left(\frac{5}{3}, 0\right)$$

$$\therefore \left(\frac{5}{3}, 0\right) \equiv \left(\frac{5}{3}, q\right) \Rightarrow q = 0.$$

14. Let the required ratio be $k : 1$.

Then, the coordinates of P are $\left(\frac{2k + \frac{1}{2}}{k+1}, \frac{-5k + \frac{3}{2}}{k+1}\right).$

$$\begin{aligned}\therefore \frac{2k + \frac{1}{2}}{k + 1} &= \frac{3}{4} \Rightarrow \frac{4k + 1}{2(k + 1)} = \frac{3}{4} \Rightarrow 16k + 4 = 6k + 6 \\ &\Rightarrow 10k = 2 \Rightarrow k = \frac{1}{5}.\end{aligned}$$

$\therefore$ required ratio is $\frac{1}{5} : 1$, i.e., $1 : 5$.

15. Let the required ratio be $k : 1$.

Then, the coordinates of P are $\left(\frac{2k - 4}{k + 1}, \frac{8k + 3}{k + 1}\right)$.

$$\therefore \frac{8k + 3}{k + 1} = 6 \Rightarrow 8k + 3 = 6k + 6 \Rightarrow 2k = 3 \Rightarrow k = \frac{3}{2}.$$

So, the required ratio is $\frac{3}{2} : 1$, i.e., $3 : 2$.

$$\therefore m = \frac{2k - 4}{k + 1} = \frac{\left(2 \times \frac{3}{2} - 4\right)}{\left(\frac{3}{2} + 1\right)} = \frac{-2}{5}.$$

17. Let the x -axis be cut by the join of $A(2, -3)$ and $B(5, 6)$ in the ratio $k : 1$ at the point $P(x, 0)$.

$$\text{Then, } \frac{6k - 3}{k + 1} = 0 \Rightarrow 6k - 3 = 0 \Rightarrow 6k = 3 \Rightarrow k = \frac{3}{6} = \frac{1}{2}.$$

$\therefore$ required ratio is $\frac{1}{2} : 1$, i.e., $1 : 2$.

Point of division is $P\left(\frac{1 \times 5 + 2 \times 2}{1 + 2}, 0\right)$, i.e., $(3, 0)$.

19. Let the required ratio be $k : 1$.

Then, the point $P\left(\frac{8k + 3}{k + 1}, \frac{9k - 1}{k + 1}\right)$ lies on the line $x - y - 2 = 0$.

$$\begin{aligned}\therefore \frac{8k + 3}{k + 1} - \frac{9k - 1}{k + 1} - 2 &= 0 \Rightarrow (8k + 3) - (9k - 1) - 2(k + 1) = 0 \\ &\Rightarrow (8k - 9k - 2k) + (3 + 1 - 2) = 0 \\ &\Rightarrow 3k = 2 \Rightarrow k = \frac{2}{3}.\end{aligned}$$

Hence, the required ratio is $\left(\frac{2}{3} : 1\right)$, i.e., $2 : 3$.

27. Let the required ratio be $k : 1$.

Then, x -coordinate of the point of division is 0.

$$\therefore \frac{3k - 4}{k + 1} = 0 \Rightarrow 3k - 4 = 0 \Rightarrow k = \frac{4}{3}.$$

Hence, the required ratio is $\left(\frac{4}{3} : 1\right)$, i.e., 4 : 3.

28. Let the required ratio be $k : 1$.

Then, we get $P\left(\frac{-7k+3}{k+1}, \frac{9k-5}{k+1}\right)$.

$$\begin{aligned}\therefore \frac{-7k+3}{k+1} &= \frac{1}{2} \Rightarrow -14k+6 = k+1 \\ &\Rightarrow 15k=5 \Rightarrow k = \frac{1}{3}.\end{aligned}$$

So, the required ratio is $\left(\frac{1}{3} : 1\right)$, i.e., 1 : 3.

Putting $k = \frac{1}{3}$, we get

$$y = \frac{9k-5}{k+1} = \frac{\left(9 \times \frac{1}{3}\right)-5}{\left(\frac{1}{3}+1\right)} = \frac{-6}{4} = \frac{-3}{2}.$$

29. Let the required ratio be $k : 1$.

Then, the point of division is $P\left(\frac{-2k+3}{k+1}, \frac{7k-3}{k+1}\right)$.

Since this point lies on the x -axis, its y -coordinates must be 0.

$$\therefore \frac{7k-3}{k+1} = 0 \Rightarrow 7k-3=0 \Rightarrow k = \frac{3}{7}.$$

So, the required ratio is $\left(\frac{3}{7} : 1\right)$, i.e., 3 : 7.

Now, put $k = \frac{3}{7}$, to get

$$P\left(\frac{-2k+3}{k+1}, \frac{7k-3}{k+1}\right) \text{ as } P\left(\frac{3}{2}, 0\right).$$

30. Since R lies on the x -axis, its coordinates are of the form $R(a, 0)$ and clearly O , the origin is the midpoint of QR . So,

$$\frac{-4+a}{2} = 0 \Rightarrow -4+a=0 \Rightarrow a=4.$$

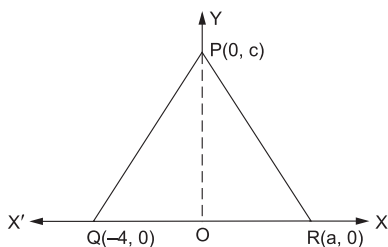
So, the coordinates of R are $R(4, 0)$.

Clearly, P lies on the y -axis. Let the coordinates of P be $P(0, c)$.

Since $\triangle PQR$ is equilateral, we have

$$PQ^2 = PR^2 = QR^2.$$

$$\therefore 16 + c^2 = 64 \Rightarrow c^2 = 48 \Rightarrow c = \pm\sqrt{48} = \pm 4\sqrt{3}.$$



31. Since B lies on the y -axis, its coordinates are of the form $B(0, b)$. Since $O(0, 0)$ is the midpoint of BC , we have

$$\frac{-3 + b}{2} = 0 \Rightarrow -3 + b = 0 \Rightarrow b = 3.$$

So, the coordinates of B are $B(0, 3)$.

Clearly, the vertex A lies on the x -axis.

Let this point be $A(a, 0)$.

Since $\triangle ABC$ is equilateral, we have

$$AC^2 = AB^2 = BC^2.$$

$$\begin{aligned} \therefore a^2 + 9 = 36 &\Rightarrow a^2 = 27 \\ &\Rightarrow a = \pm 3\sqrt{3}. \end{aligned}$$

So, the coordinates of A are $(3\sqrt{3}, 0)$ or $(-3\sqrt{3}, 0)$.

32. Let the required ratio be $k : 1$.

Then, we get $P\left(\frac{6k-3}{k+1}, \frac{-8k+10}{k+1}\right)$.

$$\text{But, } \frac{6k-3}{k+1} = -1 \Rightarrow 6k-3 = -k-1 \Rightarrow 7k=2 \Rightarrow k = \frac{2}{7}.$$

So, the required ratio is $\left(\frac{2}{7} : 1\right) = (2 : 7)$.

34. The point P is $P\left(\frac{-10-2}{2}, \frac{4+0}{2}\right)$, i.e., $(-6, 2)$.

Let $P(-6, 2)$ divide CD in the ratio $k : 1$.

$$\text{Then, } \frac{-4k-9}{k+1} = -6 \text{ and } \frac{ky-4}{k+1} = 2$$

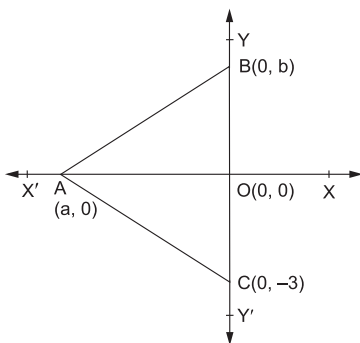
$$\Rightarrow -4k-9 = -6k-6 \text{ and } ky-4 = 2k+2$$

$$\Rightarrow 2k = 3 \text{ and } k(y-2) = 6$$

$$\Rightarrow k = \frac{3}{2} \text{ and } (y-2) = \frac{6}{k} = \left(6 \times \frac{2}{3}\right) = 4$$

$$\Rightarrow k = \frac{3}{2} \text{ and } y = 4 + 2 = 6.$$

Hence, the ratio is $3 : 2$ and $y = 6$.

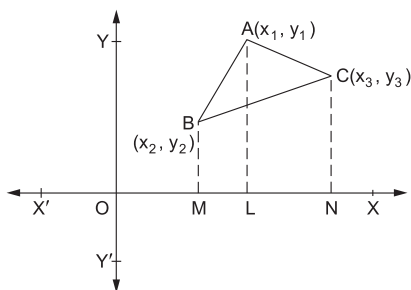


AREA OF A TRIANGLE

THEOREM 1 The area of a $\triangle ABC$ with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is given by

$$\text{area}(\triangle ABC) = \left| \frac{1}{2} \{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\} \right|.$$

PROOF Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of the given $\triangle ABC$.



Draw AL , BM and CN perpendiculars to the x -axis.

Then, $ML = (x_1 - x_2)$, $LN = (x_3 - x_1)$ and $MN = (x_3 - x_2)$.

$\therefore$ area of $\triangle ABC$

$$\begin{aligned}
 &= \text{ar(trap. } BMLA) + \text{ar(trap. } ALNC) - \text{ar(trap. } BMNC) \\
 &= \left\{ \frac{1}{2} (AL + BM) \times ML \right\} + \left\{ \frac{1}{2} (AL + CN) \times LN \right\} \\
 &\quad - \left\{ \frac{1}{2} (BM + CN) \times MN \right\} \\
 &= \frac{1}{2} (y_1 + y_2)(x_1 - x_2) + \frac{1}{2} (y_1 + y_3)(x_3 - x_1) - \frac{1}{2} (y_2 + y_3)(x_3 - x_2) \\
 &= \frac{1}{2} [x_1(y_1 + y_2 - y_1 - y_3) + x_2(y_2 + y_3 - y_1 - y_2) \\
 &\quad + x_3(y_1 + y_3 - y_2 - y_3)] \\
 &= \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)].
 \end{aligned}$$

Since the area is never negative, we have

$$\text{area}(\triangle ABC) = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

Condition for Collinearity of Three Points

Let the given points be $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$.

Then, A , B and C are collinear

$$\Rightarrow \text{area of } \triangle ABC = 0$$

$$\Rightarrow \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] = 0$$

$$\Rightarrow x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0.$$

SUMMARY OF THE RESULTS

1. Area of $\triangle ABC$ with vertices $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$ is given by

$$\text{ar}(\triangle ABC) = \frac{1}{2} | [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] |.$$

2. Three points $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$ are collinear only when $x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$.

SOLVED EXAMPLES

EXAMPLE 1 Find the area of $\triangle ABC$ whose vertices are $A(2, 7)$, $B(3, -1)$ and $C(-5, 6)$.

SOLUTION The coordinates of the vertices of the given triangle are $A(x_1 = 2, y_1 = 7)$, $B(x_2 = 3, y_2 = -1)$ and $C(x_3 = -5, y_3 = 6)$.

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) | \\ &= \frac{1}{2} | 2 \cdot (-1 - 6) + 3 \cdot (6 - 7) - 5 \cdot (7 + 1) | \\ &= \frac{1}{2} | 2 \times (-7) + 3 \times (-1) - 5 \times 8 | \\ &= \frac{1}{2} | -14 - 3 - 40 | = \frac{1}{2} | -57 | = \frac{57}{2} = 28.5 \text{ sq units.} \end{aligned}$$

Hence, the area of the given triangle is 28.5 sq units.

EXAMPLE 2 Find the values of k so that the area of the triangle with vertices $(1, -1)$, $(-4, 2k)$ and $(-k, -5)$ is 24 square units. [CBSE 2015]

SOLUTION Let $A(1, -1)$, $B(-4, 2k)$ and $C(-k, -5)$ be the vertices of the given $\triangle ABC$.

Then, $A(x_1 = 1, y_1 = -1)$, $B(x_2 = -4, y_2 = 2k)$, and $C(x_3 = -k, y_3 = -5)$.

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) | \\ &= \frac{1}{2} | 1 \cdot (2k + 5) - 4 \cdot (-5 + 1) - k(-1 - 2k) | \\ &= \frac{1}{2} | 2k + 5 + 16 + k + 2k^2 | \\ &= \frac{1}{2} | 2k^2 + 3k + 21 | \text{ sq units.} \end{aligned}$$

But, it is given that $\text{ar}(\triangle ABC) = 24$ sq units.

$$\therefore \frac{1}{2} \left| 2k^2 + 3k + 21 \right| = 24 \Rightarrow \left| k^2 + \frac{3}{2}k + \frac{21}{2} \right| = 24. \quad \dots (i)$$

$$\begin{aligned} \text{But } \left\{ k^2 + \frac{3}{2}k + \frac{21}{2} \right\} &= \left(k^2 + \frac{3}{2}k + \frac{9}{16} \right) + \left(\frac{21}{2} - \frac{9}{16} \right) \\ &= \left\{ \left(k + \frac{3}{4} \right)^2 + \frac{159}{16} \right\} > 0. \end{aligned}$$

So, we may write (i) as

$$\begin{aligned} k^2 + \frac{3}{2}k + \frac{21}{2} &= 24 \Rightarrow 2k^2 + 3k + 21 = 48 \\ \Rightarrow 2k^2 + 3k - 27 &= 0 \Rightarrow 2k^2 + 9k - 6k - 27 = 0 \\ \Rightarrow k(2k + 9) - 3(2k + 9) &= 0 \Rightarrow (2k + 9)(k - 3) = 0 \\ \Rightarrow k - 3 = 0 \text{ or } 2k + 9 = 9 &\Rightarrow k = 3 \text{ or } k = -\frac{9}{2}. \end{aligned}$$

$$\text{Hence, } k = 3 \text{ or } k = -\frac{9}{2}.$$

EXAMPLE 3

If $A(4, -6)$, $B(3, -2)$ and $C(5, 2)$ are the vertices of a $\triangle ABC$ and AD is its median. Prove that the median AD divides $\triangle ABC$ into two triangles of equal areas. [CBSE 2014]

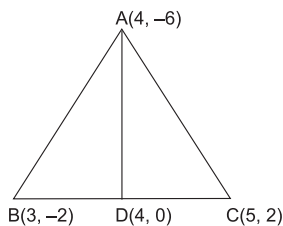
SOLUTION

It is being given that $A(4, -6)$, $B(3, -2)$ and $C(5, 2)$ are the vertices of a $\triangle ABC$ and AD is its median.

Clearly, D is the midpoint of BC .

$\therefore$ the coordinates of D are

$$\left(\frac{3+5}{2}, \frac{-2+2}{2} \right) = (4, 0).$$



For $\triangle ABD$, we have

$$A(x_1 = 4, y_1 = -6), B(x_2 = 3, y_2 = -2) \text{ and } D(x_3 = 4, y_3 = 0).$$

$$\begin{aligned} \text{ar}(\triangle ABD) &= \frac{1}{2} \left| x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right| \\ &= \frac{1}{2} \left| 4 \cdot (-2 - 0) + 3 \cdot (0 + 6) + 4 \cdot (-6 + 2) \right| \\ &= \frac{1}{2} \left| 4 \times (-2) + 3 \times 6 + 4 \times (-4) \right| \\ &= \frac{1}{2} \left| -8 + 18 - 16 \right| = \frac{1}{2} \left| -6 \right| \\ &= \left(\frac{1}{2} \times 6 \right) = 3 \text{ sq units.} \end{aligned}$$

For $\triangle ADC$, we have

$$(x_1 = 4, y_1 = -6), (x_2 = 4, y_2 = 0) \text{ and } (x_3 = 5, y_3 = 2).$$

$$\begin{aligned}\therefore \text{ar}(\triangle ADC) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |4 \cdot (0 - 2) + 4 \cdot (2 + 6) + 5 \cdot (-6 - 0)| \\ &= \frac{1}{2} |4 \times (-2) + 4 \times 8 + 5 \times (-6)| \\ &= \frac{1}{2} |-8 + 32 - 30| = \frac{1}{2} |-6| \\ &= \left(\frac{1}{2} \times 6\right) = 3 \text{ sq units.}\end{aligned}$$

$$\therefore \text{ar}(\triangle ABD) = \text{ar}(\triangle ADC).$$

Hence, the median AD divides $\triangle ABC$ into two triangles of equal areas.

EXAMPLE 4

Find the area of the triangle formed by joining the midpoints of the sides of the triangle whose vertices are $A(2, 2)$, $B(4, 4)$ and $C(2, 6)$.

[CBSE 2009C]

SOLUTION

Let $A(2, 2)$, $B(4, 4)$ and $C(2, 6)$ be the vertices of the given $\triangle ABC$. Let D , E and F be the midpoints of AB , BC and CA respectively.

Then, the coordinates of D , E and F are

$$D\left(\frac{2+4}{2}, \frac{2+4}{2}\right), E\left(\frac{4+2}{2}, \frac{4+6}{2}\right) \text{ and } F\left(\frac{2+2}{2}, \frac{2+6}{2}\right),$$

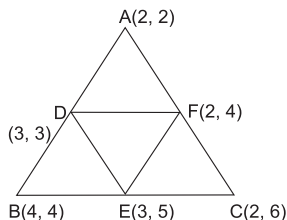
i.e., $D(3, 3)$, $E(3, 5)$ and $F(2, 4)$.

For $\triangle DEF$, we have

$$(x_1 = 3, y_1 = 3), (x_2 = 3, y_2 = 5) \text{ and } (x_3 = 2, y_3 = 4).$$

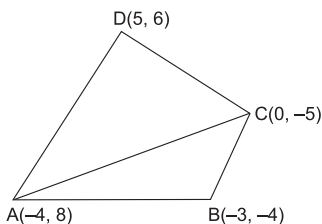
$$\begin{aligned}\therefore \text{ar}(\triangle DEF) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |3 \cdot (5 - 4) + 3 \cdot (4 - 3) + 2 \cdot (3 - 5)| \\ &= \frac{1}{2} |(3 \times 1) + (3 \times 1) + 2 \times (-2)| \\ &= \frac{1}{2} |3 + 3 - 4| = \left(\frac{1}{2} \times 2\right) = 1 \text{ sq unit.}\end{aligned}$$

Hence, the area of $\triangle DEF$ is 1 sq unit.



EXAMPLE 5 Find the area of a quadrilateral $ABCD$ whose vertices are $A(-4, 8)$, $B(-3, -4)$, $C(0, -5)$ and $D(5, 6)$. [CBSE 2015]

SOLUTION Area of quad. $ABCD = \text{ar}(\triangle ABC) + \text{ar}(\triangle ACD)$.



For $\triangle ABC$, we have

$$(x_1 = -4, y_1 = 8), (x_2 = -3, y_2 = -4) \text{ and } (x_3 = 0, y_3 = -5).$$

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |(-4) \cdot (-4 - 5) + (-3) \cdot (-5 - 8) + 0 \cdot (8 + 4)| \\ &= \frac{1}{2} |(-4) \times 1 + (-3) \times (-13) + 0| \\ &= \frac{1}{2} |-4 + 39 + 0| = \frac{35}{2} \text{ sq units.} \end{aligned}$$

For $\triangle ACD$, we have

$$(x_1 = -4, y_1 = 8), (x_2 = 0, y_2 = -5) \text{ and } (x_3 = 5, y_3 = 6).$$

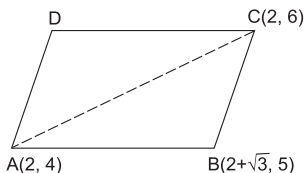
$$\begin{aligned} \therefore \text{ar}(\triangle ACD) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |(-4) \cdot (-5 - 6) + 0 \cdot (6 - 8) + 5 \cdot (8 + 5)| \\ &= \frac{1}{2} |(-4) \times (-11) + 0 \times (-2) + 5 \times 13| \\ &= \frac{1}{2} |44 + 0 + 65| = \frac{109}{2} \text{ sq unit.} \end{aligned}$$

$$\begin{aligned} \therefore \text{ar}(\text{quad. } ABCD) &= \text{ar}(\triangle ABC) + \text{ar}(\triangle ACD) \\ &= \left(\frac{35}{2} + \frac{109}{2} \right) \text{ sq units} = \frac{144}{2} \text{ sq units} \\ &= 72 \text{ sq units.} \end{aligned}$$

Hence, the area of the given quadrilateral $ABCD$ is 72 sq units.

EXAMPLE 6 Find the area of a parallelogram $ABCD$ if three of its vertices are $A(2, 4)$, $B(2 + \sqrt{3}, 5)$ and $C(2, 6)$. [CBSE 2013]

SOLUTION Let $ABCD$ be the given parallelogram, three of whose vertices are $A(2, 4)$, $B(2 + \sqrt{3}, 5)$ and $C(2, 6)$.



In $\triangle ABC$, we have

$$(x_1 = 2, y_1 = 4), (x_2 = 2 + \sqrt{3}, y_2 = 5), (x_3 = 2, y_3 = 6).$$

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |2(5 - 6) + (2 + \sqrt{3})(6 - 4) + 2(4 - 5)| \\ &= \frac{1}{2} |2 \times (-1) + (4 + 2\sqrt{3}) - 2| \\ &= \frac{1}{2} (2\sqrt{3}) = \sqrt{3} \text{ sq unit} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{ar}(\text{||gm } ABCD) &= 2 \times \text{ar}(\triangle ABC) = (2 \times \sqrt{3}) \text{ sq units} \\ &= (2\sqrt{3}) \text{ sq units.} \end{aligned}$$

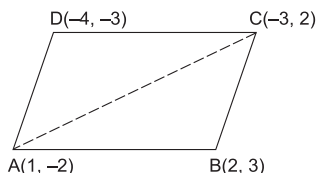
Hence, the area of the given parallelogram is $2\sqrt{3}$ sq units.

EXAMPLE 7

If the points $A(1, -2)$, $B(2, 3)$, $C(-3, 2)$ and $D(-4, -3)$ are the vertices of a parallelogram $ABCD$ then taking AB as the base, find the height of the parallelogram. [CBSE 2013]

SOLUTION Let $A(1, -2)$, $B(2, 3)$, $C(-3, 2)$ and $D(-4, -3)$ be the vertices of the given ||gm $ABCD$. Join AC . Then,

$$\text{ar}(\text{||gm } ABCD) = 2 \times \text{ar}(\triangle ABC).$$



In $\triangle ABC$, we have

$$(x_1 = 1, y_1 = -2), (x_2 = 2, y_2 = 3) \text{ and } (x_3 = -3, y_3 = 2).$$

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |1 \cdot (3 - 2) + 2 \cdot (2 + 2) - 3 \cdot (-2 - 3)| \end{aligned}$$

$$= \frac{1}{2} |1 + 8 + 15| = 12 \text{ sq units.}$$

$$\therefore \text{ar}(\text{||gm } ABCD) = 2 \times \text{ar}(\triangle ABC) = 2 \times 12 = 24 \text{ sq units.}$$

$$\text{Base } AB = \sqrt{(2-1)^2 + (3+2)^2} = \sqrt{1^2 + 5^2} = \sqrt{26} \text{ units.}$$

Let h be the height of the ||gm $ABCD$. Then,

$$\text{ar}(\text{||gm } ABCD) = (\text{base} \times \text{height}) = (\sqrt{26} \times h) \text{ sq units.}$$

$$\therefore \sqrt{26} \times h = 24 \Rightarrow h = \left\{ \frac{24}{\sqrt{26}} \times \frac{\sqrt{26}}{\sqrt{26}} \right\} = \left(\frac{24}{26} \times \sqrt{26} \right) \text{ units}$$

$$\Rightarrow h = \left(\frac{12}{13} \times 5.09 \right) = \frac{61.08}{13} = 4.69 \text{ units.}$$

Hence, the height of the parallelogram is 4.69 units.

EXAMPLE 8 Show that the points $A(-1, 1)$, $B(5, 7)$ and $C(8, 10)$ are collinear.

SOLUTION Let $A(-1, 1)$, $B(5, 7)$ and $C(8, 10)$ be the given points. Then,

$$(x_1 = -1, y_1 = 1), (x_2 = 5, y_2 = 7) \text{ and } (x_3 = 8, y_3 = 10).$$

$$\begin{aligned} \therefore x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \\ = (-1)(7 - 10) + 5(10 - 1) + 8(1 - 7) \\ = (3 + 45 - 48) = 0. \end{aligned}$$

Hence, the given points are collinear.

EXAMPLE 9 Show that the points $A(a, b + c)$, $B(b, c + a)$ and $C(c, a + b)$ are collinear. [CBSE 2010]

SOLUTION Let $A(a, b + c)$, $B(b, c + a)$ and $C(c, a + b)$ be the given points.

Then, $(x_1 = a, y_1 = b + c)$, $(x_2 = b, y_2 = c + a)$ and $(x_3 = c, y_3 = a + b)$.

$$\begin{aligned} \therefore x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \\ = a(c + a - a - b) + b(a + b - b - c) + c(b + c - c - a) \\ = a(c - b) + b(a - c) + c(b - a) = 0. \end{aligned}$$

Hence, the given points are collinear.

EXAMPLE 10 If the area of $\triangle ABC$ with vertices $A(x, y)$, $B(1, 2)$ and $C(2, 1)$ is 6 sq units then prove that $x + y = 15$ or $x + y + 9 = 0$. [CBSE 2013]

SOLUTION The vertices of $\triangle ABC$ are $A(x, y)$, $B(1, 2)$ and $C(2, 1)$.

Here, $(x_1 = x, y_1 = y)$, $(x_2 = 1, y_2 = 2)$ and $(x_3 = 2, y_3 = 1)$.

$$\therefore \text{ar}(\triangle ABC) = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\begin{aligned}
 &= \frac{1}{2} |x \cdot (2-1) + 1 \cdot (1-y) + 2 \cdot (y-2)| \\
 &= \frac{1}{2} |x + 1 - y + 2y - 4| = \frac{1}{2} |x + y - 3|.
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{1}{2} |x + y - 3| &= 6 \Rightarrow |x + y - 3| = 12 \\
 &\Rightarrow x + y - 3 = 12 \text{ or } x + y - 3 = -12 \\
 &\Rightarrow x + y = 15 \text{ or } x + y + 9 = 0.
 \end{aligned}$$

Hence, $x + y = 15$ or $x + y + 9 = 0$.

EXAMPLE 11 Find the value of k for which the points $A(k+1, 2k)$, $B(3k, 2k+3)$ and $C(5k-1, 5k)$ are collinear. [CBSE 2015, '17]

SOLUTION The given points are $A(k+1, 2k)$, $B(3k, 2k+3)$ and $C(5k-1, 5k)$.

Here, $(x_1 = k+1, y_1 = 2k)$, $(x_2 = 3k, y_2 = 2k+3)$ and $(x_3 = 5k-1, y_3 = 5k)$.

Let the given points be collinear. Then,

$$\begin{aligned}
 &x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0 \\
 \Rightarrow &(k+1)(2k+3-5k) + 3k(5k-2k) + (5k-1)[2k-(2k+3)] = 0 \\
 \Rightarrow &(k+1)(3-3k) + 3k \times 3k + (5k-1) \times (-3) = 0 \\
 \Rightarrow &3 - 3k^2 + 9k^2 - 15k + 3 = 0 \Rightarrow 6k^2 - 15k + 6 = 0 \\
 \Rightarrow &2k^2 - 5k + 2 = 0 \Rightarrow 2k^2 - 4k - k + 2 = 0 \\
 \Rightarrow &2k(k-2) - (k-2) = 0 \Rightarrow (k-2)(2k-1) = 0 \\
 \Rightarrow &k-2 = 0 \text{ or } 2k-1 = 0 \Rightarrow k = 2 \text{ or } k = \frac{1}{2}.
 \end{aligned}$$

Hence, $k = 2$ or $k = \frac{1}{2}$.

EXAMPLE 12 If the points $A(-1, -4)$, $B(b, c)$ and $C(5, -1)$ are collinear and $2b + c = 4$, find the values of b and c . [CBSE 2014]

SOLUTION Let $A(x_1 = -1, y_1 = -4)$, $B(x_2 = b, y_2 = c)$ and $C(x_3 = 5, y_3 = -1)$ be the given points.

Since these points A, B and C are collinear, we have

$$\begin{aligned}
 &x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0 \\
 \Rightarrow &(-1) \cdot (c+1) + b \cdot (-1+4) + 5 \cdot (-4-c) = 0 \\
 \Rightarrow &-c-1+3b-20-5c = 0 \\
 \Rightarrow &3b-6c = 21 \Rightarrow b-2c = 7. \quad \dots (i)
 \end{aligned}$$

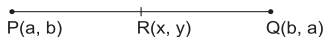
Also, it is given that $2b + c = 4$ (ii)

On solving (i) and (ii), we get $b = 3$ and $c = -2$.

Hence, $b = 3$ and $c = -2$.

EXAMPLE 13 If $R(x, y)$ is a point on the line segment joining the points $P(a, b)$ and $Q(b, a)$ then prove that $x + y = a + b$. [CBSE 2010]

SOLUTION It is being given that the points $P(a, b)$, $R(x, y)$ and $Q(b, a)$ lie on the same line segment and, therefore, these points are collinear.



Let $P(x_1 = a, y_1 = b)$, $R(x_2 = x, y_2 = y)$ and $Q(x_3 = b, y_3 = a)$ be the given collinear points. Then,

$$\begin{aligned} & x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0 \\ \Rightarrow & a(y - a) + x(a - b) + b(b - y) = 0 \\ \Rightarrow & ay - a^2 + ax - bx + b^2 - by = 0 \\ \Rightarrow & (a - b)y - (a^2 - b^2) + (a - b)x = 0 \\ \Rightarrow & y - (a + b) + x = 0 \Rightarrow x + y = a + b. \end{aligned}$$

Hence, $x + y = a + b$.

EXERCISE 6C

- Find the area of $\triangle ABC$ whose vertices are:
 - $A(1, 2)$, $B(-2, 3)$ and $C(-3, -4)$
 - $A(-5, 7)$, $B(-4, -5)$ and $C(4, 5)$ [CBSE 2008]
 - $A(3, 8)$, $B(-4, 2)$ and $C(5, -1)$ [CBSE 2008C]
 - $A(10, -6)$, $B(2, 5)$ and $C(-1, 3)$
- Find the area of quadrilateral $ABCD$ whose vertices are $A(3, -1)$, $B(9, -5)$, $C(14, 0)$ and $D(9, 19)$. [CBSE 2012]
- Find the area of quadrilateral $PQRS$ whose vertices are $P(-5, -3)$, $Q(-4, -6)$, $R(2, -3)$ and $S(1, 2)$. [CBSE 2015]
- Find the area of quadrilateral $ABCD$ whose vertices are $A(-3, -1)$, $B(-2, -4)$, $C(4, -1)$ and $D(3, 4)$. [CBSE 2013C]
- If $A(-7, 5)$, $B(-6, -7)$, $C(-3, -8)$ and $D(2, 3)$ are the vertices of a quadrilateral $ABCD$ then find the area of the quadrilateral. [CBSE 2017]
- Find the area of the triangle formed by joining the midpoints of the sides of the triangle whose vertices are $A(2, 1)$, $B(4, 3)$ and $C(2, 5)$. [CBSE 2011]

7. $A(7, -3)$, $B(5, 3)$ and $C(3, -1)$ are the vertices of a $\triangle ABC$ and AD is its median. Prove that the median AD divides $\triangle ABC$ into two triangles of equal areas.
8. Find the area of $\triangle ABC$ with $A(1, -4)$ and midpoints of sides through A being $(2, -1)$ and $(0, -1)$. [CBSE 2015]
9. $A(6, 1)$, $B(8, 2)$ and $C(9, 4)$ are the vertices of a parallelogram $ABCD$. If E is the midpoint of DC , find the area of $\triangle ADE$. [CBSE 2013C]
10. (i) If the vertices of $\triangle ABC$ be $A(1, -3)$, $B(4, p)$ and $C(-9, 7)$ and its area is 15 square units, find the values of p . [CBSE 2012]
- (ii) The area of a triangle is 5 sq units. Two of its vertices are $(2, 1)$ and $(3, -2)$. If the third vertex is $\left(\frac{7}{2}, y\right)$, find the value of y . [CBSE 2017]
11. Find the value of k so that the area of the triangle with vertices $A(k+1, 1)$, $B(4, -3)$ and $C(7, -k)$ is 6 square units. [CBSE 2015]
12. For what value of $k(k > 0)$ is the area of the triangle with vertices $(-2, 5)$, $(k, -4)$ and $(2k+1, 10)$ equal to 53 square units? [CBSE 2012]
13. Show that the following points are collinear:
- (i) $A(2, -2)$, $B(-3, 8)$ and $C(-1, 4)$
- (ii) $A(-5, 1)$, $B(5, 5)$ and $C(10, 7)$
- (iii) $A(5, 1)$, $B(1, -1)$ and $C(11, 4)$
- (iv) $A(8, 1)$, $B(3, -4)$ and $C(2, -5)$
14. Find the value of x for which the points $A(x, 2)$, $B(-3, -4)$ and $C(7, -5)$ are collinear. [CBSE 2015]
15. For what value of x are the points $A(-3, 12)$, $B(7, 6)$ and $C(x, 9)$ collinear?
16. For what value of y are the points $P(1, 4)$, $Q(3, y)$ and $R(-3, 16)$ are collinear?
17. Find the value of y for which the points $A(-3, 9)$, $B(2, y)$ and $C(4, -5)$ are collinear.
18. For what values of k are the points $A(8, 1)$, $B(3, -2k)$ and $C(k, -5)$ collinear. [CBSE 2015]
19. Find a relation between x and y , if the points $A(2, 1)$, $B(x, y)$ and $C(7, 5)$ are collinear. [CBSE 2009C]
20. Find a relation between x and y , if the points $A(x, y)$, $B(-5, 7)$ and $C(-4, 5)$ are collinear. [CBSE 2015]
21. Prove that the points $A(a, 0)$, $B(0, b)$ and $C(1, 1)$ are collinear, if $\frac{1}{a} + \frac{1}{b} = 1$.

22. If the points $P(-3, 9)$, $Q(a, b)$ and $R(4, -5)$ are collinear and $a + b = 1$, find the values of a and b . [CBSE 2014]
23. Find the area of $\triangle ABC$ with vertices $A(0, -1)$, $B(2, 1)$ and $C(0, 3)$. Also, find the area of the triangle formed by joining the midpoints of its sides. Show that the ratio of the areas of two triangles is $4 : 1$. [CBSE 2014]
24. If $a \neq b \neq c$, prove that (a, a^2) , (b, b^2) , $(0, 0)$ will not be collinear. [CBSE 2017]

ANSWERS (EXERCISE 6C)

1. (i) 11 sq units (ii) 53 sq units (iii) 37.5 sq units (iv) 24.5 sq units
2. 132 sq units 3. 28 sq units 4. 28 sq units 5. 77 sq units 6. 1 sq unit
8. 12 sq units 9. $\frac{3}{4}$ sq unit 10. (i) $p = -3$ or $p = -9$ (ii) $y = \frac{13}{2}$
11. $k = 3$ 12. $k = 3$ 14. $x = -63$ 15. $x = 2$
16. $y = -2$ 17. $y = -1$ 18. $k = 2$ or $k = \frac{11}{2}$ 19. $4x - 5y - 3 = 0$
20. $2x + y + 3 = 0$ 22. $a = 2, b = -1$ 23. 4 sq units, 1 sq unit

HINTS TO SOME SELECTED QUESTIONS

8. Let the vertices of $\triangle ABC$ be $A(1, -4)$, $B(x_2, y_2)$ and $C(x_3, y_3)$.

Let $D(2, -1)$ and $E(0, -1)$ be the midpoints of AB and AC respectively. Then,

$$\frac{1+x_2}{2} = 2, \frac{-4+y_2}{2} = -1$$

$$\Rightarrow 1+x_2 = 4, -4+y_2 = -2$$

$$\Rightarrow x_2 = 3, y_2 = 2.$$

$$\text{And, } \frac{1+x_3}{2} = 0, \frac{-4+y_3}{2} = -1$$

$$\Rightarrow 1+x_3 = 0, -4+y_3 = -2$$

$$\Rightarrow x_3 = -1, y_3 = 2.$$

In $\triangle ABC$, we have

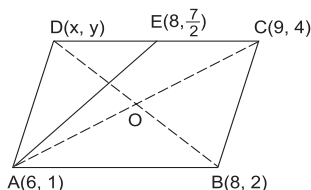
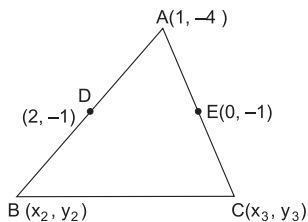
$$A(x_1 = 1, y_1 = -4), B(x_2 = 3, y_2 = 2) \text{ and } C(x_3 = -1, y_3 = 2).$$

Now, find $\text{ar}(\triangle ABC)$.

9. Let the fourth vertex be $D(x, y)$.

$$\text{Midpoint of } AC \text{ is } \left(\frac{6+9}{2}, \frac{4+1}{2} \right), \text{ i.e., } \left(\frac{15}{2}, \frac{5}{2} \right).$$

$$\text{Midpoint of } BD \text{ is } \left(\frac{8+x}{2}, \frac{2+y}{2} \right).$$



$$\therefore \frac{8+x}{2} = \frac{15}{2} \text{ and } \frac{2+y}{2} = \frac{5}{2} \Rightarrow x=7 \text{ and } y=3.$$

So, we get the point $D(7, 3)$.

$$\text{Midpoint of } DC \text{ is } E\left(\frac{7+9}{2}, \frac{3+4}{2}\right), \text{ i.e., } E\left(8, \frac{7}{2}\right).$$

Now, $A(6, 1)$, $D(7, 3)$ and $E\left(8, \frac{7}{2}\right)$ are vertices of $\triangle ADE$. Now find its area.

10. (i) Taking $A(x_1 = 1, y_1 = -3)$, $B(x_2 = 4, y_2 = p)$ and $C(x_3 = -9, y_3 = 7)$, we get
 $\text{ar}(\triangle ABC) = 15 \text{ sq units}$

$$\Rightarrow \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 15$$

$$\Rightarrow |1 \cdot (p - 7) + 4 \cdot (7 + 3) - 9 \cdot (-3 - p)| = 30$$

$$\Rightarrow |(p - 7) + 40 + 27 + 9p| = 30 \Rightarrow |10p + 60| = 30$$

$$\Rightarrow 10p + 60 = 30 \text{ or } 10p + 60 = -30 \Rightarrow p = -3 \text{ or } p = -9.$$

11. Let $A(x_1 = k + 1, y_1 = 1)$, $B(x_2 = 4, y_2 = -3)$ and $C(x_3 = 7, y_3 = -k)$ be the vertices of $\triangle ABC$. Then,

$$\begin{aligned} \text{ar}(\triangle ABC) &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |(k + 1)(-3 + k) + 4(-k - 1) + 7(1 + 3)| \\ &= \frac{1}{2} |(k^2 - 6k + 21)| = \frac{1}{2} |(k^2 - 6k + 9) + 12| \\ &= \frac{1}{2} |(k - 3)^2 + 12| = \frac{1}{2} [(k - 3)^2 + 12]. \end{aligned}$$

$$\therefore \frac{1}{2} [(k - 3)^2 + 12] = 6 \Rightarrow (k - 3)^2 + 12 = 12 \Rightarrow (k - 3)^2 = 0.$$

Hence, $k = 3$.

12. Let $A(x_1 = -2, y_1 = 5)$, $B(x_2 = k, y_2 = -4)$ and $C(x_3 = 2k + 1, y_3 = 10)$. Then,
 $\text{ar}(\triangle ABC) = 53 \text{ sq units}$

$$\Rightarrow \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 53$$

$$\Rightarrow \frac{1}{2} |(-2)(-4 - 10) + k(10 - 5) + (2k + 1)(5 + 4)| = 53$$

$$\Rightarrow |28 + 5k + 18k + 9| = 106$$

$$\Rightarrow |23k + 37| = 106$$

$$\Rightarrow 23k + 37 = 106 \text{ or } 23k + 37 = -106$$

$$\Rightarrow k = 3 \text{ or } k = -\frac{143}{23}.$$

14. Here, $A(x_1 = x, y_1 = 2)$, $B(x_2 = -3, y_2 = -4)$ and $C(x_3 = 7, y_3 = -5)$.

$$\therefore \Delta = 0 \Rightarrow x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow x(-4+5) - 3(-5-2) + 7(2+4) = 0$$

$$\Rightarrow x + 21 + 42 = 0 \Rightarrow x = -63.$$

20. Here, $A(x_1 = x, y_1 = y)$, $B(x_2 = -5, y_2 = 7)$ and $C(x_3 = -4, y_3 = 5)$.

$$\therefore \Delta = 0 \Rightarrow x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow x(7-5) - 5(5-y) - 4(y-7) = 0$$

$$\Rightarrow 2x - 25 + 5y - 4y + 28 = 0$$

$$\Rightarrow 2x + y + 3 = 0.$$

21. Here, $A(x_1 = a, y_1 = 0)$, $B(x_2 = 0, y_2 = b)$ and $C(x_3 = 1, y_3 = 1)$.

$$\therefore \Delta = 0 \Rightarrow x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow a(b-1) + 0 \cdot (1-0) + 1 \cdot (0-b) = 0$$

$$\Rightarrow ab - a - b = 0 \Rightarrow a + b = ab$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} = 1.$$

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EXERCISE 6D

Very-Short-Answer Questions

- Points $A(-1, y)$ and $B(5, 7)$ lie on a circle with centre $O(2, -3y)$. Find the values of y . [CBSE 2014]
- If the point $A(0, 2)$ is equidistant from the points $B(3, p)$ and $C(p, 5)$, find p . [CBSE 2014]
- $ABCD$ is a rectangle whose three vertices are $B(4, 0)$, $C(4, 3)$ and $D(0, 3)$. Find the length of one of its diagonal. [CBSE 2014]
- If the point $P(k-1, 2)$ is equidistant from the points $A(3, k)$ and $B(k, 5)$, find the values of k . [CBSE 2014]
- Find the ratio in which the point $P(x, 2)$ divides the join of $A(12, 5)$ and $B(4, -3)$. [CBSE 2014]
- Prove that the diagonals of a rectangle $ABCD$ with vertices $A(2, -1)$, $B(5, -1)$, $C(5, 6)$ and $D(2, 6)$ are equal and bisect each other. [CBSE 2014]
- Find the lengths of the medians AD and BE of $\triangle ABC$ whose vertices are $A(7, -3)$, $B(5, 3)$ and $C(3, -1)$. [CBSE 2014]
- If the point $C(k, 4)$ divides the join of $A(2, 6)$ and $B(5, 1)$ in the ratio $2 : 3$ then find the value of k . [CBSE 2013C]
- Find the point on x -axis which is equidistant from points $A(-1, 0)$ and $B(5, 0)$. [CBSE 2013C]
- Find the distance between the points $\left(-\frac{8}{5}, 2\right)$ and $\left(\frac{2}{5}, 2\right)$. [CBSE 2009C]

11. Find the value of a , so that the point $(3, a)$ lies on the line represented by $2x - 3y = 5$. [CBSE 2009]
12. If the points $A(4, 3)$ and $B(x, 5)$ lie on the circle with centre $O(2, 3)$, find the value of x . [CBSE 2009]
13. If $P(x, y)$ is equidistant from the points $A(7, 1)$ and $B(3, 5)$, find the relation between x and y .
14. If the centroid of $\triangle ABC$ having vertices $A(a, b)$, $B(b, c)$ and $C(c, a)$ is the origin, then find the value of $(a + b + c)$.
15. Find the centroid of $\triangle ABC$ whose vertices are $A(2, 2)$, $B(-4, -4)$ and $C(5, -8)$.
16. In what ratio does the point $C(4, 5)$ divide the join of $A(2, 3)$ and $B(7, 8)$?
17. If the points $A(2, 3)$, $B(4, k)$ and $C(6, -3)$ are collinear, find the value of k .

ANSWERS (EXERCISE 6D)

1. $y = 7$ or $y = -1$ 2. $p = 1$ 3. 5 units 4. $k = 1$ or $k = 5$ 5. $3 : 5$
7. $AD = 5$ units, $BE = 5$ units 8. $k = \frac{16}{5}$ 9. $P(2, 0)$ 10. 2 units
11. $a = \frac{1}{3}$ 12. $x = 2$ 13. $x - y = 2$ 14. $a + b + c = 0$ 15. $\left(1, \frac{-10}{3}\right)$
16. $2 : 3$ 17. $k = 0$

HINTS TO SOME SELECTED QUESTIONS

1. We know that the radii of a circle are equal.

$$\therefore OA = OB \Rightarrow OA^2 = OB^2$$

$$\Rightarrow (2 + 1)^2 + (-3y - y)^2 = (2 - 5)^2 + (-3y - 7)^2$$

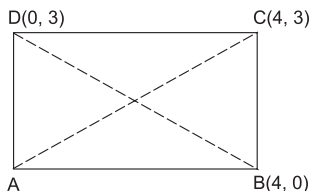
$$\Rightarrow 3^2 + 16y^2 = (-3)^2 + (9y^2 + 49 + 42y)$$

$$\Rightarrow 7y^2 - 42y - 49 = 0 \Rightarrow y^2 - 6y - 7 = 0$$

$$\Rightarrow (y - 7)(y + 1) = 0 \Rightarrow y = 7 \text{ or } y = -1.$$

3. Diagonal $AC =$ diagonal BD

$$= \sqrt{(4 - 0)^2 + (0 - 3)^2} = \sqrt{25} = 5 \text{ units.}$$



4. $PA = PB \Rightarrow PA^2 = PB^2$

$$\Rightarrow (k-1-3)^2 + (2-k)^2 = (k-1-k)^2 + (2-5)^2$$

$$\Rightarrow (k-4)^2 + (2-k)^2 = 1 + 9 = 10$$

$$\Rightarrow 2k^2 - 12k + 10 = 0 \Rightarrow k^2 - 6k + 5 = 0$$

$$\Rightarrow (k-1)(k-5) = 0$$

$$\Rightarrow k = 1 \text{ or } k = 5.$$

5. Let the required ratio be $k : 1$. Then,

$$\frac{-3k+5}{k+1} = 2 \Rightarrow -3k+5 = 2k+2 \Rightarrow 5k = 3 \Rightarrow k = \frac{3}{5}.$$

$$\text{Required ratio} = \left(\frac{3}{5} : 1\right) = 3 : 5.$$

6. Diagonal $AC = \sqrt{(5-2)^2 + (6+1)^2}$

$$= \sqrt{3^2 + 7^2} = \sqrt{9 + 49} = \sqrt{58}$$

Diagonal $BD = \sqrt{(5-2)^2 + (-1-6)^2}$

$$= \sqrt{3^2 + (-7)^2} = \sqrt{9 + 49} = \sqrt{58}.$$

$$\therefore \text{Diag. } AC = \text{Diag. } BD.$$

$$\text{Midpoint of } AC = \left(\frac{5+2}{2}, \frac{6+1}{2}\right) = \left(\frac{7}{2}, \frac{5}{2}\right).$$

$$\text{Midpoint of } BD = \left(\frac{5+2}{2}, \frac{-1+6}{2}\right) = \left(\frac{7}{2}, \frac{5}{2}\right).$$

7. Midpoint of BC is $D\left(\frac{5+3}{2}, \frac{3-1}{2}\right)$, i.e., $D(4, 1)$.

$$\text{Midpoint of } AC \text{ is } E\left(\frac{7+3}{2}, \frac{-3-1}{2}\right), \text{ i.e., } E(5, -2).$$

$$AD = \sqrt{(7-4)^2 + (-3-1)^2} = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5 \text{ units.}$$

$$BE = \sqrt{(5-5)^2 + (3+2)^2} = \sqrt{0 + 25} = \sqrt{25} = 5 \text{ units.}$$

8. $k = \frac{(2 \times 5 + 3 \times 2)}{(2+3)} = \frac{16}{5}.$

9. Let the required point be $P(x, 0)$. Then,

$$PA^2 = PB^2 \Rightarrow (x+1)^2 + (0-0)^2 = (x-5)^2 + (0-0)^2$$

$$\Rightarrow x^2 + 1 + 2x = x^2 + 25 - 10x$$

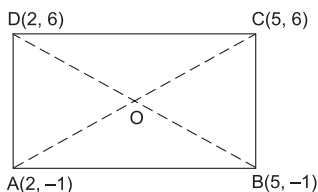
$$\Rightarrow 12x = 24 \Rightarrow x = 2.$$

$\therefore$ the required point is $P(2, 0)$.

10. Given distance $= \sqrt{\left(\frac{-8}{5} - \frac{2}{5}\right)^2 - (2-2)^2} = \sqrt{(-2)^2 - 0^2} = \sqrt{4} = 2 \text{ units.}$

11. Since $(3, a)$ lies on the line $2x - 3y = 5$, we have

$$2 \times 3 - 3a = 5 \Rightarrow 3a = 1 \Rightarrow a = \frac{1}{3}.$$



12. $(x-2)^2 + (5-3)^2 = (4-2)^2 + (3-3)^2$
 $\Rightarrow (x-2)^2 + 2^2 = 2^2 + 0^2$
 $\Rightarrow (x-2)^2 = 0 \Rightarrow x-2=0 \Rightarrow x=2.$
13. $(7-x)^2 + (1-y)^2 = (3-x)^2 + (5-y)^2$
 $\Rightarrow (49 + x^2 - 14x) + (1 + y^2 - 2y) = (9 + x^2 - 6x) + (25 + y^2 - 10y)$
 $\Rightarrow x^2 + y^2 - 14x - 2y + 50 = x^2 + y^2 - 6x - 10y + 34$
 $\Rightarrow 8x - 8y = 16 \Rightarrow x - y = 2.$
14. Centroid is $\left(\frac{a+b+c}{3}, \frac{a+b+c}{3}\right).$
 So, $\frac{a+b+c}{3} = 0 \Rightarrow a+b+c=0.$
16. Let the required ratio be $k:1$. Then, C is $\left(\frac{7k+2}{k+1}, \frac{8k+3}{k+1}\right).$
 $\therefore \frac{7k+2}{k+1} = 4 \Rightarrow 7k+2=4k+4 \Rightarrow 3k=2 \Rightarrow k=\frac{2}{3}.$
 Required ratio is $\frac{2}{3}:1$, i.e., $2:3.$
17. Let $(x_1=2, y_1=3), (x_2=4, y_2=k)$ and $(x_3=6, y_3=-3).$
 $\therefore \Delta=0 \Rightarrow x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)=0$
 $\Rightarrow 2(k+3) + 4(-3-3) + 6(3-k)=0$
 $\Rightarrow -4k=0 \Rightarrow k=0.$

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- The distance of the point $P(-6, 8)$ from the origin is [CBSE 2013C]
 (a) 8 (b) $2\sqrt{7}$ (c) 6 (d) 10
- The distance of the point $(-3, 4)$ from x -axis is [CBSE 2012]
 (a) 3 (b) -3 (c) 4 (d) 5
- The point on x -axis which is equidistant from points $A(-1, 0)$ and $B(5, 0)$ is [CBSE 2013]
 (a) $(0, 2)$ (b) $(2, 0)$ (c) $(3, 0)$ (d) $(0, 3)$
- If $R(5, 6)$ is the midpoint of the line segment AB joining the points $A(6, 5)$ and $B(4, 4)$ then y equals [CASE 2014]
 (a) 5 (b) 7 (c) 12 (d) 6
- If the point $C(k, 4)$ divides the join of the points $A(2, 6)$ and $B(5, 1)$ in the ratio $2:3$ then the value of k is [CBSE 2013C]
 (a) 16 (b) $\frac{28}{5}$ (c) $\frac{16}{5}$ (d) $\frac{8}{5}$

6. The perimeter of the triangle with vertices $(0, 4)$, $(0, 0)$ and $(3, 0)$ is [CBSE 2014]
 (a) $(7 + \sqrt{5})$ (b) 5 (c) 10 (d) 12
7. If $A(1, 3)$, $B(-1, 2)$, $C(2, 5)$ and $D(x, 4)$ are the vertices of a ||gm $ABCD$ then the value of x is [CBSE 2012]
 (a) 3 (b) 4 (c) 0 (d) $\frac{3}{2}$
8. If the points $A(x, 2)$, $B(-3, -4)$ and $C(7, -5)$ are collinear then the value of x is [CBSE 2014]
 (a) -63 (b) 63 (c) 60 (d) -60
9. The area of a triangle with vertices $A(5, 0)$, $B(8, 0)$ and $C(8, 4)$ in square units is [CBSE 2012]
 (a) 20 (b) 12 (c) 6 (d) 16
10. The area of $\triangle ABC$ with vertices $A(a, 0)$, $O(0, 0)$ and $B(0, b)$ in square units is [CBSE 2011]
 (a) ab (b) $\frac{1}{2}ab$ (c) $\frac{1}{2}a^2b^2$ (d) $\frac{1}{2}b^2$
11. If $P\left(\frac{a}{2}, 4\right)$ is the midpoint of the line segment joining the points $A(-6, 5)$ and $B(-2, 3)$ then the value of a is [CBSE 2011]
 (a) -8 (b) 3 (c) -4 (d) 4
12. $ABCD$ is a rectangle whose three vertices are $B(4, 0)$, $C(4, 3)$ and $D(0, 3)$. The length of one of its diagonals is [CBSE 2014]
 (a) 5 (b) 4 (c) 3 (d) 25
13. The coordinates of the point P dividing the line segment joining the points $A(1, 3)$ and $B(4, 6)$ in the ratio $2 : 1$ is [CBSE 2012]
 (a) $(2, 4)$ (b) $(3, 5)$ (c) $(4, 2)$ (d) $(5, 3)$
14. If the coordinates of one end of a diameter of a circle are $(2, 3)$ and the coordinates of its centre are $(-2, 5)$, then the coordinates of the other end of the diameter are [CBSE 2012]
 (a) $(-6, 7)$ (b) $(6, -7)$ (c) $(4, 2)$ (d) $(5, 3)$
15. In the given figure $P(5, -3)$ and $Q(3, y)$ are the points of trisection of the line segment joining $A(7, -2)$ and $B(1, -5)$. Then, y equals [CBSE 2012]
 (a) 2 (b) 4
 (c) -4 (d) $-\frac{5}{2}$



16. The midpoint of segment AB is $P(0, 4)$. If the coordinates of B are $(-2, 3)$, then the coordinates of A are [CBSE 2011]
(a) $(2, 5)$ (b) $(-2, -5)$ (c) $(2, 9)$ (d) $(-2, 11)$
17. The point P which divides the line segment joining the points $A(2, -5)$ and $B(5, 2)$ in the ratio $2 : 3$ lies in the quadrant [CBSE 2011]
(a) I (b) II (c) III (d) IV
18. If $A(-6, 7)$ and $B(-1, -5)$ are two given points then the distance $2AB$ is [CBSE 2011]
(a) 13 (b) 26 (c) 169 (d) 238
19. Which point on x -axis is equidistant from the points $A(7, 6)$ and $B(-3, 4)$?
(a) $(0, 4)$ (b) $(-4, 0)$ (c) $(3, 0)$ (d) $(0, 3)$
20. The distance of $P(3, 4)$ from the x -axis is
(a) 3 units (b) 4 units (c) 5 units (d) 1 unit
21. In what ratio does the x -axis divide the join of $A(2, -3)$ and $B(5, 6)$?
(a) $2 : 3$ (b) $3 : 5$ (c) $1 : 2$ (d) $2 : 1$
22. In what ratio does the y -axis divide the join of $P(-4, 2)$ and $Q(8, 3)$?
(a) $3 : 1$ (b) $1 : 3$ (c) $2 : 1$ (d) $1 : 2$
23. If $P(-1, 1)$ is the midpoint of the line segment joining $A(-3, b)$ and $B(1, b + 4)$ then $b = ?$
(a) 1 (b) -1 (c) 2 (d) 0
24. The line $2x + y - 4 = 0$ divides the line segment joining $A(2, -2)$ and $B(3, 7)$ in the ratio
(a) $2 : 5$ (b) $2 : 9$ (c) $2 : 7$ (d) $2 : 3$
25. If $A(4, 2)$, $B(6, 5)$ and $C(1, 4)$ be the vertices of $\triangle ABC$ and AD is a median, then the coordinates of D are
(a) $\left(\frac{5}{2}, 3\right)$ (b) $\left(5, \frac{7}{2}\right)$ (c) $\left(\frac{7}{2}, \frac{9}{2}\right)$ (d) none of these
26. If $A(-1, 0)$, $B(5, -2)$ and $C(8, 2)$ are the vertices of a $\triangle ABC$ then its centroid is
(a) $(12, 0)$ (b) $(6, 0)$ (c) $(0, 6)$ (d) $(4, 0)$
27. Two vertices of $\triangle ABC$ are $A(-1, 4)$ and $B(5, 2)$ and its centroid is $G(0, -3)$. Then, the coordinates of C are
(a) $(4, 3)$ (b) $(4, 15)$ (c) $(-4, -15)$ (d) $(-15, -4)$
28. The points $A(-4, 0)$, $B(4, 0)$ and $C(0, 3)$ are the vertices of a triangle, which is
(a) isosceles (b) equilateral (c) scalene (d) right angled

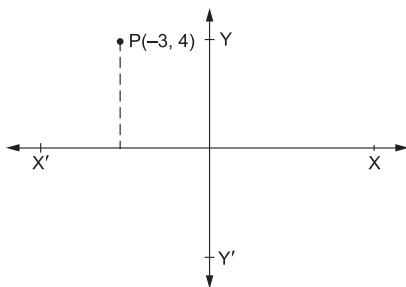
29. The points $P(0, 6)$, $Q(-5, 3)$ and $R(3, 1)$ are the vertices of a triangle, which is
 (a) equilateral (b) isosceles (c) scalene (d) right angled
30. If the points $A(2, 3)$, $B(5, k)$ and $C(6, 7)$ are collinear then
 (a) $k = 4$ (b) $k = 6$ (c) $k = \frac{-3}{2}$ (d) $k = \frac{11}{4}$
31. If the points $A(1, 2)$, $O(0, 0)$ and $C(a, b)$ are collinear then
 (a) $a = b$ (b) $a = 2b$ (c) $2a = b$ (d) $a + b = 0$
32. The area of $\triangle ABC$ with vertices $A(3, 0)$, $B(7, 0)$ and $C(8, 4)$ is
 (a) 14 sq units (b) 28 sq units
 (c) 8 sq units (d) 6 sq units
33. $AOBC$ is a rectangle whose three vertices are $A(0, 3)$, $O(0, 0)$ and $B(5, 0)$. The length of each of its diagonals is
 (a) 5 units (b) 3 units (c) 4 units (d) $\sqrt{34}$ units
34. If the distance between the points $A(4, p)$ and $B(1, 0)$ is 5 then
 (a) $p = 4$ only (b) $p = -4$ only (c) $p = \pm 4$ (d) $p = 0$

ANSWERS (MCQ)

1. (d) 2. (c) 3. (b) 4. (b) 5. (c) 6. (d) 7. (b) 8. (a)
 9. (c) 10. (b) 11. (a) 12. (a) 13. (b) 14. (a) 15. (c) 16. (a)
 17. (d) 18. (b) 19. (c) 20. (b) 21. (c) 22. (d) 23. (b) 24. (b)
 25. (c) 26. (d) 27. (c) 28. (a) 29. (d) 30. (b) 31. (c) 32. (c)
 33. (d) 34. (c)

HINTS TO SOME SELECTED QUESTIONS

1. $OP^2 = (-6)^2 + 8^2 = 36 + 64 = 100 \Rightarrow OP = \sqrt{100} = 10$ units.
 2. Clearly, the distance of the point $P(-3, 4)$ from x -axis is 4 units.



3. Let the required point be $P(x, 0)$. Then,

$$\begin{aligned} PA^2 &= PB^2 \Rightarrow (x+1)^2 = (x-5)^2 \\ &\Rightarrow x^2 + 2x + 1 = x^2 - 10x + 25 \\ &\Rightarrow 12x = 24 \Rightarrow x = 2. \end{aligned}$$

So, the required point is $P(2, 0)$.

4. $\frac{5+y}{2} = 6 \Rightarrow 5+y = 12 \Rightarrow y = 7.$

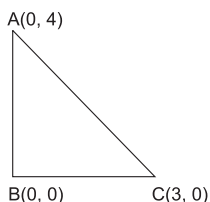
5. $k = \frac{(2 \times 5) + (3 \times 2)}{(2+3)} = \frac{16}{5}.$

6. $AB = 4$ units, $BC = 3$ units.

$$\begin{aligned} AC^2 &= (3-0)^2 + (0-4)^2 \\ &= (9+16) = 25. \end{aligned}$$

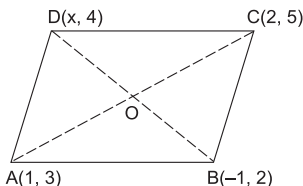
$$\Rightarrow AC = 5 \text{ units.}$$

$$\begin{aligned} \therefore \text{perimeter} &= (4+3+5) \text{ units} \\ &= 12 \text{ units.} \end{aligned}$$



7. Midpoint of BD = midpoint of AC

$$\Rightarrow \frac{-1+x}{2} = \frac{1+2}{2} \Rightarrow -1+x = 3 \Rightarrow x = 4.$$



8. Here, $(x_1 = x, y_1 = 2)$, $(x_2 = -3, y_2 = -4)$ and $(x_3 = 7, y_3 = -5)$.

$$\therefore x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow x(-4+5) - 3(-5-2) + 7(2+4) = 0$$

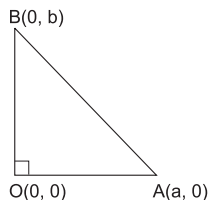
$$\Rightarrow x + 21 + 42 = 0 \Rightarrow x = -63.$$

9. Here, $(x_1 = 5, y_1 = 0)$, $(x_2 = 8, y_2 = 0)$ and $(x_3 = 8, y_3 = 4)$.

$$\therefore \Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |5(0-4) + 8(4-0) + 8(0-0)|$$

$$= \frac{1}{2} |-20 + 32| = \left(\frac{1}{2} \times 12\right) = 6 \text{ sq units.}$$



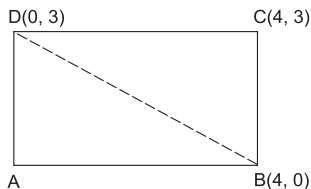
10. Here, base = a units and height = b units.

$$\therefore \text{area} = \left(\frac{1}{2} \times \text{base} \times \text{height}\right)$$

$$= \frac{1}{2} ab \text{ sq units.}$$

$$11. \frac{a}{2} = \frac{(-6-2)}{2} = -4 \Rightarrow a = -8.$$

$$12. \text{ Diagonal } BD = \sqrt{(4-0)^2 + (0-3)^2} = \sqrt{16+9} \\ = \sqrt{25} = 5 \text{ units.}$$



$$13. \text{ Coordinates of } P \text{ are } \left(\frac{(2 \times 4 + 1 \times 1)}{2+1}, \frac{(2 \times 6 + 1 \times 3)}{2+1} \right) = (3, 5).$$

14. Let $C(-2, 5)$ be the centre of the given circle and $A(2, 3)$ and $B(x, y)$ be the end points of a diameter ACB .

Then, C is the midpoint of AB .

$$\therefore \frac{2+x}{2} = -2 \text{ and } \frac{3+y}{2} = 5$$

$$\Rightarrow 2+x = -4 \text{ and } 3+y = 10$$

$$\Rightarrow x = -6 \text{ and } y = 7.$$

So, the coordinates of B are $(-6, 7)$.

15. $Q(3, y)$ divides AB in the ratio $2 : 1$.

$$\text{So, } Q \text{ is } \left(\frac{2 \times 1 + 1 \times 7}{2+1}, \frac{2 \times (-5) + 1 \times (-2)}{2+1} \right), \text{ i.e., } (3, -4).$$

Hence, $y = -4$.

16. Let the point A be (a, b) . Then,

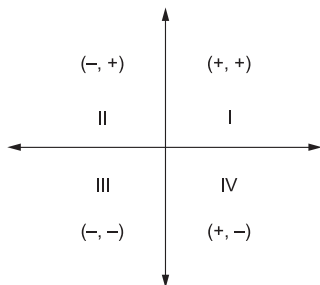
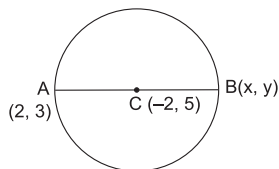
$$\frac{a+(-2)}{2} = 0 \text{ and } \frac{b+3}{2} = 4$$

$$\Rightarrow a-2=0 \text{ and } b+3=8 \Rightarrow a=2, b=5.$$

$\therefore$ the point A is $(2, 5)$.

$$17. \text{ The point } P \text{ is given by } P \left(\frac{2 \times 5 + 3 \times 2}{2+3}, \frac{2 \times 2 - 3 \times 5}{2+3} \right) = P \left(3, \frac{-11}{5} \right).$$

So, P lies in IV quadrant.



$$18. 2AB = 2 \times \sqrt{(-1+6)^2 + (-5-7)^2} = 2 \times \sqrt{(5)^2 + (-12)^2} \\ = 2 \times \sqrt{169} = 2 \times 13 = 26.$$

19. Let the required point be $P(x, 0)$. Then,

$$AP^2 = BP^2 \Rightarrow (x-7)^2 + (0-6)^2 = (x+3)^2 + (0-4)^2 \\ \Rightarrow x^2 - 14x + 85 = x^2 + 6x + 25 \\ \Rightarrow 20x = 60 \Rightarrow x = 3.$$

$\therefore$ the required point is $P(3, 0)$.

21. Let the x -axis cut AB at $P(x, 0)$ in the ratio $k : 1$.

$$\text{Then, } \frac{6k-3}{k+1} = 0 \Rightarrow 6k-3-0 \Rightarrow 6k=3 \Rightarrow k = \frac{1}{2}.$$

$$\therefore \text{ required ratio} = \left(\frac{1}{2} : 1\right) = 1 : 2.$$

22. Let the y -axis cut AB at $P(0, y)$ in the ratio $k : 1$. Then,

$$P\left(\frac{8k-4}{k+1}, \frac{3k+2}{k+1}\right) = P(0, y) \Rightarrow \frac{8k-4}{k+1} = 0 \\ \Rightarrow 8k-4=0 \Rightarrow k = \frac{1}{2}.$$

$$\therefore \text{ required ratio} = \left(\frac{1}{2} : 1\right) = 1 : 2.$$

$$23. \text{ We have } \frac{b+(b+4)}{2} = 1 \Rightarrow 2b+4=2 \Rightarrow 2b=-2 \Rightarrow b=-1.$$

24. Let the required ratio be $k : 1$.

$$\text{Then, the point of division is } P\left(\frac{3k+2}{k+1}, \frac{7k-2}{k+1}\right).$$

This point lies on the line $2x + y - 4 = 0$.

$$\therefore \frac{2(3k+2)}{k+1} + \frac{(7k-2)}{k+1} - 4 = 0 \Rightarrow 6k+4+7k-2-4k-4=0 \\ \Rightarrow 9k=2 \Rightarrow k = \frac{2}{9}.$$

So, the required ratio is $\left(\frac{2}{9} : 1\right)$, i.e., $(2 : 9)$.

$$25. \text{ Midpoint of } BC \text{ is } D\left(\frac{6+1}{2}, \frac{5+4}{2}\right) = D\left(\frac{7}{2}, \frac{9}{2}\right).$$

$$26. \text{ Centroid is } G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right) = G\left(\frac{-1+5+8}{3}, \frac{0-2+2}{3}\right) = (4, 0).$$

27. Let the vertex C be $C(x, y)$. Then,

$$\frac{-1+5+x}{3} = 0 \text{ and } \frac{4+2+y}{3} = -3 \Rightarrow x+4=0 \text{ and } 6+y=-9$$

$$\therefore x = -4 \text{ and } y = -15.$$

So, the coordinates of C are $(-4, -15)$.

$$28. AB^2 = (4 + 4)^2 + (0 - 0)^2 = 8^2 + 0^2 = 64 + 0 = 64 \Rightarrow AB = \sqrt{64} = 8 \text{ units.}$$

$$BC^2 = (0 - 4)^2 + (3 - 0)^2 = (-4)^2 + 3^2 = 16 + 9 = 25 \Rightarrow BC = \sqrt{25} = 5 \text{ units.}$$

$$AC^2 = (0 + 4)^2 + (3 - 0)^2 = 4^2 + 3^2 = 16 + 9 = 25 \Rightarrow AC = \sqrt{25} = 5 \text{ units.}$$

$\therefore \triangle ABC$ is isosceles.

$$29. PQ^2 = (-5 - 0)^2 + (3 - 6)^2 = (-5)^2 + (-3)^2 = 25 + 9 = 34,$$

$$QR^2 = (3 + 5)^2 + (1 - 3)^2 = 8^2 + (-2)^2 = 64 + 4 = 68,$$

$$PR^2 = (3 - 0)^2 + (1 - 6)^2 = 3^2 + (-5)^2 = (9 + 25) = 34.$$

$$\therefore PQ^2 + PR^2 = QR^2.$$

Hence, $\triangle PQR$ is right-angled.

$$30. \text{ Here, } (x_1 = 2, y_1 = 3), (x_2 = 5, y_2 = k) \text{ and } (x_3 = 6, y_3 = 7).$$

Since the given points are collinear, we must have:

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow 2(k - 7) + 5(7 - 3) + 6(3 - k) = 0 \Rightarrow 2k - 14 + 20 + 18 - 6k = 0$$

$$\Rightarrow 4k = 24 \Rightarrow k = 6.$$

$$31. \text{ Here, } (x_1 = 1, y_1 = 2), (x_2 = 0, y_2 = 0) \text{ and } (x_3 = a, y_3 = b).$$

Since the given points are collinear, we have

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow 1 \cdot (0 - b) + 0 \cdot (b - 2) + a \cdot (2 - 0) = 0 \Rightarrow -b + 0 + 2a = 0 \Rightarrow 2a = b.$$

$$32. \text{ Here, } (x_1 = 3, y_1 = 0), (x_2 = 7, y_2 = 0) \text{ and } (x_3 = 8, y_3 = 4).$$

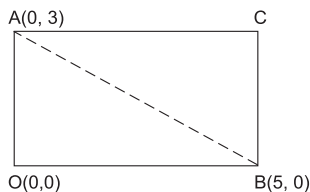
$$\begin{aligned} \therefore \text{ area of } \triangle ABC &= \frac{1}{2} \{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\} \\ &= \frac{1}{2} \{3(0 - 4) + 7(4 - 0) + 8(0 - 0)\} = \frac{1}{2} \{-12 + 28 + 0\} \\ &= \left(\frac{1}{2} \times 16\right) = 8 \text{ sq units.} \end{aligned}$$

$$33. \text{ Diagonal OC}$$

$$= \text{diagonal AB}$$

$$= \sqrt{(5 - 0)^2 + (0 - 3)^2}$$

$$= \sqrt{25 + 9} = \sqrt{34} \text{ units.}$$



$$34. (4 - 1)^2 + (p - 0)^2 = 5^2 \Rightarrow p^2 = (25 - 9) = 16 \Rightarrow p = \pm 4.$$





Triangles

CONGRUENT FIGURES *Two geometric figures which have the same shape and size are known as congruent figures.*

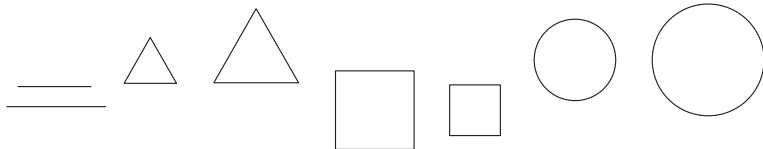
Congruent figures are alike in every respect.

SIMILAR FIGURES *Geometric figures which have the same shape but different sizes are known as similar figures.*

Two congruent figures are always similar but two similar figures need not be congruent.

Examples

- (i) Any two line segments are similar.
- (ii) Any two equivalent triangles are similar.
- (iii) Any two squares are similar.
- (iv) Any two circles are similar.



SIMILAR POLYGONS

Two polygons having the same number of sides are said to be similar, if

- (i) *their corresponding angles are equal, and*
- (ii) *the lengths of their corresponding sides are proportional.*

If two polygons $ABCDE$ and $PQRST$ are similar, we write, $ABCDE \sim PQRST$, where the symbol ' $\sim$ ' stands for 'is similar to'.

The constant ratio between the corresponding sides of two similar figures is known as the *scale factor*, or the *representative fraction*. Since triangles are also polygons, so the same set of conditions apply for the similarity of triangles.

EQUIANGULAR TRIANGLES

Two triangles are said to be equiangular if their corresponding angles are equal.

SIMILAR TRIANGLES

Two triangles are said to be similar to each other if

- (i) their corresponding angles are equal, and
- (ii) their corresponding sides are proportional.

RESULTS ON SIMILAR TRIANGLES**(BASIC-PROPORTIONALITY THEOREM) OR (THALES' THEOREM)**

THEOREM 1 If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points then the other two sides are divided in the same ratio. [CBSE 2002C, '04C, '05, '06C, '07, '09, '10]

GIVEN A $\triangle ABC$ in which $DE \parallel BC$ and DE intersects AB and AC at D and E respectively.

TO PROVE $\frac{AD}{DB} = \frac{AE}{EC}$.

CONSTRUCTION Join BE and CD .

Draw $EL \perp AB$ and $DM \perp AC$.

PROOF We have

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AD \times EL \quad [\because \Delta = \frac{1}{2} \times \text{base} \times \text{height}]$$

$$\text{and } \text{ar}(\triangle DBE) = \frac{1}{2} \times DB \times EL.$$

$$\therefore \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DBE)} = \frac{\frac{1}{2} \times AD \times EL}{\frac{1}{2} \times DB \times EL} = \frac{AD}{DB}. \quad \dots (i)$$

$$\text{Again, } \text{ar}(\triangle ADE) = \text{ar}(\triangle AED) = \frac{1}{2} \times AE \times DM$$

$$\text{and } \text{ar}(\triangle ECD) = \frac{1}{2} \times EC \times DM.$$

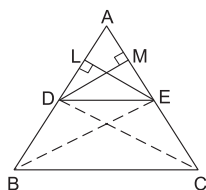
$$\therefore \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ECD)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}. \quad \dots (ii)$$

Now, $\triangle DBE$ and $\triangle ECD$ being on the same base DE and between the same parallels DE and BC , we have

$$\text{ar}(\triangle DBE) = \text{ar}(\triangle ECD) \quad \dots (iii)$$

From (i), (ii) and (iii), we have

$$\frac{AD}{DB} = \frac{AE}{EC}.$$



COROLLARY In a $\triangle ABC$, a line $DE \parallel BC$ intersects AB in D and AC in E , then prove that

$$(i) \frac{AD}{DB} = \frac{AE}{EC} \quad (ii) \frac{AD}{AB} = \frac{AE}{AC}.$$

PROOF (i) From Basic-Proportionality theorem, we have

$$\begin{aligned} \frac{AD}{DB} = \frac{AE}{EC} &\Rightarrow \frac{AD}{DB} + 1 = \frac{AE}{EC} + 1 \\ &\Rightarrow \frac{AD + DB}{DB} = \frac{AE + EC}{EC} \Rightarrow \frac{AB}{DB} = \frac{AC}{EC}. \end{aligned}$$

(ii) From Basic-Proportionality theorem, we have

$$\begin{aligned} \frac{AD}{DB} = \frac{AE}{EC} &\Rightarrow \frac{DB}{AD} = \frac{EC}{AE} \\ &\Rightarrow \left(1 + \frac{DB}{AD}\right) = \left(1 + \frac{EC}{AE}\right) \\ &\Rightarrow \frac{(AD + DB)}{AD} = \frac{(AE + EC)}{AE} \\ &\Rightarrow \frac{AB}{AD} = \frac{AC}{AE} \Rightarrow \frac{AD}{AB} = \frac{AE}{AC}. \end{aligned}$$

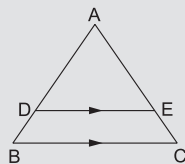
SUMMARY

In $\triangle ABC$, let $DE \parallel BC$. Then,

$$(i) \frac{AD}{DB} = \frac{AE}{EC} \text{ (B.P.T.)}$$

$$(ii) \frac{AB}{DB} = \frac{AC}{EC}$$

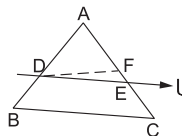
$$(iii) \frac{AD}{AB} = \frac{AE}{AC}$$



THEOREM 2 (Converse of Thales' theorem) If a line divides any two sides of a triangle in the same ratio then the line must be parallel to the third side.

GIVEN A $\triangle ABC$ and a line l intersecting AB at D and

AC at E , such that $\frac{AD}{DB} = \frac{AE}{EC}$.



TO PROVE $DE \parallel BC$.

PROOF If possible, let DE not be parallel to BC . Then, there must be another line through D , which is parallel to BC . Let $DF \parallel BC$.

Then, by Thales' theorem, we have

$$\frac{AD}{DB} = \frac{AF}{FC} \quad \dots (i)$$

$$\text{But, } \frac{AD}{DB} = \frac{AE}{EC} \text{ (given).} \quad \dots (ii)$$

From (i) and (ii), we get

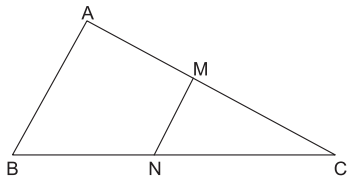
$$\begin{aligned}\frac{AF}{FC} = \frac{AE}{EC} &\Rightarrow \frac{AF}{FC} + 1 = \frac{AE}{EC} + 1 \Rightarrow \frac{AF+FC}{FC} = \frac{AE+EC}{EC} \\ &\Rightarrow \frac{AC}{FC} = \frac{AC}{EC} \Rightarrow \frac{1}{FC} = \frac{1}{EC} \Rightarrow FC = EC.\end{aligned}$$

This is possible only when E and F coincide.

Hence, $DE \parallel BC$.

SOLVED EXAMPLES

EXAMPLE 1 In the given figure, $MN \parallel AB$, $BC = 7.5$ cm, $AM = 4$ cm and $MC = 2$ cm. Find the length of BN . [CBSE 2010]



SOLUTION In $\triangle ABC$, $MN \parallel AB$.

$$\therefore \frac{MC}{AC} = \frac{NC}{BC} \quad [\text{by Thales' theorem}]$$

$$\Rightarrow \frac{MC}{AM+MC} = \frac{NC}{BC}$$

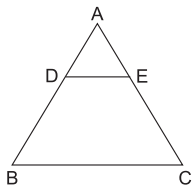
$$\Rightarrow \frac{2}{4+2} = \frac{x}{7.5}, \text{ where } NC = x \text{ cm}$$

$$\Rightarrow x = \frac{2 \times 7.5}{6} = \frac{15}{6} = 2.5$$

$$\Rightarrow NC = 2.5 \text{ cm.}$$

$$\text{Hence, } BN = BC - NC = (7.5 - 2.5) \text{ cm} = 5 \text{ cm.}$$

EXAMPLE 2 In the given figure, $DE \parallel BC$ and $\frac{AD}{DB} = \frac{3}{5}$. If $AC = 4.8$ cm, find the length of AE . [CBSE 2008C]



SOLUTION Let $AE = x$ cm.

$$\text{Then, } EC = (AC - AE) = (4.8 - x) \text{ cm.}$$

Now, in $\triangle ABC$, $DE \parallel BC$.

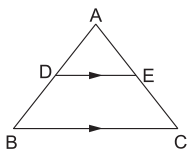
$$\therefore \frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{3}{5} = \frac{x}{(4.8 - x)}$$

$$\Rightarrow 3(4.8 - x) = 5x \Rightarrow 8x = 14.4$$

$$\Rightarrow x = 1.8.$$

$$\text{Hence, } AE = 1.8 \text{ cm.}$$

EXAMPLE 3 In the given figure, in $\triangle ABC$, $DE \parallel BC$ so that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm and $CE = (5x - 3)$ cm. Find the value of x .
[CBSE 2002C]



SOLUTION In $\triangle ABC$, $DE \parallel BC$.

$$\therefore \frac{AD}{BD} = \frac{AE}{CE} \quad [\text{by Thales' theorem}]$$

$$\Rightarrow \frac{4x-3}{3x-1} = \frac{8x-7}{5x-3} \Rightarrow (4x-3)(5x-3) = (3x-1)(8x-7)$$

$$\Rightarrow 20x^2 - 27x + 9 = 24x^2 - 29x + 7$$

$$\Rightarrow 4x^2 - 2x - 2 = 0 \Rightarrow 2x^2 - x - 1 = 0$$

$$\Rightarrow 2x^2 - 2x + x - 1 = 0 \Rightarrow 2x(x-1) + (x-1) = 0$$

$$\Rightarrow (x-1)(2x+1) = 0$$

$$\Rightarrow (x-1) = 0 \quad \text{or} \quad (2x+1) = 0$$

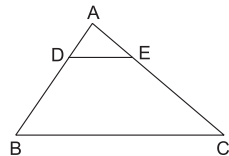
$$\Rightarrow x = 1 \quad \text{or} \quad x = -\frac{1}{2}.$$

$$\text{But, } x = -\frac{1}{2} \Rightarrow AD = \left[4 \times \left(-\frac{1}{2} \right) - 3 \right] = -5.$$

And, distance can never be negative. So, $x \neq -\frac{1}{2}$.

Hence, $x = 1$.

EXAMPLE 4 If D and E are points on the sides AB and AC respectively of $\triangle ABC$ such that $AB = 5.6$ cm, $AD = 1.4$ cm, $AC = 7.2$ cm and $AE = 1.8$ cm, show that $DE \parallel BC$.



SOLUTION Given, $AB = 5.6$ cm, $AD = 1.4$ cm,

$AC = 7.2$ cm and $AE = 1.8$ cm.

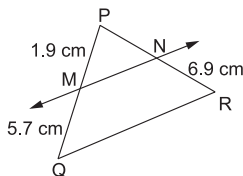
$$\therefore \frac{AD}{AB} = \frac{1.4}{5.6} = \frac{1}{4} \quad \text{and} \quad \frac{AE}{AC} = \frac{1.8}{7.2} = \frac{1}{4}$$

$$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}.$$

Hence, by the converse of Thales' theorem, $DE \parallel BC$.

EXAMPLE 5 In the adjoining figure, $MN \parallel QR$.

Find (i) PN and (ii) PR .



SOLUTION In $\triangle PQR$, $MN \parallel QR$.

$$\therefore \frac{PM}{MQ} = \frac{PN}{NR} \quad [\text{by Thales' theorem}]$$

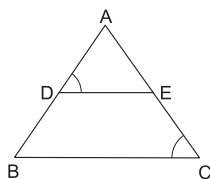
$$\Rightarrow \frac{1.9}{5.7} = \frac{x}{6.9}, \text{ where } PN = x \text{ cm}$$

$$\Rightarrow x = \frac{1.9 \times 6.9}{5.7} = 2.3.$$

Hence, (i) $PN = x \text{ cm} = 2.3 \text{ cm}$

and (ii) $PR = PN + NR = (2.3 + 6.9) \text{ cm} = 9.2 \text{ cm}$.

EXAMPLE 6 In the given figure, $\frac{AD}{DB} = \frac{AE}{EC}$ and $\angle ADE = \angle ACB$. Prove that $\triangle ABC$ is an isosceles triangle.



SOLUTION We have

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow DE \parallel BC \quad [\text{by the converse of Thales' theorem}]$$

$$\therefore \angle ADE = \angle ABC \quad (\text{corresponding } \angle).$$

But, $\angle ADE = \angle ACB$ (given).

$$\therefore \angle ABC = \angle ACB.$$

So, $AB = AC$ [sides opposite to equal angles].

Hence, $\triangle ABC$ is an isosceles triangle.

EXAMPLE 7 M and N are points on the sides AC and BC respectively of a $\triangle ABC$. In each of the following cases, state whether $MN \parallel AB$.

(i) $CM = 4.2 \text{ cm}$, $MA = 2.8 \text{ cm}$, $NB = 3.6 \text{ cm}$, $CN = 5.7 \text{ cm}$

(ii) $CB = 6.92 \text{ cm}$, $CN = 1.04 \text{ cm}$, $CA = 1.73 \text{ cm}$, $CM = 0.26 \text{ cm}$

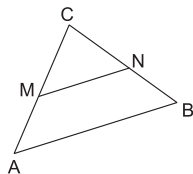
(iii) $CM = 5.1 \text{ cm}$, $CA = 6.8 \text{ cm}$, $CB = 5.6 \text{ cm}$, $NB = 1.4 \text{ cm}$

SOLUTION (i) We have

$$\frac{CM}{MA} = \frac{4.2}{2.8} = \frac{3}{2} \text{ and } \frac{CN}{NB} = \frac{5.7}{3.6} = \frac{19}{12}.$$

$$\text{Since } \frac{CM}{MA} \neq \frac{CN}{NB}.$$

So, MN is not parallel to AB .



(ii) We have

$$MA = CA - CM = (1.73 - 0.26) \text{ cm} = 1.47 \text{ cm}$$

$$\text{and } NB = CB - CN = (6.92 - 1.04) \text{ cm} = 5.88 \text{ cm}.$$

$$\therefore \frac{CM}{MA} = \frac{0.26}{1.47} = \frac{26}{147} \text{ and } \frac{CN}{NB} = \frac{1.04}{5.88} = \frac{26}{147}.$$

Clearly, $\frac{CM}{MA} = \frac{CN}{NB}$ and so $MN \parallel AB$

[by the converse of Thales' theorem].

(iii) We have

$$MA = CA - CM = (6.8 - 5.1) \text{ cm} = 1.7 \text{ cm}$$

$$\text{and } CN = CB - NB = (5.6 - 1.4) \text{ cm} = 4.2 \text{ cm}.$$

$$\therefore \frac{CM}{MA} = \frac{5.1}{1.7} = \frac{3}{1} \text{ and } \frac{CN}{NB} = \frac{4.2}{1.4} = \frac{3}{1}.$$

Clearly, $\frac{CM}{MA} = \frac{CN}{NB}$ and so $MN \parallel AB$

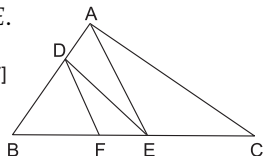
[by the converse of Thales' theorem].

EXAMPLE 8

In the given figure, $DE \parallel AC$ and $DF \parallel AE$.

Prove that $\frac{BF}{FE} = \frac{BE}{EC}$.

[CBSE 2005, '07]



SOLUTION

In $\triangle BAE$, $DF \parallel AE$.

$$\therefore \frac{BD}{DA} = \frac{BF}{FE}. \quad \dots \text{(i) [by Thales' theorem]}$$

In $\triangle BAC$, $DE \parallel AC$.

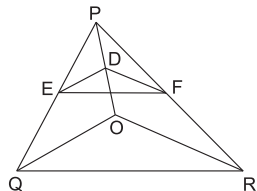
$$\therefore \frac{BD}{DA} = \frac{BE}{EC}. \quad \dots \text{(ii) [by Thales' theorem]}$$

From (i) and (ii), we get

$$\frac{BF}{FE} = \frac{BE}{EC} \quad \left[\text{each equal to } \frac{BD}{DA} \right].$$

EXAMPLE 9

In the figure given along side, $DE \parallel OQ$ and $DF \parallel OR$. Show that $EF \parallel QR$.



SOLUTION

In $\triangle POQ$, $DE \parallel OQ$.

$$\therefore \frac{PD}{DO} = \frac{PE}{EQ}. \quad \dots \text{(i) [by Thales' theorem]}$$

In $\triangle POR$, $DF \parallel OR$.

$$\therefore \frac{PD}{DO} = \frac{PF}{FR}. \quad \dots (ii)$$

From (i) and (ii), we get

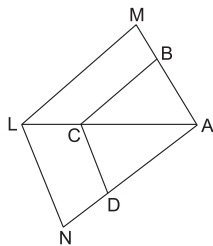
$$\frac{PE}{EQ} = \frac{PF}{FR} \quad \left[\text{each equal to } \frac{PD}{DO} \right].$$

Thus, in $\triangle PQR$, E and F are points on PQ and PR respectively such that $\frac{PE}{EQ} = \frac{PF}{FR}$.

Hence, $EF \parallel QR$ [by the converse of Thales' theorem].

EXAMPLE 10 In the given figure, $LM \parallel CB$ and $LN \parallel CD$.

Prove that $\frac{AM}{AB} = \frac{AN}{AD}$.



SOLUTION In $\triangle ALM$, $LM \parallel CB$.

$$\therefore \frac{AB}{AM} = \frac{AC}{AL} \Rightarrow \frac{AM}{AB} = \frac{AL}{AC}. \quad \dots (i) \quad [\text{by Thales' theorem}]$$

In $\triangle ALN$, $LN \parallel CD$.

$$\therefore \frac{AC}{AL} = \frac{AD}{AN} \Rightarrow \frac{AL}{AC} = \frac{AN}{AD}. \quad \dots (ii) \quad [\text{by Thales' theorem}]$$

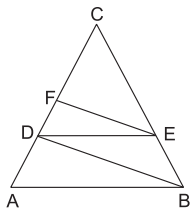
From (i) and (ii), we get

$$\frac{AM}{AB} = \frac{AN}{AD}.$$

EXAMPLE 11 In the given figure, $AB \parallel DE$ and $BD \parallel EF$.

Prove that $DC^2 = CF \times AC$.

[CBSE 2004C, '10]



SOLUTION In $\triangle ABC$, $AB \parallel DE$.

$$\therefore \frac{CD}{DA} = \frac{CE}{EB}. \quad \dots (i) \quad [\text{by Thales' theorem}]$$

In $\triangle CDB$, $BD \parallel EF$.

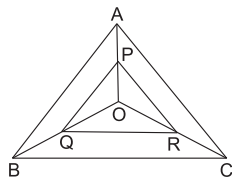
$$\therefore \frac{CF}{FD} = \frac{CE}{EB}. \quad \dots (ii) \quad [\text{by Thales' theorem}]$$

From (i) and (ii), we get

$$\frac{CD}{DA} = \frac{CF}{FD}$$

$$\begin{aligned} \Rightarrow \frac{DA}{DC} &= \frac{FD}{CF} && \text{[taking reciprocals]} \\ \Rightarrow \frac{DA}{DC} + 1 &= \frac{FD}{CF} + 1 \\ \Rightarrow \frac{DA+DC}{DC} &= \frac{FD+CF}{CF} \\ \Rightarrow \frac{AC}{DC} &= \frac{DC}{CF} \\ \Rightarrow DC^2 &= CF \times AC. \end{aligned}$$

EXAMPLE 12 In the given figure, $PQ \parallel AB$ and $PR \parallel AC$.
Prove that $QR \parallel BC$. [CBSE 2002, '05C]



SOLUTION In $\triangle OAB$, $PQ \parallel AB$.

$$\therefore \frac{OP}{PA} = \frac{OQ}{QB} \quad \dots \text{(i) [by Thales' theorem]}$$

In $\triangle AOC$, $PR \parallel AC$.

$$\therefore \frac{OP}{PA} = \frac{OR}{RC} \quad \dots \text{(ii) [by Thales' theorem]}$$

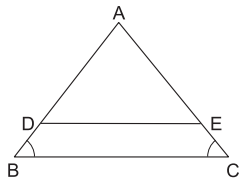
From (i) and (ii), we get

$$\frac{OQ}{QB} = \frac{OR}{RC} \text{ in } \triangle OBC.$$

Thus, in $\triangle OBC$, Q and R are points on OB and OC respectively such that $\frac{OQ}{QB} = \frac{OR}{RC}$.

Hence, by the converse of Thales' theorem, $QR \parallel BC$.

EXAMPLE 13 In the given figure, in $\triangle ABC$, $\angle B = \angle C$ and $BD = CE$. Prove that $DE \parallel BC$.



SOLUTION GIVEN A $\triangle ABC$ in which $\angle B = \angle C$ and $BD = CE$.

TO PROVE $DE \parallel BC$.

PROOF In $\triangle ABC$, $\angle B = \angle C \Rightarrow AB = AC$

[sides opposite equal $\angle$ s are equal].

Now, $AB = AC \Rightarrow (AD + BD) = (AE + CE)$

$$\Rightarrow AD = AE \quad [\because BD = CE \text{ (given)}]$$

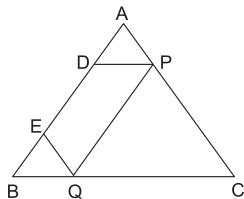
$$\Rightarrow \frac{AD}{BD} = \frac{AE}{CE} \quad [\because BD = CE].$$

Thus, $\frac{AD}{BD} = \frac{AE}{CE}$.

Hence, $DE \parallel BC$ [by the converse of Thales' theorem].

EXAMPLE 14 In $\triangle ABC$, D and E are two points on AB such that $AD = BE$. If $DP \parallel BC$ and $EQ \parallel AC$, prove that $PQ \parallel AB$.

SOLUTION GIVEN A $\triangle ABC$ and D, E are two points on AB such that $AD = BE$. Also, $DP \parallel BC$, and $EQ \parallel AC$.



TO PROVE $PQ \parallel AB$.

PROOF In $\triangle ABC$, $DP \parallel BC$.

$$\therefore \frac{AD}{DB} = \frac{AP}{PC} \quad \dots (i) \quad [\text{by Thales' theorem}]$$

In $\triangle CBA$, $EQ \parallel AC$.

$$\therefore \frac{BE}{EA} = \frac{BQ}{QC} \quad [\text{by Thales' theorem}]$$

$$\Rightarrow \frac{AD}{DB} = \frac{BQ}{QC} \quad \dots (ii)$$

$$[\because BE = AD, EA = (ED + DA) = (DE + EB) = DB]$$

From (i) and (ii), we get

$$\frac{AP}{PC} = \frac{BQ}{QC}$$

$$\therefore PQ \parallel AB \quad [\text{by the converse of Thales' theorem}].$$

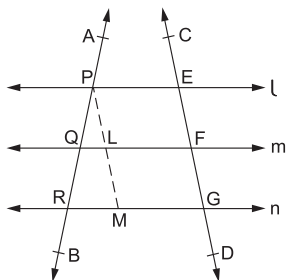
EXAMPLE 15 If three or more parallel lines are intersected by two transversals, prove that the intercepts made by them on the transversals are proportional.

SOLUTION GIVEN Three lines l, m, n such that $l \parallel m \parallel n$.

These lines are cut by the transversals AB and CD in P, Q, R and E, F, G respectively.

$$\text{TO PROVE } \frac{PQ}{QR} = \frac{EF}{FG}$$

CONSTRUCTION Draw $PM \parallel CD$, meeting the lines m and n at L and M respectively.



PROOF $PL \parallel EF$ and $PE \parallel LF \Rightarrow PLFE$ is a $\parallel gm$
 $\Rightarrow PL = EF.$... (i)

(opp. sides of a $\parallel gm$)

Also, $LM \parallel FG$ and $LF \parallel MG \Rightarrow LMFG$ is a $\parallel gm$
 $\Rightarrow LM = FG.$... (ii)

(opp. sides of a $\parallel gm$)

In $\triangle PRM$, $QL \parallel RM$ and therefore, by Thales' theorem, we have

$$\frac{PQ}{QR} = \frac{PL}{LM} = \frac{EF}{FG} \quad [\text{using (i) and (ii)}].$$

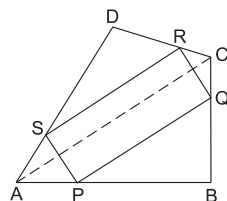
$$\therefore \frac{PQ}{QR} = \frac{EF}{FG}.$$

EXAMPLE 16 $ABCD$ is a quadrilateral and P, Q, R, S are the points of trisection of the sides AB, BC, CD and DA respectively and are adjacent to A and C . Prove that $PQRS$ is a parallelogram.

SOLUTION **GIVEN** A quadrilateral $ABCD$ in which P, Q, R, S are the points of trisection of AB, BC, CD and DA respectively (as shown).

TO PROVE $PQRS$ is a parallelogram.

CONSTRUCTION Join AC .



PROOF In $\triangle BAC$, we have $\frac{BP}{PA} = \frac{BQ}{QC} = \frac{2}{1}.$

$\therefore PQ \parallel AC.$... (i) [by the converse of Thales' theorem]

Also, in $\triangle DAC$, we have

$$\frac{DS}{SA} = \frac{DR}{RC} = \frac{2}{1}.$$

$\therefore SR \parallel AC.$... (ii) [by the converse of Thales' theorem]

Thus, $PQ \parallel SR$ [from (i) and (ii)].

Similarly, by joining BD we can prove that $SP \parallel RQ$.

Hence, $PQRS$ is a parallelogram.

EXAMPLE 17 $ABCD$ is a trapezium with $AB \parallel DC$. E and F are points on non-parallel sides AD and BC respectively such that $EF \parallel AB$. Show that $\frac{AE}{ED} = \frac{BF}{FC}.$

SOLUTION **GIVEN** A trap. $ABCD$ in which $AB \parallel DC$. E and F are points on AD and BC respectively such that $EF \parallel AB$.

TO PROVE $\frac{AE}{ED} = \frac{BF}{FC}$.

CONSTRUCTION Join AC , intersecting EF at G .

PROOF $EF \parallel AB$ and $AB \parallel DC \Rightarrow EF \parallel DC$.

Now, in $\triangle ADC$, $EG \parallel DC$.

$$\therefore \frac{AE}{ED} = \frac{AG}{GC} \quad \dots (i) \text{ [by Thales' theorem]}$$

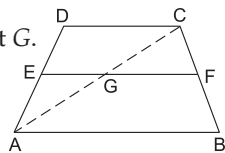
Similarly, in $\triangle CAB$, $GF \parallel AB$.

$$\therefore \frac{CG}{GA} = \frac{CF}{FB} \quad \text{[by Thales' theorem]}$$

$$\Rightarrow \frac{AG}{GC} = \frac{BF}{FC} \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{AE}{ED} = \frac{BF}{FC}.$$



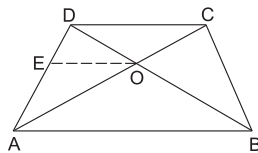
EXAMPLE 18 $ABCD$ is a trapezium in which $AB \parallel DC$ and its diagonals intersect each other at the point O .

Prove that $\frac{AO}{OC} = \frac{BO}{OD}$. [CBSE 2004]

SOLUTION **GIVEN** A trapezium $ABCD$ in which $AB \parallel DC$ and its diagonals AC and BD intersect at O .

TO PROVE $\frac{AO}{OC} = \frac{BO}{OD}$.

CONSTRUCTION Through O , draw $EO \parallel AB$, meeting AD at E .



PROOF In $\triangle ADC$, $EO \parallel DC$ [$\because EO \parallel AB \parallel DC$].

$$\therefore \frac{AE}{ED} = \frac{AO}{OC} \quad \dots (i) \text{ [by Thales' theorem]}$$

In $\triangle DAB$, $EO \parallel AB$.

$$\therefore \frac{DE}{EA} = \frac{DO}{OB} \quad \text{[by Thales' theorem]}$$

$$\Rightarrow \frac{AE}{ED} = \frac{BO}{OD} \quad \dots (ii)$$

From (i) and (ii), we get

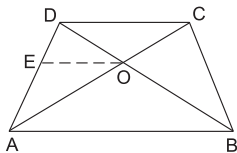
$$\frac{AO}{OC} = \frac{BO}{OD}.$$

EXAMPLE 19 The diagonals of a quadrilateral $ABCD$ intersect each other at the point O such that $\frac{AO}{OC} = \frac{BO}{OD}$. Show that $ABCD$ is a trapezium.

[CBSE 2005, '08]

SOLUTION GIVEN A quadrilateral $ABCD$ whose diagonals AC and BD intersect at a point O such that

$$\frac{AO}{OC} = \frac{BO}{OD}.$$



TO PROVE $ABCD$ is a trapezium, i.e., $AB \parallel DC$.

CONSTRUCTION Draw $EO \parallel DC$, meeting AD at E .

PROOF In $\triangle ACD$, $EO \parallel DC$.

$$\therefore \frac{AO}{OC} = \frac{AE}{ED} \quad [\text{by Thales' theorem}].$$

$$\text{But, } \frac{AO}{OC} = \frac{BO}{OD} \text{ (given)}$$

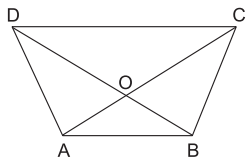
$$\therefore \frac{BO}{OD} = \frac{AE}{ED} \Rightarrow \frac{DO}{OB} = \frac{DE}{EA} \text{ in } \triangle DAB.$$

So, $EO \parallel AB$ [by the converse of Thales' theorem].

But, $EO \parallel DC$.

Hence, $AB \parallel DC$, i.e., $ABCD$ is a trapezium.

EXAMPLE 20 In the given figure, $ABCD$ is a trapezium in which $AB \parallel DC$ and its diagonals intersect at O . If $AO = (3x - 1)$ cm, $OC = (5x - 3)$ cm, $BO = (2x + 1)$ cm and $OD = (6x - 5)$ cm, find the value of x .



SOLUTION We know that $AB \parallel DC$ in trapezium $ABCD$ and its diagonals intersect at O . Then, we have

$$\frac{AO}{OC} = \frac{BO}{OD} \Rightarrow \frac{3x-1}{5x-3} = \frac{2x+1}{6x-5}$$

$$\Rightarrow (3x-1)(6x-5) = (2x+1)(5x-3)$$

$$\Rightarrow 18x^2 - 21x + 5 = 10x^2 - x - 3$$

$$\Rightarrow 8x^2 - 20x + 8 = 0 \Rightarrow 2x^2 - 5x + 2 = 0$$

$$\Rightarrow (x-2)(2x-1) = 0$$

$$\Rightarrow x = 2 \text{ or } x = \frac{1}{2}.$$

But, $x = \frac{1}{2}$ will make $OC = (5x - 3)$ cm $= \left(5 \times \frac{1}{2} - 3\right)$ cm $= -\frac{1}{2}$ cm.

And, the distance cannot be negative.

$$\therefore x \neq \frac{1}{2}.$$

Hence, $x = 2$.

MIDPOINT THEOREM

EXAMPLE 21 Prove that the line segment joining the midpoints of any two sides of a triangle is parallel to the third side.

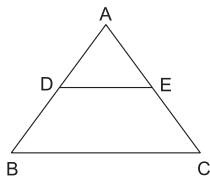
SOLUTION GIVEN $\triangle ABC$ in which D and E are the midpoints of AB and AC respectively.

TO PROVE $DE \parallel BC$.

PROOF Since D and E are the midpoints of AB and AC respectively, we have $AD = DB$ and $AE = EC$.

$$\therefore \frac{AD}{DB} = \frac{AE}{EC} \quad [\text{each equal to } 1].$$

Hence, by the converse of Thales' theorem, $DE \parallel BC$.



EXAMPLE 22 Prove that a line drawn through the midpoint of one side of a triangle parallel to another side bisects the third side.

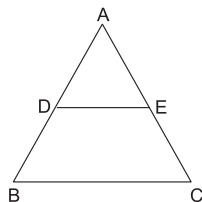
SOLUTION GIVEN $\triangle ABC$ in which D is the midpoint of AB and $DE \parallel BC$, meeting AC at E .

TO PROVE $AE = EC$.

PROOF Since $DE \parallel BC$, by Thales' theorem, we have

$$\frac{AE}{EC} = \frac{AD}{DB} = 1 \quad [\because AD = DB \text{ (given)}]$$

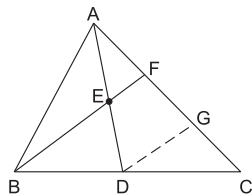
$$\Rightarrow \frac{AE}{EC} = 1 \Rightarrow AE = EC.$$



EXAMPLE 23 In $\triangle ABC$, AD is a median and E is the midpoint of AD . If BE is produced, it meets AC in F . Show that $AF = \frac{1}{3}AC$. [CBSE 2006C]

SOLUTION GIVEN $\triangle ABC$ in which AD is a median and E is the midpoint of AD . Also, BE is produced to meet AC at F .

TO PROVE $AF = \frac{1}{3}AC$.



CONSTRUCTION From D , draw $DG \parallel EF$, meeting AC at G .

PROOF In $\triangle BCF$, D is the midpoint of BC and $DG \parallel BF$.

$\therefore G$ is the midpoint of CF .

So, $FG = GC$.

In $\triangle ADG$, E is the midpoint of AD and $EF \parallel DG$.

$\therefore F$ is the midpoint of AG .

So, $AF = FG$.

Thus, $AF = FG = GC$.

$\therefore AC = (AF + FG + GC) = 3AF$.

Hence, $AF = \frac{1}{3}AC$.

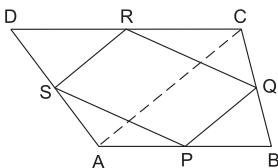
EXAMPLE 24 Prove that the line segments joining the midpoints of the adjacent sides of a quadrilateral form a parallelogram.

SOLUTION GIVEN A quadrilateral $ABCD$ in which P, Q, R, S are the midpoints of AB, BC, CD and DA respectively.

TO PROVE $PQRS$ is a parallelogram.

CONSTRUCTION Join AC .

PROOF In $\triangle ABC$, P and Q are the midpoints of AB and BC respectively.



$\therefore PQ \parallel AC$ (i) [by midpoint theorem]

In $\triangle DAC$, S and R are the midpoints of AD and CD respectively.

$\therefore SR \parallel AC$ (ii) [by midpoint theorem]

From (i) and (ii), we get $PQ \parallel SR$.

Similarly, by joining BD , we can prove that $PS \parallel QR$.

Hence, $PQRS$ is a parallelogram.

ANGLE-BISECTOR THEOREM

EXAMPLE 25 Prove that the internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle.

SOLUTION GIVEN A $\triangle ABC$ in which AD , the bisector of $\angle A$, meets BC in D .

TO PROVE $\frac{BD}{DC} = \frac{AB}{AC}$.

CONSTRUCTION Draw $CE \parallel DA$, meeting BA produced at E .

PROOF Since $DA \parallel CE$, we have

$$\angle 2 = \angle 3 \quad [\text{alternate int. } \angle]$$

and $\angle 1 = \angle 4$ [corresponding $\angle$].

But, $\angle 1 = \angle 2$ [$\because AD$ is the bisector of $\angle A$]

$$\therefore \angle 3 = \angle 4.$$

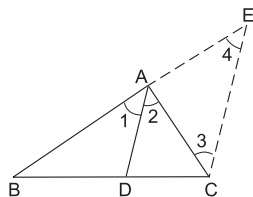
So, $AE = AC$.

Now, in $\triangle BCE$, $DA \parallel CE$.

$$\therefore \frac{BD}{DC} = \frac{AB}{AE}$$

$$\Rightarrow \frac{BD}{DC} = \frac{AB}{AC} \quad [\because AE = AC].$$

$$\text{Hence, } \frac{BD}{DC} = \frac{AB}{AC}.$$



EXAMPLE 26 In a $\triangle ABC$, let D be a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$. Prove that AD is the bisector of $\angle A$.

SOLUTION GIVEN A $\triangle ABC$ in which D is a point on BC such that

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

TO PROVE AD is the bisector of $\angle A$.

CONSTRUCTION Produce BA to E such that $AE = AC$. Join EC .

$$\text{PROOF } \frac{BD}{DC} = \frac{AB}{AC} \quad (\text{given})$$

$$\Rightarrow \frac{BD}{DC} = \frac{AB}{AE} \quad [\because AC = AE]$$

$$\Rightarrow DA \parallel CE \quad [\text{by the converse of Thales' theorem}].$$

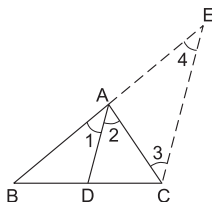
$$\therefore \angle 2 = \angle 3 \quad \dots (i) \quad [\text{alternate int. } \angle]$$

$$\text{and } \angle 1 = \angle 4 \quad \dots (ii) \quad [\text{corresponding } \angle]$$

$$\text{Also, } AE = AC \Rightarrow \angle 3 = \angle 4. \quad \dots (iii)$$

$$\therefore \angle 1 = \angle 2 \quad [\text{from (i), (ii) and (iii)}].$$

Hence, AD is the bisector of $\angle A$.



EXAMPLE 27 In the given figure, AD is the bisector of $\angle BAC$. If $AB = 10$ cm, $AC = 6$ cm and $BC = 12$ cm, find BD and DC . [CBSE 2001]

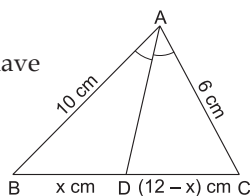
SOLUTION Let $BD = x$ cm. Then,

$$DC = (BC - BD) = (12 - x) \text{ cm.}$$

In $\triangle ABC$, AD is the bisector of $\angle BAC$.

So, by the angle-bisector theorem, we have

$$\begin{aligned}\frac{BD}{DC} &= \frac{AB}{AC} \Rightarrow \frac{x}{12-x} = \frac{10}{6} \\ &\Rightarrow 6x = 10(12-x) \\ &\Rightarrow 16x = 120 \Rightarrow x = 7.5.\end{aligned}$$



Hence, $BD = 7.5$ cm, and $DC = (12 - 7.5)$ cm = 4.5 cm.

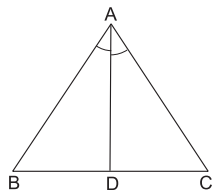
EXAMPLE 28 *If the bisector of an angle of a triangle bisects the opposite side, prove that the triangle is isosceles.* [CBSE 2000, '01, '02]

SOLUTION GIVEN A $\triangle ABC$ in which AD is the bisector of $\angle BAC$ such that $BD = DC$.

TO PROVE $AB = AC$.

PROOF Since AD is the bisector of $\angle A$, by the angle-bisector theorem, we have

$$\begin{aligned}\frac{AB}{AC} &= \frac{BD}{DC} = 1 \quad [\because BD = DC] \\ \Rightarrow \frac{AB}{AC} &= 1 \Rightarrow AB = AC.\end{aligned}$$



Hence, $\triangle ABC$ is an isosceles triangle.

EXAMPLE 29 *If the diagonal BD of a quadrilateral $ABCD$ bisects both $\angle B$ and $\angle D$, prove that $\frac{AB}{BC} = \frac{AD}{CD}$.*

SOLUTION GIVEN A quad. $ABCD$ in which diagonal BD bisects both $\angle B$ and $\angle D$.

TO PROVE $\frac{AB}{BC} = \frac{AD}{CD}$.

CONSTRUCTION Join AC , intersecting BD at E .

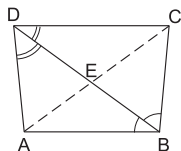
PROOF In $\triangle CBA$, BE is the bisector of $\angle ABC$.

$$\therefore \frac{AE}{EC} = \frac{AB}{BC} \quad \dots (i) \quad [\text{by the angle-bisector theorem}]$$

In $\triangle ADC$, DE is the bisector of $\angle ADC$.

$$\therefore \frac{AE}{EC} = \frac{AD}{CD} \quad \dots (ii) \quad [\text{by the angle-bisector theorem}]$$

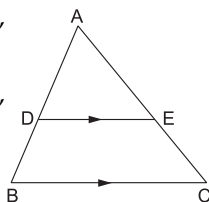
From (i) and (ii), we get $\frac{AB}{BC} = \frac{AD}{CD}$.



EXERCISE 7A

1. D and E are points on the sides AB and AC respectively of a $\triangle ABC$ such that $DE \parallel BC$.

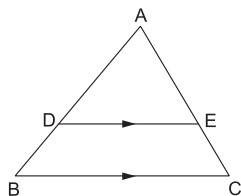
- If $AD = 3.6$ cm, $AB = 10$ cm and $AE = 4.5$ cm, find EC and AC .
- If $AB = 13.3$ cm, $AC = 11.9$ cm and $EC = 5.1$ cm, find AD .
- If $\frac{AD}{DB} = \frac{4}{7}$ and $AC = 6.6$ cm, find AE .
- If $\frac{AD}{AB} = \frac{8}{15}$ and $EC = 3.5$ cm, find AE .



2. D and E are points on the sides AB and AC respectively of a $\triangle ABC$ such that $DE \parallel BC$.

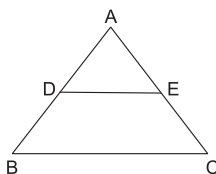
Find the value of x , when

- $AD = x$ cm, $DB = (x - 2)$ cm, $AE = (x + 2)$ cm and $EC = (x - 1)$ cm.
- $AD = 4$ cm, $DB = (x - 4)$ cm, $AE = 8$ cm and $EC = (3x - 19)$ cm.
- $AD = (7x - 4)$ cm, $AE = (5x - 2)$ cm, $DB = (3x + 4)$ cm and $EC = 3x$ cm.



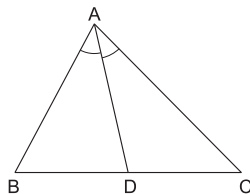
3. D and E are points on the sides AB and AC respectively of a $\triangle ABC$. In each of the following cases, determine whether $DE \parallel BC$ or not.

- $AD = 5.7$ cm, $DB = 9.5$ cm, $AE = 4.8$ cm and $EC = 8$ cm.
- $AB = 11.7$ cm, $AC = 11.2$ cm, $BD = 6.5$ cm and $AE = 4.2$ cm.
- $AB = 10.8$ cm, $AD = 6.3$ cm, $AC = 9.6$ cm and $EC = 4$ cm.
- $AD = 7.2$ cm, $AE = 6.4$ cm, $AB = 12$ cm and $AC = 10$ cm.



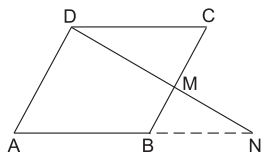
4. In a $\triangle ABC$, AD is the bisector of $\angle A$.

- If $AB = 6.4$ cm, $AC = 8$ cm and $BD = 5.6$ cm, find DC .
- If $AB = 10$ cm, $AC = 14$ cm and $BC = 6$ cm, find BD and DC .
- If $AB = 5.6$ cm, $BD = 3.2$ cm and $BC = 6$ cm, find AC . [CBSE 2001C]
- If $AB = 5.6$ cm, $AC = 4$ cm and $DC = 3$ cm, find BC . [CBSE 1999C]



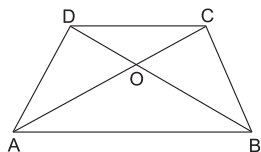
5. M is a point on the side BC of a parallelogram $ABCD$. DM when produced meets AB produced at N . Prove that

$$(i) \frac{DM}{MN} = \frac{DC}{BN}, \quad (ii) \frac{DN}{DM} = \frac{AN}{DC}.$$



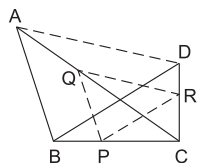
6. Show that the line segment which joins the midpoints of the oblique sides of a trapezium is parallel to the parallel sides.

7. In the adjoining figure, $ABCD$ is a trapezium in which $CD \parallel AB$ and its diagonals intersect at O . If $AO = (2x + 1)$ cm, $OC = (5x - 7)$ cm, $DO = (7x - 5)$ cm and $OB = (7x + 1)$ cm, find the value of x .

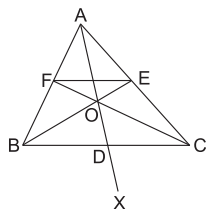


8. In a $\triangle ABC$, M and N are points on the sides AB and AC respectively such that $BM = CN$. If $\angle B = \angle C$ then show that $MN \parallel BC$.

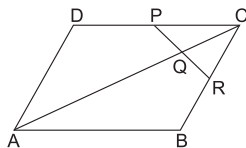
9. $\triangle ABC$ and $\triangle DBC$ lie on the same side of BC , as shown in the figure. From a point P on BC , $PQ \parallel AB$ and $PR \parallel BD$ are drawn, meeting AC at Q and CD at R respectively. Prove that $QR \parallel AD$.



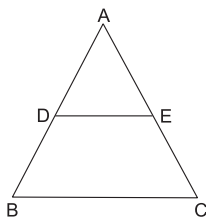
10. In the given figure, side BC of $\triangle ABC$ is bisected at D and O is any point on AD . BO and CO produced meet AC and AB at E and F respectively, and AD is produced to X so that D is the midpoint of OX . Prove that $AO : AX = AF : AB$ and show that $EF \parallel BC$.



11. $ABCD$ is a parallelogram in which P is the midpoint of DC and Q is a point on AC such that $CQ = \frac{1}{4}AC$. If PQ produced meets BC at R , prove that R is the midpoint of BC .

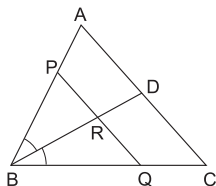


12. In the adjoining figure, ABC is a triangle in which $AB = AC$. If D and E are points on AB and AC respectively such that $AD = AE$, show that the points B, C, E and D are concyclic.



13. In $\triangle ABC$, the bisector of $\angle B$ meets AC at D .
A line $PQ \parallel AC$ meets AB , BC and BD at P , Q and R respectively.

Show that $PR \times BQ = QR \times BP$.



ANSWERS (EXERCISE 7A)

- (i) $EC = 8$ cm, $AC = 12.5$ cm (ii) $AD = 7.6$ cm (iii) $AE = 2.4$ cm
(iv) $AE = 4$ cm
- (i) $x = 4$ (ii) $x = 11$ (iii) $x = 4$
- (i) Yes (ii) No (iii) Yes (iv) No
- (i) $DC = 7$ cm (ii) $BD = 2.5$ cm, $DC = 3.5$ cm (iii) $AC = 4.9$ cm
(iv) $BC = 7.2$ cm
- $x = 2$

HINTS TO SOME SELECTED QUESTIONS

5. (i) In $\triangle NDA$, $MB \parallel DA$

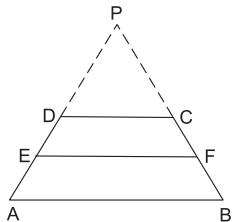
$$\Rightarrow \frac{NM}{MD} = \frac{NB}{BA} \Rightarrow \frac{DM}{MN} = \frac{AB}{BN} \Rightarrow \frac{DM}{MN} = \frac{DC}{BN} \quad [\because AB = DC].$$

$$\begin{aligned} \text{(ii)} \quad \frac{NM}{MD} = \frac{NB}{BA} &\Rightarrow \frac{NM}{MD} + 1 = \frac{NB}{BA} + 1 \Rightarrow \frac{NM + MD}{MD} = \frac{NB + BA}{BA} \\ &\Rightarrow \frac{DN}{DM} = \frac{AN}{AB} \Rightarrow \frac{DN}{DM} = \frac{AN}{DC} \quad [\because AB = DC]. \end{aligned}$$

6. Let E and F be the midpoints of the sides AD and BC of a trapezium $ABCD$ having $AB \parallel CD$. Produce AD and BC to meet at P .

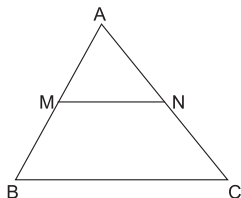
In $\triangle PAB$, $DC \parallel AB$.

$$\begin{aligned} \therefore \frac{PD}{DA} = \frac{PC}{CB} &\Rightarrow \frac{PD}{2DE} = \frac{PC}{2CF} \Rightarrow \frac{PD}{DE} = \frac{PC}{CF} \\ &\Rightarrow DC \parallel EF. \end{aligned}$$



8. $\angle B = \angle C \Rightarrow AB = AC \Rightarrow AM + BM = AN + CN \Rightarrow AM = AN \quad [\because BM = CN]$.

$$\begin{aligned} \text{Now, } AM = AN, BM = CN &\Rightarrow \frac{AM}{BM} = \frac{AN}{CN} \\ &\Rightarrow \frac{AM}{MB} = \frac{AN}{NC} \Rightarrow MN \parallel BC. \end{aligned}$$



9. In $\triangle CAB$, $QP \parallel AB \Rightarrow \frac{CP}{PB} = \frac{CQ}{QA}$.

In $\triangle CDB$, $RP \parallel DB \Rightarrow \frac{CP}{PB} = \frac{CR}{RD}$.

$\therefore \frac{CQ}{QA} = \frac{CR}{RD} \Rightarrow QR \parallel AD$ (in $\triangle CDA$).

10. Join BX and CX .

Clearly, $BD = DC$ and $OD = DX$ (given).

Thus, the diagonals of quad. $OBXC$ bisect each other.

$\therefore OBXC$ is a parallelogram.

$\therefore BX \parallel CF$ and so, $OF \parallel BX$.

Similarly, $OE \parallel XC$.

In $\triangle ABX$, $OF \parallel BX$.

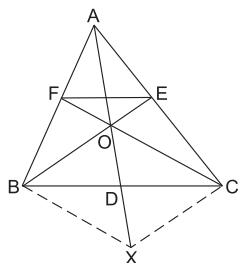
$$\therefore \frac{AO}{AX} = \frac{AF}{AB} \quad \dots (i)$$

In $\triangle ACX$, $OE \parallel XC$.

$$\therefore \frac{AO}{AX} = \frac{AE}{AC} \quad \dots (ii)$$

From (i) and (ii), we get $\frac{AF}{AB} = \frac{AE}{AC}$.

Hence, $FE \parallel BC$.



11. Join BD . Suppose it meets AC at S .

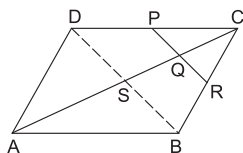
Since the diagonals of ||gm bisect each other, $CS = \frac{1}{2}AC$.

Now, $CS = \frac{1}{2}AC$ and $CQ = \frac{1}{4}AC \Rightarrow CQ = \frac{1}{2}CS$.

$\therefore Q$ is the midpoint of CS .

So, $PQ \parallel DS$ and therefore, $QR \parallel SB$.

In $\triangle CSB$, Q is the midpoint of CS and $QR \parallel SB$, so R is the midpoint of BC .



12. $AB = AC$ and $AD = AE$

$$\Rightarrow AB - AD = AC - AE \Rightarrow DB = EC$$

$$\Rightarrow \frac{AD}{AE} = \frac{DB}{EC} \quad [\text{each equal to } 1]$$

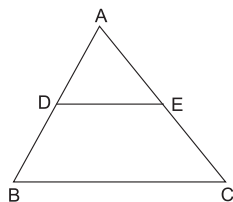
$$\Rightarrow DE \parallel BC \quad [\text{by the converse of Thales' theorem}]$$

$$\Rightarrow \angle DEC + \angle ECB = 180^\circ$$

$$\Rightarrow \angle DEC + \angle CBD = 180^\circ \quad [\because AB = AC \Rightarrow \angle C = \angle B]$$

$$\Rightarrow \text{quad. } BCED \text{ is cyclic.}$$

Hence, the points B, C, E, D are concyclic.



13. In $\triangle BQP$, BR is the bisector of $\angle B$.

$$\therefore \frac{QR}{PR} = \frac{BQ}{BP} \quad [\text{by angle-bisector theorem}].$$

CRITERIA FOR SIMILARITY OF TWO TRIANGLES

Two triangles are similar, if (i) their corresponding angles are equal and (ii) their corresponding sides are proportional.

Thus, $\triangle ABC \sim \triangle DEF$, if

$$(i) \angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

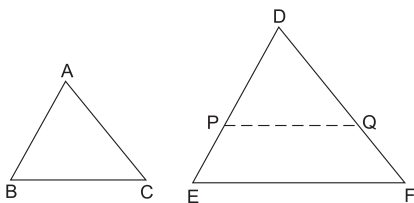
$$\text{and (ii) } \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}.$$

THEOREM 1 (AAA-similarity) *If in two triangles, the corresponding angles are equal, then their corresponding sides are proportional and hence the triangles are similar.*

GIVEN $\triangle ABC$ and $\triangle DEF$ such that $\angle A = \angle D, \angle B = \angle E$ and $\angle C = \angle F$.

TO PROVE $\triangle ABC \sim \triangle DEF$.

CONSTRUCTION Cut $DP = AB$ and $DQ = AC$. Join PQ .



PROOF In $\triangle ABC$ and $\triangle DPQ$, we have

$$AB = DP \quad [\text{by construction}]$$

$$AC = DQ \quad [\text{by construction}]$$

$$\angle A = \angle D \quad [\text{given}]$$

$$\therefore \triangle ABC \cong \triangle DPQ \quad [\text{by SAS-congruence}]$$

$$\Rightarrow \angle B = \angle P$$

$$\Rightarrow \angle E = \angle P \quad [\because \angle B = \angle E \text{ (given)}]$$

$$\Rightarrow PQ \parallel EF \quad [\because \text{corresponding } \angle \text{ are equal}]$$

$$\Rightarrow \frac{DP}{DE} = \frac{DQ}{DF}$$

$$\Rightarrow \frac{AB}{DE} = \frac{CA}{FD} \quad [\because DP = AB \text{ and } DQ = AC]$$

$$\text{Similarly, } \frac{AB}{DE} = \frac{BC}{EF}.$$

$$\therefore \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}.$$

$$\text{Thus, } \angle A = \angle D, \angle B = \angle E, \angle C = \angle F \text{ and } \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}.$$

Hence, $\triangle ABC \sim \triangle DEF$.

IMPORTANT REMARK Two $\triangle$ s are similar $\Leftrightarrow$ they are equiangular.

COROLLARY (AA-similarity) *If two angles of one triangle are respectively equal to two angles of another triangle then the two triangles are similar.*

PROOF In $\triangle ABC$ and $\triangle DEF$, let $\angle A = \angle D$ and $\angle B = \angle E$.

Then, 3rd $\angle C = 3rd \angle F$.

Thus, the two triangles are equiangular and hence similar.

REMARK AA-similarity is the same as AAA-similarity.

THEOREM 2 (SSS-similarity) *If the corresponding sides of two triangles are proportional then their corresponding angles are equal, and hence the two triangles are similar.*

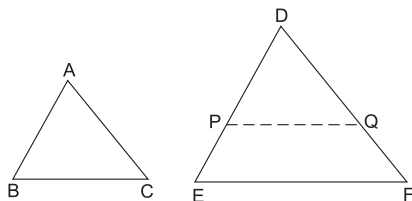
GIVEN $\triangle ABC$ and $\triangle DEF$ in which $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$.

TO PROVE $\triangle ABC \sim \triangle DEF$.

CONSTRUCTION Let us take $\triangle ABC$ and $\triangle DEF$ such that

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} (< 1).$$

Cut $DP = AB$ and $DQ = AC$. Join PQ .



PROOF $\frac{AB}{DE} = \frac{AC}{DF} \Rightarrow \frac{DP}{DE} = \frac{DQ}{DF}$ [$\because AB = DP$ and $AC = DQ$].

So, by the converse of Thales' theorem, $PQ \parallel EF$.

$\therefore \angle P = \angle E$ [corresponding $\angle$ s]

$\angle Q = \angle F$ [corresponding $\angle$ s]

$\therefore \triangle DPQ \sim \triangle DEF$ [by AA-similarity]

$$\Rightarrow \frac{DP}{DE} = \frac{PQ}{EF}$$

$$\Rightarrow \frac{AB}{DE} = \frac{PQ}{EF} \quad \dots (i) \quad [\because DP = AB]$$

$$\text{But, } \frac{AB}{DE} = \frac{BC}{EF} \quad \dots (ii) \quad [\text{given}]$$

$$\therefore \frac{PQ}{EF} = \frac{BC}{EF} \quad [\text{from (i) and (ii)}]$$

$$\Rightarrow BC = PQ.$$

Thus, $AB = DP$, $AC = DQ$ and $BC = PQ$.

$\therefore \triangle ABC \cong \triangle DPQ$ [by SSS-congruence]

$\therefore \angle A = \angle D, \angle B = \angle P = \angle E$ and $\angle C = \angle Q = \angle F$

$\Rightarrow \angle A = \angle D, \angle B = \angle E$ and $\angle C = \angle F$.

Thus, the given triangles are equiangular and hence similar.

THEOREM 3 (SAS-similarity) *If one angle of a triangle is equal to one angle of the other triangle and the sides including these angles are proportional then the two triangles are similar.*

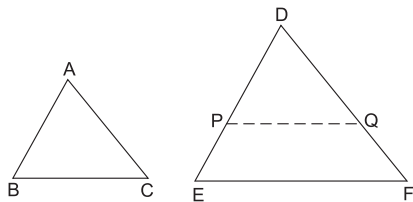
GIVEN $\triangle ABC$ and $\triangle DEF$ in which $\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$.

TO PROVE $\triangle ABC \sim \triangle DEF$.

CONSTRUCTION Let us take $\triangle ABC$ and $\triangle DEF$ such that

$$\frac{AB}{DE} = \frac{AC}{DF} (< 1) \text{ and } \angle A = \angle D.$$

Cut $DP = AB$ and $DQ = AC$. Join PQ .



PROOF In $\triangle ABC$ and $\triangle DPQ$, we have

$$AB = DP \quad [\text{by construction}]$$

$$\angle A = \angle D \quad (\text{given})$$

$$AC = DQ \quad [\text{by construction}]$$

$\therefore \triangle ABC \cong \triangle DPQ$ [by SAS-congruence]

$\therefore \angle A = \angle D, \angle B = \angle P$ and $\angle C = \angle Q$.

$$\text{Now, } \frac{AB}{DE} = \frac{AC}{DF} \quad (\text{given})$$

$$\Rightarrow \frac{DP}{DE} = \frac{DQ}{DF} \quad [\because AB = DP \text{ and } AC = DQ]$$

$$\Rightarrow PQ \parallel EF \quad [\text{by the converse of Thales' theorem}]$$

$$\Rightarrow \angle P = \angle E \text{ and } \angle Q = \angle F \quad [\text{corresponding } \angle]$$

$$\therefore \angle A = \angle D, \angle B = \angle P = \angle E \text{ and } \angle C = \angle Q = \angle F.$$

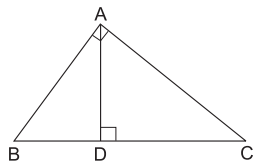
Thus, $\angle A = \angle D, \angle B = \angle E$ and $\angle C = \angle F$.

So, the given triangles are equiangular and hence similar.

THEOREM 4 *If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse then the triangles on both sides of the perpendicular are similar to the whole triangle and also to each other.*

GIVEN A $\triangle ABC$ in which $\angle BAC = 90^\circ$ and $AD \perp BC$.

TO PROVE (i) $\triangle DBA \sim \triangle ABC$
 (ii) $\triangle DAC \sim \triangle ABC$
 (iii) $\triangle DBA \sim \triangle DAC$.



PROOF (i) In $\triangle DBA$ and $\triangle ABC$, we have

$$\angle BDA = \angle BAC = 90^\circ$$

$$\angle DBA = \angle ABC \quad (\text{common})$$

$$\therefore \triangle DBA \sim \triangle ABC \quad [\text{by AA-similarity}].$$

(ii) In $\triangle DAC$ and $\triangle ABC$, we have

$$\angle CDA = \angle CAB = 90^\circ$$

$$\angle DCA = \angle ACB \quad (\text{common})$$

$$\therefore \triangle DAC \sim \triangle ABC \quad [\text{by AA-similarity}].$$

(iii) In $\triangle DBA$ and $\triangle DAC$, we have

$$\angle ADB = \angle CDA = 90^\circ.$$

$$\therefore \left. \begin{aligned} \angle B + \angle BAD &= 90^\circ \\ \angle C + \angle CAD &= 90^\circ \\ \angle BAD + \angle CAD &= 90^\circ \end{aligned} \right\} \Rightarrow \angle B = \angle CAD \text{ and } \angle C = \angle BAD.$$

Thus, in $\triangle DBA$ and $\triangle DAC$, we have:

$$\angle ADB = \angle CDA \quad [\text{each equal to } 90^\circ]$$

$$\angle B = \angle CAD$$

$$\angle BAD = \angle C$$

$$\therefore \triangle DBA \sim \triangle DAC \quad [\text{by AAA-similarity}].$$

SUMMARY

(i) If $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$, then $\triangle ABC \sim \triangle DEF$.

(AAA-similarity)

(ii) If $\angle A = \angle D$, $\angle B = \angle E$, then $\triangle ABC \sim \triangle DEF$.

(AA-similarity)

(iii) If $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$, then $\triangle ABC \sim \triangle DEF$.

(SSS-similarity)

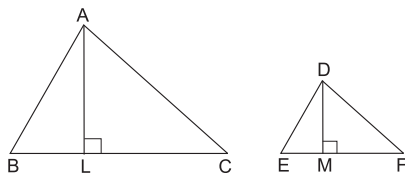
(iv) If $\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$, then $\triangle ABC \sim \triangle DEF$.

(SAS-similarity)

SOME MORE RESULTS

THEOREM 1 *If two triangles are equiangular, prove that the ratio of their corresponding sides is the same as the ratio of the corresponding altitudes.*

GIVEN $\triangle ABC$ and $\triangle DEF$ in which $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$ and $AL \perp BC$ and $DM \perp EF$.



TO PROVE $\frac{BC}{EF} = \frac{AL}{DM}$.

PROOF Since $\triangle ABC$ and $\triangle DEF$ are equiangular, $\triangle ABC \sim \triangle DEF$.

$$\therefore \frac{AB}{DE} = \frac{BC}{EF}. \quad \dots (i)$$

In $\triangle ALB$ and $\triangle DME$, we have

$$\angle ALB = \angle DME = 90^\circ \text{ and } \angle B = \angle E \text{ (given).}$$

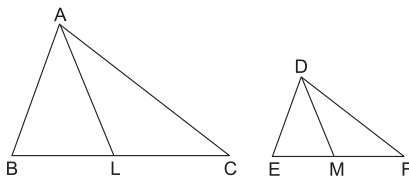
$$\therefore \triangle ALB \sim \triangle DME \text{ [by AA-similarity]}$$

$$\therefore \frac{AB}{DE} = \frac{AL}{DM}. \quad \dots (ii)$$

From (i) and (ii), we get $\frac{BC}{EF} = \frac{AL}{DM}$.

THEOREM 2 *If two triangles are equiangular, prove that the ratio of their corresponding sides is the same as the ratio of the corresponding medians.*

GIVEN $\triangle ABC$ and $\triangle DEF$ in which $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$ and AL and DM are the medians.



TO PROVE $\frac{BC}{EF} = \frac{AL}{DM}$.

PROOF Since $\triangle ABC$ and $\triangle DEF$ are equiangular, we have $\triangle ABC \sim \triangle DEF$.

$$\therefore \frac{AB}{DE} = \frac{BC}{EF}. \quad \dots (i)$$

$$\text{But, } \frac{AB}{DE} = \frac{BC}{EF} = \frac{2BL}{2EM} = \frac{BL}{EM}.$$

Now, in $\triangle ABL$ and $\triangle DEM$, we have

$$\frac{AB}{DE} = \frac{BL}{EM} \text{ and } \angle B = \angle E \text{ (given).}$$

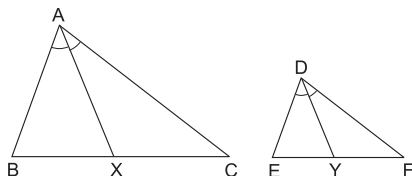
$$\therefore \triangle ABL \sim \triangle DEM \text{ [by SAS-similarity]}$$

$$\Rightarrow \frac{AB}{DE} = \frac{AL}{DM}. \quad \dots (ii)$$

$$\text{From (i) and (ii), we get } \frac{BC}{EF} = \frac{AL}{DM}.$$

THEOREM 3 *If two triangles are equiangular, show that the ratio of the corresponding sides is the same as the ratio of the corresponding angle-bisector segments.*

GIVEN $\triangle ABC$ and $\triangle DEF$ in which $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$, and AX and DY are the bisectors of $\angle A$ and $\angle D$ respectively.



$$\text{TO PROVE } \frac{BC}{EF} = \frac{AX}{DY}.$$

PROOF Since $\triangle ABC$ and $\triangle DEF$ are equiangular, we have $\triangle ABC \sim \triangle DEF$.

$$\therefore \frac{AB}{DE} = \frac{BC}{EF}. \quad \dots (i)$$

$$\text{Now, } \angle A = \angle D \Rightarrow \frac{1}{2}\angle A = \frac{1}{2}\angle D \Rightarrow \angle BAX = \angle EDY.$$

Thus, in $\triangle ABX$ and $\triangle DEY$, we have

$$\angle BAX = \angle EDY \quad (\text{proved})$$

$$\angle B = \angle E \quad (\text{given})$$

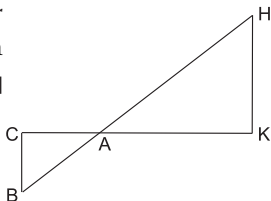
$$\therefore \triangle ABX \sim \triangle DEY \text{ [by AA-similarity]}$$

$$\Rightarrow \frac{AB}{DE} = \frac{AX}{DY}. \quad \dots (ii)$$

$$\text{From (i) and (ii), we get } \frac{BC}{EF} = \frac{AX}{DY}.$$

SOLVED EXAMPLES

EXAMPLE 1 In the adjoining figure, $\triangle AHK$ is similar to $\triangle ABC$. If $AK = 10$ cm, $BC = 3.5$ cm and $HK = 7$ cm, find AC . [CBSE 2010]



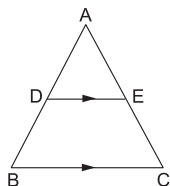
SOLUTION $\triangle AHK \sim \triangle ABC$

$$\Rightarrow \frac{AK}{AC} = \frac{HK}{BC} \Rightarrow \frac{10}{x} = \frac{7}{3.5}, \text{ where } AC = x \text{ cm}$$

$$\Rightarrow x = \frac{10 \times 3.5}{7} = 5.$$

$$\therefore AC = 5 \text{ cm.}$$

EXAMPLE 2 In the given figure, $DE \parallel BC$, $AD = 2$ cm, $BD = 2.5$ cm, $AE = 3.2$ cm and $DE = 4$ cm. Find AC and BC . [CBSE 2001C]



SOLUTION Since $DE \parallel BC$, we have

$$\angle ADE = \angle ABC \quad (\text{corresponding } \angle)$$

$$\text{and } \angle AED = \angle ACB \quad (\text{corresponding } \angle).$$

$$\therefore \triangle ADE \sim \triangle ABC \quad [\text{by AA-similarity}].$$

So, the corresponding sides of $\triangle ADE$ and $\triangle ABC$ are proportional.

$$\therefore \frac{AD}{AB} = \frac{DE}{BC} = \frac{AE}{AC}. \quad \dots (i)$$

$$\text{Now, } \frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{2}{4.5} = \frac{4}{BC} \quad [\because AB = AD + BD = 4.5 \text{ cm}]$$

$$\Rightarrow BC = \left(\frac{4 \times 4.5}{2} \right) \text{ cm} = 9 \text{ cm.}$$

$$\text{Again, } \frac{DE}{BC} = \frac{AE}{AC} \quad [\text{from (i)}]$$

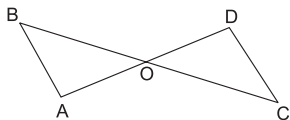
$$\Rightarrow \frac{4}{9} = \frac{3.2}{AC} \quad [\because BC = 9 \text{ cm}]$$

$$\Rightarrow AC = \left(\frac{9 \times 3.2}{4} \right) \text{ cm} = 7.2 \text{ cm.}$$

Hence, $AC = 7.2$ cm and $BC = 9$ cm.

EXAMPLE 3 In the given figure, $AB \parallel CD$. Prove that $\triangle AOB \sim \triangle DOC$.

SOLUTION $AB \parallel CD$ (given).



$\therefore \angle OAB = \angle ODC$ (alternate angles)
 $\angle OBA = \angle OCD$ (alternate angles)
 $\angle AOB = \angle DOC$ (vertical opposite $\sphericalangle$ s)
 $\therefore \triangle AOB \sim \triangle DOC$ [by AAA-similarity].

EXAMPLE 4 In the given figure, $\triangle AOB \sim \triangle DOC$.

Prove that $AB \parallel CD$.

SOLUTION

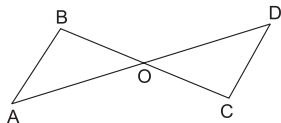
$\triangle AOB \sim \triangle DOC$.

So, the given triangles are equiangular.

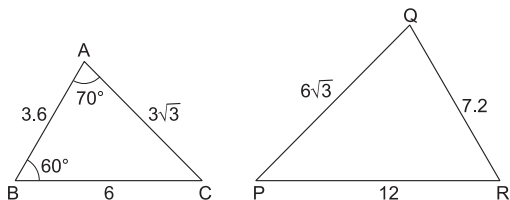
$\therefore \angle OAB = \angle ODC$.

But, these are alternate angles.

$\therefore AB \parallel CD$.



EXAMPLE 5 Find $\angle P$ in the adjoining figure.



SOLUTION

In $\triangle ABC$ and $\triangle QRP$, we have

$$\frac{AB}{QR} = \frac{3.6}{7.2} = \frac{1}{2}, \quad \frac{BC}{RP} = \frac{6}{12} = \frac{1}{2} \quad \text{and} \quad \frac{CA}{PQ} = \frac{3\sqrt{3}}{6\sqrt{3}} = \frac{1}{2}$$

Thus, $\frac{AB}{QR} = \frac{BC}{RP} = \frac{CA}{PQ}$ and so

$\triangle ABC \sim \triangle QRP$ [by SSS-similarity].

$\therefore \angle C = \angle P$ [corresponding angles of similar triangles].

But, $\angle C = 180^\circ - (\angle A + \angle B) = 180^\circ - (70^\circ + 60^\circ) = 50^\circ$.

$\therefore \angle P = 50^\circ$.

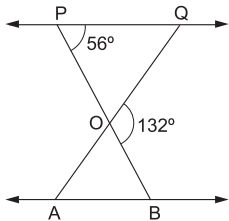
EXAMPLE 6

In the given figure, $\triangle OQP \sim \triangle OAB$, $\angle OPQ = 56^\circ$ and $\angle BOQ = 132^\circ$. Find $\angle OAB$.

SOLUTION

In $\triangle OPQ$, we have

$$\angle BOQ = \angle OPQ + \angle OQP$$



[$\therefore$ the exterior angle is equal to the sum of the two interior opposite angles]

$$\Rightarrow \angle OQP = \angle BOQ - \angle OPQ = 132^\circ - 56^\circ = 76^\circ.$$

Now, $\triangle OQP \sim \triangle OAB$

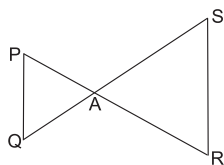
$$\Rightarrow \angle OQP = \angle OAB \text{ [corresponding angles of similar triangles].}$$

$$\therefore \angle OAB = \angle OQP = 76^\circ.$$

EXAMPLE 7

In the given figure, $AP \cdot AR = AS \cdot AQ$.

Prove that $\angle P = \angle S$ and $\angle Q = \angle R$.

**SOLUTION**

We have

$$\angle PAQ = \angle SAR. \quad \dots \text{ (i) [vertically opposite angles]}$$

Also, $AP \cdot AR = AS \cdot AQ$ (given)

$$\Rightarrow \frac{AP}{AS} = \frac{AQ}{AR}. \quad \dots \text{ (ii)}$$

From (i) and (ii), we have

$$\triangle PAQ \sim \triangle SAR \text{ [by SAS-similarity]}$$

$$\therefore \angle P = \angle S \text{ and } \angle Q = \angle R$$

[corresponding angles of similar triangles].

EXAMPLE 8

A vertical stick which is 15 cm long casts a 12-cm-long shadow on the ground. At the same time, a vertical tower casts a 50-m-long shadow on the ground. Find the height of the tower.

SOLUTION

Let AB be the vertical stick and let AC be its shadow.

Then, $AB = 0.15$ m and $AC = 0.12$ m.

Let DE be the vertical tower and let DF be its shadow.

Then, $DF = 50$ m. Let $DE = x$ m.

Now, in $\triangle BAC$ and $\triangle EDF$,
we have

$$\angle BAC = \angle EDF = 90^\circ$$

$$\angle ACB = \angle DFE$$

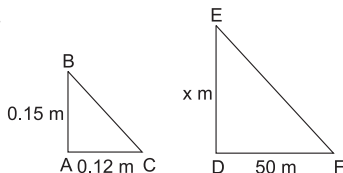
[angular elevation of the sun at the same time]

$$\therefore \triangle BAC \sim \triangle EDF$$

$$\Rightarrow \frac{AB}{DE} = \frac{AC}{DF} \Rightarrow \frac{0.15}{x} = \frac{0.12}{50}$$

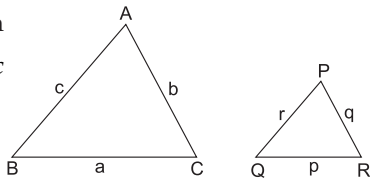
$$\Rightarrow x = \frac{(0.15 \times 50)}{0.12} = 62.5.$$

Hence, the height of the tower is 62.5 m.



EXAMPLE 9 Prove that the ratio of the perimeters of two similar triangles is the same as the ratio of their corresponding sides.

SOLUTION GIVEN $\triangle ABC$ and $\triangle PQR$ in which $BC = a, CA = b, AB = c$ and $QR = p, RP = q, PQ = r$.
Also, $\triangle ABC \sim \triangle PQR$.



TO PROVE $\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{a+b+c}{p+q+r}$.

PROOF Since $\triangle ABC$ and $\triangle PQR$ are similar, therefore their corresponding sides are proportional.

$$\therefore \frac{a}{p} = \frac{b}{q} = \frac{c}{r} = k \text{ (say)} \quad \dots (i)$$

$$\Rightarrow a = kp, b = kq \text{ and } c = kr.$$

$$\begin{aligned} \therefore \frac{\text{perimeter of } \triangle ABC}{\text{perimeter of } \triangle PQR} &= \frac{a+b+c}{p+q+r} = \frac{kp+kq+kr}{p+q+r} \\ &= \frac{k(p+q+r)}{(p+q+r)} = k. \quad \dots (ii) \end{aligned}$$

From (i) and (ii), we get

$$\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{a+b+c}{p+q+r} = \frac{\text{perimeter of } \triangle ABC}{\text{perimeter of } \triangle PQR} \quad [\text{each equal to } k].$$

EXAMPLE 10 The perimeters of two similar triangles are 25 cm and 15 cm respectively. If one side of the first triangle is 9 cm, find the corresponding side of the second triangle. [CBSE 2002C]

SOLUTION Let $\triangle ABC \sim \triangle DEF$ given in such a way that perimeter of $\triangle ABC = 25$ cm, perimeter of $\triangle DEF = 15$ cm and $AB = 9$ cm. Then, we have to find DE . Let $DE = x$ cm.

We know that the ratio of the perimeters of two similar triangles is the same as the ratio of their corresponding sides.

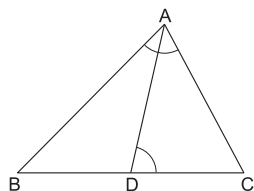
$$\therefore \frac{\text{perimeter of } \triangle ABC}{\text{perimeter of } \triangle DEF} = \frac{AB}{DE}$$

$$\Rightarrow \frac{25}{15} = \frac{9}{x} \Rightarrow x = \left(\frac{9 \times 15}{25} \right) = 5.4.$$

$$\therefore DE = 5.4 \text{ cm.}$$

Hence, the corresponding side of the second triangle is 5.4 cm.

EXAMPLE 11 In the given figure, D is a point on the side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \times CD$. [CBSE 2004]



SOLUTION GIVEN $\triangle ABC$ in which D is a point on BC such that $\angle ADC = \angle BAC$.

TO PROVE $CA^2 = CB \times CD$.

PROOF In $\triangle ABC$ and $\triangle DAC$, we have

$$\angle BAC = \angle ADC \quad (\text{given})$$

$$\angle ACB = \angle DCA \quad (\text{common})$$

$$\therefore \triangle ABC \sim \triangle DAC \quad [\text{by AA-similarity}].$$

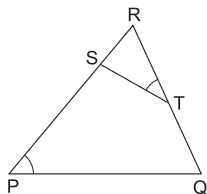
So, the sides of $\triangle ABC$ and $\triangle DAC$ are proportional.

$$\therefore \frac{CA}{CB} = \frac{CD}{CA}.$$

Hence, $CA^2 = CB \times CD$.

EXAMPLE 12 In the given figure, S and T are points on sides PR and QR of $\triangle PQR$ such that $\angle P = \angle RTS$. Show that $\triangle RPQ \sim \triangle RTS$.

SOLUTION GIVEN $\triangle RPQ$ and $\triangle RTS$ in which $\angle P = \angle RTS$.



TO PROVE $\triangle RPQ \sim \triangle RTS$.

PROOF In $\triangle RPQ$ and $\triangle RTS$, we have

$$\angle P = \angle RTS \quad (\text{given})$$

$$\angle R = \angle R \quad (\text{common})$$

$$\therefore \triangle RPQ \sim \triangle RTS \quad [\text{by AA-similarity}].$$

EXAMPLE 13 In the given figure, if $\triangle ABE \cong \triangle ACD$, show that $\triangle ADE \sim \triangle ABC$.

SOLUTION $\triangle ABE \cong \triangle ACD$ (given)

$$\therefore AE = AD \quad \dots (i) \quad [\text{cpct}]$$

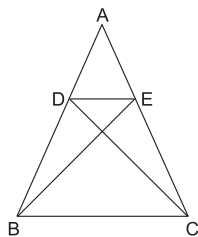
$$\text{and } AB = AC. \quad \dots (ii) \quad [\text{cpct}]$$

In $\triangle ADE$ and $\triangle ABC$, we have

$$\angle DAE = \angle BAC \quad (\text{common})$$

$$\text{and } \frac{AD}{AB} = \frac{AE}{AC} \quad [\text{using (i) and (ii)}].$$

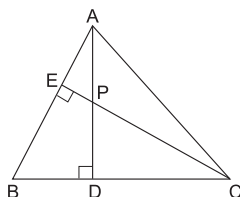
$$\therefore \triangle ADE \sim \triangle ABC \quad [\text{by SAS-similarity}].$$



EXAMPLE 14 In the given figure, altitudes AD and CE of $\triangle ABC$ intersect each other at the point P .

Show that:

- (i) $\triangle AEP \sim \triangle CDP$
- (ii) $\triangle ABD \sim \triangle CBE$
- (iii) $\triangle AEP \sim \triangle ADB$
- (iv) $\triangle PDC \sim \triangle BEC$



SOLUTION

- (i) In $\triangle AEP$ and $\triangle CDP$, we have

$$\angle AEP = \angle CDP \quad [\text{each equal to } 90^\circ]$$

$$\angle APE = \angle CPD \quad [\text{vertically opposite } \angle s]$$

$$\therefore \triangle AEP \sim \triangle CDP \quad [\text{by AA-similarity}].$$

- (ii) In $\triangle ABD$ and $\triangle CBE$, we have

$$\angle ADB = \angle CEB = 90^\circ$$

$$\angle B = \angle B \quad [\text{common}]$$

$$\therefore \triangle ABD \sim \triangle CBE \quad [\text{by AA-similarity}].$$

- (iii) In $\triangle AEP$ and $\triangle ADB$, we have

$$\angle AEP = \angle ADB = 90^\circ$$

$$\angle EAP = \angle DAB \quad (\text{common})$$

$$\text{Hence, } \triangle AEP \sim \triangle ADB \quad [\text{by AA-similarity}].$$

- (iv) In $\triangle PDC$ and $\triangle BEC$, we have

$$\angle PDC = \angle BEC = 90^\circ$$

$$\angle PCD = \angle BCE \quad (\text{common})$$

$$\therefore \triangle PDC \sim \triangle BEC \quad [\text{by AA-similarity}].$$

EXAMPLE 15 Diagonals AC and BD of a trapezium $ABCD$ with $AB \parallel DC$ intersect each other at point O . Using a similarity criterion for two triangles, show that $\frac{OA}{OC} = \frac{OB}{OD}$.

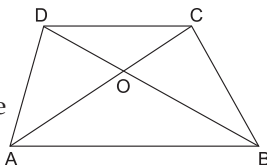
SOLUTION GIVEN A trapezium $ABCD$ in which $AB \parallel DC$. The diagonals AC and BD intersect at O .

TO PROVE $\frac{OA}{OC} = \frac{OB}{OD}$.

PROOF In $\triangle OAB$ and $\triangle OCD$, we have

$$\angle OAB = \angle OCD$$

$$[\text{alternate angles, since } AB \parallel DC]$$



and $\angle OBA = \angle ODC$ [alternate angles, since $AB \parallel DC$].

$\therefore \triangle OAB \sim \triangle OCD$ [by AA-similarity].

And so, $\frac{OA}{OC} = \frac{OB}{OD}$.

EXAMPLE 16 In $\triangle ABC$, $AD \perp BC$

and $AD^2 = BD \cdot CD$.

Prove that $\angle BAC = 90^\circ$.

SOLUTION GIVEN A $\triangle ABC$ in which
 $AD \perp BC$ and $AD^2 = BD \cdot CD$.

TO PROVE $\angle BAC = 90^\circ$.

PROOF $AD^2 = BD \cdot CD \Rightarrow \frac{BD}{AD} = \frac{AD}{CD}$.

Now, in $\triangle DBA$ and $\triangle DAC$, we have

$$\angle BDA = \angle ADC = 90^\circ \text{ and } \frac{BD}{AD} = \frac{AD}{CD}.$$

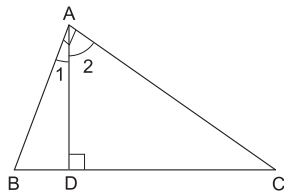
$\therefore \triangle DBA \sim \triangle DAC$ [by SAS-similarity]

$\therefore \angle B = \angle 2$ and $\angle 1 = \angle C$

$\therefore \angle 1 + \angle 2 = \angle B + \angle C \Rightarrow \angle A = \angle B + \angle C$

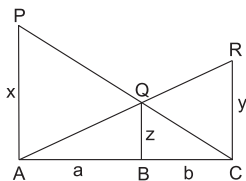
$$\Rightarrow 2\angle A = \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle A = \angle BAC = 90^\circ.$$



EXAMPLE 17 In the given figure PA , QB and RC each is perpendicular to AC such that $PA = x$, $RC = y$, $QB = z$, $AB = a$ and $BC = b$.

Prove that $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$. [CBSE 2000C]



SOLUTION $PA \perp AC$ and $QB \perp AC \Rightarrow QB \parallel PA$.

Thus, in $\triangle PAC$, $QB \parallel PA$. So, $\triangle QBC \sim \triangle PAC$.

$$\therefore \frac{QB}{PA} = \frac{BC}{AC} \Rightarrow \frac{z}{x} = \frac{b}{a+b} \dots (i) \text{ [by the property of similar } \triangle]$$

In $\triangle RAC$, $QB \parallel RC$. So, $\triangle QBA \sim \triangle RCA$.

$$\therefore \frac{QB}{RC} = \frac{AB}{AC} \Rightarrow \frac{z}{y} = \frac{a}{a+b} \dots (ii) \text{ [by the property of similar } \triangle]$$

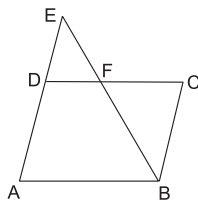
From (i) and (ii), we get

$$\frac{z}{x} + \frac{z}{y} = \left(\frac{b}{a+b} + \frac{a}{a+b} \right) = 1$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = 1 \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}.$$

$$\text{Hence, } \frac{1}{x} + \frac{1}{y} = \frac{1}{z}.$$

EXAMPLE 18 The side AD of a parallelogram $ABCD$ is produced to a point E . BE intersects CD at F . Show that $\triangle ABE \sim \triangle CFB$.



SOLUTION We have

$$AD \parallel BC \Rightarrow AE \parallel BC \quad [ABCD \text{ is a parallelogram}]$$

$$\therefore \angle AEB = \angle CBF \quad [\text{alternate angles}].$$

In $\triangle ABE$ and $\triangle CFB$, we have

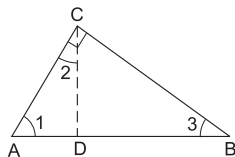
$$\angle AEB = \angle CBF \quad [\text{proved above}]$$

$$\angle EAB = \angle BCF \quad [\because \angle DAB = \angle BCD, \text{ being opposite angles of a parallelogram}]$$

$$\therefore \triangle ABE \sim \triangle CFB \quad [\text{by AA-similarity}].$$

EXAMPLE 19 In the given figure, $\angle ACB = 90^\circ$ and $CD \perp AB$. Prove that

$$CD^2 = BD \cdot AD. \quad [\text{CBSE 2006}]$$



SOLUTION GIVEN A $\triangle ABC$ in which

$$\angle ACB = 90^\circ \text{ and } CD \perp AB.$$

$$\text{TO PROVE } CD^2 = BD \cdot AD.$$

PROOF In right $\triangle ADC$, we have $\angle 1 + \angle 2 = 90^\circ$.

In right $\triangle ACB$, we have

$$\angle 1 + \angle 3 = 90^\circ.$$

$$\therefore \angle 1 + \angle 2 = \angle 1 + \angle 3 \Rightarrow \angle 2 = \angle 3.$$

In $\triangle ADC$ and $\triangle CDB$, we have

$$\angle 2 = \angle 3 \text{ (proved)}$$

$$\text{and } \angle ADC = \angle CDB = 90^\circ.$$

$$\therefore \triangle ADC \sim \triangle CDB \quad [\text{by AA-similarity}]$$

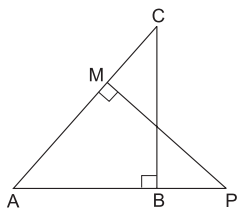
$$\therefore \frac{AD}{CD} = \frac{CD}{BD}.$$

$$\text{Hence, } CD^2 = BD \cdot AD.$$

EXAMPLE 20 In the given figure, $\triangle ABC$ and $\triangle AMP$ are right-angled at B and M respectively.

Prove that: (i) $\triangle ABC \sim \triangle AMP$

$$(ii) \frac{CA}{PA} = \frac{BC}{MP}$$



SOLUTION GIVEN $\triangle ABC$ and $\triangle AMP$ such that $\angle B = 90^\circ$ and $\angle M = 90^\circ$.

TO PROVE (i) $\triangle ABC \sim \triangle AMP$ (ii) $\frac{CA}{PA} = \frac{BC}{MP}$.

PROOF (i) In $\triangle ABC$ and $\triangle AMP$, we have

$$\angle ABC = \angle AMP = 90^\circ$$

$$\angle A = \angle A \quad (\text{common})$$

$$\therefore \triangle ABC \sim \triangle AMP \quad [\text{by AA-similarity}].$$

(ii) Since $\triangle ABC \sim \triangle AMP$, their corresponding sides are proportional.

$$\therefore \frac{CA}{PA} = \frac{BC}{MP}.$$

EXAMPLE 21 In a $\triangle ABC$, $AB = AC$ and D is a point on AC such that $BC^2 = AC \times DC$. Prove that $BD = BC$.

SOLUTION GIVEN A $\triangle ABC$ in which $AB = AC$ and D is a point on AC such that $BC^2 = AC \times DC$.

TO PROVE $BD = BC$.

PROOF $BC^2 = AC \times DC$ (given)

$$\Rightarrow \frac{BC}{DC} = \frac{AC}{BC}.$$

Thus, in $\triangle ABC$ and $\triangle BDC$, we have

$$\frac{BC}{DC} = \frac{AC}{BC} \text{ and } \angle C = \angle C \quad (\text{common}).$$

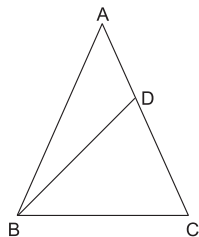
$$\therefore \triangle ABC \sim \triangle BDC \quad [\text{by SAS-similarity}]$$

$$\Rightarrow \frac{AC}{BC} = \frac{AB}{BD}$$

$$\Rightarrow \frac{AC}{BC} = \frac{AC}{BD} \quad [\because AB = AC \text{ (given)}]$$

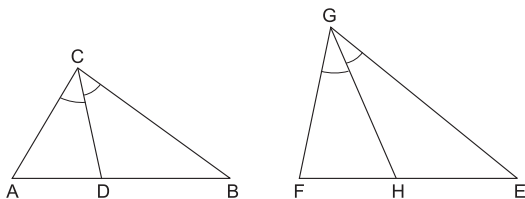
$$\Rightarrow BD = BC.$$

Hence, $BD = BC$.



EXAMPLE 22 In the given figure, CD and GH are respectively the bisectors of $\angle ACB$ and $\angle FGE$ of $\triangle ABC$ and $\triangle FEG$ respectively. If $\triangle ABC \sim \triangle FEG$, prove that:

$$(a) \triangle ADC \sim \triangle FHG \quad (b) \triangle BCD \sim \triangle EGH \quad (c) \frac{CD}{GH} = \frac{AC}{FG}$$



SOLUTION

$$\triangle ABC \sim \triangle FEG \quad (\text{given})$$

$$\therefore \angle ACB = \angle FGE$$

[corresponding angles of similar triangles are equal]

$$\Rightarrow \frac{1}{2} \angle ACB = \frac{1}{2} \angle FGE$$

$$\therefore \angle ACD = \angle FGH \text{ and } \angle DCB = \angle HGE.$$

(a) In $\triangle ADC$ and $\triangle FHG$, we have

$$\angle DAC = \angle HFG \quad [\because \angle A = \angle F \text{ since } \triangle ABC \sim \triangle FEG]$$

$$\text{and } \angle ACD = \angle FGH \quad [\text{proved above}]$$

$$\therefore \triangle ADC \sim \triangle FHG \quad [\text{by AA-similarity}].$$

(b) In $\triangle BCD$ and $\triangle EGH$, we have

$$\angle DBC = \angle HEG \quad [\because \angle B = \angle E \text{ since } \triangle ABC \sim \triangle FEG]$$

$$\text{and } \angle DCB = \angle HGE \quad [\text{proved above}]$$

$$\therefore \triangle BCD \sim \triangle EGH.$$

(c) We have

$$\triangle ADC \sim \triangle FHG \quad [\text{proved above}].$$

$$\text{And so, } \frac{CD}{GH} = \frac{AC}{FG}$$

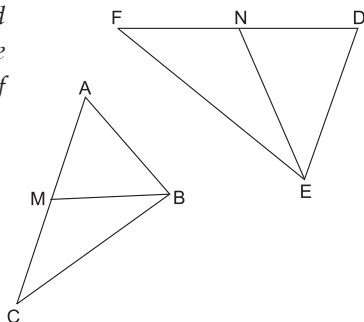
[corresponding sides of similar triangles are proportional].

EXAMPLE 23 In the given figure, BM and EN are respectively the medians of $\triangle ABC$ and $\triangle DEF$. If $\triangle ABC \sim \triangle DEF$, prove that:

(a) $\triangle AMB \sim \triangle DNE$

(b) $\triangle CMB \sim \triangle FNE$

(c) $\frac{BM}{EN} = \frac{AC}{DF}$



SOLUTION $\triangle ABC \sim \triangle DEF$ (given)

$$\therefore \angle A = \angle D, \angle B = \angle E \text{ and } \angle C = \angle F \quad \dots (i)$$

$$\text{and } \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}. \quad \dots (ii)$$

Since BM and EN are medians, we have

$$CA = 2AM = 2CM \text{ and } FD = 2DN = 2FN.$$

$\therefore$ from (ii), we have

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2AM}{2DN} = \frac{2CM}{2FN}$$

$$\Rightarrow \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{AM}{DN} = \frac{CM}{FN}. \quad \dots (iii)$$

(a) In $\triangle AMB$ and $\triangle DNE$, we have

$$\angle BAM = \angle EDN \quad [\because \angle A = \angle D \text{ from (i)}]$$

$$\text{and } \frac{AB}{DE} = \frac{AM}{DN} \quad [\text{from (iii)}].$$

$$\therefore \triangle AMB \sim \triangle DNE \quad [\text{by SAS-similarity}].$$

(b) In $\triangle CMB$ and $\triangle FNE$, we have

$$\angle BCM = \angle FEN \quad [\because \angle C = \angle F \text{ from (i)}]$$

$$\text{and } \frac{BC}{EF} = \frac{CM}{FN} \quad [\text{from (iii)}].$$

$$\therefore \triangle CMB \sim \triangle FNE \quad [\text{by SAS-similarity}].$$

(c) As proved above, $\triangle AMB \sim \triangle DNE$ and so

$$\frac{AB}{DE} = \frac{BM}{EN}. \quad \dots (iv)$$

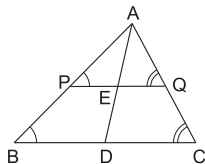
From (ii) and (iv), we get

$$\frac{BM}{EN} = \frac{AC}{FD}.$$

EXAMPLE 24 In a $\triangle ABC$, P and Q are points on AB and AC respectively such that $PQ \parallel BC$. Prove that the median AD , drawn from A to BC , bisects PQ .

SOLUTION GIVEN A $\triangle ABC$ in which P and Q are points on AB and AC respectively such that $PQ \parallel BC$ and AD is the median, cutting PQ at E .

TO PROVE $PE = EQ$.



PROOF In $\triangle APE$ and $\triangle ABD$, we have

$$\angle PAE = \angle BAD \quad [\text{common}]$$

$$\angle APE = \angle ABD \quad [\text{corresponding } \angle\text{s}]$$

$$\therefore \triangle APE \sim \triangle ABD \quad [\text{by AA-similarity}].$$

But, in similar triangles, the corresponding sides are proportional.

$$\therefore \frac{AE}{AD} = \frac{PE}{BD}. \quad \dots (i)$$

In $\triangle AEQ$ and $\triangle ADC$, we have

$$\angle QAE = \angle CAD \quad [\text{common}]$$

$$\angle AQE = \angle ACD \quad [\text{corresponding angles}]$$

$$\therefore \triangle AEQ \sim \triangle ADC \quad [\text{by AA-similarity}].$$

But, in similar triangles, the corresponding sides are proportional.

$$\therefore \frac{AE}{AD} = \frac{EQ}{DC}. \quad \dots (ii)$$

$$\text{From (i) and (ii), we get } \frac{PE}{BD} = \frac{EQ}{DC} \quad \left[\text{each equal to } \frac{AE}{AD} \right].$$

$$\text{But, } BD = DC \quad [\because AD \text{ is the median}]$$

$$\therefore PE = EQ.$$

EXAMPLE 25 In the given figure, E is a point on side CB produced of an isosceles $\triangle ABC$ with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that $\triangle ABD \sim \triangle ECF$.

SOLUTION GIVEN A $\triangle ABC$ in which $AB = AC$ and $AD \perp BC$. Side CB is produced to E and $EF \perp AC$.

TO PROVE $\triangle ABD \sim \triangle ECF$.

PROOF We know that the angles opposite to equal sides of a triangle are equal.

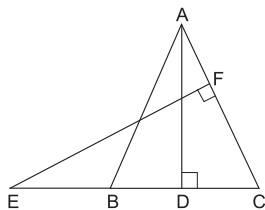
$$\therefore \angle B = \angle C \quad [\because AB = AC].$$

Now, in $\triangle ABD$ and $\triangle ECF$, we have

$$\angle B = \angle C \quad [\text{proved above}]$$

$$\angle ADB = \angle EFC = 90^\circ.$$

$$\therefore \triangle ABD \sim \triangle ECF \quad [\text{by AA-similarity}].$$

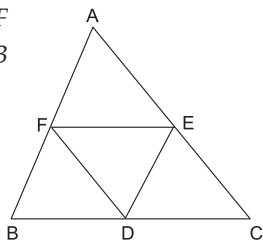


EXAMPLE 26 Prove that the line segments joining the midpoints of the sides of a triangle form four triangles, each of which is similar to the original triangle.

SOLUTION

GIVEN A $\triangle ABC$ in which D, E, F are the midpoints of BC, CA and AB respectively.

TO PROVE $\triangle AFE \sim \triangle ABC$,
 $\triangle FBD \sim \triangle ABC$,
 $\triangle EDC \sim \triangle ABC$.
 and $\triangle DEF \sim \triangle ABC$.



PROOF We shall first show that $\triangle AFE \sim \triangle ABC$.

Since F and E are the midpoints of AB and AC respectively, so by the midpoint theorem, we have $FE \parallel BC$.

$$\therefore \angle AFE = \angle B \quad [\text{corresponding } \angle s]$$

Now, in $\triangle AFE$ and $\triangle ABC$, we have

$$\angle AFE = \angle B \quad [\text{corresponding } \angle s]$$

$$\text{and } \angle A = \angle A \quad (\text{common}).$$

$$\therefore \triangle AFE \sim \triangle ABC \quad [\text{by AA-similarity}].$$

Similarly, $\triangle FBD \sim \triangle ABC$ and $\triangle EDC \sim \triangle ABC$.

Now, we shall show that $\triangle DEF \sim \triangle ABC$.

In the same manner as above, we can prove that

$$ED \parallel AF \text{ and } DF \parallel EA.$$

$$\therefore AFDE \text{ is a } \parallel\text{gm}.$$

$$\therefore \angle EDF = \angle A \quad [\text{opposite angles of a } \parallel\text{gm}].$$

Similarly, $BDEF$ is a $\parallel\text{gm}$.

$$\therefore \angle DEF = \angle B \quad [\text{opposite angles of a } \parallel\text{gm}].$$

Thus, in $\triangle DEF$ and $\triangle ABC$, we have

$$\angle EDF = \angle A \text{ and } \angle DEF = \angle B.$$

$$\therefore \triangle DEF \sim \triangle ABC \quad [\text{by AA-similarity}].$$

Hence, the result follows.

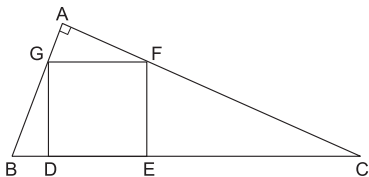
EXAMPLE 27 In the given figure, $DEFG$ is a square and $\angle BAC = 90^\circ$. Prove that

$$(i) \triangle AGF \sim \triangle DBG$$

$$(ii) \triangle AGF \sim \triangle EFC$$

$$(iii) \triangle DBG \sim \triangle EFC$$

$$(iv) DE^2 = BD \times EC$$



SOLUTION GIVEN $\triangle ABC$ in which $\angle BAC = 90^\circ$ and $DEFG$ is a square.

TO PROVE (i) $\triangle AGF \sim \triangle DBG$ (ii) $\triangle AGF \sim \triangle EFC$

(iii) $\triangle DBG \sim \triangle EFC$ (iv) $DE^2 = BD \times EC$

PROOF (i) In $\triangle AGF$ and $\triangle DBG$, we have

$$\angle GAF = \angle BDG = 90^\circ$$

$$\angle AGF = \angle DBG \quad [\text{corresponding } \angle s]$$

$[\because GF \parallel BC \text{ and } AB \text{ is the transversal}]$

$$\therefore \triangle AGF \sim \triangle DBG.$$

(ii) In $\triangle AGF$ and $\triangle EFC$, we have

$$\angle FAG = \angle CEF = 90^\circ$$

$$\angle GFA = \angle FCE \quad [\text{corresponding } \angle s]$$

$[\because GF \parallel BC \text{ and } AC \text{ is the transversal}]$

$$\therefore \triangle AGF \sim \triangle EFC.$$

(iii) $\triangle DBG \sim \triangle AGF$ and $\triangle AGF \sim \triangle EFC$

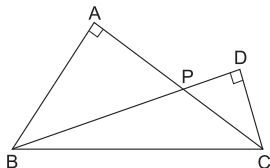
$$\Rightarrow \triangle DBG \sim \triangle EFC.$$

$$(iv) \triangle DBG \sim \triangle EFC \Rightarrow \frac{BD}{FE} = \frac{DG}{EC}$$

$$\Rightarrow \frac{BD}{DE} = \frac{DE}{EC} \quad [\because DG = DE \text{ and } FE = DE].$$

$$\text{Hence, } DE^2 = BD \times EC.$$

EXAMPLE 28 Two right triangles ABC and DBC are drawn on the same hypotenuse BC and on the same side of BC . If AC and BD intersect at P , prove that $AP \times PC = BP \times PD$. [CBSE 2000C]



SOLUTION GIVEN Right triangles $\triangle ABC$ and $\triangle DBC$ are drawn on the same hypotenuse BC and on the same side of BC . Also, AC and BD intersect at P .

TO PROVE $AP \times PC = BP \times PD$.

PROOF In $\triangle BAP$ and $\triangle CDP$, we have

$$\angle BAP = \angle CDP = 90^\circ$$

$$\angle BPA = \angle CPD \quad (\text{ver. opp. } \angle s)$$

$\therefore \triangle BAP \sim \triangle CDP$ [by AA-similarity]

$$\therefore \frac{AP}{DP} = \frac{BP}{CP}$$

$$\Rightarrow AP \times CP = BP \times DP \Rightarrow AP \times PC = BP \times PD.$$

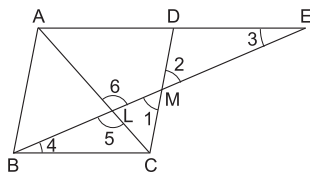
Hence, $AP \times PC = BP \times PD$.

EXAMPLE 29 Through the midpoint M of the side CD of a parallelogram $ABCD$, the line BM is drawn, intersecting AC in L and AD produced in E . Prove that $EL = 2BL$. [CBSE 2006C, '08, '09]

SOLUTION

A ||gm $ABCD$ and M is the midpoint of CD . Line BM is drawn, intersecting AC in L and AD produced in E .

TO PROVE $EL = 2BL$.



PROOF In $\triangle BMC$ and $\triangle EMD$, we have

$$\angle 1 = \angle 2 \quad (\text{vert. opp. } \angle)$$

$$MC = MD \quad [M \text{ is the midpoint of } CD]$$

$$\angle BCM = \angle EDM \quad [\text{alternate interior } \angle]$$

$$\therefore \triangle BMC \cong \triangle EMD$$

$$\therefore BC = DE.$$

But, $BC = AD$ [opposite sides of a ||gm]

$$\therefore BC = AD = DE \Rightarrow AE = (AD + DE) = 2BC. \quad \dots (i)$$

Now, in $\triangle AEL$ and $\triangle CBL$, we have

$$\angle 6 = \angle 5 \quad (\text{vert. opp. } \angle)$$

$$\angle 3 = \angle 4 \quad [\text{alternate interior } \angle]$$

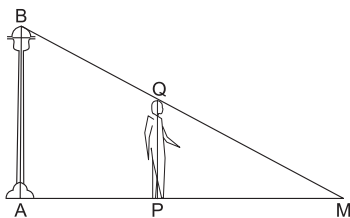
$$\therefore \triangle AEL \sim \triangle CBL \quad [\text{AA-similarity}]$$

$$\Rightarrow \frac{EL}{BL} = \frac{AE}{BC} = \frac{2BC}{BC} = 2 \quad [\text{using (i)}]$$

$$\Rightarrow EL = 2BL.$$

Hence, $EL = 2BL$.

EXAMPLE 30 A lamp is 3.3 m above the ground. A boy 110 cm tall walks away from the base of this lamp post at a speed of 0.8 m/s. Find the length of the shadow of the boy after 4 seconds.



SOLUTION Let AB be the lamp post and PQ be the boy, where P is the position of the boy after 4 seconds.

AP = distance moved in 4 s at 0.8 m/s = (4×0.8) m = 3.2 m.

PM is the length of shadow of the boy.

Let $PM = x$ m.

In $\triangle AMB$ and $\triangle PMQ$, we have

$$\angle MAB = \angle MPQ = 90^\circ$$

[$\because$ both the lamp post and the boy stand vertically erect]

$$\angle AMB = \angle PMQ \text{ (common)}$$

$\therefore \triangle AMB \sim \triangle PMQ$ [by AA-similarity].

$$\text{And so, } \frac{AM}{PM} = \frac{AB}{PQ}$$

[corres. sides of similar triangles are proportional]

$$\Rightarrow \frac{AP + PM}{PM} = \frac{AB}{PQ} \Rightarrow \frac{3.2 + x}{x} = \frac{3.3}{1.1}$$

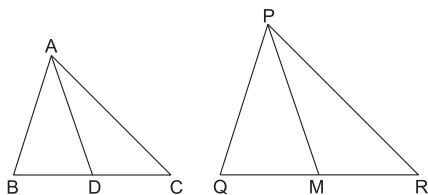
$$[\because AB = 3.3 \text{ m, } PQ = 110 \text{ cm} = 1.1 \text{ m}]$$

$$\Rightarrow 3.2 + x = 3x \Rightarrow 2x = 3.2 \Rightarrow x = 1.6.$$

$\therefore$ the length of the shadow of the boy after 4 seconds is 1.6 m.

EXAMPLE 31

Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of another triangle PQR , as shown in the figure. Prove that $\triangle ABC \sim \triangle PQR$.



SOLUTION

We have

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$$

$$\Rightarrow \frac{AB}{PQ} = \frac{AD}{PM} = \frac{BC}{QR} = \frac{\frac{1}{2}BC}{\frac{1}{2}QR} = \frac{BD}{QM} \quad \dots (i)$$

In $\triangle ABD$ and $\triangle PQM$, we have

$$\frac{AB}{PQ} = \frac{AD}{PM} = \frac{BD}{QM} \quad [\text{from (i)}]$$

$\therefore \triangle ABD \sim \triangle PQM$ [by SSS-similarity].

And so, $\angle B = \angle Q$ [corres. angles of similar triangles are equal].

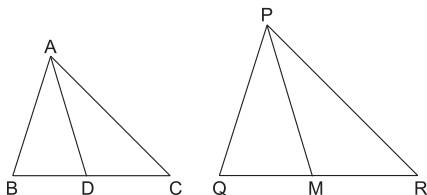
Now, in $\triangle ABC$ and $\triangle PQR$, we have

$$\angle B = \angle Q \quad [\text{proved above}]$$

$$\text{and } \frac{AB}{PQ} = \frac{BD}{QM} \quad [\text{from (i)}].$$

$\therefore \triangle ABC \sim \triangle PQR$ [by SAS-similarity].

EXAMPLE 32 Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR . Prove that $\triangle ABC \sim \triangle PQR$.



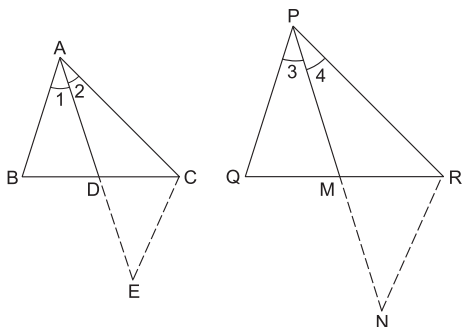
SOLUTION GIVEN AD and PM are medians of $\triangle ABC$ and $\triangle PQR$ respectively such that

$$\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}.$$

TO PROVE $\triangle ABC \sim \triangle PQR$.

CONSTRUCTION Produce AD to E such that $AD = DE$ and produce PM to N such that $PM = MN$.

Join EC and NR .



PROOF In $\triangle ABD$ and $\triangle ECD$, we have

$$BD = CD \quad [\because D \text{ is the midpoint of } BC]$$

$$AD = ED \quad [\text{by construction}]$$

$$\angle BDA = \angle CDE \quad [\text{vertically opposite angles}]$$

$\therefore \triangle ABD \cong \triangle ECD$ [by SAS-congruency].

And so, $AB = EC$ (i) [cpct]

Similarly, $\triangle PQM \cong \triangle NRM$

and so, $PQ = NR$ (ii) [cpct]

Now, $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}$ (given)

$\Rightarrow \frac{EC}{NR} = \frac{AC}{PR} = \frac{AD}{PM}$ [using (i) and (ii)]

$\Rightarrow \frac{EC}{NR} = \frac{AC}{PR} = \frac{2AD}{2PM} = \frac{AE}{PN}$ [$\because 2AD = AE, 2PM = PN$]

$\Rightarrow \triangle ACE \sim \triangle PNR$ [SSS-similarity].

$\therefore \angle 2 = \angle 4$

[corresponding angles of similar triangles are equal].

Similarly, $\angle 1 = \angle 3$ [it can be proved by joining BE and QN and showing $\triangle ABE \sim \triangle PQN$].

$\therefore \angle 1 + \angle 2 = \angle 3 + \angle 4$, i.e., $\angle BAC = \angle QPR$ (iii)

Now, in $\triangle ABC$ and $\triangle PQR$, we have

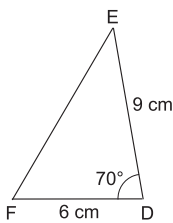
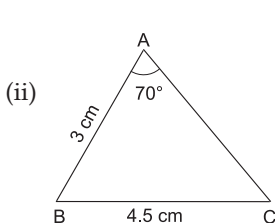
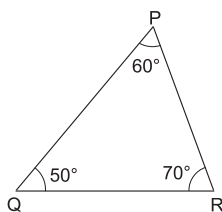
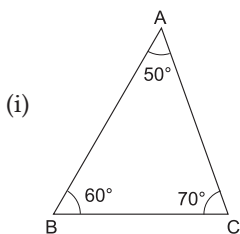
$\frac{AB}{PQ} = \frac{AC}{PR}$ [given]

$\angle BAC = \angle QPR$ [from (iii)]

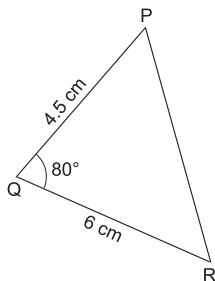
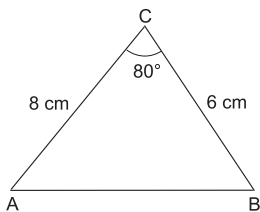
$\therefore \triangle ABC \sim \triangle PQR$ [by SAS-similarity].

EXERCISE 7B

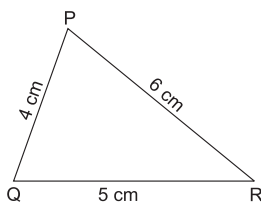
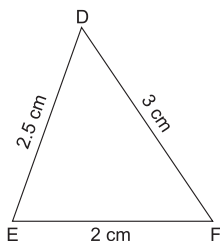
1. In each of the given pairs of triangles, find which pair of triangles are similar. State the similarity criterion and write the similarity relation in symbolic form.



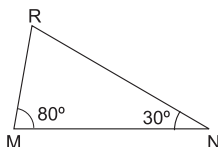
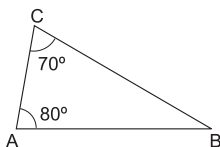
(iii)



(iv)

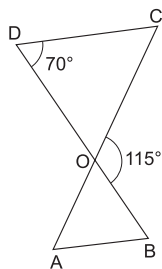


(v)

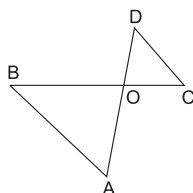


2. In the given figure, $\triangle ODC \sim \triangle OBA$, $\angle BOC = 115^\circ$ and $\angle CDO = 70^\circ$.

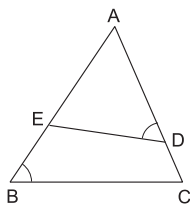
Find (i) $\angle DOC$ (ii) $\angle DCO$ (iii) $\angle OAB$
(iv) $\angle OBA$.



3. In the given figure, $\triangle OAB \sim \triangle OCD$. If $AB = 8$ cm, $BO = 6.4$ cm, $OC = 3.5$ cm and $CD = 5$ cm, find (i) OA (ii) DO .

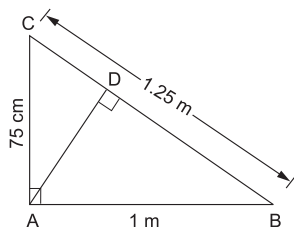


4. In the given figure, if $\angle ADE = \angle B$, show that $\triangle ADE \sim \triangle ABC$. If $AD = 3.8$ cm, $AE = 3.6$ cm, $BE = 2.1$ cm and $BC = 4.2$ cm, find DE .

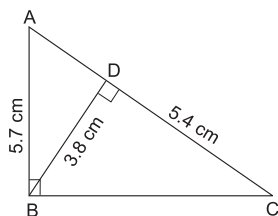


5. The perimeters of two similar triangles ABC and PQR are 32 cm and 24 cm respectively. If $PQ = 12$ cm, find AB . [CBSE 2001]
6. The corresponding sides of two similar triangles ABC and DEF are $BC = 9.1$ cm and $EF = 6.5$ cm. If the perimeter of $\triangle DEF$ is 25 cm, find the perimeter of $\triangle ABC$.

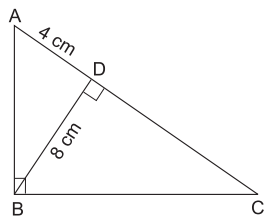
7. In the given figure, $\angle CAB = 90^\circ$ and $AD \perp BC$. Show that $\triangle BDA \sim \triangle BAC$. If $AC = 75$ cm, $AB = 1$ m and $BC = 1.25$ m, find AD .



8. In the given figure, $\angle ABC = 90^\circ$ and $BD \perp AC$. If $AB = 5.7$ cm, $BD = 3.8$ cm and $CD = 5.4$ cm, find BC .

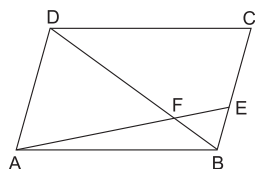


9. In the given figure, $\angle ABC = 90^\circ$ and $BD \perp AC$. If $BD = 8$ cm, $AD = 4$ cm, find CD .



10. P and Q are points on the sides AB and AC respectively of a $\triangle ABC$. If $AP = 2$ cm, $PB = 4$ cm, $AQ = 3$ cm and $QC = 6$ cm, show that $BC = 3PQ$.

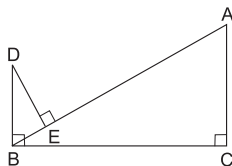
11. $ABCD$ is a parallelogram and E is a point on BC . If the diagonal BD intersects AE at F , prove that $AF \times FB = EF \times FD$.



12. In the given figure, $DB \perp BC$, $DE \perp AB$ and $AC \perp BC$.

Prove that $\frac{BE}{DE} = \frac{AC}{BC}$.

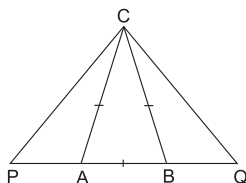
[CBSE 2008]



13. A vertical pole of length 7.5 m casts a shadow 5 m long on the ground and at the same time a tower casts a shadow 24 m long. Find the height of the tower.

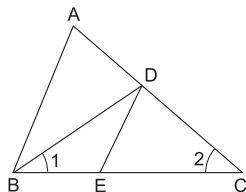
14. In an isosceles $\triangle ABC$, the base AB is produced both ways in P and Q such that $AP \times BQ = AC^2$.

Prove that $\triangle ACP \sim \triangle BCQ$.

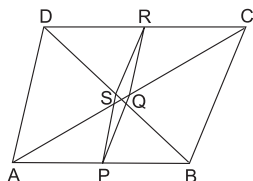


15. In the given figure, $\angle 1 = \angle 2$ and $\frac{AC}{BD} = \frac{CB}{CE}$.

Prove that $\triangle ACB \sim \triangle DCE$.

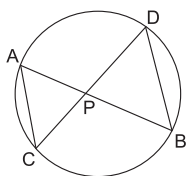


16. $ABCD$ is a quadrilateral in which $AD = BC$. If P, Q, R, S be the midpoints of AB, AC, CD and BD respectively, show that $PQRS$ is a rhombus.



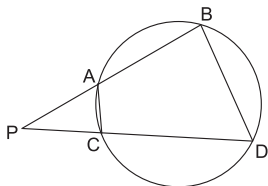
17. In a circle, two chords AB and CD intersect at a point P inside the circle. Prove that

(a) $\triangle PAC \sim \triangle PDB$ (b) $PA \cdot PB = PC \cdot PD$.



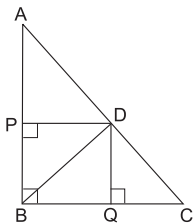
18. Two chords AB and CD of a circle intersect at a point P outside the circle. Prove that

(a) $\triangle PAC \sim \triangle PDB$ (b) $PA \cdot PB = PC \cdot PD$.



19. In a right triangle ABC , right-angled at B , D is a point on hypotenuse such that $BD \perp AC$. If $DP \perp AB$ and $DQ \perp BC$ then prove that

(a) $DQ^2 = DP \cdot QC$ (b) $DP^2 = DQ \cdot AP$.



ANSWERS (EXERCISE 7B)

1. (i) $\triangle ABC \sim \triangle QPR$ (AAA-similarity)
 (ii) not similar
 (iii) $\triangle CAB \sim \triangle QRP$ (SAS-similarity)
 (iv) $\triangle FED \sim \triangle PQR$ (SSS-similarity)
 (v) $\triangle ABC \sim \triangle MNR$ (AA-similarity)
2. (i) $\angle DOC = 65^\circ$ (ii) $\angle DCO = 45^\circ$ (iii) $\angle OAB = 45^\circ$ (iv) $\angle OBA = 70^\circ$
3. (i) $OA = 5.6$ cm (ii) $DO = 4$ cm 4. 2.8 cm 5. 16 cm
6. 35 cm 7. $AD = 60$ cm 8. $BC = 8.1$ cm 9. $CD = 16$ cm
13. 36 m

HINTS TO SOME SELECTED QUESTIONS

1. (i) $\angle A = \angle Q$, $\angle B = \angle P$ and $\angle C = \angle R$.
 $\therefore \triangle ABC \sim \triangle QPR$ (AAA-similarity).
 (ii) SAS-similarity is not satisfied as included angles are not equal.
 (iii) $\triangle CAB \sim \triangle QRP$ (SAS-similarity), as $\frac{CA}{QR} = \frac{CB}{QP}$ and $\angle C = \angle Q$.
 (iv) $FE = 2$ cm, $FD = 3$ cm, $ED = 2.5$ cm; $PQ = 4$ cm, $PR = 6$ cm, $QR = 5$ cm.
 $\therefore \triangle FED \sim \triangle PQR$ (SSS-similarity).
 (v) $\angle B = 180^\circ - (\angle A + \angle C) = 180^\circ - (80^\circ + 70^\circ) = 30^\circ$.
 $\therefore \angle A = \angle M$ and $\angle B = \angle N$, and so $\triangle ABC \sim \triangle MNR$ (AA-similarity).
2. $\angle DOC = (180^\circ - 115^\circ) = 65^\circ$, $\angle DCO = 180^\circ - (70^\circ + 65^\circ) = 45^\circ$.
 $\triangle ODC \sim \triangle OBA \Rightarrow \angle OAB = \angle DCO = 45^\circ$, $\angle OBA = \angle ODC = 70^\circ$.

$$3. \triangle OAB \sim \triangle OCD \Rightarrow \frac{OA}{OC} = \frac{AB}{CD} = \frac{BO}{DO} \Rightarrow \frac{OA}{3.5} = \frac{8}{5} = \frac{6.4}{DO}.$$

Find OA and DO.

$$4. \angle ADE = \angle B, \angle A = \angle A.$$

$$\triangle ADE \sim \triangle ABC \Rightarrow \frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{3.8}{(3.6 + 2.1)} = \frac{x}{4.2}.$$

Find x.

$$5. \frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{AB}{PQ}.$$

$$7. \triangle BDA \sim \triangle BAC \Rightarrow \frac{AD}{AC} = \frac{BA}{BC}.$$

$$8. \triangle ABC \sim \triangle BDC \Rightarrow \frac{AB}{BD} = \frac{BC}{DC}.$$

9. In $\triangle DBA$ and $\triangle DCB$, we have

$$\angle BDA = \angle CDB$$

$$\text{and } \angle DBA = \angle DCB \quad [\text{each} = 90^\circ - \angle A].$$

$$\therefore \triangle DBA \sim \triangle DCB.$$

$$\therefore \frac{BD}{CD} = \frac{AD}{BD} \Rightarrow CD = \frac{BD^2}{AD}.$$

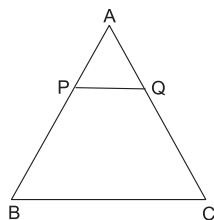
$$10. \frac{AP}{AB} = \frac{2}{6} = \frac{1}{3} \text{ and } \frac{AQ}{AC} = \frac{3}{9} = \frac{1}{3}.$$

In $\triangle APQ$ and $\triangle ABC$, we have

$$\angle A = \angle A \text{ and } \frac{AP}{AB} = \frac{AQ}{AC}.$$

$$\therefore \triangle APQ \sim \triangle ABC \quad [\text{by SAS-similarity}].$$

$$\frac{PQ}{BC} = \frac{AP}{AB} = \frac{1}{3} \Rightarrow BC = 3PQ.$$



11. $\triangle AFD \sim \triangle EFB$, as $\angle AFD = \angle EFB$ (vert. opp. $\angle$ s) and $\angle DAF = \angle BEF$ (alt. $\angle$ s).

$$\therefore \frac{AF}{EF} = \frac{FD}{FB}.$$

$$12. \left. \begin{array}{l} \angle BED = \angle ACB = 90^\circ \\ \angle EBD = \angle CAB = (90^\circ - \angle B) \end{array} \right\}; \therefore \triangle BED \sim \triangle ACB.$$

$$14. CA = CB \Rightarrow \angle CAB = \angle CBA$$

$$\Rightarrow 180^\circ - \angle CAB = 180^\circ - \angle CBA$$

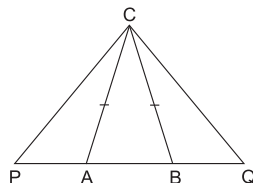
$$\Rightarrow \angle CAP = \angle CBQ.$$

$$\text{Now, } AP \times BQ = AC^2 \Rightarrow \frac{AP}{AC} = \frac{AC}{BQ}$$

$$\Rightarrow \frac{AP}{AC} = \frac{BC}{BQ} \quad [\because AC = BC].$$

$$\text{Thus, } \angle CAP = \angle CBQ \text{ and } \frac{AP}{AC} = \frac{BC}{BQ}.$$

$$\therefore \triangle ACP \sim \triangle BCQ.$$



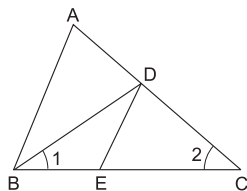
$$15. \frac{AC}{BD} = \frac{CB}{CE} \Rightarrow \frac{AC}{CB} = \frac{BD}{CE}.$$

$$\text{Also, } \angle 2 = \angle 1 \Rightarrow BD = DC.$$

$$\text{Thus, } \frac{AC}{CB} = \frac{DC}{CE} \Rightarrow \frac{AC}{DC} = \frac{CB}{CE}$$

$$\text{and } \angle ACB = \angle DCE = \angle C.$$

$$\therefore \triangle ACB \sim \triangle DCE \text{ [SAS-similarity].}$$



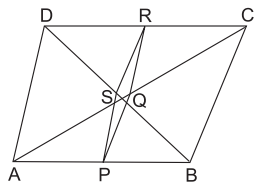
16. In $\triangle ABC$, P and Q are midpoints of AB and AC respectively.

$$\therefore PQ \parallel BC \text{ [by midpoint theorem].}$$

$$\text{And so, } \triangle APQ \sim \triangle ABC$$

$$\Rightarrow \frac{AP}{AB} = \frac{AQ}{AC} = \frac{PQ}{BC} \Rightarrow \frac{PQ}{BC} = \frac{1}{2} \Rightarrow PQ = \frac{1}{2}BC = \frac{1}{2}DA$$

$$\left[\because AP = \frac{1}{2}AB \text{ and } BC = DA \right].$$



$$\text{In } \triangle CDA, RQ \parallel DA \text{ and } RQ = \frac{1}{2}DA.$$

$$\text{In } \triangle BDA, SP \parallel DA \text{ and } SP = \frac{1}{2}DA.$$

$$\text{In } \triangle CDB, SR \parallel BC \text{ and } SR = \frac{1}{2}BC = \frac{1}{2}DA.$$

$$\therefore SP \parallel RQ \text{ and } PQ \parallel SR \text{ and } PQ = RQ = SP = SR.$$

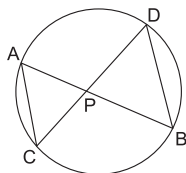
17. In $\triangle PAC$ and $\triangle PDB$, we have

$$\angle APC = \angle DPB \text{ [vertically opposite angles]}$$

$$\angle PAC = \angle PDB \text{ [angles in the same segment]}$$

$$\therefore \triangle PAC \sim \triangle PDB \text{ [by AA-similarity].}$$

$$\text{And so, } \frac{PA}{PD} = \frac{PC}{PB} \Rightarrow PA \cdot PB = PC \cdot PD.$$



18. Clearly, $ABDC$ is a cyclic quadrilateral.

$$\therefore \angle 1 + \angle 2 = 180^\circ. \quad \dots (i)$$

$$\text{In } \triangle PAC \text{ and } \triangle PDB, \text{ we have:}$$

$$\angle APC = \angle DPB \text{ [common]}$$

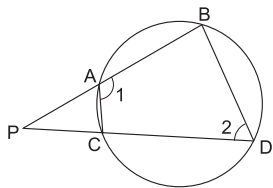
$$\angle PAC = \angle PDB \text{ [each equal to } (180^\circ - \angle 1)]$$

$$\text{[Note } \angle PAC + \angle 1 = 180^\circ \text{ (linear pair)}]$$

$$\text{and } \angle PDB + \angle 1 = 180^\circ. \text{ [using (i)]}$$

$$\therefore \triangle PAC \sim \triangle PDB \text{ [by AA-similarity]}$$

$$\text{And so, } \frac{PA}{PD} = \frac{PC}{PB} \Rightarrow PA \cdot PB = PC \cdot PD.$$



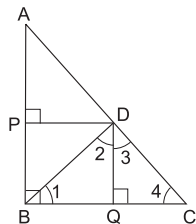
19. (a) $AB \perp BC$ and $DP \perp AB \Rightarrow DP \parallel BC \Rightarrow DP \parallel BQ$

$$BC \perp AB \text{ and } DQ \perp BC \Rightarrow DQ \parallel AB \Rightarrow DQ \parallel PB$$

$$\therefore BQDP \text{ is a rectangle.}$$

$$\text{And so, } BQ = DP \text{ and } BP = DQ$$

$$\text{In rt. } \triangle BQD: \angle 1 + \angle 2 = 90^\circ \quad \dots (i)$$



In rt. $\triangle DQC$: $\angle 3 = \angle 4 = 90^\circ$... (ii)

In rt. $\triangle BDC$: $\angle 2 + \angle 3 = 90^\circ$... (iii) [$\because BDC = 90^\circ$]

From (i) and (iii), we get $\angle 1 = \angle 3$.

From (ii) and (iii), we get $\angle 2 = \angle 4$.

$\therefore \triangle BQD \sim \triangle DQC$ [by AA-similarity]

And so, $\frac{BQ}{DQ} = \frac{DQ}{QC} \Rightarrow DQ^2 = BQ \cdot QC \Rightarrow DQ^2 = DP \cdot QC$ [$\because BQ = DP$]

(b) By proving $\triangle PDA \sim \triangle PBD$ (in a similar way), we get

$$\frac{PD}{PB} = \frac{AP}{DP} \Rightarrow DP^2 = BP \cdot AP \Rightarrow DP^2 = DQ \cdot AP \quad [\because BP = DQ].$$

RATIO OF THE AREAS OF TWO SIMILAR TRIANGLES

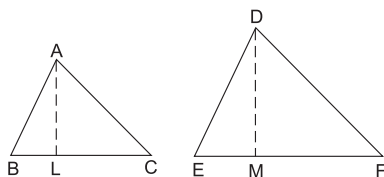
THEOREM 1 *The ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.*

[CBSE 2002, '02C, '04, '05, '05C, '06, '06C, '07, '08, '08C, '09C, '10]

GIVEN $\triangle ABC \sim \triangle DEF$.

TO PROVE $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}$.

CONSTRUCTION Draw $AL \perp BC$ and $DM \perp EF$.



PROOF Since $\triangle ABC \sim \triangle DEF$, it follows that they are equiangular and their sides are proportional.

$$\therefore \angle A = \angle D, \angle B = \angle E, \angle C = \angle F \text{ and } \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}. \quad \dots (i)$$

$$\text{Now, ar}(\triangle ABC) = \left(\frac{1}{2} \times BC \times AL \right)$$

$$\text{and ar}(\triangle DEF) = \left(\frac{1}{2} \times EF \times DM \right).$$

$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{\frac{1}{2} \times BC \times AL}{\frac{1}{2} \times EF \times DM} = \frac{BC}{EF} \times \frac{AL}{DM}. \quad \dots (ii)$$

Also, $\frac{AL}{DM} = \frac{BC}{EF}$... (iii) [$\because$ in similar triangles, the ratio of the corres. sides is the same as the ratio of corresponding altitudes]

Using (iii) in (ii), we get

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \left(\frac{BC}{EF} \times \frac{BC}{EF}\right) = \frac{BC^2}{EF^2}.$$

Similarly, $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2}$ and $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AC^2}{DF^2}$.

$$\text{Hence, } \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}.$$

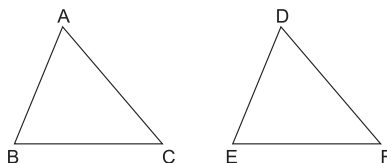
THEOREM 2 *If the areas of two similar triangles are equal then prove that the triangles are congruent.* [CBSE 2010]

GIVEN $\triangle ABC \sim \triangle DEF$ such that $\text{ar}(\triangle ABC) = \text{ar}(\triangle DEF)$.

TO PROVE $\triangle ABC \cong \triangle DEF$.

PROOF $\triangle ABC \sim \triangle DEF$ (given)

$$\Rightarrow \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}. \quad \dots (i)$$



Now, $\text{ar}(\triangle ABC) = \text{ar}(\triangle DEF)$ [given]

$$\Rightarrow \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = 1. \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2} = 1$$

$$\Rightarrow AB^2 = DE^2, AC^2 = DF^2 \text{ and } BC^2 = EF^2$$

$$\Rightarrow AB = DE, AC = DF \text{ and } BC = EF$$

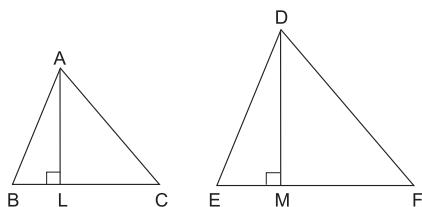
$$\Rightarrow \triangle ABC \cong \triangle DEF \quad [\text{by SSS-congruency}].$$

THEOREM 3 *Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding altitudes.*

GIVEN $\triangle ABC \sim \triangle DEF$, $AL \perp BC$ and $DM \perp EF$.

TO PROVE $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AL^2}{DM^2}$.

PROOF We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of the corresponding sides.



$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} \quad \dots (i)$$

Now, in $\triangle ALB$ and $\triangle DME$, we have

$$\angle ALB = \angle DME = 90^\circ \text{ and } \angle B = \angle E \quad [\because \triangle ABC \sim \triangle DEF].$$

$\therefore \triangle ALB \sim \triangle DME$ [by AA-similarity]

$$\Rightarrow \frac{AB}{DE} = \frac{AL}{DM}$$

$$\Rightarrow \frac{AB^2}{DE^2} = \frac{AL^2}{DM^2} \quad \dots (ii)$$

From (i) and (ii), we get

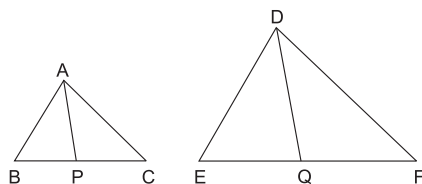
$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AL^2}{DM^2}.$$

THEOREM 4 Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding medians.

GIVEN $\triangle ABC \sim \triangle DEF$, AP and DQ are the medians of $\triangle ABC$ and $\triangle DEF$ respectively.

TO PROVE $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AP^2}{DQ^2}$.

PROOF We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.



$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} \quad \dots (i)$$

Now, $\triangle ABC \sim \triangle DEF$

$$\Rightarrow \frac{AB}{DE} = \frac{BC}{EF} = \frac{2BP}{2EQ} = \frac{BP}{EQ}$$

$$\Rightarrow \frac{AB}{DE} = \frac{BP}{EQ} \text{ and } \angle B = \angle E \quad [\because \triangle ABC \sim \triangle DEF]$$

$$\Rightarrow \triangle APB \sim \triangle DQE \quad [\text{by SAS-similarity}]$$

$$\Rightarrow \frac{AB}{DE} = \frac{AP}{DQ}$$

$$\Rightarrow \frac{AB^2}{DE^2} = \frac{AP^2}{DQ^2} \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AP^2}{DQ^2}.$$

THEOREM 5 *Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding angle-bisector segments.*

GIVEN $\triangle ABC \sim \triangle DEF$ in which AX and DY are the bisectors of $\angle A$ and $\angle D$ respectively.

TO PROVE $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AX^2}{DY^2}.$

PROOF We know that ratio of the areas of two similar triangles is equal to ratio of the squares of their corresponding sides.

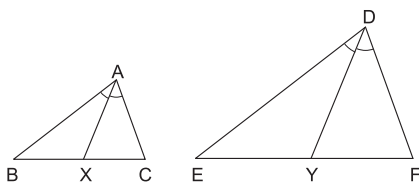
$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AB^2}{DE^2} \quad \dots (i)$$

$$\triangle ABC \sim \triangle DEF$$

$$\Rightarrow \angle A = \angle D$$

$$\Rightarrow \frac{1}{2} \angle A = \frac{1}{2} \angle D$$

$$\Rightarrow \angle BAX = \angle EDY.$$



Now, in $\triangle ABX$ and $\triangle DEY$, we have

$$\angle BAX = \angle EDY \text{ and } \angle B = \angle E \quad [\because \triangle ABC \sim \triangle DEF].$$

$$\therefore \triangle ABX \sim \triangle DEY \quad [\text{by AA-similarity}]$$

$$\frac{AB}{DE} = \frac{AX}{DY}$$

$$\Rightarrow \frac{AB^2}{DE^2} = \frac{AX^2}{DY^2} \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AX^2}{DY^2}.$$

SOLVED EXAMPLES

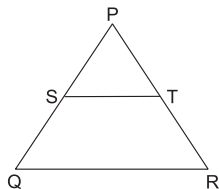
EXAMPLE 1 *If the areas of two similar triangles are in the ratio 25 : 64, find the ratio of their corresponding sides.* [CBSE 2009]

SOLUTION Let $\triangle ABC$ and $\triangle DEF$ be similar. Then,

$$\begin{aligned}\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} &= \frac{AB^2}{DE^2} = \frac{BC^2}{EF^2} = \frac{AC^2}{DF^2} \\ \Rightarrow \left(\frac{AB}{DE}\right)^2 &= \left(\frac{BC}{EF}\right)^2 = \left(\frac{AC}{DF}\right)^2 = \frac{25}{64} = \left(\frac{5}{8}\right)^2 \\ \Rightarrow \frac{AB}{DE} &= \frac{BC}{EF} = \frac{AC}{DF} = \frac{5}{8}.\end{aligned}$$

Hence, the ratio of their corresponding sides is 5 : 8.

EXAMPLE 2 *In the adjoining figure, S and T are points on the sides PQ and PR respectively of $\triangle PQR$ such that $PT = 2$ cm, $TR = 4$ cm and ST is parallel to QR . Find the ratio of the areas of $\triangle PST$ and $\triangle PQR$.*



[CBSE 2010]

SOLUTION $ST \parallel QR$ (given)

$\therefore \angle PST = \angle PQR$ [corresponding angles]

and $\angle PTS = \angle PRQ$ [corresponding angles].

And so, $\triangle PST \sim \triangle PQR$ [by AA-similarity].

$$\begin{aligned}\therefore \frac{\text{ar}(\triangle PST)}{\text{ar}(\triangle PQR)} &= \frac{PT^2}{PR^2} = \frac{PT^2}{(PT + TR)^2} \\ &= \frac{2^2}{(2 + 4)^2} = \frac{4}{36} = \frac{1}{9}.\end{aligned}$$

EXAMPLE 3 *The areas of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 25 cm^2 and 49 cm^2 respectively. If $QR = 9.8$ cm, find BC . [CBSE 2006]*

SOLUTION We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

$$\begin{aligned}\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} &= \frac{BC^2}{QR^2} \\ \Rightarrow \left(\frac{BC}{QR}\right)^2 &= \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{25}{49} = \left(\frac{5}{7}\right)^2 \\ \Rightarrow \frac{BC}{QR} &= \frac{5}{7} \Rightarrow \frac{BC}{9.8} = \frac{5}{7}\end{aligned}$$

$$\Rightarrow BC = \frac{5 \times 9.8}{7} \text{ cm} = 5 \times 1.4 \text{ cm} = 7 \text{ cm}.$$

Hence, $BC = 7 \text{ cm}$.

EXAMPLE 4 *The areas of two similar triangles are 81 cm^2 and 49 cm^2 respectively. If the altitude of the bigger triangle is 4.5 cm , find the corresponding altitude of the smaller triangle.* [CBSE 2002]

SOLUTION Let the given triangles be $\triangle ABC$ and $\triangle DEF$ such that $\text{ar}(\triangle ABC) = 81 \text{ cm}^2$ and $\text{ar}(\triangle DEF) = 49 \text{ cm}^2$. Let AL and DM be the corresponding altitudes of $\triangle ABC$ and $\triangle DEF$ respectively. Then, $AL = 4.5 \text{ cm}$.

We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding altitudes.

$$\begin{aligned} \therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} &= \frac{AL^2}{DM^2} \\ \Rightarrow \left(\frac{AL}{DM}\right)^2 &= \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{81}{49} = \left(\frac{9}{7}\right)^2 \\ \Rightarrow \frac{AL}{DM} &= \frac{9}{7} \Rightarrow \frac{4.5}{DM} = \frac{9}{7} \\ \Rightarrow DM &= \frac{4.5 \times 7}{9} \text{ cm} = \frac{7}{2} \text{ cm} = 3.5 \text{ cm}. \end{aligned}$$

Hence, the altitude of the smaller triangle is 3.5 cm .

EXAMPLE 5 *The areas of two similar triangles are 121 cm^2 and 64 cm^2 respectively. If the median of first triangle is 12.1 cm , find the corresponding median of the other.* [CBSE 2001]

SOLUTION Let $\triangle ABC \sim \triangle DEF$ such that $\text{ar}(\triangle ABC) = 121 \text{ cm}^2$ and $\text{ar}(\triangle DEF) = 64 \text{ cm}^2$.

Let AP and DQ be the corresponding medians of $\triangle ABC$ and $\triangle DEF$ respectively such that $AP = 12.1 \text{ cm}$.

We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding medians.

$$\begin{aligned} \therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} &= \frac{AP^2}{DQ^2} \\ \Rightarrow \left(\frac{AP}{DQ}\right)^2 &= \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{121}{64} = \left(\frac{11}{8}\right)^2 \\ \Rightarrow \frac{AP}{DQ} &= \frac{11}{8} \Rightarrow \frac{12.1}{DQ} = \frac{11}{8} \quad [\because AP = 12.1 \text{ cm}] \end{aligned}$$

$$\Rightarrow DQ = \frac{(12.1 \times 8)}{11} \text{ cm} = (1.1 \times 8) \text{ cm} = 8.8 \text{ cm}.$$

Hence, the corresponding median is 8.8 cm.

EXAMPLE 6

$\triangle ABC \sim \triangle DEF$ in which AX and DY are the bisectors of $\angle A$ and $\angle D$ respectively. If $AX = 6.5$ cm and $DY = 5.2$ cm, find the ratio of the areas of $\triangle ABC$ and $\triangle DEF$.

SOLUTION

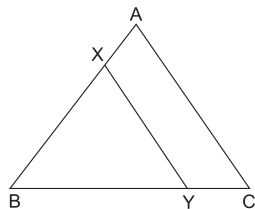
We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of the corresponding angle-bisector segments.

$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{AX^2}{DY^2} = \frac{(6.5)^2}{(5.2)^2} = \left(\frac{6.5}{5.2}\right)^2 = \left(\frac{5}{4}\right)^2 = \frac{25}{16}.$$

$$\therefore \text{ar}(\triangle ABC) : \text{ar}(\triangle DEF) = 25 : 16.$$

EXAMPLE 7

In the given figure, the line segment XY is parallel to side AC of $\triangle ABC$ and it divides the triangle into two parts of equal area. Prove that $AX : AB = (2 - \sqrt{2}) : 2$.

**SOLUTION**

Since $XY \parallel AC$, we have

$$\angle A = \angle BXY \text{ and } \angle C = \angle BYX \text{ [corres. } \angle\text{s]}.$$

$$\therefore \triangle ABC \sim \triangle XBY$$

$$\Rightarrow \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle XBY)} = \frac{AB^2}{XB^2}. \quad \dots (i)$$

But, $\text{ar}(\triangle ABC) = 2 \times \text{ar}(\triangle XBY)$ [given]

$$\Rightarrow \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle XBY)} = 2. \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{AB^2}{XB^2} = 2 \Rightarrow \left(\frac{AB}{XB}\right)^2 = 2$$

$$\Rightarrow \frac{AB}{XB} = \sqrt{2} \Rightarrow AB = \sqrt{2}(XB)$$

$$\Rightarrow AB = \sqrt{2}(AB - AX)$$

$$\Rightarrow \sqrt{2}AX = (\sqrt{2} - 1)AB$$

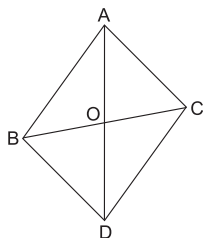
$$\Rightarrow \frac{AX}{AB} = \frac{(\sqrt{2} - 1)}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{(2 - \sqrt{2})}{2}.$$

Hence, $AX : AB = (2 - \sqrt{2}) : 2$.

EXAMPLE 8

In the given figure, $\triangle ABC$ and $\triangle DBC$ are on the same base BC . If AD intersects BC at O , prove that

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBC)} = \frac{AO}{DO} \quad [\text{CBSE 2000, '05, '07C}]$$



SOLUTION

GIVEN $\triangle ABC$ and $\triangle DBC$ are on the same base BC and AD intersects BC at O .

TO PROVE $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBC)} = \frac{AO}{DO}$

CONSTRUCTION Draw $AL \perp BC$ and $DM \perp BC$.

PROOF In $\triangle ALO$ and $\triangle DMO$, we have

$$\angle ALO = \angle DMO = 90^\circ$$

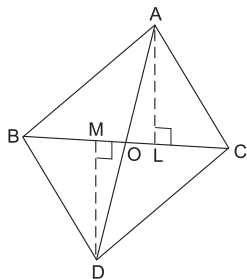
and $\angle AOL = \angle DOM$ (vert. opp. $\angle$ s).

$$\therefore \triangle ALO \sim \triangle DMO \text{ [by AA-similarity]}$$

$$\Rightarrow \frac{AL}{DM} = \frac{AO}{DO} \quad \dots (i)$$

$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBC)} = \frac{\frac{1}{2} \times BC \times AL}{\frac{1}{2} \times BC \times DM} = \frac{AL}{DM} = \frac{AO}{DO} \quad [\text{using (i)}]$$

$$\text{Hence, } \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBC)} = \frac{AO}{DO}.$$



EXAMPLE 9

$ABCD$ is a trapezium in which $AB \parallel DC$ and $AB = 2DC$. If the diagonals of the trapezium intersect each other at a point O , find the ratio of the areas of $\triangle AOB$ and $\triangle COD$.

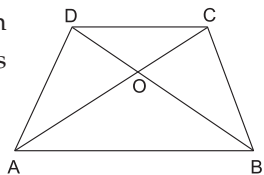
SOLUTION

GIVEN A trapezium $ABCD$ in which $AB \parallel DC$ and $AB = 2DC$. Its diagonals intersect each other at the point O .

TO FIND $\frac{\text{ar}(\triangle AOB)}{\text{ar}(\triangle COD)}$

METHOD OF SOLUTION In $\triangle AOB$ and $\triangle COD$, we have

$$\angle AOB = \angle COD \quad [\text{vert. opp. } \angle\text{s}]$$



$$\angle OAB = \angle OCD \quad [\text{alt. int. } \angle]$$

$$\therefore \triangle AOB \sim \triangle COD \quad [\text{by AA-similarity}].$$

We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of the corresponding sides.

$$\begin{aligned} \therefore \frac{\text{ar}(\triangle AOB)}{\text{ar}(\triangle COD)} &= \frac{AB^2}{DC^2} = \frac{(2 \times DC)^2}{DC^2} \quad [\because AB = 2DC] \\ &= \frac{4 \times DC^2}{DC^2} = \frac{4}{1}. \end{aligned}$$

$$\text{Hence, } \text{ar}(\triangle AOB) : \text{ar}(\triangle COD) = 4 : 1.$$

EXAMPLE 10 In a trapezium $ABCD$, O is the point of intersection of AC and BD , $AB \parallel CD$ and $AB = 2 \times CD$. If the area of $\triangle AOB = 84 \text{ cm}^2$, find the area of $\triangle COD$. [CBSE 2005, '09]

SOLUTION

In $\triangle AOB$ and $\triangle COD$, we have

$$\angle OAB = \angle OCD \quad [\text{alt. int. } \angle]$$

$$\text{and } \angle OBA = \angle ODC \quad [\text{alt. int. } \angle].$$

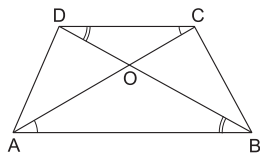
$$\therefore \triangle AOB \sim \triangle COD$$

[by AA-similarity]

$$\begin{aligned} \Rightarrow \frac{\text{ar}(\triangle AOB)}{\text{ar}(\triangle COD)} &= \frac{AB^2}{CD^2} = \frac{(2CD)^2}{CD^2} \quad [\because AB = 2 \times CD] \\ &= \frac{4 \times CD^2}{CD^2} = 4 \end{aligned}$$

$$\Rightarrow \text{ar}(\triangle COD) = \frac{1}{4} \times \text{ar}(\triangle AOB) = \left(\frac{1}{4} \times 84\right) \text{ cm}^2 = 21 \text{ cm}^2.$$

Hence, the area of $\triangle COD$ is 21 cm^2 .



EXAMPLE 11 D , E and F are respectively the midpoints of sides AB , BC and CA of $\triangle ABC$. Find the ratio of the areas of $\triangle DEF$ and $\triangle ABC$.

SOLUTION

In $\triangle ABC$, D and F are the midpoints of sides AB and CA respectively.

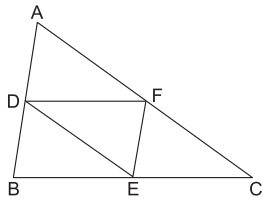
$$\therefore DF \parallel BC \quad [\text{by midpoint theorem}]$$

$$\Rightarrow DF \parallel BE.$$

Similarly, $EF \parallel BD$.

$$\therefore BEFD \text{ is a parallelogram}$$

$$\Rightarrow \angle B = \angle EFD, EF = BD = \frac{1}{2}AB$$



and $DF = BE = \frac{1}{2}BC$.

Also, $ECFD$ is a parallelogram

$$\Rightarrow \angle EDF = \angle C.$$

Now, in $\triangle DEF$ and $\triangle CAB$, we have

$$\angle EFD = \angle B$$

and $\angle EDF = \angle C$.

$\therefore \triangle DEF \sim \triangle CAB$ [by AA-similarity].

$$\text{And so, } \frac{\text{ar}(\triangle DEF)}{\text{ar}(\triangle ABC)} = \frac{\text{ar}(\triangle DEF)}{\text{ar}(\triangle CAB)} = \frac{DF^2}{BC^2} = \frac{\left(\frac{1}{2}BC\right)^2}{BC^2} = \frac{1}{4}.$$

EXAMPLE 12 Prove that the area of an equilateral triangle described on one side of a square is equal to half the area of an equilateral triangle described on one of its diagonals. [CBSE 2005C]

SOLUTION GIVEN A square $ABCD$ and equilateral $\triangle BCE$ and $\triangle ACF$ have been described on side BC and diagonal AC respectively.

TO PROVE $\text{ar}(\triangle BCE) = \frac{1}{2} \text{ar}(\triangle ACF)$.

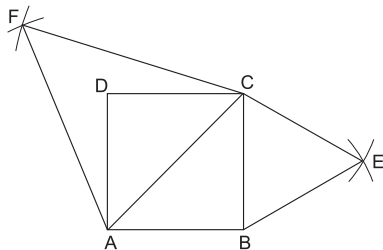
PROOF Since each of the $\triangle BCE$ and $\triangle ACF$ is an equilateral triangle, so each angle of each one of them is 60° . So, the triangles are equiangular, and hence similar.

$\therefore \triangle BCE \sim \triangle ACF$.

We know that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

$$\begin{aligned} \therefore \frac{\text{ar}(\triangle BCE)}{\text{ar}(\triangle ACF)} &= \frac{BC^2}{AC^2} = \frac{BC^2}{2(BC)^2} \quad [\because AC = \sqrt{2}BC] \\ &= \frac{1}{2}. \end{aligned}$$

Hence, $\text{ar}(\triangle BCE) = \frac{1}{2} \times \text{ar}(\triangle ACF)$.



EXAMPLE 13 Prove that the area of an equilateral triangle described on a side of a right-angled isosceles triangle is half the area of the equilateral triangle described on its hypotenuse. [CBSE 2006]

SOLUTION GIVEN A $\triangle ABC$ in which $\angle ABC = 90^\circ$ and $AB = BC$. $\triangle ABD$ and $\triangle CAE$ are equilateral triangles.

TO PROVE $\text{ar}(\triangle ABD) = \frac{1}{2} \times \text{ar}(\triangle CAE)$.

PROOF Let $AB = BC = x$ units.

$\therefore$ hyp. $CA = \sqrt{x^2 + x^2} = x\sqrt{2}$ units.

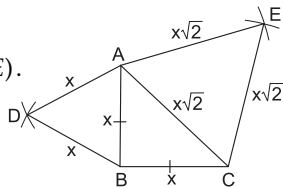
Each of the $\triangle ABD$ and $\triangle CAE$ being equilateral, each angle of each one of them is 60° .

$\therefore \triangle ABD \sim \triangle CAE$ [by AAA-similarity].

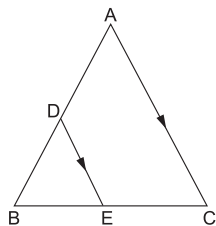
But, the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

$$\therefore \frac{\text{ar}(\triangle ABD)}{\text{ar}(\triangle CAE)} = \frac{AB^2}{CA^2} = \frac{x^2}{(x\sqrt{2})^2} = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Hence, $\text{ar}(\triangle ABD) = \frac{1}{2} \times \text{ar}(\triangle CAE)$.



EXAMPLE 14 If D is a point on the side AB of $\triangle ABC$ such that $AD : DB = 3 : 2$ and E is a point on BC such that $DE \parallel AC$, find the ratio of the areas of $\triangle ABC$ and $\triangle DBE$. [CBSE 2008C]



SOLUTION Let $AD = 3x$ cm and $DB = 2x$ cm.

Then,

$$AB = (AD + DB) = (3x + 2x) \text{ cm} = 5x \text{ cm}.$$

In $\triangle ABC$ and $\triangle DBE$, we have

$$\angle CAB = \angle EDB \quad [\text{corresponding } \angle]$$

$$\text{and } \angle ACB = \angle DEB \quad [\text{corresponding } \angle].$$

$\therefore \triangle ABC \sim \triangle DBE$ [by AA-similarity]

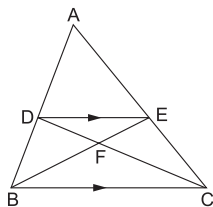
$$\Rightarrow \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBE)} = \frac{(AB)^2}{(DB)^2} = \frac{(5x)^2}{(2x)^2} = \frac{25x^2}{4x^2} = \frac{25}{4}$$

$$\Rightarrow \text{ar}(\triangle ABC) : \text{ar}(\triangle DBE) = 25 : 4.$$

EXAMPLE 15 In the given figure, $DE \parallel BC$ and $AD : DB = 5 : 4$.

Find the ratio $\text{ar}(\triangle DFE) : \text{ar}(\triangle CFB)$.

[CBSE 2000]



SOLUTION Let $AD = 5x$ cm and $DB = 4x$ cm. Then,
 $AB = AD + DB = 5x$ cm + $4x$ cm = $9x$ cm.

In $\triangle ADE$ and $\triangle ABC$, we have

$$\angle ADE = \angle ABC \quad [\text{corres. } \angle]$$

$$\angle AED = \angle ACB \quad [\text{corres. } \angle].$$

$$\therefore \triangle ADE \sim \triangle ABC \quad [\text{by AA-similarity}]$$

$$\Rightarrow \frac{DE}{BC} = \frac{AD}{AB} = \frac{5x}{9x} = \frac{5}{9}. \quad \dots (i)$$

In $\triangle DFE$ and $\triangle CFB$, we have

$$\angle EDF = \angle BCF \quad [\text{alt. int. } \angle]$$

$$\text{and } \angle DEF = \angle CBF \quad [\text{alt. int. } \angle].$$

$$\therefore \triangle DFE \sim \triangle CFB$$

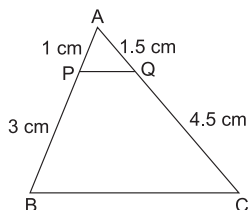
$$\Rightarrow \frac{\text{ar}(\triangle DFE)}{\text{ar}(\triangle CFB)} = \frac{DE^2}{CB^2} = \frac{DE^2}{BC^2} = \left(\frac{DE}{BC}\right)^2 = \left(\frac{5}{9}\right)^2 = \frac{25}{81}$$

$$\Rightarrow \text{ar}(\triangle DFE) : \text{ar}(\triangle CFB) = 25 : 81.$$

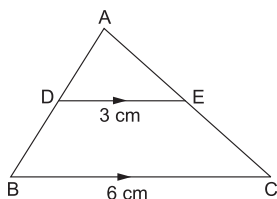
EXERCISE 7C

- $\triangle ABC \sim \triangle DEF$ and their areas are respectively 64 cm^2 and 121 cm^2 . If $EF = 15.4$ cm, find BC .
- The areas of two similar triangles ABC and PQR are in the ratio $9 : 16$. If $BC = 4.5$ cm, find the length of QR .
[CBSE 2004]
- $\triangle ABC \sim \triangle PQR$ and $\text{ar}(\triangle ABC) = 4\text{ar}(\triangle PQR)$. If $BC = 12$ cm, find QR .
- The areas of two similar triangles are 169 cm^2 and 121 cm^2 respectively. If the longest side of the larger triangle is 26 cm, find the longest side of the smaller triangle.
- $\triangle ABC \sim \triangle DEF$ and their areas are respectively 100 cm^2 and 49 cm^2 . If the altitude of $\triangle ABC$ is 5 cm, find the corresponding altitude of $\triangle DEF$.
[CBSE 2002]
- The corresponding altitudes of two similar triangles are 6 cm and 9 cm respectively. Find the ratio of their areas.

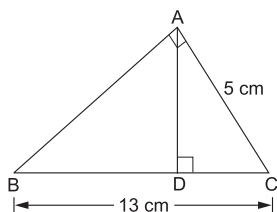
7. The areas of two similar triangles are 81 cm^2 and 49 cm^2 respectively. If the altitude of the first triangle is 6.3 cm , find the corresponding altitude of the other. [CBSE 2001]
8. The areas of two similar triangles are 100 cm^2 and 64 cm^2 respectively. If a median of the smaller triangle is 5.6 cm , find the corresponding median of the other.
9. In the given figure, ABC is a triangle and PQ is a straight line meeting AB in P and AC in Q . If $AP = 1 \text{ cm}$, $PB = 3 \text{ cm}$, $AQ = 1.5 \text{ cm}$, $QC = 4.5 \text{ cm}$, prove that area of $\triangle APQ$ is $\frac{1}{16}$ of the area of $\triangle ABC$. [CBSE 2005]



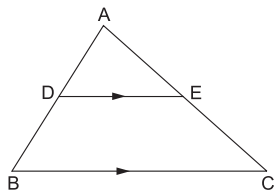
10. In the given figure, $DE \parallel BC$. If $DE = 3 \text{ cm}$, $BC = 6 \text{ cm}$ and $\text{ar}(\triangle ADE) = 15 \text{ cm}^2$, find the area of $\triangle ABC$.



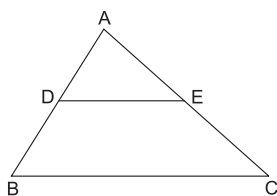
11. $\triangle ABC$ is right-angled at A and $AD \perp BC$. If $BC = 13 \text{ cm}$ and $AC = 5 \text{ cm}$, find the ratio of the areas of $\triangle ABC$ and $\triangle ADC$. [CBSE 2000C]



12. In the given figure, $DE \parallel BC$ and $DE : BC = 3 : 5$. Calculate the ratio of the areas of $\triangle ADE$ and the trapezium $BCED$.



13. In $\triangle ABC$, D and E are the midpoints of AB and AC respectively. Find the ratio of the areas of $\triangle ADE$ and $\triangle ABC$.



ANSWERS (EXERCISE 7C)

1. 11.2 cm 2. 6 cm 3. 6 cm 4. 22 cm 5. 3.5 cm 6. 4 : 9
 7. 4.9 cm 8. 7 cm 10. 60 cm^2 11. 169 : 25 12. 9 : 16 13. 1 : 4

HINTS TO SOME SELECTED QUESTIONS

6. Required ratio = $\frac{(6)^2}{(9)^2} = \frac{36}{81} = \frac{4}{9} = 4 : 9$.

9. $\frac{AP}{AB} = \frac{1}{(1+3)} = \frac{1}{4}$, $\frac{AQ}{AC} = \frac{1.5}{(1.5+4.5)} = \frac{1.5}{6} = \frac{1}{4}$.

$\therefore \triangle APQ \sim \triangle ABC$ [by SAS-similarity]

$$\Rightarrow \frac{\text{ar}(\triangle APQ)}{\text{ar}(\triangle ABC)} = \frac{AP^2}{AB^2} = \left(\frac{AP}{AB}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$\Rightarrow \text{ar}(\triangle APQ) = \frac{1}{16} \cdot \text{ar}(\triangle ABC).$$

10. $\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ABC)} = \frac{DE^2}{BC^2} = \frac{3^2}{6^2} = \frac{9}{36} = \frac{1}{4}$ [$\because \triangle ADE \sim \triangle ABC$]

$$\Rightarrow \text{ar}(\triangle ABC) = 4 \times \text{ar}(\triangle ADE) = (4 \times 15) \text{ cm}^2 = 60 \text{ cm}^2.$$

11. In $\triangle BAC$ and $\triangle ADC$, we have

$$\angle BAC = \angle ADC = 90^\circ$$

and $\angle ACB = \angle DCA = \angle C$.

$\therefore \triangle BAC \sim \triangle ADC$. [by AA-similarity].

$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle ADC)} = \frac{\text{ar}(\triangle BAC)}{\text{ar}(\triangle ADC)} = \frac{BC^2}{AC^2}.$$

12. $\triangle ADE \sim \triangle ABC$.

$$\therefore \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ABC)} = \frac{DE^2}{BC^2} = \left(\frac{DE}{BC}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}.$$

Let $\text{ar}(\triangle ADE) = 9x$ sq units.

Then, $\text{ar}(\triangle ABC) = 25x$ sq units.

$$\begin{aligned} \therefore \text{ar}(\text{trap. } BCED) &= \text{ar}(\triangle ABC) - \text{ar}(\triangle ADE) \\ &= (25x - 9x) \text{ sq units} = (16x) \text{ sq units.} \end{aligned}$$

$$\therefore \frac{\text{ar}(\triangle ADE)}{\text{ar}(\text{trap. } BCED)} = \frac{9x}{16x} = \frac{9}{16}.$$

13. Clearly, $DE \parallel BC$.

$\therefore \triangle ADE \sim \triangle ABC$.

$$\therefore \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ABC)} = \frac{AD^2}{AB^2} = \frac{AD^2}{(2AD)^2} = \frac{1}{4}.$$

.....

PYTHAGORAS' THEOREM

We have proved earlier in this chapter that:

"If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse then the triangles on both sides of the perpendicular are similar to the whole triangle and also to each other."

Now, we shall use this theorem to prove the Pythagoras' theorem.

THEOREM 1 (Pythagoras' theorem) *In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.*

[CBSE 2001, '02, '03, '04, '04C, '05, '06, '06C, '07, '07C, '09]

GIVEN A $\triangle ABC$ in which $\angle ABC = 90^\circ$.

TO PROVE $AC^2 = AB^2 + BC^2$.

CONSTRUCTION Draw $BD \perp AC$.

PROOF In $\triangle ADB$ and $\triangle ABC$, we have

$$\angle A = \angle A \quad (\text{common})$$

$$\angle ADB = \angle ABC \quad [\text{each equal to } 90^\circ]$$

$$\therefore \triangle ADB \sim \triangle ABC \quad [\text{by AA-similarity}]$$

$$\Rightarrow \frac{AD}{AB} = \frac{AB}{AC}$$

$$\Rightarrow AD \times AC = AB^2. \quad \dots (i)$$

In $\triangle BDC$ and $\triangle ABC$, we have

$$\angle C = \angle C \quad (\text{common})$$

$$\angle BDC = \angle ABC \quad [\text{each equal to } 90^\circ]$$

$$\therefore \triangle BDC \sim \triangle ABC \quad [\text{by AA-similarity}]$$

$$\Rightarrow \frac{DC}{BC} = \frac{BC}{AC}$$

$$\Rightarrow DC \times AC = BC^2. \quad \dots (ii)$$

From (i) and (ii), we get

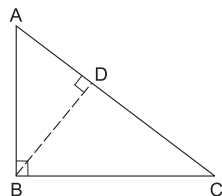
$$AD \times AC + DC \times AC = AB^2 + BC^2$$

$$\Rightarrow (AD + DC) \times AC = AB^2 + BC^2$$

$$\Rightarrow AC \times AC = AB^2 + BC^2 \Rightarrow AC^2 = AB^2 + BC^2.$$

THEOREM 2 (Converse of Pythagoras' theorem) *In a triangle, if the square of one side is equal to the sum of the squares of the other two sides then the angle opposite to the first side is a right angle.*

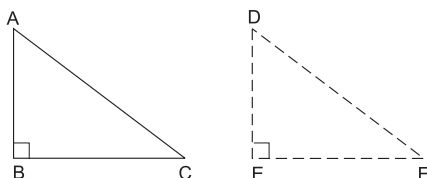
[CBSE 2001, '03, '05C, '06, '06C, '07, '09, '09C]



GIVEN A $\triangle ABC$ in which $AC^2 = AB^2 + BC^2$.

TO PROVE $\angle B = 90^\circ$.

CONSTRUCTION Draw a $\triangle DEF$ such that $DE = AB$, $EF = BC$ and $\angle E = 90^\circ$.



PROOF In $\triangle DEF$, we have $\angle E = 90^\circ$.

So, by Pythagoras' theorem, we have

$$DF^2 = DE^2 + EF^2$$

$$\Rightarrow DF^2 = AB^2 + BC^2. \quad \dots (i) \quad [\because DE = AB \text{ and } EF = BC]$$

$$\text{But, } AC^2 = AB^2 + BC^2. \quad \dots (ii) \quad [\text{given}]$$

From (i) and (ii), we get $AC^2 = DF^2 \Rightarrow AC = DF$.

Now, in $\triangle ABC$ and $\triangle DEF$, we have

$$AB = DE, BC = EF \text{ and } AC = DF.$$

$$\therefore \triangle ABC \cong \triangle DEF.$$

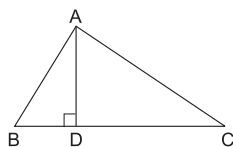
Hence, $\angle B = \angle E = 90^\circ$.

SOME IMPORTANT RESULTS BASED UPON PYTHAGORAS' THEOREM

THEOREM 1 In a $\triangle ABC$, AD is perpendicular to BC .

Prove that $(AB^2 + CD^2) = (AC^2 + BD^2)$.

[CBSE 2003, '05C, '09]



GIVEN A $\triangle ABC$ in which $AD \perp BC$.

TO PROVE $(AB^2 + CD^2) = (AC^2 + BD^2)$.

PROOF From right $\triangle ADB$, we have

$$AB^2 = AD^2 + BD^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow (AB^2 - BD^2) = AD^2. \quad \dots (i)$$

From right $\triangle ADC$, we have

$$AC^2 = AD^2 + CD^2$$

$$\Rightarrow (AC^2 - CD^2) = AD^2. \quad \dots (ii)$$

From (i) and (ii), we get $(AB^2 - BD^2) = (AC^2 - CD^2)$.

Hence, $(AB^2 + CD^2) = (AC^2 + BD^2)$.

THEOREM 2 Given a $\triangle ABC$ in which $\angle A = 90^\circ$ and $AD \perp BC$. Prove that

$$AD^2 = BD \cdot CD.$$

[CBSE 2004C, '06, '09]

GIVEN A $\triangle ABC$ in which $\angle A = 90^\circ$ and $AD \perp BC$.

TO PROVE $AD^2 = BD \cdot CD$.

PROOF In $\triangle BAC$, $\angle A = 90^\circ$.

$$\therefore BC^2 = AB^2 + AC^2 \quad \dots (i)$$

[by Pythagoras' theorem].

In $\triangle ADB$, $\angle ADB = 90^\circ$.

$$\therefore AB^2 = AD^2 + BD^2. \quad \dots (ii) \text{ [by Pythagoras' theorem]}$$

In $\triangle ADC$, $\angle ADC = 90^\circ$.

$$\therefore AC^2 = AD^2 + CD^2. \quad \dots (iii) \text{ [by Pythagoras' theorem]}$$

From (ii) and (iii), we get

$$(AB^2 + AC^2) = 2AD^2 + BD^2 + CD^2$$

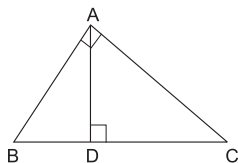
$$\Rightarrow BC^2 = 2AD^2 + BD^2 + CD^2 \quad \text{[using (i)]}$$

$$\Rightarrow (BD + CD)^2 = 2AD^2 + BD^2 + CD^2$$

$$\Rightarrow BD^2 + CD^2 + 2BD \cdot CD = 2AD^2 + BD^2 + CD^2$$

$$\Rightarrow 2AD^2 = 2BD \cdot CD \Rightarrow AD^2 = BD \cdot CD.$$

Hence, $AD^2 = BD \cdot CD$.



THEOREM 3 In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\triangle ABC$ is right-angled at A. [CBSE 2006]

GIVEN A $\triangle ABC$ in which $AD \perp BC$ and $AD^2 = BD \cdot CD$.

TO PROVE $\angle A = 90^\circ$.

PROOF In right $\triangle ADB$, $\angle ADB = 90^\circ$.

$$\therefore AB^2 = AD^2 + BD^2. \quad \dots (i)$$

[by Pythagoras' theorem]

In right $\triangle ADC$, $\angle ADC = 90^\circ$.

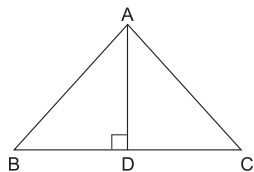
$$\therefore AC^2 = AD^2 + CD^2. \quad \dots (ii) \text{ [by Pythagoras' theorem]}$$

Adding (i) and (ii), we get

$$\begin{aligned} (AB^2 + AC^2) &= BD^2 + CD^2 + 2AD^2 \\ &= BD^2 + CD^2 + 2BD \cdot CD \quad [\because AD^2 = BD \cdot CD] \\ &= (BD + CD)^2 = BC^2. \end{aligned}$$

Thus, $(AB^2 + AC^2) = BC^2$.

Hence, $\triangle ABC$ is right-angled at A.



THEOREM 4 In a $\triangle ABC$, $\angle ABC < 90^\circ$ (i.e., $\angle B$ is acute) and $AD \perp BC$. Prove that

$$AC^2 = AB^2 + BC^2 - 2BC \cdot BD.$$

GIVEN A $\triangle ABC$ in which $\angle ABC < 90^\circ$

and $AD \perp BC$.

TO PROVE $AC^2 = AB^2 + BC^2 - 2BC \cdot BD$.

PROOF In $\triangle ADB$, $\angle ADB = 90^\circ$.

$$\therefore AB^2 = AD^2 + BD^2. \quad \dots (i) \quad [\text{by Pythagoras' theorem}]$$

In $\triangle ADC$, $\angle ADC = 90^\circ$.

$$\therefore AC^2 = AD^2 + CD^2 \quad [\text{by Pythagoras' theorem}]$$

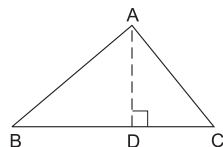
$$= AD^2 + (BC - BD)^2$$

$$= AD^2 + BC^2 + BD^2 - 2BC \cdot BD$$

$$= (AD^2 + BD^2) + BC^2 - 2BC \cdot BD$$

$$= AB^2 + BC^2 - 2BC \cdot BD \quad [\text{using (i)}].$$

Hence, $AC^2 = AB^2 + BC^2 - 2BC \cdot BD$.



Note BD is known as the projection of AB on BC . So, this theorem is stated as:

"In an acute-angled triangle, the square of the side opposite to an acute angle is equal to the sum of the squares of the other two sides minus twice the product of one side and the projection of the other on the first."

THEOREM 5 In a $\triangle ABC$, $\angle ABC > 90^\circ$ (i.e., $\angle B$ is obtuse) and $AD \perp (CB \text{ produced})$. Prove that $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$.

GIVEN A $\triangle ABC$ in which $\angle ABC > 90^\circ$ and

$AD \perp (CB \text{ produced})$.

TO PROVE $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$.

PROOF In $\triangle ADB$, $\angle ADB = 90^\circ$.

$$\therefore AB^2 = AD^2 + BD^2. \quad \dots (i) \quad [\text{by Pythagoras' theorem}]$$

In $\triangle ADC$, $\angle ADC = 90^\circ$.

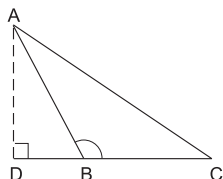
$$\therefore AC^2 = AD^2 + CD^2 \quad [\text{by Pythagoras' theorem}]$$

$$= AD^2 + (BC + BD)^2 \quad [\because CD = BC + BD]$$

$$= AD^2 + BC^2 + BD^2 + 2BC \cdot BD$$

$$= (AD^2 + BD^2) + BC^2 + 2BC \cdot BD$$

$$= AB^2 + BC^2 + 2BC \cdot BD \quad [\text{using (i)}].$$



Note BD is the projection of AB on BC . So, this theorem is stated as:

"In an obtuse-angled triangle, the square of the side opposite to the obtuse angle is equal to the sum of the squares of the other two sides plus twice the product of one side and the projection of the other on the first."

THEOREM 6 In $\triangle ABC$, if AD is the median, then prove that

$$(AB^2 + AC^2) = 2(AD^2 + BD^2).$$

GIVEN A $\triangle ABC$ in which AD is the median.

TO PROVE $(AB^2 + AC^2) = 2(AD^2 + BD^2)$.

CONSTRUCTION Draw $AL \perp BC$.

PROOF In $\triangle ALD$, $\angle ALD = 90^\circ$.

$$\therefore \angle ADL < 90^\circ \text{ and therefore,} \\ \angle ADB > 90^\circ.$$

Thus, in $\triangle ADB$, $\angle ADB > 90^\circ$ and $AL \perp (BD \text{ produced})$.

$$\therefore AB^2 = AD^2 + BD^2 + 2BD \cdot DL. \quad \dots (i)$$

In $\triangle ADC$, $\angle ADC < 90^\circ$ and $AL \perp DC$.

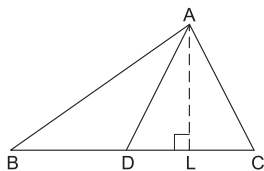
$$\therefore AC^2 = AD^2 + CD^2 - 2CD \cdot DL$$

$$\Rightarrow AC^2 = AD^2 + BD^2 - 2BD \cdot DL. \quad \dots (ii) \quad [\because CD = BD]$$

Adding (i) and (ii), we get $(AB^2 + AC^2) = 2(AD^2 + BD^2)$.

Note This theorem can be stated as:

"In any triangle, the sum of the squares of any two sides is equal to twice the square of half of the third side together with twice the square of the median which bisects the third side."



SOLVED EXAMPLES

EXAMPLE 1 Sides of some triangles are given below. Determine which of them are right triangles.

- (i) 8 cm, 15 cm, 17 cm (ii) 9 cm, 11 cm, 6 cm
 (iii) $(2a - 1)$ cm, $2\sqrt{2a}$ cm and $(2a + 1)$ cm

SOLUTION For the given triangle to be right-angled, the sum of the squares of the two smaller sides must be equal to the square of the largest side.

- (i) Let $a = 8$ cm, $b = 15$ cm and $c = 17$ cm. Then,

$$(a^2 + b^2) = \{(8)^2 + (15)^2\} \text{ cm}^2 = (64 + 225) \text{ cm}^2 = 289 \text{ cm}^2 \\ \text{and } c^2 = (17)^2 \text{ cm}^2 = 289 \text{ cm}^2.$$

$$\therefore (a^2 + b^2) = c^2.$$

Hence, the given triangle is right-angled.

- (ii) Let $a = 9$ cm, $b = 6$ cm and $c = 11$ cm. Then,

$$(a^2 + b^2) = \{(9)^2 + (6)^2\} \text{ cm}^2 = (81 + 36) \text{ cm}^2 = 117 \text{ cm}^2$$

$$\text{and } c^2 = (11)^2 \text{ cm}^2 = 121 \text{ cm}^2.$$

$$\therefore (a^2 + b^2) \neq c^2.$$

Hence, the given triangle is not right-angled.

(iii) Let $p = (2a - 1)$ cm, $q = 2\sqrt{2a}$ cm and $r = (2a + 1)$ cm. Then,

$$\begin{aligned}(p^2 + q^2) &= (2a - 1)^2 \text{ cm}^2 + (2\sqrt{2a})^2 \text{ cm}^2 \\ &= \{(4a^2 + 1 - 4a) + 8a\} \text{ cm}^2 = (4a^2 + 4a + 1) \text{ cm}^2 \\ &= (2a + 1)^2 \text{ cm}^2 = r^2.\end{aligned}$$

$$\therefore (p^2 + q^2) = r^2.$$

Hence, the given triangle is right-angled.

EXAMPLE 2 A man goes 15 m due west and then 8 m due north. How far is he from the starting point?

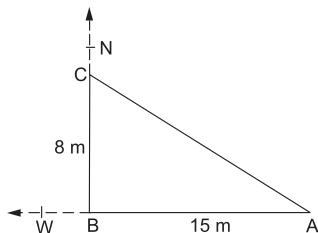
SOLUTION Starting from A , let the man go from A to B and then from B to C , as shown in the figure. Then,

$$AB = 15 \text{ m}, BC = 8 \text{ m and } \angle ABC = 90^\circ.$$

From right $\triangle ABC$, we have

$$\begin{aligned}AC^2 &= AB^2 + BC^2 \\ &= \{(15)^2 + (8)^2\} \text{ m}^2 \\ &= (225 + 64) \text{ m}^2 \\ &= 289 \text{ m}^2\end{aligned}$$

$$\therefore AC = \sqrt{289} \text{ m} = 17 \text{ m}.$$



Hence, the man is 17 m away from the starting position.

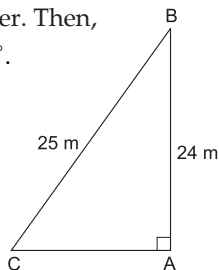
EXAMPLE 3 A ladder 25 m long just reaches the top of a building 24 m high from the ground. Find the distance of the foot of the ladder from the building.

SOLUTION Let AB be the building and CB be the ladder. Then,

$$AB = 24 \text{ m}, CB = 25 \text{ m and } \angle CAB = 90^\circ.$$

By Pythagoras' theorem, we have

$$\begin{aligned}CB^2 &= AB^2 + AC^2 \\ \Rightarrow AC^2 &= CB^2 - AB^2 = \{(25)^2 - (24)^2\} \text{ m}^2 \\ &= (625 - 576) \text{ m}^2 = 49 \text{ m}^2 \\ \Rightarrow AC &= \sqrt{49} \text{ m} = 7 \text{ m}.\end{aligned}$$



Hence, the distance of the foot of the ladder from the building is 7 m.

EXAMPLE 4 A ladder 15 m long reaches a window which is 9 m above the ground on one side of a street. Keeping its foot at the same point, the ladder is turned to the other side of the street to reach a window 12 m high. Find the width of the street.

SOLUTION Let AB be the street and let C be the foot of the ladder. Let D and E be the given windows such that $AD = 9$ m and $BE = 12$ m.

Then, CD and CE are the two positions of the ladder.

Clearly, $\angle CAD = 90^\circ$, $\angle CBE = 90^\circ$

and $CD = CE = 15$ m.

From right $\triangle CAD$, we have

$$CD^2 = AC^2 + AD^2$$

[by Pythagoras' theorem]

$$\Rightarrow AC^2 = CD^2 - AD^2$$

$$= [(15)^2 - (9)^2] \text{ m}^2 = (225 - 81) \text{ m}^2 = 144 \text{ m}^2.$$

$$\Rightarrow AC = \sqrt{144} \text{ m} = 12 \text{ m}.$$

From right $\triangle CBE$, we have

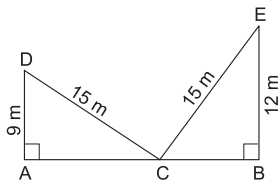
$$CE^2 = CB^2 + BE^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow CB^2 = CE^2 - BE^2$$

$$= [(15)^2 - (12)^2] \text{ m}^2 = (225 - 144) \text{ m}^2 = 81 \text{ m}^2$$

$$\Rightarrow CB = \sqrt{81} \text{ m} = 9 \text{ m}.$$

Width of the street = $AC + CB = 12 \text{ m} + 9 \text{ m} = 21 \text{ m}$.



EXAMPLE 5 Two poles of heights 6 metres and 11 metres stand vertically on a plane ground. If the distance between their feet is 12 metres, find the distance between their tops. [CBSE 2002]

SOLUTION Let AB and CD be the given vertical poles. Then,

$AB = 6$ m, $CD = 11$ m and $AC = 12$ m.

Draw $BE \parallel AC$. Then,

$CE = AB = 6$ m, $BE = AC = 12$ m.

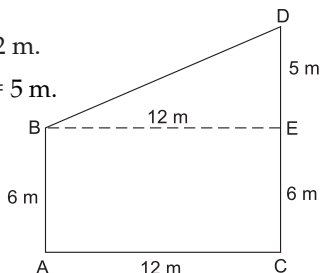
$$\therefore DE = CD - CE = 11 \text{ m} - 6 \text{ m} = 5 \text{ m}.$$

In right $\triangle BED$, we have:

$$BD^2 = BE^2 + DE^2$$

$$= \{(12)^2 + (5)^2\} \text{ m}^2$$

$$= (144 + 25) \text{ m}^2 = 169 \text{ m}^2$$



$$\Rightarrow BD = \sqrt{169} \text{ m} = 13 \text{ m}.$$

Hence, the distance between the tops of the poles = 13 m.

EXAMPLE 6

In a rhombus of side 10 cm, one of the diagonals is 12 cm long. Find the length of the second diagonal.

[CBSE 2001]

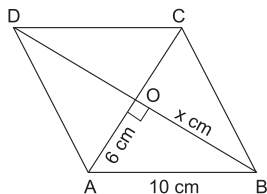
SOLUTION

Let $ABCD$ be the given rhombus whose diagonals intersect at O .

Then, $AB = 10$ cm.

Let $AC = 12$ cm and $BD = 2x$ cm.

We know that the diagonals of a rhombus bisect each other at right angles.



$$\therefore OA = \frac{1}{2}AC = 6 \text{ cm}, OB = \frac{1}{2}BD = x \text{ cm}, \text{ and } \angle AOB = 90^\circ.$$

From right $\triangle AOB$, we have

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow OB^2 = AB^2 - OA^2$$

$$= \{(10)^2 - (6)^2\} \text{ cm}^2 = (100 - 36) \text{ cm}^2 = 64 \text{ cm}^2$$

$$\Rightarrow x^2 = 64 \Rightarrow x = \sqrt{64} = 8.$$

$$\therefore OB = 8 \text{ cm}.$$

$$\therefore BD = 2 \times OB = 2 \times 8 \text{ cm} = 16 \text{ cm}.$$

Hence, the length of the second diagonal is 16 cm.

EXAMPLE 7

$\triangle ABD$ is a right triangle in which $\angle A = 90^\circ$ and $AC \perp BD$.

Prove that:

$$(i) AB^2 = BC \cdot BD \quad (ii) AC^2 = BC \cdot DC \quad (iii) AD^2 = BD \cdot CD$$

SOLUTION

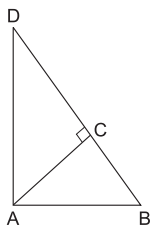
We know that:

If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse then the triangles on both sides of the perpendicular are similar to the whole triangle and also to each other.

$$\therefore \triangle ABC \sim \triangle DBA$$

$$\triangle ABC \sim \triangle DAC$$

$$\triangle DBA \sim \triangle DAC.$$



$$(i) \triangle ABC \sim \triangle DBA$$

$$\Rightarrow \frac{AB}{DB} = \frac{BC}{BA} \Rightarrow \frac{AB}{BD} = \frac{BC}{AB} \Rightarrow AB^2 = BC \cdot BD.$$

$$(ii) \triangle ABC \sim \triangle DAC$$

$$\Rightarrow \frac{AC}{DC} = \frac{BC}{AC} \Rightarrow AC^2 = BC \cdot DC.$$

$$(iii) \triangle DBA \sim \triangle DAC$$

$$\Rightarrow \frac{AD}{CD} = \frac{BD}{AD} \Rightarrow AD^2 = BD \cdot CD.$$

EXAMPLE 8 *BL and CM are medians of a $\triangle ABC$, right-angled at A. Prove that*

$$4(BL^2 + CM^2) = 5BC^2.$$
 [CBSE 2006C, '10]

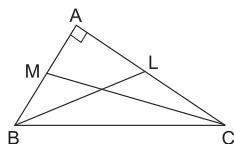
SOLUTION **GIVEN** A $\triangle ABC$ in which BL and CM are medians and $\angle A = 90^\circ$.

TO PROVE $4(BL^2 + CM^2) = 5BC^2$.

PROOF In $\triangle BAC$, $\angle A = 90^\circ$.

$$\therefore BC^2 = AB^2 + AC^2.$$

... (i)



[by Pythagoras' theorem]

In $\triangle BAL$, $\angle A = 90^\circ$.

$$\therefore BL^2 = AL^2 + AB^2$$

[by Pythagoras' theorem]

$$\Rightarrow BL^2 = \left(\frac{1}{2}AC\right)^2 + AB^2 \Rightarrow BL^2 = \frac{1}{4}AC^2 + AB^2$$

$$\Rightarrow 4BL^2 = AC^2 + 4AB^2.$$

... (ii)

In $\triangle CAM$, $\angle A = 90^\circ$.

$$\therefore CM^2 = AM^2 + AC^2$$

$$\Rightarrow CM^2 = \left(\frac{1}{2}AB\right)^2 + AC^2 \Rightarrow CM^2 = \frac{1}{4}AB^2 + AC^2$$

$$\Rightarrow 4CM^2 = AB^2 + 4AC^2.$$

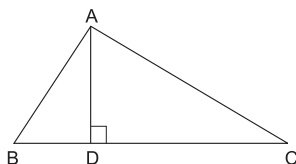
... (iii)

On adding (ii) and (iii), we get $4(BL^2 + CM^2) = 5(AB^2 + AC^2)$.

Hence, $4(BL^2 + CM^2) = 5BC^2$ [using (i)].

EXAMPLE 9 *In the given figure, the perpendicular from A on side BC of a $\triangle ABC$, intersects BC at D such that $DB = 3CD$. Prove that*

$$2AB^2 = 2AC^2 + BC^2.$$



[CBSE 2003, '05, '09]

SOLUTION GIVEN $\triangle ABC$ in which $AD \perp BC$ and $BD = 3CD$.

TO PROVE $2AB^2 = 2AC^2 + BC^2$.

PROOF We have $BD = 3CD$.

$$\therefore BC = BD + CD = 3CD + CD = 4CD$$

$$\Rightarrow CD = \frac{1}{4}BC. \quad \dots (i)$$

In $\triangle ADB$, $\angle ADB = 90^\circ$.

$$\therefore AB^2 = AD^2 + BD^2. \quad \dots (ii) \text{ [by Pythagoras' theorem]}$$

In $\triangle ADC$, $\angle ADC = 90^\circ$.

$$\therefore AC^2 = AD^2 + CD^2. \quad \dots (iii) \text{ [by Pythagoras' theorem]}$$

On subtracting (iii) from (ii), we get

$$\begin{aligned} AB^2 - AC^2 &= BD^2 - CD^2 \\ &= [(3CD)^2 - (CD^2)] = 8CD^2 \quad [\because BD = 3CD] \\ &= 8 \times \frac{1}{16}BC^2 = \frac{1}{2}BC^2 \quad \text{[using (i)]}. \end{aligned}$$

$$\therefore 2AB^2 - 2AC^2 = BC^2.$$

$$\text{Hence, } 2AB^2 = 2AC^2 + BC^2.$$

EXAMPLE 10 In $\triangle ABC$, $\angle B = 90^\circ$ and D is the midpoint of BC . Prove that $AC^2 = AD^2 + 3CD^2$.

SOLUTION GIVEN $\triangle ABC$ in which $\angle B = 90^\circ$ and D is the midpoint of BC .

TO PROVE $AC^2 = AD^2 + 3CD^2$.

CONSTRUCTION Join AD .

PROOF In $\triangle ABC$, $\angle B = 90^\circ$.

$$\therefore AC^2 = AB^2 + BC^2. \quad \dots (i) \text{ [by Pythagoras' theorem]}$$

In $\triangle ABD$, $\angle B = 90^\circ$.

$$\therefore AD^2 = AB^2 + BD^2 \quad \dots (ii) \text{ [by Pythagoras' theorem]}$$

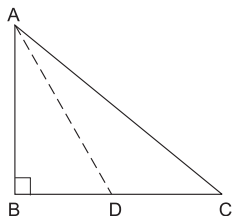
$$\Rightarrow AB^2 = AD^2 - BD^2.$$

$$\therefore AC^2 = (AD^2 - BD^2) + BC^2 \quad \text{[using (i)]}$$

$$\Rightarrow AC^2 = AD^2 - CD^2 + (2CD)^2 \quad [\because BD = CD \text{ and } BC = 2CD]$$

$$\Rightarrow AC^2 = AD^2 + 3CD^2.$$

$$\text{Hence, } AC^2 = AD^2 + 3CD^2.$$



EXAMPLE 11 $\triangle ABC$ is right-angled at B and D is the midpoint of BC . Prove that $AC^2 = (4AD^2 - 3AB^2)$. [CBSE 2008C, '10]

SOLUTION **GIVEN** A $\triangle ABC$ in which $\angle B = 90^\circ$ and D is the midpoint of BC .

TO PROVE $AC^2 = (4AD^2 - 3AB^2)$.

PROOF In $\triangle ABC$, $\angle B = 90^\circ$.

$$\therefore AC^2 = AB^2 + BC^2$$

[by Pythagoras' theorem]

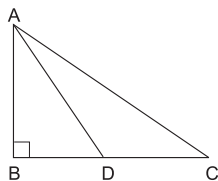
$$= AB^2 + (2BD)^2 \quad [\because BC = 2BD]$$

$$= AB^2 + 4BD^2$$

$$= AB^2 + 4(AD^2 - AB^2) \quad [\because AB^2 + BD^2 = AD^2]$$

$$= (4AD^2 - 3AB^2).$$

Hence, $AC^2 = (4AD^2 - 3AB^2)$.



EXAMPLE 12 In an isosceles $\triangle ABC$, $AB = AC$ and $BD \perp AC$. Prove that $(BD^2 - CD^2) = 2CD \cdot AD$.

SOLUTION **GIVEN** A $\triangle ABC$ in which $AB = AC$ and $BD \perp AC$.

TO PROVE $(BD^2 - CD^2) = 2CD \cdot AD$.

PROOF From right $\triangle ADB$, we have

$$AB^2 = AD^2 + BD^2$$

[by Pythagoras' theorem]

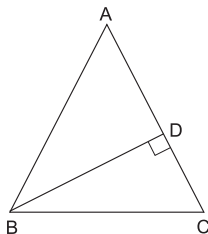
$$\Rightarrow AC^2 = AD^2 + BD^2 \quad [\because AB = AC]$$

$$\Rightarrow (CD + AD)^2 = AD^2 + BD^2 \quad [\because AC = CD + AD]$$

$$\Rightarrow CD^2 + AD^2 + 2CD \cdot AD = AD^2 + BD^2$$

$$\Rightarrow (BD^2 - CD^2) = 2CD \cdot AD.$$

Hence, $(BD^2 - CD^2) = 2CD \cdot AD$.



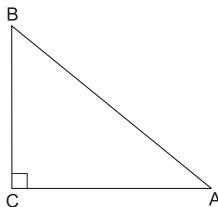
EXAMPLE 13 $\triangle ABC$ is an isosceles triangle, right-angled at C . Prove that $AB^2 = 2AC^2$.

SOLUTION $\triangle ABC$ is an isosceles triangle, right-angled at C

$$\Rightarrow BC = AC \quad \dots (i)$$

Now, by Pythagoras' theorem,

$$AB^2 = BC^2 + AC^2$$



$$\Rightarrow AB^2 = AC^2 + AC^2 \text{ [using (i)]}$$

$$\Rightarrow AB^2 = 2AC^2.$$

EXAMPLE 14 $\triangle ABC$ is an isosceles triangle with $AC = BC$. If $AB^2 = 2AC^2$, prove that $\triangle ABC$ is a right triangle. [CBSE 2000C]

SOLUTION In $\triangle ABC$, we have

$$AC = BC.$$

$$\text{Now, } AB^2 = 2AC^2 \quad (\text{given})$$

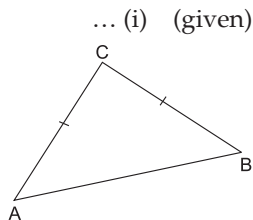
$$\Rightarrow AB^2 = AC^2 + AC^2$$

$$\Rightarrow AB^2 = AC^2 + BC^2 \quad [\text{using (i)}]$$

$$\Rightarrow \angle C = 90^\circ$$

[by converse of Pythagoras' theorem]

Hence, $\triangle ABC$ is a right triangle.



EXAMPLE 15 $\triangle ABC$ is a right triangle in which $\angle C = 90^\circ$ and $CD \perp AB$. If $BC = a$, $CA = b$, $AB = c$ and $CD = p$ then prove that

$$(i) \ cp = ab \quad (ii) \ \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}. \quad [\text{CBSE 1997C, '98, '99, '02}]$$

SOLUTION (i) We have

$$\text{ar}(\triangle ABC) = \frac{1}{2} \times AB \times CD = \frac{1}{2} cp$$

[taking AB as base]

$$\text{and } \text{ar}(\triangle ABC) = \frac{1}{2} \times BC \times AC = \frac{1}{2} ab$$

[taking BC as base].

$$\therefore \frac{1}{2} cp = \frac{1}{2} ab \Rightarrow cp = ab.$$

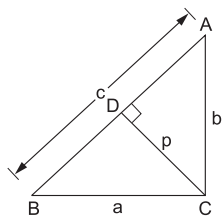
Hence, $cp = ab$.

$$(ii) \ cp = ab \Rightarrow \frac{1}{p} = \frac{c}{ab}.$$

$$\therefore \frac{1}{p^2} = \frac{c^2}{a^2 b^2} = \frac{b^2 + a^2}{a^2 b^2} \quad [\because AB^2 = AC^2 + BC^2]$$

$$= \left(\frac{b^2}{a^2 b^2} + \frac{a^2}{a^2 b^2} \right) = \left(\frac{1}{a^2} + \frac{1}{b^2} \right).$$

$$\text{Hence, } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}.$$



EXAMPLE 16 In an equilateral triangle, prove that three times the square of one side is equal to four times the square of one of its altitudes.

[CBSE 2002, '07C]

SOLUTION GIVEN $\triangle ABC$ in which $AB = BC = CA$ and $AD \perp BC$.

TO PROVE $3AB^2 = 4AD^2$.

PROOF In $\triangle ADB$ and $\triangle ADC$, we have

$$AB = AC \text{ (given), } \angle B = \angle C = 60^\circ$$

and $\angle ADB = \angle ADC = 90^\circ$.

$\therefore \triangle ADB \cong \triangle ADC$ [AAS-congruence]

$$\therefore BD = DC = \frac{1}{2}BC.$$

From right $\triangle ADB$, we have

$$AB^2 = AD^2 + BD^2 \text{ [by Pythagoras' theorem]}$$

$$= AD^2 + \left(\frac{1}{2}BC\right)^2 = AD^2 + \frac{1}{4}BC^2$$

$$\Rightarrow 4AB^2 = 4AD^2 + BC^2$$

$$\Rightarrow 3AB^2 = 4AD^2 \text{ [}\because BC = AB\text{].}$$

Hence, $3AB^2 = 4AD^2$.

EXAMPLE 17 In an equilateral triangle with side a , prove that

$$(i) \text{ altitude} = \frac{\sqrt{3}}{2}a \quad (ii) \text{ area} = \frac{\sqrt{3}}{4}a^2. \quad [\text{CBSE 1997, '99, '01C, '02C}]$$

SOLUTION Let $\triangle ABC$ be an equilateral triangle with side a .

Then, $AB = AC = BC = a$.

Draw $AD \perp BC$.

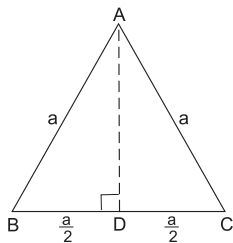
In $\triangle ADB$ and $\triangle ADC$, we have

$$AB = AC \text{ (given), } \angle B = \angle C = 60^\circ$$

and $\angle ADB = \angle ADC = 90^\circ$.

$\therefore \triangle ADB \cong \triangle ADC$.

$$\therefore BD = DC = \frac{a}{2}.$$



(i) From right $\triangle ADB$, we have

$$AB^2 = AD^2 + BD^2 \text{ [by Pythagoras' theorem]}$$

$$\Rightarrow AD = \sqrt{AB^2 - BD^2}$$

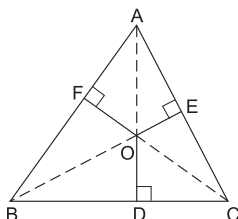
$$= \sqrt{a^2 - \left(\frac{a}{2}\right)^2} = \sqrt{a^2 - \frac{a^2}{4}} = \sqrt{\frac{3a^2}{4}} = \frac{\sqrt{3}}{2}a.$$

$$\text{Hence, altitude} = \frac{\sqrt{3}}{2}a.$$

$$\begin{aligned}
 \text{(ii) Area of } \triangle ABC &= \frac{1}{2} \times \text{base} \times \text{altitude} = \left(\frac{1}{2} \times BC \times AD \right) \\
 &= \left(\frac{1}{2} \times a \times \frac{\sqrt{3}}{2} a \right) \quad [\text{using (i)}] \\
 &= \left(\frac{\sqrt{3}}{4} a^2 \right) \text{ sq units.}
 \end{aligned}$$

$$\text{Hence, area}(\triangle ABC) = \left(\frac{\sqrt{3}}{4} a^2 \right) \text{ sq units.}$$

EXAMPLE 18 *O is a point in the interior of $\triangle ABC$, $OD \perp BC$, $OE \perp AC$ and $OF \perp AB$, as shown in the figure.*



Prove that:

- (i) $OA^2 + OB^2 + OC^2 - OD^2 - OE^2 - OF^2 = AF^2 + BD^2 + CE^2$
- (ii) $AF^2 + BD^2 + CE^2 = AE^2 + BF^2 + CD^2$

SOLUTION

- (i) Using Pythagoras' theorem for each of the right triangles namely $\triangle OFA$, $\triangle ODB$ and $\triangle OEC$, we get

$$OA^2 = OF^2 + AF^2 \quad \dots \text{ (i)}$$

$$OB^2 = OD^2 + BD^2 \quad \dots \text{ (ii)}$$

$$OC^2 = OE^2 + CE^2 \quad \dots \text{ (iii)}$$

Adding (i), (ii) and (iii), we get

$$\begin{aligned}
 OA^2 + OB^2 + OC^2 &= OD^2 + OE^2 + OF^2 + AF^2 + BD^2 + CE^2.
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence, } OA^2 + OB^2 + OC^2 - OD^2 - OE^2 - OF^2 &= AF^2 + BD^2 + CE^2.
 \end{aligned}$$

- (ii) Using Pythagoras' theorem for each of the right triangles, namely $\triangle ODB$ and $\triangle ODC$, we get

$$OB^2 = OD^2 + BD^2 \text{ and } OC^2 = OD^2 + CD^2.$$

$$\therefore OB^2 - OC^2 = BD^2 - CD^2 \quad \dots \text{ (iv)}$$

Similarly, we have

$$OC^2 - OA^2 = CE^2 - AE^2 \quad \dots \text{ (v)}$$

$$\text{and } OA^2 - OB^2 = AF^2 - BF^2. \quad \dots \text{ (vi)}$$

Adding the corresponding sides of (iv), (v) and (vi), we get

$$AF^2 + BD^2 + CE^2 - AE^2 - BF^2 - CD^2 = 0.$$

$$\text{Hence, } AF^2 + BD^2 + CE^2 = AE^2 + BF^2 + CD^2.$$

EXAMPLE 19 *O is any point inside a rectangle ABCD.*

Prove that $OB^2 + OD^2 = OA^2 + OC^2$.

[CBSE 2006C]

DEDUCTION *In the given figure, O is a point inside a rectangle ABCD such that $OB = 6$ cm, $OD = 8$ cm and $OA = 5$ cm, find the length of OC.*

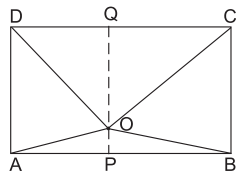
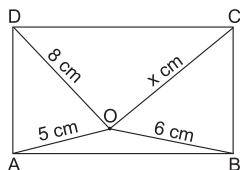
[CBSE 2009C]

SOLUTION

GIVEN O is a point inside a rectangle ABCD.

TO PROVE $OB^2 + OD^2 = OA^2 + OC^2$.

CONSTRUCTION Through O, draw $POQ \parallel BC$ so that P lies on AB and Q lies on DC.



PROOF we have

$$POQ \parallel BC \Rightarrow PQ \perp AB \text{ and } QP \perp DC$$

$$\Rightarrow \angle BPQ = 90^\circ \text{ and } \angle CQP = 90^\circ.$$

$\therefore$ BPQC and APQD are both rectangles.

$$\therefore BP = CQ \quad [\text{opposite sides of a rectangle}]$$

$$DQ = AP \quad [\text{opposite sides of a rectangle}]$$

From right $\triangle OPB$, we have $OB^2 = OP^2 + BP^2$.

From right $\triangle OQD$, we have $OD^2 = OQ^2 + DQ^2$.

From right $\triangle OPA$, we have $OA^2 = OP^2 + AP^2$.

From right $\triangle OQC$, we have $OC^2 = OQ^2 + CQ^2$.

$$\therefore OB^2 + OD^2 = OP^2 + OQ^2 + BP^2 + DQ^2$$

$$= OP^2 + OQ^2 + CQ^2 + AP^2$$

$$[\because BP = CQ \text{ and } DQ = AP]$$

$$= (OP^2 + AP^2) + (OQ^2 + CQ^2)$$

$$= OA^2 + OC^2.$$

Hence, $OB^2 + OD^2 = OA^2 + OC^2$.

DEDUCTION Let $OC = x$ cm. Then,

$$OB^2 + OD^2 = OA^2 + OC^2$$

$$\Rightarrow 6^2 + 8^2 = 5^2 + x^2$$

$$\Rightarrow x^2 = 36 + 64 - 25 = 75$$

$$\Rightarrow x = \sqrt{75} = 5\sqrt{3} = (5 \times 1.732) = 8.66 \Rightarrow OC = 8.66 \text{ cm.}$$

EXAMPLE 20 Prove that the sum of the squares on the sides of a rhombus is equal to the sum of the squares on its diagonals. [CBSE 2005, '06, '08C]

SOLUTION GIVEN A rhombus $ABCD$ whose diagonals AC and BD intersect at O .

$$\begin{aligned} \text{TO PROVE } (AB^2 + BC^2 + CD^2 + DA^2) \\ = (AC^2 + BD^2). \end{aligned}$$

PROOF We know that the diagonals of a rhombus bisect each other at right angles.

$$\therefore \angle AOB = \angle BOC = \angle COD = \angle DOA = 90^\circ,$$

$$OA = \frac{1}{2}AC \text{ and } OB = \frac{1}{2}BD.$$

From right $\triangle AOB$, we have

$$AB^2 = OA^2 + OB^2 \text{ [by Pythagoras' theorem]}$$

$$= \left(\frac{1}{2}AC\right)^2 + \left(\frac{1}{2}BD\right)^2 = \frac{1}{4}(AC^2 + BD^2)$$

$$\Rightarrow 4AB^2 = (AC^2 + BD^2). \quad \dots (i)$$

Similarly, we have:

$$4BC^2 = (AC^2 + BD^2) \quad \dots (ii)$$

$$4CD^2 = (AC^2 + BD^2) \quad \dots (iii)$$

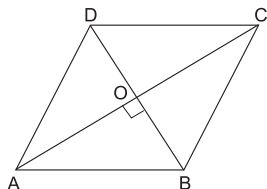
$$4DA^2 = (AC^2 + BD^2) \quad \dots (iv)$$

On adding (i), (ii), (iii) and (iv), we get

$$(AB^2 + BC^2 + CD^2 + DA^2) = (AC^2 + BD^2).$$

REMARK In a rhombus $ABCD$, we have $AB = BC = CD = DA$, so the above result may be given as $4AB^2 = (AC^2 + BD^2)$.

EXAMPLE 21 P and Q are points on the sides CA and CB of a $\triangle ABC$, right-angled at C . Prove that $(AQ^2 + BP^2) = (AB^2 + PQ^2)$. [CBSE 2007, '08]



SOLUTION

GIVEN A $\triangle ABC$ in which $\angle C = 90^\circ$.
 P and Q are points on CA and CB respectively.

TO PROVE $(AQ^2 + BP^2) = (AB^2 + PQ^2)$.

PROOF From right $\triangle ACQ$, we have

$$AQ^2 = (AC^2 + CQ^2). \quad \dots \text{(i) [by Pythagoras' theorem]}$$

From right $\triangle BCP$, we have

$$BP^2 = (BC^2 + CP^2). \quad \dots \text{(ii) [by Pythagoras' theorem]}$$

From right $\triangle ACB$, we have

$$AB^2 = AC^2 + BC^2. \quad \dots \text{(iii) [by Pythagoras' theorem]}$$

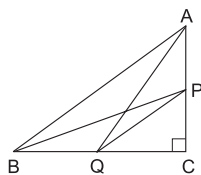
From right $\triangle PCQ$, we have

$$PQ^2 = (CQ^2 + CP^2). \quad \dots \text{(iv) [by Pythagoras' theorem]}$$

From (i) and (ii), we get

$$\begin{aligned} (AQ^2 + BP^2) &= (AC^2 + BC^2) + (CQ^2 + CP^2) \\ &= (AB^2 + PQ^2) \text{ [using (iii) and (iv)]}. \end{aligned}$$

Hence, $(AQ^2 + BP^2) = (AB^2 + PQ^2)$.

**EXAMPLE 22**

In the figure given below, $\triangle PQR$ is right-angled at Q and the points S and T trisect the side QR . Prove that $8PT^2 = 3PR^2 + 5PS^2$.

[CBSE 2006C]

SOLUTION

GIVEN A $\triangle PQR$ in which $\angle PQR = 90^\circ$,
 S and T are the points of trisection
of QR .

TO PROVE $8PT^2 = 3PR^2 + 5PS^2$.

PROOF Let $QS = ST = TR = x$. Then,

$$QS = x, QT = 2x \text{ and } QR = 3x.$$

From right triangles PQS , PQT and PQR , by Pythagoras' theorem, we have

$$PS^2 = PQ^2 + QS^2, PT^2 = PQ^2 + QT^2 \text{ and } PR^2 = PQ^2 + QR^2.$$

$$\therefore 3PR^2 + 5PS^2 - 8PT^2$$

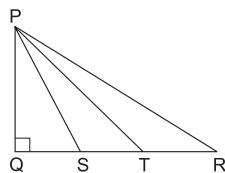
$$= 3(PQ^2 + QR^2) + 5(PQ^2 + QS^2) - 8(PQ^2 + QT^2)$$

$$= 3QR^2 + 5QS^2 - 8QT^2$$

$$= 3 \times (3x)^2 + 5(x)^2 - 8 \times (2x)^2$$

$$[\because QR = 3x, QS = x \text{ and } QT = 2x]$$

$$= (27x^2 + 5x^2 - 32x^2) = 0.$$



Thus, $3PR^2 + 5PS^2 - 8PT^2 = 0$.

Hence, $8PT^2 = 3PR^2 + 5PS^2$.

EXAMPLE 23 In an isosceles $\triangle ABC$, $AB = AC$ and D is a point on BC . Prove that $AB^2 - AD^2 = BD \cdot CD$.

SOLUTION GIVEN A $\triangle ABC$ in which $AB = AC$ and D is a point on BC .

TO PROVE $(AB^2 - AD^2) = BD \cdot CD$.

CONSTRUCTION Draw $AL \perp BC$.

PROOF In right $\triangle ALB$ and ALC , we have:

hyp. $AB = AC$ (given)

$AL = AL$ (common)

$\therefore \triangle ALB \cong \triangle ALC$ [by RHS-congruence]

$\therefore BL = CL$.

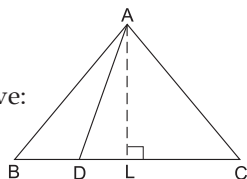
From right $\triangle ALB$ and ALD , by Pythagoras' theorem, we have:

$$AB^2 = AL^2 + BL^2 \quad \dots (i)$$

$$AD^2 = AL^2 + DL^2 \quad \dots (ii)$$

$$\begin{aligned} \therefore AB^2 - AD^2 &= BL^2 - DL^2 \\ &= (BL - DL)(BL + DL) \\ &= BD \cdot (CL + DL) \quad [\because BL = CL] \\ &= BD \cdot (CL + DL) \\ &= BD \cdot BC \quad [\because BL - DL = BD \text{ and } BL = CL] \\ &= BD \cdot CD. \end{aligned}$$

Hence, $AB^2 - AD^2 = BD \cdot CD$.



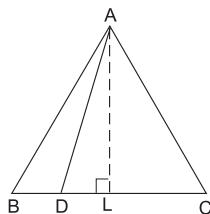
EXAMPLE 24 In an equilateral $\triangle ABC$, D is a point on side BC such that $BD = \frac{1}{3}BC$. Prove that $9AD^2 = 7AB^2$.

SOLUTION GIVEN A $\triangle ABC$ in which $AB = BC = CA$

and D is a point on BC such that $BD = \frac{1}{3}BC$.

TO PROVE $9AD^2 = 7AB^2$.

CONSTRUCTION Draw $AL \perp BC$.



PROOF In right triangles ALB and ALC , we have

$$AB = AC \text{ (given) and } AL = AL \text{ (common).}$$

$$\therefore \triangle ALB \cong \triangle ALC \quad [\text{by RHS axiom}]$$

$$\text{So, } BL = CL.$$

$$\text{Thus, } BD = \frac{1}{3}BC \text{ and } BL = \frac{1}{2}BC.$$

$$\text{In } \triangle ALB, \angle ALB = 90^\circ.$$

$$\therefore AB^2 = AL^2 + BL^2. \quad \dots (i) \quad [\text{by Pythagoras' theorem}]$$

$$\text{In } \triangle ALD, \angle ALD = 90^\circ.$$

$$\therefore AD^2 = AL^2 + DL^2 \quad [\text{by Pythagoras' theorem}]$$

$$= AL^2 + (BL - BD)^2$$

$$= AL^2 + BL^2 + BD^2 - 2BL \cdot BD$$

$$= (AL^2 + BL^2) + BD^2 - 2BL \cdot BD$$

$$= AB^2 + BD^2 - 2BL \cdot BD \quad [\text{using (i)}]$$

$$= BC^2 + \left(\frac{1}{3}BC\right)^2 - 2\left(\frac{1}{2}BC\right) \cdot \frac{1}{3}BC$$

$$\left[\because AB = BC, BD = \frac{1}{3}BC \text{ and } BL = \frac{1}{2}BC \right]$$

$$= BC^2 + \frac{1}{9}BC^2 - \frac{1}{3}BC^2$$

$$= \frac{7}{9}BC^2 = \frac{7}{9}AB^2 \quad [\because BC = AB].$$

$$\text{Hence, } 9AD^2 = 7AB^2.$$

EXAMPLE 25 In a quadrilateral $ABCD$, $\angle B = 90^\circ$. If $AD^2 = AB^2 + BC^2 + CD^2$, prove that $\angle ACD = 90^\circ$.

SOLUTION GIVEN A quad. $ABCD$ in which

$$\angle B = 90^\circ \text{ and}$$

$$AD^2 = AB^2 + BC^2 + CD^2.$$

TO PROVE $\angle ACD = 90^\circ$.

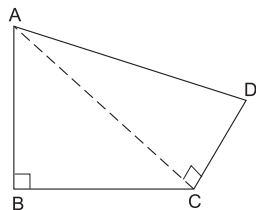
CONSTRUCTION Join AC .

PROOF In $\triangle ABC$, $\angle B = 90^\circ$.

$$\therefore AC^2 = AB^2 + BC^2. \quad \dots (i) \quad [\text{by Pythagoras' theorem}]$$

$$\text{Now, } AD^2 = AB^2 + BC^2 + CD^2 \text{ (given)}$$

$$\Rightarrow AD^2 = AC^2 + CD^2 [\text{using (i)}].$$



Thus, in $\triangle ACD$, we have $AD^2 = AC^2 + CD^2$.

Hence, $\angle ACD = 90^\circ$ [by converse of Pythagoras' theorem].

EXAMPLE 26 *Equilateral triangles are drawn on the sides of a right triangle. Prove that the area of the triangle on the hypotenuse is equal to the sum of the areas of the triangles on the other two sides.* [CBSE 2002]

SOLUTION GIVEN A $\triangle ABC$ in which $\angle B = 90^\circ$. Equilateral triangles $\triangle BCD$, $\triangle CAE$ and $\triangle ABF$ are drawn on the sides BC , CA and AB respectively.

TO PROVE $\text{ar}(\triangle CAE) = \text{ar}(\triangle BCD) + \text{ar}(\triangle ABF)$.

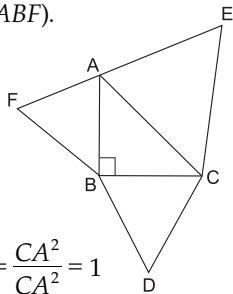
PROOF $\triangle BCD$, $\triangle CAE$ and $\triangle ABF$ are equiangular and hence similar.

We have

$$\begin{aligned} \frac{\text{ar}(\triangle BCD)}{\text{ar}(\triangle CAE)} + \frac{\text{ar}(\triangle ABF)}{\text{ar}(\triangle CAE)} &= \frac{BC^2}{CA^2} + \frac{AB^2}{CA^2} \\ \Rightarrow \frac{\text{ar}(\triangle BCD) + \text{ar}(\triangle ABF)}{\text{ar}(\triangle CAE)} &= \frac{BC^2 + AB^2}{CA^2} = \frac{CA^2}{CA^2} = 1 \end{aligned}$$

[by Pythagoras' theorem]

$$\Rightarrow \text{ar}(\triangle BCD) + \text{ar}(\triangle ABF) = \text{ar}(\triangle CAE).$$



EXAMPLE 27 *Given a right-angled $\triangle ABC$. The lengths of the sides containing the right angle are 6 cm and 8 cm. A circle is inscribed in $\triangle ABC$. Find the radius of the circle.*

SOLUTION In $\triangle ABC$, we have

$\angle B = 90^\circ$, $AB = 6$ cm
and $BC = 8$ cm.

A circle is inscribed in $\triangle ABC$. Let O be its centre and M , N and P be the points where it touches the sides AB , BC and CA respectively. Then,

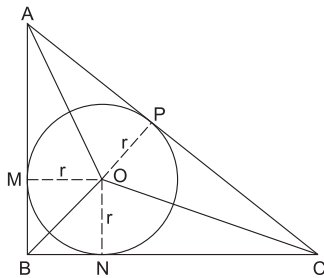
$$OM \perp AB, ON \perp BC, OP \perp CA.$$

Let r cm be the radius of the circle.

Then, $OM = ON = OP = r$ cm.

Now, $AB^2 + BC^2 = CA^2$ [by Pythagoras' theorem]

$$\Rightarrow 6^2 \text{ cm}^2 + 8^2 \text{ cm}^2 = CA^2 \Rightarrow CA = 10 \text{ cm}.$$



Now, $\text{ar}(\triangle ABC) = \text{ar}(\triangle AOB) + \text{ar}(\triangle BOC) + \text{ar}(\triangle COA)$

$$\Rightarrow \frac{1}{2} \times AB \times BC = \left(\frac{1}{2} \times AB \times OM \right) + \left(\frac{1}{2} \times BC \times ON \right) + \left(\frac{1}{2} \times CA \times OP \right)$$

$$\Rightarrow \frac{1}{2} \times 6 \times 8 = \left(\frac{1}{2} \times 6 \times r \right) + \left(\frac{1}{2} \times 8 \times r \right) + \left(\frac{1}{2} \times 10 \times r \right)$$

$$\Rightarrow r = 2 \Rightarrow \text{radius} = 2 \text{ cm.}$$

EXAMPLE 28 Prove that the sum of the squares of the diagonals of a parallelogram is equal to the sum of the squares of its sides.

SOLUTION GIVEN A parallelogram PQRS.

TO PROVE $PR^2 + QS^2$
 $= PQ^2 + QR^2 + RS^2 + SP^2$.

PROOF We have proved earlier that if AD is a median of a $\triangle ABC$, then

$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

$$\text{i.e., } AB^2 + AC^2 = 2AD^2 + \frac{1}{2}BC^2. \quad \dots (i)$$

Now, let O be the point of intersection of the diagonal PR and QS.

The diagonals of a parallelogram bisect each other.

$\therefore$ O is the midpoint of PR as well as QS.

Applying result (i) to $\triangle PQR$ and $\triangle RSP$, we get

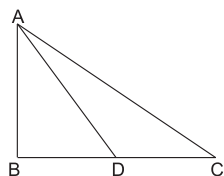
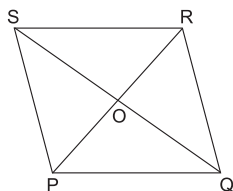
$$PQ^2 + QR^2 = 2OQ^2 + \frac{1}{2}PR^2 \quad \dots (ii)$$

$$\text{and } RS^2 + SP^2 = 2OS^2 + \frac{1}{2}PR^2. \quad \dots (iii)$$

Adding (ii) and (iii), we get

$$PQ^2 + QR^2 + RS^2 + SP^2 = 2\left(\frac{1}{2}QS\right)^2 + 2\left(\frac{1}{2}QS\right)^2 + PR^2$$

$$\Rightarrow PQ^2 + QR^2 + RS^2 + SP^2 = PR^2 + QS^2.$$



EXAMPLE 29 Given a $\triangle ABC$ in which $\angle B = 90^\circ$ and $AB = \sqrt{3}BC$. Prove that $\angle C = 60^\circ$.

SOLUTION Let D be the midpoint of the hypotenuse AC.
 Join BD.

We have

$$AC^2 = AB^2 + BC^2 \text{ [by Pythagoras' theorem]}$$

$$\Rightarrow AC^2 = (\sqrt{3} BC)^2 + BC^2$$

$$[\because AB = \sqrt{3} BC \text{ (given)}]$$

$$\Rightarrow AC^2 = 4BC^2 \Rightarrow AC = 2BC$$

$$\Rightarrow 2CD = 2BC \text{ } [\because D \text{ is the midpoint of } AC]$$

$$\Rightarrow CD = BC. \quad \dots (i)$$

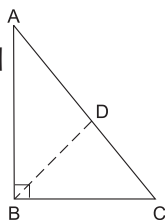
Also, we know that the midpoint of the hypotenuse of a right triangle is equidistant from the vertices.

$$\therefore BD = CD \quad \dots (ii)$$

From (i) and (ii), we get

$$BC = BD = CD.$$

$$\therefore \triangle BCD \text{ is equilateral and hence } \angle C = 60^\circ.$$



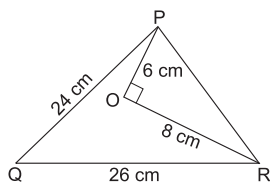
EXERCISE 7D

- The sides of certain triangles are given below. Determine which of them are right triangles.
 - 9 cm, 16 cm, 18 cm
 - 7 cm, 24 cm, 25 cm
 - 1.4 cm, 4.8 cm, 5 cm
 - 1.6 cm, 3.8 cm, 4 cm
 - $(a-1)$ cm, $2\sqrt{a}$ cm, $(a+1)$ cm
- A man goes 80 m due east and then 150 m due north. How far is he from the starting point?
- A man goes 10 m due south and then 24 m due west. How far is he from the starting point?
- A 13-m-long ladder reaches a window of a building 12 m above the ground. Determine the distance of the foot of the ladder from the building.
- A ladder is placed in such a way that its foot is at a distance of 15 m from a wall and its top reaches a window 20 m above the ground. Find the length of the ladder.
- Two vertical poles of height 9 m and 14 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

7. A guy wire attached to a vertical pole of height 18 m is 24 m long and has a stake attached to the other end. How far from the base of the pole should the stake be driven so that the wire will be taut?

8. In the given figure, O is a point inside $\triangle PQR$ such that $\angle POR = 90^\circ$, $OP = 6$ cm and $OR = 8$ cm. If $PQ = 24$ cm and $QR = 26$ cm, prove that $\triangle PQR$ is right-angled.

[CBSE 2006, '09C]



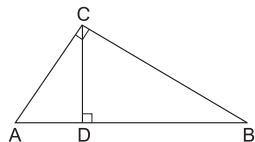
9. $\triangle ABC$ is an isosceles triangle with $AB = AC = 13$ cm. The length of altitude from A on BC is 5 cm. Find BC . [CBSE 2000C]
10. Find the length of altitude AD of an isosceles $\triangle ABC$ in which $AB = AC = 2a$ units and $BC = a$ units.
11. $\triangle ABC$ is an equilateral triangle of side $2a$ units. Find each of its altitudes.
12. Find the height of an equilateral triangle of side 12 cm.
13. Find the length of a diagonal of a rectangle whose adjacent sides are 30 cm and 16 cm.
14. Find the length of each side of a rhombus whose diagonals are 24 cm and 10 cm long.
15. In $\triangle ABC$, D is the midpoint of BC and $AE \perp BC$. If $AC > AB$, show that

$$AB^2 = AD^2 - BC \cdot DE + \frac{1}{4}BC^2.$$

[CBSE 2006C]

16. In the given figure, $\angle ACB = 90^\circ$ and $CD \perp AB$.

Prove that $\frac{BC^2}{AC^2} = \frac{BD}{AD}$.



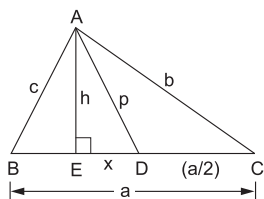
17. In the given figure, D is the midpoint of side BC and $AE \perp BC$. If $BC = a$, $AC = b$, $AB = c$, $ED = x$, $AD = p$ and $AE = h$, prove that

(i) $b^2 = p^2 + ax + \frac{a^2}{4}$

(ii) $c^2 = p^2 - ax + \frac{a^2}{4}$

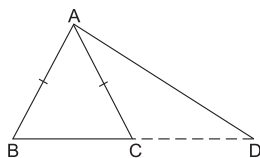
(iii) $(b^2 + c^2) = 2p^2 + \frac{1}{2}a^2$

(iv) $(b^2 - c^2) = 2ax$

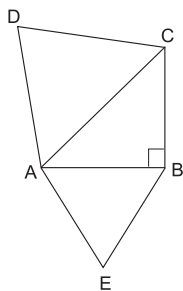


18. In $\triangle ABC$, $AB = AC$. Side BC is produced to D . Prove that

$$(AD^2 - AC^2) = BD \cdot CD.$$



19. ABC is an isosceles triangle, right-angled at B . Similar triangles ACD and ABE are constructed on sides AC and AB . Find the ratio between the areas of $\triangle ABE$ and $\triangle ACD$. [CBSE 2002]



20. An aeroplane leaves an airport and flies due north at a speed of 1000 km per hour. At the same time, another aeroplane leaves the same airport and flies due west at a speed of 1200 km per hour. How far apart will be the two planes after $1\frac{1}{2}$ hours?

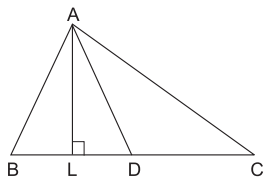
21. In a $\triangle ABC$, AD is a median and $AL \perp BC$.

Prove that:

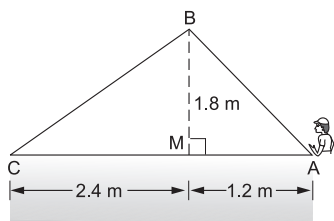
(a) $AC^2 = AD^2 + BC \cdot DL + \left(\frac{BC}{2}\right)^2$

(b) $AB^2 = AD^2 - BC \cdot DL + \left(\frac{BC}{2}\right)^2$

(c) $AC^2 + AB^2 = 2AD^2 + \frac{1}{2}BC^2$



22. Naman is doing fly-fishing in a stream. The tip of his fishing rod is 1.8 m above the surface of the water and the fly at the end of the string rests on the water 3.6 m away from him and 2.4 m from the point directly under the tip of the rod. Assuming that the string (from the tip of his rod to the fly) is taut, how much string does he have out (see the adjoining figure)? If he pulls in the string at the rate of 5 cm per second, what will be the horizontal distance of the fly from him after 12 seconds?



ANSWERS (EXERCISE 7D)

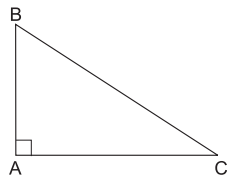
1. (ii), (iii), (v) 2. 170 m 3. 26 m 4. 5 m 5. 25 m
 6. 13 m 7. $6\sqrt{7}$ m 8. 24 m 9. $\frac{a\sqrt{15}}{2}$ units
 11. $a\sqrt{3}$ units 12. $6\sqrt{3}$ cm 13. 34 m 14. 13 cm 19. 1 : 2
 20. $300\sqrt{61}$ km 22. 2.8 m (approx.)

HINTS TO SOME SELECTED QUESTIONS

7. Let AB be the vertical pole and C be the position of the stake so that BC is a taut wire.

In rt. $\triangle ABC$, $\angle A = 90^\circ$, $AB = 18$ m, $BC = 24$ m.

$$\begin{aligned}\text{Now, } AC^2 + AB^2 &= BC^2 \Rightarrow AC^2 = (24^2 - 18^2) \text{ m}^2 = 252 \text{ m}^2 \\ \Rightarrow AC &= \sqrt{252} \text{ m} = 6\sqrt{7} \text{ m}.\end{aligned}$$



$$8. PR = \sqrt{OP^2 + OR^2} = \sqrt{6^2 + 8^2} = 10.$$

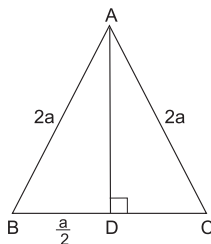
Now, $24^2 + 10^2 = 26^2$, i.e., $PQ^2 + PR^2 = QR^2 \Rightarrow \angle QPR = 90^\circ$.

10. In $\triangle ABC$, $AB = AC = 2a$ and $BC = a$.

Draw $AD \perp BC$. Then, D is the midpoint of BC

and so $BD = \frac{a}{2}$.

$$AD^2 = AB^2 - BD^2 = 4a^2 - \frac{a^2}{4} = \frac{15a^2}{4} \Rightarrow AD = \frac{a\sqrt{15}}{2}.$$



15. In $\triangle AEB$, $\angle AEB = 90^\circ$.

$$\therefore AB^2 = AE^2 + BE^2 \quad \dots (i)$$

In $\triangle AED$, $\angle AED = 90^\circ$.

$$\therefore AD^2 = (AE^2 + DE^2)$$

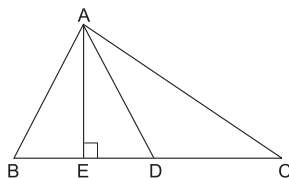
$$\Rightarrow AE^2 = (AD^2 - DE^2).$$

$$\therefore AB^2 = (AD^2 - DE^2) + BE^2 \quad [\text{using (i)}]$$

$$= (AD^2 - DE^2) + (BD - DE)^2$$

$$= (AD^2 - DE^2) + \left(\frac{1}{2}BC - DE\right)^2$$

$$= AD^2 + \frac{1}{4}BC^2 - BC \cdot DE.$$



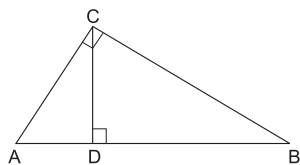
16. $\triangle ABC \sim \triangle CBD$

$$\Rightarrow \frac{BC}{BD} = \frac{AB}{CB} \Rightarrow BC^2 = AB \cdot BD. \quad \dots (i)$$

$$\triangle ABC \sim \triangle ACD$$

$$\Rightarrow \frac{AC}{AD} = \frac{AB}{AC} \Rightarrow AC^2 = AB \cdot AD. \dots (ii)$$

$$\text{Dividing (i) by (ii), we get } \frac{BC^2}{AC^2} = \frac{BD}{AD}.$$



$$17. (i) AC^2 = AE^2 + EC^2$$

$$\Rightarrow b^2 = h^2 + \left(x + \frac{a}{2}\right)^2 = (h^2 + x^2) + ax + \frac{a^2}{4}$$

$$= p^2 + ax + \frac{a^2}{4}.$$

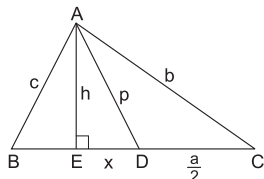
$$(ii) AB^2 = AE^2 + BE^2$$

$$\Rightarrow c^2 = h^2 + \left(\frac{a}{2} - x\right)^2 = (h^2 + x^2) - ax + \frac{a^2}{4}$$

$$= p^2 - ax + \frac{a^2}{4}.$$

(iii) Add (i) and (ii).

(iv) Subtract (ii) from (i).



18. Draw $AE \perp BC$. Then, $BE = CE$.

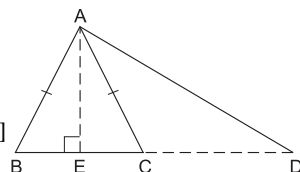
$$AD^2 = AE^2 + DE^2 \text{ and } AC^2 = AE^2 + CE^2$$

$$\Rightarrow (AD^2 - AC^2) = DE^2 - CE^2$$

$$= (DE + CE)(DE - CE)$$

$$= (DE + BE)(DE - CE) \quad [\because CE = BE]$$

$$= BD \cdot CD.$$



$$19. \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle ACD)} = \frac{AB^2}{AC^2} = \frac{AB^2}{2AB^2} = \frac{1}{2} \quad [\because AC^2 = AB^2 + BC^2 = 2AB^2].$$

20. Distance covered by first plane in $1\frac{1}{2}$ hours

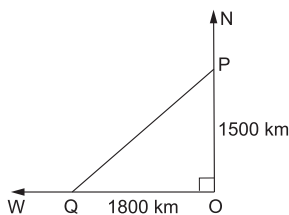
$$= OP = 1500 \text{ km (north).}$$

Distance covered by second plane in $1\frac{1}{2}$ hours

$$= OQ = 1800 \text{ km (west).}$$

Distance between the two planes after $1\frac{1}{2}$ hours

$$= PQ = \sqrt{OP^2 + OQ^2} = 300\sqrt{61} \text{ km.}$$



$$21. (a) \text{ In right } \triangle ACL, AC^2 = AL^2 + LC^2. \dots (i)$$

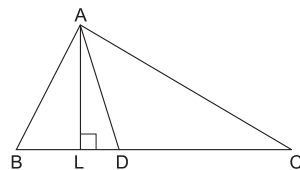
$$\text{In right } \triangle ALD, AL^2 = AD^2 - DL^2. \dots (ii)$$

$$\therefore AC^2 = (AD^2 - DL^2) + LC^2 \text{ [from (i) and (ii)]}$$

$$= (AD^2 - DL^2) + (DL + DC)^2$$

$$= (AD^2 - DL^2) + \left(DL + \frac{1}{2}BC\right)^2$$

$$= AD^2 + BC \cdot DL + \frac{1}{4}BC^2 = AD^2 + BC \cdot DL + \left(\frac{BC}{2}\right)^2.$$



(b) In right $\triangle ABL$, $AB^2 = AL^2 + LB^2$ (iii)

In right $\triangle ALD$, $AL^2 = AD^2 - DL^2$ (iv)

$$\begin{aligned}\therefore AB^2 &= (AD^2 - DL^2) + LB^2 \quad [\text{from (iii) and (iv)}] \\ &= (AD^2 - DL^2) + (BD - DL)^2 \\ &= (AD^2 - DL^2) + \left(\frac{1}{2}BC - DL\right)^2 = AD^2 - BC \cdot DL + \left(\frac{BC}{2}\right)^2.\end{aligned}$$

(c) Adding the results of (a) and (b), we get

$$AC^2 + AB^2 = 2AD^2 + \frac{1}{2}BC^2.$$

22. Let of the string that was out of the rod

$$= BC = \sqrt{BM^2 + CM^2} = \sqrt{(1.8)^2 + (2.4)^2} \text{ m}$$

$$= \sqrt{3.24 + 5.76} \text{ m} = \sqrt{9} \text{ m} = 3 \text{ m}.$$

He pulls the string at a rate of 5 cm per second.

$$\therefore \text{length of string pulled in 12 s}$$

$$= (5 \times 12) \text{ cm} = 60 \text{ cm} = 0.6 \text{ m}.$$

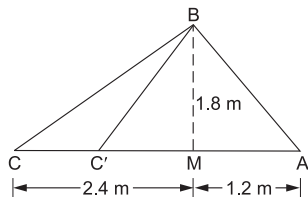
So, after 12 s, we have $BC' = (3 - 0.6) \text{ m} = 2.4 \text{ m}$ and $BM = 1.8 \text{ m}$.

$$\therefore C'M = \sqrt{(BC')^2 - BM^2} = \sqrt{(2.4)^2 - (1.8)^2} \text{ m} = \sqrt{2.52} \text{ m} \approx 1.6 \text{ m}.$$

Horizontal distance of the fly from him after 12 s = $C'A$

$$= C'M + MA = (1.6 + 1.2) \text{ m}$$

$$= 2.8 \text{ m}.$$



EXERCISE 7E

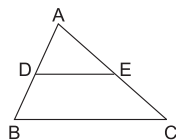
Very-Short-Answer and Short-Answer Questions:

1. State the two properties which are necessary for given two triangles to be similar.
2. State the basic proportionality theorem.
3. State the converse of Thales' theorem.
4. State the midpoint theorem.
5. State the AAA-similarity criterion.
6. State the AA-similarity criterion.
7. State the SSS-criterion for similarity of triangles.
8. State the SAS-similarity criterion.
9. State Pythagoras' theorem.
10. State the converse of Pythagoras' theorem.
11. If D , E and F are respectively the midpoints of sides AB , BC and CA of $\triangle ABC$ then what is the ratio of the areas of $\triangle DEF$ and $\triangle ABC$?

12. Two triangles ABC and PQR are such that $AB = 3$ cm, $AC = 6$ cm, $\angle A = 70^\circ$, $PR = 9$ cm, $\angle P = 70^\circ$ and $PQ = 4.5$ cm. Show that $\triangle ABC \sim \triangle PQR$ and state the similarity criterion.

13. If $\triangle ABC \sim \triangle DEF$ such that $2AB = DE$ and $BC = 6$ cm, find EF .

14. In the given figure, $DE \parallel BC$ such that $AD = x$ cm, $DB = (3x + 4)$ cm, $AE = (x + 3)$ cm and $EC = (3x + 19)$ cm. Find the value of x .



15. A ladder 10 m long reaches the window of a house 8 m above the ground. Find the distance of the foot of the ladder from the base of the wall.

16. Find the length of the altitude of an equilateral triangle of side $2a$ cm.

17. $\triangle ABC \sim \triangle DEF$ such that $\text{ar}(\triangle ABC) = 64$ cm² and $\text{ar}(\triangle DEF) = 169$ cm². If $BC = 4$ cm, find EF .

18. In a trapezium $ABCD$, it is given that $AB \parallel CD$ and $AB = 2CD$. Its diagonals AC and BD intersect at the point O such that $\text{ar}(\triangle AOB) = 84$ cm². Find $\text{ar}(\triangle COD)$.

19. The corresponding sides of two similar triangles are in the ratio $2 : 3$. If the area of the smaller triangle is 48 cm², find the area of the larger triangle.

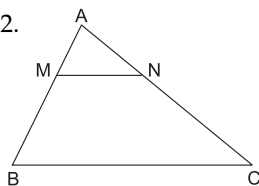
20. In an equilateral triangle with side a , prove that $\text{area} = \frac{\sqrt{3}}{4} a^2$.

21. Find the length of each side of a rhombus whose diagonals are 24 cm and 10 cm long.

22. Two triangles DEF and GHK are such that $\angle D = 48^\circ$ and $\angle H = 57^\circ$. If $\triangle DEF \sim \triangle GHK$ then find the measure of $\angle F$.

23. In the given figure, $MN \parallel BC$ and $AM : MB = 1 : 2$.

Find $\frac{\text{area}(\triangle AMN)}{\text{area}(\triangle ABC)}$.

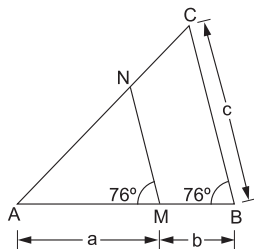


24. In triangles BMP and CNR it is given that $PB = 5$ cm, $MP = 6$ cm, $BM = 9$ cm and $NR = 9$ cm. If $\triangle BMP \sim \triangle CNR$ then find the perimeter of $\triangle CNR$.

25. Each of the equal sides of an isosceles triangle is 25 cm. Find the length of its altitude if the base is 14 cm.

26. A man goes 12 m due south and then 35 m due west. How far is he from the starting point?
27. If the lengths of the sides BC , CA and AB of a $\triangle ABC$ are a , b and c respectively and AD is the bisector of $\angle A$ then find the lengths of BD and DC .

28. In the given figure, $\angle AMN = \angle MBC = 76^\circ$.
If p , q and r are the lengths of AM , MB and BC respectively then express the length of MN in terms of p , q and r .



29. The lengths of the diagonals of a rhombus are 40 cm and 42 cm. Find the length of each side of the rhombus.

For each of the following statements state whether true (T) or false (F):

30. (i) Two circles with different radii are similar.
(ii) Any two rectangles are similar.
(iii) If two triangles are similar then their corresponding angles are equal and their corresponding sides are equal.
(iv) The length of the line segment joining the midpoints of any two sides of a triangle is equal to half the length of the third side.
(v) In a $\triangle ABC$, $AB = 6$ cm, $\angle A = 45^\circ$ and $AC = 8$ cm and in a $\triangle DEF$, $DF = 9$ cm, $\angle D = 45^\circ$ and $DE = 12$ cm, then $\triangle ABC \sim \triangle DEF$.
(vi) The polygon formed by joining the midpoints of the sides of a quadrilateral is a rhombus.
(vii) The ratio of the areas of two similar triangles is equal to the ratio of their corresponding angle-bisector segments.
(viii) The ratio of the perimeters of two similar triangles is the same as the ratio of their corresponding medians.
(ix) If O is any point inside a rectangle $ABCD$ then $OA^2 + OC^2 = OB^2 + OD^2$.
(x) The sum of the squares on the sides of a rhombus is equal to the sum of the squares on its diagonals.

ANSWERS (EXERCISE 7E)

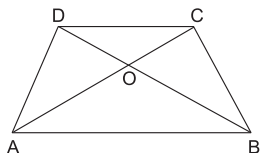
11. 1 : 4 12. SAS-similarity 13. 12 cm 14. $x = 2$ 15. 6 m
 16. $\sqrt{3}a$ cm 17. 6.5 cm 18. 21 cm² 19. 108 cm² 21. 13 cm
 22. 75° 23. $\frac{1}{9}$ 24. 30 cm 25. 24 cm 26. 37 cm
 27. $BD = \frac{ac}{b+c}$; $DC = \frac{ab}{b+c}$ 28. $MN = \frac{ac}{a+b}$ 29. 29 cm
 30. (i) T (ii) F (iii) F (iv) T (v) F (vi) F (vii) T (viii) T (ix) T (x) T

HINTS TO SOME SELECTED QUESTIONS

14. $\frac{AD}{DB} = \frac{AE}{EC}$ [by Thales' theorem]

17. $\frac{\text{ar}(\triangle AOB)}{\text{ar}(\triangle COD)} = \frac{AB^2}{CD^2} = \left(\frac{AB}{CD}\right)^2 = \frac{4}{1}$ [$\because \triangle AOB \sim \triangle COD$]

$\Rightarrow \text{ar}(\triangle COD) = \frac{1}{4} \times \text{ar}(\triangle AOB).$



22. $\angle D = 48^\circ$, $\angle E = \angle H = 57^\circ$ [$\because \triangle DEF \sim \triangle GHK$]

$\therefore \angle F = 180^\circ - (\angle D + \angle E) = 180^\circ - (48^\circ + 57^\circ) = 75^\circ.$

23. $MN \parallel BC \Rightarrow \triangle AMN \sim \triangle ABC.$

$\therefore \frac{\text{area}(\triangle AMN)}{\text{area}(\triangle ABC)} = \frac{AM^2}{AB^2} = \left(\frac{AM}{AM+MB}\right)^2 = \left(\frac{x}{x+2x}\right)^2 = \frac{1}{9}$

[$AM : MB = 1 : 2$, so, let $AM = x$, then $MB = 2x$].

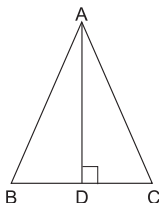
24. $\frac{\text{Perimeter of } \triangle BMP}{\text{Perimeter of } \triangle CNR} = \frac{MP}{NR}$ [$\because \triangle BMP \sim \triangle CNR$]

$\Rightarrow \text{perimeter of } \triangle CNR = \frac{NR}{MP} \times \text{perimeter of } \triangle BMP = \frac{NR}{MP} \times (PB + MP + BM)$
 $= \frac{9}{6} \times (5 + 6 + 9) \text{ cm} = 30 \text{ cm}.$

25. Let the given triangle be $\triangle ABC$ having $AB = AC = 25$ cm,
 base $BC = 14$ cm.

Let $AD \perp BC$. Then, D is the midpoint of BC .

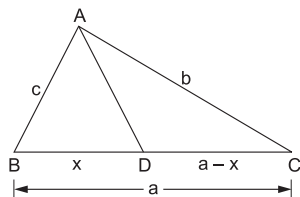
Altitude $AD = \sqrt{AC^2 - DC^2} = \sqrt{25^2 - 7^2} \text{ cm} = 24 \text{ cm}.$



27. Let $BD = x$. Then, $DC = BC - BD = a - x$.

Now, $\frac{BD}{DC} = \frac{AB}{AC}$ [by angle-bisector theorem]

$$\Rightarrow \frac{x}{a-x} = \frac{c}{b} \Rightarrow x = \frac{ac}{b+c} \text{ and so, } a-x = \frac{ab}{b+c}.$$



28. $\angle AMN = \angle MBC \Rightarrow MN \parallel BC \Rightarrow \triangle AMN \sim \triangle ABC$.

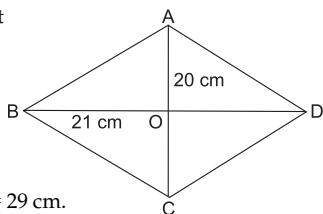
$$\therefore \frac{AM}{AB} = \frac{MN}{BC} \Rightarrow \frac{a}{a+b} = \frac{MN}{c} \Rightarrow MN = \frac{ac}{a+b}.$$

29. The diagonals of a rhombus intersect at right angles and bisect each other.

In the figure,

$$OA = \frac{40}{2} \text{ cm} = 20 \text{ cm}, OB = \frac{42}{2} \text{ cm} = 21 \text{ cm}.$$

$$AB = \sqrt{OA^2 + OB^2} = \sqrt{20^2 + 21^2} \text{ cm} = \sqrt{841} \text{ cm} = 29 \text{ cm}.$$



30. (ii) Two rectangles are similar only if their corresponding sides are proportional.

(iii) If two triangles are similar, then

(a) their corresponding angles are equal

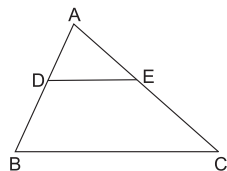
(b) their corresponding sides are proportional (but not necessarily equal)

(iv) Let D and E be the midpoints of sides AB and AC respectively of a $\triangle ABC$.

Then, by the midpoint theorem,

$$DE \parallel BC \Rightarrow \triangle ADE \sim \triangle ABC.$$

$$\therefore \frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{1}{2} = \frac{DE}{BC} \Rightarrow DE = \frac{1}{2}BC.$$



(v) We have $\frac{AB}{DE} = \frac{6}{12} = \frac{1}{2}$, $\frac{AC}{DF} = \frac{8}{9}$.

Clearly, $\frac{AB}{DE} \neq \frac{AC}{DF}$ and so $\triangle ABC$ is not similar to $\triangle DEF$.

Note Here $\triangle ABC \sim \triangle DFE$.

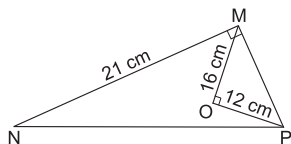
(vi) The polygon formed by joining the midpoints of the sides of a quadrilateral is a parallelogram (not necessarily a rhombus).

(vii) The ratio of the perimeters of two similar triangles is the same as the ratio of their corresponding sides which is the same as the ratio of the corresponding medians.

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- A man goes 24 m due west and then 10 m due north. How far is he from the starting point?
(a) 34 m (b) 17 m (c) 26 m (d) 28 m
- Two poles of height 13 m and 7 m respectively stand vertically on a plane ground at a distance of 8 m from each other. The distance between their tops is
(a) 9 m (b) 10 m (c) 11 m (d) 12 m
- A vertical stick 1.8 m long casts a shadow 45 cm long on the ground. At the same time, what is the length of the shadow of a pole 6 m high?
(a) 2.4 m (b) 1.35 m (c) 1.5 m (d) 13.5 m
- A vertical pole 6 m long casts a shadow of length 3.6 m on the ground. What is the height of a tower which casts a shadow of length 18 m at the same time?
(a) 10.8 m (b) 28.8 m (c) 32.4 m (d) 30 m
- The shadow of a 5-m-long stick is 2 m long. At the same time the length of the shadow of a 12.5-m-high tree (in m) is [CBSE 2011]
(a) 3.0 (b) 3.5 (c) 4.5 (d) 5.0
- A ladder 25 m long just reaches the top of a building 24 m high from the ground. What is the distance of the foot of the ladder from the building?
(a) 7 m (b) 14 m (c) 21 m (d) 24.5 m
- In the given figure, O is a point inside a $\triangle MNP$ such that $\angle MOP = 90^\circ$, $OM = 16$ cm and $OP = 12$ cm. If $MN = 21$ cm and $\angle NMP = 90^\circ$ then $NP = ?$
(a) 25 cm (b) 29 cm (c) 33 cm (d) 35 cm
- The hypotenuse of a right triangle is 25 cm. The other two sides are such that one is 5 cm longer than the other. The lengths of these sides are
(a) 10 cm, 15 cm (b) 15 cm, 20 cm
(c) 12 cm, 17 cm (d) 13 cm, 18 cm
- The height of an equilateral triangle having each side 12 cm, is
(a) $6\sqrt{2}$ cm (b) $6\sqrt{3}$ cm (c) $3\sqrt{6}$ cm (d) $6\sqrt{6}$ cm

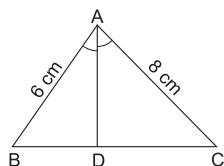


10. $\triangle ABC$ is an isosceles triangle with $AB = AC = 13$ cm and the length of altitude from A on BC is 5 cm. Then, $BC = ?$

(a) 12 cm (b) 16 cm (c) 18 cm (d) 24 cm

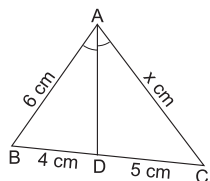
11. In a $\triangle ABC$ it is given that $AB = 6$ cm, $AC = 8$ cm and AD is the bisector of $\angle A$. Then, $BD : DC = ?$

(a) 3 : 4 (b) 9 : 16
(c) 4 : 3 (d) $\sqrt{3} : 2$



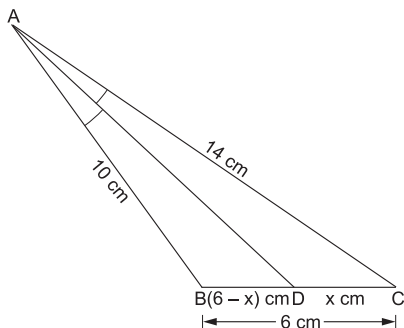
12. In a $\triangle ABC$ it is given that AD is the internal bisector of $\angle A$. If $BD = 4$ cm, $DC = 5$ cm and $AB = 6$ cm, then $AC = ?$

(a) 4.5 cm (b) 8 cm
(c) 9 cm (d) 7.5 cm



13. In a $\triangle ABC$, it is given that AD is the internal bisector of $\angle A$. If $AB = 10$ cm, $AC = 14$ cm and $BC = 6$ cm, then $CD = ?$

(a) 4.8 cm
(b) 3.5 cm
(c) 7 cm
(d) 10.5 cm

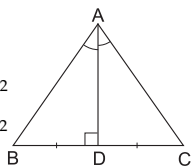


14. In a triangle, the perpendicular from the vertex to the base bisects the base. The triangle is

(a) right-angled (b) isosceles
(c) scalene (d) obtuse-angled

15. In an equilateral triangle ABC , if $AD \perp BC$ then which of the following is true?

(a) $2AB^2 = 3AD^2$ (b) $4AB^2 = 3AD^2$
(c) $3AB^2 = 4AD^2$ (d) $3AB^2 = 2AD^2$



16. In a rhombus of side 10 cm, one of the diagonals is 12 cm long. The length of the second diagonal is

(a) 20 cm (b) 18 cm (c) 16 cm (d) 22 cm

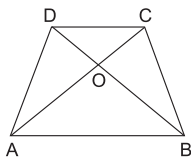
17. The lengths of the diagonals of a rhombus are 24 cm and 10 cm. The length of each side of the rhombus is

(a) 12 cm (b) 13 cm (c) 14 cm (d) 17 cm

18. If the diagonals of a quadrilateral divide each other proportionally then it is a

- (a) parallelogram (b) trapezium
(c) rectangle (d) square

19. In the given figure, $ABCD$ is a trapezium whose diagonals AC and BD intersect at O such that $OA = (3x - 1)$ cm, $OB = (2x + 1)$ cm, $OC = (5x - 3)$ cm and $OD = (6x - 5)$ cm. Then, $x = ?$



- (a) 2 (b) 3 (c) 2.5 (d) 4

20. The line segments joining the midpoints of the adjacent sides of a quadrilateral form

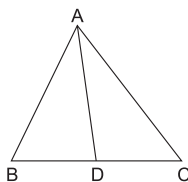
- (a) a parallelogram (b) a rectangle
(c) a square (d) a rhombus

21. If the bisector of an angle of a triangle bisects the opposite side then the triangle is

- (a) scalene (b) equilateral
(c) isosceles (d) right-angled

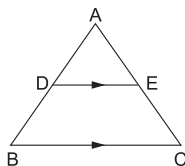
22. In $\triangle ABC$ it is given that $\frac{AB}{AC} = \frac{BD}{DC}$. If $\angle B = 70^\circ$ and $\angle C = 50^\circ$ then $\angle BAD = ?$

- (a) 30° (b) 40°
(c) 45° (d) 50°



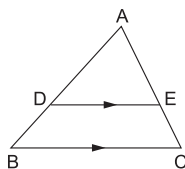
23. In $\triangle ABC$, $DE \parallel BC$ so that $AD = 2.4$ cm, $AE = 3.2$ cm and $EC = 4.8$ cm. Then, $AB = ?$

- (a) 3.6 cm (b) 6 cm
(c) 6.4 cm (d) 7.2 cm



24. In a $\triangle ABC$, if DE is drawn parallel to BC , cutting AB and AC at D and E respectively such that $AB = 7.2$ cm, $AC = 6.4$ cm and $AD = 4.5$ cm. Then, $AE = ?$

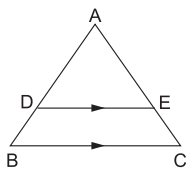
- (a) 5.4 cm (b) 4 cm
(c) 3.6 cm (d) 3.2 cm



25. In $\triangle ABC$, $DE \parallel BC$ so that $AD = (7x - 4)$ cm, $AE = (5x - 2)$ cm, $DB = (3x + 4)$ cm and $EC = 3x$ cm.

Then, we have

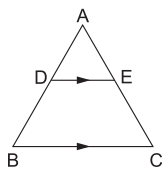
- (a) $x = 3$ (b) $x = 5$
(c) $x = 4$ (d) $x = 2.5$



26. In $\triangle ABC$, $DE \parallel BC$ such that $\frac{AD}{DB} = \frac{3}{5}$. If $AC = 5.6$ cm

then $AE = ?$

- (a) 4.2 cm (b) 3.1 cm
(c) 2.8 cm (d) 2.1 cm



27. $\triangle ABC \sim \triangle DEF$ and the perimeters of $\triangle ABC$ and $\triangle DEF$ are 30 cm and 18 cm respectively. If $BC = 9$ cm then $EF = ?$

- (a) 6.3 cm (b) 5.4 cm (c) 7.2 cm (d) 4.5 cm

28. $\triangle ABC \sim \triangle DEF$ such that $AB = 9.1$ cm and $DE = 6.5$ cm. If the perimeter of $\triangle DEF$ is 25 cm, what is the perimeter of $\triangle ABC$?

- (a) 35 cm (b) 28 cm (c) 42 cm (d) 40 cm

29. In $\triangle ABC$, it is given that $AB = 9$ cm, $BC = 6$ cm and $CA = 7.5$ cm. Also, $\triangle DEF$ is given such that $EF = 8$ cm and $\triangle DEF \sim \triangle ABC$. Then, perimeter of $\triangle DEF$ is

- (a) 22.5 cm (b) 25 cm (c) 27 cm (d) 30 cm

30. ABC and BDE are two equilateral triangles such that D is the midpoint of BC . Ratio of the areas of triangles ABC and BDE is

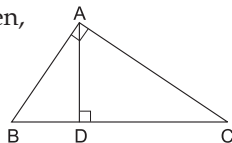
- (a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1

31. It is given that $\triangle ABC \sim \triangle DFE$. If $\angle A = 30^\circ$, $\angle C = 50^\circ$, $AB = 5$ cm, $AC = 8$ cm and $DF = 7.5$ cm then which of the following is true?

- (a) $DE = 12$ cm, $\angle F = 50^\circ$ (b) $DE = 12$ cm, $\angle F = 100^\circ$
(c) $EF = 12$ cm, $\angle D = 100^\circ$ (d) $EF = 12$ cm, $\angle D = 30^\circ$

32. In the given figure, $\angle BAC = 90^\circ$ and $AD \perp BC$. Then,

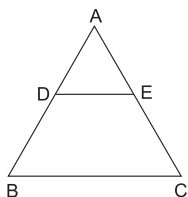
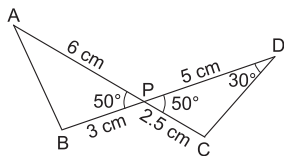
- (a) $BC \cdot CD = BC^2$
(b) $AB \cdot AC = BC^2$
(c) $BD \cdot CD = AD^2$
(d) $AB \cdot AC = AD^2$



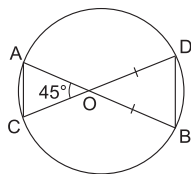
33. In $\triangle ABC$, $AB = 6\sqrt{3}$ cm, $AC = 12$ cm and $BC = 6$ cm. Then, $\angle B$ is

- (a) 45° (b) 60° (c) 90° (d) 120°

34. In $\triangle ABC$ and $\triangle DEF$, it is given that $\frac{AB}{DE} = \frac{BC}{FD}$ then
 (a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$
35. In $\triangle DEF$ and $\triangle PQR$, it is given that $\angle D = \angle Q$ and $\angle R = \angle E$, then which of the following is not true?
 (a) $\frac{EF}{PR} = \frac{DF}{PQ}$ (b) $\frac{DE}{PQ} = \frac{EF}{RP}$ (c) $\frac{DE}{QR} = \frac{DF}{PQ}$ (d) $\frac{EF}{RP} = \frac{DE}{QR}$
36. If $\triangle ABC \sim \triangle EDF$ and $\triangle ABC$ is not similar to $\triangle DEF$ then which of the following is not true?
 (a) $BC \cdot EF = AC \cdot FD$ (b) $AB \cdot EF = AC \cdot DE$
 (c) $BC \cdot DE = AB \cdot EF$ (d) $BC \cdot DE = AB \cdot FD$
37. In $\triangle ABC$ and $\triangle DEF$, it is given that $\angle B = \angle E$, $\angle F = \angle C$ and $AB = 3DE$, then the two triangles are
 (a) congruent but not similar (b) similar but not congruent
 (c) neither congruent nor similar (d) similar as well as congruent
38. If in $\triangle ABC$ and $\triangle PQR$, we have $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$ then
 (a) $\triangle PQR \sim \triangle CAB$ (b) $\triangle PQR \sim \triangle ABC$
 (c) $\triangle CBA \sim \triangle PQR$ (d) $\triangle BCA \sim \triangle PQR$
39. In the given figure, two line segments AC and BD intersect each other at the point P such that $PA = 6$ cm, $PB = 3$ cm, $PC = 2.5$ cm, $PD = 5$ cm, $\angle APB = 50^\circ$ and $\angle CDP = 30^\circ$ then $\angle PBA = ?$
 (a) 50° (b) 30° (c) 60° (d) 100°
40. Corresponding sides of two similar triangles are in the ratio 4 : 9. Areas of these triangles are in the ratio
 (a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) 16 : 81
41. It is given that $\triangle ABC \sim \triangle PQR$ and $\frac{BC}{QR} = \frac{2}{3}$ then $\frac{\text{ar}(\triangle PQR)}{\text{ar}(\triangle ABC)} = ?$
 (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{4}{9}$ (d) $\frac{9}{4}$
42. In an equilateral $\triangle ABC$, D is the midpoint of AB and E is the midpoint of AC . Then, $\text{ar}(\triangle ABC) : \text{ar}(\triangle ADE) = ?$
 (a) 2 : 1 (b) 4 : 1
 (c) 1 : 2 (d) 1 : 4



43. In $\triangle ABC$ and $\triangle DEF$, we have $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = \frac{5}{7}$, then
 $\text{ar}(\triangle ABC) : \text{ar}(\triangle DEF) = ?$
 (a) 5 : 7 (b) 25 : 49 (c) 49 : 25 (d) 125 : 343
44. $\triangle ABC \sim \triangle DEF$ such that $\text{ar}(\triangle ABC) = 36 \text{ cm}^2$ and $\text{ar}(\triangle DEF) = 49 \text{ cm}^2$.
 Then, the ratio of their corresponding sides is
 (a) 36 : 49 (b) 6 : 7 (c) 7 : 6 (d) $\sqrt{6} : \sqrt{7}$
45. Two isosceles triangles have their corresponding angles equal and their areas are in the ratio 25 : 36. The ratio of their corresponding heights is
 (a) 25 : 36 (b) 36 : 25 (c) 5 : 6 (d) 6 : 5
46. The line segments joining the midpoints of the sides of a triangle form four triangles, each of which is
 (a) congruent to the original triangle
 (b) similar to the original triangle
 (c) an isosceles triangle
 (d) an equilateral triangle
47. If $\triangle ABC \sim \triangle QRP$, $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{9}{4}$, $AB = 18 \text{ cm}$ and $BC = 15 \text{ cm}$ then
 $PR = ?$
 (a) 8 cm (b) 10 cm (c) 12 cm (d) $\frac{20}{3} \text{ cm}$
48. In the given figure, O is the point of intersection of two chords AB and CD such that $OB = OD$ and $\angle AOC = 45^\circ$. Then, $\triangle OAC$ and $\triangle ODB$ are
 (a) equilateral and similar
 (b) equilateral but not similar
 (c) isosceles and similar
 (d) isosceles but not similar



49. In an isosceles $\triangle ABC$, if $AC = BC$ and $AB^2 = 2AC^2$ then $\angle C = ?$
 (a) 30° (b) 45° (c) 60° (d) 90°
50. In $\triangle ABC$, if $AB = 16 \text{ cm}$, $BC = 12 \text{ cm}$ and $AC = 20 \text{ cm}$, then $\triangle ABC$ is
 (a) acute-angled (b) right-angled
 (c) obtuse-angled (d) not possible

True/False Type

51. Which of the following is a true statement?
 (a) Two similar triangles are always congruent.
 (b) Two figures are similar if they have the same shape and size.

- (c) Two triangles are similar if their corresponding sides are proportional.
- (d) Two polygons are similar if their corresponding sides are proportional.

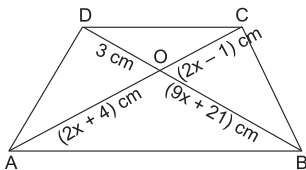
52. Which of the following is a false statement?

- (a) If the areas of two similar triangles are equal then the triangles are congruent.
- (b) The ratio of the areas of two similar triangles is equal to the ratio of their corresponding sides.
- (c) The ratio of the areas of two similar triangles is equal to the ratio of squares of their corresponding medians.
- (d) The ratio of the areas of two similar triangles is equal to the ratio of squares of their corresponding altitudes.

Matching of columns

53. Match the following columns:

Column I	Column II
(a) In a given $\triangle ABC$, $DE \parallel BC$ and $\frac{AD}{DB} = \frac{3}{5}$. If $AC = 5.6$ cm then $AE = \dots\dots$ cm.	(p) 6
(b) If $\triangle ABC \sim \triangle DEF$ such that $2AB = 3DE$ and $BC = 6$ cm then $EF = \dots\dots$ cm.	(q) 4
(c) If $\triangle ABC \sim \triangle PQR$ such that $\text{ar}(\triangle ABC) : \text{ar}(\triangle PQR) = 9 : 16$ and $BC = 4.5$ cm then $QR = \dots\dots$ cm.	(r) 3
(d) In the given figure, $AB \parallel CD$ and $OA = (2x + 4)$ cm, $OB = (9x - 21)$ cm, $OC = (2x - 1)$ cm and $OD = 3$ cm. Then $x = ?$	(s) 2.1



The correct answer is

- (a) —....., (b) —....., (c) —....., (d) —.....

54. Match the following columns:

Column I	Column II
(a) A man goes 10 m due east and then 20 m due north. His distance from the starting point is m.	(p) $25\sqrt{3}$
(b) In an equilateral triangle with each side 10 cm, the altitude is cm.	(q) $5\sqrt{3}$
(c) The area of an equilateral triangle having each side 10 cm is cm^2 .	(r) $10\sqrt{5}$
(d) The length of diagonal of a rectangle having length 8 m and breadth 6 m is m.	(s) 10

The correct answer is

(a) —.....,

(b) —.....,

(c) —.....,

(d) —.....

ANSWERS (MCQ)

1. (c) 2. (b) 3. (c) 4. (d) 5. (d) 6. (a) 7. (b) 8. (b) 9. (b)
 10. (d) 11. (a) 12. (d) 13. (b) 14. (b) 15. (c) 16. (c) 17. (b) 18. (b)
 19. (a) 20. (a) 21. (c) 22. (a) 23. (b) 24. (b) 25. (c) 26. (d) 27. (b)
 28. (a) 29. (d) 30. (d) 31. (b) 32. (c) 33. (c) 34. (c) 35. (b) 36. (c)
 37. (b) 38. (a) 39. (d) 40. (d) 41. (d) 42. (b) 43. (b) 44. (b) 45. (c)
 46. (b) 47. (b) 48. (c) 49. (d) 50. (b) 51. (c) 52. (b)
 53. (a)–(s), (b)–(q), (c)–(p), (d)–(r) 54. (a)–(r), (b)–(q), (c)–(p), (d)–(s)

HINTS TO SOME SELECTED QUESTIONS

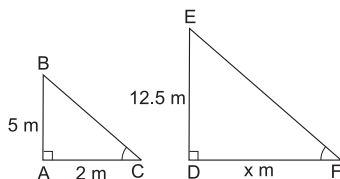
5. Let AB be the stick and AC be its shadow.

Let DE be the tree and DF be its shadow.

$\triangle ABC \sim \triangle DEF$.

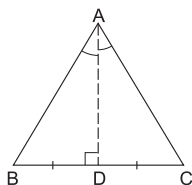
$$\therefore \frac{AB}{DE} = \frac{AC}{DF} \Rightarrow \frac{5}{12.5} = \frac{2}{x}$$

$$\Rightarrow x = \left(\frac{12.5 \times 2}{5} \right) = 5.$$



11. $\frac{BD}{DC} = \frac{AB}{AC} = \frac{6}{8} = \frac{3}{4}$ [by angle-bisector theorem]

$$\begin{aligned}
 15. \quad AB^2 &= BD^2 + AD^2 \\
 &= \left(\frac{1}{2}AB\right)^2 + AD^2 = \frac{1}{4}AB^2 + AD^2 \\
 \Rightarrow \quad \frac{3}{4}AB^2 &= AD^2 \Rightarrow 3AB^2 = 4AD^2.
 \end{aligned}$$



19. We know that the diagonals of a trapezium divide each other proportionally.

$$\begin{aligned}
 \therefore \quad \frac{OA}{OC} &= \frac{OB}{OD} \Rightarrow \frac{3x-1}{5x-3} = \frac{2x+1}{6x-5} \\
 \Rightarrow \quad (3x-1)(6x-5) &= (5x-3)(2x+1) \\
 \Rightarrow \quad 18x^2 - 21x + 5 &= 10x^2 - x - 3 \Rightarrow 8x^2 - 20x + 8 = 0 \\
 \Rightarrow \quad 2x^2 - 5x + 2 &= 0 \Rightarrow 2x^2 - 4x - x + 2 = 0 \Rightarrow 2x(x-2) - (x-2) = 0 \\
 \Rightarrow \quad (x-2)(2x-1) &= 0 \Rightarrow x = 2 \text{ or } x = \frac{1}{2}.
 \end{aligned}$$

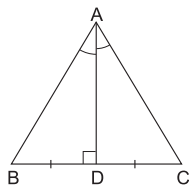
But, $x = \frac{1}{2}$ gives $(6x-5) < 0$ and the distance cannot be negative.

$$\therefore x = 2.$$

21. Let AD be the bisector of $\angle A$ of $\triangle ABC$ such that $BD = DC$.

$$\text{Then, } \frac{AB}{AC} = \frac{BD}{DC} = 1 \Rightarrow AB = AC.$$

So, the given triangle is isosceles.



$$22. \quad \angle A = 180^\circ - (70^\circ + 50^\circ) = 60^\circ.$$

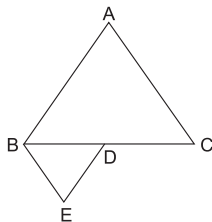
Since $\frac{BD}{DC} = \frac{AB}{AC}$, it means AD is the bisector of $\angle A$.

$$\therefore \quad \angle BAD = \left(\frac{1}{2} \times 60^\circ\right) = 30^\circ.$$

30. $\triangle ABC \sim \triangle BDE$ [$\because$ both are equilateral].

$$\text{Also, } BD = \frac{1}{2}BC = \frac{1}{2}AB \quad [\because D \text{ is the midpoint of } BC].$$

$$\text{Now, } \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle BDE)} = \frac{AB^2}{BD^2} = \frac{AB^2}{\left(\frac{1}{2}AB\right)^2} = \frac{4}{1}.$$



$$31. \quad \angle B = 180^\circ - (30^\circ + 50^\circ) = 100^\circ.$$

Since $\triangle ABC \sim \triangle DFE$, we have $\angle D = \angle A = 30^\circ$, $\angle F = \angle B = 100^\circ$

and $\angle E = \angle C = 50^\circ$. Let $DE = x$ cm. Then,

$$\frac{AB}{DF} = \frac{AC}{DE} \Rightarrow \frac{5}{7.5} = \frac{8}{x}$$

$$\therefore \quad 5x = 8 \times 7.5 \Rightarrow x = \frac{8 \times 7.5}{5} = 12.$$

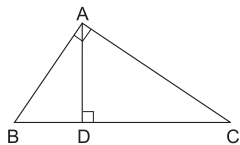
Hence, $DE = 12$ cm and $\angle F = 100^\circ$.

32. In
- $\triangle DBA$
- and
- $\triangle DAC$
- , we have

$$\angle ADB = \angle CDA = 90^\circ, \angle ABD = \angle CAD = 90^\circ - \angle C$$

and $\angle BAD = \angle ACD = 90^\circ - \angle B$.

$$\begin{aligned}\therefore \triangle DBA &\sim \triangle DAC \Rightarrow \frac{BD}{AD} = \frac{AD}{CD} \\ &\Rightarrow BD \cdot CD = AD^2.\end{aligned}$$



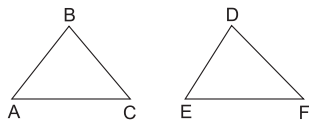
33. In
- $\triangle ABC$
- ,
- AC
- is the longest side.

$$AB^2 + BC^2 = \{(6\sqrt{3})^2 + 6^2\} \text{ cm}^2 = (108 + 36) \text{ cm}^2 = 144 \text{ cm}^2 = (12 \text{ cm})^2 = AC^2.$$

$\therefore$ by the converse of Pythagoras' theorem, we have $\angle B = 90^\circ$.

34. Clearly,
- $B \leftrightarrow D$
- ,
- $A \leftrightarrow E$
- and
- $C \leftrightarrow F$

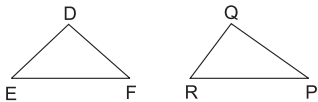
$$\therefore \angle B = \angle D.$$



- 35.
- $\angle D = \angle Q$
- ,
- $\angle E = \angle R$
- and
- $\angle F = \angle P$
- .

$$\therefore \triangle DEF \sim \triangle QRP \Rightarrow \frac{DE}{QR} = \frac{DF}{PQ} = \frac{EF}{RP}.$$

$$\therefore \frac{DE}{PQ} = \frac{EF}{RP} \text{ is not true.}$$



36. Since
- $\triangle ABC \sim \triangle EDF$
- and
- $\triangle ABC$
- is not similar to
- $\triangle DEF$
- , so

$$\frac{AB}{DE} \neq \frac{BC}{EF} \Rightarrow BC \cdot DE = AB \cdot EF \text{ is not true.}$$

- 37.
- $\triangle ABC \sim \triangle DEF$
- [by AA-similarity].

But $\triangle ABC$ and $\triangle DEF$ are not congruent since their corresponding sides AB and DE are not equal.

- 38.
- $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ} \Rightarrow B \leftrightarrow R$
- ,
- $A \leftrightarrow Q$
- and
- $C \leftrightarrow P$
- .

$$\therefore \triangle PQR \sim \triangle CAB.$$

- 41.
- $\frac{\text{ar}(\triangle PQR)}{\text{ar}(\triangle ABC)} = \frac{QR^2}{BC^2} = \left(\frac{QR}{BC}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}.$

42. Clearly,
- $\frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{2}$
- and
- $\angle A = \angle A$
- (common).

$$\therefore \triangle ABC \sim \triangle ADE \text{ [by SAS-similarity].}$$

$$\therefore \text{ar}(\triangle ABC) : \text{ar}(\triangle ADE) = \left(\frac{AB}{AD}\right)^2 = \left(\frac{2}{1}\right)^2 = \frac{4}{1} = 4 : 1.$$

- 44.
- $\left(\frac{AB}{DE}\right)^2 = \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{36}{49} = \left(\frac{6}{7}\right)^2 \Rightarrow \frac{AB}{DE} = \frac{6}{7}.$

$\therefore$ the ratio of the corresponding sides is $6 : 7$.

45. The two triangles have corresponding angles equal and so they are similar.

$\therefore$ the ratio of their areas is equal to the ratio of the squares of their corresponding sides but the ratio of their corresponding sides is equal to the ratio of their corresponding altitudes (or heights).

So, the ratio of their areas is equal to the ratio of the squares of their heights.

$$\text{Now, ratio of their areas} = \frac{25}{36} = \frac{5^2}{6^2}$$

$$\Rightarrow \text{ratio of their heights} = \frac{5}{6}.$$

$$46. \text{ar}(\triangle QRP) = \text{ar}(\triangle PQR).$$

$$\therefore \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle QRP)} = \frac{9}{4} = \frac{3^2}{2^2} \Rightarrow \frac{BC}{RP} = \frac{3}{2} \quad [\because \triangle ABC \sim \triangle QRP]$$

$$\Rightarrow PR = \frac{2}{3} \times BC = \frac{2}{3} \times 15 \text{ cm} = 10 \text{ cm}.$$

48. In $\triangle OAC$ and $\triangle ODB$, we have

$\angle AOC = \angle DOB$ (ver. opp. $\angle$ s) and $\angle OAC = \angle ODB$ ($\angle$ s in the same segment).

$$\therefore \triangle OAC \sim \triangle ODB \Rightarrow \frac{OC}{OB} = \frac{OA}{OD} = \frac{AC}{BD}$$

$$\therefore \frac{OC}{OA} = \frac{OB}{OD} = 1 \Rightarrow OC = OA \quad [\because OB = OD \text{ (given)}].$$

Clearly, $OA \neq OD$.

$$\therefore \frac{OA}{OD} \neq 1 \Rightarrow \frac{AC}{BD} \neq 1 \Rightarrow AC \neq BD.$$

$\therefore \triangle OAC$ and $\triangle ODB$ are isosceles and similar.

$$49. AB^2 = 2AC^2 = AC^2 + AC^2 = BC^2 + AC^2 \quad [\because AC = BC].$$

$\therefore$ by converse of Pythagoras' theorem, we have $\angle C = 90^\circ$.

50. AC is the longest side of $\triangle ABC$.

$$AB^2 + BC^2 = 16^2 + 12^2 = 256 + 144 = 400 = 20^2 = AC^2.$$

$\therefore \angle B = 90^\circ$ [by the converse of Pythagoras' theorem].

51. (a) Two similar triangles need not be congruent.

(b) Similar figures need not be of the same size.

(c) It is clearly true.

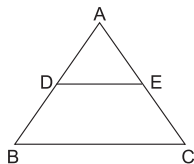
(d) Two polygons are similar only when their corresponding angles are equal and their corresponding sides are proportional.

53. (a) Let $AE = x$ cm. Then, $EC = (5.6 - x)$ cm.

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{3}{5} = \frac{x}{5.6 - x}$$

$$\therefore 3(5.6 - x) = 5x \Rightarrow 8x = 3 \times 5.6$$

$$\therefore x = \frac{3 \times 5.6}{8} = \frac{16.8}{8} = 2.1.$$



$$(b) \frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{3}{2} = \frac{6}{x} \Rightarrow 3x = 12 \Rightarrow x = 4.$$

$$(c) \frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{BC^2}{QR^2} \Rightarrow \frac{9}{16} = \frac{BC^2}{QR^2} \Rightarrow \left(\frac{3}{4}\right)^2 = \left(\frac{BC}{QR}\right)^2$$

$$\Rightarrow \frac{BC}{QR} = \frac{3}{4} \Rightarrow QR = \frac{4}{3} \times BC = \left(\frac{4}{3} \times 4.5\right) \text{ cm} = 6 \text{ cm}.$$

$$(d) \triangle OAB \sim \triangle OCD \Rightarrow \frac{OA}{OC} = \frac{OB}{OD} \Rightarrow \frac{2x+4}{2x-1} = \frac{9x-21}{3}$$

$$\Rightarrow 6x + 12 = 18x^2 - 51x + 21 \Rightarrow 18x^2 - 57x + 9 = 0$$

$$\Rightarrow 6x^2 - 19x + 3 = 0 \Rightarrow (x-3)(6x-1) = 0 \Rightarrow x = 3 \text{ or } x = \frac{1}{6}$$

But, $x = \frac{1}{6}$ makes $(2x-1) < 0$. So, we reject it.

$$\therefore x = 3.$$

The correct answer is: (a)-(s), (b)-(q), (c)-(p), (d)-(r).

54. (a) Let $OA = 10$ m and $AB = 20$ m. Then,

$$OB^2 = OA^2 + AB^2 = \{(10)^2 + (20)^2\} \text{ m}^2 = 500 \text{ m}^2$$

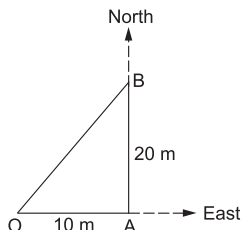
$$\Rightarrow OB = \sqrt{500} \text{ m} = \sqrt{5 \times 100} \text{ m} = 10\sqrt{5} \text{ m}.$$

$$(b) \text{Altitude} = \frac{\sqrt{3}}{2}a = \left(\frac{\sqrt{3}}{2} \times 10\right) \text{ cm} = 5\sqrt{3} \text{ cm}.$$

$$(c) \text{Area} = \frac{\sqrt{3}}{4}a^2 = \left(\frac{\sqrt{3}}{4} \times 10 \times 10\right) \text{ cm}^2 = 25\sqrt{3} \text{ cm}^2.$$

$$(d) d^2 = (8^2 + 6^2) \text{ m}^2 = (64 + 36) \text{ m}^2 = 100 \text{ m}^2 \Rightarrow d = \sqrt{100} \text{ m} = 10 \text{ m}.$$

Then, the correct answer is: (a)-(r), (b)-(q), (c)-(p), (d)-(s).



TEST YOURSELF

MCQ

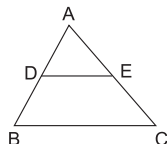
1. $\triangle ABC \sim \triangle DEF$ and their perimeters are 32 cm and 24 cm respectively.

If $AB = 10$ cm then $DE = ?$

- (a) 8 cm (b) 7.5 cm (c) 15 cm (d) $5\sqrt{3}$ cm

2. In the given figure, $DE \parallel BC$. If $DE = 5$ cm, $BC = 8$ cm and $AD = 3.5$ cm then $AB = ?$

- (a) 5.6 cm (b) 4.8 cm
(c) 5.2 cm (d) 6.4 cm



3. Two poles of height 6 m and 11 m stand vertically upright on a plane ground. If the distance between their feet is 12 m then the distance between their tops is

- (a) 12 m (b) 13 m (c) 14 m (d) 15 m

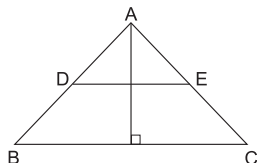
4. The areas of two similar triangles are 25 cm^2 and 36 cm^2 respectively. If the altitude of the first triangle is 3.5 cm then the corresponding altitude of the other triangle is

- (a) 5.6 cm (b) 6.3 cm (c) 4.2 cm (d) 7 cm

Short-Answer Questions

5. If $\triangle ABC \sim \triangle DEF$ such that $2AB = DE$ and $BC = 6$ cm, find EF .

6. In the given figure, $DE \parallel BC$ such that $AD = x$ cm, $DB = (3x + 4)$ cm, $AE = (x + 3)$ cm and $EC = (3x + 19)$ cm. Find the value of x .



7. A ladder 10 m long reaches the window of a house 8 m above the ground. Find the distance of the foot of the ladder from the base of the wall.

8. Find the length of the altitude of an equilateral triangle of side $2a$ cm.

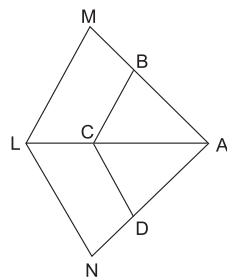
9. $\triangle ABC \sim \triangle DEF$ such that $\text{ar}(\triangle ABC) = 64 \text{ cm}^2$ and $\text{ar}(\triangle DEF) = 169 \text{ cm}^2$. If $BC = 4$ cm, find EF .

10. In a trapezium $ABCD$, it is given that $AB \parallel CD$ and $AB = 2CD$. Its diagonals AC and BD intersect at the point O such that $\text{ar}(\triangle AOB) = 84 \text{ cm}^2$. Find $\text{ar}(\triangle COD)$.

11. The corresponding sides of two similar triangles are in the ratio $2 : 3$. If the area of the smaller triangle is 48 cm^2 , find the area of the larger triangle.

12. In the given figure, $LM \parallel CB$ and $LN \parallel CD$.

Prove that $\frac{AM}{AB} = \frac{AN}{AD}$.



13. Prove that the internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle.

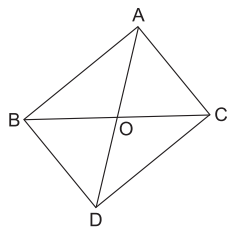
14. In an equilateral triangle with side a , prove that $\text{area} = \frac{\sqrt{3}}{4} a^2$.

15. Find the length of each side of a rhombus whose diagonals are 24 cm and 10 cm long.

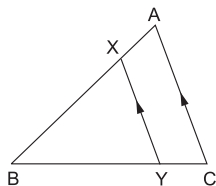
16. Prove that the ratio of the perimeters of two similar triangles is the same as the ratio of their corresponding sides.

Long-Answer Questions

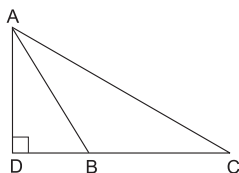
17. In the given figure, $\triangle ABC$ and $\triangle DBC$ have the same base BC . If AD and BC intersect at O , prove that $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DBC)} = \frac{AO}{DO}$.



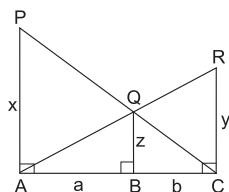
18. In the given figure, $XY \parallel AC$ and XY divides $\triangle ABC$ into two regions, equal in area. Show that $\frac{AX}{AB} = \frac{(2 - \sqrt{2})}{2}$.



19. In the given figure, $\triangle ABC$ is an obtuse triangle, obtuse-angled at B . If $AD \perp CB$ (produced) prove that $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$.



20. In the given figure, each one of PA , QB and RC is perpendicular to AC . If $AP = x$, $QB = z$, $RC = y$, $AB = a$ and $BC = b$, show that $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$.

**ANSWERS (TEST YOURSELF)**

1. (b) 2. (a) 3. (b) 4. (c) 5. 12 cm 6. $x = 2$ 7. 6 m
8. $\sqrt{3}a$ cm 9. 6.5 cm 10. 21 cm² 11. 108 cm² 15. 13 cm



CIRCLE A circle is a collection of all points in a plane which are at a constant distance from a fixed point. The constant distance is called the radius and the fixed point is called the centre of the circle.

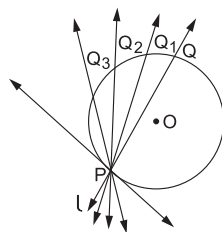
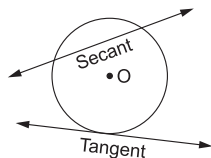
SECANT A line which intersects a circle in two distinct points is called a secant to the circle.

TANGENT A line meeting a circle only in one point is called a tangent to the circle at that point.

The point at which the tangent line intersects the circle is called the point of contact.

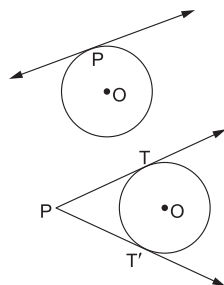
NOTE The tangent to a circle is a special case of the secant, when the two end points of its corresponding chord coincide.

Consider a line l which is a secant to a circle with centre O . Let PQ be the corresponding chord of this secant. If we rotate the secant anticlockwise about the point P then point Q comes closer to the point P . Let Q_1, Q_2, Q_3 etc., be the positions of Q as the secant goes on rotating then we find these positions getting closer to P sequentially. Finally, as the position of Q coincides with P , the secant is reduced to a tangent at point P .



NUMBER OF TANGENTS TO A CIRCLE

- (i) There is no tangent passing through a point lying inside the circle.
- (ii) There is one and only one tangent passing through a point lying on a circle.
- (iii) There are exactly two tangents through a point lying outside a circle. In the figure, PT and PT' are two tangents from a point P lying outside the circle.



LENGTH OF A TANGENT The length of the segment of the tangent from the external point P to the point of contact with the circle is called the length of the tangent from the point P to the circle.

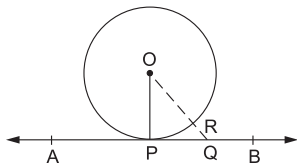
RESULTS ON TANGENTS

THEOREM 1 The tangent at any point of a circle is perpendicular to the radius through the point of contact. [CBSE 2007, '09, '11, '12, '13, '14, '15]

GIVEN A circle with centre O and a tangent AB at a point P of the circle.

TO PROVE $OP \perp AB$.

CONSTRUCTION Take a point Q , other than P , on AB . Join OQ .



PROOF Q is a point on the tangent AB , other than the point of contact P .

$\therefore Q$ lies outside the circle.

Let OQ intersect the circle at R .

Then, $OR < OQ$ [a part is less than the whole]. ... (i)

But, $OP = OR$ [radii of the same circle]. ... (ii)

$\therefore OP < OQ$ [from (i) and (ii)].

Thus, OP is shorter than any other line segment joining O to any point of AB , other than P .

In other words, OP is the shortest distance between the point O and the line AB .

But, the shortest distance between a point and a line is the perpendicular distance.

$\therefore OP \perp AB$.

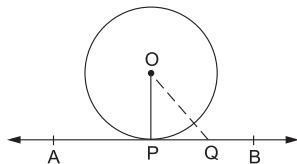
REMARKS (i) From this theorem we also conclude that at any point on a circle, one and only one tangent can be drawn to the circle.

(ii) The line containing the radius through the point of contact is called the 'normal' to the circle at the point of contact.

THEOREM 2 (Converse of Theorem 1) A line drawn through the end of a radius and perpendicular to it is a tangent to the circle.

GIVEN A circle with centre O in which OP is a radius and AB is a line through P such that $OP \perp AB$.

TO PROVE AB is a tangent to the circle at the point P .



CONSTRUCTION Take a point Q , different from P , on AB . Join OQ .

PROOF We know that the perpendicular distance from a point to a line is the shortest distance between them.

$$\therefore OP \perp AB \Rightarrow OP \text{ is the shortest distance from } O \text{ to } AB.$$

$$\therefore OP < OQ.$$

$$\therefore Q \text{ lies outside the circle } [\because OP \text{ is the radius and } OP < OQ].$$

Thus, every point on AB , other than P , lies outside the circle.

$$\therefore AB \text{ meets the circle at the point } P \text{ only.}$$

Hence, AB is the tangent to the circle at the point P .

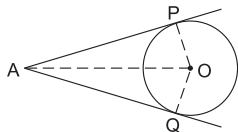
REMARK This theorem gives us a method of constructing a tangent at a point P on the circle. We first draw the radius OP and then draw $PT \perp OP$. Then, PT is the tangent at P .

THEOREM 3 *The lengths of tangents drawn from an external point to a circle are equal.* [CBSE 2007, '08, '08C, '09, '09C, '10, '11, '12, '13, '13C, '14, '15, '17]

GIVEN Two tangents AP and AQ are drawn from a point A to a circle with centre O .

TO PROVE $AP = AQ$.

CONSTRUCTION Join OP , OQ and OA .



PROOF AP is a tangent at P and OP is the radius through P .

$$\therefore OP \perp AP.$$

Similarly, $OQ \perp AQ$.

In the right $\triangle OPA$ and OQA , we have:

$$OP = OQ \quad [\text{radii of the same circle}]$$

$$OA = OA \quad [\text{common}]$$

$$\therefore \triangle OPA \cong \triangle OQA \quad [\text{by RHS-congruence}].$$

Hence, $AP = AQ$ [cpct].

REMARK The above theorem can also be proved by using the Pythagoras' theorem:

$$AP^2 = OA^2 - OP^2 = OA^2 - OQ^2 = AQ^2 \quad [\because OP = OQ = \text{radius}]$$

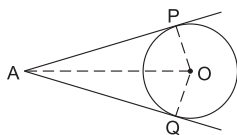
$$\Rightarrow AP = AQ.$$

THEOREM 4 *If two tangents are drawn from an external point then*

(i) *they subtend equal angles at the centre, and*

(ii) *they are equally inclined to the line segment joining the centre to that point.*

GIVEN A circle with centre O and a point A outside it. Also, AP and AQ are the two tangents to the circle.



TO PROVE $\angle AOP = \angle AOQ$ and $\angle OAP = \angle OAQ$.

PROOF In $\triangle AOP$ and $\triangle AOQ$, we have

$$AP = AQ \quad [\text{tangents from an external point are equal}]$$

$$OP = OQ \quad [\text{radii of the same circle}]$$

$$OA = OA \quad [\text{common}]$$

$$\therefore \triangle AOP \cong \triangle AOQ \quad [\text{by SSS-congruence}].$$

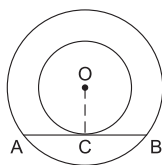
Hence, $\angle AOP = \angle AOQ$ and $\angle OAP = \angle OAQ$.

REMARK Since $\angle OAP = \angle OAQ$, i.e., AO is the bisector of $\angle PAQ$, so the centre lies on the bisector of the angle between the two tangents.

THEOREM 5 *Prove that in two concentric circles, the chord of the larger circle, which touches the smaller circle, is bisected at the point of contact.*

[CBSE 2009, '12]

GIVEN Two circles with the same centre O and AB is a chord of the larger circle touching the smaller circle at C .



TO PROVE $AC = BC$.

CONSTRUCTION Join OC .

PROOF AB is a tangent to the smaller circle at the point C and OC is the radius through C .

$$\therefore OC \perp AB.$$

But, the perpendicular drawn from the centre of a circle to a chord bisects the chord.

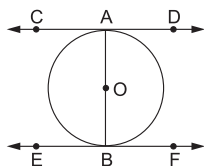
$$\therefore OC \text{ bisects } AB \quad [\because AB \text{ is a chord of larger circle}].$$

Hence, $AC = BC$.

THEOREM 6 *Prove that the tangents drawn at the ends of a diameter of a circle are parallel.*

[CBSE 2011, '12, '14, '17]

GIVEN CD and EF are the tangents at the end points A and B of the diameter AB of a circle with centre O .



TO PROVE $CD \parallel EF$.

PROOF CD is the tangent to the circle at the point A .

$$\therefore CD \perp OA \Rightarrow \angle OAD = 90^\circ \Rightarrow \angle BAD = 90^\circ.$$

EF is the tangent to the circle at the point B .

$$\therefore EF \perp OB \Rightarrow \angle OBE = 90^\circ \Rightarrow \angle ABE = 90^\circ.$$

Thus, $\angle BAD = \angle ABE$ (each equal to 90°).

But these are alternate interior angles.

$$\therefore CD \parallel EF.$$

THEOREM 7 Prove that the line segment joining the points of contact of two parallel tangents to a circle is a diameter of the circle.

GIVEN CD and EF are two parallel tangents at the points A and B of a circle with centre O .

TO PROVE AOB is a diameter of the circle.

CONSTRUCTION Join OA and OB . Draw $OG \parallel CD$.

PROOF $OG \parallel CD$ and AO cuts them.

$$\therefore \angle CAO + \angle GOA = 180^\circ$$

$$\Rightarrow 90^\circ + \angle GOA = 180^\circ \quad [\because OA \perp CD]$$

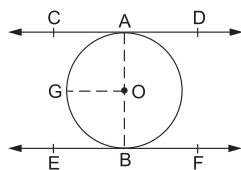
$$\Rightarrow \angle GOA = 90^\circ.$$

Similarly, $\angle GOB = 90^\circ$.

$$\therefore \angle GOA + \angle GOB = 90^\circ + 90^\circ = 180^\circ$$

$$\Rightarrow AOB \text{ is a straight line.}$$

Hence, AOB is a diameter of the circle with centre O .



THEOREM 8 Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segments joining the points of contact to the centre.

[CBSE 2008C, '14]

GIVEN PA and PB are the tangents drawn from a point P to a circle with centre O . Also, the line segments OA and OB are drawn.

TO PROVE $\angle APB + \angle AOB = 180^\circ$.

PROOF We know that the tangent to a circle is perpendicular to the radius through the point of contact.

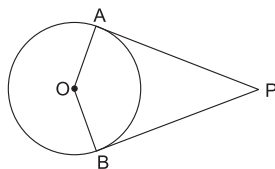
$$\therefore PA \perp OA \Rightarrow \angle OAP = 90^\circ, \text{ and}$$

$$PB \perp OB \Rightarrow \angle OBP = 90^\circ.$$

$$\therefore \angle OAP + \angle OBP = 90^\circ + 90^\circ = 180^\circ.$$

... (i)

But, we know that the sum of all the angles of a quadrilateral is 360° .



$$\therefore \angle OAP + \angle OBP + \angle APB + \angle AOB = 360^\circ. \quad \dots (ii)$$

From (i) and (ii), we get

$$\angle APB + \angle AOB = 180^\circ.$$

THEOREM 9 *Prove that there is one and only one tangent at any point on the circumference of a circle.*

PROOF Let P be a point on the circumference of a circle with centre O .

If possible, let PT and PT' be two tangents at a point P of the circle.

Now, the tangent at any point of a circle is perpendicular to the radius through the point of contact.

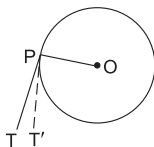
$$\therefore OP \perp PT \text{ and similarly, } OP \perp PT'$$

$$\Rightarrow \angle OPT = 90^\circ \text{ and } \angle OPT' = 90^\circ.$$

$$\Rightarrow \angle OPT = \angle OPT'$$

This is possible only when PT and PT' coincide.

Hence, there is one and only one tangent at any point on the circumference of a circle.



THEOREM 10 *Prove that the perpendicular at the point of contact of the tangent to a circle passes through the centre.*

PROOF Let PT be a tangent to a circle with centre O , where P is the point of contact.

Let $PQ \perp PT$, where Q lies on the circle, i.e., $\angle QPT = 90^\circ$.

If possible, let PQ not pass through the centre O .

Join PO and produce it to meet the circle at R .

Then PO being the radius through the point of contact, we have

$$PO \perp PT \quad [\because \text{the tangent is perpendicular to the radius through the point of contact}]$$

$$\therefore \angle OPT = 90^\circ \Rightarrow \angle RPT = 90^\circ.$$

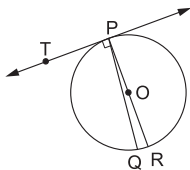
Thus, we have $\angle QPT = \angle RPT = 90^\circ$.

This is possible only if P , Q and R are collinear.

But a straight line cuts a circle in at the most two points.

So, the points Q and R coincide.

Hence, PQ passes through the centre O , i.e., the perpendicular at the point of contact to the tangent passes through the centre.



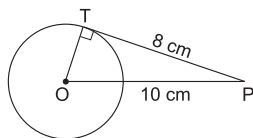
SOLVED EXAMPLES

EXAMPLE 1 From a point P , 10 cm away from the centre of a circle, a tangent PT of length 8 cm is drawn. Find the radius of the circle. [CBSE 2002]

SOLUTION Let O be the centre of the given circle.

Then, $OP = 10$ cm. Also, $PT = 8$ cm.

Join OT .



Now, PT is a tangent at T and OT is the radius through the point of contact T .

$$\therefore OT \perp PT.$$

In the right $\triangle OTP$, we have

$$OP^2 = OT^2 + PT^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow OT = \sqrt{OP^2 - PT^2} = \sqrt{(10)^2 - (8)^2} \text{ cm} = \sqrt{36} \text{ cm} = 6 \text{ cm}.$$

Hence, the radius of the circle is 6 cm.

EXAMPLE 2 A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. Find the length of PQ . [CBSE 2010]

SOLUTION We have $OP = \text{radius} = 5$ cm, $OQ = 13$ cm.

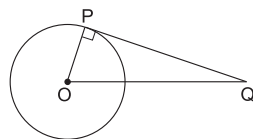
PQ is a tangent at P and OP is the radius through the point of contact P .

$$\therefore OP \perp PQ.$$

In right $\triangle OPQ$, we have

$$OQ^2 = OP^2 + PQ^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow PQ = \sqrt{OQ^2 - OP^2} = \sqrt{13^2 - 5^2} \text{ cm} = \sqrt{144} \text{ cm} = 12 \text{ cm}.$$

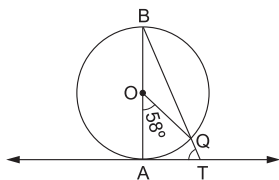


EXAMPLE 3 In the given figure, AB is the diameter of a circle with centre O and AT is a tangent. If $\angle AOQ = 58^\circ$, find $\angle ATQ$. [CBSE 2015]

SOLUTION $\angle AOQ = 58^\circ$

$$\Rightarrow \angle ABQ = \frac{1}{2} \angle AOQ = 29^\circ$$

[$\because$ the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle]



$$\Rightarrow \angle ABT = 29^\circ.$$

Now, AT is a tangent at A and OA is the radius through the point of contact A .

$$\therefore OA \perp AT, \text{ i.e., } \angle OAT = 90^\circ \Rightarrow \angle BAT = 90^\circ.$$

In $\triangle BAT$, we have

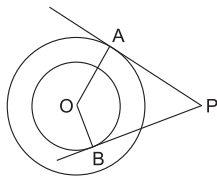
$$\angle BAT + \angle ABT + \angle ATB = 180^\circ$$

$$\Rightarrow 90^\circ + 29^\circ + \angle ATB = 180^\circ \Rightarrow \angle ATB = 61^\circ.$$

$$\therefore \angle ATQ = \angle ATB = 61^\circ.$$

EXAMPLE 4

Tangents PA and PB are drawn from an external point P to two concentric circles with centre O and radii 8 cm and 5 cm respectively, as shown in the figure. If $AP = 15$ cm then find the length of BP .



[CBSE 2012]

SOLUTION

We have

$$OA \perp AP \text{ and } OB \perp BP$$

[$\because$ the tangent at any point of a circle is perpendicular to the radius through the point of contact].

Join OP .

In right $\triangle OAP$, we have

$$OA = 8 \text{ cm}, AP = 15 \text{ cm}.$$

$$\therefore OP^2 = OA^2 + AP^2$$

[by Pythagoras' theorem]

$$\Rightarrow OP = \sqrt{OA^2 + AP^2} = \sqrt{8^2 + 15^2} \text{ cm} = \sqrt{289} \text{ cm} = 17 \text{ cm}.$$

In right $\triangle OBP$, we have

$$OB = 5 \text{ cm}, OP = 17 \text{ cm}.$$

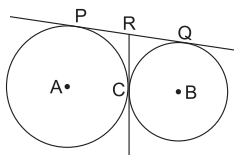
$$\therefore OP^2 = OB^2 + BP^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow BP = \sqrt{OP^2 - OB^2} = \sqrt{17^2 - 5^2} \text{ cm} = \sqrt{264} \text{ cm}.$$

Thus, the length of $BP = \sqrt{264} \text{ cm} = 16.25 \text{ cm}$ (approx).

EXAMPLE 5

In the given figure, two circles touch each other at the point C . Prove that the common tangent to the circles at C , bisects the common tangent at P and Q .



[CBSE 2013]

SOLUTION In the given figure, PR and CR are both tangents drawn to the same circle from an external point R .

$$\therefore PR = CR. \quad \dots (i)$$

Also, QR and CR are both tangents drawn to the same circle (second circle) from an external point R .

$$\therefore QR = CR. \quad \dots (ii)$$

From (i) and (ii), we get

$$PR = QR \quad [\text{each equal to } CR].$$

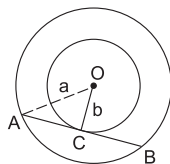
$$\therefore R \text{ is the midpoint of } PQ,$$

i.e., the common tangent to the circles at C , bisects the common tangent at P and Q .

EXAMPLE 6 Two concentric circles of radii a and b ($a > b$) are given. Find the length of the chord of the larger circle which touches the smaller circle.

[CBSE 2015]

SOLUTION Let O be the common centre of the two circles and AB be the chord of the larger circle which touches the smaller circle at C . Join OA and OC . Then, $OA = a$ and $OC = b$.



Now, $OC \perp AB$ and OC bisects AB

[$\because$ the chord of the larger circle touching the smaller circle, is bisected at the point of contact].

In right $\triangle ACO$, we have

$$OA^2 = OC^2 + AC^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow AC = \sqrt{OA^2 - OC^2} = \sqrt{a^2 - b^2}.$$

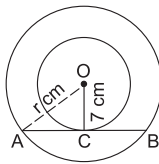
$$\therefore AB = 2AC = 2\sqrt{a^2 - b^2} \quad [\because C \text{ is the midpoint of } AB]$$

i.e., required length of the chord $AB = 2\sqrt{a^2 - b^2}$.

EXAMPLE 7 Two concentric circles are of radii 7 cm and r cm respectively, where $r > 7$. A chord of the larger circle of length 46 cm, touches the smaller circle. Find the value of r .

[CBSE 2011]

SOLUTION Let O be the common centre of the two circles and AB be the chord of the larger circle which touches the smaller circle at C . Then, $AB = 46$ cm.



Join OA and OC . Then, $OA = r$ cm and $OC = 7$ cm.

Now, $OC \perp AB$ and OC bisects AB . [See Theorem 5.]

In right $\triangle ACO$, we have

$$OA^2 = OC^2 + AC^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow OA = \sqrt{OC^2 + AC^2} = \sqrt{OC^2 + \left(\frac{1}{2}AB\right)^2}$$

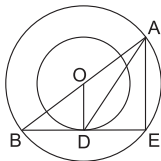
[$\because$ C is the midpoint of AB]

$$\Rightarrow r \text{ cm} = \sqrt{7^2 + 23^2} \text{ cm} = \sqrt{578} \text{ cm} = 17\sqrt{2} \Rightarrow r = 17\sqrt{2} \text{ cm}.$$

EXAMPLE 8

In the given figure, the radii of two concentric circles are 13 cm and 8 cm. AB is a diameter of the bigger circle and BD is a tangent to the smaller circle touching it at D . Find the length of AD .

[HOTS] [CBSE 2014]



SOLUTION

We have $\angle AEB = 90^\circ$ [angle in a semicircle].

Also, $OD \perp BE$ and OD bisects BE . [See Theorem 5.]

In right $\triangle OBD$, we have

$$OB^2 = OD^2 + BD^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow BD = \sqrt{OB^2 - OD^2} = \sqrt{13^2 - 8^2} \text{ cm}$$

$$= \sqrt{105} \text{ cm} \quad [\because OB = 13 \text{ cm}, OD = 8 \text{ cm}].$$

$$\therefore BE = 2BD = 2\sqrt{105} \text{ cm} \quad [\because D \text{ is the midpoint of } BE].$$

In right $\triangle AEB$, we have

$$AB^2 = AE^2 + BE^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow AE = \sqrt{AB^2 - BE^2} = \sqrt{26^2 - (2\sqrt{105})^2} \text{ cm} = \sqrt{256} \text{ cm} = 16 \text{ cm}$$

$$[\because AB = \text{diameter} = 2 \times OB = 2 \times 13 \text{ cm} = 26 \text{ cm}].$$

In right $\triangle AED$, we have

$$AD^2 = AE^2 + DE^2 \quad [\text{by Pythagoras' theorem}]$$

$$\Rightarrow AD = \sqrt{AE^2 + DE^2} = \sqrt{16^2 + (\sqrt{105})^2} \text{ cm}$$

$$= 19 \text{ cm} \quad [\because DE = BD = \sqrt{105} \text{ cm}].$$

[Note] We can also find AE by using midpoint theorem, since in $\triangle ABE$, O is the midpoint of AB and D is the midpoint of BE and so $OD \parallel AE$ and $AE = 2 \times OD = 16 \text{ cm}$.]

EXAMPLE 9

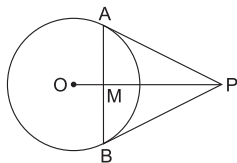
From a point P outside a circle with centre O , tangents PA and PB are drawn to the circle. Prove that OP is the right bisector of the line segment AB .

[CBSE 2015]

SOLUTION GIVEN PA and PB are tangents to a circle with centre O from an external point P .

TO PROVE OP is the right bisector of AB .

CONSTRUCTION Join AB . Let AB intersect OP at M .



PROOF In $\triangle MAP$ and $\triangle MBP$, we have

$$PA = PB \quad [\because \text{tangents to a circle from an external point are equal}]$$

$$MP = MP \quad [\text{common}]$$

$$\angle MPA = \angle MPB \quad [\because \text{tangents from an external point are equally inclined to the line segment joining the centre to that point, i.e.,} \\ \angle OPA = \angle OPB]$$

$$\therefore \triangle MAP \cong \triangle MBP \quad [\text{by SAS-congruence}].$$

$$\text{And so, } MA = MB \quad [\text{cpct}]$$

$$\text{and } \angle AMP = \angle BMP \quad [\text{cpct}].$$

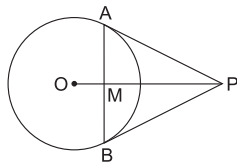
$$\text{But, } \angle AMP + \angle BMP = 180^\circ \quad [\text{linear pair}]$$

$$\therefore \angle AMP = \angle BMP = 90^\circ.$$

Hence, OP is the right bisector of AB .

EXAMPLE 10 Prove that the tangents at the extremities of any chord of a circle, make equal angles with the chord. [CBSE 2014, '17]

SOLUTION GIVEN AB is any chord of a circle with centre O . Tangents at the extremities A and B of this chord meet at an external point P . Chord AB intersects the line segment OP at M .



TO PROVE $\angle MAP = \angle MBP$.

PROOF In $\triangle MAP$ and $\triangle MBP$, we have

$$PA = PB \quad [\because \text{tangents from an external point are equal}]$$

$$MP = MP \quad [\text{common}]$$

$$\angle MPA = \angle MPB \quad [\because \angle OPA = \angle OPB \text{ since tangents from an external point are equally inclined to the line segment joining the point to the centre}]$$

$\therefore \triangle MAP \cong \triangle MBP$ [by SAS-congruence].

And so, $\angle MAP = \angle MBP$ [cpct].

EXAMPLE 11 Prove that the tangent drawn at the midpoint of an arc of a circle is parallel to the chord joining the end points of the arc. [CBSE 2015]

SOLUTION GIVEN Point P is the midpoint of arc $\widehat{QR}$ of a circle with centre O . ST is the tangent to the circle at point P .

TO PROVE Chord $QR \parallel ST$.

PROOF P is the midpoint of $\widehat{QR}$

$$\Rightarrow \widehat{QP} = \widehat{PR}$$

$$\Rightarrow \text{chord } QP = \text{chord } PR$$

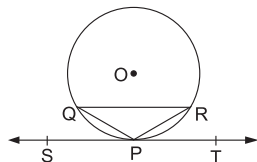
[$\because$ in a circle, if two arcs are equal, then their corresponding chords are equal]

$$\therefore \angle PQR = \angle PRQ$$

$$\Rightarrow \angle TPR = \angle PRQ$$

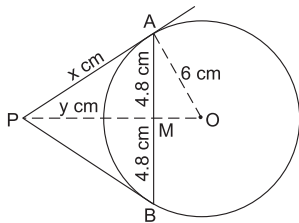
[Note $\angle PQR = \angle TPR$, angles in alternate segments]

$$\Rightarrow QR \parallel ST. \quad [\because \angle TPR \text{ and } \angle PRQ \text{ are alternate int. } \angle s]$$



EXAMPLE 12 In the given figure, AB is a chord of length 9.6 cm of a circle with centre O and radius 6 cm. The tangents at A and B intersect at P . Find the length of PA .

[CBSE 2009C, '13C, '15]



SOLUTION GIVEN A circle with centre O and radius 6 cm. AB is a chord of length 9.6 cm. The tangents at A and B intersect at P .

TO FIND The length PA .

CONSTRUCTION Join OP and OA . Let OP and AB intersect at M . Let $PA = x$ cm and $PM = y$ cm.

Now, $PA = PB$ [$\because$ tangents from an external point are equal] and OP is the bisector of $\angle APB$

[$\because$ two tangents to a circle from an external point are equally inclined to the line segment joining the centre to that point].

Also, $OP \perp AB$ and OP bisects AB at M

[$\because OP$ is the right bisector of AB].

$$\therefore AM = MB = \frac{9.6}{2} \text{ cm} = 4.8 \text{ cm}.$$

In right $\triangle AMO$, we have $OA = 6 \text{ cm}$ and $AM = 4.8 \text{ cm}$.

$$\therefore OM = \sqrt{OA^2 - AM^2} = \sqrt{6^2 - 4.8^2} = \sqrt{12.96} = 3.6 \text{ cm}.$$

In right $\triangle AMP$, we have

$$AP^2 = PM^2 + AM^2 \Rightarrow x^2 = y^2 + (4.8)^2$$

$$\Rightarrow x^2 = y^2 + 23.04. \quad \dots (i)$$

In right $\triangle PAO$, we have

$$OP^2 = PA^2 + OA^2 \quad [\text{Note } \angle PAO = 90^\circ, \text{ since } AO \text{ is the radius at the point of contact}]$$

$$\Rightarrow (y + 3.6)^2 = x^2 + 6^2 \quad [\because OP = PM + MO = (y + 3.6) \text{ cm}]$$

$$\Rightarrow y^2 + 7.2y + 12.96 = x^2 + 36 \Rightarrow 7.2y = 46.08 \quad [\text{using (i)}]$$

$$\Rightarrow y = 6.4 \text{ cm}.$$

Putting this value of y in (i), we get

$$x^2 = (6.4)^2 + 23.04 = 40.96 + 23.04 = 64 \Rightarrow x = \sqrt{64} = 8.$$

$$\therefore PA = 8 \text{ cm}.$$

EXAMPLE 13 Two tangents PA and PB are drawn to a circle with centre O from an external point P . Prove that $\angle APB = 2\angle OAB$. [CBSE 2009, '14]

SOLUTION GIVEN A circle with centre O and PA, PB are the tangents on it from a point P outside it.

TO PROVE $\angle APB = 2\angle OAB$.

PROOF Let $\angle APB = x^\circ$.

We know that the tangents to a circle from an external point are equal. So, $PA = PB$.

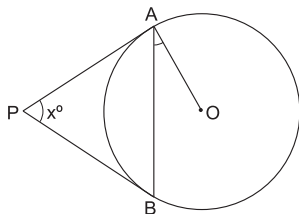
Since the angles opposite to the equal sides of a triangle are equal, so

$$PA = PB \Rightarrow \angle PBA = \angle PAB.$$

Also, the sum of the angles of a triangle is 180° .

$$\therefore \angle APB + \angle PAB + \angle PBA = 180^\circ$$

$$\Rightarrow x^\circ + 2\angle PAB = 180^\circ \quad [\because \angle PBA = \angle PAB]$$



$$\Rightarrow \angle PAB = \frac{1}{2}(180^\circ - x^\circ) = \left(90^\circ - \frac{1}{2}x^\circ\right).$$

But, PA is a tangent and OA is the radius of the given circle.

$$\therefore \angle OAB + \angle PAB = 90^\circ$$

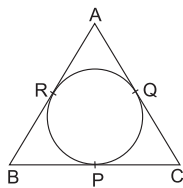
$$\Rightarrow \angle OAB = 90^\circ - \left(90^\circ - \frac{1}{2}x^\circ\right) \Rightarrow \angle OAB = \frac{1}{2}x^\circ = \frac{1}{2} \angle APB$$

$$\Rightarrow \angle APB = 2\angle OAB.$$

EXAMPLE 14 In the given figure, the incircle of $\triangle ABC$ touches the sides BC , CA and AB at P , Q and R respectively. Prove that

$$(AR + BP + CQ) = (AQ + BR + CP)$$

$$= \frac{1}{2} (\text{perimeter of } \triangle ABC).$$



SOLUTION We know that the lengths of tangents from an exterior point to a circle are equal.

$$\therefore AR = AQ, \quad \dots \text{(i) [tangents from A]}$$

$$BP = BR, \quad \dots \text{(ii) [tangents from B]}$$

$$CQ = CP. \quad \dots \text{(iii) [tangents from C]}$$

$$\therefore (AR + BP + CQ) = (AQ + BR + CP) = k \text{ (say).}$$

$$\text{Perimeter of } \triangle ABC = (AB + BC + CA)$$

$$= (AR + BR) + (BP + CP) + (CQ + AQ)$$

$$= (AR + BP + CQ) + (AQ + BR + CP)$$

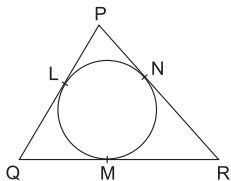
$$= (k + k) = 2k$$

$$\Rightarrow k = \frac{1}{2} (\text{perimeter of } \triangle ABC).$$

$$\therefore (AR + BP + CQ) = (AQ + BR + CP)$$

$$= \frac{1}{2} (\text{perimeter of } \triangle ABC).$$

EXAMPLE 15 In the given figure, a circle is inscribed in a triangle PQR . If $PQ = 10$ cm, $QR = 8$ cm and $PR = 12$ cm, find the lengths of QM , RN and PL . [CBSE 2012]



SOLUTION We know that the lengths of the tangents drawn from an external point to a circle are equal.

Let $PL = PN = x$; $QL = QM = y$; $RM = RN = z$.

$$\text{Now, } PL + QL = PQ \Rightarrow x + y = 10, \quad \dots \text{ (i)}$$

$$QM + RM = QR \Rightarrow y + z = 8, \quad \dots \text{ (ii)}$$

$$RN + PN = PR \Rightarrow z + x = 12. \quad \dots \text{ (iii)}$$

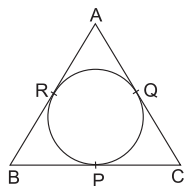
$$\text{Subtracting (ii) from (iii), we get } x - y = 4. \quad \dots \text{ (iv)}$$

Solving (i) and (iv), we get $x = 7$, $y = 3$.

Substituting $y = 3$ in (ii), we get $z = 5$.

$$\therefore QM = y = 3 \text{ cm, } RN = z = 5 \text{ cm, } PL = x = 7 \text{ cm.}$$

EXAMPLE 16 A circle is inscribed in a $\triangle ABC$, touching BC , CA and AB at P , Q and R respectively, as shown in the given figure. If $AB = 10$ cm, $AQ = 7$ cm and $CQ = 5$ cm then find the length of BC .
[CBSE 2009C]



SOLUTION We know that the lengths of tangents drawn from an external point to a circle are equal.

$$\therefore AR = AQ = 7 \text{ cm.}$$

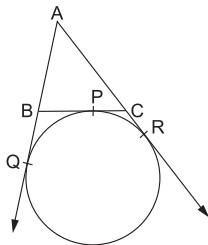
$$BR = (AB - AR) = (10 - 7) \text{ cm} = 3 \text{ cm.}$$

$$\therefore BP = BR = 3 \text{ cm,}$$

$$CP = CQ = 5 \text{ cm.}$$

$$\therefore BC = (BP + CP) = (3 + 5) \text{ cm} = 8 \text{ cm.}$$

EXAMPLE 17 A circle is touching the side BC of $\triangle ABC$ at P and touching AB and AC produced at Q and R respectively. Prove that $AQ = \frac{1}{2}$ (perimeter of $\triangle ABC$).
[CBSE 2001, '02, '06, '14, '17]



SOLUTION We know that the lengths of tangents drawn from an external point to a circle are equal.

$$\therefore AQ = AR, \quad \dots \text{ (i)} \quad [\text{tangents from } A]$$

$$BP = BQ, \quad \dots \text{ (ii) [tangents from B]}$$

$$CP = CR. \quad \dots \text{ (iii) [tangents from C]}$$

Perimeter of $\triangle ABC$

$$= AB + BC + AC$$

$$= AB + BP + CP + AC$$

$$= AB + BQ + CR + AC \quad [\text{using (ii) and (iii)}]$$

$$= AQ + AR$$

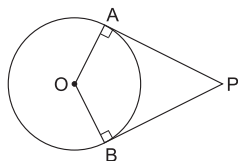
$$= 2AQ \quad [\text{using (i)}].$$

$$\therefore AQ = \frac{1}{2} (\text{perimeter of } \triangle ABC).$$

EXAMPLE 18 *PA and PB are tangents to the circle with centre O from an external point P, touching the circle at A and B respectively. Show that the quadrilateral AOBP is cyclic.* [CBSE 2014]

SOLUTION **GIVEN** PA and PB are tangents to the circle with centre O from an external point P.

TO PROVE Quadrilateral AOBP is cyclic.



PROOF We know that the tangent at any point of a circle is perpendicular to radius through the point of contact.

$$\therefore PA \perp OA, \text{ i.e., } \angle OAP = 90^\circ \quad \dots \text{ (i)}$$

$$\text{and } PB \perp OB, \text{ i.e., } \angle OBP = 90^\circ \quad \dots \text{ (ii)}$$

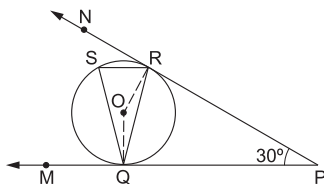
Now, the sum of all the angles of a quadrilateral is 360° .

$$\therefore \angle AOB + \angle OAP + \angle APB + \angle OBP = 360^\circ$$

$$\Rightarrow \angle AOB + \angle APB = 180^\circ \quad [\text{using (i) and (ii)}]$$

$\therefore$ quadrilateral OAPB is cyclic [since both pairs of opposite angles have the sum 180° .].

EXAMPLE 19 *In the given figure, tangents PQ and PR are drawn from an external point P to a circle with centre O, such that $\angle RPQ = 30^\circ$. A chord RS is drawn parallel to the tangent PQ. Find $\angle RQS$.*



SOLUTION Join OQ and OR . Also, produce PQ and PR to M and N respectively.

We know that the angle between two tangents from an external point is supplementary to the angle subtended by the radii at the points of contact.

$$\therefore \angle RPQ + \angle ROQ = 180^\circ$$

$$\Rightarrow \angle ROQ = 180^\circ - \angle RPQ = 180^\circ - 30^\circ = 150^\circ.$$

$$\text{Now, } \angle RSQ = \frac{1}{2} \angle ROQ = \frac{1}{2} \times 150^\circ = 75^\circ$$

[$\because$ the angle subtended by an arc at the centre is twice the angle subtended on the remaining part of the circle].

$$\therefore \angle SQM = \angle RSQ = 75^\circ \quad [\text{alternate int. } \angle; \text{ since } RS \parallel PQ]$$

$$\text{Also, } \angle PQR = \angle RSQ = 75^\circ \quad [\text{angles in alternate segments}].$$

$$\text{Now, } \angle SQM + \angle RQS + \angle PQR = 180^\circ \quad [\text{angles on a straight line}]$$

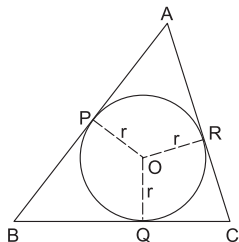
$$\therefore \angle RQS = 180^\circ - (\angle SQM + \angle PQR) = 180^\circ - (75^\circ + 75^\circ) = 30^\circ.$$

EXAMPLE 20 In the given figure, the sides AB , BC and CA of a triangle ABC touch a circle with centre O and radius r at P , Q and R respectively.

Prove that

$$(a) \quad AB + CQ = AC + BQ$$

$$(b) \quad \text{area}(\triangle ABC) = \frac{1}{2}(\text{perimeter of } \triangle ABC) \times r.$$



[CBSE 2013]

SOLUTION We know that the lengths of tangents from an exterior point to a circle are equal.

$$\therefore AP = AR, \quad \dots (i) \quad [\text{tangents from } A]$$

$$BP = BQ, \quad \dots (ii) \quad [\text{tangents from } B]$$

$$CQ = CR. \quad \dots (iii) \quad [\text{tangents from } C]$$

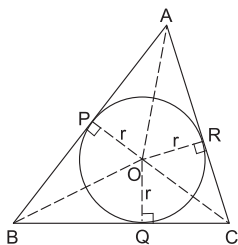
$$(a) \quad AB + CQ = AP + BP + CQ$$

$$= AR + BQ + CR \quad [\text{using (i), (ii) and (iii)}]$$

$$= (AR + CR) + BQ = AC + BQ.$$

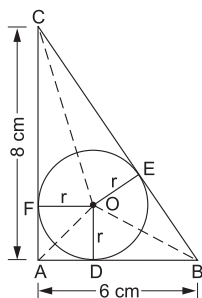
(b) Join OA , OB and OC .

$$\begin{aligned}
 \text{Area } (\triangle ABC) &= \text{area } (\triangle OAB) \\
 &\quad + \text{area } (\triangle OBC) \\
 &\quad + \text{area } (\triangle OCA) \\
 &= \left(\frac{1}{2} \times AB \times OP\right) \\
 &\quad + \left(\frac{1}{2} \times BC \times OQ\right) \\
 &\quad + \left(\frac{1}{2} \times CA \times OR\right) \\
 &= \left(\frac{1}{2} \times AB \times r\right) + \left(\frac{1}{2} \times BC \times r\right) + \left(\frac{1}{2} \times CA \times r\right) \\
 &= \frac{1}{2} (AB + BC + CA) \times r \\
 &= \frac{1}{2} (\text{perimeter of } \triangle ABC) \times r.
 \end{aligned}$$



EXAMPLE 21 In the given figure, ABC is a right-angled triangle with $AB = 6$ cm and $AC = 8$ cm. A circle with centre O has been inscribed inside the triangle. Calculate the value of r , the radius of the inscribed circle.

[CBSE 2006C, '13]



SOLUTION Join OA , OB and OC .

Draw $OD \perp AB$, $OE \perp BC$ and $OF \perp CA$.

Then, $OD = OE = OF = r$ cm.

$$\therefore \text{ar}(\triangle ABC) = \frac{1}{2} \times AB \times AC = \left(\frac{1}{2} \times 6 \times 8\right) \text{cm}^2 = 24 \text{cm}^2.$$

$$\text{Now, ar}(\triangle ABC) = \frac{1}{2} \times (\text{perimeter of } \triangle ABC) \times r$$

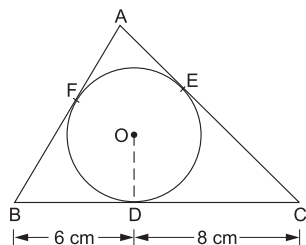
$$\Rightarrow 24 = \frac{1}{2} \times (AB + BC + CA) \times r$$

$$\Rightarrow 24 = \frac{1}{2} \times (6 + 10 + 8) \times r \Rightarrow r = 2$$

$$[\because BC^2 = AB^2 + AC^2 \Rightarrow BC = \sqrt{6^2 + 8^2} = 10].$$

Hence, the radius of the inscribed circle is 2 cm.

EXAMPLE 22 A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 6 cm and 8 cm respectively. Find the lengths of the sides AB and AC . [HOTS] [CBSE 2014]



SOLUTION We know that the lengths of tangents drawn from an exterior point to a circle are equal.

$$\therefore AE = AF = x \text{ cm (say)}, BD = BF = 6 \text{ cm}, CD = CE = 8 \text{ cm}.$$

$$\text{And so, } AB = AF + BF = (x + 6) \text{ cm}, BC = BD + CD = 14 \text{ cm},$$

$$AC = CE + AE = (x + 8) \text{ cm}.$$

$$\begin{aligned} \text{Perimeter, } 2s &= AB + BC + AC = [(x + 6) + 14 + (x + 8)] \text{ cm} \\ &= (2x + 28) \text{ cm} \end{aligned}$$

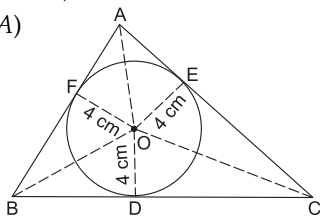
$$\Rightarrow s = (x + 14) \text{ cm}.$$

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \sqrt{s(s-AB)(s-BC)(s-AC)} \\ &= \sqrt{(x+14)\{(x+14)-(x+6)\}\{(x+14)-14\}\{(x+14)-(x+8)\}} \text{ cm}^2 \\ &= \sqrt{48x(x+14)} \text{ cm}^2. \end{aligned} \quad \dots (i)$$

Join OE and OF and also OA , OB and OC .

$$\begin{aligned} \therefore \text{ar}(\triangle ABC) &= \text{ar}(\triangle OAB) + \text{ar}(\triangle OBC) \\ &\quad + \text{ar}(\triangle OCA) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{1}{2} \times AB \times OF\right) \\ &\quad + \left(\frac{1}{2} \times BC \times OD\right) \\ &\quad + \left(\frac{1}{2} \times AC \times OE\right) \\ &= \left[\frac{1}{2} \times (x+6) \times 4\right] + \left[\frac{1}{2} \times 14 \times 4\right] + \left[\frac{1}{2} \times (x+8) \times 4\right] \\ &= 2[(x+6) + 14 + (x+8)] = 4(x+14) \text{ cm}^2. \end{aligned} \quad \dots (ii)$$



From (i) and (ii), we get

$$\sqrt{48x(x+14)} = 4(x+14)$$

$$\Rightarrow 48x(x+14) = 16(x+14)^2 \quad [\text{on squaring both sides}]$$

$$\Rightarrow 48x = 16(x+14) \Rightarrow x = \frac{16 \times 14}{32} = 7.$$

$$\therefore AB = (x+6) \text{ cm} = (7+6) \text{ cm} = 13 \text{ cm}$$

$$\text{and } AC = (x+8) \text{ cm} = (7+8) \text{ cm} = 15 \text{ cm}.$$

EXAMPLE 23 In the given figure, a triangle ABC is drawn to circumscribe a circle of radius 3 cm, such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 6 cm and 8 cm respectively. Find the side AB , if the area of $\triangle ABC$ is 63 cm^2 .

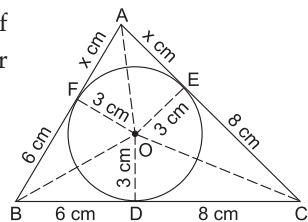
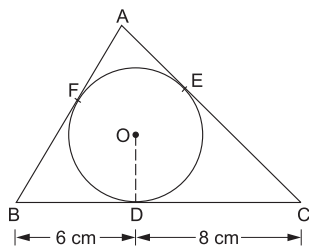
[CBSE 2010, '11, '15]

SOLUTION We know that the lengths of tangents drawn from an exterior point to a circle are equal.

$$\therefore AE = AF = x \text{ cm (say);}$$

$$BD = BF = 6 \text{ cm;}$$

$$CD = CE = 8 \text{ cm.}$$



And so, $AB = AF + BF = (x + 6) \text{ cm}$; $BC = BD + CD = 14 \text{ cm}$;

$$CA = CE + AE = (x + 8) \text{ cm.}$$

Join OE and OF and also OA , OB and OC .

$$\therefore \text{ar}(\triangle ABC) = \text{ar}(\triangle OAB) + \text{ar}(\triangle OBC) + \text{ar}(\triangle OCA)$$

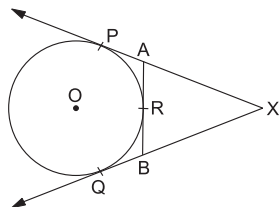
$$\Rightarrow 63 = \left(\frac{1}{2} \times AB \times OF\right) + \left(\frac{1}{2} \times BC \times OD\right) + \left(\frac{1}{2} \times CA \times OE\right)$$

$$\Rightarrow 63 = \left\{\frac{1}{2} \times (x + 6) \times 3\right\} + \left\{\frac{1}{2} \times 14 \times 3\right\} + \left\{\frac{1}{2} \times (x + 8) \times 3\right\}$$

$$\Rightarrow 63 = \frac{3}{2} \times (2x + 28) \Rightarrow x = 7.$$

$$\therefore AB = (x + 6) \text{ cm} = (7 + 6) \text{ cm} = 13 \text{ cm.}$$

EXAMPLE 24 In the given figure, XP and XQ are two tangents to the circle with centre O , drawn from an external point X . ARB is another tangent, touching the circle at R . Prove that $XA + AR = XB + BR$. [CBSE 2014]



SOLUTION We know that the lengths of tangents drawn from an exterior point to a circle are equal.

$$\therefore XP = XQ, \quad \dots \text{(i)} \quad [\text{tangents from } X]$$

$$AP = AR, \quad \dots \text{(ii)} \quad [\text{tangents from } A]$$

$$BR = BQ. \quad \dots \text{(iii)} \quad [\text{tangents from } B]$$

Now, $XP = XQ \Rightarrow XA + AP = XB + BQ$

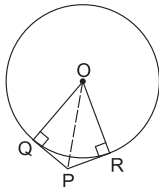
$\Rightarrow XA + AR = XB + BR$ [using (ii) and (iii)].

EXAMPLE 25 If from an external point P of a circle with centre O , two tangents PQ and PR are drawn such that $\angle QPR = 120^\circ$, prove that $2PQ = PO$.

[HOTS] [CBSE 2014]

SOLUTION In $\triangle OPQ$, we have

$\angle PQO = 90^\circ$ [$\because$ the tangent at any point is perpendicular to the radius through the point of contact]



and $\angle QPO = \frac{1}{2} \times \angle QPR = \frac{1}{2} \times 120^\circ = 60^\circ$.

[$\because$ the two tangents drawn from an external point are equally inclined to the line segment joining the centre to that point and so $\angle QPO = \angle RPO$]

In right $\triangle OPQ$, we have

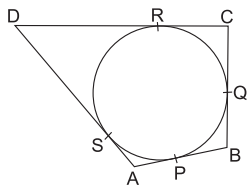
$$\cos(\angle QPO) = \frac{PQ}{PO}$$

$$\Rightarrow \cos 60^\circ = \frac{PQ}{PO} \Rightarrow \frac{1}{2} = \frac{PQ}{PO} \Rightarrow 2PQ = PO.$$

EXAMPLE 26 A quadrilateral $ABCD$ is drawn to circumscribe a circle, as shown in the figure. Prove that $AB + CD = AD + BC$.

[CBSE 2008, '08C, '09, '12, '13, '14, '17]

SOLUTION We know that the lengths of tangents drawn from an exterior point to a circle are equal.



$$\therefore AP = AS, \quad \dots \text{(i)} \quad [\text{tangents from } A]$$

$$BP = BQ, \quad \dots \text{(ii)} \quad [\text{tangents from } B]$$

$$CR = CQ, \quad \dots \text{(iii)} \quad [\text{tangents from } C]$$

$$DR = DS, \quad \dots \text{(iv)} \quad [\text{tangents from } D]$$

$$\therefore AB + CD = (AP + BP) + (CR + DR)$$

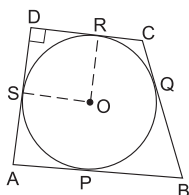
$$= (AS + BQ) + (CQ + DS) \quad [\text{using (i), (ii), (iii), (iv)}]$$

$$= (AS + DS) + (BQ + CQ)$$

$$= AD + BC.$$

Hence, $AB + CD = AD + BC$.

EXAMPLE 27 In the given figure, $ABCD$ is a quadrilateral such that $\angle D = 90^\circ$. A circle with centre O and radius r , touches the sides AB , BC , CD and DA at P , Q , R and S respectively. If $BC = 40$ cm, $CD = 25$ cm and $BP = 28$ cm, find r .



SOLUTION It is given that $\angle D = 90^\circ$.

Also, $\angle ORD = \angle OSD = 90^\circ$. [$\because$ tangent at a point is perpendicular to the radius through the point of contact]

$\therefore \angle ROS = 180^\circ - \angle D = 90^\circ$. [$\because$ angle between the tangents from an external point is supplementary to the angle subtended by the line segments joining the points of contact to the centre]

And, $OR = OS = r$.

$\therefore ROSD$ is a square and so $OR = DR$, i.e., $r = DR$ (i)

We know that the lengths of tangents drawn from an exterior point to a circle are equal.

$\therefore BP = BQ$ and $CQ = CR$.

Now, $CQ = BC - BQ = BC - BP = (40 - 28)$ cm = 12 cm.

And so, $DR = CD - CR = CD - CQ = (25 - 12)$ cm = 13 cm.

$\therefore r = DR = 13$ cm [using (i)].

EXAMPLE 28 Prove that the parallelogram circumscribing a circle is a rhombus.

[CBSE 2008, '09, '10, '12, '13, '14]

SOLUTION GIVEN A parallelogram $ABCD$ circumscribes a circle with centre O .

TO PROVE $AB = BC = CD = AD$.

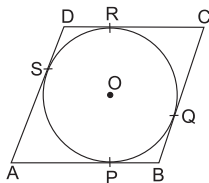
PROOF We know that the lengths of tangents drawn from an exterior point to a circle are equal.

$\therefore AP = AS$, ... (i) [tangents from A]

$BP = BQ$, ... (ii) [tangents from B]

$CR = CQ$, ... (iii) [tangents from C]

$DR = DS$ (iv) [tangents from D]



$$\begin{aligned}
 \therefore AB + CD &= AP + BP + CR + DR \\
 &= AS + BQ + CQ + DS \quad [\text{from (i), (ii), (iii) and (iv)}] \\
 &= (AS + DS) + (BQ + CQ) \\
 &= AD + BC.
 \end{aligned}$$

Thus, $AB + CD = AD + BC$

$$\Rightarrow 2AB = 2AD \quad [\because \text{opposite sides of a ||gm are equal}]$$

$$\Rightarrow AB = AD.$$

$$\therefore CD = AB = AD = BC.$$

Hence, $ABCD$ is a rhombus.

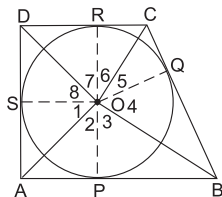
EXAMPLE 29 Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

[CBSE 2012, '13C, '14, '17]

SOLUTION **GIVEN** A quad. $ABCD$ circumscribes a circle with centre O .

TO PROVE $\angle AOB + \angle COD = 180^\circ$,
and $\angle AOD + \angle BOC = 180^\circ$.

CONSTRUCTION Join OP, OQ, OR and OS .



PROOF We know that the tangents drawn from an external point of a circle subtend equal angles at the centre.

$$\therefore \angle 1 = \angle 2, \angle 3 = \angle 4, \angle 5 = \angle 6 \text{ and } \angle 7 = \angle 8.$$

$$\text{And, } \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 = 360^\circ$$

[$\angle$ at a point]

$$\Rightarrow 2(\angle 2 + \angle 3) + 2(\angle 6 + \angle 7) = 360^\circ, \text{ and}$$

$$2(\angle 1 + \angle 8) + 2(\angle 4 + \angle 5) = 360^\circ$$

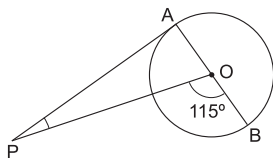
$$\Rightarrow \angle 2 + \angle 3 + \angle 6 + \angle 7 = 180^\circ \text{ and } \angle 1 + \angle 8 + \angle 4 + \angle 5 = 180^\circ$$

$$\Rightarrow \angle AOB + \angle COD = 180^\circ \text{ and } \angle AOD + \angle BOC = 180^\circ.$$

EXAMPLE 30 In the given figure, PA is a tangent from an external point P to a circle with centre O . If $\angle POB = 115^\circ$, find $\angle APO$.

[CBSE 2009C]

SOLUTION We know that the tangent at a point to a circle is perpendicular to the radius passing through the point of contact.



$$\therefore \angle OAP = 90^\circ.$$

$$\text{Now, } \angle AOP + \angle BOP = 180^\circ$$

$$\Rightarrow \angle AOP = 180^\circ - \angle BOP = 180^\circ - 115^\circ = 65^\circ.$$

$$\text{Now, } \angle OAP + \angle AOP + \angle APO = 180^\circ$$

[sum of angles of a triangle is 180°]

$$\Rightarrow \angle APO = 180^\circ - (\angle OAP + \angle AOP) = 180^\circ - (90^\circ + 65^\circ) = 25^\circ.$$

EXAMPLE 31 From a point P , two tangents PA and PB are drawn to a circle $C(O, r)$. If $OP = 2r$, show that $\triangle APB$ is equilateral.

[CBSE 2008, '11, '12]

SOLUTION Let OP meet the circle at Q .

Join OA and AQ .

Clearly, $OA \perp AP \Rightarrow \angle OAP = 90^\circ$ [radius through the point of contact is perpendicular to the tangent].

$$\text{Now, } OQ = QP = r.$$

Thus, Q is the midpoint of the hypotenuse OP of $\triangle OAP$.

So, Q is equidistant from O, A and P .

$$\therefore QA = OQ = QP = r$$

$$\Rightarrow OA = OQ = QA = r$$

$$\Rightarrow \triangle AOQ \text{ is equilateral}$$

$$\Rightarrow \angle AOQ = 60^\circ \text{ [}\because \text{each angle of an equilateral triangle is } 60^\circ\text{]}$$

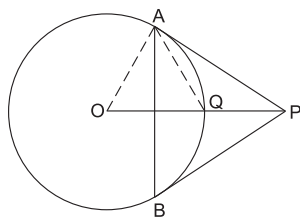
$$\Rightarrow \angle AOP = 60^\circ$$

$$\Rightarrow \angle APO = 30^\circ \text{ [}\because \angle AOP + \angle OAP + \angle APO = 180^\circ\text{]}$$

$$\Rightarrow \angle APB = 2\angle APO = 60^\circ.$$

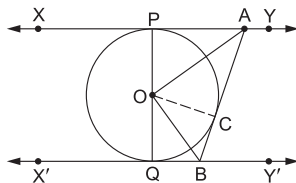
$$\text{Also, } PA = PB \Rightarrow \angle PAB = \angle PBA = 60^\circ.$$

Hence, $\triangle PAB$ is an equilateral triangle.



EXAMPLE 32 In the given figure, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersects XY at A and $X'Y'$ at B . Prove that $\angle AOB = 90^\circ$.

[HOTS] [CBSE 2012, '13, '14, '17]



SOLUTION In quad. $APQB$, we have

$$\angle APO = 90^\circ \text{ and } \angle BQO = 90^\circ$$

[$\because$ tangent at any point is perpendicular to the radius through the point of contact].

$$\text{Now, } \angle APO + \angle BQO + \angle QBC + \angle PAC = 360^\circ$$

$$\Rightarrow \angle PAC + \angle QBC = 360^\circ - (\angle APO + \angle BQO) = 180^\circ. \quad \dots (i)$$

We have

$$\angle CAO = \frac{1}{2} \angle PAC \text{ and } \angle CBO = \frac{1}{2} \angle QBC$$

[$\because$ tangents from an external point are equally inclined to the line segment joining the centre to that point].

$$\therefore \angle CAO + \angle CBO = \frac{1}{2}(\angle PAC + \angle QBC) = \frac{1}{2} \times 180^\circ = 90^\circ \dots (ii)$$

In $\triangle AOB$, we have

$$\angle CAO + \angle AOB + \angle CBO = 180^\circ$$

$$\Rightarrow \angle AOB = 180^\circ - (\angle CAO + \angle CBO) = 90^\circ \quad [\text{using (ii)}].$$

EXAMPLE 33 The incircle of an isosceles triangle ABC , with $AB = AC$, touches the sides AB , BC , CA at D , E and F respectively. Prove that E bisects BC .
[CBSE 2008, '12, '13C, '14]

SOLUTION We know that the tangents drawn from an external point to a circle are equal.

$$\therefore AD = AF, \dots (i) \quad [\text{tangents from } A]$$

$$BD = BE, \dots (ii) \quad [\text{tangents from } B]$$

$$CE = CF. \dots (iii) \quad [\text{tangents from } C]$$

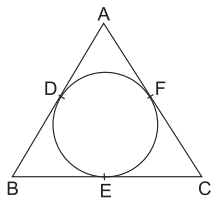
$$\text{Now, } AB = AC \quad [\text{given}]$$

$$\Rightarrow AD + BD = AF + CF$$

$$\Rightarrow BD = CF$$

$$\Rightarrow BE = CE \quad [\text{using (ii) and (iii)}]$$

$$\Rightarrow E \text{ bisects } BC.$$



EXERCISE 8A

1. A point P is at a distance of 29 cm from the centre of a circle of radius 20 cm. Find the length of the tangent drawn from P to the circle.

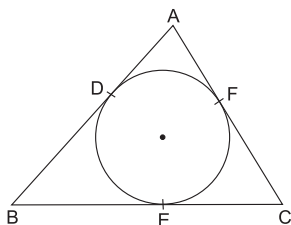
[CBSE 2017]

- A point P is 25 cm away from the centre of a circle and the length of tangent drawn from P to the circle is 24 cm. Find the radius of the circle.
- Two concentric circles are of radii 6.5 cm and 2.5 cm. Find the length of the chord of the larger circle which touches the smaller circle.

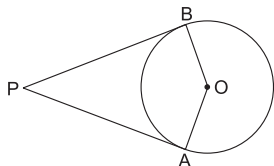
[CBSE 2011]

- In the given figure, a circle inscribed in a triangle ABC , touches the sides AB , BC and AC at points D , E and F respectively. If $AB = 12$ cm, $BC = 8$ cm and $AC = 10$ cm, find the lengths of AD , BE and CF .

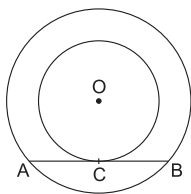
[CBSE 2013]



- In the given figure, PA and PB are the tangent segments to a circle with centre O . Show that the points A , O , B and P are concyclic.

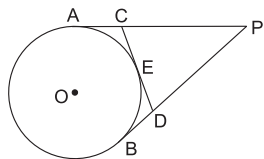


- In the given figure, the chord AB of the larger of the two concentric circles, with centre O , touches the smaller circle at C . Prove that $AC = CB$.



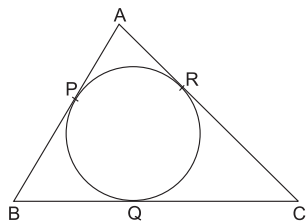
- From an external point P , tangents PA and PB are drawn to a circle with centre O . If CD is the tangent to the circle at a point E and $PA = 14$ cm, find the perimeter of $\triangle PCD$.

[CBSE 2002]

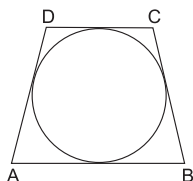


- A circle is inscribed in a $\triangle ABC$ touching AB , BC and AC at P , Q and R respectively. If $AB = 10$ cm, $AR = 7$ cm and $CR = 5$ cm, find the length of BC .

[CBSE 2009C]

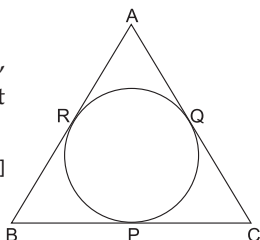


9. In the given figure, a circle touches all the four sides of a quadrilateral $ABCD$ whose three sides are $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm. Find AD .

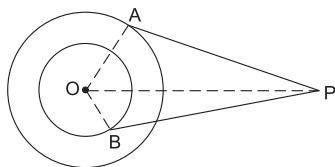


10. In the given figure, an isosceles triangle ABC , with $AB = AC$, circumscribes a circle. Prove that the point of contact P bisects the base BC .

[CBSE 2012, '17]

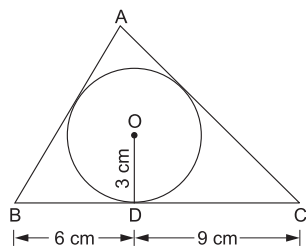


11. In the given figure, O is the centre of two concentric circles of radii 4 cm and 6 cm respectively. PA and PB are tangents to the outer and inner circle respectively. If $PA = 10$ cm, find the length of PB up to one place of decimal.



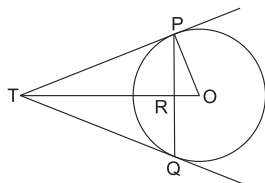
12. In the given figure, a triangle ABC is drawn to circumscribe a circle of radius 3 cm such that the segments BD and DC into which BC is divided by the point of contact D , are of lengths 6 cm and 9 cm respectively. If the area of $\triangle ABC = 54$ cm² then find the lengths of sides AB and AC .

[CBSE 2011, '15]



13. PQ is a chord of length 4.8 cm of a circle of radius 3 cm. The tangents at P and Q intersect at a point T as shown in the figure. Find the length of TP .

[CBSE 2013C]

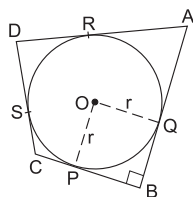


14. Prove that the line segment joining the points of contact of two parallel tangents of a circle, passes through its centre.

[CBSE 2014]

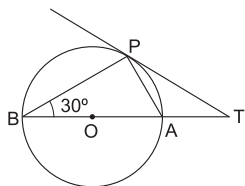
15. In the given figure, a circle with centre O , is inscribed in a quadrilateral $ABCD$ such that it touches the side BC , AB , AD and CD at points P , Q , R and S respectively. If $AB = 29$ cm, $AD = 23$ cm, $\angle B = 90^\circ$ and $DS = 5$ cm then find the radius of the circle.

[CBSE 2008, '13]



16. In the given figure, O is the centre of the circle and TP is the tangent to the circle from an external point T . If $\angle PBT = 30^\circ$, prove that $BA : AT = 2 : 1$.

[HOTS] [CBSE 2015]



ANSWERS (EXERCISE 8A)

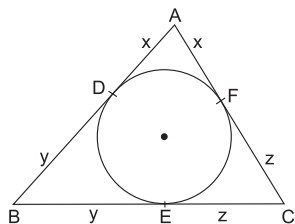
1. 21 cm 2. 7 cm 3. 12 cm
 4. $AD = 7$ cm, $BE = 5$ cm, $CF = 3$ cm 7. 28 cm 8. $BC = 8$ cm
 9. $AD = 3$ cm 11. 10.9 cm 12. $AB = 10$ cm, $AC = 12$ cm
 13. $TP = 4$ cm 15. 11 cm

HINTS TO SOME SELECTED QUESTIONS

4. $x + y = 12$, $y + z = 8$, $z + x = 10$.

Solving, we get $x = 7$, $y = 5$, $z = 3$.

$\therefore AD = x = 7$ cm, $BE = y = 5$ cm, $CF = z = 3$ cm.

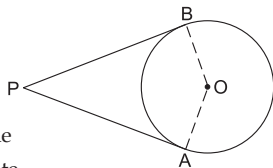


5. $OA \perp AP$ and $OB \perp BP$

$$\Rightarrow \angle OAP = 90^\circ \text{ and } \angle OBP = 90^\circ$$

$$\Rightarrow \angle OAP + \angle OBP = 180^\circ$$

$\Rightarrow$ quad. $AOBP$ is cyclic [$\because$ a quad. is cyclic if the sum of a pair of opposite angles is 180°].



7. $PA = PB, CA = CE, DB = DE$

[$\because$ tangents from an external point are equal].

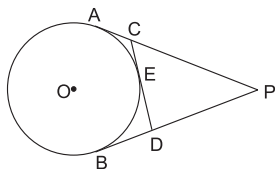
Perimeter of $\triangle PCD = PC + CD + PD$

$$= PC + CE + DE + PD$$

$$= PC + CA + DB + PD$$

$$= PA + PB = 2PA \quad [\because PB = PA]$$

$$= (2 \times 14) \text{ cm} = 28 \text{ cm}.$$



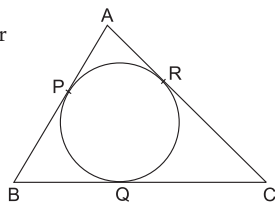
8. We know that the tangents to a circle from an exterior point are equal.

$$\therefore AP = AR = 7 \text{ cm};$$

$$BQ = BP = AB - AP = (10 - 7) \text{ cm} = 3 \text{ cm};$$

$$CQ = CR = 5 \text{ cm}.$$

$$\text{And so, } BC = BQ + CQ = (3 + 5) \text{ cm} = 8 \text{ cm}.$$



9. When a quadrilateral $ABCD$ is drawn to circumscribe a circle then $AB + CD = AD + BC$.

$$\therefore AD = AB + CD - BC = (6 + 4 - 7) \text{ cm} = 3 \text{ cm}.$$

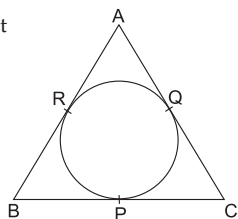
10. We know that the tangents to a circle from an external point are equal.

$$\therefore AQ = AR, BP = BR, CP = CQ.$$

$$\text{Now, } AB = AC \Rightarrow AR + BR = AQ + CQ$$

$$\Rightarrow AR + BP = AR + CP \Rightarrow BP = CP$$

$$\Rightarrow P \text{ bisects the base } BC.$$



13. R is the midpoint of PQ , i.e., $PR = QR = \left(\frac{1}{2} \times 4.8\right) \text{ cm} = 2.4 \text{ cm}.$

Also, $TO \perp PQ$.

In right $\triangle PRO$, we have

$$\begin{aligned} PO^2 &= PR^2 + RO^2 \Rightarrow RO = \sqrt{PO^2 - PR^2} \\ &= \sqrt{3^2 - 2.4^2} = \sqrt{3.24} = 1.8. \end{aligned}$$

Let $TR = x$ and $TP = y$.

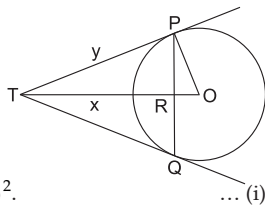
$$\text{In right } \triangle PTR, \text{ we have } TP^2 = TR^2 + PR^2 \Rightarrow y^2 = x^2 + (2.4)^2.$$

In right $\triangle OTP$, we have

$$\begin{aligned} TO^2 &= TP^2 + PO^2 \Rightarrow (TR + RO)^2 = TP^2 + PO^2 \\ \Rightarrow (x + 1.8)^2 &= y^2 + 3^2 \Rightarrow x^2 + 3.6x + 3.24 = y^2 + 9. \end{aligned} \quad \dots \text{ (ii)}$$

Solving (i) and (ii), we get $x = 3.2, y = 4$.

$$\therefore TP = 4 \text{ cm}.$$

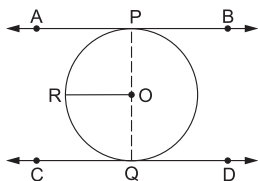


14. Let APB and CQD be two parallel tangents to a circle with centre O .

Join OP and OQ . Draw $RO \parallel AB$.

$$\text{Now, } \angle APO + \angle ROP = 180^\circ \quad [\text{co-interior angles}]$$

$$\Rightarrow \angle ROP = 90^\circ \quad [\because \angle APO = 90^\circ].$$



Similarly, $\angle CQO + \angle ROQ = 180^\circ$ [co-interior angles]

$$\Rightarrow \angle ROQ = 90^\circ \quad [\because \angle CQO = 90^\circ].$$

$$\therefore \angle POQ = \angle ROP + \angle ROQ = 180^\circ.$$

Hence, POQ is a straight line passing through O .

15. $POQB$ is a square, since $\angle PBQ = \angle BQO = \angle BPO = 90^\circ$, $OP = OQ = r$.

$$\therefore r = OQ = QB.$$

We know that tangents to a circle from an exterior point are equal.

$$\therefore DS = DR, AR = AQ.$$

$$\text{Now, } AR = AD - DR = AD - DS = (23 - 5) \text{ cm} = 18 \text{ cm}$$

$$\text{and } r = QB = AB - AQ = AB - AR = (29 - 18) \text{ cm} = 11 \text{ cm}.$$

16. $\angle APB = 90^\circ$ [angle in a semicircle]

$$\therefore \angle PAB = 90^\circ - \angle PBA = 90^\circ - 30^\circ = 60^\circ \quad [\because \angle PBA = \angle PBT = 30^\circ].$$

$$\angle PAT + \angle PAB = 180^\circ \quad [\text{linear pair}]$$

$$\Rightarrow \angle PAT = 180^\circ - \angle PAB = 180^\circ - 60^\circ = 120^\circ.$$

$$\angle APT = \angle PBA = 30^\circ \quad [\text{angles in alternate segments}].$$

In $\triangle PAT$, we have

$$\angle APT + \angle PAT + \angle PTA = 180^\circ$$

$$\Rightarrow \angle PTA = 180^\circ - (\angle APT + \angle PAT) = 180^\circ - (30^\circ + 120^\circ) = 30^\circ.$$

$$\text{Now, } \angle APT = \angle PTA = 30^\circ \Rightarrow AT = AP$$

... (i)

[sides opposite equal angles are equal]

In right $\triangle APB$, we have

$$\cos(\angle PAB) = \frac{AP}{BA} \Rightarrow \cos 60^\circ = \frac{AP}{BA} \Rightarrow BA = 2AP.$$

... (ii)

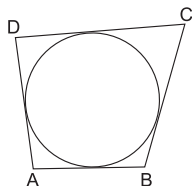
From (i) and (ii), we get $BA : AT = 2AP : AP = 2 : 1$.

EXERCISE 8B

Very-Short-Answer Questions

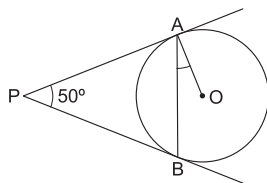
1. In the adjoining figure, a circle touches all the four sides of a quadrilateral $ABCD$ whose sides are $AB = 6$ cm, $BC = 9$ cm and $CD = 8$ cm. Find the length of side AD .

[CBSE 2011]



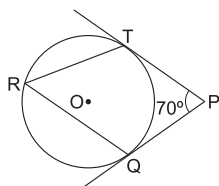
2. In the given figure, PA and PB are two tangents to the circle with centre O . If $\angle APB = 50^\circ$ then what is the measure of $\angle OAB$.

[CBSE 2015]



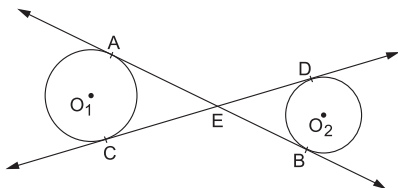
3. In the given figure, O is the centre of a circle. PT and PQ are tangents to the circle from an external point P . If $\angle TPQ = 70^\circ$, find $\angle TRQ$.

[CBSE 2015]

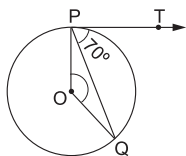


4. In the given figure, common tangents AB and CD to the two circles with centres O_1 and O_2 intersect at E . Prove that $AB = CD$.

[CBSE 2014]



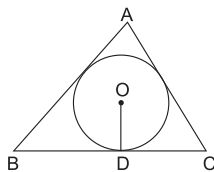
5. If PT is a tangent to a circle with centre O and PQ is a chord of the circle such that $\angle QPT = 70^\circ$, then find the measure of $\angle POQ$.



Short-Answer Questions

6. In the given figure, a triangle ABC is drawn to circumscribe a circle of radius 2 cm such that the segments BD and DC into which BC is divided by the point of contact D , are of lengths 4 cm and 3 cm respectively. If the area of $\triangle ABC = 21 \text{ cm}^2$ then find the lengths of sides AB and AC .

[CBSE 2011]



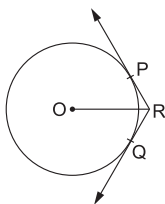
7. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle (in cm) which touches the smaller circle.

[CBSE 2012, '14]

8. Prove that the perpendicular at the point of contact of the tangent to a circle passes through the centre.

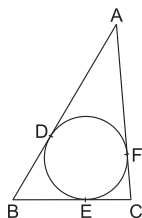
9. In the given figure, two tangents RQ and RP are drawn from an external point R to the circle with centre O . If $\angle PRQ = 120^\circ$, then prove that $OR = PR + RQ$.

[CBSE 2015]

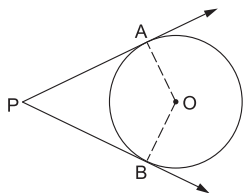


10. In the given figure, a circle inscribed in a triangle ABC touches the sides AB , BC and CA at points D , E and F respectively. If $AB = 14$ cm, $BC = 8$ cm and $CA = 12$ cm. Find the lengths AD , BE and CF .

[CBSE 2013]

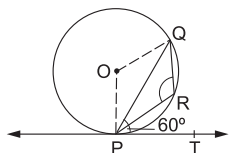


11. In the given figure, O is the centre of the circle. PA and PB are tangents. Show that $AOBP$ is a cyclic quadrilateral. [CBSE 2014]

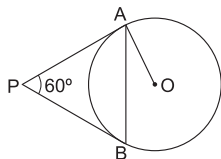


12. In two concentric circles, a chord of length 8 cm of the larger circle touches the smaller circle. If the radius of the larger circle is 5 cm then find the radius of the smaller circle. [CBSE 2013C]

13. In the given figure, PQ is a chord of a circle with centre O and PT is a tangent. If $\angle QPT = 60^\circ$, find $\angle PRQ$. [HOTS] [CBSE 2015]



14. In the given figure, PA and PB are two tangents to the circle with centre O . If $\angle APB = 60^\circ$ then find the measure of $\angle OAB$.



15. If the angle between two tangents drawn from an external point P to a circle of radius a and centre O , is 60° then find the length of OP .

[CBSE 2017]

ANSWERS (EXERCISE 8B)

- | | | | |
|---|-----------------|-----------------|----------------|
| 1. $AD = 5$ cm | 2. 25° | 3. 55° | 5. 140° |
| 6. $AB = 7.5$ cm, $AC = 6.5$ cm | 7. 8 cm | | |
| 10. $AD = 9$ cm, $BE = 5$ cm, $CF = 3$ cm | 12. 3 cm | 13. 120° | |
| 14. 30° | 15. $a\sqrt{3}$ | | |

HINTS TO SOME SELECTED QUESTIONS

1. When a quadrilateral $ABCD$ circumscribes a circle then $AB + CD = AD + BC$.
 $\therefore AD = AB + CD - BC = (6 + 8 - 9)$ cm = 5 cm.

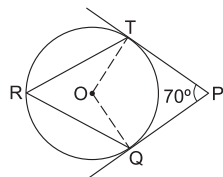
2. $\angle APB = 2\angle OAB \Rightarrow \angle OAB = \frac{1}{2} \times \angle APB = \frac{50^\circ}{2} = 25^\circ$. [See Solved Example 13.]

3. Join OT and OQ .

Then, $\angle TOQ + \angle TPQ = 180^\circ \Rightarrow \angle TOQ = 180^\circ - 70^\circ = 110^\circ$.

[See Theorem 8.]

Now, $\angle TRQ = \frac{1}{2} \times \angle TOQ = \frac{1}{2} \times 110^\circ = 55^\circ$



[$\because$ the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle].

4. We know that the tangents to a circle from an external point are equal.

$\therefore AE = CE, BE = DE$

$\Rightarrow AE + BE = CE + DE \Rightarrow AB = CD$.

5. Mark a point R in the alternate segment.

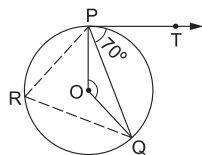
Join RP and RQ .

Then, $\angle PRQ = \angle QPT = 70^\circ$

[angles in the alternate segments].

$\therefore \angle POQ = 2\angle PRQ = 2 \times 70^\circ = 140^\circ$.

[$\because$ angle subtended by an arc on the centre is double the angle subtended at any point on the remaining part of the circle.]



6. Mark the points of contact E and F of tangents AC and AB respectively.

We know that the tangents to a circle from an external point are equal.

$\therefore AE = AF = x$ cm (say);

$BD = BF = 4$ cm; $CD = CE = 3$ cm.

Now, $ar(\triangle ABC) = \frac{1}{2} \times (\text{perimeter of } \triangle ABC) \times r$

$\Rightarrow 21 = \frac{1}{2} \times \{(x+4) + (4+3) + (3+x)\} \times 2 \Rightarrow x = 3.5$.

$\therefore AB = (x+4)$ cm = 7.5 cm; $AC = (x+3)$ cm = 6.5 cm.

7. Let AB be the chord, P be the point where it touches the smaller circle and O be the common centre.

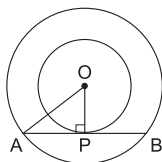
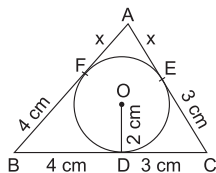
AB is tangent to smaller circle and so, $OP \perp AB$.

$\therefore P$ is the midpoint of AB , i.e., $AP = PB$

[$\because$ a perpendicular from the centre on any chord bisects the chord].

Now, $OP = 3$ cm, $OA = 5$ cm $\Rightarrow AP = \sqrt{OA^2 - OP^2} = \sqrt{5^2 - 3^2} = 4$.

$\therefore AB = 2AP = 8$ cm.

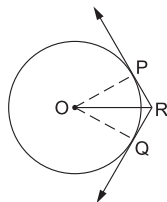


9. Join OP and OQ .

$$\angle PRQ = 120^\circ \Rightarrow \angle PRO = \angle QRO = \frac{1}{2} \times \angle PRQ = 60^\circ.$$

[$\because$ two tangents from an external point are equally inclined to the line segment joining the centre to that point.]

$$\text{And so, } \angle PRO = \angle QRO = \frac{1}{2} \angle PRQ.]$$



In right $\triangle OPR$, we have

$$\cos \angle PRO = \frac{PR}{OR} \Rightarrow \cos 60^\circ = \frac{PR}{OR} \Rightarrow \frac{1}{2} = \frac{PR}{OR} \Rightarrow PR = \frac{1}{2} \times OR.$$

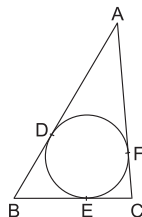
$$\text{Similarly, } RQ = \frac{1}{2} \times OR.$$

$$\therefore PR + RQ = OR.$$

10. $x + y = 14$, $y + z = 8$, $z + x = 12$.

Solving, $x = 9$, $y = 5$, $z = 3$.

$$\therefore AD = x = 9 \text{ cm}, BE = y = 5 \text{ cm}, CF = z = 3 \text{ cm}.$$



12. Let AB be the chord, P be the point where it touches the smaller circle and O be the common centre.

Then, AB is the tangent to smaller circle at P .

$$\therefore OP \perp AB.$$

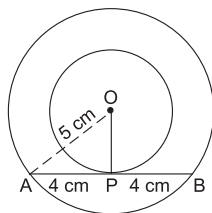
And so, P is the midpoint of AB ,

$$\text{i.e., } AP = PB = \frac{1}{2} AB = \left(\frac{1}{2} \times 8\right) \text{ cm} = 4 \text{ cm}$$

[$\because$ perpendicular from the centre on any chord bisects the chord].

$$\text{Now, } OA = 5 \text{ cm}, AP = 4 \text{ cm} \Rightarrow OP = \sqrt{OA^2 - AP^2} = \sqrt{5^2 - 4^2} = 3.$$

$$\therefore \text{radius of smaller circle} = OP = 3 \text{ cm}.$$



13. Mark a point M in the alternate segment.

Join MP and MQ .

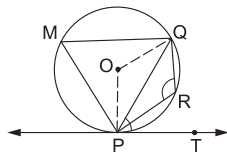
$$\text{Then, } \angle PMQ = \angle QPT = 60^\circ$$

[angles in the alternate segments].

$$\text{Now, } \angle PMQ + \angle PRQ = 180^\circ$$

[$\because PMQR$ is a cyclic quadrilateral]

$$\Rightarrow \angle PRQ = 180^\circ - \angle PMQ = 180^\circ - 60^\circ = 120^\circ.$$



14. $PA = PB$ [$\because$ tangents from an external point are equal]

$$\Rightarrow \angle PAB = \angle PBA = \frac{1}{2} \times (180^\circ - \angle APB) = \frac{1}{2} \times (180^\circ - 60^\circ) = 60^\circ.$$

Now, $\angle PAO = 90^\circ$ [$\because$ radius through the point of contact is perpendicular to the tangent]

$$\Rightarrow \angle OAB + \angle PAB = 90^\circ \Rightarrow \angle OAB = 90^\circ - \angle PAB = 90^\circ - 60^\circ = 30^\circ.$$

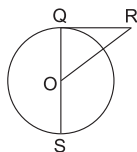
MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

1. The number of tangents that can be drawn from an external point to a circle is [CBSE 2011, '12]

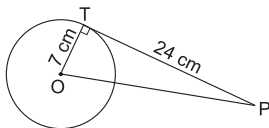
- (a) 1 (b) 2 (c) 3 (d) 4

2. In the given figure, RQ is a tangent to the circle with centre O . If $SQ = 6$ cm and $QR = 4$ cm, then OR is equal to [CBSE 2014]



- (a) 2.5 cm (b) 3 cm
(c) 5 cm (d) 8 cm

3. In a circle of radius 7 cm, tangent PT is drawn from a point P such that $PT = 24$ cm. If O is the centre of the circle, then length $OP = ?$



- (a) 30 cm (b) 28 cm (c) 25 cm (d) 18 cm

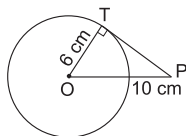
4. Which of the following pairs of lines in a circle cannot be parallel? [CBSE 2011]

- (a) Two chords (b) A chord and a tangent
(c) Two tangents (d) Two diameters

5. The chord of a circle of radius 10 cm subtends a right angle at its centre. The length of the chord (in cm) is [CBSE 2014]

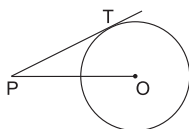
- (a) $\frac{5}{\sqrt{2}}$ (b) $5\sqrt{2}$ (c) $10\sqrt{2}$ (d) $10\sqrt{3}$

6. In the given figure, PT is a tangent to the circle with centre O . If $OT = 6$ cm and $OP = 10$ cm, then the length of tangent PT is



- (a) 8 cm (b) 10 cm
(c) 12 cm (d) 16 cm

7. In the given figure, point P is 26 cm away from the centre O of a circle and the length PT of the tangent drawn from P to the circle is 24 cm. Then, the radius of the circle is [CBSE 2011, '12]

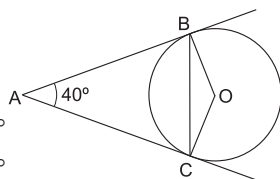


- (a) 10 cm (b) 12 cm
(c) 13 cm (d) 15 cm

8. PQ is a tangent to a circle with centre O at the point P . If $\triangle OPQ$ is an isosceles triangle, then $\angle OQP$ is equal to [CBSE 2014]

- (a) 30° (b) 45° (c) 60° (d) 90°

9. In the given figure, AB and AC are tangents to the circle with centre O such that $\angle BAC = 40^\circ$. Then, $\angle BOC$ is equal to [CBSE 2011, '14]

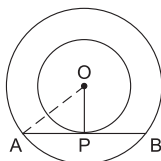


- (a) 80° (b) 100°
(c) 120° (d) 140°

10. If a chord AB subtends an angle of 60° at the centre of a circle, then the angle between the tangents to the circle drawn from A and B is [CBSE 2013C]

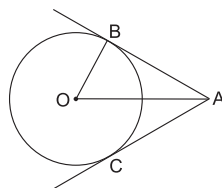
- (a) 30° (b) 60° (c) 90° (d) 120°

11. In the given figure, O is the centre of two concentric circles of radii 6 cm and 10 cm. AB is a chord of outer circle which touches the inner circle. The length of chord AB is



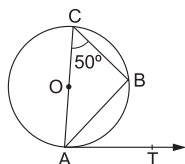
- (a) 8 cm (b) 14 cm
(c) 16 cm (d) $\sqrt{136}$ cm

12. In the given figure, AB and AC are tangents to a circle with centre O and radius 8 cm. If $OA = 17$ cm, then the length of AC (in cm) is [CBSE 2012]



- (a) 9 (b) 15
(c) $\sqrt{353}$ (d) 25

13. In the given figure, O is the centre of a circle, AOC is its diameter such that $\angle ACB = 50^\circ$. If AT is the tangent to the circle at the point A then $\angle BAT = ?$

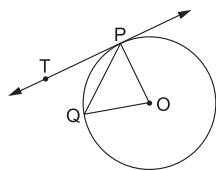


- (a) 40° (b) 50°
(c) 60° (d) 65°

14. In the given figure, O is the centre of a circle, PQ is a chord and PT is the tangent at P . If $\angle POQ = 70^\circ$, then $\angle TPQ$ is equal to

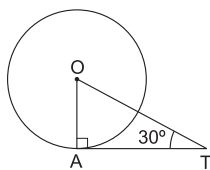
[CBSE 2011]

- (a) 35° (b) 45°
(c) 55° (d) 70°



15. In the given figure, AT is a tangent to the circle with centre O such that $OT = 4$ cm and $\angle OTA = 30^\circ$. Then, $AT = ?$

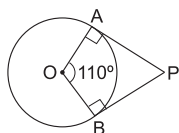
- (a) 4 cm (b) 2 cm
(c) $2\sqrt{3}$ cm (d) $4\sqrt{3}$ cm



16. If PA and PB are two tangents to a circle with centre O such that $\angle AOB = 110^\circ$ then $\angle APB$ is equal to

[CBSE 2011, '14]

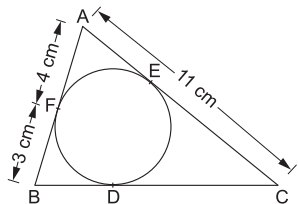
- (a) 55° (b) 60°
(c) 70° (d) 90°



17. In the given figure, the length of BC is

[CBSE 2012, '14]

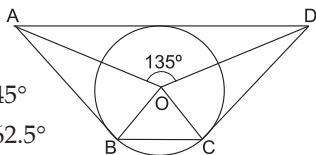
- (a) 7 cm
(b) 10 cm
(c) 14 cm
(d) 15 cm



18. In the given figure, if $\angle AOD = 135^\circ$ then $\angle BOC$ is equal to

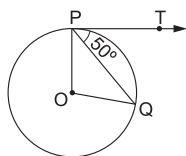
[CBSE 2013C]

- (a) 25° (b) 45°
(c) 52.5° (d) 62.5°



19. In the given figure, O is the centre of a circle and PT is the tangent to the circle. If PQ is a chord such that $\angle QPT = 50^\circ$ then $\angle POQ = ?$

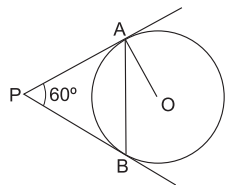
- (a) 100° (b) 90°
(c) 80° (d) 75°



20. In the given figure, PA and PB are two tangents to the circle with centre O . If $\angle APB = 60^\circ$ then $\angle OAB$ is

[CBSE 2011]

- (a) 15° (b) 30°
(c) 60° (d) 90°



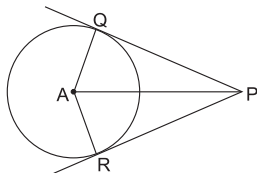
21. If two tangents inclined at an angle of 60° are drawn to a circle of radius 3 cm then the length of each tangent is

(a) 3 cm (b) $\frac{3\sqrt{3}}{2}$ cm (c) $3\sqrt{3}$ cm (d) 6 cm

22. In the given figure, PQ and PR are tangents to a circle with centre A . If $\angle QPA = 27^\circ$ then $\angle QAR$ equals

[CBSE 2012]

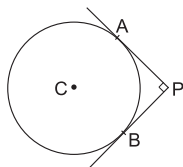
(a) 63° (b) 117°
(c) 126° (d) 153°



23. In the given figure, PA and PB are two tangents drawn from an external point P to a circle with centre C and radius 4 cm. If $PA \perp PB$, then the length of each tangent is

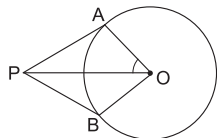
[CBSE 2013]

(a) 3 cm (b) 4 cm
(c) 5 cm (d) 6 cm



24. If PA and PB are two tangents to a circle with centre O such that $\angle APB = 80^\circ$. Then, $\angle AOP = ?$

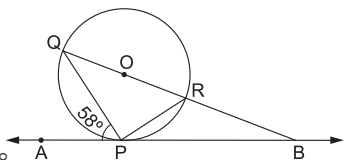
(a) 40° (b) 50°
(c) 60° (d) 70°



25. In the given figure, O is the centre of the circle. AB is the tangent to the circle at the point P . If $\angle APQ = 58^\circ$ then the measure of $\angle PQB$ is

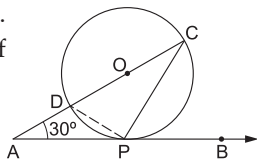
[CBSE 2014]

(a) 32° (b) 58°
(c) 122° (d) 132°



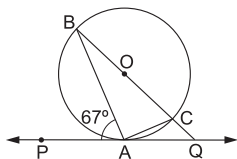
26. In the given figure, O is the centre of the circle. AB is the tangent to the circle at the point P . If $\angle PAO = 30^\circ$ then $\angle CPB + \angle ACP$ is equal to

(a) 60° (b) 90°
(c) 120° (d) 150°



27. In the given figure, PQ is a tangent to a circle with centre O . A is the point of contact. If $\angle PAB = 67^\circ$, then the measure of $\angle AQB$ is

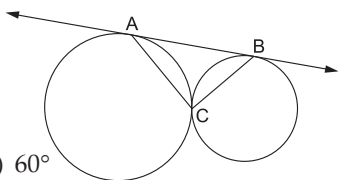
(a) 73° (b) 64°
(c) 53° (d) 44°



28. In the given figure, two circles touch each other at C and AB is a tangent to both the circles. The measure of $\angle ACB$ is

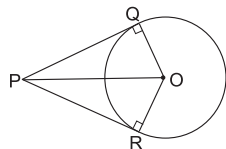
[HOTS] [CBSE 2013C]

- (a) 45° (b) 60°
(c) 90° (d) 120°



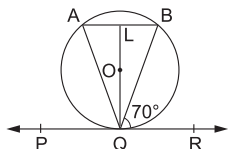
29. O is the centre of a circle of radius 5 cm. At a distance of 13 cm from O , a point P is taken. From this point, two tangents PQ and PR are drawn to the circle. Then, the area of quad. $PQOR$ is

- (a) 60 cm^2 (b) 32.5 cm^2
(c) 65 cm^2 (d) 30 cm^2



30. In the given figure, PQR is a tangent to the circle at Q , whose centre is O and AB is a chord parallel to PR such that $\angle BQR = 70^\circ$. Then, $\angle AQB = ?$

- (a) 20° (b) 35°
(c) 40° (d) 45°

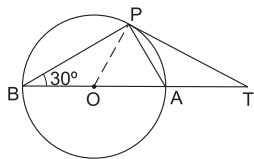


31. The length of the tangent from an external point P to a circle of radius 5 cm is 10 cm. The distance of the point from the centre of the circle is
- (a) 8 cm (b) $\sqrt{104}$ cm (c) 12 cm (d) $\sqrt{125}$ cm

[CBSE 2013C]

32. In the given figure, O is the centre of a circle, BOA is its diameter and the tangent at the point P meets BA extended at T . If $\angle PBO = 30^\circ$ then $\angle PTA = ?$

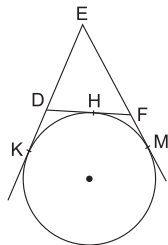
- (a) 60° (b) 30°
(c) 15° (d) 45°



33. In the given figure, a circle touches the side DF of $\triangle EDF$ at H and touches ED and EF produced at K and M respectively. If $EK = 9$ cm then the perimeter of $\triangle EDF$ is

[CBSE 2012]

- (a) 9 cm (b) 12 cm
(c) 13.5 cm (d) 18 cm



34. To draw a pair of tangents to a circle, which are inclined to each other at an angle of 45° , we have to draw tangents at the end points of those

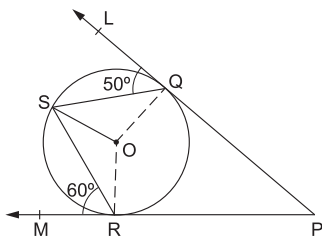
two radii, the angle between which is

[CBSE 2011]

- (a) 105° (b) 135° (c) 140° (d) 145°

35. In the given figure, O is the centre of a circle; PQL and PRM are the tangents at the points Q and R respectively and S is a point on the circle such that $\angle SQL = 50^\circ$ and $\angle SRM = 60^\circ$. Then, $\angle QSR = ?$

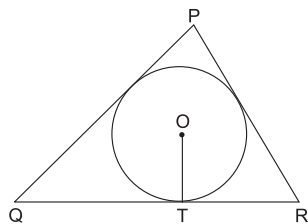
- (a) 40° (b) 50°
(c) 60° (d) 70°



36. In the given figure, a triangle PQR is drawn to circumscribe a circle of radius 6 cm such that the segments QT and TR into which QR is divided by the point of contact T , are of lengths 12 cm and 9 cm respectively. If the area of $\triangle PQR = 189 \text{ cm}^2$ then the length of side PQ is

[CBSE 2011]

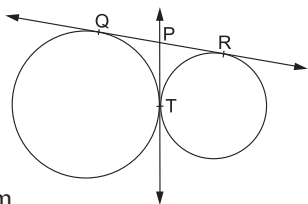
- (a) 17.5 cm (b) 20 cm (c) 22.5 cm (d) 25 cm



37. In the given figure, QR is a common tangent to the given circles, touching externally at the point T . The tangent at T meets QR at P . If $PT = 3.8 \text{ cm}$ then the length of QR is

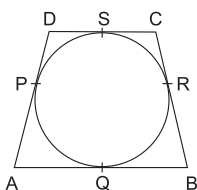
[CBSE 2014]

- (a) 1.9 cm (b) 3.8 cm
(c) 5.7 cm (d) 7.6 cm



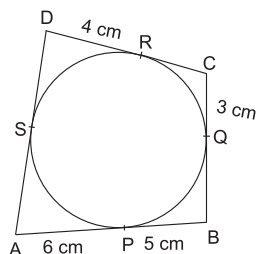
38. In the given figure, quad. $ABCD$ is circumscribed, touching the circle at P, Q, R and S . If $AP = 5 \text{ cm}$, $BC = 7 \text{ cm}$ and $CS = 3 \text{ cm}$. Then, the length $AB = ?$

- (a) 9 cm (b) 10 cm
(c) 12 cm (d) 8 cm



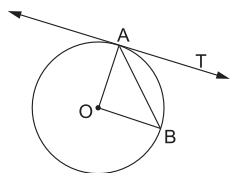
39. In the given figure, quad. $ABCD$ is circumscribed, touching the circle at P, Q, R and S . If $AP = 6 \text{ cm}$, $BP = 5 \text{ cm}$, $CQ = 3 \text{ cm}$ and $DR = 4 \text{ cm}$ then perimeter of quad. $ABCD$ is

- (a) 18 cm (b) 27 cm
(c) 36 cm (d) 32 cm



40. In the given figure, O is the centre of a circle, AB is a chord and AT is the tangent at A . If $\angle AOB = 100^\circ$ then $\angle BAT$ is equal to [CBSE 2011]

(a) 40° (b) 50°
(c) 90° (d) 100°

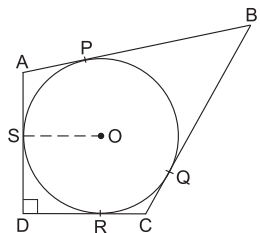


41. In a right triangle ABC , right-angled at B , $BC = 12$ cm and $AB = 5$ cm. The radius of the circle inscribed in the triangle is [CBSE 2014]

(a) 1 cm (b) 2 cm
(c) 3 cm (d) 4 cm

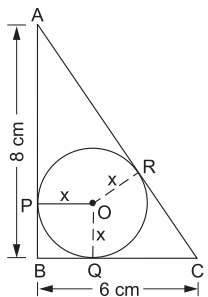
42. In the given figure, a circle is inscribed in a quadrilateral $ABCD$ touching its sides AB , BC , CD and AD at P , Q , R and S respectively. If the radius of the circle is 10 cm, $BC = 38$ cm, $PB = 27$ cm and $AD \perp CD$ then the length of CD is [CBSE 2013]

(a) 11 cm (b) 15 cm
(c) 20 cm (d) 21 cm



43. In the given figure, $\triangle ABC$ is right-angled at B such that $BC = 6$ cm and $AB = 8$ cm. A circle with centre O has been inscribed inside the triangle. $OP \perp AB$, $OQ \perp BC$ and $OR \perp AC$. If $OP = OQ = OR = x$ cm then $x = ?$

(a) 2 cm (b) 2.5 cm
(c) 3 cm (d) 3.5 cm

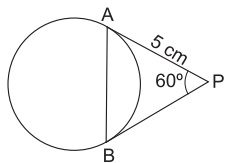


44. Quadrilateral $ABCD$ is circumscribed to a circle. If $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm then the length of AD is [CBSE 2012]

(a) 3 cm (b) 4 cm
(c) 6 cm (d) 7 cm

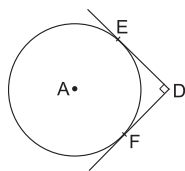
45. In the given figure, PA and PB are tangents to the given circle such that $PA = 5$ cm and $\angle APB = 60^\circ$. The length of chord AB is

(a) $5\sqrt{2}$ cm (b) 5 cm
(c) $5\sqrt{3}$ cm (d) 7.5 cm



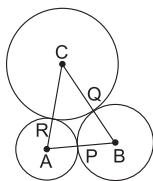
46. In the given figure, DE and DF are tangents from an external point D to a circle with centre A . If $DE = 5$ cm and $DE \perp DF$ then the radius of the circle is

[CBSE 2013]



- (a) 3 cm (b) 4 cm
(c) 5 cm (d) 6 cm

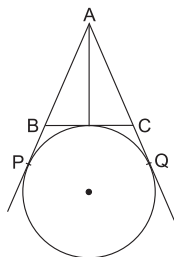
47. In the given figure, three circles with centres A , B , C respectively touch each other externally. If $AB = 5$ cm, $BC = 7$ cm and $CA = 6$ cm then the radius of the circle with centre A is



- (a) 1.5 cm (b) 2 cm
(c) 2.5 cm (d) 3 cm

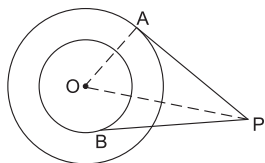
48. In the given figure, AP , AQ and BC are tangents to the circle. If $AB = 5$ cm, $AC = 6$ cm and $BC = 4$ cm then the length of AP is

[CBSE 2012]



- (a) 15 cm (b) 10 cm
(c) 9 cm (d) 7.5 cm

49. In the given figure, O is the centre of two concentric circles of radii 5 cm and 3 cm. From an external point P tangents PA and PB are drawn to these circles. If $PA = 12$ cm then PB is equal to



- (a) $5\sqrt{2}$ cm (b) $3\sqrt{5}$ cm
(c) $4\sqrt{10}$ cm (d) $5\sqrt{10}$ cm

True/False Type

50. Which of the following statements is not true?

- (a) If a point P lies inside a circle, no tangent can be drawn to the circle, passing through P .
(b) If a point P lies on the circle, then one and only one tangent can be drawn to the circle at P .
(c) If a point P lies outside the circle, then only two tangents can be drawn to the circle from P .
(d) A circle can have more than two parallel tangents, parallel to a given line.

51. Which of the following statements is not true?
- A tangent to a circle intersects the circle exactly at one point.
 - The point common to the circle and its tangent is called the point of contact.
 - The tangent at any point of a circle is perpendicular to the radius of the circle through the point of contact.
 - A straight line can meet a circle at one point only.
52. Which of the following statements is not true?
- A line which intersects a circle in two points, is called a secant of the circle.
 - A line intersecting a circle at one point only, is called a tangent to the circle.
 - The point at which a line touches the circle, is called the point of contact.
 - A tangent to the circle can be drawn from a point inside the circle.

Assertion-and-Reason Type

Each question consists of two statements, namely, Assertion (A) and Reason (R). For selecting the correct answer, use the following code:

- Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- Assertion (A) is true and Reason (R) is false.
- Assertion (A) is false and Reason (R) is true.

53.	Assertion (A)	Reason (R)
	At a point P of a circle with centre O and radius 12 cm, a tangent PQ of length 16 cm is drawn. Then, $OQ = 20$ cm.	The tangent at any point of a circle is perpendicular to the radius through the point of contact.

The correct answer is (a)/(b)/(c)/(d).

54.	Assertion (A)	Reason (R)
	If two tangents are drawn to a circle from an external point then they subtend equal angles at the centre.	A parallelogram circumscribing a circle is a rhombus.

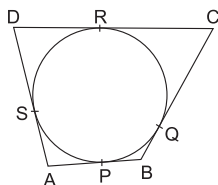
The correct answer is (a)/(b)/(c)/(d).

55.

Assertion (A)

In the given figure, a quad. $ABCD$ is drawn to circumscribe a given circle, as shown.

Then, $AB + BC = AD + DC$.

**Reason (R)**

In two concentric circles, the chord of the larger circle, which touches the smaller circle, is bisected at the point of contact.

The correct answer is (a)/(b)/(c)/(d).

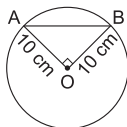
ANSWERS (MCQ)

1. (b) 2. (c) 3. (c) 4. (d) 5. (c) 6. (a) 7. (a) 8. (b) 9. (d)
 10. (d) 11. (c) 12. (b) 13. (b) 14. (a) 15. (c) 16. (c) 17. (b) 18. (b)
 19. (a) 20. (b) 21. (c) 22. (c) 23. (b) 24. (b) 25. (a) 26. (b) 27. (d)
 28. (c) 29. (a) 30. (c) 31. (d) 32. (b) 33. (d) 34. (b) 35. (d) 36. (c)
 37. (d) 38. (a) 39. (c) 40. (b) 41. (b) 42. (d) 43. (a) 44. (a) 45. (b)
 46. (c) 47. (b) 48. (d) 49. (c) 50. (d) 51. (d) 52. (d) 53. (a) 54. (b)
 55. (d)

HINTS TO SOME SELECTED QUESTIONS

4. Every diameter passes through the centre and so no two diameters of a circle can be parallel.

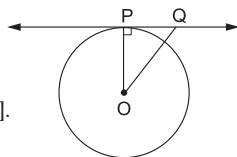
$$5. AB = \sqrt{OA^2 + OB^2} = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ cm.}$$



8. We have $OP \perp PQ$, i.e., $\angle OPQ = 90^\circ$.

$$\triangle OPQ \text{ is isosceles} \Rightarrow OP = PQ \Rightarrow \angle OQP = \angle POQ = 45^\circ$$

[$\because$ in a triangle, angles opposite equal sides are equal].



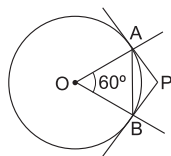
9. $ABOC$ is a cyclic quadrilateral [see Solved Example 18].

$$\therefore \angle BAC + \angle BOC = 180^\circ \Rightarrow \angle BOC = 180^\circ - \angle BAC = 180^\circ - 40^\circ = 140^\circ.$$

10. Let P be the point of intersection of tangents at A and B .

Then, $OAPB$ is a cyclic quadrilateral. [see Solved Example 18]

$$\therefore \angle AOB + \angle APB = 180^\circ \Rightarrow \angle APB = 180^\circ - 60^\circ = 120^\circ.$$



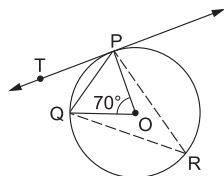
13. $\angle BAT = \angle ACB = 50^\circ$ [angles in alternate segments].

14. Mark a point R in the alternate segment.

Join PR and QR .

$$\text{Then, } \angle PRQ = \frac{1}{2} \angle POQ = \frac{1}{2} \times 70^\circ = 35^\circ.$$

Now, $\angle TPQ = \angle PRQ = 35^\circ$ [angles in alternate segments].



15. In right $\triangle OAT$ (having right angle at A), we have

$$\cos(\angle OTA) = \frac{AT}{OT} \Rightarrow \cos 30^\circ = \frac{AT}{4} \Rightarrow AT = \left(\frac{\sqrt{3}}{2} \times 4\right) \text{ cm} = 2\sqrt{3} \text{ cm}.$$

17. We have $AF = AE$, $BD = BF$, $CD = CE$. [$\because$ tangents from an external point are equal]

$$\therefore CE = AC - AE = AC - AF = (11 - 4) \text{ cm} = 7 \text{ cm};$$

$$BD = BF = 3 \text{ cm}; CD = CE = 7 \text{ cm}.$$

$$\text{And so, } BC = BD + CD = (3 + 7) \text{ cm} = 10 \text{ cm}.$$

18. $\angle AOD + \angle BOC = 180^\circ$ [see Solved Example 29]

$$\Rightarrow \angle BOC = 180^\circ - \angle AOD = 180^\circ - 135^\circ = 45^\circ.$$

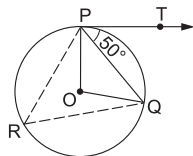
19. Mark a point R in the alternate segment.

Join PR and QR .

Then, $\angle PRQ = \angle QPT = 50^\circ$ [angles in alternate segments].

Now, $\angle POQ = 2\angle PRQ = 100^\circ$.

[$\because$ angle at the centre is double the angle on the circle]



20. $\angle APB = 2\angle OAB \Rightarrow \angle OAB = \frac{1}{2} \angle APB = 30^\circ$. [See Solved Example 13.]

21. Let PQ and PR be tangents to a circle with centre O from a point P such that $\angle QPR = 60^\circ$.

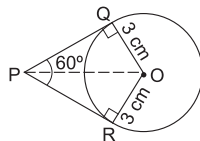
$$\text{Then, } \angle QPO = \angle RPO = \frac{1}{2} \angle QPR = 30^\circ$$

[$\because$ tangents from an external point are equally inclined to the line joining the point and the centre].

In right $\triangle OPQ$, we have

$$\tan(\angle OPQ) = \frac{OQ}{PQ} \Rightarrow \tan 30^\circ = \frac{3}{PQ} \Rightarrow \frac{1}{\sqrt{3}} = \frac{3}{PQ} \Rightarrow PQ = 3\sqrt{3} \text{ cm}.$$

$$\text{Now, } PR = PQ = 3\sqrt{3} \text{ cm}.$$



$$22. \angle QPR = 2\angle QPA = 2 \times 27^\circ = 54^\circ \quad [\because AP \text{ bisects } \angle QPR].$$

Now, $QPRI$ is a cyclic quadrilateral [see Solved Example 18]

$$\Rightarrow \angle QAR + \angle QPR = 180^\circ$$

$$\Rightarrow \angle QAR = 180^\circ - 54^\circ = 126^\circ.$$

23. Join AC and BC . Then, $ACBP$ is a square.

$$\therefore PA = PB = AC = \text{radius of the circle} = 4 \text{ cm.}$$

$$25. \angle QPR = 90^\circ \quad [\text{angle in a semicircle}]$$

$$\text{and } \angle QRP = \angle APQ = 58^\circ \quad [\text{angles in alternate segments}].$$

$$\text{In } \triangle PQR, \text{ we have } \angle PQR + \angle QRP + \angle QPR = 180^\circ \Rightarrow \angle PQR = 32^\circ.$$

$$\therefore \angle PQB = \angle PQR = 32^\circ.$$

$$26. \angle DPC = 90^\circ \quad [\text{angle in a semicircle}]$$

$$\text{and } \angle DPA = \angle DCP \quad [\text{angles in alternate segments}].$$

$$\text{Now, } \angle CPB + \angle DPA + \angle DPC = 180^\circ$$

$$\Rightarrow \angle CPB + \angle DPA = 90^\circ \Rightarrow \angle CPB + \angle DCP = 90^\circ$$

$$\Rightarrow \angle CPB + \angle ACP = 90^\circ.$$

$$27. \angle BAC = 90^\circ \quad [\text{angle in a semicircle}].$$

$$\angle PAB + \angle BAC + \angle CAQ = 180^\circ \Rightarrow \angle CAQ = 180^\circ - (90^\circ + 67^\circ) = 23^\circ.$$

$$\angle ACB = \angle PAB = 67^\circ \quad [\text{angles in alternate segments}]$$

$$\angle ACB + \angle ACQ = 180^\circ \quad [\text{linear pair}]$$

$$\Rightarrow \angle ACQ = 180^\circ - 67^\circ = 113^\circ.$$

$$\text{Now, } \angle CAQ + \angle ACQ + \angle AQC = 180^\circ \quad [\text{in } \triangle ACQ]$$

$$\Rightarrow \angle AQC = 180^\circ - (23^\circ + 113^\circ) = 44^\circ \Rightarrow \angle AQB = 44^\circ.$$

28. Draw a tangent to the circles at point C . Let it meet AB at P .

Then, $PA = PC$ and $PB = PC$.

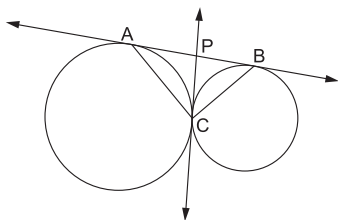
$$PA = PC \Rightarrow \angle PAC = \angle PCA$$

$$PB = PC \Rightarrow \angle PBC = \angle PCB$$

$$\therefore \angle PAC + \angle PBC = \angle PCA + \angle PCB = \angle ACB$$

$$\Rightarrow \angle PAC + \angle PBC + \angle ACB = 2\angle ACB$$

$$\Rightarrow 180^\circ = 2\angle ACB \Rightarrow \angle ACB = 90^\circ.$$



29. Clearly, $OQ = OR = 5 \text{ cm}$, $\angle OQP = \angle ORP = 90^\circ$ and $OP = 13 \text{ cm}$.

$$\therefore PQ^2 = OP^2 - OQ^2 = (13)^2 - (5)^2 = 169 - 25 = 144 \Rightarrow PQ = \sqrt{144} = 12 \text{ cm.}$$

$$\therefore \text{ar}(\triangle OQP) = \frac{1}{2} \times PQ \times OQ = \left(\frac{1}{2} \times 12 \times 5\right) \text{ cm}^2 = 30 \text{ cm}^2.$$

Similarly, $\text{ar}(\triangle ORP) = 30 \text{ cm}^2$.

$$\therefore \text{ar}(\text{quad. } PQOR) = (30 + 30) \text{ cm}^2 = 60 \text{ cm}^2.$$

30. $\angle QAB = \angle BQR = 70^\circ$ [angles in alternate segments]

$$\angle ABQ = \angle BQR = 70^\circ \quad [\text{alternate int. angles}].$$

In $\triangle AQB$, we have

$$\angle AQB = 180^\circ - (\angle QAB + \angle ABQ) = 180^\circ - (70^\circ + 70^\circ) = 40^\circ.$$

32. $\angle APB = 90^\circ$ [angle in a semicircle].

$$\therefore \angle PAB = 90^\circ - \angle PBA = 90^\circ - 30^\circ = 60^\circ.$$

$$\angle PAT + \angle PAB = 180^\circ \text{ [linear pair]}$$

$$\Rightarrow \angle PAT = 180^\circ - \angle PAB = 180^\circ - 60^\circ = 120^\circ.$$

$$\angle APT = \angle PBA = 30^\circ \text{ [angles in alternate segments].}$$

$$\text{Now, } \angle APT + \angle PAT + \angle PTA = 180^\circ \text{ [in } \triangle PAT]$$

$$\Rightarrow \angle PTA = 180^\circ - (30^\circ + 120^\circ) = 30^\circ.$$

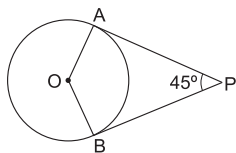
34. Let PA and PB be the desired tangents to a circle with centre O from an exterior point P .

Then, $\angle APB = 45^\circ$.

$BOAP$ must be a cyclic quadrilateral.

$$\therefore \angle AOB + \angle APB = 180^\circ \Rightarrow \angle AOB = 180^\circ - 45^\circ = 135^\circ.$$

So, the angle between the two radii must be 135° .



35. Since PQL is a tangent and OQ is a radius, so $\angle OQL = 90^\circ$.

$$\therefore \angle OQS = 90^\circ - 50^\circ = 40^\circ.$$

$$\text{Now, } OQ = OS \Rightarrow \angle OSQ = \angle OQS = 40^\circ.$$

$$\text{Similarly, } \angle ORS = 90^\circ - 60^\circ = 30^\circ.$$

$$\text{And, } OR = OS \Rightarrow \angle OSR = \angle ORS = 30^\circ.$$

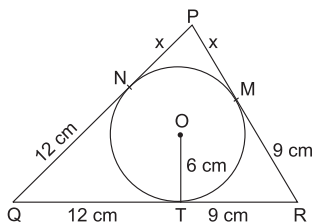
$$\therefore \angle QSR = \angle OSQ + \angle OSR = 40^\circ + 30^\circ = 70^\circ.$$

36. $\text{ar}(\triangle PQR) = \frac{1}{2} \times (\text{perimeter of } \triangle PQR) \times r$

$$\Rightarrow 189 = \frac{1}{2} \times \{(x+12) + (12+9) + (9+x)\} \times 6$$

$$\Rightarrow x = 10.5$$

$$\therefore PQ = (x+12) \text{ cm} = (10.5+12) \text{ cm} = 22.5 \text{ cm}.$$



37. $PQ = PT = 3.8 \text{ cm}$, $PR = PT = 3.8 \text{ cm}$,

$$QR = PQ + PR = (3.8 + 3.8) \text{ cm} = 7.6 \text{ cm}.$$

38. Since the lengths of tangents drawn from an external point to a circle are equal, we have $AQ = AP = 5 \text{ cm}$.

$$CR = CS = 3 \text{ cm and } BR = (BC - CR) = (7 - 3) \text{ cm} = 4 \text{ cm}.$$

$$BQ = BR = 4 \text{ cm}.$$

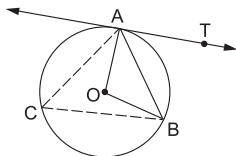
$$\therefore AB = (BR + BQ) = (5 + 4) \text{ cm} = 9 \text{ cm}.$$

40. Mark a point C in the alternate segment. Join AC and BC .

$$\angle ACB = \frac{1}{2} \angle AOB = \frac{1}{2} \times 100^\circ = 50^\circ$$

[$\because$ angle at the centre is double the angle on the circle].

$$\text{Now, } \angle BAT = \angle ACB = 50^\circ \text{ [angles in alternate segments].}$$



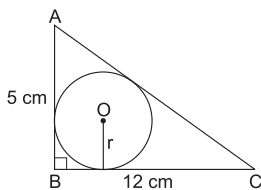
41. In right
- $\triangle ABC$
- , we have

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{5^2 + 12^2} \text{ cm} = 13 \text{ cm.}$$

$$\text{ar}(\triangle ABC) = \frac{1}{2} \times (\text{perimeter of } \triangle ABC) \times r$$

$$\Rightarrow \frac{1}{2} \times 5 \times 12 = \frac{1}{2} \times (5 + 12 + 13) \times r \Rightarrow r = 2 \text{ cm.}$$

$$\left[\because \text{ar}(\triangle ABC) = \frac{1}{2} \times BC \times AB \right]$$



- 42.
- $BQ = PB = 27 \text{ cm}$
- ;
- $CQ = BC - BQ = (38 - 27) \text{ cm} = 11 \text{ cm}$
- ;
- $CR = CQ = 11 \text{ cm}$
- .

Join OR . Then, $SDRO$ is a square.

$$\therefore DR = SO = \text{radius} = 10 \text{ cm.}$$

And so, $CD = DR + CR = (10 + 11) \text{ cm} = 21 \text{ cm}$.

- 43.
- $AC^2 = AB^2 + BC^2 = 8^2 + 6^2 = 64 + 36 = 100$

$$\Rightarrow AC = \sqrt{100} \text{ cm} = 10 \text{ cm.}$$

$$CR = CQ = (BC - BQ) = (6 - x) \text{ cm.}$$

$$AR = AP = (AB - BP) = (8 - x) \text{ cm.}$$

$$AC = (AR + CR) = [(8 - x) + (6 - x)] \text{ cm} = (14 - 2x) \text{ cm}$$

$$\Rightarrow (14 - 2x) = 10 \Rightarrow 2x = 4 \Rightarrow x = 2 \text{ cm.}$$

- 44.
- $AB + CD = AD + BC$
- [see Solved Example 26]

$$\Rightarrow AD = AB + CD - BC = (6 + 4 - 7) \text{ cm} = 3 \text{ cm.}$$

- 45.
- $PA = PB \Rightarrow \angle PBA = \angle PAB = x^\circ$
- (say)

$$\text{Then, } x^\circ + x^\circ + 60^\circ = 180 \Rightarrow x = 60^\circ.$$

$\therefore \triangle APB$ is equilateral and so, $AB = PA = 5 \text{ cm}$.

46. Join
- AE
- and
- AF
- . Then,
- $AFDE$
- is a square.

$\therefore$ radius of circle = $AE = DE = 5 \text{ cm}$.

47. Let the radii of the three circles be
- x, y, z
- respectively. Then,

$$x + y = 5, y + z = 7 \text{ and } z + x = 6 \Rightarrow 2(x + y + z) = 18 \Rightarrow x + y + z = 9.$$

$$\therefore x = (x + y + z) - (y + z) = (9 - 7) = 2 \text{ cm.}$$

49. Join
- OB
- . Then,
- $OA = 5 \text{ cm}$
- ,
- $OB = 3 \text{ cm}$
- .

$$OP = \sqrt{OA^2 + PA^2} = \sqrt{5^2 + 12^2} = 13 \text{ cm.}$$

$$PB = \sqrt{OP^2 - OB^2} = \sqrt{13^2 - 3^2} = \sqrt{160} \text{ cm} = 4\sqrt{10} \text{ cm.}$$

- 53.
- $OQ = \sqrt{OP^2 + PQ^2} = \sqrt{12^2 + 16^2} = 20 \text{ cm.}$

$\therefore$ A is true. Also, R is true and is a correct explanation of A.

Hence, the correct answer is (a).

54. A and R are both true (see Theorem 4(i) and Solved Example 28) but R is not a correct explanation of A.

Hence, the correct answer is (b).

55. A is false. Correct relation is $AB + CD = AD + BC$.

R is true (see Theorem 5).

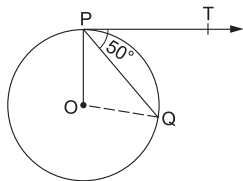
Hence, the correct answer is (d).

TEST YOURSELF

MCQ

1. In the given figure, O is the centre of a circle, PQ is a chord and the tangent PT at P makes an angle of 50° with PQ . Then, $\angle POQ = ?$

- (a) 130° (b) 100°
(c) 90° (d) 75°

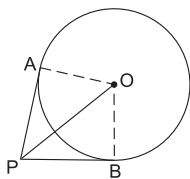


2. If the angle between two radii of a circle is 130° then the angle between the tangents at the ends of the radii is

- (a) 65° (b) 40° (c) 50° (d) 90°

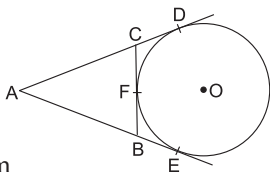
3. If tangents PA and PB from a point P to a circle with centre O are drawn so that $\angle APB = 80^\circ$ then $\angle POA = ?$

- (a) 40° (b) 50°
(c) 80° (d) 60°



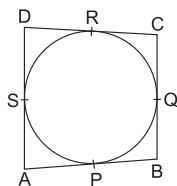
4. In the given figure, AD and AE are the tangents to a circle with centre O and BC touches the circle at F . If $AE = 5$ cm then perimeter of $\triangle ABC$ is

- (a) 15 cm (b) 10 cm
(c) 22.5 cm (d) 20 cm

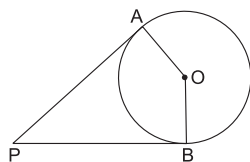


Short-Answer Questions

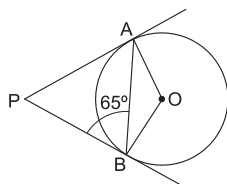
5. In the given figure, a quadrilateral $ABCD$ is drawn to circumscribe a circle such that its sides AB , BC , CD and AD touch the circle at P , Q , R and S respectively. If $AB = x$ cm, $BC = 7$ cm, $CR = 3$ cm and $AS = 5$ cm, find x .



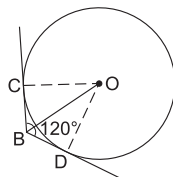
6. In the given figure, PA and PB are the tangents to a circle with centre O . Show that the points A, O, B, P are concyclic.



7. In the given figure, PA and PB are two tangents from an external point P to a circle with centre O . If $\angle PBA = 65^\circ$, find $\angle OAB$ and $\angle APB$.



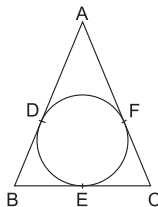
8. Two tangent segments BC and BD are drawn to a circle with centre O such that $\angle CBD = 120^\circ$. Prove that $OB = 2BC$.



9. Fill in the blanks.

- (i) A line intersecting a circle in two distinct points is called a
- (ii) A circle can have parallel tangents at the most.
- (iii) The common point of a tangent to a circle and the circle is called the
- (iv) A circle can have tangents.

10. Prove that the lengths of two tangents drawn from an external point to a circle are equal.
11. Prove that the tangents drawn at the ends of the diameter of a circle are parallel.
12. In the given figure, if $AB = AC$, prove that $BE = CE$.

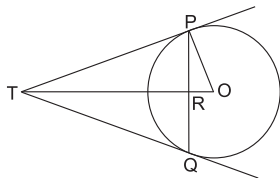


13. If two tangents are drawn to a circle from an external point, show that they subtend equal angles at the centre.
14. Prove that the tangents drawn at the ends of a chord of a circle make equal angles with the chord.

15. Prove that the parallelogram circumscribing a circle, is a rhombus.
16. Two concentric circles are of radii 5 cm and 3 cm respectively. Find the length of the chord of the larger circle which touches the smaller circle.

Long-Answer Questions

17. A quadrilateral is drawn to circumscribe a circle. Prove that the sums of opposite sides are equal.
18. Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.
19. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segments joining the points of contact at the centre.
20. PQ is a chord of length 16 cm of a circle of radius 10 cm. The tangents at P and Q intersect at a point T as shown in the figure. Find the length of TP .



ANSWERS (TEST YOURSELF)

1. (b) 2. (c) 3. (b) 4. (b) 5. 9 cm
7. $\angle OAB = 25^\circ$, $\angle APB = 50^\circ$
9. (i) secant (ii) two (iii) point of contact (iv) infinitely many
16. 8 cm 20. 10.7 cm (approx.)



DIVISION OF A LINE SEGMENT IN A GIVEN RATIO

EXAMPLE 1 Divide a line segment 6 cm long in the ratio 4 : 3. Prove your assertion.

STEPS OF CONSTRUCTION

Step 1. Draw a line segment $AB = 6$ cm.

Step 2. Draw a ray AX , making an acute angle $\angle BAX$.

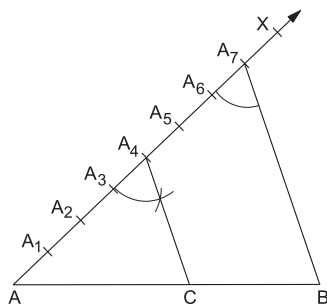
Step 3. Along AX , mark $(4 + 3) = 7$ points $A_1, A_2, A_3, A_4, A_5, A_6$ and A_7 such that

$$AA_1 = A_1A_2 = A_2A_3 = A_3A_4 = A_4A_5 = A_5A_6 = A_6A_7.$$

Step 4. Join A_7B .

Step 5. From A_4 , draw A_4C parallel to A_7B (by making an angle equal to $\angle AA_7B$), meeting AB at C .

Then, C is the point on AB , which divides it in the ratio 4 : 3.



Thus, $AC : CB = 4 : 3$.

PROOF Let $AA_1 = A_1A_2 = \dots = A_6A_7 = x$.

In $\triangle ABA_7$, we have $A_4C \parallel A_7B$.

$$\therefore \frac{AC}{CB} = \frac{AA_4}{A_4A_7} = \frac{4x}{3x} = \frac{4}{3} \quad [\text{by Thales' theorem}].$$

Hence, $AC : CB = 4 : 3$.

ALTERNATIVE METHOD

STEPS OF CONSTRUCTION

Step 1. Draw a line segment $AB = 6$ cm.

Step 2. Draw a ray AX , making an acute $\angle BAX$.

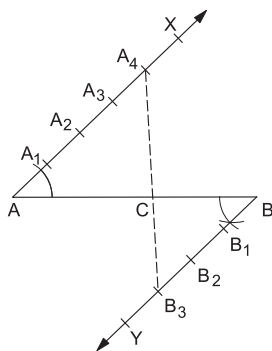
Step 3. Draw a ray BY parallel to AX by making $\angle ABY = \angle BAX$.

Step 4. Locate the points A_1, A_2, A_3, A_4 on AX and B_1, B_2, B_3 on BY such that

$$AA_1 = A_1A_2 = A_2A_3 = A_3A_4 = BB_1 = B_1B_2 = B_2B_3.$$

Step 5. Join A_4B_3 , intersecting AB at a point C .

Then, $AC : CB = 4 : 3$.



PROOF Here, $AA_4 \parallel BB_3$.

$$\therefore \angle CAA_4 = \angle CBB_3 \quad (\text{alt. int. } \angle).$$

$$\text{Also, } \angle ACA_4 = \angle BCB_3 \quad (\text{vert. opp. } \angle).$$

$$\therefore \triangle CAA_4 \sim \triangle CBB_3 \quad [\text{by AA-similarity}].$$

$$\text{And so, } \frac{AC}{CB} = \frac{AA_4}{BB_3} = \frac{4}{3}.$$

Hence, $AC : CB = 4 : 3$.

TO CONSTRUCT A TRIANGLE SIMILAR TO A GIVEN TRIANGLE AS PER GIVEN SCALE FACTOR

EXAMPLE 2 Construct a triangle with sides 5 cm, 5.5 cm and 6.5 cm. Then construct another triangle whose sides are $\frac{3}{5}$ times the corresponding sides of the first triangle. [CBSE 2014]

STEPS OF CONSTRUCTION

Step 1. Draw a line segment $BC = 5.5$ cm.

Step 2. With B as centre and radius equal to 5 cm, draw an arc.

Step 3. With C as centre and radius equal to 6.5 cm, draw another arc cutting the previous arc at A .

Step 4. Join AB and AC .

Thus, $\triangle ABC$ with given lengths of sides is obtained.

Step 5. Below BC , make an acute angle $\angle CBX$.

Step 6. Along BX , mark off five points B_1, B_2, B_3, B_4 and B_5 such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4 = B_4B_5$.

Step 7. Join B_5C .

Step 8. From B_3 , draw $B_3D \parallel B_5C$ (by making an angle $\angle BB_3D = \angle BB_5C$), meeting BC at D .

Step 9. From D , draw $DE \parallel CA$ (by making an angle $\angle BDE = \angle BCA$), meeting AB at E .

Then, $\triangle EBD$ is the required triangle, each of whose sides is $\frac{3}{5}$ of the corresponding side of $\triangle ABC$.

PROOF Since $DE \parallel CA$, we have $\triangle ABC \sim \triangle EBD$.

$$\therefore \frac{EB}{AB} = \frac{DE}{CA} = \frac{BD}{BC} = \frac{3}{5}.$$

EXAMPLE 3 Draw a triangle ABC in which $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$. Then, construct a triangle whose sides are $\frac{5}{7}$ times the corresponding sides of $\triangle ABC$. [CBSE 2011, '15]

STEPS OF CONSTRUCTION

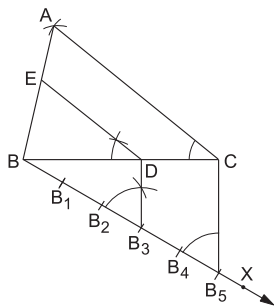
Step 1. Draw a line segment $BC = 6$ cm.

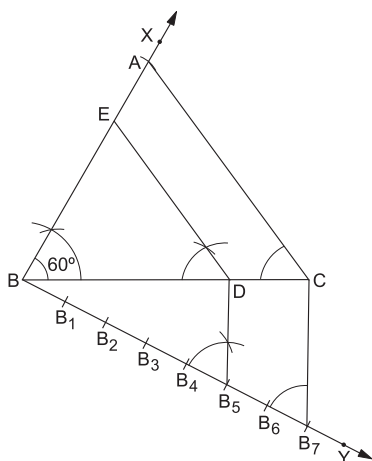
Step 2. At B , construct $\angle CBX = 60^\circ$.

Step 3. With B as centre and radius 5 cm, draw an arc cutting ray BX at A .

Step 4. Join AC .

Thus, $\triangle ABC$ is obtained.





Step 5. Below BC , make an acute angle $\angle CBY$.

Step 6. Along BY , mark off seven points $B_1, B_2, B_3, B_4, B_5, B_6$ and B_7 such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4 = B_4B_5 = B_5B_6 = B_6B_7$.

Step 7. Join B_7C .

Step 8. From B_5 , draw $B_5D \parallel B_7C$, meeting BC at D .

Step 9. From D , draw $DE \parallel CA$, meeting AB at E .

Then, $\triangle EBD$ is the required triangle, each of whose sides is $\frac{5}{7}$ of the corresponding side of $\triangle ABC$.

PROOF Since $DE \parallel CA$, we have $\triangle ABC \sim \triangle EBD$.

$$\therefore \frac{EB}{AB} = \frac{DE}{CA} = \frac{BD}{BC} = \frac{5}{7}.$$

EXAMPLE 4 Draw a triangle ABC with side $BC = 7$ cm, $\angle B = 45^\circ$, and $\angle A = 105^\circ$. Then, construct a triangle whose sides are $\frac{4}{3}$ times the corresponding sides of $\triangle ABC$.

STEPS OF CONSTRUCTION

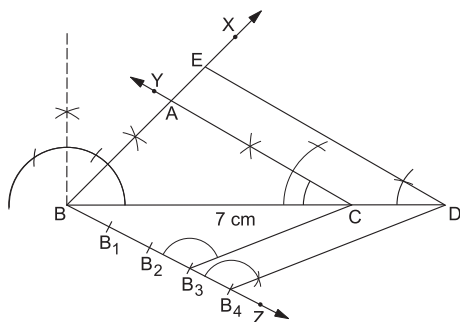
Step 1. Draw a line segment $BC = 7$ cm.

Step 2. At B , construct $\angle CBX = 45^\circ$ and at C , construct

$$\angle BCY = 180^\circ - (\angle B + \angle A) = 180^\circ - (45^\circ + 105^\circ) = 30^\circ.$$

Suppose BX and CY intersect at A .

$\triangle ABC$ is thus obtained.



Step 3. Below BC , make an acute angle $\angle CBZ$.

Step 4. Along BZ , mark off four points B_1, B_2, B_3 and B_4 such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4$.

Step 5. Join B_3C .

Step 6. From B_4 , draw $B_4D \parallel B_3C$, meeting BC (produced) at D .

Step 7. From D , draw $DE \parallel CA$, meeting ray BX at E .

Then, $\triangle EBD$ is the required triangle, each of whose sides is $\frac{4}{3}$ times the corresponding side of $\triangle ABC$.

PROOF Since $CA \parallel DE$, we have $\triangle ABC \sim \triangle EBD$.

$$\therefore \frac{EB}{AB} = \frac{DE}{CA} = \frac{BD}{BC} = \frac{4}{3}.$$

EXAMPLE 5 Construct an isosceles triangle whose base is 6 cm and altitude 4 cm. Then, construct another triangle whose sides are $\frac{3}{4}$ times the corresponding sides of the first triangle. [CBSE 2010, '15]

STEPS OF CONSTRUCTION

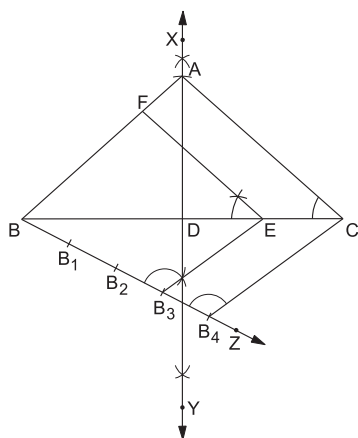
Step 1. Draw a line segment $BC = 6$ cm.

Step 2. Draw the perpendicular bisector XY of BC , cutting BC at D .

Step 3. With D as centre and radius 4 cm, draw an arc cutting XY at A .

Step 4. Join AB and AC .

Thus, isosceles $\triangle ABC$ having base 6 cm and altitude 4 cm is obtained.



Step 5. Below BC , make an acute angle $\angle CBZ$.

Step 6. Along BZ , mark off four points B_1, B_2, B_3 and B_4 such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4$.

Step 7. Join B_4C .

Step 8. From B_3 , draw $B_3E \parallel B_4C$, meeting BC at E .

Step 9. From E , draw $EF \parallel CA$, meeting BA at F .

Then, $\triangle FBE$ is the required triangle, each of whose sides is $\frac{3}{4}$ times the corresponding side of $\triangle ABC$.

PROOF Since $EF \parallel CA$, we have $\triangle FBE \sim \triangle ABC$.

$$\therefore \frac{FB}{AB} = \frac{EF}{CA} = \frac{BE}{BC} = \frac{3}{4}.$$

EXAMPLE 6 Construct a $\triangle ABC$ in which $AB = 4$ cm, $\angle B = 60^\circ$ and altitude $CL = 3$ cm. Construct a $\triangle ADE$ similar to $\triangle ABC$ such that each side of $\triangle ADE$ is $\frac{3}{2}$ times that of the corresponding side of $\triangle ABC$.

STEPS OF CONSTRUCTION

Step 1. Draw a line segment $AB = 4$ cm.

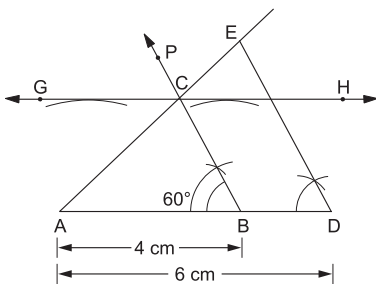
Step 2. Construct $\angle ABP = 60^\circ$.

Step 3. Draw a line $GH \parallel AB$ at a distance of 3 cm from AB , intersecting BP at C .

Step 4. Join CA .

Thus, $\triangle ABC$ is obtained.

Step 5. Extend AB to D such that $AD = \frac{3}{2}AB = \left(\frac{3}{2} \times 4\right) \text{ cm} = 6 \text{ cm}$.



Step 6. Draw $DE \parallel BC$, cutting AC produced at E .

Then, $\triangle ADE$ is the required triangle similar to $\triangle ABC$ such that each side of $\triangle ADE$ is $\frac{3}{2}$ times the corresponding side of $\triangle ABC$.

PROOF Since $DE \parallel BC$, we have $\triangle ADE \sim \triangle ABC$.

$$\therefore \frac{AD}{AB} = \frac{DE}{BC} = \frac{AE}{AC} = \frac{3}{2}.$$

EXERCISE 9A

1. Draw a line segment AB of length 7 cm. Using ruler and compasses, find a point P on AB such that $\frac{AP}{AB} = \frac{3}{5}$. [CBSE 2011]
2. (i) Draw a line segment of length 8 cm and divide it internally in the ratio 4 : 5. [CBSE 2017]
(ii) Draw a line segment of length 7.6 cm and divide it in the ratio 5 : 8. Measure the two parts.
3. Construct a $\triangle PQR$, in which $PQ = 6 \text{ cm}$, $QR = 7 \text{ cm}$ and $PR = 8 \text{ cm}$. Then, construct another triangle whose sides are $\frac{4}{5}$ times the corresponding sides of $\triangle PQR$. [CBSE 2013, '14]
4. Construct a triangle with sides 5 cm, 6 cm and 7 cm and then another triangle whose sides are $\frac{7}{5}$ of the corresponding sides of the first triangle. [CBSE 2008C]

5. Construct a $\triangle ABC$ with $BC = 7$ cm, $\angle B = 60^\circ$ and $AB = 6$ cm. Construct another triangle whose sides are $\frac{3}{4}$ times the corresponding sides of $\triangle ABC$. [CBSE 2008, '09, '15]
6. Construct a $\triangle ABC$ in which $AB = 6$ cm, $\angle A = 30^\circ$ and $\angle B = 60^\circ$. Construct another $\triangle AB'C'$ similar to $\triangle ABC$ with base $AB' = 8$ cm. [CBSE 2015]
7. Construct a $\triangle ABC$ in which $BC = 8$ cm, $\angle B = 45^\circ$ and $\angle C = 60^\circ$. Construct another triangle similar to $\triangle ABC$ such that its sides are $\frac{3}{5}$ of the corresponding sides of $\triangle ABC$. [CBSE 2010, '12, '14]
8. To construct a triangle similar to $\triangle ABC$ in which $BC = 4.5$ cm, $\angle B = 45^\circ$ and $\angle C = 60^\circ$, using a scale factor of $\frac{3}{7}$, BC will be divided in the ratio
 (a) 3 : 4 (b) 4 : 7 (c) 3 : 10 (d) 3 : 7 [CBSE 2012]
9. Construct an isosceles triangle whose base is 8 cm and altitude 4 cm and then another triangle whose sides are $1\frac{1}{2}$ times the corresponding sides of the isosceles triangle.
10. Draw a right triangle in which the sides (other than hypotenuse) are of lengths 4 cm and 3 cm. Then, construct another triangle whose sides are $\frac{5}{3}$ times the corresponding sides of the given triangle. [CBSE 2011]

HINTS TO SOME SELECTED QUESTIONS

1. Divide the line segment AB in the ratio 3 : 2.
6. Construct $\triangle ABC$. Produce AB to B' such that $AB' = 8$ cm.
 Draw $B'C' \parallel BC$ such that $B'C'$ meets AC produced at C' .

CONSTRUCTION OF TANGENTS TO A CIRCLE

CONSTRUCTION OF A TANGENT TO A CIRCLE AT A GIVEN POINT ON THE CIRCLE

EXAMPLE 1 *Draw a circle of radius 2.5 cm. Take a point P on it. Construct a tangent at the point P .*

STEPS OF CONSTRUCTION

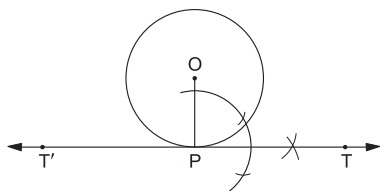
- Step 1. Draw a circle of radius 2.5 cm taking a point O as its centre.
- Step 2. Mark a point P on this circle.

Step 3. Join OP .

Step 4. Construct $\angle OPT = 90^\circ$.

Step 5. Produce TP to T' .

Then $T'PT$ is the required tangent.



CONSTRUCTION OF A TANGENT TO A CIRCLE AT A POINT ON IT WITHOUT USING THE CENTRE

EXAMPLE 2 Draw a circle of radius 5 cm. Take a point P on it. Without using the centre of the circle construct a tangent at the point P . [CBSE 2002]

STEPS OF CONSTRUCTION

Step 1. Draw a circle of radius 5 cm.

Step 2. Mark a point P on it.

Step 3. Draw any chord PQ .

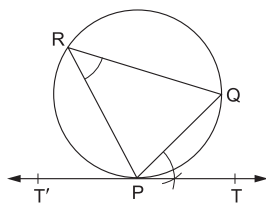
Step 4. Take a point R on the major arc QP .

Step 5. Join PR and RQ .

Step 6. Make $\angle QPT = \angle PRQ$.

Step 7. Produce TP to T' , as shown in the figure.

Then, $T'PT$ is the required tangent at P .



CONSTRUCTION OF TANGENTS TO A CIRCLE FROM A POINT OUTSIDE IT WHEN THE CENTRE OF THE CIRCLE IS KNOWN

EXAMPLE 3 Draw a circle of radius 2 cm with centre O and take a point P outside the circle such that $OP = 6.5$ cm. From P , draw two tangents to the circle.

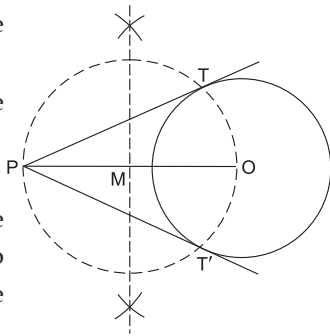
STEPS OF CONSTRUCTION

Step 1. Draw a circle with O as centre and radius 2 cm.

Step 2. Mark a point P outside the circle such that $OP = 6.5$ cm.

Step 3. Join OP and bisect it at M .

Step 4. Draw a circle with M as centre and radius equal to MP , to intersect the given circle at the points T and T' .



Step 5. Join PT and PT' .

Then, PT and PT' are the required tangents.

EXAMPLE 4 Write the steps of construction for drawing two tangents to a circle of radius 6 cm from a point 10 cm away from its centre. [CBSE 2013C]

STEPS OF CONSTRUCTION

Step 1. Draw a circle with O as centre and radius 6 cm.

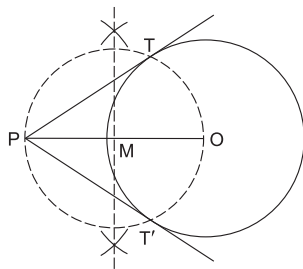
Step 2. Mark a point P outside the circle such that $OP = 10$ cm.

Step 3. Join OP and bisect it at a point M .

Step 4. Draw a circle with M as the centre and radius equal to MP , to intersect the given circle at the points T and T' .

Step 5. Join PT and PT' .

Then, PT and PT' are the required tangents.



EXAMPLE 5 Draw a line segment AB of length 7 cm. Taking A as centre, draw a circle of radius 3 cm and taking B as centre, draw another circle of radius 2 cm. Construct tangents to each circle from the centre of the other circle. [CBSE 2011, '15]

STEPS OF CONSTRUCTION

Step 1. Draw a line segment $AB = 7$ cm.

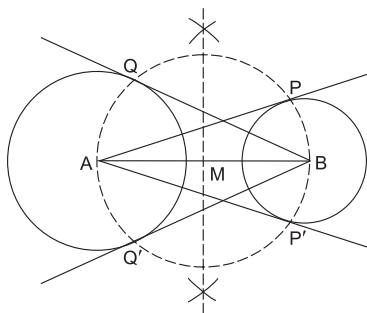
Step 2. With A as centre and radius 3 cm, draw a circle.

Step 3. With B as centre and radius 2 cm, draw another circle.

Step 4. Bisect the line segment AB at M .

Step 5. With M as centre and radius equal to AM , draw a circle intersecting the first circle at Q and Q' and the second circle at P and P' .

Step 6. Join AP and AP' . Also, join BQ and BQ' .



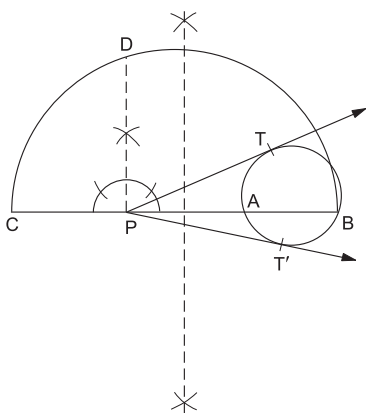
Then, BQ and BQ' are the required tangents to the first circle and AP and AP' are the required tangents to the second circle.

CONSTRUCTION OF TANGENTS TO A CIRCLE FROM A POINT OUTSIDE IT WITHOUT USING THE CENTRE

EXAMPLE 6 Draw a circle of radius 2.5 cm and take a point P outside it. Without using the centre of the circle, draw two tangents to the circle from the point P .

STEPS OF CONSTRUCTION

- Step 1. Draw a circle of radius 2.5 cm and take a point p outside it.
 - Step 2. Through P draw a secant PAB to intersect the circle at A and B .
 - Step 3. Produce AP to a point C such that $PA = PC$.
 - Step 4. Draw a semicircle with CB as diameter.
 - Step 5. Draw $PD \perp CB$, intersecting the semicircle at D .
 - Step 6. With P as centre and PD as radius, draw arcs to intersect the circle at T and T' .
 - Step 7. Join PT and PT' .
- Then, PT and PT' are the required tangents.



CONSTRUCTION OF TANGENTS TO A CIRCLE INCLINED TO EACH OTHER AT A GIVEN ANGLE

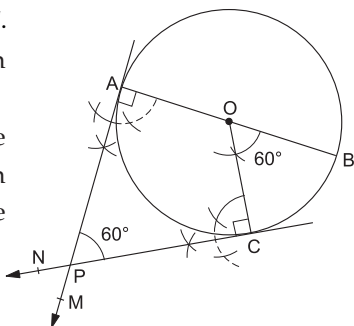
EXAMPLE 7 Draw a circle of radius 3 cm. Draw a pair of tangents to this circle, which are inclined to each other at an angle of 60° .

STEPS OF CONSTRUCTION

- Step 1. Draw a circle with O as centre and radius = 3 cm.
- Step 2. Draw any diameter AOB of this circle.
- Step 3. Construct $\angle BOC = 60^\circ$ such that radius OC meets the circle at C .

Step 4. Draw $AM \perp AB$ and $CN \perp OC$.
Let AM and CN intersect each other at P .

Then, PA and PC are the desired tangents to the given circle, inclined at an angle of 60° .



PROOF $\angle AOC = (180^\circ - 60^\circ) = 120^\circ$.

In quad. $OAPC$, we have

$$\angle OAP = 90^\circ, \angle AOC = 120^\circ, \angle OCP = 90^\circ.$$

$$\therefore \angle APC = [360^\circ - (90^\circ + 120^\circ + 90^\circ)] = 60^\circ.$$

EXAMPLE 8 Draw a circle of radius 3.5 cm. Draw two tangents to the circle which are perpendicular to each other. [CBSE 2015]

STEPS OF CONSTRUCTION

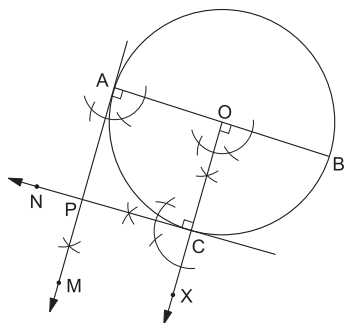
Step 1. Draw a circle with O as centre and radius = 3.5 cm.

Step 2. Draw any diameter AOB of this circle.

Step 3. Construct $\angle BOX = 90^\circ$ such that ray OX meets the circle at C .

Step 4. Draw $AM \perp AB$ and $CN \perp OC$.
Let AM and CN intersect each other at a point P .

Then, PA and PC are the required tangents (which are perpendicular to each other).



PROOF $\angle AOC = (180^\circ - \angle BOC) = 180^\circ - 90^\circ = 90^\circ$.

In quad. $OAPC$, we have

$$\angle OAP = 90^\circ, \angle AOC = 90^\circ, \angle OCP = 90^\circ.$$

$$\therefore \angle APC = [360^\circ - (90^\circ + 90^\circ + 90^\circ)] = 90^\circ.$$

CONSTRUCTION OF TANGENTS TO A CIRCLE FROM A POINT ON A LARGER CIRCLE CONCENTRIC WITH THE FIRST ONE

EXAMPLE 9 Draw two concentric circles of radii 2 cm and 5 cm. Taking a point on the outer circle, construct a pair of tangents to the other. Measure

the lengths of the two tangents. Also, verify the measurement by actual calculation.

[CBSE 2013C]

STEPS OF CONSTRUCTION

Step 1. Mark a point O on the paper.

Step 2. With O as centre and radii 2 cm and 5 cm, draw two concentric circles.

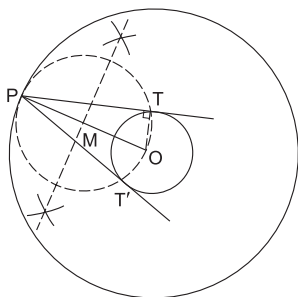
Step 3. Mark a point P on the outer circle.

Step 4. Join OP and bisect it at a point M .

Step 5. Draw a circle with M as the centre and radius equal to MP , to intersect the inner circle in points T and T' .

Step 6. Join PT and PT' .

Then, PT and PT' are the required tangents.



Measurement Upon measuring the two tangents, we find $PT = PT' = 4.6$ cm.

Calculation of the length of tangent

Join OT to form a right $\triangle OTP$ [$\because PT$ is a tangent at point T].

Then, $PT = \sqrt{OP^2 - OT^2} = \sqrt{5^2 - 2^2} = \sqrt{25 - 4} = \sqrt{21} \approx 4.58$ cm.

[$\because OP$ = radius of larger circle = 5 cm,

OT = radius of smaller circle = 2 cm.]

EXAMPLE 10 Construct a right triangle ABC with $AB = 6$ cm, $BC = 8$ cm and $\angle B = 90^\circ$. Draw BD , the perpendicular from B on AC . Draw the circle through B , C and D and construct the tangents from A to this circle.

[CBSE 2014, '15]

STEPS OF CONSTRUCTION

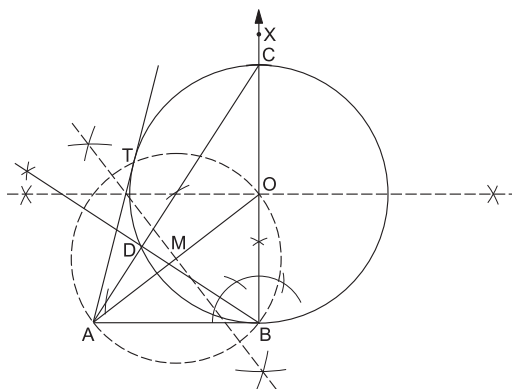
Step 1. Draw a line segment $AB = 6$ cm.

Step 2. At B , construct $\angle ABX = 90^\circ$.

Step 3. With B as centre and radius 8 cm, draw an arc cutting ray BX at C .

Step 4. Join AC .

Thus, $\triangle ABC$ is obtained.



Step 5. From B , draw $BD \perp AC$.

Step 6. Bisect BC at point O .

Step 7. With O as centre and radius OB , draw a circle. This circle passes through B, C and D .

Thus, the required circle is obtained.

Step 8. Join AO and bisect it at M .

Step 9. With M as the centre and radius equal to AM , draw a circle cutting the previous circle at the points T and B .

Step 10. Join AT .

Then, AT and AB are the required tangents.

EXERCISE 9B

1. Draw a circle of radius 3 cm. From a point P , 7 cm away from the centre of the circle, draw two tangents to the circle. Also, measure the lengths of the tangents. [CBSE 2010]
2. Draw two tangents to a circle of radius 3.5 cm from a point P at a distance of 6.2 cm from its centre. [CBSE 2013]
3. Draw a circle of radius 3 cm. Take two points P and Q on one of its diameters extended on both sides, each at a distance of 7 cm on opposite sides of its centre. Draw tangents to the circle from these two points P and Q . [CBSE 2017]
4. Draw a circle with centre O and radius 4 cm. Draw any diameter AB of this circle. Construct tangents to the circle at each of the two end points of the diameter AB .

5. Draw a circle with the help of a bangle. Take any point P outside the circle. Construct the pair of tangents from the point P to the circle.

HINT See Example 6.

6. Draw a line segment AB of length 8 cm. Taking A as centre, draw a circle of radius 4 cm and taking B as centre, draw another circle of radius 3 cm. Construct tangents to each circle from the centre of the other circle. [CBSE 2014]

7. Draw a circle of radius 4.2 cm. Draw a pair of tangents to this circle inclined to each other at an angle of 45° .

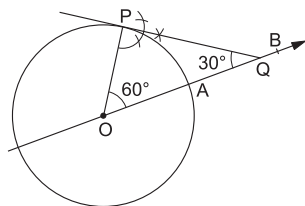
8. Write the steps of construction for drawing a pair of tangents to a circle of radius 3 cm, which are inclined to each other at an angle of 60° .

[CBSE 2011, '12, '14]

9. Draw a circle of radius 3 cm. Draw a tangent to the circle making an angle of 30° with a line passing through the centre.

HINT Draw a circle with centre O and radius 3 cm. Draw a radius OA and produce it to B . Make $\angle AOP = 60^\circ$. Draw $PQ \perp OP$, meeting OB at Q .

Then, PQ is the desired tangent such that $\angle OQP = 30^\circ$.



10. Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length. Also, verify the measurement by actual calculation.

11. Draw two concentric circles of radii 3 cm and 5 cm. Taking a point on the outer circle, construct the pair of tangents to the inner circle.

[CBSE 2017]

12. Write the steps of construction to construct the tangents to a circle from an external point.

[CBSE 2017]

TEST YOURSELF

Long-Answer Questions

- Draw a line segment AB of length 5.4 cm. Divide it into six equal parts. Write the steps of construction.
- Draw a line segment AB of length 6.5 cm and divide it in the ratio 4 : 7. Measure each of the two parts.

3. Construct a $\triangle ABC$ in which $BC = 6.5$ cm, $AB = 4.5$ cm and $\angle ABC = 60^\circ$.
Construct a triangle similar to this triangle whose sides are $\frac{3}{4}$ of the corresponding sides of $\triangle ABC$.
[CBSE 2009]
4. Construct a $\triangle ABC$ in which $BC = 5$ cm, $\angle C = 60^\circ$ and altitude from A equal to 3 cm. Construct a $\triangle ADE$ similar to $\triangle ABC$ such that each side of $\triangle ADE$ is $\frac{3}{2}$ times the corresponding side of $\triangle ABC$. Write the steps of construction.
5. Construct an isosceles triangle whose base is 9 cm and altitude 5 cm. Construct another triangle whose sides are $\frac{3}{4}$ of the corresponding sides of the first isosceles triangle.
[CBSE 2009C]
6. Draw a $\triangle ABC$, right-angled at B such that $AB = 3$ cm and $BC = 4$ cm. Now, construct a triangle similar to $\triangle ABC$, each of whose sides is $\frac{7}{5}$ times the corresponding side of $\triangle ABC$.
7. Draw a circle of radius 4.8 cm. Take a point P on it. Without using the centre of the circle, construct a tangent at the point P . Write the steps of construction.
8. Draw a circle of radius 3.5 cm. Draw a pair of tangents to this circle which are inclined to each other at an angle of 60° . Write the steps of construction.
9. Draw a circle of radius 4 cm. Draw tangent to the circle making an angle of 60° with a line passing through the centre.
10. Draw two concentric circles of radii 4 cm and 6 cm. Construct a tangent to the smaller circle from a point on the larger circle. Measure the length of this tangent.





Trigonometric Ratios

TRIGONOMETRY It is that branch of mathematics which deals with the measurement of angles and the problems allied with angles.

TRIGONOMETRIC RATIOS (T-RATIOS) OF AN ACUTE ANGLE OF A RIGHT TRIANGLE

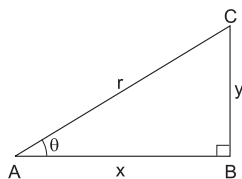
Let $\angle BAC = \theta$ be an acute angle of a right-angled $\triangle ABC$.

In right-angled $\triangle ABC$, let

base = $AB = x$ units,

perpendicular = $BC = y$ units

and hypotenuse = $AC = r$ units.



We define the following ratios, known as Trigonometric Ratios for θ .

(i) $\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{y}{r}$, and is written as $\sin \theta$.

(ii) $\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{x}{r}$, and is written as $\cos \theta$.

(iii) $\tan \theta = \frac{\text{perpendicular}}{\text{base}} = \frac{y}{x}$, and is written as $\tan \theta$.

(iv) $\text{cosec} \theta = \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{r}{y}$, and is written as $\text{cosec} \theta$.

(v) $\sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{r}{x}$, and is written as $\sec \theta$.

(vi) $\cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{x}{y}$, and is written as $\cot \theta$.

RECIPROCAL RELATION

Clearly, we have

(i) $\text{cosec} \theta = \frac{1}{\sin \theta}$

(ii) $\sec \theta = \frac{1}{\cos \theta}$

(iii) $\cot \theta = \frac{1}{\tan \theta}$

SUMMARY

Consider a $\triangle ABC$ in which $\angle B = 90^\circ$ and $\angle A = \theta$.

Let $AB = x$, $BC = y$ and $AC = r$. Then,

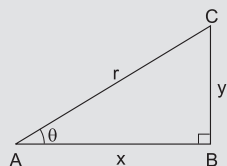
$$(i) \sin \theta = \frac{y}{r} \quad (ii) \cos \theta = \frac{x}{r} \quad (iii) \tan \theta = \frac{y}{x}$$

$$(iv) \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{r}{y} \quad (v) \sec \theta = \frac{1}{\cos \theta} = \frac{r}{x}$$

$$(vi) \cot \theta = \frac{1}{\tan \theta} = \frac{x}{y}$$

Also, we have

$$(i) \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad (ii) \sec \theta = \frac{1}{\cos \theta} \quad (iii) \cot \theta = \frac{1}{\tan \theta}$$

**T-RATIOS OF AN ANGLE ARE WELL-DEFINED**

THEOREM *Show that the value of each of the trigonometric ratios of an angle does not depend on the size of the triangle. It only depends on the angle.*

PROOF Consider a $\triangle ABC$ in which $\angle B = 90^\circ$ and $\angle A = \theta^\circ$.

Take a point P on AC and draw $PQ \perp AB$.

Then, $\triangle AQP$ is similar to $\triangle ABC$.

$$\therefore \frac{AQ}{AB} = \frac{AP}{AC} = \frac{PQ}{CB}$$

$$\Rightarrow \frac{PQ}{AP} = \frac{CB}{AC} = \sin \theta.$$

$$\text{Similarly, } \frac{AQ}{AP} = \frac{AB}{AC} = \cos \theta$$

$$\text{and } \frac{PQ}{AQ} = \frac{CB}{AB} = \tan \theta.$$

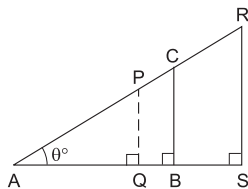
Similarly, if we produce AC to R and draw $RS \perp AB$ produced then $\triangle ASR$ is similar to $\triangle ABC$.

$$\therefore \frac{AS}{AB} = \frac{AR}{AC} = \frac{RS}{CB}$$

$$\Rightarrow \frac{RS}{AR} = \frac{CB}{AC} = \sin \theta.$$

$$\text{Similarly, } \frac{AS}{AR} = \frac{AB}{AC} = \cos \theta$$

$$\text{and } \frac{RS}{AS} = \frac{CB}{AB} = \tan \theta.$$



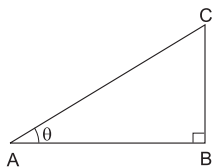
Hence, the trigonometric ratios of an angle do not depend on the size of the triangle. They only depend on the angle.

REMARK Consider a $\triangle ABC$ in which $\angle B = 90^\circ$ and $\angle A = \theta^\circ$.

We know that in a right triangle, the hypotenuse is the longest side.

$$\therefore \sin \theta = \frac{BC}{AC} < 1 \text{ and } \cos \theta = \frac{AB}{AC} < 1.$$

Thus, $\sin \theta < 1$ and $\cos \theta < 1$.



POWER OF T-RATIOS We write

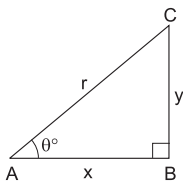
$$(\sin \theta)^2 = \sin^2 \theta; \quad (\sin \theta)^3 = \sin^3 \theta; \quad (\cos \theta)^3 = \cos^3 \theta; \text{ etc.}$$

QUOTIENT RELATION OF T-RATIOS

THEOREM 1 For any acute angle θ , prove that

$$(i) \tan \theta = \frac{\sin \theta}{\cos \theta}; \quad (ii) \cot \theta = \frac{\cos \theta}{\sin \theta}; \quad (iii) \tan \theta \cdot \cot \theta = 1.$$

PROOF Consider a right-angled $\triangle ABC$ in which $\angle B = 90^\circ$ and $\angle A = \theta^\circ$. Let $AB = x$ units, $BC = y$ units and $AC = r$ units. Then,



$$(i) \tan \theta = \frac{y}{x} = \frac{(y/r)}{(x/r)}$$

[dividing num. and denom. by r]

$$= \frac{\sin \theta}{\cos \theta}.$$

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta}.$$

$$(ii) \cot \theta = \frac{x}{y} = \frac{(x/r)}{(y/r)} \quad [\text{dividing num. and denom. by } r]$$

$$= \frac{\cos \theta}{\sin \theta}.$$

$$\therefore \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

$$(iii) \tan \theta \cdot \cot \theta = \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} = 1.$$

SUMMARY

$$(i) \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$(ii) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$(iii) \tan \theta \cdot \cot \theta = 1$$

SQUARE RELATION**THEOREM 2** For any acute angle θ , prove that

(i) $\sin^2\theta + \cos^2\theta = 1$;

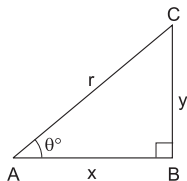
(ii) $1 + \tan^2\theta = \sec^2\theta$;

(iii) $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

PROOF Consider a right-angled $\triangle ABC$ in which $\angle B = 90^\circ$ and $\angle A = \theta^\circ$. Let $AB = x$ units, $BC = y$ units and $AC = r$ units.

Then, by Pythagoras' theorem, we have

$$x^2 + y^2 = r^2.$$



$$(i) \sin^2\theta + \cos^2\theta = \left(\frac{y}{r}\right)^2 + \left(\frac{x}{r}\right)^2 = \left(\frac{y^2}{r^2} + \frac{x^2}{r^2}\right)$$

$$= \frac{(x^2 + y^2)}{r^2} = \frac{r^2}{r^2}$$

$$[\because x^2 + y^2 = r^2]$$

$$= 1.$$

$$\therefore \sin^2\theta + \cos^2\theta = 1.$$

$$(ii) 1 + \tan^2\theta = 1 + \left(\frac{y}{x}\right)^2 = 1 + \frac{y^2}{x^2} = \frac{y^2 + x^2}{x^2} = \frac{r^2}{x^2}$$

$$[\because x^2 + y^2 = r^2]$$

$$= \left(\frac{r}{x}\right)^2 = \sec^2\theta.$$

$$\therefore 1 + \tan^2\theta = \sec^2\theta.$$

$$(iii) 1 + \cot^2\theta = 1 + \left(\frac{x}{y}\right)^2 = 1 + \frac{x^2}{y^2} = \frac{y^2 + x^2}{y^2} = \frac{r^2}{y^2}$$

$$[\because x^2 + y^2 = r^2]$$

$$= \left(\frac{r}{y}\right)^2 = \operatorname{cosec}^2\theta.$$

$$\therefore 1 + \cot^2\theta = \operatorname{cosec}^2\theta.$$

SUMMARY

(i) $\sin^2\theta + \cos^2\theta = 1$

(ii) $1 + \tan^2\theta = \sec^2\theta$

(iii) $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

SOLVED EXAMPLES

EXAMPLE 1 If $\sin A = \frac{8}{17}$, find other trigonometric ratios of $\angle A$.

SOLUTION Let us draw a $\triangle ABC$ in which $\angle B = 90^\circ$.

$$\text{Then, } \sin A = \frac{BC}{AC} = \frac{8}{17}.$$

Let $BC = 8k$ and $AC = 17k$, where k is positive.

By Pythagoras' theorem, we have

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow AB^2 = AC^2 - BC^2$$

$$\begin{aligned}\Rightarrow AB^2 &= (17k)^2 - (8k)^2 = 289k^2 - 64k^2 \\ &= 225k^2\end{aligned}$$

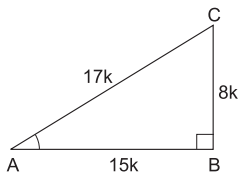
$$\Rightarrow AB = \sqrt{225k^2} = 15k.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{8k}{17k} = \frac{8}{17}; \cos A = \frac{AB}{AC} = \frac{15k}{17k} = \frac{15}{17};$$

$$\tan A = \frac{\sin A}{\cos A} = \left(\frac{8}{17} \times \frac{17}{15}\right) = \frac{8}{15};$$

$$\operatorname{cosec} A = \frac{1}{\sin A} = \frac{17}{8}; \sec A = \frac{1}{\cos A} = \frac{17}{15}$$

$$\text{and } \cot A = \frac{1}{\tan A} = \frac{15}{8}.$$



EXAMPLE 2 If $\cos A = \frac{9}{41}$, find other trigonometric ratios of $\angle A$.

SOLUTION Let us draw a $\triangle ABC$ in which $\angle B = 90^\circ$.

$$\text{Then, } \cos A = \frac{AB}{AC} = \frac{9}{41}.$$

Let $AB = 9k$ and $AC = 41k$, where k is positive.

By Pythagoras' theorem, we have

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow BC^2 = AC^2 - AB^2$$

$$\Rightarrow BC^2 = (41k)^2 - (9k)^2 = 1681k^2 - 81k^2 = 1600k^2$$

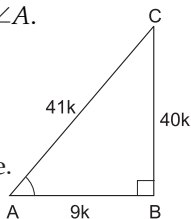
$$\Rightarrow BC = \sqrt{1600k^2} = 40k.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{40k}{41k} = \frac{40}{41}; \cos A = \frac{9}{41} \text{ (given);}$$

$$\tan A = \frac{\sin A}{\cos A} = \left(\frac{40}{41} \times \frac{41}{9}\right) = \frac{40}{9};$$

$$\operatorname{cosec} A = \frac{1}{\sin A} = \frac{41}{40}; \sec A = \frac{1}{\cos A} = \frac{41}{9}$$

$$\text{and } \cot A = \frac{1}{\tan A} = \frac{9}{40}.$$



EXAMPLE 3 If $\tan A = \sqrt{3}$, find other trigonometric ratios of $\angle A$.

SOLUTION Let us draw a $\triangle ABC$ in which $\angle B = 90^\circ$.

$$\text{Then, } \tan A = \frac{BC}{AB} = \frac{\sqrt{3}}{1}.$$

Let $BC = \sqrt{3}k$ and $AB = k$, where k is positive.

By Pythagoras' theorem, we have

$$\begin{aligned} AC^2 &= AB^2 + BC^2 = k^2 + (\sqrt{3}k)^2 \\ &= k^2 + 3k^2 = 4k^2. \end{aligned}$$

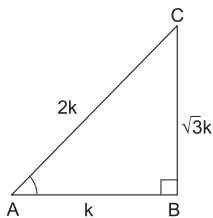
$$\therefore AC = \sqrt{4k^2} = 2k.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{\sqrt{3}k}{2k} = \frac{\sqrt{3}}{2};$$

$$\cos A = \frac{AB}{AC} = \frac{k}{2k} = \frac{1}{2};$$

$$\tan A = \sqrt{3} \text{ (given); } \operatorname{cosec} A = \frac{1}{\sin A} = \frac{2}{\sqrt{3}};$$

$$\sec A = \frac{1}{\cos A} = 2 \text{ and } \cot A = \frac{1}{\tan A} = \frac{1}{\sqrt{3}}.$$



EXAMPLE 4 If $\sec \theta = \frac{25}{7}$, find all trigonometric ratios of θ .

SOLUTION Let us draw a $\triangle ABC$ in which $\angle B = 90^\circ$. Let $\angle A = \theta^\circ$.

$$\text{Then, } \sec \theta = \frac{AC}{AB} = \frac{25}{7}.$$

Let $AC = 25k$ and $AB = 7k$, where k is positive.

By Pythagoras' theorem, we have

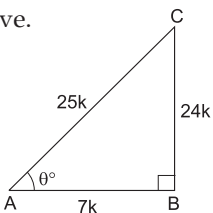
$$\begin{aligned} AC^2 &= AB^2 + BC^2 \\ \Rightarrow BC^2 &= AC^2 - AB^2 = (25k)^2 - (7k)^2 \\ &= 625k^2 - 49k^2 = 576k^2 \end{aligned}$$

$$\Rightarrow BC = \sqrt{576k^2} = 24k.$$

$$\therefore \sin \theta = \frac{BC}{AC} = \frac{24k}{25k} = \frac{24}{25}; \quad \cos \theta = \frac{1}{\sec \theta} = \frac{7}{25};$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \left(\frac{24}{25} \times \frac{25}{7} \right) = \frac{24}{7}; \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{25}{24};$$

$$\sec \theta = \frac{25}{7} \text{ (given) and } \cot \theta = \frac{1}{\tan \theta} = \frac{7}{24}.$$



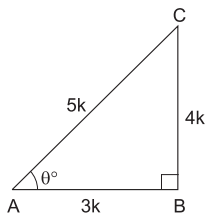
EXAMPLE 5 If $\cos \theta = \frac{3}{5}$, find the value of $\left(\frac{5 \operatorname{cosec} \theta - 4 \tan \theta}{\sec \theta + \cot \theta} \right)$.

SOLUTION Let us draw a $\triangle ABC$ in which $\angle B = 90^\circ$.

Let $\angle A = \theta^\circ$.

$$\text{Then, } \cos \theta = \frac{AB}{AC} = \frac{3}{5}.$$

Let $AB = 3k$ and $AC = 5k$, where k is positive.



By Pythagoras' theorem, we have

$$\begin{aligned}
 AC^2 &= AB^2 + BC^2 \\
 \Rightarrow BC^2 &= AC^2 - AB^2 = (5k)^2 - (3k)^2 = 25k^2 - 9k^2 = 16k^2 \\
 \Rightarrow BC &= \sqrt{16k^2} = 4k. \\
 \therefore \sec \theta &= \frac{1}{\cos \theta} = \frac{5}{3}; \tan \theta = \frac{BC}{AB} = \frac{4k}{3k} = \frac{4}{3}; \\
 \cot \theta &= \frac{1}{\tan \theta} = \frac{3}{4}; \text{ and cosec } \theta = \frac{AC}{BC} = \frac{5k}{4k} = \frac{5}{4}. \\
 \therefore \left(\frac{5 \operatorname{cosec} \theta - 4 \tan \theta}{\sec \theta + \cot \theta} \right) &= \frac{\left(5 \times \frac{5}{4} - 4 \times \frac{4}{3} \right)}{\left(\frac{5}{3} + \frac{3}{4} \right)} = \frac{\left(\frac{25}{4} - \frac{16}{3} \right)}{\left(\frac{5}{3} + \frac{3}{4} \right)} \\
 &= \frac{\frac{75 - 64}{12}}{\frac{20 + 9}{12}} = \left(\frac{11}{12} \times \frac{12}{29} \right) = \frac{11}{29}.
 \end{aligned}$$

EXAMPLE 6 If $\sec \theta = \frac{5}{4}$, show that $\frac{(2\cos \theta - \sin \theta)}{(\cot \theta - \tan \theta)} = \frac{12}{7}$.

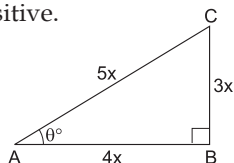
SOLUTION Consider a $\triangle ABC$ in which $\angle A = \theta$ and $\angle B = 90^\circ$.

$$\sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{AC}{AB} = \frac{5}{4} = \frac{5x}{4x} \text{ (say).}$$

$\therefore AC = 5x$ and $AB = 4x$, where x is positive.

By Pythagoras' theorem we have

$$\begin{aligned}
 AC^2 &= AB^2 + BC^2 \\
 \Rightarrow BC^2 &= AC^2 - AB^2 = (5x)^2 - (4x)^2 = 9x^2 \\
 \Rightarrow BC &= 3x.
 \end{aligned}$$



$$\therefore \cos \theta = \frac{AB}{AC} = \frac{4x}{5x} = \frac{4}{5}; \sin \theta = \frac{BC}{AC} = \frac{3x}{5x} = \frac{3}{5};$$

$$\cot \theta = \frac{AB}{BC} = \frac{4x}{3x} = \frac{4}{3}; \text{ and } \tan \theta = \frac{BC}{AB} = \frac{3x}{4x} = \frac{3}{4}.$$

$$\therefore \frac{(2\cos \theta - \sin \theta)}{(\cot \theta - \tan \theta)} = \frac{\left(2 \times \frac{4}{5} - \frac{3}{5} \right)}{\left(\frac{4}{3} - \frac{3}{4} \right)} = \frac{\left(\frac{8}{5} - \frac{3}{5} \right)}{\left(\frac{4}{3} - \frac{3}{4} \right)} = \frac{\left(\frac{5}{5} \right)}{\left(\frac{7}{12} \right)} = \frac{1}{\left(\frac{7}{12} \right)} = \frac{12}{7}.$$

EXAMPLE 7 In a $\triangle ABC$ it is given that $\angle B = 90^\circ$ and $AB : AC = 1 : \sqrt{2}$. Find the value of $\left(\frac{2 \tan A}{1 + \tan^2 A} \right)$.

SOLUTION Consider a $\triangle ABC$ in which $\angle B = 90^\circ$ and $AB : AC = 1 : \sqrt{2}$.

Let $AB = x$. Then, $AC = \sqrt{2}x$.

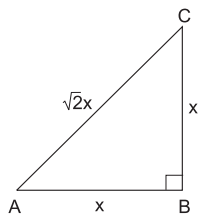
By Pythagoras' theorem, we have

$$AC^2 = AB^2 + BC^2 \Rightarrow BC^2 = AC^2 - AB^2$$

$$\Rightarrow BC^2 = (\sqrt{2}x)^2 - (x)^2 = 2x^2 - x^2 = x^2$$

$$\Rightarrow BC = x.$$

$$\therefore \tan A = \frac{BC}{AB} = \frac{x}{x} = 1.$$



$$\text{So, the given expression} = \left(\frac{2 \tan A}{1 + \tan^2 A} \right) = \left(\frac{2 \times 1}{1 + 1} \right) = \frac{2}{2} = 1.$$

EXAMPLE 8 If $3 \tan \theta = 4$, evaluate $\frac{3 \sin \theta + 2 \cos \theta}{3 \sin \theta - 2 \cos \theta}$.

SOLUTION $3 \tan \theta = 4 \Rightarrow \tan \theta = \frac{4}{3}.$

Given expression

$$= \frac{3 \sin \theta + 2 \cos \theta}{3 \sin \theta - 2 \cos \theta}$$

$$= \frac{3 \tan \theta + 2}{3 \tan \theta - 2} \quad [\text{dividing num. and denom. by } \cos \theta]$$

$$= \frac{\left(3 \times \frac{4}{3} + 2 \right)}{\left(3 \times \frac{4}{3} - 2 \right)} = \frac{6}{2} = 3. \quad \left[\because \tan \theta = \frac{4}{3} \right]$$

EXAMPLE 9 If $5 \cot \theta = 3$, find the value of $\left(\frac{5 \sin \theta - 3 \cos \theta}{4 \sin \theta + 3 \cos \theta} \right)$.

SOLUTION $5 \cot \theta = 3 \Rightarrow \cot \theta = \frac{3}{5}.$

$$\text{Given expression} = \left(\frac{5 \sin \theta - 3 \cos \theta}{4 \sin \theta + 3 \cos \theta} \right) = \frac{(5 - 3 \cot \theta)}{(4 + 3 \cot \theta)}$$

[dividing num. and denom. by $\sin \theta$]

$$= \frac{\left(5 - 3 \times \frac{3}{5} \right)}{\left(4 + 3 \times \frac{3}{5} \right)} = \frac{\left(5 - \frac{9}{5} \right)}{\left(4 + \frac{9}{5} \right)} = \left(\frac{16}{5} \times \frac{5}{29} \right) = \frac{16}{29}.$$

EXAMPLE 10 If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, show that $\tan \theta = \frac{1}{\sqrt{3}}$. [CBSE 2008]

SOLUTION $7 \sin^2 \theta + 3 \cos^2 \theta = 4$

$$\Rightarrow 4 \sin^2 \theta + 3 \sin^2 \theta + 3 \cos^2 \theta = 4$$

$$\Rightarrow 4 \sin^2 \theta + 3(\sin^2 \theta + \cos^2 \theta) = 4$$

$$\Rightarrow 4\sin^2\theta + 3 \times 1 = 4 \quad [\because \sin^2\theta + \cos^2\theta = 1]$$

$$\Rightarrow 4\sin^2\theta = 1 \Rightarrow \sin^2\theta = \frac{1}{4}.$$

$$\therefore \cos^2\theta = (1 - \sin^2\theta) = \left(1 - \frac{1}{4}\right) = \frac{3}{4}.$$

$$\therefore \tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta} = \left(\frac{1}{4} \times \frac{4}{3}\right) = \frac{1}{3}.$$

$$\text{Hence, } \tan\theta = \frac{1}{\sqrt{3}}.$$

EXAMPLE 11 If $\cot\theta = \frac{15}{8}$ then evaluate $\frac{(2 + 2\sin\theta)(1 - \sin\theta)}{(1 + \cos\theta)(2 - 2\cos\theta)}$. [CBSE 2009]

SOLUTION Given expression = $\frac{(2 + 2\sin\theta)(1 - \sin\theta)}{(1 + \cos\theta)(2 - 2\cos\theta)}$

$$= \frac{2(1 + \sin\theta)(1 - \sin\theta)}{2(1 + \cos\theta)(1 - \cos\theta)}$$

$$= \frac{(1 - \sin^2\theta)}{(1 - \cos^2\theta)} = \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta$$

$$= (\cot\theta)^2 = \left(\frac{15}{8}\right)^2 = \frac{225}{64}.$$

Hence, the value of the given expression is $\frac{225}{64}$.

EXAMPLE 12 In $\triangle ABC$, right-angled at B , $AB = 5$ cm and $BC = 12$ cm. Find the values of $\sin A$, $\sec A$, $\sin C$ and $\sec C$.

SOLUTION In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 5$ cm and $BC = 12$ cm.

By Pythagoras' theorem, we have

$$AC^2 = (AB^2 + BC^2) = \{(5)^2 + (12)^2\} \text{ cm}^2$$

$$= (25 + 144) \text{ cm}^2 = 169 \text{ cm}^2.$$

$$\therefore AC = \sqrt{169 \text{ cm}^2} = 13 \text{ cm}.$$

For T-ratios of $\angle A$, we have

$$\text{base} = AB = 5 \text{ cm},$$

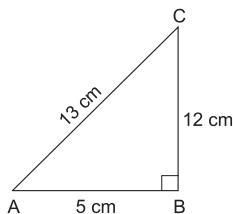
$$\text{perpendicular} = BC = 12 \text{ cm}$$

$$\text{and hypotenuse} = AC = 13 \text{ cm}.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{12}{13} \text{ and } \sec A = \frac{AC}{AB} = \frac{13}{5}.$$

For T-ratios of $\angle C$, we have

$$\text{base} = BC = 12 \text{ cm},$$



perpendicular = $AB = 5$ cm

and hypotenuse = $AC = 13$ cm.

$$\therefore \sin C = \frac{AB}{AC} = \frac{5}{13} \text{ and } \sec C = \frac{AC}{BC} = \frac{13}{12}.$$

EXAMPLE 13 In a $\triangle ABC$, $\angle B = 90^\circ$, $AB = 5$ cm and $(BC + AC) = 25$ cm. Find the values of $\sin A$, $\cos A$, $\operatorname{cosec} C$ and $\sec C$.

SOLUTION Let $BC = x$ cm. Then, $AC = (25 - x)$ cm.

By Pythagoras' theorem, we have

$$AB^2 + BC^2 = AC^2$$

$$\Rightarrow 5^2 + x^2 = (25 - x)^2$$

$$\Rightarrow 25 + x^2 = 625 - 50x + x^2$$

$$\Rightarrow 50x = 600$$

$$\Rightarrow x = 12.$$

$\therefore BC = 12$ cm, $AC = 13$ cm and $AB = 5$ cm.

For T-ratios of $\angle A$, we have

$$\sin A = \frac{BC}{AC} = \frac{12}{13} \text{ and } \cos A = \frac{AB}{AC} = \frac{5}{13}.$$

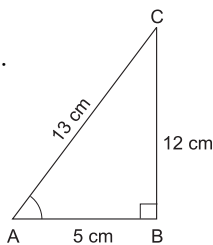
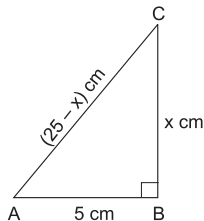
For T-ratios of $\angle C$, we have

base, $BC = 12$ cm,

perpendicular, $AB = 5$ cm

and hypotenuse, $AC = 13$ cm.

$$\therefore \operatorname{cosec} C = \frac{AC}{AB} = \frac{13}{5} \text{ and } \sec C = \frac{AC}{BC} = \frac{13}{12}.$$



EXAMPLE 14 In a $\triangle ABC$, $\angle B = 90^\circ$, $AB = 7$ cm and $(AC - BC) = 1$ cm. Find the values of $\sin A$, $\cos A$, $\sin C$ and $\cos C$.

SOLUTION Let $BC = x$ cm. Then, $AC = (x + 1)$ cm.

By Pythagoras' theorem, we have

$$AB^2 + BC^2 = AC^2$$

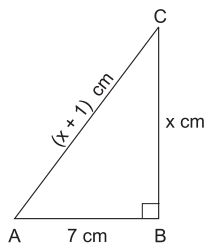
$$\Rightarrow 7^2 + x^2 = (x + 1)^2$$

$$\Rightarrow 49 + x^2 = x^2 + 2x + 1$$

$$\Rightarrow 2x = 48$$

$$\Rightarrow x = 24.$$

$\therefore BC = 24$ cm, $AC = 25$ cm and $AB = 7$ cm.

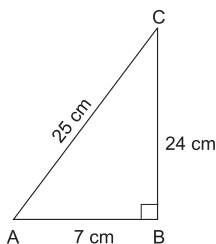


For T-ratios of $\angle A$, we have

$$\sin A = \frac{BC}{AC} = \frac{24}{25} \text{ and } \cos A = \frac{AB}{AC} = \frac{7}{25}.$$

For T-ratios of $\angle C$, we have

$$\sin C = \frac{AB}{AC} = \frac{7}{25} \text{ and } \cos C = \frac{BC}{AC} = \frac{24}{25}.$$



EXAMPLE 15 In a $\triangle ABC$, $\angle C = 90^\circ$ and $\tan A = \frac{1}{\sqrt{3}}$. Find the values of:

- (i) $(\sin A \cdot \cos B + \cos A \cdot \sin B)$ (ii) $(\cos A \cdot \cos B - \sin A \cdot \sin B)$

[CBSE 2008]

SOLUTION Consider a $\triangle ABC$ in which $\angle C = 90^\circ$ and $\tan A = \frac{1}{\sqrt{3}}$.

$$\text{Then, } \tan A = \frac{1}{\sqrt{3}} \Rightarrow \frac{BC}{AC} = \frac{1}{\sqrt{3}}.$$

Let $BC = x$. Then, $AC = \sqrt{3}x$.

By Pythagoras' theorem, we have

$$\begin{aligned} AB^2 &= AC^2 + BC^2 \\ &= (\sqrt{3}x)^2 + x^2 = (3x^2 + x^2) = 4x^2 \end{aligned}$$

$$\Rightarrow AB = \sqrt{4x^2} = 2x.$$

For T-ratios of $\angle A$, we have

base = $AC = \sqrt{3}x$, perpendicular = $BC = x$ and
hypotenuse = $AB = 2x$.

$$\therefore \sin A = \frac{BC}{AB} = \frac{x}{2x} = \frac{1}{2} \text{ and } \cos A = \frac{AC}{AB} = \frac{\sqrt{3}x}{2x} = \frac{\sqrt{3}}{2}.$$

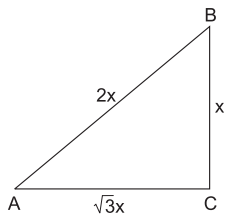
For T-ratios of $\angle B$, we have

base = $BC = x$, perpendicular = $AC = \sqrt{3}x$ and
hypotenuse = $AB = 2x$.

$$\therefore \sin B = \frac{AC}{AB} = \frac{\sqrt{3}x}{2x} = \frac{\sqrt{3}}{2} \text{ and } \cos B = \frac{BC}{AB} = \frac{x}{2x} = \frac{1}{2}.$$

$$\begin{aligned} \text{(i) } (\sin A \cdot \cos B + \cos A \cdot \sin B) &= \left(\frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} \right) \\ &= \left(\frac{1}{4} + \frac{3}{4} \right) = 1. \end{aligned}$$

$$\therefore (\sin A \cdot \cos B + \cos A \cdot \sin B) = 1.$$



$$(ii) (\cos A \cos B - \sin A \sin B) = \left(\frac{\sqrt{3}}{2} \times \frac{1}{2} - \frac{1}{2} \times \frac{\sqrt{3}}{2}\right) = 0.$$

$$\therefore (\cos A \cos B - \sin A \sin B) = 0.$$

EXAMPLE 16 If $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$ then prove that $\angle A = \angle B$.

SOLUTION Let $\triangle ACD$ and $\triangle BEF$ be two right triangles given in such a way that $\cos A = \cos B$. Then,

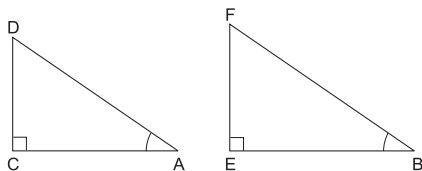
$$\cos A = \cos B$$

$$\Rightarrow \frac{AC}{AD} = \frac{BE}{BF}$$

$$\Rightarrow \frac{AC}{BE} = \frac{AD}{BF} = k \text{ (say)}$$

$$\Rightarrow AC = k(BE)$$

$$\text{and } AD = k(BF).$$



... (i)

$$\therefore \frac{CD}{EF} = \frac{\sqrt{AD^2 - AC^2}}{\sqrt{BF^2 - BE^2}} \quad [\text{using Pythagoras' theorem}]$$

$$= \frac{k\sqrt{BF^2 - BE^2}}{\sqrt{BF^2 - BE^2}} = k \quad [\text{using (i)}].$$

$$\text{Thus, we have: } \frac{AC}{BE} = \frac{AD}{BF} = \frac{CD}{EF}.$$

$$\therefore \triangle ACD \sim \triangle BEF \text{ and hence, } \angle A = \angle B.$$

EXERCISE 10

1. If $\sin \theta = \frac{\sqrt{3}}{2}$, find the value of all T-ratios of θ .
2. If $\cos \theta = \frac{7}{25}$, find the values of all T-ratios of θ .
3. If $\tan \theta = \frac{15}{8}$, find the values of all T-ratios of θ .
4. If $\cot \theta = 2$, find the values of all T-ratios of θ .
5. If $\operatorname{cosec} \theta = \sqrt{10}$, find the values of all T-ratios of θ .
6. If $\sin \theta = \frac{a^2 - b^2}{a^2 + b^2}$, find the values of all T-ratios of θ .
7. If $15 \cot A = 8$, find the values of $\sin A$ and $\sec A$.
8. If $\sin A = \frac{9}{41}$, find the values of $\cos A$ and $\tan A$.

9. If $\cos \theta = 0.6$, show that $(5\sin \theta - 3\tan \theta) = 0$.
10. If $\operatorname{cosec} \theta = 2$, show that $\left(\cot \theta + \frac{\sin \theta}{1 + \cos \theta}\right) = 2$.
11. If $\tan \theta = \frac{1}{\sqrt{7}}$, show that $\frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)} = \frac{3}{4}$.
12. If $\tan \theta = \frac{20}{21}$, show that $\frac{(1 - \sin \theta + \cos \theta)}{(1 + \sin \theta + \cos \theta)} = \frac{3}{7}$.
13. If $\sec \theta = \frac{5}{4}$, show that $\frac{(\sin \theta - 2\cos \theta)}{(\tan \theta - \cot \theta)} = \frac{12}{7}$.
14. If $\cot \theta = \frac{3}{4}$, show that $\sqrt{\frac{\sec \theta - \operatorname{cosec} \theta}{\sec \theta + \operatorname{cosec} \theta}} = \frac{1}{\sqrt{7}}$.
15. If $\sin \theta = \frac{3}{4}$, show that $\sqrt{\frac{\operatorname{cosec}^2 \theta - \cot^2 \theta}{\sec^2 \theta - 1}} = \frac{\sqrt{7}}{3}$.
16. If $\sin \theta = \frac{a}{b}$, show that $(\sec \theta + \tan \theta) = \sqrt{\frac{b+a}{b-a}}$.
17. If $\cos \theta = \frac{3}{5}$, show that $\frac{(\sin \theta - \cot \theta)}{2\tan \theta} = \frac{3}{160}$.
18. If $\tan \theta = \frac{4}{3}$, show that $(\sin \theta + \cos \theta) = \frac{7}{5}$.
19. If $\tan \theta = \frac{a}{b}$, show that $\frac{(a\sin \theta - b\cos \theta)}{(a\sin \theta + b\cos \theta)} = \frac{(a^2 - b^2)}{(a^2 + b^2)}$.
20. If $3\tan \theta = 4$, show that $\frac{(4\cos \theta - \sin \theta)}{(2\cos \theta + \sin \theta)} = \frac{4}{5}$.
21. If $3\cot \theta = 2$, show that $\frac{(4\sin \theta - 3\cos \theta)}{(2\sin \theta + 6\cos \theta)} = \frac{1}{3}$.
22. If $3\cot \theta = 4$, show that $\frac{(1 - \tan^2 \theta)}{(1 + \tan^2 \theta)} = (\cos^2 \theta - \sin^2 \theta)$.
23. If $\sec \theta = \frac{17}{8}$, verify that $\frac{(3 - 4\sin^2 \theta)}{(4\cos^2 \theta - 3)} = \frac{(3 - \tan^2 \theta)}{(1 - 3\tan^2 \theta)}$.
24. In the adjoining figure, $\angle B = 90^\circ$, $\angle BAC = \theta^\circ$, $BC = CD = 4$ cm and $AD = 10$ cm. Find (i) $\sin \theta$ and (ii) $\cos \theta$.

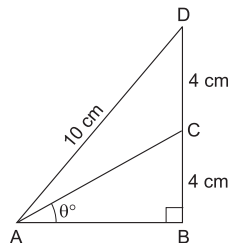
HINT $AB^2 = (AD^2 - BD^2) = 36$ cm²

$\therefore AB = 6$ cm.

$AC^2 = (AB^2 + BC^2) = 52$ cm²

$\therefore AC = 2\sqrt{13}$ cm.

Thus, $AB = 6$ cm and $AC = 2\sqrt{13}$ cm.



25. In a $\triangle ABC$, $\angle B = 90^\circ$, $AB = 24$ cm and $BC = 7$ cm.
Find (i) $\sin A$ (ii) $\cos A$ (iii) $\sin C$ (iv) $\cos C$.

26. In a $\triangle ABC$, $\angle C = 90^\circ$, $\angle ABC = \theta^\circ$, $BC = 21$ units and $AB = 29$ units.

Show that $(\cos^2 \theta - \sin^2 \theta) = \frac{41}{841}$.

27. In a $\triangle ABC$, $\angle B = 90^\circ$, $AB = 12$ cm and $BC = 5$ cm.

Find (i) $\cos A$ (ii) $\operatorname{cosec} A$ (iii) $\cos C$ (iv) $\operatorname{cosec} C$.

28. If $\sin \alpha = \frac{1}{2}$, prove that $(3\cos \alpha - 4\cos^3 \alpha) = 0$.

29. In a $\triangle ABC$, $\angle B = 90^\circ$ and $\tan A = \frac{1}{\sqrt{3}}$. Prove that

(i) $\sin A \cdot \cos C + \cos A \cdot \sin C = 1$ (ii) $\cos A \cdot \cos C - \sin A \cdot \sin C = 0$

30. If $\angle A$ and $\angle B$ are acute angles such that $\sin A = \sin B$ then prove that $\angle A = \angle B$.

31. If $\angle A$ and $\angle B$ are acute angles such that $\tan A = \tan B$ then prove that $\angle A = \angle B$.

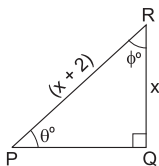
32. In a right $\triangle ABC$, right-angled at B , if $\tan A = 1$ then verify that $2\sin A \cdot \cos A = 1$.

33. In the figure of $\triangle PQR$, $\angle P = \theta^\circ$ and $\angle R = \phi^\circ$.

Find (i) $(\sqrt{x+1})\cot \phi$

(ii) $(\sqrt{x^3+x^2})\tan \theta$

(iii) $\cos \theta$



34. If $x = \operatorname{cosec} A + \cos A$ and $y = \operatorname{cosec} A - \cos A$ then prove that

$$\left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1 = 0.$$

35. If $x = \cot A + \cos A$ and $y = \cot A - \cos A$, prove that

$$\left(\frac{x-y}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 = 1.$$

ANSWERS (EXERCISE 10)

1. $\sin \theta = \frac{\sqrt{3}}{2}$, $\cos \theta = \frac{1}{2}$, $\tan \theta = \sqrt{3}$, $\cot \theta = \frac{1}{\sqrt{3}}$, $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$, $\sec \theta = 2$

2. $\sin \theta = \frac{24}{25}$, $\cos \theta = \frac{7}{25}$, $\tan \theta = \frac{24}{7}$, $\cot \theta = \frac{7}{24}$, $\operatorname{cosec} \theta = \frac{25}{24}$, $\sec \theta = \frac{25}{7}$

3. $\sin \theta = \frac{15}{17}$, $\cos \theta = \frac{8}{17}$, $\tan \theta = \frac{15}{8}$, $\cot \theta = \frac{8}{15}$, $\operatorname{cosec} \theta = \frac{17}{15}$, $\sec \theta = \frac{17}{8}$

4. $\sin \theta = \frac{1}{\sqrt{5}}$, $\cos \theta = \frac{2}{\sqrt{5}}$, $\tan \theta = \frac{1}{2}$, $\cot \theta = 2$, $\operatorname{cosec} \theta = \sqrt{5}$, $\sec \theta = \frac{\sqrt{5}}{2}$

5. $\sin \theta = \frac{1}{\sqrt{10}}, \cos \theta = \frac{3}{\sqrt{10}}, \tan \theta = \frac{1}{3}, \cot \theta = 3, \operatorname{cosec} \theta = \sqrt{10}, \sec \theta = \frac{\sqrt{10}}{3}$
6. $\sin \theta = \frac{a^2 - b^2}{a^2 + b^2}, \cos \theta = \frac{2ab}{a^2 + b^2}, \tan \theta = \frac{a^2 - b^2}{2ab},$
 $\operatorname{cosec} \theta = \frac{a^2 + b^2}{a^2 - b^2}, \sec \theta = \frac{a^2 + b^2}{2ab}, \cot \theta = \frac{2ab}{a^2 - b^2}$
7. $\sin A = \frac{15}{17}, \sec A = \frac{17}{8}$ 8. $\cos A = \frac{40}{41}, \tan A = \frac{9}{40}$
24. (i) $\sin \theta = \frac{2\sqrt{13}}{13}$ (ii) $\cos \theta = \frac{3\sqrt{13}}{13}$
25. (i) $\frac{7}{25}$ (ii) $\frac{24}{25}$ (iii) $\frac{24}{25}$ (iv) $\frac{7}{25}$
27. (i) $\frac{12}{13}$ (ii) $\frac{13}{5}$ (iii) $\frac{5}{13}$ (iv) $\frac{13}{12}$
33. (i) $\frac{x}{2}$ (ii) $\frac{x^2}{2}$ (iii) $\frac{2\sqrt{x+1}}{(x+2)}$

HINTS TO SOME SELECTED QUESTIONS

6. In right $\triangle ABC$, $\angle B = 90^\circ$ and $\angle BAC = \theta$.

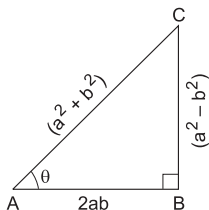
Given: $\sin \theta = \frac{a^2 - b^2}{a^2 + b^2} = \frac{BC}{AC}$.

Let $BC = (a^2 - b^2)$ and $AC = (a^2 + b^2)$. Then,

$$AB^2 = (AC^2 - BC^2) = (a^2 + b^2)^2 - (a^2 - b^2)^2 = 4a^2b^2$$

$$\Rightarrow AB = 2ab.$$

Now, we can find all the T-ratios of θ .

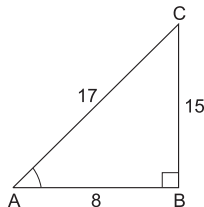


7. $\cot A = \frac{8}{15} \Rightarrow \tan A = \frac{15}{8} = \frac{BC}{AB}$.

$$\therefore AC^2 = AB^2 + BC^2 = (8)^2 + (15)^2 = 64 + 225 = 289$$

$$\Rightarrow AC = \sqrt{289} = 17.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{15}{17} \text{ and } \sec A = \frac{AC}{AB} = \frac{17}{8}.$$

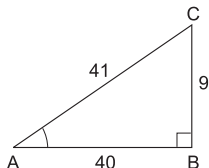


8. $\sin A = \frac{9}{41} = \frac{BC}{AC}$.

$$\therefore AB^2 = AC^2 - BC^2 = (41)^2 - (9)^2 = 1681 - 81 = 1600$$

$$\Rightarrow AB = \sqrt{1600} = 40.$$

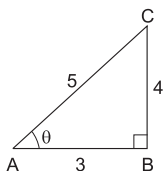
$$\therefore \cos A = \frac{AB}{AC} = \frac{40}{41} \text{ and } \tan A = \frac{BC}{AB} = \frac{9}{40}.$$



$$9. \cos \theta = \frac{6}{10} = \frac{3}{5} = \frac{AB}{AC}.$$

$$BC^2 = AC^2 - AB^2 = (5)^2 - (3)^2 = 25 - 9 = 16 \Rightarrow BC = 4.$$

$$\therefore \sin \theta = \frac{4}{5} \text{ and } \tan \theta = \frac{4}{3}.$$

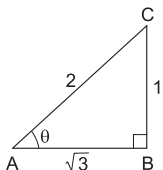


$$10. \operatorname{cosec} \theta = 2 \Rightarrow \sin \theta = \frac{1}{2} = \frac{BC}{AC}.$$

$$AB^2 = AC^2 - BC^2 = (2)^2 - (1)^2 = 4 - 1 = 3 \Rightarrow AB = \sqrt{3}.$$

$$\therefore \sin \theta = \frac{1}{2}, \cos \theta = \frac{\sqrt{3}}{2} \text{ and } \cot \theta = \sqrt{3}.$$

$$\text{Given expression} = \left\{ \sqrt{3} + \frac{1}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})} \right\} = 2.$$

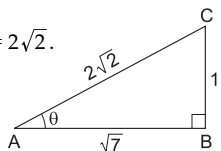


$$11. \tan \theta = \frac{1}{\sqrt{7}} = \frac{BC}{AB}.$$

$$\therefore AC^2 = AB^2 + BC^2 = (\sqrt{7})^2 + (1)^2 = (7+1) = 8 \Rightarrow AC = \sqrt{8} = 2\sqrt{2}.$$

$$\therefore \operatorname{cosec} \theta = \frac{AC}{BC} = \frac{2\sqrt{2}}{1} = 2\sqrt{2} \Rightarrow \operatorname{cosec}^2 \theta = 8$$

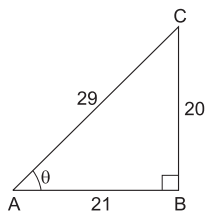
$$\text{and } \sec \theta = \frac{AC}{AB} = \frac{2\sqrt{2}}{\sqrt{7}} \Rightarrow \sec^2 \theta = \frac{8}{7}.$$



$$12. \tan \theta = \frac{20}{21} = \frac{BC}{AB}.$$

$$\therefore AC^2 = AB^2 + BC^2 = (21)^2 + (20)^2 = 441 + 400 = 841$$

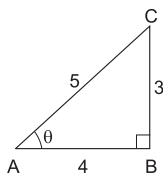
$$\Rightarrow AC = \sqrt{841} = 29.$$



$$13. \sec \theta = \frac{5}{4} \Rightarrow \cos \theta = \frac{4}{5} = \frac{AB}{AC}.$$

$$\therefore BC^2 = AC^2 - AB^2 = (5)^2 - (4)^2 = 25 - 16 = 9 \Rightarrow BC = 3.$$

$$\therefore \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}, \cot \theta = \frac{4}{3}.$$



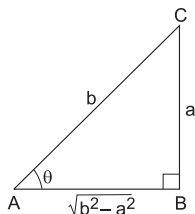
$$16. \sin \theta = \frac{a}{b} = \frac{BC}{AC}.$$

$$\therefore AB^2 = AC^2 - BC^2 = b^2 - a^2$$

$$\Rightarrow AB = \sqrt{b^2 - a^2}.$$

$$\therefore \sec \theta = \frac{b}{\sqrt{b^2 - a^2}} \text{ and } \tan \theta = \frac{a}{\sqrt{b^2 - a^2}}$$

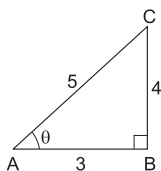
$$\begin{aligned} \therefore (\sec \theta + \tan \theta) &= \frac{(b+a)}{\sqrt{b^2 - a^2}} = \frac{(b+a)}{\sqrt{(b-a)} \cdot \sqrt{(b+a)}} \\ &= \frac{\sqrt{b+a} \cdot \sqrt{b+a}}{\sqrt{(b-a)} \cdot \sqrt{(b+a)}} = \sqrt{\frac{b+a}{b-a}}. \end{aligned}$$



$$18. \tan \theta = \frac{4}{3} = \frac{BC}{AB}.$$

$$\therefore AC^2 = AB^2 + BC^2 = 9 + 16 = 25 \Rightarrow AC = 5.$$

$$\therefore (\sin \theta + \cos \theta) = \left(\frac{4}{5} + \frac{3}{5}\right) = \frac{7}{5}.$$



$$19. \frac{(a \sin \theta - b \cos \theta)}{(a \sin \theta + b \cos \theta)} = \frac{(a \tan \theta - b)}{(a \tan \theta + b)} \quad [\text{dividing num. and denom. by } \cos \theta]$$

$$= \frac{\left(a \times \frac{a}{b} - b\right)}{\left(a \times \frac{a}{b} + b\right)} = \frac{(a^2 - b^2)}{(a^2 + b^2)}.$$

$$20. \frac{(4 \cos \theta - \sin \theta)}{(2 \cos \theta + \sin \theta)} = \frac{(4 - \tan \theta)}{(2 + \tan \theta)} \quad [\text{dividing num. and denom. by } \cos \theta]$$

$$= \frac{\left(4 - \frac{4}{3}\right)}{\left(2 + \frac{4}{3}\right)} = \frac{8}{10} = \frac{4}{5}.$$

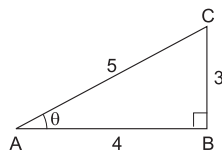
$$21. \frac{(4 \sin \theta - 3 \cos \theta)}{(2 \sin \theta + 6 \cos \theta)} = \frac{(4 - 3 \cot \theta)}{(2 + 6 \cot \theta)} \quad [\text{dividing num. and denom. by } \sin \theta]$$

$$= \frac{\left(4 - 3 \times \frac{2}{3}\right)}{\left(2 + 6 \times \frac{2}{3}\right)} = \frac{2}{6} = \frac{1}{3}.$$

$$22. \cot \theta = \frac{4}{3} = \frac{AB}{BC}.$$

$$\therefore AC^2 = AB^2 + BC^2 = (4)^2 + (3)^2 = 16 + 9 = 25 \Rightarrow AC = 5.$$

$$\therefore \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5} \text{ and } \tan \theta = \frac{3}{4}.$$



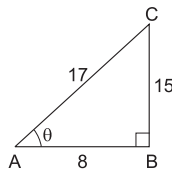
$$23. \sec \theta = \frac{17}{8} \Rightarrow \cos \theta = \frac{8}{17} = \frac{AB}{AC}.$$

$$\therefore BC^2 = AC^2 - AB^2 = (17)^2 - (8)^2 = (289 - 64) = 225$$

$$\Rightarrow BC = \sqrt{225} = 15.$$

$$\therefore \sin \theta = \frac{15}{17}, \cos \theta = \frac{8}{17} \text{ and } \tan \theta = \frac{15}{8}.$$

$$\text{Then, LHS} = \text{RHS} = \frac{33}{611}.$$



25. Given, $\triangle ABC$ in which $\angle B = 90^\circ$, $BC = 7$ cm and $AB = 24$ cm.

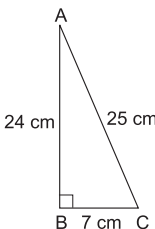
$$\therefore AC^2 = (24)^2 + (7)^2 = 576 + 49 = 625 \Rightarrow AC = 25 \text{ cm.}$$

$$(i) \sin A = \frac{BC}{AC} = \frac{7}{25}, \quad (ii) \cos A = \frac{AB}{AC} = \frac{24}{25}, \quad (iii) \sin C = \frac{AB}{AC} = \frac{24}{25}.$$

$$(iv) \cos C = \frac{BC}{AC} = \frac{7}{25}.$$

$$28. \cos^2 \alpha = (1 - \sin^2 \alpha) = \left(1 - \frac{1}{4}\right) = \frac{3}{4} \Rightarrow \cos \alpha = \frac{\sqrt{3}}{2}.$$

$$\therefore (3 \cos \alpha - 4 \cos^3 \alpha) = \left(3 \times \frac{\sqrt{3}}{2} - 4 \times \frac{3\sqrt{3}}{8}\right) = \left(\frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2}\right) = 0.$$



30. Let $\triangle ACD$ and $\triangle BEF$ be two right triangles such that $\sin A = \sin B$.

Then, $\sin A = \sin B$

$$\Rightarrow \frac{CD}{AD} = \frac{EF}{BF}$$

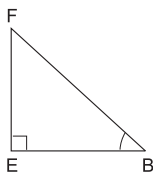
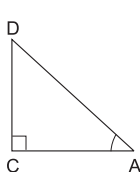
$$\Rightarrow \frac{CD}{EF} = \frac{AD}{BF} = k \text{ (say)}$$

$$\Rightarrow CD = k(EF) \text{ and } AD = k(BF).$$

$$\begin{aligned} \therefore \frac{AC}{BE} &= \frac{\sqrt{AD^2 - CD^2}}{\sqrt{BF^2 - EF^2}} \quad [\text{using Pythagoras' theorem}] \\ &= \frac{k\sqrt{BF^2 - EF^2}}{\sqrt{BF^2 - EF^2}} = k \quad [\text{using (i)}]. \end{aligned}$$

$$\therefore \frac{CD}{EF} = \frac{AD}{BF} = \frac{AC}{BE}.$$

$$\Rightarrow \triangle ACD \sim \triangle BEF \text{ and hence } \angle A = \angle B.$$



... (i)

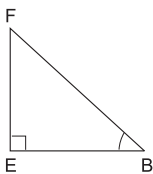
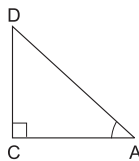
31. Let $\triangle ACD$ and $\triangle BEF$ be two right triangles such that $\tan A = \tan B$.

Then, $\tan A = \tan B$

$$\Rightarrow \frac{CD}{AC} = \frac{EF}{BE}$$

$$\Rightarrow \frac{CD}{EF} = \frac{AC}{BE} \text{ and } \angle C = \angle E = 90^\circ.$$

$$\therefore \triangle ACD \sim \triangle BEF \text{ and hence, } \angle A = \angle B.$$



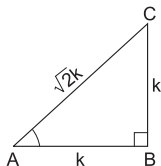
32. $\tan A = 1 \Rightarrow \frac{BC}{AB} = 1 \Rightarrow BC = AB = k$ (say).

$$\therefore AC^2 = (AB^2 + BC^2) = (k^2 + k^2) = 2k^2 \Rightarrow AC = \sqrt{2}k.$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{k}{\sqrt{2}k} = \frac{1}{\sqrt{2}}$$

$$\text{and } \cos A = \frac{AB}{AC} = \frac{k}{\sqrt{2}k} = \frac{1}{\sqrt{2}}.$$

$$\therefore 2 \sin A \cos A = \left(2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) = 1.$$



33. $PQ^2 = (PR^2 - QR^2) = (x+2)^2 - x^2 = 4(x+1)$

$$\therefore PQ = 2\sqrt{x+1}.$$

$$(i) \cot \phi = \frac{QR}{PQ} = \frac{x}{2\sqrt{x+1}} \Rightarrow (\sqrt{x+1}) \cot \phi = \frac{x}{2}.$$

$$(ii) \tan \theta = \frac{QR}{PQ} = \frac{x}{2\sqrt{x+1}} = \frac{x^2}{2\sqrt{x^3+x^2}} \Rightarrow (\sqrt{x^3+x^2}) \tan \theta = \frac{x^2}{2}.$$

$$(iii) \cos \theta = \frac{PQ}{PR} = \frac{2\sqrt{x+1}}{(x+2)}.$$

34. Adding, we get, $\operatorname{cosec} A = \frac{(x+y)}{2} \Rightarrow \sin A = \frac{2}{(x+y)}.$

$$\text{Subtracting, we get, } \cos A = \frac{x-y}{2}.$$

$$\therefore \sin^2 A + \cos^2 A = 1 \Rightarrow \left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1 = 0.$$



T-Ratios of Some Particular Angles

TRIGONOMETRIC RATIOS OF 45° , 60° AND 30° (GEOMETRICALLY)

TRIGONOMETRIC RATIOS OF 45°

Let $\triangle ABC$ be a right-angled triangle in which $\angle B = 90^\circ$ and $\angle A = 45^\circ$. Then, clearly, $\angle C = 45^\circ$.

$$\angle A = \angle C \Rightarrow AB = BC.$$

Let $AB = BC = a$ units. Then,

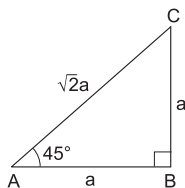
$$AC = \sqrt{AB^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}a \text{ units.}$$

Base $AB = a$; perpendicular $BC = a$ and hypotenuse $AC = \sqrt{2}a$.

$$\therefore \sin 45^\circ = \frac{BC}{AC} = \frac{a}{\sqrt{2}a} = \frac{1}{\sqrt{2}}; \quad \cos 45^\circ = \frac{AB}{AC} = \frac{a}{\sqrt{2}a} = \frac{1}{\sqrt{2}};$$

$$\tan 45^\circ = \frac{BC}{AB} = \frac{a}{a} = 1; \quad \operatorname{cosec} 45^\circ = \frac{1}{\sin 45^\circ} = \sqrt{2};$$

$$\sec 45^\circ = \frac{1}{\cos 45^\circ} = \sqrt{2}; \quad \cot 45^\circ = \frac{1}{\tan 45^\circ} = 1.$$



TRIGONOMETRIC RATIOS OF 60° AND 30°

Consider an equilateral $\triangle ABC$ with each side equal to $2a$.

Then, each angle of $\triangle ABC$ is 60° .

From A , draw $AD \perp BC$.

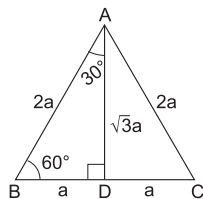
Then, clearly, $BD = DC = a$.

Also, $\angle ADB = 90^\circ$.

$$\therefore \angle BAD = 30^\circ.$$

From right-angled $\triangle ADB$, we have

$$AD = \sqrt{AB^2 - BD^2} = \sqrt{(2a)^2 - a^2} = \sqrt{4a^2 - a^2} = \sqrt{3a^2} = \sqrt{3}a.$$



T-RATIOS OF 60°

In right-angled $\triangle ADB$, we have

base $BD = a$, perpendicular $AD = \sqrt{3}a$ and hypotenuse $AB = 2a$.

$$\therefore \sin 60^\circ = \frac{AD}{AB} = \frac{\sqrt{3}a}{2a} = \frac{\sqrt{3}}{2};$$

$$\tan 60^\circ = \frac{AD}{BD} = \frac{\sqrt{3}a}{a} = \sqrt{3};$$

$$\sec 60^\circ = \frac{1}{\cos 60^\circ} = 2;$$

$$\cos 60^\circ = \frac{BD}{AB} = \frac{a}{2a} = \frac{1}{2};$$

$$\operatorname{cosec} 60^\circ = \frac{1}{\sin 60^\circ} = \frac{2}{\sqrt{3}};$$

$$\cot 60^\circ = \frac{1}{\tan 60^\circ} = \frac{1}{\sqrt{3}}.$$

T-RATIOS OF 30°

In right-angled $\triangle ADB$, we have

base $AD = \sqrt{3}a$, perpendicular $BD = a$ and hypotenuse $AB = 2a$.

$$\therefore \sin 30^\circ = \frac{BD}{AB} = \frac{a}{2a} = \frac{1}{2};$$

$$\tan 30^\circ = \frac{BD}{AD} = \frac{a}{\sqrt{3}a} = \frac{1}{\sqrt{3}};$$

$$\sec 30^\circ = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}};$$

$$\cos 30^\circ = \frac{AD}{AB} = \frac{\sqrt{3}a}{2a} = \frac{\sqrt{3}}{2};$$

$$\operatorname{cosec} 30^\circ = \frac{1}{\sin 30^\circ} = 2;$$

$$\cot 30^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}.$$

AXIOMS FOR T-RATIOS OF 0°

We define:

$$(i) \sin 0^\circ = 0 \quad (ii) \cos 0^\circ = 1 \quad (iii) \tan 0^\circ = 0 \quad (iv) \sec 0^\circ = 1$$

NOTE $\operatorname{cosec} 0^\circ$ and $\cot 0^\circ$ are not defined.

AXIOMS FOR T-RATIOS OF 90°

We define:

$$(i) \sin 90^\circ = 1 \quad (ii) \cos 90^\circ = 0 \quad (iii) \operatorname{cosec} 90^\circ = 1 \quad (iv) \cot 90^\circ = 0$$

NOTE $\tan 90^\circ$ and $\sec 90^\circ$ are not defined.

TABLE FOR T-RATIOS OF 0°, 30°, 45°, 60°, 90°

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\operatorname{cosec} \theta$	$\sec \theta$	$\cot \theta$
0°	0	1	0	not defined	1	not defined
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$	$\sqrt{3}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2	$\frac{1}{\sqrt{3}}$
90°	1	0	not defined	1	not defined	0

SOLVED EXAMPLES**EXAMPLE 1** *Evaluate:*

(i) $\sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ$

(ii) $\tan 30^\circ \cdot \operatorname{cosec} 60^\circ + \tan 60^\circ \cdot \sec 30^\circ$

SOLUTION On substituting the values of various T-ratios, we get

(i) $\sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ$

$$= \left(\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{2} \right) = \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{2}{4} = \frac{1}{2}.$$

(ii) $\tan 30^\circ \cdot \operatorname{cosec} 60^\circ + \tan 60^\circ \cdot \sec 30^\circ$

$$= \left(\frac{1}{\sqrt{3}} \times \frac{2}{\sqrt{3}} + \sqrt{3} \times \frac{2}{\sqrt{3}} \right) = \left(\frac{2}{3} + 2 \right) = 2\frac{2}{3}.$$

EXAMPLE 2 *Evaluate:*

(i) $\sin^2 30^\circ \cdot \cos^2 45^\circ + 4 \tan^2 30^\circ + \frac{1}{2} \sin^2 90^\circ + \frac{1}{8} \cot^2 60^\circ$

(ii)
$$\frac{\tan^2 60^\circ + 4 \sin^2 45^\circ + 3 \sec^2 30^\circ + 5 \cos^2 90^\circ}{\operatorname{cosec} 30^\circ + \sec 60^\circ - \cot^2 30^\circ}$$

SOLUTION On substituting the values of various T-ratios, we get

(i) $\sin^2 30^\circ \cdot \cos^2 45^\circ + 4 \tan^2 30^\circ + \frac{1}{2} \sin^2 90^\circ + \frac{1}{8} \cot^2 60^\circ$

$$= \left(\frac{1}{2} \right)^2 \cdot \left(\frac{1}{\sqrt{2}} \right)^2 + 4 \times \left(\frac{1}{\sqrt{3}} \right)^2 + \frac{1}{2} \times (1)^2 + \frac{1}{8} \times \left(\frac{1}{\sqrt{3}} \right)^2$$

$$= \left(\frac{1}{4} \times \frac{1}{2} + 4 \times \frac{1}{3} + \frac{1}{2} \times 1 + \frac{1}{8} \times \frac{1}{3} \right)$$

$$= \left(\frac{1}{8} + \frac{4}{3} + \frac{1}{2} + \frac{1}{24} \right) = \frac{48}{24} = 2.$$

(ii)
$$\frac{\tan^2 60^\circ + 4 \sin^2 45^\circ + 3 \sec^2 30^\circ + 5 \cos^2 90^\circ}{\operatorname{cosec} 30^\circ + \sec 60^\circ - \cot^2 30^\circ}$$

$$= \frac{(\sqrt{3})^2 + 4 \times \left(\frac{1}{\sqrt{2}} \right)^2 + 3 \times \left(\frac{2}{\sqrt{3}} \right)^2 + 5 \times 0^2}{2 + 2 - (\sqrt{3})^2}$$

$$= \left(\frac{3 + 4 \times \frac{1}{2} + 3 \times \frac{4}{3} + 5 \times 0}{4 - 3} \right) = \left(\frac{3 + 2 + 4 + 0}{1} \right) = 9.$$

EXAMPLE 3 *Show that*

(i) $\cos 60^\circ \cdot \cos 30^\circ - \sin 60^\circ \cdot \sin 30^\circ = \cos 90^\circ$

(ii) $\cos 60^\circ = 1 - 2 \sin^2 30^\circ = 2 \cos^2 30^\circ - 1$

(iii)
$$\frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \cdot \tan 30^\circ} = \tan 30^\circ$$

SOLUTION We have

$$(i) \cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$$

$$= \left(\frac{1}{2} \times \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \times \frac{1}{2} \right) = \left(\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \right) = 0.$$

$$\text{Also, } \cos 90^\circ = 0.$$

$$\therefore \cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ = \cos 90^\circ.$$

$$(ii) \cos 60^\circ = \frac{1}{2};$$

$$1 - 2\sin^2 30^\circ = \left[1 - 2 \times \left(\frac{1}{2} \right)^2 \right] = \left(1 - 2 \times \frac{1}{4} \right) = \left(1 - \frac{1}{2} \right) = \frac{1}{2};$$

$$2\cos^2 30^\circ - 1 = \left[2 \times \left(\frac{\sqrt{3}}{2} \right)^2 - 1 \right] = \left(2 \times \frac{3}{4} - 1 \right) = \left(\frac{3}{2} - 1 \right) = \frac{1}{2}.$$

$$\therefore \cos 60^\circ = 1 - 2\sin^2 30^\circ = 2\cos^2 30^\circ - 1.$$

$$(iii) \frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ} = \frac{\left(\sqrt{3} - \frac{1}{\sqrt{3}} \right)}{\left(1 + \sqrt{3} \times \frac{1}{\sqrt{3}} \right)} = \frac{(3-1)}{\sqrt{3} \times 2} = \frac{1}{\sqrt{3}}.$$

$$\text{Also, } \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

$$\therefore \frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ} = \tan 30^\circ.$$

EXAMPLE 4 Taking $\theta = 30^\circ$, verify each of the following:

$$(i) \sin 2\theta = 2 \sin \theta \cos \theta$$

$$(ii) \cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$(iii) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

SOLUTION When $\theta = 30^\circ$, we have $2\theta = 60^\circ$.

$$(i) \sin 2\theta = \sin 60^\circ = \frac{\sqrt{3}}{2}.$$

$$2 \sin \theta \cos \theta = 2 \sin 30^\circ \cos 30^\circ = \left(2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{2}.$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta.$$

$$(ii) \cos 2\theta = \cos 60^\circ = \frac{1}{2}.$$

$$2 \cos^2 \theta - 1 = 2 \times \cos^2 30^\circ - 1$$

$$= \left[2 \times \left(\frac{\sqrt{3}}{2} \right)^2 - 1 \right] = \left(2 \times \frac{3}{4} - 1 \right) = \left(\frac{3}{2} - 1 \right) = \frac{1}{2}.$$

$$1 - 2\sin^2\theta = 1 - 2\sin^2 30^\circ$$

$$= \left[1 - 2 \times \left(\frac{1}{2}\right)^2\right] = \left(1 - 2 \times \frac{1}{4}\right) = \left(1 - \frac{1}{2}\right) = \frac{1}{2}.$$

$$\therefore \cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta.$$

$$(iii) \tan 2\theta = \tan 60^\circ = \sqrt{3}.$$

$$\frac{2\tan\theta}{1-\tan^2\theta} = \frac{2\tan 30^\circ}{1-\tan^2 30^\circ} = \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\left(\frac{2}{\sqrt{3}}\right)}{\left(1 - \frac{1}{3}\right)}$$

$$= \left(\frac{2}{\sqrt{3}} \times \frac{3}{2}\right) = \sqrt{3}.$$

$$\therefore \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}.$$

EXAMPLE 5 Taking $\theta = 30^\circ$, verify that:

$$(i) \sin 3\theta = 3\sin\theta - 4\sin^3\theta \quad (ii) \cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

SOLUTION When $\theta = 30^\circ$, we have $3\theta = 90^\circ$.

$$(i) \sin 3\theta = \sin 90^\circ = 1.$$

$$\text{And, } 3\sin\theta - 4\sin^3\theta = 3\sin 30^\circ - 4\sin^3 30^\circ$$

$$= \left[3 \times \frac{1}{2} - 4 \times \left(\frac{1}{2}\right)^3\right] = \left(\frac{3}{2} - 4 \times \frac{1}{8}\right)$$

$$= \left(\frac{3}{2} - \frac{1}{2}\right) = 1.$$

$$\therefore \sin 3\theta = 3\sin\theta - 4\sin^3\theta.$$

$$(ii) \cos 3\theta = \cos 90^\circ = 0.$$

$$\text{And, } 4\cos^3\theta - 3\cos\theta = 4\cos^3 30^\circ - 3\cos 30^\circ$$

$$= \left[4 \times \left(\frac{\sqrt{3}}{2}\right)^3 - 3 \times \frac{\sqrt{3}}{2}\right]$$

$$= \left(4 \times \frac{3\sqrt{3}}{8} - \frac{3\sqrt{3}}{2}\right)$$

$$= \left(\frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2}\right) = 0.$$

$$\therefore \cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

EXAMPLE 6 Taking $A = 60^\circ$ and $B = 30^\circ$, verify that:

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

SOLUTION $A = 60^\circ$ and $B = 30^\circ \Rightarrow A - B = (60^\circ - 30^\circ) = 30^\circ$.

$$\therefore \sin(A - B) = \sin 30^\circ = \frac{1}{2}.$$

$$\begin{aligned}
 \text{And, } \sin A \cdot \cos B - \cos A \cdot \sin B \\
 &= \sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ \\
 &= \left(\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{2} \right) \\
 &= \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{2}{4} = \frac{1}{2}.
 \end{aligned}$$

$$\therefore \sin(A - B) = \sin A \cos B - \cos A \sin B.$$

EXAMPLE 7 If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, find A and B .

SOLUTION $\sin(A + B) = 1 \Rightarrow \sin(A + B) = \sin 90^\circ \quad [\because \sin 90^\circ = 1]$
 $\Rightarrow A + B = 90^\circ. \quad \dots (i)$

$$\begin{aligned}
 \cos(A - B) = \frac{\sqrt{3}}{2} \Rightarrow \cos(A - B) = \cos 30^\circ \\
 \Rightarrow A - B = 30^\circ. \quad \dots (ii)
 \end{aligned}$$

Solving (i) and (ii), we get $A = 60^\circ$ and $B = 30^\circ$.

Hence, $A = 60^\circ$ and $B = 30^\circ$.

EXAMPLE 8 If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, $0^\circ < A + B < 90^\circ$ and $A > B$, find the values of A and B . [CBSE 2008C]

SOLUTION $\tan(A + B) = \sqrt{3} \Rightarrow \tan(A + B) = \tan 60^\circ \quad [\because \tan 60^\circ = \sqrt{3}]$
 $\Rightarrow A + B = 60^\circ. \quad \dots (i)$

$$\begin{aligned}
 \tan(A - B) = \frac{1}{\sqrt{3}} \Rightarrow \tan(A - B) = \tan 30^\circ \quad \left[\because \tan 30^\circ = \frac{1}{\sqrt{3}} \right] \\
 \Rightarrow A - B = 30^\circ. \quad \dots (ii)
 \end{aligned}$$

Solving (i) and (ii), we get $A = 45^\circ$ and $B = 15^\circ$.

Hence, $A = 45^\circ$ and $B = 15^\circ$.

EXAMPLE 9 If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1$, $0^\circ < (A + B) < 90^\circ$ and $A > B$ then find A and B . [CBSE 2009C]

SOLUTION $\tan(A + B) = \sqrt{3} \Rightarrow \tan(A + B) = \tan 60^\circ \quad [\because \sqrt{3} = \tan 60^\circ]$
 $\Rightarrow A + B = 60^\circ. \quad \dots (i)$

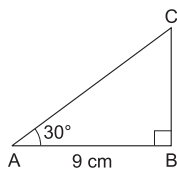
$$\begin{aligned}
 \tan(A - B) = 1 \Rightarrow \tan(A - B) = \tan 45^\circ \quad [\because 1 = \tan 45^\circ] \\
 \Rightarrow A - B = 45^\circ. \quad \dots (ii)
 \end{aligned}$$

Solving (i) and (ii), we get $A = (52.5)^\circ$ and $B = (7.5)^\circ$.

Hence, $A = (52.5)^\circ$ and $B = (7.5)^\circ$.

EXAMPLE 10 In the adjoining figure, $\triangle ABC$ is right-angled at B. If $\angle A = 30^\circ$ and $AB = 9$ cm, find (i) BC and (ii) AC .

SOLUTION From right-angled $\triangle ABC$, we have



$$(i) \frac{BC}{AB} = \tan 30^\circ \Rightarrow \frac{BC}{9 \text{ cm}} = \frac{1}{\sqrt{3}}$$

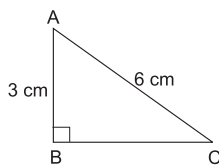
$$\Rightarrow BC = \left(\frac{9}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) \text{ cm} = \frac{9\sqrt{3}}{3} \text{ cm} = 3\sqrt{3} \text{ cm}.$$

$$(ii) \frac{AC}{BC} = \operatorname{cosec} 30^\circ = 2 \Rightarrow \frac{AC}{3\sqrt{3} \text{ cm}} = 2 \Rightarrow AC = 6\sqrt{3} \text{ cm}.$$

Hence, $BC = 3\sqrt{3}$ cm and $AC = 6\sqrt{3}$ cm.

EXAMPLE 11 In right $\triangle ABC$, $\angle B = 90^\circ$, $AB = 3$ cm and $AC = 6$ cm. Find $\angle C$ and $\angle A$.

SOLUTION $\frac{AB}{AC} = \sin C \Rightarrow \sin C = \frac{3}{6} = \frac{1}{2}$
 $\Rightarrow \sin C = \sin 30^\circ \Rightarrow \angle C = 30^\circ$
 $\therefore \angle A = 180^\circ - (90^\circ + 30^\circ) = (180^\circ - 120^\circ) = 60^\circ.$



EXAMPLE 12 If $6x = \sec \theta$ and $\frac{6}{x} = \tan \theta$, find the value of $9\left(x^2 - \frac{1}{x^2}\right)$. [CBSE 2010]

SOLUTION We have

$$6x = \sec \theta \quad \dots (i) \quad \text{and} \quad \frac{6}{x} = \tan \theta \quad \dots (ii)$$

Adding (i) and (ii), we get

$$6\left(x + \frac{1}{x}\right) = (\sec \theta + \tan \theta). \quad \dots (iii)$$

Subtracting (ii) from (i), we get

$$6\left(x - \frac{1}{x}\right) = (\sec \theta - \tan \theta). \quad \dots (iv)$$

Multiplying the corresponding sides of (iii) and (iv), we get

$$36\left(x^2 - \frac{1}{x^2}\right) = (\sec^2 \theta - \tan^2 \theta) = 1 \Rightarrow 9\left(x^2 - \frac{1}{x^2}\right) = \frac{1}{4}.$$

$$\text{Hence, } 9\left(x^2 - \frac{1}{x^2}\right) = \frac{1}{4}.$$

EXERCISE 11

Evaluate each of the following:

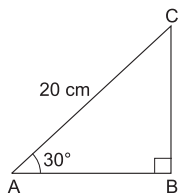
1. $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ 2. $\cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$

3. $\cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$ 4. $\frac{\sin 30^\circ}{\cos 45^\circ} + \frac{\cot 45^\circ}{\sec 60^\circ} - \frac{\sin 60^\circ}{\tan 45^\circ} + \frac{\cos 30^\circ}{\sin 90^\circ}$

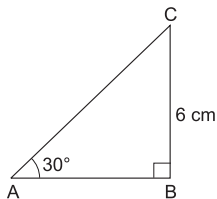
5. $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$
6. $2 \cos^2 60^\circ + 3 \sin^2 45^\circ - 3 \sin^2 30^\circ + 2 \cos^2 90^\circ$
7. $\cot^2 30^\circ - 2 \cos^2 30^\circ - \frac{3}{4} \sec^2 45^\circ + \frac{1}{4} \operatorname{cosec}^2 30^\circ$
8. $(\sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ)(\operatorname{cosec}^2 45^\circ \sec^2 30^\circ)$
9. $\frac{4}{\cot^2 30^\circ} + \frac{1}{\sin^2 30^\circ} - 2 \cos^2 45^\circ - \sin^2 0^\circ$
10. Show that:
 - (i) $\frac{1 - \sin 60^\circ}{\cos 60^\circ} = \frac{\tan 60^\circ - 1}{\tan 60^\circ + 1}$
 - (ii) $\frac{\cos 30^\circ + \sin 60^\circ}{1 + \sin 30^\circ + \cos 60^\circ} = \cos 30^\circ$
11. Verify each of the following:
 - (i) $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ = \sin 30^\circ$
 - (ii) $\cos 60^\circ \cos 30^\circ + \sin 60^\circ \sin 30^\circ = \cos 30^\circ$
 - (iii) $2 \sin 30^\circ \cos 30^\circ = \sin 60^\circ$
 - (iv) $2 \sin 45^\circ \cos 45^\circ = \sin 90^\circ$
12. If $A = 45^\circ$, verify that:
 - (i) $\sin 2A = 2 \sin A \cos A$
 - (ii) $\cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$
13. If $A = 30^\circ$, verify that:
 - (i) $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$
 - (ii) $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
 - (iii) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
14. If $A = 60^\circ$ and $B = 30^\circ$, verify that:
 - (i) $\sin (A + B) = \sin A \cos B + \cos A \sin B$
 - (ii) $\cos (A + B) = \cos A \cos B - \sin A \sin B$
15. If $A = 60^\circ$ and $B = 30^\circ$, verify that:
 - (i) $\sin (A - B) = \sin A \cos B - \cos A \sin B$
 - (ii) $\cos (A - B) = \cos A \cos B + \sin A \sin B$
 - (iii) $\tan (A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
16. If A and B are acute angles such that $\tan A = \frac{1}{3}$, $\tan B = \frac{1}{2}$ and $\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$, show that $A + B = 45^\circ$.
17. Using the formula, $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$, find the value of $\tan 60^\circ$, it being given that $\tan 30^\circ = \frac{1}{\sqrt{3}}$.
18. Using the formula, $\cos A = \sqrt{\frac{1 + \cos 2A}{2}}$, find the value of $\cos 30^\circ$, it being given that $\cos 60^\circ = \frac{1}{2}$.

19. Using the formula, $\sin A = \sqrt{\frac{1 - \cos 2A}{2}}$, find the value of $\sin 30^\circ$, it being given that $\cos 60^\circ = \frac{1}{2}$.

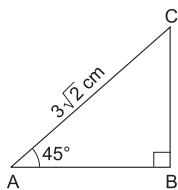
20. In the adjoining figure, $\triangle ABC$ is a right-angled triangle in which $\angle B = 90^\circ$, $\angle A = 30^\circ$ and $AC = 20$ cm. Find (i) BC , (ii) AB .



21. In the adjoining figure, $\triangle ABC$ is right-angled at B and $\angle A = 30^\circ$. If $BC = 6$ cm, find (i) AB , (ii) AC .



22. In the adjoining figure, $\triangle ABC$ is right-angled at B and $\angle A = 45^\circ$. If $AC = 3\sqrt{2}$ cm, find (i) BC , (ii) AB .



23. If $\sin(A+B) = 1$ and $\cos(A-B) = 1$, $0^\circ \leq (A+B) \leq 90^\circ$ and $A > B$ then find A and B .

24. If $\sin(A-B) = \frac{1}{2}$ and $\cos(A+B) = \frac{1}{2}$, $0^\circ < (A+B) < 90^\circ$ and $A > B$ then find A and B .

25. If $\tan(A-B) = \frac{1}{\sqrt{3}}$ and $\tan(A+B) = \sqrt{3}$, $0^\circ < (A+B) < 90^\circ$ and $A > B$ then find A and B .

26. If $3x = \operatorname{cosec} \theta$ and $\frac{3}{x} = \cot \theta$, find the value of $3\left(x^2 - \frac{1}{x^2}\right)$. [CBSE 2010]

27. If $\sin(A+B) = \sin A \cos B + \cos A \sin B$
and $\cos(A-B) = \cos A \cos B + \sin A \sin B$,
find the values of (i) $\sin 75^\circ$ and (ii) $\cos 15^\circ$.

ANSWERS (EXERCISE 11)

1. 1 2. 0 3. $\frac{(\sqrt{3}+1)}{2\sqrt{2}}$ 4. $\frac{(\sqrt{2}+1)}{2}$ 5. $\frac{67}{12}$ 6. $\frac{5}{4}$

7. 1 8. $\frac{2}{3}$ 9. $\frac{13}{3}$ 17. $\sqrt{3}$ 18. $\frac{\sqrt{3}}{2}$ 19. $\frac{1}{2}$
20. (i) $BC = 10$ cm (ii) $AB = 10\sqrt{3}$ cm
21. (i) $AB = 6\sqrt{3}$ cm (ii) $AC = 12$ cm
22. (i) $BC = 3$ cm (ii) $AB = 3$ cm 23. $\angle A = \angle B = 45^\circ$
24. $\angle A = 45^\circ, \angle B = 15^\circ$ 25. $\angle A = 45^\circ, \angle B = 15^\circ$ 26. $\frac{1}{3}$
27. (i) $\frac{(\sqrt{3}+1)}{2\sqrt{2}}$ (ii) $\frac{(\sqrt{3}+1)}{2\sqrt{2}}$

HINTS TO SOME SELECTED QUESTIONS

20. Let $BC = x$ cm. Then, $\frac{x}{20} = \sin 30^\circ = \frac{1}{2} \Rightarrow x = 10$.
 $AB^2 = (AC^2 - BC^2) = \{(20)^2 - (10)^2\} \text{ cm}^2 = 300 \text{ cm}^2 \Rightarrow AB = \sqrt{300} \text{ cm} = 10\sqrt{3} \text{ cm}$.
 $\therefore BC = 10$ cm and $AB = 10\sqrt{3}$ cm.
21. Let $AB = x$ cm. Then, $\frac{x}{6} = \cot 30^\circ = \sqrt{3} \Rightarrow x = 6\sqrt{3}$.
 $AC^2 = (AB^2 + BC^2) = \{(6\sqrt{3})^2 + (6)^2\} \text{ cm}^2 = (108 + 36) \text{ cm}^2 = 144 \text{ cm}^2 \Rightarrow AC = 12$ cm.
22. Let $BC = x$ cm. Then, $\frac{x}{3\sqrt{2}} = \sin 45^\circ = \frac{1}{\sqrt{2}} \Rightarrow x = 3$.
 $\frac{AB}{BC} = \tan 45^\circ = 1 \Rightarrow \frac{AB}{3 \text{ cm}} = 1 \Rightarrow AB = 3$ cm.
23. $\sin(A+B) = 1 \Rightarrow \sin(A+B) = \sin 90^\circ$.
 $\cos(A-B) = 1 \Rightarrow \cos(A-B) = \cos 0^\circ$.
 Solve $A+B = 90^\circ, A-B = 0^\circ$ for A and B .
24. $\sin(A-B) = \frac{1}{2} \Rightarrow \sin(A-B) = \sin 30^\circ$.
 $\cos(A+B) = \frac{1}{2} \Rightarrow \cos(A+B) = \cos 60^\circ$.
 Solve $A-B = 30^\circ$ and $A+B = 60^\circ$ for A and B .
25. $\tan(A-B) = \frac{1}{\sqrt{3}} \Rightarrow \tan(A-B) = \tan 30^\circ$.
 $\tan(A+B) = \sqrt{3} \Rightarrow \tan(A+B) = \tan 60^\circ$.
 Solve $A-B = 30^\circ$ and $A+B = 60^\circ$ for A and B .
26. Adding, we get $3\left(x + \frac{1}{x}\right) = (\operatorname{cosec} \theta + \cot \theta)$.
 Subtracting, we get $3\left(x - \frac{1}{x}\right) = (\operatorname{cosec} \theta - \cot \theta)$.
 $\therefore 9\left(x^2 - \frac{1}{x^2}\right) = (\operatorname{cosec}^2 \theta - \cot^2 \theta) = 1 \Rightarrow 3\left(x^2 - \frac{1}{x^2}\right) = \frac{1}{3}$.
27. Take $A = 45^\circ$ and $B = 30^\circ$.



Trigonometric Ratios of Complementary Angles

TRIGONOMETRIC RATIOS OF COMPLEMENTARY ANGLES

COMPLEMENTARY ANGLES Two angles are said to be complementary if their sum is 90° .

Thus, θ and $(90^\circ - \theta)$ are complementary angles.

TRIGONOMETRIC RATIOS OF COMPLEMENTARY ANGLES

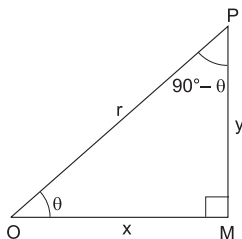
Consider a $\triangle OMP$, right-angled at M in which

$\angle MOP = \theta$ and therefore, $\angle OPM = (90^\circ - \theta)$.

Let $OM = x$, $MP = y$ and $OP = r$. Then,

$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}, \tan \theta = \frac{y}{x},$$

$$\operatorname{cosec} \theta = \frac{r}{y}, \sec \theta = \frac{r}{x}, \cot \theta = \frac{x}{y}.$$



For considering the trigonometric ratios of $(90^\circ - \theta)$, we have

base = MP , perpendicular = OM and hypotenuse = OP .

$$\therefore \sin (90^\circ - \theta) = \frac{OM}{OP} = \frac{x}{r} = \cos \theta$$

$$\cos (90^\circ - \theta) = \frac{PM}{OP} = \frac{y}{r} = \sin \theta$$

$$\tan (90^\circ - \theta) = \frac{OM}{PM} = \frac{x}{y} = \cot \theta$$

$$\therefore \operatorname{cosec} (90^\circ - \theta) = \frac{1}{\sin (90^\circ - \theta)} = \frac{1}{\cos \theta} = \sec \theta$$

$$\sec (90^\circ - \theta) = \frac{1}{\cos (90^\circ - \theta)} = \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

$$\cot (90^\circ - \theta) = \frac{1}{\tan (90^\circ - \theta)} = \frac{1}{\cot \theta} = \tan \theta$$

SUMMARY

(i) $\sin (90^\circ - \theta) = \cos \theta$

(ii) $\cos (90^\circ - \theta) = \sin \theta$

(iii) $\tan (90^\circ - \theta) = \cot \theta$

(iv) $\cot (90^\circ - \theta) = \tan \theta$

(v) $\sec (90^\circ - \theta) = \operatorname{cosec} \theta$

(vi) $\operatorname{cosec} (90^\circ - \theta) = \sec \theta$

SOLVED EXAMPLES**EXAMPLE 1** *Without using trigonometric tables, evaluate:*

$$(i) \frac{\cos 53^\circ}{\sin 37^\circ} \quad (ii) \frac{\tan 68^\circ}{\cot 22^\circ} \quad (iii) \frac{\sec 49^\circ}{\operatorname{cosec} 41^\circ} \quad (iv) \frac{\sin 30^\circ 17'}{\cos 59^\circ 43'}$$

SOLUTION

$$(i) \frac{\cos 53^\circ}{\sin 37^\circ} = \frac{\cos (90^\circ - 37^\circ)}{\sin 37^\circ} = \frac{\sin 37^\circ}{\sin 37^\circ} = 1$$

$$[\because \cos (90^\circ - \theta) = \sin \theta].$$

$$(ii) \frac{\tan 68^\circ}{\cot 22^\circ} = \frac{\tan (90^\circ - 22^\circ)}{\cot 22^\circ} = \frac{\cot 22^\circ}{\cot 22^\circ} = 1$$

$$[\because \tan (90^\circ - \theta) = \cot \theta].$$

$$(iii) \frac{\sec 49^\circ}{\operatorname{cosec} 41^\circ} = \frac{\sec (90^\circ - 41^\circ)}{\operatorname{cosec} 41^\circ} = \frac{\operatorname{cosec} 41^\circ}{\operatorname{cosec} 41^\circ} = 1$$

$$[\because \sec (90^\circ - \theta) = \operatorname{cosec} \theta].$$

$$(iv) \frac{\sin 30^\circ 17'}{\cos 59^\circ 43'} = \frac{\sin [90^\circ - (59^\circ 43')]}{\cos 59^\circ 43'} = \frac{\cos 59^\circ 43'}{\cos 59^\circ 43'} = 1.$$

EXAMPLE 2 *Without using trigonometric tables, evaluate:*

$$(i) \cos 48^\circ - \sin 42^\circ \quad (ii) \operatorname{cosec} 31^\circ - \sec 59^\circ$$

$$(iii) \cot 34^\circ - \tan 56^\circ \quad (iv) \cos^2 13^\circ - \sin^2 77^\circ$$

SOLUTION We have

$$(i) \cos 48^\circ - \sin 42^\circ = \cos (90^\circ - 42^\circ) - \sin 42^\circ$$

$$= \sin 42^\circ - \sin 42^\circ = 0$$

$$[\because \cos (90^\circ - \theta) = \sin \theta].$$

$$(ii) \operatorname{cosec} 31^\circ - \sec 59^\circ = \operatorname{cosec} (90^\circ - 59^\circ) - \sec 59^\circ$$

$$= \sec 59^\circ - \sec 59^\circ = 0$$

$$[\because \operatorname{cosec} (90^\circ - \theta) = \sec \theta].$$

$$(iii) \cot 34^\circ - \tan 56^\circ = \cot (90^\circ - 56^\circ) - \tan 56^\circ$$

$$= \tan 56^\circ - \tan 56^\circ = 0$$

$$[\because \cot (90^\circ - \theta) = \tan \theta].$$

$$(iv) \cos^2 13^\circ - \sin^2 77^\circ = \cos^2 (90^\circ - 77^\circ) - \sin^2 77^\circ$$

$$= \sin^2 77^\circ - \sin^2 77^\circ = 0$$

$$[\because \cos (90^\circ - \theta) = \sin \theta].$$

EXAMPLE 3 *Without using trigonometric tables, prove that:*

$$(i) \sin 43^\circ \cos 47^\circ + \cos 43^\circ \sin 47^\circ = 1$$

$$(ii) \cos 38^\circ \cos 52^\circ - \sin 38^\circ \sin 52^\circ = 0$$

$$(iii) \sec 50^\circ \sin 40^\circ + \cos 40^\circ \operatorname{cosec} 50^\circ = 2$$

$$(iv) \sec 70^\circ \sin 20^\circ - \cos 20^\circ \operatorname{cosec} 70^\circ = 0$$

SOLUTION

We have

$$(i) \sin 43^\circ \cos 47^\circ + \cos 43^\circ \sin 47^\circ$$

$$= \sin (90^\circ - 47^\circ) \cos 47^\circ + \cos (90^\circ - 47^\circ) \sin 47^\circ$$

$$= \cos^2 47^\circ + \sin^2 47^\circ = 1$$

$$[\because \sin (90^\circ - \theta) = \cos \theta, \cos (90^\circ - \theta) = \sin \theta].$$

$$(ii) \cos 38^\circ \cos 52^\circ - \sin 38^\circ \sin 52^\circ$$

$$= \cos (90^\circ - 52^\circ) \cos 52^\circ - \sin (90^\circ - 52^\circ) \sin 52^\circ$$

$$= \sin 52^\circ \cos 52^\circ - \sin 52^\circ \cos 52^\circ = 0$$

$$[\because \cos (90^\circ - \theta) = \sin \theta \text{ and } \sin (90^\circ - \theta) = \cos \theta].$$

$$(iii) \sec 50^\circ \sin 40^\circ + \cos 40^\circ \operatorname{cosec} 50^\circ$$

$$= \sec (90^\circ - 40^\circ) \sin 40^\circ + \cos 40^\circ \operatorname{cosec} (90^\circ - 40^\circ)$$

$$= \operatorname{cosec} 40^\circ \sin 40^\circ + \cos 40^\circ \sec 40^\circ$$

$$\left[\because \begin{array}{l} \sec (90^\circ - \theta) = \operatorname{cosec} \theta, \\ \operatorname{cosec} (90^\circ - \theta) = \sec \theta \end{array} \right]$$

$$= \frac{\sin 40^\circ}{\sin 40^\circ} + \frac{\cos 40^\circ}{\cos 40^\circ} = 1 + 1 = 2.$$

$$(iv) \sec 70^\circ \sin 20^\circ - \cos 20^\circ \operatorname{cosec} 70^\circ$$

$$= \sec (90^\circ - 20^\circ) \sin 20^\circ - \cos 20^\circ \operatorname{cosec} (90^\circ - 20^\circ)$$

$$= \operatorname{cosec} 20^\circ \sin 20^\circ - \cos 20^\circ \sec 20^\circ$$

$$\left[\because \begin{array}{l} \sec (90^\circ - \theta) = \operatorname{cosec} \theta, \\ \operatorname{cosec} (90^\circ - \theta) = \sec \theta \end{array} \right]$$

$$= \frac{\sin 20^\circ}{\sin 20^\circ} - \frac{\cos 20^\circ}{\cos 20^\circ} = 1 - 1 = 0.$$

EXAMPLE 4

Without using trigonometric tables, prove that:

$$(i) \tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ = 1$$

$$(ii) \tan 7^\circ \tan 23^\circ \tan 60^\circ \tan 67^\circ \tan 83^\circ = \sqrt{3}$$

$$(iii) \cot 12^\circ \cot 38^\circ \cot 52^\circ \cot 60^\circ \cot 78^\circ = \frac{1}{\sqrt{3}}$$

SOLUTION

We have

$$(i) \tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ$$

$$\begin{aligned}
 &= (\tan 48^\circ \tan 42^\circ)(\tan 23^\circ \tan 67^\circ) \\
 &= \{\tan 48^\circ \cdot \tan (90^\circ - 48^\circ)\} \{\tan 23^\circ \cdot \tan (90^\circ - 23^\circ)\} \\
 &= (\tan 48^\circ \cot 48^\circ)(\tan 23^\circ \cot 23^\circ) \quad [\because \tan (90^\circ - \theta) = \cot \theta] \\
 &= 1 \times 1 = 1.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad &\tan 7^\circ \tan 23^\circ \tan 60^\circ \tan 67^\circ \tan 83^\circ \\
 &= \tan 7^\circ \cdot \tan 83^\circ \cdot \tan 23^\circ \cdot \tan 67^\circ \cdot \tan 60^\circ \\
 &= \tan 7^\circ \cdot \tan (90^\circ - 7^\circ) \cdot \tan 23^\circ \cdot \tan (90^\circ - 23^\circ) \cdot \tan 60^\circ \\
 &= (\tan 7^\circ \cdot \cot 7^\circ) \cdot (\tan 23^\circ \cdot \cot 23^\circ) \cdot \tan 60^\circ \\
 &\quad [\because \tan (90^\circ - \theta) = \cot \theta] \\
 &= 1 \times 1 \times \sqrt{3} = \sqrt{3} \quad [\because \tan \theta \cdot \cot \theta = 1 \text{ and } \tan 60^\circ = \sqrt{3}].
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad &\cot 12^\circ \cdot \cot 38^\circ \cdot \cot 52^\circ \cdot \cot 60^\circ \cdot \cot 78^\circ \\
 &= (\cot 12^\circ \cdot \cot 78^\circ) \cdot (\cot 38^\circ \cdot \cot 52^\circ) \cdot \cot 60^\circ \\
 &= \{\cot 12^\circ \cdot \cot (90^\circ - 12^\circ)\} \{\cot 38^\circ \cdot \cot (90^\circ - 38^\circ)\} \cdot \cot 60^\circ \\
 &= (\cot 12^\circ \cdot \tan 12^\circ) \cdot (\cot 38^\circ \cdot \tan 38^\circ) \cdot \cot 60^\circ \\
 &\quad [\because \cot (90^\circ - \theta) = \tan \theta] \\
 &= 1 \times 1 \times \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \quad \left[\because \cot \theta \cdot \tan \theta = 1 \text{ and } \cot 60^\circ = \frac{1}{\sqrt{3}} \right].
 \end{aligned}$$

EXAMPLE 5 Prove that

$$\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ = 1.$$

SOLUTION We have

$$\begin{aligned}
 &\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ \\
 &= \tan 1^\circ \tan 2^\circ \dots \tan 44^\circ \tan 45^\circ \tan 46^\circ \dots \tan 88^\circ \tan 89^\circ \\
 &= (\tan 1^\circ \cdot \tan 89^\circ)(\tan 2^\circ \cdot \tan 88^\circ) \dots (\tan 44^\circ \cdot \tan 46^\circ) \cdot \tan 45^\circ \\
 &= \{\tan 1^\circ \cdot \tan (90^\circ - 1^\circ)\} \cdot \{\tan 2^\circ \cdot \tan (90^\circ - 2^\circ)\} \\
 &\quad \dots \{\tan 44^\circ \cdot \tan (90^\circ - 44^\circ)\} \cdot \tan 45^\circ \\
 &= (\tan 1^\circ \cdot \cot 1^\circ)(\tan 2^\circ \cdot \cot 2^\circ) \dots (\tan 44^\circ \cot 44^\circ) \cdot 1 \\
 &\quad [\because \tan (90^\circ - \theta) = \cot \theta \text{ and } \tan 45^\circ = 1] \\
 &= 1 \times 1 \times \dots \times 1 \times 1 = 1.
 \end{aligned}$$

EXAMPLE 6 Evaluate:

$$\frac{\cos 58^\circ}{\sin 32^\circ} + \frac{\sin 22^\circ}{\cos 68^\circ} - \frac{\cos 38^\circ \operatorname{cosec} 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ}.$$

[CBSE 2009]

SOLUTION Given expression

$$\begin{aligned}
 &= \frac{\cos 58^\circ}{\sin 32^\circ} + \frac{\sin 22^\circ}{\cos 68^\circ} - \frac{\cos 38^\circ \operatorname{cosec} 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ} \\
 &= \frac{\cos 58^\circ}{\sin (90^\circ - 58^\circ)} + \frac{\sin 22^\circ}{\cos (90^\circ - 22^\circ)} \\
 &\quad - \frac{\cos 38^\circ \operatorname{cosec} (90^\circ - 38^\circ)}{(\tan 18^\circ \tan 72^\circ)(\tan 35^\circ \tan 55^\circ) \tan 60^\circ} \\
 &= \frac{\cos 58^\circ}{\cos 58^\circ} + \frac{\sin 22^\circ}{\sin 22^\circ} \\
 &\quad - \frac{\cos 38^\circ \sec 38^\circ}{\{\tan 18^\circ \tan (90^\circ - 18^\circ)\} \{\tan 35^\circ \tan (90^\circ - 35^\circ)\} \tan 60^\circ} \\
 &= (1 + 1) - \frac{1}{(\tan 18^\circ \cot 18^\circ)(\tan 35^\circ \cot 35^\circ) \sqrt{3}} \\
 &= \left(2 - \frac{1}{\sqrt{3}}\right) = \frac{2\sqrt{3} - 1}{\sqrt{3}} = \frac{2\sqrt{3} - 1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{1}{3}(6 - \sqrt{3}).
 \end{aligned}$$

EXAMPLE 7 Without using trigonometric tables, evaluate each of the following:

- (i) $\sin^2 65^\circ + \sin^2 25^\circ$ (ii) $\cos^2 17^\circ - \sin^2 73^\circ$
 (iii) $\operatorname{cosec}^2 67^\circ - \tan^2 23^\circ$ (iv) $\sec^2 36^\circ - \cot^2 54^\circ$

SOLUTION We have

- (i) $\sin^2 65^\circ + \sin^2 25^\circ = \{\sin (90^\circ - 25^\circ)\}^2 + \sin^2 25^\circ$
 $= (\cos 25^\circ)^2 + \sin^2 25^\circ$
 $[\because \sin (90^\circ - \theta) = \cos \theta]$
 $= (\cos^2 25^\circ + \sin^2 25^\circ) = 1.$
- (ii) $\sin^2 17^\circ - \sin^2 73^\circ = [\cos (90^\circ - 73^\circ)]^2 - \sin^2 73^\circ$
 $= (\sin 73^\circ)^2 - \sin^2 73^\circ$
 $[\because \cos (90^\circ - \theta) = \sin \theta]$
 $= (\sin^2 73^\circ - \sin^2 73^\circ) = 0.$
- (iii) $\operatorname{cosec}^2 67^\circ - \tan^2 23^\circ = [\operatorname{cosec} (90^\circ - 23^\circ)]^2 - \tan^2 23^\circ$
 $= (\sec 23^\circ)^2 - \tan^2 23^\circ$
 $[\because \operatorname{cosec} (90^\circ - \theta) = \sec \theta]$
 $= (\sec^2 23^\circ - \tan^2 23^\circ) = 1$
 $[\because \sec^2 \theta - \tan^2 \theta = 1].$
- (iv) $\sec^2 36^\circ - \cot^2 54^\circ = [\sec (90^\circ - 54^\circ)]^2 - \cot^2 54^\circ$
 $= (\operatorname{cosec} 54^\circ)^2 - \cot^2 54^\circ$
 $[\because \sec (90^\circ - \theta) = \operatorname{cosec} \theta]$

$$= (\operatorname{cosec}^2 54^\circ - \cot^2 54^\circ) = 1$$

$$[\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1].$$

EXAMPLE 8 Evaluate $\frac{2 \sin^2 63^\circ + 1 + 2 \sin^2 27^\circ}{3 \cos^2 17^\circ - 2 + 3 \cos^2 73^\circ}$. [CBSE 2009C]

SOLUTION Given expression

$$\begin{aligned} &= \frac{2 \sin^2 63^\circ + 1 + 2 \sin^2 27^\circ}{3 \cos^2 17^\circ - 2 + 3 \cos^2 73^\circ} = \frac{2(\sin^2 63^\circ + \sin^2 27^\circ) + 1}{3(\cos^2 17^\circ + \cos^2 73^\circ) - 2} \\ &= \frac{2[\sin^2 63^\circ + \sin^2 (90^\circ - 63^\circ)] + 1}{3[\cos^2 17^\circ + \cos^2 (90^\circ - 17^\circ)] - 2} \\ &= \frac{2[\sin^2 63^\circ + \cos^2 63^\circ] + 1}{3[\cos^2 17^\circ + \sin^2 17^\circ] - 2} \\ &\quad [\because \sin (90^\circ - \theta) = \cos \theta \text{ and } \cos (90^\circ - \theta) = \sin \theta] \\ &= \frac{(2 \times 1) + 1}{(3 \times 1) - 2} = \frac{2+1}{3-2} = \frac{3}{1} = 3. \end{aligned}$$

EXAMPLE 9 Without using trigonometric tables, evaluate the following:

$$\frac{\cos^2 20^\circ + \cos^2 70^\circ}{\sec^2 50^\circ - \cot^2 40^\circ} + 2 \operatorname{cosec}^2 58^\circ - 2 \cot 58^\circ \tan 32^\circ - 4 \tan 13^\circ \tan 37^\circ \tan 45^\circ \tan 53^\circ \tan 77^\circ. \quad [\text{CBSE 2006C}]$$

SOLUTION Given expression

$$\begin{aligned} &= \frac{\cos^2 20^\circ + \cos^2 70^\circ}{\sec^2 50^\circ - \cot^2 40^\circ} + 2 \operatorname{cosec}^2 58^\circ - 2 \cot 58^\circ \tan 32^\circ \\ &\quad - 4 \tan 13^\circ \tan 37^\circ \tan 45^\circ \tan 53^\circ \tan 77^\circ \\ &= \frac{\cos^2 20^\circ + [\cos (90^\circ - 20^\circ)]^2}{\sec^2 50^\circ - [\cot (90^\circ - 50^\circ)]^2} \\ &\quad + 2[\operatorname{cosec}^2 58^\circ - \cot 58^\circ \tan (90^\circ - 58^\circ)] \\ &\quad - 4(\tan 13^\circ \tan 77^\circ)(\tan 37^\circ \tan 53^\circ) \tan 45^\circ \\ &= \frac{\cos^2 20^\circ + \sin^2 20^\circ}{\sec^2 50^\circ - \tan^2 50^\circ} + 2[\operatorname{cosec}^2 58^\circ - \cot^2 58^\circ] \\ &\quad - 4 \tan 13^\circ \tan (90^\circ - 13^\circ) \cdot \tan 37^\circ \tan (90^\circ - 37^\circ) \cdot \tan 45^\circ \\ &\quad [\because \cos (90^\circ - \theta) = \sin \theta, \cot (90^\circ - \theta) = \tan \theta \text{ and } \tan (90^\circ - \theta) = \cot \theta] \\ &= \frac{1}{1} + (2 \times 1) - 4(\tan 13^\circ \cot 13^\circ) \cdot (\tan 37^\circ \cot 37^\circ) \cdot 1 \\ &\quad [\because \cos^2 \theta + \sin^2 \theta = 1, \sec^2 \theta - \tan^2 \theta = 1, \operatorname{cosec}^2 \theta - \cot^2 \theta = 1] \\ &= (3 - 4 \times 1 \times 1 \times 1) = (3 - 4) = -1. \end{aligned}$$

EXAMPLE 10 If $\sin 3A = \cos (A - 10^\circ)$, where $3A$ is an acute angle then find the value of A . [CBSE 2009C]

SOLUTION $\sin 3A = \cos (A - 10^\circ)$
 $\Rightarrow \cos (90^\circ - 3A) = \cos (A - 10^\circ)$ [$\because \sin \theta = \cos (90^\circ - \theta)$]
 $\Rightarrow 90^\circ - 3A = A - 10^\circ \Rightarrow 4A = 100^\circ \Rightarrow A = 25^\circ$.
Hence, $A = 25^\circ$.

EXAMPLE 11 If $\tan 2A = \cot (A - 21^\circ)$, where $2A$ is an acute angle then find the value of A . [CBSE 2009C]

SOLUTION $\tan 2A = \cot (A - 21^\circ)$
 $\Rightarrow \cot (90^\circ - 2A) = \cot (A - 21^\circ)$ [$\because \tan 2A = \cot (90^\circ - 2A)$]
 $\Rightarrow 90^\circ - 2A = A - 21^\circ \Rightarrow 3A = 111^\circ \Rightarrow A = 37^\circ$.
Hence, $A = 37^\circ$.

EXAMPLE 12 If $\sec 5A = \operatorname{cosec} (A - 30^\circ)$, where $5A$ is an acute angle then find the value of A .

SOLUTION $\sec 5A = \operatorname{cosec} (A - 30^\circ)$
 $\Rightarrow \operatorname{cosec} (90^\circ - 5A) = \operatorname{cosec} (A - 30^\circ)$
 $\quad \quad \quad [\because \sec \theta = \operatorname{cosec} (90^\circ - \theta)]$
 $\Rightarrow 90^\circ - 5A = A - 30^\circ \Rightarrow 6A = 120^\circ \Rightarrow A = 20^\circ$.
Hence, $A = 20^\circ$.

EXAMPLE 13 If $\sin (\theta + 34^\circ) = \cos \theta$ and $(\theta + 34^\circ)$ is acute show that $\theta = 28^\circ$.

SOLUTION $\sin (\theta + 34^\circ) = \cos \theta$
 $\Rightarrow \sin (\theta + 34^\circ) = \sin (90^\circ - \theta)$ [$\because \cos \theta = \sin (90^\circ - \theta)$]
 $\Rightarrow \theta + 34^\circ = 90^\circ - \theta \Rightarrow 2\theta = (90^\circ - 34^\circ) = 56^\circ \Rightarrow \theta = 28^\circ$.
Hence, $\theta = 28^\circ$.

EXAMPLE 14 Express each of the following in terms of trigonometric ratios of angles between 0° and 45° .

(i) $\sin 75^\circ + \operatorname{cosec} 75^\circ$ (ii) $\tan 65^\circ + \cot 49^\circ$

(iii) $\sec 67^\circ + \operatorname{cosec} 58^\circ$ (iv) $\cos 83^\circ - \sec 76^\circ$

SOLUTION (i) $\sin 75^\circ + \operatorname{cosec} 75^\circ = \sin (90^\circ - 15^\circ) + \operatorname{cosec} (90^\circ - 15^\circ)$
 $\quad \quad \quad = (\cos 15^\circ + \sec 15^\circ)$.
(ii) $\tan 65^\circ + \cot 49^\circ = \tan (90^\circ - 25^\circ) + \cot (90^\circ - 41^\circ)$
 $\quad \quad \quad = (\cot 25^\circ + \tan 41^\circ)$.

$$\begin{aligned} \text{(iii) } \sec 67^\circ + \operatorname{cosec} 58^\circ &= \sec (90^\circ - 23^\circ) + \operatorname{cosec} (90^\circ - 32^\circ) \\ &= (\operatorname{cosec} 23^\circ + \sec 32^\circ). \end{aligned}$$

$$\begin{aligned} \text{(iv) } \cos 83^\circ - \sec 76^\circ &= \cos (90^\circ - 7^\circ) - \sec (90^\circ - 14^\circ) \\ &= (\sin 7^\circ - \operatorname{cosec} 14^\circ). \end{aligned}$$

EXAMPLE 15 If A and B are acute angles such that $\sin A = \cos B$, prove that $(A + B) = 90^\circ$.

SOLUTION $\sin A = \cos B \Rightarrow \sin A = \sin (90^\circ - B)$
 $\Rightarrow A = 90^\circ - B \quad [\because A \text{ and } (90^\circ - B) \text{ are acute}]$
 $\Rightarrow A + B = 90^\circ$.

EXAMPLE 16 If A, B, C are the angles of a $\triangle ABC$, show that $\sin \left(\frac{B+C}{2} \right) = \cos \frac{A}{2}$.
 [CBSE 2008C]

SOLUTION We know that the sum of the angles of a triangle is 180° .
 $\therefore A + B + C = 180^\circ \Rightarrow B + C = 180^\circ - A$
 $\Rightarrow \frac{B+C}{2} = 90^\circ - \frac{A}{2}$
 $\Rightarrow \sin \left(\frac{B+C}{2} \right) = \sin \left(90^\circ - \frac{A}{2} \right)$
 $\Rightarrow \sin \left(\frac{B+C}{2} \right) = \cos \frac{A}{2} \quad [\because \sin (90^\circ - \theta) = \cos \theta].$
 Hence, $\sin \left(\frac{B+C}{2} \right) = \cos \frac{A}{2}$.

EXERCISE 12

1. Without using trigonometric tables, evaluate:

$$\begin{array}{lll} \text{(i) } \frac{\sin 16^\circ}{\cos 74^\circ} & \text{(ii) } \frac{\sec 11^\circ}{\operatorname{cosec} 79^\circ} & \text{(iii) } \frac{\tan 27^\circ}{\cot 63^\circ} \\ \text{(iv) } \frac{\cos 35^\circ}{\sin 55^\circ} & \text{(v) } \frac{\operatorname{cosec} 42^\circ}{\sec 48^\circ} & \text{(vi) } \frac{\cot 38^\circ}{\tan 52^\circ} \end{array}$$

2. Without using trigonometric tables, prove that:

$$\begin{array}{ll} \text{(i) } \cos 81^\circ - \sin 9^\circ = 0 & \text{(ii) } \tan 71^\circ - \cot 19^\circ = 0 \\ \text{(iii) } \operatorname{cosec} 80^\circ - \sec 10^\circ = 0 & \text{(iv) } \operatorname{cosec}^2 72^\circ - \tan^2 18^\circ = 1 \\ \text{(v) } \cos^2 75^\circ + \cos^2 15^\circ = 1 & \text{(vi) } \tan^2 66^\circ - \cot^2 24^\circ = 0 \\ \text{(vii) } \sin^2 48^\circ + \sin^2 42^\circ = 1 & \text{(viii) } \cos^2 57^\circ - \sin^2 33^\circ = 0 \\ \text{(ix) } (\sin 65^\circ + \cos 25^\circ)(\sin 65^\circ - \cos 25^\circ) = 0 \end{array}$$

3. Without using trigonometric tables, prove that:

$$(i) \sin 53^\circ \cos 37^\circ + \cos 53^\circ \sin 37^\circ = 1$$

$$(ii) \cos 54^\circ \cos 36^\circ - \sin 54^\circ \sin 36^\circ = 0$$

$$(iii) \sec 70^\circ \sin 20^\circ + \cos 20^\circ \operatorname{cosec} 70^\circ = 2$$

$$(iv) \sin 35^\circ \sin 55^\circ - \cos 35^\circ \cos 55^\circ = 0$$

$$(v) (\sin 72^\circ + \cos 18^\circ)(\sin 72^\circ - \cos 18^\circ) = 0$$

$$(vi) \tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ = 1$$

4. Prove that:

$$(i) \frac{\sin 70^\circ}{\cos 20^\circ} + \frac{\operatorname{cosec} 20^\circ}{\sec 70^\circ} - 2 \cos 70^\circ \operatorname{cosec} 20^\circ = 0$$

$$(ii) \frac{\cos 80^\circ}{\sin 10^\circ} + \cos 59^\circ \operatorname{cosec} 31^\circ = 2$$

$$(iii) \frac{2 \sin 68^\circ}{\cos 22^\circ} - \frac{2 \cot 15^\circ}{5 \tan 75^\circ} - \frac{3 \tan 45^\circ \tan 20^\circ \tan 40^\circ \tan 50^\circ \tan 70^\circ}{5} = 1$$

$$(iv) \frac{\sin 18^\circ}{\cos 72^\circ} + \sqrt{3} (\tan 10^\circ \tan 30^\circ \tan 40^\circ \tan 50^\circ \tan 80^\circ) = 2 \quad [\text{CBSE 2008}]$$

$$(v) \frac{7 \cos 55^\circ}{3 \sin 35^\circ} - \frac{4(\cos 70^\circ \operatorname{cosec} 20^\circ)}{3(\tan 5^\circ \tan 25^\circ \tan 45^\circ \tan 65^\circ \tan 85^\circ)} = 1$$

5. Prove that:

$$(i) \sin \theta \cos (90^\circ - \theta) + \sin (90^\circ - \theta) \cos \theta = 1$$

$$(ii) \frac{\sin \theta}{\cos (90^\circ - \theta)} + \frac{\cos \theta}{\sin (90^\circ - \theta)} = 2$$

$$(iii) \frac{\sin \theta \cos (90^\circ - \theta) \cos \theta}{\sin (90^\circ - \theta)} + \frac{\cos \theta \sin (90^\circ - \theta) \sin \theta}{\cos (90^\circ - \theta)} = 1$$

$$(iv) \frac{\cos (90^\circ - \theta) \sec (90^\circ - \theta) \tan \theta}{\operatorname{cosec} (90^\circ - \theta) \sin (90^\circ - \theta) \cot (90^\circ - \theta)} + \frac{\tan (90^\circ - \theta)}{\cot \theta} = 2$$

$$(v) \frac{\cos (90^\circ - \theta)}{1 + \sin (90^\circ - \theta)} + \frac{1 + \sin (90^\circ - \theta)}{\cos (90^\circ - \theta)} = 2 \operatorname{cosec} \theta$$

$$(vi) \frac{\sec (90^\circ - \theta) \operatorname{cosec} \theta - \tan (90^\circ - \theta) \cot \theta + \cos^2 25^\circ + \cos^2 65^\circ}{3 \tan 27^\circ \tan 63^\circ} = \frac{2}{3}$$

[CBSE 2010]

$$(vii) \cot \theta \tan (90^\circ - \theta) - \sec (90^\circ - \theta) \operatorname{cosec} \theta + \sqrt{3} \tan 12^\circ \tan 60^\circ \tan 78^\circ = 2$$

[CBSE 2010]

6. Prove that:

$$(i) \tan 5^\circ \tan 25^\circ \tan 30^\circ \tan 65^\circ \tan 85^\circ = \frac{1}{\sqrt{3}}$$

$$(ii) \cot 12^\circ \cot 38^\circ \cot 52^\circ \cot 60^\circ \cot 78^\circ = \frac{1}{\sqrt{3}}$$

$$(iii) \cos 15^\circ \cos 35^\circ \operatorname{cosec} 55^\circ \cos 60^\circ \operatorname{cosec} 75^\circ = \frac{1}{2}$$

$$(iv) \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 180^\circ = 0$$

$$(v) \left(\frac{\sin 49^\circ}{\cos 41^\circ} \right)^2 + \left(\frac{\cos 41^\circ}{\sin 49^\circ} \right)^2 = 2$$

7. Prove that:

$$(i) \sin (70^\circ + \theta) - \cos (20^\circ - \theta) = 0$$

$$(ii) \tan (55^\circ - \theta) - \cot (35^\circ + \theta) = 0$$

$$(iii) \operatorname{cosec} (67^\circ + \theta) - \sec (23^\circ - \theta) = 0$$

$$(iv) \operatorname{cosec} (65^\circ + \theta) - \sec (25^\circ - \theta) - \tan (55^\circ - \theta) + \cot (35^\circ + \theta) = 0$$

$$(v) \sin (50^\circ + \theta) - \cos (40^\circ - \theta) + \tan 1^\circ \tan 10^\circ \tan 80^\circ \tan 89^\circ = 1$$

8. Express each of the following in terms of T-ratios of angles lying between 0° and 45° :

$$(i) \sin 67^\circ + \cos 75^\circ$$

$$(ii) \cot 65^\circ + \tan 49^\circ$$

$$(iii) \sec 78^\circ + \operatorname{cosec} 56^\circ$$

$$(iv) \operatorname{cosec} 54^\circ + \sin 72^\circ$$

9. If A , B and C are the angles of a $\triangle ABC$, prove that $\tan \left(\frac{C+A}{2} \right) = \cot \frac{B}{2}$.

10. If $\cos 2\theta = \sin 4\theta$, where 2θ and 4θ are acute angles, find the value of θ .

11. If $\sec 2A = \operatorname{cosec} (A - 42^\circ)$, where $2A$ is an acute angle, find the value of A . [CBSE 2008]

12. If $\sin 3A = \cos (A - 26^\circ)$, where $3A$ is an acute angle, find the value of A .

13. If $\tan 2A = \cot (A - 12^\circ)$, where $2A$ is an acute angle, find the value of A .

14. If $\sec 4A = \operatorname{cosec} (A - 15^\circ)$, where $4A$ is an acute angle, find the value of A .

15. Prove that:

$$\frac{2}{3} \operatorname{cosec}^2 58^\circ - \frac{2}{3} \cot 58^\circ \tan 32^\circ$$

$$- \frac{5}{3} \tan 13^\circ \tan 37^\circ \tan 45^\circ \tan 53^\circ \tan 77^\circ = -1. \quad [\text{CBSE 2009}]$$

ANSWERS (EXERCISE 12)

$$1. (i) 1 \quad (ii) 1 \quad (iii) 1 \quad (iv) 1 \quad (v) 1 \quad (vi) 1$$

$$8. (i) \cos 23^\circ + \sin 15^\circ \quad (ii) \tan 25^\circ + \cot 41^\circ \quad (iii) \operatorname{cosec} 12^\circ + \sec 34^\circ$$

$$(iv) \sec 36^\circ + \cos 18^\circ$$

$$10. \theta = 15^\circ$$

$$11. A = 44^\circ$$

$$12. A = 29^\circ$$

$$13. A = 34^\circ$$

$$14. A = 21^\circ$$

HINTS TO SOME SELECTED QUESTIONS

$$2. \text{ (iv) LHS} = \operatorname{cosec}^2 72^\circ - \tan^2 (90^\circ - 72^\circ) = \operatorname{cosec}^2 72^\circ - \cot^2 72^\circ = 1.$$

$$\text{ (viii) LHS} = \cos^2 57^\circ - \sin^2 (90^\circ - 57^\circ) = \cos^2 57^\circ - \cos^2 57^\circ = 0.$$

$$\begin{aligned} \text{ (ix) LHS} &= \sin^2 65^\circ - \cos^2 25^\circ = \sin^2 65^\circ - \cos^2 (90^\circ - 65^\circ) \\ &= \sin^2 65^\circ - \sin^2 65^\circ = 0. \end{aligned}$$

$$\begin{aligned} 3. \text{ (iv) LHS} &= \sin 35^\circ \sin (90^\circ - 35^\circ) - \cos 35^\circ \cos (90^\circ - 35^\circ) \\ &= \sin 35^\circ \cos 35^\circ - \sin 35^\circ \cos 35^\circ = 0. \end{aligned}$$

$$\begin{aligned} \text{ (v) LHS} &= \sin^2 72^\circ - \cos^2 18^\circ = \sin^2 72^\circ - \cos^2 (90^\circ - 72^\circ) \\ &= \sin^2 72^\circ - \sin^2 72^\circ = 0. \end{aligned}$$

$$6. \text{ (i) LHS} = (\tan 5^\circ \tan 85^\circ)(\tan 25^\circ \tan 65^\circ) \tan 30^\circ.$$

$$\text{ (ii) LHS} = (\cot 12^\circ \cot 78^\circ)(\cot 38^\circ \cot 52^\circ) \cot 60^\circ.$$

$$\text{ (iii) LHS} = (\cos 15^\circ \operatorname{cosec} 75^\circ)(\cos 35^\circ \operatorname{cosec} 55^\circ) \cos 60^\circ.$$

$$\text{ (iv) LHS} = \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 89^\circ \cos 90^\circ \cos 91^\circ \dots \cos 180^\circ = 0, \text{ since } \cos 90^\circ = 0.$$

$$\text{ (v) LHS} = \left\{ \frac{\sin 49^\circ}{\cos (90^\circ - 49^\circ)} \right\}^2 + \left\{ \frac{\cos 41^\circ}{\sin (90^\circ - 41^\circ)} \right\}^2 = \left(\frac{\sin 49^\circ}{\sin 49^\circ} \right)^2 + \left(\frac{\cos 41^\circ}{\cos 41^\circ} \right)^2 = 1 + 1 = 2.$$

$$7. \text{ (i) LHS} = \sin \{90^\circ - (20^\circ - \theta)\} - \cos (20^\circ - \theta) = \cos (20^\circ - \theta) - \cos (20^\circ - \theta) = 0.$$

$$\begin{aligned} \text{ (v) LHS} &= \sin \{90^\circ - (40^\circ - \theta)\} - \cos (40^\circ - \theta) + (\tan 1^\circ \tan 89^\circ)(\tan 10^\circ \tan 80^\circ) \\ &= \cos (40^\circ - \theta) - \cos (40^\circ - \theta) + (1 \times 1) = 1. \end{aligned}$$

$$8. \text{ (i) } \sin 67^\circ + \cos 75^\circ = \sin (90^\circ - 23^\circ) + \cos (90^\circ - 15^\circ) = \cos 23^\circ + \sin 15^\circ.$$

$$\text{ (ii) } \cot 65^\circ + \tan 49^\circ = \cot (90^\circ - 25^\circ) + \tan (90^\circ - 41^\circ) = \tan 25^\circ + \cot 41^\circ.$$

$$9. A + B + C = 180^\circ \Rightarrow C + A = 180^\circ - B \Rightarrow \frac{C+A}{2} = \left(90^\circ - \frac{B}{2}\right)$$

$$\Rightarrow \tan \left(\frac{C+A}{2} \right) = \tan \left(90^\circ - \frac{B}{2} \right) = \cot \frac{B}{2}.$$

$$10. \cos 2\theta = \sin 4\theta \Rightarrow \sin (90^\circ - 2\theta) = \sin 4\theta \Rightarrow 90^\circ - 2\theta = 4\theta \Rightarrow 6\theta = 90^\circ \Rightarrow \theta = 15^\circ.$$

$$11. \sec 2A = \operatorname{cosec} (A - 42^\circ) \Rightarrow \operatorname{cosec} (90^\circ - 2A) = \operatorname{cosec} (A - 42^\circ) \Rightarrow 90^\circ - 2A = A - 42^\circ.$$

$$12. \sin 3A = \cos (A - 26^\circ) \Rightarrow \cos (90^\circ - 3A) = \cos (A - 26^\circ) \Rightarrow 90^\circ - 3A = A - 26^\circ.$$

$$13. \tan 2A = \cot (A - 12^\circ) \Rightarrow \cot (90^\circ - 2A) = \cot (A - 12^\circ) \Rightarrow 90^\circ - 2A = A - 12^\circ.$$

$$14. \sec 4A = \operatorname{cosec} (A - 15^\circ) \Rightarrow \operatorname{cosec} (90^\circ - 4A) = \operatorname{cosec} (A - 15^\circ) \Rightarrow 90^\circ - 4A = A - 15^\circ.$$



TRIGONOMETRIC IDENTITY

An equation involving trigonometric ratios of an angle is called a trigonometric identity if it is true for all values of the angle.

SOME TRIGONOMETRIC IDENTITIES In Chapter 10, we proved some identities, summarised below.

SUMMARY

$$(i) \sin^2 \theta + \cos^2 \theta = 1$$

$$(ii) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(iii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$(iv) \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$(v) \tan \theta \times \cot \theta = 1$$

SOLVED EXAMPLES

EXAMPLE 1 Prove that

$$(i) (1 - \sin^2 \theta) \sec^2 \theta = 1$$

$$(ii) (1 + \tan^2 \theta) \cos^2 \theta = 1$$

$$(iii) (1 + \tan^2 \theta)(1 - \sin \theta)(1 + \sin \theta) = 1$$

$$(iv) 2\cos^2 \theta + \frac{2}{(1 + \cot^2 \theta)} = 2 \quad [\text{CBSE 2009C}]$$

SOLUTION We have

$$(i) \text{ LHS} = (1 - \sin^2 \theta) \sec^2 \theta$$

$$= \cos^2 \theta \cdot \sec^2 \theta$$

$$[\because 1 - \sin^2 \theta = \cos^2 \theta]$$

$$= \cos^2 \theta \cdot \frac{1}{\cos^2 \theta} = 1 = \text{RHS.}$$

$$\therefore \text{ LHS} = \text{RHS.}$$

$$(ii) \text{ LHS} = (1 + \tan^2 \theta) \cos^2 \theta$$

$$= \sec^2 \theta \cdot \cos^2 \theta$$

$$[\because 1 + \tan^2 \theta = \sec^2 \theta]$$

$$= \frac{1}{\cos^2 \theta} \cdot \cos^2 \theta = 1 = \text{RHS.}$$

$$\therefore \text{ LHS} = \text{RHS.}$$

$$\begin{aligned}
 \text{(iii) LHS} &= (1 + \tan^2 \theta)(1 - \sin \theta)(1 + \sin \theta) \\
 &= (1 + \tan^2 \theta)(1 - \sin^2 \theta) = \sec^2 \theta \cdot \cos^2 \theta \\
 &\quad [\because (1 + \tan^2 \theta) = \sec^2 \theta, (1 - \sin^2 \theta) = \cos^2 \theta] \\
 &= \frac{1}{\cos^2 \theta} \cdot \cos^2 \theta \quad \left[\because \sec^2 \theta = \frac{1}{\cos^2 \theta} \right] \\
 &= 1 = \text{RHS.} \\
 \therefore \text{ LHS} &= \text{RHS.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) LHS} &= 2 \cos^2 \theta + \frac{2}{(1 + \cot^2 \theta)} \\
 &= 2 \cos^2 \theta + \frac{2}{\operatorname{cosec}^2 \theta} \quad [\because 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta] \\
 &= 2 \cos^2 \theta + 2 \sin^2 \theta \quad \left[\because \frac{1}{\operatorname{cosec}^2 \theta} = \sin^2 \theta \right] \\
 &= 2(\cos^2 \theta + \sin^2 \theta) = 2 \times 1 \quad [\because \cos^2 \theta + \sin^2 \theta = 1] \\
 &= 2 = \text{RHS.} \\
 \therefore \text{ LHS} &= \text{RHS.}
 \end{aligned}$$

EXAMPLE 2 *Prove that*

$$(i) (\operatorname{cosec}^2 \theta - 1) \tan^2 \theta = 1 \quad (ii) (\sec^2 \theta - 1)(1 - \operatorname{cosec}^2 \theta) = -1$$

SOLUTION We have

$$\begin{aligned}
 \text{(i) LHS} &= (\operatorname{cosec}^2 \theta - 1) \tan^2 \theta \\
 &= (1 + \cot^2 \theta - 1) \tan^2 \theta \quad [\because \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta] \\
 &= \cot^2 \theta \cdot \tan^2 \theta \\
 &= \frac{1}{\tan^2 \theta} \cdot \tan^2 \theta \quad \left[\because \cot \theta = \frac{1}{\tan \theta} \right] \\
 &= 1 = \text{RHS.} \\
 \text{(ii) LHS} &= (\sec^2 \theta - 1)(1 - \operatorname{cosec}^2 \theta) \\
 &= (1 + \tan^2 \theta - 1)[1 - (1 + \cot^2 \theta)] \\
 &\quad [\because \sec^2 \theta = 1 + \tan^2 \theta \text{ and } \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta] \\
 &= \tan^2 \theta \cdot (-\cot^2 \theta) \\
 &= \tan^2 \theta \cdot \left(\frac{-1}{\tan^2 \theta} \right) \quad \left[\because \cot \theta = \frac{1}{\tan \theta} \right] \\
 &= -1 = \text{RHS.}
 \end{aligned}$$

EXAMPLE 3 *Prove that*

$$\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1. \quad [\text{CBSE 2009}]$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta \\ &= \frac{(\sin \theta + \cos \theta)(\sin^2 \theta + \cos^2 \theta - \sin \theta \cos \theta)}{(\sin \theta + \cos \theta)} + \sin \theta \cos \theta \\ &\quad [\because (a^3 + b^3) = (a + b)(a^2 + b^2 - ab)] \\ &= (1 - \sin \theta \cos \theta) + \sin \theta \cos \theta = 1 \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\ &= \text{RHS.} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

EXAMPLE 4 *Prove that*

$$(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta) = \frac{1}{\tan \theta + \cot \theta} \quad [\text{CBSE 2009C, '10}]$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= (\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta) \\ &= \left(\sin \theta - \frac{1}{\sin \theta} \right) \left(\cos \theta - \frac{1}{\cos \theta} \right) = \frac{(\sin^2 \theta - 1)}{\sin \theta} \cdot \frac{(\cos^2 \theta - 1)}{\cos \theta} \\ &= \frac{\{-(1 - \sin^2 \theta)\} \cdot \{-(1 - \cos^2 \theta)\}}{\sin \theta \cos \theta} \\ &= \frac{(-\cos^2 \theta)(-\sin^2 \theta)}{\sin \theta \cos \theta} \\ &\quad [\because 1 - \sin^2 \theta = \cos^2 \theta \text{ and } 1 - \cos^2 \theta = \sin^2 \theta] \\ &= \frac{(\cos^2 \theta \sin^2 \theta)}{\sin \theta \cos \theta} = \cos \theta \sin \theta. \\ \text{RHS} &= \frac{1}{\tan \theta + \cot \theta} \\ &= \frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} = \frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}} = \frac{\cos \theta \sin \theta}{(\sin^2 \theta + \cos^2 \theta)} \\ &= \cos \theta \sin \theta \quad [\because \sin^2 \theta + \cos^2 \theta = 1]. \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

EXAMPLE 5 *Prove that*

$$(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2. \quad [\text{CBSE 2005, '07, '08}]$$

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= (1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) \\
 &= \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right) \\
 &= \frac{(\sin \theta + \cos \theta - 1)}{\sin \theta} \cdot \frac{(\cos \theta + \sin \theta + 1)}{\cos \theta} \\
 &= \frac{[(\sin \theta + \cos \theta) - 1] \cdot [(\sin \theta + \cos \theta) + 1]}{\sin \theta \cos \theta} \\
 &= \frac{(\sin \theta + \cos \theta)^2 - 1}{\sin \theta \cos \theta} \\
 &= \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} \\
 &= \frac{1 + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\
 &= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} = 2 = \text{RHS}.
 \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 6 Prove that

$$\frac{\cos \theta}{(1 - \tan \theta)} + \frac{\sin \theta}{(1 - \cot \theta)} = (\cos \theta + \sin \theta). \quad [\text{CBSE 2007}]$$

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= \frac{\cos \theta}{(1 - \tan \theta)} + \frac{\sin \theta}{(1 - \cot \theta)} = \frac{\cos \theta}{\left(1 - \frac{\sin \theta}{\cos \theta}\right)} + \frac{\sin \theta}{\left(1 - \frac{\cos \theta}{\sin \theta}\right)} \\
 &= \frac{\cos^2 \theta}{(\cos \theta - \sin \theta)} + \frac{\sin^2 \theta}{(\sin \theta - \cos \theta)} \\
 &= \frac{\cos^2 \theta}{\cos \theta - \sin \theta} - \frac{\sin^2 \theta}{\cos \theta - \sin \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{(\cos \theta - \sin \theta)} = \frac{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)}{(\cos \theta - \sin \theta)} \\
 &= (\cos \theta + \sin \theta) = \text{RHS}.
 \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 7 Prove that

$$\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta. \quad [\text{CBSE 2010}]$$

SOLUTION We have

$$\text{LHS} = \frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)}$$

$$\begin{aligned}
 &= \frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \\
 &= \tan \theta \cdot \frac{[1 - 2(1 - \cos^2 \theta)]}{(2 \cos^2 \theta - 1)} \quad [\because \sin^2 \theta = 1 - \cos^2 \theta] \\
 &= \tan \theta \cdot \frac{(2 \cos^2 \theta - 1)}{(2 \cos^2 \theta - 1)} = \tan \theta = \text{RHS.}
 \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 8 Prove that

$$\frac{\tan A}{(1 - \cot A)} + \frac{\cot A}{(1 - \tan A)} = (1 + \tan A + \cot A). \quad [\text{CBSE 2010}]$$

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= \frac{\tan A}{(1 - \cot A)} + \frac{\cot A}{(1 - \tan A)} = \frac{\frac{\sin A}{\cos A}}{\left(1 - \frac{\cos A}{\sin A}\right)} + \frac{\frac{\cos A}{\sin A}}{\left(1 - \frac{\sin A}{\cos A}\right)} \\
 &= \frac{\sin^2 A}{\cos A (\sin A - \cos A)} + \frac{\cos^2 A}{\sin A (\cos A - \sin A)} \\
 &= \frac{\sin^2 A}{\cos A (\sin A - \cos A)} - \frac{\cos^2 A}{\sin A (\sin A - \cos A)} \\
 &= \frac{\sin^3 A - \cos^3 A}{\sin A \cos A (\sin A - \cos A)} \\
 &= \frac{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}{\sin A \cos A (\sin A - \cos A)} \\
 &\quad [\because (a^3 - b^3) = (a - b)(a^2 + b^2 + ab)] \\
 &= \frac{1 + \sin A \cos A}{\sin A \cos A}. \\
 \text{RHS} &= (1 + \tan A + \cot A) \\
 &= \left(1 + \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}\right) = \frac{\sin A \cos A + \sin^2 A + \cos^2 A}{\sin A \cos A} \\
 &= \frac{(1 + \sin A \cos A)}{\sin A \cos A}.
 \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 9 Prove that

$$(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \frac{1}{(\tan A + \cot A)}.$$

[CBSE 2010]

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= (\operatorname{cosec} A - \sin A)(\sec A - \cos A) \\
 &= \left(\frac{1}{\sin A} - \sin A\right)\left(\frac{1}{\cos A} - \cos A\right) = \left(\frac{1 - \sin^2 A}{\sin A}\right) \cdot \left(\frac{1 - \cos^2 A}{\cos A}\right) \\
 &= \frac{\cos^2 A \sin^2 A}{\cos A \sin A} = \cos A \sin A. \\
 \text{RHS} &= \frac{1}{(\tan A + \cot A)} = \frac{1}{\left(\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}\right)} \\
 &= \frac{\cos A \sin A}{(\sin^2 A + \cos^2 A)} = \cos A \sin A \quad [\because \sin^2 A + \cos^2 A = 1].
 \end{aligned}$$

Hence, LHS = RHS.

EXAMPLE 10 Prove that

$$\left(\frac{1 + \tan^2 A}{1 + \cot^2 A}\right) = \left(\frac{1 - \tan A}{1 - \cot A}\right)^2 = \tan^2 A. \quad [\text{CBSE 2008C}]$$

SOLUTION We have

$$\begin{aligned}
 \left(\frac{1 + \tan^2 A}{1 + \cot^2 A}\right) &= \frac{\sec^2 A}{\operatorname{cosec}^2 A} \\
 &[\because 1 + \tan^2 A = \sec^2 A \text{ and } 1 + \cot^2 A = \operatorname{cosec}^2 A] \\
 &= \frac{1}{\cos^2 A} \cdot \sin^2 A = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A. \\
 \text{And, } \left(\frac{1 - \tan A}{1 - \cot A}\right)^2 &= \frac{\left(1 - \frac{\sin A}{\cos A}\right)^2}{\left(1 - \frac{\cos A}{\sin A}\right)^2} \\
 &= \frac{(\cos A - \sin A)^2}{\frac{\cos^2 A}{(\sin A - \cos A)^2}} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A. \\
 \therefore \left(\frac{1 + \tan^2 A}{1 + \cot^2 A}\right) &= \left(\frac{1 - \tan A}{1 - \cot A}\right)^2 = \tan^2 A.
 \end{aligned}$$

EXAMPLE 11 Prove that

$$\begin{aligned}
 (i) \quad \frac{\tan A + \sin A}{\tan A - \sin A} &= \frac{\sec A + 1}{\sec A - 1} \\
 (ii) \quad \frac{\cot A - \cos A}{\cot A + \cos A} &= \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1} \quad [\text{CBSE 2008}]
 \end{aligned}$$

SOLUTION We have

$$\begin{aligned}
 \text{(i) LHS} &= \frac{\tan A + \sin A}{\tan A - \sin A} \\
 &= \frac{\frac{\sin A}{\cos A} + \sin A}{\frac{\sin A}{\cos A} - \sin A} = \frac{\sin A \left(\frac{1}{\cos A} + 1 \right)}{\sin A \left(\frac{1}{\cos A} - 1 \right)} \\
 &= \frac{\left(\frac{1}{\cos A} + 1 \right)}{\left(\frac{1}{\cos A} - 1 \right)} = \frac{\sec A + 1}{\sec A - 1} = \text{RHS.}
 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS.}$

$$\begin{aligned}
 \text{(ii) LHS} &= \frac{\cot A - \cos A}{\cot A + \cos A} \\
 &= \frac{\frac{\cos A}{\sin A} - \cos A}{\frac{\cos A}{\sin A} + \cos A} = \frac{\cos A \left(\frac{1}{\sin A} - 1 \right)}{\cos A \left(\frac{1}{\sin A} + 1 \right)} \\
 &= \frac{\left(\frac{1}{\sin A} - 1 \right)}{\left(\frac{1}{\sin A} + 1 \right)} = \frac{(\operatorname{cosec} A - 1)}{(\operatorname{cosec} A + 1)} = \text{RHS.}
 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS.}$

EXAMPLE 12 Prove that

$$(i) \sec^2 \theta + \operatorname{cosec}^2 \theta = \sec^2 \theta \operatorname{cosec}^2 \theta$$

$$(ii) \tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$$

$$(iii) \tan^2 \theta + \cot^2 \theta + 2 = \sec^2 \theta \operatorname{cosec}^2 \theta$$

SOLUTION We have

$$\begin{aligned}
 \text{(i) LHS} &= \sec^2 \theta + \operatorname{cosec}^2 \theta \\
 &= \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \sin^2 \theta} \\
 &= \frac{1}{\cos^2 \theta \sin^2 \theta} \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\
 &= \sec^2 \theta \operatorname{cosec}^2 \theta = \text{RHS.}
 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS.}$

$$\begin{aligned}
 \text{(ii) LHS} &= \tan^2 \theta - \sin^2 \theta \\
 &= \frac{\sin^2 \theta}{\cos^2 \theta} - \sin^2 \theta = \frac{\sin^2 \theta - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta}
 \end{aligned}$$

$$= \frac{\sin^2 \theta (1 - \cos^2 \theta)}{\cos^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \sin^2 \theta$$

$$= \tan^2 \theta \sin^2 \theta = \text{RHS.}$$

$$\therefore \text{LHS} = \text{RHS.}$$

$$\begin{aligned} \text{(iii) LHS} &= \tan^2 \theta + \cot^2 \theta + 2 \\ &= (1 + \tan^2 \theta) + (1 + \cot^2 \theta) = \sec^2 \theta + \operatorname{cosec}^2 \theta \\ &= \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \sin^2 \theta} \\ &= \frac{1}{\cos^2 \theta \sin^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta = \text{RHS.} \\ \therefore \text{LHS} &= \text{RHS.} \end{aligned}$$

EXAMPLE 13 *Prove that*

$$(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}. \quad [\text{CBSE 2000C}]$$

SOLUTION

$$\begin{aligned} \text{LHS} &= (\operatorname{cosec} \theta - \cot \theta)^2 \\ &= \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2 = \left(\frac{1 - \cos \theta}{\sin \theta} \right)^2 \\ &= \frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 - \cos^2 \theta)} \quad [\because \sin^2 \theta = 1 - \cos^2 \theta] \\ &= \frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)} = \frac{(1 - \cos \theta)}{(1 + \cos \theta)} = \text{RHS.} \\ \therefore \text{LHS} &= \text{RHS.} \end{aligned}$$

EXAMPLE 14 *Prove that*

$$(\sec^4 \theta - \sec^2 \theta) = (\tan^2 \theta + \tan^4 \theta).$$

SOLUTION

We have

$$\begin{aligned} \text{LHS} &= (\sec^4 \theta - \sec^2 \theta) = \sec^2 \theta (\sec^2 \theta - 1) \\ &= (1 + \tan^2 \theta)(1 + \tan^2 \theta - 1) = (1 + \tan^2 \theta) \tan^2 \theta \\ &= (\tan^2 \theta + \tan^4 \theta) = \text{RHS.} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

EXAMPLE 15 *Prove that*

$$\left(1 + \frac{1}{\tan^2 A} \right) \left(1 + \frac{1}{\cot^2 A} \right) = \frac{1}{(\sin^2 A - \sin^4 A)}. \quad [\text{CBSE 2006C}]$$

SOLUTION We have

$$\begin{aligned}\text{LHS} &= \left(1 + \frac{1}{\tan^2 A}\right) \left(1 + \frac{1}{\cot^2 A}\right) \\ &= (1 + \cot^2 A)(1 + \tan^2 A) = \operatorname{cosec}^2 A \cdot \sec^2 A \\ &= \frac{1}{\sin^2 A} \cdot \frac{1}{\cos^2 A} = \frac{1}{\sin^2 A \cos^2 A} \\ &= \frac{1}{\sin^2 A (1 - \sin^2 A)} = \frac{1}{(\sin^2 A - \sin^4 A)} = \text{RHS}.\end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 16 Prove that

$$(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = (7 + \tan^2 \theta + \cot^2 \theta).$$

[CBSE 2008, '09C]

SOLUTION We have

$$\begin{aligned}\text{LHS} &= (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 \\ &= (\sin^2 \theta + \operatorname{cosec}^2 \theta + 2 \sin \theta \operatorname{cosec} \theta) \\ &\quad + (\cos^2 \theta + \sec^2 \theta + 2 \cos \theta \sec \theta) \\ &= (\sin^2 \theta + \operatorname{cosec}^2 \theta + 2) + (\cos^2 \theta + \sec^2 \theta + 2) \\ &\quad [\because \sin \theta \operatorname{cosec} \theta = 1 \text{ and } \cos \theta \sec \theta = 1] \\ &= (\sin^2 \theta + \cos^2 \theta) + 4 + (\operatorname{cosec}^2 \theta + \sec^2 \theta) \\ &= 1 + 4 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) \\ &\quad [\because \sin^2 \theta + \cos^2 \theta = 1, \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta \\ &\quad \text{and } \sec^2 \theta = 1 + \tan^2 \theta] \\ &= (7 + \tan^2 \theta + \cot^2 \theta) = \text{RHS}.\end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 17 Prove that

$$(i) \frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2 \quad [\text{CBSE 2008C}]$$

$$(ii) \frac{(1 + \cos \theta)}{(1 - \cos \theta)} = (\operatorname{cosec} \theta + \cot \theta)^2 \quad [\text{CBSE 2007C}]$$

SOLUTION We have

$$\begin{aligned}(i) \text{ RHS} &= (\sec \theta - \tan \theta)^2 \\ &= \left(\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}\right)^2 = \left(\frac{1 - \sin \theta}{\cos \theta}\right)^2 = \frac{(1 - \sin \theta)^2}{\cos^2 \theta} \\ &= \frac{(1 - \sin \theta)^2}{(1 - \sin^2 \theta)} = \frac{(1 - \sin \theta)(1 - \sin \theta)}{(1 + \sin \theta)(1 - \sin \theta)}\end{aligned}$$

$$= \frac{(1 - \sin \theta)}{(1 + \sin \theta)} = \text{LHS.}$$

$$\therefore \text{RHS} = \text{LHS.}$$

$$\begin{aligned} \text{(ii) RHS} &= (\operatorname{cosec} \theta + \cot \theta)^2 = \left(\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} \right)^2 \\ &= \left(\frac{1 + \cos \theta}{\sin \theta} \right)^2 = \frac{(1 + \cos \theta)^2}{\sin^2 \theta} = \frac{(1 + \cos \theta)(1 + \cos \theta)}{1 - \cos^2 \theta} \\ &= \frac{(1 + \cos \theta)(1 + \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)} = \frac{(1 + \cos \theta)}{(1 - \cos \theta)} = \text{LHS.} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

EXAMPLE 18 Prove that

$$\begin{aligned} \text{(i)} \quad & \frac{1}{(\operatorname{cosec} \theta - \cot \theta)} = (\operatorname{cosec} \theta + \cot \theta) \\ \text{(ii)} \quad & \frac{(\sec \theta - \tan \theta)}{(\sec \theta + \tan \theta)} = (1 + 2 \tan^2 \theta - 2 \sec \theta \tan \theta) \end{aligned}$$

SOLUTION We have

$$\begin{aligned} \text{(i) LHS} &= \frac{1}{(\operatorname{cosec} \theta - \cot \theta)} = \frac{1}{(\operatorname{cosec} \theta - \cot \theta)} \times \frac{(\operatorname{cosec} \theta + \cot \theta)}{(\operatorname{cosec} \theta + \cot \theta)} \\ &= \frac{(\operatorname{cosec} \theta + \cot \theta)}{(\operatorname{cosec}^2 \theta - \cot^2 \theta)} = \operatorname{cosec} \theta + \cot \theta = \text{RHS} \\ & \quad [\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1]. \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

$$\begin{aligned} \text{(ii) LHS} &= \frac{(\sec \theta - \tan \theta)}{(\sec \theta + \tan \theta)} = \frac{(\sec \theta - \tan \theta)}{(\sec \theta + \tan \theta)} \times \frac{(\sec \theta - \tan \theta)}{(\sec \theta - \tan \theta)} \\ &= \frac{(\sec \theta - \tan \theta)^2}{(\sec^2 \theta - \tan^2 \theta)} = (\sec \theta - \tan \theta)^2 \\ & \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\ &= \sec^2 \theta + \tan^2 \theta - 2 \sec \theta \tan \theta \\ &= (1 + \tan^2 \theta) + \tan^2 \theta - 2 \sec \theta \tan \theta \\ &= 1 + 2 \tan^2 \theta - 2 \sec \theta \tan \theta = \text{RHS.} \\ \therefore \text{LHS} &= \text{RHS.} \end{aligned}$$

EXAMPLE 19 Prove that

$$\frac{1}{(\operatorname{cosec} \theta - \cot \theta)} - \frac{1}{\sin \theta} = \frac{1}{\sin \theta} - \frac{1}{(\operatorname{cosec} \theta + \cot \theta)}.$$

SOLUTION

$$\begin{aligned}
 \text{LHS} &= \frac{1}{(\operatorname{cosec} \theta - \cot \theta)} - \frac{1}{\sin \theta} \\
 &= \frac{1}{(\operatorname{cosec} \theta - \cot \theta)} \times \frac{(\operatorname{cosec} \theta + \cot \theta)}{(\operatorname{cosec} \theta + \cot \theta)} - \frac{1}{\sin \theta} \\
 &= \frac{(\operatorname{cosec} \theta + \cot \theta)}{(\operatorname{cosec}^2 \theta - \cot^2 \theta)} - \operatorname{cosec} \theta \quad \left[\because \frac{1}{\sin \theta} = \operatorname{cosec} \theta \right] \\
 &= (\operatorname{cosec} \theta + \cot \theta) - \operatorname{cosec} \theta \quad [\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1] \\
 &= \cot \theta. \\
 \text{RHS} &= \frac{1}{\sin \theta} - \frac{1}{(\operatorname{cosec} \theta + \cot \theta)} \\
 &= \operatorname{cosec} \theta - \frac{1}{(\operatorname{cosec} \theta + \cot \theta)} \times \frac{(\operatorname{cosec} \theta - \cot \theta)}{(\operatorname{cosec} \theta - \cot \theta)} \\
 &= \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - \cot \theta)}{(\operatorname{cosec}^2 \theta - \cot^2 \theta)} \\
 &= \operatorname{cosec} \theta - (\operatorname{cosec} \theta - \cot \theta) \quad [\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1] \\
 &= \cot \theta. \\
 \therefore \quad \text{LHS} &= \text{RHS}.
 \end{aligned}$$

EXAMPLE 20 *Prove that*

$$\frac{1}{(\sec \theta - \tan \theta)} - \frac{1}{\cos \theta} = \frac{1}{\cos \theta} - \frac{1}{(\sec \theta + \tan \theta)} \quad [\text{CBSE 2005, '08}]$$

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= \frac{1}{(\sec \theta - \tan \theta)} - \frac{1}{\cos \theta} \\
 &= \frac{1}{(\sec \theta - \tan \theta)} \times \frac{(\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} - \sec \theta \\
 &= \frac{(\sec \theta + \tan \theta)}{(\sec^2 \theta - \tan^2 \theta)} - \sec \theta \\
 &= (\sec \theta + \tan \theta) - \sec \theta \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\
 &= \tan \theta. \\
 \text{RHS} &= \frac{1}{\cos \theta} - \frac{1}{(\sec \theta + \tan \theta)} \\
 &= \sec \theta - \frac{1}{(\sec \theta + \tan \theta)} \times \frac{(\sec \theta - \tan \theta)}{(\sec \theta - \tan \theta)} \\
 &= \sec \theta - \frac{(\sec \theta - \tan \theta)}{(\sec^2 \theta - \tan^2 \theta)} \\
 &= \sec \theta - (\sec \theta - \tan \theta) \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\
 &= \tan \theta. \\
 \therefore \quad \text{LHS} &= \text{RHS}.
 \end{aligned}$$

EXAMPLE 21 *Prove that*

$$(i) \frac{\sin \theta}{(1 - \cos \theta)} = (\operatorname{cosec} \theta + \cot \theta)$$

$$(ii) \frac{1}{(\sec \theta - \tan \theta)} = (\sec \theta + \tan \theta)$$

SOLUTION We have

$$\begin{aligned} (i) \text{ LHS} &= \frac{\sin \theta}{(1 - \cos \theta)} \\ &= \frac{\sin \theta}{(1 - \cos \theta)} \times \frac{(1 + \cos \theta)}{(1 + \cos \theta)} \\ &\quad \text{[multiplying num. and denom. by } (1 + \cos \theta)] \\ &= \frac{\sin \theta(1 + \cos \theta)}{(1 - \cos^2 \theta)} = \frac{\sin \theta(1 + \cos \theta)}{\sin^2 \theta} = \frac{(1 + \cos \theta)}{\sin \theta} \\ &= \left(\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} \right) = (\operatorname{cosec} \theta + \cot \theta) = \text{RHS.} \\ \therefore \text{ LHS} &= \text{RHS.} \end{aligned}$$

$$\begin{aligned} (ii) \text{ LHS} &= \frac{1}{(\sec \theta - \tan \theta)} \\ &= \frac{1}{(\sec \theta - \tan \theta)} \times \frac{(\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} \\ &\quad \text{[multiplying num. and denom. by } (\sec \theta + \tan \theta)] \\ &= \frac{(\sec \theta + \tan \theta)}{(\sec^2 \theta - \tan^2 \theta)} \\ &= (\sec \theta + \tan \theta) \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\ &= \text{RHS.} \\ \therefore \text{ LHS} &= \text{RHS.} \end{aligned}$$

EXAMPLE 22 *Prove that*

$$\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{\cos \theta}{(1 - \sin \theta)}. \quad [\text{CBSE 2002, '03, '05C}]$$

$$\begin{aligned} \text{SOLUTION} \quad \text{LHS} &= \frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} \\ &= \frac{(\sec \theta + \tan \theta) - (\sec^2 \theta - \tan^2 \theta)}{(\tan \theta - \sec \theta + 1)} \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\ &= \frac{(\sec \theta + \tan \theta)[1 - (\sec \theta - \tan \theta)]}{(\tan \theta - \sec \theta + 1)} \\ &= \frac{(\sec \theta + \tan \theta)(\tan \theta - \sec \theta + 1)}{(\tan \theta - \sec \theta + 1)} = (\sec \theta + \tan \theta) \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \right) = \frac{(1 + \sin \theta)}{\cos \theta} = \frac{(1 + \sin \theta)}{\cos \theta} \times \frac{(1 - \sin \theta)}{(1 - \sin \theta)} \\
 &= \frac{(1 - \sin^2 \theta)}{\cos \theta (1 - \sin \theta)} = \frac{\cos^2 \theta}{\cos \theta (1 - \sin \theta)} = \frac{\cos \theta}{(1 - \sin \theta)} = \text{RHS.}
 \end{aligned}$$

Hence, $\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{\cos \theta}{(1 - \sin \theta)}.$

EXAMPLE 23 *Prove that*

$$\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{(\sec \theta - \tan \theta)}.$$

SOLUTION

$$\begin{aligned}
 \text{LHS} &= \frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} \\
 &= \frac{\frac{\sin \theta}{\cos \theta} - 1 + \frac{1}{\cos \theta}}{\frac{\sin \theta}{\cos \theta} + 1 - \frac{1}{\cos \theta}}
 \end{aligned}$$

[on dividing num. and denom. by $\cos \theta$]

$$\begin{aligned}
 &= \frac{\tan \theta - 1 + \sec \theta}{\tan \theta + 1 - \sec \theta} = \frac{(\sec \theta + \tan \theta - 1)}{(\tan \theta - \sec \theta + 1)} \\
 &= \frac{(\sec \theta + \tan \theta) - (\sec^2 \theta - \tan^2 \theta)}{(\tan \theta - \sec \theta + 1)} \quad [\because 1 = \sec^2 \theta - \tan^2 \theta] \\
 &= \frac{(\sec \theta + \tan \theta)[1 - (\sec \theta - \tan \theta)]}{(\tan \theta - \sec \theta + 1)} \\
 &= \frac{(\sec \theta + \tan \theta)(\tan \theta - \sec \theta + 1)}{(\tan \theta - \sec \theta + 1)} = (\sec \theta + \tan \theta).
 \end{aligned}$$

$$\text{RHS} = \frac{1}{(\sec \theta - \tan \theta)}$$

$$\begin{aligned}
 &= \frac{1}{(\sec \theta - \tan \theta)} \times \frac{(\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} = \frac{(\sec \theta + \tan \theta)}{(\sec^2 \theta - \tan^2 \theta)} \\
 &= (\sec \theta + \tan \theta) \quad [\because \sec^2 \theta - \tan^2 \theta = 1].
 \end{aligned}$$

Hence, LHS = RHS.

EXAMPLE 24 *Prove that*

$$\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \frac{1 + \cos \theta}{\sin \theta}.$$

SOLUTION

We have

$$\text{LHS} = \frac{(\cot \theta + \operatorname{cosec} \theta) - 1}{(\cot \theta - \operatorname{cosec} \theta + 1)}$$

$$\begin{aligned}
 &= \frac{(\operatorname{cosec} \theta + \cot \theta) - (\operatorname{cosec}^2 \theta - \cot^2 \theta)}{(\cot \theta - \operatorname{cosec} \theta + 1)} \\
 &\quad [\because 1 = \operatorname{cosec}^2 \theta - \cot^2 \theta] \\
 &= \frac{(\operatorname{cosec} \theta + \cot \theta)[1 - (\operatorname{cosec} \theta - \cot \theta)]}{(\cot \theta - \operatorname{cosec} \theta + 1)} \\
 &= \frac{(\operatorname{cosec} \theta + \cot \theta)(\cot \theta - \operatorname{cosec} \theta + 1)}{(\cot \theta - \operatorname{cosec} \theta + 1)} \\
 &= (\operatorname{cosec} \theta + \cot \theta) = \left(\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} \right) = \frac{1 + \cos \theta}{\sin \theta} = \text{RHS.} \\
 \text{Hence, } \frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} &= \frac{1 + \cos \theta}{\sin \theta}.
 \end{aligned}$$

EXAMPLE 25 Prove that

$$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \tan \theta + \cot \theta.$$

SOLUTION We have

$$\begin{aligned}
 \text{LHS} &= \sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \\
 &= \sqrt{(1 + \tan^2 \theta) + (1 + \cot^2 \theta)} = \sqrt{\tan^2 \theta + \cot^2 \theta + 2} \\
 &= \sqrt{\tan^2 \theta + \cot^2 \theta + 2 \tan \theta \cdot \cot \theta} \quad [\because \tan \theta \cdot \cot \theta = 1] \\
 &= \sqrt{(\tan \theta + \cot \theta)^2} = \tan \theta + \cot \theta = \text{RHS.}
 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

EXAMPLE 26 Prove that

$$(i) \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sec \theta - \tan \theta$$

$$(ii) \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \operatorname{cosec} \theta + \cot \theta$$

SOLUTION We have

$$\begin{aligned}
 (i) \text{ LHS} &= \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}} \times \frac{\sqrt{1 - \sin \theta}}{\sqrt{1 - \sin \theta}} \\
 &= \frac{1 - \sin \theta}{\sqrt{1 - \sin^2 \theta}} = \frac{1 - \sin \theta}{\cos \theta} \\
 &= \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} = \sec \theta - \tan \theta = \text{RHS.}
 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS.}$$

$$\begin{aligned}
 (ii) \text{ LHS} &= \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \frac{\sqrt{1 + \cos \theta}}{\sqrt{1 - \cos \theta}} \times \frac{\sqrt{1 + \cos \theta}}{\sqrt{1 + \cos \theta}} \\
 &= \frac{1 + \cos \theta}{\sqrt{1 - \cos^2 \theta}} = \frac{1 + \cos \theta}{\sin \theta}
 \end{aligned}$$

$$= \frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} = \operatorname{cosec} \theta + \cot \theta = \text{RHS.}$$

$\therefore$ LHS = RHS.

EXAMPLE 27 Prove that

$$\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = 2 \sec \theta. \quad [\text{CBSE 2001C}]$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} \\ &= \frac{\sqrt{1+\sin \theta}}{\sqrt{1-\sin \theta}} + \frac{\sqrt{1-\sin \theta}}{\sqrt{1+\sin \theta}} = \frac{1+\sin \theta + 1-\sin \theta}{\sqrt{(1-\sin \theta)(1+\sin \theta)}} \\ &= \frac{2}{\sqrt{(1-\sin^2 \theta)}} = \frac{2}{\sqrt{\cos^2 \theta}} = \frac{2}{\cos \theta} = 2 \sec \theta = \text{RHS.} \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 28 Prove that

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} + \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} = 2 \operatorname{cosec} \theta. \quad [\text{CBSE 2006C}]$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= \sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} + \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} = \frac{\sqrt{\sec \theta - 1}}{\sqrt{\sec \theta + 1}} + \frac{\sqrt{\sec \theta + 1}}{\sqrt{\sec \theta - 1}} \\ &= \frac{\sec \theta - 1 + \sec \theta + 1}{\sqrt{(\sec \theta + 1)(\sec \theta - 1)}} = \frac{2 \sec \theta}{\sqrt{\sec^2 \theta - 1}} = \frac{2 \sec \theta}{\tan \theta} \\ &\quad [\because \sec^2 \theta - 1 = \tan^2 \theta] \\ &= 2 \sec \theta \cot \theta = \frac{2}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{2}{\sin \theta} = 2 \operatorname{cosec} \theta = \text{RHS.} \end{aligned}$$

$\therefore$ LHS = RHS.

EXAMPLE 29 Prove that

$$\frac{\operatorname{cosec} \theta}{(\operatorname{cosec} \theta - 1)} + \frac{\operatorname{cosec} \theta}{(\operatorname{cosec} \theta + 1)} = 2 \sec^2 \theta. \quad [\text{CBSE 2004C, '09}]$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= \frac{\operatorname{cosec} \theta}{(\operatorname{cosec} \theta - 1)} + \frac{\operatorname{cosec} \theta}{(\operatorname{cosec} \theta + 1)} \\ &= \frac{\operatorname{cosec} \theta (\operatorname{cosec} \theta + 1) + \operatorname{cosec} \theta (\operatorname{cosec} \theta - 1)}{(\operatorname{cosec}^2 \theta - 1)} \\ &= \frac{2 \operatorname{cosec}^2 \theta}{(1 + \cot^2 \theta - 1)} \quad [\because \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta] \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \operatorname{cosec}^2 \theta}{\cot^2 \theta} = 2 \operatorname{cosec}^2 \theta \tan^2 \theta \\
 &= 2 \times \frac{1}{\sin^2 \theta} \times \frac{\sin^2 \theta}{\cos^2 \theta} = \frac{2}{\cos^2 \theta} = 2 \sec^2 \theta = \text{RHS.}
 \end{aligned}$$

AN IMPORTANT REMARK

In order to show that a given trigonometric equation is not an identity, it is sufficient to find by hit and trial, a single value of θ which does not satisfy it.

EXAMPLE 30 Show that $(\cos^2 \theta - \sin^2 \theta) = \frac{2 \tan \theta}{(1 - \tan^2 \theta)}$ is not an identity.

SOLUTION Putting $\theta = 30^\circ$, we find:

$$\begin{aligned}
 \text{LHS} &= (\cos^2 30^\circ - \sin^2 30^\circ) \\
 &= \left\{ \left(\frac{\sqrt{3}}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right\} = \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{2}{4} = \frac{1}{2}, \text{ and} \\
 \text{RHS} &= \frac{2 \tan 30^\circ}{(1 - \tan^2 30^\circ)} = \frac{2 \times \frac{1}{\sqrt{3}}}{\left[1 - \left(\frac{1}{\sqrt{3}} \right)^2 \right]} = \frac{\frac{2}{\sqrt{3}}}{\left(1 - \frac{1}{3} \right)} = \left(\frac{2}{\sqrt{3}} \times \frac{3}{2} \right) = \sqrt{3}.
 \end{aligned}$$

$\therefore$ LHS $\neq$ RHS.

Hence, the given equation is not an identity.

EXAMPLE 31 Prove that

$$(i) (\sin^2 A \cos^2 B - \cos^2 A \sin^2 B) = (\sin^2 A - \sin^2 B)$$

$$(ii) (\tan^2 A \sec^2 B - \sec^2 A \tan^2 B) = (\tan^2 A - \tan^2 B)$$

SOLUTION We have

$$\begin{aligned}
 (i) \text{ LHS} &= (\sin^2 A \cos^2 B - \cos^2 A \sin^2 B) \\
 &= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \\
 &= \sin^2 A - \sin^2 B = \text{RHS.}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \text{ LHS} &= (\tan^2 A \sec^2 B - \sec^2 A \tan^2 B) \\
 &= \tan^2 A (1 + \tan^2 B) - (1 + \tan^2 A) \tan^2 B \\
 &= (\tan^2 A - \tan^2 B) = \text{RHS.}
 \end{aligned}$$

EXAMPLE 32 Prove that

$$(\tan^2 A - \tan^2 B) = \frac{(\sin^2 A - \sin^2 B)}{\cos^2 A \cos^2 B} = \frac{(\cos^2 B - \cos^2 A)}{\cos^2 B \cos^2 A}.$$

SOLUTION We have

$$\begin{aligned}
 (\tan^2 A - \tan^2 B) &= \frac{\sin^2 A}{\cos^2 A} - \frac{\sin^2 B}{\cos^2 B} \\
 &= \frac{\sin^2 A \cos^2 B - \cos^2 A \sin^2 B}{\cos^2 A \cos^2 B} \\
 &= \frac{\sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B}{\cos^2 A \cos^2 B} \\
 &= \frac{(\sin^2 A - \sin^2 B)}{\cos^2 A \cos^2 B}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } (\tan^2 A - \tan^2 B) &= \frac{\sin^2 A}{\cos^2 A} - \frac{\sin^2 B}{\cos^2 B} \\
 &= \frac{\sin^2 A \cos^2 B - \cos^2 A \sin^2 B}{\cos^2 A \cos^2 B} \\
 &= \frac{(1 - \cos^2 A) \cos^2 B - \cos^2 A (1 - \cos^2 B)}{\cos^2 B \cos^2 A} \\
 &= \frac{(\cos^2 B - \cos^2 A)}{\cos^2 B \cos^2 A}.
 \end{aligned}$$

$$\text{Hence, } (\tan^2 A - \tan^2 B) = \frac{(\sin^2 A - \sin^2 B)}{\cos^2 A \cos^2 B} = \frac{(\cos^2 B - \cos^2 A)}{\cos^2 B \cos^2 A}.$$

EXERCISE 13A

Prove each of the following identities:

1. (i) $(1 - \cos^2 \theta) \operatorname{cosec}^2 \theta = 1$ (ii) $(1 + \cot^2 \theta) \sin^2 \theta = 1$
2. (i) $(\sec^2 \theta - 1) \cot^2 \theta = 1$ (ii) $(\sec^2 \theta - 1)(\operatorname{cosec}^2 \theta - 1) = 1$
 (iii) $(1 - \cos^2 \theta) \sec^2 \theta = \tan^2 \theta$
3. (i) $\sin^2 \theta + \frac{1}{(1 + \tan^2 \theta)} = 1$ (ii) $\frac{1}{(1 + \tan^2 \theta)} + \frac{1}{(1 + \cot^2 \theta)} = 1$
4. (i) $(1 + \cos \theta)(1 - \cos \theta)(1 + \cot^2 \theta) = 1$
 (ii) $\operatorname{cosec} \theta (1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta) = 1$
5. (i) $\cot^2 \theta - \frac{1}{\sin^2 \theta} = -1$ (ii) $\tan^2 \theta - \frac{1}{\cos^2 \theta} = -1$
 (iii) $\cos^2 \theta + \frac{1}{(1 + \cot^2 \theta)} = 1$
6. $\frac{1}{(1 + \sin \theta)} + \frac{1}{(1 - \sin \theta)} = 2 \sec^2 \theta$

7. (i) $\sec \theta(1 - \sin \theta)(\sec \theta + \tan \theta) = 1$
 (ii) $\sin \theta(1 + \tan \theta) + \cos \theta(1 + \cot \theta) = (\sec \theta + \operatorname{cosec} \theta)$ [CBSE 2008]
8. (i) $1 + \frac{\cot^2 \theta}{(1 + \operatorname{cosec} \theta)} = \operatorname{cosec} \theta$
 (ii) $1 + \frac{\tan^2 \theta}{(1 + \sec \theta)} = \sec \theta$
9. $\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$
10. $\frac{\tan^2 \theta}{(1 + \tan^2 \theta)} + \frac{\cot^2 \theta}{(1 + \cot^2 \theta)} = 1$
11. $\frac{\sin \theta}{(1 + \cos \theta)} + \frac{(1 + \cos \theta)}{\sin \theta} = 2 \operatorname{cosec} \theta$
12. $\frac{\tan \theta}{(1 - \cot \theta)} + \frac{\cot \theta}{(1 - \tan \theta)} = (1 + \sec \theta \operatorname{cosec} \theta)$ [CBSE 2008C]
13. $\frac{\cos^2 \theta}{(1 - \tan \theta)} + \frac{\sin^3 \theta}{(\sin \theta - \cos \theta)} = (1 + \sin \theta \cos \theta)$
14. $\frac{\cos \theta}{(1 - \tan \theta)} - \frac{\sin^2 \theta}{(\cos \theta - \sin \theta)} = (\cos \theta + \sin \theta)$
15. $(1 + \tan^2 \theta)(1 + \cot^2 \theta) = \frac{1}{(\sin^2 \theta - \sin^4 \theta)}$
16. $\frac{\tan \theta}{(1 + \tan^2 \theta)^2} + \frac{\cot \theta}{(1 + \cot^2 \theta)^2} = \sin \theta \cos \theta$
17. (i) $\sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$ [CBSE 2007]
 (ii) $\sin^2 \theta + \cos^4 \theta = \cos^2 \theta + \sin^4 \theta$
 (iii) $\operatorname{cosec}^4 \theta - \operatorname{cosec}^2 \theta = \cot^4 \theta + \cot^2 \theta$
18. (i) $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = (\cos^2 \theta - \sin^2 \theta)$
 (ii) $\frac{1 - \tan^2 \theta}{\cot^2 - 1} = \tan^2 \theta$
19. (i) $\frac{\tan \theta}{(\sec \theta - 1)} + \frac{\tan \theta}{(\sec \theta + 1)} = 2 \operatorname{cosec} \theta$
 (ii) $\frac{\cot \theta}{(\operatorname{cosec} \theta + 1)} + \frac{(\operatorname{cosec} \theta + 1)}{\cot \theta} = 2 \sec \theta$
20. (i) $\frac{\sec \theta - 1}{\sec \theta + 1} = \frac{\sin^2 \theta}{(1 + \cos \theta)^2}$
 (ii) $\frac{\sec \theta - \tan \theta}{\sec \theta + \tan \theta} = \frac{\cos^2 \theta}{(1 + \sin \theta)^2}$

$$21. (i) \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = (\sec \theta + \tan \theta)$$

$$(ii) \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} = (\operatorname{cosec} \theta - \cot \theta)$$

$$(iii) \sqrt{\frac{1+\cos \theta}{1-\cos \theta}} + \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} = 2 \operatorname{cosec} \theta$$

$$22. \frac{\cos^3 \theta + \sin^3 \theta}{\cos \theta + \sin \theta} + \frac{\cos^3 \theta - \sin^3 \theta}{\cos \theta - \sin \theta} = 2 \quad [\text{CBSE 2009C}]$$

$$23. \frac{\sin \theta}{(\cot \theta + \operatorname{cosec} \theta)} - \frac{\sin \theta}{(\cot \theta - \operatorname{cosec} \theta)} = 2$$

$$24. (i) \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} + \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{2}{(2 \sin^2 \theta - 1)}$$

$$(ii) \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{2}{(1 - 2 \cos^2 \theta)}$$

$$25. \frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} = \cot \theta \quad [\text{CBSE 2003}]$$

$$26. (i) \frac{\operatorname{cosec} \theta + \cot \theta}{\operatorname{cosec} \theta - \cot \theta} = (\operatorname{cosec} \theta + \cot \theta)^2 = 1 + 2 \cot^2 \theta + 2 \operatorname{cosec} \theta \cot \theta$$

[CBSE 2003]

$$(ii) \frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = (\sec \theta + \tan \theta)^2 = 1 + 2 \tan^2 \theta + 2 \sec \theta \tan \theta$$

$$27. (i) \frac{1 + \cos \theta + \sin \theta}{1 + \cos \theta - \sin \theta} = \frac{1 + \sin \theta}{\cos \theta}$$

$$(ii) \frac{\sin \theta + 1 - \cos \theta}{\cos \theta - 1 + \sin \theta} = \frac{1 + \sin \theta}{\cos \theta}$$

[CBSE 2001C]

$$28. \frac{\sin \theta}{(\sec \theta + \tan \theta - 1)} + \frac{\cos \theta}{(\operatorname{cosec} \theta + \cot \theta - 1)} = 1$$

$$29. \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{2}{(\sin^2 \theta - \cos^2 \theta)} = \frac{2}{(2 \sin^2 \theta - 1)} \quad [\text{CBSE 2007}]$$

$$30. \frac{\cos \theta \operatorname{cosec} \theta - \sin \theta \sec \theta}{\cos \theta + \sin \theta} = \operatorname{cosec} \theta - \sec \theta$$

$$31. (1 + \tan \theta + \cot \theta)(\sin \theta - \cos \theta) = \left(\frac{\sec \theta}{\operatorname{cosec}^2 \theta} - \frac{\operatorname{cosec} \theta}{\sec^2 \theta} \right)$$

$$32. \frac{\cot^2 \theta (\sec \theta - 1)}{(1 + \sin \theta)} + \frac{\sec^2 \theta (\sin \theta - 1)}{(1 + \sec \theta)} = 0$$

$$33. \left\{ \frac{1}{(\sec^2 \theta - \cos^2 \theta)} + \frac{1}{(\operatorname{cosec}^2 \theta - \sin^2 \theta)} \right\} (\sin^2 \theta \cos^2 \theta) = \frac{1 - \sin^2 \theta \cos^2 \theta}{2 + \sin^2 \theta \cos^2 \theta}$$

$$34. \frac{(\sin A - \sin B)}{(\cos A + \cos B)} + \frac{(\cos A - \cos B)}{(\sin A + \sin B)} = 0$$

$$35. \frac{\tan A + \tan B}{\cot A + \cot B} = \tan A \tan B$$

36. Show that none of the following is an identity:

$$(i) \cos^2 \theta + \cos \theta = 1 \quad (ii) \sin^2 \theta + \sin \theta = 2 \quad (iii) \tan^2 \theta + \sin \theta = \cos^2 \theta$$

$$37. \text{ Prove that: } (\sin \theta - 2\sin^3 \theta) = (2\cos^3 \theta - \cos \theta) \tan \theta.$$

$$38. \text{ If } 1 + \sin^2 \theta = 3 \sin \theta \cos \theta \text{ then prove that } \tan \theta = 1 \text{ or } \frac{1}{2}.$$

HINTS TO SOME SELECTED QUESTIONS

$$8. (i) \text{ Write } \cot^2 \theta = (\operatorname{cosec}^2 \theta - 1).$$

$$(ii) \text{ Write } \tan^2 \theta = (\sec^2 \theta - 1).$$

$$10. \text{ LHS} = \frac{\tan^2 \theta}{\sec^2 \theta} + \frac{\cot^2 \theta}{\operatorname{cosec}^2 \theta} = \left(\frac{\sin^2 \theta}{\cos^2 \theta} \times \cos^2 \theta \right) + \left(\frac{\cos^2 \theta}{\sin^2 \theta} \times \sin^2 \theta \right) = (\sin^2 \theta + \cos^2 \theta) = 1.$$

$$12. \text{ Write } \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

$$13. \text{ Write } \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ and simplify.}$$

$$14. \text{ Write } \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ and simplify.}$$

$$15. \text{ LHS} = \sec^2 \theta \cdot \operatorname{cosec}^2 \theta = \frac{1}{\cos^2 \theta \sin^2 \theta}$$

$$= \frac{1}{\sin^2 \theta (1 - \sin^2 \theta)} = \frac{1}{(\sin^2 \theta - \sin^4 \theta)}.$$

$$16. \text{ LHS} = \frac{\tan \theta}{\sec^4 \theta} + \frac{\cot \theta}{\operatorname{cosec}^4 \theta} = \left(\frac{\sin \theta}{\cos \theta} \times \cos^4 \theta \right) + \left(\frac{\cos \theta}{\sin \theta} \times \sin^4 \theta \right)$$

$$= (\sin \theta \cos^3 \theta + \sin^3 \theta \cos \theta) = \sin \theta \cos \theta (\cos^2 \theta + \sin^2 \theta) = \sin \theta \cos \theta.$$

$$17. (i) \sin^6 \theta + \cos^6 \theta = (\sin^2 \theta)^3 + (\cos^2 \theta)^3$$

$$= (\sin^2 \theta + \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta)$$

$$= (\sin^2 \theta + \cos^2 \theta)^2 - 3\sin^2 \theta \cos^2 \theta = (1 - 3\sin^2 \theta \cos^2 \theta).$$

$$(ii) (\cos^4 \theta - \sin^4 \theta) = (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)$$

$$\Rightarrow (\cos^4 \theta - \sin^4 \theta) = (\cos^2 \theta - \sin^2 \theta)$$

$$\Rightarrow (\sin^2 \theta + \cos^4 \theta) = (\cos^2 \theta + \sin^4 \theta).$$

$$(iii) (\operatorname{cosec}^4 \theta - \cot^4 \theta) = (\operatorname{cosec}^2 \theta + \cot^2 \theta)(\operatorname{cosec}^2 \theta - \cot^2 \theta)$$

$$\Rightarrow (\operatorname{cosec}^4 \theta - \cot^4 \theta) = (\operatorname{cosec}^2 \theta + \cot^2 \theta) \quad [\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1]$$

$$\Rightarrow (\operatorname{cosec}^4 \theta - \operatorname{cosec}^2 \theta) = (\cot^4 \theta + \cot^2 \theta).$$

$$20. (i) \text{ LHS} = \frac{\left(\frac{1}{\cos \theta} - 1 \right)}{\left(\frac{1}{\cos \theta} + 1 \right)} = \frac{(1 - \cos \theta)}{(1 + \cos \theta)} \times \frac{(1 + \cos \theta)}{(1 + \cos \theta)} = \frac{(1 - \cos^2 \theta)}{(1 + \cos \theta)^2} = \frac{\sin^2 \theta}{(1 + \cos \theta)^2}.$$

$$(ii) \text{ LHS} = \frac{\left(\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \right)}{\left(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \right)} = \frac{(1 - \sin \theta)}{(1 + \sin \theta)} \times \frac{(1 + \sin \theta)}{(1 + \sin \theta)} = \frac{1 - \sin^2 \theta}{(1 + \sin \theta)^2} = \frac{\cos^2 \theta}{(1 + \sin \theta)^2}.$$

$$21. \text{ (i) LHS} = \frac{\sqrt{1+\sin\theta}}{\sqrt{1-\sin\theta}} \times \frac{\sqrt{1+\sin\theta}}{\sqrt{1+\sin\theta}} = \frac{(1+\sin\theta)}{\sqrt{1-\sin^2\theta}} = \frac{(1+\sin\theta)}{\cos\theta} = \left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right) \\ = (\sec\theta + \tan\theta).$$

$$22. (\cos^3\theta + \sin^3\theta) = (\cos\theta + \sin\theta)(\cos^2\theta + \sin^2\theta - \cos\theta\sin\theta).$$

$$\text{And, } (\cos^3\theta - \sin^3\theta) = (\cos\theta - \sin\theta)(\cos^2\theta + \sin^2\theta + \cos\theta\sin\theta).$$

$$23. \text{ Write } \cot\theta = \frac{\cos\theta}{\sin\theta} \text{ and } \operatorname{cosec}\theta = \frac{1}{\sin\theta}.$$

$$25. \text{ Write } \sin^2\theta = (1 - \cos^2\theta).$$

$$26. \text{ (i) LHS} = \frac{(\operatorname{cosec}\theta + \cot\theta)}{(\operatorname{cosec}\theta - \cot\theta)} \times \frac{(\operatorname{cosec}\theta + \cot\theta)}{(\operatorname{cosec}\theta + \cot\theta)} = (\operatorname{cosec}\theta + \cot\theta)^2.$$

$$\text{And, } (\operatorname{cosec}\theta + \cot\theta)^2 = \operatorname{cosec}^2\theta + \cot^2\theta + 2\operatorname{cosec}\theta\cot\theta \\ = (1 + \cot^2\theta) + \cot^2\theta + 2\operatorname{cosec}\theta\cot\theta.$$

$$\text{(ii) LHS} = \frac{(\sec\theta + \tan\theta)}{(\sec\theta - \tan\theta)} \times \frac{(\sec\theta + \tan\theta)}{(\sec\theta + \tan\theta)}.$$

$$27. \text{ (i) On dividing num. and denom. by } \cos\theta, \text{ we get}$$

$$\text{LHS} = \frac{(\sec\theta + \tan\theta) + 1}{(\sec\theta + 1 - \tan\theta)} = \frac{(\sec\theta + \tan\theta) + (\sec^2\theta - \tan^2\theta)}{(\sec\theta + 1 - \tan\theta)} \\ = \frac{(\sec\theta + \tan\theta)(1 + \sec\theta - \tan\theta)}{(1 + \sec\theta - \tan\theta)} = (\sec\theta + \tan\theta) = \left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right).$$

$$\text{(ii) On dividing num. and denom. by } \cos\theta, \text{ we get}$$

$$\text{LHS} = \frac{\tan\theta + \sec\theta - 1}{1 - \sec\theta + \tan\theta} = \frac{(\sec\theta + \tan\theta) - (\sec^2\theta - \tan^2\theta)}{(\tan\theta - \sec\theta + 1)} \\ = \frac{(\sec\theta + \tan\theta)[1 - (\sec\theta - \tan\theta)]}{(\tan\theta - \sec\theta + 1)} = (\sec\theta + \tan\theta).$$

$$28. \text{ LHS} = \frac{\sin\theta}{\left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} - 1\right)} + \frac{\cos\theta}{\left(\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} - 1\right)} \\ = \left\{ \frac{\sin\theta\cos\theta}{1 + \sin\theta - \cos\theta} + \frac{\sin\theta\cos\theta}{1 + \cos\theta - \sin\theta} \right\} \\ = (\sin\theta\cos\theta) \cdot \left\{ \frac{1}{1 + (\sin\theta - \cos\theta)} + \frac{1}{1 - (\sin\theta - \cos\theta)} \right\} \\ = (\sin\theta\cos\theta) \cdot \left\{ \frac{(1 - \sin\theta + \cos\theta) + (1 + \sin\theta - \cos\theta)}{1 - (\sin\theta - \cos\theta)^2} \right\} \\ = (\sin\theta\cos\theta) \times \frac{2}{2\sin\theta\cos\theta} = 1.$$

$$31. \text{ LHS} = \left(1 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)(\sin\theta - \cos\theta) = \frac{(\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta)(\sin\theta - \cos\theta)}{\sin\theta\cos\theta} \\ = \frac{(\sin^3\theta - \cos^3\theta)}{\sin\theta\cos\theta} = \left(\frac{\sin^3\theta}{\sin\theta\cos\theta} - \frac{\cos^3\theta}{\sin\theta\cos\theta}\right) = \left(\frac{\sin^2\theta}{\cos\theta} - \frac{\cos^2\theta}{\sin\theta}\right) \\ = \left(\frac{\sec\theta}{\operatorname{cosec}^2\theta} - \frac{\operatorname{cosec}\theta}{\sec^2\theta}\right) = \text{RHS.}$$

$$32. \text{ LHS} = \frac{\cot^2\theta(\sec^2\theta - 1) + \sec^2\theta(\sin^2\theta - 1)}{(1 + \sin\theta)(1 + \sec\theta)} = \frac{(\cot^2\theta\tan^2\theta) - (\sec^2\theta\cos^2\theta)}{(1 + \sin\theta)(1 + \sec\theta)} = 0.$$

$$\begin{aligned}
 33. \text{ LHS} &= \left\{ \frac{1}{\left(\frac{1}{\cos^2 \theta} - \cos^2 \theta\right)} + \frac{1}{\left(\frac{1}{\sin^2 \theta} - \sin^2 \theta\right)} \right\} (\sin^2 \theta \cos^2 \theta) \\
 &= \left\{ \frac{\cos^2 \theta}{(1 - \cos^4 \theta)} + \frac{\sin^2 \theta}{(1 - \sin^4 \theta)} \right\} \sin^2 \theta \cos^2 \theta \\
 &= \left\{ \frac{\cos^2 \theta}{\sin^2 \theta (1 + \cos^2 \theta)} + \frac{\sin^2 \theta}{\cos^2 \theta (1 + \sin^2 \theta)} \right\} \sin^2 \theta \cos^2 \theta \\
 &= \frac{\cos^4 \theta (1 + \sin^2 \theta) + \sin^4 \theta (1 + \cos^2 \theta)}{(1 + \cos^2 \theta)(1 + \sin^2 \theta)} \\
 &= \frac{\cos^4 \theta + \sin^4 \theta + \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)}{1 + (\sin^2 \theta + \cos^2 \theta) + \sin^2 \theta \cos^2 \theta} \\
 &= \frac{(\cos^2 \theta + \sin^2 \theta)^2 - \sin^2 \theta \cos^2 \theta}{2 + \sin^2 \theta \cos^2 \theta} = \frac{1 - \sin^2 \theta \cos^2 \theta}{2 + \sin^2 \theta \cos^2 \theta}.
 \end{aligned}$$

$$\begin{aligned}
 34. \text{ LHS} &= \frac{(\sin A - \sin B)(\sin A + \sin B) + (\cos A - \cos B)(\cos A + \cos B)}{(\cos A + \cos B)(\sin A + \sin B)} \\
 &= \frac{(\sin^2 A - \sin^2 B) + (\cos^2 A - \cos^2 B)}{(\cos A + \cos B)(\sin A + \sin B)} = \frac{(\sin^2 A + \cos^2 A) - (\sin^2 B + \cos^2 B)}{(\cos A + \cos B)(\sin A + \sin B)} \\
 &= \frac{(1 - 1)}{(\cos A + \cos B)(\sin A + \sin B)} = 0.
 \end{aligned}$$

$$35. \text{ LHS} = \frac{\tan A + \tan B}{\frac{1}{\tan A} + \frac{1}{\tan B}} = \frac{\tan A \tan B (\tan A + \tan B)}{(\tan A + \tan B)} = \tan A \tan B.$$

$$36. \text{ (i) } \cos^2 30^\circ + \cos 30^\circ = \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{\sqrt{3}}{2} = \frac{3}{4} + \frac{\sqrt{3}}{2} = \frac{3 + 2\sqrt{3}}{4} \neq 1.$$

$$\text{(ii) } \sin^2 30^\circ + \sin 30^\circ = \left(\frac{1}{2}\right)^2 + \frac{1}{2} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \neq 2.$$

$$\text{(iii) LHS} = \tan^2 45^\circ + \sin 45^\circ = (1)^2 + \frac{1}{\sqrt{2}} = 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2} + 1}{\sqrt{2}}.$$

$$\text{RHS} = \cos^2 45^\circ = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}.$$

$\therefore \text{ LHS} \neq \text{RHS}.$

$$\begin{aligned}
 38. 1 + \sin^2 \theta &= 3 \sin \theta \cos \theta \Rightarrow \sec^2 \theta + \tan^2 \theta = 3 \tan \theta \quad [\text{dividing throughout by } \cos^2 \theta] \\
 &\Rightarrow (1 + \tan^2 \theta) + \tan^2 \theta = 3 \tan \theta \\
 &\Rightarrow 2 \tan^2 \theta - 3 \tan \theta + 1 = 0 \\
 &\Rightarrow 2 \tan^2 \theta - 2 \tan \theta - \tan \theta + 1 = 0 \\
 &\Rightarrow 2 \tan \theta (\tan \theta - 1) - (\tan \theta - 1) = 0 \\
 &\Rightarrow (\tan \theta - 1)(2 \tan \theta - 1) = 0 \\
 &\Rightarrow \tan \theta = 1 \text{ or } \frac{1}{2}.
 \end{aligned}$$

ELIMINATION OF TRIGONOMETRIC RATIOS

In order to eliminate the T-ratios from given relations, we make use of the fundamental trigonometrical identities, as shown in the examples given below.

SOLVED EXAMPLES

EXAMPLE 1 If $x = a \sin \theta + b \cos \theta$ and $y = a \cos \theta - b \sin \theta$, prove that $x^2 + y^2 = a^2 + b^2$.

SOLUTION We have

$$\begin{aligned} x^2 + y^2 &= (a \sin \theta + b \cos \theta)^2 + (a \cos \theta - b \sin \theta)^2 \\ &= a^2(\sin^2 \theta + \cos^2 \theta) + b^2(\cos^2 \theta + \sin^2 \theta) \\ &= a^2 + b^2 \quad [\because \sin^2 \theta + \cos^2 \theta = 1]. \end{aligned}$$

Hence, $x^2 + y^2 = a^2 + b^2$.

EXAMPLE 2 If $x = a \sin \theta$ and $y = b \tan \theta$, prove that $\left(\frac{a^2}{x^2} - \frac{b^2}{y^2}\right) = 1$.

SOLUTION We have

$$x = a \sin \theta \Rightarrow \frac{a}{x} = \frac{1}{\sin \theta} \Rightarrow \frac{a}{x} = \operatorname{cosec} \theta \quad \dots (i)$$

$$\text{and } y = b \tan \theta \Rightarrow \frac{b}{y} = \frac{1}{\tan \theta} \Rightarrow \frac{b}{y} = \cot \theta. \quad \dots (ii)$$

Squaring (i) and (ii) and subtracting, we get

$$\left(\frac{a^2}{x^2} - \frac{b^2}{y^2}\right) = (\operatorname{cosec}^2 \theta - \cot^2 \theta) = 1.$$

Hence, $\left(\frac{a^2}{x^2} - \frac{b^2}{y^2}\right) = 1$.

EXAMPLE 3 If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$, prove that

$$(m^2 - n^2) = 4\sqrt{mn}. \quad \text{[CBSE 2010]}$$

SOLUTION We have

$$\begin{aligned} \text{LHS} &= (m^2 - n^2) \\ &= (\tan \theta + \sin \theta)^2 - (\tan \theta - \sin \theta)^2 = 4 \tan \theta \sin \theta \\ &\quad [\because (a+b)^2 - (a-b)^2 = 4ab]. \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 4\sqrt{mn} = 4\sqrt{(\tan \theta + \sin \theta)(\tan \theta - \sin \theta)} \\ &= 4\sqrt{(\tan^2 \theta - \sin^2 \theta)} = 4 \cdot \sqrt{\left(\frac{\sin^2 \theta}{\cos^2 \theta} - \sin^2 \theta\right)} \end{aligned}$$

$$\begin{aligned}
 &= 4 \cdot \frac{\sqrt{\sin^2 \theta - \sin^2 \theta \cos^2 \theta}}{\cos \theta} = 4 \cdot \frac{\sin \theta}{\cos \theta} \cdot \sqrt{1 - \cos^2 \theta} \\
 &= 4 \tan \theta \cdot \sqrt{\sin^2 \theta} = 4 \tan \theta \sin \theta.
 \end{aligned}$$

Thus, LHS = RHS.

Hence, $(m^2 - n^2) = 4\sqrt{mn}$.

EXAMPLE 4 If $\sec \theta + \tan \theta = m$, show that $\frac{(m^2 - 1)}{(m^2 + 1)} = \sin \theta$. [CBSE 2004]

SOLUTION We have

$$\begin{aligned}
 (m^2 - 1) &= (\sec \theta + \tan \theta)^2 - 1 \\
 &= \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta - 1 \\
 &= (\sec^2 \theta - 1) + \tan^2 \theta + 2 \sec \theta \tan \theta \\
 &= 2 \tan^2 \theta + 2 \sec \theta \tan \theta \quad [\because \sec^2 \theta - 1 = \tan^2 \theta] \\
 &= 2 \tan \theta (\tan \theta + \sec \theta). \quad \dots (i)
 \end{aligned}$$

$$\begin{aligned}
 (m^2 + 1) &= (\sec \theta + \tan \theta)^2 + 1 \\
 &= \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta + 1 \\
 &= (1 + \tan^2 \theta) + \sec^2 \theta + 2 \sec \theta \tan \theta \\
 &= 2 \sec^2 \theta + 2 \sec \theta \tan \theta \quad [\because 1 + \tan^2 \theta = \sec^2 \theta] \\
 &= 2 \sec \theta (\sec \theta + \tan \theta). \quad \dots (ii)
 \end{aligned}$$

From (i) and (ii), we get

$$\frac{(m^2 - 1)}{(m^2 + 1)} = \frac{\tan \theta}{\sec \theta} = \left(\frac{\sin \theta}{\cos \theta} \times \cos \theta \right) = \sin \theta.$$

Hence, $\frac{(m^2 - 1)}{(m^2 + 1)} = \sin \theta$.

EXAMPLE 5 If $\sin \theta + \cos \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$, prove that $n(m^2 - 1) = 2m$.

SOLUTION We have

$$\begin{aligned}
 n(m^2 - 1) &= (\sec \theta + \operatorname{cosec} \theta)[(\sin \theta + \cos \theta)^2 - 1] \\
 &= \left(\frac{1}{\cos \theta} + \frac{1}{\sin \theta} \right)[(\sin^2 \theta + \cos^2 \theta) + 2 \sin \theta \cos \theta - 1] \\
 &= \frac{(\sin \theta + \cos \theta)}{\sin \theta \cos \theta} \cdot 2 \sin \theta \cos \theta \\
 &= 2(\sin \theta + \cos \theta) = 2m.
 \end{aligned}$$

Hence, $n(m^2 - 1) = 2m$.

EXAMPLE 6 If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, show that $(\cos \theta - \sin \theta) = \sqrt{2} \sin \theta$.

[CBSE 2008C]

SOLUTION We have

$$\begin{aligned}
 \cos \theta + \sin \theta &= \sqrt{2} \cos \theta \\
 \Rightarrow (\cos \theta + \sin \theta)^2 &= 2 \cos^2 \theta \\
 \Rightarrow \cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta &= 2 \cos^2 \theta \\
 \Rightarrow \cos^2 \theta - 2 \cos \theta \sin \theta &= \sin^2 \theta \\
 \Rightarrow \cos^2 \theta - 2 \cos \theta \sin \theta + \sin^2 \theta &= \sin^2 \theta + \sin^2 \theta \\
 &\quad \text{[adding } \sin^2 \theta \text{ on both sides]} \\
 \Rightarrow (\cos \theta - \sin \theta)^2 &= 2 \sin^2 \theta \\
 \Rightarrow (\cos \theta - \sin \theta) &= \sqrt{2} \sin \theta. \\
 \text{Hence, } (\cos \theta - \sin \theta) &= \sqrt{2} \sin \theta.
 \end{aligned}$$

EXAMPLE 7 If $\sin \theta + \sin^2 \theta = 1$, prove that $\cos^2 \theta + \cos^4 \theta = 1$.

SOLUTION

$$\begin{aligned}
 \sin \theta + \sin^2 \theta &= 1 \\
 \Rightarrow \sin \theta &= 1 - \sin^2 \theta \\
 \Rightarrow \sin \theta &= \cos^2 \theta & [\because 1 - \sin^2 \theta = \cos^2 \theta] \\
 \Rightarrow \sin^2 \theta &= \cos^4 \theta \\
 \Rightarrow 1 - \cos^2 \theta &= \cos^4 \theta & [\because \sin^2 \theta = 1 - \cos^2 \theta] \\
 \Rightarrow \cos^2 \theta + \cos^4 \theta &= 1. \\
 \text{Hence, } \cos^2 \theta + \cos^4 \theta &= 1.
 \end{aligned}$$

EXAMPLE 8 If $\cos \theta + \sin \theta = 1$, prove that $\cos \theta - \sin \theta = \pm 1$.

SOLUTION We have

$$\begin{aligned}
 (\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 &= 2(\cos^2 \theta + \sin^2 \theta) \\
 \Rightarrow (\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 &= 2 \quad [\because (\cos^2 \theta + \sin^2 \theta) = 1] \\
 \Rightarrow 1 + (\cos \theta - \sin \theta)^2 &= 2 & [\because \cos \theta + \sin \theta = 1 \text{ (given)}] \\
 \Rightarrow (\cos \theta - \sin \theta)^2 &= 1 \\
 \Rightarrow (\cos \theta - \sin \theta) &= \pm 1 & \text{[taking square root on both sides].} \\
 \text{Hence, } (\cos \theta - \sin \theta) &= \pm 1.
 \end{aligned}$$

EXAMPLE 9 If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, prove that $x^2 + y^2 = 1$.

SOLUTION

$$\begin{aligned}
 x \sin^3 \theta + y \cos^3 \theta &= \sin \theta \cos \theta \\
 \Rightarrow (x \sin \theta) \sin^2 \theta + (y \cos \theta) \cos^2 \theta &= \sin \theta \cos \theta \\
 \Rightarrow (x \sin \theta) \sin^2 \theta + (x \sin \theta) \cos^2 \theta &= \sin \theta \cos \theta \\
 &\quad [\because y \cos \theta = x \sin \theta]
 \end{aligned}$$

$$\Rightarrow (x \sin \theta)(\sin^2 \theta + \cos^2 \theta) = \sin \theta \cos \theta$$

$$\Rightarrow x \sin \theta = \sin \theta \cos \theta \quad [\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$\Rightarrow x = \cos \theta. \quad \dots (i)$$

$$\text{Now, } x \sin \theta = y \cos \theta$$

$$\Rightarrow \cos \theta \sin \theta = y \cos \theta \quad [\because x = \cos \theta]$$

$$\Rightarrow y = \sin \theta. \quad \dots (ii)$$

On squaring (i) and (ii) and adding, we get $x^2 + y^2 = 1$.

Hence, $x^2 + y^2 = 1$.

EXAMPLE 10 If $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$ and $z = c \tan \theta$ then prove that

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = \left(1 + \frac{z^2}{c^2} \right).$$

SOLUTION We have

$$\frac{x}{a} = \sec \theta \cos \phi \quad \dots (i) \quad \text{and} \quad \frac{y}{b} = \sec \theta \sin \phi \quad \dots (ii).$$

Squaring (i) and (ii) and adding, we get

$$\begin{aligned} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) &= \sec^2 \theta (\cos^2 \phi + \sin^2 \phi) \\ &= \sec^2 \theta \quad [\because \cos^2 \phi + \sin^2 \phi = 1] \\ &= (1 + \tan^2 \theta) = \left(1 + \frac{z^2}{c^2} \right) \quad \left[\because \tan \theta = \frac{z}{c} \right] \end{aligned}$$

$$\text{Hence, } \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = \left(1 + \frac{z^2}{c^2} \right).$$

EXAMPLE 11 If $x = r \sin \alpha \cos \beta$, $y = r \sin \alpha \sin \beta$ and $z = r \cos \alpha$, prove that

$$x^2 + y^2 + z^2 = r^2.$$

SOLUTION We have

$$\begin{aligned} x^2 + y^2 + z^2 &= r^2 \sin^2 \alpha \cos^2 \beta + r^2 \sin^2 \alpha \sin^2 \beta + r^2 \cos^2 \alpha \\ &= r^2 \sin^2 \alpha (\cos^2 \beta + \sin^2 \beta) + r^2 \cos^2 \alpha \\ &= r^2 \sin^2 \alpha + r^2 \cos^2 \alpha \quad [\because \cos^2 \beta + \sin^2 \beta = 1] \\ &= r^2 (\sin^2 \alpha + \cos^2 \alpha) = r^2 \quad [\because \sin^2 \alpha + \cos^2 \alpha = 1]. \end{aligned}$$

$$\text{Hence, } (x^2 + y^2 + z^2) = r^2.$$

EXAMPLE 12 If $\operatorname{cosec} \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$, prove that

$$(m^2 n)^{2/3} + (mn^2)^{2/3} = 1.$$

SOLUTION We have

$$\begin{aligned} m^2 n &= (\operatorname{cosec} \theta - \sin \theta)^2 \cdot (\sec \theta - \cos \theta) \\ &= \left(\frac{1}{\sin \theta} - \sin \theta \right)^2 \cdot \left(\frac{1}{\cos \theta} - \cos \theta \right) \\ &= \frac{(1 - \sin^2 \theta)^2}{\sin^2 \theta} \cdot \frac{(1 - \cos^2 \theta)}{\cos \theta} = \frac{\cos^4 \theta}{\sin^2 \theta} \times \frac{\sin^2 \theta}{\cos \theta} = \cos^3 \theta \end{aligned}$$

$$\therefore (m^2 n)^{1/3} = \cos \theta. \quad \dots (i)$$

$$\text{Again, } mn^2 = (\operatorname{cosec} \theta - \sin \theta) \cdot (\sec \theta - \cos \theta)^2$$

$$\begin{aligned} &= \left(\frac{1}{\sin \theta} - \sin \theta \right) \cdot \left(\frac{1}{\cos \theta} - \cos \theta \right)^2 \\ &= \frac{(1 - \sin^2 \theta)}{\sin \theta} \cdot \frac{(1 - \cos^2 \theta)^2}{\cos^2 \theta} \\ &= \left(\frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^4 \theta}{\cos^2 \theta} \right) = \sin^3 \theta \end{aligned}$$

$$\therefore (mn^2)^{1/3} = \sin \theta. \quad \dots (ii)$$

On squaring (i) and (ii) and adding the results, we get

$$(m^2 n)^{2/3} + (mn^2)^{2/3} = 1 \quad [\because \cos^2 \theta + \sin^2 \theta = 1].$$

$$\text{Hence, } (m^2 n)^{2/3} + (mn^2)^{2/3} = 1.$$

EXAMPLE 13 If $a \cos \theta - b \sin \theta = c$, prove that

$$(a \sin \theta + b \cos \theta) = \pm \sqrt{a^2 + b^2 - c^2}. \quad [\text{CBSE 2006C}]$$

SOLUTION Given, $a \cos \theta - b \sin \theta = c$ (i)

$$\begin{aligned} \text{Now, } (a \cos \theta - b \sin \theta)^2 + (a \sin \theta + b \cos \theta)^2 \\ = a^2(\cos^2 \theta + \sin^2 \theta) + b^2(\sin^2 \theta + \cos^2 \theta) = (a^2 + b^2). \end{aligned}$$

$$\text{Thus, } (a \cos \theta - b \sin \theta)^2 + (a \sin \theta + b \cos \theta)^2 = (a^2 + b^2)$$

$$\Rightarrow c^2 + (a \sin \theta + b \cos \theta)^2 = (a^2 + b^2)$$

$$\Rightarrow (a \sin \theta + b \cos \theta)^2 = (a^2 + b^2 - c^2)$$

$$\Rightarrow (a \sin \theta + b \cos \theta) = \pm \sqrt{a^2 + b^2 - c^2}.$$

$$\text{Hence, } (a \sin \theta + b \cos \theta) = \pm \sqrt{a^2 + b^2 - c^2}.$$

EXAMPLE 14 If $(3 \sin \theta + 5 \cos \theta) = 5$, prove that $(5 \sin \theta - 3 \cos \theta) = \pm 3$.

SOLUTION We have

$$\begin{aligned}(3 \sin \theta + 5 \cos \theta)^2 + (5 \sin \theta - 3 \cos \theta)^2 \\&= 9(\sin^2 \theta + \cos^2 \theta) + 25(\sin^2 \theta + \cos^2 \theta) \\&= (9 + 25) = 34.\end{aligned}$$

$$\therefore (3 \sin \theta + 5 \cos \theta)^2 + (5 \sin \theta - 3 \cos \theta)^2 = 34$$

$$\Rightarrow 5^2 + (5 \sin \theta - 3 \cos \theta)^2 = 34 \quad [\because 3 \sin \theta + 5 \cos \theta = 5]$$

$$\Rightarrow (5 \sin \theta - 3 \cos \theta) = \pm 3 \quad [\text{taking square root on each side}]$$

$$\text{Hence, } (5 \sin \theta - 3 \cos \theta) = \pm 3.$$

EXERCISE 13B

1. If $a \cos \theta + b \sin \theta = m$ and $a \sin \theta - b \cos \theta = n$, prove that

$$(m^2 + n^2) = (a^2 + b^2).$$

2. If $x = a \sec \theta + b \tan \theta$ and $y = a \tan \theta + b \sec \theta$, prove that

$$(x^2 - y^2) = (a^2 - b^2).$$

3. If $\left(\frac{x}{a} \sin \theta - \frac{y}{b} \cos \theta\right) = 1$ and $\left(\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta\right) = 1$, prove that

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) = 2.$$

4. If $(\sec \theta + \tan \theta) = m$ and $(\sec \theta - \tan \theta) = n$, show that $mn = 1$.

5. If $(\operatorname{cosec} \theta + \cot \theta) = m$ and $(\operatorname{cosec} \theta - \cot \theta) = n$, show that $mn = 1$.

6. If $x = a \cos^3 \theta$ and $y = b \sin^3 \theta$, prove that $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$.

7. If $(\tan \theta + \sin \theta) = m$ and $(\tan \theta - \sin \theta) = n$, prove that

$$(m^2 - n^2)^2 = 16mn.$$

8. If $(\cot \theta + \tan \theta) = m$ and $(\sec \theta - \cos \theta) = n$, prove that

$$(m^2 n)^{2/3} - (mn^2)^{2/3} = 1.$$

9. If $(\operatorname{cosec} \theta - \sin \theta) = a^3$ and $(\sec \theta - \cos \theta) = b^3$, prove that

$$a^2 b^2 (a^2 + b^2) = 1.$$

10. If $(2 \sin \theta + 3 \cos \theta) = 2$, prove that $(3 \sin \theta - 2 \cos \theta) = \pm 3$.

11. If $(\sin \theta + \cos \theta) = \sqrt{2} \cos \theta$, show that $\cot \theta = (\sqrt{2} + 1)$.

12. If $(\cos \theta + \sin \theta) = \sqrt{2} \sin \theta$, prove that $(\sin \theta - \cos \theta) = \sqrt{2} \cos \theta$.

13. If $\sec \theta + \tan \theta = p$, prove that

$$(i) \sec \theta = \frac{1}{2} \left(p + \frac{1}{p} \right) \quad (ii) \tan \theta = \frac{1}{2} \left(p - \frac{1}{p} \right) \quad (iii) \sin \theta = \frac{p^2 - 1}{p^2 + 1}.$$

14. If $\tan A = n \tan B$ and $\sin A = m \sin B$, prove that $\cos^2 A = \frac{(m^2 - 1)}{(n^2 - 1)}$.

15. If $m = (\cos \theta - \sin \theta)$ and $n = (\cos \theta + \sin \theta)$ then show that

$$\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} = \frac{2}{\sqrt{1 - \tan^2 \theta}}.$$

HINTS TO SOME SELECTED QUESTIONS

7. See Example 3.

$$8. m = \left(\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right) = \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} \right) = \frac{1}{\sin \theta \cos \theta}.$$

$$n = \left(\frac{1}{\cos \theta} - \cos \theta \right) = \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right) = \frac{\sin^2 \theta}{\cos \theta}.$$

$$\therefore m^2 n = \left(\frac{1}{\sin^2 \theta \cos^2 \theta} \times \frac{\sin^2 \theta}{\cos \theta} \right) = \frac{1}{\cos^3 \theta} = \sec^3 \theta$$

$$\text{and } mn^2 = \left(\frac{1}{\sin \theta \cos \theta} \times \frac{\sin^4 \theta}{\cos^2 \theta} \right) = \frac{\sin^3 \theta}{\cos^3 \theta} = \tan^3 \theta.$$

$$\therefore (m^2 n)^{2/3} - (mn^2)^{2/3} = (\sec^3 \theta)^{2/3} - (\tan^3 \theta)^{2/3} = (\sec^2 \theta - \tan^2 \theta) = 1.$$

$$9. a^3 = \left(\frac{1}{\sin \theta} - \sin \theta \right) = \left(\frac{1 - \sin^2 \theta}{\sin \theta} \right) = \frac{\cos^2 \theta}{\sin \theta} \Rightarrow a = \frac{\cos^{2/3} \theta}{\sin^{1/3} \theta}.$$

$$b^3 = \left(\frac{1}{\cos \theta} - \cos \theta \right) = \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right) = \frac{\sin^2 \theta}{\cos \theta} \Rightarrow b = \frac{\sin^{2/3} \theta}{\cos^{1/3} \theta}.$$

$$\begin{aligned} \therefore a^2 b^2 (a^2 + b^2) &= a^4 b^2 + a^2 b^4 = a^3 (ab^2) + (a^2 b) b^3 \\ &= \frac{\cos^2 \theta}{\sin \theta} \cdot \left[\frac{\cos^{2/3} \theta}{\sin^{1/3} \theta} \cdot \frac{\sin^{4/3} \theta}{\cos^{2/3} \theta} \right] + \left[\frac{\cos^{4/3} \theta}{\sin^{2/3} \theta} \cdot \frac{\sin^{2/3} \theta}{\cos^{1/3} \theta} \right] \cdot \frac{\sin^2 \theta}{\cos \theta} \\ &= \frac{\cos^2 \theta}{\sin \theta} \cdot \sin \theta + \cos \theta \cdot \frac{\sin^2 \theta}{\cos \theta} = (\cos^2 \theta + \sin^2 \theta) = 1. \end{aligned}$$

$$10. (2 \sin \theta + 3 \cos \theta)^2 + (3 \sin \theta - 2 \cos \theta)^2 = 13(\sin^2 \theta + \cos^2 \theta) = 13$$

$$\Rightarrow 2^2 + (3 \sin \theta - 2 \cos \theta)^2 = 13 \Rightarrow (3 \sin \theta - 2 \cos \theta)^2 = 9$$

$$\Rightarrow (3 \sin \theta - 2 \cos \theta) = \pm 3.$$

$$11. \sin \theta + \cos \theta = \sqrt{2} \cos \theta$$

$$\Rightarrow 1 + \cot \theta = \sqrt{2} \cot \theta \quad [\text{dividing both side by } \sin \theta]$$

$$\Rightarrow (\sqrt{2} - 1) \cot \theta = 1 \Rightarrow \cot \theta = \frac{1}{(\sqrt{2} - 1)} = \frac{1}{(\sqrt{2} - 1)} \times \frac{(\sqrt{2} + 1)}{(\sqrt{2} + 1)} = (\sqrt{2} + 1).$$

$$12. (\cos \theta + \sin \theta)^2 = 2 \sin^2 \theta \Rightarrow \cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta = 2 \sin^2 \theta$$

$$\Rightarrow \sin^2 \theta - 2 \cos \theta \sin \theta = \cos^2 \theta \Rightarrow \sin^2 \theta - 2 \cos \theta \sin \theta + \cos^2 \theta = 2 \cos^2 \theta$$

$$\Rightarrow (\sin \theta - \cos \theta)^2 = 2 \cos^2 \theta \Rightarrow (\sin \theta - \cos \theta) = \sqrt{2} \cos \theta.$$

$$13. \sec \theta + \tan \theta = p. \quad \dots (i)$$

$$\sec^2 \theta - \tan^2 \theta = 1. \quad \dots (ii)$$

On dividing (ii) by (i), we get

$$\sec \theta - \tan \theta = \frac{1}{p}. \quad \dots (iii)$$

$$\therefore \text{ from (i) and (iii), we get } \sec \theta = \frac{1}{2} \left(p + \frac{1}{p} \right), \tan \theta = \frac{1}{2} \left(p - \frac{1}{p} \right).$$

$$\text{Also, } \sin \theta = \frac{\tan \theta}{\sec \theta} = \frac{\frac{1}{2} \left(p - \frac{1}{p} \right)}{\frac{1}{2} \left(p + \frac{1}{p} \right)} = \frac{(p^2 - 1)}{(p^2 + 1)}.$$

14. We have to eliminate B . So, we find cosec B and cot B from the given relations and use the identity $\operatorname{cosec}^2 B - \cot^2 B = 1$.

$$\sin A = m \sin B \Rightarrow \operatorname{cosec} B = \frac{m}{\sin A}. \quad \dots (i)$$

$$\tan A = n \tan B \Rightarrow \cot B = \frac{n}{\tan A}. \quad \dots (ii)$$

Squaring (i) and (ii), and subtracting the results, we get

$$\begin{aligned} \frac{m^2}{\sin^2 A} - \frac{n^2}{\tan^2 A} &= 1 \Rightarrow m^2 - n^2 \cos^2 A = \sin^2 A \Rightarrow m^2 - n^2 \cos^2 A = 1 - \cos^2 A \\ &\Rightarrow (n^2 - 1) \cos^2 A = (m^2 - 1) \Rightarrow \cos^2 A = \frac{(m^2 - 1)}{(n^2 - 1)}. \end{aligned}$$

$$15. \text{LHS} = \frac{(m+n)}{\sqrt{mn}}.$$

$$\text{Now, } (m+n) = 2 \cos \theta \text{ and } mn = (\cos^2 \theta - \sin^2 \theta) = \cos^2 \theta (1 - \tan^2 \theta).$$

$$\therefore \sqrt{mn} = \cos \theta \sqrt{1 - \tan^2 \theta}.$$

$$\text{Hence, LHS} = \frac{(m+n)}{\sqrt{mn}} = \frac{2 \cos \theta}{\cos \theta \sqrt{1 - \tan^2 \theta}} = \frac{2}{\sqrt{1 - \tan^2 \theta}} = \text{RHS}.$$

EXERCISE 13C

Very-Short and Short-Answer Questions

- Write the value of $(1 - \sin^2 \theta) \sec^2 \theta$.
- Write the value of $(1 - \cos^2 \theta) \operatorname{cosec}^2 \theta$.
- Write the value of $(1 + \tan^2 \theta) \cos^2 \theta$.
- Write the value of $(1 + \cot^2 \theta) \sin^2 \theta$.
- Write the value of $\left(\sin^2 \theta + \frac{1}{1 + \tan^2 \theta} \right)$.
- Write the value of $\left(\cot^2 \theta - \frac{1}{\sin^2 \theta} \right)$.
- Write the value of $\sin \theta \cos (90^\circ - \theta) + \cos \theta \sin (90^\circ - \theta)$.
- Write the value of $\operatorname{cosec}^2 (90^\circ - \theta) - \tan^2 \theta$.

9. Write the value of $\sec^2 \theta (1 + \sin \theta)(1 - \sin \theta)$. [CBSE 2009]
10. Write the value of $\operatorname{cosec}^2 \theta (1 + \cos \theta)(1 - \sin \theta)$.
11. Write the value of $\sin^2 \theta \cos^2 \theta (1 + \tan^2 \theta)(1 + \cot^2 \theta)$.
12. Write the value of $(1 + \tan^2 \theta)(1 + \sin \theta)(1 - \sin \theta)$. [CBSE 2008]
13. Write the value of $3 \cot^2 \theta - 3 \operatorname{cosec}^2 \theta$.
14. Write the value of $4 \tan^2 \theta - \frac{4}{\cos^2 \theta}$.
15. Write the value of $\frac{\tan^2 \theta - \sec^2 \theta}{\cot^2 \theta - \operatorname{cosec}^2 \theta}$.
16. If $\sin \theta = \frac{1}{2}$, write the value of $(3 \cot^2 \theta + 3)$. [CBSE 2009]
17. If $\cos \theta = \frac{2}{3}$, write the value of $(4 + 4 \tan^2 \theta)$.
18. If $\cos \theta = \frac{7}{25}$, write the value of $(\tan \theta + \cot \theta)$. [CBSE 2008]
19. If $\cos \theta = \frac{2}{3}$, write the value of $\frac{(\sec \theta - 1)}{(\sec \theta + 1)}$.
20. If $5 \tan \theta = 4$, write the value of $\frac{(\cos \theta - \sin \theta)}{(\cos \theta + \sin \theta)}$.
21. If $3 \cot \theta = 4$, write the value of $\frac{(2 \cos \theta + \sin \theta)}{(4 \cos \theta - \sin \theta)}$.
22. If $\cot \theta = \frac{1}{\sqrt{3}}$, write the value of $\frac{(1 - \cos^2 \theta)}{(2 - \sin^2 \theta)}$.
23. If $\tan \theta = \frac{1}{\sqrt{5}}$, write the value of $\frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)}$.
24. If $\cot A = \frac{4}{3}$ and $(A + B) = 90^\circ$, what is the value of $\tan B$?
25. If $\cos B = \frac{3}{5}$ and $(A + B) = 90^\circ$, find the value of $\sin A$.
26. If $\sqrt{3} \sin \theta = \cos \theta$ and θ is an acute angle, find the value of θ .
27. Write the value of $\tan 10^\circ \tan 20^\circ \tan 70^\circ \tan 80^\circ$.
28. Write the value of $\tan 1^\circ \tan 2^\circ \dots \tan 89^\circ$.
29. Write the value of $\cos 1^\circ \cos 2^\circ \dots \cos 180^\circ$.
30. If $\tan A = \frac{5}{12}$, find the value of $(\sin A + \cos A) \sec A$. [CBSE 2008]
31. If $\sin \theta = \cos (\theta - 45^\circ)$, where θ is acute, find the value of θ .
32. Find the value of $\frac{\sin 50^\circ}{\cos 40^\circ} + \frac{\operatorname{cosec} 40^\circ}{\sec 50^\circ} - 4 \cos 50^\circ \operatorname{cosec} 40^\circ$.
33. Find the value of $\sin 48^\circ \sec 42^\circ + \cos 48^\circ \operatorname{cosec} 42^\circ$.

34. If $x = a \sin \theta$ and $y = b \cos \theta$, write the value of $(b^2 x^2 + a^2 y^2)$.
35. If $5x = \sec \theta$ and $\frac{5}{x} = \tan \theta$, find the value of $5\left(x^2 - \frac{1}{x^2}\right)$. [CBSE 2010]
36. If $\operatorname{cosec} \theta = 2x$ and $\cot \theta = \frac{2}{x}$, find the value of $2\left(x^2 - \frac{1}{x^2}\right)$. [CBSE 2010]
37. If $\sec \theta + \tan \theta = x$, find the value of $\sec \theta$.
38. Find the value of $\frac{\cos 38^\circ \operatorname{cosec} 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ}$.
39. If $\sin \theta = x$, write the value of $\cot \theta$.
40. If $\sec \theta = x$, write the value of $\tan \theta$.

ANSWERS (EXERCISE 13C)

1. 1 2. 1 3. 1 4. 1 5. 1 6. -1 7. 1 8. 1 9. 1
10. 1 11. 1 12. 1 13. -3 14. -4 15. 1 16. 12 17. 9 18. $\frac{625}{168}$
19. $\frac{1}{5}$ 20. $\frac{1}{9}$ 21. $\frac{11}{13}$ 22. $\frac{3}{5}$ 23. $\frac{2}{3}$ 24. $\frac{4}{3}$ 25. $\frac{3}{5}$ 26. 30° 27. 1
28. 1 29. 0 30. $\frac{17}{12}$ 31. $(67.5)^\circ$ 32. -2 33. 2 34. $a^2 b^2$ 35. $\frac{1}{5}$
36. $\frac{1}{2}$ 37. $\frac{x^2+1}{2x}$ 38. $\frac{1}{\sqrt{3}}$ 39. $\frac{\sqrt{1-x^2}}{x}$ 40. $\sqrt{x^2-1}$

HINTS TO SOME SELECTED QUESTIONS

2. $(1 - \cos^2 \theta) \operatorname{cosec}^2 \theta = \sin^2 \theta \cdot \operatorname{cosec}^2 \theta = 1$.
4. $(1 + \cot^2 \theta) \sin^2 \theta = \operatorname{cosec}^2 \theta \sin^2 \theta = 1$.
5. $\left(\sin^2 \theta + \frac{1}{1 + \tan^2 \theta}\right) = \left(\sin^2 \theta + \frac{1}{\sec^2 \theta}\right) = (\sin^2 \theta + \cos^2 \theta) = 1$.
6. $\left(\cot^2 \theta - \frac{1}{\sin^2 \theta}\right) = (\cot^2 \theta - \operatorname{cosec}^2 \theta) = \cot^2 \theta - (1 + \cot^2 \theta) = -1$.
7. $\sin \theta \cos (90^\circ - \theta) + \cos \theta \sin (90^\circ - \theta) = \sin \theta \cdot \sin \theta + \cos \theta \cdot \cos \theta = \sin^2 \theta + \cos^2 \theta = 1$.
8. $\operatorname{cosec}^2 (90^\circ - \theta) - \tan^2 \theta = \sec^2 \theta - \tan^2 \theta = 1$.
9. Given expression $= \sec^2 \theta (1 - \sin^2 \theta) = \sec^2 \theta \cos^2 \theta = 1$.
10. Given expression $= \operatorname{cosec}^2 \theta (1 - \cos^2 \theta) = \operatorname{cosec}^2 \theta \sin^2 \theta = 1$.
11. Given expression $= \sin^2 \theta (1 + \cot^2 \theta) \cos^2 \theta (1 + \tan^2 \theta)$
 $= (\sin^2 \theta \operatorname{cosec}^2 \theta) (\cos^2 \theta \sec^2 \theta) = (1 \times 1) = 1$.
12. Given expression $= (1 + \tan^2 \theta) (1 - \sin^2 \theta) = \sec^2 \theta \cos^2 \theta = 1$.

13. Given expression $= 3 \cot^2 \theta - 3(1 + \cot^2 \theta) = -3$.

14. Given expression $= 4 \tan^2 \theta - 4 \sec^2 \theta = 4 \tan^2 \theta - 4(1 + \tan^2 \theta) = -4$.

15. Given expression $= \frac{(\sec^2 \theta - \tan^2 \theta)}{(\operatorname{cosec}^2 \theta - \cot^2 \theta)} = \frac{1}{1} = 1$.

16. $3 \cot^2 \theta + 3 = 3(\cot^2 \theta + 1) = 3 \operatorname{cosec}^2 \theta = (3 \times 4) = 12$ [$\because \operatorname{cosec} \theta = 2$].

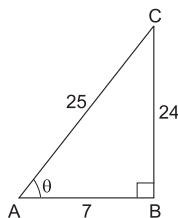
17. $(4 + 4 \tan^2 \theta) = 4(1 + \tan^2 \theta) = 4 \sec^2 \theta = \left(4 \times \frac{9}{4}\right) = 9$ [$\because \sec \theta = \frac{3}{2}$].

18. Consider a right $\triangle ABC$ in which $\angle B = 90^\circ$, $AB = 7$ and $AC = 25$.

$$BC^2 = AC^2 - AB^2 = (25)^2 - (7)^2 = (625 - 49) = 576.$$

$$\therefore BC = \sqrt{576} = 24.$$

$$\therefore (\tan \theta + \cot \theta) = \left(\frac{24}{7} + \frac{7}{24}\right) = \frac{576 + 49}{168} = \frac{625}{168}.$$



19. $\frac{(\sec \theta - 1)}{(\sec \theta + 1)} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{3}{2} + 1\right)} = \left(\frac{1}{2} \times \frac{2}{5}\right) = \frac{1}{5}$.

20. $\frac{(\cos \theta - \sin \theta)}{(\cos \theta + \sin \theta)} = \frac{(1 - \tan \theta)}{(1 + \tan \theta)}$ [dividing num. and denom. by $\cos \theta$]

$$= \frac{\left(1 - \frac{4}{5}\right)}{\left(1 + \frac{4}{5}\right)} = \frac{1}{9} \quad \left[\because \tan \theta = \frac{4}{5}\right].$$

21. $\frac{(2 \cos \theta + \sin \theta)}{(4 \cos \theta - \sin \theta)} = \frac{(2 \cot \theta + 1)}{(4 \cot \theta - 1)}$ [dividing num. and denom. by $\sin \theta$]

$$= \frac{\left(2 \times \frac{4}{3} + 1\right)}{\left(4 \times \frac{4}{3} - 1\right)} = \frac{11}{13}.$$

22. $\frac{(1 - \cos^2 \theta)}{(2 - \sin^2 \theta)} = \frac{\sin^2 \theta}{(2 - \sin^2 \theta)} = \frac{\frac{3}{4}}{\left(2 - \frac{3}{4}\right)} \quad \left[\operatorname{cosec}^2 \theta = (1 + \cot^2 \theta) = \left(1 + \frac{1}{3}\right) = \frac{4}{3} \Rightarrow \sin^2 \theta = \frac{3}{4}\right]$

$$= \left(\frac{3}{4} \times \frac{4}{5}\right) = \frac{3}{5}.$$

23. $\operatorname{cosec}^2 \theta = (1 + \cot^2 \theta) = (1 + 5) = 6$, $\sec^2 \theta = (1 + \tan^2 \theta) = \left(1 + \frac{1}{5}\right) = \frac{6}{5}$.

$$\therefore \frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)} = \frac{\left(6 - \frac{6}{5}\right)}{\left(6 + \frac{6}{5}\right)} = \frac{24}{36} = \frac{2}{3}.$$

24. $A = 90^\circ - B \Rightarrow \cot A = \cot (90^\circ - B) = \tan B$.

$$\therefore \tan B = \cot A = \frac{4}{3}.$$

25. $B = 90^\circ - A \Rightarrow \cos B = \cos (90^\circ - A) = \sin A$

$$\therefore \sin A = \cos B = \frac{3}{5}.$$

$$26. \sqrt{3} \sin \theta = \cos \theta \Rightarrow \tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ.$$

$$\begin{aligned} 27. \text{ Given expression} &= (\tan 10^\circ \tan 80^\circ)(\tan 20^\circ \tan 70^\circ) \\ &= \{\tan 10^\circ \tan (90^\circ - 10^\circ)\} \{\tan 20^\circ \tan (90^\circ - 20^\circ)\} \\ &= (\tan 10^\circ \cot 10^\circ)(\tan 20^\circ \cot 20^\circ) = (1 \times 1) = 1. \end{aligned}$$

$$\begin{aligned} 28. \text{ Given expression} &= (\tan 1^\circ \tan 89^\circ)(\tan 2^\circ \tan 88^\circ) \dots (\tan 44^\circ \tan 46^\circ) \tan 45^\circ \\ &= \{\tan 1^\circ \tan (90^\circ - 1^\circ)\} \{\tan 2^\circ \tan (90^\circ - 2^\circ)\} \dots \{\tan 44^\circ \tan (90^\circ - 44^\circ)\} \tan 45^\circ \\ &= (\tan 1^\circ \cot 1^\circ)(\tan 2^\circ \cot 2^\circ) \dots (\tan 44^\circ \cot 44^\circ) \times 1 \\ &= (1 \times 1 \times 1 \times \dots \times 1 \times 1) = 1. \end{aligned}$$

$$\begin{aligned} 29. \text{ Given expression} &= (\cos 1^\circ \cos 2^\circ \dots \cos 89^\circ \cos 90^\circ \cos 91^\circ \dots \cos 180^\circ) = 0, \text{ since } \cos 90^\circ = 0. \end{aligned}$$

$$\begin{aligned} 30. (\sin A + \cos A) \sec A &= (\sin A + \cos A) \times \frac{1}{\cos A} = \left(\frac{\sin A}{\cos A} + \frac{\cos A}{\cos A} \right) \\ &= (\tan A + 1) = \left(\frac{5}{12} + 1 \right) = \frac{17}{12}. \end{aligned}$$

$$\begin{aligned} 31. \sin \theta &= \cos (\theta - 45^\circ) \Rightarrow \cos (90^\circ - \theta) = \cos (\theta - 45^\circ). \\ \Rightarrow 90^\circ - \theta &= \theta - 45^\circ \Rightarrow 2\theta = 135^\circ \Rightarrow \theta = (67.5)^\circ. \end{aligned}$$

$$\begin{aligned} 32. \text{ Given expression} &= \frac{\sin 50^\circ}{\cos (90^\circ - 50^\circ)} + \frac{\operatorname{cosec} 40^\circ}{\sec (90^\circ - 40^\circ)} - 4 \cos 50^\circ \operatorname{cosec} (90^\circ - 50^\circ) \\ &= \frac{\sin 50^\circ}{\sin 50^\circ} + \frac{\operatorname{cosec} 40^\circ}{\operatorname{cosec} 40^\circ} - 4 \cos 50^\circ \sec 50^\circ = 1 + 1 - 4 = -2. \end{aligned}$$

$$\begin{aligned} 33. \sin 48^\circ \sec 42^\circ + \cos 48^\circ \operatorname{cosec} 42^\circ &= \sin 48^\circ \sec (90^\circ - 48^\circ) + \cos 48^\circ \operatorname{cosec} (90^\circ - 48^\circ) \\ &= \sin 48^\circ \operatorname{cosec} 48^\circ + \cos 48^\circ \sec 48^\circ = 1 + 1 = 2. \end{aligned}$$

$$34. \sin \theta = \frac{x}{a} \text{ and } \cos \theta = \frac{y}{b}.$$

$$\text{Squaring and adding, we get } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow b^2 x^2 + a^2 y^2 = a^2 b^2.$$

35. Squaring and subtracting, we get

$$25 \left(x^2 - \frac{1}{x^2} \right) = (\sec^2 \theta - \tan^2 \theta) = 1 \Rightarrow 5 \left(x^2 - \frac{1}{x^2} \right) = \frac{1}{5}.$$

36. Squaring and subtracting, we get

$$4 \left(x^2 - \frac{1}{x^2} \right) = (\operatorname{cosec}^2 \theta - \cot^2 \theta) = 1 \Rightarrow 2 \left(x^2 - \frac{1}{x^2} \right) = \frac{1}{2}.$$

$$37. \sec \theta + \tan \theta = x \Rightarrow \sec \theta - \tan \theta = \frac{1}{x}.$$

$$\text{Adding, we get } 2 \sec \theta = \left(x + \frac{1}{x} \right) \Rightarrow \sec \theta = \left(\frac{x^2 + 1}{2x} \right).$$

38. Given expression

$$= \frac{\cos 38^\circ \operatorname{cosec} (90^\circ - 38^\circ)}{(\tan 18^\circ \tan 72^\circ)(\tan 35^\circ \tan 55^\circ) \tan 60^\circ}$$

$$= \frac{\cos 38^\circ \sec 38^\circ}{\{\tan 18^\circ \tan (90^\circ - 18^\circ)\} \{\tan 35^\circ \tan (90^\circ - 35^\circ)\} \cdot \sqrt{3}}$$

$$= \frac{1}{(\tan 18^\circ \cot 18^\circ)(\tan 35^\circ \cot 35^\circ)\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

$$39. \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - x^2}.$$

$$\therefore \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{1 - x^2}}{x}.$$

$$40. \sec^2 \theta = (1 + \tan^2 \theta) \Rightarrow \tan^2 \theta = (\sec^2 \theta - 1)$$

$$\Rightarrow \tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{x^2 - 1}.$$

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

$$1. \frac{\sec 30^\circ}{\operatorname{cosec} 60^\circ} = ?$$

$$(a) \frac{2}{\sqrt{3}}$$

$$(b) \frac{\sqrt{3}}{2}$$

$$(c) \sqrt{3}$$

$$(d) 1$$

$$2. \frac{\tan 35^\circ}{\cot 55^\circ} + \frac{\cot 78^\circ}{\tan 12^\circ} = ?$$

$$(a) 0$$

$$(b) 1$$

$$(c) 2$$

$$(d) \text{ none of these}$$

$$3. \tan 10^\circ \tan 15^\circ \tan 75^\circ \tan 80^\circ = ?$$

$$(a) \sqrt{3}$$

$$(b) \frac{1}{\sqrt{3}}$$

$$(c) -1$$

$$(d) 1$$

$$4. \tan 5^\circ \tan 25^\circ \tan 30^\circ \tan 65^\circ \tan 85^\circ = ?$$

$$(a) \sqrt{3}$$

$$(b) \frac{1}{\sqrt{3}}$$

$$(c) 1$$

$$(d) \text{ none of these}$$

$$5. \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 180^\circ = ?$$

$$(a) -1$$

$$(b) 1$$

$$(c) 0$$

$$(d) \frac{1}{2}$$

$$6. \frac{2 \sin^2 63^\circ + 1 + 2 \sin^2 27^\circ}{3 \cos^2 17^\circ - 2 + 3 \cos^2 73^\circ} = ?$$

$$(a) \frac{3}{2}$$

$$(b) \frac{2}{3}$$

$$(c) 2$$

$$(d) 3$$

$$7. \sin 47^\circ \cos 43^\circ + \cos 47^\circ \sin 43^\circ = ?$$

$$(a) \sin 4^\circ$$

$$(b) \cos 4^\circ$$

$$(c) 1$$

$$(d) 0$$

8. $\sec 70^\circ \sin 20^\circ + \cos 20^\circ \operatorname{cosec} 70^\circ = ?$
 (a) 0 (b) 1 (c) -1 (d) 2
9. If $\sin 3A = \cos (A - 10^\circ)$ and $3A$ is acute then $\angle A = ?$
 (a) 35° (b) 25° (c) 20° (d) 45°
10. If $\sec 4A = \operatorname{cosec} (A - 10^\circ)$ and $4A$ is acute then $\angle A = ?$
 (a) 20° (b) 30° (c) 40° (d) 50°
11. If A and B are acute angles such that $\sin A = \cos B$ then $(A + B) = ?$
 (a) 45° (b) 60° (c) 90° (d) 180°
12. If $\cos (\alpha + \beta) = 0$ then $\sin (\alpha - \beta) = ?$
 (a) $\sin \alpha$ (b) $\cos \beta$ (c) $\sin 2\alpha$ (d) $\cos 2\beta$
13. $\sin (45^\circ + \theta) - \cos (45^\circ - \theta) = ?$
 (a) $2 \sin \theta$ (b) $2 \cos \theta$ (c) 0 (d) 1
14. $\sec^2 10^\circ - \cot^2 80^\circ = ?$
 (a) 1 (b) 0 (c) $\frac{3}{2}$ (d) $\frac{1}{2}$
15. $\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ = ?$
 (a) 0 (b) 1 (c) -1 (d) 2
16. $\frac{2 \tan^2 30^\circ \sec^2 52^\circ \sin^2 38^\circ}{\operatorname{cosec}^2 70^\circ - \tan^2 20^\circ} = ?$
 (a) 2 (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$
17. $\left\{ \frac{(\sin^2 22^\circ + \sin^2 68^\circ)}{(\cos^2 22^\circ + \cos^2 68^\circ)} + \sin^2 63^\circ + \cos 63^\circ \sin 27^\circ \right\} = ?$
 (a) 0 (b) 1 (c) 2 (d) 3
18. $\frac{\cot(90^\circ - \theta) \cdot \sin(90^\circ - \theta)}{\sin \theta} + \frac{\cot 40^\circ}{\tan 50^\circ} - (\cos^2 20^\circ + \cos^2 70^\circ) = ?$
 (a) 0 (b) 1 (c) -1 (d) none of these
19. $\frac{\cos 38^\circ \operatorname{cosec} 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ} = ?$
 (a) $\sqrt{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$
20. If $2 \sin 2\theta = \sqrt{3}$ then $\theta = ?$
 (a) 30° (b) 45° (c) 60° (d) 90°
21. If $2 \cos 3\theta = 1$ then $\theta = ?$
 (a) 10° (b) 15° (c) 20° (d) 30°

22. If $\sqrt{3} \tan 2\theta - 3 = 0$ then $\theta = ?$
(a) 15° (b) 30° (c) 45° (d) 60°
23. If $\tan x = 3 \cot x$ then $x = ?$
(a) 45° (b) 60° (c) 30° (d) 15°
24. If $x \tan 45^\circ \cos 60^\circ = \sin 60^\circ \cot 60^\circ$ then $x = ?$
(a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{3}$
25. If $\tan^2 45^\circ - \cos^2 30^\circ = x \sin 45^\circ \cos 45^\circ$ then $x = ?$
(a) 2 (b) -2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
26. $\sec^2 60^\circ - 1 = ?$
(a) 2 (b) 3 (c) 4 (d) 0
27. $(\cos 0^\circ + \sin 30^\circ + \sin 45^\circ)(\sin 90^\circ + \cos 60^\circ - \cos 45^\circ) = ?$
(a) $\frac{5}{6}$ (b) $\frac{5}{8}$ (c) $\frac{3}{5}$ (d) $\frac{7}{4}$
28. $\sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ = ?$
(a) 0 (b) $\frac{1}{4}$ (c) 4 (d) 1
29. $3 \cos^2 60^\circ + 2 \cot^2 30^\circ - 5 \sin^2 45^\circ = ?$
(a) $\frac{13}{6}$ (b) $\frac{17}{4}$ (c) 1 (d) 4
30. $\cos^2 30^\circ \cos^2 45^\circ + 4 \sec^2 60^\circ + \frac{1}{2} \cos^2 90^\circ - 2 \tan^2 60^\circ = ?$
(a) $\frac{73}{8}$ (b) $\frac{75}{8}$ (c) $\frac{81}{8}$ (d) $\frac{83}{8}$
31. If $\operatorname{cosec} \theta = \sqrt{10}$ then $\sec \theta = ?$
(a) $\frac{3}{\sqrt{10}}$ (b) $\frac{\sqrt{10}}{3}$ (c) $\frac{1}{\sqrt{10}}$ (d) $\frac{2}{\sqrt{10}}$
32. If $\tan \theta = \frac{8}{15}$ then $\operatorname{cosec} \theta = ?$
(a) $\frac{17}{8}$ (b) $\frac{8}{17}$ (c) $\frac{17}{15}$ (d) $\frac{15}{17}$
33. If $\sin \theta = \frac{a}{b}$ then $\cos \theta = ?$
(a) $\frac{b}{\sqrt{b^2 - a^2}}$ (b) $\frac{\sqrt{b^2 - a^2}}{b}$ (c) $\frac{a}{\sqrt{b^2 - a^2}}$ (d) $\frac{b}{a}$
34. If $\tan \theta = \sqrt{3}$ then $\sec \theta = ?$
(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) 2

35. If $\sec \theta = \frac{25}{7}$ then $\sin \theta = ?$

- (a) $\frac{7}{24}$ (b) $\frac{24}{7}$ (c) $\frac{24}{25}$ (d) none of these

36. If $\sin \theta = \frac{1}{2}$ then $\cot \theta = ?$

- (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) 1

37. If $\cos \theta = \frac{4}{5}$ then $\tan \theta = ?$

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{3}{5}$ (d) $\frac{5}{3}$

38. If $3x = \operatorname{cosec} \theta$ and $\frac{3}{x} = \cot \theta$ then $3\left(x^2 - \frac{1}{x^2}\right) = ?$

- (a) $\frac{1}{27}$ (b) $\frac{1}{81}$ (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

39. If $2x = \sec A$ and $\frac{2}{x} = \tan A$ then $2\left(x^2 - \frac{1}{x^2}\right) = ?$

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$

40. If $\tan \theta = \frac{4}{3}$ then $(\sin \theta + \cos \theta) = ?$

- (a) $\frac{7}{3}$ (b) $\frac{7}{4}$ (c) $\frac{7}{5}$ (d) $\frac{5}{7}$

41. If $(\tan \theta + \cot \theta) = 5$ then $(\tan^2 \theta + \cot^2 \theta) = ?$

- (a) 27 (b) 25 (c) 24 (d) 23

42. If $(\cos \theta + \sec \theta) = \frac{5}{2}$ then $(\cos^2 \theta + \sec^2 \theta) = ?$

- (a) $\frac{21}{4}$ (b) $\frac{17}{4}$ (c) $\frac{29}{4}$ (d) $\frac{33}{4}$

43. If $\tan \theta = \frac{1}{\sqrt{7}}$ then $\frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)} = ?$

- (a) $\frac{-2}{3}$ (b) $\frac{-3}{4}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

44. If $7 \tan \theta = 4$ then $\frac{(7 \sin \theta - 3 \cos \theta)}{(7 \sin \theta + 3 \cos \theta)} = ?$

- (a) $\frac{1}{7}$ (b) $\frac{5}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{14}$

45. If $3 \cot \theta = 4$ then $\frac{(5 \sin \theta + 3 \cos \theta)}{(5 \sin \theta - 3 \cos \theta)} = ?$
 (a) $\frac{1}{3}$ (b) 3 (c) $\frac{1}{9}$ (d) 9
46. If $\tan \theta = \frac{a}{b}$ then $\frac{(a \sin \theta - b \cos \theta)}{(a \sin \theta + b \cos \theta)} = ?$
 (a) $\frac{(a^2 + b^2)}{(a^2 - b^2)}$ (b) $\frac{(a^2 - b^2)}{(a^2 + b^2)}$ (c) $\frac{a^2}{(a^2 + b^2)}$ (d) $\frac{b^2}{(a^2 + b^2)}$
47. If $\sin A + \sin^2 A = 1$ then $\cos^2 A + \cos^4 A = ?$
 (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 3
48. If $\cos A + \cos^2 A = 1$ then $\sin^2 A + \sin^4 A = ?$
 (a) 1 (b) 2 (c) 4 (d) 3
49. $\sqrt{\frac{1 - \sin A}{1 + \sin A}} = ?$
 (a) $\sec A + \tan A$ (b) $\sec A - \tan A$
 (c) $\sec A \tan A$ (d) none of these
50. $\sqrt{\frac{1 + \cos A}{1 - \cos A}} = ?$
 (a) $\operatorname{cosec} A - \cot A$ (b) $\operatorname{cosec} A + \cot A$
 (c) $\operatorname{cosec} A \cot A$ (d) none of these
51. If $\tan \theta = \frac{a}{b}$ then $\frac{(\cos \theta + \sin \theta)}{(\cos \theta - \sin \theta)} = ?$
 (a) $\frac{a+b}{a-b}$ (b) $\frac{a-b}{a+b}$ (c) $\frac{b+a}{b-a}$ (d) $\frac{b-a}{b+a}$
52. $(\operatorname{cosec} \theta - \cot \theta)^2 = ?$
 (a) $\frac{1 + \cos \theta}{1 - \cos \theta}$ (b) $\frac{1 - \cos \theta}{1 + \cos \theta}$ (c) $\frac{1 + \sin \theta}{1 - \sin \theta}$ (d) none of these
53. $(\sec A + \tan A)(1 - \sin A) = ?$
 (a) $\sin A$ (b) $\cos A$ (c) $\sec A$ (d) $\operatorname{cosec} A$

ANSWERS (MCQ)

1. (d) 2. (c) 3. (d) 4. (b) 5. (c) 6. (d) 7. (c) 8. (d) 9. (b)
 10. (a) 11. (c) 12. (d) 13. (c) 14. (a) 15. (b) 16. (c) 17. (c) 18. (b)
 19. (c) 20. (a) 21. (c) 22. (b) 23. (b) 24. (a) 25. (c) 26. (b) 27. (d)

28. (b) 29. (b) 30. (d) 31. (b) 32. (a) 33. (b) 34. (d) 35. (c) 36. (b)
 37. (a) 38. (c) 39. (a) 40. (c) 41. (d) 42. (b) 43. (d) 44. (a) 45. (d)
 46. (b) 47. (b) 48. (a) 49. (b) 50. (b) 51. (c) 52. (b) 53. (b)

HINTS TO SOME SELECTED QUESTIONS

2. Given expression = $\frac{\tan 35^\circ}{\cot(90^\circ - 35^\circ)} + \frac{\cot 78^\circ}{\tan(90^\circ - 78^\circ)} = \frac{\tan 35^\circ}{\tan 35^\circ} + \frac{\cot 78^\circ}{\cot 78^\circ} = (1 + 1) = 2.$
3. Given expression = $(\tan 10^\circ \tan 80^\circ)(\tan 15^\circ \tan 75^\circ)$
 $= \{\tan 10^\circ \tan(90^\circ - 10^\circ)\} \{\tan 15^\circ \tan(90^\circ - 15^\circ)\}$
 $= (\tan 10^\circ \cdot \cot 10^\circ)(\tan 15^\circ \cdot \cot 15^\circ) = 1 \times 1 = 1.$
4. Given expression = $(\tan 5^\circ \tan 85^\circ)(\tan 25^\circ \tan 65^\circ) \tan 30^\circ$
 $= \{\tan 5^\circ \cdot \tan(90^\circ - 5^\circ)\} \{\tan 25^\circ \cdot \tan(90^\circ - 25^\circ)\} \cdot \tan 30^\circ$
 $= (\tan 5^\circ \cdot \cot 5^\circ)(\tan 25^\circ \cdot \cot 25^\circ) \cdot \frac{1}{\sqrt{3}} = 1 \times 1 \times \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}.$
6. Given expression = $\frac{2(\sin^2 63^\circ + \sin^2 27^\circ) + 1}{3(\cos^2 17^\circ + \cos^2 73^\circ) - 2} = \frac{2(\sin^2 63^\circ + \cos^2 63^\circ) + 1}{3(\cos^2 17^\circ + \sin^2 17^\circ) - 2}$
 $= \frac{(2 \times 1) + 1}{(3 \times 1) - 2} = \frac{3}{1} = 3. \quad \left[\begin{array}{l} \because \sin 27^\circ = \sin(90^\circ - 63^\circ) = \cos 63^\circ \\ \cos 73^\circ = \cos(90^\circ - 17^\circ) = \sin 17^\circ \end{array} \right]$
7. Given expression = $\sin 47^\circ \cos(90^\circ - 47^\circ) + \cos 47^\circ \sin(90^\circ - 47^\circ)$
 $= (\sin^2 47^\circ + \cos^2 47^\circ) = 1.$
8. Given expression = $\sec 70^\circ \sin(90^\circ - 70^\circ) + \cos 20^\circ \operatorname{cosec}(90^\circ - 20^\circ)$
 $= \sec 70^\circ \cos 70^\circ + \cos 20^\circ \sec 20^\circ = 1 + 1 = 2.$
9. $\sin 3A = \cos(A - 10^\circ) \Rightarrow \cos(90^\circ - 3A) = \cos(A - 10^\circ)$
 $\Rightarrow 90^\circ - 3A = A - 10^\circ \Rightarrow 4A = 100^\circ \Rightarrow A = 25^\circ.$
10. $\sec 4A = \operatorname{cosec}(A - 10^\circ) \Rightarrow \operatorname{cosec}(90^\circ - 4A) = \operatorname{cosec}(A - 10^\circ)$
 $\Rightarrow 90^\circ - 4A = A - 10^\circ \Rightarrow 5A = 100^\circ \Rightarrow A = 20^\circ.$
11. $\sin A = \cos B \Rightarrow \cos(90^\circ - A) = \cos B \Rightarrow 90^\circ - A = B \Rightarrow A + B = 90^\circ.$
12. $\cos(\alpha + \beta) = 0 \Rightarrow \alpha + \beta = 90^\circ \Rightarrow \alpha = 90^\circ - \beta$
 $\therefore (\alpha - \beta) = (90^\circ - 2\beta) \Rightarrow \sin(\alpha - \beta) = \sin(90^\circ - 2\beta) = \cos 2\beta.$
13. $\sin(45^\circ + \theta) - \cos(45^\circ - \theta) = \sin(45^\circ + \theta) - \sin\{90^\circ - (45^\circ - \theta)\}$
 $= \sin(45^\circ + \theta) - \sin(45^\circ + \theta) = 0.$
14. $\sec^2 10^\circ - \cot^2 80^\circ = \sec^2 10^\circ - \cot^2(90^\circ - 10^\circ) = \sec^2 10^\circ - \tan^2 10^\circ = 1.$
15. $\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ = \operatorname{cosec}^2 57^\circ - \tan^2(90^\circ - 57^\circ) = \operatorname{cosec}^2 57^\circ - \cot^2 57^\circ = 1.$
16. Given expression = $\frac{2 \times \left(\frac{1}{\sqrt{3}}\right)^2 \cdot \sec^2 52^\circ \sin^2(90^\circ - 52^\circ)}{\operatorname{cosec}^2 70^\circ - \tan^2(90^\circ - 70^\circ)} = \frac{2}{3} \cdot \frac{\sec^2 52^\circ \cos^2 52^\circ}{(\operatorname{cosec}^2 70^\circ - \cot^2 70^\circ)}$
 $= \frac{2}{3} \times \frac{1}{1} = \frac{2}{3}.$

$$\begin{aligned}
 17. \text{ Given expression} &= \frac{\sin^2 22^\circ + \sin^2(90^\circ - 22^\circ)}{\cos^2 22^\circ + \cos^2(90^\circ - 22^\circ)} + \sin^2 63^\circ + \cos 63^\circ \sin(90^\circ - 63^\circ) \\
 &= \frac{\sin^2 22^\circ + \cos^2 22^\circ}{\cos^2 22^\circ + \sin^2 22^\circ} + \sin^2 63^\circ + \cos^2 63^\circ = \frac{1}{1} + 1 = 1 + 1 = 2.
 \end{aligned}$$

$$\begin{aligned}
 18. \text{ Given expression} &= \frac{\tan \theta \cdot \cos \theta}{\sin \theta} + \frac{\cot 40^\circ}{\tan(90^\circ - 40^\circ)} - \{\cos^2 20^\circ + \cos^2(90^\circ - 20^\circ)\} \\
 &= \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} + \frac{\cot 40^\circ}{\cot 40^\circ} - \{\cos^2 20^\circ + \sin^2 20^\circ\} = 1 + 1 - 1 = 1.
 \end{aligned}$$

$$\begin{aligned}
 19. \frac{\cos 38^\circ \operatorname{cosec} 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ} &= \frac{\cos 38^\circ \operatorname{cosec}(90^\circ - 38^\circ)}{(\tan 18^\circ \tan 72^\circ)(\tan 35^\circ \tan 55^\circ) \tan 60^\circ} \\
 &= \frac{\cos 38^\circ \sec 38^\circ}{\tan 18^\circ \tan(90^\circ - 18^\circ) \tan 35^\circ \tan(90^\circ - 35^\circ) \sqrt{3}} \quad [\because \tan 60^\circ = \sqrt{3}] \\
 &= \frac{1}{(\tan 18^\circ \cot 18^\circ)(\tan 35^\circ \cot 35^\circ) \sqrt{3}} = \frac{1}{\sqrt{3}}.
 \end{aligned}$$

$$20. 2 \sin 2\theta = \sqrt{3} \Rightarrow \sin 2\theta = \frac{\sqrt{3}}{2} = \sin 60^\circ \Rightarrow 2\theta = 60^\circ \Rightarrow \theta = 30^\circ.$$

$$21. 2 \cos 3\theta = 1 \Rightarrow \cos 3\theta = \frac{1}{2} = \cos 60^\circ \Rightarrow 3\theta = 60^\circ \Rightarrow \theta = 20^\circ.$$

$$22. \sqrt{3} \tan 2\theta = 3 \Rightarrow \tan 2\theta = \frac{3}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ \Rightarrow 2\theta = 60^\circ \Rightarrow \theta = 30^\circ.$$

$$23. \tan x = 3 \cot x \Rightarrow \tan^2 x = 3 \Rightarrow \tan x = \sqrt{3} = \tan 60^\circ \Rightarrow x = 60^\circ.$$

$$24. x = \frac{\sin 60^\circ \cot 60^\circ}{\tan 45^\circ \cos 60^\circ} = \frac{\tan 60^\circ \cot 60^\circ}{\tan 45^\circ} = 1.$$

$$\begin{aligned}
 25. x &= \frac{\tan^2 45^\circ - \cos^2 30^\circ}{\sin 45^\circ \cos 45^\circ} = \frac{1 - \left(\frac{\sqrt{3}}{2}\right)^2}{\left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right)} = \frac{1 - \frac{3}{4}}{\frac{1}{2}} = \frac{1}{4} \times \frac{2}{1} = \frac{1}{2}.
 \end{aligned}$$

$$26. \sec^2 60^\circ - 1 = 2^2 - 1 = 4 - 1 = 3.$$

$$\begin{aligned}
 27. (\cos 0^\circ + \sin 30^\circ + \sin 45^\circ)(\sin 90^\circ + \cos 60^\circ - \cos 45^\circ) \\
 = \left(1 + \frac{1}{2} + \frac{1}{\sqrt{2}}\right) \left(1 + \frac{1}{2} - \frac{1}{\sqrt{2}}\right) = \left(\frac{3}{2} + \frac{1}{\sqrt{2}}\right) \left(\frac{3}{2} - \frac{1}{\sqrt{2}}\right) = \left(\frac{9}{4} - \frac{1}{2}\right) = \frac{7}{4}.
 \end{aligned}$$

$$28. \sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ = \left(\frac{1}{2}\right)^2 + 4 \times (1)^2 - 2^2 = \frac{1}{4} + 4 - 4 = \frac{1}{4}.$$

$$\begin{aligned}
 29. 3 \cos^2 60^\circ + 2 \cot^2 30^\circ - 5 \sin^2 45^\circ \\
 = 3 \times \left(\frac{1}{2}\right)^2 + 2 \times (\sqrt{3})^2 - 5 \times \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{3}{4} + 6 - \frac{5}{2} = \frac{17}{4}.
 \end{aligned}$$

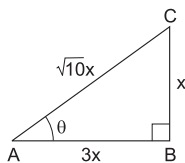
$$\begin{aligned}
 30. \cos^2 30^\circ \cos^2 45^\circ + 4 \sec^2 60^\circ + \frac{1}{2} \cos^2 90^\circ - 2 \tan^2 60^\circ \\
 = \left(\frac{\sqrt{3}}{2}\right)^2 \cdot \left(\frac{1}{\sqrt{2}}\right)^2 + (4 \times 2^2) + \left(\frac{1}{2} \times 0^2\right) - 2 \times (\sqrt{3})^2 \\
 = \left(\frac{3}{4} \times \frac{1}{2}\right) + 16 + 0 - 6 = \frac{3}{8} + 10 = \frac{83}{8}.
 \end{aligned}$$

$$31. \operatorname{cosec} \theta = \frac{AC}{BC} = \frac{\sqrt{10}}{1} = \frac{\sqrt{10}x}{x} \Rightarrow AC = \sqrt{10}x \text{ and } BC = x.$$

$$\therefore AB^2 = AC^2 - BC^2 = (\sqrt{10}x)^2 - (x)^2 = 10x^2 - x^2 = 9x^2$$

$$\Rightarrow AB = \sqrt{9x^2} = 3x.$$

$$\therefore \sec \theta = \frac{AC}{AB} = \frac{\sqrt{10}x}{3x} = \frac{\sqrt{10}}{3}.$$

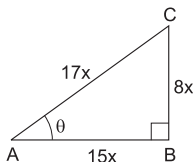


$$32. \tan \theta = \frac{BC}{AB} = \frac{8}{15} = \frac{8x}{15x} \Rightarrow BC = 8x \text{ and } AB = 15x.$$

$$\therefore AC^2 = AB^2 + BC^2 = (15x)^2 + (8x)^2 = 225x^2 + 64x^2 = 289x^2$$

$$\Rightarrow AC = \sqrt{289x^2} = 17x.$$

$$\therefore \operatorname{cosec} \theta = \frac{AC}{BC} = \frac{17x}{8x} = \frac{17}{8}.$$

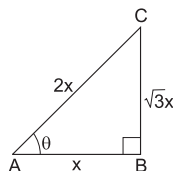


$$33. \cos^2 \theta = (1 - \sin^2 \theta) = \left(1 - \frac{a^2}{b^2}\right) = \frac{b^2 - a^2}{b^2} \Rightarrow \cos \theta = \frac{\sqrt{b^2 - a^2}}{b}.$$

$$34. \tan \theta = \frac{BC}{AB} = \frac{\sqrt{3}}{1} = \frac{\sqrt{3}x}{x} \Rightarrow BC = \sqrt{3}x \text{ and } AB = x.$$

$$\therefore AC^2 = AB^2 + BC^2 = x^2 + 3x^2 = 4x^2 \Rightarrow AC = 2x.$$

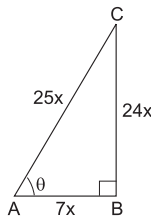
$$\therefore \sec \theta = \frac{AC}{AB} = \frac{2x}{x} = 2.$$



$$35. \sec \theta = \frac{AC}{AB} = \frac{25}{7} = \frac{25x}{7x} \Rightarrow AC = 25x \text{ and } AB = 7x.$$

$$\therefore BC^2 = AC^2 - AB^2 = 625x^2 - 49x^2 = 576x^2 \Rightarrow BC = 24x.$$

$$\therefore \sin \theta = \frac{BC}{AC} = \frac{24x}{25x} = \frac{24}{25}.$$

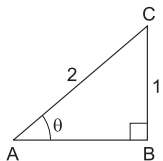


$$36. \sin \theta = \frac{1}{2} = \frac{BC}{AC}.$$

$$\therefore AB^2 = AC^2 - BC^2 = 2^2 - 1^2 = 3$$

$$\therefore AB = \sqrt{3}.$$

$$\cot \theta = \frac{AB}{BC} = \frac{\sqrt{3}}{1} = \sqrt{3}.$$

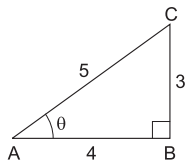


$$37. \cos \theta = \frac{4}{5} = \frac{AB}{AC}.$$

$$\therefore BC^2 = AC^2 - AB^2 = 25 - 16 = 9$$

$$\Rightarrow BC = 3.$$

$$\therefore \tan \theta = \frac{BC}{AB} = \frac{3}{4}.$$



38. We know that $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$.

$$\therefore (3x)^2 - \left(\frac{3}{x}\right)^2 = 1 \Rightarrow 9x^2 - \frac{9}{x^2} = 1 \Rightarrow 9\left(x^2 - \frac{1}{x^2}\right) = 1$$

$$\Rightarrow \left(x^2 - \frac{1}{x^2}\right) = \frac{1}{9} \Rightarrow 3\left(x^2 - \frac{1}{x^2}\right) = 3 \times \frac{1}{9} = \frac{1}{3}.$$

39. We know that $\sec^2 A - \tan^2 A = 1$.

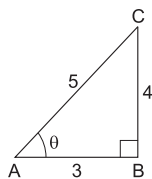
$$\begin{aligned} \therefore (2x)^2 - \left(\frac{2}{x}\right)^2 &= 1 \Rightarrow 4x^2 - \frac{4}{x^2} = 1 \Rightarrow 4\left(x^2 - \frac{1}{x^2}\right) = 1 \\ &\Rightarrow \left(x^2 - \frac{1}{x^2}\right) = \frac{1}{4} \Rightarrow 2\left(x^2 - \frac{1}{x^2}\right) = 2 \times \frac{1}{4} = \frac{1}{2}. \end{aligned}$$

40. $\tan \theta = \frac{4}{3} = \frac{BC}{AB}$.

$$\therefore AC^2 = AB^2 + BC^2 = (3)^2 + (4)^2 = 25$$

$$\Rightarrow AC = \sqrt{25} = 5.$$

$$\therefore (\sin \theta + \cos \theta) = \left(\frac{4}{5} + \frac{3}{5}\right) = \frac{7}{5}.$$



41. $(\tan \theta + \cot \theta)^2 = 5^2 \Rightarrow \tan^2 \theta + \cot^2 \theta + 2 = 25$

$$\therefore (\tan^2 \theta + \cot^2 \theta) = 23.$$

42. $(\cos \theta + \sec \theta)^2 = \frac{25}{4} \Rightarrow \cos^2 \theta + \sec^2 \theta + 2 = \frac{25}{4} \Rightarrow \cos^2 \theta + \sec^2 \theta = \frac{25}{4} - 2 = \frac{17}{4}.$

43. $\tan \theta = \frac{1}{\sqrt{7}} \Rightarrow \cot \theta = \sqrt{7}.$

$$\sec^2 \theta = (1 + \tan^2 \theta) = \left(1 + \frac{1}{7}\right) = \frac{8}{7} \text{ and } \operatorname{cosec}^2 \theta = (1 + \cot^2 \theta) = (1 + 7) = 8.$$

$$\therefore \frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta} = \frac{\left(8 - \frac{8}{7}\right)}{\left(8 + \frac{8}{7}\right)} = \frac{48}{64} = \frac{3}{4}.$$

44. Given, $\tan \theta = \frac{4}{7}.$

$$\begin{aligned} \therefore \frac{(7 \sin \theta - 3 \cos \theta)}{(7 \sin \theta + 3 \cos \theta)} &= \frac{(7 \tan \theta - 3)}{(7 \tan \theta + 3)} \quad [\text{dividing num. and denom. by } \cos \theta] \\ &= \frac{\left(7 \times \frac{4}{7} - 3\right)}{\left(7 \times \frac{4}{7} + 3\right)} = \frac{(4 - 3)}{(4 + 3)} = \frac{1}{7}. \end{aligned}$$

45. Given, $\cot \theta = \frac{4}{3}.$

$$\begin{aligned} \therefore \frac{(5 \sin \theta + 3 \cos \theta)}{(5 \sin \theta - 3 \cos \theta)} &= \frac{5 + 3 \cot \theta}{5 - 3 \cot \theta} \quad [\text{dividing num. and denom. by } \sin \theta] \\ &= \frac{\left(5 + 3 \times \frac{4}{3}\right)}{\left(5 - 3 \times \frac{4}{3}\right)} = \frac{(5 + 4)}{(5 - 4)} = \frac{9}{1} = 9. \end{aligned}$$

46. Dividing num. and denom. by $\cos \theta$, we get

$$\frac{(a \sin \theta - b \cos \theta)}{(a \sin \theta + b \cos \theta)} = \frac{(a \tan \theta - b)}{(a \tan \theta + b)} = \frac{\left(a \times \frac{a}{b} - b\right)}{\left(a \times \frac{a}{b} + b\right)} = \frac{(a^2 - b^2)}{(a^2 + b^2)}.$$

$$47. \sin A + \sin^2 A = 1 \Rightarrow \sin A = (1 - \sin^2 A) \Rightarrow \sin A = \cos^2 A.$$

$$\therefore (\cos^2 A + \cos^4 A) = (\sin A + \sin^2 A) = 1.$$

$$48. \cos A + \cos^2 A = 1 \Rightarrow \cos A = (1 - \cos^2 A) \Rightarrow \cos A = \sin^2 A.$$

$$\therefore (\sin^2 A + \sin^4 A) = (\cos A + \cos^2 A) = 1.$$

$$49. \sqrt{\frac{1-\sin A}{1+\sin A}} = \sqrt{\frac{(1-\sin A)}{(1+\sin A)} \times \frac{(1-\sin A)}{(1-\sin A)}} = \frac{(1-\sin A)}{\sqrt{1-\sin^2 A}} = \frac{(1-\sin A)}{\sqrt{\cos^2 A}}$$

$$= \frac{(1-\sin A)}{\cos A} = \left(\frac{1}{\cos A} - \frac{\sin A}{\cos A} \right) = (\sec A - \tan A).$$

$$50. \sqrt{\frac{1+\cos A}{1-\cos A}} = \sqrt{\frac{(1+\cos A)}{(1-\cos A)} \times \frac{(1+\cos A)}{(1+\cos A)}} = \frac{(1+\cos A)}{\sqrt{1-\cos^2 A}} = \frac{(1+\cos A)}{\sqrt{\sin^2 A}}$$

$$= \frac{(1+\cos A)}{\sin A} = \left(\frac{1}{\sin A} + \frac{\cos A}{\sin A} \right) = (\operatorname{cosec} A + \cot A).$$

$$51. \frac{(\cos \theta + \sin \theta)}{(\cos \theta - \sin \theta)} = \frac{(1 + \tan \theta)}{(1 - \tan \theta)} \quad [\text{dividing num. and denom. by } \cos \theta]$$

$$= \frac{\left(1 + \frac{a}{b}\right)}{\left(1 - \frac{a}{b}\right)} = \frac{(b+a)}{(b-a)}.$$

$$52. (\operatorname{cosec} \theta - \cot \theta)^2 = \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2 = \frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)^2}{(1 - \cos^2 \theta)} = \frac{(1 - \cos \theta)}{(1 + \cos \theta)}.$$

$$53. (\sec A + \tan A)(1 - \sin A) = \left(\frac{1}{\cos A} + \frac{\sin A}{\cos A} \right)(1 - \sin A)$$

$$= \frac{(1 + \sin A)(1 - \sin A)}{\cos A} = \frac{1 - \sin^2 A}{\cos A} = \frac{\cos^2 A}{\cos A} = \cos A.$$

TEST YOURSELF

MCQ

$$1. \frac{\cos^2 56^\circ + \cos^2 34^\circ}{\sin^2 56^\circ + \sin^2 34^\circ} + 3 \tan^2 56^\circ \tan^2 34^\circ = ?$$

$$(a) 3\frac{1}{2}$$

$$(b) 4$$

$$(c) 6$$

$$(d) 5$$

$$2. \text{The value of } \left(\sin^2 30^\circ \cos^2 45^\circ + 4 \tan^2 30^\circ + \frac{1}{2} \sin^2 90^\circ + \frac{1}{8} \cot^2 60^\circ \right) = ?$$

$$(a) \frac{3}{8}$$

$$(b) \frac{5}{8}$$

$$(c) 6$$

$$(d) 2$$

$$3. \text{If } \cos A + \cos^2 A = 1 \text{ then } (\sin^2 A + \sin^4 A) = ?$$

$$(a) \frac{1}{2}$$

$$(b) 2$$

$$(c) 1$$

$$(d) 4$$

4. If $\sin \theta = \frac{\sqrt{3}}{2}$ then $(\operatorname{cosec} \theta + \cot \theta) = ?$
 (a) $(2 + \sqrt{3})$ (b) $2\sqrt{3}$ (c) $\sqrt{2}$ (d) $\sqrt{3}$

Short-Answer Questions

5. If $\cot A = \frac{4}{5}$, prove that $\frac{(\sin A + \cos A)}{(\sin A - \cos A)} = 9$.
6. If $2x = \sec A$ and $\frac{2}{x} = \tan A$, prove that $\left(x^2 - \frac{1}{x^2}\right) = \frac{1}{4}$.
7. If $\sqrt{3} \tan \theta = 3 \sin \theta$, prove that $(\sin^2 \theta - \cos^2 \theta) = \frac{1}{3}$.
8. Prove that $\frac{(\sin^2 73^\circ + \sin^2 17^\circ)}{(\cos^2 28^\circ + \cos^2 62^\circ)} = 1$.
9. If $2 \sin 2\theta = \sqrt{3}$, prove that $\theta = 30^\circ$.
10. Prove that $\sqrt{\frac{1 + \cos A}{1 - \cos A}} = (\operatorname{cosec} A + \cot A)$.
11. If $\operatorname{cosec} \theta + \cot \theta = p$, prove that $\cos \theta = \frac{(p^2 - 1)}{(p^2 + 1)}$.
12. Prove that $(\operatorname{cosec} A - \cot A)^2 = \frac{(1 - \cos A)}{(1 + \cos A)}$.
13. If $5 \cot \theta = 3$, show that the value of $\left(\frac{5 \sin \theta - 3 \cos \theta}{4 \sin \theta + 3 \cos \theta}\right)$ is $\frac{16}{29}$.
14. Prove that $(\sin 32^\circ \cos 58^\circ + \cos 32^\circ \sin 58^\circ) = 1$.
15. If $x = a \sin \theta + b \cos \theta$ and $y = a \cos \theta - b \sin \theta$, prove that $x^2 + y^2 = a^2 + b^2$.
16. Prove that $\frac{(1 + \sin \theta)}{(1 - \sin \theta)} = (\sec \theta + \tan \theta)^2$.

Long-Answer Questions

17. Prove that $\frac{1}{(\sec \theta - \tan \theta)} - \frac{1}{\cos \theta} = \frac{1}{\cos \theta} - \frac{1}{(\sec \theta + \tan \theta)}$.
18. Prove that $\frac{(\sin A - 2 \sin^3 A)}{(2 \cos^3 A - \cos A)} = \tan A$.
19. Prove that $\frac{\tan A}{(1 - \cot A)} + \frac{\cot A}{(1 - \tan A)} = (1 + \tan A + \cot A)$.
20. If $\sec 5A = \operatorname{cosec} (A - 36^\circ)$ and $5A$ is an acute angle, show that $A = 21^\circ$.

ANSWERS (TEST YOURSELF)

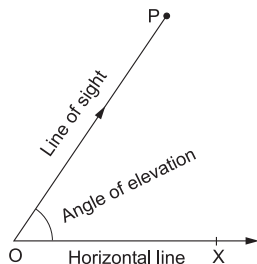
1. (b) 2. (d) 3. (c) 4. (d)



LINE OF SIGHT When an observer looks from a point O at an object P then the line OP is called the line of sight.

ANGLE OF ELEVATION

Suppose that from a point O , you look up at an object P , placed above the level of your eye. Then, the angle which the line of sight makes with the horizontal line through O is called the *angle of elevation* of P , as seen from O .



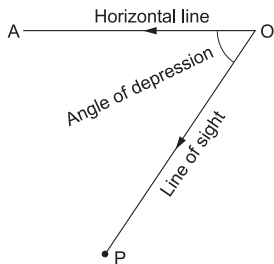
Example Let OX be a horizontal line on the level ground and let a person at O be looking up towards an object P , say an aeroplane or the top of a tree or the top of a tower or a flag at the top of a house.

Then, $\angle XOP$ is the angle of elevation of P from O .

ANGLE OF DEPRESSION

Now, suppose that from a point O , you look down at an object P , placed below the level of your eye.

Then, the angle which the line of sight makes with the horizontal line through O is called the *angle of depression* of P , as seen from O .



SOLVED EXAMPLES

EXAMPLE 1 A vertical pole stands on the level ground. From a point on the ground, 25 m away from the foot of the pole, the angle of elevation of its top is found to be 60° . Find the height of the pole. [Take $\sqrt{3} = 1.732$.]

SOLUTION Let AB be the pole standing on a level ground and let O be the position of the observer. Then, $OA = 25$ m, $\angle OAB = 90^\circ$ and $\angle AOB = 60^\circ$.

Let $AB = h$ metres.

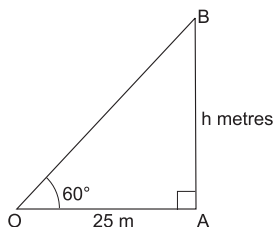
From the right $\triangle OAB$, we have

$$\frac{AB}{OA} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow \frac{h}{25} = \sqrt{3}$$

$$\Rightarrow h = 25 \times \sqrt{3} = 25 \times 1.732 = 43.3.$$

Hence, the height of the pole is 43.3 m.



EXAMPLE 2

A kite is flying, attached to a thread which is 165 m long. The thread makes an angle of 30° with the ground. Find the height of the kite from the ground, assuming that there is no slack in the thread.

SOLUTION

Let OX be the horizontal ground and let A be the position of the kite. Let O be the position of the observer and OA be the thread. Draw $AB \perp OX$.

Then, $\angle BOA = 30^\circ$, $OA = 165$ m and $\angle OBA = 90^\circ$.

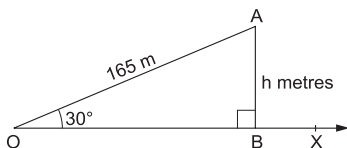
Height of the kite from the ground = AB .

Let $AB = h$ metres.

From right $\triangle OBA$, we have

$$\frac{AB}{OA} = \sin 30^\circ = \frac{1}{2}$$

$$\Rightarrow \frac{h}{165} = \frac{1}{2} \Rightarrow h = \frac{165}{2} = 82.5.$$



Hence, the height of the kite from the ground = 82.5 m.

EXAMPLE 3

The length of a string between a kite and a point on the ground is 85 m. If the string makes an angle θ with the ground level such that $\tan \theta = \frac{15}{8}$ then find the height of the kite from the ground. Assume that there is no slack in the string. [CBSE 2014]

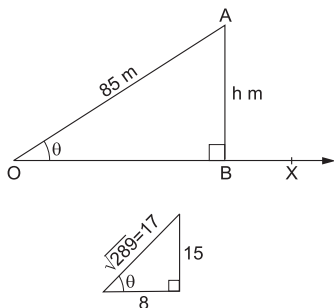
SOLUTION.

Let OX be the horizontal ground and let A be the position of the kite. Let O be the position of the observer and OA be the string. Draw $AB \perp OX$.

Then, $\angle BOA = \theta$ such that $\tan \theta = \frac{15}{8}$, $OA = 85$ m and

$\angle OBA = 90^\circ$.

Let $AB = h$ metres.



From right $\triangle OBA$, we have

$$\frac{AB}{OA} = \sin \theta = \frac{15}{17} \quad \left[\because \tan \theta = \frac{15}{8} \Rightarrow \sin \theta = \frac{15}{17} \right]$$

$$\Rightarrow \frac{h}{85} = \frac{15}{17} \Rightarrow h = \frac{15}{17} \times 85 = 75.$$

Hence, the height of the kite from the ground is 75 m.

EXAMPLE 4

A ladder 15 m long just reaches the top of a vertical wall. If the ladder makes an angle of 60° with the wall, find the height of the wall.

[CBSE 2013]

SOLUTION

Let OX be the horizontal ground and let OA be the ladder leaning against the wall AB . Then,

$OA = 15$ m, $\angle OAB = 60^\circ$ and $\angle OBA = 90^\circ$.

Let $AB = h$ metres.

Now, $\angle AOB = (90^\circ - 60^\circ) = 30^\circ$.

From right $\triangle OBA$, we have

$$\frac{AB}{OA} = \sin 30^\circ = \frac{1}{2} \Rightarrow \frac{h}{15} = \frac{1}{2}$$

$$\Rightarrow h = 15 \times \frac{1}{2} = 7.5.$$

Hence, the height of the wall is 7.5 metres.

EXAMPLE 5

If a tower 30 m high, casts a shadow $10\sqrt{3}$ m long on the ground, then what is the angle of elevation of the sun?

[CBSE 2017]

SOLUTION

Let AB be the pole and let AC be its shadow.

Let the angle of elevation of the sun be θ° .

Then, $\angle ACB = \theta$, $\angle CAB = 90^\circ$,

$AB = 30$ m and $AC = 10\sqrt{3}$ m.

From right $\triangle CAB$, we have

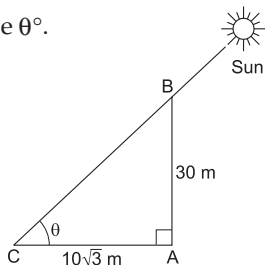
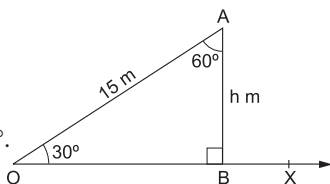
$$\tan \theta = \frac{AB}{AC} = \frac{30}{10\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ.$$

Hence, the angular elevation of the sun is 60° .

EXAMPLE 6

If a 1.5-m-tall girl stands at a distance of 3 m from a lamp-post and casts a shadow of length 4.5 m on the ground then find the height of the lamp-post.



SOLUTION Let AB be the lamp-post and CD be the girl.

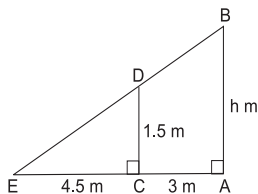
Let CE be the shadow of CD . Then,

$$CD = 1.5 \text{ m}, CE = 4.5 \text{ m and } AC = 3 \text{ m.}$$

Let $AB = h$ metres.

Now, $\triangle AEB$ and $\triangle CED$ are similar.

$$\begin{aligned} \therefore \frac{AB}{AE} &= \frac{CD}{CE} \Rightarrow \frac{h}{(3 + 4.5)} = \frac{1.5}{4.5} = \frac{1}{3} \\ \Rightarrow h &= \frac{1}{3} \times 7.5 = 2.5. \end{aligned}$$



Hence, the height of the lamp-post is 2.5 metres.

EXAMPLE 7

The shadow of a tower, when the angle of elevation of the sun is 45° , is found to be 10 metres longer than when the angle of elevation is 60° . Find the height of the tower. [Given $\sqrt{3} = 1.732$.]

SOLUTION

Let AB be the tower and let AC and AD be its shadows when the angles of elevation of the sun are 60° and 45° respectively.

$$\therefore \angle ACB = 60^\circ, \angle ADB = 45^\circ,$$

$$\angle DAB = 90^\circ \text{ and } CD = 10 \text{ m.}$$

Let $AB = h$ metres and $AC = x$ metres.

From right $\triangle CAB$, we get

$$\frac{AC}{AB} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{x}{h} = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h}{\sqrt{3}}. \quad \dots (i)$$

From right $\triangle DAB$, we get

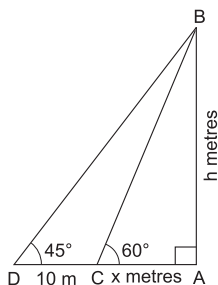
$$\frac{AD}{AB} = \cot 45^\circ = 1 \Rightarrow \frac{x + 10}{h} = 1$$

$$\Rightarrow x + 10 = h \Rightarrow x = h - 10. \quad \dots (ii)$$

Equating the values of x from (i) and (ii), we get

$$\frac{h}{\sqrt{3}} = h - 10 \Rightarrow h = \sqrt{3}h - 10\sqrt{3} \Rightarrow (\sqrt{3} - 1)h = 10\sqrt{3}$$

$$\begin{aligned} \Rightarrow h &= \frac{10\sqrt{3}}{(\sqrt{3} - 1)} = \left\{ \frac{10\sqrt{3}}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)} \right\} \\ &= 5\sqrt{3}(\sqrt{3} + 1) = 15 + 5\sqrt{3} \end{aligned}$$



$$\Rightarrow h = (5 + 5 \times 1.732) = (15 + 8.66) = 23.66.$$

Hence, the height of the tower is 23.66 m.

SUN'S ALTITUDE The angle of elevation of the sun from the earth is called the sun's altitude.

The sun's altitudes are different at different times of the day.

EXAMPLE 8 A tower is 50 m high. Its shadow is x metres shorter when the sun's altitude is 45° than when it is 30° . Find the value of x . [Given $\sqrt{3} = 1.732$.]

SOLUTION Let PQ be the tower and let PA and PB be its shadows when the altitudes of the sun are 45° and 30° respectively. Then,

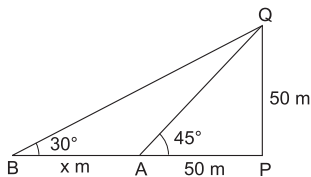
$$\angle PAQ = 45^\circ, \angle PBQ = 30^\circ, \angle BPQ = 90^\circ, PQ = 50 \text{ m.}$$

Let $AB = x$ metres.

From right $\triangle APQ$, we have

$$\frac{AP}{PQ} = \cot 45^\circ = 1$$

$$\Rightarrow \frac{AP}{50 \text{ m}} = 1 \Rightarrow AP = 50 \text{ m.}$$



From right $\triangle BPQ$, we have

$$\frac{BP}{PQ} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x+50}{50} = \sqrt{3} \Rightarrow x = 50(\sqrt{3} - 1).$$

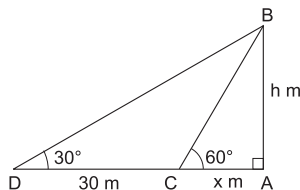
$$\Rightarrow x = 50(1.732 - 1) = (50 \times 0.732) = 36.6$$

Hence, $x = 36.6$.

EXAMPLE 9 The shadow of a tower standing on a level ground is found to be 30 m longer when the sun's altitude is 30° , than when it was 60° . Find the height of the tower. [Take $\sqrt{3} = 1.732$.] [CBSE 2012]

SOLUTION Let AB be the tower and let AC and AD be the lengths of its shadows when $\angle ACB = 60^\circ$ and $\angle ADB = 30^\circ$.

Let $AB = h$ metres, and
 $AC = x$ metres.



From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{x}{h} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = \left(h \times \frac{1}{\sqrt{3}} \right) = \frac{h}{\sqrt{3}}. \quad \dots (i)$$

From right $\triangle DAB$, we have

$$\frac{AD}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x + 30}{h} = \sqrt{3}$$

$$\Rightarrow x = (\sqrt{3}h - 30). \quad \dots (ii)$$

Equating the values of x from (i) and (ii), we get

$$\frac{h}{\sqrt{3}} = (\sqrt{3}h - 30) \Rightarrow h = 3h - 30\sqrt{3}$$

$$\Rightarrow 2h = 30\sqrt{3} \Rightarrow h = 15\sqrt{3} = (15 \times 1.732) = 25.98.$$

Hence, the height of the tower is 25.98 m.

EXAMPLE 10

From a point O on the ground, the angle of elevation of the top of a tower is 30° and that of the top of the flagstaff on the top of the tower is 60° . If the length of the flagstaff is 5 metres, find the height of the tower. [CBSE 2015]

SOLUTION

Let AB be the tower and BC be the flagstaff.

Let O be a point on the ground such that $\angle AOB = 30^\circ$ and $\angle AOC = 60^\circ$
 $BC = 5$ m (given).

Let $AB = h$ metres and $OA = x$ metres.

From right $\triangle OAB$, we have

$$\frac{OA}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x}{h} = \sqrt{3}$$

$$\Rightarrow x = h\sqrt{3}. \quad \dots (i)$$

From right $\triangle OAC$, we have

$$\frac{OA}{AC} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{x}{h+5} = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h+5}{\sqrt{3}}. \quad \dots (ii)$$

Equating the values of x from (i) and (ii), we get

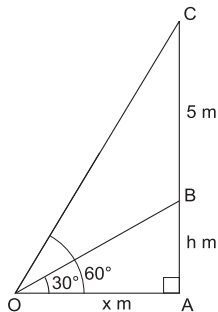
$$h\sqrt{3} = \frac{h+5}{\sqrt{3}} \Rightarrow 3h = h+5 \Rightarrow 2h = 5 \Rightarrow h = 2.5.$$

Hence, the height of the tower is 2.5 metres.

EXAMPLE 11

Two pillars of equal heights stand on either side of a road which is 100 m wide. At a point on the road between the pillars, the angles of elevation of the tops of the pillars are 60° and 30° . Find the height of each pillar and position of the point on the road. [Take $\sqrt{3} = 1.732$.]

[CBSE 2011, '13]



SOLUTION Let AB and CD be two pillars, each of height h metres and let AC be the road such that $AC = 100$ m.

Let O be the point of observation on AC .

Let $OA = x$ metres and $OC = (100 - x)$ m.

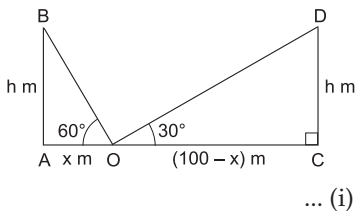
Also, $\angle AOB = 60^\circ$ and $\angle COD = 30^\circ$.

$AB \perp AC$ and $CD \perp AC$.

From right $\triangle OAB$, we have

$$\frac{AB}{OA} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow \frac{h}{x} = \sqrt{3} \Rightarrow h = \sqrt{3}x.$$



... (i)

From right $\triangle OCD$, we have

$$\frac{CD}{OC} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{h}{(100 - x)} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{(100 - x)}{\sqrt{3}}. \quad \dots \text{(ii)}$$

Equating the values of h from (i) and (ii), we get

$$\sqrt{3}x = \frac{(100 - x)}{\sqrt{3}} \Rightarrow 3x = (100 - x) \Rightarrow 4x = 100 \Rightarrow x = 25.$$

Putting $x = 25$ m in (i), we get

$$h = (25 \times \sqrt{3}) = (25 \times 1.732) = 43.3.$$

Hence, the height of each pillar is 43.3 m and the point of observation is 25 m away from the first pillar.

EXAMPLE 12

The angle of elevation of an aeroplane from a point on the ground is 60° . After a flight of 15 seconds, the angle of elevation changes to 30° . If the aeroplane is flying at a constant height of $1500\sqrt{3}$ m, find the speed of the plane in km/hr. [CBSE 2015]

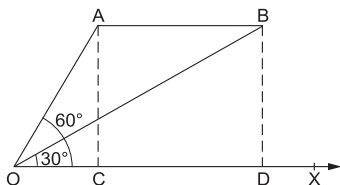
SOLUTION Let OX be the horizontal ground and let O be the point of observation. Let A and B be the two positions of the aeroplane. Let $AC \perp OX$ and $BD \perp OX$. Then,

$$\angle COA = 60^\circ, \angle DOB = 30^\circ$$

and $AC = BD = 1500\sqrt{3}$ m.

From right $\triangle OCA$, we have

$$\frac{OC}{AC} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$



$$\Rightarrow \frac{OC}{1500\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow OC = 1500 \text{ m.}$$

From right $\triangle ODB$, we have

$$\frac{OD}{BD} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{OD}{1500\sqrt{3} \text{ m}} = \sqrt{3}$$

$$\Rightarrow OD = (1500 \times 3) \text{ m} = 4500 \text{ m.}$$

$$\therefore CD = (OD - OC) = (4500 - 1500) \text{ m} = 3000 \text{ m.}$$

Thus, the aeroplane covers 3000 m in 15 seconds.

$$\begin{aligned} \therefore \text{ speed of the aeroplane} &= \left(\frac{3000}{15} \times \frac{60 \times 60}{1000} \right) \text{ km/hr} \\ &= 720 \text{ km/hr.} \end{aligned}$$

Hence, the speed of the aeroplane is 720 km/hr.

EXAMPLE 13 The angles of elevation and depression of the top and bottom of a tower from the top of a building 60 m high are 30° and 60° respectively. Find the difference between the heights of the building and the tower and also the distance between them. [CBSE 2013C, '14]

SOLUTION

Let AB be the tower and CD be the building. Then, $CD = 60 \text{ m}$.

Let CAX be the horizontal ground.

Draw $DE \perp AB$. Now, $\angle EDB = 30^\circ$

and $\angle CAD = \angle ADE = 60^\circ$.

Also, $AE = CD = 60 \text{ m}$.

Let $BE = h$ metres.

From right $\triangle ACD$, we have

$$\begin{aligned} \frac{CA}{CD} &= \cot 60^\circ = \frac{1}{\sqrt{3}} \\ \Rightarrow \frac{CA}{60 \text{ m}} &= \frac{1}{\sqrt{3}} \Rightarrow CA = \frac{60}{\sqrt{3}} \dots (i) \end{aligned}$$

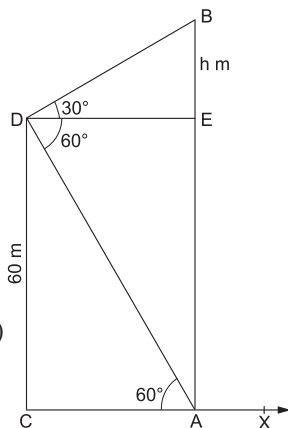
From right $\triangle EDB$, we have

$$\begin{aligned} \frac{DE}{BE} &= \cot 30^\circ = \sqrt{3} \Rightarrow \frac{DE}{h \text{ m}} = \sqrt{3} \\ \Rightarrow DE &= h\sqrt{3} \text{ m.} \dots (ii) \end{aligned}$$

But, $CA = DE$.

$\therefore$ from (i) and (ii), we get

$$\frac{60}{\sqrt{3}} = h\sqrt{3} \Rightarrow 3h = 60 \Rightarrow h = 20.$$



Hence, the difference between the heights of the tower and the building is 20 metres.

From (i), we get $CA = \left(\frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) \text{ m} = 20\sqrt{3} \text{ metres}$.

Hence, the distance between the tower and the building is $20\sqrt{3} \text{ m}$.

EXAMPLE 14

A man on a cliff observes a boat at an angle of depression of 30° which is approaching the shore to the point immediately beneath the observer with a uniform speed. Six minutes later, the angle of depression of the boat is found to be 60° . Find the total time taken by the boat to reach the shore.

[CBSE 2014]

SOLUTION

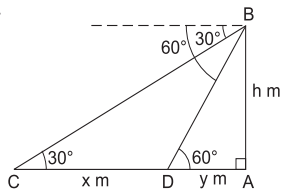
Let AB be the cliff with the man at B .

Let C and D be the two positions of the boat approaching the shore at A .

Then,

$$\angle ACB = 30^\circ \text{ and } \angle ADB = 60^\circ.$$

Let $AB = h$ metres, $CD = x$ metres and $DA = y$ metres.



From right $\triangle DAB$, we have

$$\frac{AD}{AB} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{y}{h} = \frac{1}{\sqrt{3}} \Rightarrow y = \frac{h}{\sqrt{3}}. \quad \dots (i)$$

From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x+y}{h} = \sqrt{3} \Rightarrow x+y = h\sqrt{3}. \quad \dots (ii)$$

On subtracting (i) from (ii), we get

$$x = \left(h\sqrt{3} - \frac{h}{\sqrt{3}} \right) \Rightarrow x = \frac{2h}{\sqrt{3}}. \quad \dots (iii)$$

From (iii) and (i), we get

$$x : y = \frac{2h}{\sqrt{3}} : \frac{h}{\sqrt{3}} = 2 : 1.$$

Let the time taken to cover y units be t minutes. Then,

ratio of distances = ratio of times taken to cover them.

$$\text{So, } 2 : 1 = 6 : t \Rightarrow \frac{2}{1} = \frac{6}{t} \Rightarrow 2t = 6 \Rightarrow t = 3.$$

Thus, the boat takes 3 minutes to cover the distance DA .

Hence, the total time taken by the boat to reach the shore is $(6 + 3) \text{ minutes} = 9 \text{ minutes}$.

EXAMPLE 15 Observed from the top of a 75-m-high lighthouse (from sea level), the angles of depression of two ships are 30° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the ships. (Use $\sqrt{3} = 1.73$.) [CBSE 2014]

SOLUTION Let AB be the lighthouse and C and D be the positions of two ships. Then,

$$AB = 75 \text{ m}, \angle ACB = 30^\circ$$

$$\text{and } \angle ADB = 45^\circ.$$

Let $CD = x$ metres.

From right $\triangle DAB$, we have

$$\frac{AD}{AB} = \cot 45^\circ = 1$$

$$\Rightarrow \frac{AD}{75 \text{ m}} = 1 \Rightarrow AD = 75 \text{ m}.$$

From right $\triangle CAB$, we have

$$\frac{CA}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x+75}{75} = \sqrt{3}$$

$$\Rightarrow x + 75 = 75\sqrt{3} \Rightarrow x = 75(\sqrt{3} - 1) = 75(1.73 - 1)$$

$$\Rightarrow x = (75 \times 0.73) = 54.75.$$

Hence, the distance between the ships is 54.75 metres.

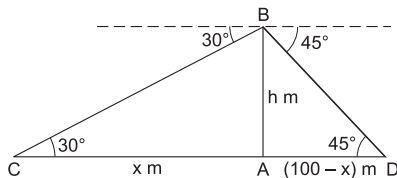
EXAMPLE 16 Two ships are approaching a lighthouse from opposite directions. The angles of depression of the two ships from the top of a lighthouse are 30° and 45° . If the distance between the two ships is 100 metres, find the height of the lighthouse. (Use $\sqrt{3} = 1.732$.) [CBSE 2014]

SOLUTION Let AB be the lighthouse and C and D be the positions of the two ships.

$$\text{Then, } \angle ACB = 30^\circ \text{ and } \angle ADB = 45^\circ.$$

$$\text{Let } AB = h \text{ metres, } CA = x \text{ metres.}$$

$$\text{Then, } AD = (100 - x) \text{ m}.$$



From right $\triangle BAD$, we have

$$\frac{AD}{AB} = \cot 45^\circ = 1$$

$$\Rightarrow \frac{(100 - x)}{h} = 1 \Rightarrow x = (100 - h). \quad \dots (i)$$

From right $\triangle BAC$, we have

$$\frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x}{h} = \sqrt{3} \Rightarrow x = h\sqrt{3}. \quad \dots (ii)$$

Equating the values of x from (i) and (ii), we get

$$100 - h = h\sqrt{3} \Rightarrow h(\sqrt{3} + 1) = 100$$

$$\Rightarrow h = \left\{ \frac{100}{(\sqrt{3} + 1)} \times \frac{(\sqrt{3} - 1)}{(\sqrt{3} - 1)} \right\} = 50(\sqrt{3} - 1)$$

$$\Rightarrow h = 50(1.732 - 1) = (50 \times 0.732) = 36.6.$$

Hence, the height of the lighthouse is 36.6 m.

EXAMPLE 17 From the top of a lighthouse, the angles of depression of two ships on the opposite sides of it are observed to be α and β . If the height of the lighthouse be h metres and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{h(\tan \alpha + \tan \beta)}{\tan \alpha \tan \beta}$ metres. [HOTS]

SOLUTION

Let AB be the lighthouse and C and D be the positions of the two ships. Then, $AB = h$ metres.

Clearly, $\angle ACB = \alpha$ and $\angle ADB = \beta$.

Let $AC = x$ metres and

$AD = y$ metres.

From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot \alpha \Rightarrow \frac{x}{h} = \cot \alpha \Rightarrow x = h \cot \alpha. \quad \dots (i)$$

From right $\triangle DAB$, we have

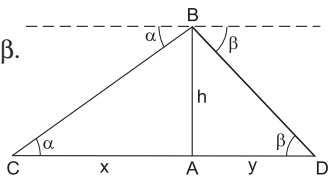
$$\frac{AD}{AB} = \cot \beta \Rightarrow \frac{y}{h} = \cot \beta \Rightarrow y = h \cot \beta. \quad \dots (ii)$$

Adding the corresponding sides of (i) and (ii), we get

$$x + y = h(\cot \alpha + \cot \beta) = h \left(\frac{1}{\tan \alpha} + \frac{1}{\tan \beta} \right)$$

$$\Rightarrow x + y = \frac{h(\tan \alpha + \tan \beta)}{\tan \alpha \tan \beta}.$$

Hence, the distance between the ships is $\frac{h(\tan \alpha + \tan \beta)}{\tan \alpha \tan \beta}$ m.



EXAMPLE 18 The angle of elevation of a cloud from a point h metres above a lake is α and the angle of depression of its reflection in the lake is β . Prove that the height of the cloud is $\frac{h(\tan \beta + \tan \alpha)}{(\tan \beta - \tan \alpha)}$ metres. [HOTS]

[CBSE 2008]

SOLUTION Let AB be the surface of the lake and let P be a point vertically above A such that $AP = h$ metres.

Let C be the position of the cloud and let D be its reflection in the lake.

Draw $PQ \perp CD$. Then,

$$\angle QPC = \alpha, \angle QPD = \beta,$$

$$BQ = AP = h \text{ metres.}$$

Let $CQ = x$ metres. Then,

$$BD = BC = (x + h) \text{ metres.}$$

From right $\triangle PQC$, we have

$$\frac{PQ}{CQ} = \cot \alpha \Rightarrow \frac{PQ}{x \text{ m}} = \cot \alpha$$

$$\Rightarrow PQ = x \cot \alpha \text{ metres.} \quad \dots (i)$$

From right $\triangle PQD$, we have

$$\frac{PQ}{QD} = \cot \beta \Rightarrow \frac{PQ}{(x + 2h) \text{ m}} = \cot \beta$$

$$\Rightarrow PQ = (x + 2h) \cot \beta \text{ metres.} \quad \dots (ii)$$

From (i) and (ii), we get

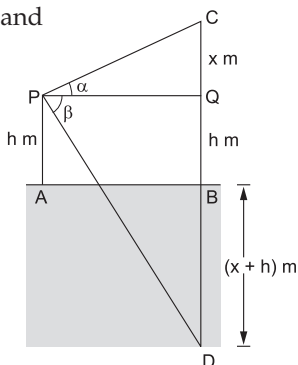
$$x \cot \alpha = (x + 2h) \cot \beta$$

$$\Rightarrow x(\cot \alpha - \cot \beta) = 2h \cot \beta \Rightarrow x \left(\frac{1}{\tan \alpha} - \frac{1}{\tan \beta} \right) = \frac{2h}{\tan \beta}$$

$$\Rightarrow x \left(\frac{\tan \beta - \tan \alpha}{\tan \alpha + \tan \beta} \right) = \frac{2h}{\tan \beta} \Rightarrow x = \frac{2h \tan \alpha}{(\tan \beta - \tan \alpha)}.$$

$\therefore$ height of the cloud from the surface of the lake

$$\begin{aligned} &= (x + h) = \left\{ \frac{2h \tan \alpha}{(\tan \beta - \tan \alpha)} + h \right\} \text{ m} \\ &= \frac{h(\tan \alpha + \tan \beta)}{(\tan \beta - \tan \alpha)} \text{ metres.} \end{aligned}$$



EXAMPLE 19 The angle of elevation of a cloud from a point 60 m above the surface of the water of a lake is 30° and the angle of depression of its shadow

in water of lake is 60° . Find the height of the cloud from the surface of water. [CBSE 2010, '15, '17]

SOLUTION

Let AB be the surface of the lake and let P be a point vertically above A such that $AP = 60$ m.

Let C be the position of the cloud and let D be its reflection in the lake.

Draw $PQ \perp CD$. Then,

$$\angle QPC = 30^\circ, \angle QPD = 60^\circ,$$

$$BQ = AP = 60 \text{ m.}$$

Let $CQ = x$ metres. Then,

$$BD = BC = (x + 60) \text{ m.}$$

From right $\triangle PQC$, we have

$$\frac{PQ}{CQ} = \cot 30^\circ = \sqrt{3}$$

$$\Rightarrow \frac{PQ}{x \text{ m}} = \sqrt{3} \Rightarrow PQ = x\sqrt{3} \text{ m.} \quad \dots (i)$$

From right $\triangle PQD$, we have

$$\frac{PQ}{QD} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{PQ}{(x + 60 + 60) \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow PQ = \frac{(x + 120)}{\sqrt{3}} \text{ m.} \quad \dots (ii)$$

Equating the values of PQ from (i) and (ii), we get

$$x\sqrt{3} = \frac{(x + 120)}{\sqrt{3}}$$

$$\Rightarrow 3x = x + 120 \Rightarrow 2x = 120 \Rightarrow x = 60.$$

$\therefore$ height of the cloud from the surface of the lake

$$= BC = (60 + x) \text{ m} = (60 + 60) \text{ m} = 120 \text{ m.}$$

Hence, the height of the cloud from the surface of the lake is 120 metres.

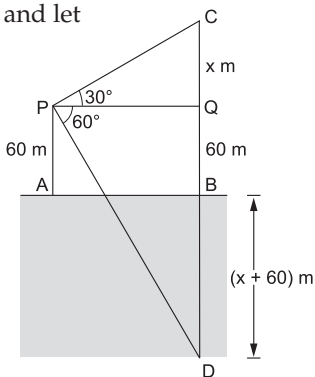
EXAMPLE 20

A round balloon of radius r subtends an angle α at the eye of the observer while the angle of elevation of its centre is β . Prove that the height of the centre of the balloon is

$$\left(r \sin \beta \operatorname{cosec} \frac{\alpha}{2} \right). \quad [\text{HOTS}]$$

SOLUTION

Let us represent the balloon by a circle with centre C and radius r . Let OX be the horizontal ground and let O be the



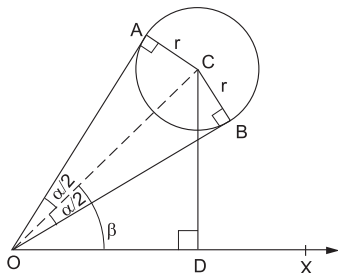
point of observation. From O , draw tangents OA and OB to the circle. Join CA , CB and CO . Draw $CD \perp OX$.

$\therefore \angle AOB = \alpha, \angle DOC = \beta$ and

$$\angle AOC = \angle BOC = \frac{\alpha}{2}.$$

From right $\triangle OAC$, we have

$$\begin{aligned} \frac{OC}{AC} &= \operatorname{cosec} \frac{\alpha}{2} \\ \Rightarrow \frac{OC}{r} &= \operatorname{cosec} \frac{\alpha}{2} \\ \Rightarrow OC &= r \operatorname{cosec} \frac{\alpha}{2}. \end{aligned}$$



... (i)

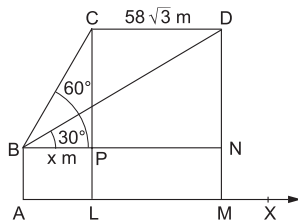
From right $\triangle ODC$, we have

$$\begin{aligned} \frac{CD}{OC} &= \sin \beta \Rightarrow CD = (OC) \times \sin \beta \\ \Rightarrow CD &= r \sin \beta \operatorname{cosec} \frac{\alpha}{2} \quad [\text{using (i)}]. \end{aligned}$$

Hence, the height of the centre of the balloon from the ground is $r \sin \beta \operatorname{cosec} \frac{\alpha}{2}$.

EXAMPLE 21 A boy whose eye level is 1.3 m from the ground, spots a balloon moving with the wind in a horizontal line at some height from the ground. The angle of elevation of the balloon from the eyes of the boy at an instant is 60° . After 2 seconds, the angle of elevation reduces to 30° . If the speed of the wind is $29\sqrt{3}$ m/s then find the height of the balloon from the ground. [HOTS] [CBSE 2009C]

SOLUTION Let AB be the position of the boy and AX be the horizontal ground. Let C and D be the two positions of the balloon. Draw $CL \perp AX$, $DM \perp AX$ and $BN \perp DM$, intersecting CL at P .



Then, $\angle CBN = 60^\circ$ and $\angle DBN = 30^\circ$.

Distance covered by the balloon in 2 seconds

$$= (29\sqrt{3} \times 2) \text{ m} = 58\sqrt{3} \text{ m}.$$

$\therefore CD = 58\sqrt{3} \text{ m}.$

Let $BP = x$ metres. Then,

$$BN = (BP + PN) = (BP + CD) = (x + 58\sqrt{3}) \text{ m and } DN = CP.$$

From right $\triangle BND$, we have

$$\frac{DN}{BN} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{DN}{(x + 58\sqrt{3}) \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow DN = \frac{(x + 58\sqrt{3})}{\sqrt{3}} \text{ m.} \quad \dots (i)$$

From right $\triangle BPC$, we have

$$\frac{CP}{BP} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow \frac{CP}{x \text{ m}} = \sqrt{3} \Rightarrow CP = (x\sqrt{3}) \text{ m.} \quad \dots (ii)$$

$$\text{Now, } DN = CP \Rightarrow \frac{(x + 58\sqrt{3})}{\sqrt{3}} = (x\sqrt{3})$$

$$\Rightarrow (3x - x) = 58\sqrt{3} \Rightarrow 2x = 58\sqrt{3} \Rightarrow x = 29\sqrt{3}.$$

From (ii), we get

$$CP = (29\sqrt{3} \times \sqrt{3}) \text{ m} = (29 \times 3) \text{ m} = 87 \text{ m.}$$

$\therefore$ height of the balloon from the ground

$$= CL = CP + PL = CP + AB = 87 \text{ m} + 1.3 \text{ m} = 88.3 \text{ m.}$$

Hence, the height of the balloon from the ground is 88.3 metres.

EXAMPLE 22 *An aeroplane when flying at a height of 3000 metres from the ground passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Find the vertical distance between the aeroplanes at that instant. [Take $\sqrt{3} = 1.73$.] [CBSE 2008]*

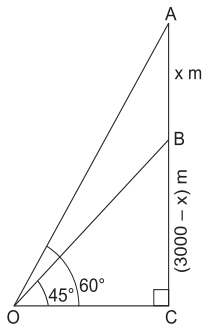
SOLUTION

Let O be the point of observation. Let A and B be the positions of the two planes at the given instant when A is vertically above B .

Let AB when produced meet the ground at C .

Then, $\angle COA = 60^\circ$, $\angle COB = 45^\circ$, $\angle OCB = \angle OCA = 90^\circ$ and $AC = 3000 \text{ m}$.

Let $AB = x$ metres. Then, $BC = (3000 - x) \text{ m}$.



From right $\triangle OCB$, we have

$$\frac{OC}{BC} = \cot 45^\circ = 1 \Rightarrow \frac{OC}{(3000 - x) \text{ m}} = 1$$

$$\Rightarrow OC = (3000 - x) \text{ m.} \quad \dots (i)$$

From right $\triangle OCA$, we have

$$\frac{OC}{AC} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{OC}{3000 \text{ m}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow OC = \left(\frac{3000}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) \text{ m} = 1000\sqrt{3} \text{ m.} \quad \dots (ii)$$

From (i) and (ii), we get

$$(3000 - x) = 1000\sqrt{3}$$

$$\begin{aligned} \Rightarrow x &= (3000 - 1000\sqrt{3}) = (3000 - 1000 \times 1.73) \\ &= (3000 - 1730) = 1270. \end{aligned}$$

Hence, the required distance between the two aeroplanes is 1270 metres.

EXAMPLE 23 A man standing on the deck of a ship, which is 10 m above the water level, observes the angle of elevation of the top of a hill as 60° , and the angle of depression of the base of the hill as 30° . Find the distance of the hill from the ship and the height of the hill. [CBSE 2006, '10]

SOLUTION

Let AB be the deck and CD be the hill.

Let the man be at B .

Then, $AB = 10 \text{ m}$.

Let $BE \perp CD$ and $AC \perp CD$.

Then, $\angle EBD = 60^\circ$ and $\angle EBC = 30^\circ$.

$\therefore \angle ACB = \angle EBC = 30^\circ$.

Let $CD = h$ metres.

Then, $CE = AB = 10 \text{ m}$ and

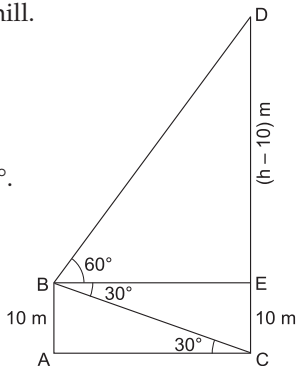
$ED = (h - 10) \text{ m}$.

From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{AC}{10 \text{ m}} = \sqrt{3}$$

$$\Rightarrow AC = 10\sqrt{3} \text{ m.} \quad \dots (i)$$

$$\therefore BE = AC = 10\sqrt{3} \text{ m.}$$



From right $\triangle BED$, we have

$$\frac{DE}{BE} = \tan 60^\circ = \sqrt{3} \Rightarrow \frac{h-10}{10\sqrt{3}} = \sqrt{3} \quad [\text{using (i)}]$$

$$\Rightarrow h - 10 = 30 \Rightarrow h = 40.$$

Hence, the distance of the ship from the hill is $10\sqrt{3}$ metres and the height of the hill is 40 metres.

EXAMPLE 24 From a window (h metres high above the ground) of a house in a street, the angles of elevation and depression of the top and the foot of another house on the opposite side of the street are θ and ϕ respectively. Show that the height of the opposite house is $h(1 + \tan \theta \cot \phi)$ metres. [CBSE 2006]

SOLUTION Let AB be the house with window at B and let CD be the another house. Then, $AB = h$ metres.

Draw $BE \parallel AC$, meeting CD at E . Then,

$$\angle EBD = \theta \text{ and } \angle ACB = \angle EBC = \phi.$$

Let $CD = H$ metres. Then,

$$CE = AB = h \text{ metres and}$$

$$ED = (H - h) \text{ m.}$$

From right $\triangle ACB$, we have

$$\frac{AC}{AB} = \cot \phi \Rightarrow \frac{AC}{h} = \cot \phi \Rightarrow AC = h \cot \phi \text{ metres.}$$

From right $\triangle BED$, we have

$$\frac{DE}{BE} = \tan \theta \Rightarrow \frac{(H - h)}{h \cot \phi} = \tan \theta \quad [\because BE = AC = h \cot \phi \text{ m}]$$

$$\Rightarrow (H - h) = h \tan \theta \cot \phi$$

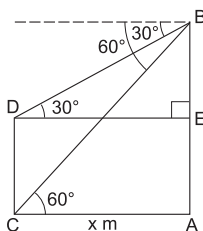
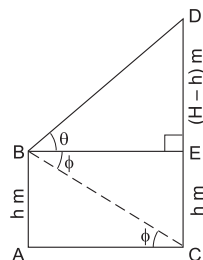
$$\Rightarrow H = h(1 + \tan \theta \cot \phi).$$

Hence, the height of the opposite house is $h(1 + \tan \theta \cot \phi)$ metres.

EXAMPLE 25 From the top of a building 60 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 60° . Find the height of the tower. [CBSE 2005]

SOLUTION Let AB be the building and CD be the tower such that $\angle BDE = 30^\circ$, $\angle BCA = 60^\circ$ and $AB = 60$ m.

Let $CA = DE = x$ metres.



From right $\triangle CAB$, we have

$$\begin{aligned}\frac{CA}{AB} &= \cot 60^\circ = \frac{1}{\sqrt{3}} \\ \Rightarrow \frac{x}{60} &= \frac{1}{\sqrt{3}} \Rightarrow x = 60 \times \frac{1}{\sqrt{3}} \\ \Rightarrow x &= 60 \times \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3} \\ \Rightarrow CA &= DE = 20\sqrt{3} \text{ m.} \quad \dots (i)\end{aligned}$$

From right $\triangle BED$, we have

$$\begin{aligned}\frac{BE}{DE} &= \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{BE}{20\sqrt{3} \text{ m}} = \frac{1}{\sqrt{3}} \quad [\text{using (i)}] \\ \Rightarrow BE &= 20\sqrt{3} \times \frac{1}{\sqrt{3}} \text{ m} = 20 \text{ m.}\end{aligned}$$

$$\therefore CD = AE = AB - BE = 60 \text{ m} - 20 \text{ m} = 40 \text{ m.}$$

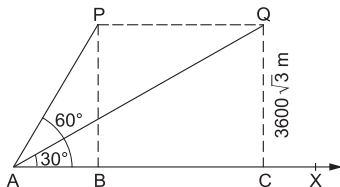
Hence, the height of the tower is 40 m.

EXAMPLE 26

The angle of elevation of a jet plane from a point A on the ground is 60° . After a flight of 30 seconds, the angle of elevation changes to 30° . If the jet plane is flying at a constant height of $3600\sqrt{3}$ metres, find the speed of the jet plane. [CBSE 2008]

SOLUTION

Let A be the point of observation and let AX be a horizontal line through A and $QC \perp AX$. Let P and Q be the two positions of the plane. Let $PB \perp AX$.



Then, $PB = QC = 3600\sqrt{3} \text{ m}$, $\angle BAP = 60^\circ$ and $\angle BAQ = 30^\circ$.

From right $\triangle ABP$, we have

$$\begin{aligned}\frac{AB}{BP} &= \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{AB}{3600\sqrt{3} \text{ m}} = \frac{1}{\sqrt{3}} \\ \Rightarrow AB &= 3600\sqrt{3} \times \frac{1}{\sqrt{3}} \text{ m} = 3600 \text{ m.} \quad \dots (i)\end{aligned}$$

Let $BC = PQ = x$ metres.

Then, $AC = AB + BC = (x + 3600) \text{ m}$ [using (i)].

From right $\triangle ACQ$, we have

$$\frac{AC}{CQ} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{x + 3600}{3600\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow x + 3600 = (3600 \times 3) = 10800$$

$$\Rightarrow x = 10800 - 3600 = 7200.$$

Thus, $PQ = 7200$ metres.

Now, 7200 m is covered in 30 seconds.

$$\begin{aligned} \therefore \text{ speed of the jet plane} &= \left(\frac{7200}{30} \times \frac{60 \times 60}{1000} \right) \text{ km/hr} \\ &= 864 \text{ km/hr.} \end{aligned}$$

Hence, the speed of the jet plane is 864 km/hr.

EXAMPLE 27

The angle of elevation of a jet fighter from a point A on the ground is 60° . After a flight of 15 seconds, the angle of elevation changes to 30° . If the jet is flying at a speed of 720 km/hour, find the constant height at which the jet is flying. [Use $\sqrt{3} = 1.732$.] [CBSE 2008]

SOLUTION

Let O be the point of observation on the ground OX.

Let A and B be the two positions of the jet.

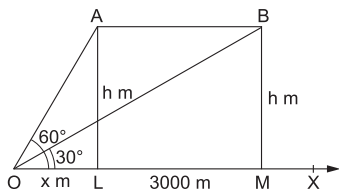
Then, $\angle XO A = 60^\circ$ and $\angle XO B = 30^\circ$.

Draw $AL \perp OX$ and $BM \perp OX$.

Let $AL = BM = h$ metres.

Speed of the jet

$$\begin{aligned} &= \left(720 \times \frac{5}{18} \right) \text{ m/s} \\ &= 200 \text{ m/s.} \end{aligned}$$



Time taken to cover the distance $AB = 15$ s.

Distance covered = speed $\times$ time

$$= 200 \text{ m/s} \times 15 \text{ s}$$

$$= 200 \times 15 \text{ m} = 3000 \text{ m.}$$

$$\therefore LM = AB = 3000 \text{ m.}$$

Let $OL = x$ metres.

From right $\triangle OLA$, we have

$$\frac{OL}{AL} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{x}{h} = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots (i)$$

From right $\triangle OMB$, we have

$$\begin{aligned}\frac{OM}{BM} &= \cot 30^\circ = \sqrt{3} \\ \Rightarrow \frac{(x + 3000)}{h} &= \sqrt{3} \Rightarrow x = (h\sqrt{3} - 3000). \quad \dots (ii)\end{aligned}$$

Equating the values of x from (i) and (ii), we get

$$\begin{aligned}\frac{h}{\sqrt{3}} &= h\sqrt{3} - 3000 \Rightarrow h = 3h - 3000\sqrt{3} \\ \Rightarrow 2h &= 3000\sqrt{3} \Rightarrow h = 1500\sqrt{3} = 1500 \times 1.732 = 2598.\end{aligned}$$

Hence, the required height is 2598 m.

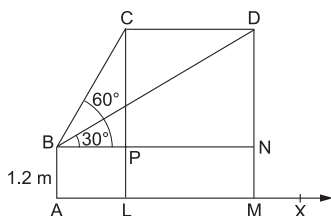
EXAMPLE 28

A 1.2-m-tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2 metres from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60° . After some time, the angle of elevation reduces to 30° . Find the distance travelled by the balloon during the interval.

SOLUTION

Let AB be the position of the girl and AX be the horizontal ground. Let C and D be the two positions of the balloon.

Draw $CL \perp AX$, $DM \perp AX$ and $BN \perp DM$, intersecting CL at P .



Then, $\angle CBP = 60^\circ$, $\angle DBN = 30^\circ$, $AB = PL = NM = 1.2$ m and $CL = DM = 88.2$ m.

$$\therefore CP = 88.2 \text{ m} - 1.2 \text{ m} = 87 \text{ m}.$$

From right $\triangle BPC$, we have

$$\begin{aligned}\frac{BP}{CP} &= \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{BP}{87 \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow BP = \frac{87 \text{ m}}{\sqrt{3}} \\ \Rightarrow BP &= \frac{87 \text{ m}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 29\sqrt{3} \text{ m}.\end{aligned}$$

From right $\triangle BND$, we have

$$\begin{aligned}\frac{BN}{DN} &= \cot 30^\circ = \sqrt{3} \Rightarrow \frac{BP + PN}{CP} = \sqrt{3} \\ \Rightarrow \frac{29\sqrt{3} \text{ m} + CD}{87 \text{ m}} &= \sqrt{3} \quad [\because PN = CD \text{ and } DN = CP]\end{aligned}$$

$$\Rightarrow 29\sqrt{3} \text{ m} + CD = 87\sqrt{3} \text{ m}$$

$$\Rightarrow CD = 87\sqrt{3} \text{ m} - 29\sqrt{3} \text{ m} = 58\sqrt{3} \text{ m}.$$

Hence, the required distance travelled by the balloon is $58\sqrt{3} \text{ m}$.

EXAMPLE 29 A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 9 m. Find the height of the tree.

SOLUTION

Let AB be the original height of the tree. Suppose it got bent at a point C and let the part CB take the position CD , meeting the ground at D . Then,

$$AD = 9 \text{ m}, \angle ADC = 30^\circ \text{ and } CD = CB.$$

Let $AC = x$ metres and $CD = CB = y$ metres.

From right $\triangle DAC$, we have

$$\frac{AC}{AD} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{x}{9} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = \frac{9}{\sqrt{3}} \Rightarrow x = \frac{9}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 3\sqrt{3}.$$

Also, from right $\triangle DAC$, we have

$$\frac{CD}{AD} = \sec 30^\circ = \frac{2}{\sqrt{3}} \Rightarrow \frac{y}{9} = \frac{2}{\sqrt{3}}$$

$$\Rightarrow y = \frac{18}{\sqrt{3}} \Rightarrow y = \frac{18}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 6\sqrt{3}.$$

$$\therefore AC = 3\sqrt{3} \text{ m and } CB = 6\sqrt{3} \text{ m}.$$

$$\text{Total height of the tree} = 3\sqrt{3} \text{ m} + 6\sqrt{3} \text{ m} = 9\sqrt{3} \text{ metres.}$$

EXAMPLE 30 A person standing on the bank of a river observes that the angle of elevation of the top of a tree standing on the opposite bank is 60° . When he moves 30 metres away from the bank, he finds the angle of elevation to be 30° . Find the height of the tree and the width of the river. [Take $\sqrt{3} = 1.732$.] [CBSE 2008C]

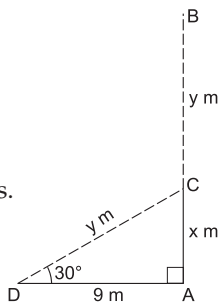
SOLUTION

Let AB be the tree and AC be the river.

Let C and D be the two positions of the person.

Then, $\angle ACB = 60^\circ$, $\angle ADB = 30^\circ$, $\angle DAB = 90^\circ$ and $CD = 30 \text{ m}$.

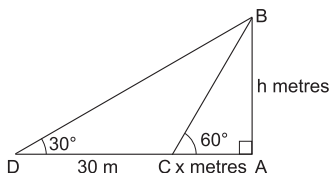
Let $AB = h$ metres and $AC = x$ metres.



From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{x}{h} = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots (i)$$



From right $\triangle DAB$, we have

$$\frac{AD}{AB} = \cot 30^\circ = \sqrt{3}$$

$$\Rightarrow \frac{x+30}{h} = \sqrt{3} \Rightarrow x = \sqrt{3}h - 30. \quad \dots (ii)$$

Equating the values of x from (i) and (ii), we get

$$\frac{h}{\sqrt{3}} = \sqrt{3}h - 30 \Rightarrow h = 3h - 30\sqrt{3}$$

$$\Rightarrow 2h = 30\sqrt{3} \Rightarrow h = 15\sqrt{3} = 15 \times 1.732 = 25.98.$$

Putting $h = 15\sqrt{3}$ in (i), we get $x = \frac{15\sqrt{3}}{\sqrt{3}} = 15.$

Hence, the height of the tree is 25.98 m and the width of the river is 15 metres.

EXAMPLE 31

The angles of elevation of the top of a tower from two points on the ground at distances a metres and b metres from the base of the tower and in the same straight line are complementary. Prove that the height of the tower is $\sqrt{ab}$ metres. [HOTS] [CBSE 2000C, '05C]

SOLUTION

Let AB be the tower and let C and D be the two positions of the observer. Then,

$$AC = a \text{ metres and } AD = b \text{ metres.}$$

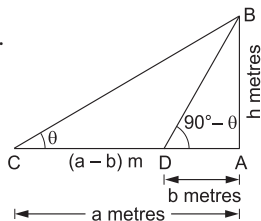
Let $\angle ACB = \theta$. Then, $\angle ADB = (90^\circ - \theta)$.

Let $AB = h$ metres.

From right $\triangle DAB$, we have

$$\frac{AB}{AD} = \tan (90^\circ - \theta) \Rightarrow \frac{h}{b} = \cot \theta$$

$$\Rightarrow h = b \tan \theta. \quad \dots (i)$$



From right $\triangle CAB$, we have

$$\frac{AB}{AC} = \tan \theta \Rightarrow \frac{h}{a} = \tan \theta \Rightarrow h = a \tan \theta. \quad \dots (ii)$$

From (i) and (ii), we get $h^2 = ab \Rightarrow h = \sqrt{ab}.$

Hence, the height of the tower is $\sqrt{ab}$ metres.

EXAMPLE 32 A boy standing on a horizontal plane finds a bird flying at a distance of 100 m from him at an elevation of 30° . A girl standing on the roof of a 20-m-high building, finds the angle of elevation of the same bird to be 45° . The boy and the girl are on the opposite sides of the bird. Find the distance of the bird from the girl. [Given $\sqrt{2} = 1.41$.] [HOTS]

[CBSE 2007]

SOLUTION Let O be the position of the bird, B be the position of the boy and FG be the building at which G is the position of the girl.

Let $OL \perp BF$ and $GM \perp OL$. Then,

$$BO = 100 \text{ m}, \angle OBL = 30^\circ,$$

$$FG = 20 \text{ m and } \angle OGM = 45^\circ.$$

From right $\triangle OLB$, we have

$$\frac{OL}{BO} = \sin 30^\circ \Rightarrow \frac{OL}{100 \text{ m}} = \frac{1}{2}$$

$$\Rightarrow OL = 100 \text{ m} \times \frac{1}{2} = 50 \text{ m}.$$

$$OM = OL - ML = OL - FG = 50 \text{ m} - 20 \text{ m} = 30 \text{ m}.$$

From right $\triangle OMG$, we have

$$\frac{OM}{OG} = \sin 45^\circ = \frac{1}{\sqrt{2}} \Rightarrow OG = \sqrt{2} \times OM = \sqrt{2} \times 30 \text{ m}$$

$$\Rightarrow OG = 30 \times 1.41 \text{ m} = 42.3 \text{ m}.$$

Distance of the bird from the girl = 42.3 m.

EXAMPLE 33 A 1.5-m-tall boy is standing at some distance from a 30-m-tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building. [HOTS]

SOLUTION Let AB be the building and let CD and EF be the two positions of the boy. Draw $DFG \parallel CEA$. Then,

$$CD = EF = 1.5 \text{ m}, \angle GDB = 30^\circ \text{ and } \angle GFB = 60^\circ,$$

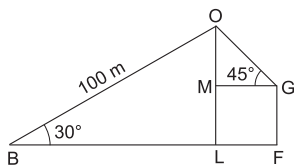
$$AB = 30 \text{ m}, GB = 30 \text{ m} - 1.5 \text{ m} = 28.5 \text{ m}.$$

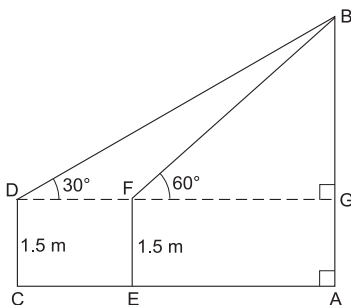
From right $\triangle DGB$, we have

$$\frac{DG}{GB} = \cot 30^\circ \Rightarrow \frac{DG}{28.5 \text{ m}} = \sqrt{3} \Rightarrow DG = \frac{57\sqrt{3}}{2} \text{ m}.$$

From right $\triangle FGB$, we have

$$\frac{FG}{GB} = \cot 60^\circ \Rightarrow \frac{FG}{28.5 \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow FG = \frac{57}{2\sqrt{3}} \text{ m}.$$





$$\begin{aligned}\therefore DF &= DG - FG = \left(\frac{57\sqrt{3}}{2} - \frac{57}{2\sqrt{3}} \right) \text{ m} = \left(\frac{171 - 57}{2\sqrt{3}} \right) \text{ m} \\ &= \frac{114}{2\sqrt{3}} \text{ m} = \frac{57}{\sqrt{3}} \text{ m} = \frac{57}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \text{ m} = 19\sqrt{3} \text{ m}.\end{aligned}$$

Hence, the required distance is $19\sqrt{3}$ m.

EXERCISE 14

1. A tower stands vertically on the ground. From a point on the ground which is 20 m away from the foot of the tower, the angle of elevation of its top is found to be 60° . Find the height of the tower. [Take $\sqrt{3} = 1.732$.]
2. A kite is flying at a height of 75 m from the level ground, attached to a string inclined at 60° to the horizontal. Find the length of the string, assuming that there is no slack in it. [Take $\sqrt{3} = 1.732$.]
3. An observer 1.5 m tall is 30 m away from a chimney. The angle of elevation of the top of the chimney from his eye is 60° . Find the height of the chimney. [CBSE 2013C]
4. The angles of elevation of the top of a tower from two points at distances of 5 metres and 20 metres from the base of the tower and in the same straight line with it, are complementary. Find the height of the tower. [CBSE 2014]
5. The angle of elevation of the top of a tower at a distance of 120 m from a point A on the ground is 45° . If the angle of elevation of the top of a flagstaff fixed at the top of the tower, at A is 60° , then find the height of the flagstaff. [Use $\sqrt{3} = 1.732$.] [CBSE 2014]
6. From a point on the ground 40 m away from the foot of a tower, the angle of elevation of the top of the tower is 30° . The angle of elevation of

the top of a water tank (on the top of the tower) is 45° . Find (i) the height of the tower, (ii) the depth of the tank.

7. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 m. At a point on the plane, the angle of elevation of the bottom of the flagstaff is 30° and that of the top of the flagstaff is 60° . Find the height of the tower. [Use $\sqrt{3} = 1.732$.] [CBSE 2011]
8. A statue 1.46 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point, the angle of elevation of the top of the pedestal is 45° . Find the height of the pedestal. [Use $\sqrt{3} = 1.73$.] [CBSE 2008]
9. The angle of elevation of the top of an unfinished tower at a distance of 75 m from its base is 30° . How much higher must the tower be raised so that the angle of elevation of its top at the same point may be 60° ? [Take $\sqrt{3} = 1.732$.]
10. On a horizontal plane there is a vertical tower with a flagpole on the top of the tower. At a point, 9 metres away from the foot of the tower, the angle of elevation of the top and bottom of the flagpole are 60° and 30° respectively. Find the height of the tower and the flagpole mounted on it. [Take $\sqrt{3} = 1.73$.] [CBSE 2005]
11. Two poles of equal heights are standing opposite to each other on either side of the road which is 80 m wide. From a point P between them on the road, the angle of elevation of the top of one pole is 60° and the angle of depression from the top of another pole at P is 30° . Find the height of each pole and distances of the point P from the poles. [CBSE 2015]
12. Two men are on opposite sides of a tower. They measure the angles of elevation of the top of the tower as 30° and 45° respectively. If the height of the tower is 50 metres, find the distance between the two men. [Take $\sqrt{3} = 1.732$.]
13. From the top of a tower 100 m high, a man observes two cars on the opposite sides of the tower and in same straight line with its base, with angles of depression 30° and 45° respectively. Find the distance between the cars. [Take $\sqrt{3} = 1.732$.] [CBSE 2011, '17]
14. A straight highway leads to the foot of a tower. A man standing on the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.

[HINT See Solved Example 14.]

15. A TV tower stands vertically on a bank of canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60° . From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30° . Find the height of the tower and the width of the canal.
16. The angle of elevation of the top of a building from the foot of a tower is 30° . The angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 60 m high, find the height of the building.
[CBSE 2013]
17. The horizontal distance between two towers is 60 metres. The angle of depression of the top of the first tower when seen from the top of the second tower is 30° . If the height of the second tower is 90 metres, find the height of the first tower. [Use $\sqrt{3} = 1.732$.]
18. The angle of elevation of the top of a chimney from the foot of a tower is 60° and the angle of depression of the foot of the chimney from the top of the tower is 30° . If the height of the tower is 40 metres, find the height of the chimney.
- According to pollution control norms, the minimum height of a smoke-emitting chimney should be 100 metres. State if the height of the above-mentioned chimney meets the pollution norms. What value is discussed in this question?
[CBSE 2014]
19. From the top of a 7-metre-high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower. [Use $\sqrt{3} = 1.732$.]
[CBSE 2017]
20. The angle of depression from the top of a tower of a point A on the ground is 30° . On moving a distance of 20 metres from the point A towards the foot of the tower to a point B , the angle of elevation of the top of the tower from the point B is 60° . Find the height of the tower and its distance from the point A .
[CBSE 2012]
21. The angle of elevation of the top of a vertical tower from a point on the ground is 60° . From another point 10 m vertically above the first, its angle of elevation is 30° . Find the height of the tower.
[CBSE 2011]
22. The angles of depression of the top and bottom of a tower as seen from the top of a $60\sqrt{3}$ -m-high cliff are 45° and 60° respectively. Find the height of the tower.
[CBSE 2012]
23. A man on the deck of a ship, 16 m above water level, observes that the angles of elevation and depression respectively of the top and bottom

- of a cliff are 60° and 30° . Calculate the distance of the cliff from the ship and height of the cliff. [Take $\sqrt{3} = 1.732$.] [CBSE 2007C]
24. The angle of elevation of the top Q of a vertical tower PQ from a point X on the ground is 60° . At a point Y , 40 m vertically above X , the angle of elevation is 45° . Find the height of tower PQ . [Take $\sqrt{3} = 1.73$.] [CBSE 2003C]
25. The angle of elevation of an aeroplane from a point on the ground is 45° . After flying for 15 seconds, the elevation changes to 30° . If the aeroplane is flying at a height of 2500 metres, find the speed of the aeroplane.
26. The angle of elevation of the top of a tower from a point on the same level as the foot of the tower is 30° . On advancing 150 m towards the foot of the tower, the angle of elevation becomes 60° . Show that the height of the tower is 129.9 metres. [Given $\sqrt{3} = 1.732$.] [CBSE 2006C]
27. As observed from the top of a lighthouse, 100 m above sea level, the angle of depression of a ship, sailing directly towards it, changes from 30° to 60° . Determine the distance travelled by the ship during the period of observation. [Use $\sqrt{3} = 1.732$.] [CBSE 2004, '08C]
28. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45° respectively. If the bridge is at a height of 2.5 m from the banks, find the width of the river. [Take $\sqrt{3} = 1.732$.]
29. The angles of elevation of the top of a tower from two points at distances of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Show that the height of the tower is 6 metres.
30. A ladder of length 6 metres makes an angle of 45° with the floor while leaning against one wall of a room. If the foot of the ladder is kept fixed on the floor and it is made to lean against the opposite wall of the room, it makes an angle of 60° with the floor. Find the distance between two walls of the room. [CBSE 2011]
31. From the top of a vertical tower, the angles of depression of two cars in the same straight line with the base of the tower, at an instant are found to be 45° and 60° . If the cars are 100 m apart and are on the same side of the tower, find the height of the tower. [CBSE 2011]
32. An electrician has to repair an electric fault on a pole of height 4 metres. He needs to reach a point 1 metre below the top of the pole to undertake the repair work. What should be the length of the ladder that he should use, which when inclined at an angle of 60° to the horizontal would enable him to reach the required position? [Use $\sqrt{3} = 1.73$.]

33. From the top of a building AB , 60 m high, the angles of depression of the top and bottom of a vertical lamp-post CD are observed to be 30° and 60° respectively. Find
- the horizontal distance between AB and CD ,
 - the height of the lamp-post,
 - the difference between the heights of the building and the lamp-post. [CBSE 2009]
34. A man observes a car from the top of a tower, which is moving towards the tower with a uniform speed. If the angle of depression of the car changes from 30° to 45° in 12 minutes, find the time taken by the car now to reach the tower. [CBSE 2017]
35. An aeroplane is flying at a height of 300 m above the ground. Flying at this height the angles of depression from the aeroplane of two points on both banks of a river in opposite directions are 45° and 60° respectively. Find the width of the river. [Use $\sqrt{3} = 1.732$] [CBSE 2017]
36. From a point on the ground the angles of elevation of the bottom and top of a communication tower fixed on the top of a 20-m-high building are 45° and 60° respectively. Find the height of the tower. [Take $\sqrt{3} = 1.732$.] [CBSE 2017]
37. From the top of a hill, the angles of depression of two consecutive kilometre stones due east are found to be 45° and 30° respectively. Find the height of the hill. [CBSE 2017]
38. If at some time of the day the ratio of the height of a vertically standing pole to the length of its shadow on the ground is $\sqrt{3} : 1$ then find the angle of elevation of the sun at that time. [CBSE 2017]

ANSWERS (EXERCISE 14)

- | | | | | |
|--|-------------------|---------------|--------------------------|------------|
| 1. 34.64 m | 2. 86.6 m | 3. 53.46 m | 4. 10 m | 5. 87.84 m |
| 6. (i) 23.1 m | (ii) 16.9 m | 7. 3 m | 8. 2 m | 9. 86.6 m |
| 10. 5.19 m, 10.38 m | | | | |
| 11. $20\sqrt{3}$ m; 20 m from left pole and 60 m from right pole | | | | |
| 12. 136.6 m | 13. 273.2 m | 14. 3 seconds | 15. $10\sqrt{3}$ m, 10 m | 16. 20 m |
| 17. 55.36 m | 18. 120 m | 19. 19.12 m | 20. 17.32 m, 30 m | 21. 15 m |
| 22. 43.92 m | 23. 27.71 m, 64 m | 24. 94.6 m | 25. 439.2 km/hr | |
| 27. 115.46 m | 28. 6.83 m | 30. 7.24 m | 31. 136.6 m | 32. 3.46 m |

33. (i) 34.64 m (ii) 40 m (iii) 20 m

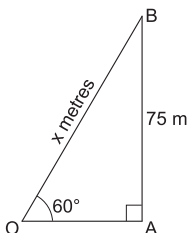
34. $6(\sqrt{3} + 1)$ minutes35. 473.2 m 36. 14.64 m 37. $\frac{\sqrt{3}+1}{2}$ km 38. 60° **HINTS TO SOME SELECTED QUESTIONS**

2. Let the length of the string be
- x
- metres.

$$\text{Then, } \frac{x}{75} = \operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}$$

$$\Rightarrow x = 75 \times \frac{2}{\sqrt{3}}$$

$$\therefore x = 75 \times \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ = 50\sqrt{3}.$$

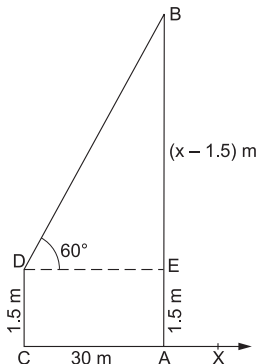


3. Let
- AB
- be the chimney and
- CD
- be the observer.

Draw $DE \perp AB$.Then, $AE = CD = 1.5$ m.Let $AB = x$ metres. Then, $BE = (x - 1.5)$ m. $\angle EDB = 60^\circ$.Also, $DE = CA = 30$ m.From right $\triangle BED$, we have

$$\frac{BE}{DE} = \tan 60^\circ$$

$$\Rightarrow \frac{x - 1.5}{30} = \sqrt{3}.$$



$$4. \frac{AB}{AC} = \tan \theta \Rightarrow \frac{h}{5} = \tan \theta$$

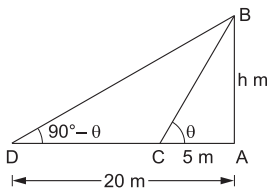
$$\Rightarrow h = 5 \tan \theta. \quad \dots (i)$$

$$\frac{AB}{AD} = \tan(90^\circ - \theta) = \cot \theta.$$

$$\Rightarrow \frac{h}{20} = \cot \theta \Rightarrow h = 20 \cot \theta \quad \dots (ii)$$

Multiplying (i) and (ii), we get

$$\therefore h^2 = 100 \tan \theta \times \cot \theta = 100 \Rightarrow h = 10.$$

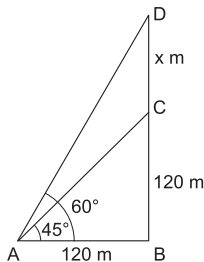


5. Let
- BC
- be the tower and
- CD
- be the flagstaff and let
- AB
- be the horizontal ground such that
- $AB = 120$
- m,
- $\angle BAC = 45^\circ$
- and
- $\angle BAD = 60^\circ$
- . Then,

$$\frac{BC}{AB} = \tan 45^\circ = 1 \Rightarrow \frac{BC}{120 \text{ m}} = 1 \Rightarrow BC = 120 \text{ m}.$$

Let $CD = x$ metres. Then,

$$\frac{BD}{AB} = \tan 60^\circ = \sqrt{3} \Rightarrow \frac{120 + x}{120} = \sqrt{3}. \text{ Find } x.$$



6. Let BC be the tower and CD be the water tank.

Let A be the point of observation. Then, $\angle BAC = 30^\circ$, $\angle BAD = 45^\circ$ and $AB = 40$ m.

From right $\triangle ABD$, we have

$$\frac{BD}{AB} = \tan 45^\circ = 1 \Rightarrow \frac{BD}{40 \text{ m}} = 1 \Rightarrow BD = 40 \text{ m}.$$

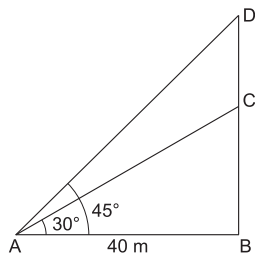
From right $\triangle ABC$, we have

$$\frac{BC}{AB} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{BC}{40 \text{ m}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow BC = \frac{40 \text{ m}}{\sqrt{3}} \Rightarrow BC = \frac{40 \text{ m}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{40\sqrt{3}}{3} \text{ m}.$$

$$(i) \text{ Height of the tower} = BC = \frac{40\sqrt{3}}{3} \text{ m} = 23.1 \text{ m}.$$

$$(ii) \text{ Depth of the tank} = CD = (BD - BC) = (40 - 23.1) \text{ m} = 16.9 \text{ m}.$$

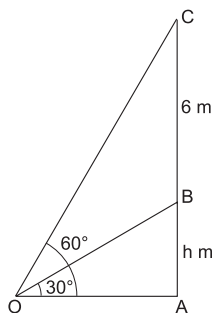


$$7. \frac{OA}{AB} = \cot 30^\circ \Rightarrow \frac{OA}{h \text{ m}} = \sqrt{3} \Rightarrow OA = h\sqrt{3} \text{ m}. \quad \dots (i)$$

$$\frac{OA}{AC} = \cot 60^\circ \Rightarrow \frac{OA}{(h+6) \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow OA = \frac{(h+6)}{\sqrt{3}} \text{ m}. \quad \dots (ii)$$

$$\therefore h\sqrt{3} = \frac{(h+6)}{\sqrt{3}} \Rightarrow 3h = h+6.$$

$$\Rightarrow 2h = 6 \Rightarrow h = 3.$$



9. Let AB be the unfinished tower and let AC be the complete tower. Let O be the point of observation.

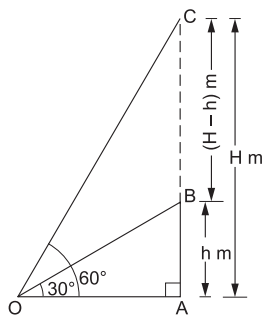
Then, $OA = 75$ m, $\angle AOB = 30^\circ$ and $\angle AOC = 60^\circ$.

Let $AB = h$ metres and $AC = H$ metres.

$$\frac{AB}{OA} = \tan 30^\circ \Rightarrow \frac{h}{75 \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow h = 25\sqrt{3} \text{ m}.$$

$$\text{And, } \frac{AC}{OA} = \tan 60^\circ \Rightarrow \frac{H}{75 \text{ m}} = \sqrt{3} \Rightarrow H = 75\sqrt{3} \text{ m}.$$

Hence, the required height is $(H - h) \text{ m} = (50\sqrt{3}) \text{ m}$.



11. From right $\triangle PAB$, we have

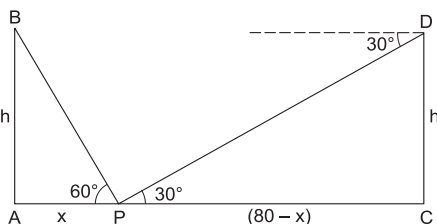
$$\frac{AB}{AP} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow h = x\sqrt{3}. \quad \dots (i)$$

From right $\triangle PCD$, we have

$$\frac{CD}{PC} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow h = \frac{(80-x)}{\sqrt{3}}. \quad \dots (ii)$$



$$\therefore x\sqrt{3} = \frac{(80-x)}{\sqrt{3}} \Rightarrow 3x + x = 80$$

$$\Rightarrow 4x = 80 \Rightarrow x = 20$$

$$\therefore \text{height} = 20\sqrt{3} \text{ m.}$$

P is 20 m from left pole and 60 m from right pole.

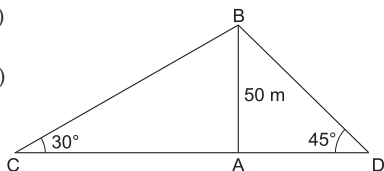
$$12. \frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow AC = 50\sqrt{3} \text{ m.} \quad \dots (i)$$

$$\frac{AD}{AB} = \cot 45^\circ = 1 \Rightarrow AD = 50 \text{ m.} \quad \dots (ii)$$

Required distance

$$= AC + AD = 50(\sqrt{3} + 1) \text{ m}$$

$$= (50 \times 2.732) \text{ m} = 136.6 \text{ m.}$$



13. From right $\triangle BAD$, we have

$$\frac{AD}{AB} = \cot 45^\circ = 1 \Rightarrow \frac{AD}{100 \text{ m}} \Rightarrow AD = 100 \text{ m.}$$

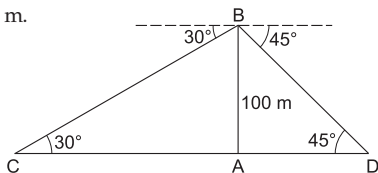
From right $\triangle BAC$, we have

$$\frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow \frac{AC}{100 \text{ m}} = \sqrt{3}$$

$$\Rightarrow AC = (100 \times 1.732) \text{ m} = 173.2 \text{ m.}$$

$\therefore$ distance between the cars

$$= (173.2 + 100) \text{ m} = 273.2 \text{ m.}$$



15. Let AB be the tower and AC be the canal.

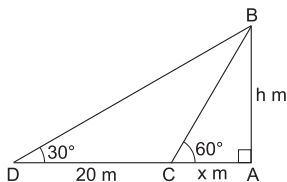
$$\frac{x}{h} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h}{\sqrt{3}}. \quad \dots (i)$$

$$\frac{(x+20)}{h} = \cot 30^\circ = \sqrt{3} \Rightarrow x = (h\sqrt{3} - 20). \quad \dots (ii)$$

From (i) and (ii), we get $\frac{h}{\sqrt{3}} = (h\sqrt{3} - 20)$.

$$\therefore h = (3h - 20\sqrt{3}) \Rightarrow 2h = 20\sqrt{3} \Rightarrow h = 10\sqrt{3}.$$

Putting $h = 10\sqrt{3}$ in (i), we get $x \text{ m} = 10 \text{ m}$ (width of the canal).



16. Let AB be the building and CD be the tower.

From right $\triangle ACD$, we have

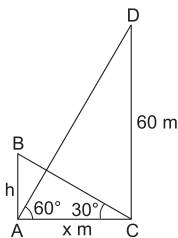
$$\frac{AC}{CD} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{x}{60} = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{60}{\sqrt{3}}$$

$$\Rightarrow x = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3}.$$

From right $\triangle BAC$, we have

$$\frac{AB}{AC} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{h}{x} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{x}{\sqrt{3}}$$

$$\Rightarrow h = \frac{20\sqrt{3}}{\sqrt{3}} = 20.$$



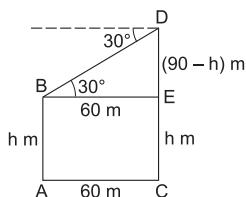
17. From right
- $\triangle BED$
- , we have

$$\frac{90 - h}{60} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow 90 - h = \frac{60}{\sqrt{3}}$$

$$\Rightarrow 90 - h = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3}$$

$$\Rightarrow h = 90 - 20\sqrt{3} = 90 - 20 \times 1.732$$

$$\Rightarrow h = 90 - 34.64 = 55.36.$$



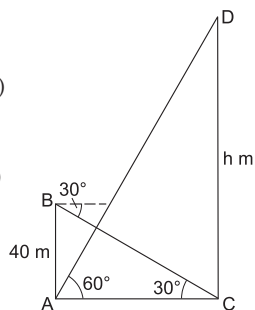
18. Let
- AB
- be the tower and
- CD
- be the chimney.

 From right $\triangle ACD$, we have

$$\frac{AC}{CD} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{AC}{h} = \frac{1}{\sqrt{3}} \Rightarrow AC = \frac{h}{\sqrt{3}} \quad \dots (i)$$

 From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 30^\circ = \sqrt{3} \Rightarrow AC = 40\sqrt{3}. \quad \dots (ii)$$

 From (i) and (ii), we get $\frac{h}{\sqrt{3}} = 40\sqrt{3} \Rightarrow h = 120$.


Clearly, the height of the given chimney meets the pollution norms.

We should comply with the prescribed rules and contribute to the cleanliness of the environment.

19. Let
- AB
- be the building and
- CD
- be the cable tower.

 Draw $BE \perp CD$. Let $CD = h$ metres.

 Then, $CE = AB = 7$ m and $DE = (h - 7)$ m.

 From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 45^\circ = 1 \Rightarrow \frac{AC}{7} = 1$$

$$\Rightarrow AC = 7.$$

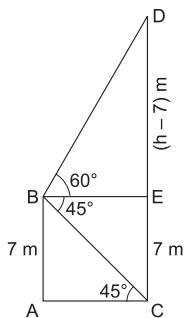
$$\therefore BE = AC = 7 \text{ m.}$$

 From right $\triangle BED$, we have

$$\frac{DE}{BE} = \tan 60^\circ = \sqrt{3} \Rightarrow \frac{h - 7}{7} = \sqrt{3}.$$

$$\therefore h = 7\sqrt{3} + 7 = 7(\sqrt{3} + 1) = 7(1.732 + 1)$$

$$\Rightarrow h = (7 \times 2.732) = 19.12.$$



20. From right
- $\triangle APQ$
- , we have

$$\frac{PQ}{PA} = \tan 30^\circ \Rightarrow \frac{h}{x + 20} = \frac{1}{\sqrt{3}}$$

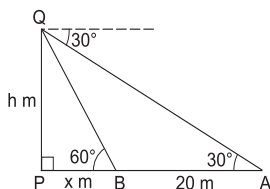
$$\Rightarrow h = \frac{x + 20}{\sqrt{3}}. \quad \dots (i)$$

 From right $\triangle BPQ$, we have

$$\frac{PQ}{PB} = \tan 60^\circ \Rightarrow \frac{h}{x} = \sqrt{3} \Rightarrow h = x\sqrt{3}. \quad \dots (ii)$$

From (i) and (ii), we get

$$\frac{(x + 20)}{\sqrt{3}} = x\sqrt{3} \Rightarrow x = 10.$$



$$\therefore h = 10\sqrt{3} = 10 \times 1.732 = 17.32.$$

$$\text{Distance } AP = (20 + x) \text{ m} = (20 + 10) \text{ m} = 30 \text{ m}.$$

21. Let the height of the tower be h metres.

From right $\triangle CAB$, we have

$$\begin{aligned}\frac{CA}{AB} &= \cot 60^\circ \Rightarrow \frac{CA}{h} = \frac{1}{\sqrt{3}} \\ \Rightarrow CA &= \frac{h}{\sqrt{3}}.\end{aligned}$$

From right $\triangle BED$, we have

$$\begin{aligned}\frac{DE}{BE} &= \cot 30^\circ \Rightarrow \frac{DE}{(h-10)} = \sqrt{3} \\ \Rightarrow DE &= \sqrt{3}(h-10).\end{aligned}$$

But, $CA = DE$.

$$\therefore \frac{h}{\sqrt{3}} = \sqrt{3}(h-10) \Rightarrow 3h - 30 = h$$

$$\Rightarrow 2h = 30 \Rightarrow h = 15.$$

22. Let AB be the cliff and CD be the tower.

Let $AE = CD = h$ metres. Then, $BE = (60\sqrt{3} - h)$ m.

From right $\triangle BED$, we have

$$\begin{aligned}\frac{DE}{BE} &= \cot 45^\circ \Rightarrow \frac{DE}{60\sqrt{3} - h} = 1 \\ \Rightarrow DE &= 60\sqrt{3} - h.\end{aligned}$$

From right $\triangle CAB$, we have

$$\begin{aligned}\frac{CA}{AB} &= \cot 60^\circ \Rightarrow \frac{CA}{60\sqrt{3}} = \frac{1}{\sqrt{3}} \\ \Rightarrow CA &= \frac{60\sqrt{3}}{\sqrt{3}} = 60.\end{aligned}$$

But, $CA = DE$.

$$\begin{aligned}\therefore 60\sqrt{3} - h &= 60 \Rightarrow h = 60(\sqrt{3} - 1) = 60 \times (1.732 - 1) \\ &= (60 \times 0.732) = 43.92.\end{aligned}$$

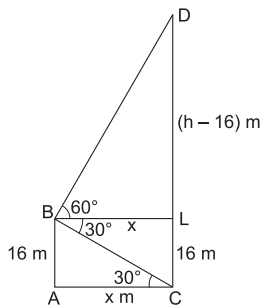
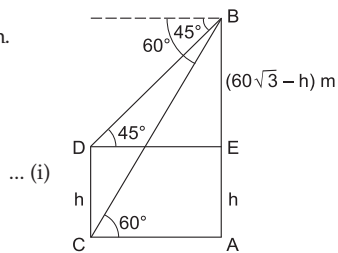
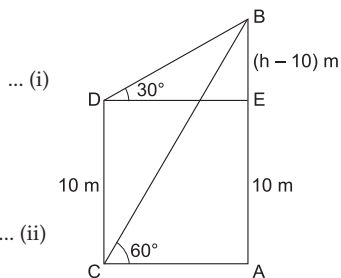
23. Let the height of the cliff be h metres and the distance of the cliff from the ship be x metres.

From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 30^\circ \Rightarrow \frac{x}{16} = \sqrt{3} \Rightarrow x = 16\sqrt{3}.$$

From right $\triangle BLD$, we have

$$\begin{aligned}\frac{DL}{BL} &= \tan 60^\circ = \sqrt{3} \Rightarrow \frac{h-16}{16\sqrt{3}} = \sqrt{3} \\ [\because BL &= x = 16\sqrt{3}] \\ \Rightarrow h - 16 &= 48 \Rightarrow h = 64.\end{aligned}$$



24. Let the height of tower PQ be h metres.

From right $\triangle XPQ$, we have

$$\frac{XP}{PQ} = \cot 60^\circ \Rightarrow \frac{XP}{h} = \frac{1}{\sqrt{3}} \Rightarrow XP = \frac{h}{\sqrt{3}} \quad \dots (i)$$

From right $\triangle YRQ$, we have

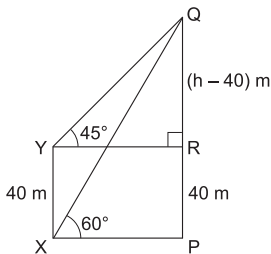
$$\frac{YR}{QR} = \cot 45^\circ \Rightarrow \frac{YR}{h-40} = 1 \Rightarrow YR = (h-40). \quad \dots (ii)$$

But, $YR = XP$.

$$\therefore h-40 = \frac{h}{\sqrt{3}} \Rightarrow h(\sqrt{3}-1) = 40\sqrt{3}$$

$$\Rightarrow h = \left\{ \frac{40\sqrt{3}}{(\sqrt{3}-1)} \times \frac{(\sqrt{3}+1)}{(\sqrt{3}+1)} \right\} = 20(3+\sqrt{3})$$

$$\Rightarrow h = 20(3+1.73) = (20 \times 4.73) = 94.6.$$



25. Let A and B be the two positions of the aeroplane and let O be the point of observation.

From right $\triangle OCA$, we have

$$\frac{OC}{AC} = \cot 45^\circ \Rightarrow \frac{OC}{2500} = 1 \Rightarrow OC = 2500 \text{ m.}$$

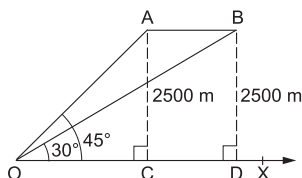
From right $\triangle ODB$, we have

$$\frac{OD}{BD} = \cot 30^\circ \Rightarrow \frac{OD}{2500} = \sqrt{3} \Rightarrow OD = 2500\sqrt{3} \text{ m.}$$

$$\therefore AB = CD = OD - OC = 2500(\sqrt{3} - 1) \text{ m} = (25000 \times 0.732) \text{ m} = 1830 \text{ m.}$$

Thus, the aeroplane covers 1830 m in 15 seconds.

$$\therefore \text{its speed} = \left(\frac{1830}{15} \times \frac{60 \times 60}{1000} \right) \text{ km/hr} = 439.2 \text{ km/hr.}$$



27. Let AB be the lighthouse and C and D be the two positions of the ship.

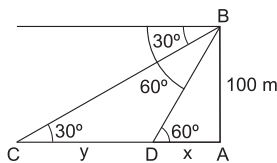
$$\text{Then, } \frac{x}{100 \text{ m}} = \cot 60^\circ \Rightarrow x = \frac{100}{\sqrt{3}} \text{ m.}$$

$$\frac{x+y}{100 \text{ m}} = \cot 30^\circ = \sqrt{3} \Rightarrow x+y = (100\sqrt{3}) \text{ m.}$$

$$\therefore y = x+y-x = 100\sqrt{3} \text{ m} - \frac{100}{\sqrt{3}} \text{ m} = \frac{200}{\sqrt{3}} \text{ m}$$

$$= \frac{200 \text{ m}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{200 \text{ m}}{3} \times \sqrt{3}$$

$$\Rightarrow y = \frac{200 \text{ m}}{3} \times 1.732 = \frac{346.4 \text{ m}}{3} = 115.46 \text{ m.}$$



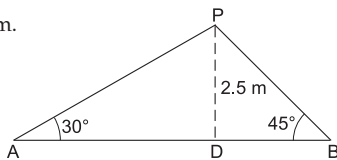
28. Let A and B be two points on the banks on opposite sides of the river.

Let P be a point on the bridge at a height of 2.5 m.

Then, $DP = 2.5$ m.

From right $\triangle PDB$, we have

$$\frac{DB}{PD} = \cot 45^\circ \Rightarrow \frac{DB}{2.5 \text{ m}} = 1 \Rightarrow DB = 2.50 \text{ m.}$$



From right $\triangle PDA$, we have

$$\frac{AD}{PD} = \cot 30^\circ \Rightarrow \frac{AD}{2.5 \text{ m}} = \sqrt{3} \Rightarrow AD = 2.5\sqrt{3} \text{ m.}$$

$$\therefore \text{Width of the river} = AB = AD + DB = 2.5\sqrt{3} \text{ m} + 2.5 \text{ m}$$

$$= 2.5(\sqrt{3} + 1) \text{ m} = \frac{5}{2} \times (1.732 + 1) \text{ m} = 6.83 \text{ m.}$$

30. Let AB and CD be the two opposite walls and let the foot of the ladder be fixed at the point O on the ground. Let OB and OD be the two positions of the ladder.

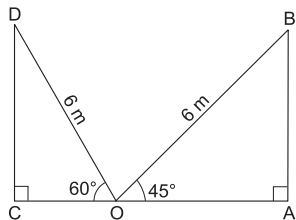
From right $\triangle OAB$, we have

$$\frac{OA}{OB} = \cos 45^\circ \Rightarrow \frac{OA}{6 \text{ m}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow OA = \frac{6 \text{ m}}{\sqrt{2}} \Rightarrow OA = \frac{6 \text{ m}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = 3\sqrt{2} \text{ m.}$$

From right $\triangle OCD$, we have

$$\frac{OC}{OD} = \cos 60^\circ \Rightarrow \frac{OC}{6 \text{ m}} = \frac{1}{2} \Rightarrow OC = 3 \text{ m.}$$



$$\text{Distance between two walls} = AC = OA + OC = 3(\sqrt{2} + 1) \text{ m} = 3(1.414 + 1) \text{ m}$$

$$= (3 \times 2.414) \text{ m} = 7.242 \text{ m} \approx 7.24 \text{ m.}$$

32. Let AB be the electric pole such that $AB = 4 \text{ m}$. Let C be a point 1 m below B .

Then, $AC = 4 \text{ m} - 1 \text{ m} = 3 \text{ m}$.

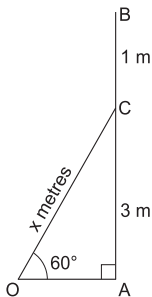
Let OC be the ladder. Then, $\angle AOC = 60^\circ$.

Let $OC = x$ metres.

From right $\triangle OAC$, we have

$$\begin{aligned} \frac{OC}{AC} &= \operatorname{cosec} 60^\circ \Rightarrow \frac{x}{3} = \frac{2}{\sqrt{3}} \Rightarrow x = \frac{6}{\sqrt{3}} \Rightarrow x = \frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 2\sqrt{3}. \\ &\Rightarrow x = (2 \times 1.73) = 3.46. \end{aligned}$$

Hence, the length of the ladder is 3.46 m.

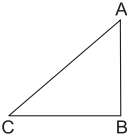


MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- If the height of a vertical pole is equal to the length of its shadow on the ground, the angle of elevation of the sun is [CBSE 2014]
 (a) 0° (b) 30° (c) 45° (d) 60°
- If the height of a vertical pole is $\sqrt{3}$ times the length of its shadow on the ground then the angle of elevation of the sun at that time is [CBSE 2012, '14]
 (a) 30° (b) 45° (c) 60° (d) 75°

3. If the length of the shadow of a tower is $\sqrt{3}$ times its height then the angle of elevation of the sun is [CBSE 2012]
(a) 45° (b) 30° (c) 60° (d) 90°
4. If a pole 12 m high casts a shadow $4\sqrt{3}$ m long on the ground then the sun's elevation is [CBSE 2013C]
(a) 60° (b) 45° (c) 30° (d) 90°
5. The shadow of a 5-m-long stick is 2 m long. At the same time, the length of the shadow of a 12.5-m-high tree is [CBSE 2011]
(a) 3 m (b) 3.5 m (c) 4.5 m (d) 5 m
6. A ladder makes an angle of 60° with the ground when placed against a wall. If the foot of the ladder is 2 m away from the wall, the length of the ladder is [CBSE 2014]
(a) $\frac{4}{\sqrt{3}}$ m (b) $4\sqrt{3}$ m (c) $2\sqrt{2}$ m (d) 4 m
7. A ladder 15 m long makes an angle of 60° with the wall. Find the height of the point, where the ladder touches the wall. [CBSE 2017]
(a) $15\sqrt{3}$ m (b) $\frac{15\sqrt{3}}{2}$ m (c) $\frac{15}{2}$ m (d) 15 m
8. From a point on the ground, 30 m away from the foot of a tower, the angle of elevation of the top of the tower is 30° . The height of the tower is [CBSE 2014]
(a) 30 m (b) $10\sqrt{3}$ m (c) 10 m (d) $30\sqrt{3}$ m
9. The angle of depression of a car parked on the road from the top of a 150-m-high tower is 30° . The distance of the car from the tower is [CBSE 2014]
(a) $50\sqrt{3}$ m (b) $150\sqrt{3}$ m (c) $150\sqrt{2}$ m (d) 75 m
10. A kite is flying at a height of 30 m from the ground. The length of string from the kite to the ground is 60 m. Assuming that there is no slack in the string, the angle of elevation of the kite at the ground is [CBSE 2012]
(a) 45° (b) 30° (c) 60° (d) 90°
11. From the top of a cliff 20 m high, the angle of elevation of the top of a tower is found to be equal to the angle of depression of the foot of the tower. The height of the tower is [CBSE 2013C]
(a) 20 m (b) 40 m (c) 60 m (d) 80 m
12. If a 1.5-m-tall girl stands at a distance of 3 m from a lamp-post and casts a shadow of length 4.5 m on the ground, then the height of the lamp-post is
(a) 1.5 m (b) 2 m (c) 2.5 m (d) 2.8 m

13. The length of the shadow of a tower standing on level ground is found to be $2x$ metres longer when the sun's elevation is 30° than when it was 45° . The height of the tower is
 (a) $(2\sqrt{3}x)$ m (b) $(3\sqrt{2}x)$ m (c) $(\sqrt{3} - 1)x$ m (d) $(\sqrt{3} + 1)x$ m
14. The lengths of a vertical rod and its shadow are in the ratio $1 : \sqrt{3}$. The angle of elevation of the sun is
 (a) 30° (b) 45° (c) 60° (d) 90°
15. A pole casts a shadow of length $2\sqrt{3}$ m on the ground when the sun's elevation is 60° . The height of the pole is [CBSE 2015]
 (a) $4\sqrt{3}$ m (b) 6 m (c) 12 m (d) 3 m
16. In the given figure, a tower AB is 20 m high and BC , its shadow on the ground is $20\sqrt{3}$ m long. The sun's altitude is [CBSE 2015]
 (a) 30° (b) 45°
 (c) 60° (d) none of these
- 
17. The tops of two towers of heights x and y , standing on a level ground subtend angles of 30° and 60° respectively at the centre of the line joining their feet. Then, $x : y$ is [CBSE 2015]
 (a) $1 : 2$ (b) $2 : 1$ (c) $1 : 3$ (d) $3 : 1$
18. The angle of elevation of the top of a tower from a point on the ground 30 m away from the foot of the tower is 30° . The height of the tower is
 (a) 30 m (b) $10\sqrt{3}$ m (c) 20 m (d) $10\sqrt{2}$ m
19. The string of a kite is 100 m long and it makes an angle of 60° with the horizontal. If there is no slack in the string, the height of the kite from the ground is
 (a) $50\sqrt{3}$ m (b) $100\sqrt{3}$ m (c) $50\sqrt{2}$ m (d) 100 m
20. If the angles of elevation of the top of a tower from two points at distances a and b from the base and in the same straight line with it are complementary then the height of the tower is
 (a) $\sqrt{\frac{a}{b}}$ (b) $\sqrt{ab}$ (c) $\sqrt{a+b}$ (d) $\sqrt{a-b}$
21. On the level ground, the angle of elevation of a tower is 30° . On moving 20 m nearer, the angle of elevation is 60° . The height of the tower is
 (a) 10 m (b) $10\sqrt{3}$ m (c) 15 m (d) 20 m
22. In a rectangle, the angle between a diagonal and a side is 30° and the length of this diagonal is 8 cm. The area of the rectangle is
 (a) 16 cm^2 (b) $\frac{16}{\sqrt{3}} \text{ cm}^2$ (c) $16\sqrt{3} \text{ cm}^2$ (d) $8\sqrt{3} \text{ cm}^2$

23. From the top of a hill, the angles of depression of two consecutive km stones due east are found to be 30° and 45° . The height of the hill is
- (a) $\frac{1}{2}(\sqrt{3} - 1)$ km (b) $\frac{1}{2}(\sqrt{3} + 1)$ km
- (c) $(\sqrt{3} - 1)$ km (d) $(\sqrt{3} + 1)$ km
24. If the elevation of the sun changes from 30° to 60° then the difference between the lengths of shadows of a pole 15 m high, is
- (a) 7.5 m (b) 15 m (c) $10\sqrt{3}$ m (d) $5\sqrt{3}$ m
25. An observer 1.5 m tall is 28.5 m away from a tower and the angle of elevation of the top of the tower from the eye of the observer is 45° . The height of the tower is
- (a) 27 m (b) 30 m (c) 28.5 m (d) none of these

ANSWERS (MCQ)

1. (c) 2. (c) 3. (b) 4. (a) 5. (d) 6. (d) 7. (c) 8. (b) 9. (b) 10. (b)
11. (a) 12. (c) 13. (d) 14. (a) 15. (b) 16. (a) 17. (c) 18. (b) 19. (a) 20. (b)
21. (b) 22. (c) 23. (b) 24. (c) 25. (b)

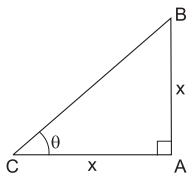
HINTS TO SOME SELECTED QUESTIONS

1. Let AB be the pole and AC be its shadow such that $AB = AC$.

Let $\angle ACB = \theta$. Then,

$$\tan \theta = \frac{AB}{AC} = 1 \Rightarrow \tan \theta = \tan 45^\circ$$

$$\Rightarrow \theta = 45^\circ.$$

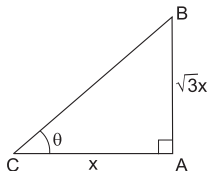


2. Let AB be the pole and AC be its shadow.

Let $AC = x$ m. Then, $AB = \sqrt{3}x$ m.

Let $\angle ACB = \theta$. Then,

$$\tan \theta = \frac{AB}{AC} = \frac{\sqrt{3}x}{x} = \sqrt{3} \Rightarrow \theta = 60^\circ.$$



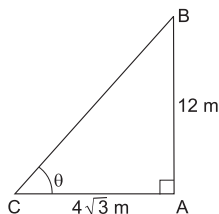
4. Let AB be the pole and AC be its shadow.

$AB = 12$ m and $AC = 4\sqrt{3}$ m.

Let $\angle ACB = \theta$. Then, $\tan \theta = \frac{AB}{AC} = \frac{12}{4\sqrt{3}}$

$$\Rightarrow \tan \theta = \frac{12}{4\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ$$

$$\Rightarrow \theta = 60^\circ.$$



5. Ratio of lengths of objects = ratio of lengths of their shadows.

Let the length of shadow of the tree be x m. Then,

$$\frac{5}{12.5} = \frac{2}{x} \Rightarrow 5x = 2 \times 12.5 = 25 \Rightarrow x = 5$$

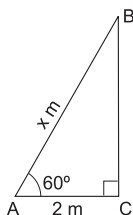
6. Let AB be the ladder and BC be the wall.

Then, $\angle CAB = 60^\circ$ and $CA = 2$ m.

Let $AB = x$ metres. Then,

$$\frac{AC}{AB} = \cos 60^\circ = \frac{1}{2} \Rightarrow \frac{2}{x} = \frac{1}{2}$$

$$\Rightarrow x = 4.$$



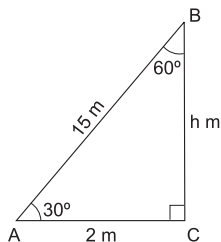
7. Let AB be the ladder and BC be the wall.

Then, $\angle ABC = 60^\circ \Rightarrow \angle CAB = (90^\circ - 60^\circ) = 30^\circ$.

Let $BC = h$ m. Then,

$$\frac{BC}{AB} = \sin 30^\circ \Rightarrow \frac{h}{15} = \frac{1}{2}$$

$$\Rightarrow h = \frac{15}{2}.$$

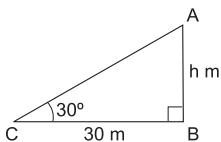


8. Let AB be the tower and $BC = 30$ m be the ground such that $\angle BCA = 30^\circ$.

Let $AB = h$ metres. Then,

$$\frac{AB}{BC} = \tan 30^\circ \Rightarrow \frac{h}{30} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{30}{\sqrt{3}}$$

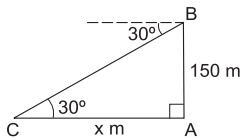
$$\Rightarrow h = \left(\frac{30}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) = 10\sqrt{3}.$$



9. Let AB be the tower and C be the position of the car on the ground such that $\angle ACB = 30^\circ$.

Let $AC = x$ metres. Then, $\frac{AC}{AB} = \cot 30^\circ \Rightarrow \frac{x}{150} = \sqrt{3}$

$$\Rightarrow x = 150\sqrt{3}.$$



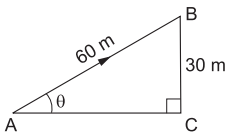
10. Let AB be the string and B be the kite.

Let AC be the horizontal and let $BC \perp AC$.

Let $\angle CAB = \theta$.

$BC = 30$ m and $AB = 60$ m. Then,

$$\frac{BC}{AB} = \sin \theta \Rightarrow \sin \theta = \frac{30}{60} = \frac{1}{2} \Rightarrow \sin \theta = \sin 30^\circ \Rightarrow \theta = 30^\circ.$$



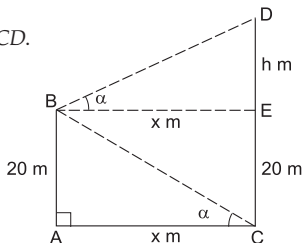
11. Let AB be the cliff and CD be the tower. Draw $BE \perp CD$.

Let $\angle ACB = \angle EBD = \alpha$ and let $DE = h$ metres.

Also, $AB = 20$ m. Let $AC = BE = x$ m. Then

$$\frac{x}{h} = \cot \alpha \text{ and } \frac{x}{20} = \cot \alpha$$

$$\text{Thus, } \frac{x}{h} = \frac{x}{20} \Rightarrow h = 20 \text{ m.}$$

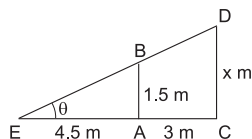


12. Let AB be the position of the girl and let CD be the lamp-post. Let AE be the shadow of AB . Then, $AB = 1.5$ m, $AC = 3$ m and $AE = 4.5$ m.

Let $CD = x$ metres. Now, $\triangle AEB$ and $\triangle CED$ are similar.

$$\therefore \frac{CD}{EC} = \frac{AB}{AE} \Rightarrow \frac{x}{7.5} = \frac{1.5}{4.5} = \frac{1}{3} \Rightarrow x = \left(7.5 \times \frac{1}{3}\right) = 2.5.$$

Hence, the height of the lamp-post is 2.5 m.



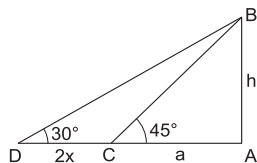
13. Let h m be the height of the tower and let a m and $(a + 2x)$ m be the lengths of the shadows of the tower when sun's elevation is 45° and 30° respectively. Then,

$$\frac{h}{a} = \tan 45^\circ = 1 \Rightarrow a = h.$$

$$\frac{h}{a + 2x} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{h}{h + 2x} = \frac{1}{\sqrt{3}}.$$

$$\therefore \sqrt{3}h = h + 2x \Rightarrow 2x = (\sqrt{3} - 1)h \Rightarrow h = \frac{2x}{(\sqrt{3} - 1)}$$

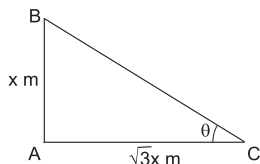
$$\therefore h = \left\{ \frac{2x}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)} \right\} = x(\sqrt{3} + 1).$$



14. Let AB be the rod of length x metres and let AC be its shadow of length $(\sqrt{3}x)$ m. Let $\angle ACB = \theta$.

$$\text{Then, } \frac{AB}{AC} = \tan \theta \Rightarrow \tan \theta = \frac{x}{\sqrt{3}x} = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

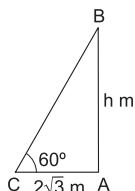
$$\Rightarrow \theta = 30^\circ.$$



15. Let the height of the pole be h metres.

$$\text{Then, } \frac{h}{2\sqrt{3}} = \tan 60^\circ = \sqrt{3}$$

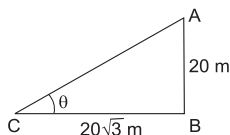
$$\Rightarrow h = (2\sqrt{3} \times \sqrt{3}) = 6.$$



16. Let $AB = 20$ m be the tower and $BC = 20\sqrt{3}$ m be the length of its shadow. Let $\angle ACB = \theta$.

$$\text{Then, } \tan \theta = \frac{AB}{BC} = \frac{20}{20\sqrt{3}} = \frac{1}{\sqrt{3}} = \tan 30^\circ.$$

$$\therefore \theta = 30^\circ.$$



17. Let AB and CD be the given pillars and O be the midpoint of AC .

Then, $AB = x$, $CD = y$, $\angle AOB = 30^\circ$ and $\angle COD = 60^\circ$.

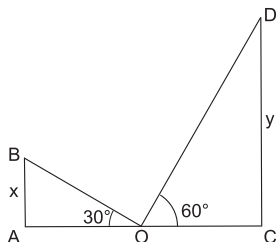
From right $\triangle OAB$, we have

$$\frac{OA}{AB} = \cot 30^\circ \Rightarrow \frac{OA}{x} = \sqrt{3}$$

$$\Rightarrow OA = x\sqrt{3}. \quad \dots (i)$$

From right $\triangle OCD$, we have

$$\frac{OC}{OD} = \cot 60^\circ \Rightarrow \frac{OC}{y} = \frac{1}{\sqrt{3}} \Rightarrow OC = \frac{y}{\sqrt{3}}. \quad \dots (ii)$$

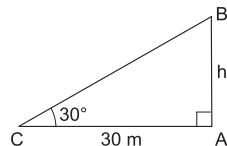


But, $OA = OC$.

$$\therefore x\sqrt{3} = \frac{y}{\sqrt{3}} \Rightarrow 3x = y \Rightarrow \frac{x}{y} = \frac{1}{3} \Rightarrow x : y = 1 : 3.$$

$$18. \frac{h}{30} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{30 \text{ m}}{\sqrt{3}}$$

$$\Rightarrow h = \frac{30 \text{ m}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 10\sqrt{3} \text{ m}.$$



$$20. \frac{h}{a} = \tan \theta \Rightarrow h = a \tan \theta. \quad \dots (i)$$

$$\frac{h}{b} = \tan(90^\circ - \theta) = \cot \theta$$

$$\Rightarrow h = b \cot \theta \quad \dots (ii)$$

From (i) and (ii), we get

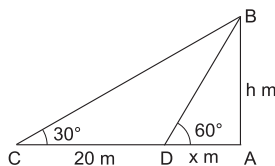
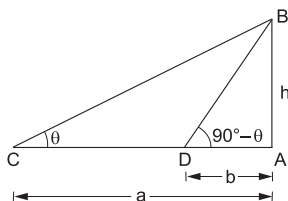
$$h^2 = ab \text{ and hence } h = \sqrt{ab}.$$

$$21. \frac{x}{h} = \cot 60^\circ = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots (i)$$

$$\frac{x + 20}{h} = \cot 30^\circ = \sqrt{3} \Rightarrow x = (h\sqrt{3} - 20) \quad \dots (ii)$$

$$\therefore \frac{h}{\sqrt{3}} = (h\sqrt{3} - 20) \Rightarrow h = 3h - 20\sqrt{3}$$

$$\Rightarrow 2h = 20\sqrt{3} \Rightarrow h = 10\sqrt{3}.$$



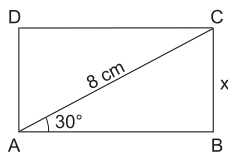
22. Let $ABCD$ be the rectangle.

$$\frac{BC}{AC} = \sin 30^\circ = \frac{1}{2} \Rightarrow \frac{x}{8 \text{ cm}} = \frac{1}{2} \Rightarrow x = 4 \text{ cm}.$$

$$\therefore AB^2 = (8)^2 \text{ cm}^2 - (4)^2 = (64 - 16) \text{ cm}^2 = 48 \text{ cm}^2$$

$$\Rightarrow AB = \sqrt{48 \text{ cm}^2} = 4\sqrt{3} \text{ cm}.$$

$$\therefore \text{ar}(ABCD) = (4\sqrt{3} \times 4) \text{ cm}^2 = 16\sqrt{3} \text{ cm}^2.$$



23. Let AB be the hill.

From right $\triangle BAD$, we have

$$\frac{AD}{AB} = \cot 45^\circ \Rightarrow \frac{x}{h} = 1 \Rightarrow x = h. \quad \dots (i)$$

From right $\triangle BAC$, we have

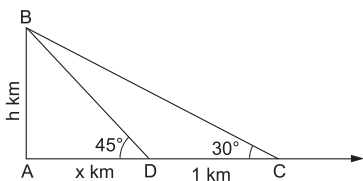
$$\frac{AC}{AD} = \cot 30^\circ \Rightarrow \frac{x + 1}{h} = \sqrt{3} \Rightarrow x = (h\sqrt{3} - 1) \quad \dots (ii)$$

From (i) and (ii), we have

$$h = (h\sqrt{3} - 1) \Rightarrow h(\sqrt{3} - 1) = 1$$

$$\Rightarrow h = \frac{1}{(\sqrt{3} - 1)}$$

$$\Rightarrow h = \left\{ \frac{1}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)} \right\} = \frac{1}{2}(\sqrt{3} + 1).$$



24. Let AB be the pole and AC and AD be its shadows when $\angle ACB = 30^\circ$ and $\angle ADB = 60^\circ$.

From right $\triangle CAB$, we have

$$\frac{AC}{AB} = \cot 30^\circ \Rightarrow \frac{AC}{15 \text{ m}} = \sqrt{3}$$

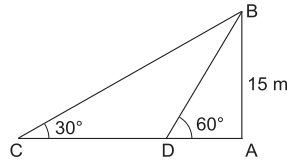
$$\Rightarrow AC = 15\sqrt{3} \text{ m.} \quad \dots (i)$$

From right $\triangle DAB$, we have

$$\frac{AD}{AB} = \cot 60^\circ \Rightarrow \frac{AD}{15 \text{ m}} = \frac{1}{\sqrt{3}} \Rightarrow AD = \frac{15 \text{ m}}{\sqrt{3}}$$

$$\Rightarrow AD = \frac{15 \text{ m}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 5\sqrt{3} \text{ m} \quad \dots (ii)$$

$$\text{Required difference} = (15\sqrt{3} - 5\sqrt{3}) \text{ m} = 10\sqrt{3} \text{ m.}$$



25. Let AB be the observer and $CD = h$ metres be the tower.

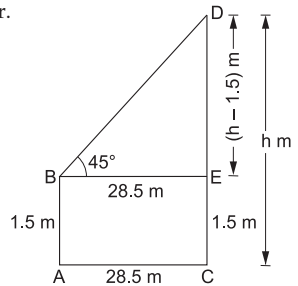
$$BE = AC = 28.5 \text{ m.}$$

From right $\triangle BED$, we have

$$\frac{DE}{BE} = \tan 45^\circ \Rightarrow \frac{DE}{28.5 \text{ m}} = 1$$

$$\Rightarrow DE = 28.5 \text{ m}$$

$$\therefore h - 1.5 = 28.5 \Rightarrow h = 30.$$



PERIMETER *The perimeter of a plane figure is the length of its boundary.*

In case of a triangle or a polygon, the perimeter is the sum of the lengths of its sides.

The unit of perimeter is same as the unit of length.

AREA *The area of a plane figure is the measure of the surface enclosed by its boundary.*

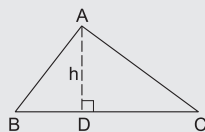
The area of a triangle or a polygon is the measure of the surface enclosed by its sides.

Area is measured in square units such as square centimetres and square metres, written as cm^2 and m^2 respectively.

PERIMETER AND AREA OF TRIANGLES [FORMULAE]

1. (i) Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{corresponding height}$.

$$\text{ar}(\triangle ABC) = \left(\frac{1}{2} \times BC \times AD \right) \text{ sq units.}$$



- (ii) Heron's formula:

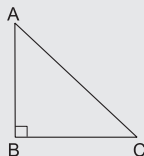
Let a, b, c be the sides of a $\triangle ABC$. Then,

$s = \frac{1}{2}(a + b + c)$ is called its semiperimeter.

$$\text{ar}(\triangle ABC) = \sqrt{s(s-a)(s-b)(s-c)}.$$

2. In a right $\triangle ABC$, let $\angle B = 90^\circ$. Then,

$$\text{ar}(\triangle ABC) = \left(\frac{1}{2} \times BC \times AB \right) \text{ sq units.}$$

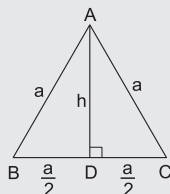


3. In an equilateral triangle of side a , we have

(i) Height $= \left(\frac{\sqrt{3}}{2} a \right)$ units.

(ii) Area $= \left(\frac{\sqrt{3}}{4} a^2 \right)$ sq units.

(iii) Perimeter $= 3a$ units.

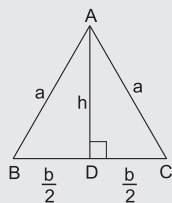


4. For an isosceles $\triangle ABC$ in which $AB = AC = a$ and $BC = b$, we have

(i) Height = $\frac{\sqrt{4a^2 - b^2}}{2}$ units.

(ii) Area = $\left(\frac{1}{4} b \sqrt{4a^2 - b^2}\right)$ sq units.

(iii) Perimeter = $(2a + b)$ units.



SOLVED EXAMPLES

EXAMPLE 1 Find the area of a triangle having base 25 cm and height 10.8 cm.

SOLUTION Area of the given triangle = $\left(\frac{1}{2} \times \text{base} \times \text{height}\right)$

$$= \left(\frac{1}{2} \times 25 \times 10.8\right) \text{ cm}^2 = 135 \text{ cm}^2.$$

EXAMPLE 2 The base of a triangular field is three times its altitude. If the cost of sowing the field at ₹ 960 per hectare is ₹ 12960, find its base and height.

SOLUTION Total cost of sowing the field = ₹ 12960.

Rate per hectare = ₹ 960.

$$\text{Area} = \frac{\text{total cost}}{\text{rate}} = \left(\frac{12960}{960}\right) \text{ hectares} = 13.5 \text{ hectares}$$

$$= (13.5 \times 10000) \text{ m}^2 = 135000 \text{ m}^2.$$

Let the altitude of the field be x metres.

Then, its base = $3x$ metres.

$$\text{Area} = \left(\frac{1}{2} \times \text{base} \times \text{height}\right) = \left(\frac{1}{2} \times 3x \times x\right) \text{ m}^2 = \left(\frac{3x^2}{2}\right) \text{ m}^2.$$

$$\therefore \frac{3x^2}{2} = 135000 \Rightarrow x^2 = \left(135000 \times \frac{2}{3}\right) = 90000$$

$$\Rightarrow x = \sqrt{90000} = 300.$$

$$\therefore \text{base} = (3 \times 300) \text{ m} = 900 \text{ m and altitude} = 300 \text{ m.}$$

EXAMPLE 3 The lengths of the sides of a triangle are in the ratio 3 : 4 : 5, and its perimeter is 144 cm. Find (i) the area of the triangle, and (ii) the height corresponding to the longest side.

SOLUTION On dividing 144 cm in the ratio 3 : 4 : 5, we get

$$a = \left(144 \times \frac{3}{12}\right) \text{ cm} = 36 \text{ cm}, b = \left(144 \times \frac{4}{12}\right) \text{ cm} = 48 \text{ cm}$$

$$\text{and } c = \left(144 \times \frac{5}{12}\right) \text{ cm} = 60 \text{ cm.}$$

$$\therefore s = \frac{1}{2} (36 + 48 + 60) \text{ cm} = 72 \text{ cm.}$$

$$(s - a) = (72 - 36) \text{ cm} = 36 \text{ cm,}$$

$$(s - b) = (72 - 48) \text{ cm} = 24 \text{ cm}$$

$$\text{and } (s - c) = (72 - 60) \text{ cm} = 12 \text{ cm.}$$

$$\begin{aligned} \text{(i) Area of the triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{72 \times 36 \times 24 \times 12} \text{ cm}^2 \\ &= 72 \times 12 \text{ cm}^2 = 864 \text{ cm}^2. \end{aligned}$$

(ii) Let base = 60 cm and the corresponding height = h cm.

$$\text{Then, area of the triangle} = \left(\frac{1}{2} \times 60 \times h\right) \text{ cm}^2 = (30h) \text{ cm}^2.$$

$$\therefore 30h = 864 \Rightarrow h = \frac{864}{30} = 28.8.$$

Longest side = 60 cm, corresponding height = 28.8 cm.

EXAMPLE 4 Each side of an equilateral triangle measures 10 cm. Calculate (i) the area of the triangle, and (ii) the height of the triangle. [Given, $\sqrt{3} = 1.732$.]

SOLUTION Here, $a = 10$ cm.

$$\begin{aligned} \text{(i) Area of the triangle} &= \left(\frac{\sqrt{3}}{4} \times a^2\right) \text{ sq units} \\ &= \left(\frac{\sqrt{3}}{4} \times 10 \times 10\right) \text{ cm}^2 = (25 \times \sqrt{3}) \text{ cm}^2 \\ &= (25 \times 1.732) \text{ cm}^2 = 43.3 \text{ cm}^2. \end{aligned}$$

$$\begin{aligned} \text{(ii) Height of the triangle} &= \left(\frac{\sqrt{3}}{2} \times a\right) \text{ units} \\ &= \left(\frac{\sqrt{3}}{2} \times 10\right) \text{ cm} = (5 \times \sqrt{3}) \text{ cm} \\ &= (5 \times 1.732) \text{ cm} = 8.66 \text{ cm.} \end{aligned}$$

EXAMPLE 5 The height of an equilateral triangle is 15 cm. Find its area. [Given, $\sqrt{3} = 1.73$.]

SOLUTION Let each side of the triangle be a cm. Then,

$$\text{its height} = \left(\frac{\sqrt{3}}{2} \times a\right) \text{ cm.}$$

$$\therefore \frac{\sqrt{3}}{2} \times a = 15 \Rightarrow a = \left(\frac{30}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) = 10\sqrt{3}.$$

Thus, each side of the triangle = a cm = $10\sqrt{3}$ cm.

$$\begin{aligned} \text{Area of the triangle} &= \left(\frac{\sqrt{3}}{4} \times a^2 \right) = \left(\frac{\sqrt{3}}{4} \times 10\sqrt{3} \times 10\sqrt{3} \right) \text{ cm}^2 \\ &= (75\sqrt{3}) \text{ cm}^2 = \left(\frac{75 \times 173}{100} \right) \text{ cm}^2 \\ &= \frac{519}{4} \text{ cm}^2 = 129.75 \text{ cm}^2. \end{aligned}$$

EXAMPLE 6 Find the area of an isosceles triangle, each of whose equal sides is 13 cm and whose base is 24 cm.

SOLUTION Here, each equal side, $a = 13$ cm and base, $b = 24$ cm.

$$\begin{aligned} \therefore \text{ area of the triangle} &= \left(\frac{1}{4} b \cdot \sqrt{4a^2 - b^2} \right) \text{ cm}^2 \\ &= \left[\frac{1}{4} \times 24 \times \sqrt{4 \times 169 - 24 \times 24} \right] \text{ cm}^2 \\ &= 60 \text{ cm}^2. \end{aligned}$$

EXAMPLE 7 The base of an isosceles triangle measures 24 cm and its area is 192 cm^2 . Find its perimeter.

SOLUTION Here, base, $b = 24$ cm, and let each equal side be a cm. Then,

$$\begin{aligned} \text{area} &= \frac{1}{4} b \sqrt{4a^2 - b^2} \text{ sq units} = \frac{1}{4} \times 24 \times \sqrt{4a^2 - 576} \text{ cm}^2 \\ &= 12 \times \sqrt{a^2 - 144} \text{ cm}^2. \end{aligned}$$

But, area = 192 cm^2 [given].

$$\therefore 12 \times \sqrt{a^2 - 144} = 192 \Rightarrow \sqrt{a^2 - 144} = 16$$

$$\Rightarrow a^2 - 144 = 256 \Rightarrow a^2 = 400 \Rightarrow a = 20.$$

$$\begin{aligned} \therefore \text{ perimeter of the triangle} &= (2a + b) \text{ cm} \\ &= (2 \times 20 + 24) \text{ cm} = 64 \text{ cm}. \end{aligned}$$

EXAMPLE 8 The difference between the sides at right angles in a right-angled triangle is 14 cm. The area of the triangle is 120 cm^2 . Calculate the perimeter of the triangle.

SOLUTION Let the sides containing the right angle be x cm and $(x - 14)$ cm.

$$\text{Then, its area} = \left[\frac{1}{2} \times x \times (x - 14) \right] \text{ cm}^2.$$

But, area = 120 cm^2 [given].

$$\therefore \frac{1}{2} x(x - 14) = 120 \Rightarrow x^2 - 14x - 240 = 0$$

$$\Rightarrow x^2 - 24x + 10x - 240 = 0 \Rightarrow x(x - 24) + 10(x - 24) = 0$$

$$\Rightarrow (x - 24)(x + 10) = 0 \Rightarrow x = 24 \quad [\text{neglecting } x = -10].$$

$\therefore$ one side = 24 cm, and other side = $(24 - 14)$ cm = 10 cm.

$$\text{Hypotenuse} = \sqrt{(24)^2 + (10)^2} \text{ cm} = \sqrt{576 + 100} \text{ cm}$$

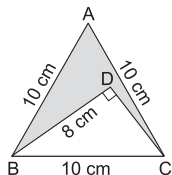
$$= \sqrt{676} \text{ cm} = 26 \text{ cm.}$$

$\therefore$ perimeter of the triangle = $(24 + 10 + 26)$ cm = 60 cm.

EXERCISE 15A

- Find the area of the triangle whose base measures 24 cm and the corresponding height measures 14.5 cm.
- Find the area of the triangle whose sides are 42 cm, 34 cm and 20 cm. Also, find the height corresponding to the longest side.
- Find the area of the triangle whose sides are 18 cm, 24 cm and 30 cm. Also, find the height corresponding to the smallest side.
- The sides of a triangle are in the ratio 5 : 12 : 13, and its perimeter is 150 m. Find the area of the triangle.
- The perimeter of a triangular field is 540 m, and its sides are in the ratio 25 : 17 : 12. Find the area of the field. Also, find the cost of ploughing the field at ₹ 40 per 100 m².
- The perimeter of a right triangle is 40 cm and its hypotenuse measures 17 cm. Find the area of the triangle.
- The difference between the sides at right angles in a right-angled triangle is 7 cm. The area of the triangle is 60 cm². Find its perimeter.
- The lengths of the two sides of a right triangle containing the right angle differ by 2 cm. If the area of the triangle is 24 cm², find the perimeter of the triangle.
- Each side of an equilateral triangle is 10 cm. Find (i) the area of the triangle and (ii) the height of the triangle.
- The height of an equilateral triangle is 6 cm. Find its area. [Take $\sqrt{3} = 1.73$]
- If the area of an equilateral triangle is $36\sqrt{3}$ cm², find its perimeter.
- If the area of an equilateral triangle is $81\sqrt{3}$ cm², find its height.
- The base of a right-angled triangle measures 48 cm and its hypotenuse measures 50 cm. Find the area of the triangle.
- The hypotenuse of a right-angled triangle is 65 cm and its base is 60 cm. Find the length of perpendicular and the area of the triangle.

15. Find the area of a right-angled triangle, the radius of whose circumcircle measures 8 cm and the altitude drawn to the hypotenuse measures 6 cm.
16. Find the length of the hypotenuse of an isosceles right-angled triangle whose area is 200 cm^2 . Also, find its perimeter. [Given, $\sqrt{2} = 1.41$.]
17. The base of an isosceles triangle measures 80 cm and its area is 360 cm^2 . Find the perimeter of the triangle.
18. Each of the equal sides of an isosceles triangle measures 2 cm more than its height, and the base of the triangle measures 12 cm. Find the area of the triangle.
19. Find the area and perimeter of an isosceles right triangle, each of whose equal sides measures 10 cm. [Take $\sqrt{2} = 1.41$.]
20. In the given figure, $\triangle ABC$ is an equilateral triangle the length of whose side is equal to 10 cm, and $\triangle DBC$ is right-angled at D and $BD = 8 \text{ cm}$. Find the area of the shaded region. [Take $\sqrt{3} = 1.732$.]



ANSWERS (EXERCISE 15A)

1. 174 cm^2 2. 336 cm^2 , 16 cm 3. 216 cm^2 , 24 cm 4. 750 m^2
5. 9000 m^2 , ₹ 3600 6. 60 cm^2 7. 40 cm 8. 24 cm
9. (i) 43.3 cm^2 (ii) 8.66 cm 10. 20.76 cm^2 11. 36 cm 12. $9\sqrt{3} \text{ cm}$
13. 336 cm^2 14. 25 cm, 750 cm^2 15. 48 cm^2 16. 28.2 cm, 68.2 cm
17. 162 cm 18. 48 cm^2 19. 50 cm^2 , 34.1 cm 20. 19.3 cm^2

HINTS TO SOME SELECTED QUESTIONS

6. Sum of two sides = $(40 - 17) \text{ cm} = 23 \text{ cm}$.

Let these sides be $x \text{ cm}$ and $(23 - x) \text{ cm}$.

$$\begin{aligned} \therefore x^2 + (23 - x)^2 &= (17)^2 \Rightarrow x^2 + x^2 - 46x + 529 = 289 \\ &\Rightarrow 2x^2 - 46x + 240 = 0 \Rightarrow x^2 - 23x + 120 = 0 \\ &\Rightarrow (x - 15)(x - 8) = 0 \Rightarrow x = 15 \text{ or } x = 8. \end{aligned}$$

$\therefore$ base = 15 cm and height = 8 cm. Find the area of the triangle.

7. Let these sides be $x \text{ cm}$ and $(x - 7) \text{ cm}$.

$$\begin{aligned} \therefore \frac{1}{2} \times x \times (x - 7) &= 60 \Rightarrow x^2 - 7x - 120 = 0 \Rightarrow (x - 15)(x + 8) = 0 \\ &\Rightarrow x = 15 \quad [\because x \neq -8]. \end{aligned}$$

$\therefore$ these sides are 15 cm and 8 cm.

$$\text{Hypotenuse} = \sqrt{(15)^2 + 8^2} \text{ cm} = \sqrt{289} \text{ cm} = 17 \text{ cm}.$$

9. (i) Area of the triangle = $\left(\frac{\sqrt{3}}{4} \times a^2\right) = \left(\frac{1.732}{4} \times 10 \times 10\right) \text{ cm}^2 = 43.3 \text{ cm}^2$.

(ii) $\frac{1}{2} \times a \times \text{height} = 43.3 \text{ cm}^2 \Rightarrow \frac{1}{2} \times 10 \text{ cm} \times h = 43.3 \text{ cm}^2 \Rightarrow h = \frac{43.3 \text{ cm}^2}{5 \text{ cm}} = 8.66 \text{ cm}$.

10. Height = $\frac{\sqrt{3}a}{2} \Rightarrow \frac{\sqrt{3}a}{2} = 6 \text{ cm} \Rightarrow a = \frac{12}{\sqrt{3}} \text{ cm} = \frac{12}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \text{ cm} = 4\sqrt{3} \text{ cm}$.

$\therefore \text{area} = \frac{\sqrt{3}}{4} \times a^2 = \frac{\sqrt{3}}{4} \times (4\sqrt{3})^2 \text{ cm}^2 = \left(\frac{\sqrt{3}}{4} \times 48\right) \text{ cm}^2 = (12 \times 1.73) \text{ cm}^2$.

11. $\frac{\sqrt{3}}{4} a^2 = 36\sqrt{3} \text{ cm}^2 \Rightarrow a^2 = 144 \text{ cm}^2 \Rightarrow a = \sqrt{144} \text{ cm} = 12 \text{ cm}$.

12. $\frac{\sqrt{3}}{4} a^2 = 81\sqrt{3} \text{ cm}^2 \Rightarrow a^2 = (81 \times 4) \text{ cm}^2 \Rightarrow a = \sqrt{196} \text{ cm} = 18 \text{ cm}$.

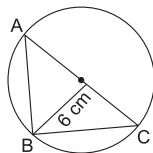
Height = $\frac{\sqrt{3}a}{2} = \left(\frac{\sqrt{3} \times 18}{2}\right) \text{ cm} = 9\sqrt{3} \text{ cm}$.

13. Height = $\sqrt{(50)^2 - (48)^2} \text{ cm} = \sqrt{98 \times 2} \text{ cm} = (7 \times 2) \text{ cm} = 14 \text{ cm}$.

15. The circumcentre of a right triangle is the midpoint of its hypotenuse.

$\therefore \text{hypotenuse} = 2 \times (\text{radius of the circumcircle})$
 $= (2 \times 8) \text{ cm} = 16 \text{ cm}$.

$\therefore \text{base} = 16 \text{ cm}, \text{height} = 6 \text{ cm}$.



16. Let base = height = $a \text{ cm}$. Then,

$\frac{1}{2} \times a \times a = 200 \text{ cm}^2 \Rightarrow a^2 = 400 \text{ cm}^2 \Rightarrow a = \sqrt{400} \text{ cm} = 20 \text{ cm}$.

$\therefore \text{hypotenuse} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2 \times 400} \text{ cm}$
 $= 20\sqrt{2} \text{ cm} = (20 \times 1.41) \text{ cm}$.

17. $\frac{1}{2} \times 80 \times \sqrt{4a^2 - 6400} = 360 \Rightarrow 80 \times \sqrt{a^2 - 1600} = 360 \Rightarrow \sqrt{a^2 - 1600} = 9$
 $\Rightarrow a^2 - 1600 = 81 \Rightarrow a^2 = 1681 \Rightarrow a = \sqrt{1681} = 41 \text{ cm}$.

Perimeter = $(41 + 41 + 80) \text{ cm} = 162 \text{ cm}$.

18. Let the height be $h \text{ cm}$. Then, $a = (h + 2) \text{ cm}$ and $b = 12 \text{ cm}$.

$\therefore \frac{1}{2} \times 12 \times h = \frac{1}{4} \times 12 \times \sqrt{4(h + 2)^2 - 144}$.

20. $CD = \sqrt{(10)^2 - 8^2} \text{ cm} = \sqrt{36} \text{ cm} = 6 \text{ cm}$.

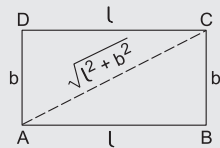
Required area = $\text{ar}(\triangle ABC) - \text{ar}(\triangle DBC)$

$= \left\{ \frac{\sqrt{3}}{4} \times (10)^2 - \frac{1}{2} \times 8 \times 6 \right\} \text{ cm}^2$.

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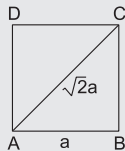
PERIMETER AND AREA OF QUADRILATERALS [FORMULAE]

1. Area and perimeter of a rectangle:

(i) Area = (length $\times$ breadth) sq units.(ii) Length = $\frac{\text{area}}{\text{breadth}}$ and Breadth = $\frac{\text{area}}{\text{length}}$.(iii) Perimeter = $2(l + b)$ units.(iv) Diagonal = $\sqrt{l^2 + b^2}$ units.

2. Area of 4 walls of a room =
- $[2(l + b) \times h]$
- sq units.

3. Area and perimeter of a square:

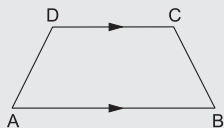
(i) Area = a^2 sq units.(ii) Area = $\left\{ \frac{1}{2} \times (\text{diagonal})^2 \right\}$ sq units.(iii) Perimeter = $4a$ units.(iv) Diagonal = $\sqrt{2}a$ units.

4. (i) Area of a parallelogram = {(base)
- $\times$
- (height)} sq units.

(ii) Area of a rhombus = $\left(\frac{1}{2} \times \text{product of diagonals} \right)$ sq units.

5. Area of a trapezium

$$= \frac{1}{2} \times (\text{sum of parallel sides})$$

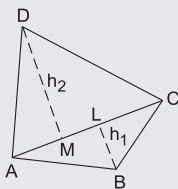
$$\times (\text{distance between them}).$$


6. Area of a quadrilateral:

(i) Let ABCD be a quadrilateral with diagonal AC.

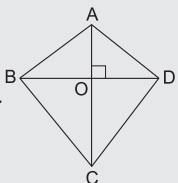
Let $BL \perp AC$ and $DM \perp AC$. Let $BL = h_1$ and $DM = h_2$.

$$\text{ar}(\text{quad. } ABCD) = \left\{ \frac{1}{2} (h_1 + h_2) \times AC \right\} \text{ sq units.}$$



(ii) When diagonals of a quadrilateral are perpendicular to each other:

$$\text{ar}(\text{quad. } ABCD) = \left(\frac{1}{2} \times \text{product of diagonals} \right) \text{ sq units.}$$

**SOLVED EXAMPLES****EXAMPLE 1**

The length of a rectangular field exceeds its breadth by 8 m and the area of the field is 240 m^2 . Find the dimensions of the field.

[CBSE 2014]

SOLUTION Let the breadth of the field be x metres.

Then, its length = $(x + 8)$ m.

$$\therefore \text{area} = (x + 8) \cdot x \text{ m}^2.$$

$$\therefore (x + 8)x = 240 \Rightarrow x^2 + 8x - 240 = 0$$

$$\Rightarrow x^2 + 20x - 12x - 240 = 0 \Rightarrow x(x + 20) - 12(x + 20) = 0$$

$$\Rightarrow (x + 20)(x - 12) = 0 \Rightarrow x = 12 \quad [\because x \neq -20].$$

$\therefore$ breadth = 12 m and length = 20 m.

EXAMPLE 2 Sum of the areas of two squares is 400 cm^2 . If the difference of their perimeters is 16 cm, find the sides of the two squares. [CBSE 2013]

SOLUTION Let the sides of the given squares be a cm and b cm respectively. Then,

$$a^2 + b^2 = 400 \quad \dots (i)$$

$$\text{and } 4a - 4b = 16 \Rightarrow a - b = 4. \quad \dots (ii)$$

Squaring (ii) on both sides, we get

$$(a - b)^2 = 16 \Rightarrow (a^2 + b^2) - 2ab = 16$$

$$\Rightarrow 400 - 2ab = 16 \text{ [using (i)]}$$

$$\Rightarrow 2ab = 384 \Rightarrow ab = 192. \quad \dots (iii)$$

$$\text{Now, } (a + b)^2 - (a - b)^2 = 4ab$$

$$\Rightarrow (a + b)^2 - 16 = 4 \times 192.$$

$$\Rightarrow (a + b)^2 = 768 + 16 = 784 \Rightarrow a + b = \sqrt{784} = 28.$$

On solving $a + b = 28$ and $a - b = 4$, we get $a = 16$ and $b = 12$.

Hence, the sides of the given squares are 16 cm and 12 cm.

EXAMPLE 3 The length of the diagonal of a square is 24 cm. Find (i) the area of the square and (ii) its perimeter. [Given, $\sqrt{2} = 1.41$.]

SOLUTION (i) Area of the square = $\left\{ \frac{1}{2} \times (\text{diagonal})^2 \right\}$ sq units.

$$= \left(\frac{1}{2} \times 24 \times 24 \right) \text{ cm}^2 = 288 \text{ cm}^2.$$

$$(ii) \text{ Side of the square} = \sqrt{288} \text{ cm} = \sqrt{144 \times 2} \text{ cm}$$

$$= 12\sqrt{2} \text{ cm} = (12 \times 1.41) \text{ cm} = 16.92 \text{ cm}.$$

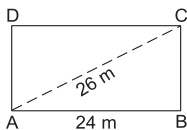
$$\therefore \text{perimeter of the square} = (4 \times 16.92) \text{ cm} = 67.68 \text{ cm}.$$

EXAMPLE 4 The longer side of a rectangular hall is 24 m, and the length of its diagonal is 26 m. Find the area of the hall.

SOLUTION Let ABCD be the hall in which $AB = 24$ m and $AC = 26$ m.

By Pythagoras' theorem, we have

$$BC = \sqrt{AC^2 - AB^2} \text{ units} = \sqrt{(26)^2 - (24)^2} \text{ m}$$



$$\begin{aligned}
 &= \sqrt{(26-24)(26+24)} \text{ m} = \sqrt{50 \times 2} \text{ m} \\
 &= \sqrt{100} \text{ m} = 10 \text{ m}.
 \end{aligned}$$

Thus, length = 24 m, breadth = 10 m.

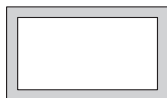
$$\therefore \text{ area of the hall} = (24 \times 10) \text{ m}^2 = 240 \text{ m}^2.$$

EXAMPLE 5

The length and the breadth of a rectangular park are in the ratio 8 : 5. A path, 1.5 m wide, running all around the outside of the park has an area of 594 m^2 . Find the dimensions of the park.

SOLUTION

Let the length and the breadth of the park be $8x$ metres and $5x$ metres respectively.



$$\begin{aligned}
 \text{Then, area of the park} &= (8x \times 5x) \text{ m}^2 \\
 &= (40x^2) \text{ m}^2.
 \end{aligned}$$

$$\text{Length of the park including the path} = (8x + 3) \text{ m}.$$

$$\text{Breadth of the park including the path} = (5x + 3) \text{ m}.$$

$$\text{Area of the park including the path} = (8x + 3)(5x + 3) \text{ m}^2.$$

$$\begin{aligned}
 \text{Area of the path} &= [(8x + 3)(5x + 3) - 40x^2] \text{ m}^2 \\
 &= (39x + 9) \text{ m}^2.
 \end{aligned}$$

$$\therefore 39x + 9 = 594 \Rightarrow 39x = 585$$

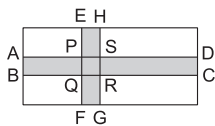
$$\Rightarrow x = \frac{585}{39} = 15.$$

$$\therefore \text{ length} = (8 \times 15) \text{ m} = 120 \text{ m},$$

$$\text{breadth} = (5 \times 15) \text{ m} = 75 \text{ m}.$$

EXAMPLE 6

A rectangular lawn, 75 m by 60 m, has two roads, each 4 m wide, running through the middle of the lawn, one parallel to length and the other parallel to breadth, as shown in the figure. Find the cost of gravelling the roads at ₹ 50 per m^2 .

**SOLUTION**

$$\text{Area of the road } ABCD = (75 \times 4) \text{ m}^2 = 300 \text{ m}^2.$$

$$\text{Area of the road } EFGH = (60 \times 4) \text{ m}^2 = 240 \text{ m}^2.$$

Clearly, area $PQRS$ is common to both the roads.

$$\text{Area } PQRS = (4 \times 4) \text{ m}^2 = 16 \text{ m}^2.$$

Area of the roads to be gravelled

$$= (300 + 240 - 16) \text{ m}^2 = 524 \text{ m}^2.$$

Cost of gravelling the roads

$$= ₹ (524 \times 50) = ₹ 26200.$$

EXAMPLE 7 Find the area of the quadrilateral whose sides measure 9 cm, 40 cm, 28 cm and 15 cm, and in which the angle between the first two sides is a right angle.

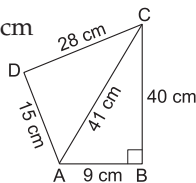
SOLUTION Let $ABCD$ be the given quadrilateral in which $AB = 9$ cm, $BC = 40$ cm, $CD = 28$ cm, $DA = 15$ cm and $\angle ABC = 90^\circ$.

By Pythagoras' theorem,

$$AC = \sqrt{AB^2 + BC^2} \text{ units} = \sqrt{(9)^2 + (40)^2} \text{ cm} \\ = \sqrt{1681} \text{ cm} = 41 \text{ cm}.$$

$$\text{Area of } \triangle ABC = \left(\frac{1}{2} \times AB \times BC \right)$$

$$= \left(\frac{1}{2} \times 9 \times 40 \right) \text{ cm}^2 = 180 \text{ cm}^2.$$



In $\triangle ACD$,

let $a = AC = 41$ cm, $b = CD = 28$ cm and $c = DA = 15$ cm.

$$\therefore s = \frac{1}{2}(41 + 28 + 15) \text{ cm} = 42 \text{ cm}.$$

$$\therefore (s - a) = 1 \text{ cm}, (s - b) = 14 \text{ cm and } (s - c) = 27 \text{ cm}.$$

$$\begin{aligned} \text{Area of } \triangle ACD &= \sqrt{s(s-a)(s-b)(s-c)} \text{ sq units} \\ &= \sqrt{42 \times 1 \times 14 \times 27} \text{ cm}^2 \\ &= (14 \times 3 \times 3) \text{ cm}^2 = 126 \text{ cm}^2. \end{aligned}$$

Area of the quadrilateral $ABCD$

$$\begin{aligned} &= \text{area of } \triangle ABC + \text{area of } \triangle ACD \\ &= (180 + 126) \text{ cm}^2 = 306 \text{ cm}^2. \end{aligned}$$

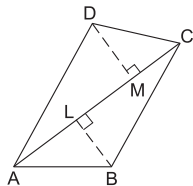
EXAMPLE 8 In a four-sided field, the length of the longer diagonal is 120 m. The lengths of perpendiculars from the opposite vertices on this diagonal are 20 m and 15 m. Find the area of the field.

SOLUTION Let $ABCD$ be the field and AC be its longer diagonal.

Let $BL \perp AC$ and $DM \perp AC$. Then,

$$AC = 120 \text{ m}, BL = 20 \text{ m}$$

and $DM = 15 \text{ m}.$



$$\begin{aligned} \text{Area of the field} &= \left\{ \frac{1}{2} \times (BL + DM) \times AC \right\} \text{ sq units} \\ &= \left\{ \frac{1}{2} \times (20 + 15) \times 120 \right\} \text{ m}^2 = 2100 \text{ m}^2. \end{aligned}$$

EXAMPLE 9 Find the area of a parallelogram one of whose sides measures 40 cm and the corresponding height measures 19.5 cm.

SOLUTION Area of the given parallelogram

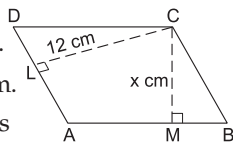
$$\begin{aligned}
 &= (\text{base} \times \text{height}) \text{ sq units} \\
 &= (40 \times 19.5) \text{ cm}^2 = \left(40 \times \frac{195}{10}\right) \text{ cm}^2 \\
 &= 780 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 10 The adjacent sides of a parallelogram are 36 cm and 27 cm in length. If the distance between the shorter sides is 12 cm, find the distance between the longer sides.

SOLUTION Longer side = 36 cm, shorter side = 27 cm.

Distance between the shorter sides = 12 cm.

Let the distance between the longer sides be x cm.



Area of the ||gm

$$= (\text{longer side} \times \text{distance between the longer sides})$$

$$= (\text{shorter side} \times \text{distance between the shorter sides})$$

$$\therefore 36 \times x = 27 \times 12 \Rightarrow x = \frac{27 \times 12}{36} = 9.$$

$$\therefore \text{distance between the longer sides} = 9 \text{ cm.}$$

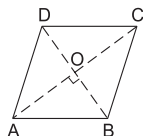
EXAMPLE 11 The diagonals of a rhombus are 48 cm and 20 cm long. Find (i) the area of the rhombus, and (ii) the perimeter of the rhombus.

SOLUTION (i) Area of the rhombus

$$= \frac{1}{2} \times (\text{product of diagonals}) \text{ sq units}$$

$$= \left(\frac{1}{2} \times 48 \times 20\right) \text{ cm}^2$$

$$= 480 \text{ cm}^2.$$



(ii) We know that the diagonals of a rhombus bisect each other at right angles.

$$\therefore OA = OC = 24 \text{ cm}, OB = OD = 10 \text{ cm and } \angle AOB = 90^\circ.$$

$\therefore$ by Pythagoras' theorem, we have

$$AB = \sqrt{OA^2 + OB^2} \text{ units} = \sqrt{(24)^2 + (10)^2} \text{ cm}$$

$$= \sqrt{676} \text{ cm} = 26 \text{ cm.}$$

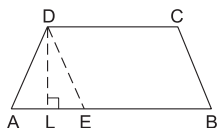
$$\therefore \text{perimeter of the rhombus} = (4 \times 26) \text{ cm} = 104 \text{ cm.}$$

EXAMPLE 12 Find the area of a trapezium whose parallel sides are 35 cm and 23 cm long and the distance between them is 15 cm.

SOLUTION Area of the trapezium

$$\begin{aligned}
 &= \frac{1}{2} \times (\text{sum of parallel sides}) \times (\text{distance between them}) \\
 &= \left\{ \frac{1}{2} \times (35 + 23) \times 15 \right\} \text{ cm}^2 = 435 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 13 Find the area of a trapezium ABCD in which $AB \parallel DC$, $AB = 77$ cm, $BC = 25$ cm, $CD = 60$ cm and $DA = 26$ cm. [HOTS]



SOLUTION Draw $DE \parallel BC$ and $DL \perp AB$.

Then, $DE = BC = 25$ cm.

$AE = (AB - EB) = (AB - DC) = (77 - 60)$ cm = 17 cm.

In $\triangle DAE$, we have

$a = AE = 17$ cm, $b = DE = 25$ cm and $c = DA = 26$ cm.

$$\therefore s = \frac{1}{2}(17 + 25 + 26) \text{ cm} = 34 \text{ cm},$$

$(s - a) = 17$ cm, $(s - b) = 9$ cm and $(s - c) = 8$ cm.

$$\begin{aligned}
 \therefore \text{ area of } \triangle DAE &= \sqrt{s(s-a)(s-b)(s-c)} \\
 &= \sqrt{34 \times 17 \times 9 \times 8} \text{ cm}^2 \\
 &= (17 \times 3 \times 4) \text{ cm}^2 = 204 \text{ cm}^2.
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, area of } \triangle DAE &= \frac{1}{2} \times AE \times DL \text{ sq units} \\
 &= \frac{1}{2} \times 17 \text{ cm} \times DL.
 \end{aligned}$$

$$\therefore \frac{1}{2} \times 17 \text{ cm} \times DL = 204 \text{ cm}^2 \Rightarrow DL = \left(\frac{204 \times 2}{17} \right) \text{ cm} = 24 \text{ cm}.$$

$$\begin{aligned}
 \text{ar(trap. } ABCD) &= \left\{ \frac{1}{2} (AB + DC) \times DL \right\} \text{ sq units} \\
 &= \left\{ \frac{1}{2} (77 + 60) \times 24 \right\} \text{ cm}^2 = 1644 \text{ cm}^2.
 \end{aligned}$$

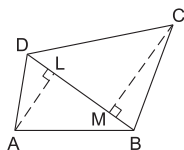
EXERCISE 15B

1. The perimeter of a rectangular plot of land is 80 m and its breadth is 16 m. Find the length and area of the plot.
2. The length of a rectangular park is twice its breadth and its perimeter is 840 m. Find the area of the park.

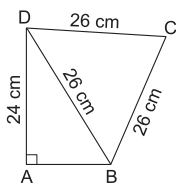
3. One side of a rectangle is 12 cm long and its diagonal measures 37 cm. Find the other side and the area of the rectangle.
4. The area of a rectangular plot is 462 m^2 and its length is 28 m. Find its perimeter.
5. A lawn is in the form of a rectangle whose sides are in the ratio 5 : 3. The area of the lawn is 3375 m^2 . Find the cost of fencing the lawn at ₹ 65 per metre.
6. A room is 16 m long and 13.5 m broad. Find the cost of covering its floor with 75-m-wide carpet at ₹ 60 per metre.
7. The floor of a rectangular hall is 24 m long and 18 m wide. How many carpets, each of length 2.5 m and breadth 80 cm, will be required to cover the floor of the hall?
8. A 36-m-long, 15-m-broad verandah is to be paved with stones, each measuring 6 dm by 5 dm. How many stones will be required?
9. The area of a rectangle is 192 cm^2 and its perimeter is 56 cm. Find the dimensions of the rectangle.
10. A rectangular park 35 m long and 18 m wide is to be covered with grass, leaving 2.5 m uncovered all around it. Find the area to be laid with grass.
11. A rectangular plot measures 125 m by 78 m. It has a gravel path 3 m wide all around on the outside. Find the area of the path and the cost of gravelling it at ₹ 75 per m^2 .
12. (i) A footpath of uniform width runs all around the inside of a rectangular field 54 m long and 35 m wide. If the area of the path is 420 m^2 , find the width of the path.
(ii) A carpet is laid on the floor of a room 8 m by 5 m. There is a border of constant width all around the carpet. If the area of the border is 12 m^2 , find its width.
13. The length and the breadth of a rectangular garden are in the ratio 9 : 5. A path 3.5 m wide, running all around inside it has an area of 1911 m^2 . Find the dimensions of the garden.
14. A room 4.9 m long and 3.5 m broad is covered with carpet, leaving an uncovered margin of 25 cm all around the room. If the breadth of the carpet is 80 cm, find its cost at ₹ 80 per metre.
15. In a rectangular park of dimensions 50 m $\times$ 40 m, a rectangular pond is constructed so that the area of grass strip of uniform width surrounding the pond would be 1184 m^2 . Find the length and breadth of the pond. [CBSE 2017]
16. A 80 m by 64 m rectangular lawn has two roads, each 5 m wide, running through its middle, one parallel to its length and the other parallel to its breadth. Find the cost of gravelling the roads at ₹ 40 per m^2 .

17. The dimensions of a room are $14 \text{ m} \times 10 \text{ m} \times 6.5 \text{ m}$. There are two doors and 4 windows in the room. Each door measures $2.5 \text{ m} \times 1.2 \text{ m}$ and each window measures $1.5 \text{ m} \times 1 \text{ m}$. Find the cost of painting the four walls of the room at ₹ 35 per m^2 .
18. The cost of painting the four walls of a room 12 m long at ₹ 30 per m^2 is ₹ 7560 and the cost of covering the floor with mat at ₹ 25 per m^2 is ₹ 2700. Find the dimensions of the room.
19. Find the area and perimeter of a square plot of land whose diagonal is 24 m long. [Take $\sqrt{2} = 1.41$.]
20. Find the length of the diagonal of a square whose area is 128 cm^2 . Also, find its perimeter.
21. The area of a square field is 8 hectares. How long would a man take to cross it diagonally by walking at the rate of 4 km per hour?
22. The cost of harvesting a square field at ₹ 900 per hectare is ₹ 8100. Find the cost of putting a fence around it at ₹ 18 per metre.
23. The cost of fencing a square lawn at ₹ 14 per metre is ₹ 28000. Find the cost of mowing the lawn at ₹ 54 per 100 m^2 .

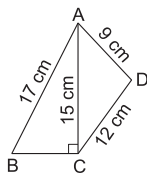
24. In the given figure, $ABCD$ is a quadrilateral in which diagonal $BD = 24 \text{ cm}$, $AL \perp BD$ and $CM \perp BD$ such that $AL = 9 \text{ cm}$ and $CM = 12 \text{ cm}$. Calculate the area of the quadrilateral.



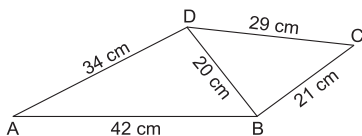
25. Find the area of the quadrilateral $ABCD$ in which $AD = 24 \text{ cm}$, $\angle BAD = 90^\circ$ and $\triangle BCD$ is an equilateral triangle having each side equal to 26 cm. Also, find the perimeter of the quadrilateral. [Given, $\sqrt{3} = 1.73$.]

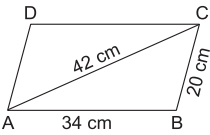


26. Find the perimeter and area of the quadrilateral $ABCD$ in which $AB = 17 \text{ cm}$, $AD = 9 \text{ cm}$, $CD = 12 \text{ cm}$, $\angle ACB = 90^\circ$ and $AC = 15 \text{ cm}$.



27. Find the area of the quadrilateral $ABCD$ in which $AB = 42 \text{ cm}$, $BC = 21 \text{ cm}$, $CD = 29 \text{ cm}$, $DA = 34 \text{ cm}$ and diagonal $BD = 20 \text{ cm}$.



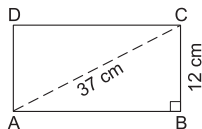
28. Find the area of a parallelogram with base equal to 25 cm and the corresponding height measuring 16.8 cm.
29. The adjacent sides of a parallelogram are 32 cm and 24 cm. If the distance between the longer sides is 17.4 cm, find the distance between the shorter sides.
30. The area of a parallelogram is 392 m^2 . If its altitude is twice the corresponding base, determine the base and the altitude.
31. The adjacent sides of a parallelogram $ABCD$ measure 34 cm and 20 cm, and the diagonal AC measures 42 cm. Find the area of the parallelogram.
- 
32. Find the area of the rhombus, the lengths of whose diagonals are 30 cm and 16 cm. Also, find the perimeter of the rhombus.
33. The perimeter of a rhombus is 60 cm. If one of its diagonals is 18 cm long, find (i) the length of the other diagonal, and (ii) the area of the rhombus.
34. The area of a rhombus is 480 cm^2 , and one of its diagonals measures 48 cm. Find (i) the length of the other diagonal, (ii) the length of each of its sides, and (iii) its perimeter.
35. The parallel sides of a trapezium are 12 cm and 9 cm and the distance between them is 8 cm. Find the area of the trapezium.
36. The shape of the cross section of a canal is a trapezium. If the canal is 10 m wide at the top, 6 m wide at the bottom and the area of its cross section is 640 m^2 , find the depth of the canal.
37. Find the area of a trapezium whose parallel sides are 11 m and 25 m long, and the nonparallel sides are 15 m and 13 m long. [HOTS]

ANSWERS (EXERCISE 15B)

1. Length = 24 m, area = 384 m^2 2. 39200 m^2 3. 35 cm, 420 cm^2
 4. 89 m 5. ₹ 15600 6. ₹ 17280 7. 216 8. 1800
 9. Length = 16 cm, breadth = 12 cm 10. 240 m^2 11. 1254 m^2 , ₹ 94050
 12. (i) 2.5 m (ii) 50 cm 13. Length = 180 m, breadth = 100 m
 14. ₹ 1320 15. length = 34 m, breadth = 24 m 16. ₹ 27800 17. ₹ 10500
 18. Length = 12 m, breadth = 9 m and height = 6 m 19. 288 m^2 , 67.68 m
 20. 16 cm, 45.12 cm 21. 6 minutes 22. ₹ 21600 23. ₹ 135000 24. 252 cm^2
 25. 412.37 cm^2 , 86 cm 26. 46 cm, 114 cm^2 27. 546 cm^2
 28. 420 cm^2 29. 23.2 cm 30. Base = 14 m, altitude = 28 m
 31. 672 cm^2 32. 240 cm^2 , 68 cm 33. (i) 24 cm (ii) 216 cm^2
 34. (i) 20 cm (ii) 26 cm (iii) 104 cm 35. 84 cm^2 36. 80 m 37. 216 m^2

HINTS TO SOME SELECTED QUESTIONS

$$\begin{aligned}
 3. \text{ Other side} &= \sqrt{(37)^2 - (12)^2} \text{ cm} \\
 &= \sqrt{(37 + 12)(37 - 12)} \text{ cm} \\
 &= \sqrt{49 \times 25} \text{ cm} = (7 \times 5) \text{ cm} \\
 &= 35 \text{ cm}.
 \end{aligned}$$



$$6. \text{ Length of the carpet} = \frac{\text{area of the floor}}{\text{width of the carpet}} = \left(\frac{16 \times 13.5}{0.75} \right) \text{ m} = 288 \text{ m}.$$

$$7. \text{ Area of the floor} = (24 \times 18) \text{ m}^2.$$

$$\text{Area of each carpet} = \left(\frac{25}{10} \times \frac{80}{100} \right) \text{ m}^2 = 2 \text{ m}^2.$$

$$\text{Number of carpets} = \frac{\text{area of the floor}}{\text{area of each carpet}}.$$

$$8. \text{ Number of stones} = \frac{\text{area of the verandah}}{\text{area of each stone}} = \left(\frac{36 \times 15}{0.6 \times 0.5} \right).$$

$$9. l + b = 28 \text{ and } lb = 192.$$

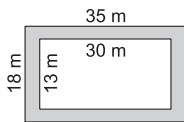
$$(l - b)^2 = (l + b)^2 - 4lb = (28)^2 - 4 \times 192 = 16 \Rightarrow l - b = 4.$$

On solving $l + b = 28$ and $l - b = 4$, we get $l = 16$, $b = 12$.

$$10. \text{ Outer area} = (35 \times 18) \text{ m}^2 = 630 \text{ m}^2.$$

$$\text{Inner area} = (30 \times 13) \text{ m}^2 = 390 \text{ m}^2.$$

$$\text{Grass area} = (630 - 390) \text{ m}^2 = 240 \text{ m}^2.$$



$$11. \text{ Area of the path} = [(125 + 6) \times (78 + 6) - (125 \times 78)] \text{ m}^2.$$

$$12. \text{ (i) Let the width of the path be } x \text{ metres. Then,}$$

$$(54 \times 35) - \{(54 - 2x) \times (35 - 2x)\} = 420$$

$$\Rightarrow 2x^2 - 89x + 210 = 0 \Rightarrow 2x^2 - 84x - 5x + 210 = 0$$

$$\Rightarrow (x - 42)(2x - 5) = 0 \Rightarrow x = 42 \text{ or } x = 2.5 \quad [\because \text{width of the path} \neq 42 \text{ m}].$$

$$\text{(ii) Let the width of the border be } x \text{ metres. Then,}$$

$$(8 \times 5) - (8 - 2x)(5 - 2x) = 12 \Rightarrow 2x^2 - 13x + 6 = 0 \Rightarrow (x - 6)(2x - 1) = 0.$$

$$\therefore x = \frac{1}{2} \text{ m} = 50 \text{ cm}.$$

$$13. \text{ Let the length and breadth be } (9x) \text{ m and } (5x) \text{ m respectively.}$$

$$\text{Then, } (9x \times 5x) - (9x - 7)(5x - 7) = 1911.$$

$$14. \text{ Area to be carpeted} = (4.9 - 0.5)(3.5 - 0.5) \text{ m}^2 = 13.2 \text{ m}^2.$$

$$\text{Length of the carpet} = \left(\frac{13.2}{0.80} \right) \text{ m} = 16.5 \text{ m}.$$

$$16. \text{ Area of the roads} = \{(80 \times 5) + (64 \times 5) - (5 \times 5)\} \text{ m}^2 = 695 \text{ m}^2.$$

$$17. \text{ Area of 4 walls} = [2(l + b) \times h] = \left(2 \times 24 \times \frac{13}{2} \right) \text{ m}^2 = 312 \text{ m}^2.$$

$$\text{Area not to be painted} = \left\{ \left(2 \times \frac{5}{2} \times \frac{12}{10} \right) + \left(4 \times \frac{3}{2} \times 1 \right) \right\} \text{ m}^2 = 12 \text{ m}^2.$$

$$18. \text{Area of 4 walls} = \frac{\text{total cost of painting}}{\text{rate per m}^2} = \left(\frac{7560}{30}\right) \text{ m}^2 = 252 \text{ m}^2.$$

$$\text{Area of the floor} = \frac{\text{total cost of matting}}{\text{rate per m}^2} = \left(\frac{2700}{25}\right) \text{ m}^2 = 108 \text{ m}^2.$$

Let breadth = b metres and height = h metres. Then,

$$12 \times b = 108 \Rightarrow \text{breadth} = \frac{108}{12} \text{ m} = 9 \text{ m}.$$

$$\text{Also, } 2(l + b) \times h = 252 \Rightarrow 2(12 + 9) \times h = 252 \Rightarrow h = 6 \text{ m}.$$

$$19. \text{Area} = \frac{1}{2}d^2 = \left(\frac{1}{2} \times 24 \times 24\right) \text{ m}^2 = 288 \text{ m}^2.$$

$$\text{Now, } a^2 = 288 \Rightarrow a = \sqrt{288} = 12\sqrt{2} \Rightarrow 4a = 48\sqrt{2} \text{ m}.$$

$$20. \frac{1}{2}d^2 = 128 \Rightarrow d^2 = 256 = (16)^2 \Rightarrow d = 16.$$

$$a^2 = 128 \Rightarrow a = \sqrt{128} = 8\sqrt{2} \Rightarrow 4a = 32\sqrt{2}.$$

$$21. \text{Area} = (8 \times 10000) \text{ m}^2 \Rightarrow \frac{1}{2} \times (\text{diagonal})^2 = 8 \times 10000$$

$$\Rightarrow (\text{diagonal})^2 = 160000 \Rightarrow \text{diagonal} = 400 \text{ m}.$$

$$24. \text{ar}(\text{quad. } ABCD) = \text{ar}(\triangle ABD) + \text{ar}(\triangle CBD) = \left(\frac{1}{2} \times BD \times AL\right) + \left(\frac{1}{2} \times BD \times CM\right)$$

$$= \frac{1}{2} \times BD \times (AL + CM) \text{ cm}^2.$$

$$25. AB = \sqrt{BD^2 - AD^2} = \sqrt{(26)^2 - (24)^2} \text{ cm} = \sqrt{(26 + 24)(26 - 24)} \text{ cm}$$

$$= \sqrt{50 \times 2} \text{ cm} = \sqrt{100} \text{ cm} = 10 \text{ cm}.$$

$$\text{ar}(\text{quad. } ABCD) = \text{ar}(\text{right } \triangle BAD) + \text{ar}(\text{equilateral } \triangle BCD).$$

$$29. \text{Area of the parallelogram} = (32 \times 17.4) \text{ cm}^2.$$

$$24 \times h = 32 \times 17.4. \text{ Find } h.$$

$$31. \text{ar}(\text{llgm } ABCD) = 2 \times \text{ar}(\triangle ABC).$$

32. We know that the diagonals of a rhombus bisect each other at right angles.

$$\text{So, } OA = 15 \text{ cm, } OB = 8 \text{ cm and } \angle AOB = 90^\circ.$$

$$\therefore AB^2 = (OA^2 + OB^2) = (15)^2 + (8)^2 = 225 + 64 = 289$$

$$\Rightarrow AB = \sqrt{289} = 17 \text{ cm}.$$

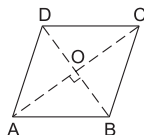
$$\text{So, perimeter} = (4 \times 17) \text{ cm} = 68 \text{ cm}.$$

$$33. \text{Each side} = 15 \text{ cm}.$$

$$(15)^2 = 9^2 + x^2 \Rightarrow x^2 = (225 - 81) = 144 \Rightarrow x = \sqrt{144} = 12 \text{ cm}.$$

$$\therefore \text{other diagonal} = 2x = 24 \text{ cm. Area} = \left(\frac{1}{2} \times d_1 \times d_2\right) \text{ cm}^2.$$

$$36. \frac{1}{2}(10 + 6) \times d = 640 \Rightarrow d = 80 \text{ cm}.$$



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MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

1. The length of a rectangular hall is 5 m more than its breadth. If the area of the hall is 750 m^2 then its length is
(a) 15 m (b) 20 m (c) 25 m (d) 30 m
2. The length of a rectangular field is 23 m more than its breadth. If the perimeter of the field is 206 m, then its area is
(a) 2420 m^2 (b) 2520 m^2 (c) 2480 m^2 (d) 2620 m^2
3. The length of a rectangular field is 12 m and the length of its diagonal is 15 m. The area of the field is
(a) 108 m^2 (b) 180 m^2 (c) $30\sqrt{3} \text{ m}^2$ (d) $12\sqrt{15} \text{ m}^2$
4. The cost of carpeting a room 15 m long with a carpet 75 cm wide, at ₹ 70 per metre, is ₹ 8400. The width of the room is
(a) 9 m (b) 8 m (c) 6 m (d) 12 m
5. The length of a rectangle is thrice its breadth and the length of its diagonal is $8\sqrt{10}$ cm. The perimeter of the rectangle is
(a) $15\sqrt{10}$ cm (b) $16\sqrt{10}$ cm (c) $24\sqrt{10}$ cm (d) 64 cm
6. On increasing the length of a rectangle by 20% and decreasing its breadth by 20%, what is the change in its area?
(a) 20% increase (b) 20% decrease
(c) No change (d) 4% decrease
7. A rectangular ground $80 \text{ m} \times 50 \text{ m}$ has a path 1 m wide outside around it. The area of the path is
(a) 264 m^2 (b) 284 m^2 (c) 400 m^2 (d) 464 m^2
8. The length of the diagonal of a square is $10\sqrt{2}$ cm. Its area is
(a) 200 cm^2 (b) 100 cm^2 (c) 150 cm^2 (d) $100\sqrt{2} \text{ cm}^2$
9. The area of a square field is 6050 m^2 . The length of its diagonal is
(a) 135 m (b) 120 m (c) 112 m (d) 110 m
10. The area of a square field is 0.5 hectare. The length of its diagonal is
(a) 150 m (b) $100\sqrt{2}$ m (c) 100 m (d) $50\sqrt{2}$ m
11. The area of an equilateral triangle is $4\sqrt{3} \text{ cm}^2$. Its perimeter is
(a) 9 cm (b) 12 cm (c) $12\sqrt{3}$ cm (d) $6\sqrt{3}$ cm
12. Each side of an equilateral triangle is 8 cm. Its area is
(a) 24 cm^2 (b) $24\sqrt{3} \text{ cm}^2$ (c) $16\sqrt{3} \text{ cm}^2$ (d) $8\sqrt{3} \text{ cm}^2$

13. Each side of an equilateral triangle is $6\sqrt{3}$ cm. The altitude of the triangle is
(a) 8 cm (b) 9 cm (c) $3\sqrt{3}$ cm (d) 6 cm
14. The height of an equilateral triangle is $3\sqrt{3}$ cm. Its area is
(a) $6\sqrt{3}$ cm² (b) 27 cm² (c) $9\sqrt{3}$ cm² (d) $27\sqrt{3}$ cm²
15. The base and height of a triangle are in the ratio 3 : 4 and its area is 216 cm². The height of the triangle is
(a) 18 cm (b) 24 cm (c) 21 cm (d) 28 cm
16. The lengths of the sides of a triangular field are 20 m, 21 m and 29 m. The cost of cultivating the field at ₹ 9 per m² is
(a) ₹ 2610 (b) ₹ 3780 (c) ₹ 1890 (d) ₹ 1800
17. The side of a square is equal to the side of an equilateral triangle. The ratio of their areas is
(a) 4 : 3 (b) $2 : \sqrt{3}$ (c) $4 : \sqrt{3}$ (d) none of these
18. The side of an equilateral triangle is equal to the radius of a circle whose area is 154 cm². The area of the triangle is
(a) 49 cm² (b) $\frac{49\sqrt{3}}{4}$ cm² (c) $\frac{7\sqrt{3}}{4}$ cm² (d) 77 cm²
19. The area of a rhombus is 480 cm² and the length of one of its diagonals is 20 cm. The length of each side of the rhombus is
(a) 24 cm (b) 30 cm (c) 26 cm (d) 28 cm
20. One side of a rhombus is 20 cm long and one of its diagonals measures 24 cm. The area of the rhombus is
(a) 192 cm² (b) 480 cm² (c) 240 cm² (d) 384 cm²

ANSWERS (MCQ)

1. (d) 2. (b) 3. (a) 4. (c) 5. (d) 6. (d) 7. (a) 8. (b) 9. (d) 10. (c)
11. (b) 12. (c) 13. (b) 14. (c) 15. (b) 16. (c) 17. (c) 18. (b) 19. (c) 20. (d)

HINTS TO SOME SELECTED QUESTIONS

1. Let the length be x metres. Then, breadth = $(x - 5)$ m.
 $\therefore x(x - 5) = 750 \Rightarrow x^2 - 5x - 750 = 0 \Rightarrow (x - 30)(x + 25) = 0 \Rightarrow x = 30$ [$\because x \neq -25$].
Hence, length = 30 m.
2. Let the breadth be x metres. Then, length = $(x + 23)$ m.
 $\therefore 2[(x + 23) + x] = 206 \Rightarrow 4x = 160 \Rightarrow x = 40$.
 $\therefore b = 40$ m and $l = 63$ m.
Hence, area = (63×40) m² = 2520 m².

3. $l = 12$ m and diagonal (d) = 15 m.

$$b^2 = (d^2 - l^2) = (15)^2 - (12)^2 = (15 + 12)(15 - 12) = 27 \times 3 = 81$$

$$\Rightarrow b = \sqrt{81} \text{ m} = 9 \text{ m}.$$

$$\therefore \text{area} = (12 \times 9) \text{ m}^2 = 108 \text{ m}^2.$$

4. Length of the carpet = $\frac{\text{total cost}}{\text{rate per m}} = \left(\frac{8400}{70}\right) \text{ m} = 120 \text{ m}.$

$$\text{Area of the room} = \text{area of the carpet} = \left(120 \times \frac{3}{4}\right) \text{ m}^2 = 90 \text{ m}^2.$$

$$\text{Width of the room} = \frac{\text{area}}{\text{length}} = \left(\frac{90}{15}\right) \text{ m} = 6 \text{ m}.$$

5. Let the breadth be x cm. Then, length = $3x$ cm.

$$\therefore \text{diagonal} = \sqrt{(3x)^2 + x^2} \text{ cm} = x\sqrt{10} \text{ cm}.$$

$$\text{Thus, } x\sqrt{10} = 8\sqrt{10} \Rightarrow x = 8.$$

$$\text{Perimeter} = 2(3x + x) \text{ cm} = 8x \text{ cm} = (8 \times 8) \text{ cm} = 64 \text{ cm}.$$

6. Let the length be l m and breadth be b m.

$$\text{Then, original area} = (l \times b) \text{ m}^2 = lb \text{ m}^2.$$

$$\text{New length} = \left(l \times \frac{120}{100}\right) \text{ m} = \frac{6l}{5} \text{ m, new breadth} = \left(b \times \frac{80}{100}\right) \text{ m} = \frac{4b}{5} \text{ m}.$$

$$\text{New area} = \left(\frac{24lb}{25}\right) \text{ m}^2 = 96 \text{ m}^2. \text{ Change in area} = \left(ab - \frac{24lb}{25}\right) \text{ m}^2 = \frac{lb}{25} \text{ m}^2.$$

$$\therefore \% \text{ change} = \frac{lb \times 100}{25 \times lb} \% = 4\%.$$

7. Area of the ground without path = $(80 \times 50) \text{ m}^2 = 4000 \text{ m}^2.$

$$\text{Area of the ground with path} = (82 \times 52) \text{ m}^2 = 4264 \text{ m}^2.$$

$$\text{Area of the path} = (4264 - 4000) \text{ m}^2 = 264 \text{ m}^2.$$

8. Area = $\frac{1}{2} \times (\text{diagonal})^2 = \frac{1}{2} \times (10\sqrt{2})^2 \text{ cm}^2 = 100 \text{ cm}^2.$

9. $\frac{1}{2} \times (\text{diagonal})^2 = 6050 \Rightarrow (\text{diagonal})^2 = 12100 = (110)^2 \Rightarrow \text{diagonal} = 110 \text{ m}.$

10. Area = $(0.5 \times 10000) \text{ m}^2 = 5000 \text{ m}^2.$

$$\frac{1}{2} \times d^2 = 5000 \Rightarrow d^2 = 10000 = (100)^2 \Rightarrow d = 100 \text{ m}.$$

11. $\frac{\sqrt{3}}{4} \times a^2 = 4\sqrt{3} \Rightarrow a^2 = 16 \Rightarrow a = 4.$

$$\therefore \text{perimeter} = 3a = (3 \times 4) \text{ cm} = 12 \text{ cm}.$$

13. Area of the triangle = $\left(\frac{\sqrt{3}}{4} \times a^2\right) = \left\{\frac{\sqrt{3}}{4} \times (6\sqrt{3})^2\right\} \text{ cm}^2$

$$= \left(\frac{\sqrt{3}}{4} \times 108\right) \text{ cm}^2 = (27\sqrt{3}) \text{ cm}^2.$$

$$\frac{1}{2} \times 6\sqrt{3} \times h = 27\sqrt{3} \Rightarrow h = \frac{27\sqrt{3}}{3\sqrt{3}} = 9 \text{ cm}.$$

$$14. \frac{1}{2} \times a \times h = \frac{\sqrt{3}}{4} a^2 \Rightarrow a = \frac{2h}{\sqrt{3}} = \left(\frac{2}{\sqrt{3}} \times 3\sqrt{3} \right) \text{ cm} = 6 \text{ cm}.$$

$$\therefore \text{area} = \left(\frac{\sqrt{3}}{4} \times 6 \times 6 \right) \text{ cm}^2 = 9\sqrt{3} \text{ cm}^2.$$

15. Let the base be $3x$ cm and height be $4x$ cm. Then,

$$\frac{1}{2} \times 3x \times 4x = 216 \Rightarrow x^2 = 36 \Rightarrow x = 6.$$

$$\therefore \text{height} = (4 \times 6) \text{ cm} = 24 \text{ cm}.$$

$$16. s = \frac{1}{2}(20 + 21 + 29) = 35, (s - a) = 15, (s - b) = 14 \text{ and } (s - c) = 6.$$

$$\therefore \text{area} = \sqrt{35 \times 15 \times 14 \times 6} \text{ m}^2 = (7 \times 5 \times 3 \times 2) \text{ m}^2 = 210 \text{ m}^2.$$

$$\text{Required cost} = ₹(210 \times 9) = ₹1890.$$

17. Let the side of the square = side of the triangle = a . Then,

$$\frac{\text{area of the square}}{\text{area of the triangle}} = \frac{a^2}{\frac{\sqrt{3}}{4} a^2} = \frac{4}{\sqrt{3}} = 4 : \sqrt{3}.$$

18. Area of the circle = 154 cm^2 .

$$\therefore \pi R^2 = 154 \Rightarrow \frac{22}{7} \times R^2 = 154$$

$$\Rightarrow R^2 = \left(154 \times \frac{7}{22} \right) = 7^2 \Rightarrow R = 7 \text{ cm}.$$

$$\therefore \text{side of the triangle} = \text{radius of the circle} = 7 \text{ cm}.$$

$$\text{Area of the triangle} = \left(\frac{\sqrt{3}}{4} \times 7 \times 7 \right) \text{ cm}^2 = \frac{49\sqrt{3}}{4} \text{ cm}^2.$$

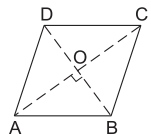
$$19. \frac{1}{2} \times d_1 \times d_2 = 480 \text{ cm}^2 \Rightarrow \frac{1}{2} \times 20 \times d_2 = 480 \text{ cm}^2 \Rightarrow d_2 = 48 \text{ cm}.$$

$$\therefore OB = 10 \text{ cm}, OA = 24 \text{ cm} \text{ and } \angle AOB = 90^\circ.$$

$$\therefore AB^2 = OA^2 + OB^2 = (24)^2 + (10)^2 = 576 + 100 = 676$$

$$\Rightarrow AB = \sqrt{676} \text{ cm} = 26 \text{ cm}.$$

Hence, each side of the rhombus is 26 cm.



20. Let $ABCD$ be the rhombus in which $AB = 20$ cm and

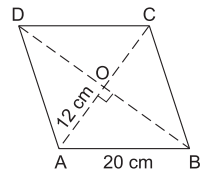
$$OA = \left(\frac{1}{2} \times 24 \right) \text{ cm} = 12 \text{ cm}.$$

$$\therefore OB^2 = AB^2 - OA^2 = (20)^2 - (12)^2 = 256 = (16)^2$$

$$\Rightarrow OB = 16 \text{ cm}.$$

$$\therefore BD = (2 \times 16) \text{ cm} = 32 \text{ cm}.$$

$$\therefore \text{area} = \left(\frac{1}{2} \times 24 \times 32 \right) \text{ cm}^2 = 384 \text{ cm}^2.$$



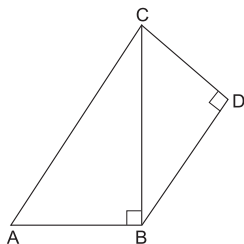
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TEST YOURSELF

MCQ

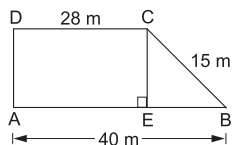
1. In the given figure $ABDC$ is a quadrilateral in which $\angle ABC = 90^\circ$, $\angle BDC = 90^\circ$, $AC = 17$ cm, $BC = 15$ cm, $BD = 12$ cm and $CD = 9$ cm. The area of quad. $ABDC$ is

- (a) 102 cm^2 (b) 114 cm^2
(c) 95 cm^2 (d) 57 cm^2



2. In the given figure $ABCD$ is a trapezium in which $AB = 40$ m, $BC = 15$ m, $CD = 28$ m, $AD = 9$ m and $CE \perp AB$. Area of trap. $ABCD$ is

- (a) 306 m^2 (b) 316 m^2
(c) 296 m^2 (d) 284 m^2

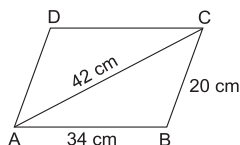


3. The sides of a triangle are in the ratio $12 : 14 : 25$ and its perimeter is 25.5 cm. The largest side of the triangle is
(a) 7 cm (b) 14 cm (c) 12.5 cm (d) 18 cm
4. The parallel sides of a trapezium are 9.7 cm and 6.3 cm, and the distance between them is 6.5 cm. The area of the trapezium is
(a) 104 cm^2 (b) 78 cm^2 (c) 52 cm^2 (d) 65 cm^2

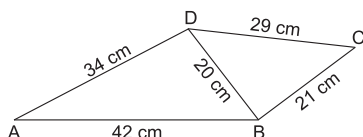
Short-Answer Questions

- Find the area of an equilateral triangle having each side of length 10 cm. [Take $\sqrt{3} = 1.732$.]
- Find the area of an isosceles triangle each of whose equal sides is 13 cm and whose base is 24 cm.
- The longer side of a rectangular hall is 24 m and the length of its diagonal is 26 m. Find the area of the hall.
- The length of the diagonal of a square is 24 cm. Find its area.
- Find the area of a rhombus whose diagonals are 48 cm and 20 cm long.
- Find the area of a triangle whose sides are 42 cm, 34 cm and 20 cm.
- A lawn is in the form of a rectangle whose sides are in the ratio $5 : 3$ and its area is 3375 m^2 . Find the cost of fencing the lawn at ₹ 20 per metre.
- Find the area of a rhombus each side of which measures 20 cm and one of whose diagonals is 24 cm.

13. Find the area of a trapezium whose parallel sides are 11 cm and 25 cm long and nonparallel sides are 15 cm and 13 cm.
14. The adjacent sides of a $\parallel\text{gm}$ $ABCD$ measure 34 cm and 20 cm and the diagonal AC is 42 cm long. Find the area of the $\parallel\text{gm}$.



15. The cost of fencing a square lawn at ₹ 14 per metre is ₹ 2800. Find the cost of mowing the lawn at ₹ 54 per 100 m^2 .
16. Find the area of quad. $ABCD$ in which $AB = 42$ cm, $BC = 21$ cm, $CD = 29$ cm, $DA = 34$ cm and diag. $BD = 20$ cm.



Long-Answer Questions

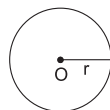
17. A parallelogram and a rhombus are equal in area. The diagonals of the rhombus measure 120 m and 44 m. If one of the sides of the $\parallel\text{gm}$ is 66 m long, find its corresponding altitude.
18. The diagonals of a rhombus are 48 cm and 20 cm long. Find the perimeter of the rhombus.
19. The adjacent sides of a parallelogram are 36 cm and 27 cm in length. If the distance between the shorter sides is 12 cm, find the distance between the longer sides.
20. In a four-sided field, the length of the longer diagonal is 128 m. The lengths of perpendiculars from the opposite vertices upon this diagonal are 22.7 m and 17.3 m. Find the area of the field.

ANSWERS (TEST YOURSELF)

- | | | | | |
|------------------------|------------------------|------------------------|------------------------|------------------------|
| 1. (b) | 2. (a) | 3. (c) | 4. (c) | 5. 43.3 cm^2 |
| 6. 60 cm^2 | 7. 240 m^2 | 8. 288 cm^2 | 9. 480 cm^2 | 10. 336 cm^2 |
| 11. ₹ 4800 | 12. 384 cm^2 | 13. 216 cm^2 | 14. 672 cm^2 | 15. ₹ 1350 |
| 16. 546 cm^2 | 17. 40 m | 18. 104 cm | 19. 9 cm | 20. 2560 m^2 |



CIRCLE The set of points which are at a constant distance of r units from a fixed point O is called a circle with *centre* O and *radius* $= r$ units. The circle is denoted by $C(O, r)$.



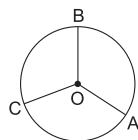
In other words, a circle is the locus of a point which moves in such a way that its distance from a fixed point O remains constant at r units.

The fixed point O is called the *centre* and the constant distance r units is called its *radius*.

CIRCUMFERENCE The perimeter (or length of boundary) of a circle is called its circumference.

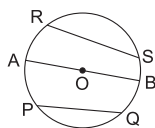
RADIUS A line segment joining the centre of a circle and a point on the circle is called a radius of the circle.

Plural of radius is *radii*. In the given figure, OA, OB, OC are three radii of the circle.



CHORD A line segment joining any two points on a circle is called a chord of the circle.

In the given figure, PQ, RS and AOB are three chords of a circle with centre O .



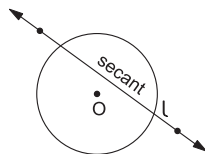
DIAMETER A chord of a circle passing through its centre is called a diameter of the circle.

Diameter is the longest chord of a circle. In the above figure, AOB is a diameter.

$$\text{Diameter} = 2 \times \text{radius.}$$

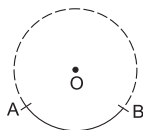
SECANT A line which intersects a circle at two points is called a secant of the circle.

In the given figure, line l is a secant of the circle with centre O .



ARC A continuous piece of a circle is called an arc of the circle.

In the given figure, AB is an arc of a circle, with centre O , denoted by $\widehat{AB}$. The remaining part of the circle, shown by the dotted lines, represents $\widehat{BA}$.



CENTRAL ANGLE An angle subtended by an arc at the centre of a circle is called its central angle.

In the given figure of a circle with centre O ,

central angle of $\widehat{AB} = \angle AOB = \theta^\circ$.

If $\theta^\circ < 180^\circ$ then $\widehat{AB}$ is called the *minor arc* and $\widehat{BA}$ is called the *major arc*.

SEMICIRCLE A diameter divides a circle into two equal arcs. Each of these two arcs is called a semicircle.

In the given figure of a circle with centre O , $\widehat{ACB}$ and $\widehat{BDA}$ are semicircles.

An arc whose length is less than the arc of a semicircle is called a *minor arc*. An arc whose length is more than the arc of a semicircle is called a *major arc*.

SEGMENT A segment of a circle is the region bounded by an arc and a chord, including the arc and the chord.

The segment containing the minor arc is called a *minor segment*, while the segment containing the major arc is the *major segment*.

The centre of the circle lies in the major segment.

SECTOR OF A CIRCLE The region enclosed by an arc of a circle and its two bounding radii is called a sector of the circle.

In the given figure, $OACBO$ is a sector of the circle with centre O .

If arc AB is a minor arc then $OACBO$ is called the *minor sector* of the circle.

The remaining part of the circle is called the *major sector* of the circle.

QUADRANT One-fourth of a circular disc is called a quadrant. The central angle of a quadrant is 90° .

FORMULAE ON AREA OF CIRCLE, SECTOR AND SEGMENT

1. For a circle of radius r , we have

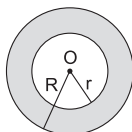
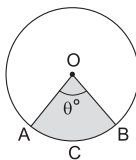
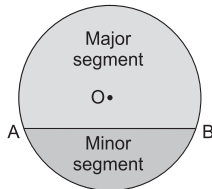
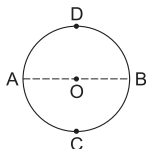
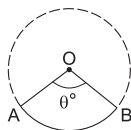
(i) Circumference $= 2\pi r$ (ii) Area $= \pi r^2$.

2. For a semicircle of radius r , we have

(i) Perimeter $= (\pi r + 2r)$ (ii) Area $= \frac{1}{2}\pi r^2$.

3. For a ring having outer radius $= R$ and inner radius $= r$, we have

Area of the ring $= \pi(R^2 - r^2)$.



4. Let an arc ACB make an angle θ° at the centre of a circle of radius r . Then, we have

(i) Length of minor arc $ACB = \frac{2\pi r\theta}{360}$.

Length of major arc $BDA = \left(2\pi r - \frac{2\pi r\theta}{360}\right)$.

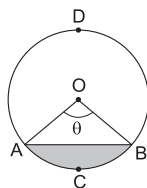
(ii) Area of minor sector $OACBO = \frac{\pi r^2\theta}{360} = \left(\frac{1}{2} \times \text{radius} \times \text{arc length}\right)$.

Area of major sector $OADBO = \left(\pi r^2 - \frac{\pi r^2\theta}{360}\right)$.

(iii) Area of minor segment $ACBA = \left(\frac{\pi r^2\theta}{360} - \frac{1}{2}r^2 \sin \theta\right)$.

Area of major segment $BDAB = [\pi r^2 - (\text{area of minor segment})]$.

(iv) Perimeter of sector $OACBO = \left(2r + \frac{2\pi r\theta}{360}\right)$.



5. For Rotation of the Hands of a Clock:

(i) Angle described by minute hand in 60 minutes = 360° .

(ii) Angle described by hour hand in 12 hours = 360° .

6. For Rotating Wheels:

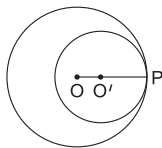
(i) Distance moved by a wheel in 1 rotation = its circumference.

(ii) Number of rotations in 1 minute = $\frac{\text{distance moved in 1 minute}}{\text{circumference}}$.

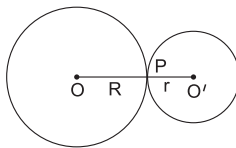
7. Touching Circles:

(i) When two circles touch internally [see fig. (i)], then distance between their centres = difference of their radii.

(ii) When two circles touch externally [see fig. (ii)], then distance between their centres = sum of their radii.



(i)



(ii)

SOLVED EXAMPLES

EXAMPLE 1 Find the circumference and area of a circle of diameter 28 cm.

SOLUTION Diameter = 28 cm $\Rightarrow$ radius $r = 14$ cm.

$$\begin{aligned}\therefore \text{circumference of the circle} &= 2\pi r \\ &= \left(2 \times \frac{22}{7} \times 14\right) \text{cm} = 88 \text{cm}.\end{aligned}$$

$$\begin{aligned}\text{Area of the circle} &= \pi r^2 = \left(\frac{22}{7} \times 14 \times 14\right) \text{cm}^2 \\ &= 616 \text{cm}^2.\end{aligned}$$

EXAMPLE 2 Find the area of a circle whose circumference is 66 cm.

SOLUTION Let the radius of the circle be r cm.

Then, its circumference = $(2\pi r)$ cm.

$$\begin{aligned}\therefore 2\pi r &= 66 \Rightarrow 2 \times \frac{22}{7} \times r = 66 \\ \Rightarrow r &= \left(66 \times \frac{7}{44}\right) = \frac{21}{2}.\end{aligned}$$

$$\begin{aligned}\therefore \text{area of the circle} &= \pi r^2 \text{cm}^2 = \left(\frac{22}{7} \times \frac{21}{2} \times \frac{21}{2}\right) \text{cm}^2 \\ &= 346.5 \text{cm}^2.\end{aligned}$$

EXAMPLE 3 A steel wire, when bent in the form of a square, encloses an area of 121 cm^2 . The same wire is bent in the form of a circle. Find the area of the circle.

SOLUTION Area of the square = 121 cm^2 .

Side of the square = $\sqrt{121} \text{ cm} = 11 \text{ cm}$.

Perimeter of the square = $4 \times 11 \text{ cm} = 44 \text{ cm}$.

$\therefore$ length of the wire = 44 cm.

$\therefore$ circumference of the circle = length of the wire = 44 cm.

Let the radius of the circle be r cm.

$$\text{Then, } 2\pi r = 44 \Rightarrow 2 \times \frac{22}{7} \times r = 44 \Rightarrow r = 7.$$

$$\begin{aligned}\therefore \text{area of the circle} &= \pi r^2 \\ &= \left(\frac{22}{7} \times 7 \times 7\right) \text{cm}^2 = 154 \text{cm}^2.\end{aligned}$$

EXAMPLE 4 A wire is looped in the form of a circle of radius 28 cm. It is rebent into a square form. Determine the length of the side of the square.

SOLUTION Length of the wire = circumference of the circle

$$= \left(2 \times \frac{22}{7} \times 28 \right) \text{ cm} = 176 \text{ cm.}$$

$\therefore$ perimeter of the square = length of the wire = 176 cm.

Hence, the side of the square = $\left(\frac{176}{4} \right) \text{ cm} = 44 \text{ cm.}$

EXAMPLE 5 A circular park, 42 m in diameter, has a path 3.5 m wide running round it on the outside. Find the cost of gravelling the path at ₹ 20 per m^2 .

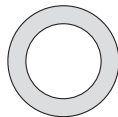
SOLUTION Radius of the circular park = 21 m.

Radius of the outer circle = 21 m + 3.5 m = 24.5 m.

Area of the path = $\pi \times [(24.5)^2 - (21)^2] \text{ m}^2$

$$= \frac{22}{7} \times (24.5 + 21)(24.5 - 21) \text{ m}^2$$

$$= \left(\frac{22}{7} \times 45.5 \times 3.5 \right) \text{ m}^2 = 500.5 \text{ m}^2.$$



$\therefore$ cost of gravelling the path = ₹ (500.5×20) = ₹ 10010.

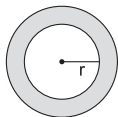
EXAMPLE 6 A road which is 7 m wide surrounds a circular park whose circumference is 352 m. Find the area of the road.

SOLUTION Let the radius of the park be r m.

Then, its circumference = $2\pi r$ m.

$$\therefore 2\pi r = 352 \Rightarrow 2 \times \frac{22}{7} \times r = 352$$

$$\Rightarrow r = 352 \times \frac{7}{44} = 56.$$



Thus, inner radius = 56 m,

outer radius = $(56 + 7) \text{ m} = 63 \text{ m.}$

Area of the road = $\pi[(63)^2 - (56)^2] \text{ m}^2$

$$= \frac{22}{7} \times (63 + 56)(63 - 56) \text{ m}^2$$

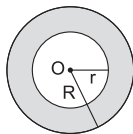
$$= \left(\frac{22}{7} \times 119 \times 7 \right) \text{ m}^2 = 2618 \text{ m}^2.$$

EXAMPLE 7 A racetrack is in the form of a ring whose inner and outer circumferences are 437 m and 503 m respectively. Find the width of the track and also its area.

SOLUTION Let r m and R m be the radii of inner and outer boundaries.

Then, $2\pi r = 437$ and $2\pi R = 503$

$$\Rightarrow r = \frac{437}{2\pi} \text{ and } R = \frac{503}{2\pi}.$$



Width of the track = $(R - r)$ m

$$\begin{aligned} &= \left(\frac{503}{2\pi} - \frac{437}{2\pi} \right) \text{ m} \\ &= \frac{1}{2\pi} \times (503 - 437) \text{ m} \\ &= \left(\frac{1}{2} \times \frac{7}{22} \times 66 \right) \text{ m} = 10.5 \text{ m}. \end{aligned}$$

Area of the track = $\pi(R^2 - r^2) \text{ m}^2 = \pi(R + r)(R - r) \text{ m}^2$

$$\begin{aligned} &= \left[\pi \left(\frac{503}{2\pi} + \frac{437}{2\pi} \right) \times 10.5 \right] \text{ m}^2 \\ &\quad [\because (R - r) = 10.5] \\ &= \left[\left(\pi \times \frac{940}{2\pi} \right) \times 10.5 \right] \text{ m}^2 = \left(470 \times \frac{21}{2} \right) \text{ m}^2 \\ &= (235 \times 21) \text{ m}^2 = 4935 \text{ m}^2. \end{aligned}$$

EXAMPLE 8 If the perimeter of a semicircular protractor is 36 cm, find its diameter.

SOLUTION Let the radius of the protractor be r cm.

Then, perimeter = $(\pi r + 2r)$ cm = $(\pi + 2)r$ cm

$$= \left(\frac{22}{7} + 2 \right) r \text{ cm} = \frac{36}{7} r \text{ cm}.$$

$$\therefore \frac{36}{7} r = 36 \Rightarrow r = \left(36 \times \frac{7}{36} \right) = 7.$$

Hence, diameter = $2r$ cm = 14 cm.

EXAMPLE 9 A bicycle wheel makes 5000 revolutions in moving 11 km. Find the diameter of the wheel.

SOLUTION Distance covered by the wheel in 1 revolution

$$= \left(\frac{11 \times 1000 \times 100}{5000} \right) \text{ cm} = 220 \text{ cm}.$$

$\therefore$ the circumference of the wheel = 220 cm.

Let the diameter of the wheel be d cm.

$$\text{Then, } \pi d = 220 \Rightarrow \frac{22}{7} \times d = 220 \Rightarrow d = 220 \times \frac{7}{22} = 70.$$

Hence, the diameter of the wheel is 70 cm.

EXAMPLE 10 The diameter of each wheel of a bus is 140 cm. How many revolutions per minute must a wheel make in order to move at a speed of 66 km per hour?

SOLUTION Distance covered by a wheel in 1 minute

$$= \left(\frac{66 \times 1000 \times 100}{60} \right) \text{ cm} = 110000 \text{ cm}.$$

$$\text{Circumference of a wheel} = \left(2 \times \frac{22}{7} \times 70 \right) \text{ cm} = 440 \text{ cm}.$$

$$\text{Number of revolutions in 1 min} = \left(\frac{110000}{440} \right) = 250.$$

EXAMPLE 11 Two circles touch externally. The sum of their areas is 130π sq cm and the distance between their centres is 14 cm. Find the radii of the circles.

SOLUTION Since the given circles touch externally, we have

sum of their radii = distance between their centres = 14 cm.

Let the radii of the given circles be x cm and $(14 - x)$ cm.

$$\text{Sum of their areas} = [\pi x^2 + \pi(14 - x)^2] \text{ cm}^2.$$

$$\therefore \pi x^2 + \pi(14 - x)^2 = 130\pi$$

$$\Rightarrow x^2 + (14 - x)^2 = 130$$

$$\Rightarrow 2x^2 - 28x + 66 = 0$$

$$\Rightarrow x^2 - 14x + 33 = 0$$

$$\Rightarrow (x - 11)(x - 3) = 0$$

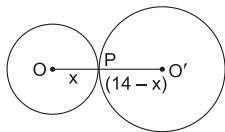
$$\Rightarrow x - 11 = 0 \text{ or } x - 3 = 0$$

$$\Rightarrow x = 11 \text{ or } x = 3.$$

$$\text{Now, } x = 11 \Rightarrow (14 - x) = (14 - 11) = 3.$$

$$\text{And, } x = 3 \Rightarrow (14 - x) = (14 - 3) = 11.$$

Hence, the radii of the given circles are 11 cm and 3 cm.



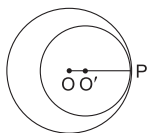
EXAMPLE 12 Two circles touch internally. The sum of their areas is (116π) cm² and the distance between their centres is 6 cm. Find the radii of the circles.

[CBSE 2017]

SOLUTION The circles touch internally.

$\therefore$ difference of their radii

= distance between their centres = 6 cm.



Let the radii of given circles be r cm and $(r + 6)$ cm.

Sum of their areas = $[\pi r^2 + \pi(r + 6)^2]$ cm² = $\pi[r^2 + (r + 6)^2]$ cm².

$$\therefore \pi[r^2 + (r + 6)^2] = 116\pi$$

$$\Rightarrow r^2 + (r + 6)^2 = 116$$

$$\Rightarrow 2r^2 + 12r - 80 = 0$$

$$\Rightarrow r^2 + 6r - 40 = 0$$

$$\Rightarrow (r + 10)(r - 4) = 0$$

$$\Rightarrow r + 10 = 0 \text{ or } r - 4 = 0$$

$$\Rightarrow r = 4 \text{ [neglecting } r = -10, \text{ as radius cannot be negative]}$$

$\therefore$ the radii of the given circles are 4 cm and 10 cm.

EXAMPLE 13 Find the area of a right-angled triangle, if the radius of its circum-circle is 7.5 cm and the altitude drawn to the hypotenuse is 6 cm long.

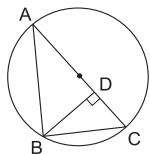
SOLUTION Let $\triangle ABC$ be right-angled at B .

Hypotenuse AC

= diameter of its circumcircle

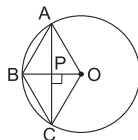
$$= (2 \times 7.5) \text{ cm} = 15 \text{ cm.}$$

Let $BD \perp AC$. Then, $BD = 6$ cm.



$$\therefore \text{ar}(\triangle ABC) = \left(\frac{1}{2} AC \times BD \right) = \left(\frac{1}{2} \times 15 \times 6 \right) \text{ cm}^2 = 45 \text{ cm}^2.$$

EXAMPLE 14 In the given figure, $OABC$ is a rhombus whose three vertices A, B, C lie on a circle of radius 10 cm and centre O . Find the area of the rhombus. [Take $\sqrt{3} = 1.732$.]



SOLUTION Clearly, $OA = OB = OC = 10$ cm. Let OB and AC intersect at P .

Since the diagonals of a rhombus bisect each other at right angles, we have $OP = 5$ cm and $\angle OPC = 90^\circ$.

Now, $AC = 2CP$ and

$$CP = \sqrt{OC^2 - OP^2} \text{ cm} = \sqrt{(10)^2 - 5^2} \text{ cm}$$

$$= \sqrt{75} \text{ cm} = 5\sqrt{3} \text{ cm.}$$

$$\therefore AC = (2 \times 5\sqrt{3}) \text{ cm} = 10\sqrt{3} \text{ cm}$$

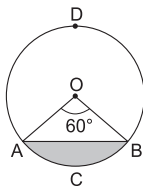
$$= (10 \times 1.732) \text{ cm} = 17.32 \text{ cm.}$$

$$\begin{aligned}
 \therefore \text{ar}(\text{rhombus } OABC) &= \left(\frac{1}{2} \times OB \times AC \right) \\
 &= \left(\frac{1}{2} \times 10 \times 17.32 \right) \text{cm}^2 \\
 &= 86.6 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 15 In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find (i) the length of the arc, (ii) the area of the sector, (iii) the area of the minor segment, and (iv) the area of the major segment. [Given, $\sqrt{3} = 1.72$.]

SOLUTION Let ACB be the given arc subtending an angle of 60° at the centre. Then, $r = 21$ cm and $\theta = 60^\circ$.

$$\begin{aligned}
 \text{(i) Length of the arc } ACB &= \frac{2\pi r\theta}{360} \\
 &= \left(2 \times \frac{22}{7} \times 21 \times \frac{60}{360} \right) \text{cm} \\
 &= 22 \text{ cm.}
 \end{aligned}$$



$$\begin{aligned}
 \text{(ii) Area of the sector } OACBO &= \frac{\pi r^2 \theta}{360} \\
 &= \left(\frac{22}{7} \times 21 \times 21 \times \frac{60}{360} \right) \text{cm}^2 \\
 &= 231 \text{ cm}^2.
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) Area of the minor segment } ACBA &= (\text{area of the sector } OACB) - (\text{area of the } \triangle OAB) \\
 &= \left(231 - \frac{1}{2} r^2 \sin \theta \right) \text{cm}^2 = \left[231 - \left(\frac{1}{2} \times 21 \times 21 \times \sin 60^\circ \right) \right] \text{cm}^2 \\
 &= \left(231 - \frac{1}{2} \times 21 \times 21 \times \frac{\sqrt{3}}{2} \right) \text{cm}^2 \\
 &= \left(231 - \frac{1}{4} \times 441 \times 1.72 \right) \text{cm}^2 \\
 &= (231 - 189.63) \text{cm}^2 = 41.37 \text{ cm}^2.
 \end{aligned}$$

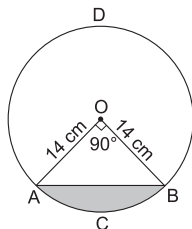
$$\begin{aligned}
 \text{(iv) Area of the major segment } BDAB &= (\text{area of the circle}) - (\text{area of the minor segment}) \\
 &= \left\{ \left(\frac{22}{7} \times 21 \times 21 \right) - 41.37 \right\} \text{cm}^2 \\
 &= (1386 - 41.37) \text{cm}^2 = 1344.63 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 16 A chord of a circle of radius 14 cm makes a right angle at the centre. Find the areas of the minor and the major segments of the circle.

SOLUTION Let AB be the chord of a circle of centre O and radius = 14 cm such that $\angle AOB = 90^\circ$.

$$\begin{aligned}\therefore \text{ area of the sector } OACBO &= \frac{\pi r^2 \theta}{360} \text{ cm}^2 \\ &= \left(\frac{22}{7} \times 14 \times 14 \times \frac{90}{360} \right) \text{ cm}^2 \\ &= 154 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}\text{Area of } \triangle OAB &= \frac{1}{2} r^2 \sin \theta \\ &= \left(\frac{1}{2} \times 14 \times 14 \times \sin 90^\circ \right) \text{ cm}^2 \\ &= 98 \text{ cm}^2.\end{aligned}$$



$$\begin{aligned}\text{Area of the minor segment } ACBA &= (\text{area of the sector } OACBO) - (\text{area of the } \triangle OAB) \\ &= (154 - 98) \text{ cm}^2 = 56 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}\text{Area of the major segment } BDAB &= (\text{area of the circle}) - (\text{area of the minor segment}) \\ &= \left[\left(\frac{22}{7} \times 14 \times 14 \right) - 56 \right] \text{ cm}^2 = (616 - 56) \text{ cm}^2 = 560 \text{ cm}^2.\end{aligned}$$

EXAMPLE 17 The perimeter of a sector of a circle of radius 14 cm is 68 cm. Find the area of the sector.

SOLUTION Let O be the centre of a circle of radius 14 cm.

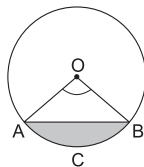
Let $OACBO$ be the sector whose perimeter is 68 cm.

$$\text{Then, } OA + OB + \text{arc } ACB = 68 \text{ cm}$$

$$\Rightarrow 14 \text{ cm} + 14 \text{ cm} + \text{arc } ACB = 68 \text{ cm}$$

$$\Rightarrow \text{arc } ACB = (68 - 28) \text{ cm} = 40 \text{ cm}.$$

$$\begin{aligned}\therefore \text{ ar(sector } OACBO) &= \left(\frac{1}{2} \times \text{radius} \times \text{arc length} \right) \\ &= \left(\frac{1}{2} \times 14 \times 40 \right) \text{ cm}^2 \\ &= 280 \text{ cm}^2.\end{aligned}$$



EXAMPLE 18 *The minute hand of a clock is 12 cm long. Find the area of the face of the clock described by the minute hand in 35 minutes.*

SOLUTION Angle described by the minute hand in 60 minutes = 360° .

Angle described by the minute hand in 35 minutes

$$= \left(\frac{360}{60} \times 35 \right)^\circ = 210^\circ.$$

$\therefore \theta = 210^\circ$ and $r = 12$ cm.

Area swept by the minute hand in 35 minutes

$$= \left(\frac{\pi r^2 \theta}{360} \right) = \left(\frac{22}{7} \times 12 \times 12 \times \frac{210}{360} \right) \text{ cm}^2 = 264 \text{ cm}^2.$$

EXAMPLE 19 *A car has wheels which are 80 cm in diameter. How many complete revolutions does each wheel make in 10 minutes, when the car is travelling at a speed of 66 km an hour?*

SOLUTION Distance covered by wheel in 1 minute

$$= \left(\frac{66 \times 1000 \times 100}{60} \right) \text{ cm} = 110000 \text{ cm}.$$

Circumference of a wheel = $2\pi R = \left(2 \times \frac{22}{7} \times 40 \right) \text{ cm}$

$$= \frac{1760}{7} \text{ cm}.$$

Number of revolutions made by a wheel in 1 minute

$$= \left(110000 \times \frac{7}{1760} \right) \text{ cm}.$$

Number of revolutions made by a wheel in 10 minutes

$$= \left(110000 \times \frac{7}{1760} \times 10 \right) = 4375.$$

EXAMPLE 20 *A car has two wipers which do not overlap. Each wiper has a blade of length 21 cm, sweeping through an angle of 120° . Find the total area cleaned at each sweep of the blades.*

SOLUTION Clearly, each wiper sweeps a sector of a circle of radius 21 cm, making an angle of 120° at the centre of the circle.

So, the required area swept by two wipers

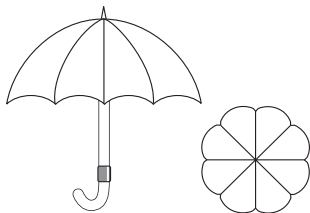
$$\begin{aligned} &= \left(2 \times \frac{\pi r^2 \theta}{360} \right) = \left(2 \times \frac{22}{7} \times 21 \times 21 \times \frac{120}{360} \right) \text{ cm}^2 \\ &= 924 \text{ cm}^2. \end{aligned}$$

EXAMPLE 21 To warn ships for underwater rocks, a lighthouse spreads a red-coloured light over a sector of angle 72° to a distance of 15 km. Find the area of the sea over which the ships are warned. [Use $\pi = 3.14$.]

SOLUTION Required area = area of the sector in which, $r = 15$ km
and $\theta = 72^\circ$

$$\begin{aligned} &= \left(\frac{\pi r^2 \theta}{360} \right) = \left(3.14 \times 15 \times 15 \times \frac{72}{360} \right) \text{ km}^2 \\ &= \left(\frac{314 \times 45}{100} \right) \text{ km}^2 = \frac{1413}{10} \text{ km}^2 \\ &= 141.3 \text{ km}^2. \end{aligned}$$

EXAMPLE 22 An umbrella has 8 ribs which are equally spaced (as shown in the figure). Assuming the umbrella to be a flat circle of radius 42 cm, find the area between the two consecutive ribs of the umbrella.

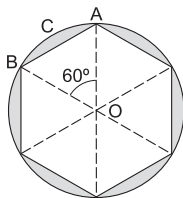


SOLUTION Since the ribs are equally spaced, so the angle made by two consecutive ribs at the centre $= \left(\frac{360}{8} \right)^\circ = 45^\circ$.

Area between two consecutive ribs

$$\begin{aligned} &= \text{area of a sector of a circle with } r = 42 \text{ cm and } \theta = 45^\circ \\ &= \left(\frac{45}{360} \times \frac{22}{7} \times 42 \times 42 \right) \text{ cm}^2 \\ &= 693 \text{ cm}^2. \end{aligned}$$

EXAMPLE 23 A round table cover has six equal designs, as shown in the figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of ₹ 0.35 per cm^2 . [Use $\sqrt{3} = 1.7$.] [CBSE 2009]



SOLUTION

Let O be the centre of the table cover and let it be divided into six equal designs, each being a segment. Let one of these segments be $ACBA$. Clearly, $\angle AOB = \left(\frac{360}{6}\right)^\circ = 60^\circ$.

$\text{ar}(\text{segment } ACBA) = \text{ar}(\text{sector } OACBO) - \text{ar}(\text{equilateral } \triangle OAB)$

$$\begin{aligned}
 &= \left(\frac{\pi r^2 \theta}{360} - \frac{\sqrt{3}}{4} a^2 \right) \\
 &= \left[\left(\frac{22}{7} \times 28 \times 28 \times \frac{60}{360} \right) - \left(\frac{\sqrt{3}}{4} \times 28 \times 28 \right) \right] \text{cm}^2 \\
 &\quad [\because a = OA = 28 \text{ cm}] \\
 &= \left(\frac{1232}{3} - \frac{17}{10} \times 7 \times 28 \right) \text{cm}^2 \\
 &= \left(\frac{1232}{3} - \frac{1666}{5} \right) \text{cm}^2 \\
 &= \frac{(6160 - 4998)}{15} \text{cm}^2 \\
 &= \frac{1162}{15} \text{cm}^2.
 \end{aligned}$$

$$\text{ar}(\text{all the six segments}) = \left(\frac{1162}{15} \times 6 \right) \text{cm}^2 = \frac{2324}{5} \text{cm}^2.$$

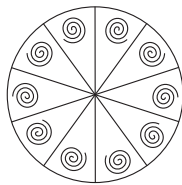
$$\text{Cost of designs} = ₹ \left(\frac{2324}{5} \times \frac{35}{100} \right) = ₹ 162.68.$$

EXAMPLE 24

A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into ten equal sectors, as shown in the figure.

Find (i) the total length of the silver wire required,

(ii) the area of each sector of the brooch.



SOLUTION

(i) Total length of the wire required

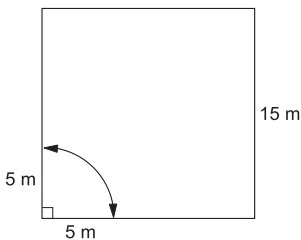
$$\begin{aligned}
 &= (\text{circumference of given circle}) \\
 &\quad + (\text{length of five diameters}) \\
 &= \left(2 \times \frac{22}{7} \times \frac{35}{2} \right) \text{mm} + (5 \times 35) \text{mm} = (110 + 175) \text{mm} \\
 &= 285 \text{mm} = 28.5 \text{cm}.
 \end{aligned}$$

(ii) The given circle has been divided into 10 equal sectors.

$$\begin{aligned}
 \text{Area of each sector} &= \frac{1}{10} \times \text{area of the given circle} \\
 &= \left(\frac{1}{10} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} \right) \text{ mm}^2 \\
 &= \frac{385}{4} \text{ mm}^2 = 96.25 \text{ mm}^2.
 \end{aligned}$$

EXAMPLE 25 A horse is tied to a peg at one corner of a square-shaped grass field of side 15 m by means of a 5-m-long rope (as shown in the figure).

Find (i) the area of that part of the field in which the horse can graze,
 (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. [Use $\pi = 3.14$.]



SOLUTION

(i) Clearly, the required area is the area of a quadrant of a circle of radius 5 m.

$$\therefore \text{required area} = \frac{1}{4} \times \pi r^2 = \left(\frac{1}{4} \times 3.14 \times 5^2 \right) \text{ m}^2 = 19.625 \text{ m}^2.$$

(ii) Let the length of the rope be 10 m. Then,

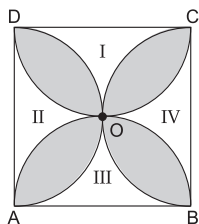
$$\text{area grazed} = \frac{1}{4} \times \pi R^2 = \left(\frac{1}{4} \times 3.14 \times 10^2 \right) \text{ m}^2 = 78.50 \text{ m}^2.$$

$$\begin{aligned}
 \text{Increase in grazing area} &= (78.50 - 19.625) \text{ m}^2 \\
 &= 58.875 \text{ m}^2.
 \end{aligned}$$

AREAS OF COMBINATIONS OF PLANE FIGURES

SOLVED EXAMPLES

EXAMPLE 1 In the given figure, ABCD is a square of side 10 cm and semicircles are drawn with each side of the square as diameter. Find the area of the shaded region. [Use $\pi = 3.14$.]



SOLUTION Let us mark the unshaded regions as I, II, III and IV as shown in the given figure.

Let these regions meet at a common point O . Then,

$$\begin{aligned}
 & (\text{area of I}) + (\text{area of III}) \\
 &= \text{ar(sq } ABCD) - \{\text{ar(semicircle } AOD) \\
 &\quad + \text{ar(semicircle } BOC)\} \\
 &= \left\{ (10 \times 10) - \left(\frac{1}{2} \times 3.14 \times 5^2 + \frac{1}{2} \times 3.14 \times 5^2 \right) \right\} \text{ cm}^2 \\
 &= (100 - 78.5) \text{ cm}^2 = 21.5 \text{ cm}^2.
 \end{aligned}$$

Similarly, $(\text{area of II}) + (\text{area of IV}) = 21.5 \text{ cm}^2$.

$$\begin{aligned}
 \text{Area of shaded region} &= \text{ar(sq } ABCD) - \text{ar(I + II + III + IV)} \\
 &= \{(10 \times 10) - (2 \times 21.5)\} \text{ cm}^2 \\
 &= (100 - 43) \text{ cm}^2 = 57 \text{ cm}^2.
 \end{aligned}$$

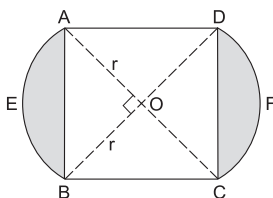
EXAMPLE 2

In the given figure, two circular flower beds have been shown on two sides of a square lawn $ABCD$ of side $AB = 42$ m. If the centre of each circular flower bed is the point of intersection O of the diagonals of the square lawn, find

(i) *the sum of the areas of the lawn and the flower beds,*

(ii) *the sum of the areas of two flower beds.*

[CBSE 2015]



SOLUTION

Area of the square lawn $ABCD = (42 \times 42) \text{ m}^2$.

Let $OA = OB = r$ metres. Then, $\angle AOB = 90^\circ$.

$$\begin{aligned}
 \therefore OA^2 + OB^2 &= AB^2 \Rightarrow r^2 + r^2 = (42)^2 \\
 &\Rightarrow 2r^2 = 1764 \Rightarrow r^2 = 882.
 \end{aligned}$$

(i) Sum of the areas of the lawn and the flower beds

$$\begin{aligned}
 &= \text{ar(sector } OAEBO) + \text{ar(sector } OCFDO) \\
 &\quad + \text{ar}(\triangle OAD) + \text{ar}(\triangle OBC) \\
 &= \left(\frac{22}{7} \times 882 \times \frac{90}{360} \right) \text{ m}^2 + \left(\frac{22}{7} \times 882 \times \frac{90}{360} \right) \text{ m}^2 \\
 &\quad + \left(\frac{1}{2} \times r \times r \right) \text{ m}^2 + \left(\frac{1}{2} \times r \times r \right) \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 &= 2 \left\{ \left(\frac{22}{7} \times 882 \times \frac{90}{360} \right) + \frac{1}{2} r^2 \right\} \text{ m}^2 \\
 &= 2 \left(693 + \frac{1}{2} \times 882 \right) \text{ m}^2 = 2(693 + 441) \text{ m}^2 \\
 &= 2268 \text{ m}^2.
 \end{aligned}$$

Hence, the sum of the areas of the lawn and the two flower beds is 2268 m^2 .

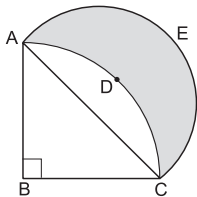
(ii) Sum of the areas of the two flower beds

$$\begin{aligned}
 &= 2 \{ \text{ar}(\text{sector } OAEBO) - \text{ar}(\triangle OAB) \} \\
 &= 2 \left\{ \left(\frac{22}{7} \times 882 \times \frac{90}{360} \right) - \frac{1}{2} r^2 \right\} \text{ m}^2 \\
 &= 2 \left(693 - \frac{1}{2} \times 882 \right) \text{ m}^2 = 2(693 - 441) \text{ m}^2 \\
 &= 504 \text{ m}^2.
 \end{aligned}$$

EXAMPLE 3

In the given figure, $ABCD$ is a quadrant of a circle of radius 28 cm and a semicircle $AECA$ is drawn with AC as diameter. Find the area of the shaded region.

[CBSE 2007, '08, '12, '14]



SOLUTION

Since $ABCD$ is a quadrant of a circle, we have $\angle ABC = 90^\circ$.

$$\begin{aligned}
 \therefore AC^2 &= AB^2 + BC^2 = [(28)^2 + (28)^2] \text{ cm}^2 \\
 &= (784 + 784) \text{ cm}^2 = 1568 \text{ cm}^2
 \end{aligned}$$

$$\Rightarrow AC = \sqrt{1568} \text{ cm} = 28\sqrt{2} \text{ cm}.$$

$$\therefore \text{radius of the semicircle} = \frac{1}{2} AC = 14\sqrt{2} \text{ cm}.$$

Area of the shaded region

$$\begin{aligned}
 &= \text{ar}(\triangle ABC) + \text{ar}(\text{semicircle } ACEA) - \text{ar}(\text{quadrant } BCDA) \\
 &= \left[\left(\frac{1}{2} \times 28 \times 28 \right) + \left\{ \frac{1}{2} \times \frac{22}{7} \times (14\sqrt{2})^2 \right\} \right. \\
 &\quad \left. - \left\{ \frac{22}{7} \times 28 \times 28 \times \frac{90}{360} \right\} \right] \text{ cm}^2 \\
 &= (392 + 616 - 616) \text{ cm}^2 = 392 \text{ cm}^2.
 \end{aligned}$$

Hence, the area of the shaded region is 392 cm^2 .

EXAMPLE 4

Find the area of the shaded region in the given figure, where a circular arc of radius 6 cm has been drawn with vertex O of an equilateral triangle OAB of side 12 cm as centre.

[Take $\sqrt{3} = 1.73$ and $\pi = 3.14$]

**SOLUTION**

Since $\triangle OAB$ is equilateral, we have $\angle AOB = 60^\circ$.

Required area

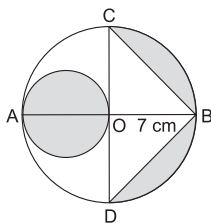
$$\begin{aligned}
 &= \{(\text{area of equilateral } \triangle OAB) + (\text{area of circle with } r = 6 \text{ cm})\} \\
 &\quad - (\text{area of sector of a circle with } r = 6 \text{ cm and } \theta = 60^\circ) \\
 &= \left\{ \left(\frac{\sqrt{3}}{4} \times 12 \times 12 \right) + (\pi \times 6 \times 6) - \left(\pi \times 6 \times 6 \times \frac{60}{360} \right) \right\} \text{ cm}^2 \\
 &= \{(1.73 \times 36) + 36\pi - 6\pi\} \text{ cm}^2 = \{(1.73 \times 36) + (30\pi)\} \text{ cm}^2 \\
 &= \left\{ \left(\frac{173}{100} \times 36 \right) + \left(30 \times \frac{314}{100} \right) \right\} \text{ cm}^2 = \left(\frac{1557}{25} + \frac{471}{5} \right) \text{ cm}^2 \\
 &= \left(\frac{1557 + 2355}{25} \right) \text{ cm}^2 = \frac{3912}{25} \text{ cm}^2 = 156.48 \text{ cm}^2.
 \end{aligned}$$

Hence, the area of the shaded region is 156.48 cm^2 .

EXAMPLE 5

In the given figure, AB and CD are the diameters of a circle with centre O , perpendicular to each other. OA is the diameter of the smaller circle. If $OB = 7 \text{ cm}$, find the area of the shaded region.

[CBSE 2010, '13]

**SOLUTION**

Clearly, the diameter of the larger circle is 14 cm and the diameter of the smaller circle is 7 cm.

So, the radius of the larger circle is 7 cm and that of the smaller circle is 3.5 cm.

Area of the shaded region

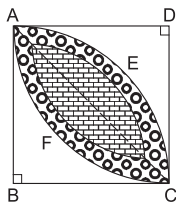
$$\begin{aligned}
 &= \{(\text{area of smaller circle}) + (\text{area of larger semicircle})\} \\
 &\quad - (\text{area of } \triangle CBD)
 \end{aligned}$$

$$\begin{aligned}
 &= \left[\left\{ \pi \times \left(\frac{7}{2} \right)^2 \right\} + \left\{ \frac{1}{2} \times \pi \times 7 \times 7 \right\} - \left\{ \frac{1}{2} \times CD \times OB \right\} \right] \text{cm}^2 \\
 &= \left\{ \left(\frac{22}{7} \times \frac{49}{4} \right) + \left(\frac{1}{2} \times \frac{22}{7} \times 49 \right) - \left(\frac{1}{2} \times 14 \times 7 \right) \right\} \text{cm}^2 \\
 &= \left(\frac{77}{2} + 77 - 49 \right) \text{cm}^2 = (38.5 + 28) \text{cm}^2 = 66.5 \text{cm}^2.
 \end{aligned}$$

Hence, the area of the shaded region is 66.5 cm^2 .

EXAMPLE 6

Calculate the area of the designed region in the given figure, common between two quadrants of circles of radius 7 cm each.

**SOLUTION**

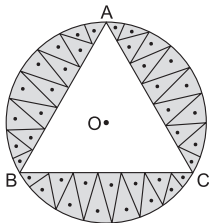
Area of the designed region

$$\begin{aligned}
 &= 2[\text{ar}(\text{quadrant } ABCE) - \text{ar}(\triangle ABC)] \\
 &= 2 \left[\frac{1}{4} \pi \times (7)^2 - \frac{1}{2} \times 7 \times 7 \right] \text{cm}^2 \\
 &= 2 \left[\left(\frac{1}{4} \times \frac{22}{7} \times 49 \right) - \left(\frac{49}{2} \right) \right] \text{cm}^2 \\
 &= 2 \left(\frac{77}{2} - \frac{49}{2} \right) \text{cm}^2 = \left(2 \times \frac{28}{2} \right) \text{cm}^2 = 28 \text{cm}^2.
 \end{aligned}$$

Hence, the area of the designed region is 28 cm^2 .

EXAMPLE 7

In a circular table cover of radius 42 cm, a design is formed, leaving an equilateral triangle ABC in the middle, as shown in the figure. Find the area of the design. [Use $\sqrt{3} = 1.73$.] [CBSE 2013C]

**SOLUTION**

Let O be the centre of the circle. Join BO and CO.

Now, $\angle BAC = 60^\circ \Rightarrow \angle BOC = 2(\angle BAC) = 120^\circ$.

Draw $OD \perp BC$. Then, $\angle BOD = \angle COD = 60^\circ$.

From right $\triangle BDO$, we have

$$\frac{OD}{OB} = \cos 60^\circ = \frac{1}{2} \Rightarrow \frac{OD}{42 \text{ cm}} = \frac{1}{2}$$

$$\Rightarrow OD = \left(42 \times \frac{1}{2} \right) \text{ cm} = 21 \text{ cm},$$

$$\text{and } \frac{BD}{OB} = \sin 60^\circ = \frac{\sqrt{3}}{2} \Rightarrow \frac{BD}{42 \text{ cm}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow BD = \left(42 \times \frac{\sqrt{3}}{2} \right) \text{ cm} = 21\sqrt{3} \text{ cm}.$$

$$\therefore BC = 2 \times BD = (2 \times 21\sqrt{3}) \text{ cm} = (42\sqrt{3}) \text{ cm}.$$

Area of the designed region

$$= (\text{area of the circle with } r = 42 \text{ cm})$$

$$- (\text{area of equilateral } \triangle ABC \text{ with } a = 42\sqrt{3} \text{ cm})$$

$$= \left(\pi r^2 - \frac{\sqrt{3}}{4} a^2 \right) = \left\{ \left(\frac{22}{7} \times 42 \times 42 \right) - \left(\frac{1.73}{4} \times (42\sqrt{3})^2 \right) \right\} \text{ cm}^2$$

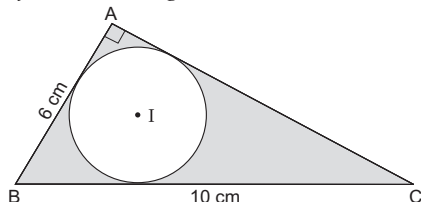
$$= \left\{ 5544 - \left(\frac{1.73}{4} \times 1764 \times 3 \right) \right\} \text{ cm}^2 = (5544 - 2288.79) \text{ cm}^2$$

$$= 3255.21 \text{ cm}^2.$$

Hence, the required area is 3255.21 cm^2 .

EXAMPLE 8

In the given figure, ABC is a right-angled triangle, right-angled at A , in which $AB = 6 \text{ cm}$, $BC = 10 \text{ cm}$ and I is the incentre of $\triangle ABC$. Find the area of the shaded region. [Take $\pi = 3.14$.] [CBSE 2009]



SOLUTION

From right $\triangle BAC$, we have

$$AC^2 = BC^2 - AB^2 = \{(10)^2 - 6^2\} \text{ cm}^2$$

$$= (100 - 36) \text{ cm}^2 = 64 \text{ cm}^2$$

$$\Rightarrow AC = \sqrt{64} \text{ cm} = 8 \text{ cm}.$$

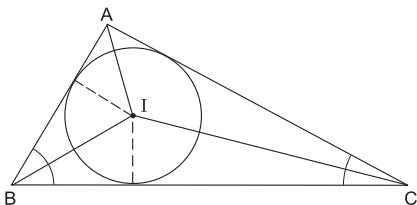
$$\text{ar}(\triangle ABC) = \left(\frac{1}{2} \times \text{base} \times \text{height} \right) = \left(\frac{1}{2} \times AB \times AC \right)$$

$$= \left(\frac{1}{2} \times 6 \times 8 \right) \text{ cm}^2 = 24 \text{ cm}^2.$$

Let the radius of the incircle be r . Then,

$$\text{ar}(\triangle ABC) = \text{ar}(\triangle IAB) + \text{ar}(\triangle IBC) + \text{ar}(\triangle ICA)$$

$$\begin{aligned} \Rightarrow 24 \text{ cm}^2 &= \left(\frac{1}{2} \times AB \times r \right) + \left(\frac{1}{2} \times BC \times r \right) + \left(\frac{1}{2} \times CA \times r \right) \\ &= \frac{1}{2} r (AB + BC + CA) = \frac{1}{2} r (6 \text{ cm} + 10 \text{ cm} + 8 \text{ cm}) = (12 \text{ cm})r \\ \Rightarrow r &= 2 \text{ cm}. \end{aligned}$$



$\therefore$ area of the shaded region

$$\begin{aligned} &= \text{ar}(\triangle ABC) - (\text{area of incircle with } r = 2 \text{ cm}) \\ &= (24 - \pi r^2) \text{ cm}^2 = (24 - 3.14 \times 2 \times 2) \text{ cm}^2 \\ &= (24 - 12.56) \text{ cm}^2 = 11.44 \text{ cm}^2. \end{aligned}$$

Hence, the required area is 11.44 cm^2 .

EXAMPLE 9

In an equilateral triangle of side 12 cm, a circle is inscribed touching its sides. Find the area of the portion of the triangle not included in the circle. [Take $\sqrt{3} = 1.73$ and $\pi = 3.14$]

SOLUTION

Let ABC be an equilateral triangle of side 12 cm.

Let $AD \perp BC$. Then, D is the midpoint of BC .

$$\therefore BD = DC = 6 \text{ cm and } AB = 12 \text{ cm}.$$

$$\therefore AD = \sqrt{AB^2 - BD^2} \text{ [by Pythagoras' theorem]}$$

$$= \sqrt{(12)^2 - 6^2} \text{ cm} = \sqrt{(144 - 36)} \text{ cm}$$

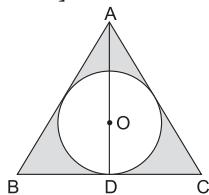
$$= \sqrt{108} \text{ cm} = 6\sqrt{3} \text{ cm}.$$

Let O be the centre of the inscribed circle.

Then, O is the centroid of $\triangle ABC$.

$$\therefore AO : OD = 2 : 1 \text{ and } OD = \frac{1}{3} AD = \frac{1}{3} \times 6\sqrt{3} \text{ cm} = 2\sqrt{3} \text{ cm}.$$

$$\therefore r = OD = 2\sqrt{3} \text{ cm}.$$

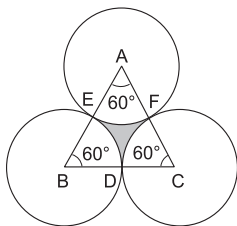


$$\begin{aligned}
 \text{Required area} &= \text{area of the shaded region} \\
 &= \text{ar}(\triangle ABC) - (\text{area of the incircle}) \\
 &= \left[\left(\frac{\sqrt{3}}{4} \times 12 \times 12 \right) - \{3.14 \times (2\sqrt{3})^2\} \right] \text{cm}^2 \\
 &= [36\sqrt{3} - (3.14 \times 12)] \text{cm}^2 \\
 &= (36 \times 1.73 - 37.68) \text{cm}^2 \\
 &= (62.28 - 37.68) \text{cm}^2 = 24.6 \text{cm}^2.
 \end{aligned}$$

Hence, the required area is 24.6 cm^2 .

EXAMPLE 10

The area of an equilateral triangle is $100\sqrt{3} \text{ cm}^2$. Taking each vertex as centre, a circle is described with radius equal to half the length of the side of the triangle, as shown in the figure. Find the area of that part of the triangle which is not included in the circles. [Take $\pi = 3.14$ and $\sqrt{3} = 1.732$.]

**SOLUTION**

Let each side of the triangle be $a \text{ cm}$. Then,

$$\text{area of the triangle} = \left(\frac{\sqrt{3}}{4} a^2 \right) \text{cm}^2.$$

$$\therefore \frac{\sqrt{3}}{4} a^2 = 100\sqrt{3} \Rightarrow a^2 = 400 \Rightarrow a = 20.$$

Thus, the length of each side of $\triangle ABC$ is 20 cm .

$\therefore$ radius of each circle = 10 cm .

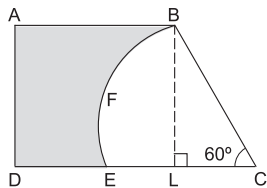
Required area

$$\begin{aligned}
 &= (\text{area of } \triangle ABC) - 3(\text{area of a sector with } r = 10 \text{ cm}, \theta = 60^\circ) \\
 &= \left(100\sqrt{3} - 3 \times 3.14 \times 10 \times 10 \times \frac{60}{360} \right) \text{cm}^2 \\
 &= \{(100 \times 1.732) - 157\} \text{cm}^2 = (173.2 - 157) \text{cm}^2 = 16.2 \text{cm}^2.
 \end{aligned}$$

Hence, the area of the required part is 16.2 cm^2 .

EXAMPLE 11

In the given figure, $ABCD$ is a trapezium with $AB \parallel CD$ and $\angle BCD = 60^\circ$. If $BFEC$ is a sector of a circle with centre C and $AB = BC = 7 \text{ cm}$ and $DE = 4 \text{ cm}$, then find the area of the shaded region. [Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$]



[CBSE 2010]

SOLUTION Draw $BL \perp CD$.

Clearly, $CE = CB = 7$ cm.

And, $CD = CE + DE = 7$ cm + 4 cm = 11 cm.

From right $\triangle CLB$, we have

$$\begin{aligned}\frac{BL}{BC} &= \sin 60^\circ = \frac{\sqrt{3}}{2} \Rightarrow \frac{BL}{7 \text{ cm}} = \frac{\sqrt{3}}{2} \\ \Rightarrow BL &= \frac{7\sqrt{3}}{2} \text{ cm} = \frac{7 \times 1.73}{2} \text{ cm} = \frac{12.11}{2} \text{ cm} = 6.05 \text{ cm}.\end{aligned}$$

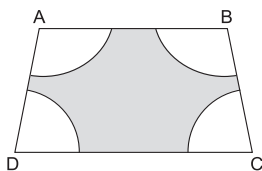
Required shaded area

$$\begin{aligned}&= \text{ar(trapezium } ABCD) - \text{ar(sector } BCEFB) \\ &= \left\{ \frac{1}{2} \times (AB + CD) \times BL \right\} - \left\{ \frac{\pi r^2 \theta}{360} \right\}, \text{ where } r = CB = 7 \text{ cm} \\ &= \left[\left\{ \frac{1}{2} (7 + 11) \times 6.05 \right\} - \left\{ \frac{22}{7} \times 7 \times 7 \times \frac{60}{360} \right\} \right] \text{ cm}^2 \\ &= (54.45 - 25.66) \text{ cm}^2 = 28.79 \text{ cm}^2.\end{aligned}$$

Hence, the area of the shaded region is 28.79 cm^2 .

EXAMPLE 12

In the given figure, $ABCD$ is a trapezium in which $AB \parallel DC$, $AB = 18$ cm, $DC = 32$ cm and the distance between AB and DC is 14 cm. If arcs of equal radii 7 cm have been drawn with centres A, B, C and D then find the area of the shaded region. [CBSE 2015, '17]



$$\begin{aligned}\text{SOLUTION } \text{ar(trap. } ABCD) &= \frac{1}{2} (\text{sum of parallel sides}) \\ &\quad \times (\text{distance between them}) \\ &= \left\{ \frac{1}{2} (18 + 32) \times 14 \right\} \text{ cm}^2 = 350 \text{ cm}^2.\end{aligned}$$

Sum of the areas of the 4 sectors

$$\begin{aligned}&= \text{area of a circle of radius } 7 \text{ cm} \\ &= \left(\frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 = 154 \text{ cm}^2.\end{aligned}$$

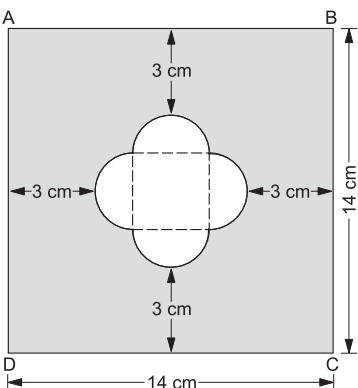
Area of the shaded region = $(350 - 154) \text{ cm}^2 = 196 \text{ cm}^2$.

Hence, the area of the shaded region is 196 cm^2 .

EXAMPLE 13 Find the area of the shaded region in the figure, given below.

[Take $\pi = 3.14$.]

[CBSE 2015]



SOLUTION

We have

$$\text{ar(sq } ABCD) = (14 \times 14) \text{ cm}^2 = 196 \text{ cm}^2.$$

Let the radius of each semicircle be a cm.

Then, $(a + 2a + a) = \{14 - (3 + 3)\} \text{ cm} = 8 \text{ cm}$.

$$\therefore 4a = 8 \Rightarrow a = 2.$$

So, the radius of each semicircle = 2 cm.

$$\text{Area of 4 semicircles} = \left(4 \times \frac{1}{2} \times 3.14 \times 2 \times 2\right) \text{ cm}^2 = 25.12 \text{ cm}^2.$$

Length of each side of the smaller square

$$= 2a \text{ cm} = (2 \times 2) \text{ cm} = 4 \text{ cm}.$$

$$\text{Area of smaller square} = (4 \times 4) \text{ cm}^2 = 16 \text{ cm}^2.$$

$$\text{Area of unshaded region} = (25.12 + 16) \text{ cm}^2 = 41.12 \text{ cm}^2$$

$$\text{Area of shaded region} = (196 - 41.12) \text{ cm}^2 = 154.88 \text{ cm}^2$$

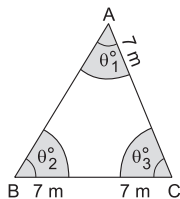
Hence, the required area is 154.88 cm^2 .

EXAMPLE 14 Three horses are tethered with 7-metre-long ropes at the three corners of a triangular field having sides 20 m, 34 m and 42 m. Find the area of the plot which can be grazed by the horses. Also, find the area of the plot which remains ungrazed. [CBSE 2009]

SOLUTION Let $\angle A = \theta_1^\circ$, $\angle B = \theta_2^\circ$ and $\angle C = \theta_3^\circ$.

Area which can be grazed by the horses

= sum of the areas of three sectors
with central angles θ_1° , θ_2° and θ_3° ,
and each with $r = 7$ m



$$\begin{aligned}
 &= \left(\frac{\pi r^2 \theta_1}{360} + \frac{\pi r^2 \theta_2}{360} + \frac{\pi r^2 \theta_3}{360} \right) \text{ m}^2, \text{ where } r = 7 \text{ m} \\
 &= \frac{\pi r^2}{360} (\theta_1 + \theta_2 + \theta_3) \text{ m}^2 = \left(\frac{\pi r^2 \times 180}{360} \right) \text{ m}^2 \\
 &\quad [\because \theta_1 + \theta_2 + \theta_3 = \angle A + \angle B + \angle C = 180^\circ] \\
 &= \left(\frac{22}{7} \times 7 \times 7 \times \frac{1}{2} \right) \text{ m}^2 = 77 \text{ m}^2.
 \end{aligned}$$

Sides of the plot are $a = 20 \text{ m}$, $b = 34 \text{ m}$, $c = 42 \text{ m}$.

$$\therefore s = \frac{1}{2}(a + b + c) = 48 \text{ m}, (s - a) = 28 \text{ m}, (s - b) = 14 \text{ m and}$$

$$(s - c) = 6 \text{ m}.$$

$$\therefore \text{ area of the whole plot} = \text{area of } \triangle ABC$$

$$= \sqrt{s(s-a)(s-b)(s-c)} \text{ sq units}$$

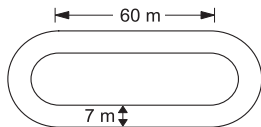
$$= \sqrt{48 \times 28 \times 14 \times 6} \text{ m}^2 = 336 \text{ m}^2.$$

$$\text{Area ungrazed} = (336 - 77) \text{ m}^2 = 259 \text{ m}^2.$$

EXAMPLE 15

The inside perimeter of a running track, as shown in the figure, is 340 m. The length of each straight portion is 60 m, and the curved portions are semicircles. If the track is 7 m wide, find the area of the track. Also, find the outer perimeter of the track.

[CBSE 2008C]

**SOLUTION**

$$\text{Length of inner curved portion} = (340 - 2 \times 60) \text{ m} = 220 \text{ m}.$$

$$\therefore \text{ the length of each inner curved part} = 110 \text{ m}.$$

Let the radius of each inner curved part be r .

$$\text{Then, } \pi r = 110 \text{ m} \Rightarrow \frac{22}{7} \times r = 110 \text{ m}$$

$$\Rightarrow r = \left(110 \times \frac{7}{22} \right) \text{ m} = 35 \text{ m}.$$

$$\therefore \text{ inner radius} = 35 \text{ m}, \text{ outer radius} = (35 + 7) \text{ m} = 42 \text{ m}.$$

$$\therefore \text{ area of the track}$$

$$= (\text{area of 2 rectangles, each } 60 \text{ m} \times 7 \text{ m})$$

$$+ (\text{area of the circular ring with } R = 42 \text{ m}, r = 35 \text{ m})$$

$$= \left[(2 \times 60 \times 7) + \frac{22}{7} \times \{(42)^2 - (35)^2\} \right] \text{ m}^2$$

$$= \left[840 + \frac{22}{7} \times (42 + 35)(42 - 35) \right] \text{ m}^2$$

$$= (840 + 1694) \text{ m}^2 = 2534 \text{ m}^2.$$

Length of the outer boundary of the track

$$= \left(2 \times 60 + 2 \times \frac{22}{7} \times 42 \right) \text{ m} = 384 \text{ m}.$$

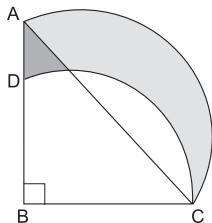
Hence, the outer perimeter of the track is 384 m.

EXAMPLE 16

In the given figure, $\triangle ABC$ is a right-angled triangle with $\angle B = 90^\circ$, $AB = 48 \text{ cm}$ and $BC = 14 \text{ cm}$. With AC as diameter a semicircle is drawn and with BC as radius, a quadrant of a circle is drawn. Find the area of the shaded region.

[Use $\pi = \frac{22}{7}$.]

[CBSE 2009C]

**SOLUTION**

In right $\triangle ABC$, we have

$$\begin{aligned} AC^2 &= AB^2 + BC^2 = [(48)^2 + (14)^2] \text{ cm}^2 \\ &= (2304 + 196) \text{ cm}^2 = 2500 \text{ cm}^2 \end{aligned}$$

$$\Rightarrow AC = \sqrt{2500} \text{ cm} = 50 \text{ cm} \Rightarrow R = \frac{50}{2} \text{ cm} = 25 \text{ cm}.$$

Area of the shaded region

$$= \text{ar}(\triangle ABC) + (\text{area of semicircle on } AC) - \text{ar}(\text{quadrant } BCD)$$

$$= \left(\frac{1}{2} \times 14 \times 48 \right) + \frac{1}{2} \pi \times 25 \times 25 - \frac{1}{4} \pi \times BC^2$$

$$= \left[336 + \frac{1}{2} \times \frac{22}{7} \times 625 - \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 \right] \text{ cm}^2$$

$$= \left(336 + \frac{6875}{7} - 154 \right) \text{ cm}^2 = (182 + 982.14) \text{ cm}^2$$

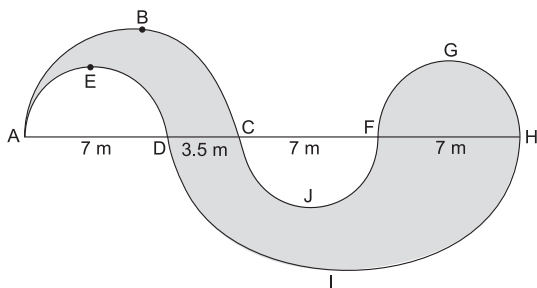
$$= 1164.14 \text{ cm}^2.$$

Hence, the area of the shaded region is 1164.14 cm^2 .

EXAMPLE 17

Find the area of the shaded region of the figure given below.

[CBSE 2011]



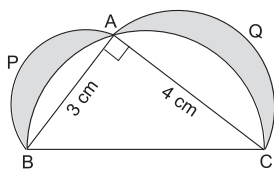
SOLUTION

Area of the shaded region

$$\begin{aligned}
 &= \text{ar}(ABCDEA) + \text{ar}(FGHF) + \text{ar}(HIDCJFH) \\
 &= \text{ar}(\text{semicircle } ABC) - \text{ar}(\text{semicircle } AED) \\
 &\quad + \text{ar}(\text{semicircle } FGH) - \text{ar}(\text{semicircle } HID) \\
 &\quad - \text{ar}(\text{semicircle } FJC) \\
 &= \left[\left\{ \frac{1}{2} \pi \times \frac{21}{4} \times \frac{21}{4} \right\} - \left\{ \frac{1}{2} \pi \times \frac{7}{2} \times \frac{7}{2} \right\} + \left\{ \frac{1}{2} \pi \times \frac{7}{2} \times \frac{7}{2} \right\} \right. \\
 &\quad \left. + \left\{ \frac{1}{2} \pi \times \frac{35}{4} \times \frac{35}{4} \right\} - \left\{ \frac{1}{2} \pi \times \frac{7}{2} \times \frac{7}{2} \right\} \right] \text{m}^2 \\
 &= \left\{ \left(\frac{22}{7} \times \frac{441}{32} \right) + \left(\frac{1}{2} \times \frac{22}{7} \times \frac{35}{4} \times \frac{35}{4} \right) - \left(\frac{22}{7} \times \frac{49}{8} \right) \right\} \text{m}^2 \\
 &= \left\{ \left(\frac{11 \times 63}{16} \right) + \left(\frac{55 \times 35}{16} \right) - \frac{77}{4} \right\} \text{m}^2 = \left(\frac{693}{16} + \frac{1925}{16} - \frac{77}{4} \right) \text{m}^2 \\
 &= \left(\frac{693 + 1925 - 308}{16} \right) \text{m}^2 = \frac{1155}{8} \text{m}^2 = 144.38 \text{m}^2.
 \end{aligned}$$

Hence, the required area is 144.38 m^2 .**EXAMPLE 18**

In the given figure, $\triangle ABC$ is right angled at A . Semicircles are drawn on AB , AC and BC as diameters. It is given that $AB = 3 \text{ cm}$ and $AC = 4 \text{ cm}$. Find the area of the shaded region. [CBSE 2017]



SOLUTION

From right $\triangle BAC$, we get

$$\begin{aligned}
 BC &= \sqrt{AB^2 + AC^2} = \sqrt{3^2 + 4^2} \text{ cm} \\
 &= \sqrt{9 + 16} \text{ cm} = \sqrt{25} \text{ cm} = 5 \text{ cm}.
 \end{aligned}$$

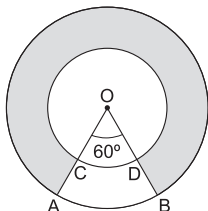
Area of the shaded region

$$\begin{aligned}
 &= \{ \text{ar}(\triangle ABC) + \text{ar}(\text{semicircle } APB) + \text{ar}(\text{semicircle } AQC) \} \\
 &\quad - \text{ar}(\text{semicircle } BAC) \\
 &= \left[\left(\frac{1}{2} \times 3 \times 4 \right) + \left(\frac{1}{2} \pi \times \frac{3}{2} \times \frac{3}{2} \right) + \left(\frac{1}{2} \pi \times 2 \times 2 \right) - \left(\frac{1}{2} \pi \times \frac{5}{2} \times \frac{5}{2} \right) \right] \text{cm}^2 \\
 &= \left\{ 6 + \frac{1}{2} \pi \left(\frac{9}{4} + 4 - \frac{25}{4} \right) \right\} \text{cm}^2 = (6 + 0) \text{cm}^2 = 6 \text{cm}^2.
 \end{aligned}$$

Hence, the area of the shaded region is 6 cm^2 .

EXAMPLE 19 In the given figure, two concentric circles with centre O , have radii 21 cm and 42 cm. If $\angle AOB = 60^\circ$ find the area of the shaded region.

[CBSE 2014]



SOLUTION

$$\begin{aligned} \text{ar}(\text{region } ACDB) &= \text{ar}(\text{sector } AOB) - \text{ar}(\text{sector } COD) \\ &= \left\{ \left(\frac{22}{7} \times 42 \times 42 \times \frac{60}{360} \right) - \left(\frac{22}{7} \times 21 \times 21 \times \frac{60}{360} \right) \right\} \text{cm}^2 \\ &= (924 - 231) \text{cm}^2 = 693 \text{cm}^2. \end{aligned}$$

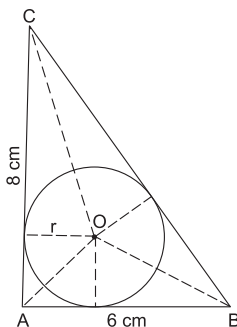
$$\begin{aligned} \text{ar}(\text{circular ring}) &= \left\{ \left(\frac{22}{7} \times 42 \times 42 \right) - \left(\frac{22}{7} \times 21 \times 21 \right) \right\} \text{cm}^2 \\ &= (5544 - 1386) \text{cm}^2 = 4158 \text{cm}^2. \end{aligned}$$

$$\text{Required area} = \text{ar}(\text{circular ring}) - \text{ar}(\text{region } ACDB)$$

$$= (4158 - 693) \text{cm}^2 = 3465 \text{cm}^2.$$

Hence, the area of the shaded region is 3465cm^2 .

EXAMPLE 20 In the given figure, $\triangle ABC$ is right-angled at A with $AB = 6 \text{ cm}$ and $AC = 8 \text{ cm}$. A circle with centre O has been inscribed inside the triangle. Find the value of r , the radius of the inscribed circle.



SOLUTION

$$\begin{aligned} BC &= \sqrt{AB^2 + AC^2} = \sqrt{6^2 + 8^2} \\ &= \sqrt{36 + 64} = \sqrt{100} = 10 \text{ cm}. \end{aligned}$$

Let O be the centre of the inscribed circle and r be its radius. Join OA , OB and OC .

$$\text{ar}(\triangle OAC) + \text{ar}(\triangle OAB) + \text{ar}(\triangle BOC) = \text{ar}(\triangle ABC)$$

$$\Rightarrow \left(\frac{1}{2} \times 8 \times r\right) + \left(\frac{1}{2} \times 6 \times r\right) + \left(\frac{1}{2} \times 10 \times r\right) = \frac{1}{2} \times 6 \times 8.$$

$$\Rightarrow (4r + 3r + 5r) = 24 \Rightarrow 12r = 24 \Rightarrow r = 2 \text{ cm.}$$

Hence, $r = 2 \text{ cm.}$

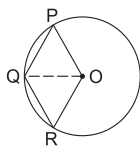
EXERCISE 16A

Use $\pi = \frac{22}{7}$, unless stated otherwise.

- The circumference of a circle is 39.6 cm. Find its area.
- The area of a circle is 98.56 cm^2 . Find its circumference.
- The circumference of a circle exceeds its diameter by 45 cm. Find the circumference of the circle.
- A copper wire when bent in the form of a square encloses an area of 484 cm^2 . The same wire is now bent in the form of a circle. Find the area enclosed by the circle.
- A wire when bent in the form of an equilateral triangle encloses an area of $121\sqrt{3} \text{ cm}^2$. The same wire is bent to form a circle. Find the area enclosed by the circle.
- The length of a chain used as the boundary of a semicircular park is 108 m. Find the area of the park.
- The sum of the radii of two circles is 7 cm, and the difference of their circumferences is 8 cm. Find the circumferences of the circles.
- Find the area of a ring whose outer and inner radii are respectively 23 cm and 12 cm.
- A path of 8 m width runs around the outside of a circular park whose radius is 17 m. Find the area of the path.
 - A park is of the shape of a circle of diameter 7 m. It is surrounded by a path of width of 0.7 m. Find the expenditure of cementing the path, if its cost is ₹ 110 per sq m. [CBSE 2017]
- A racetrack is in the form of a ring whose inner circumference is 352 m and outer circumference is 396 m. Find the width and the area of the track.
- A sector is cut from a circle of radius 21 cm. The angle of the sector is 150° . Find the length of the arc and the area of the sector.

12. A chord PQ of a circle of radius 10 cm subtends an angle of 60° at the centre of the circle. Find the area of major and minor segments of the circle. [CBSE 2017]
13. The length of an arc of a circle, subtending an angle of 54° at the centre is 16.5 cm. Calculate the radius, circumference and area of the circle.
14. The radius of a circle with centre O is 7 cm. Two radii OA and OB are drawn at right angles to each other. Find the areas of minor and major segments.
15. Find the lengths of the arcs cut off from a circle of radius 12 cm by a chord 12 cm long. Also, find the area of the minor segment. [Take $\pi = 3.14$ and $\sqrt{3} = 1.73$.]
16. A chord 10 cm long is drawn in a circle whose radius is $5\sqrt{2}$ cm. Find the areas of both the segments. [Take $\pi = 3.14$.]
17. Find the area of both the segments of a circle of radius 42 cm with central angle 120° . [Given, $\sin 120^\circ = \frac{\sqrt{3}}{2}$ and $\sqrt{3} = 1.73$.]
18. A chord of a circle of radius 30 cm makes an angle of 60° at the centre of the circle. Find the areas of the minor and major segments. [Take $\pi = 3.14$ and $\sqrt{3} = 1.732$.]
19. In a circle of radius 10.5 cm, the minor arc is one-fifth of the major arc. Find the area of the sector corresponding to the major arc.
20. The short and long hands of a clock are 4 cm and 6 cm long respectively. Find the sum of distances travelled by their tips in 2 days. [Take $\pi = 3.14$.]
21. Find the area of a quadrant of a circle whose circumference is 88 cm.
22. A rope by which a cow is tethered is increased from 16 m to 23 m. How much additional ground does it have now to graze?
23. A horse is placed for grazing inside a rectangular field 70 m by 52 m. It is tethered to one corner by a rope 21 m long. On how much area can it graze? How much area is left ungrazed?
24. A horse is tethered to one corner of a field which is in the shape of an equilateral triangle of side 12 m. If the length of the rope is 7 m, find the area of the field which the horse cannot graze. Take $\sqrt{3} = 1.732$. Write the answer correct to 2 places of decimal.
25. Four cows are tethered at the four corners of a square field of side 50 m such that each can graze the maximum unshared area. What area will be left ungrazed? [Take $\pi = 3.14$.]

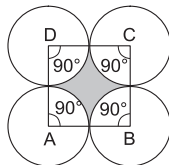
26. In the given figure, $OPQR$ is a rhombus, three of whose vertices lie on a circle with centre O . If the area of the rhombus is $32\sqrt{3} \text{ cm}^2$, find the radius of the circle.



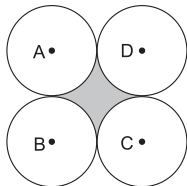
27. The side of a square is 10 cm. Find (i) the area of the inscribed circle, and (ii) the area of the circumscribed circle. [Take $\pi = 3.14$.]
28. If a square is inscribed in a circle, find the ratio of the areas of the circle and the square.
29. The area of a circle inscribed in an equilateral triangle is 154 cm^2 . Find the perimeter of the triangle. [Take $\sqrt{3} = 1.73$.]
30. The radius of the wheel of a vehicle is 42 cm. How many revolutions will it complete in a 19.8-km-long journey?
31. The wheels of the locomotive of a train are 2.1 m in radius. They make 75 revolutions in one minute. Find the speed of the train in km per hour.
32. The wheels of a car make 2500 revolutions in covering a distance of 4.95 km. Find the diameter of a wheel.
33. A boy is cycling in such a way that the wheels of his bicycle are making 140 revolutions per minute. If the diameter of a wheel is 60 cm, calculate the speed (in km/hr) at which the boy is cycling.
34. The wheel of a motorcycle is of radius 35 cm. How many revolutions per minute must the wheel make so as to keep a speed of 66 km/hr?
35. The diameters of the front and rear wheels of a tractor are 80 cm and 2 m respectively. Find the number of revolutions that a rear wheel makes to cover the distance which the front wheel covers in 800 revolutions.

36. Four equal circles are described about the four corners of a square so that each touches two of the others, as shown in the figure. Find the area of the shaded region, if each side of the square measures 14 cm.

[CBSE 2007]

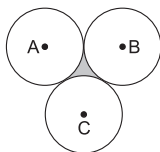


37. Four equal circles, each of radius 5 cm, touch each other, as shown in the figure. Find the area included between them. [Take $\pi = 3.14$.]

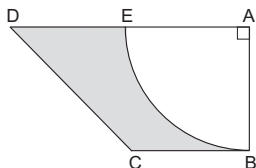


38. Four equal circles, each of radius a units, touch each other. Show that the area between them is $\left(\frac{6}{7}a^2\right)$ sq units.

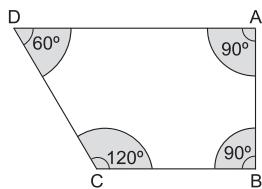
39. Three equal circles, each of radius 6 cm, touch one another as shown in the figure. Find the area enclosed between them. [Take $\pi = 3.14$ and $\sqrt{3} = 1.732$.]



40. If three circles of radius a each, are drawn such that each touches the other two, prove that the area included between them is equal to $\frac{4}{25}a^2$. [Take $\sqrt{3} = 1.73$ and $\pi = 3.14$.]
41. In the given figure, $ABCD$ is a trapezium of area 24.5 cm^2 . If $AD \parallel BC$, $\angle DAB = 90^\circ$, $AD = 10 \text{ cm}$, $BC = 4 \text{ cm}$ and ABE is quadrant of a circle then find the area of the shaded region. [CBSE 2014]



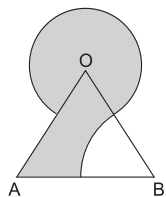
42. $ABCD$ is a field in the shape of a trapezium, $AD \parallel BC$, $\angle ABC = 90^\circ$ and $\angle ADC = 60^\circ$. Four sectors are formed with centres A, B, C and D , as shown in the figure. The radius of each sector is 14 m. Find the following:



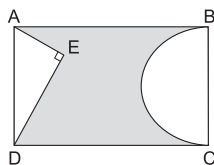
- (i) total area of the four sectors,
- (ii) area of the remaining portion, given that $AD = 55 \text{ m}$, $BC = 45 \text{ m}$ and $AB = 30 \text{ m}$. [CBSE 2013C]

43. Find the area of the shaded region in the given figure, where a circular arc of radius 6 cm has been drawn with vertex of an equilateral triangle of side 12 cm as centre and a sector of circle of radius 6 cm with centre B is made. [Use $\sqrt{3} = 1.73$ and $\pi = 3.14$.]

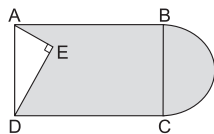
[CBSE 2014]



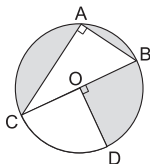
44. In the given figure, $ABCD$ is a rectangle with $AB = 80 \text{ cm}$ and $BC = 70 \text{ cm}$, $\angle AED = 90^\circ$ and $DE = 42 \text{ cm}$. A semicircle is drawn, taking BC as diameter. Find the area of the shaded region. [CBSE 2014]



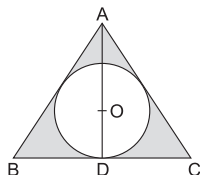
45. In the given figure, from a rectangular region $ABCD$ with $AB = 20$ cm, a right triangle AED with $AE = 9$ cm and $DE = 12$ cm, is cut off. On the other end, taking BC as diameter, a semicircle is added on outside the region. Find the area of the shaded region. [Use $\pi = 3.14$.] [CBSE 2014]



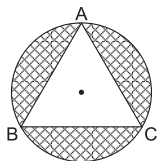
46. In the given figure, O is the centre of the circle with $AC = 24$ cm, $AB = 7$ cm and $\angle BOD = 90^\circ$. Find the area of shaded region. [Use $\pi = 3.14$.] [CBSE 2012, '17]



47. In the given figure, a circle is inscribed in an equilateral triangle ABC of side 12 cm. Find the radius of inscribed circle and the area of the shaded region. [Use $\sqrt{3} = 1.73$ and $\pi = 3.14$.] [CBSE 2014]

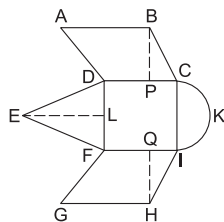


48. On a circular table cover of radius 42 cm, a design is formed by a girl leaving an equilateral triangle ABC in the middle, as shown in the figure. Find the covered area of the design. [Use $\sqrt{3} = 1.73$ and $\pi = \frac{22}{7}$.] [CBSE 2013C]

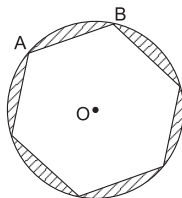


49. The perimeter of the quadrant of a circle is 25 cm. Find its area. [CBSE 2012]
50. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the minor segment. [Use $\pi = 3.14$.] [CBSE 2012]
51. The radius of a circular garden is 100 m. There is a road 10 m wide, running all around it. Find the area of the road and the cost of levelling it at ₹ 20 per m^2 . [Use $\pi = 3.14$.] [CBSE 2011]
52. The area of an equilateral triangle is $49\sqrt{3} \text{ cm}^2$. Taking each angular point as centre, circles are drawn with radius equal to half the length of the side of the triangle. Find the area of the triangle not included in the circles. [Take $\sqrt{3} = 1.73$.] [CBSE 2009]

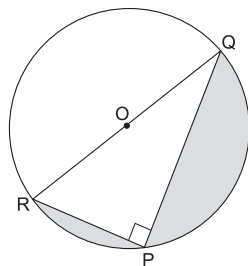
53. A child draws the figure of an aeroplane as shown. Here, the wings $ABCD$ and $FGHI$ are parallelograms, the tail DEF is an isosceles triangle, the cockpit CKI is a semicircle and $CDFI$ is a square. In the given figure, $BP \perp CD$, $HQ \perp FI$ and $EL \perp DF$. If $CD = 8$ cm, $BP = HQ = 4$ cm and $DE = EF = 5$ cm, find the area of the whole figure. [Take $\pi = 3.14$.]



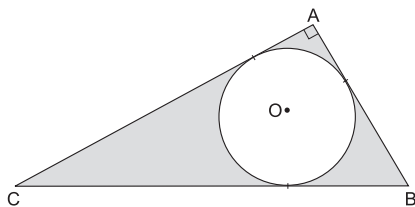
54. A circular disc of radius 6 cm is divided into three sectors with central angles 90° , 120° and 150° . What part of the whole circle is the sector with central angle 150° ? Also, calculate the ratio of the areas of the three sectors.
55. A round table cover has six equal designs as shown in the given figure. If the radius of the cover is 35 cm then find the total area of the design. [Use $\sqrt{3} = 1.732$ and $\pi = 3.14$.] [CBSE 2009, '14]



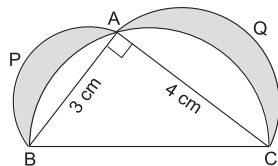
56. In the given figure, $PQ = 24$ cm, $PR = 7$ cm and O is the centre of the circle. Find the area of the shaded region. [Take $\pi = 3.14$.] [CBSE 2009]



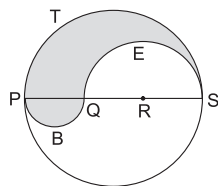
57. In the given figure, $\triangle ABC$ is right-angled at A . Find the area of the shaded region if $AB = 6$ cm, $BC = 10$ cm and O is the centre of the incircle of $\triangle ABC$. [Take $\pi = 3.14$.] [CBSE 2009]



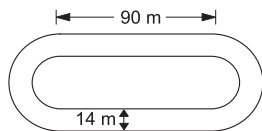
58. In the given figure, $\triangle ABC$ is right-angled at A . Semicircles are drawn on AB , AC and BC as diameters. It is given that $AB = 3$ cm and $AC = 4$ cm. Find the area of the shaded region. [CBSE 2017]



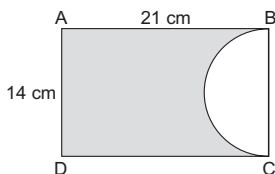
59. $PQRS$ is a diameter of a circle of radius 6 cm. The lengths PQ , QR and RS are equal. Semicircles are drawn with PQ and QS as diameters, as shown in the given figure. If $PS = 12$ cm, find the perimeter and area of the shaded region. [Take $\pi = 3.14$.]



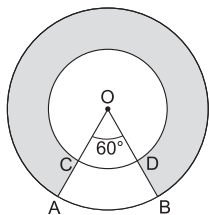
60. The inside perimeter of a running track shown in the figure is 400 m. The length of each of the straight portions is 90 m, and the ends are semicircles. If the track is 14 m wide everywhere, find the area of the track. Also, find the length of the outer boundary of the track.



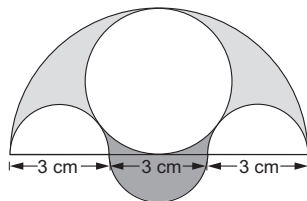
61. In the given figure, $ABCD$ is a rectangle of dimensions 21 cm $\times$ 14 cm. A semicircle is drawn with BC as diameter. Find the area and the perimeter of the shaded region in the figure. [CBSE 2017]



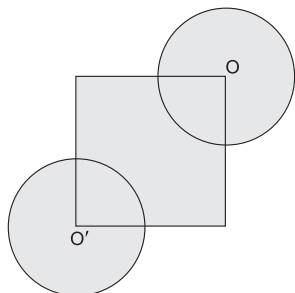
62. In the given figure, two concentric circles with centre O have radii 21 cm and 42 cm. If $\angle AOB = 60^\circ$, find the area of the shaded region. [Use $\pi = \frac{22}{7}$.] [CBSE 2017]



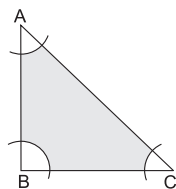
63. Three semicircles each of diameter 3 cm, a circle of diameter 4.5 cm and a semicircle of radius 4.5 cm are drawn in the given figure. Find the area of the shaded region. [CBSE 2017]



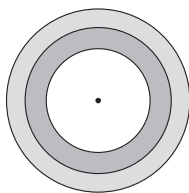
64. In the given figure, the side of square is 28 cm and radius of each circle is half of the length of the side of the square, where O and O' are centres of the circles. Find the area of shaded region. [CBSE 2017]



65. With the vertices A , B and C of a triangle ABC as centres, arcs are drawn with radii 5 cm each as shown in the given figure. If $AB = 14$ cm, $BC = 48$ cm and $CA = 50$ cm then find the area of the shaded region. [Use $\pi = 3.14$.]



66. If the diameters of the concentric circles shown in the figure below are in the ratio 1 : 2 : 3 then find the ratio of the areas of three regions.



ANSWERS (EXERCISE 16A)

1. 124.74 cm^2
2. 35.2 cm
3. 66 cm
4. 616 cm^2
5. 346.5 cm^2
6. 693 m^2
7. $26 \text{ cm}, 18 \text{ cm}$
8. 1210 cm^2
9. (i) 1056 m^2 (ii) ₹ 1863.40
10. $7 \text{ m}, 2618 \text{ m}^2$
11. $55 \text{ cm}, 577.5 \text{ cm}^2$
12. $9.08 \text{ cm}^2, 305.2 \text{ cm}^2$
13. $17.5 \text{ cm}, 110 \text{ cm}, 962.5 \text{ cm}^2$
14. $14 \text{ cm}^2, 140 \text{ cm}^2$
15. $(12.56 \text{ cm}, 62.8 \text{ cm}), 13.08 \text{ cm}$
16. $14.25 \text{ cm}^2, 142.75 \text{ cm}^2$
17. $1085.07 \text{ cm}^2, 4458.93 \text{ cm}^2$
18. $81.3 \text{ cm}^2, 2744.7 \text{ cm}^2$
19. 288.75 cm^2
20. 1909.12 cm^2
21. 154 cm^2
22. 858 m^2
23. $346.5 \text{ m}^2, 3293.5 \text{ m}^2$
24. 36.69 m^2
25. 537.5 m^2
26. 8 cm
27. (i) 78.5 cm^2 (ii) 157 cm^2
28. $\pi : 2$
29. 72.66 cm
30. 7500
31. 59.4 km/hr
32. 63 cm
33. 15.84 km/hr
34. 500
35. 320
36. 42 cm^2
37. 21.5 cm^2
39. 5.76 cm^2
41. 14.875 cm^2
42. (i) 616 m^2 (ii) 884 m^2
43. 137.64 cm^2
44. 2499 cm^2
45. 334.31 cm^2
46. 283.96 cm^2
47. $r = 2\sqrt{3} \text{ cm}$, ar(shaded region) = 24.6 cm^2
48. 3255.21 cm^2
49. 38.5 cm^2
50. 28.5 cm^2
51. 6594 m^2 , ₹ 131880
52. 7.77 cm^2
53. 65.12 cm^2
54. $\frac{5}{12}, 3 : 4 : 5$
55. 663.95 cm^2
56. 161.31 cm^2
57. 11.44 cm^2
58. 6 cm^2
59. $37.68 \text{ cm}, 37.68 \text{ cm}^2$
60. $6216 \text{ m}^2, 488 \text{ m}$
61. $217 \text{ cm}^2, 78 \text{ cm}$
62. 3465 cm^2
63. $12\frac{3}{8} \text{ cm}^2$
64. 1708 cm^2
65. 296.75 cm^2
66. $1 : 3 : 5$

HINTS TO SOME SELECTED QUESTIONS

$$3. 2\pi R - 2R = 45 \Rightarrow 2R(\pi - 1) = 45 \Rightarrow 2R\left(\frac{22}{7} - 1\right) = 45 \Rightarrow R = 10.5 \text{ cm.}$$

$$4. a^2 = 484 \Rightarrow a = \sqrt{484} = 22 \text{ cm.}$$

$$\text{Length of the wire} = 4a = (4 \times 22) \text{ cm} = 88 \text{ cm.}$$

$$2\pi R = 88.$$

$$5. \frac{\sqrt{3}}{4}a^2 = 121\sqrt{3} \Rightarrow a^2 = 484 \Rightarrow a = \sqrt{484} = 22 \text{ cm.}$$

$$\text{Length of the wire} = 3a = (3 \times 22) \text{ cm} = 66 \text{ cm.}$$

$$2\pi R = 66.$$

$$6. \pi R + 2R = 168 \Rightarrow \left(\frac{22}{7} + 2\right)R = 168 \Rightarrow R = 21 \text{ cm.}$$

$$\text{Area of the park} = \frac{1}{2}\pi R^2.$$

$$7. \text{ Let the radii of the circles be } x \text{ cm and } (7 - x) \text{ cm.}$$

$$\text{Then, } 2\pi x - 2\pi(7 - x) = 8 \Rightarrow 2\pi x = 26.$$

$$\text{Substitute this value of } 2\pi x \text{ and find } 2\pi(7 - x).$$

$$9. (i) R_1 = 17 \text{ m and } R_2 = (17 + 8) \text{ m} = 25 \text{ m.}$$

$$\text{Area of the path} = \pi\{(25)^2 - (17)^2\} \text{ m}^2.$$

$$13. 2 \times \frac{22}{7} \times r \times \frac{54}{360} = 16.5. \text{ Find } r.$$

$$14. \text{ Area of the minor segment} = \left\{ \frac{22}{7} \times 7 \times 7 \times \frac{90}{360} - \frac{1}{2} \times 7 \times 7 \times \sin 90^\circ \right\} \text{ cm}^2 = 14 \text{ cm}^2.$$

$$\text{Area of the major segment} = \left\{ \left(\frac{22}{7} \times 7 \times 7 \right) - 14 \right\} \text{ cm}^2 = 140 \text{ cm}^2.$$

$$15. \triangle OAB \text{ is equilateral. So, } \angle AOB = 60^\circ.$$

$$\text{arc } ACB = \left(\frac{2\pi R\theta}{360} \right) = \left(\frac{2\pi \times 12 \times 60}{360} \right) \text{ cm}$$

$$= 4\pi \text{ cm} = (4 \times 3.14) \text{ cm} = 12.56 \text{ cm.}$$

$$\text{arc } BDA = (2\pi r - \text{arc } ACB)$$

$$= (2 \times 3.14 \times 12 - 12.56) \text{ cm} = 62.8 \text{ cm.}$$

$$\text{ar}(\text{minor segment } ACBA) = \text{ar}(\text{sector } OACBO) - \text{ar}(\triangle OAB)$$

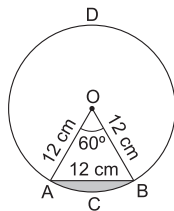
$$= \left(\frac{\pi r^2 \theta}{360} - \frac{\sqrt{3}}{4} a^2 \right)$$

$$= \left\{ \frac{3.14 \times 12 \times 12 \times 60}{360} - \frac{\sqrt{3} \times 12 \times 12}{4} \right\} \text{ cm}^2$$

$$= (75.36 - 62.8) \text{ cm}^2 = 13.08 \text{ cm}^2.$$

$$16. \text{ Let } OA = 5\sqrt{2} \text{ cm, } OB = 5\sqrt{2} \text{ cm and } AB = 10 \text{ cm.}$$

$$\text{Then, } OA^2 + OB^2 = AB^2 \Rightarrow \angle AOB = 90^\circ.$$



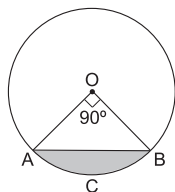
ar(minor segment ACBA)

$$= \text{ar}(\text{sector } OACBO) - \text{ar}(\triangle OAB)$$

$$= \left\{ \frac{3.14 \times (5\sqrt{2})^2 \times 90}{360} - \frac{1}{2} \times 5\sqrt{2} \times 5\sqrt{2} \right\} \text{cm}^2$$

$$= \frac{57}{4} \text{cm}^2 = 14.25 \text{cm}^2.$$

$$\begin{aligned} \text{ar}(\text{major segment}) &= \{3.14 \times (5\sqrt{2})^2 - 14.25\} \text{cm}^2 = (157 - 14.25) \text{cm}^2 \\ &= 142.75 \text{cm}^2. \end{aligned}$$



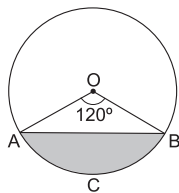
17. ar(minor segment ACBA)

$$= \text{ar}(\text{sector } OACBO) - \text{ar}(\triangle OAB)$$

$$= \left(\frac{\pi r^2 \theta}{360} - \frac{1}{2} r^2 \sin \theta \right) = (1848 - 762.93) \text{cm}^2$$

$$= 1085.07 \text{cm}^2.$$

$$\text{ar}(\text{major segment}) = \left[\left(\frac{22}{7} \times 42 \times 42 \right) - 1085.07 \right] \text{cm}^2.$$



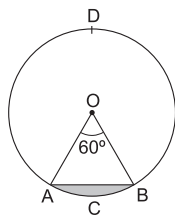
18. ar(minor segment ACBA)

$$= \text{ar}(\text{sector } OACBO) - \text{ar}(\triangle OAB) = \left(\frac{\pi r^2 \theta}{360} - \frac{1}{2} r^2 \sin \theta \right)$$

$$= \left(\frac{3.14 \times 30 \times 30 \times 60}{360} - \frac{1}{2} \times 30 \times 30 \times \sin 60^\circ \right)$$

$$= (471 - 389.7) = 81.3 \text{cm}^2.$$

$$\text{ar}(\text{major segment } BDAB) = (\pi R^2 - 81.3) \text{cm}^2 = (2826 - 81.3) = 2744.7 \text{cm}^2.$$



19. Let the length of the major arc be x cm. Then,

$$\text{the length of the minor arc} = \frac{x}{5} \text{cm}.$$

$$\text{Circumference of the circle} = \left(x + \frac{x}{5} \right) \text{cm} = \frac{6x}{5} \text{cm}.$$

$$\therefore \frac{6x}{5} = \left(2 \times \frac{22}{7} \times \frac{21}{2} \right) \Rightarrow x = 55.$$

Length of the major arc = 55 cm.

$$\text{Area of the sector corresponding to major arc} = \left(\frac{1}{2} rl \right) = \left(\frac{1}{2} \times \frac{21}{2} \times 55 \right) \text{cm}^2.$$

20. In 2 days, the short hand will complete 4 rounds.

$$\begin{aligned} \therefore \text{length moved by it} &= 4(\text{circumference of the circle with } r = 4 \text{ cm}) \\ &= (4 \times 2\pi \times 4) \text{cm} = 32\pi \text{cm}. \end{aligned}$$

In 2 days, the long hand will complete 48 rounds.

$$\begin{aligned} \therefore \text{length moved by it} &= 48(\text{circumference of the circle with } r = 6 \text{ cm}) \\ &= (48 \times 2\pi \times 6) \text{cm} = 576\pi \text{cm}. \end{aligned}$$

$$\therefore \text{sum of the lengths moved} = (32\pi + 576\pi) \text{cm} = (608\pi) \text{cm}.$$

$$21. \quad 2\pi R = 88 \Rightarrow 2 \times \frac{22}{7} \times R = 88 \Rightarrow R = 14 \text{ cm.}$$

$$\text{Required area} = \left(\frac{1}{4} \pi R^2 \right) \text{cm}^2 = 154 \text{ cm}^2.$$

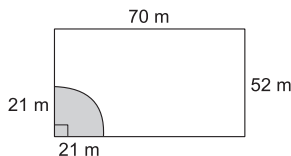
$$22. \quad \text{Additional ground} = \pi \times [(23)^2 - (16)^2] \text{m}^2.$$

23. Area which can be grazed

= area of the quadrant of radius 21 m

$$= \left(\frac{1}{4} \times \frac{22}{7} \times 21 \times 21 \right) \text{m}^2 = 346.5 \text{ m}^2.$$

$$\text{Area ungrazed} = [(70 \times 52) - 346.5] \text{m}^2.$$



24. Each angle of an equilateral triangle is 60° .

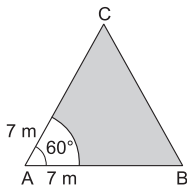
Area which cannot be grazed

= (area of the equilateral $\triangle ABC$)

– (area of the sector with $r = 7 \text{ m}, \theta = 60^\circ$)

$$= \left[\frac{\sqrt{3}}{4} \times (12)^2 - \frac{22}{7} \times (7)^2 \times \frac{60}{360} \right] \text{m}^2 = \frac{(108\sqrt{3} - 77)}{3} \text{m}^2$$

$$= 36.69 \text{ m}^2.$$

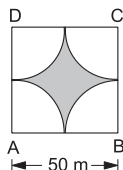


25. Ungrazed area = shaded area

$$= \left[(50 \times 50) - 4 \times \frac{\pi \times (25)^2 \times 90}{360} \right] \text{m}^2.$$

$$= [2500 - (625 \times 3.14)] \text{m}^2$$

$$= (2500 - 1962.5) \text{m}^2 = 537.5 \text{m}^2.$$



26. $OP = OR = OQ = r$.

Let OQ and PR intersect at S .

Since diagonals of a rhombus bisect each other at right angles, we have

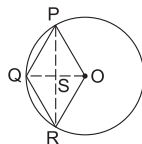
$$OS = \frac{1}{2}r \text{ and } \angle OSR = 90^\circ.$$

$$\therefore SR = \sqrt{OR^2 - OS^2} = \sqrt{r^2 - \frac{r^2}{4}} = \frac{\sqrt{3}r}{2}.$$

$$\therefore PR = 2 \times SR = \sqrt{3}r.$$

$$\text{Area of the rhombus} = \frac{1}{2} \times OQ \times PR = \frac{1}{2} \times r \times \sqrt{3}r = \frac{\sqrt{3}}{2}r^2.$$

$$\therefore \frac{\sqrt{3}}{2}r^2 = 32\sqrt{3} \Rightarrow r = 8 \text{ cm.}$$



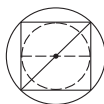
27. Diameter of the inscribed circle = side of the square = 10 cm.

$\therefore$ radius of the inscribed circle = 5 cm.

Diameter of the circumscribed circle = diagonal of the square

$$= (\sqrt{2} \times 10) \text{ cm.}$$

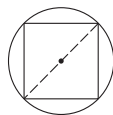
$\therefore$ radius of the circumscribed circle = $5\sqrt{2}$ cm.



28. Let the radius of the circle be
- r
- cm.

Then, diagonal of the square = diameter of the circle = $2r$ cm.Area of the circle = (πr^2) cm².

$$\text{Area of the square} = \frac{1}{2} \times (\text{diagonal})^2 = \frac{1}{2} \times 4r^2 = (2r^2) \text{ cm}^2.$$



29. Let the radius of the incircle be
- r
- cm.

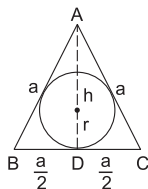
$$\text{Then, } \pi r^2 = 154 \Rightarrow r^2 = \left(154 \times \frac{7}{22}\right) \Rightarrow r = 7 \text{ cm.}$$

Let each side of the triangle be a cm and its height be h cm.

$$\text{Then, } r = \frac{h}{3} \Rightarrow h = 3r = 21 \text{ cm.}$$

$$h = \sqrt{a^2 - \frac{a^2}{4}} = \frac{\sqrt{3}a}{2} \Rightarrow \frac{\sqrt{3}a}{2} = 21.$$

$$\therefore a = \frac{42}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 14\sqrt{3}.$$



30. Circumference =
- $\left(2 \times \frac{22}{7} \times 42\right)$
- cm = 264 cm.

$$\text{Required number of revolutions} = \left(\frac{19.8 \times 1000 \times 100}{264}\right) = 7500.$$

31. Circumference =
- $\left(2 \times \frac{22}{7} \times 2.1\right)$
- m =
- $\frac{66}{5}$
- m.

$$\text{Distance covered in 60 minutes} = \left(\frac{66}{5} \times 75 \times 60\right) \text{ m,}$$

$$\text{speed in km/hr} = \left(\frac{66}{5} \times 75 \times 60 \times \frac{1}{1000}\right) \text{ km/hr.}$$

32. Distance covered in 1 revolution =
- $\left(\frac{4.95 \times 1000 \times 100}{2500}\right)$
- cm = 198 cm.

$$2\pi R = 198 \Rightarrow 2R \times \frac{22}{7} = 198 \Rightarrow 2R = \left(198 \times \frac{7}{22}\right) = 63 \text{ cm.}$$

33. Number of revolutions per hour =
- $(140 \times 60) = 8400$
- .

$$\text{Distance covered in 1 revolution} = 2\pi R = \left(\frac{22}{7} \times 60\right) \text{ cm.}$$

Distance covered in 1 hour = distance covered in 8400 revolutions

$$\begin{aligned} &= \left(\frac{22}{7} \times 60 \times 8400\right) \text{ cm} = \left(22 \times 72000 \times \frac{1}{100} \times \frac{1}{1000}\right) \text{ km} \\ &= \frac{1584}{100} \text{ km} = 15.84 \text{ km} = \text{speed per hr.} \end{aligned}$$

35. Radius of the front wheel = 40 cm =
- $\frac{2}{5}$
- m.

$$\text{Circumference of the front wheel} = \left(2\pi \times \frac{2}{5}\right) \text{ m} = \frac{4\pi}{5} \text{ m.}$$

Distance moved by it in 800 revolutions = $\left(\frac{4\pi}{5} \times 800\right) \text{ m} = (640\pi) \text{ m}$.

Circumference of the rear wheel = $(2\pi \times 1) \text{ m} = (2\pi) \text{ m}$.

Required number of revolutions = $\left(\frac{640\pi}{2\pi}\right) = 320$.

36. Required area = ar(square $ABCD$) - 4 ar(a segment with $r = 14 \text{ cm}$ and $\theta = 90^\circ$)

$$= \left\{ (14 \times 14) - 4 \left(\frac{22}{7} \times 14 \times 14 \times \frac{90}{360} \right) \right\} \text{ cm}^2$$

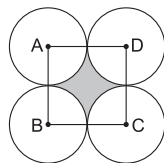
$$= (196 - 154) \text{ cm}^2 = 42 \text{ cm}^2.$$

37. Let A, B, C, D be the centres of these circles.

Join AB, BC, CD and DA .

Required area = (area of the square $ABCD$ with $a = 10 \text{ cm}$)

- 4(area of a sector with $r = 5 \text{ cm}$, $\theta = 90^\circ$).



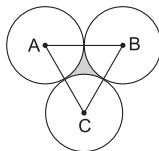
38. Required area = $\left\{ (2a)^2 - \frac{22}{7} a^2 \right\} \text{ sq units} = \left(\frac{6}{7} a^2 \right) \text{ sq units}$.

39. Let A, B, C be the centres of these circles.

Join AB, BC and CA .

Required area = (area of the $\triangle ABC$ with each side $a = 12 \text{ cm}$)

- 3(area of a sector with $r = 6 \text{ cm}$, $\theta = 60^\circ$).



40. As discussed in the previous question, we have

$$\text{required area} = \left[\frac{\sqrt{3}}{4} \times (2a)^2 - \frac{3\pi a^2 \times 60}{360} \right] = (1.73a^2 - 1.57a^2) = \left(\frac{4}{25} a^2 \right) \text{ sq units}.$$

41. Area of the trapezium $ABCD = \frac{1}{2}(AD + BC) \times AB = \frac{1}{2}AB \times (10 \text{ cm} + 4 \text{ cm})$

$$= AB \times 7 \text{ cm}.$$

$$\therefore AB \times 7 \text{ cm} = 24.5 \text{ cm}^2 \Rightarrow AB = 3.5 \text{ cm}.$$

Area of the shaded region = ar(trapezium $ABCD$) - ar(quadrant ABE)

$$= \left\{ 24.5 - \frac{1}{4} \pi \times (AB)^2 \right\} \text{ cm}^2$$

$$= \left(24.5 - \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \text{ cm}^2$$

$$= \left(24.5 - \frac{77}{8} \right) \text{ cm}^2 = (24.5 - 9.625) \text{ cm}^2$$

$$= 14.875 \text{ cm}^2.$$

42. Since $AD \parallel BC$ and $\angle ABC = 90^\circ$, so $\angle BAD = 90^\circ$.

Also, $\angle ADC = 60^\circ$ (given).

$$\therefore \angle BCD = 360^\circ - (90^\circ + 90^\circ + 60^\circ) = 120^\circ.$$

$$(i) \text{ Total area of 4 sectors} = \left\{ \pi \times (14)^2 \times \left(\frac{90}{360} + \frac{90}{360} + \frac{120}{360} + \frac{60}{360} \right) \right\} \text{ m}^2$$

$$= \left\{ \frac{22}{7} \times 14 \times 14 \times \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{3} + \frac{1}{6} \right) \right\} \text{m}^2$$

$$= (22 \times 28 \times 1) \text{m}^2 = 616 \text{m}^2.$$

(ii) Area of remaining portion = ar(trapezium $ABCD$) – total area of 4 sectors

$$= \left[\left\{ \frac{1}{2} (55 + 45) \times 30 \right\} - 616 \right] \text{m}^2$$

$$= (1500 - 616) \text{m}^2 = 884 \text{m}^2.$$

43. Since $\triangle OAB$ is equilateral, we have $\angle O = \angle A = \angle B = 60^\circ$ and

$OA = OB = AB = 12$ cm.

Radius of the circle = 6 cm.

$$\text{ar(shaded circular part)} = \left(\pi R^2 - \frac{\pi r^2 \times 60}{360} \right), \text{ where } r = 6 \text{ cm}$$

$$= \pi R^2 \left(1 - \frac{1}{6} \right) \text{cm}^2$$

$$= \left(3.14 \times 6 \times 6 \times \frac{5}{6} \right) \text{cm}^2 = 94.2 \text{cm}^2.$$

$$\text{ar(shaded triangular region)} = \left\{ \left(\frac{\sqrt{3}}{4} \times 12 \times 12 \right) - \frac{\pi \times 6 \times 6 \times 60}{360} \right\} \text{cm}^2$$

$$= \{ (36 \times 1.73) - (3.14 \times 6) \} \text{cm}^2 = \left(\frac{1557}{25} - \frac{471}{25} \right) \text{cm}^2$$

$$= \frac{1086}{25} \text{cm}^2 = 43.44 \text{cm}^2.$$

$$\text{Area of the shaded region} = (94.2 + 43.44) \text{cm}^2 = 137.64 \text{cm}^2.$$

$$44. AE^2 = (AD^2 - DE^2) = \{(70)^2 - (42)^2\} = (70 + 42)(70 - 42) = (112 \times 28) = (4 \times 28 \times 28)$$

$$\therefore AE = (2 \times 28) = 56 \text{ cm}.$$

$$\text{ar}(\triangle AED) = \frac{1}{2} \times DE \times AE = \left(\frac{1}{2} \times 42 \times 56 \right) \text{cm}^2 = 1176 \text{cm}^2.$$

$$\text{ar(semicircle)} = \frac{1}{2} \pi \times (35)^2 = \left\{ \frac{1}{2} \times \frac{22}{7} \times 35 \times 35 \right\} \text{cm}^2 = 1925 \text{cm}^2.$$

$$\text{Area of the shaded region} = \{(80 \times 70) - (1176 + 1925)\} \text{cm}^2$$

$$= (5600 - 3101) \text{cm}^2 = 2499 \text{cm}^2.$$

$$45. AD^2 = (DE^2 + AE^2) = (144 + 81) \text{cm}^2 = 225 \text{cm}^2$$

$$\Rightarrow AD = \sqrt{225} \text{ cm} = 15 \text{ cm} \Rightarrow r = \frac{15}{2} \text{ cm} = 7.5 \text{ cm}.$$

$$\text{ar(shaded region)} = \text{ar(rectangle } ABCD) + \text{ar(semicircle on } BC) - \text{ar}(\triangle ADE)$$

$$= \left\{ (20 \times 15) + \left(\frac{1}{2} \times 3.14 \times \frac{15}{2} \times \frac{15}{2} \right) - \left(\frac{1}{2} \times 9 \times 12 \right) \right\} \text{cm}^2$$

$$= (300 + 88.3125 - 54) \text{cm}^2 = (388.31 - 54) \text{cm}^2 = 334.31 \text{cm}^2.$$

46. $\angle BAC = 90^\circ$ (angle in a semicircle).

$$CB^2 = AC^2 + AB^2 = [(24)^2 + 7^2] \text{cm}^2 = (576 + 49) \text{cm}^2 = 625 \text{cm}^2$$

$$\Rightarrow CB = \sqrt{625} \text{ cm} = 25 \text{ cm}.$$

$$\therefore OC = \frac{1}{2} CB = \frac{25}{2} \text{ cm} = \text{radius of the circle}.$$

$$\begin{aligned}
 \text{Required area} &= (\text{area of the circle}) - \{\ar(\triangle BAC) + \ar(\text{quadrant } COD)\} \\
 &= \left(3.14 \times \frac{25}{2} \times \frac{25}{2} \right) - \left\{ \left(\frac{1}{2} \times 24 \times 7 \right) + \left(\frac{1}{4} \times 3.14 \times \frac{25}{2} \times \frac{25}{2} \right) \right\} \\
 &= \left\{ \left(\frac{3}{4} \times 3.14 \times \frac{25}{2} \times \frac{25}{2} \right) - 84 \right\} \text{ cm}^2 = (367.96 - 84) \text{ cm}^2 = 283.96 \text{ cm}^2.
 \end{aligned}$$

47. Draw $AOD \perp BC$. Then, D is the midpoint of BC .

$$\therefore AC = 12 \text{ cm and } DC = 6 \text{ cm.}$$

$$AD^2 = AC^2 - DC^2 = (12)^2 - 6^2 = 108 \Rightarrow AD = \sqrt{108} = 6\sqrt{3} \text{ cm.}$$

$$\therefore h = 6\sqrt{3} \text{ cm. Now, } 3r = h = 6\sqrt{3} \Rightarrow r = 2\sqrt{3} \text{ cm.}$$

So, the radius of inscribed circle is $2\sqrt{3}$ cm.

$$\begin{aligned}
 \ar(\text{shaded region}) &= \left[\left(\frac{\sqrt{3}}{4} \times 12 \times 12 \right) - \{ 3.14 \times (2\sqrt{3})^2 \} \right] \text{ cm}^2 \\
 &= [(36 \times 1.73) - (12 \times 3.14)] \text{ cm}^2 \\
 &= (62.28 - 37.68) \text{ cm}^2 \\
 &= 24.6 \text{ cm}^2.
 \end{aligned}$$

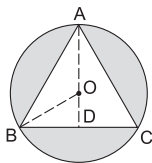
Note $h = 3r$ in case of incircle.

48. Let O be the centre of the circumcircle. Join OB and draw $OD \perp BC$. Then, $OB = 42$ cm and $\angle OBD = 30^\circ$.

$$\frac{OD}{OB} = \sin 30^\circ = \frac{1}{2} \Rightarrow \frac{OD}{42} = \frac{1}{2} \Rightarrow OD = 21 \text{ cm.}$$

$$\begin{aligned}
 BD^2 &= OB^2 - OD^2 = (42)^2 - (21)^2 \\
 &= (42 + 21)(42 - 21) = 63 \times 21.
 \end{aligned}$$

$$\therefore BD = \sqrt{63 \times 21} = 21\sqrt{3} \text{ cm} \Rightarrow BC = 2 \times 21\sqrt{3} = 42\sqrt{3} \text{ cm.}$$



Area of the shaded region

$$\begin{aligned}
 &= (\text{area of a circle with } r = 42 \text{ cm}) - \ar(\text{equilateral } \triangle ABC \text{ with } a = 42\sqrt{3} \text{ cm}) \\
 &= \left\{ \left(\frac{22}{7} \times 42 \times 42 \right) - \frac{173}{4} \times 42\sqrt{3} \times 42\sqrt{3} \right\} \text{ cm}^2 \\
 &= (5544 \times 2288.79) \text{ cm}^2 = 3255.21 \text{ cm}^2.
 \end{aligned}$$

Note : $a = r\sqrt{3}$ in case of circumcircle.

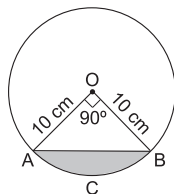
$$49. \text{ Perimeter of the quadrant} = \left(2r + \frac{2\pi r \times 90}{360} \right) \text{ cm} = \frac{25r}{7} \text{ cm.}$$

$$\therefore \frac{25r}{7} = 25 \Rightarrow r = 7.$$

$$\text{Area of the quadrant} = \frac{1}{4} \pi r^2 = \left(\frac{1}{4} \times \frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 = \frac{77}{2} \text{ cm}^2 = 38.5 \text{ cm}^2.$$

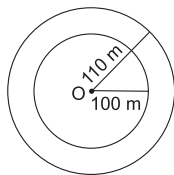
50. $\ar(\text{minor segment})$

$$\begin{aligned}
 &= \ar(\text{sector } OACBO) - \ar(\triangle OAB) \\
 &= \left\{ \left(3.14 \times 10 \times 10 \times \frac{90}{360} \right) - \left(\frac{1}{2} \times 10 \times 10 \right) \right\} \text{ cm}^2 \\
 &= (78.5 - 50) \text{ cm}^2 = 28.5 \text{ cm}^2.
 \end{aligned}$$



$$\begin{aligned}
 51. \text{ Area of the road} &= \pi(R^2 - r^2) \\
 &= 3.14 \times \{(110)^2 - (100)^2\} \text{ m}^2 \\
 &= (3.14 \times 210 \times 10) \text{ m}^2 = 6954 \text{ m}^2.
 \end{aligned}$$

Cost of levelling = ₹(6954 × 20) = ₹131880.



$$52. \frac{\sqrt{3}}{4} a^2 = 49\sqrt{3} \text{ cm}^2 \Rightarrow a^2 = 49 \times 4 \text{ cm}^2 \Rightarrow a = 7 \times 2 \text{ cm} = 14 \text{ cm}.$$

∴ radius of each circle = $\frac{14}{2} \text{ cm} = 7 \text{ cm}$.

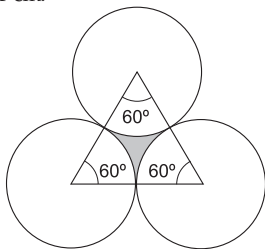
Clearly, each angle of the equilateral triangle is 60° .

Area of the shaded region

$$= \text{ar}(\triangle ABC) - 3(\text{area of a sector with } r = 7 \text{ cm} \text{ and } \theta = 60^\circ)$$

$$= \left\{ 49\sqrt{3} - 3 \left(\frac{22}{7} \times 7 \times 7 \times \frac{60}{360} \right) \right\} \text{ cm}^2$$

$$= (49 \times 1.73 - 77) \text{ cm}^2 = (84.77 - 77) \text{ cm}^2 = 7.77 \text{ cm}^2.$$



$$\begin{aligned}
 53. \text{ Required area} &= 2 \times \text{ar}(\text{llgm } ABCD) + \text{ar}(\triangle EDF) \\
 &\quad + \text{ar}(\text{square } DCIF) + \text{ar}(\text{semicircle } IKC).
 \end{aligned}$$

$$EL^2 = DE^2 - DL^2 = 5^2 - 4^2 = 25 - 16 = 9 \Rightarrow EL = 3 \text{ cm}.$$

$$\begin{aligned}
 \therefore \text{ required area} &= \left\{ 2 \times (8 \times 4) + \left(\frac{1}{2} \times 8 \times 3 \right) + (8 \times 8) + \left(\frac{1}{2} \times 3.14 \times 4 \times 4 \right) \right\} \text{ cm}^2 \\
 &= (64 + 12 + 64 + 25.12) \text{ cm}^2 = 165.12 \text{ cm}^2.
 \end{aligned}$$

$$54. \frac{\text{Area of the sector with } \theta = 150^\circ}{\text{Area of the circle}} = \frac{\pi \times (6)^2 \times \frac{150}{360}}{\pi \times (6)^2} = \frac{150}{360} = \frac{5}{12}.$$

$$\begin{aligned}
 \text{Required ratio} &= \left(36\pi \times \frac{90}{360} \right) : \left(36\pi \times \frac{120}{360} \right) : \left(36\pi \times \frac{150}{360} \right) \\
 &= \frac{1}{4} : \frac{1}{3} : \frac{5}{12} = 3 : 4 : 5.
 \end{aligned}$$

55. Area of the shaded region

$$= (\text{area of the circle}) - (\text{area of the hexagon})$$

$$= \pi r^2 - 6 \times \text{ar}(\triangle OAB \text{ with each side } r \text{ cm})$$

$$= \left[(3.14 \times 35 \times 35) - \left(6 \times \frac{\sqrt{3}}{4} \times 35 \times 35 \right) \right] \text{ cm}^2$$

$$= (35 \times 35) \left[3.14 - \frac{3}{2} \times 1.732 \right] \text{ cm}^2$$

$$= 1225 \times (3.14 - 2.598) \text{ cm}^2 = (1225 \times 0.542) \text{ cm}^2$$

$$= \left(1225 \times \frac{542}{1000} \right) \text{ cm}^2 = 663.95 \text{ cm}^2.$$

56. Clearly, $\angle QPR = 90^\circ$ [angle in a semicircle].

$$QR^2 = PQ^2 + PR^2 = [(24)^2 + 7^2] \text{ cm}^2 = 625 \text{ cm}^2 \Rightarrow QR = \sqrt{625} \text{ cm} = 25 \text{ cm}.$$

$\therefore$ radius of the circle (r) = $\frac{25}{2}$ cm.

Area of the shaded region

$$\begin{aligned}
 &= \text{ar}\left(\text{semicircle with } r = \frac{25}{2}\right) - \text{ar}(\triangle PQR) \\
 &= \left\{ \left(\frac{1}{2} \times 3.14 \times \frac{25}{2} \times \frac{25}{2} \right) - \left(\frac{1}{2} \times 7 \times 24 \right) \right\} \text{ cm}^2 \\
 &= (245.31 - 84) \text{ cm}^2 = 161.31 \text{ cm}^2.
 \end{aligned}$$

57. $AC^2 = BC^2 - AB^2 = 100 - 36 = 64 \Rightarrow AC = 8$ cm.

Draw $OD \perp AB$, $OE \perp AC$ and $OF \perp CB$.

Let $OD = OE = OF = r$ cm.

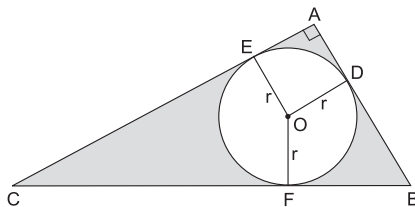
Then, $AEOD$ is a square such that $AD = AE = r$.

[$\because$ tangents to a circle from a point are equal.]

$CF = CE = (8 - r)$ cm and $BF = BD = (6 - r)$ cm.

$BF + CF = 10 \Rightarrow (6 - r) + (8 - r) = 10 \Rightarrow r = 2$.

$\text{ar}(\text{shaded region}) = \text{ar}(\triangle ABC) - (\text{area of the incircle})$



$$= \left\{ \frac{1}{2} \times AB \times AC - \pi \times 2^2 \right\} \text{ cm}^2 = \left(\frac{1}{2} \times 6 \times 8 - 3.14 \times 4 \right) \text{ cm}^2 = 11.44 \text{ cm}^2.$$

58. Area of the shaded region

$$\begin{aligned}
 &= \{ \text{ar}(\triangle ABC) + \text{ar}(\text{semicircle } APB) + \text{ar}(\text{semicircle } AQC) \} - \text{ar}(\text{semicircle } BAC) \\
 &= \left[\left(\frac{1}{2} \times 3 \times 4 \right) + \left(\frac{1}{2} \pi \times \frac{3}{2} \times \frac{3}{2} \right) + \left(\frac{1}{2} \pi \times 2 \times 2 \right) - \left(\frac{1}{2} \pi \times \frac{5}{2} \times \frac{5}{2} \right) \right] \text{ cm}^2 \\
 &= \left\{ 6 + \frac{1}{2} \pi \left(\frac{9}{4} + 4 - \frac{25}{4} \right) \right\} \text{ cm}^2 = (6 + 0) \text{ cm}^2 = 6 \text{ cm}^2.
 \end{aligned}$$

59. $PQ = QR = RS = 4$ cm, $QS = 8$ cm.

Perimeter = arc PTS + arc PBQ + arc QES

$$= (\pi \times 6 + \pi \times 2 + \pi \times 4) \text{ cm} = 12\pi \text{ cm}.$$

Area = (area of the semicircle PBQ) + (area of the semicircle PTS)
 $-$ (area of the semicircle QES)

$$= \left[\frac{1}{2} \pi \times (2)^2 + \frac{1}{2} \pi \times (6)^2 - \frac{1}{2} \pi \times (4)^2 \right] \text{ cm}^2 = (12\pi) \text{ cm}^2.$$

60. $90 + 90 + 2\pi r = 400 \Rightarrow 2 \times \frac{22}{7} \times r = 220 \Rightarrow r = 35$ m.

So, $R = (35 + 14)$ m = 49 m.

$$\text{Area of the track} = \left[(90 \times 14) + \frac{22}{7} \times \{ (49)^2 - (35)^2 \} \right] \text{ m}^2 = 6216 \text{ m}^2.$$

Length of the outer boundary $= (90 + 90 + 2\pi R)$ m, where $R = 49$ m
 $= 488$ m.

.....

EXERCISE 16B

Very-Short-Answer Questions

1. The difference between the circumference and radius of a circle is 37 cm. Using $\pi = \frac{22}{7}$, find the circumference of the circle. [CBSE 2013]
2. The circumference of a circle is 22 cm. Find the area of its quadrant. [CBSE 2012]
3. What is the diameter of a circle whose area is equal to the sum of the areas of two circles of diameters 10 cm and 24 cm? [CBSE 2012]
4. If the area of a circle is numerically equal to twice its circumference, then what is the diameter of the circle? [CBSE 2011]
5. What is the perimeter of a square which circumscribes a circle of radius a cm? [CBSE 2011]
6. Find the length of the arc of a circle of diameter 42 cm which subtends an angle of 60° at the centre. [CBSE 2012]
7. Find the diameter of the circle whose area is equal to the sum of the areas of two circles having radii 4 cm and 3 cm. [CBSE 2011]
8. Find the area of a circle whose circumference is 8π . [CBSE 2014]
9. Find the perimeter of a semicircular protractor whose diameter is 14 cm. [CBSE 2009]
10. Find the radius of a circle whose perimeter and area are numerically equal.
11. The radii of two circles are 19 cm and 9 cm. Find the radius of the circle which has circumference equal to the sum of the circumferences of the two circles.
12. The radii of two circles are 8 cm and 6 cm. Find the radius of the circle having area equal to the sum of the areas of the two circles.
13. Find the area of the sector of a circle having radius 6 cm and of angle 30° . [Take $\pi = 3.14$.]
14. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find the length of the arc.
15. The circumferences of two circles are in the ratio 2 : 3. What is the ratio between their areas?

16. The areas of two circles are in the ratio 4 : 9. What is the ratio between their circumferences?

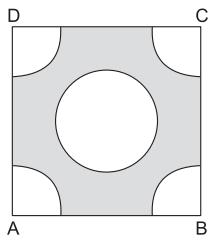
Short-Answer Questions

17. A square is inscribed in a circle. Find the ratio of the areas of the circle and the square.
18. The circumference of a circle is 8 cm. Find the area of the sector whose central angle is 72° .
19. A pendulum swings through an angle of 30° and describes an arc 8.8 cm in length. Find the length of the pendulum.
20. The minute hand of a clock is 15 cm long. Calculate the area swept by it in 20 minutes. [Take $\pi = 3.14$.]
21. A sector of 56° , cut out from a circle, contains 17.6 cm^2 . Find the radius of the circle.
22. The area of the sector of a circle of radius 10.5 cm is 69.3 cm^2 . Find the central angle of the sector.
23. The perimeter of a certain sector of a circle of radius 6.5 cm is 31 cm. Find the area of the sector.
24. The radius of a circle is 17.5 cm. Find the area of the sector enclosed by two radii and an arc 44 cm in length.
25. Two circular pieces of equal radii and maximum area, touching each other are cut out from a rectangular cardboard of dimensions $14 \text{ cm} \times 7 \text{ cm}$. Find the area of the remaining cardboard. [CBSE 2013]

26. In the given figure, $ABCD$ is a square of side 4 cm. A quadrant of a circle of radius 1 cm is drawn at each vertex of the square and a circle of diameter 2 cm is also drawn. Find the area of the shaded region.

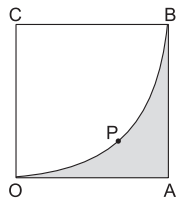
[Use $\pi = 3.14$.]

[CBSE 2012]

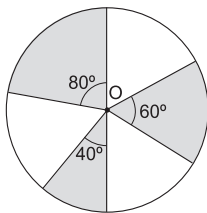


27. From a rectangular sheet of paper $ABCD$ with $AB = 40 \text{ cm}$ and $AD = 28 \text{ cm}$, a semicircular portion with BC as diameter is cut off. Find the area of the remaining paper. [CBSE 2012]

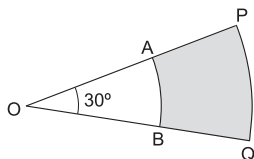
28. In the given figure, $OACB$ is a square of side 7 cm. If $COPB$ is a quadrant of a circle with centre C find the area of the shaded region. [CBSE 2012]



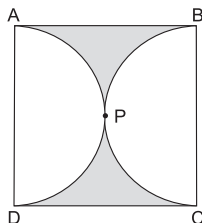
29. In the given figure, three sectors of a circle of radius 7 cm, making angles of 60° , 80° and 40° at the centre are shaded. Find the area of the shaded region. [CBSE 2012]



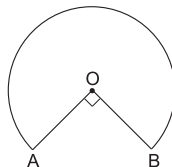
30. In the given figure, PQ and AB are respectively the arcs of two concentric circles of radii 7 cm and 3.5 cm with centre O . If $\angle POQ = 30^\circ$, find the area of the shaded region. [CBSE 2012]



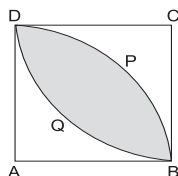
31. In the given figure, find the area of the shaded region, if $ABCD$ is a square of side 14 cm and APD and BPC are semicircles. [CBSE 2012]



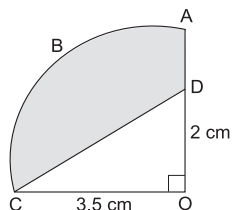
32. In the given figure, the shape of the top of a table is that of a sector of a circle with centre O and $\angle AOB = 90^\circ$. If $AO = OB = 42$ cm then find the perimeter of the top of the table. [CBSE 2012]



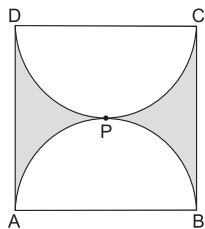
33. In the given figure, $ABCD$ is a square of side 7 cm, $DPBA$ and $DQBC$ are quadrants of circles each of the radius 7 cm. Find the area of shaded region. [CBSE 2012]



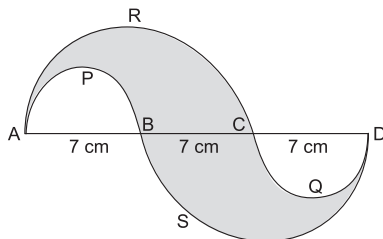
34. In the given figure, $OABC$ is a quadrant of a circle with centre O and radius 3.5 cm. If $OD = 2$ cm, find the area of the shaded region. [CBSE 2017]



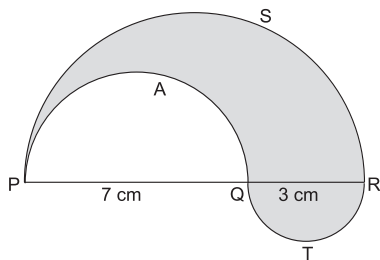
35. Find the perimeter of the shaded region in the figure, if $ABCD$ is a square of side 14 cm and APB and CPD are semicircles. [CBSE 2011]



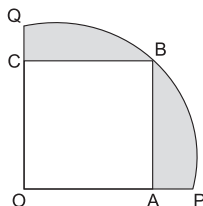
36. In a circle of radius 7 cm, a square $ABCD$ is inscribed. Find the area of the circle which is outside the square. [CBSE 2011]
37. In the given figure, APB and CQD are semicircles of diameter 7 cm each, while ARC and BSD are semicircles of diameter 14 cm each. Find the (i) perimeter, (ii) area of the shaded region. [CBSE 2011]



38. In the given figure, PSR , RTQ and PAQ are three semicircles of diameter 10 cm, 3 cm and 7 cm respectively. Find the perimeter of shaded region. [Use $\pi = 3.14$.] [CBSE 2014]

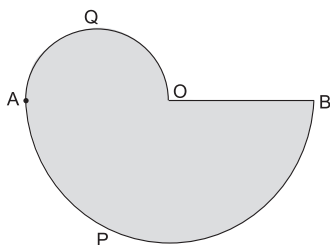


39. In the given figure, a square $OABC$ is inscribed in a quadrant $OPBQ$ of a circle. If $OA = 20$ cm, find the area of the shaded region. [Use $\pi = 3.14$.] [CBSE 2014]



40. In the given figure, APB and AQO are semicircles and $AO = OB$. If the perimeter of the figure is 40 cm, find the area of the shaded region.

[CBSE 2015]

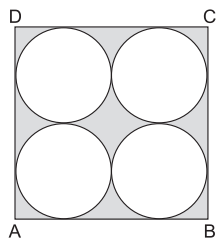


41. Find the area of a quadrant of a circle whose circumference is 44 cm.

[CBSE 2011]

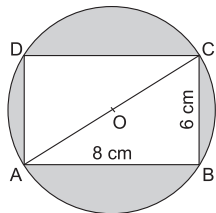
42. In the given figure, find the area of the shaded region, where $ABCD$ is a square of side 14 cm and all circles are of the same diameter.

[CBSE 2014]

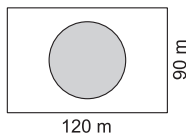


43. Find the area of the shaded region in the given figure, if $ABCD$ is a rectangle with sides 8 cm and 6 cm and O is the centre of the circle.

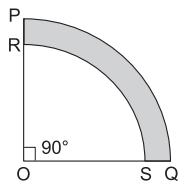
[CBSE 2014]



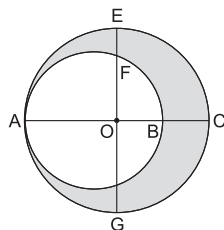
44. A wire is bent to form a square enclosing an area of 484 m^2 . Using the same wire, a circle is formed. Find the area of the circle. [CBSE 2014]
45. A square $ABCD$ is inscribed in a circle of radius r . Find the area of the square.
46. The cost of fencing a circular field at the rate of ₹ 25 per metre is ₹ 5500. The field is to be ploughed at the rate of 50 paise per m^2 . Find the cost of ploughing the field. [Take $\pi = \frac{22}{7}$.]
47. A park is in the form of a rectangle 120 m by 90 m. At the centre of the park, there is a circular lawn as shown in the figure. The area of the park excluding the lawn is 2950 m^2 . Find the radius of the circular lawn. [Given, $\pi = 3.14$.]



48. In the given figure, $PQSR$ represents a flower bed. If $OP = 21$ m and $OR = 14$ m, find the area of the flower bed.

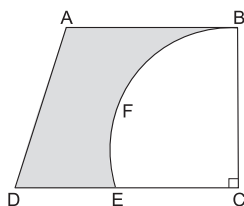


49. In the given figure, O is the centre of the bigger circle, and AC is its diameter. Another circle with AB as diameter is drawn. If $AC = 54$ cm and $BC = 10$ cm, find the area of the shaded region.

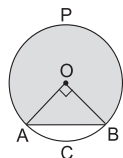


50. From a thin metallic piece in the shape of a trapezium $ABCD$ in which $AB \parallel CD$ and $\angle BCD = 90^\circ$, a quarter circle $BFEC$ is removed. Given, $AB = BC = 3.5$ cm and $DE = 2$ cm, calculate the area of remaining (shaded) part of metal sheet.

[CBSE 2011]



51. Find the area of the major segment APB of a circle of radius 35 cm and $\angle AOB = 90^\circ$, as shown in the given figure.



ANSWERS (EXERCISE 16B)

- | | | | | |
|--------------------------|---|--------------------------|--------------------------|---------------------------------|
| 1. 44 cm | 2. $\frac{77}{8} \text{ cm}^2$ | 3. 26 cm | 4. 8 cm | 5. $8a$ cm |
| 6. 22 cm | 7. 10 cm | 8. 16π | 9. 36 cm | 10. 2 units |
| 11. 28 cm | 12. 10 cm | 13. 9.42 cm^2 | 14. 22 cm | 15. 4 : 9 |
| 16. 2 : 3 | 17. $\pi : 2$ | 18. 123.2 cm^2 | 19. 16.8 cm | 20. 235.5 cm^2 |
| 21. 6 cm | 22. 72° | 23. 58.5 cm^2 | 24. 385 cm^2 | 25. 21 cm^2 |
| 26. 9.72 cm^2 | 27. 812 cm^2 | 28. 10.5 cm^2 | 29. 77 cm^2 | 30. $\frac{77}{8} \text{ cm}^2$ |
| 31. 42 cm^2 | 32. 282 cm | 33. 28 cm^2 | 34. 6.125 cm^2 | 35. 72 cm |
| 36. 63 cm^2 | 37. (i) 66 cm (ii) 115.5 cm^2 | 38. 31.4 cm | 39. 228 cm^2 | |
| 40. 96.25 cm^2 | 41. 38.5 cm^2 | 42. 42 cm^2 | 43. 30.57 cm^2 | 44. 616 m^2 |

45. $(2r^2)$ sq units

46. ₹ 1925

47. 50 m

48. 192.5 m^2

49. 770 cm^2

50. 6.125 cm^2

51. 3500 cm^2

HINTS TO SOME SELECTED QUESTIONS

1. $(2\pi R - R) = 37 \Rightarrow R(2\pi - 1) = 37$

$$\Rightarrow R\left(2 \times \frac{22}{7} - 1\right) = 37 \Rightarrow R = 7 \text{ cm.}$$

$$\therefore \text{circumference} = 2\pi R = \left(2 \times \frac{22}{7} \times 7\right) \text{ cm} = 44 \text{ cm.}$$

2. $2\pi R = 22 \Rightarrow 2 \times \frac{22}{7} \times R = 22 \Rightarrow R = \frac{7}{2} \text{ cm.}$

$$\text{Area of the quadrant} = \frac{1}{4} \times \pi R^2 = \left(\frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2}\right) \text{ cm}^2 = \frac{77}{8} \text{ cm}^2.$$

3. $\pi R^2 = \{\pi \times 5^2 + \pi \times (12)^2\} = (25\pi + 144\pi) = 169\pi$
 $\Rightarrow R^2 = 169 = (13)^2 \Rightarrow R = 13 \text{ cm} \Rightarrow 2R = 26 \text{ cm.}$

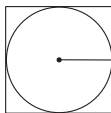
4. $\pi R^2 = 4\pi R \Rightarrow R = 4 \Rightarrow \text{diameter} = 8 \text{ cm.}$

5. Side of the square

$$= 2 \times \text{radius of circle} = 2a.$$

Perimeter of the square

$$= (4 \times 2a) = 8a \text{ cm.}$$

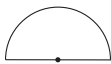


6. Length of the arc = $\frac{2\pi R\theta}{360} = \left(2 \times \frac{22}{7} \times 21 \times \frac{60}{360}\right) \text{ cm} = 22 \text{ cm.}$

7. $\pi R^2 = \pi \times (4)^2 + \pi \times (3)^2 = 25\pi \Rightarrow R^2 = 25 \Rightarrow R = 5 \Rightarrow 2R = 10 \text{ cm.}$

9. Perimeter = $(\pi R + 2R) = R(\pi + 2)$

$$= 7\left(\frac{22}{7} + 2\right) \text{ cm} = 36 \text{ cm.}$$



11. $2\pi R = (2\pi \times 19) + (2\pi \times 9) \Rightarrow 2\pi R = 2\pi \times (19 + 9) = 2\pi \times 28$
 $\Rightarrow R = 28 \text{ cm.}$

12. $\pi R^2 = (\pi \times 8^2) + (\pi \times 6^2) = \pi(64 + 36) = 100\pi$
 $\Rightarrow R^2 = 100 \Rightarrow R = 10 \text{ cm.}$

13. Area of the sector = $\frac{\pi R^2\theta}{360} = \left(3.14 \times 36 \times \frac{30}{360}\right) \text{ cm}^2 = 9.42 \text{ cm}^2.$

14. Length of the arc = $\frac{2\pi R\theta}{360} = \left(2 \times \frac{22}{7} \times 21 \times \frac{60}{360}\right) \text{ cm} = 22 \text{ cm.}$

15. $\frac{2\pi R_1}{2\pi R_2} = \frac{2}{3} \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}.$

$$\therefore \frac{\pi R_1^2}{\pi R_2^2} = \frac{R_1^2}{R_2^2} = \left(\frac{R_1}{R_2}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}.$$

Required ratio = 4 : 9.

$$16. \frac{\pi R_1^2}{\pi R_2^2} = \frac{4}{9} \Rightarrow \frac{R_1^2}{R_2^2} = \frac{4}{9} \Rightarrow \left(\frac{R_1}{R_2}\right)^2 = \left(\frac{2}{3}\right)^2 \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}.$$

$$\frac{2\pi R_1}{2\pi R_2} = \frac{R_1}{R_2} = \frac{2}{3}.$$

Required ratio = 2 : 3.

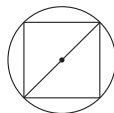
17. Let the radius of the circle be r cm.

Then, diagonal of the square = diameter of the circle = $2r$ cm.

Area of the circle = (πr^2) cm².

Area of the square = $\frac{1}{2} \times (\text{diagonal})^2 = \frac{1}{2} \times 4r^2 = (2r^2)$ cm².

$\therefore$ required ratio = $\pi r^2 : 2r^2 = \pi : 2$.



$$18. 2\pi R = 88 \Rightarrow 2 \times \frac{22}{7} \times R = 88 \Rightarrow R = 14 \text{ cm.}$$

$$\text{Area of the sector} = \frac{\pi R^2 \theta}{360} = \left(\frac{22}{7} \times 14 \times 14 \times \frac{72}{360}\right) \text{ cm}^2 = 123.2 \text{ cm}^2.$$

19. Length of the pendulum = radius of the sector = r cm.

$$\text{Arc length} = 8.8 \Rightarrow 2 \times \frac{22}{7} \times r \times \frac{30}{360} = 8.8 \Rightarrow r = \frac{168}{10} = 16.8 \text{ cm.}$$

$$20. \text{Angle described by the minute hand in 20 minutes} = \left(\frac{360}{60} \times 20\right)^\circ = 120^\circ.$$

Now, $\theta = 120^\circ$ and $r = 15$ cm.

$$\therefore \text{area swept by minute hand in 20 min} = \left(\frac{\pi r^2 \theta}{360}\right) \text{ cm}^2$$

$$= \left(3.14 \times 15 \times 15 \times \frac{120}{360}\right) \text{ cm}^2$$

$$= 235.5 \text{ cm}^2.$$

$$21. \frac{\pi r^2 \theta}{360} = 17.6 \Rightarrow \frac{22}{7} \times r^2 \times \frac{56}{360} = 17.6 \Rightarrow r^2 = 36 \Rightarrow r = 6 \text{ cm.}$$

$$22. \frac{\pi r^2 \theta}{360} = 69.3 \Rightarrow \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \theta = \frac{693}{10} \times 360 \Rightarrow \theta = 72^\circ.$$

$$23. \text{Perimeter of the sector} = \left(2r + \frac{2\pi r \theta}{360}\right) = \left(2 \times 6.5 + \frac{2\pi \times 6.5 \times \theta}{360}\right) \text{ cm}$$

$$\therefore 13 + \frac{13\pi\theta}{360} = 31 \Rightarrow \frac{13\pi\theta}{360} = 18 \Rightarrow \theta = \left(\frac{18 \times 360}{13\pi}\right)^\circ.$$

$$\text{Area} = \frac{\pi r^2 \theta}{360} = \left(\pi \times \frac{13}{2} \times \frac{13}{2} \times \frac{18 \times 360}{13\pi} \times \frac{1}{360}\right) \text{ cm}^2 = 58.5 \text{ cm}^2.$$

$$24. \frac{2\pi r \theta}{360} = 44 \Rightarrow 2\pi \times \frac{35}{2} \times \frac{\theta}{360} = 44 \Rightarrow \theta = \frac{44 \times 72}{7\pi}.$$

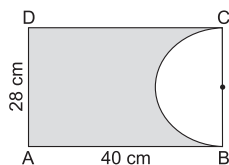
$$\text{Area} = \frac{\pi r^2 \theta}{360} = \left(\pi \times \frac{35}{2} \times \frac{35}{2} \times \frac{44 \times 72}{7\pi} \times \frac{1}{360}\right) \text{ cm}^2 = 385 \text{ cm}^2.$$

25. Clearly, the diameter of each circle is 7 cm.

$$\begin{aligned}\text{Required area} &= \left\{ (14 \times 7) - \left(2 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \right\} \text{ cm}^2 \\ &= (98 - 77) \text{ cm}^2 = 21 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}26. \text{ Required area} &= \left\{ (4 \times 4) - 4 \times \frac{1}{4} \pi \times (1)^2 + \pi \times (1)^2 \right\} \text{ cm}^2 \\ &= (16 - 2 \times 3.14 \times 1) \text{ cm}^2 = 9.72 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}27. \text{ Required area} &= \left\{ (40 \times 28) - \left(\frac{1}{2} \times \frac{22}{7} \times 14 \times 14 \right) \right\} \text{ cm}^2 \\ &= (1120 - 308) \text{ cm}^2 = 812 \text{ cm}^2.\end{aligned}$$



$$\begin{aligned}28. \text{ Required area} &= \left\{ (7 \times 7) - \left(\frac{1}{4} \times \frac{22}{7} \times 7 \times 7 \right) \right\} \text{ cm}^2 \\ &= \left(49 - \frac{77}{2} \right) \text{ cm}^2 = \frac{21}{2} \text{ cm}^2 = 10.5 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}29. \text{ Required area} &= \left\{ \pi \times 7^2 \times \frac{(60^\circ + 80^\circ \times 40^\circ)}{360^\circ} \right\} \text{ cm}^2 \\ &= \left(\frac{22}{7} \times 49 \times \frac{180}{360} \right) = 77 \text{ cm}^2.\end{aligned}$$

$$\begin{aligned}30. \text{ Required area} &= (\text{area of sector } OQP) - (\text{area of sector } OAB) \\ &= \left[\left(\frac{\pi \times 7^2 \times 30}{360} \right) - \left(\pi \times \frac{7}{2} \times \frac{7}{2} \times \frac{30}{360} \right) \right] \text{ cm}^2 \\ &= \left[\left(\frac{22}{7} \times \frac{49}{12} \right) - \left(\frac{22}{7} \times \frac{49}{48} \right) \right] \text{ cm}^2 = \left(\frac{77}{6} - \frac{77}{24} \right) \text{ cm}^2 = \frac{77}{8} \text{ cm}^2.\end{aligned}$$

31. Required area = (area of sq ABCD) – (area of 2 semicircles)

$$\begin{aligned}&= \left[(14 \times 14) - \left(\frac{22}{7} \times 7 \times 7 \right) \right] \text{ cm}^2 \\ &= (196 - 154) \text{ cm}^2 = 42 \text{ cm}^2.\end{aligned}$$

32. Required perimeter = (circumference – arc AB) + (OA + OB)

$$\begin{aligned}&= \left\{ \left(2 \times \frac{22}{7} \times 42 \right) - \left(2 \times \frac{22}{7} \times 42 \times \frac{90}{360} \right) + (42 + 42) \right\} \text{ cm} \\ &= (264 - 66 + 83) \text{ cm} = 282 \text{ cm}.\end{aligned}$$

33. Required area = (area of quadrant ABPD + area of quadrant CDQB) – ar(area of square ABCD)

$$\begin{aligned}&= \left\{ \left(\frac{1}{4} \times \frac{22}{7} \times 7 \times 7 \right) + \left(\frac{1}{4} \times \frac{22}{7} \times 7 \times 7 \right) - (7 \times 7) \right\} \text{ cm}^2 \\ &= \left(\frac{77}{2} + \frac{77}{2} - 49 \right) \text{ cm}^2 = (77 - 49) \text{ cm}^2 = 28 \text{ cm}^2.\end{aligned}$$

34. Required area = ar(quadrant $OABC$) - ar($\triangle COD$)

$$\begin{aligned}
 &= \left\{ \left(\frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) - \left(\frac{1}{2} \times \frac{7}{2} \times 2 \right) \right\} \text{ cm}^2 \\
 &= \left(\frac{77}{8} - \frac{7}{2} \right) \text{ cm}^2 = \frac{(77 - 28)}{8} \text{ cm}^2 \\
 &= \frac{49}{8} \text{ cm}^2 = 6.125 \text{ cm}^2.
 \end{aligned}$$

35. Perimeter of the shaded region

= perimeter of semicircle DPC + perimeter of semicircle APB + AD + BC

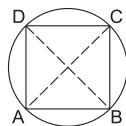
$$= \left\{ \left(\frac{22}{7} \times 7 \right) + \left(\frac{22}{7} \times 7 \right) + 14 + 14 \right\} \text{ cm} = (22 + 22 + 28) \text{ cm} = 72 \text{ cm}.$$

36. Diagonal of the square = $(2 \times 7) \text{ cm} = 14 \text{ cm}$.

$$\text{Area of the square} = \frac{1}{2} d^2 = \left(\frac{1}{2} \times 14 \times 14 \right) \text{ cm}^2 = 91 \text{ cm}^2.$$

$$\text{Area of the circle} = \left(\frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 = 154 \text{ cm}^2.$$

$$\text{Required area} = (154 - 91) \text{ cm}^2 = 63 \text{ cm}^2.$$



$$\begin{aligned}
 37. \text{ (i) Perimeter of the shaded region} &= \left\{ 2 \left(\frac{22}{7} \times \frac{7}{2} \right) + 2 \left(\frac{22}{7} \times 7 \right) \right\} \text{ cm} \\
 &= (22 + 44) \text{ cm} = 66 \text{ cm}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) Area of the shaded region} &= 2 \left\{ \left(\frac{1}{2} \times \frac{22}{7} \times 7 \times 7 \right) - \left(\frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \right\} \text{ cm}^2 \\
 &= 2 \left(77 - \frac{77}{4} \right) \text{ cm}^2 = 115.5 \text{ cm}^2.
 \end{aligned}$$

38. PSR is a semicircle of radius 5 cm, PAQ is a semicircle of radius 3.5 cm and QRT is a semicircle of radius 1.5 cm.

$$\text{Perimeter} = (5\pi + 3.5\pi + 1.5\pi) \text{ cm} = (10\pi) \text{ cm} = (10 \times 3.14) \text{ cm} = 31.4 \text{ cm}.$$

39. Side of the square = 20 cm.

$$\therefore \text{ area of the square} = (20 \times 20) \text{ cm}^2 = 400 \text{ cm}^2.$$

$$\text{Diagonal of the square} = \sqrt{(20)^2 + (20)^2} \text{ cm} = \sqrt{800} \text{ cm} = 20\sqrt{2} \text{ cm}.$$

$$\text{Radius of the quadrant} = 20\sqrt{2} \text{ cm}.$$

$$\begin{aligned}
 \text{Area of the shaded region} &= \left\{ \frac{1}{4} \times 3.14 \times (20\sqrt{2})^2 - 400 \right\} \text{ cm}^2 \\
 &= (628 - 400) \text{ cm}^2 = 228 \text{ cm}^2.
 \end{aligned}$$

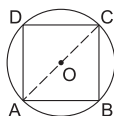
40. Let $AO = OB = r$.

$$\text{Perimeter} = (\text{arc } APB + OB + \text{arc } OQA)$$

$$= \left(\pi r + r + \frac{\pi r}{2} \right) = \left(\frac{3\pi}{2} + 1 \right) r = \frac{40r}{7}.$$

$$\therefore \frac{40r}{7} = 40 \text{ cm} \Rightarrow r = 7 \text{ cm}.$$

$$\begin{aligned}\text{Area of the shaded region} &= \left[\left(\frac{1}{2} \times \frac{22}{7} \times 7 \times 7 \right) + \left(\frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \right] \text{ cm}^2 \\ &= \left(77 + \frac{77}{4} \right) \text{ cm}^2 = \frac{385}{4} \text{ cm}^2 = 96.25 \text{ cm}^2.\end{aligned}$$



$$41. 2\pi R = 44 \Rightarrow 2 \times \frac{22}{7} \times R = 44 \Rightarrow R = 7 \text{ cm}.$$

$$\begin{aligned}\text{Area of the quadrant} &= \left(\frac{1}{4} \times \frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 \\ &= \frac{77}{2} \text{ cm}^2 = 38.5 \text{ cm}^2.\end{aligned}$$

$$42. \text{Radius of each circle} = \frac{7}{2} \text{ cm} = 3.5 \text{ cm}.$$

$$\begin{aligned}\text{Shaded area} &= \text{ar}(\text{square } ABCD) - 4\text{ar}(\text{circle with } r = 3.5 \text{ cm}) \\ &= \left\{ (14 \times 14) - \left(4 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \right\} \text{ cm}^2 \\ &= (196 - 154) \text{ cm}^2 = 42 \text{ cm}^2.\end{aligned}$$

$$43. AC^2 = AB^2 + BC^2 = (8^2 + 6^2) = (64 + 36) = 100 \Rightarrow AC = \sqrt{100} = 10 \text{ cm}.$$

$$\therefore r = 5 \text{ cm}.$$

$$\begin{aligned}\text{Shaded area} &= \left\{ \left(\frac{22}{7} \times 5 \times 5 \right) - (8 \times 6) \right\} \text{ cm}^2 = \left(\frac{550}{7} - 48 \right) \text{ cm}^2 \\ &= \frac{(550 - 336)}{7} \text{ cm}^2 = \frac{214}{7} \text{ cm}^2 \\ &= 30.57 \text{ cm}^2.\end{aligned}$$

$$44. \text{Side of the square} = \sqrt{484} \text{ m} = 22 \text{ m}.$$

$$\text{Circumference of the circle} = \text{perimeter of square} = (22 \times 4) \text{ m} = 88 \text{ m}$$

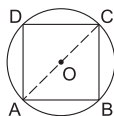
$$\Rightarrow 2 \times \frac{22}{7} \times R = 88 \Rightarrow R = \left(88 \times \frac{7}{44} \right) = 14 \text{ m}.$$

$$\text{Area of the circle} = \pi R^2 = \left(\frac{22}{7} \times 14 \times 14 \right) \text{ m}^2 = 616 \text{ m}^2.$$

$$45. \text{Let } O \text{ be the centre of the given circle and let } ABCD \text{ be the square inscribed in it.}$$

$$\text{So, diag. } AC = 2r.$$

$$\text{ar}(\text{square } ABCD) = \frac{1}{2} \times (\text{diagonal})^2 = \left\{ \frac{1}{2} \times (2r)^2 \right\} \text{ sq units} = (2r^2) \text{ sq units}.$$



$$46. \text{Rate of fencing} = ₹ 25 \text{ per metre. Cost of fencing} = ₹ 5500.$$

$$\text{Perimeter} = \frac{\text{total cost}}{\text{rate/metre}} = \left(\frac{5500}{25} \right) \text{ m} = 220 \text{ m}.$$

Let R be the radius of the field. Then,

$$2\pi R = 220 \Rightarrow 2 \times \frac{22}{7} \times R = 220 \Rightarrow R = \left(220 \times \frac{7}{44} \right) \text{ m} = 35 \text{ m}.$$

$$\therefore \text{area of the field} = \pi R^2 = \left(\frac{22}{7} \times 35 \times 35 \right) = 3850 \text{ m}^2.$$

$$\text{Cost of ploughing at 50 P per m}^2 = ₹ \left(3850 \times \frac{50}{100} \right) = ₹ 1925.$$

$$47. \text{Area of the lawn} = [(120 \times 90) - 2950] \text{ m}^2 = 7850 \text{ m}^2.$$

$$\therefore 3.14 \times r^2 = 7850. \text{ Find } r.$$

$$48. \text{Area of flower bed} = \text{ar}(\text{quadrant } OPQ) - \text{ar}(\text{quadrant } ORS)$$

$$= \left\{ \left(\frac{1}{4} \times \frac{22}{7} \times 21 \times 21 \right) - \left(\frac{1}{4} \times \frac{22}{7} \times 14 \times 14 \right) \right\} \text{ m}^2 = 192.5 \text{ m}^2.$$

$$49. \text{Diameter of the larger circle} = AC = 54 \text{ cm}.$$

$$\text{Diameter of the smaller circle} = AB = (AC - BC) = 44 \text{ cm}.$$

$$\begin{aligned} \text{Area of the shaded region} &= \{\pi \times (27)^2 - \pi \times (22)^2\} \text{ cm}^2 \\ &= \{\pi \times (27 + 22)(27 - 22)\} \text{ cm}^2 \\ &= \left(\frac{22}{7} \times 49 \times 5 \right) \text{ cm}^2. \end{aligned}$$

$$50. \text{Clearly, } AB = BC = CE = 3 \text{ cm and } DE = 2 \text{ cm}.$$

$$\begin{aligned} \text{Area of the shaded part} &= \text{ar}(\text{trapezium } ABCD) - \text{ar}(\text{quadrant } BCE) \\ &= \left[\left\{ \frac{1}{2} (AB + CD) \times BC \right\} - \left\{ \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right\} \right] \text{ cm}^2 \\ &= \left[\left\{ \frac{1}{2} (3.5 + 5.5) \times 3.5 \right\} - \frac{77}{8} \right] = \frac{49}{8} = 6.125 \text{ cm}^2. \end{aligned}$$

$$51. \text{Area of the minor segment } ACBA = \text{ar}(\text{sector } OACBO) - \text{ar}(\triangle OAB)$$

$$= \left[\left(\frac{22}{7} \times 35 \times 35 \times \frac{90}{360} \right) - \left(\frac{1}{2} \times 35 \times 35 \right) \right] \text{ cm}^2 = 350 \text{ cm}^2.$$

$$\begin{aligned} \text{Area of the major segment} &= \left\{ \left(\frac{22}{7} \times 35 \times 35 \right) - 350 \right\} \text{ cm}^2 \\ &= (3850 - 350) \text{ cm}^2 = 3500 \text{ cm}^2. \end{aligned}$$

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- The area of a circle is 38.5 cm^2 . The circumference of the circle is
(a) 6.2 cm (b) 12.1 cm (c) 11 cm (d) 22 cm
- The area of a circle is $49\pi \text{ cm}^2$. Its circumference is
(a) $7\pi \text{ cm}$ (b) $14\pi \text{ cm}$ (c) $21\pi \text{ cm}$ (d) $28\pi \text{ cm}$
- The difference between the circumference and radius of a circle is 37 cm. The area of the circle is
(a) 111 cm^2 (b) 184 cm^2 (c) 154 cm^2 (d) 259 cm^2

4. The perimeter of a circular field is 242 m. The area of the field is
(a) 9317 m^2 (b) 18634 m^2 (c) 4658.5 m^2 (d) none of these
5. On increasing the diameter of a circle by 40%, its area will be increased by
(a) 40% (b) 80% (c) 96% (d) 82%
6. On decreasing the radius of a circle by 30%, its area is decreased by
(a) 30% (b) 60% (c) 45% (d) none of these
7. The area of a circle is the same as the area of a square. Their perimeters are in the ratio
(a) $1 : 1$ (b) $2 : \pi$ (c) $\pi : 2$ (d) $\sqrt{\pi} : 2$
8. The circumference of a circle is equal to the sum of the circumferences of two circles having diameters 36 cm and 20 cm. The radius of the new circle is
(a) 16 cm (b) 28 cm (c) 42 cm (d) 56 cm
9. The area of a circle is equal to the sum of the areas of two circles of radii 24 cm and 7 cm. The diameter of the new circle is
(a) 25 cm (b) 31 cm (c) 50 cm (d) 62 cm
10. If the perimeter of a square is equal to the circumference of a circle then the ratio of their areas is
(a) $4 : \pi$ (b) $\pi : 4$ (c) $\pi : 7$ (d) $7 : \pi$
11. If the sum of the areas of two circles with radii R_1 and R_2 is equal to the area of a circle of radius R then
(a) $R_1 + R_2 = R$ (b) $R_1 + R_2 < R$ (c) $R_1^2 + R_2^2 < R^2$ (d) $R_1^2 + R_2^2 = R^2$
12. If the sum of the circumferences of two circles with radii R_1 and R_2 is equal to the circumference of a circle of radius R then
(a) $R_1 + R_2 = R$ (b) $R_1 + R_2 > R$ (c) $R_1 + R_2 < R$ (d) none of these
13. If the circumference of a circle and the perimeter of a square are equal then
(a) area of the circle = area of the square
(b) (area of the circle) > (area of the square)
(c) (area of the circle) < (area of the square)
(d) none of these
14. The radii of two concentric circles are 19 cm and 16 cm respectively. The area of the ring enclosed by these circles is
(a) 320 cm^2 (b) 330 cm^2
(c) 332 cm^2 (d) 340 cm^2

15. The areas of two concentric circles are 1386 cm^2 and 962.5 cm^2 . The width of the ring is
(a) 2.8 cm (b) 3.5 cm (c) 4.2 cm (d) 3.8 cm
16. The circumferences of two circles are in the ratio 3 : 4. The ratio of their areas is
(a) 3 : 4 (b) 4 : 3 (c) 9 : 16 (d) 16 : 9
17. The areas of two circles are in the ratio 9 : 4. The ratio of their circumferences is
(a) 3 : 2 (b) 4 : 9 (c) 2 : 3 (d) 81 : 16
18. The radius of a wheel is 0.25 m. How many revolutions will it make in covering 11 km?
(a) 2800 (b) 4000 (c) 5500 (d) 7000
19. The diameter of a wheel is 40 cm. How many revolutions will it make in covering 176 m?
(a) 140 (b) 150 (c) 160 (d) 166
20. In making 1000 revolutions, a wheel covers 88 km. The diameter of the wheel is
(a) 14 m (b) 24 m (c) 28 m (d) 40 m
21. The area of a sector of angle θ° of a circle with radius R is
(a) $\frac{2\pi R\theta}{180}$ (b) $\frac{\pi R^2\theta}{180}$ (c) $\frac{2\pi R\theta}{360}$ (d) $\frac{\pi R^2\theta}{360}$
22. The length of an arc of a sector of angle θ° of a circle with radius R is
(a) $\frac{2\pi R\theta}{180}$ (b) $\frac{2\pi R\theta}{360}$ (c) $\frac{\pi R^2\theta}{180}$ (d) $\frac{\pi R^2\theta}{360}$
23. The length of the minute hand of a clock is 21 cm. The area swept by the minute hand in 10 minutes is [CBSE 2012]
(a) 231 cm^2 (b) 210 cm^2 (c) 126 cm^2 (d) 252 cm^2
24. A chord of a circle of radius 10 cm subtends a right angle at the centre. The area of the minor segments (given, $\pi = 3.14$) is
(a) 32.5 cm^2 (b) 34.5 cm^2 (c) 28.5 cm^2 (d) 30.5 cm^2
25. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. The length of the arc is
(a) 21 cm (b) 22 cm (c) 18.16 cm (d) 23.5 cm
26. In a circle of radius 14 cm, an arc subtends an angle of 120° at the centre. If $\sqrt{3} = 1.73$ then the area of the segment of the circle is
(a) 120.56 cm^2 (b) 124.63 cm^2 (c) 118.24 cm^2 (d) 130.57 cm^2

ANSWERS (MCQ)

1. (d) 2. (b) 3. (c) 4. (c) 5. (c) 6. (d) 7. (d) 8. (b) 9. (c)
 10. (b) 11. (d) 12. (a) 13. (b) 14. (b) 15. (b) 16. (c) 17. (a) 18. (d)
 19. (a) 20. (c) 21. (d) 22. (b) 23. (a) 24. (c) 25. (b) 26. (a)

HINTS TO SOME SELECTED QUESTIONS

$$1. \pi R^2 = 38.5 \Rightarrow \frac{22}{7} \times R^2 = \frac{385}{10} \Rightarrow R^2 = \left(\frac{385}{10} \times \frac{7}{22} \right) = \frac{49}{4} \Rightarrow R = \frac{7}{2} \text{ cm.}$$

$$\text{Circumference} = 2\pi R = \left(2 \times \frac{22}{7} \times \frac{7}{2} \right) \text{ cm} = 22 \text{ cm.}$$

$$2. \pi R^2 = 49\pi \Rightarrow R^2 = 49 \Rightarrow R = 7 \text{ cm.}$$

$$\text{Circumference} = 2\pi R = (2\pi \times 7) \text{ cm} = 14\pi \text{ cm.}$$

$$3. 2\pi R - R = 37 \Rightarrow R \left(2 \times \frac{22}{7} - 1 \right) = 37 \Rightarrow R = \left(37 \times \frac{7}{37} \right) = 7.$$

$$\text{Area} = \pi R^2 = \left(\frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 = 154 \text{ cm}^2.$$

$$4. 2\pi R = 242 \Rightarrow 2 \times \frac{22}{7} \times R = 242 \Rightarrow R = \left(242 \times \frac{7}{44} \right) = \frac{77}{2}.$$

$$\text{Area} = \pi R^2 = \left(\frac{22}{7} \times \frac{77}{2} \times \frac{77}{2} \right) \text{ m}^2 = 4658.5 \text{ m}^2.$$

$$5. \text{ Let original diameter be 100 units. Then, original radius} = 50.$$

$$\therefore \text{ original area} = \pi \times (50)^2 = 2500\pi.$$

$$\text{New diameter} = 140. \text{ New radius} = 70.$$

$$\text{New area} = \pi \times (70)^2 = 4900\pi.$$

$$\text{Increase \%} = \left(\frac{2400}{2500} \times 100 \right) \% = 96\%.$$

$$6. \text{ Let original radius be 100 units.}$$

$$\text{Then, original area} = \pi \times (100)^2 = 10000\pi.$$

$$\text{New radius} = 70 \text{ units.}$$

$$\text{New area} = \pi \times (70)^2 = 4900\pi$$

$$\text{Decrease \%} = \left(\frac{5100}{10000} \times 100 \right) \% = 51\%.$$

$$7. a^2 = \pi R^2 \Rightarrow \frac{R^2}{a^2} = \frac{1}{\pi} \Rightarrow \frac{R}{a} = \frac{1}{\sqrt{\pi}}.$$

$$\text{Ratio of their perimeters} = \frac{2\pi R}{4a} = \frac{\pi}{2} \times \left(\frac{R}{a} \right) = \frac{\pi}{2} \times \frac{1}{\sqrt{\pi}} = \frac{\sqrt{\pi}}{2} = \sqrt{\pi} : 2.$$

$$8. \text{ Circumference of new circle} = (2\pi \times 18 + 2\pi \times 10) = 2\pi \times (18 + 10) = (2\pi \times 28) \text{ cm.}$$

$$2\pi R = 2\pi \times 28 \text{ cm} \Rightarrow R = 28 \text{ cm.}$$

$$\therefore \text{ radius of the new circle is 28 cm.}$$

$$\begin{aligned} 9. \text{ Area of the new circle} &= \{\pi \times (24)^2 + \pi \times (7)^2\} \text{ cm}^2 \\ &= \{\pi \times (576 + 49)\} \text{ cm}^2 = (625\pi) \text{ cm}^2. \end{aligned}$$

$$\therefore \pi R^2 = 625\pi \Rightarrow R^2 = 625 = (25)^2 \Rightarrow R = 25 \text{ cm.}$$

$$\therefore \text{ radius of the new circle} = 25 \text{ cm.}$$

$$10. 4a = 2\pi r \Rightarrow \frac{a}{r} = \frac{\pi}{2}.$$

$$\frac{\text{Area of square}}{\text{Area of circle}} = \frac{a^2}{\pi r^2} = \frac{1}{\pi} \times \left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{4\pi} = \frac{\pi}{4} = \pi : 4.$$

$$11. \pi R_1^2 + \pi R_2^2 = \pi R^2 \Rightarrow R_1^2 + R_2^2 = R^2.$$

$$12. 2\pi R_1 + 2\pi R_2 = 2\pi R \Rightarrow R_1 + R_2 = R.$$

$$\begin{aligned} 13. 2\pi R &= 4a \Rightarrow \frac{R}{a} = \frac{2}{\pi} \Rightarrow \frac{R^2}{a^2} = \frac{4}{\pi^2} \\ &\Rightarrow \frac{\pi R^2}{a^2} = \frac{4}{\pi} > 1 \quad [\because \pi = 3.14 < 4] \\ &\Rightarrow \pi R^2 > a^2 \\ &\Rightarrow \text{area of circle} > \text{area of square.} \end{aligned}$$

$$\begin{aligned} 14. \text{ Required area} &= \pi\{(19)^2 - (16)^2\} \text{ cm}^2 \\ &= \left(\frac{22}{7} \times 35 \times 3\right) \text{ cm}^2 = 330 \text{ cm}^2. \end{aligned}$$

$$15. \pi R^2 = 1386 \Rightarrow R^2 = \left(1386 \times \frac{7}{22}\right) = 441 = (21)^2 \Rightarrow R = 21 \text{ cm.}$$

$$\pi r^2 = 962.5 \Rightarrow r^2 = \left(\frac{9625}{10} \times \frac{7}{22}\right) = \frac{(49 \times 25)}{4} \Rightarrow r = \left(\frac{7 \times 5}{2}\right) \text{ cm} = \frac{35}{2} \text{ cm.}$$

$$\text{Width of the ring} = (R - r) = \left(21 - \frac{35}{2}\right) \text{ cm} = \frac{7}{2} \text{ cm} = 3.5 \text{ cm.}$$

$$\begin{aligned} 16. \frac{2\pi R_1}{2\pi R_2} &= \frac{3}{4} \Rightarrow \frac{R_1}{R_2} = \frac{3}{4} \Rightarrow \left(\frac{R_1}{R_2}\right)^2 = \left(\frac{3}{4}\right)^2 \\ &\Rightarrow \frac{R_1^2}{R_2^2} = \frac{9}{16} \Rightarrow \frac{\pi R_1^2}{\pi R_2^2} = \frac{9}{16}. \end{aligned}$$

$$\begin{aligned} 17. \frac{\pi R_1^2}{\pi R_2^2} &= \frac{9}{4} \Rightarrow \frac{R_1^2}{R_2^2} = \frac{9}{4} \Rightarrow \left(\frac{R_1}{R_2}\right)^2 = \left(\frac{3}{2}\right)^2 \\ &\Rightarrow \frac{R_1}{R_2} = \frac{3}{2} \Rightarrow \frac{2\pi R_1}{2\pi R_2} = \frac{3}{2}. \end{aligned}$$

$$18. \text{ Distance moved in 1 revolution} = 2\pi R = \left(2 \times \frac{22}{7} \times \frac{25}{100}\right) \text{ m} = \frac{11}{7} \text{ m.}$$

$$\text{Total distance covered} = 11 \text{ km} = 11000 \text{ m.}$$

$$\text{Number of revolutions} = \left(11000 \times \frac{7}{11}\right) = 7000.$$

$$19. \text{ Distance moved in 1 revolution} = \pi d = \left(\frac{22}{7} \times \frac{40}{100} \right) \text{ m} = \frac{44}{35} \text{ m}.$$

Total distance covered = 176 m.

$$\text{Number of revolutions} = \left(176 \times \frac{35}{44} \right) = 140.$$

$$20. \text{ Distance moved in 1 revolution} = \frac{88000}{1000} \text{ m} = 88 \text{ m}.$$

$$\pi d = 88 \Rightarrow \frac{22}{7} \times d = 88$$

$$\Rightarrow d = \left(88 \times \frac{7}{22} \right) = 28 \text{ m}.$$

$$23. \text{ Area swept by minute hand in 60 minutes} = \pi R^2.$$

Area swept by it in 10 minutes

$$= \left(\frac{\pi R^2}{60} \times 10 \right) \text{ cm}^2 = \left(\frac{22}{7} \times 21 \times 21 \times \frac{1}{6} \right) \text{ cm}^2$$

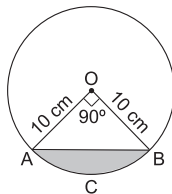
$$= 231 \text{ cm}^2.$$

$$24. \text{ ar(minor segment ACBA)} = \text{ar(sector OACBO)} - \text{ar}(\triangle OAB)$$

$$= \left(\frac{\pi r^2 \theta}{360} - \frac{1}{2} \times r \times r \right)$$

$$= \left(\frac{3.14 \times 10 \times 10 \times 90}{360} - \frac{1}{2} \times 10 \times 10 \right) \text{ cm}^2$$

$$= (78.5 - 50) \text{ cm}^2 = 28.5 \text{ cm}^2.$$



$$25. \text{ Arc length} = \frac{2\pi r \theta}{360} = \left(2 \times \frac{22}{7} \times 21 \times \frac{60}{360} \right) \text{ cm} = 22 \text{ cm}.$$

$$26. \text{ ar(segment)} = \left(\frac{\pi r^2 \theta}{360} - r^2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$$

$$= \left(\frac{22}{7} \times 14 \times 14 \times \frac{120}{360} \right) - (14 \times 14 \times \sin 60^\circ \cos 60^\circ)$$

$$= \left(\frac{616}{3} - \frac{\sqrt{3}}{2} \times \frac{1}{2} \times 14 \times 14 \right) \text{ cm}^2$$

$$= (205.33 - 49 \times 1.73) \text{ cm}^2$$

$$= (205.33 - 84.77) \text{ cm}^2$$

$$= 120.56 \text{ cm}^2.$$

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TEST YOURSELF

MCQ

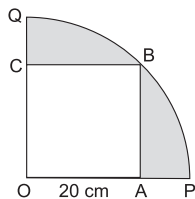
1. In the given figure, a square $OABC$ has been inscribed in the quadrant $OPBQ$. If $OA = 20$ cm then the area of the shaded region is [take $\pi = 3.14$]

(a) 214 cm^2

(b) 228 cm^2

(c) 242 cm^2

(d) 248 cm^2



2. The diameter of a wheel is 84 cm. How many revolutions will it make to cover 792 m?

(a) 200

(b) 250

(c) 300

(d) 350

3. The area of a sector of a circle with radius r , making an angle of x° at the centre is

(a) $\frac{x}{180} \times 2\pi r$

(b) $\frac{x}{180} \times \pi r^2$

(c) $\frac{x}{360} \times 2\pi r$

(d) $\frac{x}{360} \times \pi r^2$

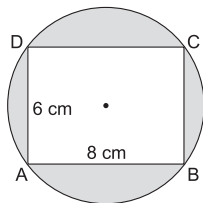
4. In the given figure, $ABCD$ is a rectangle inscribed in a circle having length 8 cm and breadth 6 cm. If $\pi = 3.14$ then the area of the shaded region is

(a) 264 cm^2

(b) 266 cm^2

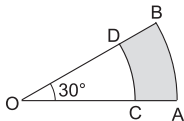
(c) 272 cm^2

(d) 254 cm^2

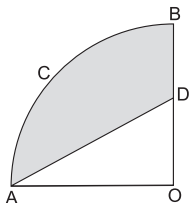


Short-Answer Questions

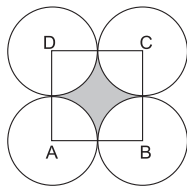
5. The circumference of a circle is 22 cm. Find its area. [Take $\pi = \frac{22}{7}$.]
6. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find the length of the arc.
7. The minute hand of a clock is 12 cm long. Find the area swept by it in 35 minutes.
8. The perimeter of a sector of a circle of radius 5.6 cm is 27.2 cm. Find the area of the sector.
9. A chord of a circle of radius 14 cm makes a right angle at the centre. Find the area of the sector.
10. In the give figure, the sectors of two concentric circles of radii 7 cm and 3.5 cm are shown. Find the area of the shaded region.



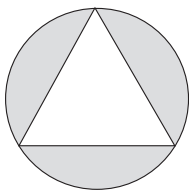
11. A wire when bent in the form of an equilateral triangle encloses an area of $121\sqrt{3} \text{ cm}^2$. If the same wire is bent into the form of a circle, what will be the area of the circle? [Take $\pi = \frac{22}{7}$.]
12. The wheel of a cart is making 5 revolutions per second. If the diameter of the wheel is 84 cm, find its speed in km per hour. [Take $\pi = \frac{22}{7}$.]
13. $OACB$ is a quadrant of a circle with centre O and its radius is 3.5 cm. If $OD = 2 \text{ cm}$, find the area of (i) the quadrant $OACB$ (ii) the shaded region. [Take $\pi = \frac{22}{7}$.]



14. In the given figure, $ABCD$ is a square each of whose sides measures 28 cm. Find the area of the shaded region. [Take $\pi = \frac{22}{7}$.]



15. In the given figure, an equilateral triangle has been inscribed in a circle of radius 4 cm. Find the area of the shaded region. [Take $\pi = 3.14$ and $\sqrt{3} = 1.73$.]



16. The minute hand of a clock is 7.5 cm long. Find the area of the face of the clock described by the minute hand in 56 minutes.

Long-Answer Questions

17. A racetrack is in the form of a ring whose inner circumference is 352 m and outer circumference is 396 m. Find the width and the area of the track.
18. A chord of a circle of radius 30 cm makes an angle of 60° at the centre of the circle. Find the areas of the minor and major segments. [Take $\pi = 3.14$ and $\sqrt{3} = 1.732$.]
19. Four cows are tethered at the four corners of a square field of side 50 m such that each can graze the maximum unshared area. What area will be left ungrazed? [Take $\pi = 3.14$.]

20. A square tank has an area of 1600 m^2 . There are four semicircular plots around it. Find the cost of turfing the plots at ₹ 12.50 per m^2 . [Take $\pi = 3.14$.]

ANSWERS (TEST YOURSELF)

- | | | | |
|------------------------------|-----------------------------|---|--------------------------|
| 1. (b) | 2. (c) | 3. (d) | 4. (b) |
| 5. 38.5 cm^2 | 6. 22 cm | 7. 264 cm^2 | 8. 44.8 cm^2 |
| 9. 154 cm^2 | 10. 9.625 cm^2 | 11. 346.5 cm^2 | 12. 47.52 km/hr |
| 13. (i) 9.625 cm^2 | (ii) 6.125 cm^2 | 14. 168 cm^2 | 15. 29.48 cm^2 |
| 16. 165 cm^2 | 17. 7 m, 2618 m^2 | 18. 81.27 cm^2 , 2744.73 cm^2 | |
| 19. 537.5 m^2 | 20. ₹ 31400 | | |

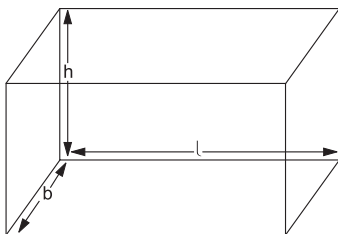


Volume and Surface Areas of Solids

SOLIDS The objects having definite shape and size are called solids.

A solid occupies a definite space.

CUBOID Solids like matchbox, chalkbox, a tile, a book, an almirah, a room, etc., are in the shape of a cuboid.



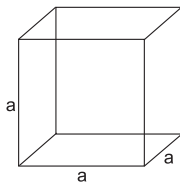
FORMULAE For a cuboid of length = l , breadth = b and height = h , we have

- (i) $\text{Volume} = (l \times b \times h)$ cubic units.
- (ii) $\text{Total surface area} = 2(lb + bh + lh)$ sq units.
- (iii) $\text{Lateral surface area} = [2(l + b) \times h]$ sq units.
- (iv) $\text{Diagonal} = \sqrt{l^2 + b^2 + h^2}$ units.

CUBE Solids like ice cubes, sugar cubes, dice, etc., are in the shape of a cube.

FORMULAE For a cube having each edge = a units, we have

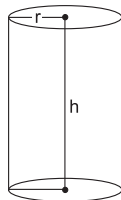
- (i) $\text{Volume} = a^3$ cubic units.
- (ii) $\text{Total surface area} = 6a^2$ sq units.
- (iii) $\text{Lateral surface area} = 4a^2$ sq units.
- (iv) $\text{Diagonal} = \sqrt{3}a$ units.



CYLINDER Solids like measuring jars, circular pillars, circular pencils, circular pipes, road rollers, gas cylinders, etc., are said to have a cylindrical shape.

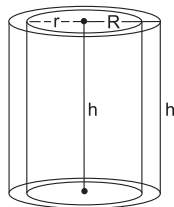
FORMULAE For a cylinder of base radius = r and height = h , we have

- (i) $\text{Volume} = (\pi r^2 h)$ cubic units.
- (ii) $\text{Curved surface area} = 2\pi r h$ sq units.
- (iii) $\text{Total surface area} = (2\pi r h + 2\pi r^2)$ sq units
 $= 2\pi r(h + r)$ sq units.



HOLLOW CYLINDERS Solids like iron pipes, rubber tubes are in the shape of hollow cylinders.

FORMULAE Consider a hollow cylinder having
external radius = R , internal radius = r
and height = h .



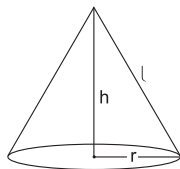
Then, we have

$$\begin{aligned} \text{(i) Volume of material} &= (\text{external volume}) - (\text{internal volume}) \\ &= (\pi R^2 h - \pi r^2 h) \text{ cubic units} \\ &= \pi h(R^2 - r^2) \text{ cubic units.} \end{aligned}$$

$$\begin{aligned} \text{(ii) Curved surface area of hollow cylinder} &= (\text{external surface area}) + (\text{internal surface area}) \\ &= (2\pi R h + 2\pi r h) \text{ sq units} \\ &= 2\pi h(R + r) \text{ sq units.} \end{aligned}$$

$$\begin{aligned} \text{(iii) Total surface area of hollow cylinder} &= (\text{curved surface area}) + (\text{area of the base rings}) \\ &= \{(2\pi R h + 2\pi r h) + 2(\pi R^2 - \pi r^2)\} \text{ sq units} \\ &= \{2\pi h(R + r) + 2\pi(R^2 - r^2)\} \text{ sq units} \\ &= \{2\pi h(R + r) + 2\pi(R + r)(R - r)\} \text{ sq units} \\ &= 2\pi(R + r)(h + R - r) \text{ sq units.} \end{aligned}$$

CONE Solids like ice-cream cones, conical tents, funnels, etc., are having the shape of a cone.



FORMULAE Consider a cone in which
base radius = r , height = h and slant height, $l = \sqrt{h^2 + r^2}$.

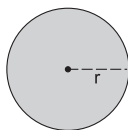
Then, we have

$$\begin{aligned} \text{(i) Volume of the cone} &= \frac{1}{3} \pi r^2 h \text{ cubic units.} \\ \text{(ii) Curved surface area of the cone} &= \pi r l = \pi r \sqrt{r^2 + h^2} \text{ sq units.} \\ \text{(iii) Total surface area of the cone} &= (\text{curved surface area}) + (\text{area of the base}) \\ &= (\pi r l + \pi r^2) = \pi r(l + r) \text{ sq units.} \end{aligned}$$

SPHERE Objects like a football, a cricket ball, etc., are said to have the shape of a sphere.

FORMULAE For a sphere of radius r , we have

$$\begin{aligned} \text{(i) Volume of the sphere} &= \left(\frac{4}{3} \pi r^3 \right) \text{ cubic units.} \\ \text{(ii) Surface area of the sphere} &= (4\pi r^2) \text{ sq units.} \end{aligned}$$



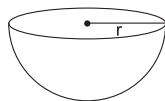
HEMISPHERE A plane through the centre of a sphere cuts it into two equal parts. Each part is called a hemisphere.

FORMULAE For a hemisphere of radius r , we have

(i) *Volume of the hemisphere* $= \frac{2}{3}\pi r^3$ cubic units.

(ii) *Curved surface area of the hemisphere* $= (2\pi r^2)$ sq units.

(iii) *Total surface area of the hemisphere* $= (3\pi r^2)$ sq units.



SPHERICAL SHELL Consider a spherical shell having external radius $= R$ and internal radius $= r$. Then, we have

$$\begin{aligned}\text{Volume of material} &= (\text{external volume}) - (\text{internal volume}) \\ &= \left(\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 \right) \text{ cubic units} \\ &= \frac{4}{3}\pi(R^3 - r^3) \text{ cubic units.}\end{aligned}$$

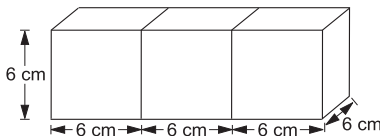
VOLUME AND SURFACE AREA OF A COMBINATION OF SOLIDS

SOLVED EXAMPLES

EXAMPLE 1 Three cubes each of volume 216 cm^3 are joined end to end to form a cuboid. Find the total surface area of the resulting cuboid.

[CBSE 2012]

SOLUTION Edge of each cube $= \sqrt[3]{216} \text{ cm} = 6 \text{ cm}$.
 Length of resulting cuboid, $l = 18 \text{ cm}$.
 Breadth of resulting cuboid, $b = 6 \text{ cm}$.
 Height of resulting cuboid, $h = 6 \text{ cm}$.



$$\begin{aligned}\therefore \text{ total surface area of the cuboid} &= 2(lb + bh + lh) = 2(18 \times 6 + 6 \times 6 + 18 \times 6) \text{ cm}^2 \\ &= 2(108 + 36 + 108) \text{ cm}^2 \\ &= (2 \times 252) \text{ cm}^2 = 504 \text{ cm}^2.\end{aligned}$$

EXAMPLE 2 A sphere and a cube have equal surface areas. Show that the ratio of the volume of sphere to that of the cube is $\sqrt{6} : \sqrt{\pi}$. [CBSE 2011]

SOLUTION Let the radius of the sphere be r and the edge of the cube be a .
Then,

surface area of the sphere = surface area of the cube

$$\Rightarrow 4\pi r^2 = 6a^2 \Rightarrow a^2 = \frac{2}{3}\pi r^2$$

$$\Rightarrow a = r\sqrt{\frac{2\pi}{3}}$$

$\therefore$ required ratio = volume of sphere : volume of cube

$$\begin{aligned} &= \frac{4}{3}\pi r^3 : a^3 = \frac{4}{3}\pi r^3 : \frac{2}{3}\pi r^3 \cdot \sqrt{\frac{2\pi}{3}} \\ &= 2 : \sqrt{\frac{2\pi}{3}} = \sqrt{2} : \sqrt{\frac{\pi}{3}} = \sqrt{6} : \sqrt{\pi}. \end{aligned}$$

EXAMPLE 3

A vessel is in the form of a hemispherical bowl mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find

(i) the capacity of the vessel,

[CBSE 2006C]

(ii) the inner surface area of the vessel.

[CBSE 2013]

SOLUTION

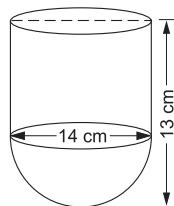
Radius of the hemisphere, $r = 7$ cm.

Total height of the vessel, $H = 13$ cm.

Height of the cylinder, $h = (13 - 7)$ cm = 6 cm.

Radius of the cylinder

= radius of the hemisphere = 7 cm.



(i) Capacity of the vessel

= volume of the hemisphere + volume of the cylinder

$$= \frac{2}{3}\pi r^3 + \pi r^2 h$$

$$= \left[\left(\frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 \right) + \left(\frac{22}{7} \times 7 \times 7 \times 6 \right) \right] \text{ cm}^3$$

$$= \left(\frac{2156}{3} + 924 \right) \text{ cm}^3 = (718.67 + 924) \text{ cm}^3$$

$$= 1642.67 \text{ cm}^3.$$

(ii) Inner surface area of the vessel

= curved surface area of the hemisphere

+ curved surface area of the cylinder

$$= 2\pi r^2 + 2\pi r h$$

$$= \left[\left(2 \times \frac{22}{7} \times 7 \times 7 \right) + \left(2 \times \frac{22}{7} \times 7 \times 6 \right) \right] \text{ cm}^2$$

$$= (308 + 264) \text{ cm}^2 = 572 \text{ cm}^2.$$

EXAMPLE 4 Due to some floods, some welfare associations jointly requested the government to get 100 tents fixed immediately and offered to contribute 50% of the cost. If the lower part of each tent is of the form of a cylinder of diameter 4.2 m and height 4 m with the conical upper part of same diameter but of height 2.8 m, and the canvas to be used costs ₹ 100 per square metre, find the amount the associations will have to pay. What values are shown by these associations?

SOLUTION Radius of the cylinder, $r = 2.1$ m.

Height of the cylinder, $h = 4$ m.

Radius of the cone = radius of the cylinder = 2.1 m.

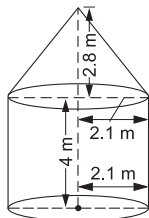
Height of the cone, $H = 2.8$ m.

Slant height of the cone, $l = \sqrt{r^2 + H^2}$

$$= \sqrt{(2.1)^2 + (2.8)^2} \text{ m}$$

$$= \sqrt{4.41 + 7.84} \text{ m}$$

$$= \sqrt{12.25} \text{ m} = 3.5 \text{ m}.$$



Area of the canvas required for each tent

= curved surface area of the cylinder

+ curved surface area of the cone

$$= 2\pi rh + \pi rl = \left[\left(2 \times \frac{22}{7} \times 2.1 \times 4 \right) + \left(\frac{22}{7} \times 2.1 \times 3.5 \right) \right] \text{ m}^2$$

$$= (52.8 + 23.1) \text{ m}^2 = 75.9 \text{ m}^2.$$

Total area of the canvas required for 100 tents

$$= (75.9 \times 100) \text{ m}^2 = 7590 \text{ m}^2.$$

Total cost of 100 tents = ₹ (7590 × 100) = ₹ 759000.

Amount to be paid by the associations

$$= 50\% \text{ of ₹ } 759000 = ₹ \left(\frac{50}{100} \times 759000 \right) = ₹ 379500.$$

Hence, the associations will have to pay ₹ 379500.

Values shown: We must

- help the people in need
- donate a part of our income to charity
- join hands together for a noble cause like community service.

EXAMPLE 5

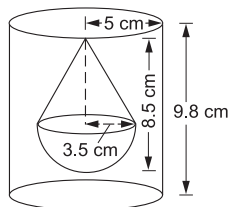
A cylindrical tub of radius 5 cm and length 9.8 cm is full of water. A solid in the form of a right circular cone mounted on a hemisphere is immersed into the tub. If the radius of the hemisphere is 3.5 cm and the total height of the solid is 8.5 cm, find the volume of water left in the tub.

SOLUTIONRadius of the cylinder, $r = 5$ cm.Height of the cylinder, $h = 9.8$ cm.Radius of the hemisphere, $R = 3.5$ cm.

Radius of the cone

= radius of the hemisphere

$$= 3.5 \text{ cm} = \frac{7}{2} \text{ cm}.$$

Height of the cone, $H = (8.5 - 3.5) \text{ cm} = 5 \text{ cm}$.

Volume of the water in the cylindrical tub

$$= \pi r^2 h = \left(\frac{22}{7} \times 5 \times 5 \times 9.8 \right) \text{ cm}^3 = 770 \text{ cm}^3.$$

Volume of the solid immersed in the tub

= volume of the hemisphere + volume of the cone

$$= \frac{2}{3} \pi R^3 + \frac{1}{3} \pi R^2 H$$

$$= \left[\left(\frac{2}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} \right) + \left(\frac{1}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 5 \right) \right] \text{ cm}^3$$

$$= \left(\frac{539}{6} + \frac{385}{6} \right) \text{ cm}^3 = \left(\frac{924}{6} \right) \text{ cm}^3 = 154 \text{ cm}^3.$$

$$\therefore \text{ volume of water left in the tub} = (770 - 154) \text{ cm}^3 = 616 \text{ cm}^3.$$

EXAMPLE 6

A solid wooden toy is in the form of a hemisphere surmounted by a cone of same radius. The radius of the hemisphere is 3.5 cm and the total wood used in the making of toy is $166\frac{5}{6} \text{ cm}^3$. Find the height of the toy. Also, find the cost of painting the hemispherical part of the toy at the rate of ₹ 10 per cm^2 .

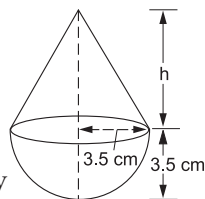
[CBSE 2015]

SOLUTIONLet the height of the conical part be h .Radius of the hemisphere, r

$$= \text{radius of the cone} = 3.5 \text{ cm} = \frac{7}{2} \text{ cm}.$$

Volume of the wood used in making the toy

= volume of the hemisphere + volume of the cone



$$\begin{aligned}
 &= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2(2r + h) \\
 &= \frac{1}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \left(2 \times \frac{7}{2} + h\right) = \frac{77}{6}(7 + h).
 \end{aligned}$$

But, the volume of the wood = $166\frac{5}{6} \text{ cm}^3 = \frac{1001}{6} \text{ cm}^3$.

$$\therefore \frac{77}{6}(7 + h) = \frac{1001}{6} \Rightarrow 7 + h = \frac{1001}{6} \times \frac{6}{77} = 13$$

$$\Rightarrow h = (13 - 7) \text{ cm} = 6 \text{ cm}.$$

So, the height of the toy = $h + r = (6 + 3.5) \text{ cm} = 9.5 \text{ cm}$.

Area to be painted = curved surface area of the hemisphere

$$\begin{aligned}
 &= 2\pi r^2 = \left(2 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2}\right) \text{ cm}^2 \\
 &= 77 \text{ cm}^2.
 \end{aligned}$$

Hence, cost of painting = ₹ (77×10) = ₹ 770.

EXAMPLE 7

A solid is in the shape of a cone mounted on a hemisphere of same base radius. If the curved surface areas of the hemispherical part and the conical part are equal then find the ratio of the radius and the height of the conical part. [CBSE 2012]

SOLUTION

Let the radius of each of the cone and the hemisphere be r . Let the height of the cone be h and its slant height be l .

Then, curved surface area of the hemisphere

= curved surface area of the cone

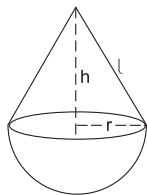
$$\Rightarrow 2\pi r^2 = \pi r l \Rightarrow 2\pi r^2 = \pi r \sqrt{r^2 + h^2}$$

$$\Rightarrow 2r = \sqrt{r^2 + h^2} \Rightarrow 4r^2 = r^2 + h^2$$

[squaring both sides]

$$\Rightarrow 3r^2 = h^2 \Rightarrow \frac{r^2}{h^2} = \frac{1}{3} \Rightarrow \frac{r}{h} = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}.$$

Hence, the required ratio is $1 : \sqrt{3}$.



EXAMPLE 8

A solid cylinder of diameter 12 cm and height 15 cm is melted and recast into 12 toys in the shape of a right circular cone mounted on a hemisphere. Find the radius of the hemisphere and total height of the toy, if the height of the cone is 3 times the radius.

SOLUTION

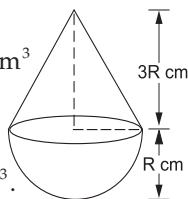
Radius of the cylinder, $r = 6 \text{ cm}$ and

height of the cylinder, $h = 15 \text{ cm}$.

$$\begin{aligned}\therefore \text{ volume of the cylinder} &= \pi r^2 h \\ &= (\pi \times 6 \times 6 \times 15) \text{ cm}^3 \\ &= (540\pi) \text{ cm}^3.\end{aligned}$$

$$\text{Volume of 12 toys} = (540\pi) \text{ cm}^3.$$

$$\therefore \text{ volume of 1 toy} = \left(\frac{540\pi}{12} \right) \text{ cm}^3 = (45\pi) \text{ cm}^3.$$



Let the radius of each of the hemisphere and cone be R cm.

Then, height of the cone, $H = (3R)$ cm.

Volume of 1 toy = volume of the hemisphere
+ volume of the cone

$$\begin{aligned}&= \frac{2}{3}\pi R^3 + \frac{1}{3}\pi R^2 H \\ &= \left(\frac{2}{3}\pi R^3 + \frac{1}{3}\pi R^2 \times 3R \right) \text{ cm}^3 \\ &= \left(\frac{5\pi R^3}{3} \right) \text{ cm}^3.\end{aligned}$$

$$\therefore \frac{5\pi R^3}{3} = 45\pi \Rightarrow R^3 = \left(45 \times \frac{3}{5} \right) = 27 = 3^3 \Rightarrow R = 3.$$

$$\begin{aligned}\text{Total height of the toy} &= (R + 3R) \text{ cm} = 4R \text{ cm} \\ &= (4 \times 3) \text{ cm} = 12 \text{ cm}.\end{aligned}$$

EXAMPLE 9

An iron pillar has some part in the form of a right circular cylinder and the remaining in the form of a right circular cone. The radius of the base of each of the cone and the cylinder is 8 cm. The cylindrical part is 240 cm high and the conical part is 36 cm high. Find the weight of the pillar if 1 cubic centimetre of iron weighs 7.5 g.

SOLUTION

Radius of the cylinder, $r = 8$ cm.

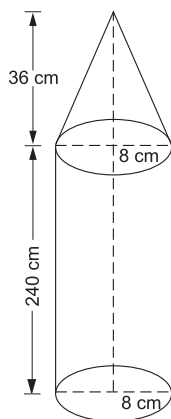
Radius of the cone, $r = 8$ cm.

Height of the cylinder, $h = 240$ cm.

Height of the cone, $H = 36$ cm.

Total volume of the iron

$$\begin{aligned}&= \text{volume of the cylinder} \\ &\quad + \text{volume of the cone} \\ &= \pi r^2 h + \frac{1}{3}\pi r^2 H = \pi r^2 \left(h + \frac{1}{3}H \right) \\ &= \left[\frac{22}{7} \times 8 \times 8 \times \left(240 + \frac{1}{3} \times 36 \right) \right] \text{ cm}^3 \\ &= 50688 \text{ cm}^3.\end{aligned}$$



$$\begin{aligned}\therefore \text{ weight of the pillar} &= \text{volume in cm}^3 \times \text{weight per cm}^3 \\ &= \left(\frac{50688 \times 7.5}{1000} \right) \text{ kg} = 380.16 \text{ kg.}\end{aligned}$$

Hence, the weight of the pillar is 380.16 kg.

EXAMPLE 10 A solid is composed of a cylinder with hemispherical ends. If the whole length of the solid is 104 cm and the radius of each of its hemispherical ends is 7 cm, find the cost of polishing its surface at the rate of ₹ 10 per dm^2 . [CBSE 2006C]

SOLUTION

Radius of each hemispherical end = 7 cm.

Height of each hemispherical part

= its radius = 7 cm.

Height of the cylindrical part

= $(104 - 2 \times 7)$ cm = 90 cm.

Area of surface to be polished

= 2 (curved surface area of the hemisphere)

+ (curved surface area of the cylinder)

= $[2(2\pi r^2) + 2\pi rh]$ sq units

$$= \left[\left(4 \times \frac{22}{7} \times 7 \times 7 \right) + \left(2 \times \frac{22}{7} \times 7 \times 90 \right) \right] \text{ cm}^2$$

$$= (616 + 3960) \text{ cm}^2 = 4576 \text{ cm}^2$$

$$= \left(\frac{4576}{10 \times 10} \right) \text{ dm}^2 = 45.76 \text{ dm}^2 \quad [\because 10 \text{ cm} = 1 \text{ dm}].$$

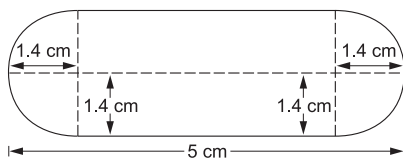
$\therefore$ cost of polishing the surface of the solid

$$= ₹ (45.76 \times 10) = ₹ 457.60.$$

EXAMPLE 11 A gulabjamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulabjamuns, each shaped like a cylinder with two hemispherical ends of length 5 cm and diameter 2.8 cm. [CBSE 2008]

SOLUTION

Radius of each hemispherical part, $r = \left(\frac{2.8}{2} \right)$ cm = 1.4 cm.



Radius of the cylindrical part, $r = 1.4$ cm.

Length of the cylindrical part, $h = (5 - 2 \times 1.4)$ cm = 2.2 cm.

Volume of one gulabjamun

$$\begin{aligned}
 &= \text{volume of 2 hemispherical parts} \\
 &\quad + \text{volume of the cylindrical part} \\
 &= \left(2 \times \frac{2}{3} \pi r^3 + \pi r^2 h \right) = \left(\frac{4}{3} \pi r^3 + \pi r^2 h \right) \\
 &= \pi r^2 \left(\frac{4}{3} r + h \right) = \left[\frac{22}{7} \times 1.4 \times 1.4 \times \left(\frac{4}{3} \times 1.4 + 2.2 \right) \right] \text{ cm}^3 \\
 &= \left(6.16 \times \frac{12.2}{3} \right) \text{ cm}^3 = \left(\frac{75.152}{3} \right) \text{ cm}^3.
 \end{aligned}$$

Volume of 45 gulabjamuns

$$= \left(\frac{75.152}{3} \times 45 \right) \text{ cm}^3 = 1127.28 \text{ cm}^3.$$

Volume of the syrup = (30% of 1127.28) cm^3

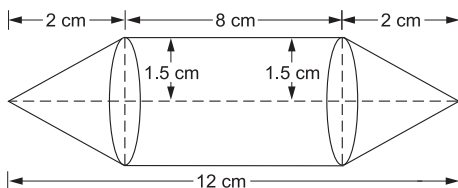
$$= \left(\frac{30}{100} \times 1127.28 \right) \text{ cm}^3 = 338.184 \text{ cm}^3.$$

Hence, 45 gulabjamuns contain approximately 338 cm^3 of syrup.

EXAMPLE 12 Rahul, an engineering student, prepared a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3 cm and its length is 12 cm. If each conical part has a height of 2 cm, find the (i) volume of air contained in Rahul's model (assume the outer and inner dimensions of the model to be nearly the same), (ii) cost of painting the outer surface of the model at ₹ 12.50 per cm^2 .

SOLUTION Radius of each conical part, $r = \frac{3}{2}$ cm.

Radius of cylindrical part, $r = \frac{3}{2}$ cm.



Height of each conical part, $h = 2$ cm.

Length of the cylindrical part, $H = (12 - 2 \times 2)$ cm = 8 cm.

Slant height of each conical part,

$$\begin{aligned} l &= \sqrt{r^2 + h^2} = \sqrt{\left(\frac{3}{2}\right)^2 + 2^2} \text{ cm} = \sqrt{\frac{9}{4} + 4} \text{ cm} \\ &= \sqrt{\frac{25}{4}} \text{ cm} = \frac{5}{2} \text{ cm}. \end{aligned}$$

(i) Volume of air contained in the model

= volume of 2 conical parts

+ volume of the cylindrical part

$$\begin{aligned} &= 2 \times \frac{1}{3} \pi r^2 h + \pi r^2 H = \pi r^2 \left(\frac{2}{3} h + H \right) \\ &= \left[\frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} \times \left(\frac{2}{3} \times 2 + 8 \right) \right] \text{ cm}^3 \\ &= \left(\frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} \times \frac{28}{3} \right) \text{ cm}^3 = 66 \text{ cm}^3. \end{aligned}$$

(ii) Total surface area of the model

= curved surface area of 2 conical parts

+ curved surface area of the cylindrical part

$$\begin{aligned} &= 2 \times \pi r l + 2 \pi r H = 2 \pi r (l + H) \\ &= \left[2 \times \frac{22}{7} \times \frac{3}{2} \times (2.5 + 8) \right] \text{ cm}^2 \\ &= \left(2 \times \frac{22}{7} \times \frac{3}{2} \times 10.5 \right) \text{ cm}^2 = 99 \text{ cm}^2. \end{aligned}$$

$$\therefore \text{ cost of painting the model} = ₹ (99 \times 12.50)$$

$$= ₹ 1237.50.$$

EXAMPLE 13 A spherical glass vessel has a cylindrical neck 8 cm long, 2 cm in diameter; the diameter of the spherical part is 8.5 cm. By measuring the amount of water it holds, Radhika finds its volume to be 345 cm^3 . Check mathematically whether she is correct, taking the above as the inside measurements and $\pi = 3.14$.

SOLUTION Radius of the spherical part, $R = \left(\frac{8.5}{2} \right)$ cm = 4.25 cm.

Radius of the cylindrical neck, $r = 1$ cm.

Height of the cylindrical neck, $h = 8$ cm.

Volume of the vessel

= volume of the spherical part
+ volume of the cylindrical part

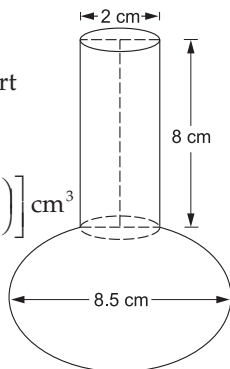
$$= \frac{4}{3}\pi R^3 + \pi r^2 h = \pi \left(\frac{4}{3}R^3 + r^2 h \right)$$

$$= \left[3.14 \times \left(\frac{4}{3} \times 4.25 \times 4.25 \times 4.25 + 1 \times 1 \times 8 \right) \right] \text{ cm}^3$$

$$= [3.14 \times (102.354 + 8)] \text{ cm}^3$$

$$= (3.14 \times 110.354) \text{ cm}^3$$

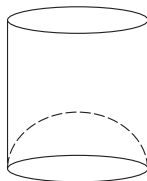
$$= 346.51156 \text{ cm}^3 \approx 346.51 \text{ cm}^3.$$



Hence, the volume found by Radhika is almost correct.

EXAMPLE 14

A juice seller was serving his customers using glasses as shown in the figure. The inner diameter of the cylindrical glass was 5 cm, but the bottom of the glass had a hemispherical raised portion which reduced the capacity of the glass. If the height of a glass was 10 cm, find the apparent capacity of the glass and its actual capacity. [Use $\pi = 3.14$.] [CBSE 2009]



SOLUTION

Radius of the cylindrical glass, $r = \frac{5}{2}$ cm.

Height of the cylindrical glass, $h = 10$ cm.

Radius of the hemispherical part, $r = \frac{5}{2}$ cm.

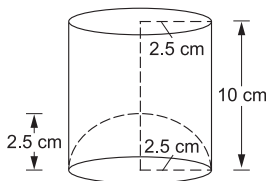
Apparent capacity of the glass

= volume of the cylinder

$$= \pi r^2 h$$

$$= \left(3.14 \times \frac{5}{2} \times \frac{5}{2} \times 10 \right) \text{ cm}^3$$

$$= 196.25 \text{ cm}^3.$$



Volume of the hemispherical part

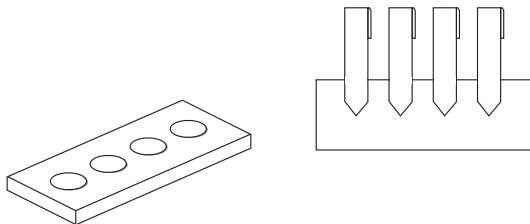
$$= \frac{2}{3}\pi r^3 = \left(\frac{2}{3} \times 3.14 \times \frac{5}{2} \times \frac{5}{2} \times \frac{5}{2} \right) \text{ cm}^3$$

$$= \left(\frac{196.25}{6} \right) \text{ cm}^3 = 32.708 \text{ cm}^3$$

$$\approx 32.71 \text{ cm}^3.$$

$$\begin{aligned}
 \therefore \text{ actual capacity of the glass} \\
 &= \text{apparent capacity of the glass} \\
 &\quad - \text{volume of the hemispherical part} \\
 &= (196.25 - 32.71) \text{ cm}^3 = 163.54 \text{ cm}^3.
 \end{aligned}$$

EXAMPLE 15 A pen stand made of wood is in the shape of a cuboid with four conical depressions to hold pens. The dimensions of the cuboid are 15 cm by 10 cm by 3.5 cm. The diameter of each of the depressions is 1 cm and the depth is 1.4 cm. Find the volume of wood in the entire stand.



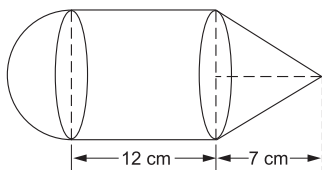
SOLUTION

Radius of each conical depression, $r = \frac{1}{2}$ cm.
 Depth of each conical depression, $h = 1.4$ cm.
 Volume of the wood in the stand

$$\begin{aligned}
 &= (\text{volume of the cuboid} \\
 &\quad - \text{volume of 4 conical depressions}) \\
 &= \left[(15 \times 10 \times 3.5) - \left(4 \times \frac{1}{3} \pi r^2 h \right) \right] \text{ cm}^3 \\
 &= \left[525 - \left(\frac{4}{3} \times \frac{22}{7} \times \frac{1}{2} \times \frac{1}{2} \times 1.4 \right) \right] \text{ cm}^3 \\
 &= \left(525 - \frac{4.4}{3} \right) \text{ cm}^3 = (525 - 1.47) \text{ cm}^3 \\
 &= 523.53 \text{ cm}^3.
 \end{aligned}$$

EXAMPLE 16 A solid toy is in the form of a right circular cylinder with a hemispherical shape at one end and a cone at the other end. Their common diameter is 4.2 cm, and the heights of the cylindrical and conical portions are 12 cm and 7 cm respectively. Find the volume of the given toy. [Take $\pi = \frac{22}{7}$.]

SOLUTION The shape of the toy is given below.



Radius of the hemispherical part, $r = \left(\frac{4.2}{2}\right)$ cm = 2.1 cm.

Radius of the cylindrical part, $r = 2.1$ cm.

Radius of the conical part, $r = 2.1$ cm.

Height of the cylindrical part, $h = 12$ cm.

Height of the conical part, $H = 7$ cm.

Volume of the toy

= volume of the hemispherical part

+ volume of the cylindrical part

+ volume of the conical part

$$= \frac{2}{3} \pi r^3 + \pi r^2 h + \frac{1}{3} \pi r^2 H = \pi r^2 \left(\frac{2}{3} r + h + \frac{1}{3} H \right)$$

$$= \left[\frac{22}{7} \times 2.1 \times 2.1 \times \left(\frac{2}{3} \times 2.1 + 12 + \frac{7}{3} \right) \right] \text{ cm}^3$$

$$= \left(13.86 \times \frac{47.2}{3} \right) \text{ cm}^3 = 218.064 \text{ cm}^3.$$

Hence, the volume of the given toy is 218.064 cm^3 .

EXAMPLE 17 The largest possible sphere is carved out from a solid wooden cube of side 7 cm. Find

(i) the volume of the sphere,

[CBSE 2011]

(ii) the percentage of wood wasted in the process.

SOLUTION

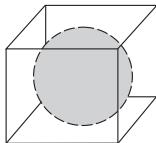
Clearly, diameter of the largest possible sphere = 7 cm.

Radius of the sphere, $r = \frac{7}{2}$ cm.

(i) Volume of the sphere

$$= \frac{4}{3} \pi r^3 = \left(\frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} \right) \text{ cm}^3$$

$$= \left(\frac{539}{3} \right) \text{ cm}^3 = 179.67 \text{ cm}^3.$$



(ii) Volume of the cube = $(7 \times 7 \times 7) \text{ cm}^3 = 343 \text{ cm}^3$.

Volume of wood wasted

= volume of cube – volume of sphere

$$= \left(343 - \frac{539}{3} \right) \text{ cm}^3 = \left(\frac{490}{3} \right) \text{ cm}^3.$$

$\therefore$ percentage of wood wasted

$$= \left(\frac{490}{3} \times \frac{1}{343} \times 100 \right) \% = \left(\frac{1000}{21} \right) \% = 47 \frac{13}{21} \%.$$

EXAMPLE 18 From a solid right circular cylinder with height 12 cm and radius of the base 5 cm, a right circular cone of the same height and the same base radius is removed. Find the volume and total surface area of the remaining solid. [Use $\pi = 3.14$.] [CBSE 2014]

SOLUTION Radius of each of the cylinder and cone, $r = 5$ cm.
Height of each of the cylinder and cone, $h = 12$ cm.
Slant height of the cone,

$$l = \sqrt{r^2 + h^2} = \sqrt{5^2 + (12)^2} \text{ cm} \\ = \sqrt{169} \text{ cm} = 13 \text{ cm}.$$

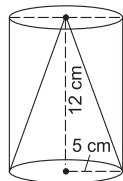
Volume of the remaining solid

= volume of the cylinder – volume of the cone.

$$= \pi r^2 h - \frac{1}{3} \pi r^2 h = \frac{2}{3} \pi r^2 h \\ = \left(\frac{2}{3} \times 3.14 \times 5 \times 5 \times 12 \right) \text{ cm}^3 = 628 \text{ cm}^3.$$

Total surface area of the remaining solid

$$\begin{aligned} &= \text{curved surface area of the cylinder} \\ &\quad + \text{curved surface area of the cone} \\ &\quad + \text{area of the upper circular face of the cylinder} \\ &= 2\pi rh + \pi rl + \pi r^2 = \pi r(2h + l + r) \\ &= [3.14 \times 5 \times (2 \times 12 + 13 + 5)] \text{ cm}^2 \\ &= (3.14 \times 5 \times 42) \text{ cm}^2 = 659.4 \text{ cm}^2. \end{aligned}$$



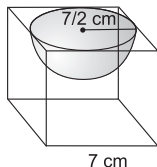
EXAMPLE 19 A hemispherical depression is cut out from one face of a cubical block of side 7 cm, such that the diameter of the hemisphere is equal to the edge of the cube. Find the surface area of the remaining solid. [CBSE 2014]

SOLUTION Edge of the cube, $a = 7$ cm.

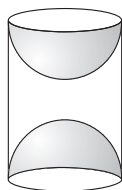
Radius of the hemisphere, $r = \frac{7}{2}$ cm.

Surface area of remaining solid

$$\begin{aligned} &= \text{total surface area of the cube} \\ &\quad - \text{area of the top of hemispherical part} \\ &\quad + \text{curved surface area of the hemisphere} \\ &= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2 \\ &= \left(6 \times 7 \times 7 + \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \text{ cm}^2 \\ &= (294 + 38.5) \text{ cm}^2 = 332.5 \text{ cm}^2. \end{aligned}$$



EXAMPLE 20 A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in the figure. If the height of the cylinder is 12 cm and its base is of radius 4.2 cm, find the total surface area of the article. Also, find the volume of the wood left in the article. [CBSE 2009C]



SOLUTION Radius of the hemisphere, $r = 4.2$ cm.

Radius of the cylinder, $r = 4.2$ cm

Height of the cylinder, $h = 12$ cm

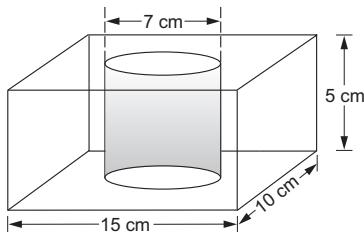
Total surface area of the article

$$\begin{aligned}
 &= \text{curved surface area of cylinder} \\
 &\quad + \text{curved surface area of 2 hemispheres} \\
 &= 2\pi rh + 2 \times 2\pi r^2 = 2\pi rh + 4\pi r^2 \\
 &= 2\pi r(h + 2r) = \left[2 \times \frac{22}{7} \times 4.2 \times (12 + 2 \times 4.2) \right] \text{ cm}^2 \\
 &= (26.4 \times 20.4) \text{ cm}^2 = 538.56 \text{ cm}^2.
 \end{aligned}$$

Total volume of the wood left in the article

$$\begin{aligned}
 &= \text{volume of the cylinder} - \text{volume of 2 hemispheres} \\
 &= \pi r^2 h - 2 \times \frac{2}{3} \pi r^3 = \pi r^2 \left(h - \frac{4}{3} r \right) \\
 &= \left[\frac{22}{7} \times 4.2 \times 4.2 \times \left(12 - \frac{4}{3} \times 4.2 \right) \right] \text{ cm}^3 \\
 &= (55.44 \times 6.4) \text{ cm}^3 = 354.816 \text{ cm}^3.
 \end{aligned}$$

EXAMPLE 21 In the given figure, from a cuboidal solid metallic block of dimensions 15 cm $\times$ 10 cm $\times$ 5 cm, a cylindrical hole of diameter 7 cm is drilled out. Find the surface area of the remaining block. [HOTS] [CBSE 2015]



SOLUTION Length of the cuboid, $l = 15$ cm.

Breadth of the cuboid, $b = 10$ cm.

Height of the cuboid, $h = 5$ cm.

Radius of the cylinder, $r = \frac{7}{2}$ cm.

Height of the cylinder, $h = 5$ cm.

Surface area of the remaining block

= total surface area of the cuboid

– area of two circular faces of the cylinder
+ curved surface area of the cylinder

$$= 2(lb + bh + lh) - 2 \times \pi r^2 + 2\pi rh$$

$$= \left[2(15 \times 10 + 10 \times 5 + 15 \times 5) - 2 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} + 2 \times \frac{22}{7} \times \frac{7}{2} \times 5 \right] \text{cm}^2$$

$$= (550 - 77 + 110) \text{cm}^2 = 583 \text{cm}^2.$$

EXAMPLE 22

A wooden toy rocket is in the shape of a cone mounted on a cylinder. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. The base of the conical portion has a diameter of 5 cm, while the base diameter of the cylindrical portion is 3 cm. If the conical portion is to be painted orange and the cylindrical portion yellow, find the area of the rocket painted with each of these colours. [Take $\pi = 3.14$.] [HOTS]

SOLUTION

Radius of the conical part, $r = \frac{5}{2}$ cm.

Height of the conical part, $h = 6$ cm.

Radius of the cylindrical part, $R = \frac{3}{2}$ cm.

Height of cylindrical part, $H = (26 - 6)$ cm
 $= 20$ cm.

Slant height of the conical part,

$$\begin{aligned} l &= \sqrt{r^2 + h^2} = \sqrt{\left(\frac{5}{2}\right)^2 + 6^2} \text{ cm} \\ &= \sqrt{\frac{25}{4} + 36} \text{ cm} = \sqrt{\frac{169}{4}} \text{ cm} = \frac{13}{2} \text{ cm}. \end{aligned}$$

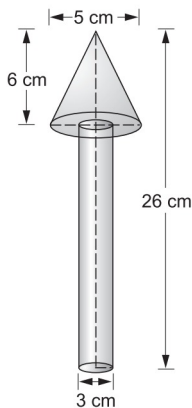
Area to be painted orange

= curved surface area of the cone

+ base area of the cone – base area of the cylinder

$$= \pi rl + \pi r^2 - \pi R^2 = \pi (rl + r^2 - R^2)$$

$$= \left[3.14 \times \left(\frac{5}{2} \times \frac{13}{2} + \frac{5}{2} \times \frac{5}{2} - \frac{3}{2} \times \frac{3}{2} \right) \right] \text{cm}^2$$



$$= \left[3.14 \times \left(\frac{65}{4} + \frac{25}{4} - \frac{9}{4} \right) \right] \text{cm}^2 = \left(3.14 \times \frac{81}{4} \right) \text{cm}^2$$

$$= (3.14 \times 20.25) \text{cm}^2 = 63.585 \text{cm}^2.$$

Area to be painted yellow

$$= \text{curved surface area of the cylinder}$$

$$+ \text{base area of the cylinder}$$

$$= 2\pi RH + \pi R^2 = \pi R(2H + R)$$

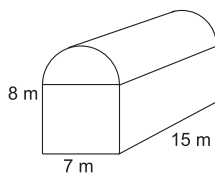
$$= \left[3.14 \times \frac{3}{2} \times \left(2 \times 20 + \frac{3}{2} \right) \right] \text{cm}^2$$

$$= \left(3.14 \times \frac{3}{2} \times \frac{83}{2} \right) \text{cm}^2 = \left(\frac{781.86}{4} \right) \text{cm}^2$$

$$= 195.465 \text{cm}^2.$$

EXAMPLE 23

Deepa runs an industry in a shed which is in the shape of a cuboid surmounted by a half-cylinder. The base of the shed is of dimensions 7 m $\times$ 15 m and the height of the cuboidal portion is 8 m.



- (i) Find the volume of air that the shed can hold.
- (ii) Suppose the machinery in the shed occupies a total space of 300 m^3 and there are 20 workers, each of whom occupy about 0.08 m^3 space on an average. Then, how much air is in the shed?
- (iii) Find the internal surface area of the shed, excluding the floor.

[HOTS]

SOLUTION The shed consists of a cuboid with dimensions $l = 15 \text{ m}$, $b = 7 \text{ m}$, $h = 8 \text{ m}$ and a half-cylinder of radius, $r = \frac{7}{2} \text{ m}$ and height, $H = 15 \text{ m}$.

- (i) Volume of air that the shed can hold

$$= \text{volume of the cuboid} + \frac{1}{2} \times \text{volume of the cylinder}$$

$$= lbh + \frac{1}{2} \pi r^2 H = \left(15 \times 7 \times 8 + \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 15 \right) \text{m}^3$$

$$= \left(840 + \frac{1155}{4} \right) \text{m}^3 = \left(\frac{4515}{4} \right) \text{m}^3$$

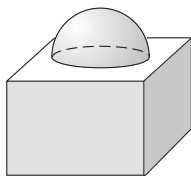
$$= 1128.75 \text{m}^3.$$

- (ii) Total space occupied by machinery and 20 workers
 $= (300 + 0.08 \times 20) \text{ m}^3 = 301.6 \text{ m}^3$.

Volume of air in the shed when there are machinery and workers inside it $= (1128.75 - 301.6) \text{ m}^3 = 827.15 \text{ m}^3$

- (iii) Total internal surface area of the shed (excluding floor)
 $= (\text{surface area of walls}) + (\text{surface area of ceiling})$
 $= (\text{area of two walls each measuring } 15 \text{ m} \times 8 \text{ m})$
 $+ (\text{area of two walls each measuring } 7 \text{ m} \times 8 \text{ m})$
 $+ (\text{area of two semicircles each of radius } 3.5 \text{ m})$
 $+ (\text{curved surface area } \pi rh \text{ of half-cylinder})$
 $= \left[(2 \times 15 \times 8) + (2 \times 7 \times 8) + \left(2 \times \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \right.$
 $\left. + \left(\frac{22}{7} \times \frac{7}{2} \times 15 \right) \right] \text{ m}^2$
 $= (240 + 112 + 38.5 + 165) \text{ m}^2 = 555.5 \text{ m}^2$.

EXAMPLE 24 A solid is made up of a cube and a hemisphere attached on its top, as shown in the figure. Each edge of the cube measures 5 cm and the hemisphere has a diameter of 4.2 cm. Find the total area to be painted.
 [Take $\pi = \frac{22}{7}$.] [CBSE 2009]



SOLUTION Edge of the cube, $a = 5 \text{ cm}$.

Radius of the hemisphere, $r = \left(\frac{4.2}{2} \right) \text{ cm} = 2.1 \text{ cm}$.

Total area to be painted

$$\begin{aligned}
 &= \text{total surface area of the cube} \\
 &\quad - \text{base area of the hemisphere} \\
 &\quad + \text{curved surface area of the hemisphere} \\
 &= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2 \\
 &= \left(6 \times 5 \times 5 + \frac{22}{7} \times 2.1 \times 2.1 \right) \text{ cm}^2 \\
 &= (150 + 13.86) \text{ cm}^2 = 163.86 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 25 A solid toy is in the form of a hemisphere surmounted by a right circular cone of height 2 cm and diameter of base 4 cm. If a right circular cylinder circumscribes the toy, find how much more space than the toy it will cover. [Use $\pi = 3.14$.]

SOLUTION

Let OAB be the cone and ACB be the hemisphere, having the same base AB . Let the right circular cylinder $DEFG$ circumscribe the given solid.

Radius of each of the cone, cylinder and hemisphere, $r = 2$ cm.

Height of the cone, $h = 2$ cm.

Height of the cylinder,

$$H = h + r = (2 + 2) \text{ cm} = 4 \text{ cm}.$$

Volume of the toy

$$= (\text{volume of the hemisphere}) + (\text{volume of the cone})$$

$$= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h$$

$$= \left(\frac{2}{3}\pi \times 2^3 + \frac{1}{3}\pi \times 2^2 \times 2 \right) \text{ cm}^3$$

$$= \left(\frac{16\pi}{3} + \frac{8\pi}{3} \right) \text{ cm}^3 = 8\pi \text{ cm}^3.$$

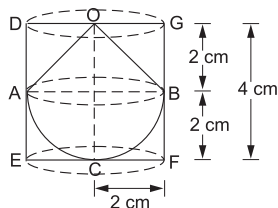
Volume of the cylinder = $\pi r^2 H$

$$= (\pi \times 2^2 \times 4) \text{ cm}^3 = (16\pi) \text{ cm}^3.$$

Required volume = volume of the cylinder – volume of the toy

$$= (16\pi - 8\pi) \text{ cm}^3 = 8\pi \text{ cm}^3$$

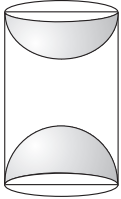
$$= (8 \times 3.14) \text{ cm}^3 = 25.12 \text{ cm}^3.$$

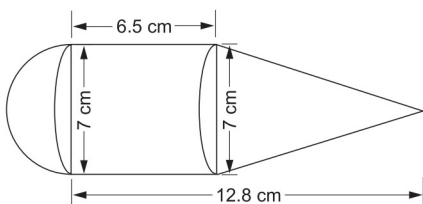


EXERCISE 17A

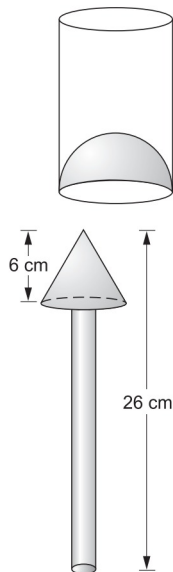
- Two cubes each of volume 27 cm^3 are joined end to end to form a solid. Find the surface area of the resulting cuboid. [CBSE 2011, '14]
- The volume of a hemisphere is $2425\frac{1}{2} \text{ cm}^3$. Find its curved surface area. [CBSE 2012]
- If the total surface area of a solid hemisphere is 462 cm^2 , find its volume. [CBSE 2014]
- A 5-m-wide cloth is used to make a conical tent of base diameter 14 m and height 24 m. Find the cost of cloth used at the rate of ₹ 25 per metre. [CBSE 2014]
 - The radius and height of a solid right-circular cone are in the ratio of 5 : 12. If its volume is 314 cm^3 , find its total surface area. [Take $\pi = 3.14$.] [CBSE 2017]

5. If the volumes of two cones are in the ratio of 1 : 4 and their diameters are in the ratio of 4 : 5, find the ratio of their heights.
6. The slant height of a conical mountain is 2.5 km and the area of its base is 1.54 km^2 . Find the height of the mountain.
7. The sum of the radius of the base and the height of a solid cylinder is 37 metres. If the total surface area of the cylinder be 1628 sq metres , find its volume.
8. The surface area of a sphere is 2464 cm^2 . If its radius be doubled, what will be the surface area of the new sphere?
9. A military tent of height 8.25 m is in the form of a right circular cylinder of base diameter 30 m and height 5.5 m surmounted by a right circular cone of same base radius. Find the length of canvas used in making the tent, if the breadth of the canvas is 1.5 m. [CBSE 2012]
10. A tent is in the shape of a right circular cylinder up to a height of 3 m and conical above it. The total height of the tent is 13.5 m and the radius of its base is 14 m. Find the cost of cloth required to make the tent at the rate of ₹ 80 per square metre. [Take $\pi = \frac{22}{7}$.] [CBSE 2005]
11. A circus tent is cylindrical to a height of 3 m and conical above it. If its base radius is 52.5 m and the slant height of the conical portion is 53 m, find the area of canvas needed to make the tent. [Take $\pi = \frac{22}{7}$.] [CBSE 2004]
12. A rocket is in the form of a circular cylinder closed at the lower end and a cone of the same radius is attached to the top. The radius of the cylinder is 2.5 m, its height is 21 m and the slant height of the cone is 8 m. Calculate the total surface area of the rocket.
13. A solid is in the shape of a cone surmounted on a hemisphere, the radius of each of them being 3.5 cm and the total height of the solid is 9.5 cm. Find the volume of the solid. [CBSE 2012]
14. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius on its circular face. The total height of the toy is 15.5 cm. Find the total surface area of the toy. [CBSE 2017]
15. A toy is in the shape of a cone mounted on a hemisphere of same base radius. If the volume of the toy is 231 cm^3 and its diameter is 7 cm, find the height of the toy. [CBSE 2012]
16. A cylindrical container of radius 6 cm and height 15 cm is filled with ice cream. The whole ice cream has to be distributed to 10 children in equal cones with hemispherical tops. If the height of the conical portion is 4 times the radius of its base, find the radius of the ice cream cone.
17. A vessel is in the form of a hemispherical bowl surmounted by a hollow cylinder. The diameter of the hemisphere is 21 cm and the total height of the vessel is 14.5 cm. Find its capacity.

18. A toy is in the form of a cylinder with hemispherical ends. If the whole length of the toy is 90 cm and its diameter is 42 cm, find the cost of painting the toy at the rate of 70 paise per sq cm. [CBSE 2014]
19. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.
20. A wooden article was made by scooping out a hemisphere from each end of a cylinder, as shown in the figure. If the height of the cylinder is 20 cm and its base is of diameter 7 cm, find the total surface area of the article when it is ready. [CBSE 2008C]
- 
21. A solid is in the form of a right circular cone mounted on a hemisphere. The radius of the hemisphere is 2.1 cm and the height of the cone is 4 cm. The solid is placed in a cylindrical tub full of water in such a way that the whole solid is submerged in water. If the radius of the cylinder is 5 cm and its height is 9.8 cm, find the volume of the water left in the tub. [CBSE 1996C, 2000C]
22. From a solid cylinder whose height is 8 cm and radius 6 cm, a conical cavity of height 8 cm and of base radius 6 cm is hollowed out. Find the volume of the remaining solid. Also, find the total surface area of the remaining solid. [Take $\pi = 3.14$.] [HOTS] [CBSE 2009]
23. From a solid cylinder of height 2.8 cm and diameter 4.2 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid. [CBSE 2014]
24. From a solid cylinder of height 14 cm and base diameter 7 cm, two equal conical holes each of radius 2.1 cm and height 4 cm are cut off. Find the volume of the remaining solid. [CBSE 2011]
25. A metallic cylinder has radius 3 cm and height 5 cm. To reduce its weight, a conical hole is drilled in the cylinder. The conical hole has a radius of $\frac{3}{2}$ cm and its depth is $\frac{8}{9}$ cm. Calculate the ratio of the volume of metal left in the cylinder to the volume of metal taken out in conical shape. [CBSE 2015]
26. A spherical glass vessel has a cylindrical neck 7 cm long and 4 cm in diameter. The diameter of the spherical part is 21 cm. Find the quantity of water it can hold. [Use $\pi = \frac{22}{7}$.] [CBSE 2009C]
27. The given figure represents a solid consisting of a cylinder surmounted by a cone at one end and a hemisphere at the other. Find the volume of the solid.



28. From a cubical piece of wood of side 21 cm, a hemisphere is carved out in such a way that the diameter of the hemisphere is equal to the side of the cubical piece. Find the surface area and volume of the remaining piece. [CBSE 2014]
29. (i) A hemisphere of maximum possible diameter is placed over a cuboidal block of side 7 cm. Find the surface area of the solid so formed. [CBSE 2017]
- (ii) A cubical block of side 10 cm is surmounted by a hemisphere. What is the largest diameter that the hemisphere can have? Find the cost of painting the total surface area of the solid so formed, at the rate of ₹ 5 per 100 sq cm. [Use $\pi = 3.14$.] [CBSE 2015]
30. A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The radius and height of the cylindrical part are 5 cm and 13 cm respectively. The radii of the hemispherical and the conical parts are the same as that of the cylindrical part. Find the surface area of the toy, if the total height of the toy is 30 cm.
31. The inner diameter of a glass is 7 cm and it has a raised portion in the bottom in the shape of a hemisphere, as shown in the figure. If the height of the glass is 16 cm, find the apparent capacity and the actual capacity of the glass.
32. A wooden toy is in the shape of a cone mounted on a cylinder, as shown in the figure. The total height of the toy is 26 cm, while the height of the conical part is 6 cm. The diameter of the base of the conical part is 5 cm and that of the cylindrical part is 4 cm. The conical part and the cylindrical part are respectively painted red and white. Find the area to be painted by each of these colours. [Take $\pi = \frac{22}{7}$.]



ANSWERS (EXERCISE 17A)

1. 90 cm^2 2. 693 cm^2 3. 718.67 cm^2 4. (i) ₹ 2750 (ii) 282.6 cm^2
 5. $25 : 64$ 6. 2.4 km 7. 4620 m^3 8. 9856 cm^2
 9. 825 m 10. ₹ 82720 11. 9735 m^2 12. 412.5 m^2
 13. 166.83 cm^3 14. 214.5 cm^2 15. 18 cm 16. 3 cm
 17. 3811.5 cm^3 18. ₹ 8316 19. 220 mm^2 20. 594 cm^2
 21. 732.116 cm^3 22. 602.88 cm^2 23. 73.92 cm^2 24. 502.04 cm^3
 25. $1463 : 22$ 26. 4939 cm^3 27. 376.016 m^3 28. 6835.5 cm^3 , 2992.5 cm^2
 29. (i) 332.5 cm^2 (ii) 10 cm , ₹ 53.93 30. 770 cm^2 31. 526.17 cm^3 , 616 cm^3
 32. Area painted red = 58.143 cm^2 , Area painted white = 264 cm^2

HINTS TO SOME SELECTED QUESTIONS

4. (i) Area of the cloth required = curved surface area of the tent = $\pi r l = \pi r \sqrt{r^2 + h^2}$.
 Length of the cloth required = $\frac{\text{area}}{\text{width}}$.
5. Let the radii of the cones be $4r$ and $5r$ and their heights be h_1 and h_2 respectively. Then,

$$\frac{\frac{1}{3} \pi \times (4r)^2 \times h_1}{\frac{1}{3} \pi \times (5r)^2 \times h_2} = \frac{1}{4} \Rightarrow \frac{16h_1}{25h_2} = \frac{1}{4} \Rightarrow \frac{h_1}{h_2} = \frac{1}{4} \times \frac{25}{16} = \frac{25}{64}$$
6. $\pi r^2 = 1.54$. Find r .
 Calculate l , using the formula, $l = \sqrt{r^2 + h^2}$.
7. $h + r = 37 \text{ m}$ and $2\pi r(h + r) = 1628$. Calculate r . Then, find h and calculate volume.
8. $4\pi r^2 = 2464$. Find r .
 Now, calculate surface area using $2r$ as radius.
9. Total area of the canvas required
 = curved surface area of the cylinder + curved surface area of the cone.
 Length of the canvas = $\frac{\text{area}}{\text{width}}$.
13. Height of the conical part = $(9.5 - 3.5) \text{ cm} = 6 \text{ cm}$.
 Volume of the solid = volume of the cone + volume of the hemisphere.
18. Length of the cylindrical part = $(90 - 2 \times 21) \text{ cm} = 48 \text{ cm}$.
 Total area to be painted
 = $2 \times$ curved surface area of a hemisphere + curved surface area of the cylinder.
20. Total surface area
 = curved surface area of the cylinder + $2 \times$ curved surface area of each hemisphere.

22. Volume of the remaining solid = volume of the cylinder – volume of the cone.

Surface area of the remaining solid

$$= \text{curved surface area of the cylinder} + \text{curved surface area of the cone} \\ + \text{area of upper circular face of the cylinder.}$$

28. Surface area of the remaining piece

$$= \text{total surface area of the cube} - \text{area of the top of the hemispherical part} \\ + \text{curved surface area of the hemisphere.}$$

Volume of the remaining piece

$$= \text{volume of the cube} - \text{volume of the hemisphere.}$$

29. (ii) Largest possible diameter = edge of the cube.

Total surface area of the solid

$$= \text{surface area of the cube} - \text{area of the top of the hemispherical part} \\ + \text{curved surface area of the hemisphere.}$$

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CONVERSION OF SOLID FROM ONE SHAPE TO ANOTHER AND MIXED PROBLEMS

EXAMPLE 1 *The dimensions of a metallic cuboid are $100 \text{ cm} \times 80 \text{ cm} \times 64 \text{ cm}$. It is melted and recast into a cube. Find the surface area of the cube.*

[CBSE 2011]

SOLUTION Let each edge of the cube be a .

Then, volume of the cube = a^3 .

$$\text{Volume of the cuboid} = (100 \times 80 \times 64) \text{ cm}^3 \\ = 512000 \text{ cm}^3.$$

Volume of the cube = volume of the cuboid

$$\Rightarrow a^3 = 512000 \text{ cm}^3$$

$$\Rightarrow a = \sqrt[3]{512 \times 1000} \text{ cm} = (8 \times 10) \text{ cm} = 80 \text{ cm.}$$

Surface area of the cube = $6a^2$

$$= (6 \times 80 \times 80) \text{ cm}^2 = 38400 \text{ cm}^2.$$

EXAMPLE 2 *How many spherical solid bullets can be made out of a solid cube of lead whose edge measures 44 cm , each bullet being 4 cm in diameter?*

[CBSE 2014]

SOLUTION Edge of the cube, $a = 44 \text{ cm}$.

$$\text{Volume of the cube} = a^3 = (44 \times 44 \times 44) \text{ cm}^3.$$

Radius of each spherical bullet, $r = 2 \text{ cm}$.

$$\text{Volume of each spherical bullet} = \frac{4}{3}\pi r^3$$

$$= \left(\frac{4}{3} \times \frac{22}{7} \times 2 \times 2 \times 2 \right) \text{cm}^3 = \left(\frac{704}{21} \right) \text{cm}^3$$

$\therefore$ required number of bullets

$$= \frac{\text{volume of the cube}}{\text{volume of each spherical bullet}}$$

$$= \frac{44 \times 44 \times 44}{704/21} = 44 \times 44 \times 44 \times \frac{21}{704} = 2541.$$

EXAMPLE 3 How many silver coins, 1.75 cm in diameter and of thickness 2 mm, must be melted to form a cuboid of dimensions

$$5.5 \text{ cm} \times 10 \text{ cm} \times 3.5 \text{ cm}?$$

SOLUTION Radius of each coin, $r = \left(\frac{1.75}{2} \right) \text{cm} = \left(\frac{175}{200} \right) \text{cm} = \frac{7}{8} \text{cm}$.

Thickness of each coin, $h = 2 \text{ mm} = \left(\frac{2}{10} \right) \text{cm} = \frac{1}{5} \text{cm}$.

Volume of each coin $= \pi r^2 h$

$$= \left(\frac{22}{7} \times \frac{7}{8} \times \frac{7}{8} \times \frac{1}{5} \right) \text{cm}^3 = \left(\frac{77}{160} \right) \text{cm}^3.$$

Volume of the cuboid $= (5.5 \times 10 \times 3.5) \text{cm}^3$
 $= 192.5 \text{cm}^3$.

$\therefore$ number of coins $= \frac{\text{volume of the cuboid}}{\text{volume of each coin}}$
 $= 192.5 \times \frac{160}{77} = 400$.

EXAMPLE 4 A metallic sphere of radius 10.5 cm is melted and then recast into smaller cones, each of radius 3.5 cm and height 3 cm. How many cones are obtained? [CBSE 2004, '12, '17]

SOLUTION Radius of the sphere, $R = \frac{21}{2} \text{cm}$.

Volume of the sphere $= \frac{4}{3} \pi R^3 = \left(\frac{4}{3} \pi \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} \right) \text{cm}^3$
 $= \left(\frac{3087 \pi}{2} \right) \text{cm}^3$.

Radius of each cone, $r = \frac{7}{2} \text{cm}$.

Height of each cone, $h = 3 \text{cm}$.

$$\begin{aligned}\text{Volume of each cone} &= \frac{1}{3} \pi r^2 h = \left(\frac{1}{3} \pi \times \frac{7}{2} \times \frac{7}{2} \times 3 \right) \text{ cm}^3 \\ &= \left(\frac{49\pi}{4} \right) \text{ cm}^3.\end{aligned}$$

$$\begin{aligned}\therefore \text{required number of cones} &= \frac{\text{volume of the sphere}}{\text{volume of each cone}} \\ &= \left(\frac{3087\pi}{2} \times \frac{4}{49\pi} \right) = 126.\end{aligned}$$

EXAMPLE 5

The internal and external radii of a hollow sphere are 3 cm and 5 cm respectively. The sphere is melted to form a solid cylinder of height $2\frac{2}{3}$ cm. Find the diameter and the curved surface area of the cylinder.

SOLUTION

External radius of the sphere, $r_1 = 5$ cm.

Internal radius of the sphere, $r_2 = 3$ cm.

Volume of metal in the hollow sphere

$$\begin{aligned}&= \frac{4}{3} \pi (r_1^3 - r_2^3) = \left[\frac{4}{3} \pi \times (5^3 - 3^3) \right] \text{ cm}^3 \\ &= \left(\frac{392\pi}{3} \right) \text{ cm}^3.\end{aligned}$$

Let the radius of the solid cylinder be R cm.

Height of the solid cylinder, $h = \frac{8}{3}$ cm.

Volume of the solid cylinder

$$= \pi R^2 h = \left(\pi R^2 \times \frac{8}{3} \right) \text{ cm}^3 = \left(\frac{8\pi R^2}{3} \right) \text{ cm}^3.$$

Volume of cylinder = volume of metal in the hollow sphere

$$\Rightarrow \frac{8\pi R^2}{3} = \frac{392\pi}{3} \Rightarrow R^2 = 49 \Rightarrow R = 7.$$

$\therefore$ diameter of the cylinder formed = (2×7) cm = 14 cm.

Curved surface area of the cylinder

$$\begin{aligned}&= 2\pi r h = \left(2 \times \frac{22}{7} \times 7 \times \frac{8}{3} \right) \text{ cm}^2 \\ &= \frac{352}{3} \text{ cm}^2 = 117\frac{1}{3} \text{ cm}^2.\end{aligned}$$

EXAMPLE 6 A hollow sphere of internal and external diameters 4 cm and 8 cm is melted to form a cone of base diameter 8 cm. Find the height and the slant height of the cone. [CBSE 2011]

SOLUTION External radius of the sphere, $r_1 = 4$ cm.

Internal radius of the sphere, $r_2 = 2$ cm.

Volume of metal in the hollow sphere

$$= \frac{4}{3}\pi(r_1^3 - r_2^3) = \left[\frac{4}{3}\pi \times (4^3 - 2^3) \right] \text{cm}^3 = \left(\frac{224\pi}{3} \right) \text{cm}^3.$$

Radius of the cone formed, $R = 4$ cm.

Let the height of the cone be h .

Volume of the cone

$$= \frac{1}{3}\pi R^2 h = \left(\frac{1}{3}\pi \times 4 \times 4 \times h \right) \text{cm}^3 = \left(\frac{16\pi h}{3} \right) \text{cm}^3.$$

Volume of the cone = volume of metal in the sphere

$$\Rightarrow \frac{16\pi h}{3} = \frac{224\pi}{3} \Rightarrow h = 14 \text{ cm}.$$

Slant height, $l = \sqrt{R^2 + h^2} = \sqrt{4^2 + 14^2}$ cm

$$= \sqrt{212} \text{ cm} = 2\sqrt{53} \text{ cm}.$$

Hence, the height of the cone is 14 cm and its slant height is $2\sqrt{53}$ cm.

EXAMPLE 7 A girl empties a cylindrical bucket full of sand, of base radius 18 cm and height 32 cm, on the floor to form a conical heap of sand. If the height of this conical heap is 24 cm then find its slant height correct to one place of decimal. [CBSE 2014]

SOLUTION Radius of the cylindrical bucket, $R = 18$ cm.

Height of the cylindrical bucket, $H = 32$ cm.

Volume of sand in the bucket = $\pi R^2 H = (\pi \times 18 \times 18 \times 32) \text{cm}^3$.

Let the radius of the conical heap be r .

Height of the conical heap, $h = 24$ cm.

$$\begin{aligned} \text{Volume of the conical heap} &= \frac{1}{3}\pi r^2 h = \left(\frac{1}{3}\pi \times r^2 \times 24 \right) \text{cm}^3 \\ &= (8\pi r^2) \text{cm}^3. \end{aligned}$$

Volume of the conical heap = volume of sand in the bucket

$$\Rightarrow 8\pi r^2 = \pi \times 18 \times 18 \times 32 \Rightarrow r^2 = \frac{18 \times 18 \times 32}{8} = 1296$$

$$\Rightarrow r = \sqrt{1296} = 36 \text{ cm}.$$

$$\begin{aligned}
 \text{Slant height, } l &= \sqrt{r^2 + h^2} \\
 &= \sqrt{(36)^2 + (24)^2} \text{ cm} = \sqrt{1296 + 576} \text{ cm} \\
 &= \sqrt{1872} \text{ cm} = 43.26 \text{ cm.} \\
 &\approx 43.3 \text{ cm (correct to one decimal place).}
 \end{aligned}$$

EXAMPLE 8 504 cones, each of diameter 3.5 cm and height 3 cm, are melted and recast into a metallic sphere. Find the diameter of the sphere and hence find its surface area. [CBSE 2015]

SOLUTION Radius of each cone, $r = \left(\frac{3.5}{2}\right) \text{ cm} = \left(\frac{35}{20}\right) \text{ cm} = \frac{7}{4} \text{ cm}$.

Height of each cone, $h = 3 \text{ cm}$.

$$\begin{aligned}
 \text{Volume of each cone} &= \frac{1}{3} \pi r^2 h \\
 &= \left(\frac{1}{3} \times \pi \times \frac{7}{4} \times \frac{7}{4} \times 3\right) \text{ cm}^3 = \left(\frac{49\pi}{16}\right) \text{ cm}^3.
 \end{aligned}$$

Total volume of 504 cones

$$= \left(\frac{49\pi}{16} \times 504\right) \text{ cm}^3 = \left(\frac{49\pi \times 63}{2}\right) \text{ cm}^3.$$

Let the radius of the sphere be R .

$$\text{Then, volume of the sphere} = \frac{4}{3} \pi R^3.$$

Volume of the sphere = total volume of 504 cones

$$\begin{aligned}
 \Rightarrow \frac{4}{3} \pi R^3 &= \frac{49\pi \times 63}{2} \Rightarrow R^3 = \left(\frac{49 \times 63}{2} \times \frac{3}{4}\right) \\
 \Rightarrow R^3 &= \left(\frac{7 \times 7 \times 7 \times 9 \times 3}{2 \times 4}\right) \Rightarrow R = \left(\frac{7 \times 3}{2}\right) \text{ cm} = \frac{21}{2} \text{ cm.}
 \end{aligned}$$

$$\text{Diameter of the sphere} = \left(\frac{21}{2} \times 2\right) \text{ cm} = 21 \text{ cm.}$$

$$\begin{aligned}
 \text{Surface area of the sphere} &= 4\pi R^2 = \left(4 \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2}\right) \text{ cm}^2 \\
 &= 1386 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 9 Two spheres of same metal weigh 1 kg and 7 kg. The radius of the smaller sphere is 3 cm. The two spheres are melted to form a single big sphere. Find the diameter of the new sphere. [HOTS] [CBSE 2015]

SOLUTION Since both the spheres are made of same metal, their weights are directly proportional to their volumes, or the ratio of their volumes is equal to the ratio of their weights.

Radius of first small sphere, $r_1 = 3$ cm.

Let the radius of second small sphere be r_2 cm.

$$\therefore \frac{\text{volume of first small sphere}}{\text{volume of second small sphere}} = \frac{1}{7}$$

$$\Rightarrow \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{1}{7} \Rightarrow \frac{3^3}{r_2^3} = \frac{1}{7} \Rightarrow r_2^3 = 27 \times 7 = 189.$$

Sum of volumes of two small spheres

$$\begin{aligned} &= \frac{4}{3}\pi (r_1^3 + r_2^3) = \left[\frac{4}{3}\pi (3^3 + 189) \right] \text{cm}^3 \\ &= \left(\frac{4}{3}\pi \times 216 \right) \text{cm}^3. \end{aligned}$$

Let the radius of the new sphere be R .

Volume of new sphere = sum of volumes of two small spheres

$$\Rightarrow \frac{4}{3}\pi R^3 = \frac{4}{3}\pi \times 216 \Rightarrow R^3 = 216$$

$$\Rightarrow R = \sqrt[3]{216} = 6 \text{ cm} \Rightarrow \text{diameter} = 2R = 12 \text{ cm}.$$

EXAMPLE 10 *The diameter of a copper sphere is 6 cm. The sphere is melted and drawn into a long wire of uniform circular cross section. If the length of the wire is 36 m, find its thickness.*

SOLUTION Radius of the sphere, $R = 3$ cm.

$$\begin{aligned} \text{Volume of the sphere} &= \frac{4}{3}\pi R^3 \\ &= \left(\frac{4}{3}\pi \times 3 \times 3 \times 3 \right) \text{cm}^3 \\ &= (36\pi) \text{cm}^3. \end{aligned}$$

Length of the wire, $h = 36 \text{ m} = 3600 \text{ cm}$.

Let the radius of the wire be r .

$$\text{Volume of the wire} = \pi r^2 h = (\pi r^2 \times 3600) \text{cm}^3.$$

But, volume of the wire = volume of the sphere

$$\Rightarrow 3600\pi r^2 = 36\pi \Rightarrow r^2 = \frac{1}{100}$$

$$\Rightarrow r = \sqrt{\frac{1}{100}} \text{ cm} = \frac{1}{10} \text{ cm} = 1 \text{ mm}.$$

Hence, thickness of the wire = its diameter = 2 mm.

EXAMPLE 11 A solid cylinder of diameter 12 cm and height 15 cm is melted and recast into 12 toys in the shape of a right circular cone mounted on a hemisphere. Find the radius of the hemisphere and the total height of the toy, if the height of the conical part is thrice its radius.

[CBSE 2005C]

SOLUTION Radius of the cylinder, $R = 6$ cm.

Height of the cylinder, $H = 15$ cm

Volume of the cylinder $= \pi R^2 H$

$$= (\pi \times 6 \times 6 \times 15) \text{ cm}^3$$

$$= (540\pi) \text{ cm}^3.$$

Let the radius of each of the conical and hemispherical parts be r .

Then, height of the conical part, $h = 3r$.

Volume of each toy

= volume of the hemisphere + volume of the cone

$$= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = \left(\frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 \times 3r \right) \text{ cm}^3$$

$$= \left(\frac{2}{3}\pi r^3 + \pi r^3 \right) \text{ cm}^3 = \left(\frac{5\pi r^3}{3} \right) \text{ cm}^3.$$

Now, total volume of 12 toys = volume of the cylinder

$$\Rightarrow 12 \times \frac{5\pi r^3}{3} = 540\pi \Rightarrow r^3 = \frac{540}{20} = 27 \Rightarrow r = \sqrt[3]{27} = 3 \text{ cm}.$$

Hence, radius of the hemisphere = 3 cm.

Total height of the toy $= (r + h) = (r + 3r) = 4r$

$$= (4 \times 3) \text{ cm} = 12 \text{ cm}.$$

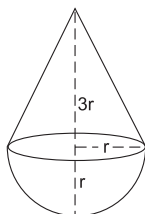
EXAMPLE 12 Find the number of coins, 1.5 cm in diameter and 2 mm thick, to be melted to form a right circular cylinder of height 10 cm and diameter 4.5 cm.

SOLUTION Radius of each cylindrical coin, $r = \frac{1.5}{2} \text{ cm} = \frac{15}{20} \text{ cm} = \frac{3}{4} \text{ cm}$.

Thickness of each cylindrical coin, $h = 2 \text{ mm} = \frac{2}{10} \text{ cm} = \frac{1}{5} \text{ cm}$.

Volume of each coin $= \pi r^2 h = \left(\pi \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{5} \right) \text{ cm}^3$

$$= \left(\frac{9\pi}{80} \right) \text{ cm}^3.$$



Radius of the required cylinder, $R = \left(\frac{4.5}{2}\right) \text{ cm} = 2.25 \text{ cm}$.

Height of the required cylinder, $H = 10 \text{ cm}$.

Volume of the required cylinder

$$= \pi R^2 H = (\pi \times 2.25 \times 2.25 \times 10) \text{ cm}^3$$

$$= \left(\pi \times \frac{225}{100} \times \frac{225}{100} \times 10\right) \text{ cm}^3 = \left(\frac{405\pi}{8}\right) \text{ cm}^3.$$

Required number of coins

$$= \frac{\text{volume of the required cylinder}}{\text{volume of each coin}} = \left(\frac{405\pi}{8} \times \frac{80}{9\pi}\right) = 450.$$

EXAMPLE 13 A solid iron rectangular block of dimensions 4.4 m, 2.6 m and 1 m is cast into a hollow cylindrical pipe of internal radius 30 cm and thickness 5 cm. Find the length of the pipe. [CBSE 2017]

SOLUTION

Volume of iron = $(440 \times 260 \times 100) \text{ cm}^3$.

Internal radius of the pipe = 30 cm.

External radius of the pipe = $(30 + 5) \text{ cm} = 35 \text{ cm}$.

Let the length of the pipe be $h \text{ cm}$.

Volume of iron in the pipe

$$= (\text{external volume}) - (\text{internal volume})$$

$$= [\pi(35)^2 h - \pi(30)^2 h] \text{ cm}^3 = \pi h[(35)^2 - (30)^2] \text{ cm}^3$$

$$= (65 \times 5)\pi h \text{ cm}^3 = (325\pi h) \text{ cm}^3.$$

$$\therefore 325\pi h = 440 \times 260 \times 100$$

$$\Rightarrow \text{length} = h \text{ cm} = \left(\frac{440 \times 260 \times 100 \times 7}{325 \times 22}\right) \text{ cm}$$

$$= 11200 \text{ cm} = 112 \text{ m}.$$

Hence, the length of the pipe is 112 m.

EXAMPLE 14 A hemispherical bowl of internal radius 9 cm is full of liquid. The liquid is to be filled into cylindrical-shaped small bottles, each of diameter 3 cm and height 4 cm. How many bottles are needed to empty the bowl? [CBSE 2005, '12]

SOLUTION

Radius of the hemispherical bowl, $R = 9 \text{ cm}$.

$$\text{Volume of liquid in the bowl} = \frac{2}{3}\pi R^3$$

$$= \left(\frac{2}{3}\pi \times 9 \times 9 \times 9\right) \text{ cm}^3$$

$$= (486\pi) \text{ cm}^3.$$

Radius of each cylindrical bottle, $r = \frac{3}{2}$ cm.

Height of each cylindrical bottle, $h = 4$ cm.

Volume of each cylindrical bottle

$$= \pi r^2 h = \left(\pi \times \frac{3}{2} \times \frac{3}{2} \times 4 \right) \text{ cm}^3 = (9\pi) \text{ cm}^3.$$

Required number of bottles

$$= \frac{\text{volume of liquid in the bowl}}{\text{volume of each bottle}} = \frac{486\pi}{9\pi} = 54.$$

EXAMPLE 15 *A hemispherical bowl of internal diameter 36 cm contains liquid. This liquid is filled into 72 cylindrical bottles of diameter 6 cm. Find the height of each bottle if 10% liquid is wasted in this transfer.*

[CBSE 2015]

SOLUTION Radius of the hemispherical bowl, $R = 18$ cm.

Volume of liquid in the bowl,

$$r = \frac{2}{3} \pi R^3 = \left(\frac{2}{3} \pi \times 18 \times 18 \times 18 \right) \text{ cm}^3$$

$$= (3888\pi) \text{ cm}^3.$$

Volume of liquid filled into bottles

$$= (90\% \text{ of } 3888\pi) \text{ cm}^3$$

$$= \left(\frac{90}{100} \times 3888\pi \right) \text{ cm}^3.$$

Let the height of each cylindrical bottle be h .

Radius of each cylindrical bottle, $r = 3$ cm.

Volume of each cylindrical bottle

$$= \pi r^2 h = (\pi \times 3^2 \times h) \text{ cm}^3 = (9\pi h) \text{ cm}^3.$$

Now, volume of 72 cylindrical bottles

$$= \text{volume of liquid filled into bottles}$$

$$\Rightarrow 72 \times 9\pi h = \frac{90}{100} \times 3888\pi$$

$$\Rightarrow h = \left(\frac{90 \times 3888}{100 \times 72 \times 9} \right) \text{ cm} = \frac{27}{5} \text{ cm} = 5.4 \text{ cm}.$$

EXAMPLE 16 *A conical vessel whose internal radius is 5 cm and height 24 cm, is full of water. The water is emptied into a cylindrical vessel with internal radius 10 cm. Find the height to which the water rises in the cylindrical vessel.*

SOLUTION Radius of the conical vessel, $r = 5$ cm.

Height of the conical vessel, $h = 24$ cm.

Volume of the conical vessel

$$= \frac{1}{3} \pi r^2 h = \left(\frac{1}{3} \pi \times 5 \times 5 \times 24 \right) \text{ cm}^3 = (200\pi) \text{ cm}^3.$$

Radius of the cylindrical vessel, $R = 10$ cm.

Let the height to which water rises in the vessel be H . Then, volume of the water in the cylindrical vessel

$$= \pi R^2 H = (\pi \times 10 \times 10 \times H) \text{ cm}^3 = (100\pi H) \text{ cm}^3.$$

Volume of the water in cylindrical vessel

$$= \text{volume of the water in the conical vessel}$$

$$\Rightarrow 100\pi H = 200\pi \Rightarrow H = \frac{200}{100} = 2 \text{ cm}.$$

Hence, the required height is 2 cm.

EXAMPLE 17 A sphere of diameter 6 cm is dropped into a right circular cylindrical vessel, partly filled with water. The diameter of the cylindrical vessel is 12 cm. If the sphere is completely submerged in water, by how much will the level of water rise in the cylindrical vessel?

SOLUTION Radius of the sphere, $R = 3$ cm.

$$\text{Volume of the sphere} = \frac{4}{3} \pi R^3 = \left(\frac{4}{3} \pi \times 3^3 \right) \text{ cm}^3 = (36\pi) \text{ cm}^3.$$

Radius of the cylindrical vessel, $r = 6$ cm.

Let the rise in water level be h .

Increase in volume of water when the sphere is submerged

$$= \pi r^2 h = (\pi \times 6 \times 6 \times h) \text{ cm}^3 = (36\pi h) \text{ cm}^3.$$

But this volume must be equal to the volume of the sphere.

$$\therefore 36\pi h = 36\pi \Rightarrow h = 1 \text{ cm}.$$

Hence, rise in the water level is 1 cm.

EXAMPLE 18 Sushant has a vessel of the form of an inverted cone, open at the top, of height 11 cm and radius of the top as 2.5 cm and is full of water. Metallic spherical balls each of diameter 0.5 cm are put in the vessel due to which two-fifths of the water in the vessel flows out. Find how many balls were put in the vessel. Sushant made the arrangement so that the water that flows out irrigates the flower beds. What value has been shown by Sushant?

[CBSE 2014]

SOLUTION Height of the conical vessel, $h = 11$ cm.

Radius of the conical vessel, $r = 2.5$ cm.

Volume of the conical vessel

$$\begin{aligned} &= \frac{1}{3} \pi r^2 h = \left(\frac{1}{3} \pi \times 2.5 \times 2.5 \times 11 \right) \text{ cm}^3 \\ &= \left(\frac{1}{3} \pi \times \frac{5}{2} \times \frac{5}{2} \times 11 \right) \text{ cm}^3 = \left(\frac{275\pi}{12} \right) \text{ cm}^3. \end{aligned}$$

Volume of water flown out after putting spherical balls

$$= \left(\frac{2}{5} \text{ of } \frac{275\pi}{12} \right) \text{ cm}^3 = \left(\frac{55\pi}{6} \right) \text{ cm}^3.$$

Radius of each spherical ball,

$$R = \frac{0.5}{2} \text{ cm} = 0.25 \text{ cm} = \left(\frac{25}{100} \right) \text{ cm} = \frac{1}{4} \text{ cm}.$$

Volume of each spherical ball

$$= \frac{4}{3} \pi R^3 = \left(\frac{4}{3} \pi \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \right) \text{ cm}^3 = \left(\frac{\pi}{48} \right) \text{ cm}^3.$$

$\therefore$ required number of balls

$$= \frac{\text{volume of water flown out}}{\text{volume of each spherical ball}} = \left(\frac{55\pi}{6} \right) \times \frac{48}{\pi} = 440.$$

Value shown: We must not waste water and try to put every drop of it to optimum use.

EXAMPLE 19 *A vessel full of water is in the form of an inverted cone of height 8 cm and the radius of its top, which is open, is 5 cm. 100 spherical lead balls are dropped into the vessel. One-fourth of the water flows out of the vessel. Find the radius of a spherical ball.* [CBSE 2015]

SOLUTION Height of the conical vessel, $h = 8$ cm.

Radius of the conical vessel, $r = 5$ cm.

Volume of the conical vessel

$$= \frac{1}{3} \pi r^2 h = \left(\frac{1}{3} \pi \times 5 \times 5 \times 8 \right) \text{ cm}^3 = \left(\frac{200\pi}{3} \right) \text{ cm}^3.$$

Volume of water flown out on dropping spherical balls

$$= \left(\frac{1}{4} \text{ of } \frac{200\pi}{3} \right) \text{ cm}^3 = \left(\frac{50\pi}{3} \right) \text{ cm}^3.$$

Let the radius of each spherical ball be R .

Volume of each spherical ball $= \frac{4}{3} \pi R^3$.

Now, volume of 100 balls = volume of water flow out

$$\Rightarrow 100 \times \frac{4}{3} \pi R^3 = \frac{50\pi}{3} \Rightarrow R^3 = \left(\frac{50}{3} \times \frac{3}{4} \times \frac{1}{100} \right) = \frac{1}{8}$$

$$\Rightarrow R = \sqrt[3]{\frac{1}{8}} = \frac{1}{2} \text{ cm} = 0.5 \text{ cm}.$$

EXAMPLE 20 *A housing society used to collect rain water from the roof of its building $22 \text{ m} \times 20 \text{ m}$ to a cylindrical vessel having diameter of base 2 m and height 3.5 m and then pump this water into the main water tank so that all members can use it. On a particular day the rain water collected from the roof just filled the cylindrical vessel. Then, find the rainfall in centimetre.* [CBSE 2013C]

SOLUTION Length of the roof = 22 m . Breadth of the roof = 20 m .

Let the rainfall be $x \text{ cm}$.

Volume of water on the roof

$$= \left(22 \times 20 \times \frac{x}{100} \right) \text{ m}^3 = \left(\frac{22x}{5} \right) \text{ m}^3.$$

Radius of the cylindrical vessel, $r = 1 \text{ m}$.

Height of the cylindrical vessel, $h = 3.5 \text{ m}$.

Volume of water in the cylindrical vessel when just full

$$= \pi r^2 h = \left(\frac{22}{7} \times 1 \times 1 \times \frac{7}{2} \right) \text{ m}^3 = 11 \text{ m}^3.$$

Now, volume of water on the roof

= volume of water in the cylindrical vessel

$$\Rightarrow \frac{22x}{5} = 11 \Rightarrow x = \left(\frac{11 \times 5}{22} \right) = 2.5.$$

Hence, the rainfall is 2.5 cm .

EXAMPLE 21 *Water flows through a circular pipe whose internal diameter is 2 cm , at the rate of 0.7 m per second into a cylindrical tank, the radius of whose base is 40 cm . By how much will the level of water rise in the tank in half an hour?* [CBSE 2006C]

SOLUTION Internal radius of the pipe, $r = 1 \text{ cm}$.

Length of water flowing in 1 second, $h = 0.7 \text{ m} = 70 \text{ cm}$.

Volume of water flowing in 1 second

$$= \pi r^2 h = (\pi \times 1 \times 1 \times 70) \text{ cm}^3 = (70\pi) \text{ cm}^3.$$

Volume of water flowing in 30 minutes

$$= (70\pi \times 60 \times 30) \text{ cm}^3 = (126000\pi) \text{ cm}^3.$$

Radius of cylindrical tank, $R = 40$ cm.

Let the rise in level of water be H cm.

Volume of water in the tank

$$\begin{aligned} &= \pi R^2 H = (\pi \times 40 \times 40 \times H) \text{ cm}^3 \\ &= (1600\pi H) \text{ cm}^3. \end{aligned}$$

Volume of water in the tank

= volume of water flown through the pipe

$$\Rightarrow 1600\pi H = 126000\pi \Rightarrow H = \frac{126000}{1600} = \frac{315}{4} = 78.75.$$

Hence, rise in level = 78.75 cm.

EXAMPLE 22 *Water is flowing at the rate of 15 km per hour through a pipe of diameter 14 cm into a rectangular tank which is 50 m long and 44 m wide. Find the time in which the level of water in the tank will rise by 21 cm.* [CBSE 2006, '11]

SOLUTION Length of the tank = 50 m.

Width of the tank = 44 m.

Desired rise in level = 21 cm = $\frac{21}{100}$ m.

Desired volume of water in the tank

$$= \left(50 \times 44 \times \frac{21}{100} \right) \text{ m}^3 = 462 \text{ m}^3.$$

Radius of the pipe, $r = 7$ cm = $\frac{7}{100}$ m.

Length of water flowing in 1 hour, $h = 15$ km = 15000 m.

Volume of water flowing from the pipe in 1 hour

$$= \pi r^2 h = \left(\frac{22}{7} \times \frac{7}{100} \times \frac{7}{100} \times 15000 \right) \text{ m}^3 = 231 \text{ m}^3.$$

$\therefore$ required time = $\frac{\text{desired volume}}{\text{volume of water flown in 1 hour}}$

$$= \left(\frac{462}{231} \right) \text{ hours} = 2 \text{ hours.}$$

EXAMPLE 23 *A farmer connects a pipe of internal diameter 20 cm from a canal into a cylindrical tank in his field, which is 10 m in diameter and 2 m deep. If water flows through the pipe at the rate of 3 km/hr, in how much time will the tank be filled?* [CBSE 2008C]

SOLUTION Radius of the pipe, $r = 10$ cm = $\left(\frac{10}{100} \right)$ m = $\frac{1}{10}$ m.

Length of water flowing through the pipe in 1 hour,

$$h = 3 \text{ km} = 3000 \text{ m}.$$

Volume of water that flows through the pipe in 1 hour

$$= \pi r^2 h = \left(\pi \times \frac{1}{10} \times \frac{1}{10} \times 3000 \right) \text{ m}^3 = (30\pi) \text{ m}^3.$$

Radius of the cylindrical tank, $R = 5 \text{ m}$.

Depth of the tank, $H = 2 \text{ m}$.

$$\begin{aligned} \text{Volume of the tank} &= \pi R^2 H = (\pi \times 5 \times 5 \times 2) \text{ m}^3 \\ &= (50\pi) \text{ m}^3. \end{aligned}$$

Time taken to fill the tank

$$\begin{aligned} &= \frac{\text{volume of the tank}}{\text{volume of water flown in 1 hour}} \\ &= \left(\frac{50\pi}{30\pi} \right) \text{ hours} = \frac{5}{3} \text{ hours} = 1 \text{ hr } 40 \text{ min}. \end{aligned}$$

Hence, the required time is 1 hr 40 min.

EXAMPLE 24 *Water is flowing at the rate of 2.52 km/hr through a cylindrical pipe into a cylindrical tank, the radius of whose base is 40 cm. If the increase in the level of water in the tank, in half an hour is 3.15 m, find the internal diameter of the pipe.* [CBSE 2015]

SOLUTION

Let the internal radius of the pipe be r .

Length of water flowing through the pipe in 1 hour,

$$h = 2.52 \text{ km} = 2520 \text{ m}.$$

Volume of water flowing through the pipe in 1 hour

$$= \pi r^2 h = (\pi \times r^2 \times 2520) \text{ m}^3.$$

Volume of water flowing through the pipe in half an hour

$$= \left(\frac{1}{2} \times \pi r^2 \times 2520 \right) \text{ m}^3 = (1260\pi r^2) \text{ m}^3.$$

$$\text{Radius of the cylindrical tank, } R = 40 \text{ cm} = \frac{40}{100} \text{ m} = \frac{2}{5} \text{ m}.$$

Increase in level of water, $H = 3.15 \text{ m}$.

Volume of water filled in the tank

$$= \pi R^2 H = \left(\pi \times \frac{2}{5} \times \frac{2}{5} \times \frac{315}{100} \right) \text{ m}^3 = \left(\frac{63\pi}{125} \right) \text{ m}^3.$$

Now, volume of water flown in half an hour

$$= \text{volume of water filled in the tank}$$

$$\Rightarrow 1260\pi r^2 = \frac{63\pi}{125} \Rightarrow r^2 = \frac{63}{125 \times 1260} = \frac{1}{2500}$$

$$\Rightarrow r = \sqrt{\frac{1}{2500}} = \frac{1}{50} \text{ m} = \left(\frac{1}{50} \times 100\right) \text{ cm} = 2 \text{ cm}.$$

Hence, the internal diameter of the pipe = $(2 \times 2) \text{ cm} = 4 \text{ cm}$.

EXAMPLE 25 A 21-m-deep well with diameter 6 m is dug and the earth from digging is evenly spread to form a platform 27 m $\times$ 11 m. Find the height of the platform. [CBSE 2015]

SOLUTION Depth of the well, $h = 21 \text{ m}$.

Radius of the well, $r = 3 \text{ m}$.

Volume of the earth taken out

$$= \pi r^2 h = \left(\frac{22}{7} \times 3 \times 3 \times 21\right) \text{ m}^3 = 594 \text{ m}^3.$$

Area of the platform = $(27 \times 11) \text{ m}^2 = 297 \text{ m}^2$.

Height of the platform

$$= \frac{\text{volume of the earth}}{\text{area of the platform}} = \left(\frac{594}{297}\right) \text{ m} = 2 \text{ m}.$$

EXAMPLE 26 A well of diameter 4 m is dug 14 m deep. The earth taken out of it is spread evenly all around the well to form a 40-cm-high embankment. Find the width of the embankment. [HOTS] [CBSE 2015]

SOLUTION Radius of the well, $r = 2 \text{ m}$.

Depth of the well, $h = 14 \text{ m}$.

Volume of the earth taken out

$$= \pi r^2 H = \left(\frac{22}{7} \times 2 \times 2 \times 14\right) \text{ m}^3 = 176 \text{ m}^3.$$

Let the width of the embankment be x metres.

Internal radius of the embankment, $r = 2 \text{ m}$.

External radius of the embankment, $R = (2 + x) \text{ m}$.

Height of the embankment, $H = 40 \text{ cm} = \left(\frac{40}{100}\right) \text{ m}$.

Volume of the embankment

$$\begin{aligned} &= \pi H (R^2 - r^2) = \left\{ \frac{40\pi}{100} [(2+x)^2 - 2^2] \right\} \text{ m}^3 \\ &= \left[\frac{2\pi}{5} (x^2 + 4x) \right] \text{ m}^3. \end{aligned}$$

Now, volume of the embankment = volume of the earth dug out

$$\Rightarrow \frac{2}{5} \times \frac{22}{7} \times (x^2 + 4x) = 176 \Rightarrow x^2 + 4x = \frac{176 \times 5 \times 7}{2 \times 22} = 140$$

$$\Rightarrow x^2 + 4x - 140 = 0 \Rightarrow x^2 + 14x - 10x - 140 = 0$$

$$\Rightarrow x(x + 14) - 10(x + 14) = 0 \Rightarrow (x + 14)(x - 10) = 0$$

$$\Rightarrow x = 10 \quad [\because x \neq -14].$$

Hence, width of the embankment = 10 m.

EXAMPLE 27 From each end of a solid metal cylinder, metal was scooped out in hemispherical form of same diameter. The height of the cylinder is 10 cm and its base is of radius 4.2 cm. The rest of the cylinder is melted and converted into a cylindrical wire of 1.4 cm thickness. Find the length of the wire. [HOTS] [CBSE 2015]

SOLUTION Height of the cylinder, $h = 10$ cm.

Radius of each of the cylinder and hemisphere, $r = 4.2$ cm.

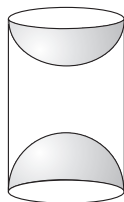
Volume of the solid

= volume of the cylinder – volume of 2 hemispheres

$$= \pi r^2 h - 2 \times \frac{2}{3} \pi r^3 = \pi r^2 \left(h - \frac{4}{3} r \right)$$

$$= \left[\pi \times 4.2 \times 4.2 \times \left(10 - \frac{4}{3} \times 4.2 \right) \right] \text{ cm}^3$$

$$= (\pi \times 4.2 \times 4.2 \times 4.4) \text{ cm}^3.$$



Let the length of the wire be H .

$$\text{Radius of the wire, } R = \left(\frac{1.4}{2} \right) \text{ cm} = 0.7 \text{ cm} = \frac{7}{10} \text{ cm}.$$

$$\text{Volume of the wire} = \pi R^2 H = \left(\pi \times \frac{7}{10} \times \frac{7}{10} \times H \right) \text{ cm}^3.$$

Now, volume of the wire = volume of the solid

$$\Rightarrow \pi \times \frac{7}{10} \times \frac{7}{10} \times H = \pi \times 4.2 \times 4.2 \times 4.4$$

$$\Rightarrow H = \left(\frac{4.2 \times 4.2 \times 4.4 \times 10 \times 10}{7 \times 7} \right) \text{ cm} = \left(\frac{42 \times 42 \times 4.4}{7 \times 7} \right) \text{ cm}$$

$$= 158.4 \text{ cm}.$$

EXAMPLE 28 50 circular plates each of radius 7 cm and thickness 5 mm are placed one above another to form a solid right circular cylinder. Find the total surface area of the cylinder so formed. [HOTS] [CBSE 2013C]

SOLUTION Clearly, we have

radius of the cylinder so formed, $r = 7$ cm, and
 height of the cylinder so formed, $h = (50 \times 5)$ mm
 $= 250$ mm $= 25$ cm.

$\therefore$ total surface area of the cylinder so formed

$$= 2\pi r(h + r) = \left[2 \times \frac{22}{7} \times 7 \times (25 + 7) \right] \text{ cm}^2$$

$$= (2 \times 22 \times 32) \text{ cm}^2 = 1408 \text{ cm}^2.$$

EXAMPLE 29 A copper wire, 3 mm in diameter, is wound about a cylinder whose length is 12 cm and diameter 10 cm, so as to cover the curved surface of the cylinder. Find the length and mass of the wire, assuming the density of copper to be 8.88 g per cm^3 . [Use $\pi = 3.14$.] [HOTS]

SOLUTION Diameter of the wire = 3 mm = 0.3 cm.

Length of the cylinder = 12 cm.

Radius of the cylinder = 5 cm.

$$\text{Number of turns} = \frac{\text{length of the cylinder}}{\text{diameter of the wire}} = \frac{12}{0.3} = 40.$$

$$\text{Length of one turn} = \text{circumference of the base of the cylinder}$$

$$= 2\pi r = (2 \times 3.14 \times 5) \text{ cm} = 31.4 \text{ cm}.$$

$$\text{Length of 40 turns} = (31.4 \times 40) \text{ cm} = 1256 \text{ cm}.$$

Total length of the wire wrapped, $l = 1256$ cm.

$$\text{Radius of the wire, } r = \left(\frac{0.3}{2} \right) \text{ cm} = 0.15 \text{ cm}.$$

$$\text{Volume of the wire} = \pi r^2 l = (3.14 \times 0.15 \times 0.15 \times 1256) \text{ cm}^3$$

$$= 88.7364 \text{ cm}^3 \approx 88.74 \text{ cm}^3.$$

$$\therefore \text{mass of the wire} = \text{volume} \times \text{density} = (88.74 \times 8.88) \text{ g}$$

$$= 788.01 \text{ g} \approx 788 \text{ g}.$$

EXAMPLE 30 A cistern, internally measuring $150 \text{ cm} \times 120 \text{ cm} \times 110 \text{ cm}$, has 129600 cm^3 of water in it. Porous bricks are placed in the water until the cistern is full to the brim. Each brick absorbs one-seventeenth of its own volume of water. How many bricks can be put in without overflowing the water, each brick being $22.5 \text{ cm} \times 7.5 \text{ cm} \times 6.5 \text{ cm}$?

SOLUTION Total volume of the cistern $= (150 \times 120 \times 110) \text{ cm}^3$
 $= 1980000 \text{ cm}^3.$

Volume of water in the cistern $= 129600 \text{ cm}^3.$

$$\text{Volume of empty space in the cistern}$$

$$= (1980000 - 129600) \text{ cm}^3 = 1850400 \text{ cm}^3.$$

$$\begin{aligned}\text{Total volume of each brick} &= (22.5 \times 7.5 \times 6.5) \text{ cm}^3 \\ &= 1096.875 \text{ cm}^3.\end{aligned}$$

Since each brick absorbs water equal to $\frac{1}{17}$ of its own volume, so the effective volume of each brick may be taken as $\left(1 - \frac{1}{17}\right)$, i.e., $\frac{16}{17}$ of its actual volume.

$$\text{So, the effective volume of each brick} = \left(\frac{16}{17} \times 1096.875\right) \text{ cm}^3.$$

$\therefore$ required number of bricks

$$\begin{aligned}&= \frac{\text{volume of empty space in the cistern}}{\text{effective volume of each brick}} \\ &= \frac{1850400}{\left(\frac{16}{17} \times 1096.875\right)} = \frac{17 \times 1850400}{16 \times 1096.875} \\ &= \frac{17 \times 1850400}{17550} = 1792.41 \approx 1792.\end{aligned}$$

EXAMPLE 31 A right triangle whose sides are 3 cm and 4 cm (other than hypotenuse) is made to revolve about its hypotenuse. Find the volume and surface area of the double cone so formed. (Choose the value of π as found appropriate). [HOTS]

SOLUTION Let $\triangle AOB$ be the given right triangle in which $\angle AOB = 90^\circ$. When $\triangle AOB$ is rotated about the hypotenuse AB , the two cones generated are $AO'O$ and BOO' .

Clearly, $OA = 3$ cm, $OB = 4$ cm.

In right-angled $\triangle AOB$, we have

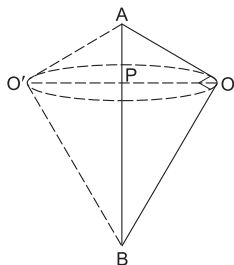
$$\begin{aligned}AB &= \sqrt{(OA)^2 + (OB)^2} = \sqrt{3^2 + 4^2} \text{ cm} \\ &= \sqrt{25} \text{ cm} = 5 \text{ cm}.\end{aligned}$$

Let $OP = x$ cm. Then,

$$\begin{aligned}\frac{1}{2} \times OA \times OB &= \frac{1}{2} \times AB \times OP \\ \Rightarrow \frac{1}{2} \times 3 \times 4 &= \frac{1}{2} \times 5 \times OP \Rightarrow OP = \left(\frac{3 \times 4}{5}\right) \text{ cm} = 2.4 \text{ cm}.\end{aligned}$$

In right-angled $\triangle APO$, we have

$$\begin{aligned}AP &= \sqrt{(OA)^2 - (OP)^2} = \sqrt{3^2 - (2.4)^2} \text{ cm} \\ &= \sqrt{9 - 5.76} \text{ cm} = \sqrt{3.24} \text{ cm} = 1.8 \text{ cm}.\end{aligned}$$



$$BP = AB - AP = (5 - 1.8) \text{ cm} = 3.2 \text{ cm}.$$

$\therefore$ radius of each of the cones $AO'O$ and BOO' , $r = OP = 2.4 \text{ cm}$.

Height of the cone $AO'O$, $h = AP = 1.8 \text{ cm}$.

Height of the cone BOO' , $H = BP = 3.2 \text{ cm}$.

Volume of the double cone

= volume of the cone $AO'O$ + volume of the cone BOO'

$$= \frac{1}{3} \pi r^2 h + \frac{1}{3} \pi r^2 H = \frac{1}{3} \pi r^2 (h + H)$$

$$= \left(\frac{1}{3} \times 3.14 \times 2.4 \times 2.4 \times 5 \right) \text{ cm}^3 = 30.144 \text{ cm}^3.$$

Slant height of the cone $AO'O$, $l_1 = OA = 3 \text{ cm}$.

Slant height of the cone $BO'O$, $l_2 = OB = 4 \text{ cm}$.

Surface area of the double cone

= curved surface area of the cone $AO'O$

+ curved surface area of the cone BOO'

$$= \pi r l_1 + \pi r l_2 = \pi r (l_1 + l_2) = \left[\frac{22}{7} \times 2.4 \times (3 + 4) \right] \text{ cm}^2$$

$$= \left(\frac{22}{7} \times 2.4 \times 7 \right) \text{ cm}^2 = 52.8 \text{ cm}^2.$$

EXERCISE 17B

1. A solid metallic cuboid of dimensions $9 \text{ m} \times 8 \text{ m} \times 2 \text{ m}$ is melted and recast into solid cubes of edge 2 m . Find the number of cubes so formed.
[CBSE 2017]
2. A cone of height 20 cm and radius of base 5 cm is made up of modelling clay. A child reshapes it in the form of a sphere. Find the diameter of the sphere.
[CBSE 2011]
3. Metallic spheres of radii 6 cm , 8 cm and 10 cm respectively are melted to form a single solid sphere. Find the radius of the resulting sphere.
[CBSE 2012]
4. A solid metal cone with radius of base 12 cm and height 24 cm is melted to form solid spherical balls of diameter 6 cm each. Find the number of balls thus formed.
[CBSE 2005C]
5. The radii of internal and external surfaces of a hollow spherical shell are 3 cm and 5 cm respectively. It is melted and recast into a solid cylinder of diameter 14 cm . Find the height of the cylinder.
[CBSE 2012]

6. The internal and external diameters of a hollow hemispherical shell are 6 cm and 10 cm respectively. It is melted and recast into a solid cone of base diameter 14 cm. Find the height of the cone so formed. [CBSE 2005C]
7. A copper rod of diameter 2 cm and length 10 cm is drawn into a wire of uniform thickness and length 10 m. Find the thickness of the wire.
[CBSE 2012]
8. A hemispherical bowl of internal diameter 30 cm contains some liquid. This liquid is to be filled into cylindrical-shaped bottles each of diameter 5 cm and height 6 cm. Find the number of bottles necessary to empty the bowl.
[CBSE 2004, '11]
9. A solid metallic sphere of diameter 21 cm is melted and recast into a number of smaller cones, each of diameter 3.5 cm and height 3 cm. Find the number of cones so formed.
[CBSE 2004]
10. A spherical cannon ball 28 cm in diameter is melted and recast into a right circular conical mould, base of which is 35 cm in diameter. Find the height of the cone.
[CBSE 2001C]
11. A spherical ball of radius 3 cm is melted and recast into three spherical balls. The radii of two of these balls are 1.5 cm and 2 cm. Find the radius of the third ball.
12. A spherical shell of lead whose external and internal diameters are respectively 24 cm and 18 cm, is melted and recast into a right circular cylinder 37 cm high. Find the diameter of the base of the cylinder.
13. A hemisphere of lead of radius 9 cm is cast into a right circular cone of height 72 cm. Find the radius of the base of the cone.
14. A spherical ball of diameter 21 cm is melted and recast into cubes, each of side 1 cm. Find the number of cubes so formed.
15. How many lead balls, each of radius 1 cm, can be made from a sphere of radius 8 cm?
16. A solid sphere of radius 3 cm is melted and then cast into small spherical balls, each of diameter 0.6 cm. Find the number of small balls so obtained.
17. The diameter of a sphere is 42 cm. It is melted and drawn into a cylindrical wire of diameter 2.8 cm. Find the length of the wire.
18. The diameter of a copper sphere is 18 cm. It is melted and drawn into a long wire of uniform cross section. If the length of the wire is 108 m, find its diameter.

19. A hemispherical bowl of internal radius 9 cm is full of water. Its contents are emptied into a cylindrical vessel of internal radius 6 cm. Find the height of water in the cylindrical vessel. [CBSE 2012]
20. A hemispherical tank, full of water, is emptied by a pipe at the rate of $25\frac{7}{7}$ litres per second. How much time will it take to empty half the tank if the diameter of the base of the tank is 3 m? [CBSE 2012]
21. The rain water from a roof of $44\text{ m} \times 20\text{ m}$ drains into a cylindrical tank having diameter of base 4 m and height 3.5 m. If the tank is just full, find the rainfall in cm. [CBSE 2014]
22. The rain water from a $22\text{ m} \times 20\text{ m}$ roof drains into a cylindrical vessel of diameter 2 m and height 3.5 m. If the rain water collected from the roof fills $\frac{4}{5}$ th of the cylindrical vessel then find the rainfall in centimetre. [CBSE 2015, '17]
23. A solid right circular cone of height 60 cm and radius 30 cm is dropped in a right circular cylinder full of water, of height 180 cm and radius 60 cm. Find the volume of water left in the cylinder, in cubic metres. [CBSE 2015]
24. Water is flowing through a cylindrical pipe of internal diameter 2 cm, into a cylindrical tank of base radius 40 cm, at the rate of 0.4 m per second. Determine the rise in level of water in the tank in half an hour. [CBSE 2013]
25. Water is flowing at the rate of 6 km/hr through a pipe of diameter 14 cm into a rectangular tank which is 60 m long and 22 m wide. Determine the time in which the level of water in the tank will rise by 7 cm. [CBSE 2011]
26. Water in a canal, 5.4 m wide and 1.8 m deep, is flowing with a speed of 25 km/hr. How much area can it irrigate in 40 minutes, if 10 cm of standing water is required for irrigation? [CBSE 2017]
27. A farmer connects a pipe of internal diameter 25 cm from a canal into a cylindrical tank in his field, which is 12 m in diameter and 2.5 m deep. If water flows through the pipe at the rate of 3.6 km/hr, in how much time will the tank be filled? Also, find the cost of water if the canal department charges at the rate of ₹ 0.07 per m^3 . [CBSE 2009C]
28. Water running in a cylindrical pipe of inner diameter 7 cm, is collected in a container at the rate of 192.5 litres per minute. Find the rate of flow of water in the pipe in km/hr. [CBSE 2013]
29. 150 spherical marbles, each of diameter 14 cm, are dropped in a cylindrical vessel of diameter 7 cm containing some water, which are completely immersed in water. Find the rise in the level of water in the vessel. [CBSE 2014]

30. Marbles of diameter 1.4 cm are dropped into a cylindrical beaker of diameter 7 cm, containing some water. Find the number of marbles that should be dropped into the beaker so that the water level rises by 5.6 cm.
31. In a village, a well with 10 m inside diameter, is dug 14 m deep. Earth taken out of it is spread all around to a width of 5 m to form an embankment. Find the height of the embankment. What value of the villagers is reflected here? [CBSE 2014]
32. In a corner of a rectangular field with dimensions 35 m $\times$ 22 m, a well with 14 m inside diameter is dug 8 m deep. The earth dug out is spread evenly over the remaining part of the field. Find the rise in the level of the field. [HOTS]
33. A copper wire of diameter 6 mm is evenly wrapped on a cylinder of length 18 cm and diameter 49 cm to cover its whole surface. Find the length and the volume of the wire. If the density of copper be 8.8 g per cu-cm, find the weight of the wire. [HOTS]
34. A right triangle whose sides are 15 cm and 20 cm (other than hypotenuse), is made to revolve about its hypotenuse. Find the volume and surface area of the double cone so formed. (Choose value of π as found appropriate) [HOTS]
35. In a hospital, used water is collected in a cylindrical tank of diameter 2 m and height 5 m. After recycling, this water is used to irrigate a park of hospital whose length is 25 m and breadth is 20 m. If the tank is filled completely then what will be the height of standing water used for irrigating the park? Write your views on recycling of water. [CBSE 2017]

ANSWERS (EXERCISE 17B)

- | | | | | |
|---------------------------|------------|---------------------------|--------------------------|----------------------|
| 1. 18 | 2. 10 cm | 3. 12 cm | 4. 32 | 5. $2\frac{2}{3}$ cm |
| 6. 4 cm | 7. 2 mm | 8. 60 | 9. 504 | 10. 35.84 cm |
| 11. 2.5 cm | 12. 12 cm | 13. 4.5 cm | 14. 4851 | 15. 512 |
| 16. 1000 | 17. 63 m | 18. 0.6 cm | 19. 13.5 cm | |
| 20. 16 minutes 30 seconds | 21. 5 cm | 22. 2 cm | 23. 1.98 m^3 | |
| 24. 180 cm | 25. 1 hour | 26. 1620000 m^2 | 27. 1 hr 36 min, ₹ 19.80 | |
| 28. 3 km/hr | 29. 56 m | 30. 150 | | |
| 31. 2.8 m; | | | | |

Value: We must labour hard to make maximum use of the available resources.

32. 2 m 33. 46.2 m, 1306.8 cm^3 , 11.5 kg 34. 3768 cm^3 , 1320 cm^2

VOLUME AND SURFACE AREA OF A FRUSTUM OF A CONE

FRUSTUM OF A CONE

When a cone is cut by a plane parallel to the base of the cone then the portion between the plane and the base is called the frustum of the cone.

FORMULAE Let R and r be the radii of the base and the top of the frustum of a cone.

Let h be its height and l be its slant height.

Then,

(i) Volume of the frustum of the cone

$$= \frac{\pi h}{3} [R^2 + r^2 + Rr] \text{ cubic units.}$$

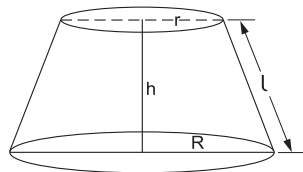
(ii) Lateral surface area of the frustum of the cone

$$= \pi l(R + r), \text{ where } l^2 = h^2 + (R - r)^2 \text{ sq units.}$$

(iii) Total surface area of the frustum of the cone

$$= (\text{area of the base}) + (\text{area of the top}) + (\text{lateral surface area})$$

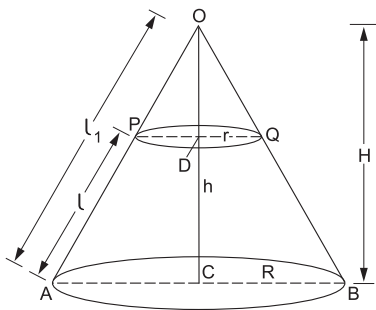
$$= [\pi R^2 + \pi r^2 + \pi l(R + r)] = \pi [R^2 + r^2 + l(R + r)] \text{ sq units.}$$



DERIVATION OF THE ABOVE FORMULAE

Let H , R and l_1 be the height, radius of the base and slant height respectively of a cone OAB . Suppose it is cut by a plane PQ parallel to its base at a height h from the base to form the frustum $ABQP$.

Let the radius of the base of cone OPQ be r and l be the slant height of the frustum so formed.



Clearly, we have $OC = H$, $DC = h$, $OD = OC - DC = (H - h)$, $DQ = r$,
 $CB = R$, $OB = l_1$, $BQ = l$, $OQ = OB - BQ = (l_1 - l)$.

Clearly, $\triangle ODQ \sim \triangle OCB$.

$$\therefore \frac{OD}{OC} = \frac{DQ}{CB} = \frac{OQ}{OB} \Rightarrow \frac{H-h}{H} = \frac{r}{R} = \frac{l_1-l}{l_1} \quad \dots (i)$$

$$\frac{H-h}{H} = \frac{r}{R} \Rightarrow R(H-h) = Hr \Rightarrow HR - hR = Hr$$

$$\Rightarrow H(R-r) = hR \Rightarrow H = \frac{hR}{R-r} \quad \dots (ii)$$

$$\text{And, } \frac{l_1-l}{l_1} = \frac{r}{R} \Rightarrow (l_1-l)R = l_1r \Rightarrow l_1R - lR = l_1r$$

$$\Rightarrow l_1(R-r) = lR \Rightarrow l_1 = \frac{lR}{R-r}. \quad \dots \text{(iii)}$$

Now, volume of the frustum $ABQP$

$$\begin{aligned} &= \text{volume of the cone } OAB - \text{volume of the cone } OPQ \\ &= \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 (H-h) = \frac{1}{3}\pi \left[R^2 \left(\frac{hR}{R-r} \right) - r^2 \left\{ \frac{hR}{R-r} - h \right\} \right] \\ &= \frac{1}{3}\pi \left[\left(\frac{hR^3}{R-r} \right) - r^2 \left\{ \frac{hR - hR + hr}{R-r} \right\} \right] = \frac{1}{3}\pi \left[\frac{hR^3}{R-r} - \frac{hr^3}{R-r} \right] \\ &= \frac{1}{3}\pi \left[\frac{h(R^3 - r^3)}{(R-r)} \right] = \frac{1}{3}\pi \left[\frac{h(R-r)(R^2 + Rr + r^2)}{(R-r)} \right] = \frac{1}{3}\pi h(R^2 + Rr + r^2). \end{aligned}$$

And, curved surface area of the frustum $ABQP$

$$\begin{aligned} &= \text{curved surface area of the cone } OAB \\ &\quad - \text{curved surface area of the cone } OPQ \\ &= \pi R l_1 - \pi r(l_1 - l) = \pi \left[R \left(\frac{lR}{R-r} \right) - r \left\{ \frac{lR}{R-r} - l \right\} \right] \\ &= \pi \left[\frac{lR^2}{R-r} - r \left\{ \frac{lR - lR + lr}{R-r} \right\} \right] = \pi \left[\frac{lR^2}{R-r} - \frac{lr^2}{R-r} \right] \\ &= \pi \left[\frac{l(R^2 - r^2)}{(R-r)} \right] = \pi \left[\frac{l(R-r)(R+r)}{(R-r)} \right] = \pi(R+r)l. \end{aligned}$$

SOLVED EXAMPLES

EXAMPLE 1 An open metal bucket is in the shape of a frustum of a cone of height 21 cm with radii of its lower and upper ends as 10 cm and 20 cm respectively. Find the cost of milk which can completely fill the bucket at ₹ 30 per litre. [CBSE 2012]

SOLUTION Let R and r be the radii of the top and the base of the bucket respectively, and let h be its height.

Then, $R = 20$ cm, $r = 10$ cm and $h = 21$ cm.

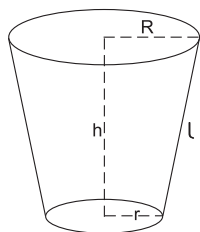
Capacity of the bucket

= volume of frustum of the cone

$$= \frac{1}{3}\pi h[R^2 + r^2 + Rr]$$

$$= \left[\left(\frac{1}{3} \times \frac{22}{7} \times 21 \right) \cdot \{(20)^2 + (10)^2 + 20 \times 10\} \right] \text{ cm}^3$$

$$= (22 \times 700) \text{ cm}^3 = 15400 \text{ cm}^3.$$



Volume of milk that the bucket can hold

$$= \left(\frac{15400}{1000} \right) \text{ litres} = 15.4 \text{ litres.}$$

$$\therefore \text{cost of milk} = ₹ (15.4 \times 30) = ₹ 462.$$

EXAMPLE 2

A bucket open at the top, and made up of a metal sheet is in the form of the frustum of a cone. The depth of the bucket is 24 cm and the diameters of its upper and lower circular ends are 30 cm and 10 cm respectively. Find the cost of metal sheet used in it at the rate of ₹ 10 per 100 m². [Use $\pi = 3.14$.] [CBSE 2013]

SOLUTION

Radius of the upper end, $R = 15$ cm.

Radius of the lower end, $r = 5$ cm.

Depth of the bucket, $h = 24$ cm.

$$\begin{aligned} \text{Slant height of the bucket, } l &= \sqrt{h^2 + (R - r)^2} \\ &= \sqrt{(24)^2 + (15 - 5)^2} \text{ cm} \\ &= \sqrt{676} \text{ cm} = 26 \text{ cm.} \end{aligned}$$

Area of metal sheet used

= curved surface area + area of the bottom

$$= \pi l(R + r) + \pi r^2$$

$$= [3.14 \times 26 \times (15 + 5) + 3.14 \times 5 \times 5] \text{ cm}^2$$

$$= (3.14 \times 26 \times 20 + 3.14 \times 25) \text{ cm}^2$$

$$= [3.14 \times (520 + 25)] \text{ cm}^2 = (3.14 \times 545) \text{ cm}^2 = 1711.3 \text{ cm}^2.$$

$$\therefore \text{cost of metal sheet used} = ₹ \left(\frac{1711.3}{100} \times 10 \right) = ₹ 171.13.$$

EXAMPLE 3

A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm³ of water. The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket and the area of the metal sheet used in its making. [Use $\pi = 3.14$.] [CBSE 2006C]

SOLUTION

Here, $R = 20$ cm, $r = 12$ cm

and volume = 12308.8 cm³

Let the height of the bucket be h cm.

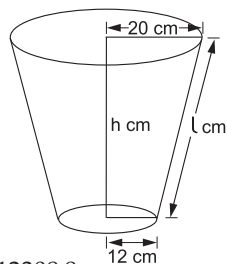
Volume of the bucket

= volume of frustum of the cone.

$$\therefore \frac{1}{3} \pi h(R^2 + r^2 + Rr) = 12308.8$$

$$\Rightarrow \frac{1}{3} \times 3.14 \times h[(20)^2 + (12)^2 + 20 \times 12] = 12308.8$$

$$\Rightarrow 784h = \frac{12308.8 \times 3}{3.14}$$



$$\Rightarrow h = \left(\frac{12308.8 \times 3}{3.14 \times 784} \right) = 15.$$

Slant height of the bucket

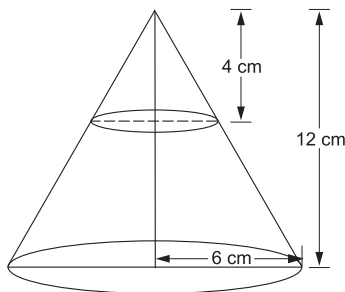
$$\begin{aligned} l &= \sqrt{h^2 + (R - r)^2} \text{ units} \\ &= \sqrt{(15)^2 + (20 - 12)^2} \text{ cm} = \sqrt{(15)^2 + 8^2} \text{ cm} \\ &= \sqrt{225 + 64} \text{ cm} = \sqrt{289} \text{ cm} = 17 \text{ cm}. \end{aligned}$$

Area of the metal sheet used

$$\begin{aligned} &= (\text{curved surface area}) + (\text{area of the bottom}) \\ &= [\pi l(R + r) + \pi r^2] \\ &= [3.14 \times 17 \times (20 + 12) + 3.14 \times 12 \times 12] \text{ cm}^2 \\ &= [3.14 \times (17 \times 32) + 3.14 \times 144] \text{ cm}^2 \\ &= [3.14 \times (544 + 144)] \text{ cm}^2 \\ &= (3.14 \times 688) \text{ cm}^2 = 2160.32 \text{ cm}^2. \end{aligned}$$

EXAMPLE 4

In the figure, from the top of a solid cone of height 12 cm and base radius 6 cm, a cone of height 4 cm is removed by a plane parallel to the base. Find the total surface area of the remaining solid. [Use $\pi = \frac{22}{7}$ and $\sqrt{5} = 2.236$.] [CBSE 2015]



SOLUTION

Clearly, the remaining solid is the frustum $CDBA$ of cone OPD .

Radius of lower end of the frustum, $R = CD = 6$ cm.

Height of the frustum,

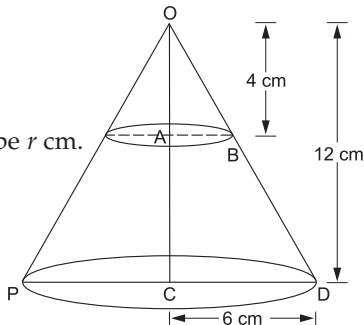
$$\begin{aligned} h &= AC = OC - OA \\ &= (12 - 4) \text{ cm} = 8 \text{ cm}. \end{aligned}$$

Let the radius of upper face be r cm.

Now, $\triangle OAB \sim \triangle OCD$.

$$\text{So, } \frac{OA}{OC} = \frac{AB}{CD} \Rightarrow \frac{4}{12} = \frac{r}{6}$$

$$\Rightarrow r = \left(\frac{4}{12} \times 6 \right) \text{ cm} = 2 \text{ cm}.$$



Slant height of the frustum,

$$l = \sqrt{h^2 + (R - r)^2} = \sqrt{8^2 + (6 - 2)^2} = \sqrt{80} \text{ cm} = 4\sqrt{5} \text{ cm}.$$

Total surface area of the frustum

= area of base + area of top + curved surface area

$$= \pi R^2 + \pi r^2 + \pi l(R + r) = \pi [R^2 + r^2 + l(R + r)]$$

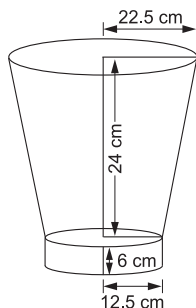
$$= \frac{22}{7} \times [6^2 + 2^2 + 4 \times 2.236 \times (6 + 2)] \text{ cm}^2$$

$$= \left[\frac{22}{7} \times (36 + 4 + 71.552) \right] \text{ cm}^2 = \left(\frac{22}{7} \times 11.552 \right) \text{ cm}^2$$

$$= (22 \times 15.936) \text{ cm}^2 = 350.592 \text{ cm}^2.$$

EXAMPLE 5

An open metallic bucket is in the shape of a frustum of a cone mounted on hollow cylindrical base made of metallic sheet. If the diameters of the two circular ends of the bucket are 45 cm and 25 cm, the total vertical height of the bucket is 30 cm and that of the cylindrical portion is 6 cm, find the area of the metallic sheet used to make the bucket. Also, find the volume of water it can hold. [Take $\pi = \frac{22}{7}$.]



SOLUTION

Here $R = \frac{45}{2} \text{ cm}$, $r = \frac{25}{2} \text{ cm} = 12.5 \text{ cm}$.

Height of the frustum of the cone, $h = (30 - 6) \text{ cm} = 24 \text{ cm}$.

$\therefore h = 24 \text{ cm}$.

Slant height of the frustum of the cone,

$$l = \sqrt{h^2 + (R - r)^2} \text{ units} = \sqrt{(24)^2 + (22.5 - 12.5)^2} \text{ cm}$$

$$= \sqrt{(24)^2 + (10)^2} \text{ cm} = \sqrt{576 + 100} \text{ cm}$$

$$= \sqrt{676} \text{ cm} = 26 \text{ cm}.$$

Area of metallic sheet used

= (curved surface area of the frustum of the cone)

+ (area of the base) + (curved surface area of the cylinder)

$$= \pi l(R + r) + \pi r^2 + 2\pi rH, \text{ where } H = 6 \text{ cm}$$

$$= \pi \cdot \{l(R + r) + r^2 + 2rH\} \text{ sq units}$$

$$= \frac{22}{7} \cdot \{26(22.5 + 12.5) + (12.5)^2 + 2 \times 12.5 \times 6\} \text{ cm}^2$$

$$= \frac{22}{7} \cdot \{(26 \times 35) + (12.5 \times 12.5) + 150\} \text{ cm}^2$$

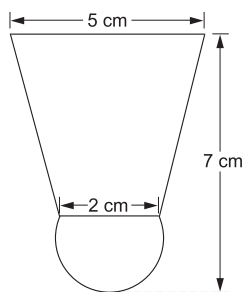
$$\begin{aligned}
 &= \frac{22}{7} \cdot \{910 + 156.25 + 150\} \text{ cm}^2 = \left(\frac{22}{7} \times 1216.25 \right) \text{ cm}^2 \\
 &= (22 \times 173.75) \text{ cm}^2 = 3822.5 \text{ cm}^2.
 \end{aligned}$$

Volume of water which the bucket can hold

$$\begin{aligned}
 &= \frac{1}{3} \pi h [R^2 + r^2 + Rr] \text{ cm}^3 \\
 &= \frac{1}{3} \times \frac{22}{7} \times 24 \times \left[\left(\frac{45}{2} \right)^2 + \left(\frac{25}{2} \right)^2 + \left(\frac{45}{2} \times \frac{25}{2} \right) \right] \text{ cm}^3 \\
 &= \frac{176}{7} \times \left(\frac{2025}{4} + \frac{625}{4} + \frac{1125}{4} \right) \text{ cm}^3 = \left(\frac{176}{7} \times \frac{3775}{4} \right) \text{ cm}^3 \\
 &= \frac{166100}{7} \text{ cm}^3 = 23728.57 \text{ cm}^3 = 23.73 \text{ litres.}
 \end{aligned}$$

EXAMPLE 6

A shuttlecock used for playing badminton has the shape of a frustum of a cone mounted on a hemisphere. The external diameters of the frustum are 5 cm and 2 cm, and the height of the entire shuttlecock is 7 cm. Find its external surface area.



SOLUTION

Here, $R = \frac{5}{2}$ cm, $r = 1$ cm and $h = (7 - 1)$ cm = 6 cm.

Let l be the slant height of the frustum of the cone.

$$\begin{aligned}
 \therefore l^2 &= h^2 + (R - r)^2 = \left\{ 6^2 + \left(\frac{5}{2} - 1 \right)^2 \right\} \text{ cm}^2 \\
 &= \left(36 + \frac{9}{4} \right) \text{ cm}^2 = \frac{153}{4} \text{ cm}^2 \\
 \Rightarrow l &= \frac{\sqrt{153}}{2} \text{ cm} = \frac{12.36}{2} \text{ cm} = 6.18 \text{ cm.}
 \end{aligned}$$

External surface area of the shuttlecock

$$\begin{aligned}
 &= (\text{its lateral surface area} + \text{area of the base}) \\
 &= [\pi l (R + r) + 2\pi r^2] \text{ sq units} \\
 &= \left\{ \frac{22}{7} \times 6.18 \times \left(\frac{5}{2} + 1 \right) + 2 \times \frac{22}{7} \times 1 \times 1 \right\} \text{ cm}^2 \\
 &= \left(11 \times 6.18 + \frac{44}{7} \right) \text{ cm}^2 = (67.98 + 6.28) \text{ cm}^2 = 74.26 \text{ cm}^2.
 \end{aligned}$$

EXAMPLE 7

The slant height of the frustum of a cone is 4 cm and the perimeters of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.

[CBSE 2017]

SOLUTION

Let the radii of its circular ends be R cm and r cm, and its slant height be l cm. Then,

$$2\pi R = 18 \Rightarrow \pi R = 9.$$

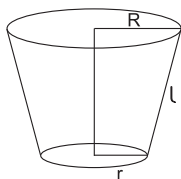
$$2\pi r = 6 \Rightarrow \pi r = 3.$$

Also, $l = 4$ cm (given).

Curved surface area of the frustum

$$= \pi l(R + r) \text{ sq units}$$

$$= l \times (\pi R + \pi r) = 4 \times (9 + 3) \text{ cm}^2 = 48 \text{ cm}^2.$$

**EXAMPLE 8**

The height of a cone is 10 cm. The cone is divided into two parts using a plane parallel to its base at the middle of its height. Find the ratio of the volumes of the two parts. [CBSE 2017]

SOLUTION

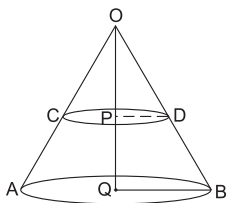
Let OAB be the cone and OQ be its axis, and let P be the midpoint of OQ .

Let $OQ = h$ cm.

Then, $OP = PQ = \frac{h}{2}$ cm and $QB = 10$ cm.

Also, $\triangle OPD \sim \triangle OQB$.

$$\therefore \frac{OP}{OQ} = \frac{PD}{QB} \Rightarrow \frac{(h/2)}{h} = \frac{PD}{10} \Rightarrow PD = 5 \text{ cm.}$$



The plane CD divides the cone into two parts, namely

- (i) a smaller cone of radius = 5 cm and height $(h/2)$ cm,
- (ii) frustum of a cone in which

$$R = 10 \text{ cm}, r = 5 \text{ cm and height} = \left(\frac{h}{2}\right) \text{ cm.}$$

Volume of the smaller cone

$$= \left(\frac{1}{3} \times \pi \times 5 \times 5 \times \frac{h}{2}\right) \text{ cm}^3 = \left(\frac{25\pi h}{6}\right) \text{ cm}^3.$$

Volume of the frustum of the cone

$$= \frac{1}{3} \pi \times \frac{h}{2} \cdot [(10)^2 + (5)^2 + 10 \times 5] \text{ cm}^3 = \left(\frac{175\pi h}{6}\right) \text{ cm}^3.$$

$$\text{Ratio of the required volumes} = \frac{25\pi h}{6} : \frac{175\pi h}{6} = 25 : 175 = 1 : 7.$$

EXAMPLE 9

The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $1/27$ of the volume of the given cone, at what height above the base is the section made?

[HOTS] [CBSE 2005C]

SOLUTION Height of the given cone = 30 cm.

Let the radius of its base be R cm.

$$\text{Volume of the given cone} = \left(\frac{1}{3} \pi R^2 \times 30 \right) \text{ cm}^3 = (10\pi R^2) \text{ cm}^3.$$

Let the radius and height of the smaller cone be r cm and h cm respectively.

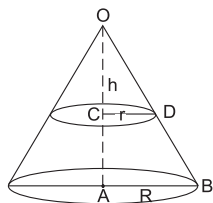
$$\text{Then, volume of the smaller cone} = \left(\frac{1}{3} \pi r^2 h \right) \text{ cm}^3.$$

$$\therefore \frac{1}{3} \pi r^2 h = \frac{1}{27} \cdot (10\pi R^2) \quad [\text{given}]$$

$$\Rightarrow \left(\frac{R}{r} \right)^2 = \frac{9h}{10} \quad \dots (i)$$

Now, $\triangle OAB \sim \triangle OCD$.

$$\therefore \frac{AB}{CD} = \frac{OA}{OC} \Rightarrow \frac{R}{r} = \frac{30}{h} \quad \dots (ii)$$



From (i) and (ii), we get

$$\left(\frac{30}{h} \right)^2 = \frac{9h}{10} \Rightarrow \frac{30 \times 30}{h \times h} = \frac{9h}{10}$$

$$\Rightarrow h^3 = \frac{30 \times 30 \times 10}{9} = 1000 \Rightarrow h^3 = (10)^3 \Rightarrow h = 10.$$

$\therefore$ height of the smaller cone = 10 cm.

Height of the section from the base = $(30 - 10)$ cm = 20 cm.

EXAMPLE 10 A hollow cone is cut by a plane parallel to the base and the upper portion is removed. If the curved surface of the remainder is $8/9$ of the curved surface of the whole cone, find the ratio of the line segments into which the altitude of the cone is divided by the plane.

[CBSE 2004, '04C]

SOLUTION Let OAB be the given hollow cone cut by the plane CD parallel to base AB and let cone OCD be removed. Then, the remainder is the frustum $CABD$ of the given cone.

Let $OE = h$ units, $OF = H$ units,

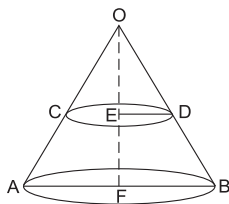
$OD = l$ units, $OB = L$ units,

$ED = r$ units, $FB = R$ units.

In $\triangle OED$ and $\triangle OFB$, we have

$\angle EOD = \angle FOB$ (common),

$\angle OED = \angle OFB = 90^\circ$.



$$\therefore \triangle OED \sim \triangle OFB$$

$$\Rightarrow \frac{OE}{OF} = \frac{OD}{OB} = \frac{ED}{FB} \Rightarrow \frac{h}{H} = \frac{l}{L} = \frac{r}{R} \dots (i) \text{ [by Thales' theorem]}$$

Now, (curved surface area of the frustum $CABD$)

$$= \frac{8}{9} (\text{curved surface area of the cone } OAB)$$

$$\Rightarrow (\text{curved surface area of the cone } OCD)$$

$$= (\text{curved surface of the cone } OAB)$$

$$- (\text{curved surface of the frustum } CABD)$$

$$= (\text{curved surface of the cone } OAB)$$

$$- \frac{8}{9} (\text{curved surface of the cone } OAB)$$

$$= \frac{1}{9} (\text{curved surface of the cone } OAB)$$

$$\Rightarrow \pi r l = \frac{1}{9} \pi R L$$

$$\Rightarrow \left(\frac{r}{R}\right) \left(\frac{l}{L}\right) = \frac{1}{9} \Rightarrow \left(\frac{h}{H} \times \frac{h}{H}\right) = \frac{1}{9} \Rightarrow \frac{h}{H} = \frac{1}{3} \quad [\text{using (i)}]$$

$$\Rightarrow H = 3h. \quad \dots (ii)$$

$$\text{Now, } EF = (OF - OE) = (H - h) = (3h - h) = 2h.$$

$$\therefore \frac{OE}{EF} = \frac{h}{2h} = \frac{1}{2}.$$

Hence, $OE : EF = 1 : 2$.

EXAMPLE 11

A metallic right circular cone is 20 cm high and has a vertical angle of 60° . This is cut into two parts at the middle of its height by a plane parallel to the base. If the frustum so obtained is drawn into a wire of diameter $\frac{1}{16}$ cm, find the length of the wire. [CBSE 2017]

SOLUTION

Let OAB be the cone in which $\angle AOB = 60^\circ$.

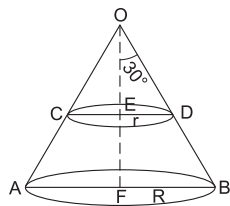
Clearly $\angle DOE = 30^\circ$, $OE = 10$ cm, $OF = 20$ cm.

Let $ED = r$ cm and $FB = R$ cm.

$$\therefore \frac{ED}{OE} = \tan 30^\circ \Rightarrow \frac{ED}{10} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow ED = \left(10 \times \frac{1}{\sqrt{3}}\right) \text{ cm} \Rightarrow r = \frac{10}{\sqrt{3}} \text{ cm}$$

$$\text{and, } \frac{FB}{OF} = \tan 30^\circ \Rightarrow \frac{FB}{20} = \frac{1}{\sqrt{3}}$$



$$\Rightarrow FB = \left(20 \times \frac{1}{\sqrt{3}} \right) \text{ cm} \Rightarrow R = \frac{20}{\sqrt{3}} \text{ cm}.$$

Also, $EF = 10$ cm.

Thus, $ABCD$ is the frustum of a cone in which

$$R = \frac{20}{\sqrt{3}} \text{ cm}, r = \frac{10}{\sqrt{3}} \text{ cm and } h = 10 \text{ cm}.$$

$$\begin{aligned} \text{Volume of this frustum} &= \frac{1}{3} \pi h (R^2 + r^2 + Rr) \\ &= \frac{1}{3} \times \pi \times 10 \left\{ \frac{400}{3} + \frac{100}{3} + \frac{200}{3} \right\} \text{ cm}^3 \\ &= \left(\frac{\pi \times 10}{3} \times \frac{700}{3} \right) \text{ cm}^3 = \left(\frac{7000\pi}{9} \right) \text{ cm}^3. \end{aligned}$$

Let the length of the wire be l .

$$\text{Radius of the wire, } r_1 = \frac{1}{32} \text{ cm}.$$

$$\text{Volume of the wire} = \pi r_1^2 l = \pi \times \left(\frac{1}{32} \right)^2 \times l.$$

$$\therefore \frac{7000\pi}{9} = \frac{\pi l}{32 \times 32} \Rightarrow l = \left(\frac{7000 \times 32 \times 32}{9} \right) \text{ cm}$$

$$\Rightarrow l = \left(\frac{7000 \times 32 \times 32}{9 \times 100} \right) \text{ m} = \left(\frac{71680}{9} \right) \text{ m}$$

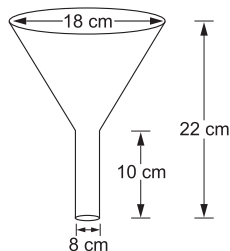
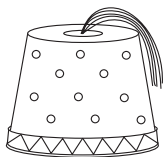
$$= 7964.44 \text{ m}.$$

EXERCISE 17C

1. A drinking glass is in the shape of a frustum of a cone of height 14 cm. The diameters of its two circular ends are 16 cm and 12 cm. Find the capacity of the glass. [CBSE 2012]
2. The radii of the circular ends of a solid frustum of a cone are 18 cm and 12 cm and its height is 8 cm. Find its total surface area. [Use $\pi = 3.14$.] [CBSE 2011]
3. A metallic bucket, open at the top, of height 24 cm is in the form of the frustum of a cone, the radii of whose lower and upper circular ends are 7 cm and 14 cm respectively. Find
 - (i) the volume of water which can completely fill the bucket;
 - (ii) the area of the metal sheet used to make the bucket. [CBSE 2014]

4. A container, open at the top, is in the form of a frustum of a cone of height 24 cm with radii of its lower and upper circular ends as 8 cm and 20 cm respectively. Find the cost of milk which can completely fill the container at the rate of ₹ 21 per litre. [CBSE 2014]
5. A container made of a metal sheet open at the top is of the form of frustum of cone, whose height is 16 cm and the radii of its lower and upper circular edges are 8 cm and 20 cm respectively. Find
- (i) the cost of metal sheet used to make the container if it costs ₹ 10 per 100 cm^2 [CBSE 2013]
 - (ii) the cost of milk at the rate of ₹ 35 per litre which can fill it completely. [CBSE 2017]
6. The radii of the circular ends of a solid frustum of a cone are 33 cm and 27 cm, and its slant height is 10 cm. Find its capacity and total surface area. [Take $\pi = \frac{22}{7}$.] [CBSE 2005]
7. A bucket is in the form of a frustum of a cone. Its depth is 15 cm and the diameters of the top and the bottom are 56 cm and 42 cm respectively. Find how many litres of water the bucket can hold. [Take $\pi = \frac{22}{7}$.] [CBSE 2005C]
8. A bucket made up of a metal sheet is in the form of a frustum of a cone of height 16 cm with radii of its lower and upper ends as 8 cm and 20 cm respectively. Find the cost of the bucket if the cost of metal sheet used is ₹ 15 per 100 cm^2 . [Use $\pi = 3.14$.] [CBSE 2006, '08, '08C]
9. A bucket made up of a metal sheet is in the form of frustum of a cone. Its depth is 24 cm and the diameters of the top and bottom are 30 cm and 10 cm respectively. Find the cost of milk which can completely fill the bucket at the rate of ₹ 20 per litre and the cost of metal sheet used if it costs ₹ 10 per 100 cm^2 . [Use $\pi = 3.14$.] [CBSE 2006, '09C]
10. A container in the shape of a frustum of a cone having diameters of its two circular faces as 35 cm and 30 cm and vertical height 14 cm, is completely filled with oil. If each cm^3 of oil has mass 1.2 g then find the cost of oil in the container if it costs ₹ 40 per kg. [CBSE 2009C]
11. A bucket is in the form of a frustum of a cone and it can hold 28.49 litres of water. If the radii of its circular ends are 28 cm and 21 cm, find the height of the bucket. [CBSE 2012]
12. The radii of the circular ends of a bucket of height 15 cm are 14 cm and r cm ($r < 14$). If the volume of bucket is 5390 cm^3 , find the value of r . [CBSE 2011]
13. The radii of the circular ends of a solid frustum of a cone are 33 cm and 27 cm and its slant height is 10 cm. Find its total surface area. [Use $\pi = 3.14$.] [CBSE 2012]

14. A tent is made in the form of a frustum of a cone surmounted by another cone. The diameters of the base and the top of the frustum are 20 m and 6 m respectively, and the height is 24 m. If the height of the tent is 28 m and the radius of the conical part is equal to the radius of the top of the frustum, find the quantity of canvas required. [Take $\pi = \frac{22}{7}$.]
15. A tent consists of a frustum of a cone, surmounted by a cone. If the diameters of the upper and lower circular ends of the frustum be 14 m and 26 m respectively, the height of the frustum be 8 m and the slant height of the surmounted conical portion be 12 m, find the area of the canvas required to make the tent. (Assume that the radii of the upper circular end of the frustum and the base of the surmounted conical portion are equal.) [CBSE 2008]
16. The perimeters of the two circular ends of a frustum of a cone are 48 cm and 36 cm. If the height of the frustum is 11 cm, find its volume and curved surface area. [CBSE 2014]
17. A solid cone of base radius 10 cm is cut into two parts through the midpoint of its height, by a plane parallel to its base. Find the ratio of the volumes of the two parts of the cone. [CBSE 2013]
18. The height of a right circular cone is 20 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{8}$ of the volume of the given cone, at what height above the base is the section made? [HOTS] [CBSE 2014]
19. A solid metallic right circular cone 20 cm high and whose vertical angle is 60° , is cut into two parts at the middle of its height by a plane parallel to its base. If the frustum so obtained be drawn into a wire of diameter $\frac{1}{12}$ cm, find the length of the wire. [HOTS] [CBSE 2014]
20. A fez, the cap used by the Turks, is shaped like the frustum of a cone. If its radius on the open side is 10 cm, radius at the upper base is 4 cm and its slant height is 15 cm, find the area of material used for making it. [Use $\pi = \frac{22}{7}$.]
21. An oil funnel made of tin sheet consists of a 10 cm long cylindrical portion attached to a frustum of a cone. If the total height is 22 cm, diameter of the cylindrical portion is 8 cm and the diameter of the top of the funnel is 18 cm, find the area of the tin sheet required to make the funnel.



22. A right circular cone is divided into three parts by trisecting its height by two planes drawn parallel to the base. Show that the volumes of the three portions starting from the top are in the ratio 1 : 7 : 19. [CBSE 2017]

ANSWERS (EXERCISE 17C)

1. 2170.67 cm^3 2. 2411.52 cm^2 3. (i) 8624 cm^3 (ii) 1804 cm^2 4. ₹ 329.49
 5. (i) ₹ 145.83 (ii) ₹ 366.08 6. 22704 cm^3 , 7599.43 cm^2 7. 28.49 litres
 8. ₹ 293.90 9. ₹ 163.28, ₹ 171.13 10. ₹ 558.80 11. 15 cm 12. $r = 7$
 13. 7592.52 cm^2 14. 1068.57 m^2 15. 892.6 m^2 16. 1554 cm^3 , 468.91 cm^2
 17. 1 : 7 18. 10 cm 19. 4480 m 20. 710.28 cm^2 21. 782.57 cm^2

HINTS TO SOME SELECTED QUESTIONS

10. $r = 15 \text{ cm}$, $R = 17.5 \text{ cm}$, $h = 14 \text{ cm}$.

$$\text{Volume of the container} = \frac{1}{3} \pi (R^2 + r^2 + Rr)h = \left(\frac{3175 \times 11}{3} \right) \text{ cm}^3.$$

$$\text{Mass of oil in the container} = \left(\frac{3175 \times 11}{3} \times 1.2 \right) \text{ g} = 13.97 \text{ kg}.$$

$$\text{Cost of oil} = ₹ (13.97 \times 40) = ₹ 558.80.$$

11. $r = 21 \text{ cm}$, $R = 28 \text{ cm}$.

$$\text{Volume of the bucket} = (28.49 \times 1000) \text{ cm}^3.$$

$$\therefore \frac{1}{3} \pi [(28)^2 + (21)^2 + 28 \times 21]h = 28490. \text{ Solve for } h.$$

14. $R = 10 \text{ m}$, $r = 3 \text{ m}$ and $h = 24 \text{ m}$.

Let l be the slant height of the frustum. Then,

$$\begin{aligned} l &= \sqrt{h^2 + (R - r)^2} = \sqrt{(24)^2 + 7^2} \text{ m} \\ &= \sqrt{576 + 49} \text{ m} = \sqrt{625} \text{ m} = 25 \text{ m}. \end{aligned}$$

Let L be the slant height of the conical part.

For this conical part, $r = 3 \text{ m}$ and $H = 4 \text{ m}$.

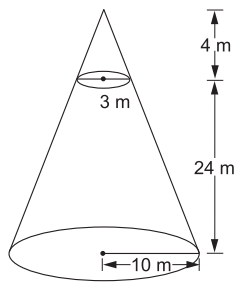
$$\therefore L = \sqrt{3^2 + 4^2} \text{ m} = \sqrt{25} \text{ m} = 5 \text{ m}.$$

Quantity of canvas

$$= (\text{lateral surface area of the frustum})$$

$$+ (\text{lateral surface area of the cone})$$

$$= \{ \pi l (R + r) + \pi r L \} \text{ m}^2 = \frac{22}{7} \times \{ (25 \times 13) + (3 \times 5) \} \text{ m}^2 = 1068.57 \text{ m}^2.$$

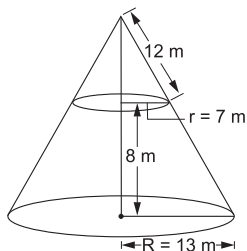


15. $R = 13 \text{ m}$, $r = 7 \text{ m}$ and $h = 8 \text{ m}$.

$$\therefore l = \sqrt{h^2 + (R - r)^2} = 10 \text{ m}.$$

Required area

$$= \left\{ \frac{22}{7} \times 10 \times (13 + 7) + \frac{22}{7} \times 7 \times 12 \right\} \text{ m}^2.$$



16. $2\pi r = 36$ and $2\pi R = 48$. Find r and R .
20. Area of material = curved surface area + area of closed, small circular face
 $= \pi l(r + R) + \pi r^2$.
21. Height of the cylindrical portion, $h = 10$ cm.
 Height of the frustum, $H = (22 - 10)$ cm = 12 cm.
 Radius of the cylindrical portion, $r = 4$ cm.
 Radius of smaller end of the frustum, $r = 4$ cm.
 Radius of bigger end of the frustum, $R = 9$ cm.
 Area of tin sheet required
 $=$ curved surface area of the frustum + curved surface area of the cylindrical portion
 $= \pi l(r + R) + 2\pi rh$.
-

EXERCISE 17D

Very-Short-Answer Questions

- A river 1.5 m deep and 36 m wide is flowing at the rate of 3.5 km/hr. Find the amount of water (in cubic metres) that runs into the sea per minute.
- The volume of a cube is 729 cm^3 . Find its surface area.
- How many cubes of 10 cm edge can be put in a cubical box of 1 m edge?
- Three cubes of iron whose edges are 6 cm, 8 cm and 10 cm respectively are melted and formed into a single cube. Find the edge of the new cube formed.
- Five identical cubes, each of edge 5 cm, are placed adjacent to each other. Find the volume of the resulting cuboid.
- The volumes of two cubes are in the ratio 8 : 27. Find the ratio of their surface areas.
- The volume of a right circular cylinder with its height equal to the radius is $25\frac{1}{7} \text{ cm}^3$. Find the height of the cylinder.
- The ratio between the radius of the base and the height of a cylinder is 2 : 3. If the volume of the cylinder is 12936 cm^3 , find the radius of the base of the cylinder.
- The radii of two cylinders are in the ratio of 2 : 3 and their heights are in the ratio of 5 : 3. Find the ratio of their volumes.
- 66 cubic cm of silver is drawn into a wire 1 mm in diameter. Calculate the length of the wire in metres.

11. If the area of the base of a right circular cone is 3850 cm^2 and its height is 84 cm, find the slant height of the cone.
12. A cylinder with base radius 8 cm and height 2 cm is melted to form a cone of height 6 cm. Calculate the radius of the base of the cone.
13. A right cylindrical vessel is full of water. How many right cones having the same radius and height as those of the right cylinder will be needed to store that water?
14. The volume of a sphere is 4851 cm^3 . Find its curved surface area.
15. The curved surface area of a sphere is 5544 cm^2 . Find its volume.
16. The surface areas of two spheres are in the ratio of 4 : 25. Find the ratio of their volumes.
17. A solid metallic sphere of radius 8 cm is melted and recast into spherical balls each of radius 2 cm. Find the number of spherical balls obtained.
18. How many lead shots each 3 mm in diameter can be made from a cuboid of dimensions $9 \text{ cm} \times 11 \text{ cm} \times 12 \text{ cm}$?
19. A metallic cone of radius 12 cm and height 24 cm is melted and made into spheres of radius 2 cm each. How many spheres are formed?
20. A hemisphere of lead of radius 6 cm is cast into a right circular cone of height 75 cm. Find the radius of the base of the cone.
21. A copper sphere of diameter 18 cm is drawn into a wire of diameter 4 mm. Find the length of the wire.
22. The radii of the circular ends of a frustum of height 6 cm are 14 cm and 6 cm respectively. Find the slant height of the frustum.
23. Find the ratio of the volume of a cube to that of a sphere which will fit inside it.
24. Find the ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height?

Short-Answer Questions

25. Two cubes each of volume 125 cm^3 are joined end to end to form a solid. Find the surface area of the resulting cuboid. [CBSE 2013C]
26. Three metallic cubes whose edges are 3 cm, 4 cm and 5 cm, are melted and recast into a single large cube. Find the edge of the new cube formed. [CBSE 2011]
27. A solid metallic sphere of diameter 8 cm is melted and drawn into a cylindrical wire of uniform width. If the length of the wire is 12 m, find its width. [CBSE 2013]

28. A 5-m-wide cloth is used to make a conical tent of base diameter 14 m and height 24 m. Find the cost of cloth used, at the rate of ₹ 25 per metre. [CBSE 2014]
29. A wooden toy was made by scooping out a hemisphere of same radius from each end of a solid cylinder. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm, find the volume of wood in the toy. [CBSE 2013]
30. Three cubes of a metal whose edges are in the ratio 3 : 4 : 5 are melted and converted into a single cube whose diagonal is $12\sqrt{3}$ cm. Find the edges of the three cubes. [CBSE 2013C]
- HINT** $\sqrt{3} \times \text{edge} = 12\sqrt{3} \Rightarrow \text{edge} = 12$ cm.
 $\therefore (3x)^3 + (4x)^3 + (5x)^3 = (12)^3$. Solve to find x .
31. A hollow sphere of external and internal diameters 8 cm and 4 cm respectively is melted into a solid cone of base diameter 8 cm. Find the height of the cone. [CBSE 2013C]
32. A bucket of height 24 cm is in the form of frustum of a cone whose circular ends are of diameter 28 cm and 42 cm. Find the cost of milk at the rate of ₹ 30 per litre, which the bucket can hold. [CBSE 2013C]
33. The interior of a building is in the form of a right circular cylinder of diameter 4.2 m and height 4 m surmounted by a cone of same diameter. The height of the cone is 2.8 m. Find the outer surface area of the building. [CBSE 2014]
34. A metallic solid right circular cone is of height 84 cm and the radius of its base is 21 cm. It is melted and recast into a solid sphere. Find the diameter of the sphere. [CBSE 2014]
35. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy. [CBSE 2012]
36. If the radii of the circular ends of a bucket 28 cm high, are 28 cm and 7 cm, find its capacity and total surface area. [CBSE 2011]
37. A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm^3 of water. The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket. (Use $\pi = 3.14$). [CBSE 2006C]
38. A milk container is made of metal sheet in the shape of frustum of a cone whose volume is $10459\frac{3}{7} \text{ cm}^3$. The radii of its lower and upper circular ends are 8 cm and 20 cm respectively. Find the cost of metal sheet used in making the container at the rate of ₹ 1.40 per cm^2 . [CBSE 2010]

39. A solid metallic sphere of diameter 28 cm is melted and recast into a number of smaller cones, each of diameter $4\frac{2}{3}$ cm and height 3 cm. Find the number of cones so formed.
40. A cylindrical vessel with internal diameter 10 cm and height 10.5 cm is full of water. A solid cone of base diameter 7 cm and height 6 cm is completely immersed in water. Find the volume of water (i) displaced out of the cylinder (ii) left in the cylinder. [CBSE 2009]

ANSWERS (EXERCISE 17D)

- | | | | | |
|-----------------------|-------------------------------|--|---------------------------|--------------------------|
| 1. 3150 m^3 | 2. 486 cm^2 | 3. 1000 | 4. 12 cm | 5. 625 cm^3 |
| 6. 4 : 9 | 7. 2 cm | 8. 14 cm | 9. 20 : 27 | 10. 84 m |
| 11. 91 cm | 12. 8 cm | 13. 3 | 14. 1386 cm^2 | 15. 38808 cm^3 |
| 16. 8 : 125 | 17. 64 | 18. 84000 | 19. 108 | 20. 2.4 cm |
| 21. 243 m | 22. 10 cm | 23. $6 : \pi$ | 24. 3 : 1 : 2 | 25. 250 cm^2 |
| 26. 6 cm | 27. $\frac{8}{15} \text{ cm}$ | 28. ₹ 2750 | 29. 205.33 cm^3 | |
| 30. 6 cm, 8 cm, 10 cm | 31. 14 cm | 32. ₹ 702.24 | 33. 75.9 cm^2 | |
| 34. 42 cm | 35. 231 cm^2 | 36. 30184 cm^3 , 6468 cm^2 | 37. 15 cm | |
| 38. ₹ 4224 | 39. 126 | 40. 77 cm^3 , 748 cm^3 | | |
-

MULTIPLE-CHOICE QUESTIONS (MCQ)

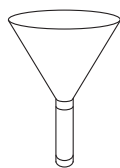
Choose the correct answer in each of the following:

- A **cylindrical pencil** sharpened at one edge is the combination of
 - a cylinder and a cone
 - a cylinder and frustum of a cone
 - a cylinder and a hemisphere
 - two cylinders
- A **shuttlecock** used for playing badminton is the combination of
 - cylinder and a hemisphere
 - frustum of a cone and a hemisphere
 - a cone and a hemisphere
 - a cylinder and a sphere



Shuttlecock

3. A **funnel** is the combination of
- a cylinder and a cone
 - a cylinder and a hemisphere
 - a cylinder and frustum of a cone
 - a cone and a hemisphere



Funnel

4. A **surahi** is a combination of
- a sphere and a cylinder
 - a hemisphere and a cylinder
 - a cylinder and a cone
 - two hemispheres



Surahi

5. The shape of a **glass (tumbler)** is usually in the form of
- a cylinder
 - frustum of a cone
 - a cone
 - a sphere



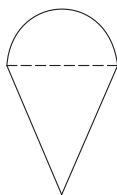
Glass

6. The shape of a **gilli** in the **gilli-danda** game is a combination of
- a cone and a cylinder
 - two cylinders
 - two cones and a cylinder
 - two cylinders and a cone



Gilli

7. A **plumbline (sahul)** is the combination of
- a hemisphere and a cone
 - a cylinder and a cone
 - a cylinder and frustum of a cone
 - a cylinder and a sphere



Plumbline

8. A cone is cut by a plane parallel to its base and the upper part is removed. The part that is left over is called
- a cone
 - a sphere
 - a cylinder
 - frustum of a cone
9. During conversion of a solid from one shape to another, the volume of the new shape will
- decrease
 - increase
 - remain unaltered
 - be doubled



10. In a right circular cone, the cross section made by a plane parallel to the base is a
(a) sphere (b) hemisphere (c) circle (d) a semicircle
11. A solid piece of iron in the form of a cuboid of dimensions $(49\text{ cm} \times 33\text{ cm} \times 24\text{ cm})$ is moulded to form a solid sphere. The radius of the sphere is
(a) 19 cm (b) 21 cm (c) 23 cm (d) 25 cm
12. The radius (in cm) of the largest right circular cone that can be cut out from a cube of edge 4.2 cm is [CBSE 2011]
(a) 2.1 (b) 4.2 (c) 8.4 (d) 1.05
13. A metallic solid sphere of radius 9 cm is melted to form a solid cylinder of radius 9 cm. The height of the cylinder is [CBSE 2014]
(a) 12 cm (b) 18 cm (c) 36 cm (d) 96 cm
14. A rectangular sheet of paper $40\text{ cm} \times 22\text{ cm}$, is rolled to form a hollow cylinder of height 40 cm. The radius of the cylinder (in cm) is [CBSE 2014]
(a) 3.5 (b) 7 (c) $\frac{80}{7}$ (d) 5
15. The number of solid spheres, each of diameter 6 cm, that can be made by melting a solid metal cylinder of height 45 cm and diameter 4 cm, is [CBSE 2014]
(a) 2 (b) 4 (c) 5 (d) 6
16. The surface areas of two spheres are in the ratio 16 : 9. The ratio of their volumes is [CBSE 2013C]
(a) 64 : 27 (b) 16 : 9 (c) 4 : 3 (d) $16^3 : 9^3$
17. If the surface area of a sphere is 616 cm^2 , its diameter (in cm) is [CBSE 2012]
(a) 7 (b) 14 (c) 28 (d) 56
18. If the radius of a sphere becomes 3 times then its volume will become [CBSE 2011]
(a) 3 times (b) 6 times (c) 9 times (d) 27 times
19. If the height of a bucket in the shape of frustum of a cone is 16 cm and the diameters of its two circular ends are 40 cm and 16 cm then its slant height is [CBSE 2013C]
(a) 20 cm (b) $12\sqrt{5}\text{ cm}$ (c) $8\sqrt{13}\text{ cm}$ (d) 16 cm
20. A sphere of diameter 18 cm is dropped into a cylindrical vessel of diameter 36 cm, partly filled with water. If the sphere is completely submerged then the water level rises by [CBSE 2011]
(a) 3 cm (b) 4 cm (c) 5 cm (d) 6 cm

21. A solid right circular cone is cut into two parts at the middle of its height by a plane parallel to its base. The ratio of the volume of the smaller cone to the whole cone is [CBSE 2012]
(a) 1 : 2 (b) 1 : 4 (c) 1 : 6 (d) 1 : 8
22. The radii of the circular ends of a bucket of height 40 cm are 24 cm and 15 cm. The slant height (in cm) of the bucket is [CBSE 2012]
(a) 41 (b) 43 (c) 49 (d) 51
23. A solid is hemispherical at the bottom and conical (of same radius) above it. If the surface areas of the two parts are equal then the ratio of its radius and the slant height of the conical part is [CBSE 2011]
(a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1
24. If the radius of the base of a right circular cylinder is halved, keeping the height the same, then the ratio of the volume of the cylinder thus obtained to the volume of original cylinder is [CBSE 2012]
(a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1
25. A cubical ice-cream brick of edge 22 cm is to be distributed among some children by filling ice-cream cones of radius 2 cm and height 7 cm up to its brim. How many children will get the ice-cream cones?
(a) 163 (b) 263 (c) 363 (d) 463
26. A mason constructs a wall of dimensions (270 cm $\times$ 300 cm $\times$ 350 cm) with bricks, each of size (22.5 cm $\times$ 11.25 cm $\times$ 8.75 cm) and it is assumed that $\frac{1}{8}$ space is covered by the mortar. Number of bricks used to construct the wall is
(a) 11000 (b) 11100 (c) 11200 (d) 11300
27. Twelve solid spheres of the same size are made by melting a solid metallic cylinder of base diameter 2 cm and height 16 cm. The diameter of each sphere is
(a) 2 cm (b) 3 cm (c) 4 cm (d) 6 cm
28. The diameters of two circular ends of a bucket are 44 cm and 24 cm, and the height of the bucket is 35 cm. The capacity of the bucket is
(a) 31.7 litres (b) 32.7 litres (c) 33.7 litres (d) 34.7 litres
29. The slant height of a bucket is 45 cm and the radii of its top and bottom are 28 cm and 7 cm respectively. The curved surface area of the bucket is
(a) 4953 cm² (b) 4952 cm² (c) 4951 cm² (d) 4950 cm²

30. The volumes of two spheres are in the ratio 64 : 27. The ratio of their surface areas is
(a) 9 : 16 (b) 16 : 9 (c) 3 : 4 (d) 4 : 3
31. A hollow cube of internal edge 22 cm is filled with spherical marbles of diameter 0.5 cm and $\frac{1}{8}$ space of the cube remains unfilled. Number of marbles required is
(a) 142296 (b) 142396 (c) 142496 (d) 142596
32. A metallic spherical shell of internal and external diameters 4 cm and 8 cm respectively, is melted and recast into the form of a cone of base diameter 8 cm. The height of the cone is
(a) 12 cm (b) 14 cm (c) 15 cm (d) 8 cm
33. A medicine capsule is in the shape of a cylinder of diameter 0.5 cm with two hemispheres stuck to each of its ends. The length of the entire capsule is 2 cm. The capacity of the capsule is
(a) 0.33 cm^3 (b) 0.34 cm^3 (c) 0.35 cm^3 (d) 0.36 cm^3
34. The length of the longest pole that can be kept in a room ($12 \text{ m} \times 9 \text{ m} \times 8 \text{ m}$) is
(a) 29 m (b) 21 m (c) 19 m (d) 17 m
35. The length of the diagonal of a cube is $6\sqrt{3}$ cm. Its total surface area is
(a) 144 cm^2 (b) 216 cm^2 (c) 180 cm^2 (d) 108 cm^2
36. The volume of a cube is 2744 cm^3 . Its surface area is
(a) 196 cm^2 (b) 1176 cm^2 (c) 784 cm^2 (d) 588 cm^2
37. The total surface area of a cube is 864 cm^2 . Its volume is
(a) 3456 cm^3 (b) 432 cm^3 (c) 1728 cm^3 (d) 3456 cm^3
38. How many bricks each measuring ($25 \text{ cm} \times 11.25 \text{ cm} \times 6 \text{ cm}$) will be required to construct a wall ($8 \text{ m} \times 6 \text{ m} \times 22.5 \text{ cm}$)?
(a) 8000 (b) 6400 (c) 4800 (d) 7200
39. The area of the base of a rectangular tank is 6500 cm^2 and the volume of water contained in it is 2.6 m^3 . The depth of water in the tank is
(a) 3.5 m (b) 4 m (c) 5 m (d) 8 m
40. The volume of a wall, 5 times as high as it is broad and 8 times as long as it is high, is 12.8 m^3 . The breadth of the wall is
(a) 30 cm (b) 40 cm (c) 22.5 cm (d) 25 cm

41. If the areas of three adjacent faces of a cuboid are x , y , z respectively then the volume of the cuboid is
(a) xyz (b) $2xyz$ (c) $\sqrt{xyz}$ (d) $3\sqrt{xyz}$
42. The sum of length, breadth and height of a cuboid is 19 cm and its diagonal is $5\sqrt{5}$ cm. Its surface area is
(a) 361 cm^2 (b) 125 cm^2 (c) 236 cm^2 (d) 486 cm^2
43. If each edge of a cube is increased by 50%, the percentage increase in the surface area is
(a) 50% (b) 75% (c) 100% (d) 125%
44. How many bags of grain can be stored in a cuboidal granary ($8 \text{ m} \times 6 \text{ m} \times 3 \text{ m}$), if each bag occupies a space of 0.64 m^3 ?
(a) 8256 (b) 90 (c) 212 (d) 225
45. A cube of side 6 cm is cut into a number of cubes each of side 2 cm. The number of cubes formed is
(a) 6 (b) 9 (c) 12 (d) 27
46. In a shower, 5 cm of rain falls. The volume of the water that falls on 2 hectares of ground, is
(a) 100 m^3 (b) 10 m^3 (c) 1000 m^3 (d) 10000 m^3
47. Two cubes have their volumes in the ratio 1 : 27. The ratio of their surface areas is
(a) 1 : 3 (b) 1 : 8 (c) 1 : 9 (d) 1 : 18
48. The diameter of the base of a cylinder is 4 cm and its height is 14 cm. The volume of the cylinder is
(a) 176 cm^3 (b) 196 cm^3 (c) 276 cm^3 (d) 352 cm^3
49. The diameter of a cylinder is 28 cm and its height is 20 cm. The total surface area of the cylinder is
(a) 2993 cm^2 (b) 2992 cm^2 (c) 2292 cm^2 (d) 2229 cm^2
50. The height of a cylinder is 14 cm and its curved surface area is 264 cm^2 . The volume of the cylinder is
(a) 308 cm^3 (b) 396 cm^3 (c) 1232 cm^3 (d) 1848 cm^3
51. The curved surface area of a cylinder is 1760 cm^2 and its base radius is 14 cm. The height of the cylinder is
(a) 10 cm (b) 15 cm (c) 20 cm (d) 40 cm
52. The ratio of the total surface area to the lateral surface area of a cylinder with base radius 80 cm and height 20 cm is
(a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 5 : 1

53. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3 . The height of the pillar is
(a) 4 m (b) 5 m (c) 6 m (d) 7 m
54. The ratio between the radius of the base and the height of the cylinder is $2 : 3$. If its volume is 1617 cm^3 , the total surface area of the cylinder is
(a) 308 cm^2 (b) 462 cm^2 (c) 540 cm^2 (d) 770 cm^2
55. The radii of two cylinders are in the ratio $2 : 3$ and their heights are in the ratio $5 : 3$. The ratio of their volumes is
(a) $27 : 20$ (b) $20 : 27$ (c) $4 : 9$ (d) $9 : 4$
56. Two circular cylinders of equal volume have their heights in the ratio $1 : 2$. The ratio of their radii is
(a) $1 : \sqrt{2}$ (b) $\sqrt{2} : 1$ (c) $1 : 2$ (d) $1 : 4$
57. The radius of the base of a cone is 5 cm and its height is 12 cm. Its curved surface area is
(a) $60\pi \text{ cm}^2$ (b) $65\pi \text{ cm}^2$ (c) $30\pi \text{ cm}^2$ (d) none of these
58. The diameter of the base of a cone is 42 cm and its volume is 12936 cm^3 . Its height is
(a) 28 cm (b) 21 cm (c) 35 cm (d) 14 cm
59. The area of the base of a right circular cone is 154 cm^2 and its height is 14 cm. Its curved surface area is
(a) $154\sqrt{5} \text{ cm}^2$ (b) $154\sqrt{7} \text{ cm}^2$ (c) $77\sqrt{7} \text{ cm}^2$ (d) $77\sqrt{5} \text{ cm}^2$
60. On increasing each of the radius of the base and the height of a cone by 20% its volume will be increased by
(a) 20% (b) 40% (c) 60% (d) 72.8%
61. The radii of the base of a cylinder and a cone are in the ratio $3 : 4$. If they have their heights in the ratio $2 : 3$, the ratio between their volumes is
(a) $9 : 8$ (b) $3 : 4$ (c) $8 : 9$ (d) $4 : 3$
62. A metallic cylinder of radius 8 cm and height 2 cm is melted and converted into a right circular cone of height 6 cm. The radius of the base of this cone is
(a) 4 cm (b) 5 cm (c) 6 cm (d) 8 cm
63. The height of a conical tent is 14 m and its floor area is 346.5 m^2 . How much canvas, 1.1 m wide, will be required for it ?
(a) 490 m (b) 525 m (c) 665 m (d) 860 m

64. The diameter of a sphere is 14 cm. Its volume is
(a) 1428 cm^3 (b) 1439 cm^3 (c) $1437\frac{1}{3} \text{ cm}^3$ (d) 1440 cm^3
65. The ratio between the volumes of two spheres is 8 : 27. What is the ratio between their surface areas?
(a) 2 : 3 (b) 4 : 5 (c) 5 : 6 (d) 4 : 9
66. A hollow metallic sphere with external diameter 8 cm and internal diameter 4 cm is melted and moulded into a cone having base radius 4 cm. The height of the cone is
(a) 12 cm (b) 14 cm (c) 15 cm (d) 18 cm
67. A metallic cone having base radius 2.1 cm and height 8.4 cm is melted and moulded into a sphere. The radius of the sphere is
(a) 2.1 cm (b) 1.05 cm (c) 1.5 cm (d) 2 cm
68. The volume of a hemisphere is 19404 cm^3 . The total surface area of the hemisphere is
(a) 4158 cm^2 (b) 16632 cm^2 (c) 8316 cm^2 (d) 3696 cm^2
69. The surface area of a sphere is 154 cm^2 . The volume of the sphere is
(a) $179\frac{2}{3} \text{ cm}^3$ (b) $359\frac{1}{3} \text{ cm}^3$ (c) $1437\frac{1}{3} \text{ cm}^3$ (d) none of these
70. The total surface area of a hemisphere of radius 7 cm is
(a) $(588\pi) \text{ cm}^2$ (b) $(392\pi) \text{ cm}^2$ (c) $(147\pi) \text{ cm}^2$ (d) $(98\pi) \text{ cm}^2$
71. The circular ends of a bucket are of radii 35 cm and 14 cm and the height of the bucket is 40 cm. Its volume is
(a) 60060 cm^3 (b) 80080 cm^3 (c) 70040 cm^3 (d) 80160 cm^3
72. If the radii of the ends of a bucket are 5 cm and 15 cm and it is 24 cm high then its surface area is
(a) 1815.3 cm^2 (b) 1711.3 cm^2
(c) 2025.3 cm^2 (d) 2360 cm^2
73. A circus tent is cylindrical to a height of 4 m and conical above it. If its diameter is 105 m and its slant height is 40 m, the total area of canvas required is
(a) 1760 m^2 (b) 2640 m^2
(c) 3960 m^2 (d) 7920 m^2

Matching of columns

74. Match the following columns:

Column I	Column II
(a) A solid metallic sphere of radius 8 cm is melted and the material is used to make solid right cones with height 4 cm and radius of the base 8 cm. How many cones are formed?	(p) 18
(b) A 20-m-deep well with diameter 14 m is dug up and the earth from digging is evenly spread out to form a platform 44 m by 14 m. The height of the platform is m.	(q) 8
(c) A sphere of radius 6 cm is melted and recast into the shape of a cylinder of radius 4 cm. Then, the height of the cylinder is cm.	(r) 16 : 9
(d) The volumes of two spheres are in the ratio 64 : 27. The ratio of their surface areas is	(s) 5

The correct answer is

(a)–.....,

(b)–.....,

(c)–.....,

(d)–..... .

75. Match the following columns:

Column I	Column II
(a) The radii of the circular ends of a bucket in the form of frustum of a cone of height 30 cm are 20 cm and 10 cm respectively. The capacity of the bucket is cm^3 . [Take $\pi = \frac{22}{7}$.]	(p) 2418π

(b) The radii of the circular ends of a conical bucket of height 15 cm are 20 cm and 12 cm respectively. The slant height of the bucket is cm.	(q) 22000
(c) The radii of the circular ends of a solid frustum of a cone are 33 cm and 27 cm and its slant height is 10 cm. The total surface area of the bucket is cm^2 .	(r) 12
(d) Three solid metallic spheres of radii 3 cm, 4 cm and 5 cm are melted to form a single solid sphere. The diameter of the resulting sphere is cm.	(s) 17

The correct answer is

(a)-....., (b)-....., (c)-....., (d)-..... .

Assertion-and-Reason Type

Each question consists of two statements, namely, Assertion (A) and Reason (R). For selecting the correct answer, use the following code:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

76.	Assertion (A)	Reason (R)
	If the radii of the circular ends of a bucket 24 cm high are 15 cm and 5 cm respectively, then the surface area of the bucket is $545\pi \text{ cm}^2$.	If the radii of the circular ends of the frustum of a cone are R and r respectively and its height is h, then its surface area is $\pi\{R^2 + r^2 + l(R - r)\},$ where $l^2 = h^2 + (R + r)^2$.

The correct answer is (a)/(b)/(c)/(d).

77.	Assertion (A)	Reason (R)
	A hemisphere of radius 7 cm is to be painted outside on the surface. The total cost of painting at ₹ 5 per cm^2 is ₹ 2300.	The total surface volume of a hemisphere is $3\pi r^2$.

The correct answer is (a)/(b)/(c)/(d).

78.	Assertion (A)	Reason (R)
	The number of coins 1.75 cm in diameter and 2 mm thick from a melted cuboid ($10 \text{ cm} \times 5.5 \text{ cm} \times 3.5 \text{ cm}$) is 400.	Volume of a cylinder of base radius r and height h is given by $V = (\pi r^2 h)$ cubic units. And, volume of a cuboid $= (l \times b \times h)$ cubic units.

The correct answer is (a)/(b)/(c)/(d).

79.	Assertion (A)	Reason (R)
	If the volumes of two spheres are in the ratio 27 : 8 then their surface areas are in the ratio 3 : 2.	Volume of a sphere $= \frac{4}{3}\pi R^3$. Surface area of a sphere $= 4\pi R^2$.

The correct answer is (a)/(b)/(c)/(d).

80.	Assertion (A)	Reason (R)
	The curved surface volume of a cone of base radius 3 cm and height 4 cm is $(15\pi) \text{ cm}^2$.	Volume of a cone $= \pi r^2 h$.

The correct answer is (a)/(b)/(c)/(d).

ANSWERS (MCQ)

1. (a) 2. (b) 3. (c) 4. (a) 5. (b) 6. (c) 7. (a) 8. (d) 9. (c)
 10. (c) 11. (b) 12. (a) 13. (a) 14. (a) 15. (c) 16. (a) 17. (b) 18. (d)
 19. (a) 20. (a) 21. (d) 22. (a) 23. (a) 24. (c) 25. (c) 26. (c) 27. (a)
 28. (b) 29. (d) 30. (b) 31. (a) 32. (b) 33. (d) 34. (d) 35. (b) 36. (b)
 37. (c) 38. (b) 39. (b) 40. (b) 41. (c) 42. (c) 43. (d) 44. (d) 45. (d)
 46. (c) 47. (c) 48. (a) 49. (b) 50. (b) 51. (c) 52. (d) 53. (c) 54. (d)
 55. (b) 56. (b) 57. (b) 58. (a) 59. (a) 60. (d) 61. (a) 62. (d) 63. (b)
 64. (c) 65. (d) 66. (b) 67. (a) 68. (a) 69. (a) 70. (c) 71. (b) 72. (b)
 73. (d) 74. (a)–(q), (b)–(s), (c)–(p), (d)–(r) 75. (a)–(q), (b)–(s), (c)–(p), (d)–(r)
 76. (d) 77. (d) 78. (a) 79. (d) 80. (c)

HINTS TO SOME SELECTED QUESTIONS

9. During conversion of a solid from one shape to another, the volume remains the same.
10. In a right circular cone, the cross section made by a plane parallel to the base is a
11. $\frac{4}{3}\pi r^3 = 49 \times 33 \times 24 \Rightarrow \frac{4}{3} \times \frac{22}{7} \times r^3 = 49 \times 33 \times 24$
 $\Rightarrow r^3 = 49 \times 33^3 \times 24^3 \times \frac{21}{88 \times 61} = 21 \times 21 \times 21 \Rightarrow r = 21 \text{ cm}.$
12. Diameter of the cone = edge of the cube = 4.2 cm.
 $\therefore$ radius = 2.1 cm.
13. $\frac{4}{3}\pi \times 9 \times 9 \times 9 = \pi \times 9 \times 9 \times h \Rightarrow h = \left(\frac{4}{3} \times 9\right) \text{ cm} = 12 \text{ cm}.$
14. $2\pi r = 22 \Rightarrow r = \left(22 \times \frac{1}{2} \times \frac{7}{22}\right) \text{ cm} = 3.5 \text{ cm}.$
15. Number of spheres = $\frac{\text{volume of the cylinder}}{\text{volume of each sphere}}$
 $= \frac{\pi \times 2 \times 2 \times 45}{\frac{4}{3} \times \pi \times 3 \times 3 \times 3} = 5.$
16. $\frac{4\pi R^2}{4\pi r^2} = \frac{16}{9} \Rightarrow \frac{R^2}{r^2} = \frac{16}{9} \Rightarrow \frac{R}{r} = \sqrt{\frac{16}{9}} = \frac{4}{3}$
 $\Rightarrow \frac{R^3}{r^3} = \frac{4^3}{3^3} = \frac{64}{27} \Rightarrow \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3} = \frac{64}{27}.$
17. $4\pi r^2 = 616 \Rightarrow r^2 = 616 \times \frac{1}{4} \times \frac{7}{22} = 49 \Rightarrow r = 7 \text{ cm}.$
18. Volume of a sphere of radius r , $V = \frac{4}{3}\pi r^3.$
 Volume of a sphere of radius $3r = \frac{4}{3}\pi \times (3r)^3 = 27 \times \frac{4}{3}\pi r^3 = 27V.$
19. $l = \sqrt{h^2 + (R - r)^2} = \sqrt{(16)^2 + (20 - 8)^2} \text{ cm}$
 $= \sqrt{256 + 144} \text{ cm} = \sqrt{400} \text{ cm} = 20 \text{ cm}.$
20. Increase in volume of water = volume of the sphere
 $\Rightarrow \pi \times 18 \times 18 \times h = \frac{4}{3}\pi \times 9 \times 9 \times 9 \Rightarrow h = \left(\frac{4}{3} \times \frac{9 \times 9 \times 9}{18 \times 18}\right) \text{ cm} = 3 \text{ cm}.$
21. Ratio of volumes = $\frac{\frac{1}{3}\pi \times \left(\frac{r}{2}\right)^2 \times \left(\frac{h}{2}\right)}{\frac{1}{3}\pi \times r^2 \times h} = \frac{1}{8}.$
22. $l = \sqrt{h^2 + (R - r)^2} = \sqrt{(40)^2 + (24 - 15)^2} \text{ cm}$
 $= \sqrt{1600 + 81} \text{ cm} = \sqrt{1681} \text{ cm} = 41 \text{ cm}.$

$$23. 2\pi r^2 = \pi r l \Rightarrow \frac{r}{l} = \frac{1}{2}.$$

$$24. \text{Ratio of volumes} = \frac{\pi \times \left(\frac{r}{2}\right)^2 \times h}{\pi \times r^2 \times h} = \frac{1}{4}.$$

25. Let the number of cones be n . Then,

$$n \times \frac{1}{3} \pi r^2 h = 22 \times 22 \times 22 \Rightarrow n \times \frac{1}{3} \times \frac{22}{7} \times 2 \times 2 \times 7 = 22 \times 22 \times 22$$

$$\therefore n = \left(22 \times 22 \times 22 \times \frac{3}{88} \right) = 363.$$

$$26. \text{Volume of the wall filled with bricks} = \left(\frac{7}{8} \times 270 \times 300 \times 350 \right) \text{ cm}^3.$$

$$\text{Volume of each brick} = \left(\frac{225^9}{10^2} \times \frac{1125^{225}}{100^4} \times \frac{875^{35}}{100^4} \right) = \left(\frac{9 \times 225 \times 35}{32} \right) \text{ cm}^3.$$

$$\text{Number of bricks} = \left\{ \frac{7}{8} \times 270 \times 300 \times 350 \times \frac{32}{9 \times 225 \times 35} \right\} = 11200.$$

27. Let the diameter of each sphere be d cm. Then,

$$12 \times \frac{4}{3} \pi r^3 = \pi R^2 h \Rightarrow 12 \times \frac{4}{3} \pi \times \left(\frac{d}{2} \right)^3 = \pi \times 1^2 \times 16$$

$$\Rightarrow 2d^3 = 16 \Rightarrow d^3 = 8 = 2^3 \Rightarrow d = 2 \text{ cm}.$$

28. Here, $R = 22$ cm, $r = 12$ cm and $h = 35$ cm.

$$\begin{aligned} \text{Capacity of the bucket} &= \frac{1}{3} \pi h (R^2 + r^2 + Rr) \text{ cm}^3 \\ &= \frac{1}{3} \times \frac{22}{7} \times 35 \times [(22)^2 + (12)^2 + 22 \times 12] \text{ cm}^3 \\ &= \left(\frac{110}{3} \times 892 \right) \text{ cm}^3 = \left(\frac{110 \times 892}{3 \times 1000} \right) \text{ litres} = 32.7 \text{ litres.} \end{aligned}$$

29. Here, $l = 45$ cm, $R = 28$ cm and $r = 7$ cm.

$$\therefore \text{curved surface area} = \pi l (R + r) = \frac{22}{7} \times 45 \times (28 + 7) \text{ cm}^2 = 4950 \text{ cm}^2.$$

$$30. \frac{\frac{4}{3} \pi R^3}{\frac{4}{3} \pi r^3} = \frac{64}{27} \Rightarrow \frac{R^3}{r^3} = \frac{64}{27} \Rightarrow \left(\frac{R}{r} \right)^3 = \left(\frac{4}{3} \right)^3 \Rightarrow \frac{R}{r} = \frac{4}{3}.$$

$$\text{Ratio of their surface areas} = \frac{4\pi R^2}{4\pi r^2} = \frac{R^2}{r^2} = \left(\frac{R}{r} \right)^2 = \left(\frac{4}{3} \right)^2 = \frac{16}{9} = 16 : 9.$$

$$31. \text{Space filled in the cube} = \left(\frac{7}{8} \times 22 \times 22 \times 22 \right) \text{ cm}^3 = (7 \times 1331) \text{ cm}^3.$$

$$\text{Radius of each marble} = \frac{0.5}{2} \text{ cm} = \frac{5}{20} \text{ cm} = \frac{1}{4} \text{ cm}.$$

$$\text{Volume of each marble} = \frac{4}{3}\pi r^3 = \left(\frac{4}{3} \times \frac{22}{7} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}\right) \text{cm}^3 = \left(\frac{11}{24 \times 7}\right) \text{cm}^3.$$

$$\text{Number of marbles} = \left(\frac{7 \times 1331 \times 24 \times 7}{11}\right) = 142296.$$

32. Here, $R = 4$ cm and $r = 2$ cm.

$$\text{Volume of spherical shell} = \frac{4}{3}\pi[(4)^3 - (2)^3] \text{cm}^3 = \left(\frac{4}{3}\pi \times 56\right) \text{cm}^3.$$

Let the height of the cone be h cm.

$$\therefore \frac{1}{3}\pi \times 4 \times 4 \times h = \frac{4}{3}\pi \times 56 \Rightarrow h = 14 \text{ cm}.$$

33. Radius of the capsule = $\frac{0.5}{2}$ cm = 0.25 cm.

Let the length of cylindrical part be x cm.

$$\text{Then, } 0.25 + x + 0.25 = 2 \Rightarrow x + 0.5 = 2 \Rightarrow x = 1.5 \text{ cm}.$$



$$\begin{aligned} \text{Capacity of the capsule} &= \left(\frac{2}{3}\pi r^3 \times 2\right) + \pi r^2 h \\ &= \frac{4}{3} \times \frac{22}{7} \times (0.25)^3 + \frac{22}{7} \times (0.25)^2 \times 1.5 \\ &= \left(\frac{4}{3} \times \frac{22}{7} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}\right) + \left(\frac{22}{7} \times \frac{1}{4} \times \frac{1}{4} \times \frac{15}{10}\right) \\ &= \frac{11}{168} + \frac{33}{112} = \left(\frac{22 + 99}{336}\right) = \frac{121}{336} = 0.36 \text{ cm}^3. \end{aligned}$$

34. Length of the longest pole = length of diagonal of the room

$$\begin{aligned} &= \sqrt{l^2 + b^2 + h^2} \\ &= \sqrt{(12)^2 + 9^2 + 8^2} \text{ m} = \sqrt{289} \text{ m} = 17 \text{ m}. \end{aligned}$$

35. $\sqrt{3}a = 6\sqrt{3} \Rightarrow a = 6.$

$$\text{Total surface area} = 6a^2 \text{ cm}^2 = (6 \times 6 \times 6) \text{ cm}^2 = 216 \text{ cm}^2.$$

36. $a^3 = 2744 = (2^3 \times 7^3) \Rightarrow a = (2 \times 7) = 14$ cm.

$$\therefore \text{surface area} = 6a^2 \text{ cm}^2 = (6 \times 14 \times 14) \text{ cm}^2 = 1176 \text{ cm}^2.$$

37. $6a^2 = 864 \Rightarrow a^2 = 144 \Rightarrow a = 12$ cm.

$$\therefore \text{volume} = (12 \times 12 \times 12) \text{ cm}^3 = 1728 \text{ cm}^3.$$

38. Volume of the wall = $(800 \times 600 \times 22.5) \text{ cm}^3.$

$$\begin{aligned} \text{Number of bricks} &= \frac{\text{volume of the wall}}{\text{volume of 1 brick}} \\ &= \left(\frac{800 \times 600 \times 22.5}{25 \times 11.25 \times 6}\right) = 6400. \end{aligned}$$

39. Area of the base = $\frac{6500}{100 \times 100} \text{ m}^2 = \frac{13}{20} \text{ m}^2.$

Let the depth of water be d metres. Then,

$$\frac{13}{20} \times d = 2.6 \Rightarrow d = \left(\frac{26}{10} \times \frac{20}{13}\right) = 4.$$

40. Let breadth = x cm. Then, height = $5x$ cm and length = $40x$ cm

$$\therefore 40x \times x \times 5x = 12.8 \times 100 \times 100 \times 100$$

$$\Rightarrow x^3 = 64000 = (40 \times 40 \times 40) \Rightarrow x = 40.$$

41. Let length = 2, breadth = b and height = h . Then, $lb = x$, $bh = y$, and $lh = z$

$$\Rightarrow lb \times bh \times lh = xyz \Rightarrow lbh = \sqrt{xyz}$$

$$\Rightarrow \text{volume} = \sqrt{xyz}.$$

42. $l + b + h = 19 \Rightarrow (l + b + h)^2 = (19)^2 = 361$

$$\Rightarrow (l^2 + b^2 + h^2) + 2(lb + bh + lh) = 361$$

$$\Rightarrow (5\sqrt{5})^2 + 2(lb + bh + lh) = 361$$

$$\Rightarrow 2(lb + bh + lh) = 361 - 125 = 236 \text{ cm}^2.$$

43. Let original edge be a . Original surface area = $6a^2$.

$$\text{New edge} = 150\% \text{ of } a = \frac{150a}{100} = \frac{3a}{2}.$$

$$\text{New surface area} = 6 \times \left(\frac{3a}{2}\right)^2 = \frac{27a^2}{2}.$$

$$\text{Increase in area} = \left(\frac{27a^2}{2} - 6a^2\right) = \frac{15a^2}{2}.$$

$$\text{Increase \%} = \left(\frac{15a^2}{2} \times \frac{1}{6a^2} \times 100\right) = 125\%.$$

44. Number of bags = $\frac{(8 \times 6 \times 3)}{0.64} = 225$.

45. Number of cubes formed = $\frac{\text{volume of given cube}}{\text{volume of each small cube}}$

$$= \frac{(6 \times 6 \times 6)}{(2 \times 2 \times 2)} = 27.$$

46. Required volume = (area $\times$ depth)

$$= \left(2 \times 10000 \times \frac{5}{100}\right) \text{ m}^3 = 1000 \text{ m}^3.$$

47. $\frac{a^3}{b^3} = \frac{1}{27} \Rightarrow \left(\frac{a}{b}\right)^3 = \left(\frac{1}{3}\right)^3 \Rightarrow \frac{a}{b} = \frac{1}{3}.$

$$\frac{S_1}{S_2} = \frac{6a^2}{6b^2} = \frac{a^2}{b^2} = \left(\frac{1}{3}\right)^2 = \frac{1}{9}.$$

Required ratio = 1 : 9.

48. Here, $r = 2$ cm and $h = 14$ cm.

$$\therefore \text{volume} = \pi r^2 h = \left(\frac{22}{7} \times 2 \times 2 \times 14\right) \text{ cm}^3 = 176 \text{ cm}^3.$$

49. Here, $r = 14$ cm and $h = 20$ cm.

$$\text{Total surface area} = 2\pi rh + 2\pi r^2$$

$$= 2\pi r (h + r)$$

$$= 2 \times \frac{22}{7} \times 14 \times (20 + 14) \text{ cm}^2 = 2992 \text{ cm}^2.$$

$$50. 2 \times \frac{22}{7} \times r \times 14 = 264 \Rightarrow r = \frac{264}{88} = 3.$$

$$\text{Volume} = \left(\frac{22}{7} \times 3 \times 3 \times 14 \right) \text{cm}^3 = 396 \text{ cm}^3.$$

$$51. 2\pi rh = 1760 \Rightarrow 2 \times \frac{22}{7} \times 14 \times h = 1760$$

$$\Rightarrow h = \frac{1760}{88} = 20 \text{ cm.}$$

$$52. \frac{\text{Total surface area}}{\text{Lateral surface area}} = \frac{2\pi r(h+r)}{2\pi rh} = \frac{(h+r)}{h}$$

$$= \frac{(20+80)}{20} = \frac{100}{20} = \frac{5}{1}.$$

$$53. 2\pi rh = 264 \text{ and } \pi r^2 h = 924$$

$$\Rightarrow \frac{\pi r^2 h}{2\pi rh} = \frac{924}{264} \Rightarrow r = \frac{2 \times 924}{264} = 7 \text{ m.}$$

$$\therefore 2 \times \frac{22}{7} \times 7 \times h = 264 \Rightarrow h = \frac{264}{44} = 6 \text{ cm.}$$

$$54. \text{ Let radius} = 2x \text{ and height} = 3x. \text{ Then,}$$

$$\pi r^2 h = 1617 \Rightarrow \frac{22}{7} \times (2x)^2 \times 3x = 1617$$

$$\Rightarrow x^3 = \left(1617 \times \frac{7}{22} \times \frac{1}{12} \right) = \left(\frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} \right) = \left(\frac{7}{2} \right)^3$$

$$\Rightarrow x = \frac{7}{2} \text{ cm.}$$

$$\therefore r = 7 \text{ cm and } h = \frac{21}{2} \text{ cm.}$$

$$\text{Total surface area} = 2\pi r(h+r)$$

$$= 2 \times \frac{22}{7} \times 7 \times \left(\frac{21}{2} + 7 \right) \text{cm}^2 = 770 \text{ cm}^2.$$

$$55. \text{ Let the radii be } 2r, 3r \text{ and heights be } 5h \text{ and } 3h. \text{ Then, ratio of their volumes}$$

$$= \frac{\pi \times (2r)^2 \times 5h}{\pi \times (3r)^2 \times 3h} = \frac{20}{27}.$$

$$56. \pi \times r^2 \times h = \pi \times R^2 \times 2h \Rightarrow \frac{r^2}{R^2} = \frac{2}{1} \Rightarrow \left(\frac{r}{R} \right)^2 = 2 \Rightarrow \frac{r}{R} = \sqrt{2}.$$

$$\therefore \text{ required ratio} = \sqrt{2} : 1.$$

$$57. \text{ Here, } r = 5 \text{ cm and } h = 12 \text{ cm.}$$

$$\therefore l^2 = (r^2 + h^2) = (5)^2 + (12)^2 = 169 \Rightarrow l = \sqrt{169} = 13 \text{ cm.}$$

$$\text{Curved surface area} = \pi r l = (\pi \times 5 \times 13) \text{ cm}^2 = (65\pi) \text{ cm}^2.$$

$$58. \frac{1}{3} \pi r^2 h = 12936 \Rightarrow \frac{1}{3} \times \frac{22}{7} \times 21 \times 21 \times h = 12936 \Rightarrow h = \left(\frac{12936}{22 \times 21} \right) \text{ cm} = 28 \text{ cm.}$$

$$59. \pi r^2 = 154 \Rightarrow \frac{22}{7} \times r^2 = 154 \Rightarrow r^2 = \left(154 \times \frac{7}{22}\right) = 49 \Rightarrow r = 7.$$

Now, $r = 7$ cm and $h = 14$ cm.

$$\therefore l = \sqrt{r^2 + h^2} = \sqrt{49 + 196} = \sqrt{245} = 7\sqrt{5} \text{ cm.}$$

Curved surface area = πrl

$$= \left(\frac{22}{7} \times 7 \times 7\sqrt{5}\right) \text{ cm}^2 = 154\sqrt{5} \text{ cm}^2.$$

60. Let the original radius be r and height be h .

$$\text{Then, original volume} = \frac{1}{3} \pi r^2 h = V.$$

$$\text{New radius} = 120\% \text{ of } r = \left(\frac{120r}{100}\right) = \frac{6r}{5}.$$

$$\text{New height} = 120\% \text{ of } h = \left(\frac{120h}{100}\right) = \frac{6h}{5}.$$

$$\text{New volume} = \frac{1}{3} \pi \times \left(\frac{6r}{5}\right)^2 \times \frac{6h}{5} = \frac{216}{125} \left(\frac{1}{3} \pi r^2 h\right) = \frac{216}{125} V.$$

$$\text{Increase} = \left(\frac{216}{125} V - V\right) = \frac{91V}{125}.$$

$$\text{Increase \%} = \left(\frac{91V}{125} \times \frac{100}{V}\right) \% = 72.8\%.$$

61. Let the radii of the cylinder and cone be $3r$ and $4r$ respectively and their heights be $2h$ and $3h$ respectively. Then,

$$\frac{V_1}{V_2} = \frac{\pi(3r)^2 \times (2h)}{\frac{1}{3} \pi (4r)^2 \times (3h)} = \frac{54}{48} = \frac{9}{8}.$$

$$62. \pi \times 8 \times 8 \times 2 = \frac{1}{3} \pi R^2 \times 6 \Rightarrow R^2 = 64 \Rightarrow R = 8 \text{ cm.}$$

$$63. \pi R^2 = 346.5 \Rightarrow R^2 = \left(\frac{3465}{10} \times \frac{7}{22}\right) = \frac{441}{4} = \left(\frac{21}{2}\right)^2 \Rightarrow R = \frac{21}{2} \text{ m.}$$

$$l = \sqrt{R^2 + h^2} = \sqrt{\frac{441}{4} + (14)^2} = \sqrt{\frac{1225}{4}} = \frac{35}{2} \text{ m.}$$

$$\text{Area of canvas} = \pi Rl = \left(\frac{22}{7} \times \frac{21}{2} \times \frac{35}{2}\right) \text{ m}^2 = 577.5 \text{ m}^2.$$

$$\text{Length of canvas} = \frac{\text{area}}{\text{width}} = \frac{577.50}{1.1} \text{ m} = 525 \text{ m.}$$

$$64. \text{Volume} = \left(\frac{4}{3} \pi r^3\right) = \left(\frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7\right) \text{ cm}^3 = 1437 \frac{1}{3} \text{ cm}^3.$$

$$65. \frac{\frac{4}{3} \pi R^3}{\frac{4}{3} \pi r^3} = \frac{8}{27} \Rightarrow \frac{R^3}{r^3} = \frac{8}{27} \Rightarrow \left(\frac{R}{r}\right)^3 = \left(\frac{2}{3}\right)^3 \Rightarrow \frac{R}{r} = \frac{2}{3}.$$

$$\text{Ratio between their surface areas} = \frac{4\pi R^2}{4\pi r^2} = \frac{R^2}{r^2} = \left(\frac{R}{r}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}.$$

$$66. \frac{4}{3}\pi \times (4)^3 - \frac{4}{3}\pi \times (2)^3 = \frac{1}{3}\pi \times (4)^2 \times h \Rightarrow 16h = 224 \Rightarrow h = 14 \text{ cm.}$$

$$67. \frac{1}{3}\pi \times (2.1)^2 \times 8.4 = \frac{4}{3}\pi \times r^3$$

$$\Rightarrow r^3 = \left(\frac{1}{3} \times 4.41 \times 8.4 \times \frac{3}{4}\right) = (2.1)^3 \Rightarrow r = 2.1 \text{ cm.}$$

$$68. \frac{2}{3} \times \frac{22}{7} \times R^3 = 19404 \Rightarrow R^3 = \left(19404 \times \frac{21}{44}\right)$$

$$\Rightarrow R^3 = (21)^3 \Rightarrow R = 21 \text{ cm.}$$

$$\text{Total surface area} = 3\pi R^2 = \left(3 \times \frac{22}{7} \times 21 \times 21\right) \text{ cm}^2 = 4158 \text{ cm}^2.$$

$$69. 4\pi R^2 = 154 \Rightarrow 4 \times \frac{22}{7} \times R^2 = 154$$

$$\Rightarrow R^2 = \left(154 \times \frac{7}{88}\right) \Rightarrow R = \frac{7}{2}.$$

$$\text{Volume of the sphere} = \frac{4}{3}\pi R^3$$

$$= \left(\frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2}\right) \text{ cm}^3 = 179\frac{2}{3} \text{ cm}^3.$$

$$70. \text{Total surface area} = 3\pi R^2 = (3\pi \times 7 \times 7) \text{ cm}^2 = (147\pi) \text{ cm}^2.$$

$$71. \text{Volume of the bucket} = \frac{\pi h}{3}(R^2 + r^2 + Rr)$$

$$= \frac{22}{7} \times \frac{1}{3} \times 40 \times [(35)^2 + (14)^2 + (35 \times 14)] \text{ cm}^3$$

$$= 80080 \text{ cm}^3.$$

$$72. l = \sqrt{h^2 + (R - r)^2} = \sqrt{(24)^2 + (15 - 5)^2} = 26 \text{ cm.}$$

$$\text{Surface area of the bucket} = \pi[l(R + r) + r^2]$$

$$= 3.14 \times [26 \times 20 + 25] \text{ cm}^2 = 1711.3 \text{ cm}^2.$$

$$73. \text{Area of canvas} = \left(2 \times \frac{22}{7} \times \frac{105}{2} \times 4 + \frac{22}{7} \times \frac{105}{2} \times 40\right) \text{ m}^2$$

$$= (1320 + 6600) \text{ m}^2 = 7920 \text{ m}^2.$$

$$74. \text{(a) Let the number of cones be } n. \text{ Then,}$$

$$\frac{4}{3}\pi R^3 = n \times \frac{1}{3}\pi r^2 h \Rightarrow 4R^3 = nr^2 h \Rightarrow 4 \times 8^3 = n \times 8^2 \times 4 \Rightarrow n = 8.$$

$$\text{(b) Volume of the earth dug out} = \pi r^2 h = \left(\frac{22}{7} \times 7 \times 7 \times 20\right) \text{ m}^3 = (22 \times 140) \text{ m}^3.$$

Let the height of the platform be h metres.

$$\text{Volume of the earth on platform} = (44 \times 14 \times h) \text{ m}^3.$$

$$\therefore 44 \times 14 \times h = 22 \times 140 \Rightarrow h = \frac{22 \times 140}{44 \times 14} \text{ m} = 5 \text{ m.}$$

(c) Volume of the sphere = $\frac{4}{3}\pi r^3$.

Volume of the cylinder = $\pi R^2 h$.

$$\therefore \frac{4}{3}\pi r^3 = \pi R^2 h \Rightarrow \frac{4}{3}r^3 = R^2 h$$

$$\therefore \frac{4}{3} \times 6 \times 6 \times 6 = 4 \times 4 \times h \Rightarrow h = \frac{8 \times 6 \times 6}{4 \times 4} = 18 \text{ cm.}$$

(d) $\frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3} = \frac{64}{27} \Rightarrow \frac{R^3}{r^3} = \frac{64}{27} \Rightarrow \left(\frac{R}{r}\right)^3 = \left(\frac{4}{3}\right)^3 \Rightarrow \frac{R}{r} = \frac{4}{3}$.

$$\text{Ratio of their surface areas} = \frac{4\pi R^2}{4\pi r^2} = \frac{R^2}{r^2} = \left(\frac{R}{r}\right)^2 = \left(\frac{4}{3}\right)^2 = \frac{16}{9} = 16:9.$$

The correct answer is (a)–(q), (b)–(s), (c)–(p), (d)–(r).

75. (a) $R = 20 \text{ cm}$, $r = 10 \text{ cm}$ and $h = 30 \text{ cm}$.

$$\begin{aligned} \text{Capacity} &= \frac{1}{3}\pi h(R^2 + r^2 + Rr) \\ &= \left\{ \frac{1}{3} \times \frac{22}{7} \times 15 \times (400 + 100 + 200) \right\} \text{ cm}^3 = 11000 \text{ cm}^3. \end{aligned}$$

(b) $R = 20 \text{ cm}$, $r = 12 \text{ cm}$ and $h = 15 \text{ cm}$.

$$\begin{aligned} l &= \sqrt{h^2 + (R - r)^2} = \sqrt{(15)^2 + (20 - 12)^2} \text{ cm} = \sqrt{(15)^2 + 8^2} \text{ cm} \\ &= \sqrt{225 + 64} \text{ cm} = \sqrt{289} \text{ cm} = 17 \text{ cm.} \end{aligned}$$

(c) $R = 33 \text{ cm}$, $r = 27 \text{ cm}$ and $l = 10 \text{ cm}$.

$$\begin{aligned} \text{Total surface area} &= \pi[R^2 + r^2 + l(R + r)] \\ &= \pi[(33)^2 + (27)^2 + 10 \times (33 + 27)] \text{ cm}^2 \\ &= \pi \times (2418) \text{ cm}^2 = (2418\pi) \text{ cm}^2. \end{aligned}$$

(d) $\frac{4}{3}\pi R^3 = \frac{4}{3}\pi \times [(3)^3 + (4)^3 + (5)^3] \Rightarrow R^3 = 216 = (6)^3 \Rightarrow R = 6$.

$$\therefore \text{diameter} = 12 \text{ cm.}$$

The correct answer is (a)–(q), (b)–(s), (c)–(p), (d)–(r).

76. The Reason (R) is clearly true.

Now, $h = 24 \text{ cm}$, $R = 15 \text{ cm}$ and $r = 5 \text{ cm}$.

$$\begin{aligned} \therefore l^2 &= h^2 + (R - r)^2 = \{(24)^2 + (15 - 5)^2\} \text{ cm}^2 = (576 + 100) \text{ cm}^2 = 676 \text{ cm}^2 \\ \Rightarrow l &= \sqrt{676} \text{ cm} = 26 \text{ cm.} \end{aligned}$$

$$\begin{aligned} \therefore \text{surface area} &= \pi\{R^2 + r^2 + l(R + r)\} \\ &= \pi\{(15)^2 + 5^2 + 26 \times 10\} \text{ cm}^2 = \pi\{225 + 25 + 260\} \text{ cm}^2 \\ &= (510\pi) \text{ cm}^2, \text{ which is true.} \end{aligned}$$

$\therefore$ Assertion (A) is true and Reason (R) gives Assertion (A).

Hence, the correct answer is (a).

77. The Reason (R) is clearly true.

$$\text{Total surface area} = 3\pi r^2 = \left(3 \times \frac{22}{7} \times 7 \times 7\right) \text{ cm}^2 = 462 \text{ cm}^2.$$

Cost of painting = ₹(462 × 5) = ₹ 2310.

∴ Assertion (A) is false and Reason (R) is true.

So, the correct answer is (d).

78. The Reason (R) is clearly true.

Radius of each coin = $\frac{1.75}{2}$ cm, thickness = $\frac{2}{10}$ cm = $\frac{1}{5}$ cm.

$$\begin{aligned}\therefore \text{ volume of each coin} &= \pi r^2 h = \left(\frac{22}{7} \times \frac{1.75}{2} \times \frac{1.75}{2} \times \frac{1}{5} \right) \text{ cm}^3 \\ &= \left(\frac{11}{70} \times \frac{175}{100} \times \frac{175}{100} \right) \text{ cm}^3 = \frac{77}{160} \text{ cm}^3.\end{aligned}$$

$$\text{Volume of cuboid} = \left(10 \times \frac{55}{10} \times \frac{35}{10} \right) \text{ cm}^3 = \frac{385}{2} \text{ cm}^3.$$

$$\text{Number of coins} = \left(\frac{385}{2} \times \frac{160}{77} \right) = 400, \text{ which is true.}$$

Clearly, Reason (R) is true and Assertion (A) gives Reason (R).

So, the correct answer is (a).

79. We know that the volume of a sphere is $\frac{4}{3}\pi R^3$ and its surface area is $4\pi R^2$. So, the

Reason (R) is clearly true.

Now, let R and r be the radii of two given spheres. Then,

$$\frac{V_1}{V_2} = \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3} = \frac{27}{8} \Rightarrow \frac{R^3}{r^3} = \frac{27}{8} \Rightarrow \left(\frac{R}{r} \right)^3 = \left(\frac{3}{2} \right)^3 \Rightarrow \frac{R}{r} = \frac{3}{2}.$$

$$\therefore \text{ ratio of their surface areas} = \frac{4\pi R^2}{4\pi r^2} = \frac{R^2}{r^2} = \left(\frac{R}{r} \right)^2 = \left(\frac{3}{2} \right)^2 = \frac{9}{4} = 9:4.$$

So, the given result is false.

Thus, Assertion (A) is false and Reason (R) is true.

So, the correct answer is (d).

80. We know that the volume of a cone is $\frac{1}{3}\pi r^2 h$.

So, the given result, i.e., Reason (R), is false.

Now, $r = 3$ cm and $h = 4$ cm.

$$\begin{aligned}\therefore \text{ curved surface area of the cone} \\ &= \pi r \times \sqrt{r^2 + h^2} = \pi \times 3 \times \sqrt{3^2 + 4^2} \\ &= 3\pi\sqrt{25} \text{ cm}^2 = (15\pi) \text{ cm}^2, \text{ which is true.}\end{aligned}$$

Thus, Assertion (A) is true and Reason (R) is false.

Hence, the correct answer is (c).

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TEST YOURSELF*Very-Short-Answer Questions*

1. Find the number of solid spheres, each of diameter 6 cm, that could be moulded to form a solid metallic cylinder of height 45 cm and diameter 4 cm.
2. Two right circular cylinders of equal volumes have their heights in the ratio 1 : 2. What is the ratio of their radii?
3. A circus tent is cylindrical to a height of 4 m and conical above it. If its diameter is 105 m and its slant height is 40 m, find the total area of the canvas required.
4. The radii of the top and bottom of a bucket of slant height 45 cm are 28 cm and 7 cm respectively. Find the curved surface area of the bucket.

Short-Answer Questions

5. A solid metal cone with radius of base 12 cm and height 24 cm is melted to form solid spherical balls of diameter 6 cm each. Find the number of balls formed.
6. A hemispherical bowl of internal diameter 30 cm is full of a liquid. This liquid is filled into cylindrical-shaped bottles each of diameter 5 cm and height 6 cm. How many bottles are required?
7. A solid metallic sphere of diameter 21 cm is melted and recast into small cones, each of diameter 3.5 cm and height 3 cm. Find the number of cones so formed.
8. The diameter of a sphere is 42 cm. It is melted and drawn into a cylindrical wire of diameter 2.8 cm. Find the length of the wire.
9. A drinking glass is in the shape of frustum of a cone of height 21 cm with 6 cm and 4 cm as the diameters of its two circular ends. Find the capacity of the glass.
10. Two cubes, each of volume 64 cm^3 , are joined end to end. Find the total surface area of the resulting cuboid.
11. The radius of the base and the height of a solid right circular cylinder are in the ratio 2 : 3 and its volume is 1617 cm^3 . Find the total surface area of the cylinder. [Take $\pi = \frac{22}{7}$.]
12. A toy is in the form of a cone mounted on a hemisphere of common base radius 7 cm. The total height of the toy is 31 cm. Find the total surface area of the toy.

13. A hemispherical bowl of internal radius 9 cm is full of water. This water is to be filled in cylindrical bottles of diameter 3 cm and height 4 cm. Find the number of bottles needed to fill the whole water of the bowl.
14. The surface areas of a sphere and a cube are equal. Find the ratio of their volumes. [Take $\pi = \frac{22}{7}$]
15. The slant height of the frustum of a cone is 4 cm and the perimeters (i.e., circumferences) of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.
16. A solid is composed of a cylinder with hemispherical ends. If the whole length of the solid is 104 cm and the radius of each hemispherical end is 7 cm, find the surface area of the solid.

Long-Answer Questions

17. From a solid cylinder whose height is 15 cm and diameter 16 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid. [Use $\pi = 3.14$.]
18. A solid rectangular block of dimensions 4.4 m, 2.6 m and 1 m is cast into a hollow cylindrical pipe of internal radius 30 cm and thickness 5 cm. Find the length of the pipe.
19. An open metal bucket is in the shape of a frustum of a cone, mounted on a hollow cylindrical base made of the same metallic sheet. The diameters of the two circular ends of the bucket are 45 cm and 25 cm, the total vertical height of the bucket is 40 cm and that of the cylindrical base is 6 cm. Find the area of the metallic sheet used to make the bucket. Also, find the volume of water the bucket can hold, in litres.
20. A farmer connects a pipe of internal diameter 20 cm from a canal into a cylindrical tank which is 10 m in diameter and 2 m deep. If the water flows through the pipe at the rate of 4 km/hr, in how much time will the tank be filled completely?

ANSWERS (TEST YOURSELF)

- | | | | | |
|-------------------------|---------------------------|-----------------------|--|--------------------------|
| 1. 5 | 2. $\sqrt{2} : 1$ | 3. 7920 m^2 | 4. 4950 cm^2 | 5. 32 |
| 6. 60 | 7. 504 | 8. 63 m | 9. 418 cm^3 | 10. 160 cm^2 |
| 11. 770 cm^2 | 12. 858 cm^2 | 13. 54 | 14. $231 : 121$ | 15. 710.3 cm^2 |
| 16. 4576 cm^2 | 17. 443.14 cm^2 | 18. 112 m | 19. 4860.9 cm^2 , 33.62 litres | |
| 20. 1 hour 15 minutes | | | | |





Mean, Median, Mode of Grouped Data, Cumulative Frequency Graph and Ogive

INTRODUCTION

Suppose we want to compare the wage distribution of workers in two factories and determine which factory pays more to its workers. If we compare on the basis of individual workers, we cannot conclude anything. However, if for the given data, we get a representative value that signifies the characteristics of the data, the comparison becomes easy.

A certain value representative of the whole data and signifying its characteristics is called an average of the data.

An average tends to lie centrally with the values of the variable arranged in ascending order of magnitude. So, we call an average a *measure of central tendency* of the data.

Three measures of central tendency are useful for analysing the data, namely

(a) Mean

(b) Median

(c) Mode.

MEAN OF GROUPED DATA

We already know that

$$\text{mean} = \frac{\text{sum of observations}}{\text{number of observations}}.$$

Thus, if $x_1, x_2, \dots, x_n$ are n observations with respective frequencies $f_1, f_2, \dots, f_n$ then the mean is given by

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum_{i=1}^n f_i x_i}{\sum f_i},$$

where the Greek letter Σ (read as, sigma) stands for summation of the terms.

There are three methods for computing the mean of grouped data.

I. DIRECT METHOD In this method, the midpoint or class mark of each class interval is taken to represent the observations falling in the class. We

proceed stepwise as follows:

Step 1. For each class interval, find the class mark x_i as

$$x_i = \frac{\text{upper class limit} + \text{lower class limit}}{2}.$$

Step 2. Calculate $f_i x_i$ for each i .

Step 3. Calculate the mean using the formula

$$\text{mean} = \frac{\sum f_i x_i}{\sum f_i}.$$

EXAMPLE 1 Find the mean of the following data:

Class interval	0–10	10–20	20–30	30–40	40–50
Frequency	8	12	10	11	9

[CBSE 2007]

SOLUTION We may prepare the table as shown:

Class interval	Frequency f_i	Class mark x_i	$(f_i \times x_i)$
0–10	8	5	40
10–20	12	15	180
20–30	10	25	250
30–40	11	35	385
40–50	9	45	405
	$\Sigma f_i = 50$		$\Sigma (f_i \times x_i) = 1260$

$$\therefore \text{mean} = \frac{\Sigma (f_i \times x_i)}{\Sigma f_i} = \frac{1260}{50} = 25.2.$$

EXAMPLE 2 The arithmetic mean of the following frequency distribution is 25. Determine the value of p . [CBSE 2006]

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	18	15	p	6

SOLUTION We have

Class interval	Frequency f_i	Midvalue x_i	$(f_i \times x_i)$
0–10	5	5	25
10–20	18	15	270

20–30	15	25	375
30–40	p	35	$35p$
40–50	6	45	270
	$\Sigma f_i = (44 + p)$		$\Sigma (f_i \times x_i) = (940 + 35p)$

$$\begin{aligned}
 \therefore \text{mean, } \bar{x} &= \frac{\Sigma (f_i \times x_i)}{\Sigma f_i} \Rightarrow \frac{(940 + 35p)}{(44 + p)} = 25 \\
 &\Rightarrow (940 + 35p) = 25(44 + p) \\
 &\Rightarrow (35p - 25p) = (1100 - 940) \\
 &\Rightarrow 10p = 160 \Rightarrow p = 16.
 \end{aligned}$$

Hence, $p = 16$.

EXAMPLE 3

If the mean of the following frequency distribution is 65.6, find the missing frequencies f_1 and f_2 . [CBSE 2010]

Class	10–30	30–50	50–70	70–90	90–110	110–130	Total
Frequency	5	8	f_1	20	f_2	2	50

SOLUTION

We have

$$5 + 8 + f_1 + 20 + f_2 + 2 = 50 \Rightarrow f_2 = (15 - f_1).$$

Now, we may prepare the table given below:

Class interval	Frequency f_i	Class mark x_i	$(f_i \times x_i)$
10–30	5	20	100
30–50	8	40	320
50–70	f_1	60	$60f_1$
70–90	20	80	1600
90–110	$15 - f_1$	100	$1500 - 100f_1$
110–130	2	120	240
	$\Sigma f_i = 50$		$\Sigma (f_i \times x_i) = (3760 - 40f_1)$

$$\therefore \text{mean} = \frac{\Sigma (f_i \times x_i)}{\Sigma f_i} = \frac{(3760 - 40f_1)}{50}.$$

But, mean = 65.6 (given).

$$\begin{aligned}
 \therefore \frac{(3760 - 40f_1)}{50} &= 65.6 \Rightarrow 3760 - 40f_1 = 3280 \\
 &\Rightarrow 40f_1 = 480 \Rightarrow f_1 = 12.
 \end{aligned}$$

Thus, $f_1 = 12$ and $f_2 = (15 - 12) = 3$.

II. ASSUMED-MEAN METHOD This method is usually used when the numerical values of x_i and f_i are large due to which calculations become tedious and time-consuming.

We proceed stepwise as follows:

Step 1. For each class interval, calculate the class mark x_i by using the formula,

$$x_i = \frac{\text{upper class limit} + \text{lower class limit}}{2}.$$

Step 2. Choose a suitable value of x_i in the middle as the assumed mean and denote it by A .

Step 3. Calculate the deviations $d_i = (x_i - A)$ for each i .

Step 4. Calculate the product $(f_i d_i)$ for each i .

Step 5. Find $n = \Sigma f_i$.

Step 6. Calculate the mean, $\bar{x}$, by using the formula,

$$\bar{x} = A + \frac{\Sigma f_i d_i}{n}.$$

MATHEMATICAL DERIVATION OF ASSUMED MEAN FORMULA

We have

$$\text{mean of the deviations, } \bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}.$$

But, $d_i = (x_i - A)$, where A is the assumed mean.

$$\therefore \bar{d} = \frac{\Sigma f_i (x_i - A)}{\Sigma f_i} = \frac{\Sigma f_i x_i}{\Sigma f_i} - \frac{\Sigma f_i A}{\Sigma f_i} = \bar{x} - A \frac{\Sigma f_i}{\Sigma f_i} = \bar{x} - A$$

$$\Rightarrow \bar{x} = A + \bar{d}$$

$$\Rightarrow \bar{x} = A + \frac{\Sigma f_i d_i}{\Sigma f_i}.$$

EXAMPLE 4 Data regarding the weights of students of Class X of a school is given below.

Weight (in kg)	50–52	52–54	54–56	56–58	58–60	60–62	62–64
Number of students	18	21	17	28	16	35	15

Compute the mean weight of the students.

[CBSE 2014]

SOLUTION Let $A = 57$ be the assumed mean. Then, we have

Class interval	Frequency f_i	Class mark x_i	Deviation $d_i = (x_i - A)$ $= (x_i - 57)$	$(f_i \times d_i)$
50–52	18	51	–6	–108
52–54	21	53	–4	–84
54–56	17	55	–2	–34
56–58	28	$57 = A$	0	0
58–60	16	59	2	32
60–62	35	61	4	140
62–64	15	63	6	90
	$\Sigma f_i = 150$			$\Sigma (f_i \times d_i) = 36$

$$\therefore \bar{x} = A + \frac{\Sigma (f_i \times d_i)}{\Sigma f_i} = \left(57 + \frac{36}{150} \right) = 57 + 0.24 = 57.24.$$

Hence, mean weight = 57.24 kg.

III. STEP-DEVIATION METHOD This method is also used to simplify calculations in the process of computing the mean. It is particularly used when the values of $(x_i - A)$ are large and divisible by the class size (= upper limit – lower limit).

We proceed stepwise as shown below.

Step 1. For each class interval, calculate the class mark x_i , where

$$x_i = \frac{\text{upper class limit} + \text{lower class limit}}{2}.$$

Step 2. Choose a suitable value of x_i in the middle as the assumed mean and denote it by A .

Step 3. Calculate h = (upper limit – lower limit), which is the same for all the classes.

Step 4. Calculate $u_i = \frac{(x_i - A)}{h}$ for each class.

Step 5. Calculate $f_i u_i$ for each class.

Step 6. Calculate Σf_i and $\Sigma f_i u_i$.

Step 7. Calculate the mean by using the formula

$$\bar{x} = A + \left\{ h \times \frac{\Sigma f_i u_i}{\Sigma f_i} \right\}.$$

MATHEMATICAL DERIVATION OF STEP-DEVIATION FORMULA

We have $\bar{u} = \frac{\Sigma(f_i u_i)}{\Sigma f_i}$.

But, $u_i = \frac{x_i - A}{h}$.

$$\therefore \bar{u} = \frac{\Sigma f_i \frac{(x_i - A)}{h}}{\Sigma f_i} = \frac{1}{h} \left[\frac{\Sigma(f_i x_i - A \Sigma f_i)}{\Sigma f_i} \right]$$

$$= \frac{1}{h} \left[\frac{\Sigma(f_i x_i)}{\Sigma f_i} - A \frac{\Sigma f_i}{\Sigma f_i} \right]$$

$$= \frac{1}{h} (\bar{x} - A)$$

$$\Rightarrow h\bar{u} = \bar{x} - A$$

$$\Rightarrow \bar{x} = A + h\bar{u}$$

$$\Rightarrow \bar{x} = A + h \left(\frac{\Sigma(f_i u_i)}{\Sigma f_i} \right).$$

EXAMPLE 5 Calculate the arithmetic mean of the following frequency distribution, using the step-deviation method: [CBSE 2001C]

Class interval	Frequency
0-50	17
50-100	35
100-150	43
150-200	40
200-250	21
250-300	24

SOLUTION Here, $h = 50$. Let the assumed mean be $A = 125$.

For calculating the mean, we prepare the table given as follows:

Class interval	Frequency f_i	Midvalue x_i	$u_i = \frac{(x_i - A)}{h}$	$(f_i \times u_i)$
0-50	17	25	-2	-34
50-100	35	75	-1	-35
100-150	43	125 = A	0	0

150–200	40	175	1	40
200–250	21	225	2	42
250–300	24	275	3	72
	$\Sigma f_i = 180$			$\Sigma (f_i u_i) = 154 - 69$ $= 85$

Thus, we have

$$A = 125, h = 50, \Sigma f_i = 180 \text{ and } \Sigma (f_i u_i) = 85.$$

$$\begin{aligned} \text{Mean, } \bar{x} &= A + \left\{ h \times \frac{\Sigma (f_i u_i)}{\Sigma f_i} \right\} \\ &= 125 + \left\{ 50 \times \frac{85}{180} \right\} = 125 + 23.61 = 148.61. \end{aligned}$$

Hence, the mean of the given frequency is 148.61.

EXAMPLE 6 *The following table gives the distribution of expenditures of different families on education. Find the mean expenditure on education of a family.* [CBSE 2004C]

Expenditure (in ₹)	Number of families
1000–1500	24
1500–2000	10
2000–2500	33
2500–3000	28
3000–3500	30
3500–4000	22
4000–4500	16
4500–5000	7

SOLUTION Here, $h = 500$.

For calculating the mean, we prepare the table given as follows:

Class interval	Frequency f_i	Midvalue x_i	$u_i = \frac{(x_i - A)}{h}$	$(f_i \times u_i)$
1000–1500	24	1250	-3	-72
1500–2000	10	1750	-2	-20
2000–2500	33	2250	-1	-33
2500–3000	28	2750 = A	0	0

3000–3500	30	3250	1	30
3500–4000	22	3750	2	44
4000–4500	16	4250	3	48
4500–5000	7	4750	4	28
	$\Sigma f_i = 170$			$\Sigma (f_i \times u_i) = 25$

Thus, $A = 2750$, $h = 500$, $\Sigma f_i = 170$ and $\Sigma (f_i \times u_i) = 25$.

$$\begin{aligned}\therefore \text{mean} &= A + \left\{ h \times \frac{\Sigma (f_i \times u_i)}{\Sigma f_i} \right\} \\ &= 2750 + \left(\frac{500 \times 25}{170} \right) = 2750 + 73.53 = 2823.53.\end{aligned}$$

Hence, the required mean expenditure = ₹ 2823.53.

EXAMPLE 7 Compute the arithmetic mean for the following data:

Marks obtained	Number of students
Less than 10	14
Less than 20	22
Less than 30	37
Less than 40	58
Less than 50	67
Less than 60	75

SOLUTION The above data may be written as:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	14	8	15	21	9	8

Here, $h = 10$.

Let assumed mean = midpoint of (30–40) = 35.

Now, we form the table as under.

Class	Frequency f_i	Midvalue x_i	$u_i = \frac{(x_i - A)}{h}$ $= \frac{(x_i - 35)}{10}$	$(f_i \times u_i)$
0–10	14	5	–3	–42
10–20	8	15	–2	–16
20–30	15	25	–1	–15

30–40	21	35 = A	0	0
40–50	9	45	1	9
50–60	8	55	2	16
	$\Sigma f_i = 75$			$\Sigma (f_i \times u_i) = -48$

$$\therefore \text{mean, } \bar{x} = \left\{ A + \frac{h \times (\Sigma f_i \times u_i)}{\Sigma f_i} \right\} = \left\{ 35 + \left(\frac{10 \times (-48)}{75} \right) \right\}$$

$$= 35 - 6.4 = 28.6.$$

MEAN FOR AN INCLUSIVE SERIES

EXAMPLE 8 Find the arithmetic mean of the following frequency distribution:

Class	25–29	30–34	35–39	40–44	45–49	50–54	55–59
Frequency	14	22	16	6	5	3	4

[CBSE 2006C]

SOLUTION The given series is an inclusive series. Making it an exclusive series, we get

Class	Frequency f_i	Midvalue x_i	$u_i = \frac{(x_i - A)}{h}$ $= \frac{(x_i - 42)}{5}$	$(f_i \times u_i)$
24.5–29.5	14	27	–3	–42
29.5–34.5	22	32	–2	–44
34.5–39.5	16	37	–1	–16
39.5–44.5	6	42 = A	0	0
44.5–49.5	5	47	1	5
49.5–54.5	3	52	2	6
54.5–59.5	4	57	3	12
	$\Sigma f_i = 70$			$\Sigma (f_i \times u_i) = -79$

Thus, $A = 42$, $h = 5$, $\Sigma f_i = 70$ and $\Sigma (f_i \times u_i) = -79$.

$$\therefore \text{mean, } \bar{x} = A + \left\{ h \times \frac{\Sigma (f_i \times u_i)}{\Sigma f_i} \right\} = 42 + \left\{ 5 \times \frac{(-79)}{70} \right\}$$

$$= 42 - 5.64 = 36.36.$$

Hence, the required arithmetic mean is 36.36.

REMEMBER The value of mean obtained by all the above three methods is the same.

EXERCISE 18A

- If the mean of 5 observations $x, x + 2, x + 4, x + 6$ and $x + 8$ is 11, find the value of x . [CBSE 2014]
- If the mean of 25 observations is 27 and each observation is decreased by 7, what will be the new mean? [CBSE 2014]
- Compute the mean of the following data:

Class	1–3	3–5	5–7	7–9
Frequency	12	22	27	19

[CBSE 2013]

- Find the mean of the following data, using direct method:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	7	5	6	12	8	2

- Calculate the mean of the following data, using direct method:

Class	25–35	35–45	45–55	55–65	65–75
Frequency	6	10	8	12	4

- Compute the mean of the following data, using direct method:

Class	0–100	100–200	200–300	300–400	400–500
Frequency	6	9	15	12	8

[CBSE 2005C]

- Using an appropriate method, find the mean of the following frequency distribution:

Class interval	84–90	90–96	96–102	102–108	108–114	114–120
Frequency	8	10	16	23	12	11

Which method did you use, and why?

- If the mean of the following frequency distribution is 24, find the value of p .

Class	0–10	10–20	20–30	30–40	40–50
Frequency	3	4	p	3	2

[CBSE 2013]

9. The following distribution shows the daily pocket allowance of children of a locality. If the mean pocket allowance is ₹ 18, find the missing frequency f .

Daily pocket allowance (in ₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Number of children	7	6	9	13	f	5	4

10. If the mean of the following frequency distribution is 54, find the value of p .

Class	0–20	20–40	40–60	60–80	80–100
Frequency	7	p	10	9	13

[CBSE 2006C]

11. The mean of the following data is 42. Find the missing frequencies x and y if the sum of frequencies is 100.

Class interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	7	10	x	13	y	10	14	9

[CBSE 2014]

12. The daily expenditure of 100 families are given below. Calculate f_1 and f_2 if the mean daily expenditure is ₹ 188.

Expenditure (in ₹)	140–160	160–180	180–200	200–220	220–240
Number of families	5	25	f_1	f_2	5

[CBSE 2014]

13. The mean of the following frequency distribution is 57.6 and the total number of observations is 50.

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	7	f_1	12	f_2	8	5

Find f_1 and f_2 .

14. During a medical check-up, the number of heartbeats per minute of 30 patients were recorded and summarised as follows:

Number of heartbeats per minute	65–68	68–71	71–74	74–77	77–80	80–83	83–86
Number of patients	2	4	3	8	7	4	2

Find the mean heartbeats per minute for these patients, choosing a suitable method.

15. Find the mean marks per student, using assumed-mean method:

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	12	18	27	20	17	6

16. Find the mean of the following frequency distribution, using the assumed-mean method:

Class	100–120	120–140	140–160	160–180	180–200
Frequency	10	20	30	15	5

17. Find the mean of the following data, using the assumed-mean method:

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	20	35	52	44	38	31

18. The following table gives the literacy rate (in percentage) in 40 cities. Find the mean literacy rate, choosing a suitable method.

Literacy rate (%)	45–55	55–65	65–75	75–85	85–95
Number of cities	4	11	12	9	4

[CBSE 2014]

19. Find the mean of the following frequency distribution using step-deviation method.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	7	10	15	8	10

[CBSE 2014]

20. Find the mean of the following data, using step-deviation method:

Class	5–15	15–25	25–35	35–45	45–55	55–65	65–75
Frequency	6	10	16	15	24	8	7

[CBSE 2013]

21. The weights of tea in 70 packets are shown in the following table:

Weight (in grams)	200–201	201–202	202–203	203–204	204–205	205–206
Number of packets	13	27	18	10	1	1

Find the mean weight of packets using step-deviation method. [CBSE 2013]

22. Find the mean of the following frequency distribution using a suitable method:

Class	20–30	30–40	40–50	50–60	60–70
Frequency	25	40	42	33	10

[CBSE 2013]

23. In an annual examination, marks (out of 90) obtained by students of Class X in mathematics are given below:

Marks obtained	0–15	15–30	30–45	45–60	60–75	75–90
Number of students	2	4	5	20	9	10

Find the mean marks.

[CBSE 2014]

24. Find the arithmetic mean of the following frequency distribution using step-deviation method:

Age (in years)	18–24	24–30	30–36	36–42	42–48	48–54
Number of workers	6	8	12	8	4	2

25. Find the mean of the following data using step-deviation method:

Class	500–520	520–540	540–560	560–580	580–600	600–620
Frequency	14	9	5	4	3	5

26. Find the mean age from the following frequency distribution:

Age (in years)	25–29	30–34	35–39	40–44	45–49	50–54	55–59
Number of persons	4	14	22	16	6	5	3

HINT Change the given series to the exclusive series.

27. The following table shows the age distribution of patients of malaria in a village during a particular month:

Age (in years)	5–14	15–24	25–34	35–44	45–54	55–64
Number of cases	6	11	21	23	14	5

Find the average age of the patients.

28. Weight of 60 eggs were recorded as given below:

Weight (in grams)	75–79	80–84	85–89	90–94	95–99	100–104	105–109
Number of eggs	4	9	13	17	12	3	2

Calculate their mean weight to the nearest gram.

29. The following table shows the marks scored by 80 students in an examination:

Marks	Less than 5	Less than 10	Less than 15	Less than 20	Less than 25	Less than 30	Less than 35	Less than 40
Number of students	3	10	25	49	65	73	78	80

Calculate the mean marks correct to 2 decimal places.

ANSWERS (EXERCISE 18A)

1. $x = 7$ 2. 20 3. 5.325 4. 28.75 5. 49.5
 6. 264 7. 103.05 8. $p = 18$ 9. $f = 20$ 10. $p = 11$
 11. $x = 12, y = 25$ 12. $f_1 = 50, f_2 = 15$ 13. $f_1 = 8, f_2 = 10$
 14. 75.9 15. 28 16. 145 17. 62.55 18. 69.5%
 19. 25.8 20. 40.81 21. 201.96 g 22. 42.53 23. 55.5
 24. 33.3 years 25. 544 26. 39.36 years 27. 34.87 years 28. 90 g
 29. 18.56

MEDIAN FOR GROUPED DATA

The *median* is a measure of central tendency which gives the value of the middlemost observation in the data.

As we have already studied, for an ungrouped data comprising n observations:

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)\text{th observation, if } n \text{ is odd} \\ \frac{\left(\frac{n}{2}\right)\text{th observation} + \left(\frac{n}{2} + 1\right)\text{th observation}}{2}, \text{ if } n \text{ is even} \end{cases}$$

METHOD FOR FINDING THE MEDIAN FOR GROUPED DATA

We proceed stepwise as follows:

Step 1. For the given frequency distribution and obtain $N = \Sigma f_i$.

Step 2. Find $\frac{N}{2}$.

Step 3. Look at the cumulative frequency just greater than $\frac{N}{2}$ and find the corresponding class, known as *median class* (as the middlemost observation lies in this class).

Step 4. Compute the median using the formula:

$$\text{Median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\},$$

where l = lower limit of median class;

h = width of median class;

f = frequency of median class;

cf = cumulative frequency of the class preceding the median class;

$$N = \Sigma f_i.$$

SOLVED EXAMPLES

EXAMPLE 1 Find the median of the following data:

Marks	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Number of students	5	15	25	20	7	8	10

[CBSE 2013]

SOLUTION We prepare the cumulative frequency table, as given below.

Class	Frequency (f_i)	Cumulative frequency
20–30	5	5
30–40	15	20
40–50	25	45
50–60	20	65
60–70	7	72
70–80	8	80
80–90	10	90
	$N = \Sigma f_i = 90$	

$$\text{Now, } N = 90 \Rightarrow \frac{N}{2} = 45.$$

The cumulative frequency just greater than 45 is 65 and the corresponding class is 50–60.

Thus, the median class is 50–60.

$$\therefore l = 50, h = 10, f = 20, cf = \text{c.f. of preceding class} = 45,$$

$$\text{and } \frac{N}{2} = 45.$$

$$\text{Median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\} = 50 + \left\{ 10 \times \frac{(45 - 45)}{20} \right\}$$

$$= 50 + 0 = 50.$$

Hence, the median mark is 50.

EXAMPLE 2 Find the median wage from the following data:

Wages (in ₹)	800–820	820–840	840–860	860–880	880–900	900–920	920–940
Number of workers	7	14	19	25	20	10	5

[CBSE 2013]

SOLUTION We prepare the cumulative frequency table, as given below.

Class	Frequency (f_i)	Cumulative frequency
800–820	7	7
820–840	14	21
840–860	19	40
860–880	25	65
880–900	20	85
900–920	10	95
920–940	5	100
	$N = \Sigma f_i = 100$	

$$\text{Now, } N = 100 \Rightarrow \frac{N}{2} = 50.$$

The cumulative frequency just greater than 50 is 65 and the corresponding class is 860–880.

Thus, the median class is 860–880.

$$\therefore l = 860, h = 20, f = 25, cf = \text{c.f. of preceding class} = 40,$$

$$\text{and } \frac{N}{2} = 50.$$

$$\begin{aligned}\text{Median, } M_e &= l + \left\{ h \times \frac{\left(\frac{N}{2} - cf\right)}{f} \right\} = 860 + \left\{ 20 \times \frac{(50 - 40)}{25} \right\} \\ &= 860 + 20 \times \frac{10}{25} = 860 + 8 = 868.\end{aligned}$$

Hence, the median wage is ₹ 868.

MEDIAN FOR INCLUSIVE SERIES

EXAMPLE 3 Find the median for the following frequency distribution:

Height (in cm)	160–162	163–165	166–168	169–171	172–174
Frequency	15	117	136	118	14

SOLUTION The given series is in inclusive form. Converting it to exclusive form and preparing the cumulative frequency table, we get

Class	Frequency (f_i)	Cumulative frequency
159.5–162.5	15	15
162.5–165.5	117	132
165.5–168.5	136	268
168.5–171.5	118	386
171.5–174.5	14	400
	$N = \Sigma f_i = 400$	

$$\text{Now, } N = 400 \Rightarrow \frac{N}{2} = 200.$$

The cumulative frequency just greater than 200 is 268 and the corresponding class is 165.5–168.5.

Thus, the median class is 165.5–168.5.

$$\therefore l = 165.5, h = 3, f = 136, cf = \text{c.f. of preceding class} = 132,$$

$$\text{and } \frac{N}{2} = 200.$$

$$\text{Median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf\right)}{f} \right\}$$

$$\begin{aligned}
 &= 165.5 + \left\{ 3 \times \frac{(200 - 132)}{136} \right\} = 165.5 + \left(\frac{3 \times 68}{136} \right) \\
 &= 165.5 + 1.5 = 167.
 \end{aligned}$$

Hence, median height is 167 cm.

EXAMPLE 4 Given below is the distribution of IQ of 100 students. Find the median IQ.

IQ	75–84	85–94	95–104	105–114	115–124	125–134	135–144
Frequency	8	11	26	31	18	4	2

SOLUTION The given series is in inclusive form. Converting it to exclusive form and preparing the cumulative frequency table, we get

Class	Frequency (f_i)	Cumulative frequency
74.5–84.5	8	8
84.5–94.5	11	19
94.5–104.5	26	45
104.5–114.5	31	76
114.5–124.5	18	94
124.5–134.5	4	98
134.5–144.5	2	100
	$N = \Sigma f_i = 100$	

$$\text{Now, } N = 100 \Rightarrow \frac{N}{2} = 50.$$

The cumulative frequency just greater than 50 is 76 and the corresponding class interval is 104.5–114.5.

Thus, the median class is 104.5–114.5.

$$\begin{aligned}
 \therefore \quad l &= 104.5, h = 10, f = 31, cf = \text{c.f. of preceding class} = 45 \\
 &\text{and } \frac{N}{2} = 50.
 \end{aligned}$$

$$\text{Median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\}$$

$$\begin{aligned}
 &= 104.5 + \left\{ 10 \times \frac{(50 - 45)}{31} \right\} = 104.5 + \frac{50}{31} \\
 &= 104.5 + 1.6 = 106.1.
 \end{aligned}$$

Hence, the median IQ is 106.1.

EXAMPLE 5 Calculate the median for the following data:

Marks obtained	No. of students
Below 10	6
Below 20	15
Below 30	29
Below 40	41
Below 50	60
Below 60	70

SOLUTION From the given table, we may get back frequencies and cumulative frequencies as shown below.

Class interval	Frequency (f_i)	Cumulative frequency
0–10	6	6
10–20	9	15
20–30	14	29
30–40	12	41
40–50	19	60
50–60	10	70
	$N = \Sigma f_i = 70$	

$$\text{Now, } N = 70 \Rightarrow \frac{N}{2} = 35.$$

The cumulative frequency just greater than 35 is 41 and the corresponding class is 30–40.

Thus, the median class is 30–40.

$$\therefore l = 30, h = 10, f = 12, cf = \text{c.f. of preceding class} = 29,$$

$$\text{and } \frac{N}{2} = 35.$$

$$\text{Median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\} = 30 + \left\{ 10 \times \frac{(35 - 29)}{12} \right\} = 35.$$

Hence, the required median is 35.

EXAMPLE 6 Find the missing frequencies in the following frequency distribution table, if $N = 100$ and median is 32.

Marks	0–10	10–20	20–30	30–40	40–50	50–60	Total
Number of students	10	?	25	30	?	10	100

[CBSE 2013]

SOLUTION Let f_1 and f_2 be the missing frequencies of class intervals 10–20 and 40–50 respectively. Then,

$$10 + f_1 + 25 + 30 + f_2 + 10 = 100 \Rightarrow f_1 + f_2 = 25.$$

Median is 32, which lies in 30–40. So, the median class is 30–40.

$$\therefore l = 30, h = 10, f = 30, N = 100 \text{ and } cf = 10 + f_1 + 25 = f_1 + 35.$$

$$\text{Now, median, } M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\}$$

$$\Rightarrow 30 + \left[10 \times \frac{\{50 - (f_1 + 35)\}}{30} \right] = 32$$

$$\Rightarrow 30 + \frac{(15 - f_1)}{3} = 32 \Rightarrow (15 - f_1) = b \Rightarrow f_1 = 9.$$

$$\therefore f_1 = 9 \text{ and } f_2 = 25 - 9 = 16.$$

Hence, $f_1 = 9$ and $f_2 = 16$.

EXERCISE 18B

1. In a hospital, the ages of diabetic patients were recorded as follows. Find the median age.

Age (in years)	0–15	15–30	30–45	45–60	60–75
Number of patients	5	20	40	50	25

[CBSE 2014]

2. Compute the median from the following data:

Marks	0–7	7–14	14–21	21–28	28–35	35–42	42–49
Number of students	3	4	7	11	0	16	9

3. The following table shows the daily wages of workers in a factory:

Daily wages (in ₹)	0–100	100–200	200–300	300–400	400–500
Number of workers	40	32	48	22	8

Find the median daily wage income of the workers.

4. Calculate the median from the following frequency distribution:

Class	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	5	6	15	10	5	4	2	2

5. Given below is the number of units of electricity consumed in a week in a certain locality:

Consumption (in units)	65-85	85-105	105-125	125-145	145-165	165-185	185-205
Number of consumers	4	5	13	20	14	7	4

Calculate the median.

6. Calculate the median from the following data:

Height (in cm)	135-140	140-145	145-150	150-155	155-160	160-165	165-170	170-175
No. of boys	6	10	18	22	20	15	6	3

7. Calculate the missing frequency from the following distribution, it being given that the median of the distribution is 24.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	5	25	?	18	7

8. The median of the following data is 16. Find the missing frequencies a and b if the total of frequencies is 70.

Class	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	12	a	12	15	b	6	6	4

[CBSE 2013]

9. In the following data the median of the runs scored by 60 top batsmen of the world in one-day international cricket matches is 5000. Find the missing frequencies x and y .

Runs scored	2500-3500	3500-4500	4500-5500	5500-6500	6500-7500	7500-8500
Number of batsmen	5	x	y	12	6	2

10. If the median of the following frequency distribution is 32.5, find the values of f_1 and f_2 .

Class interval	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
Frequency	f_1	5	9	12	f_2	3	2	40

11. Calculate the median for the following data:

Age (in years)	19–25	26–32	33–39	40–46	47–53	54–60
Frequency	35	96	68	102	35	4

HINT Convert it to exclusive form.

12. Find the median wages for the following frequency distribution:

Wages per day (in ₹)	61–70	71–80	81–90	91–100	101–110	111–120
No. of women workers	5	15	20	30	20	8

HINT Convert it to exclusive form.

13. Find the median from the following data:

Class	1–5	6–10	11–15	16–20	21–25	26–30	31–35	36–40	41–45
Frequency	7	10	16	32	24	16	11	5	2

HINT Convert it to exclusive form.

14. Find the median from the following data:

Marks	No. of students
Below 10	12
Below 20	32
Below 30	57
Below 40	80
Below 50	92
Below 60	116
Below 70	164
Below 80	200

ANSWERS (EXERCISE 18B)

1. 46.5 years 2. 28 3. ₹ 206.25 4. 19.5 5. 136.5 units
 6. 153.64 cm 7. 25 8. $a = 8, b = 7$ 9. $x = 25, y = 10$
 10. $f_1 = 3, f_2 = 6$ 11. 36 years 12. ₹ 93.50 13. 19.95 14. 53.33
-

MODE OF A GROUPED DATA

Mode is that value of a variate which occurs most often, i.e., the value of the observation having the maximum frequency.

More precisely, mode is that value of the variable at which the concentration of the data is maximum.

MODAL CLASS In a frequency distribution, the class having maximum frequency is called the modal class.

Formula for Calculating Mode:

We have

$$\text{mode, } M_o = x_k + h \cdot \left\{ \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\},$$

where x_k = lower limit of the modal class interval;

f_k = frequency of the modal class;

f_{k-1} = frequency of the class preceding the modal class;

f_{k+1} = frequency of the class succeeding the modal class;

h = width of the class interval.

SOLVED EXAMPLES

EXAMPLE 1 Find the mode of the following data:

Class	0-20	20-40	40-60	60-80	80-100	100-120	120-140
Frequency	6	8	10	12	6	5	3

[CBSE 2013]

SOLUTION Clearly, the modal class is 60-80, as it has the maximum frequency.

$$\therefore x_k = 60, h = 20, f_k = 12, f_{k-1} = 10, f_{k+1} = 6.$$

$$\begin{aligned} \text{Mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\ &= 60 + \left\{ 20 \times \frac{(12 - 10)}{(2 \times 12 - 10 - 6)} \right\} \\ &= 60 + \left\{ 20 \times \frac{2}{(24 - 16)} \right\} = 60 + \left\{ 20 \times \frac{2}{8} \right\} = 60 + 5 = 65. \end{aligned}$$

Hence, mode = 65.

EXAMPLE 2 The distribution of sale of shirts sold in a month in a departmental store is as under.

Size (in cm)	80-85	85-90	90-95	95-100	100-105	105-110	110-115
Number of shirts sold	33	27	85	155	110	45	15

Calculate the modal size of shirts sold.

[CBSE 2014]

SOLUTION Clearly, the modal class is 95–100 as it has the maximum frequency.

$$\therefore x_k = 95, h = 5, f_k = 155, f_{k-1} = 85, f_{k+1} = 110.$$

$$\begin{aligned}\text{Mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\ &= 95 + \left\{ 5 \times \frac{(155 - 85)}{(2 \times 155 - 85 - 110)} \right\} \\ &= 95 + \left\{ 5 \times \frac{70}{(310 - 195)} \right\} = 95 + \left\{ 5 \times \frac{70}{115} \right\} \\ &= 95 + \frac{70}{23} = 95 + 3.04 = 98.04.\end{aligned}$$

$$\therefore \text{modal size} = 98.04 \text{ cm.}$$

EXAMPLE 3 Given below is the frequency distribution of the heights of players in a school.

Height (in cm)	160–162	163–165	166–168	169–171	172–174
Number of students	15	118	142	127	18

Find the modal height and interpret it.

SOLUTION The given data is an inclusive series. So, we convert it into an exclusive form, as given below.

Class	159.5–162.5	162.5–165.5	165.5–168.5	168.5–171.5	171.5–174.5
Frequency	15	118	142	127	18

Clearly, the class 165.5–168.5 has maximum frequency, so it is the modal class.

$$\therefore x_k = 165.5, f_k = 142, f_{k-1} = 118, f_{k+1} = 127 \text{ and } h = 3.$$

$$\begin{aligned}\text{Mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\ &= 165.5 + \left\{ 3 \times \frac{(142 - 118)}{(2 \times 142 - 118 - 127)} \right\} \\ &= 165.5 + \left\{ \frac{3 \times 24}{39} \right\} = 165.5 + \frac{24}{13} \\ &= 165.5 + 1.85 = 167.35.\end{aligned}$$

Thus, modal height = 167.35 cm.

This means that the height of maximum number of players in the school is 167.35 cm (approx.).

EXAMPLE 4 The following table shows the ages of the patients admitted in a hospital during a month:

Age (in years)	6–15	16–25	26–35	36–45	46–55	56–65
Number of patients	6	11	21	23	14	5

Find the mode and the mean of the data given above. Compare and interpret the two measures of central tendency.

SOLUTION The given data is an inclusive series. Making it an exclusive series, we get

Class	Frequency f_i	Class mark x_i	$u_i = \frac{(x_i - A)}{h}$ $= \frac{(x_i - 40.5)}{10}$	$f_i u_i$
5.5–15.5	6	10.5	–3	–18
15.5–25.5	11	20.5	–2	–22
25.5–35.5	21	30.5	–1	–21
35.5–45.5	23	40.5 = A	0	0
45.5–55.5	14	50.5	1	14
55.5–65.5	5	60.5	2	10
	$\Sigma f_i = 80$			$\Sigma f_i u_i = -37$

Thus, $A = 40.5$, $h = 10$, $\Sigma f_i = 80$, $\Sigma f_i u_i = -37$.

$$\begin{aligned}
 \text{Mean, } \bar{x} &= A + \left\{ h \times \frac{\Sigma f_i u_i}{\Sigma f_i} \right\} \\
 &= 40.5 + \left\{ 10 \times \frac{(-37)}{80} \right\} = 40.5 - \frac{37}{8} \\
 &= 40.5 - 4.63 = 35.87.
 \end{aligned}$$

Also, the modal class of the given data is 35.5–45.5, as it has the maximum frequency.

$$\therefore x_k = 35.5, f_k = 23, f_{k-1} = 21, f_{k+1} = 14 \text{ and } h = 10.$$

$$\begin{aligned}
 \text{Mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\
 &= 35.5 + \left\{ 10 \times \frac{(23 - 21)}{(2 \times 23 - 21 - 14)} \right\} = 35.5 + \left(\frac{10 \times 2}{11} \right) \\
 &= 35.5 + 1.82 = 37.32.
 \end{aligned}$$

Clearly, mode > mean.

Interpretation The modal age is 37.32 years. This means that the maximum number of patients admitted in the hospital during the given month are of the age 37.32 years (approx.).

The mean age is 35.87 years. This means that on an average the age of a patient admitted to the hospital is 35.87 years.

EXAMPLE 5 The mode of the following series is 36. Find the missing frequency in it.

Class interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	8	10	...	16	12	6	7

SOLUTION Since the mode of the given series is 36 and maximum frequency 16 lies in the class 30–40, so the modal class is 30–40.

Let the missing frequency be x . Then,

$$\therefore x_k = 30, f_k = 16, f_{k-1} = x, f_{k+1} = 12 \text{ and } h = 10.$$

Also, $m_o = 36$.

Using the formula, $M_o = x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$, we get

$$30 + \left\{ 10 \times \frac{(16 - x)}{(32 - x - 12)} \right\} = 36$$

$$\Rightarrow \frac{10 \times (16 - x)}{(20 - x)} = 6 \Rightarrow 160 - 10x = 120 - 6x$$

$$\Rightarrow 4x = 40 \Rightarrow x = 10.$$

Hence, the missing frequency is 10.

EXAMPLE 6 Compare the modal ages of two groups of students appearing for an entrance examination.

Age (in years)	16–18	18–20	20–22	22–24	24–26
Group A	50	78	46	28	23
Group B	54	89	40	25	17

SOLUTION **Case I.** Computation of Modal Age of Group A:

In group A, class 18–20 has the maximum frequency.

So, 18–20 is the modal class and its frequency is 78.

$$\therefore x_k = 18, f_k = 78, f_{k-1} = 50, f_{k+1} = 46 \text{ and } h = 2.$$

$$\therefore \text{mode, } M_o = x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$$

$$\begin{aligned}
 &= 18 + \left\{ 2 \times \frac{(78 - 50)}{(2 \times 78 - 50 - 46)} \right\} = 18 + \left\{ \frac{2 \times 28}{60} \right\} \\
 &= 18 + \frac{14}{15} = 18 + 0.93 = 18.93.
 \end{aligned}$$

Case II. *Computation of Modal Age of Group B:*

In group B, class 18–20 has the maximum frequency.

So, 18–20 is the modal class and its frequency is 89.

$$\therefore x_k = 18, f_k = 89, f_{k-1} = 54, f_{k+1} = 40 \text{ and } h = 2.$$

$$\begin{aligned}
 \therefore \text{mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\
 &= 18 + \left\{ 2 \times \frac{(89 - 54)}{(2 \times 89 - 54 - 40)} \right\} = 18 + \frac{(2 \times 35)}{(178 - 94)} \\
 &= 18 + \frac{70}{84} = 18 + \frac{5}{6} = 18 + 0.83 = 18.83.
 \end{aligned}$$

Hence, the modal ages of students in groups A and B are 18.93 years and 18.83 years respectively.

$\therefore$ (modal age in group A) > (modal age in group B).

EXAMPLE 7 *The following table shows the marks obtained by 100 students of Class X in a school during a particular academic session. Find the mode of this distribution.*

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60	Less than 70	Less than 80
Number of students	7	21	34	46	66	77	92	100

[CBSE 2013]

SOLUTION Clearly, the above data can be written as:

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	7	14	13	12	20	11	15	8

Clearly, the modal class is 40–50, as it has the maximum frequency.

$$\therefore x_k = 40, f_k = 20, f_{k-1} = 12, f_{k+1} = 11, h = 10.$$

$$\begin{aligned}
 \text{Mode, } M_o &= x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\
 &= 40 + \left\{ 10 \times \frac{(20 - 12)}{(2 \times 20 - 12 - 11)} \right\}
 \end{aligned}$$

$$\begin{aligned}
 &= 40 + \left\{ 10 \times \frac{8}{(40 - 23)} \right\} = 40 + \left\{ 10 \times \frac{8}{17} \right\} \\
 &= 40 + \frac{80}{17} = 40 + 4.71 = 44.71.
 \end{aligned}$$

Hence, mode = 44.71.

EXERCISE 18C

1. Find the mode of the following frequency distribution:

Marks	10–20	20–30	30–40	40–50	50–60
Frequency	12	35	45	25	13

[CBSE 2014]

2. Compute the mode of the following data:

Class	0–20	20–40	40–60	60–80	80–100
Frequency	25	16	28	20	5

[CBSE 2013]

3. Heights of students of Class X are given in the following frequency distribution:

Height (in cm)	150–155	155–160	160–165	165–170	170–175
Number of students	15	8	20	12	5

Find the modal height.

[CBSE 2014]

Also, find the mean height. Compare and interpret the two measures of central tendency.

4. Find the mode of the following distribution:

Class interval	10–14	14–18	18–22	22–26	26–30	30–34	34–38	38–42
Frequency	8	6	11	20	25	22	10	4

5. Given below is the distribution of total household expenditure of 200 manual workers in a city.

Expenditure (in ₹)	No. of manual workers
1000–1500	24
1500–2000	40
2000–2500	31

2500–3000	28
3000–3500	32
3500–4000	23
4000–4500	17
4500–5000	5

Find the expenditure done by maximum number of manual workers.

6. Calculate the mode from the following data:

Monthly salary (in ₹)	No. of employees
0–5000	90
5000–10000	150
10000–15000	100
15000–20000	80
20000–25000	70
25000–30000	10

7. Compute the mode from the following data:

Age (in years)	0–5	5–10	10–15	15–20	20–25	25–30	30–35
Number of patients	6	11	18	24	17	13	5

8. Compute the mode from the following series:

Size	45–55	55–65	65–75	75–85	85–95	95–105	105–115
Frequency	7	12	17	30	32	6	10

9. Compute the mode of the following data:

Class interval	1–5	6–10	11–15	16–20	21–25	26–30	31–35	36–40	41–45	46–50
Frequency	3	8	13	18	28	20	13	8	6	4

10. The agewise participation of students in the Annual Function of a school is shown in the following distribution. [CBSE 2014]

Age (in years)	5–7	7–9	9–11	11–13	13–15	15–17	17–19
Number of students	x	15	18	30	50	48	x

Find the missing frequencies when the sum of frequencies is 181. Also, find the mode of the data.

ANSWERS (EXERCISE 18C)

1. 33.33 2. 52
3. Modal height = 163 cm. This means that the height of maximum number of students is 163 cm. Mean height = 161.17 cm. This means that on an average the height of a student of the class is 161.17 cm.
4. 28.5 5. ₹ 1820 6. ₹ 7727.27 7. 17.3 years
8. 81.5 9. 23.28
10. Each of two missing frequencies is 10; 11.18 years

COMPARATIVE STUDY OF THE THREE MEASURES OF CENTRAL TENDENCY

The mean is the most frequently used measure of central tendency because it takes into account all the observations and lies between the largest and the smallest observations of the entire data. It also enables us to compare two or more distributions, e.g., by comparing the mean marks of students of different classes in a particular examination, we can conclude which class has a better performance. However, the mean is useful when there are no extreme values in the data set, i.e., if the data are normally distributed.

However, the median is considered a better measure of central tendency when there are a few extreme values that could greatly influence the mean and distort what might be considered typical, as in finding the typical productivity rate of workers, average wage in a country, etc., e.g., in a group of employees, if the salary of one is 10 times the mean salary of others then the mean salary of the group will be high. In such a case, the median may better represent the typical salary level of the group.

The mode is particularly useful for dealing with categorical data, where most popular item is to be ascertained, e.g., to find the most popular TV programme being watched, the consumer item in greatest demand, the colour of the vehicle used by most of the people, etc. It is particularly useful for the manufacturers. Thus, a shirt manufacturer is more interested in the modal size of shirts sold, so as to produce it in large numbers.

EMPIRICAL RELATIONSHIP BETWEEN THE THREE MEASURES OF CENTRAL TENDENCY

$$3(\text{Median}) = \text{Mode} + 2(\text{Mean})$$

MISCELLANEOUS QUESTIONS ON MEAN, MODE AND MEDIAN

EXAMPLE 100 surnames were randomly picked up from a telephone directory and the distribution of the number of letters of the English alphabet in the surnames are obtained as follows:

Number of letters	1-4	4-7	7-10	10-13	13-16	16-19
Number of surnames	6	30	40	16	4	4

Determine the median and mean number of letters in the surnames.
Also, find the modal size of surnames. [CBSE 2014]

SOLUTION We have

Class	Class mark x_i	Frequency f_i	Cumulative frequency	$u_i = \frac{(x_i - A)}{h} = \frac{(x_i - 11.5)}{3}$	$(f_i u_i)$
1-4	2.5	6	6	-3	-18
4-7	5.5	30	36	-2	-60
7-10	8.5	40	76	-1	-40
10-13	11.5 = A	16	92	0	0
13-16	14.5	4	96	1	4
16-19	17.5	4	100	2	8
		$\Sigma f_i = 100$			$\Sigma (f_i u_i) = -106$

Clearly, $A = 11.5$, $h = 3$, $\Sigma f_i = 100$ and $\Sigma (f_i u_i) = -106$.

$$\begin{aligned}\text{Mean, } \bar{x} &= A + \left\{ h \times \frac{\Sigma (f_i u_i)}{\Sigma f_i} \right\} = 11.5 + \left\{ 3 \times \frac{(-106)}{100} \right\} \\ &= 11.5 - 3.18 = 8.32.\end{aligned}$$

$$\text{Here, } N = 100 \Rightarrow \frac{N}{2} = 50.$$

Cumulative frequency just greater than 50 is 76 and the corresponding class is 7-10. Thus, the median class is 7-10.

$$\therefore l = 7, h = 3, f = 40, cf = \text{c.f. of preceding class} = 36, \frac{N}{2} = 50.$$

$$\begin{aligned}\text{Median, } M_e &= l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\} \\ &= 7 + \left\{ 3 \times \frac{(50 - 36)}{40} \right\} = 7 + \left(3 \times \frac{14}{40} \right) \\ &= 7 + \frac{21}{20} = 7 + 1.05 = 8.05.\end{aligned}$$

$$\text{Mode} = 3(\text{median}) - 2(\text{mean})$$

$$= 3 \times 8.05 - 2 \times 8.32 = 24.15 - 16.64 = 7.51.$$

EXERCISE 18D

1. Find the mean, mode and median of the following frequency distribution:

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	4	7	10	12	8	5

[CBSE 2010]

2. Find the mean, median and mode of the following data:

Class	0-20	20-40	40-60	60-80	80-100	100-120	120-140
Frequency	6	8	10	12	6	5	3

[CBSE 2008]

3. Find the mean, median and mode of the following data:

Class	0-50	50-100	100-150	150-200	200-250	250-300	300-350
Frequency	2	3	5	6	5	3	1

[CBSE 2013]

4. Find the mode, median and mean for the following data:

Marks obtained	25-35	35-45	45-55	55-65	65-75	75-85
Number of students	7	31	33	17	11	1

[CBSE 2009]

5. A survey regarding the heights (in cm) of 50 girls of a class was conducted and the following data was obtained:

Height (in cm)	120-130	130-140	140-150	150-160	160-170	Total
Number of girls	2	8	12	20	8	50

Find the mean, median and mode of the above data.

[CBSE 2008]

6. The following table gives the daily income of 50 workers of a factory:

Daily income (in ₹)	100-120	120-140	140-160	160-180	180-200
Number of workers	12	14	8	6	10

Find the mean, mode and median of the above data.

[CBSE 2009]

7. The table below shows the daily expenditure on food of 30 households in a locality:

Daily expenditure (in ₹)	Number of households
100–150	6
150–200	7
200–250	12
250–300	3
300–350	2

Find the mean and median daily expenditure on food. [CBSE 2009C]

ANSWERS (EXERCISE 18D)

1. Mean = 38.2, median = 40, mode = 43.6
 2. Mean = 62.4, median = 61.67, mode = 60.21
 3. Mean = 169, median = 170.83, mode = 174.49
 4. Mean = 49.70, median = 48.64, mode = 46.52
 5. Mean = 149.8 cm, median = 152.5 cm, mode = 157.9 cm
 6. Mean = ₹ 145.2, median = ₹ 137.5, mode = ₹ 122.1
 7. Mean = ₹ 205, median = ₹ 208.33, mode = ₹ 214.99
-

CUMULATIVE FREQUENCY CURVE (OR OGIVE)

An ogive is a freehand graph showing the curve of a cumulative frequency distribution. The term 'ogive' is derived from the word 'ogee' meaning a shape consisting of a concave arc flowing into a convex arc, so forming an S-shaped curve with vertical ends.

Let a grouped frequency distribution be given to us.

FOR A 'LESS THAN' SERIES:

On a graph paper, we mark the upper class limits along the x -axis and the corresponding *cumulative frequencies* along the y -axis.

- (i) On joining these points successively by line segments, we get a polygon, called *cumulative frequency polygon*.
- (ii) On joining these points successively by smooth curves, we get a curve, known as *cumulative frequency curve*, or an *ogive*.

FOR A 'GREATER THAN' SERIES

On a graph paper, we mark the lower class limits along the x -axis and the corresponding *cumulative frequencies* along the y -axis.

- (i) On joining these points successively by line segments, we get a polygon, called *cumulative frequency polygon*.
- (ii) On joining these points successively by smooth curves, we get a curve, known as *cumulative frequency curve*, or an *ogive*.

HOW TO OBTAIN MEDIAN FROM CUMULATIVE FREQUENCY CURVE?

METHOD 1 We proceed stepwise as follows:

- Step 1. Draw either the less than type or the more than type cumulative frequency curve for the given frequency distribution.
- Step 2. Locate $\frac{N}{2}$ on the y -axis. Let it be P .
- Step 3. From point P , draw a line PQ parallel to the x -axis cutting the curve at a point Q .
- Step 4. From Q , draw QM perpendicular to the x -axis cutting the x -axis at M .

The point of intersection of this perpendicular with the x -axis, i.e., the abscissa or x -coordinate of M gives the median of the data.

METHOD 2 We proceed stepwise as follows:

- Step 1. Draw both ogives (i.e., of the 'less than' type and of the 'more than' type) for the given frequency distribution on the same axis.
- Step 2. The two ogives intersect each other at a point, say P .
- Step 3. From P , draw PQ perpendicular to the x -axis meeting the x -axis at Q .

Clearly, the abscissa or x -coordinate of point Q gives the median of the data.

SOLVED EXAMPLES

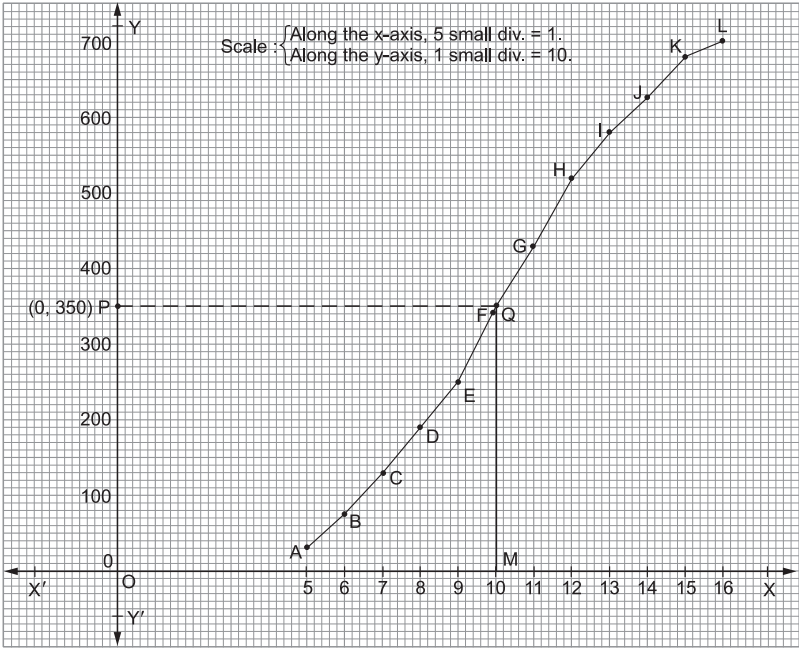
EXAMPLE 1 Following is the age distribution of a group of students. Draw the cumulative frequency curve of 'less than' type and hence obtain the median value.

<i>Age (in years)</i>	<i>Frequency</i>
4-5	36
5-6	42
6-7	52
7-8	60
8-9	68
9-10	84
10-11	96
11-12	82
12-13	66
13-14	48
14-15	50
15-16	16

SOLUTION From the given table, we may prepare the ('less than' type) cumulative frequency table as shown below:

<i>Age (in years)</i>	<i>cf</i>
Less than 5	36
Less than 6	78
Less than 7	130
Less than 8	190
Less than 9	258
Less than 10	342
Less than 11	438
Less than 12	520
Less than 13	586
Less than 14	634
Less than 15	684
Less than 16	700

On a graph paper, we take the scale.



Scale: { Along the x-axis, 5 small div. = 1.
Along the y-axis, 1 small div. = 10.

And, plot the points A(5, 36), B(6, 78), C(7, 130), D(8, 190), E(9, 258), F(10, 342), G(11, 438), H(12, 520), I(13, 586), J(14, 634), K(15, 684) and L(16, 700).

We join freehand these points successively a to get the cumulative frequency curve, or an ogive.

Here, $N = 700 \Rightarrow \frac{N}{2} = 350$.

Take a point P(0, 350) on the y-axis and draw PQ $\parallel$ x-axis, meeting the curve at Q. Draw QM $\perp$ x-axis, intersecting the x-axis at M whose coordinates are (10, 0).

Hence, median = 10 years.

EXAMPLE 2 For the following frequency distribution, draw a cumulative frequency curve of 'more than' type and hence obtain the median value.

Class interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	5	15	20	23	17	11	9

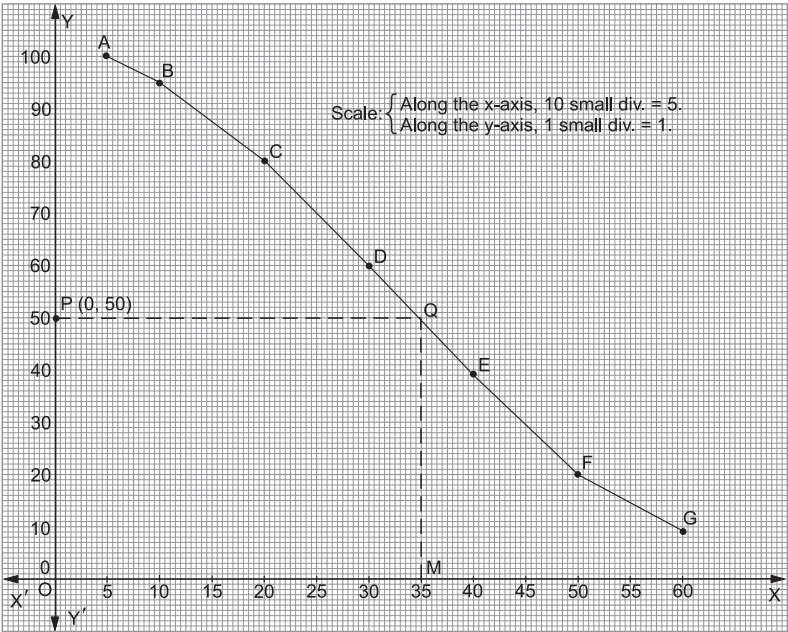
SOLUTION From the given table, we may prepare the ‘more than’ series as shown below:

More than 60	9
More than 50	20
More than 40	37
More than 30	60
More than 20	80
More than 10	95
More than 5	100

Scale: $\begin{cases} \text{Along the } x\text{-axis, 10 small div.} = 5. \\ \text{Along the } y\text{-axis, 1 small div.} = 1. \end{cases}$

Now plot the points $A(5, 100)$, $B(10, 95)$, $C(20, 80)$, $D(30, 60)$, $E(40, 37)$, $F(50, 20)$ and $G(60, 9)$ on a graph paper. Join AB , BC , CD , DE , EF and FG freehand to get the required curve, as shown below.

Here, $N = 100 \Rightarrow \frac{N}{2} = 50$.



From $P(0, 50)$ draw $PQ \parallel x\text{-axis}$, meeting the curve at Q . Draw $QM \perp OX$, meeting $x\text{-axis}$ at M whose coordinates are $(35, 0)$. Hence, median = 35.

EXAMPLE 3 The following table gives production yield per hectare of wheat of 100 farms of a village:

Production yield (kg/ha)	40–45	45–50	50–55	55–60	60–65	65–70
Number of farms	4	6	16	20	30	24

Change the distribution to a ‘more than’ type distribution and draw its ogive. [CBSE 2009C, '13]

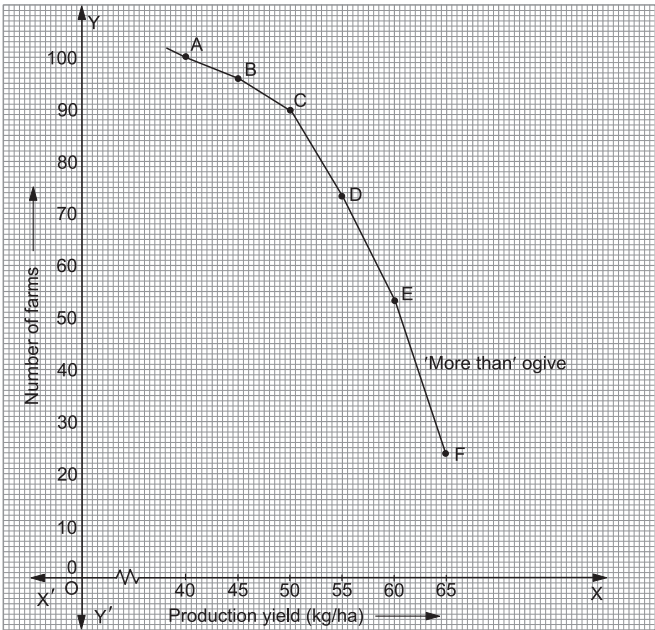
SOLUTION We may prepare the ‘more than’ series as shown below:

More than 65	24
More than 60	54
More than 55	74
More than 50	90
More than 45	96
More than 40	100

Scale: $\left\{ \begin{array}{l} \text{Along the } x\text{-axis, 1 small div.} = 1. \\ \text{Along the } y\text{-axis, 1 small div.} = 1. \end{array} \right.$

On a graph paper, we plot the points A(40, 100), B(45, 96), C(50, 90), D(55, 74), E(60, 54) and F(65, 24).

Join AB, BC, CD, DE and EF freehand to get a ‘More Than Ogive’.



EXAMPLE 4 During the medical check-up of 35 students of a class their weights were recorded as follows:

Weight (in kg)	38–40	40–42	42–44	44–46	46–48	48–50	50–52
No. of students	3	2	4	5	14	4	3

Draw a less than type and a more than type ogive from the given data. Hence, obtain the median weight from the graph. [CBSE 2009]

SOLUTION

(i) Less Than Series:

We may prepare the less than series as under.

Weight (in kg)	Number of students
Less than 40	3
Less than 42	5
Less than 44	9
Less than 46	14
Less than 48	28
Less than 50	32
Less than 52	35

Scale: $\left\{ \begin{array}{l} \text{Along the x-axis, 5 small div.} = 1 \text{ kg.} \\ \text{Along the y-axis, 10 small div.} = 5 \text{ students.} \end{array} \right.$

Now, plot the points $A(40, 3)$, $B(42, 5)$, $C(44, 9)$, $D(46, 14)$, $E(48, 28)$, $F(50, 32)$ and $G(52, 35)$ on a graph paper.

Join AB , BC , CD , DE , EF and FG freehand to get the curve representing 'Less Than Series'.

(ii) More Than Series:

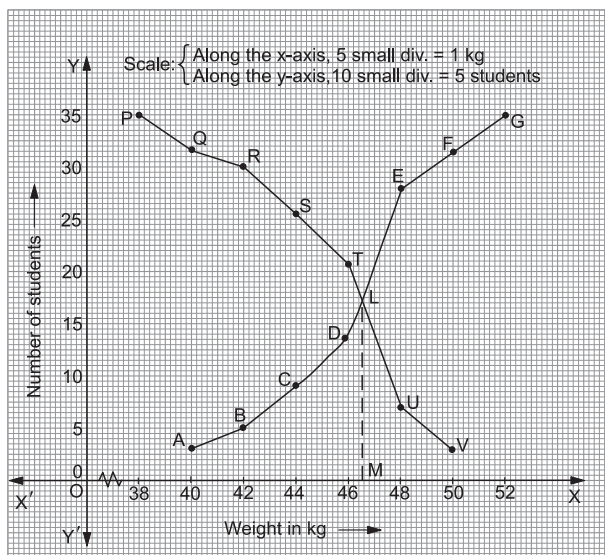
We may prepare the more than series as under.

Weight (in kg)	Number of students
More than 38	35
More than 40	32
More than 42	30
More than 44	26

More than 46	21
More than 48	7
More than 50	3

On the same graph paper as above, we plot the points $P(38, 35)$, $Q(40, 32)$, $R(42, 30)$, $S(44, 26)$, $T(46, 21)$, $U(48, 7)$ and $V(50, 3)$.

Join PQ , QR , RS , ST , TU and UV freehand to get the curve representing 'More Than Series'.



The two curves intersect at the point L . Draw $LM \perp OX$.

$\therefore$ median weight = $OM = 46.5$ kg.

EXAMPLE 5 The table given below shows the frequency distribution of the scores obtained by 200 candidates in a BCA entrance examination.

Score	200–250	250–300	300–350	350–400	400–450	450–500	500–550	550–600
No. of candidates	30	15	45	20	25	40	10	15

Draw cumulative frequency curves by using (i) 'less than' series and (ii) 'more than' series.

Hence, find the median.

SOLUTION

(i) Less Than Series:

We may prepare the less than series as under.

Score	Number of candidates
Less than 250	30
Less than 300	45
Less than 350	90
Less than 400	110
Less than 450	135
Less than 500	175
Less than 550	185
Less than 600	200

Scale: $\left\{ \begin{array}{l} \text{Along the } x\text{-axis, 1 small div.} = 5. \\ \text{Along the } y\text{-axis, 1 small div.} = 2. \end{array} \right.$

We plot the points $A(250, 30)$, $B(300, 45)$, $C(350, 90)$, $D(400, 110)$, $E(450, 135)$, $F(500, 175)$, $G(550, 185)$ and $H(600, 200)$ on a graph paper.

Join AB , BC , CD , DE , EF , FG and GH freehand to get the curve representing '*Less Than Series*'.

(ii) More Than Series:

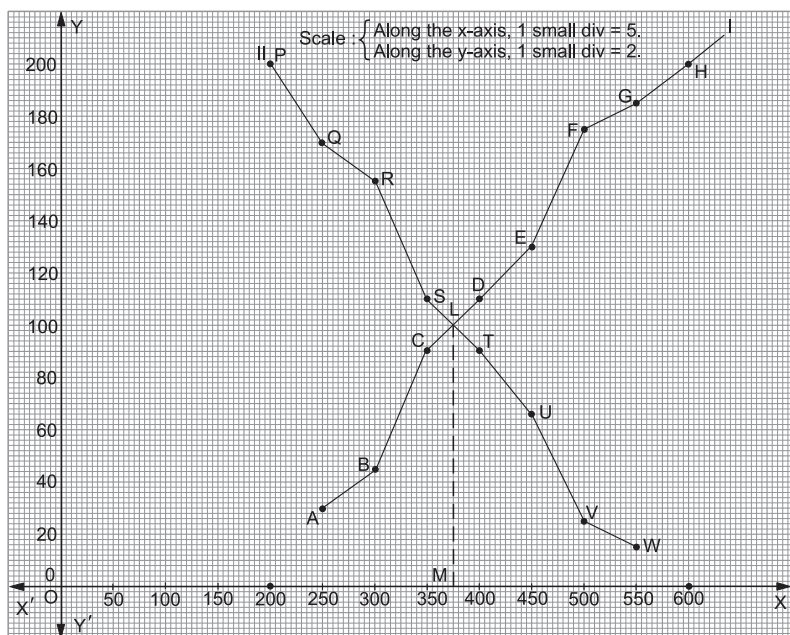
We may prepare the more than series as under.

Score	Number of candidates
Less than 200	200
More than 250	170
More than 300	155
More than 350	110
More than 400	90
More than 450	65
More than 500	25
More than 550	15

Now, on the same graph paper as above, we plot the points $P(200, 200)$, $Q(250, 170)$, $R(300, 155)$, $S(350, 110)$, $T(400, 90)$, $U(450, 65)$, $V(500, 25)$, $W(550, 15)$.

Join PQ , QR , RS , ST , TU , UV and VW freehand to get the curve representing '*More Than Series*'.

The two curves intersect at the point L . Draw $LM \perp OX$, cutting the x -axis at M .



Clearly, M represents 375.

Hence, median = 375.

EXERCISE 18E

1. Find the median of the following data by making a 'less than ogive'.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Number of students	5	3	4	3	3	4	7	9	7	8

[CBSE 2014]

2. The given distribution shows the number of wickets taken by the bowlers in one-day international cricket matches:

Number of wickets	Less than 15	Less than 30	Less than 45	Less than 60	Less than 75	Less than 90	Less than 105	Less than 120
Number of bowlers	2	5	9	17	39	54	70	80

Draw a 'less than type' ogive from the above data. Find the median.

[CBSE 2014]

3. Draw a 'more than' ogive for the data given below which gives the marks of 100 students.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Number of students	4	6	10	10	25	22	18	5

[CBSE 2013]

4. The heights of 50 girls of Class X of a school are recorded as follows:

Height (in cm)	135-140	140-145	145-150	150-155	155-160	160-165
Number of girls	5	8	9	12	14	2

Draw a 'more than type' ogive for the above data. [CBSE 2014]

5. The monthly consumption of electricity (in units) of some families of a locality is given in the following frequency distribution:

Monthly consumption (in units)	140-160	160-180	180-200	200-220	220-240	240-260	260-280
Number of families	3	8	15	40	50	30	10

Prepare a 'more than type' ogive for the given frequency distribution.

[CBSE 2014]

6. The following table gives the production yield per hectare of wheat of 100 farms of a village.

Production yield (kg/ha)	50-55	55-60	60-65	65-70	70-75	75-80
Number of farms	2	8	12	24	38	16

Change the distribution to a 'more than type' distribution and draw its ogive. Using ogive, find the median of the given data. [CBSE 2013]

7. The table given below shows the weekly expenditures on food of some households in a locality.

Weekly expenditure (in ₹)	Number of households
100-200	5
200-300	6
300-400	11
400-500	13
500-600	5

600–700	4
700–800	3
800–900	2

Draw a 'less than type ogive' and a 'more than type ogive' for this distribution.

8. From the following frequency distribution, prepare the 'more than' ogive.

Score	Number of candidates
400–450	20
450–500	35
500–550	40
550–600	32
600–650	24
650–700	27
700–750	18
750–800	34
Total	230

Also, find the median.

9. The marks obtained by 100 students of a class in an examination are given below:

Marks	Number of students
0–5	2
5–10	5
10–15	6
15–20	8
20–25	10
25–30	25
30–35	20
35–40	18
40–45	4
45–50	2

Draw cumulative frequency curves by using (i) 'less than' series and (ii) 'more than' series.

Hence, find the median.

10. From the following data, draw the two types of cumulative frequency curves and determine the median.

Height (in cm)	Frequency
140–144	3
144–148	9
148–152	24
152–156	31
156–160	42
160–164	64
164–168	75
168–172	82
172–176	86
176–180	34

SUMMARY OF THE RESULTS AND FORMULAE

1. MEAN OF THE GROUPED DATA

(i) DIRECT METHOD

$Mean(\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$, where $x_i = \frac{1}{2}$ (lower limit + upper limit) of i th class interval and f_i is its frequency.

(ii) $\sum f_i (x_i - \bar{x}) = 0$.

(iii) ASSUMED-MEAN METHOD

$Mean(\bar{x}) = A + \frac{\sum f_i d_i}{\sum f_i}$, where A is the assumed mean and $d_i = (x_i - A)$ for each i .

(iv) STEP-DEVIATION METHOD

$Mean(\bar{x}) = A + \left\{ h \times \frac{\Sigma(f_i \times u_i)}{\Sigma f_i} \right\}$, where A = assumed mean, h = class size and $u_i = \frac{(x_i - A)}{h}$.

2. (i) MEDIAN CLASS

Let $N = \Sigma f_i$. Then, the class whose cumulative frequency is just greater than $\left(\frac{N}{2}\right)$ is the median class.

(ii) MEDIAN FOR GROUPED DATA

$Median(M_e) = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf\right)}{f} \right\}$, where l = lower limit of the median class;

h = width of the median class; f = frequency of the median class; $N = \Sigma f_i$; cf = cumulative frequency of the class preceding the median class.

3. $Mode = 3(\text{median}) - 2(\text{mean})$.

4. (i) MODAL CLASS

The class having maximum frequency is called the modal class.

(ii) $Mode(M_o) = x_k + h \cdot \left\{ \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$, where x_k = lower limit of the modal class interval; f_k = frequency of the modal class; f_{k-1} = frequency of the class preceding the modal class; f_{k+1} = frequency of the class succeeding the modal class; h = width of the class interval.

5. OGIVES (OR CUMULATIVE FREQUENCY CURVES)**(i) FOR 'LESS THAN' SERIES**

On a graph paper, we mark the *upper class limits* along x -axis and the corresponding *cumulative frequencies* along y -axis. Join these points successively to obtain a curve, called *ogive for less than series*.

(ii) FOR 'GREATER THAN' SERIES

On the same graph paper as above, we mark the lower class limits along x -axis and the corresponding cumulative frequencies along y -axis. Join these points successively to obtain a curve, called *ogive for greater than series*.

(iii) The abscissa of the point of intersection of the two ogives gives us the *median*.

EXERCISE 18F

Very-Short-Answer Questions

1. Write the median class of the following distribution:

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	4	8	10	12	8	4

[CBSE 2009]

2. What is the lower limit of the modal class of the following frequency distribution?

Age (in years)	0-10	10-20	20-30	30-40	40-50	50-60
Number of patients	16	13	6	11	27	18

[CBSE 2009]

3. The monthly pocket money of 50 students of a class are given in the following distribution:

Monthly pocket money (in ₹)	0-50	50-100	100-150	150-200	200-250	250-300
Number of students	2	7	8	30	12	1

Find the modal class and also give class mark of the modal class.

[CBSE 2014]

4. A data has 25 observations arranged in a descending order. Which observation represents the median? [CBSE 2014]
5. For a certain distribution, mode and median were found to be 1000 and 1250 respectively. Find mean for this distribution using an empirical relation. [CBSE 2014]
6. In a class test, 50 students obtained marks as follows:

Marks obtained	0-20	20-40	40-60	60-80	80-100
Number of students	4	6	25	10	5

Find the modal class and the median class.

[CBSE 2014]

7. Find the class marks of classes 10-25 and 35-55.

[CBSE 2008]

8. While calculating the mean of a given data by the assumed-mean method, the following values were obtained:

$$A = 25, \Sigma f_i d_i = 110, \Sigma f_i = 50.$$

Find the mean.

9. The distributions X and Y with total number of observations 36 and 64, and mean 4 and 3 respectively are combined. What is the mean of the resulting distribution $X + Y$?
10. In a frequency distribution table with 12 classes, the class-width is 2.5 and the lowest class boundary is 8.1, then what is the upper class boundary of the highest class?
11. The observations 29, 32, 48, 50, x , $x + 2$, 72, 78, 84, 95 are arranged in ascending order. What is the value of x if the median of the data is 63?
12. The median of 19 observations is 30. Two more observations are made and the values of these are 8 and 32. Find the median of the 21 observations taken together.

HINT Since 8 is less than 30 and 32 is more than 30, so the value of median (middle value) remains unchanged.

13. If the median of $\frac{x}{5}, \frac{x}{4}, \frac{x}{2}, x$ and $\frac{x}{3}$, where $x > 0$, is 8, find the value of x .

HINT Arranging the observations in ascending order, we have $\frac{x}{5}, \frac{x}{4}, \frac{x}{3}, \frac{x}{2}, x$.

$$\text{Median} = \frac{x}{3} = 8.$$

14. What is the cumulative frequency of the modal class of the following distribution?

Class	3–6	6–9	9–12	12–15	15–18	18–21	21–24
Frequency	7	13	10	23	4	21	16

Short-Answer Questions

15. Find the mode of the given data:

Class interval	0–20	20–40	40–60	60–80
Frequency	15	6	18	10

[CBSE 2015]

16. The following are the ages of 300 patients getting medical treatment in a hospital on a particular day:

Age (in years)	10–20	20–30	30–40	40–50	50–60	60–70
Number of patients	60	42	55	70	53	20

Form a 'less than type' cumulative frequency distribution. [CBSE 2013]

17. In the following data, find the values of p and q . Also, find the median class and modal class.

Class	Frequency (f)	Cumulative frequency (cf)
100–200	11	11
200–300	12	p
300–400	10	33
400–500	q	46
500–600	20	66
600–700	14	80

[CBSE 2013]

18. The following frequency distribution gives the monthly consumption of electricity of 64 consumers of a locality.

Monthly consumption (in units)	65–85	85–105	105–125	125–145	145–165	165–185
Number of consumers	4	5	13	20	14	8

Form a 'more than type' cumulative frequency distribution.

19. The following table gives the life-time (in days) of 100 electric bulbs of a certain brand.

Life-time (in days)	Less than 50	Less than 100	Less than 150	Less than 200	Less than 250	Less than 300
Number of bulbs	7	21	52	79	91	100

From this table, construct the frequency distribution table.

20. The following table gives the frequency distribution of the percentage of marks obtained by 2300 students in a competitive examination.

Marks obtained (in per cent)	11–20	21–30	31–40	41–50	51–60	61–70	71–80
Number of students	141	221	439	529	495	322	153

- Convert the given frequency distribution into the continuous form.
- Find the median class and write its class mark.
- Find the modal class and write its cumulative frequency.

21. If the mean of the following distribution is 27, find the value of p .

Class	0–10	10–20	20–30	30–40	40–50
Frequency	8	p	12	13	10

22. Calculate the missing frequency from the following distribution, it being given that the median of the distribution is 24.

Age (in years)	0–10	10–20	20–30	30–40	40–50
Number of persons	5	25	?	18	7

ANSWERS (EXERCISE 18F)

1. 30–40 2. 40 3. 150–200, 175 4. 13th 5. 1375
 6. Modal class $\rightarrow$ 40–60, median class $\rightarrow$ 40–60 7. 17.5, 45 8. 27.2
 9. 3.36 10. 38.1 11. 62 12. 30 13. 24 14. 53 15. 52
 17. $p = 23, q = 13$; median class $\rightarrow$ 400–500, modal class $\rightarrow$ 500–600
 20. (b) 40.5–50.5, 45.5 (c) 40.5–50.5, 1330 21. $p = 7$ 22. 25

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- Which of the following is not a measure of central tendency? [CBSE 2013]
 (a) Mean (b) Mode (c) Median (d) Range
- Which of the following cannot be determined graphically?
 (a) Mean (b) Median (c) Mode (d) None of these
- Which of the following measures of central tendency is influenced by extreme values?
 (a) Mean (b) Median (c) Mode (d) None of these
- The mode of a frequency distribution is obtained graphically from
 (a) a frequency curve (b) a frequency polygon
 (c) a histogram (d) an ogive
- The median of a frequency distribution is found graphically with the help of
 (a) a histogram (b) a frequency curve
 (c) a frequency polygon (d) ogives

6. The cumulative frequency table is useful in determining the
- mean
 - median
 - mode
 - all of these
7. The abscissa of the point of intersection of the Less Than Type and of the More Than Type cumulative frequency curves of a grouped data gives its
- mean
 - median
 - mode
 - none of these
8. If x_i 's are the midpoints of the class intervals of a grouped data, f_i 's are the corresponding frequencies and $\bar{x}$ is the mean then $\sum f_i(x_i - \bar{x}) = ?$
- 1
 - 0
 - 1
 - 2
9. For finding the mean by using the formula, $\bar{x} = A + h \left(\frac{\sum f_i u_i}{\sum f_i} \right)$, we have $u_i = ?$
- $\frac{(A - x_i)}{h}$
 - $\frac{(x_i - A)}{h}$
 - $\frac{(A + x_i)}{h}$
 - $h(x_i - A)$
10. In the formula, $\bar{x} = \left\{ A + \frac{\sum f_i d_i}{\sum f_i} \right\}$ for finding the mean of the grouped data, the d_i 's are the deviations from A of
- lower limits of the classes
 - upper limits of the classes
 - midpoints of the classes
 - none of these
11. While computing the mean of the grouped data, we assume that the frequencies are
- evenly distributed over the classes
 - centred at the class marks of the classes
 - centred at the lower limits of the classes
 - centred at the upper limits of the classes
12. The relation between mean, mode and median is
- mode = $(3 \times \text{mean}) - (2 \times \text{median})$
 - mode = $(3 \times \text{median}) - (2 \times \text{mean})$
 - median = $(3 \times \text{mean}) - (2 \times \text{mode})$
 - mean = $(3 \times \text{median}) - (2 \times \text{mode})$
13. If the 'less than type' ogive and 'more than type' ogive intersect each other at (20.5, 15.5) then the median of the given data is [CBSE 2013]
- 5.5
 - 15.5
 - 20.5
 - 36.0

19. Look at the frequency distribution table given below:

Class interval	35–45	45–55	55–65	65–75
Frequency	8	12	20	10

The median of the above distribution is

- (a) 56.5 (b) 57.5 (c) 58.5 (d) 59
20. Consider the following table:

Class interval	10–14	14–18	18–22	22–26	26–30
Frequency	5	11	16	25	19

The mode of the above data is

- (a) 23.5 (b) 24 (c) 24.4 (d) 25
21. The mean and mode of a frequency distribution are 28 and 16 respectively. The median is
- (a) 22 (b) 23.5 (c) 24 (d) 24.5
22. The median and mode of a frequency distribution are 26 and 29 respectively. Then, the mean is
- (a) 27.5 (b) 24.5 (c) 28.4 (d) 25.8
23. For a symmetrical frequency distribution, we have
- (a) mean < mode < median (b) mean > mode > median
- (c) mean = mode = median (d) mode = $\frac{1}{2}$ (mean + median)
24. Look at the cumulative frequency distribution table given below:

Monthly income	Number of families
More than ₹ 10000	100
More than ₹ 14000	85
More than ₹ 18000	69
More than ₹ 20000	50
More than ₹ 25000	37
More than ₹ 30000	15

Number of families having income range 20000 to 25000 is

- (a) 19 (b) 16 (c) 13 (d) 22
25. The median of first 8 prime numbers is
- (a) 7 (b) 9 (c) 11 (d) 13
26. The mean of 20 numbers is zero. Of them, at the most, how many may be greater than zero?
- (a) 0 (b) 1 (c) 10 (d) 19

27. If the median of the data 4, 7, $x - 1$, $x - 3$, 16, 25, written in ascending order, is 13 then x is equal to
 (a) 13 (b) 14 (c) 15 (d) 16
28. The mean of 2, 7, 6 and x is 15 and the mean of 18, 1, 6, x and y is 10. What is the value of y ?
 (a) 5 (b) 10 (c) 20 (d) 30

Matching of columns

29. Match the following columns:

Column I	Column II
(a) The most frequent value in a data is known as	(p) standard deviation
(b) Which of the following cannot be determined graphically out of mean, mode and median?	(q) median
(c) An ogive is used to determine	(r) mean
(d) Out of mean, mode, median and standard deviation, which is not a measure of central tendency?	(s) mode

The correct answer is:

- (a) —……, (b) —……, (c) —……, (d) —…….

Assertion-and-Reason Type

Each question consists of two statements, namely, Assertion (A) and Reason (R). For selecting the correct answer, use the following code:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
 (c) Assertion (A) is true and Reason (R) is false.
 (d) Assertion (A) is false and Reason (R) is true.

30.

Assertion (A)	Reason (R)
If the median and mode of a frequency distribution are 150 and 154 respectively, then its mean is 148.	Mean, median and mode of a frequency distribution are related as: mode = 3 median – 2 mean.

The correct answer is: (a)/(b)/(c)/(d).

31.	Assertion (A)	Reason (R)														
	Consider the following frequency distribution: <table><tr><td>Class interval</td><td>3–6</td><td>6–9</td><td>9–12</td><td>12–15</td><td>15–18</td><td>18–21</td></tr><tr><td>Frequency</td><td>2</td><td>5</td><td>21</td><td>23</td><td>10</td><td>12</td></tr></table> The mode of the above data is 12.4.	Class interval	3–6	6–9	9–12	12–15	15–18	18–21	Frequency	2	5	21	23	10	12	The value of the variable which occurs most often is the mode.
Class interval	3–6	6–9	9–12	12–15	15–18	18–21										
Frequency	2	5	21	23	10	12										

The correct answer is: (a)/(b)/(c)/(d).

ANSWERS (MCQ)

1. (d) 2. (a) 3. (a) 4. (c) 5. (d) 6. (b) 7. (b) 8. (b) 9. (b)
 10. (c) 11. (b) 12. (b) 13. (c) 14. (b) 15. (c) 16. (b) 17. (a) 18. (c)
 19. (b) 20. (c) 21. (c) 22. (b) 23. (c) 24. (c) 25. (b) 26. (d) 27. (c)
 28. (c) 29. (a)–(s), (b)–(r), (c)–(q), (d)–(p) 30. (a) 31. (b)

HINTS TO SOME SELECTED QUESTIONS

- Range is not a measure of central tendency.
- Mean cannot be determined graphically.
- Only mean is affected by extreme values, while both median and mode remain unaffected.
- Mode of a frequency distribution can be obtained graphically from a histogram.
- The cumulative frequency table is useful in determining the median.
- Clearly, we have $\sum f_i(x_i - \bar{x}) = 0$.
- We have $u_i = \frac{(x_i - A)}{h}$.
- d_i 's are deviations from A of midpoints of the classes.
- In computing the mean of the grouped data, we assume that the frequencies are centred at the class marks of the classes.
- Clearly, the abscissa of the point of intersection of both the ogives gives the median.
- The class having maximum frequency is the modal class.
 So, the modal class is 150–155. Its lower limit is 150.

Also, $N = 60 \Rightarrow \frac{N}{2} = 30$ and the cumulative frequency just more than 30 is 37. Its class is 160–165, whose upper limit is 165.

Required sum = $(150 + 165) = 315$.

15. The class 30–40 has maximum frequency. So, the modal class is 30–40.

$$18. \text{Mode} = (3 \text{ median}) - (2 \text{ mean}) = (3 \times 9 - 2 \times 8.9) = (27 - 17.8) = 9.2.$$

19. We have:

Class interval	35–45	45–55	55–65	65–75
Frequency	8	12	20	10
Cumulative frequency	8	20	40	50

Here, $N = 50 \Rightarrow \frac{N}{2} = 25$, which lies in class interval 55–65.

$$\text{Median} = l + \left\{ h \times \frac{\left(\frac{N}{2} - cf \right)}{f} \right\} = 55 + \frac{(65 - 55)}{20} \times (25 - 20) = 57.5.$$

20. The maximum frequency is 25 and the modal class is 22–26.

$$\therefore x_k = 22, f_k = 25, f_{k-1} = 16, f_{k+1} = 19 \text{ and } h = 4.$$

$$\begin{aligned} \therefore \text{mode} &= x_k + h \cdot \left\{ \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\} \\ &= \left\{ 22 + 4 \times \frac{(25 - 16)}{(50 - 16 - 19)} \right\} = \left(22 + 4 \times \frac{9}{15} \right) = \left(22 + \frac{12}{5} \right) = (22 + 2.4) = 24.4. \end{aligned}$$

$$21. 3 \times \text{median} = (\text{mode} + 2 \text{ mean}) = (16 + 2 \times 28) = 72$$

$$\Rightarrow \text{median} = \frac{72}{3} = 24.$$

$$22. \text{Mode} = (3 \text{ median}) - (2 \text{ mean})$$

$$\Rightarrow 2 \text{ mean} = (3 \text{ median}) - (\text{mode}) = (3 \times 26) - 29 = 49$$

$$\Rightarrow \text{mean} = \frac{49}{2} = 24.5.$$

23. We must have, mean = mode = median.

25. First 8 prime numbers are 2, 3, 5, 7, 11, 13, 17, 19.

Required median = mean of fourth and fifth observations

$$= \frac{7 + 11}{2} = 9.$$

26. Mean of 20 numbers = 0.

$$\therefore \text{sum of 20 numbers} = 0 \times 20 = 0.$$

It is possible that 19 of these numbers may be positive and if their sum is a , the 20th number is $(-a)$.

27. Clearly, median of 6 observations

= mean of 3rd and 4th observations

$$= \frac{(x-1) + (x-3)}{2} = \frac{2x-4}{2} = x-2.$$

$$\therefore x-2=13 \text{ or } x=13+2=15.$$

$$28. \text{ We have } \left(\frac{2+7+6+x}{4}\right)=5 \text{ or } 15+x=20 \text{ or } x=5.$$

$$\text{Also, } \left(\frac{18+1+6+x+y}{5}\right)=10 \text{ or } 25+5+y=50 \text{ or } y=20.$$

30. Reason (R) is clearly true.

Using the relation given in Reason (R), we have

$$2 \text{ mean} = (3 \text{ median}) - (\text{mode}) = (3 \times 150) - (154) = 450 - 154 = 296.$$

$$\therefore \text{mean} = 148, \text{ which is true.}$$

Thus, Assertion (A) and Reason (R) are both true and Reason (R) is the correct explanation of Assertion (A).

Hence, the correct answer is (a).

31. Reason (R) is clearly true.

The maximum frequency is 23 and the modal class is 12–15.

$$\therefore x_k = 12, f_k = 23, f_{k-1} = 21, f_{k+1} = 10 \text{ and } h = 3.$$

$$\therefore \text{mode} = \left\{12 + 3 \times \frac{(23-21)}{(2 \times 23 - 21 - 10)}\right\} = \left(12 + 3 \times \frac{2}{15}\right) = 12.4.$$

$$\therefore \text{Assertion (A) is true.}$$

But, Reason (R) is not a correct explanation of Assertion (A).

Hence, the correct answer is (b).

TEST YOURSELF

MCQ

- Which one of the following measures is determined only after the construction of cumulative frequency distribution?
(a) Mean (b) Median (c) Mode (d) None of these
- If the mean of a data is 27 and its median is 33 then the mode is
(a) 30 (b) 43 (c) 45 (d) 47
- Consider the following distribution:

Class	0–5	5–10	10–15	15–20	20–25
Frequency	10	15	12	20	9

The sum of the lower limits of the median class and the modal class is

- (a) 15 (b) 25 (c) 30 (d) 35

4. Consider the following frequency distribution:

Class	0–5	6–11	12–17	18–23	24–29
Frequency	13	10	15	8	11

The upper limit of the median class is

- (a) 16.5 (b) 18.5 (c) 18 (d) 17.5

Very-Short-Answer Questions

5. If the mean and mode of a frequency distribution be 53.4 and 55.2 respectively, find the median.
6. In the table given below, the times taken by 120 athletes to run a 100-m-hurdle race are given.

Class	13.8–14	14–14.2	14.2–14.4	14.4–14.6	14.6–14.8	14.8–15
Frequency	2	4	15	54	25	20

Find the number of athletes who completed the race in less than 14.6 seconds.

7. Consider the following frequency distribution:

Class	0–5	6–11	12–17	18–23	24–29
Frequency	13	10	15	8	11

Find the upper limit of the median class.

8. The annual profits earned by 30 shops of a shopping complex in a locality are recorded in the table shown below:

Profit (in lakhs ₹)	Number of shops
More than or equal to 5	30
More than or equal to 10	28
More than or equal to 15	16
More than or equal to 20	14
More than or equal to 25	10
More than or equal to 30	7
More than or equal to 35	3

If we draw the frequency distribution table for the above data, find the frequency corresponding to the class 20–25.

Short-Answers Questions

9. Find the mean of the following frequency distribution:

Class	1-3	3-5	5-7	7-9
Frequency	9	22	27	18

10. The maximum bowling speeds (in km/hr) of 33 players at a cricket coaching centre are given below:

Speed in km/hr	85-100	100-115	115-130	130-145
No. of players	10	4	7	9

Calculate the median bowling speed.

11. The arithmetic mean of the following frequency distribution is 50.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	16	p	30	32	14

Find the value of p .

12. Find the median of the following frequency distribution:

Marks	0-10	10-20	20-30	30-40	40-50
Number of students	6	16	30	9	4

13. Following is the distribution of marks of 70 students in a periodical test:

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50
Number of students	3	11	28	48	70

Draw a cumulative frequency curve for the above data.

14. Find the median of the following data.

Class interval	0-10	10-20	20-30	30-40	40-50	Total
Frequency	8	16	36	34	6	100

[CBSE 2014]

15. For the following distribution draw a 'less than type' ogive and from the curve find the median.

Marks obtained	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60	Less than 70	Less than 80	Less than 90	Less than 100
Number of students	2	7	17	40	60	82	85	90	100

[CBSE 2014]

16. The median value for the following frequency distribution is 35 and the sum of all the frequencies is 170. Using the formula for median, find the missing frequencies.

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	10	20	?	40	?	25	15

17. Find the missing frequencies f_1 and f_2 in the table given below, it being given that the mean of the given frequency distribution is 50.

Class	0–20	20–40	40–60	60–80	80–100	Total
Frequency	17	f_1	32	f_2	19	120

18. Find the mean of the following frequency distribution using step-deviation method:

Class	84–90	90–96	96–102	102–108	108–114	114–120
Frequency	15	22	20	18	20	25

Long-Answer Questions

19. Find the mean, median and mode of the following data:

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	6	8	10	15	5	4	2

20. Draw 'less than ogive' and 'more than ogive' on a single graph paper and hence find the median of the following data:

Class interval	5–10	10–15	15–20	20–25	25–30	30–35	35–40
Frequency	2	12	2	4	3	4	3

[CBSE 2014]

21. The production yield per hectare of wheat of some farms of a village are given in the following table:

Production yield (in kg/ha)	40–45	45–50	50–55	55–60	60–65	65–70	70–75	75–80	80–85
Number of farms	1	9	15	18	40	26	16	14	10

Draw a less than type ogive and a more than type ogive for this data.

[CBSE 2014]

22. The following table gives the marks obtained by 50 students in a class test:

Marks	11–15	16–20	21–25	26–30	31–35	36–40	41–45	46–50
Number of students	2	3	6	7	14	12	4	2

Calculate the mean, median and mode for the above data.

ANSWERS (TEST YOURSELF)

1. (b) 2. (c) 3. (b) 4. (d) 5. 54
 6. 75 7. 17.5 8. 4 9. 5.42
 10. 117.1 km/hr 11. 28 12. 24 14. 27.22 16. 35, 25
 17. $f_1 = 28, f_2 = 24$ 18. 102.75
 19. Mean = 30, median = 30.67, mode = 32.01
 22. Mean = 32, median = 33, mode = 35



INTRODUCTION

In class IX, we have studied the concept of empirical probability. Since empirical probability is based on experiments, we also call it *experimental probability*.

Suppose we toss a coin 500 times and get a head, say, 240 times and tail 260 times. Then, we would say that in a single throw of a coin, the probability of getting a head is $\frac{240}{500}$, i.e., $\frac{12}{25}$.

Again, suppose we toss a coin 1000 times and get a head, say, 530 times and tail 470 times. Then, we would say that in a single throw of a coin, the probability of getting a head is $\frac{530}{1000}$, i.e., $\frac{53}{100}$.

Thus, in various experiments, we would get different probabilities for the same event.

However, theoretical probability overcomes the above problem. In this chapter, by probability, we shall mean theoretical probability.

PROBABILITY

Probability is a concept which numerically measures the degree of certainty of the occurrence of events.

Before defining probability, we shall define certain concepts used therein.

EXPERIMENT *An operation which can produce some well-defined outcomes is called an experiment.*

RANDOM EXPERIMENT *An experiment in which all possible outcomes are known, and the exact outcome cannot be predicted in advance, is called a random experiment.*

By a *trial*, we mean 'performing a random experiment'.

Examples (i) Tossing a fair coin

(ii) Rolling an unbiased die

(iii) Drawing a card from a pack of well-shuffled cards

(iv) Picking up a ball from a bag of balls of different colours

These are all examples of a random experiment.

SOME DETAILS ABOUT THESE EXPERIMENTS

- I. *Tossing a coin* When we throw a coin, either a *head* (H) or a *tail* (T) appears on the upper face.
- II. *Throwing a die* A die is a solid cube, having 6 faces, marked 1, 2, 3, 4, 5 and 6, or having 1, 2, 3, 4, 5 and 6 dots.

In throwing a die, the outcome is the number or number of dots appearing on the uppermost face.

The plural of die is *dice*.

- III. *Drawing a card from a well-shuffled deck of 52 cards.*

A deck of playing cards has in all 52 cards.

- (i) It has 13 cards each of four suits, namely

spades, clubs, hearts and diamonds.

- (a) Cards of *spades* and *clubs* are *black cards*.

- (b) Cards of *hearts* and *diamonds* are *red cards*.



Spades



Clubs



Hearts



Diamonds

- (ii) Kings, queens and jacks (or knaves) are known as *face cards*.

Thus, there are in all 12 face cards.



King



Queen



Jack

LOOKING AT ALL POSSIBLE OUTCOMES IN VARIOUS EXPERIMENTS

- I. When we toss a coin, we get either a *head* (H) or a *tail* (T).

Thus, all possible outcomes are H, T .

- II. Suppose two coins are tossed simultaneously.

Then, all possible outcomes are HH, HT, TH, TT .

REMARKS HH means head on first coin and head on second coin.

HT means head on first coin and tail on second coin, etc.

- III. On rolling a die, the number on the upper face is the outcome.

Thus, all possible outcomes are 1, 2, 3, 4, 5, 6.

IV. In drawing a card from a well-shuffled deck of 52 cards, total number of possible outcomes is 52.

EVENT *The collection of all or some of the possible outcomes is called an event.*

- Examples*
- (i) In throwing a coin, H is the event of getting a head.
 - (ii) Suppose we throw two coins simultaneously and let E be the event of getting at least one head. Then, E contains HT, TH, HH .

EQUALLY LIKELY EVENTS *A given number of events are said to be equally likely if none of them is expected to occur in preference to the others.*

For example, if we roll an unbiased die, each number is equally likely to occur. If, however, a die is so formed that a particular face occurs most often then the die is biased. In this case, the outcomes are not equally likely to happen.

PROBABILITY OF OCCURRENCE OF AN EVENT

Probability of occurrence of an event E , denoted by $P(E)$ is defined as:

$$P(E) = \frac{\text{number of outcomes favourable to } E}{\text{total number of possible outcomes}}.$$

SURE EVENT

It is evident that in a single toss of die, we will always get a number less than 7.

So, getting a number less than 7 is a *sure event*.

$$P(\text{getting a number less than 7}) = \frac{6}{6} = 1.$$

Thus, the probability of a sure event is 1.

IMPOSSIBLE EVENT

In a single toss of a die, what is the probability of getting a number 8?

We know that in tossing a coin, 8 will never come up.

So, getting 8 is an impossible event.

$$P(\text{getting 8 in a single throw of a die}) = \frac{0}{6} = 0.$$

Thus, the probability of an impossible event is zero.

COMPLEMENTARY EVENT

Let E be an event and (not E) be an event which occurs only when E does not occur. We denote (not E) by E' , or $\bar{E}$, called complement of event E .

The event (not E) is called the complementary event of E .

Clearly, $P(E) + P(\text{not } E) = 1$.

$\therefore P(E) = 1 - P(\text{not } E)$.

SUMMARY

- (i) For some event E , we have $0 \leq P(E) \leq 1$.
- (ii) $P(E) = 0$, when E is an impossible event.
- (iii) $P(E) = 1$, when E is a sure event.
- (iv) $P(\text{not } E) = 1 - P(E)$. Thus, $P(\bar{E}) = 1 - P(E)$.

SOLVED EXAMPLES

EXAMPLE 1 *A coin is tossed once. What is the probability of getting a head?*

SOLUTION When a coin is tossed once, all possible outcomes are H and T .

Total number of possible outcomes = 2.

The favourable outcome is H .

Number of favourable outcomes = 1.

$$\begin{aligned} \therefore P(\text{getting a head}) \\ = P(H) &= \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} = \frac{1}{2}. \end{aligned}$$

EXAMPLE 2 *A die is thrown once. What is the probability of getting a prime number?*

SOLUTION In a single throw of a die, all possible outcomes are 1, 2, 3, 4, 5, 6.

Total number of possible outcomes = 6.

Let E be the event of getting a prime number.

Then, the favourable outcomes are 2, 3, 5.

Number of favourable outcomes = 3.

$$\therefore P(\text{getting a prime number}) = P(E) = \frac{3}{6} = \frac{1}{2}.$$

EXAMPLE 3 *A die is thrown once. What is the probability that it shows (i) a '3', (ii) a '5', (iii) an odd number, (iv) a number greater than 4?*

SOLUTION When a die is thrown, all possible outcomes are 1, 2, 3, 4, 5, 6.
Total number of possible outcomes = 6.

- (i) Let E_1 be the event of getting a 3.

Then, the number of favourable outcomes = 1.

$$\therefore P(\text{getting a 3}) = P(E_1) = \frac{1}{6}.$$

- (ii) Let E_2 be the event of getting a 5.

Then, the number of favourable outcomes = 1.

$$\therefore P(\text{getting a 5}) = P(E_2) = \frac{1}{6}.$$

- (iii) Let E_3 be the event of getting an odd number.

Then, the favourable outcomes are 1, 3, 5.

Number of favourable outcomes = 3.

$$\therefore P(\text{getting an odd number}) = P(E_3) = \frac{3}{6} = \frac{1}{2}.$$

- (iv) Let E_4 be the event of getting a number greater than 4.

Then, the favourable outcomes are 5, 6.

Number of favourable outcomes = 2.

$$\therefore P(\text{getting a number greater than 4}) = P(E_4) = \frac{2}{6} = \frac{1}{3}.$$

EXAMPLE 4

A die is thrown once. Find the probability of getting (i) an even prime number, (ii) a multiple of 3.

[CBSE 2012]

SOLUTION

When a die is thrown, all possible outcomes are 1, 2, 3, 4, 5, 6.
Total number of possible outcomes = 6.

- (i) Let E_1 be the event of getting an even prime number.

Then, the favourable outcome is 2 only.

Number of favourable outcomes = 1.

$$\therefore P(\text{getting an even prime number}) = P(E_1) = \frac{1}{6}.$$

- (ii) Let E_2 be the event of getting a multiple of 3.

Then, the favourable outcomes are 3 and 6.

Number of favourable outcomes = 2.

$$\therefore P(\text{getting a multiple of 3}) = P(E_2) = \frac{2}{6} = \frac{1}{3}.$$

EXAMPLE 5

Two coins are tossed simultaneously. What is the probability of getting at least one head?

[CBSE 2014]

SOLUTION

When two coins are tossed simultaneously, all possible outcomes are HH, HT, TH, TT .

Total number of possible outcomes = 4.

Let E be the event of getting at least one head.

Then, E is the event of getting 1 head or 2 heads.

So, the favourable outcomes are HT, TH, HH .

Number of favourable outcomes = 3.

$$\therefore P(\text{getting at least one head}) = P(E) = \frac{3}{4}.$$

EXAMPLE 6 Three unbiased coins are tossed simultaneously. Find the probability of getting (i) exactly 2 heads, (ii) at least 2 heads, (iii) at most 2 heads. [CBSE 2015]

SOLUTION When 3 coins are tossed simultaneously, all possible outcomes are $HHH, HHT, HTH, THH, HTT, THT, TTH, TTT$.

Total number of possible outcomes = 8.

(i) Let E_1 be the event of getting exactly 2 heads.

Then, the favourable outcomes are HHT, HTH, THH .

Number of favourable outcomes = 3.

$$\therefore P(\text{getting exactly 2 heads}) = P(E_1) = \frac{3}{8}.$$

(ii) Let E_2 be the event of getting at least 2 heads.

Then, E_2 is the event of getting 2 or 3 heads.

So, the favourable outcomes are

HHT, HTH, THH, HHH .

Number of favourable outcomes = 4.

$$\therefore P(\text{getting at least 2 heads}) = P(E_2) = \frac{4}{8} = \frac{1}{2}.$$

(iii) Let E_3 be the event of getting at most 2 heads.

Then, E_3 is the event of getting 0 or 1 head or 2 heads.

So, the favourable outcomes are

$TTT, HTT, THT, TTH, HHT, HTH, THH$.

Number of favourable outcomes = 7.

$$\therefore P(\text{getting at most 2 heads}) = P(E_3) = \frac{7}{8}.$$

EXAMPLE 7 Cards numbered 11 to 60 are kept in a box. If a card is drawn at random from the box, find the probability that the number on the drawn card is (i) an odd number, (ii) a perfect square number, (iii) divisible by 5, (iv) a prime number less than 20. [CBSE 2014]

SOLUTION All possible outcomes are 11, 12, 13, ..., 60.

Number of all possible outcomes = $(60 - 10) = 50$.

(i) Let E_1 be the event that the number on the drawn card is an odd number.

Then, the favourable outcomes are 11, 13, 15, ..., 59.

Clearly, these numbers form an AP with $a = 11$ and $d = 2$.

Let the number of these numbers be n . Then,

$$\begin{aligned}T_n = 59 &\Rightarrow 11 + (n-1) \times 2 = 59 \Rightarrow (n-1) \times 2 = 48 \\&\Rightarrow (n-1) = 24 \Rightarrow n = 25.\end{aligned}$$

So, the number of favourable outcomes = 25.

$$\therefore P(\text{getting an odd number}) = P(E_1) = \frac{25}{50} = \frac{1}{2}.$$

- (ii) Let E_2 be the event that the number on the drawn card is a perfect square number.

Then, the favourable outcomes are 16, 25, 36, 49.

So, the number of favourable outcomes = 4.

$$\therefore P(\text{getting a perfect square number}) = P(E_2) = \frac{4}{50} = \frac{2}{25}.$$

- (iii) Let E_3 be the event that the number on the drawn card is divisible by 5.

Then, the favourable outcomes are 15, 20, 25, ..., 60.

Clearly, these numbers form an AP with $a = 15$ and $d = 5$.

Let the number of these terms be m . Then,

$$\begin{aligned}T_m = 60 &\Rightarrow 15 + (m-1) \times 5 = 60 \Rightarrow (m-1) \times 5 = 45 \\&\Rightarrow m-1 = 9 \Rightarrow m = 10.\end{aligned}$$

So, the number of favourable outcomes = 10.

$$\therefore P(\text{getting a number divisible by 5}) = P(E_3) = \frac{10}{50} = \frac{1}{5}.$$

- (iv) Let E_4 be the event that the number on the drawn card is a prime number less than 20.

Then, the favourable outcomes are 11, 13, 17, 19.

So, the number of favourable outcomes = 4.

$$\therefore P(\text{getting a prime number less than 20}) = P(E_4) = \frac{4}{50} = \frac{2}{25}.$$

EXAMPLE 8

A box contains 100 red balls, 200 yellow balls and 50 blue balls. If a ball is drawn at random from the box, then find the probability that it will be (i) a blue ball, (ii) not a yellow ball, (iii) neither yellow nor a blue ball. [CBSE 2012]

SOLUTION

Total number of all possible outcomes = total number of balls
 $= 100 + 200 + 50 = 350$.

- (i) Number of blue balls = 50.

$$\therefore P(\text{getting a blue ball}) = \frac{50}{350} = \frac{1}{7}.$$

(ii) Number of balls which are not yellow = $100 + 50 = 150$.

$$\therefore P(\text{getting a ball which is not yellow}) = \frac{150}{350} = \frac{3}{7}.$$

(iii) Number of balls which are neither yellow nor blue = 100.

$$\begin{aligned}\therefore P(\text{getting a ball which is neither yellow nor blue}) \\ = \frac{100}{350} = \frac{2}{7}.\end{aligned}$$

EXAMPLE 9 A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball from the bag is thrice that of a red ball, find the number of blue balls in the bag. [CBSE 2007]

SOLUTION Let the number of blue balls in the bag be x .

Then, total number of balls = $(5 + x)$.

Given, $P(\text{a blue ball}) = 3 \times P(\text{a red ball})$

$$\therefore \frac{x}{(5+x)} = 3 \times \frac{5}{(5+x)} \Rightarrow x = 15.$$

Hence, the number of blue balls in the bag is 15.

EXAMPLE 10 A bag contains white, black and red balls only. A ball is drawn at random from the bag. If the probability of getting a white ball is $\frac{3}{10}$ and that of a black ball is $\frac{2}{5}$ then find the probability of getting a red ball. If the bag contains 20 black balls then find the total number of balls in the bag. [CBSE 2015]

SOLUTION Let E be the event of getting a red ball. Then,

$$P(\text{getting a white ball}) + P(\text{getting a black ball}) + P(E) = 1$$

$$\Rightarrow \frac{3}{10} + \frac{2}{5} + P(E) = 1 \Rightarrow \frac{7}{10} + P(E) = 1$$

$$\Rightarrow P(E) = 1 - \frac{7}{10} = \frac{3}{10}$$

$$\Rightarrow P(\text{getting a red ball}) = \frac{3}{10}.$$

Since $P(\text{getting a white ball}) = P(\text{getting a red ball})$, so the number of white balls is equal to the number of red balls, say x .

$$\therefore P(\text{getting a red ball}) = \frac{x}{x+20+x} = \frac{x}{2x+20}.$$

$$\therefore \frac{x}{2x+20} = \frac{3}{10} \Rightarrow 10x = 6x + 60 \Rightarrow 4x = 60 \Rightarrow x = 15.$$

Hence, the total number of balls in the bag

$$= 2x + 20 = 2 \times 15 + 20 = 50.$$

EXAMPLE 11 Two different dice are rolled together. Find the probability of getting (i) the sum of numbers on two dice to be 5, (ii) even number on both dice, (iii) a doublet. [CBSE 2015]

SOLUTION When two dice are thrown simultaneously, all possible outcomes are

(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),
(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),
(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),
(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),
(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),
(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6).

Number of all possible outcomes = 36.

- (i) Let E_1 be the event of getting two numbers whose sum is 5.

Then, the favourable outcomes are (1, 4) (2, 3), (3, 2), (4, 1).

Number of favourable outcomes = 4.

$$\therefore P(\text{getting two numbers whose sum is 5}) = P(E_1) = \frac{4}{36} = \frac{1}{9}.$$

- (ii) Let E_2 be the event of getting a even numbers on both dice.

Then, the favourable outcomes are

(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6).

Number of favourable outcomes = 9.

$$\therefore P(\text{getting even number on both dice}) = P(E_2) = \frac{9}{36} = \frac{1}{4}.$$

- (iii) Let E_3 be the event of getting a doublet.

Then, the favourable outcomes are

(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6).

Number of favourable outcomes = 6.

$$\therefore P(\text{getting a doublet}) = P(E_3) = \frac{6}{36} = \frac{1}{6}.$$

EXAMPLE 12 Two dice are thrown at the same time. Find the probability that the sum of the two numbers appearing on the top of the dice is more than 9. [CBSE 2009C]

SOLUTION We know that when two dice are thrown the same time, then the number of all possible outcomes is 36.

Let E be the event that the sum of the numbers appearing on the top of the two dice is more than 9.

The favourable outcomes are

$(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)$.

Number of favourable outcomes = 6.

$$\therefore P(\text{getting a sum more than 9}) = P(E) = \frac{6}{36} = \frac{1}{6}.$$

EXAMPLE 13 A piggy bank contains hundred 50-p coins, fifty ₹ 1 coins, twenty ₹ 2 and ten ₹ 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, find the probability the coin falling out will be (i) a 50-p coin, (ii) of value more than ₹ 1, (iii) of value less than ₹ 5 (iv) a ₹ 1 or ₹ 2 coin. [CBSE 2014]

SOLUTION Total number of coins = $(100 + 50 + 20 + 10) = 180$.

So, the number of all possible outcomes is 180.

(i) Let E_1 be the event of getting a 50-p coin.

Then, the number of favourable outcomes = 100.

$$\therefore P(\text{getting a 50-p coin}) = P(E_1) = \frac{100}{180} = \frac{5}{9}.$$

(ii) Let E_2 be the event of getting a coin of value more than ₹ 1.

Then, it can be ₹ 2 or ₹ 5 coin.

Number of all such coins = $20 + 10 = 30$.

$$\begin{aligned} \therefore P(\text{getting a coin of value more than ₹ 1}) \\ = P(E_2) = \frac{30}{180} = \frac{1}{6}. \end{aligned}$$

(iii) Let E_3 be the event of getting a coin of value less than ₹ 5.

Then, it can be 50-p or ₹ 1 or ₹ 2 coin.

Number of favourable outcomes

= number of all coins of 50-p, ₹ 1 and ₹ 2

= $(100 + 50 + 20) = 170$.

$$\begin{aligned} \therefore P(\text{getting a coin of value less than ₹ 5}) \\ = P(E_3) = \frac{170}{180} = \frac{17}{18}. \end{aligned}$$

(iv) Let E_4 be the event of getting a ₹ 1 or ₹ 2 coin.

Number of all such coins = $50 + 20 = 70$.

Number of favourable outcomes = 70.

$$\therefore P(\text{getting a ₹ 1 or ₹ 2 coin}) = P(E_4) = \frac{70}{180} = \frac{7}{18}.$$

EXAMPLE 14 A game consists of tossing a one-rupee coin three times and noting its outcome each time. Hanif wins if all the tosses give the same result, i.e., three heads or three tails and loses otherwise. Calculate the probability that Hanif will lose the game. [CBSE 2009C, '11, '17]

SOLUTION In tossing a one-rupee coin three times, all possible outcomes are HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

Total number of all possible outcomes = 8.

Let E be the event of getting 3 heads or 3 tails.

Then, E consists of HHH, TTT.

Number of favourable outcomes of $E = 2$.

$P(\text{that Hanif wins the game}) = P(\text{getting 3 heads or 3 tails})$

$$= P(E) = \frac{2}{8} = \frac{1}{4}.$$

$$P(\text{that Hanif loses the game}) = 1 - P(E) = \left(1 - \frac{1}{4}\right) = \frac{3}{4}.$$

FACTS ABOUT PLAYING CARDS

1. A deck of playing cards has 52 cards.
2. There are 4 suits, namely (i) spades, (ii) clubs, (iii) hearts and (iv) diamonds. There are 13 cards of each suit.
 - I. Spades and clubs are black cards.
 - II. Hearts and diamonds are red cards.
3. There are 12 face cards, namely 4 kings, 4 queens and 4 jacks.

EXAMPLE 15 One card is drawn at random from a well-shuffled deck of 52 cards. Find the probability that the card drawn is (i) a king, (ii) a red eight, (iii) a spade, (iv) a red card, (v) the six of the clubs and (vi) a face card.

SOLUTION Total number of all possible outcomes = 52.

(i) There are 4 kings in all.

$$\therefore P(\text{drawing a king}) = \frac{4}{52} = \frac{1}{13}.$$

(ii) There are 2 red eights in all.

$$\therefore P(\text{drawing a red eight}) = \frac{2}{52} = \frac{1}{26}.$$

(iii) There are 13 cards of spades.

$$\therefore P(\text{drawing a spade}) = \frac{13}{52} = \frac{1}{4}.$$

(iv) There are 26 red cards.

$$\therefore P(\text{drawing a red card}) = \frac{26}{52} = \frac{1}{2}.$$

(v) There is one six of the clubs.

$$\therefore P(\text{drawing the six of the clubs}) = \frac{1}{52}.$$

(vi) There are 12 face cards.

$$\therefore P(\text{drawing a face card}) = \frac{12}{52} = \frac{3}{13}.$$

EXAMPLE 16 One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (i) a king of red suit, (ii) a queen of black suit, (iii) a jack of hearts, (iv) a red face card. [CBSE 2010]

SOLUTION Total number of possible outcomes = 52.

(i) Number of kings of red suit = 2.

$$\therefore P(\text{getting a king of red suit}) = \frac{2}{52} = \frac{1}{26}.$$

(ii) Number of queens of black suit = 2.

$$\therefore P(\text{getting a queen of black suit}) = \frac{2}{52} = \frac{1}{26}.$$

(iii) Number of jacks of hearts = 1.

$$\therefore P(\text{getting a jack of hearts}) = \frac{1}{52}.$$

(iv) Red face cards are 2 kings, 2 queens, 2 jacks.

Number of red face cards = 6.

$$\therefore P(\text{getting a red face card}) = \frac{6}{52} = \frac{3}{26}.$$

EXAMPLE 17 A card is drawn at random from a well-shuffled deck of 52 playing cards. Find the probability that the card drawn is (i) a card of spades or an ace, (ii) a black king, (iii) neither a jack nor a king, (iv) either a king or a queen. [CBSE 2015]

SOLUTION Total number of all possible outcomes = 52.

(i) There are 13 cards of spades including one ace and there are 3 more aces.

So, the number of favourable cases = $13 + 3 = 16$.

$$\therefore P(\text{getting a card of spades or an ace}) = \frac{16}{52} = \frac{4}{13}.$$

(ii) There are 2 black kings.

$$\therefore P(\text{getting a black king}) = \frac{2}{52} = \frac{1}{26}.$$

(iii) There are 4 jacks and 4 kings.

So, the number of cards which are neither jacks nor kings
 $= \{52 - (4 + 4)\} = 44.$

$$\therefore P(\text{getting a card which is neither a jack nor a king}) \\ = \frac{44}{52} = \frac{11}{13}.$$

(iv) There are 4 kings and 4 queens.

So, the number of cards which are either kings or queens
 $= 4 + 4 = 8.$

$$\therefore P(\text{getting a card which is either a king or a queen}) \\ = \frac{8}{52} = \frac{2}{13}.$$

EXAMPLE 18 One card is drawn at random from a well-shuffled deck of 52 playing cards. Find the probability that the card drawn is (i) either a red card or a king, (ii) neither a red card nor a queen.

SOLUTION Total number of all possible outcomes = 52.

(i) Let E_1 be the event of getting a red card or a king.

There are 26 red cards (including 2 kings) and there are 2 more kings.

So, the number of favourable outcomes = $26 + 2 = 28.$

$$\therefore P(\text{getting a red card or a king}) = P(E_1) = \frac{28}{52} = \frac{7}{13}.$$

(ii) Let E_1 be the event of getting a card which is neither a red card nor a queen.

There are 26 red cards (including 2 queens) and there are 2 more queens.

So, the number of non-favourable outcomes = $26 + 2 = 28.$

$\therefore$ the number of favourable outcomes = $52 - 28 = 24.$

$$\therefore P(\text{getting neither a red card nor a queen}) = \frac{24}{52} = \frac{6}{13}.$$

EXAMPLE 19 From a pack of 52 playing cards jacks, queens, kings and aces of red colour are removed. From the remaining, a card is drawn at random. Find the probability that the card drawn is

- (i) a black queen (ii) a red card (iii) a ten [CBSE 2006C]
 (iv) a picture card (jacks, queens and kings are picture cards).

SOLUTION Number of cards removed = $2 + 2 + 2 + 2 = 8$.

Total number of remaining cards = $52 - 8 = 44$.

Now, there are 2 jacks, 2 queens, 2 kings and 2 aces of black colour only.

(i) Number of black queens = 2.

$$\therefore P(\text{getting a black queen}) = \frac{2}{44} = \frac{1}{22}.$$

(ii) Remaining number of red cards = $26 - 8 = 18$.

$$\therefore P(\text{getting a red card}) = \frac{18}{44} = \frac{9}{22}.$$

(iii) Number of tens = 4.

$$\therefore P(\text{getting a ten}) = \frac{4}{44} = \frac{1}{11}.$$

(iv) We know that jacks, queens and kings are picture cards. Out of 12 picture cards, it is given that 6 have been removed. So, the remaining number of picture cards = $12 - 6 = 6$.

$$\therefore P(\text{getting a picture card}) = \frac{6}{44} = \frac{3}{22}.$$

EXAMPLE 20 All the black face cards are removed from a pack of 52 playing cards. The remaining cards are well shuffled and then a card is drawn at random. Find the probability of getting a (i) face card, (ii) red card, (iii) black card, (iv) king. [CBSE 2014]

SOLUTION Out of 52 playing cards; 2 black jacks, 2 black queens and 2 black kings have been removed.

Total number of remaining cards = $(52 - 6) = 46$.

(i) Now, there are 6 face cards in the remaining cards.

$$\therefore P(\text{getting a face card}) = \frac{6}{46} = \frac{3}{23}.$$

(ii) There are 26 red cards.

$$\therefore P(\text{getting a red card}) = \frac{26}{46} = \frac{13}{23}.$$

(iii) Out of 46 cards, number of black cards = $26 - 6 = 20$.

$$\therefore P(\text{getting a black card}) = \frac{20}{46} = \frac{10}{23}.$$

(iv) Now, these 46 cards have 2 kings.

$$\therefore P(\text{getting a king}) = \frac{2}{46} = \frac{1}{23}.$$

EXAMPLE 21 Red queens and black jacks are removed from a pack of 52 playing cards. A card is drawn at random from the remaining cards, after reshuffling them. Find the probability that the drawn card is (i) a king, (ii) of red colour, (iii) a face card, (iv) a queen. [CBSE 2014]

SOLUTION After removing 2 red queens and 2 black jacks, the number of remaining cards = $52 - (2 + 2) = 48$.

(i) Out of 48 cards, there are 4 kings.

$$\therefore P(\text{getting a king}) = \frac{4}{48} = \frac{1}{12}.$$

(ii) Number of cards of red colour = $26 - 2 = 24$.

Total number of cards = 48.

$$\therefore P(\text{getting a card of red colour}) = \frac{24}{48} = \frac{1}{2}.$$

(iii) Number of face cards = $12 - (2 + 2) = 8$.

Total number of cards = 48.

$$\therefore P(\text{getting a face card}) = \frac{8}{48} = \frac{1}{6}.$$

(iv) Number of queens in 48 cards = $4 - 2 = 2$.

$$\therefore P(\text{getting a queen}) = \frac{2}{48} = \frac{1}{24}.$$

EXERCISE 19A

Very-Short-Answer Questions

1. Fill in the blanks:

(i) The probability of an impossible event is

(ii) The probability of a sure event is

(iii) For any event E , $P(E) + P(\text{not } E) = \dots\dots$.

(iv) The probability of a possible but not a sure event lies between and

(v) The sum of probabilities of all the outcomes of an experiment is

2. A coin is tossed once. What is the probability of getting a tail?

3. Two coins are tossed simultaneously. Find the probability of getting

(i) exactly 1 head (ii) at most 1 head (iii) at least 1 head.

4. A die is thrown once. Find the probability of getting

(i) an even number

(ii) a number less than 5

(iii) a number greater than 2

(iv) a number between 3 and 6

(v) a number other than 3

(vi) the number 5.

Short-Answer Questions

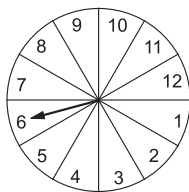
5. A letter of English alphabet is chosen at random. Determine the probability that the chosen letter is a consonant. [CBSE 2015]
6. A child has a die whose 6 faces show the letters given below:



The die is thrown once. What is the probability of getting (i) A, (ii) D?

7. It is known that a box of 200 electric bulbs contains 16 defective bulbs. One bulb is taken out at random from the box. What is the probability that the bulb drawn is (i) defective, (ii) nondefective?
8. If the probability of winning a game is 0.7, what is the probability of losing it?
9. There are 35 students in a class of whom 20 are boys and 15 are girls. From these students one is chosen at random. What is the probability that the chosen student is a (i) boy, (ii) girl?
10. In a lottery there are 10 prizes and 25 blanks. What is the probability of getting a prize?
11. 250 lottery tickets were sold and there are 5 prizes on these tickets. If Kunal has purchased one lottery ticket, what is the probability that he wins a prize?
12. 17 cards numbered 1, 2, 3, 4, ..., 17 are put in a box and mixed thoroughly. A card is drawn at random from the box. Find the probability that the card drawn bears (i) an odd number (ii) a number divisible by 5. [CBSE 2012]
13. A game of chance consists of spinning an arrow, which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 and these are equally likely outcomes. Find the probability that the arrow will point at any factor of 8. [CBSE 2015]
14. In a family of 3 children, find the probability of having at least one boy. [CBSE 2014]
15. A bag contains 4 white balls, 5 red balls, 2 black balls and 4 green balls. A ball is drawn at random from the bag. Find the probability that it is (i) black, (ii) not green, (iii) red or white, (iv) neither red nor green. [CBSE 2012]
16. A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability of getting (i) a red king, (ii) a queen or a jack. [CBSE 2012]

17. A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that the drawn card is neither a king nor a queen. [CBSE 2013]
18. A card is drawn from a well-shuffled pack of 52 cards. Find the probability of getting (i) a red face card (ii) a black king. [CBSE 2013C]
19. Two different dice are tossed together. Find the probability that (i) the number on each die is even, (ii) the sum of the numbers appearing on the two dice is 5. [CBSE 2014]
20. Two different dice are rolled simultaneously. Find the probability that the sum of the numbers on the two dice is 10. [CBSE 2014]
21. Two different dice are thrown together. Find the probability that
(i) the sum of the numbers appeared is less than 7. [CBSE 2011]
(ii) the product of the numbers appeared is less than 18. [CBSE 2017]
22. Two dice are rolled together. Find the probability of getting such numbers on two dice whose product is a perfect square. [CBSE 2011]
23. Two dice are rolled together. Find the probability of getting such numbers on the two dice whose product is 12. [CBSE 2013]
24. Cards marked with numbers 5 to 50 are placed in a box and mixed thoroughly. A card is drawn from the box at random. Find the probability that the number on the taken out card is (i) a prime number less than 10 (ii) a number which is a perfect square. [CBSE 2008]
25. A game of chance consists of spinning an arrow which is equally likely to come to rest pointing to one of the numbers 1, 2, 3, ..., 12 as shown in the figure. What is the probability that it will point to
(i) 6? (ii) an even number?
(iii) a prime number? (iv) a number which is a multiple of 5?
26. 12 defective pens are accidentally mixed with 132 good ones. It is not possible to just look at pen and tell whether or not it is defective. One pen is taken out at random from this lot. Find the probability that the pen taken out is good one.
27. A lot consists of 144 ballpoint pens of which 20 are defective and others good. Tanvy will buy a pen if it is good, but will not buy it if it is defective. The shopkeeper draws one pen at random and gives it to her. What is the probability that (i) she will buy it, (ii) she will not buy it?
28. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a



two-digit number, (ii) a perfect square number, (iii) a number divisible by 5.

29. (i) A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?
- (ii) Suppose the bulb drawn in (i) is not defective and not replaced. Now, bulb is drawn at random from the rest. What is the probability that this bulb is not defective?
30. A bag contains lemon-flavoured candies only. Hema takes out one candy without looking into the bag. What is the probability that she takes out (i) an orange-flavoured candy? (ii) a lemon-flavoured candy?
31. There are 40 students in a class of whom 25 are girls and 15 are boys. The class teacher has to select one student as a class representative. He writes the name of each student on a separate card, the cards being identical. Then she puts cards in a bag and stirs them thoroughly. She then draws one card from the bag. What is the probability that the name written on the card is the name of (i) a girl? (ii) a boy?
32. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of drawing
- (i) an ace (ii) a '4' of spades
(iii) a '9' of a black suit (iv) a red king.
33. A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability of getting [CBSE 2003]
- (i) a queen (ii) a diamond
(iii) a king or an ace (iv) a red ace.
34. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting
- (i) a king of red suit (ii) a face card
(iii) a red face card (iv) a queen of black suit
(v) a jack of hearts (vi) a spade.
35. A card is drawn at random from a well-shuffled deck of playing cards. Find the probability that the card drawn is [CBSE 2006]
- (i) a card of spades or an ace (ii) a red king
(iii) either a king or a queen (iv) neither a king nor a queen.
36. Two different dice are thrown together. Find the probability that the numbers obtained have
- (i) even sum (ii) even product. [CBSE 2017]

37. Two different dice are thrown together. Find the probability that the numbers obtained
- (i) have a sum less than 7 (ii) have a product less than 16
 (iii) is a doublet of odd numbers. [CBSE 2017]
38. The king, the jack and the 10 of spades are lost from a pack of 52 cards and a card is drawn from the remaining cards after shuffling. Find the probability of getting a
- (i) red card (ii) black jack
 (iii) red king (iv) 10 of hearts. [CBSE 2017]
39. Peter throws two different dice together and finds the product of the two numbers obtained. Rina throws a die and squares the number obtained. Who has the better chance to get the number 25? [CBSE 2017]

ANSWERS (EXERCISE 19A)

1. (i) 0 (ii) 1 (iii) 1 (iv) 0, 1 (v) 1 2. $\frac{1}{2}$ 3. (i) $\frac{1}{2}$ (ii) $\frac{3}{4}$ (iii) $\frac{3}{4}$
 4. (i) $\frac{1}{2}$ (ii) $\frac{2}{3}$ (iii) $\frac{2}{3}$ (iv) $\frac{1}{3}$ (v) $\frac{5}{6}$ (vi) $\frac{1}{6}$ 5. $\frac{21}{26}$ 6. (i) $\frac{1}{2}$ (ii) $\frac{1}{6}$
 7. (i) $\frac{2}{25}$ (ii) $\frac{23}{25}$ 8. 0.3 9. (i) $\frac{4}{7}$ (ii) $\frac{3}{7}$ 10. $\frac{2}{7}$ 11. $\frac{1}{50}$
 12. (i) $\frac{9}{17}$ (ii) $\frac{3}{17}$ 13. $\frac{3}{8}$ 14. $\frac{7}{8}$ 15. (i) $\frac{2}{15}$ (ii) $\frac{11}{15}$ (iii) $\frac{3}{5}$ (iv) $\frac{2}{5}$
 16. (i) $\frac{1}{26}$ (ii) $\frac{2}{13}$ 17. $\frac{11}{13}$ 18. (i) $\frac{3}{26}$ (ii) $\frac{1}{26}$ 19. (i) $\frac{1}{4}$ (ii) $\frac{1}{9}$ 20. $\frac{1}{12}$
 21. (i) $\frac{5}{12}$ (ii) $\frac{13}{18}$ 22. $\frac{2}{9}$ 23. $\frac{1}{9}$ 24. (i) $\frac{1}{23}$ (ii) $\frac{5}{46}$
 25. (i) $\frac{1}{12}$ (ii) $\frac{1}{2}$ (iii) $\frac{5}{12}$ (iv) $\frac{1}{6}$ 26. $\frac{11}{12}$ 27. (i) $\frac{31}{36}$ (ii) $\frac{5}{36}$
 28. (i) $\frac{9}{10}$ (ii) $\frac{1}{10}$ (iii) $\frac{1}{5}$ 29. (i) $\frac{1}{5}$ (ii) $\frac{15}{19}$ 30. (i) 0 (ii) 1
 31. (i) $\frac{5}{8}$ (ii) $\frac{3}{8}$ 32. (i) $\frac{1}{13}$ (ii) $\frac{1}{52}$ (iii) $\frac{1}{26}$ (iv) $\frac{1}{26}$
 33. (i) $\frac{1}{13}$ (ii) $\frac{1}{4}$ (iii) $\frac{2}{13}$ (iv) $\frac{1}{26}$ 34. (i) $\frac{1}{26}$ (ii) $\frac{3}{13}$ (iii) $\frac{3}{26}$ (iv) $\frac{1}{26}$ (v) $\frac{1}{52}$ (vi) $\frac{1}{4}$
 35. (i) $\frac{4}{13}$ (ii) $\frac{1}{26}$ (iii) $\frac{2}{13}$ (iv) $\frac{11}{13}$ 36. (i) $\frac{1}{2}$ (ii) $\frac{3}{4}$ 37. (i) $\frac{5}{12}$ (ii) $\frac{25}{36}$ (iii) $\frac{1}{4}$
 38. (i) $\frac{26}{49}$ (ii) $\frac{1}{49}$ (iii) $\frac{2}{49}$ (iv) $\frac{1}{49}$ 39. Rina

HINTS TO SOME SELECTED QUESTIONS

5. Out of 26 letters of English alphabet, there are 11 consonants.

$$\therefore P(\text{getting a consonant}) = \frac{21}{26}.$$

6. There are 6 letters in all consisting of 3As, 1B, 1C and 1D.

$$\therefore \quad \text{(i) } P(\text{getting } A) = \frac{3}{6} = \frac{1}{2}. \quad \text{(ii) } P(\text{getting } D) = \frac{1}{6}.$$

7. Total number of bulbs = 200.

Number of defective bulbs = 16.

Number of non-defective bulbs = $200 - 16 = 184$.

$$\text{(i) } P(\text{getting a defective bulb}) = \frac{16}{200} = \frac{2}{25}.$$

$$\text{(ii) } P(\text{getting a non-defective bulb}) = \frac{184}{200} = \frac{23}{25}.$$

8. Let E be the event of winning the game. Then, $P(E) = 0.7$.

Probability of losing the game = $1 - P(E) = (1 - 0.7) = 0.3$.

9. (i) $P(\text{choosing a boy}) = \frac{20}{35} = \frac{4}{7}$. (ii) $P(\text{choosing a girl}) = \frac{15}{35} = \frac{3}{7}$.

10. Total number of tickets = $10 + 25 = 35$.

Number of prizes = 10.

$$P(\text{getting a prize}) = \frac{10}{35} = \frac{2}{7}.$$

11. $P(\text{getting a prize}) = \frac{5}{250} = \frac{1}{50}$.

12. Total number of cards = 17.

- (i) Let E_1 be the event of choosing an odd number.

These numbers are 1, 3, 5, ..., 17.

Let their number be n . Then,

$$T_n = 17 \Rightarrow 1 + (n - 1) \times 2 = 17 \Rightarrow n = 9.$$

$$\therefore P(E_1) = \frac{9}{17}.$$

- (ii) Let E_2 be the event of choosing a number divisible by 5.

Numbers divisible by 5 are 5, 10, 15. Their number is 3.

$$\therefore P(E_2) = \frac{3}{17}.$$

13. All possible outcomes are 1, 2, 3, 4, 5, 6, 7, 8.

Number of all possible outcomes is 8.

All factors of 8 are 2, 4, 8.

Number of favourable outcomes = 3.

Probability that the arrow will point at any factor of 8 = $\frac{3}{8}$.

14. All possible outcomes are $BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG$.

Number of all possible outcomes = 8.

Let E be the event of having at least one boy.

Then, E contains $GGB, GBG, BGG, BBG, BGB, GBB, BBB$.

Number of cases favourable to $E = 7$.

$$\therefore \text{required probability} = P(E) = \frac{7}{8}.$$

15. Total number of balls = $4 + 5 + 2 + 4 = 15$.

- (i) Number of black balls = 2.

$$P(\text{getting a black ball}) = \frac{2}{15}.$$

- (ii) Number of balls which are not green = $4 + 5 + 2 = 11$.

$$P(\text{getting a ball which is not green}) = \frac{11}{15}.$$

- (iii) Number of balls which are red or white = $5 + 4 = 9$.

$$P(\text{getting a ball which is red or white}) = \frac{9}{15} = \frac{3}{5}.$$

- (iv) Number of balls which are neither red nor green = $4 + 2 = 6$.

$$P(\text{getting a ball which is neither red nor green}) = \frac{6}{15} = \frac{2}{5}.$$

16. Total number of cards = 52.

- (i) Number of red kings = 2.

$$\therefore P(\text{getting a red king}) = \frac{2}{52} = \frac{1}{26}.$$

- (ii) There are 4 queens and 4 jacks.

$$\therefore P(\text{getting a queen or a jack}) = \frac{8}{52} = \frac{2}{13}.$$

17. Total number of cards = 52.

Total number of kings and queens = $4 + 4 = 8$.

Remaining number of cards = $52 - 8 = 44$.

$$\therefore P(\text{getting a card which is neither a king nor a queen}) = \frac{44}{52} = \frac{11}{13}.$$

18. Total number of cards = 52.

- (i) 4 kings, 4 queens and 4 jacks are all face cards.

Number of red face cards = $2 + 2 + 2 = 6$.

$$\therefore P(\text{getting a red face card}) = \frac{6}{52} = \frac{3}{26}.$$

- (ii) Number of black kings = 2.

$$\therefore P(\text{getting a black king}) = \frac{2}{52} = \frac{1}{26}.$$

19. When two different dice are thrown, then total number of outcomes = 36.

- (i) Let E_1 be the event of getting an even number on each die.

These numbers are $(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6)$.

Number of favourable outcomes = 9.

$$\therefore P(\text{getting an even number on both dice}) = P(E_1) = \frac{9}{36} = \frac{1}{4}.$$

- (ii) Let E_2 be the event of getting the sum 5. Then these numbers are (1, 4), (2, 3), (4, 1), (3, 2).

Number of favourable outcomes = 4.

$$\therefore P(E_2) = \frac{4}{36} = \frac{1}{9}.$$

20. Number of all possible outcomes is 36.

Let E be the event of getting the sum 10 on the two dice.

Then, the favourable outcomes are (4, 6), (6, 4), (5, 5).

Number of favourable outcomes = 3.

$$\therefore P(E) = \frac{3}{36} = \frac{1}{12}.$$

21. (i) Number of all possible outcomes is 36.

Let E be the event of getting the sum less than 7 on the two dice.

Then, the favourable outcomes are

(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3),
(2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1).

Number of favourable outcomes = 15.

$$\therefore P(E) = \frac{15}{36} = \frac{5}{12}.$$

22. Number of all possible outcomes is 36.

Let E be the event of getting the product of numbers on the two dice, as a perfect square.

Then, the favourable outcomes are

(1, 1), (1, 4), (4, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6).

Number of favourable outcomes = 8.

$$\therefore P(E) = \frac{8}{36} = \frac{2}{9}.$$

23. Number of all possible outcomes is 36.

Let E be the event of getting all those two numbers whose product is 12.

Then, the favourable outcomes are (2, 6), (6, 2), (3, 4), (4, 3).

Number of favourable outcomes = 4.

$$\therefore P(E) = \frac{4}{36} = \frac{1}{9}.$$

24. Total number of cards = 50 - 4 = 46.

- (i) Out of the given numbers, prime numbers less than 10 are 5 and 7.

$\therefore$ number of prime numbers less than 10 = 2.

$$\therefore P(\text{getting a prime number less than 10}) = \frac{2}{46} = \frac{1}{23}.$$

- (ii) From given numbers, the perfect square numbers are 9, 16, 25, 36, 49.

Their number is 5.

$$\therefore P(\text{getting a perfect square number}) = \frac{5}{46}.$$

25. Number of all possible outcomes = 12.

(i) $P(\text{getting a 6}) = \frac{1}{12}.$

- (ii) Out of the given numbers there are 6 even numbers.

$$\therefore P(\text{getting an even number}) = \frac{6}{12} = \frac{1}{2}.$$

- (iii) Out of the given numbers, the prime numbers are 2, 3, 5, 7, 11.

Their number is 5.

$$\therefore P(\text{getting a prime number}) = \frac{5}{12}.$$

- (iv) Out of the given numbers, the multiples of 5 are 5, 10.

Their number is 2.

$$\therefore P(\text{getting a multiple of 5}) = \frac{2}{12} = \frac{1}{6}.$$

26. Total number of pens = $132 + 12 = 144$.

Number of good pens = 132.

$$\therefore P(\text{getting a good pen}) = \frac{132}{144} = \frac{11}{12}.$$

27. Total number of pens = 144.

Number of defective pens = 20.

Number of non-defective pens = $144 - 20 = 124$.

(i) $P(\text{buying the pen}) = P(\text{getting a non-defective pen}) = \frac{124}{144} = \frac{31}{36}.$

(ii) $P(\text{not buying the pen}) = P(\text{getting a defective pen}) = \frac{20}{144} = \frac{5}{36}.$

28. Total number of discs = 90.

- (i) Number of discs bearing 2-digit numbers

= number of numbers from 10 to 90 = $(90 - 9) = 81$.

$$P(\text{getting a 2-digit number}) = \frac{81}{90} = \frac{9}{10}.$$

- (ii) Perfect square numbers are $1^2, 2^2, 3^2, \dots, 9^2$. Their number is 9.

Number of discs bearing perfect square numbers = 9.

$$\therefore P(\text{getting a perfect square number}) = \frac{9}{90} = \frac{1}{10}.$$

- (iii) Numbers divisible by 5 are 5, 10, 15, ..., 90. They are 18 in number.

$$\therefore P(\text{getting a number divisible by 5}) = \frac{18}{90} = \frac{1}{5}.$$

29. (i) Total number of bulbs = 20. Number of defective bulbs = 4.

Number of non-defective bulbs = $20 - 4 = 16$.

$$\therefore P(\text{getting a defective bulb}) = \frac{4}{20} = \frac{1}{5}.$$

- (ii) After removing 1 non-defective bulb, we have

remaining number of bulbs = $20 - 1 = 19$.

Out of these, the number of non-defective bulbs = $16 - 1 = 15$.

$$\therefore P(\text{getting a non-defective bulb}) = \frac{15}{19}.$$

30. Suppose there are x candies in the bag.

Then, number of orange-flavoured candies in the bag = 0.

And, the number of lemon-flavoured candies in the bag = x .

$$\therefore \text{(i) } P(\text{getting an orange-flavoured candy}) = \frac{0}{x} = 0.$$

$$\text{(ii) } P(\text{getting a lemon-flavoured candy}) = \frac{x}{x} = 1.$$

31. Total number of students in the class = 40.

Number of girls = 25, number of boys = 15.

$$\text{(i) } P(\text{selecting the name of a girl}) = \frac{25}{40} = \frac{5}{8}.$$

$$\text{(ii) } P(\text{selecting the name of a boy}) = \frac{15}{40} = \frac{3}{8}.$$

32. (i) There are 4 aces in all.

$$\therefore P(\text{getting an ace}) = \frac{4}{52} = \frac{1}{13}.$$

- (ii) There is one '4' of spades.

$$\therefore P(\text{getting 4 of spades}) = \frac{1}{52}.$$

- (iii) There are two nines of black suits.

$$\therefore P(\text{getting a '9' of a black suit}) = \frac{2}{52} = \frac{1}{26}.$$

- (iv) There are two red kings.

$$\therefore P(\text{getting a red king}) = \frac{2}{52} = \frac{1}{26}.$$

33. (i) There are 4 queens.

$$\therefore P(\text{getting a queen}) = \frac{4}{52} = \frac{1}{13}.$$

- (ii) There are 13 diamonds.

$$\therefore P(\text{getting a diamond}) = \frac{13}{52} = \frac{1}{4}.$$

- (iii) There are 4 kings and 4 aces.

$$\therefore P(\text{getting a king or an ace}) = \frac{8}{52} = \frac{2}{13}.$$

(iv) There are 2 red aces.

$$\therefore P(\text{getting a red ace}) = \frac{2}{52} = \frac{1}{26}.$$

34. (i) There are 4 kings of red suits.

$$\therefore P(\text{getting a king of a red suit}) = \frac{2}{52} = \frac{1}{26}.$$

(ii) There are 12 face cards in all.

$$\therefore P(\text{getting a face card}) = \frac{12}{52} = \frac{3}{13}.$$

(iii) There are 6 red face cards.

$$\therefore P(\text{getting a red face card}) = \frac{6}{52} = \frac{3}{26}.$$

(iv) There are 2 queens of black suits.

$$\therefore P(\text{getting a queen of a black suit}) = \frac{2}{52} = \frac{1}{26}.$$

(v) There is 1 jack of hearts.

$$\therefore P(\text{getting a jack of hearts}) = \frac{1}{52}.$$

(vi) There are 3 cards of spades.

$$\therefore P(\text{getting a spade}) = \frac{13}{52} = \frac{1}{4}.$$

35. (i) There are 13 cards of spades including 1 ace and 3 more aces are there.

$$\therefore P(\text{getting a card of spades or an ace}) = \frac{13 + 3}{52} = \frac{16}{52} = \frac{4}{13}.$$

(ii) There are 2 red kings.

$$\therefore P(\text{getting a red king}) = \frac{2}{52} = \frac{1}{26}.$$

(iii) There are 4 kings and 4 queens.

$$\therefore P(\text{getting either a king or a queen}) = \frac{4 + 4}{52} = \frac{8}{52} = \frac{2}{13}.$$

(iv) $P(\text{neither a king nor a queen}) = 1 - P(\text{either a king or a queen})$

$$= \left(1 - \frac{2}{13}\right) = \frac{11}{13}.$$

.....

EXERCISE 19B

Long-Answer Questions

1. A box contains 25 cards numbered from 1 to 25. A card is drawn at random from the bag. Find the probability that the number on the drawn card is (i) divisible by 2 or 3, (ii) a prime number. [CBSE 2015]

2. A box contains cards numbered 3, 5, 7, 9, ... , 35, 37. A card is drawn at random from the box. Find the probability that the number on the card is a prime number. [CBSE 2013]
3. Cards numbered 1 to 30 are put in a bag. A card is drawn at random from the bag. Find the probability that the number on the drawn card is (i) not divisible by 3, (ii) a prime number greater than 7, (iii) not a perfect square number. [CBSE 2014]
4. Cards bearing numbers 1, 3, 5, ... , 35 are kept in a bag. A card is drawn at random from the bag. Find the probability of getting a card bearing (i) a prime number less than 15, (ii) a number divisible by 3 and 5. [CBSE 2010]
5. A box contains cards bearing numbers 6 to 70. If one card is drawn at random from the box, find the probability that it bears (i) a one-digit number, (ii) a number divisible by 5, (iii) an odd number less than 30, (iv) a composite number between 50 and 70. [CBSE 2015]
6. Cards marked with numbers 1, 3, 5, ..., 101 are placed in a bag and mixed thoroughly. A card is drawn at random from the bag. Find the probability that the number on the drawn card is (i) less than 19, (ii) a prime number less than 20. [CBSE 2012]
7. Tickets numbered 2, 3, 4, 5, ..., 100, 101 are placed in a box and mixed thoroughly. One ticket is drawn at random from the box. Find the probability that the number on the ticket is
- (i) an even number
 - (ii) a number less than 16
 - (iii) a number which is a perfect square
 - (iv) a prime number less than 40.
8. (i) A box contains 80 discs, which are numbered from 1 to 80. If one disc is drawn at random from the box, find the probability that it bears a perfect square number. [CBSE 2011]
- (ii) A box contains 90 discs which are numbered 1 to 90. If one disc is drawn at random from the box, find the probability that it bears
- (a) a two-digit number
 - (b) a number divisible by 5
- [CBSE 2017]
9. A piggy bank contains hundred 50-p coins, seventy ₹ 1 coin, fifty ₹ 2 coins and thirty ₹ 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a ₹ 1 coin? (ii) will not be a ₹ 5 coin (iii) will be 50-p or a ₹ 2 coin? [CBSE 2013C]

10. The probability of selecting a red ball at random from a jar that contains only red, blue and orange balls is $\frac{1}{4}$. The probability of selecting a blue ball at random from the same jar is $\frac{1}{3}$. If the jar contains 10 orange balls, find the total number of balls in the jar. [CBSE 2015]
11. A bag contains 18 balls out of which x balls are red.
- (i) If one ball is drawn at random from the bag, what is the probability that it is not red?
 - (ii) If two more red balls are put in the bag, the probability of drawing a red ball will be $\frac{9}{8}$ times the probability of drawing a red ball in the first case. Find the value of x . [CBSE 2015]
12. A jar contains 24 marbles. Some of these are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$. Find the number of blue marbles in the jar.
13. A jar contains 54 marbles, each of which some are blue, some are green and some are white. The probability of selecting a blue marble at random is $\frac{1}{3}$ and the probability of selecting a green marble at random is $\frac{4}{9}$. How many white marbles does the jar contain?
14. A carton consists of 100 shirts of which 88 are good and 8 have minor defects. Rohit, a trader, will only accept the shirts which are good. But, Kamal, an another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. What is the probability that it is acceptable to (i) Rohit and (ii) Kamal?
15. A group consists of 12 persons, of which 3 are extremely patient, other 6 are extremely honest and rest are extremely kind. A person from the group is selected at random. Assuming that each person is equally likely to be selected, find the probability of selecting a person who is (i) extremely patient, (ii) extremely kind or honest. Which of the above values you prefer more? [CBSE 2013]
16. A die is rolled twice. Find the probability that [CBSE 2014]
- (i) 5 will not come up either time,
 - (ii) 5 will come up exactly one time,
 - (iii) 5 will come up both the times.

17. Two dice are rolled once. Find the probability of getting such numbers on two dice whose product is a perfect square. [CBSE 2011]
18. A letter is chosen at random from the letters of the word 'ASSOCIATION'. Find the probability that the chosen letter is a (i) vowel (ii) consonant (iii) an S.
19. Five cards—the ten, jack, queen, king and ace of diamonds are well shuffled with their faces downwards. One card is then picked up at random. (a) What is the probability that the drawn card is the queen? (b) If the queen is drawn and put aside and a second card is drawn, find the probability that the second card is (i) an ace, (ii) a queen. [CBSE 2014]
20. A card is drawn at random from a well shuffled pack of 52 cards. Find the probability that the card drawn is neither a red card nor a queen. [CBSE 2011]
21. What is the probability that an ordinary year has 53 Mondays?
22. All red face cards are removed from a pack of playing cards. The remaining cards are well shuffled and then a card is drawn at random from them. Find the probability that the drawn card is (i) a red card, (ii) a face card, (iii) a card of clubs. [CBSE 2015]
23. All kings, queens and aces are removed from a pack of 52 cards. The remaining cards are well-shuffled and then a card is drawn from it. Find the probability that the drawn card is (i) a black face card, (ii) a red card. [CBSE 2012]
24. A game consists of tossing a one-rupee coin three times, and noting its outcome each time. Find the probability of getting (i) three heads, (ii) at least 2 tails. [CBSE 2015]
25. Find the probability that a leap year selected at random will contain 53 Sundays.
26. The probability of selecting a rotten apple randomly from a heap of 900 apples is 0.18. What is the number of rotten apples in the heap? [CBSE 2017]
27. A bag contains 15 white and some black balls. If the probability of drawing a black ball from the bag is thrice that of drawing a white ball find the number of black balls in the bag. [CBSE 2017]
28. Find the probability of getting the sum of two numbers, less than 3 or more than 11, when a pair of distinct dice is thrown together. [CBSE 2017]

ANSWERS (EXERCISE 19B)

1. (i) $\frac{16}{25}$ (ii) $\frac{9}{25}$ 2. $\frac{11}{18}$ 3. (i) $\frac{2}{3}$ (ii) $\frac{1}{5}$ (iii) $\frac{5}{6}$ 4. (i) $\frac{5}{18}$ (ii) $\frac{1}{9}$
5. (i) $\frac{4}{65}$ (ii) $\frac{1}{5}$ (iii) $\frac{12}{65}$ (iv) $\frac{3}{13}$ 6. (i) $\frac{3}{17}$ (ii) $\frac{7}{51}$
7. (i) $\frac{1}{2}$ (ii) $\frac{7}{50}$ (iii) $\frac{9}{100}$ (iv) $\frac{3}{25}$ 8. (i) $\frac{1}{10}$ (ii) (a) $\frac{9}{10}$ (b) $\frac{1}{5}$
9. (i) $\frac{7}{25}$ (ii) $\frac{22}{25}$ (iii) $\frac{3}{5}$ 10. 24 11. (i) $\frac{(18-x)}{18}$ (ii) $x = 8$
12. 8 13. 12 14. (i) $\frac{22}{25}$ (ii) $\frac{24}{25}$ 15. (i) $\frac{1}{4}$ (ii) $\frac{3}{4}$
16. (i) $\frac{25}{36}$ (ii) $\frac{5}{18}$ (iii) $\frac{1}{36}$ 17. $\frac{2}{9}$ 18. (i) $\frac{6}{11}$ (ii) $\frac{5}{11}$ (iii) $\frac{2}{11}$
19. (a) $\frac{1}{5}$ (b) (i) $\frac{1}{4}$ (ii) 0 20. $\frac{6}{13}$ 21. $\frac{1}{7}$ 22. (i) $\frac{10}{23}$ (ii) $\frac{3}{23}$ (iii) $\frac{6}{23}$
23. (i) $\frac{1}{20}$ (ii) $\frac{1}{2}$ 24. (i) $\frac{1}{8}$ (ii) $\frac{1}{2}$ 25. $\frac{2}{7}$ 26. 162
27. 45 28. $\frac{1}{18}$

HINTS TO SOME SELECTED QUESTIONS

1. Number of all possible outcomes = 25.
- (i) Numbers divisible by 2 or 3 are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 3, 9, 15, 21.
Their number is 16.
 $\therefore P(\text{getting a number divisible by 2 or 3}) = \frac{16}{25}$.
- (ii) Prime numbers from 1 to 25 are 2, 3, 5, 7, 11, 13, 17, 19, 23.
Their number is 9.
 $\therefore P(\text{getting a prime number}) = \frac{9}{25}$.
2. Given numbers 3, 5, 7, 9, ..., 35, 37 form an AP with $a = 3$ and $d = 2$.
Let $T_n = 37$. Then,
 $3 + (n-1) \times 2 = 37 \Rightarrow (n-1) \times 2 = 34 \Rightarrow n-1 = 17 \Rightarrow n = 18$.
Out of these numbers, the prime numbers are 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37.
Their number is 11.
 $\therefore P(\text{getting a prime number}) = \frac{11}{18}$.
3. Number of all possible outcomes = 30.
- (i) There are 10 numbers divisible by 3.
Number of numbers not divisible by 3 = $30 - 10 = 20$.

$$\therefore P(\text{getting a number not divisible by 3}) = \frac{20}{30} = \frac{2}{3}.$$

- (ii) Prime numbers greater than 7 are 11, 13, 17, 19, 23, 29.

Number of these prime numbers = 6.

$$\therefore P(\text{getting a prime number greater than 7}) = \frac{6}{30} = \frac{1}{5}.$$

- (iii) Perfect square numbers are 1, 4, 9, 16, 25. These are 5 in number.

Number of those numbers which are not perfect squares = $30 - 5 = 25$.

$$\therefore P(\text{getting non-perfect square numbers}) = \frac{25}{30} = \frac{5}{6}.$$

4. The numbers 1, 3, 5, ..., 35 form an AP with $a = 1$ and $d = 2$.

Let $T_n = 35$. Then,

$$1 + (n - 1) \times 2 = 35 \Rightarrow (n - 1) \times 2 = 34 \Rightarrow n - 1 = 17 \Rightarrow n = 18.$$

$\therefore$ number of all possible outcomes = 18.

- (i) Out of the given numbers, the prime numbers less than 15 are 3, 5, 7, 11, 13. Their number is 5.

$$\therefore P(\text{getting a prime number}) = \frac{5}{18}.$$

- (ii) A number is divisible by 3 and 5 means, it must be divisible by 15.

The numbers divisible by 15 are 15 and 30.

Their number is 2.

$$\therefore P(\text{getting a number divisible by both 3 and 5}) = \frac{2}{18} = \frac{1}{9}.$$

5. Given numbers 6, 7, 8, ..., 70 form an AP with $a = 6$ and $d = 1$.

Let $T_n = 70$. Then, $6 + (n - 1) \times 1 = 70 \Rightarrow n - 1 = 64 \Rightarrow n = 65$.

$\therefore$ total number of cards = 65.

- (i) Out of the given numbers, the one-digit numbers are 6, 7, 8, 9.

Number of one-digit numbers = 4.

$$\therefore P(\text{getting a one-digit number}) = \frac{4}{65}.$$

- (ii) Out of the given numbers, those divisible by 5 are 10, 15, 20, 25, ..., 70.

Let $T_n = 70$. Then,

$$10 + (n - 1) \times 5 = 70 \Rightarrow (n - 1) \times 5 = 60 \Rightarrow n - 1 = 12 \Rightarrow n = 13.$$

$$\therefore P(\text{getting a number divisible by 5}) = \frac{13}{65} = \frac{1}{5}.$$

- (iii) Out of the given numbers, odd numbers less than 30 are 7, 9, 11, 13, ..., 29.

Let $T_m = 29$. Then,

$$7 + (m - 1) \times 2 = 29 \Rightarrow (m - 1) \times 2 = 22 \Rightarrow m - 1 = 11 \Rightarrow m = 12.$$

$$\therefore P(\text{getting an odd number less than 30}) = \frac{12}{65}.$$

- (iv) Number of numbers between 50 and 70 = numbers from 51 to 69.

Their number = $(69 - 51) + 1 = 19$.

Prime numbers between 50 and 70 = 53, 59, 61, 67.

Number of prime numbers = 4.

Number of composite numbers = $19 - 4 = 15$.

$$\therefore P(\text{getting a composite number}) = \frac{15}{65} = \frac{3}{13}.$$

6. Given numbers 1, 3, 5, ..., 101 form an AP with $a = 1$ and $d = 2$.

Let $T_n = 101$. Then,

$$1 + (n - 1) \times 2 = 101 \Rightarrow (n - 1) \times 2 = 100 \Rightarrow n - 1 = 50 \Rightarrow n = 51.$$

$\therefore$ total number of cards = 51.

- (i) Out of the given numbers, those less than 19 are 1, 3, 5, ..., 17.

Let $t_m = 17$.

$$\text{Then, } 1 + (m - 1) \times 2 = 17 \Rightarrow (m - 1) \times 2 = 16 \Rightarrow m - 1 = 8 \Rightarrow m = 9.$$

$$\therefore P(\text{getting a number less than 19}) = \frac{9}{51} = \frac{3}{17}.$$

- (ii) Out of the given numbers, the prime numbers less than 20 are 3, 5, 7, 11, 17, 19.

Their number is 7.

$$\therefore P(\text{getting a prime number less than 20}) = \frac{7}{51}.$$

7. The tickets bear the numbers 2, 3, 4, ..., 100, 101. So, the number of tickets = 100.

- (i) Out of the given numbers, the even numbers are 2, 4, 6, 8, ..., 100.

Their number is 50.

$$\therefore P(\text{getting an even number}) = \frac{50}{100} = \frac{1}{2}.$$

- (ii) Out of the given numbers, the number of numbers less than 16 is 14.

$$\therefore P(\text{getting number less than 16}) = \frac{14}{100} = \frac{7}{50}.$$

- (iii) Out of the given numbers, the perfect square numbers are $2^2, 3^2, 4^2, \dots, 10^2$.

Their number is 9.

$$\therefore P(\text{getting a perfect square number}) = \frac{9}{100}.$$

- (iv) Prime numbers less than 40 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37.

Their number is 12.

$$\therefore P(\text{getting a prime number less than 40}) = \frac{12}{100} = \frac{3}{25}.$$

$$10. P(\text{getting a red ball}) = \frac{1}{4}, P(\text{getting a blue ball}) = \frac{1}{3}.$$

Let $P(\text{getting an orange ball})$ be x .

Since there are only three types of balls in the jar, the sum of probabilities of drawing these balls must be 1.

$$\therefore \frac{1}{4} + \frac{1}{3} + x = 1 \Rightarrow \frac{7}{12} + x = 1 \Rightarrow x = \left(1 - \frac{7}{12}\right) = \frac{5}{12}.$$

$$\therefore P(\text{getting an orange ball}) = \frac{5}{12}. \quad \dots (i)$$

Let the total number of balls in the jar be n .

Number of orange balls = 10.

$$\therefore P(\text{getting an orange ball}) = \frac{10}{n}.$$

$$\Rightarrow \frac{10}{n} = \frac{5}{12} \Rightarrow 5n = 120 \Rightarrow n = 24 \text{ [using (i)]}.$$

Hence, the total number of balls in the jar is 24.

11. (i) Total number of balls = 18.

Number of red balls = x .

Number of balls which are not red = $18 - x$.

$$\therefore P(\text{getting a ball which is not red}) = \frac{18 - x}{18}.$$

- (ii) Now, total number of balls = $18 + 2 = 20$.

Number of red balls now = $x + 2$.

$$\begin{aligned} P(\text{getting a red ball now}) &= \frac{x+2}{20} \text{ and } P(\text{getting a red ball in first case}) = \frac{x}{18} \\ \Rightarrow \frac{x+2}{20} &= \frac{9}{8} \times \frac{x}{18} \Rightarrow 144(x+2) = 180x \\ \Rightarrow 180x - 144x &= 288 \Rightarrow 36x = 288 \Rightarrow x = \frac{288}{36} = 8. \end{aligned}$$

Hence, $x = 8$.

12. Total number of marbles in the jar = 24.

Let the number of blue marbles be x .

Then, the number of green marbles = $24 - x$.

$$P(\text{getting a green marble}) = \frac{24 - x}{24}.$$

$$\text{But, } P(\text{getting a green marble}) = \frac{2}{3} \text{ (given).}$$

$$\therefore \frac{24 - x}{24} = \frac{2}{3} \Rightarrow 72 - 3x = 48 \Rightarrow 3x = 24 \Rightarrow x = 8.$$

Hence, the number of blue marbles in the jar is 8.

13. Total number of marbles in the jar = 54.

$$P(\text{getting a blue marble}) = \frac{1}{3} \text{ and } P(\text{getting a green marble}) = \frac{4}{9}.$$

Let $P(\text{getting a white marble})$ be x .

Since there are only three given types of marbles in the jar, we have

$$\frac{1}{3} + \frac{4}{9} + x = 1 \Rightarrow \frac{7}{9} + x = 1 \Rightarrow x = 1 - \frac{7}{9} = \frac{2}{9}.$$

$$\therefore P(\text{getting a white marble}) = \frac{2}{9}.$$

Let the number of white marbles be n .

$$\text{Then, } P(\text{getting a white marble}) = \frac{n}{54}.$$

$$\therefore \frac{n}{54} = \frac{2}{9} \Rightarrow 9n = 108 \Rightarrow n = 12.$$

Hence, there are 12 white marbles in the jar.

14. Total number of shirts = 100.

Number of good shirts = 88.

Number of shirts having minor defects = 8.

Number of shirts having major defects = $100 - (88 + 8) = 4$.

$$\therefore \text{(i) } P(\text{that the drawn shirt is acceptable to Rohit}) = \frac{88}{100} = \frac{22}{25}.$$

$$\text{(ii) } P(\text{that the drawn shirt is acceptable to Kamal}) = \frac{88 + 8}{100} = \frac{96}{100} = \frac{24}{25}.$$

15. Total number of persons in the group = 12.

- (i) Number of persons who are extremely patient = 3.

$$P(\text{selecting a person who is extremely patient}) = \frac{3}{12} = \frac{1}{4}.$$

- (ii) Number of persons who are extremely honest = 6.

Number of persons who are extremely kind = $12 - (3 + 6) = 3$.

$$\therefore P(\text{selecting a person who is extremely kind or extremely honest}) = \frac{3 + 6}{12} = \frac{9}{12} = \frac{3}{4}.$$

From the given three values, we prefer honesty. Honesty can get rid of rampant corruption which is a burning issue of the present society.

16. Number of all possible outcomes = 36.

- (i) All those cases where 5 comes up on at least one face are

(1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6).

Number of such cases = 11.

Number of those cases where 5 will not come up any time = $36 - 11 = 25$.

$$\therefore P(\text{that 5 will not come up either time}) = \frac{25}{36}.$$

- (ii) All those case where 5 comes up exactly one time are

(1, 5), (2, 5), (3, 5), (4, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6).

Number of such cases = 10.

$$\therefore P(\text{that 5 will come up exactly one time}) = \frac{10}{36} = \frac{5}{18}.$$

- (iii) There is only one case, namely (5, 5) when 5 comes up both the times.

$$\therefore P(\text{that 5 will come up both the times}) = \frac{1}{36}.$$

17. Number of all possible outcomes = 36.

Let E be the event of getting 2 numbers on the two dice whose product is a perfect square.

Then, the favourable outcomes are (1, 1), (1, 4), (4, 1), (2, 2), (3, 3), (4, 4) (5, 5), (6, 6).

Their number is 8. So, $P(E) = \frac{8}{36} = \frac{2}{9}$.

18. Total number of letters in the the given word = 11.

- (i) Number of vowels in the given word = 6.

$$\therefore P(\text{selecting a vowel}) = \frac{6}{11}.$$

- (ii) Number of consonants in the given word = 5.

$$P(\text{selecting a consonant}) = \frac{5}{11}.$$

- (iii) Number of S in the given word = 2.

$$\therefore P(\text{getting an S}) = \frac{2}{11}.$$

19. Total number of cards = 5.

(a) $P(\text{getting a queen}) = \frac{1}{5}.$

- (b) When the queen is set aside, remaining number of cards = 4.

(i) $P(\text{getting an ace now}) = \frac{1}{4}.$

(ii) $P(\text{getting a queen now}) = \frac{0}{4} = 0.$

20. There are 26 red cards containing 2 queens and 2 more queens are there.

$$P(\text{getting a red card or a queen}) = \frac{28}{52} = \frac{7}{13}.$$

$$\therefore P(\text{getting neither a red card nor a queen}) = 1 - \frac{7}{13} = \frac{6}{13}.$$

21. An ordinary year has 365 days consisting of 52 weeks 1 day.
This day can be any day of the week.

$$P(\text{of this day to be Monday}) = \frac{1}{7}.$$

22. There are 6 red face cards. These are removed.

Remaining number of cards = $52 - 6 = 46$.

- (i) Number of red cards now = $26 - 6 = 20$.

$$\therefore P(\text{getting a red card}) = \frac{20}{46} = \frac{10}{23}.$$

- (ii) Remaining number of face cards = $12 - 6 = 6$.

$$\therefore P(\text{getting a face card}) = \frac{6}{46} = \frac{3}{23}.$$

- (iii) There are 12 cards of clubs.

$$\therefore P(\text{getting a card of clubs}) = \frac{12}{46} = \frac{6}{23}.$$

23. 4 kings, 4 queens and 4 aces have been removed.

Remaining number of cards = $52 - 12 = 40$.

- (i) There are 2 black face cards in the remaining cards.
These are 2 black jacks.

$$\therefore P(\text{getting 2 black face cards}) = \frac{2}{40} = \frac{1}{20}.$$

- (ii) Remaining number of red cards = $26 - (2 + 2 + 2) = 20$.

$$\therefore P(\text{getting a red card}) = \frac{20}{40} = \frac{1}{2}.$$

24. When a coin is tossed 3 times, all possible outcomes are

HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

Number of all possible outcomes = 8.

(i) $P(\text{getting 3 heads}) = \frac{1}{8}$.

(ii) At least 2 tails means 2 or 3 tails.

All such cases are HTT, THT, TTH, TTT . Their number is 4.

$$\therefore P(\text{getting at least 2 tails}) = \frac{4}{8} = \frac{1}{2}.$$

25. A leap year has 366 days = 52 weeks and 2 days.

Now, 52 weeks contain 52 Sundays.

The remaining 2 days can be:

- | | |
|-----------------------------|-----------------------------|
| (i) Sunday and Monday | (ii) Monday and Tuesday |
| (iii) Tuesday and Wednesday | (iv) Wednesday and Thursday |
| (v) Thursday and Friday | (vi) Friday and Saturday |
| (vii) Saturday and Sunday | |

Out of these 7 cases, there are 2 cases favouring it to be Sunday.

$$\therefore P(\text{a leap year having 53 Sundays}) = \frac{2}{7}.$$

MULTIPLE-CHOICE QUESTIONS (MCQ)

Choose the correct answer in each of the following questions:

- If $P(E)$ denotes the probability of an event E then [CBSE 2013C]

(a) $P(E) < 0$ (b) $P(E) > 1$ (c) $0 \leq P(E) \leq 1$ (d) $-1 \leq P(E) \leq 1$
- If the probability of occurrence of an event is p then the probability of non-happening of this event is [CBSE 2013C]

(a) $(p - 1)$ (b) $(1 - p)$ (c) p (d) $\left(1 - \frac{1}{p}\right)$
- What is the probability of an impossible event?

(a) $\frac{1}{2}$ (b) 0 (c) 1 (d) More than 1
- What is the probability of a sure event?

(a) 0 (b) $\frac{1}{2}$ (c) 1 (d) Less than 1
- Which of the following cannot be the probability of an event? [CBSE 2011]

(a) 1.5 (b) $\frac{3}{5}$ (c) 25% (d) 0.3
- A number is selected at random from the numbers 1 to 30. What is the probability that the selected number is a prime number? [CBSE 2014]

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{11}{30}$

7. The probability that a number selected at random from the numbers 1, 2, 3, ..., 15 is a multiple of 4, is [CBSE 2014]
(a) $\frac{4}{15}$ (b) $\frac{2}{15}$ (c) $\frac{1}{5}$ (d) $\frac{1}{3}$
8. A box contains cards numbered 6 to 50. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square is [CBSE 2013]
(a) $\frac{1}{45}$ (b) $\frac{2}{15}$ (c) $\frac{4}{45}$ (d) $\frac{1}{9}$
9. A box contains 90 discs, numbered from 1 to 90. If one disc is drawn at random from the box, the probability that it bears prime number less than 23 is [CBSE 2013]
(a) $\frac{7}{90}$ (b) $\frac{1}{9}$ (c) $\frac{4}{45}$ (d) $\frac{8}{89}$
10. Cards bearing numbers 2, 3, 4, ..., 11 are kept in a bag. A card is drawn at random from the bag. The probability of getting a card with a prime number is [CBSE 2012]
(a) $\frac{1}{2}$ (b) $\frac{2}{5}$ (c) $\frac{3}{10}$ (d) $\frac{5}{9}$
11. One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number, which is a multiple of 7, is [CBSE 2013C]
(a) $\frac{1}{7}$ (b) $\frac{1}{8}$ (c) $\frac{1}{5}$ (d) $\frac{7}{40}$
12. Which of the following cannot be the probability of an event?
(a) $\frac{1}{3}$ (b) 0.3 (c) 33% (d) $\frac{7}{6}$
13. If the probability of winning a game is 0.4 then the probability of losing it, is
(a) 0.96 (b) $\frac{1}{0.4}$ (c) 0.6 (d) none of these
14. If an event cannot occur then its probability is
(a) 1 (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 0
15. There are 20 tickets numbered as 1, 2, 3, ..., 20 respectively. One ticket is drawn at random. What is the probability that the number on the ticket drawn is a multiple of 5?
(a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) $\frac{2}{5}$ (d) $\frac{3}{10}$

16. There are 25 tickets numbered as 1, 2, 3, 4, ..., 25 respectively. One ticket is drawn at random. What is the probability that the number on the ticket is a multiple of 3 or 5?

(a) $\frac{2}{5}$ (b) $\frac{11}{25}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

17. Cards, each marked with one of the numbers 6, 7, 8, ..., 15, are placed in a box and mixed thoroughly. One card is drawn at random from the box. What is the probability of getting a card with number less than 10?

[CBSE 2009C]

(a) $\frac{3}{5}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{5}$

18. A die is thrown once. The probability of getting an even number is

[CBSE 2013]

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{5}{6}$

19. The probability of throwing a number greater than 2 with a fair die is

[CBSE 2011]

(a) $\frac{2}{5}$ (b) $\frac{5}{6}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

20. A die is thrown once. The probability of getting an odd number greater than 3 is

[CBSE 2013C]

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2}$ (d) 0

21. A die is thrown once. The probability of getting a prime number is

[CBSE 2013]

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{6}$

22. Two dice are thrown together. The probability of getting the same number on both dice is

[CBSE 2012]

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{1}{12}$

23. The probability of getting 2 heads, when two coins are tossed, is

[CBSE 2012]

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

24. Two dice are thrown together. The probability of getting a doublet is

[CBSE 2013]

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{4}$ (d) $\frac{2}{3}$

25. Two coins are tossed simultaneously. What is the probability of getting at most one head?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$
26. Three coins are tossed simultaneously. What is the probability of getting exactly two heads?
(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{8}$ (d) $\frac{3}{4}$
27. In a lottery, there are 8 prizes and 16 blanks. What is the probability of getting a prize?
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) None of these
28. In a lottery, there are 6 prizes and 24 blanks. What is the probability of not getting a prize?
(a) $\frac{3}{4}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) None of these
29. A box contains 3 blue, 2 white and 4 red marbles. If a marble is drawn at random from the box, what is the probability that it will not be a white marble? [CBSE 2009C]
(a) $\frac{1}{3}$ (b) $\frac{4}{9}$ (c) $\frac{7}{9}$ (d) $\frac{2}{9}$
30. A bag contains 4 red and 6 black balls. A ball is taken out of the bag at random. What is the probability of getting a black ball? [CBSE 2008]
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{1}{10}$ (d) None of these
31. A bag contains 8 red, 2 black and 5 white balls. One ball is drawn at random. What is the probability that the ball drawn is not black?
(a) $\frac{8}{15}$ (b) $\frac{2}{15}$ (c) $\frac{13}{15}$ (d) $\frac{1}{3}$
32. A bag contains 3 white, 4 red and 5 black balls. One ball is drawn at random. What is the probability that the ball drawn is neither black nor white?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{3}{4}$
33. A card is drawn at random from a well-shuffled deck of 52 cards. What is the probability of getting a black king? [CBSE 2009C]
(a) $\frac{1}{13}$ (b) $\frac{1}{26}$ (c) $\frac{2}{39}$ (d) None of these

34. From a well-shuffled deck of 52 cards, one card is drawn at random. What is the probability of getting a queen?
(a) $\frac{1}{13}$ (b) $\frac{1}{26}$ (c) $\frac{4}{39}$ (d) None of these
35. One card is drawn at random from a well-shuffled deck of 52 cards. What is the probability of getting a face card?
(a) $\frac{1}{26}$ (b) $\frac{3}{26}$ (c) $\frac{3}{13}$ (d) $\frac{4}{13}$
36. One card is drawn at random from a well-shuffled deck of 52 cards. What is the probability of getting a black face card?
(a) $\frac{1}{26}$ (b) $\frac{3}{26}$ (c) $\frac{3}{13}$ (d) $\frac{3}{14}$
37. One card is drawn at random from a well-shuffled deck of 52 cards. What is the probability of getting a 6?
(a) $\frac{3}{26}$ (b) $\frac{1}{52}$ (c) $\frac{1}{13}$ (d) None of these

ANSWERS (MCQ)

1. (c) 2. (b) 3. (b) 4. (c) 5. (a) 6. (c) 7. (c) 8. (d) 9. (c) 10. (a)
11. (b) 12. (d) 13. (c) 14. (d) 15. (b) 16. (c) 17. (d) 18. (a) 19. (d) 20. (b)
21. (c) 22. (c) 23. (d) 24. (b) 25. (d) 26. (c) 27. (b) 28. (c) 29. (c) 30. (b)
31. (c) 32. (c) 33. (b) 34. (a) 35. (c) 36. (b) 37. (c)

HINTS TO SOME SELECTED QUESTIONS

2. $P(\text{occurrence of an event}) = p$

$$\Rightarrow P(\text{non-occurrence of this event}) = (1 - p).$$

5. The probability of an event cannot be greater than 1.

6. Prime numbers from 1 to 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

Their number is 10.

$$\therefore P(\text{getting a prime number}) = \frac{10}{30} = \frac{1}{3}.$$

7. Total number of given numbers = 15.

From given numbers, the multiples of 4 are 4, 8, 12.

They are 3 in number.

$$\therefore P(\text{getting a multiple of 4}) = \frac{3}{15} = \frac{1}{5}.$$

8. Given numbers are 6, 7, 8, 9, ..., 50.

Number of these numbers = $50 - 5 = 45$.

Perfect square numbers from these are $3^2, 4^2, 5^2, 6^2, 7^2$.

Their number is 5.

$$\therefore P(\text{getting a perfect square number}) = \frac{5}{45} = \frac{1}{9}.$$

9. Total number of discs = 90.

Prime number less than 23 are 2, 3, 5, 7, 11, 13, 17, 19.

Their number is 8.

$$P(\text{getting a prime number less than 23}) = \frac{8}{90} = \frac{4}{45}.$$

10. Total number of cards = 10.

Prime numbers from given numbers are 2, 3, 5, 7, 11.

Their number is 5.

$$\therefore P(\text{getting a prime number}) = \frac{5}{10} = \frac{1}{2}.$$

11. Total number of tickets = 40.

Tickets bearing the numbers as multiple of 7 bear the numbers 7, 14, 21, 28, 35.

Their number is 5.

$$\therefore P(\text{getting a multiple of 7}) = \frac{5}{40} = \frac{1}{8}.$$

13. $P(\text{losing the game}) = 1 - P(\text{winning the game}) = (1 - 0.4) = 0.6$.

14. If an event cannot occur then its probability is 0.

15. Total number of tickets = 20.

Multiples of 5 are 5, 10, 15, 20.

Their number is 4.

$$\therefore P(\text{getting a multiple of 5}) = \frac{4}{20} = \frac{1}{5}.$$

16. Total number of tickets = 25.

Multiples of 3 or 5 are 3, 6, 9, 12, 15, 18, 21, 24, 5, 10, 20, 25.

Number of these numbers = 12.

$$\therefore P(\text{getting a multiple of 3 or 5}) = \frac{12}{25}.$$

17. Total number of cards = $15 - 5 = 10$.

Number of cards with number less than 10 = 4.

$$\therefore P(\text{getting a card with number less than 10}) = \frac{4}{10} = \frac{2}{5}.$$

18. Number of all possible outcomes = 6.

Even numbers are 2, 4, 6. Their number is 3.

$$\therefore P(\text{getting an even number}) = \frac{3}{6} = \frac{1}{2}.$$

19. Number of all possible outcomes = 6.

Numbers greater than 2 are 3, 4, 5, 6. Their number is 4.

$$\therefore P(\text{getting a number greater than 2}) = \frac{4}{6} = \frac{2}{3}.$$

20. Number of all possible outcomes = 6.

Odd number greater than 3 is 5 only. Its number is 1.

$$\therefore P(\text{getting an odd number greater than 3}) = \frac{1}{6}.$$

21. Number of all possible outcomes = 6.

Prime numbers are 2, 3, 5. Their number is 3.

$$\therefore P(\text{getting a prime number}) = \frac{3}{6} = \frac{1}{2}.$$

22. Total number of all possible outcomes = 36.

Getting same number on both dice means getting

(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6).

Their number is 6.

$$\therefore P(\text{getting the same number on both dice}) = \frac{6}{36} = \frac{1}{6}.$$

23. All possible outcomes are HH, HT, TH, TT . Their number is 4.

Getting 2 heads, means getting HH . Its number is 1.

$$\therefore P(\text{getting 2 heads}) = \frac{1}{4}.$$

24. Number of all possible outcomes = 36.

The doublets are (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6).

Their number is 6.

$$\therefore P(\text{getting a doublet}) = \frac{6}{36} = \frac{1}{6}.$$

25. When two coins are tossed simultaneously, all possible outcomes are HH, HT, TH, TT .

Their number is 4.

All favourable outcomes are TT, TH, HT . Their number is 3.

$$\therefore P(\text{getting at most 1 head}) = \frac{3}{4}.$$

26. When 3 coins are tossed, all possible outcomes are $HHH, HHT, HTH, THH, HTT, THT, TTH, TTT$.

Their number is 8.

Favourable outcomes are HHT, HTH, THH . Their number is 3.

$$\therefore P(\text{getting exactly 2 heads}) = \frac{3}{8}.$$

27. Total number of lottery tickets = $8 + 16 = 24$.

Number of prizes = 8.

$$\therefore P(\text{getting a prize}) = \frac{8}{24} = \frac{1}{3}.$$

28. Total number of tickets = $6 + 24 = 30$.

Number of blanks = 24.

$$\therefore P(\text{not getting a prize}) = \frac{24}{30} = \frac{4}{5}.$$

29. Total number of marbles = $3 + 2 + 4 = 9$.

Number of non-white marbles = $3 + 4 = 7$.

$$\therefore P(\text{getting a non-white marble}) = \frac{7}{9}.$$

30. Total number of balls in the bag = $4 + 6 = 10$.

Number of black balls = 6.

$$\therefore P(\text{getting a black ball}) = \frac{6}{10} = \frac{3}{5}.$$

31. Total number of balls in the bag = $8 + 2 + 5 = 15$.

Number of non-black balls = $8 + 5 = 13$.

$$P(\text{getting a non-black ball}) = \frac{13}{15}.$$

32. Total number of balls in the bag = $3 + 4 + 5 = 12$.

Number of non-black and non-white balls = 4.

$$\therefore P(\text{getting a ball which is neither black nor white}) = \frac{4}{12} = \frac{1}{3}.$$

33. Number of all possible outcomes = 52.

Number of black kings = 2.

$$\therefore P(\text{getting a black king}) = \frac{2}{52} = \frac{1}{26}.$$

34. Number of all possible outcomes = 52.

Number of queens = 4.

$$\therefore P(\text{getting a queen}) = \frac{4}{52} = \frac{1}{13}.$$

35. Total number of cards = 52.

Number of face cards = 12

(4 kings + 4 queens + 4 jacks).

$$\therefore P(\text{getting a face card}) = \frac{12}{52} = \frac{3}{13}.$$

36. Total number of cards = 52.

Number of black face cards = 6

(2 kings + 2 queens + 2 jacks).

$$\therefore P(\text{getting a face card}) = \frac{6}{52} = \frac{3}{26}.$$

37. Total number of cards = 52.

Number of 6s = 4.

$$\therefore P(\text{getting a 6}) = \frac{4}{52} = \frac{1}{13}.$$



SAMPLE PAPER I

Time: 3 hours

80 Marks

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. The paper consists of 30 questions divided into four sections: A, B, C and D.
3. (i) Section A consists of 6 Very-Short-Answer (VSA) questions of 1 mark each.
(ii) Section B consists of 6 Short-Answer (SA) questions of 2 marks each.
(iii) Section C consists of 10 Short-Answer (SA) questions of 3 marks each.
(iv) Section D consists of 8 Long-Answer (LA) questions of 4 marks each.
4. Use of calculator is not permitted.

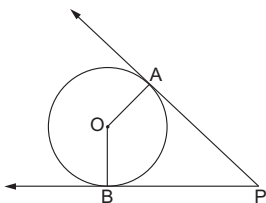
SECTION A

1. What is the distance between the points $A(\sin \theta - \cos \theta, 0)$ and $B(0, \sin \theta + \cos \theta)$?
2. The angle of depression of a car standing on the ground from the top of a 66 m tower, is 30° . Find the distance of the car from the base of the tower.
3. A rational number in its decimal expansion is 554.6023. What can you say about the prime factors of q , when this number is expressed in the form $\frac{p}{q}$?
4. If $\sin \theta = \frac{1}{3}$ then find the value of $2\cot^2 \theta + 2$.
5. In a family of 3 children, find the probability of having at least one boy.

6. If α and β are the zeros of the quadratic polynomial $f(x) = 5x^2 - 7x + 1$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

SECTION B

7. The length of the minute hand of a clock is 5 cm. Find the area swept by the minute hand during the time from 4:05 a.m. to 4:40 a.m.
8. Prove that three times the square of any side of an equilateral triangle is equal to four times the square of the altitude.
9. In the given figure, O is the centre of the circle. PA and PB are tangent segments. Show that the quadrilateral $AOBP$ is cyclic.



10. Find the lengths of the medians of a $\triangle ABC$ whose vertices are $A(7, -3)$, $B(5, 3)$ and $C(3, -1)$.
11. If p, q are prime positive integers, prove that $\sqrt{p} + \sqrt{q}$ is an irrational number.
12. Determine the value of k so that the following system of linear equations has no solution:

$$(3k + 1)x + 3y - 2 = 0,$$

$$(k^2 + 1)x + (k - 2)y - 5 = 0.$$

SECTION C

13. Show that the square of any positive integer cannot be of the form $5q + 2$ or $5q + 3$ for any integer q .
14. Prove that $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = 1 + \tan A + \cot A = 1 + \sec A \cdot \operatorname{cosec} A$.
15. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the further time taken by the car to reach the foot of the tower.

- | Classes | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | 100-120 | 120-140 |
|-----------|------|-------|-------|-------|--------|---------|---------|
| Frequency | 6 | 8 | 10 | 12 | 6 | 5 | 3 |

- ## SECTION D

23. An open metal bucket is in the shape of a frustum of a cone, mounted on a hollow cylindrical base made of the same metallic sheet. The diameters of the two circular ends of the bucket are 45 cm and 25 cm, the total vertical height of the bucket is 40 cm and that of the cylindrical base is 6 cm. Find the area of the metallic sheet used to make the bucket. Also, find the volume of water the bucket can hold, in litres.
24. It can take 12 hours to fill a swimming pool using two pipes. If the pipe of larger diameter is used for 4 hours and the pipe of smaller diameter for 9 hours, only half of the pool can be filled. How long would it take for each pipe to fill the pool separately?

20. (i) $\frac{1}{6}$ (ii) $\frac{5}{6}$ (iii) $\frac{5}{36}$

21. $\frac{20}{3}$ cm

22. $x = a + b, y = -\frac{2ab}{a+b}$

23. $4860.9 \text{ cm}^2, 33.62 \text{ litres}$

24. 20 hours, 30 hours

27. $(24\sqrt{21} - 77) \text{ m}^2$

28. 15 cm, 13 cm

30. Length = 34 m, Breadth = 24 m



SAMPLE PAPER II

Time: 3 hours

80 Marks

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. The paper consists of 30 questions divided into four sections: A, B, C and D.
3. (i) Section A consists of 6 Very-Short-Answer (VSA) questions of 1 mark each.
(ii) Section B consists of 6 Short-Answer (SA) questions of 2 marks each.
(iii) Section C consists of 10 Short-Answer (SA) questions of 3 marks each.
(iv) Section D consists of 8 Long-Answer (LA) questions of 4 marks each.
4. Use of calculator is not permitted.

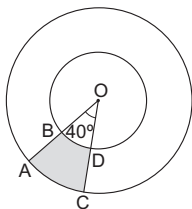
SECTION A

1. Evaluate: $\sin 35^\circ \cdot \sin 55^\circ - \cos 35^\circ \cdot \cos 55^\circ$.
2. The decimal expansion of the rational number $\frac{83}{2^3 \times 5^4}$ will terminate after how many places of decimals?
3. If the point $P(x, y)$ is equidistant from $A(5, 1)$ and $B(-1, 5)$ then find the relation between x and y .
4. If α and β are the zeros of the polynomial $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of k .
5. The tops of two towers of heights x and y , standing on level ground, subtend angles of 30° and 60° respectively at the centre of the line joining their feet. Find $x : y$.

6. A die is rolled twice. Find the probability that 4 will come up exactly one time.

SECTION B

7. Find the value of a and b for which the following system of equations has infinitely many solutions:
- $$x + 2y = 1,$$
- $$(a - b)x + (a + b)y = a + b - 2.$$
8. Use Euclid's division algorithm to find the HCF of 441, 567 and 693.
9. Find k so that the point $(-4, 6)$ lies on the line segment joining $A(k, 10)$ and $B(3, -8)$. Also, find the ratio in which P divides AB .
10. Prove that the segment joining the points of contact of two parallel tangents passes through the centre.
11. The perimeter of a right triangle is 60 cm. Its hypotenuse is 25 cm. Find the area of the triangle.
12. Find the area of the shaded region in the figure given below where radii of the two concentric circles with centre O are 7 cm and 14 cm respectively and $\angle AOC = 40^\circ$.



SECTION C

13. Three unbiased coins are tossed simultaneously. Find the probability of getting
- exactly two heads;
 - at least two heads;
 - at most 2 heads.
14. Prove that the angle between two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segments joining the points of contact at the centre.
15. Solve the following system of equations in x and y :

$$ax + by = c,$$

$$bx + ay = 1 + c.$$

16. Use Euclid's division lemma to show that the cube of any positive integer is either of the form $9m$, $9m + 1$ or $9m + 8$ for some integer m .
17. If $1 + \sin^2 \theta = 3 \sin \theta \cdot \cos \theta$, prove that $\tan \theta = 1$ or $\frac{1}{2}$.
18. From the top of a building AB , 60 m high, the angles of depression of the top and bottom of a vertical lamp-post CD are observed to be 30° and 60° respectively. Find
- the horizontal distance between AB and CD ,
 - the height of the lamp-post.
19. Find the area of a parallelogram $ABCD$ if three of its vertices are $A(2, 4)$, $B(2 + \sqrt{3}, 5)$ and $C(2, 6)$.
20. A train travelling at a uniform speed for 360 km would have taken 48 minutes less to travel the same distance if its speed were 5 km/hr more. Find the original speed of the train.
21. If two zeros of the polynomial $f(x) = x^4 - 6x^3 - 26x^2 + 138x - 35$ are $2 \pm \sqrt{3}$, find other zeros.
22. The following table shows the marks obtained by 100 students of a class in a school during a particular academic session. Find the modal marks.

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60	Less than 70	Less than 80
Number of students	7	21	34	46	66	77	92	100

SECTION D

23. There is a square field whose side is 44 m. A square flower bed is prepared in its centre leaving a gravel path all around the flower bed. The total cost of laying the flower bed and gravelling the path at ₹ 2.75 and ₹ 1.50 per square metre respectively, is ₹ 4904. Find the width of the gravel path.
24. Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares of any two corresponding sides.
25. From an external point P , a tangent PT and a line segment PAB is drawn to a circle with centre O . ON is perpendicular on the chord AB . Prove that
- $PA \cdot PB = PN^2 - AN^2$
 - $PN^2 - AN^2 = OP^2 - OT^2$
 - $PA \cdot PB = PT^2$

26. $ABCD$ is a trapezium with $AB \parallel DC$, $AB = 18$ cm, $DC = 32$ cm and the distance between AB and DC is 14 cm. If arcs of equal radii 7 cm with centres A, B, C and D are drawn find the area of the remaining region, i.e., region of the trapezium excluding the circular arcs.
27. The table given below shows the frequency distribution of the scores obtained by 200 candidates in an entrance examination:

Scores	200–250	250–300	300–350	350–400	400–450	450–500	500–550	550–600
No. of candidates	30	15	45	20	25	40	10	15

Draw cumulative frequency curve using 'less than series'.

28. A window of a house is h metres above the ground. From the window, the angles of elevation and depression of the top and bottom of another house situated on the opposite side of the lane are found to be α and β respectively. Prove that the height of the house is $h(1 + \tan \alpha \cdot \tan \beta)$ metres.
29. 8 men and 12 boys can finish a piece of work in 10 days while 6 men and 8 boys can finish it in 14 days. Find the time taken by one man alone and that by one boy alone to finish the work.
30. From a solid cylinder whose height is 15 cm and diameter 16 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid. [Use $\pi = 3.14$.]

ANSWERS

1. 0 2. 4 3. $2x = 3y$ 4. 6 5. 1 : 3
6. $\frac{5}{18}$ 7. $a = 3, b = 1$ 8. 63 9. $k = -6; 2 : 7$
11. 150 cm^2 12. $\frac{154}{3} \text{ cm}^2$ 13. (i) $\frac{3}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{7}{8}$
15. $x = \frac{c}{a+b} - \frac{b}{a^2-b^2}, y = \frac{c}{a+b} + \frac{a}{a^2-b^2}$ 18. (i) 34.64 m (ii) 40 m
19. $2\sqrt{3}$ sq units 20. 45 km/hr 21. 7, -5 22. 44.71
23. 2 metres 26. 196 cm^2
29. Time taken by one man alone = 140 days and
time taken by one boy alone = 280 days
30. 443.14 cm^2

